BiG Equity 360: User guide

BiG Equity 360 is a research terminal that runs on your own machine. It gives a 360º view of listed companies (fundamentals, valuation, scores, technicals, ownership, events) and of the macro and cross-asset context around them. This guide is the reference for every page and component. For each one it covers what it is for, how to use its controls, the method behind the numbers, where the data comes from and how often it refreshes, how far the model has been validated, a worked example, and the limits to keep in mind.

The guide is written for analysts and portfolio managers who are new to the app. It is equally useful as a methodology reference. The same text is available inside the app: open **Guide** in the menu, type `HELP` in the command bar, or click the **?** icon next to any page title to jump to that page's section.

> **Note:** The screenshots were taken on 23 October 2026, with data as of 19 October 2026 (snapshot built 22 October). Your numbers will differ. The orange numbered badges in the screenshots are added by the guide. They are not part of the app. Pages that need saved user content (theses, backtests, studies, watchlists) were captured from a separate test copy with example content.

How to read this guide

- Every page section starts with a locator line: the **route** (the address after `http://127.0.0.1:8010`), where it sits in the **menu**, the command-bar **mnemonics** that open it, and the **refresh** cadence.
- The sections then follow the same order: *Purpose*, *Screenshot* (with a table that explains each numbered callout), *Controls*, *Methodology*, *Data sources and refresh*, *Validation status*, *Correct usage and worked example*, *Limitations and pitfalls*, and *Related mnemonics and BQL fields*.
- **Labels matter.** "Model-based" means an estimate from a model, not an official statistic or a consensus figure. "Experimental" means the output failed, or has not passed, its validation gate. It is hidden unless you switch experimental outputs on. "Approximation" means a stated simplification. Probabilities are "calibrated over history", never "the chance today". Chapter 17 explains how to read them.
- Validation results are quoted from the project's validation documents. The summary table of every model's gate status collects them in one place.

Contents

Chapter	What it covers
0. Getting started	Install and launch (BiG360.exe and the tray), the interface tour, the command bar, keyboard shortcuts, themes, freshness indicators and labels, your first 15 minutes
1. Data and coverage	Sources by domain, universes, point-in-time rules, the serving snapshot and data-quality gate, refresh schedule (weekly by default, daily option), gap filler, coverage gaps, survivorship, licensing
2A. Macro	Macro cockpit (heatmap, cycle clock, surprise index, recession risk), release calendar with model-based expectations, nowcasts, themes, countries, series explorer, central banks and implied paths
2B. Markets and cross-asset	Markets overview, instruments, macro to markets (exposures, regime performance, scenario builder, episode replay), cross-asset dashboard, CFTC positioning
3. Sector Explorer	Sector and subsector drill-down with sector-specific KPIs, distributions, leaders and laggards
4. Screener and Ranking	Filters, factor weights and live re-ranking, the composite score and the ten factors

Chapter	What it covers
5A. Company 360: fundamentals	Header, Overview, Financials, Forecast, Peers (incl. the conformal peer-group panel), Moat, Insiders, Ownership, News & Sentiment
5B. Company 360: Technicals	Calibrated range fan, context chips, confluence zones, touch probabilities, the trade setup with first-passage and meta probabilities, technical score, evidence-based factors, breakouts, the experimental toggle
5C. Company 360: macro, earnings, events, network and model	Macro sensitivities, earnings history and model-based surprises, event risk, supply-chain network and revenue geography, the ML model
6. Earnings and events calendar	Earnings and corporate-events calendar and the earnings-season dashboard
7. Smart Money and Insider Radar	13F flows and crowding, short interest, EU / UK short register, activism, insider activity
8. Global events and news	GDELT world events, escalation and geopolitical risk, contagion and spillovers, event studies, and the News Terminal
9. Model Studio	The econometrics and ML workbench: builder, methods catalogue, presets, studies and runs, results, export
10. Model Lab	The nightly ML monitor and model registry
11. Backtest Lab	Point-in-time strategy backtests: builder, signals, costs, results, robustness statistics (DSR, PBO), experimental signals
12. Thesis Tracker	Investment theses with KPIs, automatic status, performance, reviews and journal
13. Watchlists and alerts	Watchlists, alert rules (every rule type) and fired alerts
14. BQL console and mnemonics	The query language, field dictionary, command bar and the full mnemonic reference
15. System, Data Health and QA	Jobs and schedule, data freshness, the Data Health supervisor, conformal coverage audit, QA guard
16. Methodology appendix	Point-in-time data, purged CV, conformal prediction, first-passage probabilities, meta-labelling, calibration, multiple testing, and the gate status of every model
17. Good practice and pitfalls	What not to infer, how to read probabilities, survivorship, licensing, a checklist for investment decisions
18. Troubleshooting and FAQ	When something does not work, logs to send to IT, frequent questions
Glossary	Terms and acronyms used across the app

Status of this edition

- Screenshots were captured between 23 and 24 October 2026 from the running app (data as of 19 October 2026). The News Terminal, the BQL console and the mnemonic command bar went live during that window and are captured from the live build.
- Pages that need saved user content (theses, backtest results, Model Studio runs, a populated watchlist, saved BQL queries) were captured from a separate test copy with example content. The "WORKER OFF" chip in those images comes from that copy.
- A few control states could not be captured (the tray menu, which is outside the browser; a hover tooltip on the range fan; some builder sections). They are marked "Screenshot pending" in the text.
- Where the guide found a label in the app that does not match what the code computes, it follows the code and says so in the relevant *Limitations and pitfalls* section.

Printing

The whole guide can be printed or saved as a PDF from the Help page (**Print / PDF**), which opens every chapter on one page in the light theme. A PDF copy may also be provided next to this file as

```
BiG_Equity_360_Guide.pdf .
```

0. Getting started

BiG Equity 360 is a research terminal for listed equities and macro data that runs on your own PC. It collects public and official data (stock prices, company filings, insider and ownership filings, macro statistics, central-bank publications, news and global events) into a local data store. It then computes scores, valuations, technical levels and statistical models, and shows everything in a browser-based dashboard at `http://127.0.0.1:8010/` .

This chapter covers installing and starting the app, the tray icon, a tour of the interface, the command bar, keyboard shortcuts, themes, the freshness indicators, and how to read the labels the app puts on its numbers. It ends with a 15-minute walkthrough.

> **Note:** Everything runs locally. Nothing you type is sent to an external service, and the data refreshes on a schedule (weekly by default, see Data and coverage). The numbers you see are as of the date in the top bar, not live market prices.

What the app is

Route: `/` (opens the Macro cockpit) · **Menu:** all · **Refresh:** weekly profile by default (Saturday 08:00 Lisbon), daily optional

Purpose

BiG Equity 360 gives a 360-degree view of a listed company (fundamentals, valuation, quality, moat, insiders, ownership, news, technicals, macro sensitivity, event risk and a machine-learning view) and of the macro and market backdrop it trades in. It is built for analysts and portfolio managers who want one place to screen, drill down and check a thesis, with every number traceable to an official or labelled source.

The app is organised in four layers:

Layer	What it covers	Main pages
Top-down	Economies, central banks, markets, cross-asset, positioning	Macro (cockpit, calendar, nowcast, themes, countries, series, central banks, markets, instruments, cross-asset, CFTC positioning)
Cross-section	Sectors, ranking, events, smart money, insiders, news	Sector Explorer, Screener / Ranking, Earnings & Events, Smart Money, Insider Radar, Global Events, News Terminal
Single name	One company in depth	Company 360 (14 tabs)
Research tools	Your own studies, backtests, queries, decisions and lists	Model Studio, Backtest Lab, BQL console, Thesis Tracker, Model Lab, Watchlist & Alerts

The **System** pages (System overview, Data health, Conformal validation) show how fresh the data is and whether the background jobs are healthy. The **Guide** page (`/help`) is this user guide, built into the app.

Limitations and pitfalls

- The app is a **research tool**, not an order-management or real-time trading system. Prices are end-of-day.
- Some sources are not licensed for use beyond one analyst's research (see Licensing caveats).
- There are **no consensus estimates and no options data**. Earnings and macro "expectations" are the app's own model-based estimates, clearly labelled.

Install and launch

Route: `http://127.0.0.1:8010/` · **Menu:** none (Windows) · **Refresh:** not applicable

Purpose

The everyday way to run the app is the Windows launcher **`BiG360.exe`**. It starts everything in the background (no terminal windows, no Node.js), puts a BiG icon in the Windows notification area and opens the dashboard in your default browser.

Behind the icon there is a single process tree:

```
BiG360.exe → tray app → supervisor → API (serves the dashboard), worker (schedule), gap filler, live news service
```

The **supervisor** restarts any part that crashes, with an increasing back-off (5 s, 15 s, 60 s, then 5 min).

Install options

Option	How	Notes
Project folder (this PC)	Double-click `BiG360.exe` in the project folder (`C:\Users\Asus\Documents\STOCK SCANNER`)	Uses the project's Python environment, its `data\` folder and its settings. Nothing is installed.
Installer	Run `BiG360Setup-<version>.exe`	Per user, no administrator rights. Installs to `%LOCALAPPDATA%\Programs\BiG360` with Start-menu and desktop shortcuts. Optional tasks: *start at login* and *copy API keys from .env*. The installer is built by IT with `packaging\build.ps1 -Installer` (needs Inno Setup 6).
Portable	Unzip `BiG360-<version>-portable.zip` and double-click `BiG360.exe`	Same files as the installer. Point it at the data folder by setting `BIG360_DATA_DIR=<folder>` in the `.env` file of that folder.

> **Note:** The data folder (tens of GB) is never copied. The installer asks for the existing data folder and writes its location to `<install folder>\.env` as `BIG360_DATA_DIR`. The trained models (`models_store`) are linked, not copied.

First run

1. **API keys (one-time, usually done by the app owner or IT).** The Windows app reads data-provider keys from **Windows Credential Manager**, where they are stored per user and protected by Windows (DPAPI). Each key appears under *Control Panel › Credential Manager › Windows Credentials* as `BiG360/<key name>`. Copy them once from the `.env` file with the credentials tool (commands below). The tool prints key **names** only, never values.
2. **Start `BiG360.exe`.** A Windows notification says *Starting... the dashboard opens at http://127.0.0.1:8010/*. The browser opens when the API answers, usually after 10 to 60 seconds.
3. **Optional:** tray menu › **Start at login** (a per-user entry; no administrator rights needed). At login the app starts in the background without opening the browser.

The credentials commands for step 1, run from the `backend` folder:

```
..\.venv\Scripts\python.exe -m app.desktop.credentials migrate   # copy keys from .env into Credential Manager
..\.venv\Scripts\python.exe -m app.desktop.credentials verify    # "match" / "differs" per key
..\.venv\Scripts\python.exe -m app.desktop.credentials status    # which keys are stored (yes / no only)
..\.venv\Scripts\python.exe -m app.desktop.credentials set <key name>   # set or change one key (hidden input)
```

In an installed or portable build the Python path is `..\python\python.exe` instead of `..\.venv\Scripts\python.exe`. The `.env` file is not modified; keep it as the development fallback or blank the keys once `verify` reports `match`.

The key names the app understands are:

Key name	Unlocks	Needed?
`fred_apikey`	US macro (FRED / ALFRED)	Yes
`polygon_apikey`	US end-of-day prices, corporate actions, US news (Massive, formerly Polygon.io)	Yes
`edinet_apikey`	Japanese filings (EDINET)	Recommended
`opendart_apikey`	Korean financial statements and insider holdings (OpenDART)	Recommended: largest coverage gain
`krx_openapi_key`	Korean prices for the whole market	Optional
`companies_house_apikey`	UK registry data	Optional

Key name	Unlocks	Needed?
`openfigi_apikey`	Faster identifier mapping (OpenFIGI)	Optional
`bok_ecos_apikey` , `banxico_token` , `estat_appid` , `kosis_apikey` , `inegi_token` , `bcch_user` , `bcch_pass` , `eia_apikey`	Extra national macro series (Korea, Mexico, Japan, Chile, US energy)	Optional
`comtrade_apikey`	Faster UN Comtrade trade-linkage refreshes	Optional

Keys are optional except FRED and Polygon: a job whose key is missing skips that source and reports "missing key". The order of precedence is: an explicit environment variable, then Credential Manager, then `.env` . The SEC also requires a contact string in the environment variable `SEC_USER_AGENT` (not a secret).

Dev mode (for IT and the app owner)

Developers can run the parts in separate PowerShell windows instead of the tray app:

```
cd "C:\Users\Asus\Documents\STOCK SCANNER"
.\run_all.ps1 -Live              # API + worker + Vite dashboard, each in its own window
.\run_all.ps1 -Live -NoWorker    # dashboard only, no scheduled jobs
```

In dev mode the dashboard is served by Vite on `http://localhost:5173` and calls the API at the URL in `frontend\.env` (`VITE_API_URL`). The app build (`npm run build:app` → `frontend\dist_app`) instead calls `/api/v1` on its own origin. Setting `VITE_USE_MOCK=true` in `frontend\.env` shows the whole interface with synthetic data and a **MOCK DATA** badge; this is for development only.

> **Caution:** Do not run a dev API and the Windows app on the same port. If a dev API already listens on 8010, the supervisor treats the API as "external" and the dashboard is not served at `/` . See Dev mode and the app on the same port.

Quit

Right-click the tray icon › **Quit BiG360**. Running jobs finish or cancel (this can take up to about 5 minutes), then the icon disappears. Closing the browser tab does **not** stop the app; it keeps running in the background so that scheduled refreshes and the gap filler can work.

Tray icon and menu

Route: Windows notification area (BiG icon) · **Menu:** right-click the icon · **Refresh:** status polled every 60 s

Purpose

The tray icon is the control panel of the background app: it shows whether the app is running, what the data date is, when the next refresh is due, and lets you run or pause background work without opening a terminal. Hover over the icon to see a tooltip such as *BiG Equity 360 - data 19 Oct 2026 - weekly*. **Double-click** opens the dashboard.

Screenshot

> Screenshot pending: the tray menu opened from the Windows notification area, with the Status submenu expanded (Windows desktop, not a browser page).

Controls

Menu item	What it does
Open dashboard	Opens `http://127.0.0.1:8010/` in the default browser. This is the default item (bold); double-clicking the icon does the same.

Menu item	What it does
Status ▸	A read-only summary: app version; supervisor running or not; *Data as of* date; schedule profile and whether the scheduler is on; *Next refresh*; gap filler state; live news state; number of error and warning alerts; *Update available* if any. **Refresh status** re-reads it immediately.
Run refresh now	Starts the active profile's refresh in the worker as if it had been scheduled (weekly profile: nightly → weekly → monthly if due). A notification confirms it.
Pause background jobs / **Resume background jobs**	Pauses or resumes the **gap filler** (the low-priority background repair of missing data). Scheduled refreshes still run. The item is ticked while paused.
Live news on/off	Starts or stops the live news service (GDELT 15-minute files, official RSS feeds, Polygon news) that feeds the News Terminal and the live strip. Data already received is kept.
Schedule ▸	Choose **Weekly** or **Daily**. *Intraday (placeholder)* is shown but disabled. The worker applies the change within seconds; no restart is needed.
News Terminal	Opens `/news` in the browser.
Data health	Opens `/system/data-health` .
Open logs folder	Opens `data\logs` in Explorer.
Start at login	Turns autostart on or off (ticked when on).
Update available: x.y.z	Only shown when a newer build is available in the update folder configured by IT. Opens that folder.
Quit BiG360	Stops everything gracefully (running jobs finish or cancel, up to about 5 minutes), then removes the icon.

Windows notifications. The tray shows a Windows notification for new operational alerts (refresh failed or overdue, data stale beyond the service level, scheduler off, worker down) and when a scheduled refresh finishes (success, partial or failed). The hourly news job never raises a notification. To turn notifications off, IT sets `notifications: false` in `config\app.yaml` .

Limitations and pitfalls

- The tray is for an interactive Windows user. On a server, IT runs the same supervisor as a Windows service instead (see `docs/it/DEPLOYMENT_SERVER.md`).
- **Pause background jobs** does not stop the scheduled Saturday refresh; to stop all activity, use **Quit BiG360**.

Interface tour

Route: every page · **Menu:** not applicable · **Mnemonics:** `HELP` · **Refresh:** freshness polled in the background

Purpose

Every page shares the same frame: a menu on the left, a top bar with search, region filter and data-status indicators, a live strip, and the page itself. Knowing the frame once makes every page familiar.

Screenshot

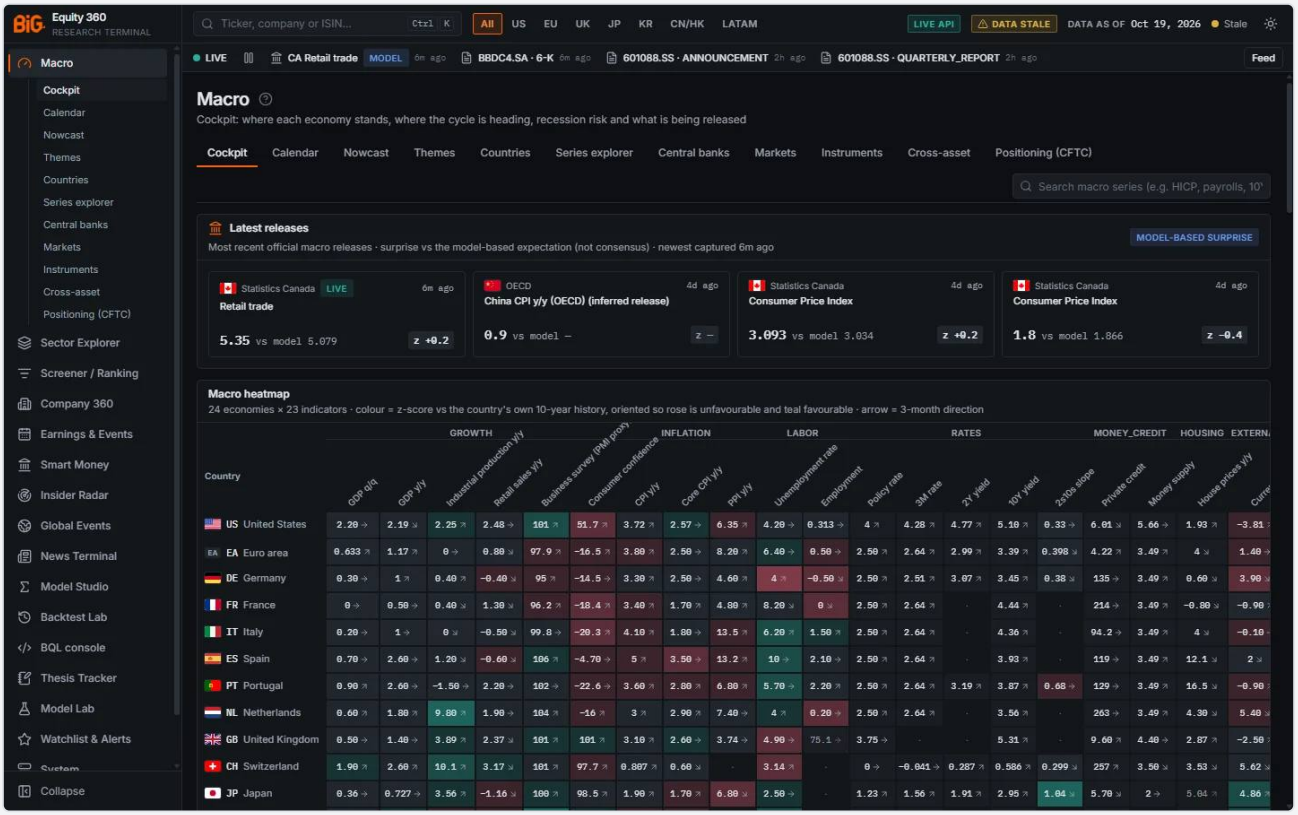


Figure 0.1: The application frame on the Macro cockpit (`/macro`), dark theme, data as of 19 Oct 2026, with the live strip connected (LIVE).

Region	Where	What it is
A	Left column	**Sidebar menu**: the BiG logo, the main sections, the sub-pages of the active section, and **Collapse** at the bottom
B	Top row	**Top bar**: search box / command bar (`Ctrl K`), region filter chips (All, US, EU, UK, JP, KR, CN/HK, LATAM), **LIVE API** badge, **DATA STALE** warning chip, **DATA AS OF** date, freshness dot ("Stale") and the theme toggle
C	Second row	**Live strip**: connection state (here **LIVE**), pause button, the latest live items (a Canadian retail-trade release with a MODEL badge, filings such as a 6-K) and the **Feed** button
D	Below the strip	**Page header**: page title ("Macro") with the **?** guide icon, and a one-line description of the page
E	Under the header	**Page tabs** ("Cockpit", "Calendar", "Nowcast" ...) and page-level tools, here the macro series search
F	Main area	**Panels**: each has a title, a subtitle that states the method and units, and status badges such as "MODEL-BASED SURPRISE"; a release that arrived through the live layer carries a LIVE badge

Sidebar menu

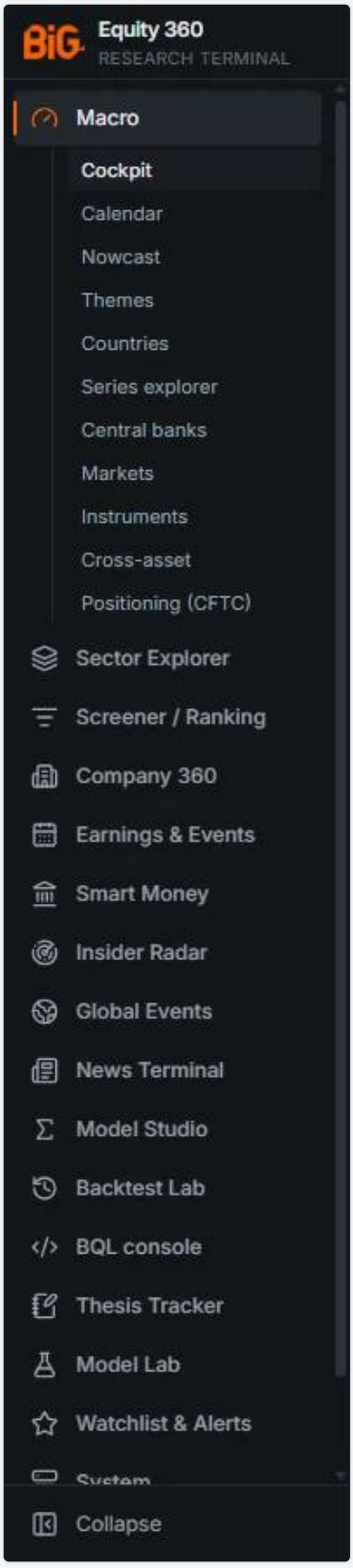


Figure 0.2: Sidebar with the Macro section expanded. The menu scrolls: System and Guide are below Watchlist & Alerts.

The sidebar lists the main sections. The section you are in is highlighted with an orange bar, and its sub-pages appear indented underneath.

Menu item	Route	Sub-pages
Macro	`/macro`	Cockpit, Calendar, Nowcast, Themes, Countries, Series explorer, Central banks, Markets, Instruments, Cross-asset, Positioning (CFTC)
Sector Explorer	`/sectors`	none

Menu item	Route	Sub-pages
Screener / Ranking	`/screener`	none
Company 360	`/company/<ticker>`	none; opens the last company you visited (Microsoft by default)
Earnings & Events	`/earnings`	Calendar, Earnings season
Smart Money	`/smart-money`	Overview, EU / UK short register, Activism (13D), Sectors
Insider Radar	`/insiders`	none
Global Events	`/events`	none (tabs inside the page)
News Terminal	`/news`	Live, Top stories
Model Studio	`/studio`	none (tabs inside the page)
Backtest Lab	`/backtest`	none (tabs inside the page)
BQL console	`/bql`	none
Thesis Tracker	`/theses`	none
Model Lab	`/models`	none
Watchlist & Alerts	`/watchlist`	none
System	`/system`	Overview, Data health, Conformal validation
Guide	`/help`	The chapters of this user guide (see User guide (Help page))

Collapse (bottom of the sidebar) shrinks the menu to icons only; hover an icon to see its name, and click **Expand** (same place) to restore it. The choice is remembered in this browser.

Top bar

Figure 0.3: Top bar, data as of 19 Oct 2026, with the "Data stale" warning.

Control	What it does
Search box ("Ticker, company or ISIN...", `Ctrl K`)	Opens the command bar (see Command bar). On a Mac the hint reads `⌘ K`.
Region chips (All, US, EU, UK, JP, KR, CN/HK, LATAM)	Sets the region filter used by the Screener, Sector Explorer, Insider Radar and Markets overview. Hover a chip for its full name ("Continental Europe (incl. CH, Nordics)", "Mainland China & Hong Kong" ...). Below 1,180 px of window width the chips become a drop-down. The choice is remembered in this browser.
LIVE API badge	The dashboard is reading the real API. Hover to see its address.
MOCK DATA badge	Shown instead of LIVE API in a development build that uses synthetic data. Figures are not real.
MOCK FALLBACK · n badge	Development builds only: n endpoints did not answer and were replaced by synthetic data. Hover for the list.
Warning chip ("Data stale", "Scheduler off", "Worker off", "Prices failed", "Analytics stale", "Analytics failed")	Appears only when something needs attention. Hover for the alerts behind it; click to open Data health ("Data stale") or the System page (the others). See Data-as-of and freshness indicators.
DATA AS OF date	The date of the most recent data in the published snapshot. Hover for the date per region.
Freshness dot and word ("Fresh", "Stale", "Error", "offline")	Overall status of the published data. Hover for the snapshot rebuild time, scheduler state and profile, and the list of stale or failing sections.
Theme toggle (sun / moon icon)	Switches between dark and light themes.

Live strip

The strip under the top bar shows the newest filings, news, macro releases and alerts as they arrive.

Element	What it means
Connection dot and text	**LIVE** (green, pulsing): the live stream is connected and the realtime service is running. **Connected · realtime layer stopped** (amber): the browser is connected, but the background service that collects live items is not running. Also: *Paused*, *Connecting...*, *Reconnecting...*, *Disconnected · retrying*.
Pause / resume button	Pauses the stream in this browser. On resume, items missed while paused are replayed.
NOT LIVE · LAST HEARTBEAT ... AGO · LAST ITEMS RECEIVED:	Shown when the realtime service is stopped. The items that follow are the last ones received, not current news; the strip is dimmed to make this clear. Hover the label for the exact time of the last heartbeat.
Latest items (up to 4)	Icon by type (filing, news, macro, alert, event), headline and age. A "model" badge marks model-based items (for example a macro surprise measured against the app's model). Click an item to open it.
Feed button ("Feed · 3 new")	Opens the **Live feed** side panel: every item received, filterable by All / Filings / News / Macro / Alerts, each tagged *streamed* (arrived live) or *nightly* (from the last refresh), with the stream counters and last heartbeat.

To turn live news on, use tray › **Live news on/off** (see Tray icon and menu).

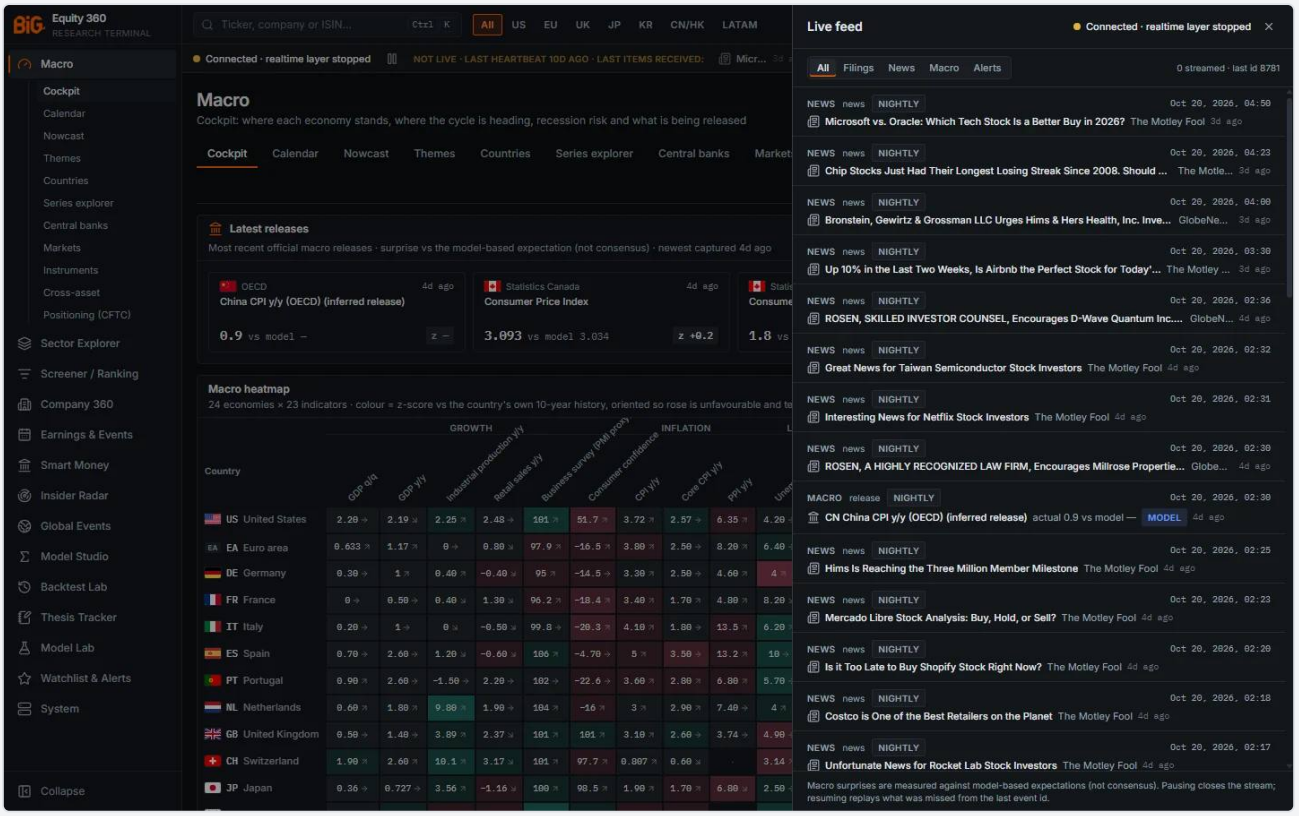


Figure 0.4: The Live feed side panel (Feed button in the live strip) with the realtime layer stopped: every item is tagged NIGHTLY (received with the last refresh) and the macro release carries a MODEL badge.

Page header, tabs and panels

Each page starts with its title and a one-line description, followed by tabs where the page has several views. A **?** icon next to the page title (on Company 360, next to **Create thesis**) opens the section of this guide that describes the page, and the tab you are on where the guide has one. The selected tab is part of the address (for example `/company/AAPL?tab=technicals`), so you can bookmark or share a view.

Panels follow one convention:

- the **title** names the object ("Macro heatmap");
- the **subtitle** states units, method and window ("colour = z-score vs the country's own 10-year history ...");
- **badges** on the right carry labels and warnings ("MODEL-BASED SURPRISE", "NO NEW RELEASE INGESTED FOR 3 DAYS");
- hover over most figures, column headers and badges for a tooltip with the definition.

When a panel cannot show data, it says why, using one of four states:

State	Text	Meaning
Loading	grey placeholder rows	The request is running.
Empty	"No data" plus a reason	The query worked but returned nothing for this selection.
Pending	"Not yet available — pipeline pending"	The view exists but the pipeline has not produced it yet (for example a model still training).
Error	"Could not load data" with a **Retry** button	The request failed; the message gives the reason. See A panel says "Could not load data".

A yellow note "Partial data — some inputs for this view are missing or stale" means the figures are what the pipeline has published, but some inputs are missing.

Related mnemonics and BQL fields

The **?** icon next to each page title opens this guide; `HELP` in the command bar opens it too and lists every command-bar function. Related chapters: Data Health for the status pages, Troubleshooting for warning chips.

Command bar

Route: every page (overlay) · **Menu:** top bar search box · **Mnemonics:** `HELP`, `HELP <function>` · **Refresh:** not applicable

Purpose

The command bar is the fastest way around the app. It finds a security by ticker, name or ISIN, jumps to any page, and understands Bloomberg-style functions such as `AAPL US Equity GP` or `WIRP US`.

Open it with `Ctrl K` (`⌘ K` on a Mac), with `/` when you are not typing in a field, or by clicking the search box in the top bar. Press `Esc` to close it.

Screenshot

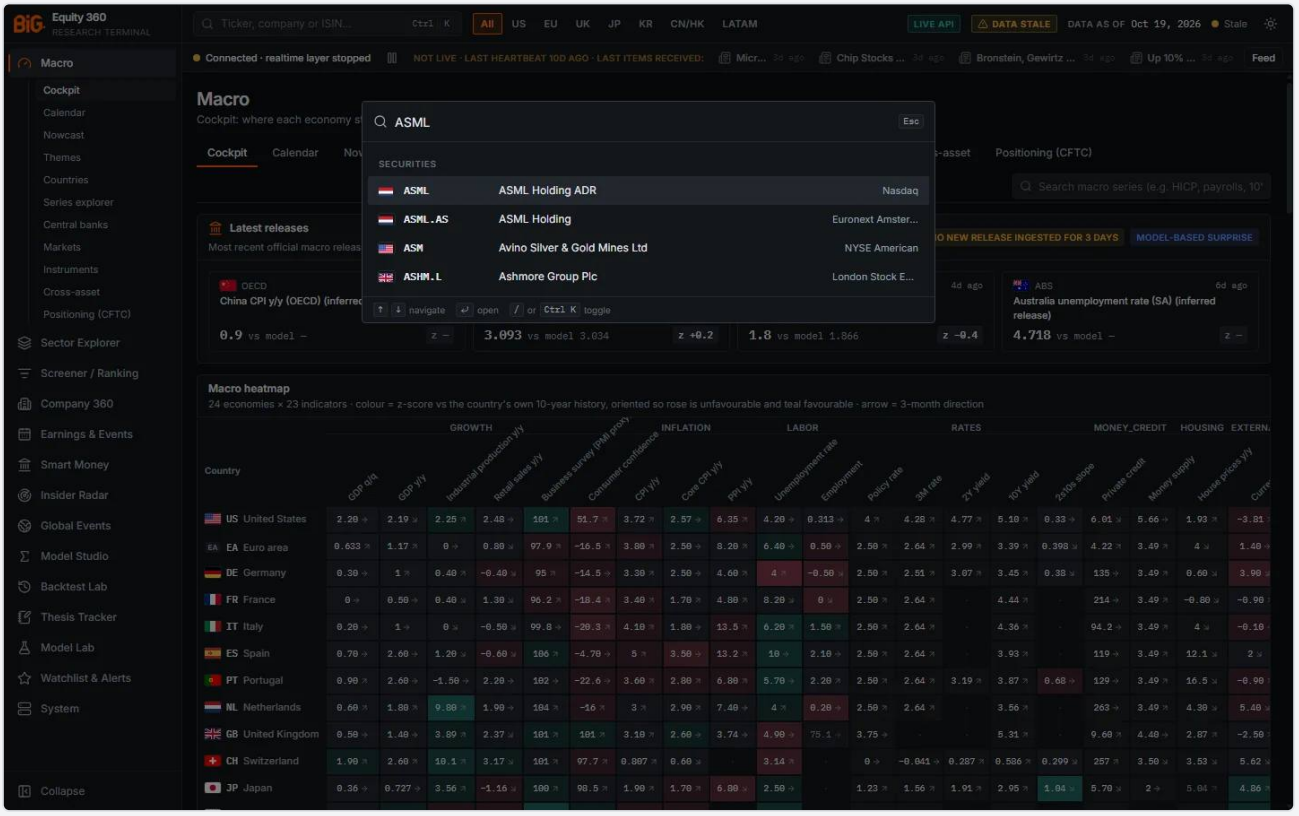


Figure 0.5: Command bar with "ASML" typed: matching securities with flag, ticker, name and exchange.

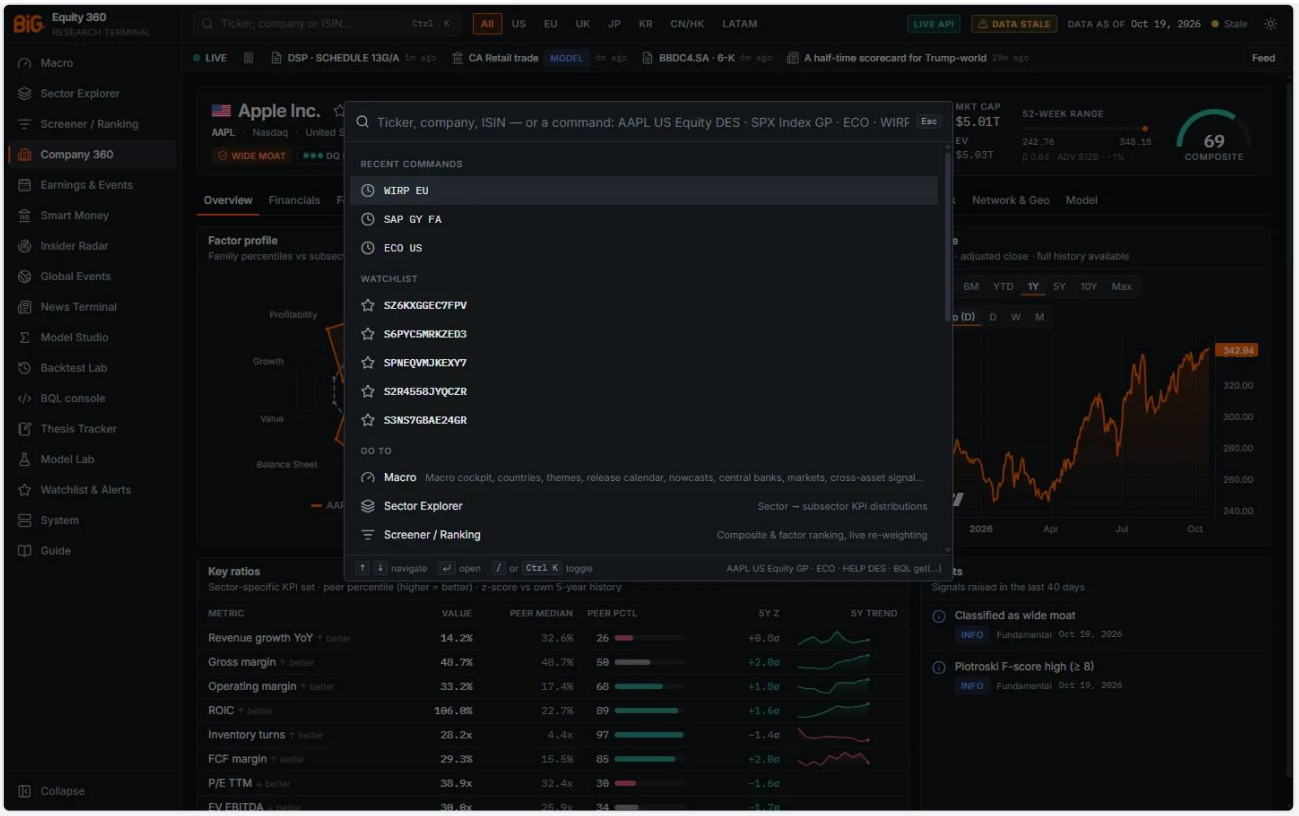


Figure 0.6: Command bar with an empty input on Company 360: recent commands (WIRP EU, SAP GY FA, ECO US), the watchlist, then "Go to" pages.

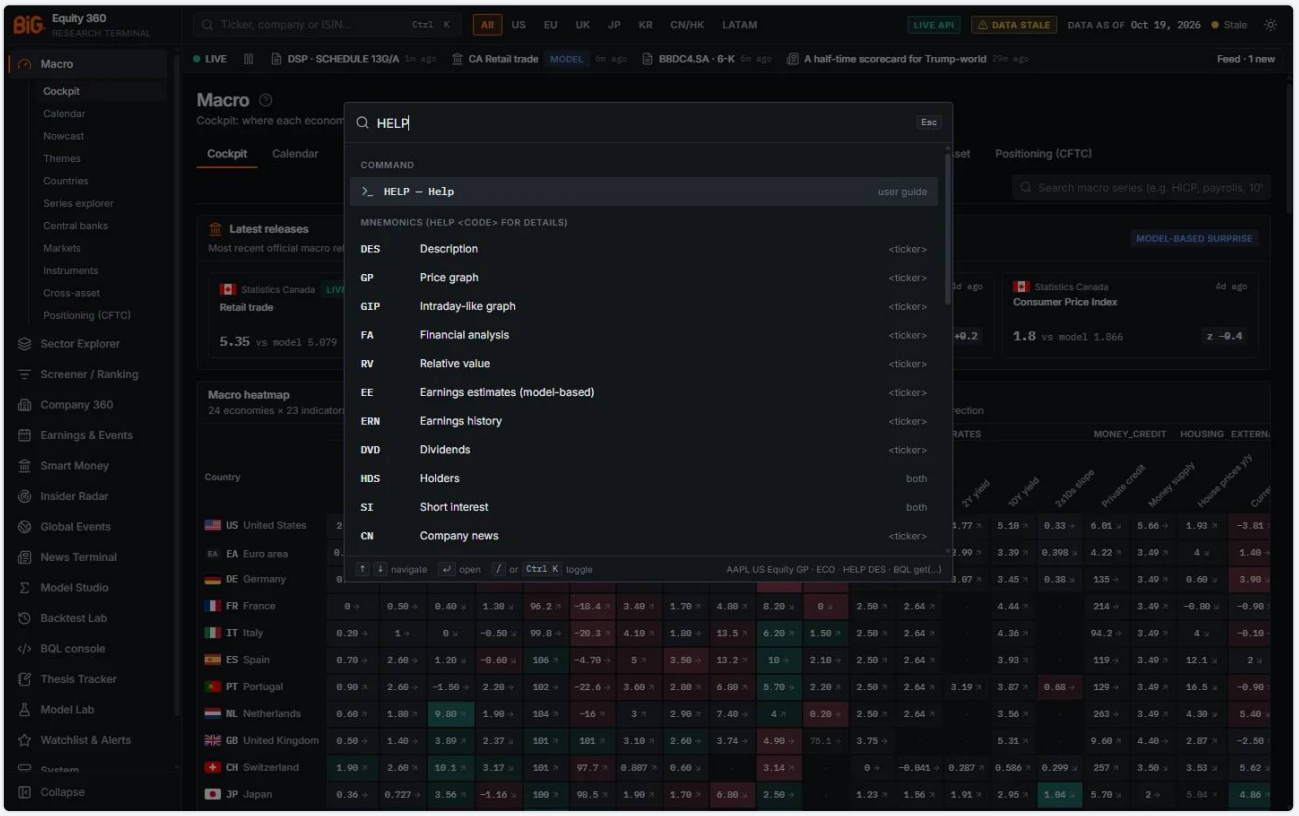


Figure 0.7: `HELP` typed: the first item "HELP — Help" opens the user guide, followed by every mnemonic with its scope (`<ticker>`, both or global).

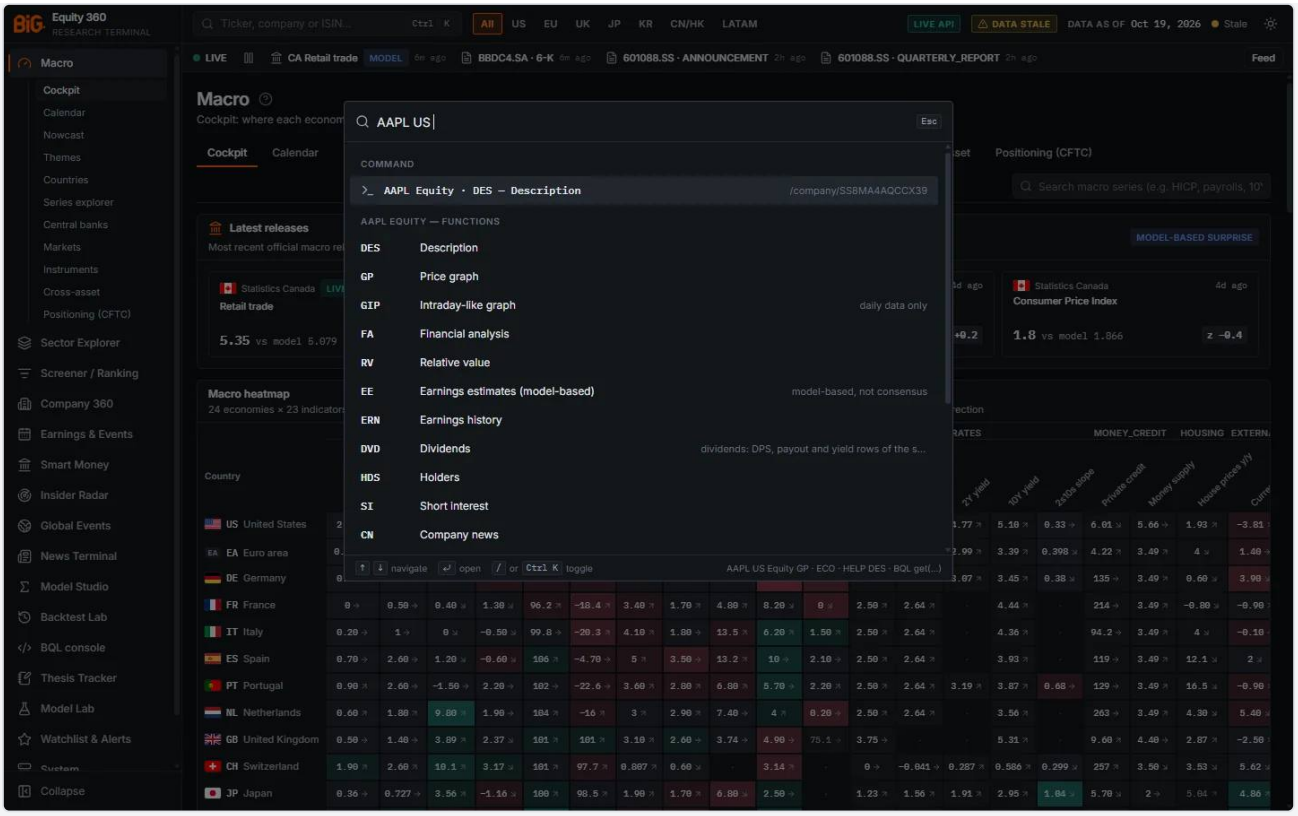


Figure 0.8: `AAPL US` typed (with a trailing space): the command "AAPL Equity · DES — Description" and the list of functions available for Apple, with notes such as "model-based, not consensus" for `EE`.

> **Note:** The watchlist group lists the watchlist's internal security ids (for example `SZ6KXGGEC7FPV`) rather than tickers; selecting one still opens the right company.

Controls

What appears depends on what you type:

Input	Groups shown	Enter does
Nothing	**Recent commands** (last 6), **Watchlist** (up to 8), **Go to** (every menu page), **Actions**	Opens the highlighted item
A ticker, name or ISIN (`asml`, `toyota`, `US0378331005`)	**Securities** (flag, ticker, name, ISIN, exchange), **Go to** pages whose name or description matches, and **Functions** whose code starts with the text	Opens the security's page: Company 360 for a stock, the instrument page for an index, FX pair or future
A security with a market or asset class (`AAPL US`, `SPX Index`, `EURUSD Curncy`)	**Command** (the security's description, `DES`)	Opens it
A security followed by a space (`AAPL US `)	**"AAPL Equity — functions"** (the security's functions): every function that applies to that asset class	Opens the highlighted function
A security and a function (`AAPL GP`, `SAP GY FA`, `7203 JT TECH`)	**Command** with its target	Runs it
A function with an optional argument (`ECO`, `ECO US`, `WIRP US`, `ECST DE`, `N`, `TOP`)	**Command**	Runs it
A security-only function on its own (`DES`, `GP`) on a Company 360 or instrument page	**Command** applied to that page's security	Runs it
`HELP`	**Command**: "HELP — Help" (first item, opens the user guide at `/help`), then **Mnemonics**: every function with its scope	Opens the guide; on a mnemonic row, shows `HELP <code>` for it
`HELP DES`	**Help**: one-line description and usage, then "Open the mnemonic reference in the user guide"	First row: fills the input with the function; second row: opens the mnemonic reference in chapter 14

Input	Groups shown	Enter does
`BQL get(...) for(...)` or `get(...)`	**Command**: "Run BQL: ..."	Opens the BQL console and runs the query

The **Actions** group (shown with an empty input, or when you type "theme") offers: **Switch to light / dark theme**, **Open example: Microsoft 360**, **Command bar help (mnemonics)** (types `HELP` for you) and **Open the user guide**.

Forms of a command:

- `<TICKER> [<MARKET>] [Equity|Index|Curncy|Comdty] <FUNCTION>`, for example `AAPL GP`, `AAPL US Equity DES`, `SAP GY FA`, `SX5E Index GP`, `EURUSD Curncy GP`, `CL1 Comdty GP`.
- `<FUNCTION> [argument]`, for example `ECO US` (US release calendar) or `WIRP EU` (ECB page with the implied policy path).
- Market codes follow Bloomberg (`US`, `UN`, `UW`, `LN`, `GY`, `FP`, `NA`, `SM`, `IM`, `SW`, `JT`, `KS`, `HK`, `CH`, `AU`, `CN`, `BZ`, `MM`, `IN` and others). `CN` is both the Canada market code and the company-news function: `RY CN` is the Canadian listing of Royal Bank, while `AAPL US CN` is Apple news.
- A function marked unavailable is shown but cannot run. `OMON` (options monitor) is listed as "no options data".
- If the input is not a command, the bar falls back to the ordinary security search.

Mnemonics overview. The 34 functions fall into four families:

Family	Functions
Security	`DES` description, `GP` / `GIP` price graph, `FA` financials, `RV` relative value, `EE` earnings estimates (model-based, not consensus), `ERN` earnings history, `DVD` dividends, `CN` company news, `TECH` technicals, `OMON` (unavailable)
Security or market-wide	`HDS` holders, `SI` short interest, `PEERS`, `NET` network and geography, `SMRT` smart money, `EVTS` events, `THES` thesis, `BQL` console
Market-wide	`WEI` world indices, `WIRP` implied policy rates (approximation), `ECO` economic calendar, `ECST` country statistics, `GC` curves, `FXIP` (alias `FX`), `CMDTY` commodities, `COT` CFTC positioning, `EQS` screener, `PORT` (alias `BT`) backtest, `TOP` top news, `N` news terminal, `ALRT` alerts, `MDL` model registry
Help	`HELP` (opens the user guide and lists the functions), `HELP <function>`

The full reference, with the page each function opens and its fallbacks, is in BQL console and mnemonics.

Recent commands. Commands you run are remembered (up to 20, the last 6 shown) in this browser only. Clearing the browser's site data clears them.

Correct usage and worked example

1. Press `Ctrl K` and type `asml`. The **Securities** group lists ASML (the Nasdaq ADR) and ASML.AS (the Amsterdam home listing). Pick **ASML.AS** to see scores and technicals: for an ADR, the composite score, technicals and fair value are computed on the home listing. An ADR page says so in its header, with a link to the home listing (Figure 0.9).
2. On the ASML.AS page, press `Ctrl K` again and type `GP`: the command applies to the page's security and opens the Technicals tab.
3. Type `ECO DE` to jump to the release calendar filtered on Germany.

Figure 0.9: Header of the Toyota Motor ADR page (`/company/TM`): the ADR note "Composite, technicals and fair value are computed on the home listing by design" and the link "See home listing → 7203.T (Tokyo Stock Exchange)"; the composite gauge is empty on the ADR.

Limitations and pitfalls

- Search matches tickers, names and ISINs in the app's universe only. A company that is not covered does not appear (see A company is not found).
- The index names `'S&P 500'` and `'STOXX 600'` in BQL map to the app's own universes, which are approximations, not official constituents.

Related mnemonics and BQL fields

All functions above; full list in chapter 14.

User guide (Help page)

Route: `/help`, `/help/<chapter>`, `/help/print` · **Menu:** Guide (last item of the sidebar) · **Mnemonics:** `HELP` · **Refresh:** not applicable

Purpose

This guide is built into the app. The **Guide** page renders every chapter with its figures, so you can look up how a page works, how a number is computed and what its validation status is without leaving the dashboard. It opens on the contents page, with every chapter listed in the sidebar.

Screenshot

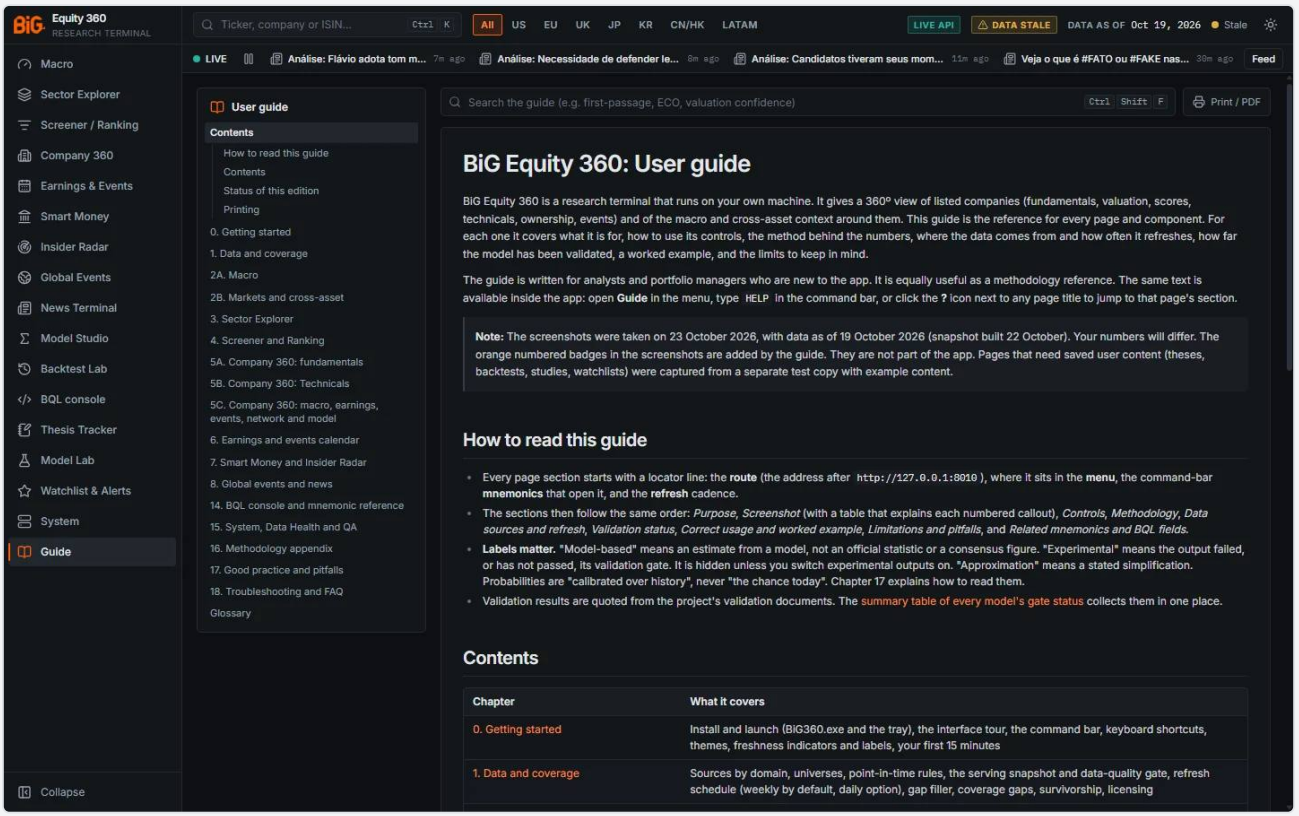


*Figure 0.10: The Guide (`/help`) on its contents page: chapter list on the left, the search box and **Print / PDF** at the top.*

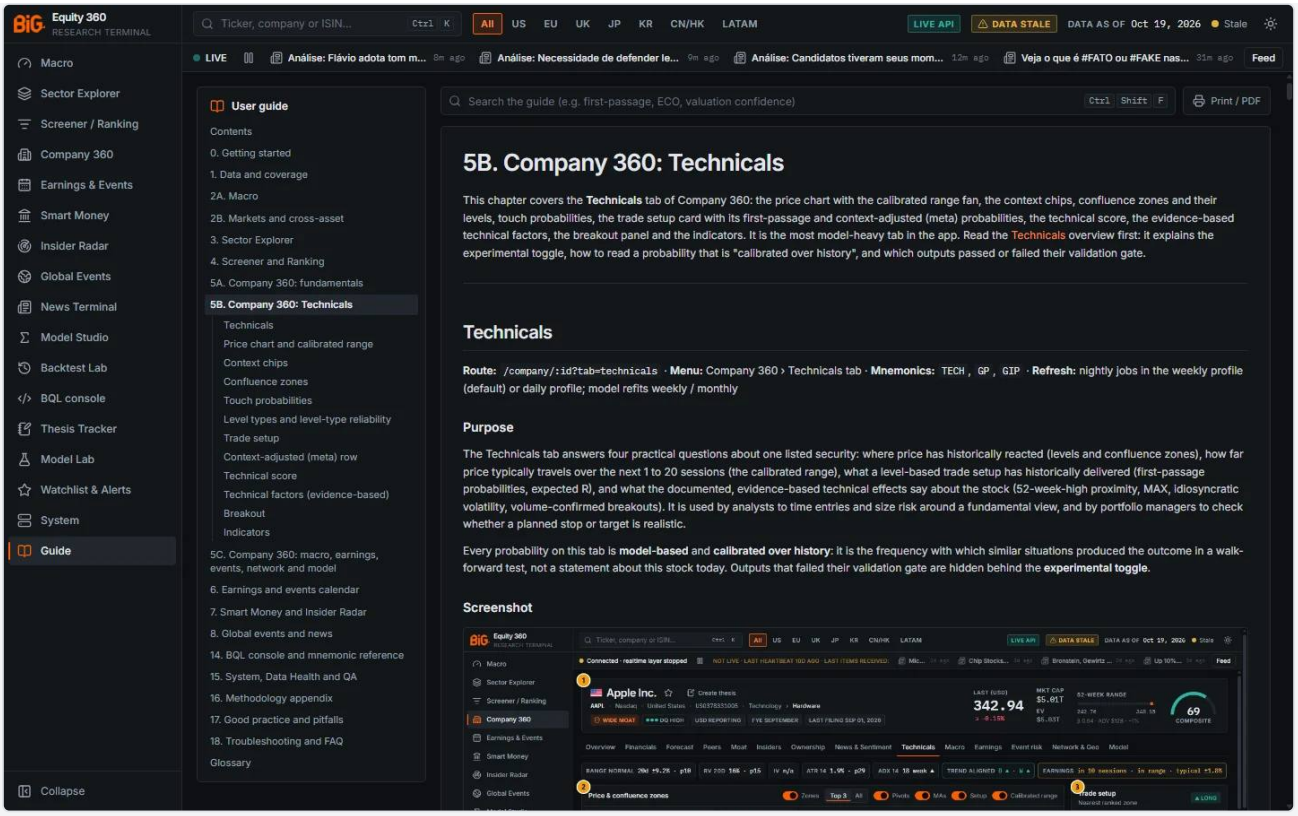


Figure 0.11: Chapter 5B open (`/help/05b-company-technicals`): the sidebar expands the open chapter into its sections.

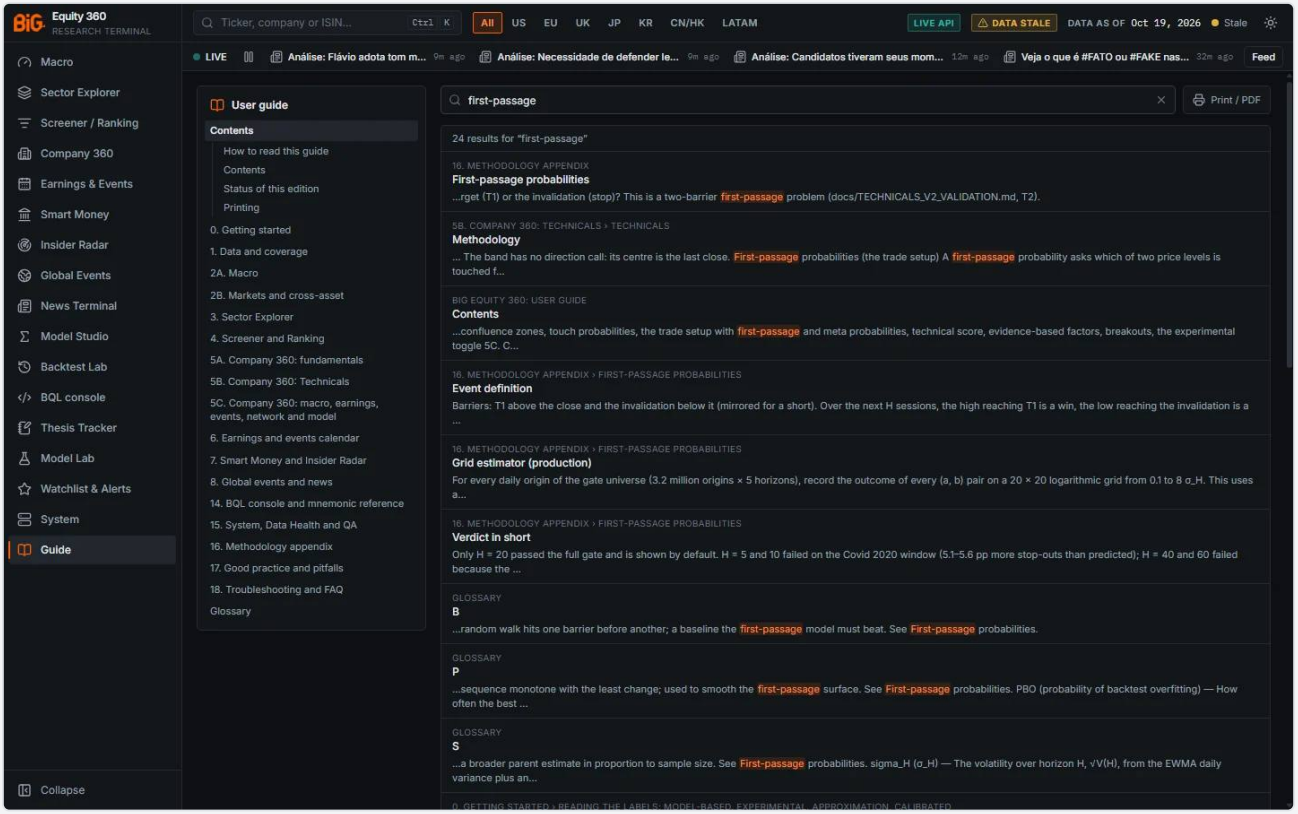


Figure 0.12: Searching the guide for "first-passage": 24 results, each with its chapter and section and the matching text highlighted.

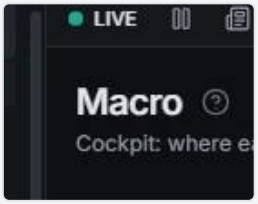


*Figure 0.13: The *?** icon next to the page title ("Macro"), which opens the guide section for the page.*

Controls

Control	What it does
Guide (sidebar)	Opens `/help` on the contents page
Chapter sidebar	Lists the chapters with their sections; click one to open it (the address becomes `/help/<chapter>#<section>`)
Search the guide box	Full-text search across every section; the results list the matching sections. `Ctrl Shift F` focuses the box, `Esc` clears it.
Heading anchor links	Each heading has its own link, so you can copy the address of one section and share it
Previous / next chapter links	At the bottom of each chapter
Print / PDF	Opens `/help/print` in a new tab: the whole guide on one page in the light theme, ready for the browser's print dialog or "Save as PDF"
? icon next to a page title	Opens the guide section for that page (and tab). Tooltip: "Guide: how this page works". On Company 360 it sits next to **Create thesis**.
`HELP` in the command bar	The first item "HELP — Help" opens the guide; **Actions › Open the user guide** does the same

Limitations and pitfalls

- The figures in the guide were captured on data as of 19 Oct 2026, so the numbers in them will differ from what you see today.

Keyboard shortcuts

Route: every page · **Menu:** not applicable

Controls

Where	Keys	Action
Anywhere	`Ctrl K` (`⌘ K` on a Mac)	Open or close the command bar
Anywhere, when not typing in a field	`/`	Open the command bar
Command bar	`↑` `↓`	Move through the results (the list wraps around)
Command bar	`Enter`	Open the highlighted item
Command bar, dialogs, side panels	`Esc`	Close
Any page, first `Tab`	`Tab` then `Enter`	"Skip to content": jump past the menu to the page
Tables with clickable rows (Screener, Data Health ...)	`Tab` to a row, then `Enter`	Open the row (same as clicking it)
User guide (`/help`)	`Ctrl Shift F` (`⌘ Shift F` on a Mac)	Focus the guide search box
User guide search box	`Esc`	Clear the search
Tabs and segmented controls	`←` `→`	Move between tabs or options once one has focus
Macro series search	`Esc`	Clear the search
Security inputs (Model Studio, Backtest Lab)	`Enter`	Accept the typed ticker or id
BQL console editor	`Ctrl Enter`	Run the query
BQL console editor, suggestion list open	`↑` `↓`, `Tab` or `Enter`, `Esc`	Move, accept the suggestion, close the list
News Terminal (not typing)	`j` / `↓`, `k` / `↑`	Next / previous headline

Where	Keys	Action
News Terminal	`/`	Focus the filter box (`Esc` leaves it)
News Terminal	`Enter`	Open the selected story's detail
News Terminal	`o`	Open the source article in a new tab
News Terminal	`s`	Mute the selected story
News Terminal	`w`	Toggle "watchlist only"
News Terminal	`g`	Toggle grouping by story

> **Note:** In the News Terminal, `/` focuses the page's own filter instead of opening the command bar; use `Ctrl K` there.

Themes and number locale

Route: top bar (theme), `/system` (locale) · **Menu:** System › Overview › Preferences & connection

Purpose

The app has a **dark** theme (default) and a **light** theme, and formats numbers and dates in the locale you choose. Both choices are stored in this browser and apply to every page and chart.

Screenshot

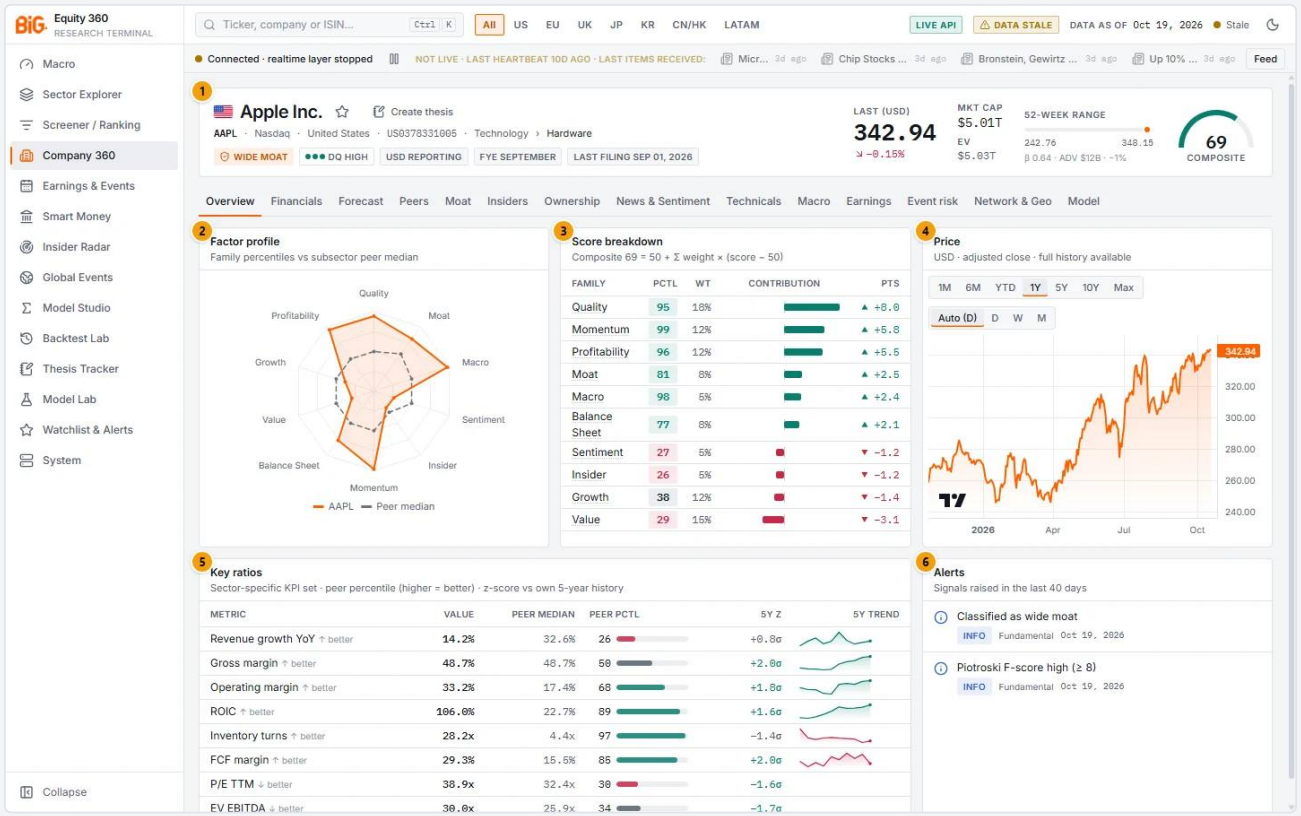


Figure 0.14: Company 360 for Apple (`/company/AAPL`), light theme.

#	Panel	What it shows
1	Apple Inc. (header)	Name, ticker, exchange, ISIN, sector › subsector, badges (WIDE MOAT, DQ HIGH, USD REPORTING, FYE SEPTEMBER, LAST FILING), last price, market cap, EV, 52-week range and the composite score gauge (69)
2	Factor profile	Radar of the ten factor-family percentiles vs the subsector peer median
3	Score breakdown	How each family moves the composite: `Composite = 50 + Σ weight × (score − 50)`

#	Panel	What it shows
4	Price	Adjusted-close chart with range buttons (1M ... Max) and bar size (Auto, D, W, M)
5	Key ratios	Sector-specific KPIs with peer percentile, 5-year z-score and trend
6	Alerts	Signals raised in the last 40 days (here "Classified as wide moat", "Piotroski F-score high")

The panels are described in Company 360.

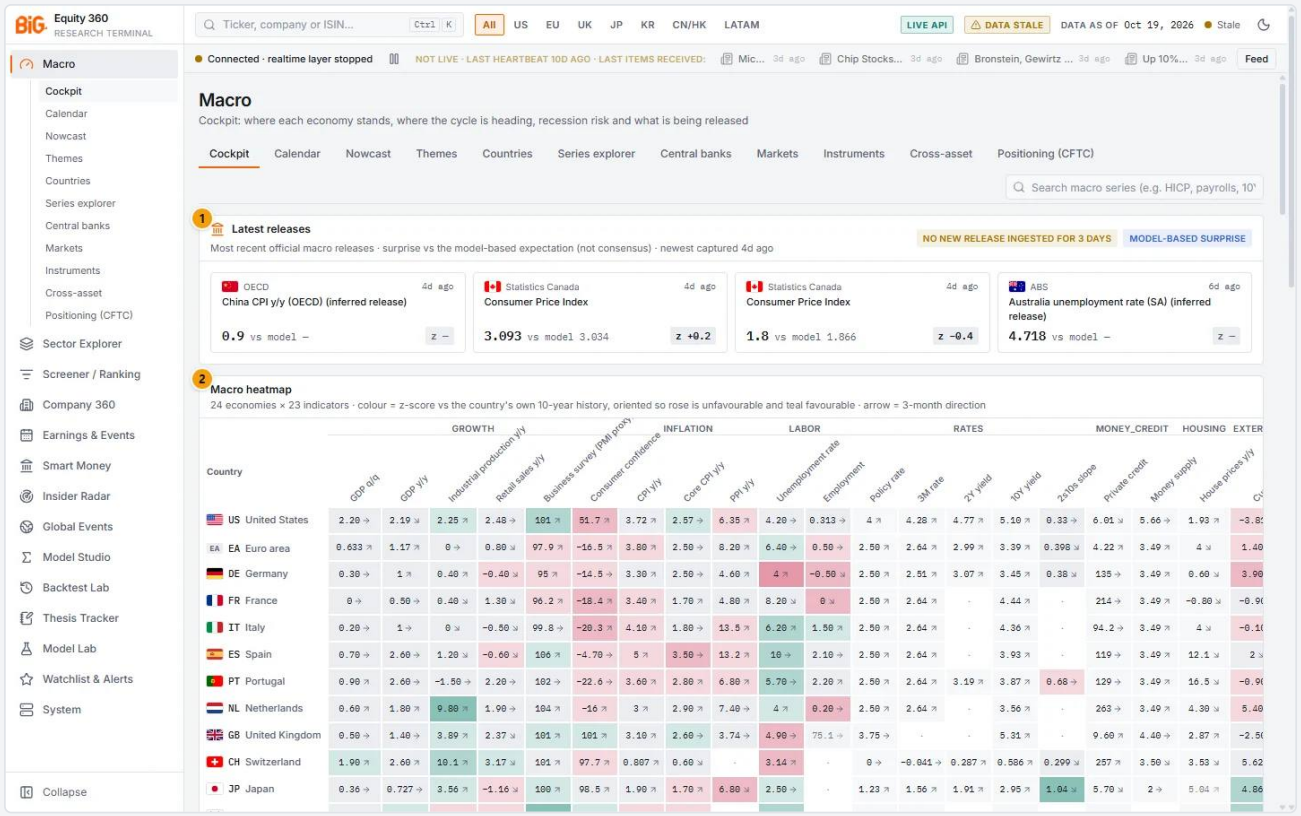


Figure 0.15: Macro cockpit (`/macro`), light theme.

#	Panel	What it shows
1	Latest releases	Most recent official macro releases and the surprise vs the model-based expectation (not consensus)
2	Macro heatmap	24 economies × 23 indicators, coloured by z-score vs each country's own 10-year history

See Macro cockpit for the full page.

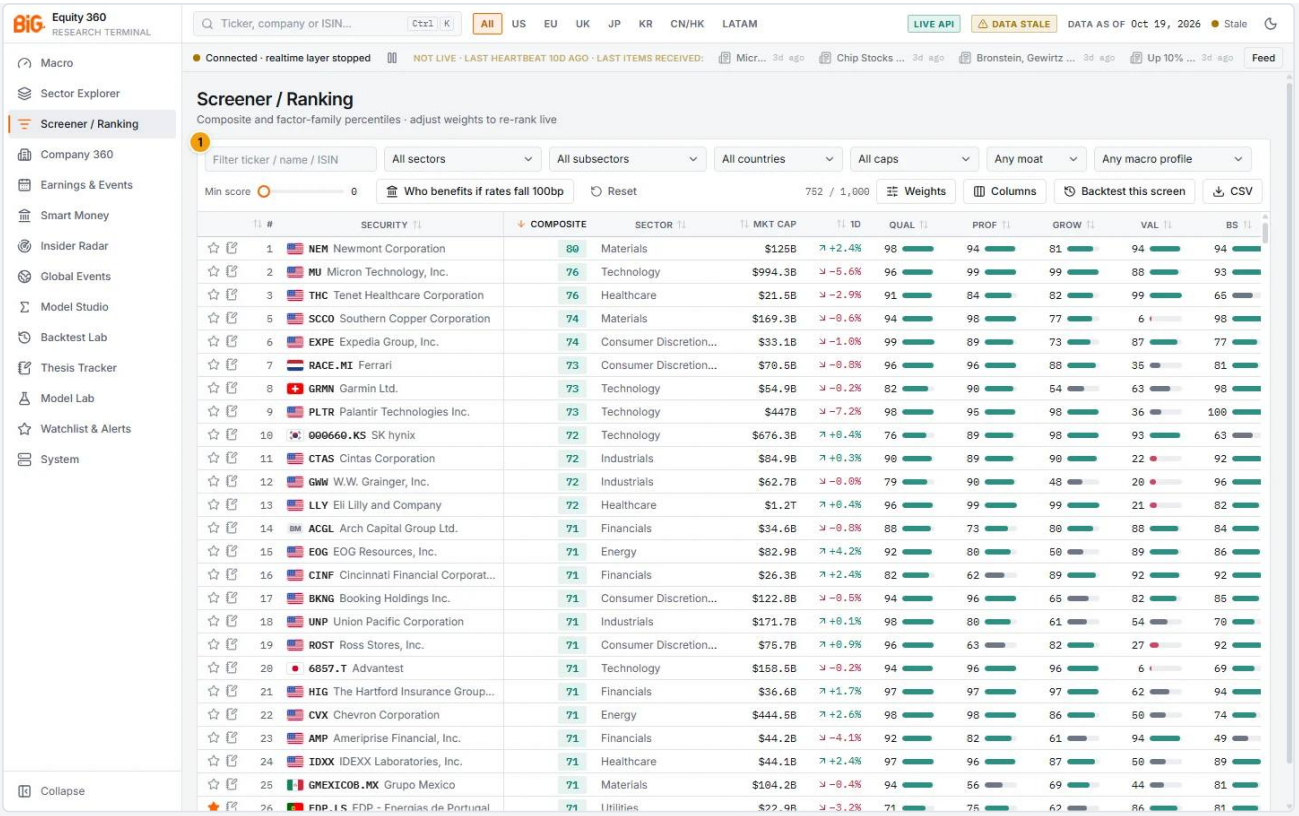


Figure 0.16: Screener / Ranking (`/screener`), light theme.

#	Panel	What it shows
1	Filters and ranking table	Filters (text, sector, subsector, country, size, moat, macro profile, minimum score), Weights, Columns, Backtest this screen, CSV export, and the ranked table

See Screener.

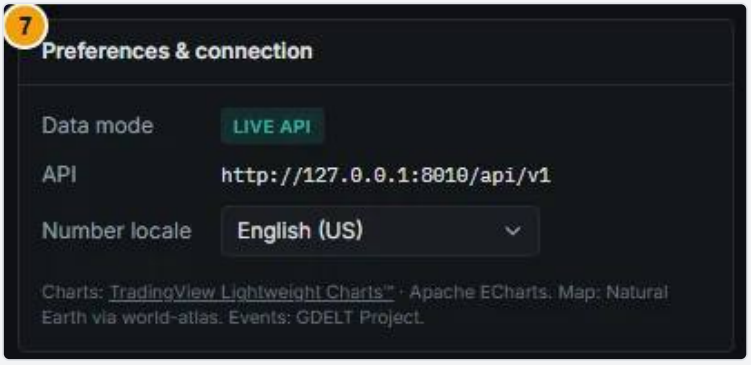


Figure 0.17: System › Overview › Preferences & connection: data mode, API address and number locale.

Controls

Control	What it does
Theme toggle (top bar, sun / moon icon)	Switches theme. The sun icon is shown in the dark theme ("Switch to light theme"), the moon in the light theme.
Command bar › Actions › **Switch to light / dark theme**	Same, from the keyboard.
Number locale (System › Overview › Preferences & connection)	Formats every number and date: English (US), English (UK), Português (PT), Deutsch (DE), Français (FR), 日本語 (JP). For example, in Português (PT) and Deutsch (DE) the decimal separator is a comma. Charts follow the same locale.

Limitations and pitfalls

- Both settings are per browser. A colleague on the same PC with another browser profile sees their own choice.

- Changing the locale changes only formatting, never the values. Units are still in the panel subtitles (for example "%", "bn", "x").
- Times (for example the next scheduled run on the System page) are shown in your browser's time zone, while schedules are defined in Lisbon time.

Data-as-of and freshness indicators

Route: top bar (every page) · **Menu:** System › Data health for details · **Refresh:** re-read in the background

Purpose

The top bar always tells you how current the numbers are. This matters because the app refreshes on a schedule (weekly by default), regions close at different times, and a failed job can leave one region behind.

Methodology

The indicators are computed from `GET /api/v1/system/status`:

- **DATA AS OF** is the latest data date in the published snapshot (`data_as_of_overall`). The tooltip lists the date per region (US, EU, UK, JP, KR, CN, HK, CA, LATAM, APAC, Other). It is the date of the data, not the time the snapshot was rebuilt.
- The **freshness status** follows the snapshot's sections:
  - *Error* (red dot) when a critical table (prices, FX, macro observations) is missing or a price section failed;
  - *Stale* (amber dot) when any market section (prices of a region, FX, macro, filings, insider, events, news, CFTC) is behind, or when an alert says core data is stale beyond the service level or the refresh is overdue;
  - *Fresh* (green dot) otherwise;
  - *offline* (red) when the status endpoint does not answer.
- The **warning chip** shows the most important problem, in this order: *Worker off* (no worker, so nothing runs), *Scheduler off* (worker running without its schedule; red after 24 hours), *Prices failed*, *Data stale*, then *Analytics failed* / *Analytics stale* (scores, valuation, technicals or models behind while the market data is fine).

What "stale" means. Each dataset has an expected cadence (exchange trading calendars for prices, official release calendars for macro, the CFTC Friday schedule and so on) and a service level. A dataset is stale when its data lags its latest expected date by more than the service level. For the weekly profile the default service level is 9 days (prices 6 days); for the daily profile it is 3 days. Stale does not mean wrong: the numbers are correct as of their date, but newer data exists at the source.

Correct usage and worked example

On 23 Oct 2026 the top bar read **DATA STALE**, **DATA AS OF Oct 19, 2026** and a **Stale** dot (Figure 0.3). Hovering the date showed US 16 Oct, EU / UK / KR / CN 19 Oct, HK and LATAM 16 Oct, and **JP 13 Oct**. The freshness tooltip listed `prices:JP (stale)`: Japanese prices were six days behind the snapshot date, which is what turned the chip on. Everything else was usable; only Japanese prices and anything derived from them that week (Japanese technicals, momentum) were older.

What to do:

1. Check the per-region date before comparing names across regions.
2. Click the chip to open Data health and see the dataset, its lag and whether the gap filler is already repairing it.
3. If you need fresher data before the next scheduled run, use tray › **Run refresh now** (it takes one to several hours).

Limitations and pitfalls

- A snapshot can be rebuilt today with data from last week: always read **DATA AS OF**, not the rebuild time.
- Per-panel dates matter as well: many panels show their own "as of" or "newest captured 4d ago" in the subtitle.

Related mnemonics and BQL fields

The System pages are described in System and Data Health; data cadences in Refresh schedule.

Reading the labels: model-based, experimental, approximation, calibrated

Route: every page · **Menu:** not applicable · **Refresh:** not applicable

Purpose

The app never presents its own estimates as official data or as consensus. Every number that comes from a model, a simplification or a test that failed carries a label. Read these labels before you use a number.

Screenshot

Figure 0.18: Trade setup card for Apple with "show experimental" switched on: experimental horizons and the meta row are marked with a flask icon, and the card states "Calibration gate failed ... model-based, calibrated on history".

Methodology

Label (as shown)	Meaning	Examples
official	Published by a statistics office, central bank, regulator or exchange, shown as published	Releases in the macro calendar, filings, policy rates
model-based (also "model", "MODEL-BASED SURPRISE", "model-based (ours)")	Estimated by the app's own model from historical data. It is not an official figure and not analyst consensus.	Macro expectations and surprises, earnings expectations (seasonal model on the company's own history, 80% band), nowcasts, recession and regime models, Taylor rule, central-bank tone, smart-money score, setup probabilities
approximation	Derived from official data with a simplified method that a market professional would compute differently	Central-bank implied paths: "Approximation from official curves (not futures/OIS-implied probabilities)"
calibrated on history, **Calibrated range**, calibrated band	A range or probability whose realised frequency has been checked against history. A 90% calibrated range contained the realised price about 90% of the time in the walk-forward test (89.6% to 90.0% at every horizon from 1 to 20 sessions).	The calibrated range fan on the Technicals chart, touch probabilities, the 20-session first-passage probabilities
experimental (flask icon, "experimental", "failed validation")	The output was tested and **failed** its validation gate (or has not passed it). It is hidden by default and only shown when you switch it on, or in Model Studio and the Backtest Lab.	Setup probabilities at 5, 10, 40 and 60 sessions; P(hold given touch); the calibrated technical score ("no out-of-sample edge"); the joint conformal envelope
descriptive · model-based · not a trading signal	Passed a calibration test as a description, but failed as a predictor	Conformal peer-group anomaly measures
n/v	Not validated	Shown in validation lists where no gate exists
Announced / Provider est. / Projection	Source of an earnings date: published by the company or exchange; a data provider's estimate; the app's projection from the reporting cadence	Earnings calendar, Company 360 › Earnings

How to read a probability: "37% T1 first" means that, over the history used to calibrate it, setups like this one reached the first target before the stop about 37% of the time. It is **calibrated over history**, not "a 37% chance today".

Controls

Control	What it does
show experimental (failed validation) switch	On the Company 360 › Technicals tab, click **calibrated on history** on the trade setup, technical score or breakout card to open the popover with the validation lines and this switch. When on, outputs that failed their gate are shown with a flask icon and an "experimental" badge. When off (default), they are hidden and the card says so, for example "Experimental (failed the validation gate) — turn on “show experimental” to see it."

The switch is stored per browser (key `big360.technicals.experimental`) and applies to the Technicals tab and to the setup probabilities shown in the Thesis Tracker.

Validation status

The gate result of every model is summarised in Methodology appendix. Examples quoted from the validation documents: the marginal calibrated range fan is validated (on target per horizon, about 40% narrower than a 2×ATR band); the first-passage probabilities pass at 20 sessions and fail at 5, 10, 40 and 60 sessions; the calibrated technical score shows no out-of-sample edge. See Technicals for details.

Limitations and pitfalls

- "Model-based" is not a quality grade: some model-based outputs passed strict tests, others did not. Check the validation line.
- Switching on experimental outputs does not make them more reliable. Use them for research, not for decisions.

Where things live

What	Where
Data (lake, serving snapshots, state)	The folder in `BIG360_DATA_DIR`; by default `<project>\data` (about 125 GB on this PC)
Your watchlists, alerts, theses, journal, studies, backtests, scenarios	`data\db\app_state.duckdb` (back it up; it cannot be rebuilt). The watchlist and alert rules are also cached in the browser.
Your display preferences (theme, region, collapsed menu, locale, experimental switch, recent commands, BQL history)	This browser only (local storage)
Logs	`data\logs\` (see Logs to send to IT)
App settings	`config\app.yaml` (port, dashboard serving, gap filler, notifications, update folder), `config\schedule.yaml` (refresh profiles), `config\ops.yaml` (gap filler limits)
API keys	Windows Credential Manager, `BiG360/<key name>`; `.env` as the development fallback
Version and updates	`VERSION` file; IT can point `update_dir` in `config\app.yaml` at a share with newer builds

Your first 15 minutes

A suggested first session, using data as of 19 Oct 2026.

1. **Start the app (1 minute).** Double-click `BiG360.exe`, wait for the browser to open on the Macro cockpit. Check the tray tooltip says the data date and "weekly".
2. **Read the top bar (1 minute).** Note **DATA AS OF**, hover it to see the per-region dates, and note any warning chip. In our example, Japan was at 13 Oct while the US was at 16 Oct and Europe at 19 Oct.
3. **Macro backdrop (3 minutes).** On the Cockpit, scan **Latest releases** (each card shows the release and "vs model": the surprise is against the app's model, not consensus) and the **Macro heatmap** (rose = unfavourable, teal = favourable vs each country's own 10-year history). Open **Markets** in the sub-menu for indices, breadth and the cross-asset tape. See Macro.
4. **Find ideas (3 minutes).** Open **Screener / Ranking**, pick a region chip (for example **EU**), then sort by **Composite**. Move a factor weight under **Weights** and watch the ranking change. Export with **CSV** if needed. See Screener.
5. **One company in depth (5 minutes).** Press `Ctrl K`, type `AAPL`, press `Enter`. On **Overview**, Apple shows a composite score of 69, a wide moat, Quality at the 95th percentile of its peers and Value at the 29th. Then visit:
   - **Financials** (point-in-time statements);
   - **Forecast** (DCF and multiples; model-based, not consensus);
   - **Technicals** (levels, confluence zones, calibrated range, trade setup; open **calibrated on history** to see what is validated);
   - **Earnings** (date source label, reactions, model-based surprises);
   - click the star next to the name to add it to your watchlist. See Company 360.
6. **Set up tracking (2 minutes).** Open **Watchlist & Alerts** and add an alert rule, or click **Create thesis** in the company header to record why you are interested. See Watchlists and alerts and Thesis Tracker.
7. **Know where to look when something seems off.** Open **System › Data health** once to see what it shows. See System and Data Health. On any page, the **?** icon next to the title opens the matching section of this guide.

Then read Data and coverage to understand what the numbers are built from, and Good practice before using any output in a decision.

1. Data and coverage

This chapter explains where the numbers come from, how the app keeps them point-in-time, when they refresh, how missing data is repaired, which regions are well covered, and which sources need a licence review before the app is used more widely in the bank.

Read it once before relying on any page: most "why does this number look odd?" questions are answered by a source, a lag or a coverage gap described here.

> **Note:** Figures in this chapter are as of the snapshot published on 22 Oct 2026 (data as of 19 Oct 2026) unless stated otherwise. Live counts are on the System and Data health pages.

How data flows through the app

Route: `/system`, `/system/data-health` · **Menu:** System · **Refresh:** weekly profile by default

Purpose

Every page reads from one read-only, internally consistent copy of the data (the **serving snapshot**), rebuilt at the end of each refresh. Knowing the pipeline helps you understand lags and labels.

Methodology

```
sources → raw (landing zone) → normalised (point-in-time tables) → features → scores / models → serving snapshot → API → dashboard
```

1. **Ingestion jobs** download from each source into the raw zone, using rate limits and per-source watermarks so that each run only fetches what is new.
2. **Normalisation** maps every record to the app's identifiers (issuer, security, listing), converts currencies and units, and stamps each fact with a **knowledge date**: the date it became public.
3. **Analytics** compute ratios, sector KPIs, valuation, scores, moat, technicals, sentiment, macro models and machine-learning predictions from the normalised tables.
4. `build_serving_db` materialises everything the dashboard needs into a new snapshot file and publishes it atomically. A data-quality gate decides whether it may be published (see Serving snapshot and data-quality gate).

The storage is a Parquet data lake (about 125 GB on this PC) and DuckDB snapshot files (about 6.6 GB each). There is no database server.

Data sources by domain

Route: System › Overview › Data sources; System › Data health › Sources · **Menu:** System · **Refresh:** see Refresh schedule

Purpose

The app uses official sources wherever they exist (statistics offices, central banks, regulators, filing systems), commercial or unofficial sources only where no official one is available, and labels its own estimates as model-based.

Prices and market data

Region / instrument	Source	Notes
United States	Massive (formerly Polygon.io) grouped daily bars (one call per day covers the whole market), splits and dividends; Yahoo Finance via `yfinance` as fallback and for history	Polygon free tier allows 5 calls per minute, so history comes from Yahoo
Europe, UK, Japan, Hong Kong, Korea, Canada, India, Australia	Yahoo Finance (`yfinance`)	Korea switches to the official KRX Open API when the `krx_openapi_key` is present
China A-shares	BaoStock, then Yahoo	BaoStock has blocked this PC's IP address; several hundred Shanghai names are priced from Yahoo only and do not update nightly
Brazil	B3 COTAHIST (official exchange file) and Yahoo	Pre-2005 B3 units need review
Taiwan	TWSE / TPEx OpenAPI	Official open data

Region / instrument	Source	Notes
Benchmarks (indices, ETFs, futures, FX, commodities)	Yahoo	Futures curves for the cross-asset dashboard are Yahoo market data and labelled as such

Prices are stored in the trading currency (pence and cents are normalised to pounds and rand) and adjusted for splits and dividends. Before publication, `prices_finalize` breaks series where a ticker was reused by a different company, repairs 100× unit flips (for example pence vs pounds), and drops unfinished intraday bars.

FX

Source	Use
ECB euro reference rates	First choice for every currency it publishes
FRED H.10 (Federal Reserve)	Currencies the ECB does not publish, and pre-1999 history. Taiwan dollar is only weekly, so it can lag up to about 7 days.
Central banks (BCRA, BCRP, Colombia)	Latin American currencies
Yahoo	Last resort

Gaps are forward-filled for up to 5 days and marked `filled`. Balance-sheet items are converted at the period-end rate, flow items (revenue, earnings) at the period-average rate, and valuation ratios are always computed in USD on both sides, which removes the trading-currency vs reporting-currency mismatch for ADRs and dual listings.

Macro and central banks

Macro data comes from **official sources only**. The app's own expectations, surprises, nowcasts and regime estimates are model-based and labelled.

Group	Sources
United States	FRED / ALFRED (vintages give the knowledge date), BLS and BEA release calendars, US Treasury, New York Fed (ACM term premium, r*, staff nowcast), Atlanta Fed GDPNow
Euro area and Europe	ECB Data Portal, Eurostat, Bundesbank, Destatis, INSEE, ISTAT, INE Spain, Banco de Portugal BPstat, SNB, Riksbank, Norges Bank, Bank of England, ONS
International organisations	OECD, IMF, BIS, World Bank (including the commodity Pink Sheet)
Asia-Pacific	Bank of Japan, Ministry of Finance Japan, e-Stat (key), Bank of Korea ECOS (key), KOSIS (key), HKMA, Census and Statistics Department (HK), CBC Taiwan, RBA, ABS, PBoC / ChinaMoney (LPR)
Americas	Statistics Canada, Bank of Canada, BCB, IBGE, BCRA / INDEC, BCRP, BanRep, INE Chile, BCCh (key), INEGI (key), Banxico (key)
Release calendars	Official calendars of BLS / BEA, FOMC, Eurostat, ECB, ONS, BoE, BoJ, StatCan, BoC, ABS, RBA, IBGE, BCB, INSEE, Destatis, ISTAT, INE

Company filings and fundamentals

Region	Source (filing system)	What it provides	Knowledge date
United States	SEC EDGAR: XBRL company facts, Financial Statement Data Sets (needed for IFRS 20-F filers), filing index and text	Point-in-time statements (XBRL from 2009), 10-K risk-factor text, 8-K Item 2.02 earnings times	SEC filing date (acceptance time kept)
Japan	EDINET (FSA) and TDnet (Tokyo Stock Exchange)	Annual and half-year XBRL, large holdings, quarterly earnings releases (kessan tanshin), JPX earnings schedule	EDINET submission time; TDnet publication
EU / EEA	ESEF reports from filings.xbrl.org and national storage mechanisms: DILA / AMF (FR), AFM (NL), FSMA (BE), CNMV (ES), eMarket Storage / 1Info (IT), Nasdaq Nordic, Oslo NewsWeb	Annual reports from FY2020/21	Deposit / publication date (conservative)
United Kingdom	FCA National Storage Mechanism; Companies House (key)	Annual reports, director dealings, UK registry data	Publication date
Korea	OpenDART (key) and DART bulk files	K-IFRS statements, disclosures, insider holdings	Receipt date
Hong Kong	HKEXnews	Announcements, results dates, disclosure of interests	Publication
China	AKShare / BaoStock (CNINFO and exchange sites)	A-share statements and announcements	Imputed where no date is given
Taiwan	MOPS (TWSE)	IFRS iXBRL	Upload time
Brazil	CVM open data	DFP / ITR statements, insider positions	`DT_RECEB` (receipt date)
Mexico, Chile, Peru	BIVA, CMF, SMV	IFRS statements	Filing date
Fallback (any region)	Yahoo fundamentals	Latest 4 to 5 annual periods	**Imputed** (not point-in-time; see below)

Insiders, ownership, short interest and positioning

Dataset	Source	Cadence
US insider trades	SEC Forms 3, 4 and 5 (open-market buys and sells only are scored)	Daily (filing date, never the trade date, is the knowledge date)
Non-US insider trades	FCA (UK), BaFin (DE), AFM (NL), AMF (FR), HKEX (HK), CVM (BR)	Daily; CVM monthly
Institutional holdings	SEC 13F	Quarterly; filed up to 45 days after the quarter (median 42 days), history from 2023-Q2
Activism	SEC 13D / 13G	Daily
US short interest	FINRA	Twice a month, published one to two weeks after the settlement date
Fails-to-deliver	SEC	Twice a month
EU / UK net short positions	FCA, AMF, CNMV, AFM, CBI, FSMA, FI (Sweden), Finanstilsynet (Norway) registers	Daily; Germany and Italy are not available (robots exclusion and captcha)
Futures positioning	CFTC Commitments of Traders (official Public Reporting Environment)	Weekly: Tuesday positions released Friday 15:30 ET; full history from 1986

News, events and text

Source	Use
Massive (Polygon) news	US company news
Curated RSS / Atom feeds (45 verified feeds)	Central banks, statistics offices, regulators, exchanges and selected financial press. Only headline, a snippet of up to 300 characters, URL, publisher, time and language are stored, never article text.
GDELT 2.0 (events, mentions, Global Knowledge Graph)	Event intensity and tone by country and company, escalation, contagion
Common Crawl CC-NEWS	Historical news metadata (headline, short description, URL, publisher); no article bodies
SEC 8-K, TDnet and HKEX announcements	Official corporate news in the story layer
Caldara–Iacoviello GPR index	Geopolitical risk

Sentiment is scored with FinBERT for English and XLM-R for other languages; central-bank tone with a RoBERTa hawkish / dovish classifier plus a lexicon (model-based).

Reference, identity and linkage

- **Identifiers:** SEC company tickers, ESMA FIRDS, GLEIF LEI, OpenFIGI, JPX and KRX listed lists, Shanghai / Shenzhen exchange lists, CVM register. ADR ratios come from a checked seed file.
- **Supply chain and geography:** UN Comtrade, OECD inter-country input-output tables, IMF trade statistics, BIS banking statistics, company filings (rules only, no language model).
- **Maps:** Natural Earth.

Limitations and pitfalls

- Yahoo data has no publication date, so Yahoo fundamentals get an **imputed** knowledge date and are never treated as truly point-in-time.
- Some sources need keys that are not installed on this PC (OpenDART, KRX, Companies House, OpenFIGI and several national macro keys); the jobs skip them with "missing key".

Universes

Route: Screener filters, BQL `members(...)`, Backtest Lab universe selector · **Menu:** Screener / Ranking, BQL console, Backtest Lab

Purpose

A universe is a named list of securities used by screens, scores, models and backtests. Membership is stored with validity dates so that history can be reconstructed.

Methodology

Universe	Contents	Count
`mvp_us`	The 500 largest US-domiciled common stocks by market cap. Excludes ADRs, SPACs, funds and REIT preferreds.	500
`mvp_global`	A curated list of global large caps: Europe, UK, Japan, Korea, China / Hong Kong, Latin America, Asia-Pacific and Canada	271 (248 when measured on 11 Oct 2026)

Universe	Contents	Count
`eu_broad`	European issuers (EU-27, UK, CH, NO, IS) with market cap of at least €300M and liquid prices. Fundamentals from official ESEF or national filing systems where available (54% had an official FY2025 report when built on 12 Oct 2026); the rest use Yahoo.	1,708 issuers
`full`	Everything priced: US above $300M market cap and $5, `eu_broad`, full exchange lists for China A-shares, Taiwan, Japan, Korea, Hong Kong and Brazil, plus `mvp_global`	24,595 securities

On 23 Oct 2026 the System status reported 24,961 securities, 24,943 with prices, 5,022 with fundamentals and **769 with a composite score**. Composite scores, valuation and the machine-learning predictions are computed for the two MVP universes (`mvp_us` and `mvp_global`); the broad universes are mostly price-only outside the US, Japan and Europe.

Benchmarks (indices, ETFs and futures) and delisted US stocks have prices but belong to no universe.

In BQL, `'S&P 500'` maps to `mvp_us` and `'STOXX 600'` to `eu_broad`. These are approximations (top names by market cap, current membership), not the official index constituents, and the query result says so.

Limitations and pitfalls

- See Survivorship and history depth: membership history only starts on 10 Oct 2026.
- `full` mixes very different data depth by region; filter by region or use `mvp_*` for score-based work.

Point-in-time rules

Route: every page (applied in the data layer) · **Menu:** not applicable

Purpose

Point-in-time (PIT) means every figure is used only from the date it became public. This prevents look-ahead bias in screens, backtests and models, and lets you see what was known on a past date (BQL `as_of=`).

Methodology

Knowledge dates. Fundamentals, filings, insider trades, news, GDELT events, macro observations, 13F holdings and short interest all carry a mandatory `knowledge_date` :

Source	Knowledge date
SEC	Filing date (acceptance time stored separately)
EDINET	Submission time
DART	Receipt date
ESEF	Date the filing was added or processed (conservative), or the publication date if present
CVM	Receipt date (`DT_RECEB`)
FRED	ALFRED vintage start date
GDELT	Date added
Form 4	Filing date, never the transaction date

Imputed knowledge dates (flag `knowledge_date_estimated`) apply to sources without a publication date (Yahoo fundamentals, some AKShare data, non-vintage macro):

Case	Lag added after period end
US quarterly	45 days
US annual	75 days
Non-US quarterly / half-year	60 days
Non-US annual	120 days
Macro	The series' usual release lag

As-of rule. For a date D, the app takes, for each item, the row with the latest knowledge date on or before D. In practice:

- fundamentals use the latest period known at D, preferring trailing twelve months (TTM);
- a **restatement** of an older period never overwrites a newer period;
- a restatement of the same period counts only from its own filing date, so the original figure stays visible before that date;
- prices use bars dated on or before D.

Snapshot-only fields. Some fields exist only for the current snapshot (for example the forward P/E `pe_fwd`, the score grade `grade`, `beta_1y`). BQL refuses them with a date range or an earlier `as_of`.

Model labels. Forward returns used to train and test models run from the close of D+1 to the close of D+1+h, so information released after the close on D never leaks into the outcome.

Limitations and pitfalls

- **Sector classification is current only.** A company reclassified last year appears in its new sector for all of history (a small, accepted look-ahead).
- Imputed knowledge dates are conservative estimates; a company that filed early is treated as having filed late.
- The Financials tab shows each figure as it was first known on its filing date; restatements appear as new versions.

Serving snapshot and data-quality gate

Route: System › Overview (status), System › Data health · **Menu:** System

Purpose

The dashboard never reads data while it is being written. Each refresh builds a complete new snapshot, checks it, and switches to it only if it passes. A failed refresh therefore never breaks the dashboard; at worst it shows older data, labelled as such.

Methodology

1. `build_serving_db` writes `data\db\serving_<run id>.duckdb`.
2. `publish` rewrites the pointer file `data\db\CURRENT`. The API checks it at most once a second, sends new requests to the new file and closes the old one after a 120-second grace period. No restart is needed.
3. Housekeeping keeps the last 2 snapshots, so IT can roll back by writing the previous file name into `CURRENT`.

Per-section status. Market data (prices per region, FX, macro, filings, insider, events, news, CFTC) is published even if an analytics job fails. Sections whose inputs failed or are behind are marked in `serving_sections` with their own data date, and the top bar shows "Data stale" or "Analytics stale" (see Data-as-of and freshness indicators).

Data-quality checks (`dq_checks`, run every refresh):

Check	Severity
Schema conformance, primary-key uniqueness	error
Point-in-time violations (knowledge date before period end)	error
FX coverage (a currency without a rate in the window)	error
Freshness beyond the expected lag	warning / error
Null rate more than twice the 30-day baseline	warning
Row count change of more than ±50% vs the previous run	warning
Value ranges (for example a daily return above 80% without a corporate action)	warning
Polygon vs Yahoo close differing by more than 1%	warning
Share of unresolved identities	warning

A failed **error** check blocks publication and the previous snapshot stays live; a **warning** is shown on the System page.

Correct usage and worked example

The data-quality run of 20 Oct 2026 (`dq summary`) reported 151 error-level checks passed and 0 failed, 301 warning-level checks passed and 4 failed, and 264 informational checks passed and 1 failed. Because no error-level check failed, the gate was not blocking and the snapshot was published.

Limitations and pitfalls

- A warning-level failure does not stop publication: read the DQ column on the System page if a number looks implausible.
- The previous snapshot stays live after a blocked refresh; the **DATA AS OF** date then stops moving forward, which is your signal.

Refresh schedule

Route: System › Overview › Settings · refresh schedule; tray › Schedule · **Menu:** System · **Refresh:** this section

Purpose

Data refreshes on a **schedule profile**. The default is **weekly** (one full refresh every Saturday morning), chosen to keep the PC quiet during the week. A **daily** profile is available, and an **intraday** profile is reserved for later.

Screenshot

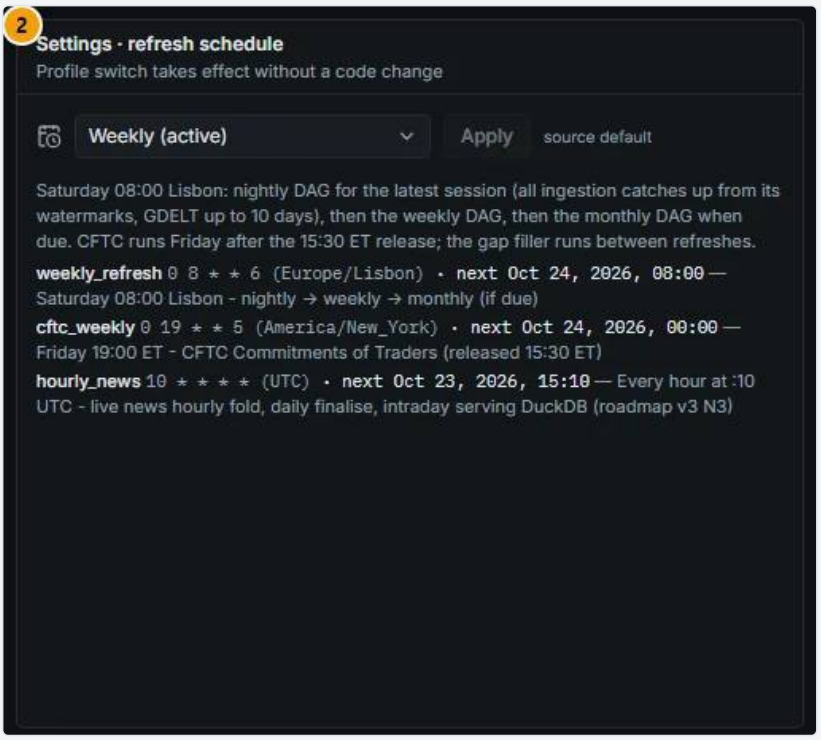


Figure 1.1: System › Overview › Settings · refresh schedule, weekly profile active, next runs as of 23 Oct 2026 (times shown in the browser's time zone).

Controls

Control	What it does
Profile drop-down and **Apply** (System page)	Switches the profile; the worker applies it within seconds without a restart
Tray › **Schedule** ▸ **Weekly / Daily**	Same, from the tray
Tray › **Run refresh now**	Runs the active profile's refresh immediately
`BIG360_SCHEDULE_PROFILE` environment variable (IT)	Overrides the stored choice

Methodology

Profiles (defined in `config\schedule.yaml` ; times in Europe/Lisbon unless noted):

Profile	Entry	When	What runs	Service level
Weekly (default)	`weekly_refresh`	Saturday 08:00	One chain: the nightly job group for the latest session (all ingestion catches up from its watermarks, GDELT up to 10 days back), then the weekly group, then the monthly group if its last run is more than 28 days old	9 days
	`cftc_weekly`	Friday 19:00 New York time	CFTC Commitments of Traders, after the 15:30 ET release	
	`hourly_news`	Every hour at :10 UTC	Folds live news into the store; never wakes the PC, no notification	
Daily	`nightly`	Tuesday to Saturday 01:30	Nightly job group (the previous US session; Asian and European sessions of that date are already closed)	3 days
	`weekly`	Saturday 06:00	Weekly job group (waits for the nightly)	
	`monthly`	First Sunday of the month 08:00	Monthly job group	

Profile	Entry	When	What runs	Service level
	`cftc_weekly`, `hourly_news`	As in weekly		
Intraday (placeholder)		Behaves like daily today	Will later turn on the realtime layer and intraday pulls	

The gap filler runs between refreshes in every profile (see Gap filler).

Job groups (defined in `contracts/jobs.yaml`):

Group	Typical jobs
Nightly (about 100 jobs)	Reference universe and identities; prices for every region, FX, corporate actions; macro (FRED, ECB, Eurostat, international and national sources, release calendar, expectations, surprises, nowcasts, recession and regime models, central-bank rates, curves, projections, tone); SEC filings index, XBRL facts, filing text, Form 4, activism; global filings (EDINET, TDnet, ESEF and national storage mechanisms, FCA, BaFin / AMF / AFM director dealings, HKEX, CVM, BIVA, CMF, SMV, China, OpenDART); EU short positions; GDELT events and knowledge graph; news and sentiment; fundamentals normalisation, ratios and KPIs, valuation, forecast, scoring, technicals (levels, confluence zones, probabilities, factors, breakouts, calibrated ranges, peer groups); ML features and predictions; earnings dates, expectations, reactions and signals; cross-asset signals; thesis and alert evaluation; data-quality checks, snapshot build, regression guard, gap detection
Weekly	SEC bulk XBRL and Financial Statement Data Sets, 13F, FINRA short interest, moat, ML labels and training, model ensemble, level-type backtests, technical probability and meta-model fits, macro betas, exposures and regime fit, conformal peer-group fit, event studies and spillovers, DART bulk, MOPS Taiwan, CVM, Yahoo fundamentals, Companies House, scheduled Model Studio studies and Backtest Lab strategies
Monthly	Universe rebalance, JPX and KRX listed lists, OpenFIGI refresh, supply-chain and geography linkage tables, news story re-clustering, technical factor evaluation
CFTC weekly	CFTC report download and positioning signals
Hourly	Live news hourly fold, daily finalisation, intraday news store
Manual	Backfills and on-demand training, run by the app owner or the gap filler

A full weekly refresh uses all CPU cores for one to three hours on Saturday morning.

Unattended running.

- **Catch-up:** if the PC was asleep or off at the scheduled time, the worker runs the missed refresh at start-up or within 15 minutes, as long as it is not older than the profile's catch-up limit (6 days for weekly, 2 for daily).
- **Keep awake:** the worker keeps the PC awake while a refresh runs and sets a Windows wake timer for the next one. Wake timers must be allowed in the Windows power plan.
- **Retries:** each job retries twice (60 s back-off by default); jobs that time out are cancelled, killed and retried.

Correct usage and worked example

On Friday 23 Oct 2026 the System page showed the weekly profile active (source "default"), with `weekly_refresh` next on Saturday 24 Oct at 08:00 Lisbon and `cftc_weekly` the same evening at 19:00 New York (shown as 00:00 on 24 Oct in a Lisbon browser). The last nightly and weekly runs had completed on 20 Oct (manual catch-up runs). Data dated 19 Oct is therefore what you see until Saturday's refresh publishes the week of 19 to 23 Oct.

If you need more current data for a meeting, switch to **Daily** for the week, or use **Run refresh now** the evening before.

Limitations and pitfalls

- With the weekly profile, prices can be up to a week old by Friday. This is by design; the top bar shows the date.
- The daily profile uses more CPU every night and needs the PC on or able to wake at 01:30.
- `contracts/jobs.yaml` describes the crons as New York time, but the worker interprets them in its scheduler time zone, Europe/Lisbon, as the System page shows.

Gap filler

Route: System › Data health (Gap filler & resource governor, Gap queue, Backfill campaigns) · **Menu:** System › Data health · **Refresh:** continuous, low priority

Purpose

The gap filler (the "Data Ops supervisor", operations roadmap O1) makes the data self-repairing. It knows what data should exist, finds what is missing, stale or partial, and fills it in the background between refreshes using the same code as the scheduled jobs, so backfilled and live data are identical.

Methodology

1. **Coverage registry** (`contracts/coverage.yaml`): for each dataset, the expected cadence (exchange trading calendars, official release calendars, the CFTC Friday schedule, 13F deadlines), the expected entities, the service level, the source, the fill action and its cost.
2. **Gap detector** (job `data_gaps_detect` , after every refresh and once a day): compares expected against actual and classifies each gap as **missing**, **stale**, **partial** (for example a GDELT day with 512 of 576 files), **suspicious** (failed a data-quality rule, unit or FX glitch) or **unfillable** (the source never published it; recorded so the detector stops retrying).
3. **Filler** (`python -m app.ops.filler`): drains the gap queue by priority, with exponential back-off per gap and per-source rate budgets (Polygon about 5 requests per minute, SEC 10 requests per second, OECD 60 downloads per hour, GDELT bandwidth, and so on).
4. **Resource governor** (`config\ops.yaml`):

Setting	Default	Effect
RAM cap	4 GB for the filler's processes	No new task above it; the least urgent task is stopped above 1.5× the cap
Minimum free RAM	3 GB (resume above 3.5 GB)	Pauses every filler task when the PC is short of memory
Concurrency	2 tasks	
CPU priority	Below normal	Your work always comes first
Pause on battery / metered connection	On	Pauses on battery power or a phone hotspot
Minimum free disk	15 GB	Pauses below it
Yield to refreshes	nightly, weekly, monthly	Heavy filler tasks stop while a refresh runs

5. **Campaigns** are long, resumable backfills that report progress and an estimated finish time. On 23 Oct 2026:

Campaign	Status
GDELT 2015-02 to 2026-07 deep backfill	Waiting, 15.9% done
GDELT partial / missing days re-run	Waiting (runs through the gap queue)
13F holdings history (2023-Q2 onwards)	Done
CFTC full history (1986 onwards)	Done
Earnings dates history (Yahoo, JPX, SEC 8-K)	Done
Yahoo price history for unpriced securities	Running, 90.9% (18 securities left)
CC-NEWS historical news metadata	Waiting
SEC 8-K, TDnet and HKEX announcements history	Waiting

Controls

Control	What it does
Tray › **Pause background jobs** / **Resume background jobs**	Pauses the filler (it stops its tasks within about 15 s). Scheduled refreshes still run.
Gap filler state on Data health	Shows *running* or *paused* with the reasons (battery, metered network, low RAM, low disk, refresh running, manual pause)

Limitations and pitfalls

- The filler cannot create data that a source never published; such gaps stay *unfillable*.
- Some gaps are marked *manual*: they need a decision or a key (for example the OpenDART key for Korea) and are listed for the app owner.

Data Health overview

Route: `/system/data-health` · **Menu:** System › Data health · **Refresh:** on load; **Refresh** button

Purpose

Data health answers "what data should the app have, what is missing or stale, and how is it being repaired?" It is the place to go when the top bar shows a warning chip. The page is described in full in System and Data Health; this section gives an overview.

Screenshot

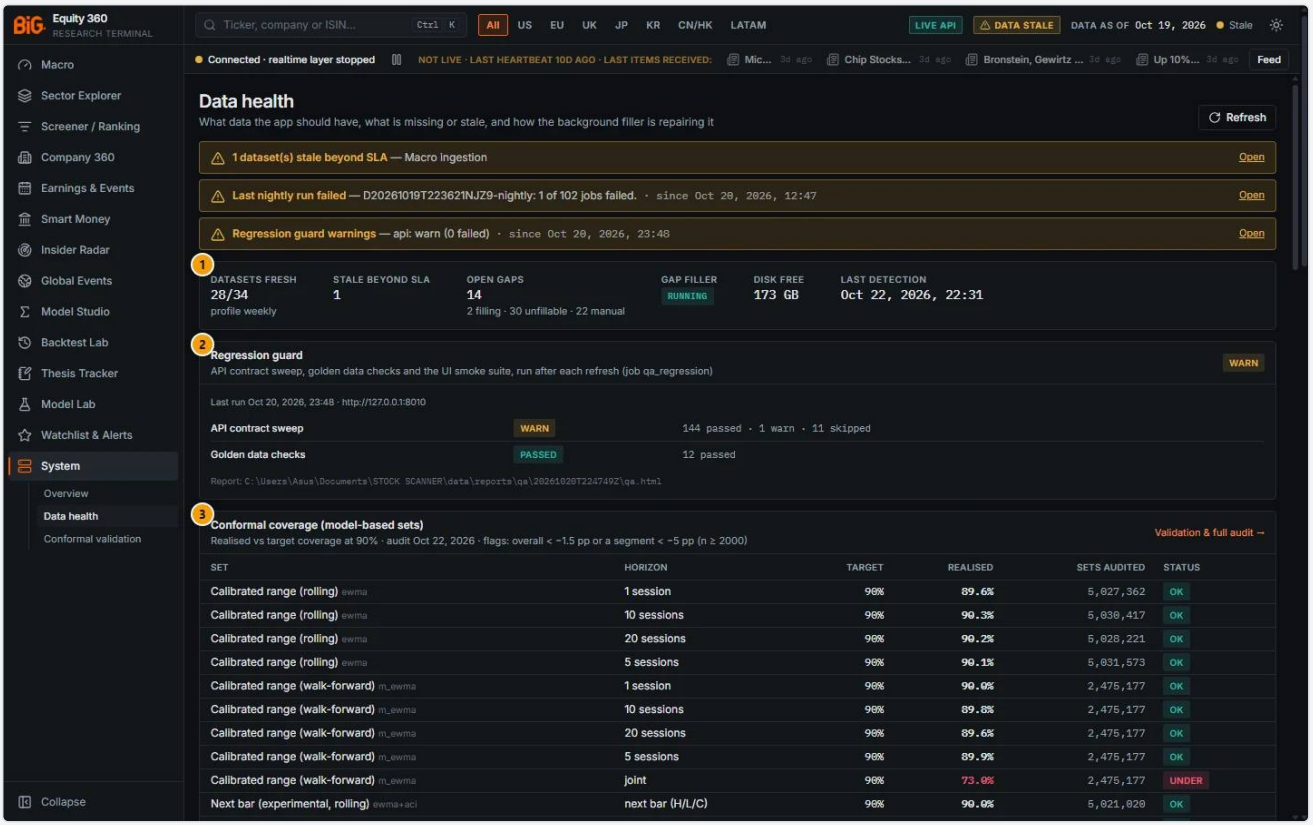


Figure 1.2: Data health (`/system/data-health`), 23 Oct 2026, with three open alerts.

#	Panel	What it shows
1	Summary tiles	Datasets fresh (28 of 34), stale beyond service level (1), open gaps (14: 2 filling, 30 unfillable, 22 manual), gap filler state (RUNNING), free disk (173 GB), last gap detection
2	Regression guard	The automatic checks run after each refresh: API contract sweep (144 passed, 1 warning, 11 skipped) and golden data checks (12 passed)
3	Conformal coverage (model-based sets)	Realised vs target coverage of the calibrated ranges at 90% (for example 89.6% at 1 session); flags any set under target
4	Freshness heatmap	Coverage per day (actual vs expected entities) for each dataset over the last 21 days
5	Backfill campaigns	Progress and estimated finish of the long backfills listed above
6	Gap filler & resource governor	Filler process, heartbeat, running tasks and the governor's readings (free RAM, power, network, disk)
7	Gap queue	Every tracked gap by class (missing, stale, partial, unfillable, suspicious) and status
8	Sources	Health of each source over 14 days (ok, degraded, failing, idle) and rate-budget use
9	SLA breaches & recent failures	Datasets beyond their service level and failed job runs of the last 7 days, with error messages

The banners above the tiles repeat the operational alerts (here "1 dataset(s) stale beyond SLA — Macro ingestion", "Last nightly run failed … 1 of 102 jobs failed", "Regression guard warnings"), each with an **Open** link.

Coverage by region and known gaps

Route: System › Overview › Data sources (per region view) · **Menu:** System

Purpose

Coverage is not uniform. The US is the deepest; some markets have prices but few fundamentals. Check this table before comparing companies across regions.

Methodology

Region	Prices (as of in the 22 Oct snapshot)	Fundamentals	Other gaps
United States	Polygon + Yahoo (16 Oct)	Deepest: SEC XBRL, point-in-time, 10 to 15 years; about 488 of the 500 `mvp_us` names had recent fundamentals	US earnings dates are Yahoo estimates ("Provider est."), as there is no free official confirmed-date source
Europe	Yahoo (19 Oct)	ESEF from FY2020/21 via filings.xbrl.org and national storage mechanisms. No free official source for Germany and Switzerland (Yahoo only). No national adapter for PL, GR, PT, LU, RO, HU, CZ. Spain partly covered.	German and Italian short positions not available
United Kingdom	Yahoo (19 Oct)	ESEF via FCA NSM and filings.xbrl.org	Companies House key not installed
Japan	Yahoo (13 Oct, stale in this snapshot)	EDINET and TDnet; EDINET history was short (about FY2025 onwards) when last measured on 11 Oct 2026	
Korea	Yahoo (19 Oct)	Thin: OpenDART key not installed (15 issuers had FY2025 statements on 19 Oct)	KRX key not installed
China A-shares	BaoStock / Yahoo (19 Oct)	Thin (21 issuers with FY2025 statements on 19 Oct); AKShare / BaoStock	BaoStock IP block: several hundred Shanghai names on Yahoo only
Hong Kong	Yahoo (16 Oct)	Thin	
Taiwan	TWSE / TPEx	Very thin (MOPS: 2 issuers with FY2025 statements on 19 Oct)	Taiwan dollar FX weekly
India, Australia, Singapore	Yahoo	None or Yahoo only	
Latin America	B3 + Yahoo (16 Oct)	Good for Brazil (CVM); Mexico, Chile and Peru from BIVA, CMF and SMV	
Canada	Yahoo (19 Oct)	SEC EDGAR for 40-F filers, else Yahoo	

On 19 Oct 2026 (audit), FY2025 or later fundamentals by source were: SEC EDGAR 2,686 issuers, Yahoo 1,073, ESEF 679, TDnet 240, AKShare / BaoStock 21, DART 15, MOPS 2.

Other known gaps:

- **History depth.** GDELT history is being backfilled (15.9% of the 2015 to 2026 target done), news sentiment starts around August 2026, 13F starts in 2023-Q2.
- **13F mapping.** Part of the 13F rows cannot be mapped to a security without an OpenFIGI batch or key (81% of 2025-26 value mapped).
- **FINRA short interest.** Symbol resolution is partial; free float is approximated by shares outstanding.
- **Macro.** No euro investment-grade credit spread from an official free source; some BIS daily policy series lag by weeks.
- **Realtime.** The live layer is off unless live news is switched on.
- **Consensus and options.** Not available: no free source.

Open data issues are tracked in `docs/PHASE3_TODO.md`.

Limitations and pitfalls

- A low peer percentile for a Korean or Chinese company may reflect missing data rather than a weak company. Check the factor coverage shown next to each score.
- Japanese prices in this snapshot are as of 13 Oct while US prices are as of 16 Oct and European prices 19 Oct: cross-region return comparisons over short windows are not like for like.

Survivorship and history depth

Route: Backtest Lab, Model Lab, Screener history · **Menu:** not applicable

Purpose

Survivorship bias arises when a backtest uses today's list of companies, which excludes those that later failed or were taken over.

Methodology

- **Universe membership history starts on 10 Oct 2026.** From that date, `universe_membership` keeps delisted securities, so results are survivorship-free going forward.
- **Before 10 Oct 2026, backtests and model training use today's company lists.** This flatters them, because failures are missing. Delisted US stocks have prices but are not in any universe.
- Prices go back to 2000 where the source has them. Fundamentals go back 10 to 15 years in the US (XBRL since 2009) and typically 5 to 10 years elsewhere (ESEF from FY2020/21).

Limitations and pitfalls

- Treat any backtest that starts before 10 Oct 2026 as an upper bound. The Backtest Lab and Model Lab repeat this warning; see Backtest Lab and Model Lab.

Licensing caveats

Route: not applicable · **Menu:** not applicable · **Source:** `docs/it/DATA_LICENSING.md`

Purpose

The app was built for one analyst's research. Several sources allow that but not wider use inside a bank (display to several employees, feeding internal systems or decisions). BiG Legal / Compliance and the market-data team must confirm each source before wider use. The assessment below is an engineering view, not legal advice.

Methodology

Status	Sources	Action before bank use
Fine with attribution	SEC EDGAR (filings, insiders, 13F, fails-to-deliver); FRED public-domain series, ECB, Eurostat, OECD, IMF, World Bank, BIS, national statistics offices and central banks; CFTC; EU / UK short registers; GDELT (with citation); EDINET, OpenDART, CVM, Companies House, filings.xbrl.org, French DILA, Dutch, Belgian and Spanish registers; Taiwan open data; GLEIF, ESMA FIRDS, OpenFIGI, Natural Earth	Add an attribution notice; set `SEC_USER_AGENT` to a bank contact mailbox
Needs review	Massive (Polygon): free and individual plans are for personal use, internal display needs a business plan; exchange websites (TDnet, HKEX, KRX, JPX, Shanghai / Shenzhen lists, Nasdaq Nordic, Oslo, B3); FCA NSM and other undocumented endpoints; FINRA short interest; Common Crawl news metadata; UN Comtrade free tier; GPR index; Loughran–McDonald word list; FinBERT and XLM-R model licences	Confirm terms or buy the relevant licence
Not licensed for this use	**Yahoo Finance / yfinance** (non-US prices, benchmarks, futures curves, earnings dates, fallback fundamentals); AKShare and BaoStock (China); Google News RSS (already off); mindicador.cl; the archived ICE BofA credit-spread series copied from FRED	Switch off and replace with a licensed vendor or the official source; delete or license the ICE archive

A **vendor adapter interface** lets a licensed feed (Bloomberg, LSEG, FactSet or the bank's own market-data service) replace Yahoo for prices and fundamentals without changing anything downstream. A file-drop adapter for vendor end-of-day files already works; the others are skeletons. See `docs/it/DATA_LICENSING.md` §4.

Limitations and pitfalls

- Until the review is done, do not distribute screenshots, exports or numbers from the app outside the research team.
- Removing Yahoo without a replacement would remove most non-US prices.

Related sections

- Getting started for the top-bar freshness indicators.
- System and Data Health for the operational pages.
- Methodology appendix for how each model uses this data.
- Troubleshooting and FAQ for stale data and failed refreshes.
- For IT: `docs/it/ARCHITECTURE.md`, `docs/it/RUNBOOK.md`, `docs/it/DATA_LICENSING.md`, `docs/CONFIGURATION.md`.

2A. Macro

The Macro section covers the world economy, from the cockpit overview down to individual official series and central banks. This chapter covers the cockpit, the release calendar, GDP nowcasts, the theme dashboards, the 24 country pages, the series explorer and the central-bank monitor. The macro-to-markets pages (Markets, Instruments, Cross-asset, Positioning) are in 2B. Markets and cross-asset.

Three rules apply to every page in this chapter. They are policy decisions recorded in `docs/MACRO_V2_PLAN.md` and `docs/MACRO_V3_PLAN.md`:

- **Official sources only.** Every input comes from a central bank, a statistical office, a Treasury, a Federal Reserve bank, the OECD, the BIS, the IMF, the World Bank or Eurostat. FRED and ALFRED (St. Louis Fed) count as official. Scraped commercial sources (for example AKShare) are excluded from every macro calculation.
- **Expectations are our model, not a consensus.** The app has no consensus feed. Every "expected" value comes from the app's own forecast and carries the label **Model-based expectation (not consensus)**. Surprises are measured against that forecast.
- **Approximations are allowed but labelled.** For example, central-bank implied paths are built from official government curves, not from OIS or futures. The label says so wherever a path appears.

All macro pages share the **macro sections strip** under the page title: Cockpit · Calendar · Nowcast · Themes · Countries · Series explorer · Central banks · Markets · Instruments · Cross-asset · Positioning (CFTC). The **Search macro series** box at its right edge searches the whole macro catalogue from any of these pages (see Series explorer).

> **Note:** The screenshots in this chapter were taken on 23 Oct 2026 against the snapshot whose data runs to 19 Oct 2026, which is why the header shows **DATA STALE**. Live values will differ.

Macro cockpit

Route: `/macro` · **Menu:** Macro › Cockpit · **Mnemonics:** none open it directly (`ECO` opens the calendar, `ECST` a country page) · **Refresh:** nightly DAG (weekly profile by default: Saturday 08:00 Lisbon; daily profile: Tue–Sat 01:30 Lisbon); the Latest releases tiles poll every 2 minutes

Purpose

The cockpit answers four questions on one screen: where each of the 24 covered economies stands against its own history, where its cycle is heading, how large recession risk is under several independent methods, and what has just been released or is about to be. Portfolio managers use it as the morning macro check, and analysts use it as the jumping-off point into country pages, series and the calendar. `/macro` is also the app's start page: the root URL redirects here.

Screenshot

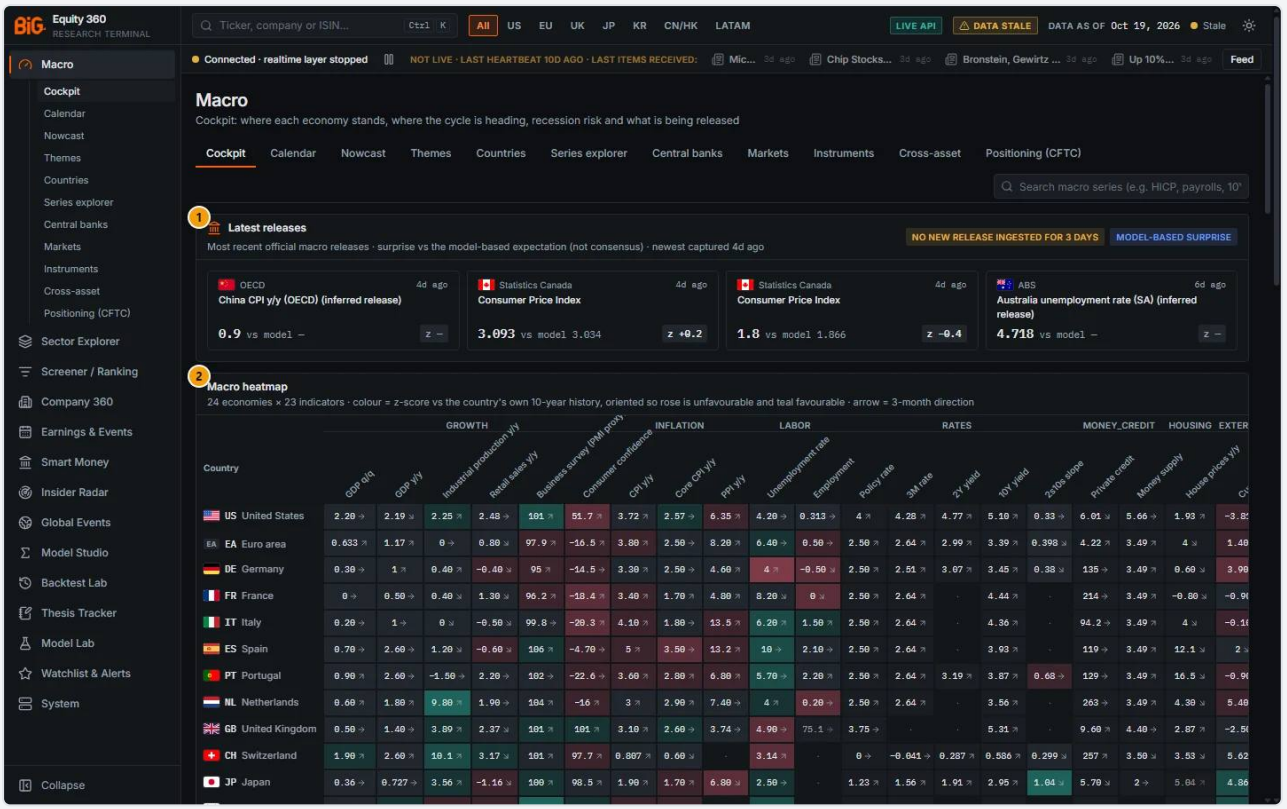


Figure 2A.1: Macro cockpit (`/macro`), data as of 19 Oct 2026.

#	Panel	What it shows
1	Latest releases	The four most recent official releases captured, each showing the actual, the model expectation ("vs model") and the surprise z. Neutral chip when z is between −1 and +1, teal or rose otherwise
2	Macro heatmap	24 economies × 23 indicators, coloured by the z-score against each country's own 10-year history and oriented so that teal is favourable and rose unfavourable
3	Cycle clock	Each country placed by growth momentum against inflation momentum, with 12-month trails
4	Surprise index	Exponentially weighted index of releases against our model expectations, one bar per country plus GLOBAL
5	Recession risk	Official and model-based recession indicators side by side, each labelled
6	Today and the next 7 days	Upcoming tier 1–2 releases from official calendars, with the model expectation and its 80% band

Panels 3 to 6 are below the fold:

Figure 2A.2: Cycle clock. 23 countries placed (Expansion 10, Slowdown 7, Recovery 3, Contraction 3); TW not placed.

Figure 2A.3: Surprise index, theme "All". IN +111 at the top, CN −85 at the bottom; US +21; GLOBAL +13.

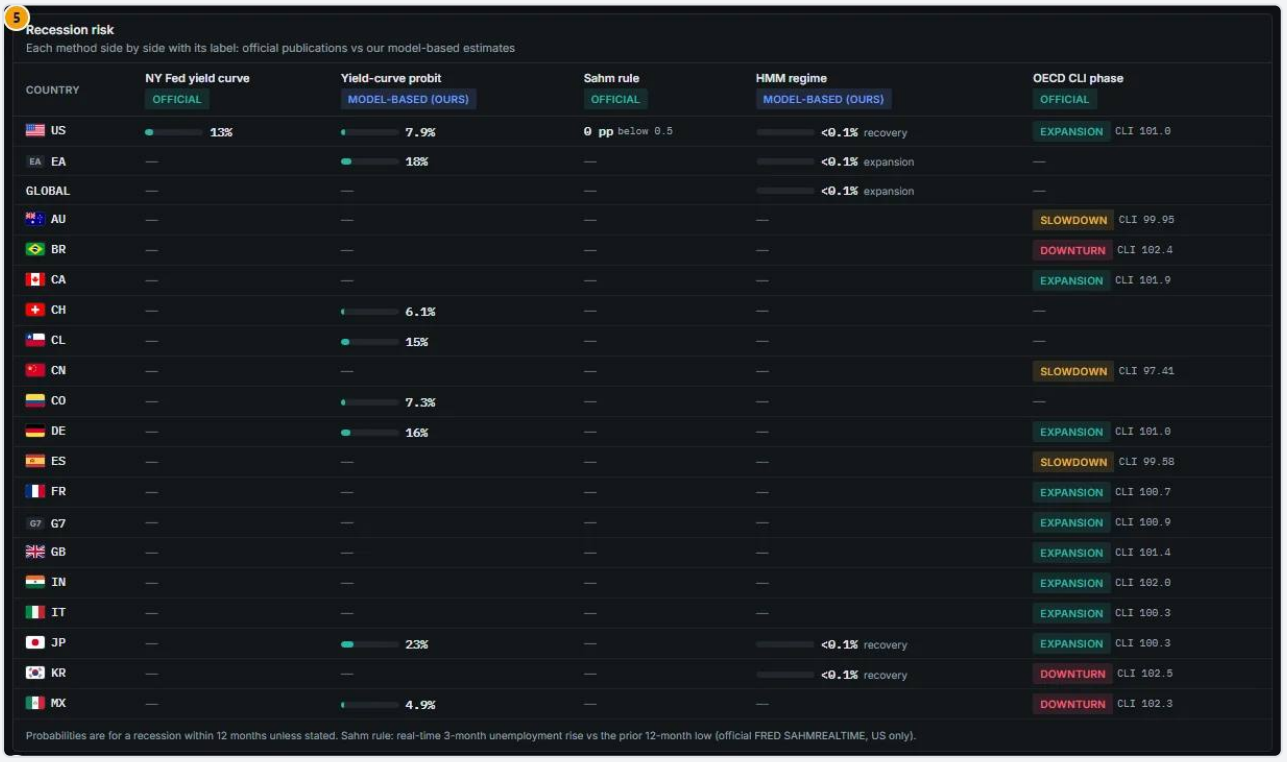


Figure 2A.4: Recession risk. Official methods carry a green OFFICIAL badge and ours a blue MODEL-BASED (OURS) badge.

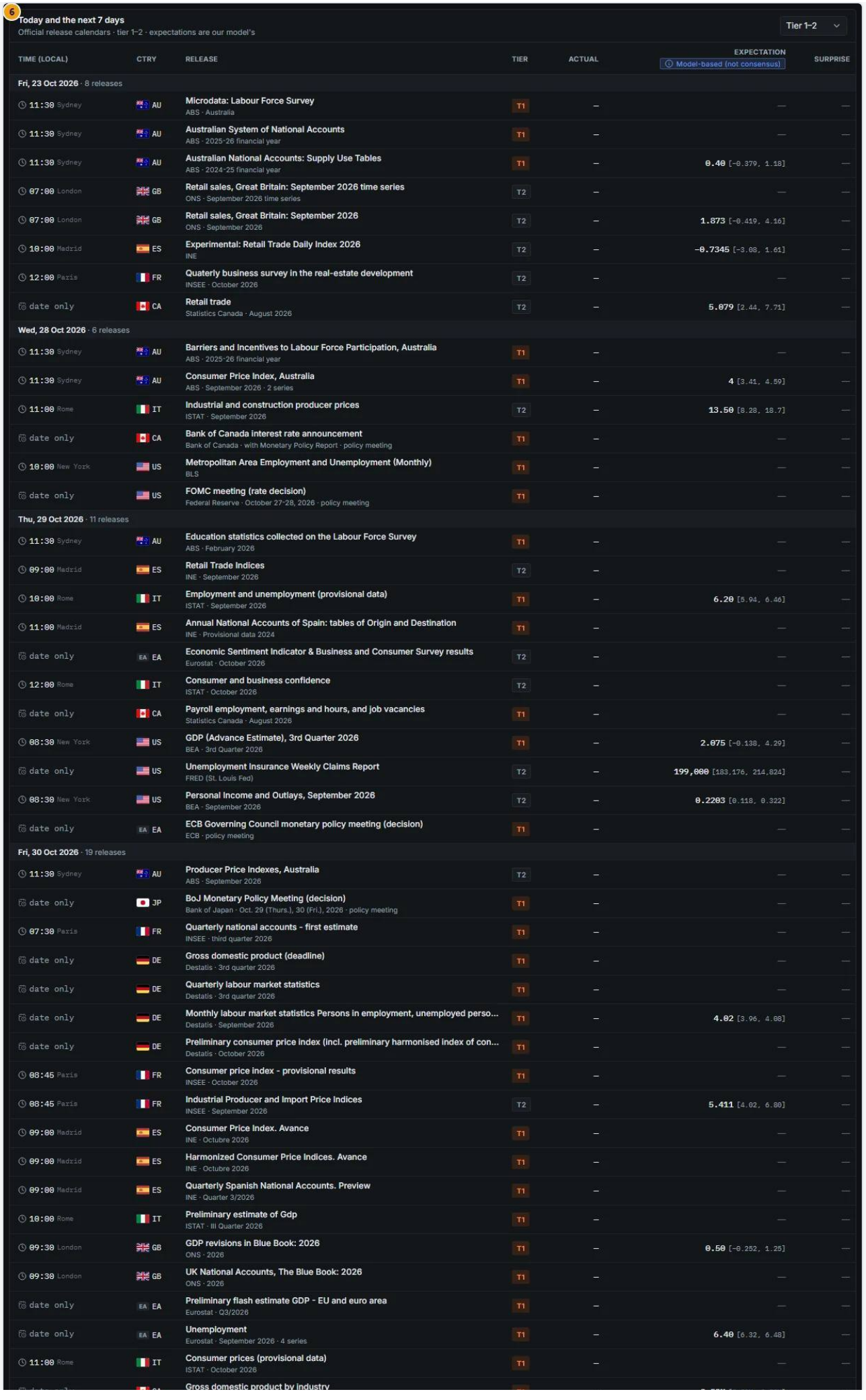

6
Today and the next 7 days
Official release calendars · tier 1–2 · expectations are our model's
Tier 1–2
TIME (LOCAL)
CTRY
RELEASE
TIER
ACTUAL
EXPECTATION
Model-based (not consensus)
SURPRISE
Fri, 23 Oct 2026 · 8 releases
11:30 Sydney AU Microdata: Labour Force Survey ABS · Australia T1 – — —
11:30 Sydney AU Australian System of National Accounts ABS · 2025-26 financial year T1 – — —
11:30 Sydney AU Australian National Accounts: Supply Use Tables ABS · 2024-25 financial year T1 – 0.40 [-0.379, 1.18] —
07:00 London GB Retail sales, Great Britain: September 2026 time series ONS · September 2026 time series T2 – — —
07:00 London GB Retail sales, Great Britain: September 2026 ONS · September 2026 T2 – 1.873 [-0.419, 4.16] —
10:00 Madrid ES Experimental: Retail Trade Daily Index 2026 INE T2 – -0.7345 [-3.08, 1.61] —
12:00 Paris FR Quaterly business survey in the real-estate development INSEE · October 2026 T2 – — —
date only CA Retail trade Statistics Canada · August 2026 T2 – 5.079 [2.44, 7.71] —
Wed, 28 Oct 2026 · 6 releases
11:30 Sydney AU Barriers and Incentives to Labour Force Participation, Australia ABS · 2025-26 financial year T1 – — —
11:30 Sydney AU Consumer Price Index, Australia ABS · September 2026 · 2 series T1 – 4 [3.41, 4.59] —
11:00 Rome IT Industrial and construction producer prices ISTAT · September 2026 T2 – 13.50 [8.28, 18.7] —
date only CA Bank of Canada interest rate announcement Bank of Canada · with Monetary Policy Report · policy meeting T1 – — —
10:00 New York US Metropolitan Area Employment and Unemployment (Monthly) BLS T1 – — —
date only US FOMC meeting (rate decision) Federal Reserve · October 27-28, 2026 · policy meeting T1 – — —
Thu, 29 Oct 2026 · 11 releases
11:30 Sydney AU Education statistics collected on the Labour Force Survey ABS · February 2026 T1 – — —
09:00 Madrid ES Retail Trade Indices INE · September 2026 T2 – — —
10:00 Rome IT Employment and unemployment (provisional data) ISTAT · September 2026 T1 – 6.20 [5.94, 6.46] —
11:00 Madrid ES Annual National Accounts of Spain: tables of Origin and Destination INE · Provisional data 2024 T1 – — —
date only EA Economic Sentiment Indicator & Business and Consumer Survey results Eurostat · October 2026 T2 – — —
12:00 Rome IT Consumer and business confidence ISTAT · October 2026 T2 – — —
date only CA Payroll employment, earnings and hours, and job vacancies Statistics Canada · August 2026 T1 – — —
08:30 New York US GDP (Advance Estimate), 3rd Quarter 2026 BEA · 3rd Quarter 2026 T1 – 2.075 [-0.138, 4.29] —
date only US Unemployment Insurance Weekly Claims Report FRED (St. Louis Fed) T2 – 199,000 [183,176, 214,824] —
08:30 New York US Personal Income and Outlays, September 2026 BEA · September 2026 T2 – 0.2203 [0.118, 0.322] —
date only EA ECB Governing Council monetary policy meeting (decision) ECB · policy meeting T1 – — —
Fri, 30 Oct 2026 · 19 releases
11:30 Sydney AU Producer Price Indexes, Australia ABS · September 2026 T2 – — —
date only JP BoJ Monetary Policy Meeting (decision) Bank of Japan · Oct. 29 (Thurs.), 30 (Fri.), 2026 · policy meeting T1 – — —
07:30 Paris FR Quarterly national accounts - first estimate INSEE · third quarter 2026 T1 – — —
date only DE Gross domestic product (deadline) Destatis · 3rd quarter 2026 T1 – — —
date only DE Quarterly labour market statistics Destatis · 3rd quarter 2026 T1 – — —
date only DE Monthly labour market statistics Persons in employment, unemployed perso... Destatis · September 2026 T1 – 4.02 [3.96, 4.08] —
date only DE Preliminary consumer price index (incl. preliminary harmonised index of con... Destatis · October 2026 T1 – — —
08:45 Paris FR Consumer price index - provisional results INSEE · October 2026 T1 – — —
08:45 Paris FR Industrial Producer and Import Price Indices INSEE · September 2026 T2 – 5.411 [4.02, 6.80] —
09:00 Madrid ES Consumer Price Index. Avance INE · Octubre 2026 T1 – — —
09:00 Madrid ES Harmonized Consumer Price Indices. Avance INE · Octubre 2026 T1 – — —
09:00 Madrid ES Quarterly Spanish National Accounts. Preview INE · Quarter 3/2026 T1 – — —
10:00 Rome IT Preliminary estimate of Gdp ISTAT · III Quarter 2026 T1 – — —
09:30 London GB GDP revisions in Blue Book: 2026 ONS · 2026 T1 – 0.50 [-0.252, 1.25] —
09:30 London GB UK National Accounts, The Blue Book: 2026 ONS · 2026 T1 – — —
date only EA Preliminary flash estimate GDP - EU and euro area Eurostat · Q3/2026 T1 – — —
date only EA Unemployment Eurostat · September 2026 · 4 series T1 – 6.40 [6.32, 6.48] —
11:00 Rome IT Consumer prices (provisional data) ISTAT · October 2026 T1 – — —
Gross domestic product by industry

Figure 2A.5: Today and the next 7 days, Tier 1–2 (23–30 Oct 2026).

Controls

Control	What it does
Macro sections strip	Moves between the macro pages. The active page is underlined
Search macro series (top right)	Type at least 2 characters to search the macro catalogue (12 best matches with flag, series id, frequency and last date). A click opens the series in the Series explorer. Esc clears the box
Latest releases tile	Click to open the country page of the release (the cockpit itself when no country is known). The **live** badge marks items streamed by the realtime layer. The z chip's tooltip reads "Surprise z vs the model-based expectation"
"no new release ingested for N days" badge	Appears when the newest captured release is more than 3 days old, so that "no new data" is not mistaken for "nothing was released"
Heatmap column header (hover)	Shows the indicator name and "higher is better", "lower is better" or "neutral (no good/bad direction)"
Heatmap country name	Opens the country page (`/macro/country/:cc`)
Heatmap cell (hover)	Tooltip with country, indicator and series title; value, units and period; release date (marked "(estimated)" when inferred) and status; 3/6/12-month changes; z (10y) and percentile; a favourable / unfavourable / neutral verdict; first release → revised value for series with vintages; source and series id with an **official** tag; a link to the official source page
Heatmap cell (click)	Opens the series drawer (see below)
Trails switch (cycle clock)	Shows or hides each country's 12-month trail
Cycle clock dot (hover / click)	Hover lists the growth and inflation momentum, every input (as the z of its 6-month change) and the as-of date. Click opens the country page
"method cycle_v1" (hover)	Shows the full method text of the cycle clock
All / Growth / Inflation / Labor (surprise index)	Theme filter: recomputes the bars from that theme's releases only
Surprise bar (hover / click)	Hover shows the index value, the number of releases in the 365-day window and the date of the last release. Click opens the country page (GLOBAL is not clickable)
Recession method header (hover)	The method's full label, e.g. "Yield-curve probit 3m10y, 12m ahead (model-based re-estimate, ours)"
Recession country code	Opens the country page (for two-letter codes; G7 and GLOBAL are aggregates)
Maximum tier (next 7 days)	Tier 1 only / Tier 1–2 (default) / All tiers
Release row (next 7 days)	Opens the release drawer (see Release calendar)

Series drawer. Clicking a heatmap cell, a country tile or a theme source line opens a side drawer for that series. It contains a **Transform** control (Canonical / Level / y/y / m/m / z-score), a 10-year chart (US series are shaded for NBER recessions), the **Provenance** box, and two buttons: **Open in series explorer** and **Open in Model Studio** (see Series explorer). Close it with the × or Esc.

Methodology

HEATMAP CELLS AND Z-SCORES

A *cell* is one (country, indicator) pair, e.g. (DE, CPI y/y). For each cell the app picks a **primary series** from the official candidates in the macro catalogue. The ranking is freshness first (fresh, then stale, then discontinued), then catalogue rank, then latest observation. Euro-area members (DE, FR, IT, ES, PT, NL) inherit the EA policy rate, 3M rate and money supply cells, marked "[euro area]".

Each cell is shown on its **canonical measure**:

- **y/y** for growth-rate indicators published as an index or level (CPI, core CPI, PPI, GDP y/y, industrial production, retail sales, house prices, money, credit, employment, REER): $100 \times (x_t / x_{t-12m} - 1)$.
- **q/q** for GDP q/q from an index: $100 \times (x_t / x_{t-3m} - 1)$.
- **level** for anything already published as a rate or percentage, and for diffusion and confidence indices.
- The **2s10s slope** is derived as `10Y yield − 2Y yield` in percentage points.

Statistics are computed on the canonical measure over the **10 years ending at the latest period** (at least 8 points):

```
z_10y          = (value − mean10y) / std10y          (sample std, ddof 1)
percentile_10y = share of window values ≤ value
change_k       = value − canonical value k months earlier (k = 3, 6, 12)
direction      = sign of the 3-month change (falls back to 6m, then 12m);
                 'flat' when |change| ≤ 0.1 × window std
```

The arrow in each cell is the 3-month direction (↗ up, ↘ down, → flat).

COLOUR ORIENTATION

The colour is based on an **oriented** z, so that teal always means "economically favourable" and rose "unfavourable":

```
oriented z = z_10y × sign(good_direction)
bucket     = clip(round(1.5 × oriented z), −3, +3)
```

good_direction	Indicators
+1 (higher is better)	GDP q/q and y/y, industrial production, retail sales, business survey (PMI proxy), consumer confidence, employment, 2s10s slope, current account % GDP, fiscal balance % GDP
−1 (lower is better)	CPI y/y, core CPI y/y, PPI y/y, unemployment rate, government debt % GDP
0 (neutral, grey)	Policy rate, 3M rate, 2Y and 10Y yields, private credit, money supply, house prices y/y, real effective FX y/y

Bucket 0 ($|\text{oriented z}| < 0.33$) is the "normal" grey. Buckets ±1, ±2 and ±3 use increasingly strong teal or rose tints. Bucket ±3 starts at $|\text{oriented z}| \approx 1.67$; the legend labels it "≥ +2σ" / "≤ −2σ". Neutral indicators never take a colour, however extreme their z. A faded cell is **stale**: its series has missed its expected update given its frequency and usual publication lag. A dot (·) means no official series is available for that cell.

POINT IN TIME

The heatmap is served from a daily **point-in-time snapshot**: each snapshot uses only observations known on its date (latest vintage per period as known then). For series with vintages (ALFRED real-time rows for the US, or first-seen revisions recorded at ingest elsewhere), the cell also stores the canonical value **as first published** and the revision since then. The footer shows the snapshot date, coverage and method version (`cells_v1`).

CYCLE CLOCK (METHOD `CYCLE_V1`)

1. **Inputs.** Growth: GDP y/y (GDP q/q when there is no y/y cell), industrial production, retail sales and the PMI proxy (OECD CLI/BCI, EC ESI, Tankan and similar). Inflation: CPI y/y, core CPI y/y and PPI y/y. Each input is

the cell's canonical measure from its primary series. If that history is shorter than 42 months, the longest current candidate series of the same cell is used instead.

2. **Monthly grid.** Monthly inputs are carried forward at most 3 months, quarterly 6 and annual 12, so a discontinued input drops out instead of freezing the clock.
3. **Momentum.** $m_t = x_t - x_{t-6}$ (6-month change).
4. **Standardisation.** $z_t = m_t / sd_t$, where sd_t is the standard deviation of m over the trailing 120 months (at least 36), and z_t is clipped to ±3.
5. **Composites.** Growth momentum G is the mean of the available growth z's; inflation momentum I is the same for the inflation inputs. Both are smoothed with a 3-month moving average.
6. **Quadrants**, from the signs of the smoothed composites:

Quadrant	Rule	Clock label
Expansion	G > 0, I > 0	growth ↑ inflation ↑
Recovery	G > 0, I ≤ 0	growth ↑ inflation ↓
Slowdown	G ≤ 0, I > 0	growth ↓ inflation ↑
Contraction	G ≤ 0, I ≤ 0	growth ↓ inflation ↓

The clock measures *momentum* (acceleration or deceleration against the country's own volatility), not levels. A country with weak but improving growth sits in a "growth ↑" quadrant. The history uses **latest revised** data, not first releases, and is recomputed nightly. The trail is the last 12 monthly points. Countries without growth inputs are listed as "not placed" (TW in Figure 2A.2).

SURPRISE INDEX

Each released value is compared with the primary **model-based expectation** made before the release (see Release calendar):

```
error       = actual − model expectation
scale       = std of the series' past forecast errors (last 36 releases, ≥ 6)
              else std of past release-to-release changes (≥ 6)
surprise_z  = clip(error / scale, −5, +5)
signed_z    = surprise_z × direction
```

`direction` is +1 for every series except **unemployment and jobless claims (−1)**. In the code's words, a positive `signed_z` means "higher = stronger growth / hotter inflation / tighter labour".

The index for each country, GLOBAL, and theme (all, growth, inflation, labour) is computed daily:

```
index_d = 100 × Σ w_i · signed_z_i / max(Σ w_i, 1)
w_i     = tier weight (T1 1.0, T2 0.6, T3 0.3) × 0.5^(age_i / 45 days)
```

It sums over releases in the last 365 days, which gives a 45-day half-life. The `max(Σw, 1)` floor pulls the index towards 0 when there is little recent information. As a scale reference, +100 corresponds roughly to recent releases averaging one standard deviation above our model.

> **Caution:** The surprise index is signed by *economic strength*, not by welfare. An inflation print above our model gives a **positive** (teal) bar, even though the panel note says "teal = positive surprises for the economy". Read the inflation theme as "hotter than expected", not "better than expected".

RECESSION RISK

Column	Method	Label	Horizon
NY Fed yield curve	The New York Fed's published probability: a probit of an NBER recession 12 months ahead on the 10y − 3m Treasury spread. Stored at the spread month	official	12 months
Yield-curve probit	Our re-estimate of the same probit per country, $P(rec_{(t+12)}) = \Phi(a + b \cdot spread_{(t)})$ on monthly 10y − 3m official rates. Recession = NBER (FRED USREC) for the US; elsewhere 2+ consecutive negative quarters of official real GDP q/q. Needs ≥ 2 recession episodes and ≥ 10 years of spreads	model-based (ours), **in-sample fit**	12 months
Sahm rule	FRED SAHMREALTIME (real-time Sahm rule, St. Louis Fed). Triggered at ≥ 0.5 pp. US only	official	current
HMM regime	P(contraction) from our 4-state Gaussian HMM on monthly macro factors: forward-filtered (no look-ahead), refitted weekly, filtered nightly. The regime label is shown next to it	model-based (ours)	current state
OECD CLI phase	The official OECD CLI (amplitude adjusted) with our turning-point rule: expansion = ≥ 100 and rising; downturn = ≥ 100 and falling; recovery = < 100 and rising; slowdown = < 100 and falling. Turning points are local extremes within ±6 months, confirmed after 6 months	official CLI; rule ours	current phase

Probability bars turn amber at ≥ 30% and rose at ≥ 50%. The panel footnote, "Probabilities are for a recession within 12 months unless stated", applies to the two yield-curve columns. The HMM value is a probability of the *current* regime.

Data sources and refresh

- Cells: the official series in `contracts/macro_catalog.yaml`, i.e. FRED/ALFRED, ECB Data Portal, Eurostat, ONS, BoE, BoJ, national statistical offices and central banks, OECD, BIS, IMF and the World Bank. Job `macro_cells` runs nightly and writes today's point-in-time snapshot. The first run backfilled 36 month-end snapshots.
- Cycle clock: job `macro_cycle` (nightly), recomputed from the cell history.
- Surprise index: job `macro_surprise` (nightly), after `macro_calendar` and `macro_expectations`.
- Recession panel: job `macro_recession` (nightly). It downloads the NY Fed `allmonth.xls`, FRED SAHMREALTIME and the OECD CLI, and reads the HMM output from `macro_regime_nowcast`, which is refitted weekly by `macro_regime_fit`.
- Latest releases: the realtime layer (`/live/macro`), polled every 2 minutes, with streamed items merged in when the realtime supervisor is running. Otherwise the panel shows the latest releases from the nightly snapshot.
- Coverage on 16 Oct 2026: 447 of 552 possible cells (TW 8, HK 12, PE 13; US, EA, DE, PT and CA complete).

Validation status

- **Heatmap:** a descriptive statistic; no model and no gate.
- **Cycle clock:** a rule-based composite (`cycle_v1`). It has not been formally validated as a predictor of returns or of turning points.
- **Surprise index:** its inputs are model-based expectations (see Release calendar). The bar chart's caption carries the API label "Surprises vs model-based expectations (not consensus)". Not formally validated as a signal.
- **Recession panel:** each column carries its own label and the panel never blends them. Our yield-curve probit is an **in-sample** fit and is labelled so in the API (`in_sample: true`). The October 2026 audit (`docs/AUDIT_2026-10-19.md`, section 4) calls the HMM over-confident (a 99.99% "Expansion" posterior on the Markets overview). Read the HMM column's "<0.1%" accordingly: the model rarely expresses uncertainty.
- **Latest releases staleness:** audit finding F41 ("does not say no new releases ingested since ...") is fixed: the panel now shows "newest captured ... ago" and the warning badge.

Correct usage and worked example

Use the heatmap to find **where** an economy is unusual, then click through to the series before drawing conclusions. A z of +1.5 on a series that was stable for 10 years is a bigger event than the same z on a volatile one, and the tooltip's 10-year percentile helps here.

Worked example (snapshot of 16 Oct 2026, US row):

- **CPI y/y** is 3.717% for September 2026 (released 14 Oct), with z (10y) +0.20 and percentile 76. Because lower inflation is better, the oriented z is −0.20, which rounds to bucket 0, so the cell is **grey (normal)**. The 10-year window contains the 2021–23 inflation spike, which is why a 3.7% print is unremarkable against this history.
- **Consumer confidence** (UMich, `fred:UMCSENT`) is 51.7 for August, with z −1.50 and percentile 6. Higher is better, so the oriented z is −1.50 and the bucket is round(−2.25) = −2: a **medium rose** cell, even though the arrow points up (+6.9 over 3 months).
- On the JP row, the policy rate cell shows z +3.9 but stays grey, because rates are a neutral indicator.

Cycle clock: the US sits in **Expansion** with growth momentum +0.20 and inflation momentum +0.30 (as of 1 Oct 2026). Its inputs are industrial production +0.17, retail sales +0.30, PMI proxy +0.33; CPI −0.04, core CPI −0.21, PPI +0.28.

Surprise index (theme All, 19 Oct 2026): US **+21** from 210 releases in the window, GLOBAL **+13** from 1,808 releases. Under the orientation above, part of the US reading comes from hotter-than-modelled inflation prints (see the CPI example under Release calendar).

Recession panel, US row: NY Fed 13% (official), our probit 7.9%, Sahm 0 pp (below 0.5), HMM <0.1% (regime "recovery"), OECD CLI 101.0 (expansion).

Limitations and pitfalls

- **z is relative to the country's own decade.** It does not compare countries. A 4% unemployment rate can be rose in one country and teal in another.
- **Stale cells** keep their last value and are only faded. Check the release date in the tooltip before treating a cell as current.
- Euro-area member cells for the policy rate, 3M rate and money supply are EA values, not national ones.
- The cycle clock uses **revised** history and 3-month smoothing: it lags turning points, and past positions on the trail can move when data are revised.
- The **surprise index measures surprise against our model, not against the market's consensus.** A large positive reading can simply mean our time-series model under-predicts a trending series. It is not a measure of what markets were positioned for.
- Recession probabilities are "calibrated over history" statements for the method in question, not "X% chance today". Our probit is fitted in-sample, and outside the US its recession definition is a technical one (2+ negative quarters).
- When the realtime layer is stopped (as in the screenshots), the Latest releases tiles can be days old. The amber badge tells you so.

Related mnemonics and BQL fields

- `ECO` (calendar), `ECST <cc>` (country page), `WIRP` (central banks). See 14. BQL and mnemonics.
- No BQL fields expose macro series directly. The `macro.beta_*` fields are security exposures to macro factors and are covered in 2B and 5C.
- Macro alert rules `macro_release`, `macro_surprise` (|z| above a threshold) and `regime_change` (cycle quadrant change, recession probability crossing a level): see 13. Watchlists and alerts.

Release calendar

Route: `/macro/calendar` · **Menu:** Macro › Calendar · **Mnemonics:** `ECO` , `ECO <cc>` · **Refresh:** nightly (`macro_calendar` , `macro_expectations` , `macro_surprise`)

Purpose

The release calendar lists scheduled data releases and central-bank policy decisions from **official publisher calendars**. For each release it shows the actual (once published), our **model-based expectation** with its 80% interval, the prior and its revision, and the surprise z. Use it to prepare for the week, to check whether a print was out of line with a simple statistical forecast, and to open the full surprise history of a release.

Screenshot

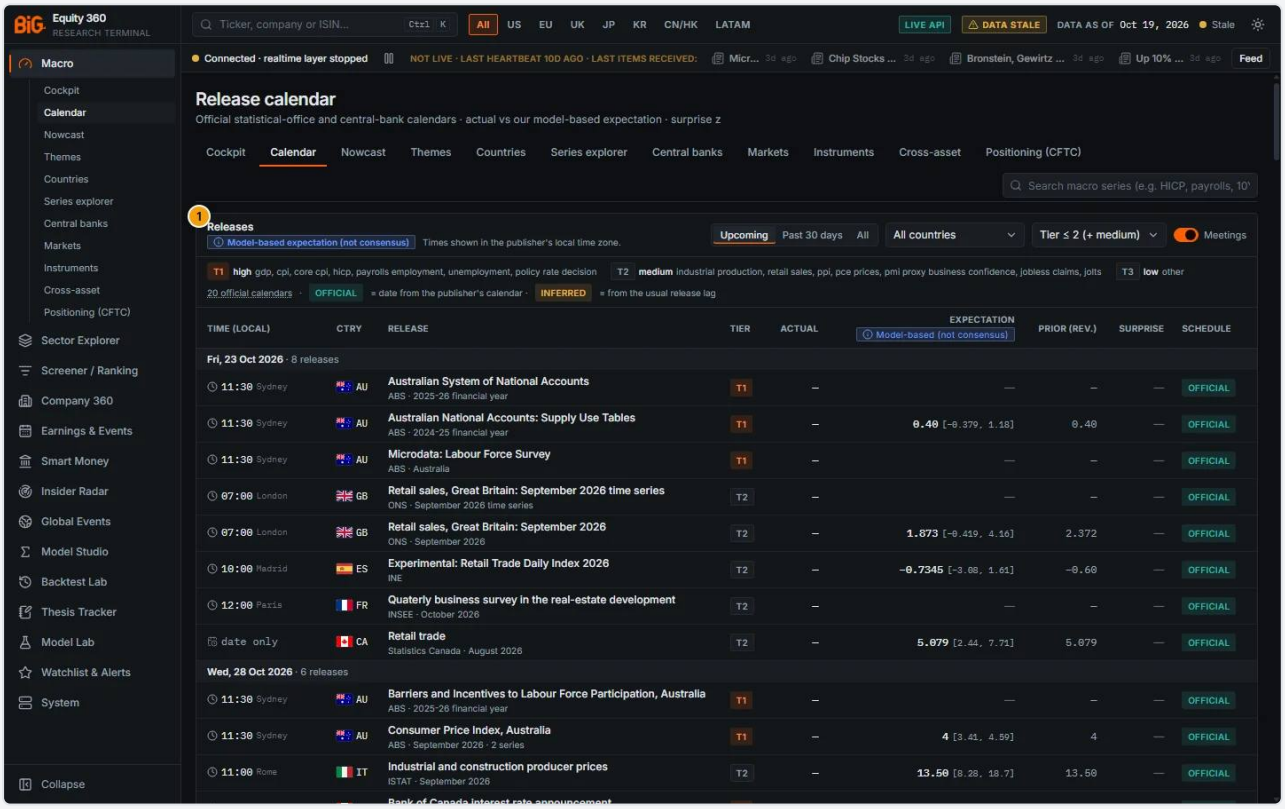


Figure 2A.6: Release calendar (`/macro/calendar`), Upcoming, all countries, Tier ≤ 2, meetings on.

#	Panel	What it shows
1	Releases	The tier legend, the official-calendar count and the schedule badge legend, then releases grouped by day. Columns: Time (local), Ctry, Release, Tier, Actual, Expectation (model-based, with 80% band), Prior (rev.), Surprise, Schedule

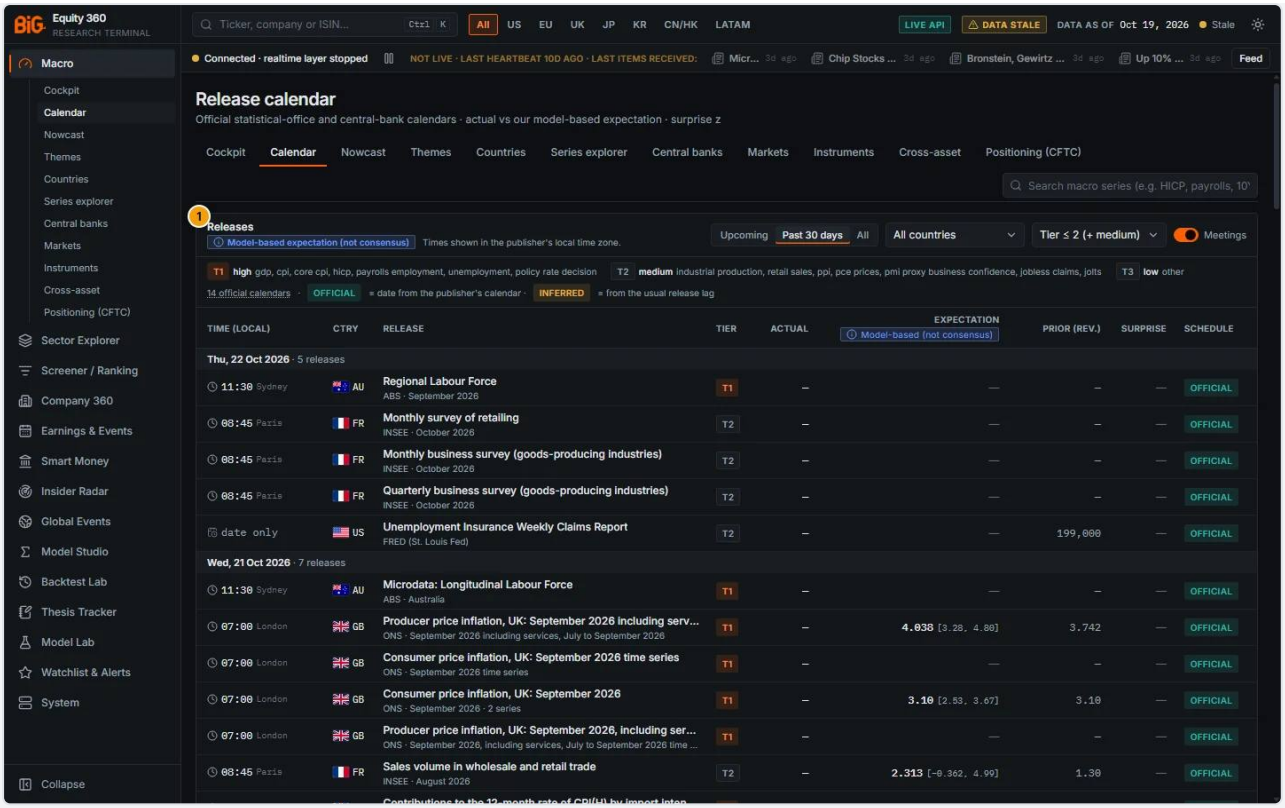


Figure 2A.7: Release calendar, Past 30 days (most recent day first; 14 official calendars contribute to this window). The top rows (21–22 Oct) fall after the 19 Oct data snapshot, so their actuals are still blank while priors and model expectations are filled in.

#	Panel	What it shows
1	Releases	The same table as Figure 2A.6 in reverse date order. Actuals and surprise chips appear once the release has been ingested

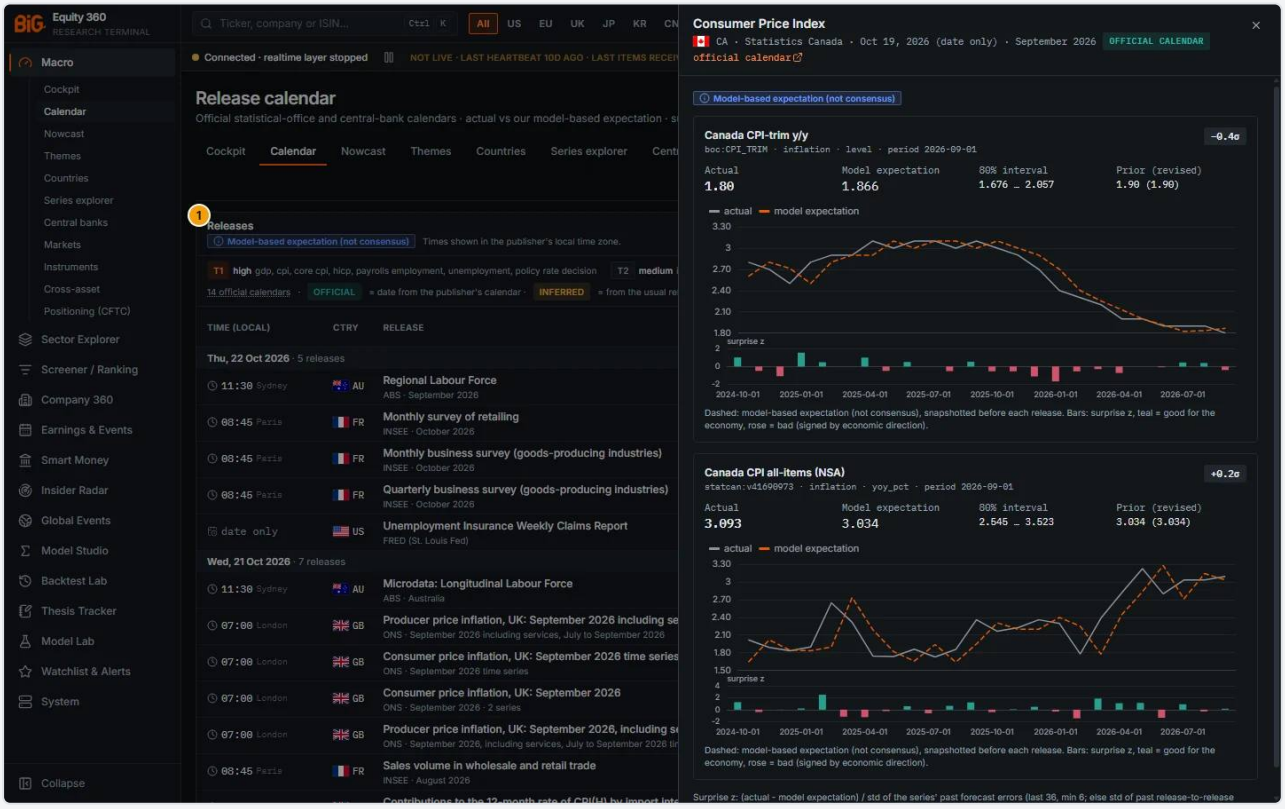


Figure 2A.8: Release drawer for Statistics Canada's Consumer Price Index (19 Oct 2026, date only, official calendar). CPI-trim y/y: actual 1.80, model 1.866 (80% interval 1.676 to 2.057), surprise −0.4σ. CPI all-items y/y:

actual 3.093, model 3.034 (2.545 to 3.523), surprise +0.2σ. Each card has the history of actual against the dashed model expectation, with surprise-z bars underneath.

#	Panel	What it shows
1	Releases	The calendar table, behind the open drawer

Controls

Control	What it does
Upcoming / Past 30 days / All	Upcoming = today to +45 days; Past 30 days = −30 days to today (most recent day first); All = −30 to +45 days
Country	All countries, or one of the 24 coverage countries
Maximum tier	Tier 1 (high) / Tier ≤ 2 (+ medium) (default) / Tier ≤ 3 (+ low)
Meetings switch	Includes or hides central-bank policy meetings
"N official calendars" (hover)	Lists the calendar sources contributing to the loaded window (20 in Figure 2A.6)
Model-based expectation tag (hover)	"Our own forecast (ARIMA/SARIMA with order search, seasonal-naive fallback, nowcast-informed where available), snapshotted before the release. It is not a market consensus."
Expectation value (hover)	The model label, model and specification (e.g. `ARIMA(2, 1, 1)`), version, 80% and 95% intervals, when the forecast was made (UTC), the data cut-off, and "point-in-time backfill" for historical rows
Surprise chip (hover)	"(actual − model expectation) / std of past forecast errors; colour = signed by economic direction"
Time icon	A clock means the publisher states an exact time, shown in the publisher's time zone (e.g. "11:30 Sydney"). A calendar icon with "date only" means only the date is published
Schedule badge	**official** = date from the publisher's calendar; **inferred** = reconstructed from the usual release lag
Release row (click)	Opens the **release drawer**

Release drawer. The drawer has a header with country, publisher, date and local time, reference period, an official-calendar or inferred-date badge, and a link to the official calendar. It repeats the model-based tag, then shows one card per linked series. Each card has the title (click to open the series drawer), series id, theme, measure and period, the surprise chip, and four figures: **Actual**, **Model expectation**, **80% interval** and **Prior (revised)**. Below them is a history chart of actual against the dashed model expectation for past releases, with surprise-z bars underneath, coloured by economic direction. The footer states the surprise method. A policy meeting with no linked series shows "No linked series".

Methodology

CALENDAR AND TIERS

- **Sources.** FRED/ALFRED release dates for US releases (BLS CPI, Employment Situation, PPI, JOLTS; BEA GDP and personal income; Census retail sales, housing starts and durable goods; Fed G.17; DOL claims; UMich). FRED supplies dates only, so their time is the publisher's documented release time. One parser per official calendar covers BLS, BEA, FOMC, Eurostat, ECB, ONS, BoE, BoJ, StatCan, BoC, ABS, RBA, IBGE, BCB, INSEE, Destatis, ISTAT and INE. A failed parser is skipped rather than failing the job.
- **Window.** The past 3 years plus 90 days ahead.
- **Point in time.** The actual of a release is the **first vintage** of that period (ALFRED `realtime_start` for US vintage series, the real-time knowledge date elsewhere), not today's revised value. A past release that no official calendar covered gets an **inferred** row dated at the period's first knowledge date, so that surprise history is complete. Inferred rows are never shown as upcoming, and the calendar API excludes them unless asked.
- **Importance tiers.** Tier 1 = GDP, CPI / core CPI / HICP, payrolls / employment, unemployment, policy-rate decisions. Tier 2 = industrial production, retail sales, PPI, PCE prices, PMI proxies / business confidence,

jobless claims, JOLTS. Tier 3 = everything else. A release linked to several series takes the best tier among them.

MODEL-BASED EXPECTATIONS (JOB `MACRO_EXPECTATIONS`, VERSION `V2`)

For every release in the next 14 days and every linked series, the job snapshots a forecast of the value about to be published, **strictly before the release**:

Model	How	Interval
`sarima`	SARIMA / ARIMA on the release measure (e.g. m/m % for US CPI). Bounded AIC order search (p, q ≤ 2; P, Q ≤ 1 for seasonal series; d chosen by ADF/KPSS, max 1), at most 16 orders, fitted with Model Studio's SARIMAX fitter on the last ≤ 15 years (monthly: 180 obs). The order is re-searched at most once a year	Gaussian 80% and 95% from the forecast standard error
`seasonal_naive` / `naive`	The value one season earlier (seasonal series) or the last value	Empirical spread of past same-horizon naive errors
`dfm_nowcast`	For GDP (US, EA, GB, JP): our DynamicFactorMQ nowcast for the target quarter (see Nowcast)	From the nowcast

- **Primary model.** One model per release is marked primary, and that is the one the calendar shows and the surprise uses. The choice is data-driven and re-evaluated quarterly. Among the candidate models, the primary is the one with the lowest RMSE over the last 12 releases dated before the start of the release's quarter on which every candidate had a forecast (at least 4). Otherwise the default ranking applies (SARIMA, then DFM nowcast, then naive).
- **Pandemic mask.** Observations from March 2020 to September 2021 are treated as missing when fitting.
- **Point in time and immutability.** Training data is the as-of series known before the release. `forecast_made_at` is always before `scheduled_at`: live forecasts are made at run time, while backfilled ones are dated 12 hours before the release using data known through the previous day. Stored rows are never rewritten.

SURPRISE Z

The surprise uses the same formula as the cockpit's surprise index: `surprise_z = clip((actual − expectation) / scale, −5, 5)`, where `scale` is the std of up to 36 past forecast errors (at least 6). The chip's colour uses `signed_z` (sign flipped for unemployment and claims), with buckets `clip(round(signed_z), −3, 3)`. The expectation must target the released period, or no surprise is computed.

Data sources and refresh

- Calendars: FRED release dates plus the official publisher calendars above. Job `macro_calendar` runs nightly; the plan also schedules a weekly horizon extension.
- Expectations: job `macro_expectations` (nightly, releases in the next 14 days). Historical expectations come from the manual, resumable `macro_expectations_backfill` job (ALFRED vintages / real-time knowledge dates).
- Surprises: job `macro_surprise` (nightly).
- An actual appears in the calendar after the next nightly run that ingests the new observation. Under the default weekly profile that can be up to a week later; the realtime layer (Latest releases tiles) can be faster when it is running.

Validation status

- Expectations are **model-based**, labelled "Model-based expectation (not consensus)" on every view, and are **not consensus**.
- Backtest result recorded in `docs/MACRO_V2_PLAN.md` (W2.2): 9,683 model-based expectations for 171 series, a point-in-time backfill of 9,595 rows, and "SARIMA beats naive on 120/168 series". This is the only accuracy statistic recorded. There is no formal pass/fail gate, and the app shows no accuracy figure on the page.

- The primary-model rule (trailing out-of-sample RMSE, quarterly) is itself a guard against using a model that is worse than the naive benchmark for that series.

Correct usage and worked example

Use the expectation as a **statistical baseline**: "what would a simple time-series model have expected?". Do not use it as a market forecast. The 80% interval is the more useful part: a print outside it is unusual relative to the series' own dynamics.

Worked example: US CPI, released 14 Oct 2026 (event `RE408e2d546e72d7cd`):

Series	Measure	Actual	Model expectation	80% interval	Model	Surprise z
CPI all items (`fred:CPIAUCSL`)	m/m %	0.649	0.355	0.074 to 0.636	ARIMA(2, 1, 1)	+1.33
Core CPI (`fred:CPILFESL`)	m/m %	0.335	0.235	0.129 to 0.341	ARIMA(0, 1, 2)	+0.91

Headline CPI printed just **above** the model's 80% band, giving a +1.33σ surprise; its chip is teal because higher inflation counts as positive (see the caution under Macro cockpit). By contrast, the US unemployment rate on 2 Oct 2026 printed 4.2% against a naive expectation of 4.1% (80% band 3.94 to 4.26). The surprise z is +0.89 but the signed z is −0.89, so its chip is rose. For a print inside the band, see Canada's CPI on 19 Oct 2026 (Figure 2A.8): both series landed inside their 80% intervals, and the surprises were small (−0.4σ and +0.2σ).

To see one country's week, open the calendar and choose the country in the **Country** list. Then click the tier-1 rows to read each release's surprise history in the drawer.

Limitations and pitfalls

- **Not consensus.** A "surprise" here is a deviation from our model. Market reactions depend on the consensus survey, which the app does not have.
- `ECO US` opens `/macro/calendar?country=US` , but the page does **not** read the `country` parameter: it opens on "All countries". Choose the country in the Country list.
- Upcoming rows from FRED-sourced US releases may show "date only". Their time is the publisher's documented time and is not confirmed for the specific date.
- Only the first linked series of a release fills the row's Actual / Expectation / Surprise columns. Open the drawer for multi-series releases ("· N series").
- Many upcoming rows have no expectation (—): only releases linked to a tracked series get one, and only within 14 days of the release.
- Policy meetings carry no expectation. For what is priced, use Central banks overview.

Related mnemonics and BQL fields

- `ECO` opens the calendar. `ECO <cc>` is meant to pre-select a country (see the limitation above).
- Alert rules `macro_release` and `macro_surprise` : 13. Watchlists and alerts.
- Company earnings expectations use the same "model-based, not consensus" convention: 6. Earnings and events.

Nowcast

Route: `/macro/nowcast` (`?cc=US|EA|GB|JP`) · **Menu:** Macro › Nowcast · **Mnemonics:** none · **Refresh:** nightly (`macro_nowcast` , re-estimated only when the input panel changed)

Purpose

The Nowcast page tracks our model-based estimate of the current and next quarter's real GDP growth, how it has moved as data arrived, and which releases moved it. For the US it shows the Atlanta Fed **GDPNow** and the New York Fed **Staff Nowcast** next to ours as official reference points. It answers "is incoming data pointing up or down for this quarter?".

Screenshot

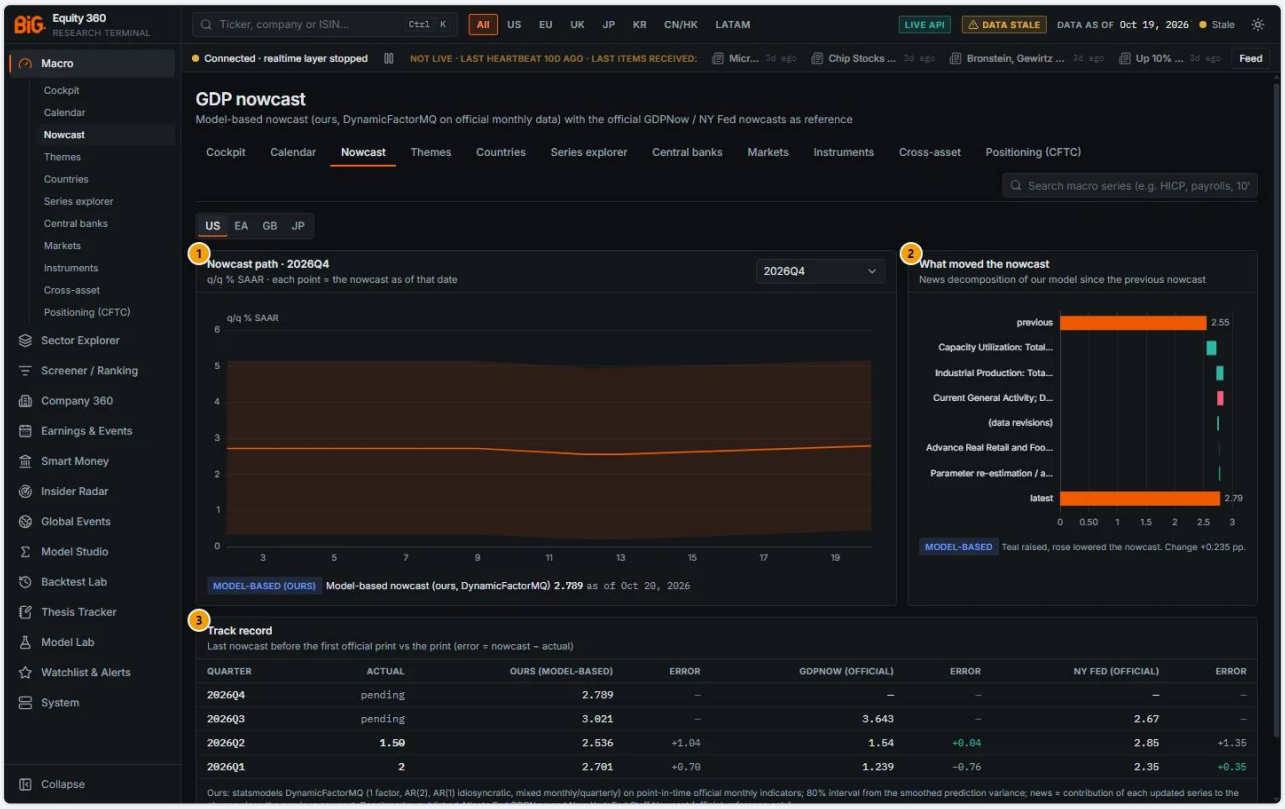


Figure 2A.9: GDP nowcast (`/macro/nowcast`), US, target 2026Q4, nowcast as of 20 Oct 2026.

#	Panel	What it shows
1	Nowcast path · 2026Q4	Our nowcast for the chosen quarter at each as-of date, with its 80% band, the official nowcasts (dashed) and the first print once released. Below: each model's latest value, badged official or model-based (ours)
2	What moved the nowcast	Waterfall from the previous nowcast to the latest: contribution of each updated series, data revisions, and parameter re-estimation
3	Track record	For each quarter, the last nowcast before the first official print against that print, with errors, for ours, GDPNow and the NY Fed. Followed by the method text and the input panel

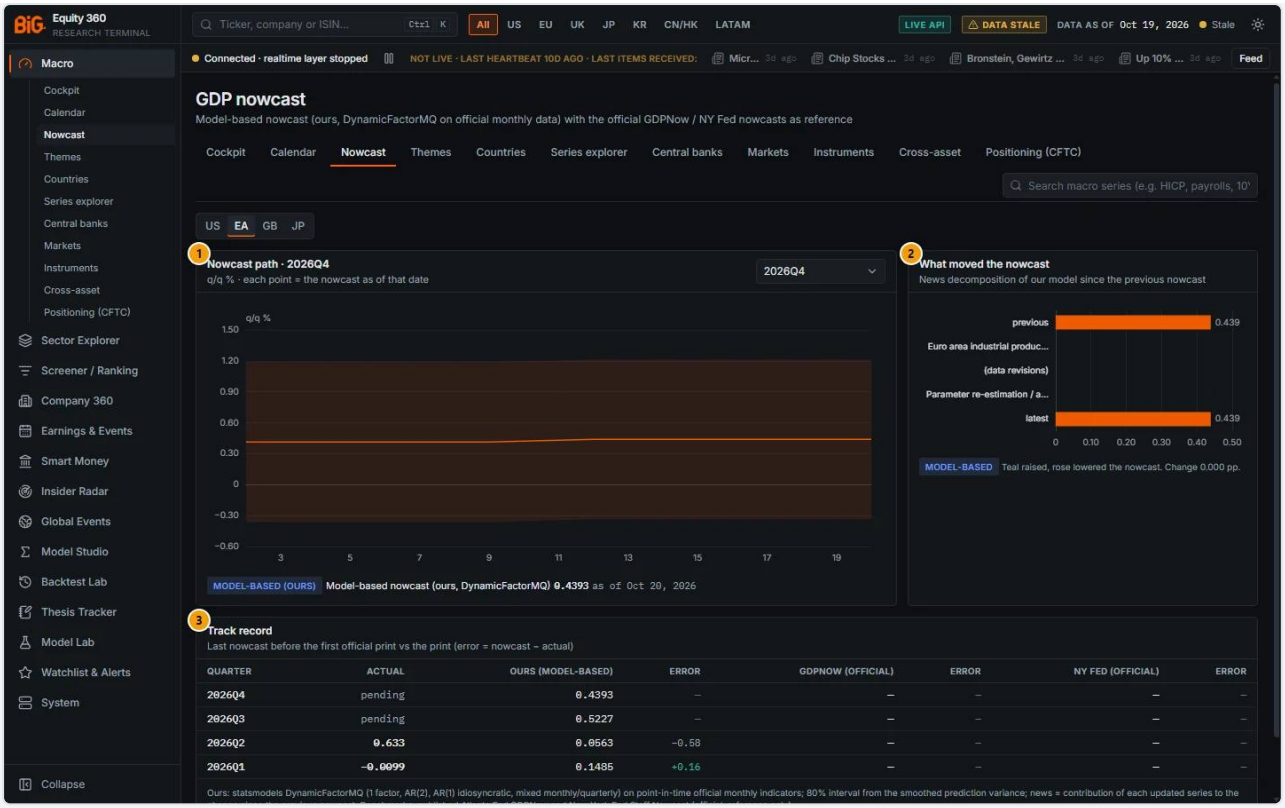


Figure 2A.10: GDP nowcast for the euro area (`/macro/nowcast?cc=EA`), target 2026Q4: 0.4393% q/q (not annualised), as of 20 Oct 2026. There are no official benchmarks, so the GDPNow and NY Fed columns are empty.

#	Panel	What it shows
1	Nowcast path · 2026Q4	Our EA nowcast (q/q %) with its 80% band; flat through October
2	What moved the nowcast	Previous 0.439 → latest 0.439: euro-area industrial production, data revisions and parameter re-estimation net to 0.000 pp
3	Track record	2026Q2: actual 0.633, ours 0.0563 (error −0.58). 2026Q1: actual −0.0099, ours 0.1485 (+0.16)

Controls

Control	What it does
US / EA / GB / JP	Country; written to the URL as `?cc=` (any other value falls back to US)
Target quarter	Every quarter from the first one without a published GDP up to the current quarter, plus released ones marked "(released)"
Model chip (hover)	The model's source (e.g. the official workbook URL)
Waterfall bar (hover)	Contribution in pp and the series id
Track-record error colour	Green when the absolute error is below 0.5 pp, rose above 1.5 pp

Methodology

Model (`dfm_mq` , model-based). statsmodels `DynamicFactorMQ` , which mixes monthly and quarterly data (Mariano–Murasawa aggregation, Bańbura–Modugno EM). It uses 1 global factor with AR(2) dynamics, AR(1) idiosyncratic errors and standardised inputs. The sample starts in January 2000. Pandemic months (March–December 2020; 2020Q2–Q4) are masked as missing.

Point in time. Inputs are as known on each nowcast date: ALFRED vintages for US vintage series, real-time knowledge dates elsewhere.

Panels (official sources only).

Country	Target	Monthly inputs
US	Real GDP (GDPC1), q/q % SAAR	IP, payrolls, real retail sales, unemployment, real PCE, real income ex transfers, housing starts, durable goods orders, initial claims (monthly mean), capacity utilisation, real manufacturing and trade sales, UMich sentiment, Philadelphia and Dallas Fed manufacturing surveys, OECD CLI (15 series)
EA	Real GDP q/q % (ECB MNA)	IP (Eurostat), retail y/y, unemployment, EC ESI / industry / consumer confidence, OECD BCI
GB	Real GDP q/q % (ONS IHYQ)	IP, retail volumes, monthly GDP index, unemployment, employment rate, OECD BCI/CCI
JP	Real GDP q/q % (OECD QNA)	IP y/y, retail y/y, unemployment (OECD KEI), OECD BCI/CCI/CLI

Interval. 80%, from the smoothed prediction variance (including idiosyncratic noise).

News decomposition. The latest results are compared with the previous vintage using the current parameters for both. This gives the contribution of each updated series, plus a "(data revisions)" item. The effect of re-estimating the parameters is reported as its own item ("Parameter re-estimation / ..."), so the bars add up exactly to `latest − previous`.

Track record. The last nowcast of each model **before** the first official print, against that print: `error = nowcast − actual`.

Official benchmarks. These are downloaded verbatim and labelled official, for reference only: Atlanta Fed GDPNow (`GDPTrackingModelDataAndForecasts.xlsx`) and the NY Fed Staff Nowcast (download workbook; 80% bands of concluded quarters). They exist for the US only.

Which countries. Models exist only for **US, EA, GB and JP**. Any other country returns 404 (e.g. DE, CN, BR), and the country pages of those countries show upcoming releases instead of a nowcast.

Data sources and refresh

Job `macro_nowcast` runs nightly but re-estimates only when the input panel's fingerprint has changed (a new observation or revision). The nowcast history is weekly and point-in-time since July 2023. The GDPNow and NY Fed workbooks are downloaded on the same job. The US first print comes from BEA via FRED.

Validation status

Not formally validated; no gate. The app shows the evidence in the **Track record** panel. The October 2026 audit (F27) states: "For 2026Q2 our nowcast error is +1.04 pp vs GDPNow +0.04 pp." It recommends showing the track record more prominently "as it currently understates how weak the model is". The model chips are labelled **model-based (ours)**, and the page title says "Model-based nowcast (ours, DynamicFactorMQ ...)".

Correct usage and worked example

Use the nowcast **direction** and the news waterfall ("payrolls and IP lifted it, sentiment lowered it") as a summary of incoming data. For the US, prefer the official GDPNow and NY Fed values as the level estimate, and treat ours as a cross-check. Always read the 80% band: for the US it is roughly ±2.3 pp wide.

Worked example (US, 20 Oct 2026, Figure 2A.9):

- **2026Q4**: ours 2.789% SAAR (80% band 0.45 to 5.17). The waterfall shows the change from the previous nowcast of 2.55: capacity utilisation and industrial production raised it, the Philadelphia Fed survey ("Current

General Activity") lowered it, and data revisions, retail sales and parameter re-estimation contributed little. Net change **+0.235 pp**.

- **2026Q3** (pending): ours 3.021, GDPNow 3.643, NY Fed 2.67.
- **Track record**: 2026Q2 actual 1.50, ours 2.536 (error +1.04), GDPNow 1.54 (+0.04), NY Fed 2.85 (+1.35). 2026Q1 actual 2.0, ours 2.701 (+0.70), GDPNow 1.239 (−0.76), NY Fed 2.35 (+0.35).

Limitations and pitfalls

- Our model has a short and, so far, unflattering track record against GDPNow (audit F27). Do not trade on it.
- The 80% band is wide, especially early in the quarter.
- EA, GB and JP have no official benchmark in the app, so there is nothing to cross-check them against.
- **Units differ.** The US is q/q % SAAR (annualised); EA, GB and JP are plain q/q % (Figure 2A.10). Multiply a non-US figure by roughly 4 before comparing it with the US.
- Inputs with publication lags (OECD CLI, quarterly-only data for JP) make the JP and EA nowcasts update in steps.
- The 2020 masking means the model has never been estimated on a pandemic-size shock.

Related mnemonics and BQL fields

- No mnemonic. The country pages for US, EA, GB and JP carry a **GDP nowcast** summary with an "Open nowcast →" link.
- GDP releases use the DFM nowcast as their model expectation in the Release calendar when it beats SARIMA on trailing RMSE.

Themes

Route: `/macro/theme/:theme` with `growth`, `inflation`, `labor`, `rates`, `credit_conditions`, `liquidity`, `commodities_fx` (`/macro/theme` redirects to growth) · **Menu:** Macro › Themes · **Mnemonics:** none · **Refresh:** nightly (`macro_cells` and source ingests)

Purpose

Theme dashboards gather the official series that matter for one macro topic, across countries and in detail for the largest economies. Use them to compare the same indicator across up to four countries, to look inside inflation or the labour market by component, or to animate yield curves through history.

Screenshot

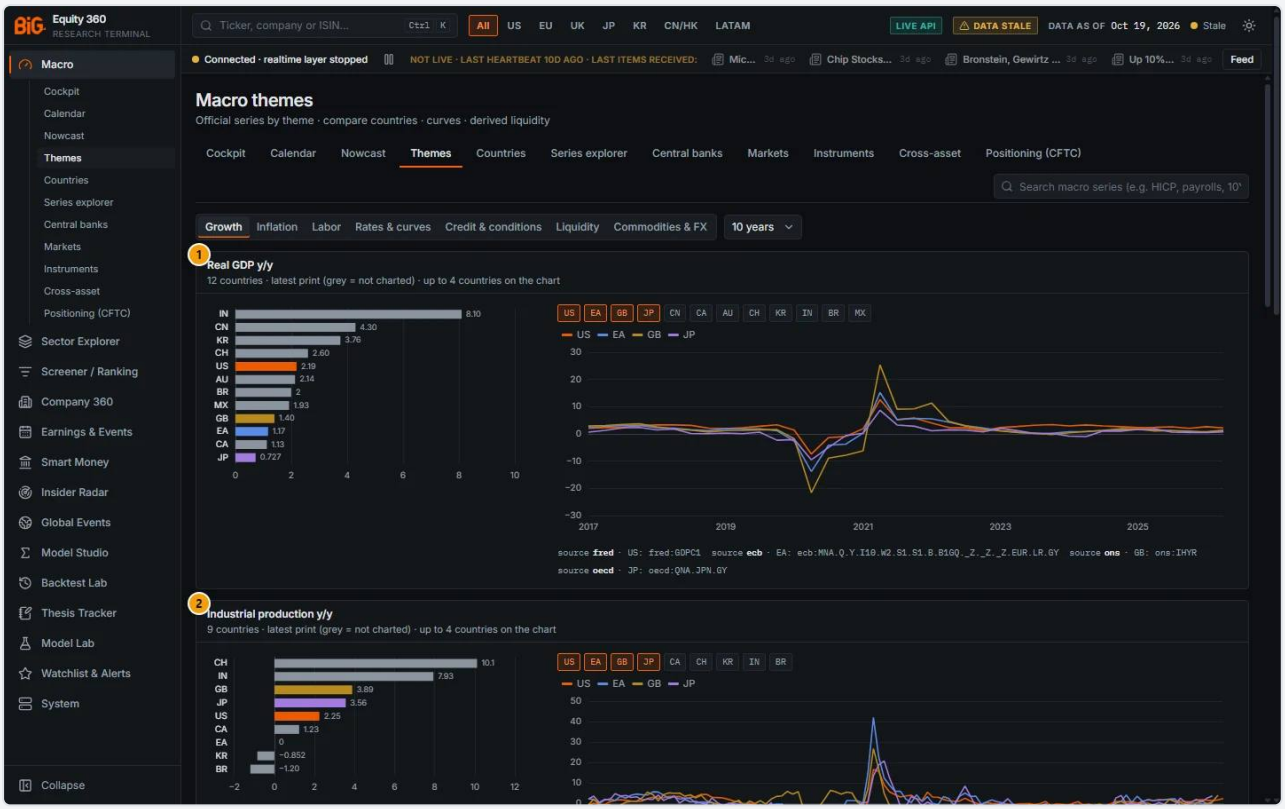


Figure 2A.11: Themes › Growth (`/macro/theme/growth`), 10 years.

#	Panel	What it shows
1	Real GDP y/y	Compare panel: latest print for 12 countries (bars), history for up to 4 chosen countries
2	Industrial production y/y	Compare panel, 9 countries
3	Retail sales y/y	Compare panel, 9 countries
4	Business surveys (PMI proxy)	Compare panel, 12 countries (OECD BCI/CLI, EC ESI, ...)
5	US activity	Series panel: 5 official US activity series
6	Euro area members: GDP y/y	Compare panel, 7 euro-area economies

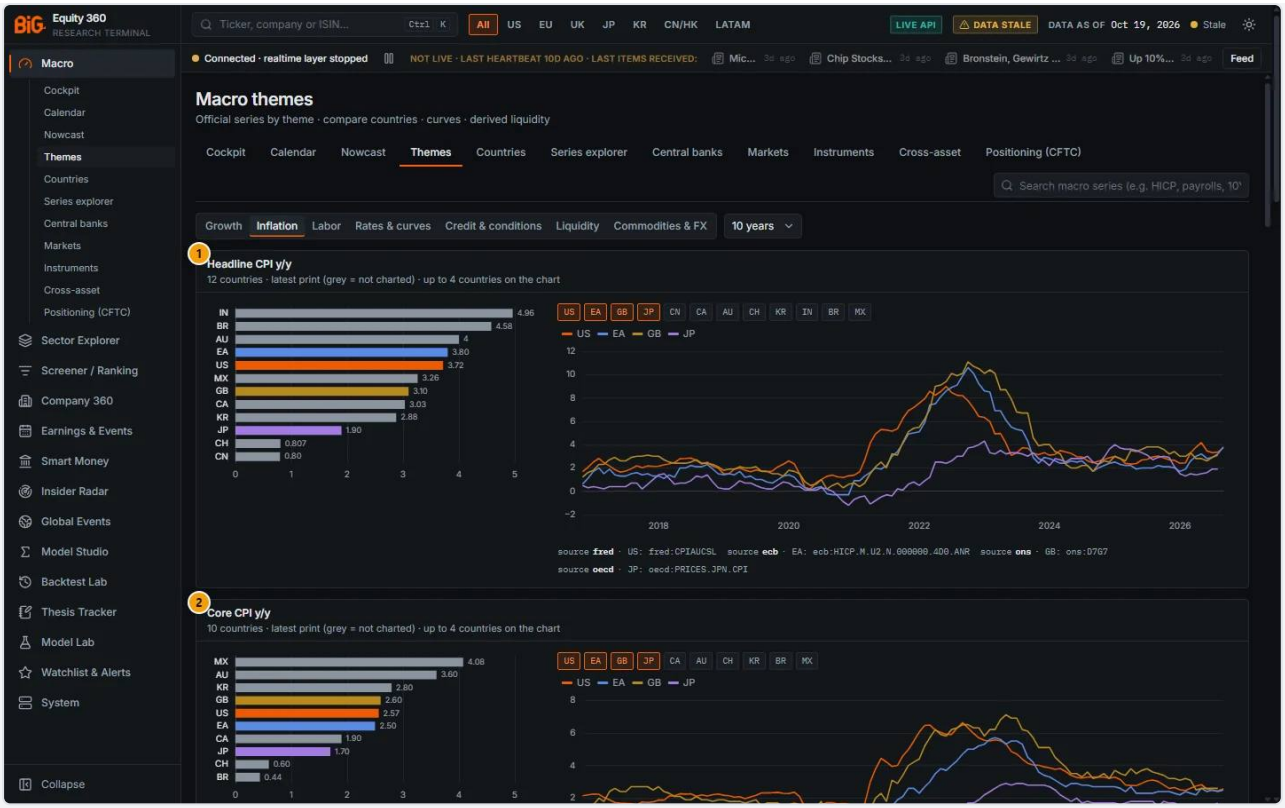


Figure 2A.12: Themes › Inflation (`/macro/theme/inflation`).

#	Panel	What it shows
1	Headline CPI y/y	Compare panel, 12 countries
2	Core CPI y/y	Compare panel, 10 countries
3	US CPI components (BLS via FRED), % y/y	9 official series (small multiples)
4	US PCE and alternative core measures, % y/y	4 official series
5	Euro area HICP components (ECB), % y/y	6 official series
6	UK CPI components (ONS), % y/y	7 official series
7	US breakevens, real yields, expectations (%)	6 official series (FRED T5YIE / T10YIE / DFII10 and related)
8	Producer prices y/y	Compare panel, 6 countries

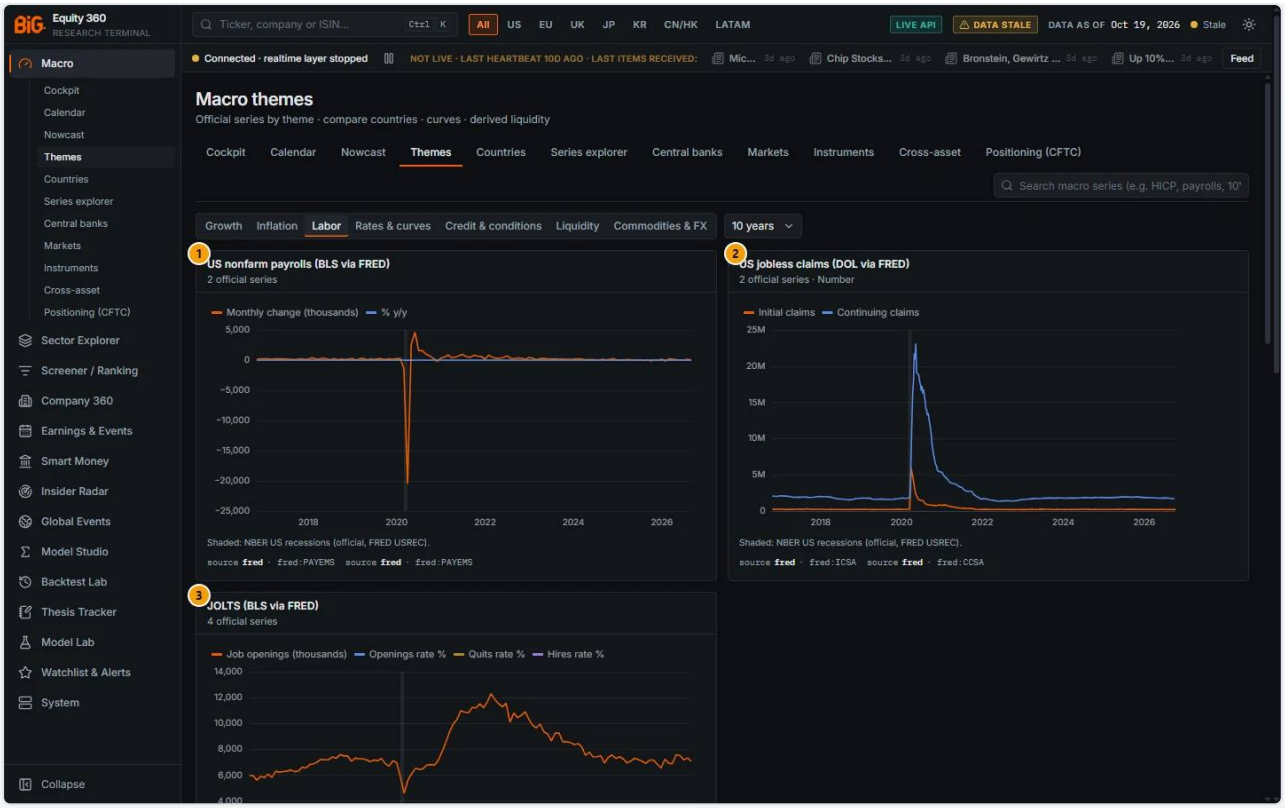


Figure 2A.13: Themes › Labor (`/macro/theme/labor`).

#	Panel	What it shows
1	US nonfarm payrolls (BLS via FRED)	Monthly change (thousands) and % y/y
2	US jobless claims (DOL via FRED)	Initial and continuing claims
3	JOLTS (BLS via FRED)	Job openings, openings rate, quits rate, hires rate
4	US unemployment, participation, wages	5 official series
5	Unemployment rate	Compare panel, 10 countries
6	Employment	Compare panel, 4 countries

Other themes. Credit & conditions, Liquidity and Commodities & FX have no screenshot in this edition. Rates & curves is shown after the table.

Theme	Panels
Credit & conditions	Corporate OAS (ICE BofA via FRED); financial conditions and stress (Chicago Fed NFCI/ANFCI, St. Louis Fed STLFSI); ECB CISS; Fed Senior Loan Officer Survey; ECB Bank Lending Survey; private credit (compare)
Liquidity	Fed balance sheet (H.4.1); **US net liquidity = Fed assets − TGA − ON RRP** (derived); Eurosystem balance sheet (ECB ILM); broad money (compare); US M2
Commodities & FX	Energy, metals and agriculture (World Bank Pink Sheet, monthly); daily spot energy (EIA via FRED); trade-weighted USD (Fed H.10); FX rates (Fed H.10); real effective exchange rates (BIS, compare)

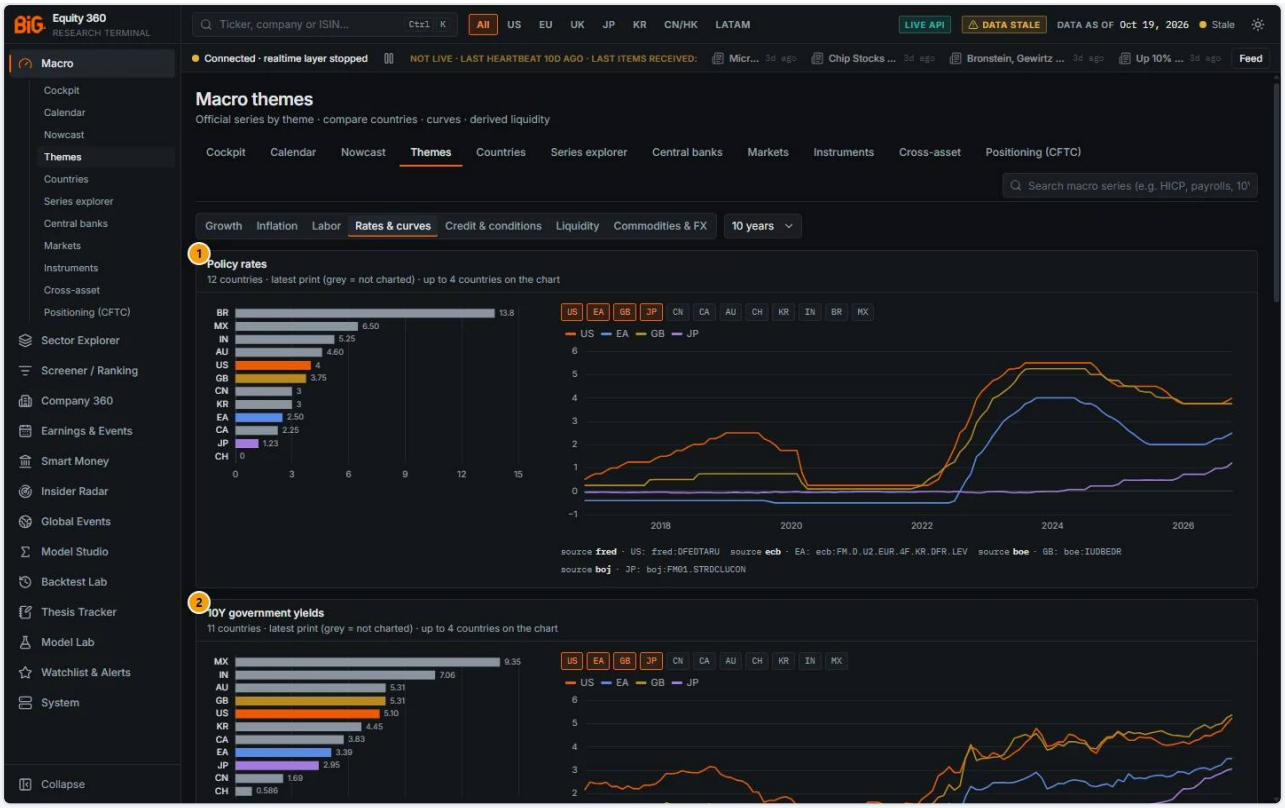


Figure 2A.14: Themes › Rates & curves (`/macro/theme/rates`), 10 years.

#	Panel	What it shows
1	Policy rates	Compare panel, 12 countries (BR 13.8 at the top, CH 0 at the bottom)
2	10Y government yields	Compare panel, 11 countries
3	2s10s slope (pp)	Compare panel, 6 countries
4	US curve spreads (pp)	3 official series
5	US real yields (TIPS, %)	2 official series
6	Treasury term premia (%)	5 official series (NY Fed ACM)
7	Sovereign curve history (month-end)	Animated month-end curve with Curve selector, Play, date slider and Pin date

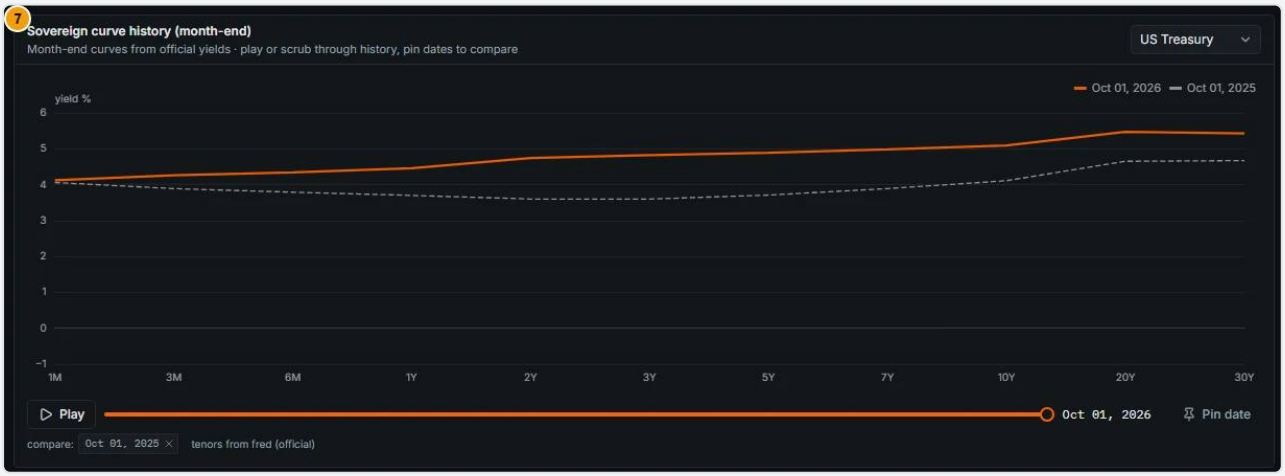


Figure 2A.15: Sovereign curve history, US Treasury. The latest month-end (1 Oct 2026, solid) is compared with the default pin one year earlier (1 Oct 2025, dashed); tenors 1M to 30Y come from FRED (official).

Controls

Control	What it does
Theme (Growth · Inflation · Labor · Rates & curves · Credit & conditions · Liquidity · Commodities & FX)	Switches the dashboard and the URL
History	3 / 5 / 10 (default) / 20 years of history
Compare panel: country chips	Toggle countries on the line chart. At most 4; adding a fifth drops the oldest. Defaults to US, EA, GB, JP. In the bar chart, charted countries take their line colour and the rest are grey
Compare panel: bar (hover)	Latest value, period and source
Source line (click)	Opens the series drawer (not available for derived series)
Series panel with more than 4 series	Drawn as small multiples (one chart per series)
Curve panel: Curve	Picks the sovereign curve
Curve panel: Play / Pause	Animates month-end curves from the start (one step per ~0.26 s)
Curve panel: date slider	Scrubs to any month-end
Curve panel: Pin date	Adds the current date as a dashed comparison curve (up to 3 pins). The date one year ago is pinned by default; × unpins

US-only series panels are shaded for NBER recessions (FRED USREC).

Methodology

- **Series panels** show explicit official series with a display transform (level, y/y, difference or m/m). Daily series are thinned to weekly (last observation per ISO week) beyond 3 years.
- **Compare panels** show one cell indicator across countries, using the canonical primary series of each country's cell, the same as the heatmap.
- **Curve panels** show month-end snapshots per tenor from official yields. The tenor sources are listed under the chart.
- **Derived panels** are computed from official inputs and badged "derived from official inputs". US net liquidity `= WALCL − WTREGEN − 1000 × RRPONTSYD` (USD millions).

These dashboards involve no models: every number is an official observation or a simple transform of one.

Data sources and refresh

FRED (copies of BLS, BEA, Fed Board, DOL, EIA and ICE BofA OAS); Chicago, Kansas City, St. Louis, Atlanta and Cleveland Feds; NY Fed (ACM term premia, official CSV); ECB Data Portal; Eurostat; ONS; World Bank Pink Sheet; OECD, BIS and IMF via the cells. Refreshed nightly with the source ingests.

Validation status

Not applicable: official data and arithmetic transforms only. The derived panel is labelled.

Correct usage and worked example

Use compare panels to rank countries on the latest print, then add up to four of them to the line chart to see whether the ranking is new or long-standing. In Figure 2A.11 the Real GDP y/y bars run from IN 8.10% and CN 4.30% down to EA 1.17%, CA 1.13% and JP 0.727%, with US at 2.19%. The line chart shows that the 2020–21 swings dwarf everything since, so pick 3 or 5 years in **History** to see recent divergence clearly.

Limitations and pitfalls

- The latest prints are **not synchronised**: one country's bar can be for Q2 and another's for Q3. Hover a bar for its period.
- Compare panels use each country's canonical measure. Where only a level or different definition exists, values are not strictly comparable (e.g. national unemployment definitions, PMI proxies on different scales).
- Lines with very different units in one series panel (e.g. payrolls change and % y/y) share one axis.
- Daily series are weekly beyond 3 years, so short spikes are smoothed.

Related mnemonics and BQL fields

- `GC` opens the central-bank monitor for policy and yield curves. See Central banks overview.
- Series from any panel can be opened in Model Studio through the series drawer: 9. Model Studio.

Countries

Route: `/macro/country/:cc` (24 codes: US, EA, DE, FR, IT, ES, PT, NL, GB, CH, JP, KR, CN, HK, TW, IN, AU, CA, BR, MX, CL, CO, PE, AR; `/macro/country` redirects to US) · **Menu:** Macro › Countries · **Mnemonics:** `ECST <cc>` (default US) · **Refresh:** nightly

Purpose

A country page is the one-stop view of one economy. It shows its cycle quadrant, headline cells, nine sections of official series, its equity index against the macro composite, its surprise index, recession indicators, a GDP nowcast where available, upcoming releases, and recent event coverage. Analysts covering a market start here.

Screenshot

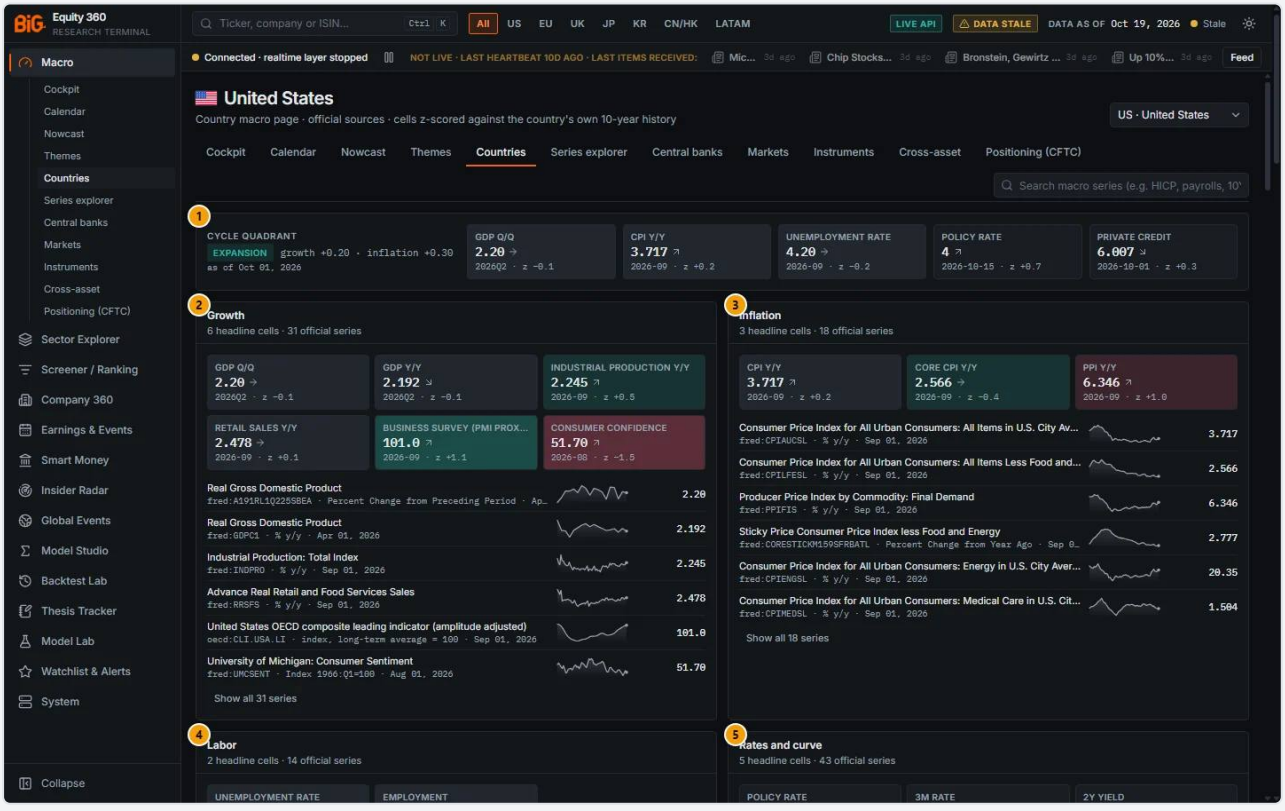


Figure 2A.16: United States (`/macro/country/US`).

#	Panel	What it shows
1	Cycle quadrant	Quadrant badge, growth and inflation momentum, as-of date, plus five headline tiles (the first cell of each of the first five sections)
2	Growth	Headline cells (GDP q/q and y/y, IP, retail, PMI proxy, consumer confidence) and the list of official growth series with 5-year sparklines
3	Inflation	CPI, core CPI and PPI cells; series list
4	Labor	Unemployment and employment cells; series list
5	Rates and curve	Policy rate, 3M, 2Y, 10Y and 2s10s cells; series list
6	Money, credit and stress	Private credit and money supply cells; series list
7	Housing	House prices cell; series list
8	External	Current account cell; series list
9	Fiscal	Fiscal balance and debt cells; series list
10	FX	Real effective FX cell; series list
11	Equities vs the macro composite	Equity index y/y (top) against growth and inflation momentum (bottom), with their correlation
12	Surprise index	The country's surprise index history (theme all), labelled model-based (not consensus)
13	Recession risk	Each method's latest reading and label, plus probability histories
14	GDP nowcast	First two target quarters: each model's latest value and band, actual when released; "Open nowcast →"
15	Upcoming releases	Next 30 days, all tiers, compact list
16	Events	GDELT coverage labelled to this country, last 90 days (see 8. Global events)

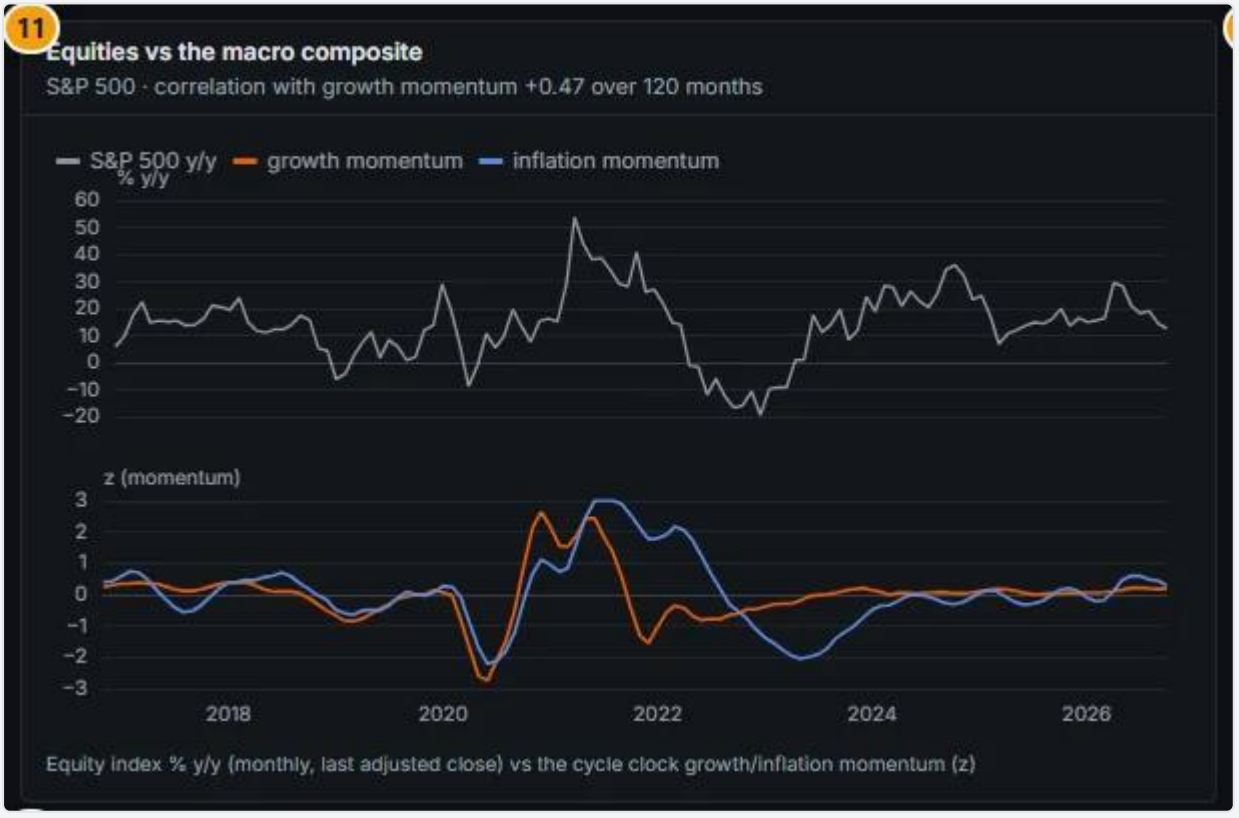


Figure 2A.17: S&P 500 y/y against US growth and inflation momentum; correlation with growth momentum +0.47 over 120 months.

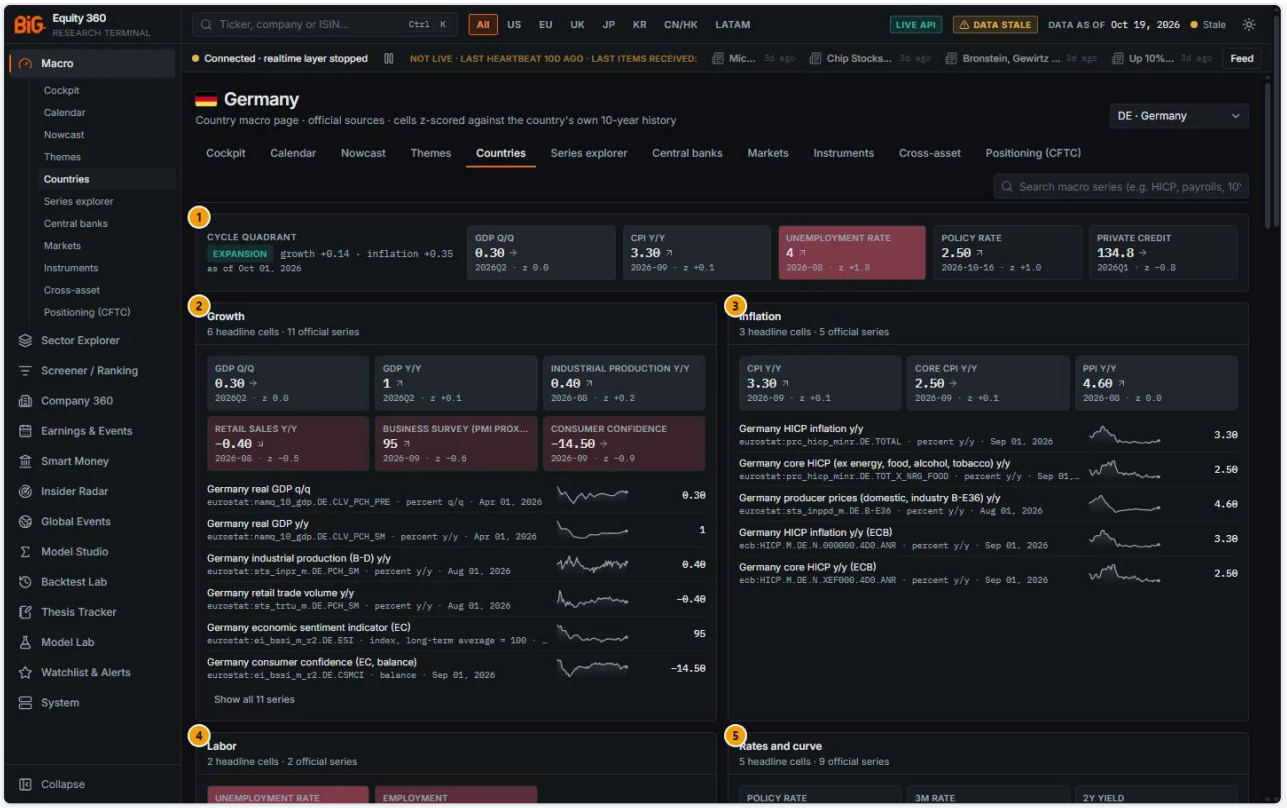


Figure 2A.18: Germany (`/macro/country/DE`). Same layout, but with no nowcast panel.

#	Panel	What it shows
1	Cycle quadrant	Expansion; growth +0.14, inflation +0.35; headline tiles (unemployment 4%, z +1.8, rose)
2	Growth	6 headline cells, 11 official series (Eurostat, EC)
3	Inflation	3 cells, 5 series (Eurostat HICP, ECB)
4	Labor	2 cells, 2 series
5	Rates and curve	5 cells, 9 series (policy rate inherited from EA)
6	Money, credit and stress	2 cells, 5 series
7	Housing	1 cell, 3 series
8	External	1 cell, 3 series
9	Fiscal	2 cells, 5 series
10	FX	1 cell, 2 series
11	Equities vs the macro composite	DAX; correlation with growth momentum +0.41 over 120 months
12	Surprise index	DE surprise history
13	Recession risk	Probit and OECD CLI phase
14	Upcoming releases	Next 30 days (shown in place of the nowcast)
15	Events	GDELT coverage for DE

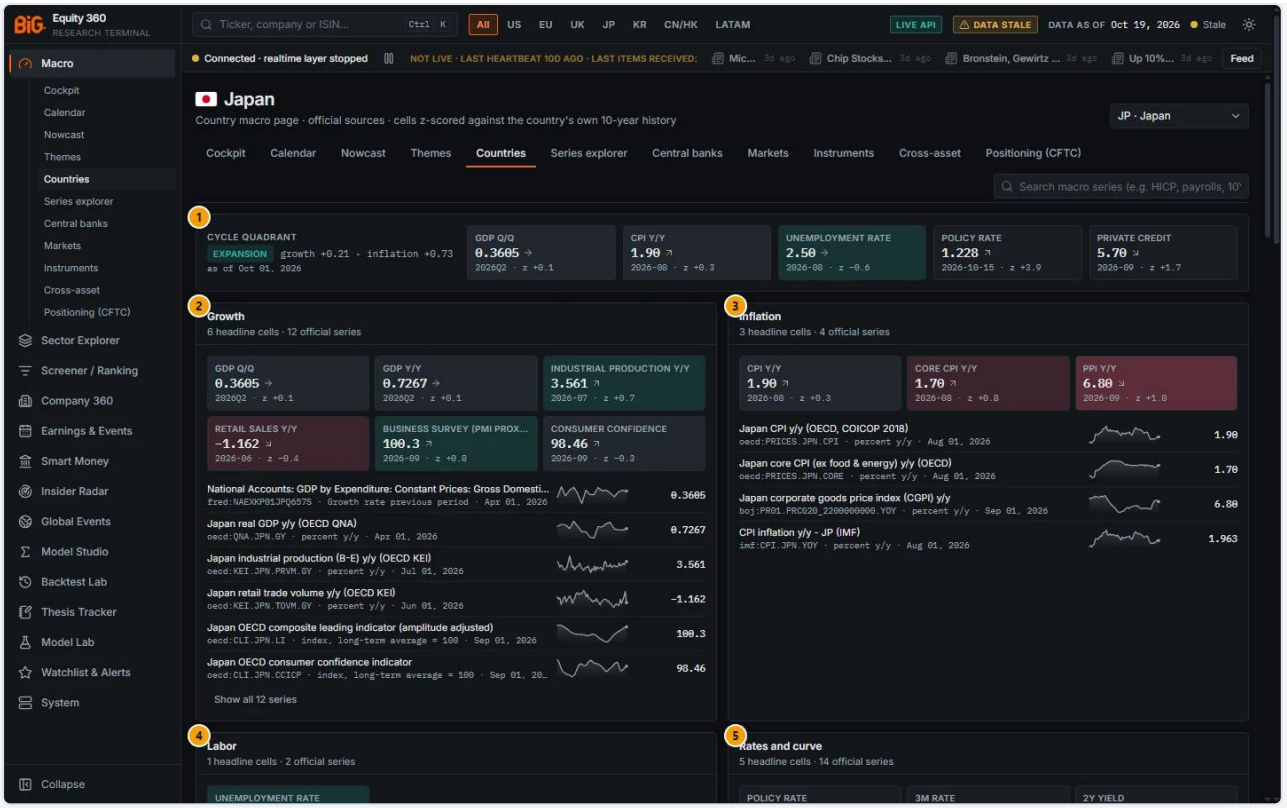


Figure 2A.19: Japan (`/macro/country/JP`).

#	Panel	What it shows
1	Cycle quadrant	Expansion; growth +0.21, inflation +0.73; policy rate tile z +3.9 (neutral, grey)
2	Growth	6 cells, 12 series (OECD QNA/KEI, FRED)
3	Inflation	3 cells, 4 series (OECD, BoJ CGPI, IMF)
4	Labor	1 cell, 2 series
5	Rates and curve	5 cells, 14 series
6	Money, credit and stress	2 cells, 4 series
7	Housing	1 cell, 1 series
8	External	1 cell, 2 series
9	Fiscal	2 cells, 3 series
10	FX	1 cell, 2 series
11	Equities vs the macro composite	Nikkei 225; correlation +0.38 over 120 months
12	Surprise index	JP surprise history
13	Recession risk	Probit, HMM regime, OECD CLI phase
14	GDP nowcast	JP model (ours)
15	Upcoming releases	Next 30 days
16	Events	GDELT coverage for JP

Controls

Control	What it does
Country (top right)	Switches to another of the 24 countries
Cell tile (hover / click)	The same tooltip as a heatmap cell; click opens the series drawer

Control	What it does
Series row (click)	Opens the series drawer. The sparkline is 5 years
Show all N series / Show fewer	Expands a section beyond its first 6 series (primary series first)
Open nowcast →	Opens `/macro/nowcast?cc=<cc>` (US, EA, GB, JP only)
Upcoming release row	Opens the release drawer
Events panel	See 8. Global events

Methodology

- **Cells and quadrant** are computed exactly as on the cockpit: 10-year z, oriented colour, the cycle clock `cycle_v1`.
- **Equities vs the macro composite** puts the official equity index's y/y return (monthly, last adjusted close, from the price lake) above the cycle clock's growth and inflation momentum on a shared time axis. There is no dual axis. The subtitle gives the correlation of index y/y with growth momentum over the overlapping months. CL, CO and PE have no official-index price series in the lake.
- **Surprise index** is the country's daily index, theme "all", over the 365-day window.
- **Recession risk** shows the latest row of each method with its full label. Probability histories (NY Fed, our probit, HMM) are drawn on a 0–1 scale. The note "50% is the conventional signal level" refers to these lines.

Data sources and refresh

The country's official series from the macro catalogue (national statistical offices, central banks, Eurostat/ECB for euro members, OECD/IMF/BIS where national APIs are unavailable), refreshed nightly. Equity indices come from the price lake. Events come from GDELT (chapter 8).

Validation status

Descriptive page. Its model-based components (surprise index, probit, HMM, nowcast) carry the statuses given in Macro cockpit and Nowcast.

Correct usage and worked example

Start at the cycle quadrant and the headline tiles, then open the sections whose tiles are coloured. In Figure 2A.18, Germany is in **Expansion** (growth +0.14, inflation +0.35, as of 1 Oct 2026). The unemployment tile is rose: 4% for August 2026 with z +1.8, high for Germany's own decade. Retail sales y/y (−0.40%, z −0.5) and consumer confidence (−14.5, z −0.9) are also rose. The page therefore shows positive momentum on surveys and IP but a weak labour market, which the Labor section's two series let you verify.

For Japan (Figure 2A.19), the policy-rate tile shows 1.228% with z +3.9 but stays grey, because rates are neutral. The z simply says Japanese rates are far above their decade average.

Limitations and pitfalls

- Coverage varies widely: TW has 8 cells and no cycle position; HK has 12. Check the "N headline cells · M official series" subtitle.
- Euro-area members show EA monetary cells (policy rate, 3M, money supply).
- A correlation over 120 months is descriptive and is dominated by large episodes (2020). It is not a beta to trade on.
- The Upcoming releases panel shows all tiers for the next 30 days and only officially scheduled releases.

Related mnemonics and BQL fields

- `ECST <cc>` opens a country page (two-letter code, default US). `ECST DE` opens Germany.

- `WIRP <cc>` opens the country's central bank (e.g. `WIRP DE` → ECB).
- Company-level macro exposures: 5C. Company 360: Macro.

Series explorer

Route: `/macro/series` (search) and `/macro/series/:id` (e.g. `/macro/series/fred:CPIAUCSL`) · **Menu:** Macro › Series explorer · **Mnemonics:** none · **Refresh:** nightly (series ingest)

Purpose

The series explorer shows any official macro series in the lake with full transparency: transforms, first release against latest revised values, revision statistics, the heatmap cells that use it, the same indicator in other countries, release history and provenance. It is also the bridge to modelling: **Open in Model Studio** starts a time-series study with the series as the target.

Screenshot

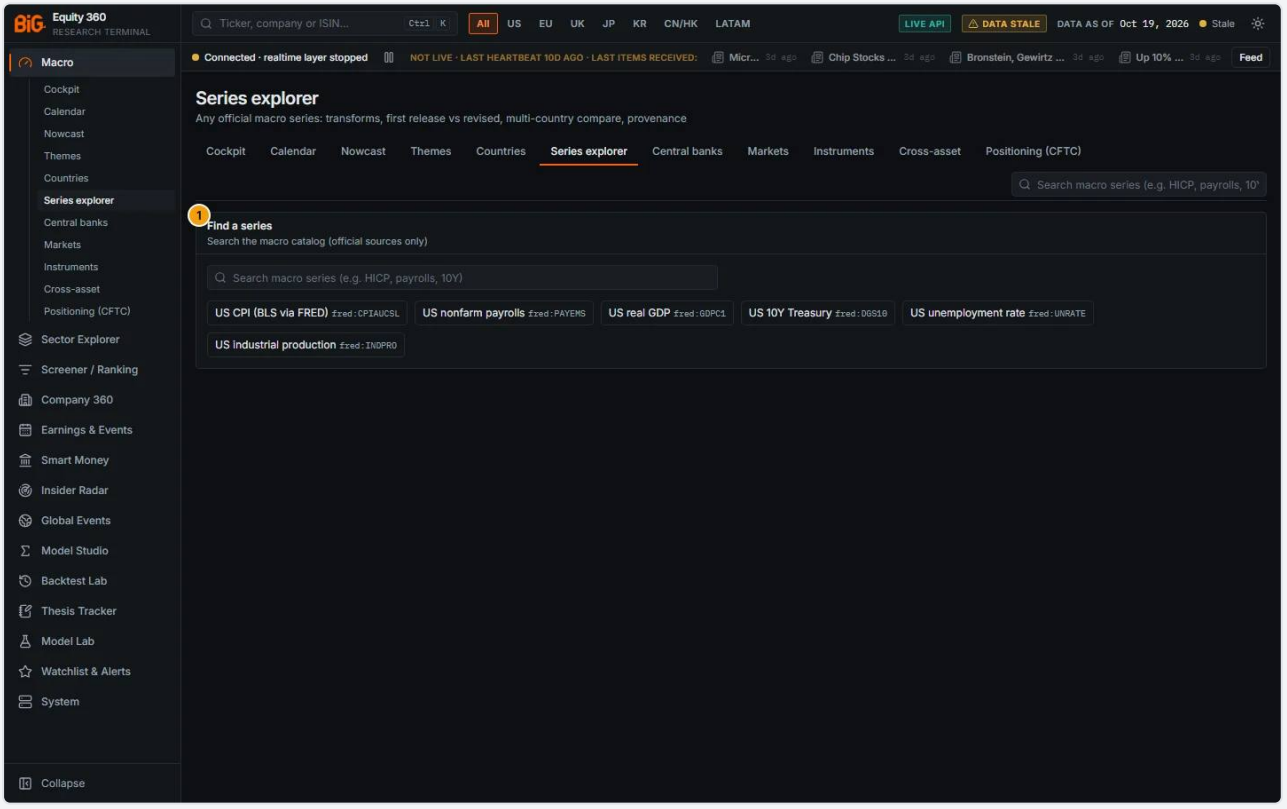


Figure 2A.20: Series explorer without a series (`/macro/series`).

#	Panel	What it shows
1	Find a series	Catalogue search box ("official sources only") and six suggested US series: CPI, nonfarm payrolls, real GDP, 10Y Treasury, unemployment rate, industrial production

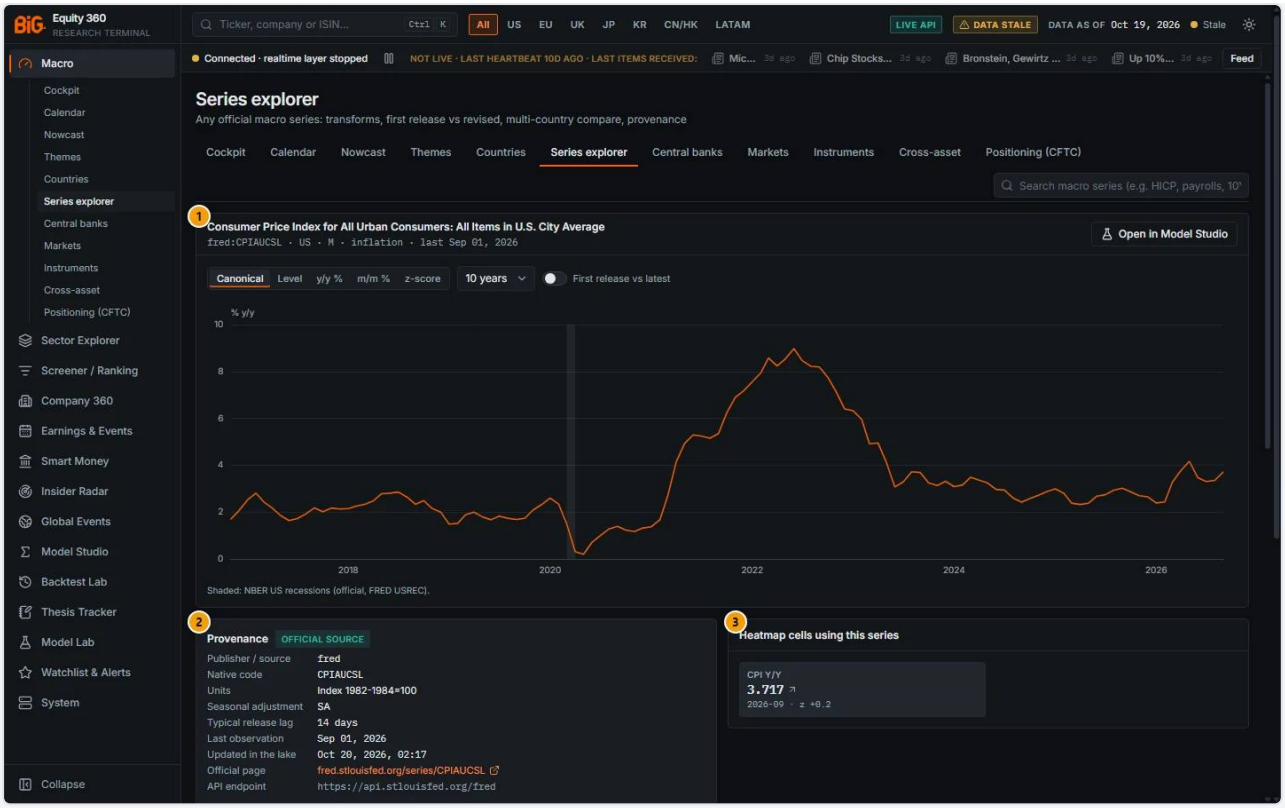


Figure 2A.21: Series explorer for US CPI (`/macro/series/fred:CPIAUCSL`), canonical measure (% y/y), 10 years.

#	Panel	What it shows
1	Series title	Chart with transform, range and vintage controls; subtitle with id · country · frequency · category · last observation; **Open in Model Studio**
2	Provenance	Publisher / source, native code, units, seasonal adjustment, typical release lag, last observation, time updated in the lake, official page, API endpoint; badge **official source** or **derived**
3	Heatmap cells using this series	The cockpit cells built on this series (here: US CPI y/y 3.717, z +0.2)
4	Compare countries	The same indicator (`cpi_yoy`) across up to 4 countries
5	Release history	Per period: first released, latest vintage, number of vintages

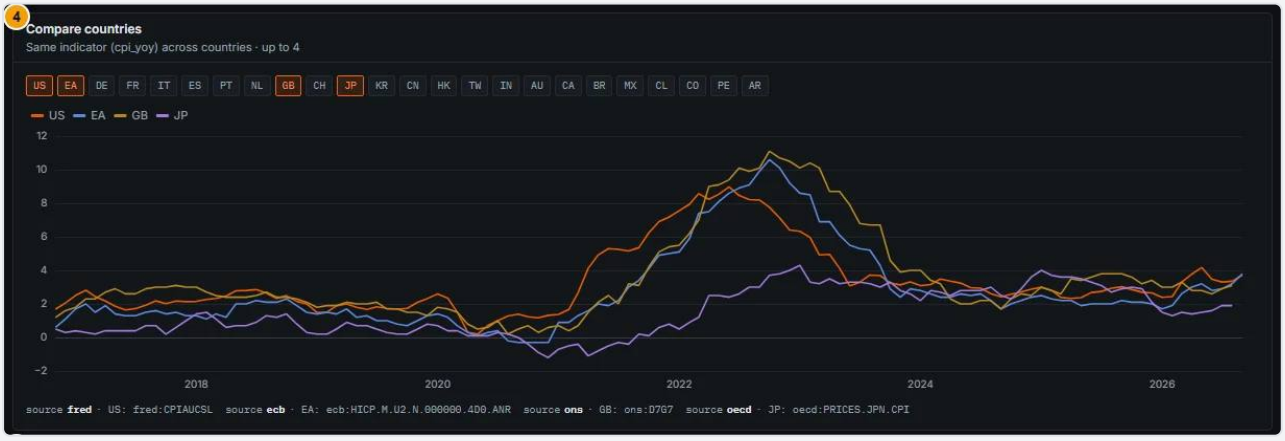


Figure 2A.22: Compare countries for `cpi_yoy` (below the fold).

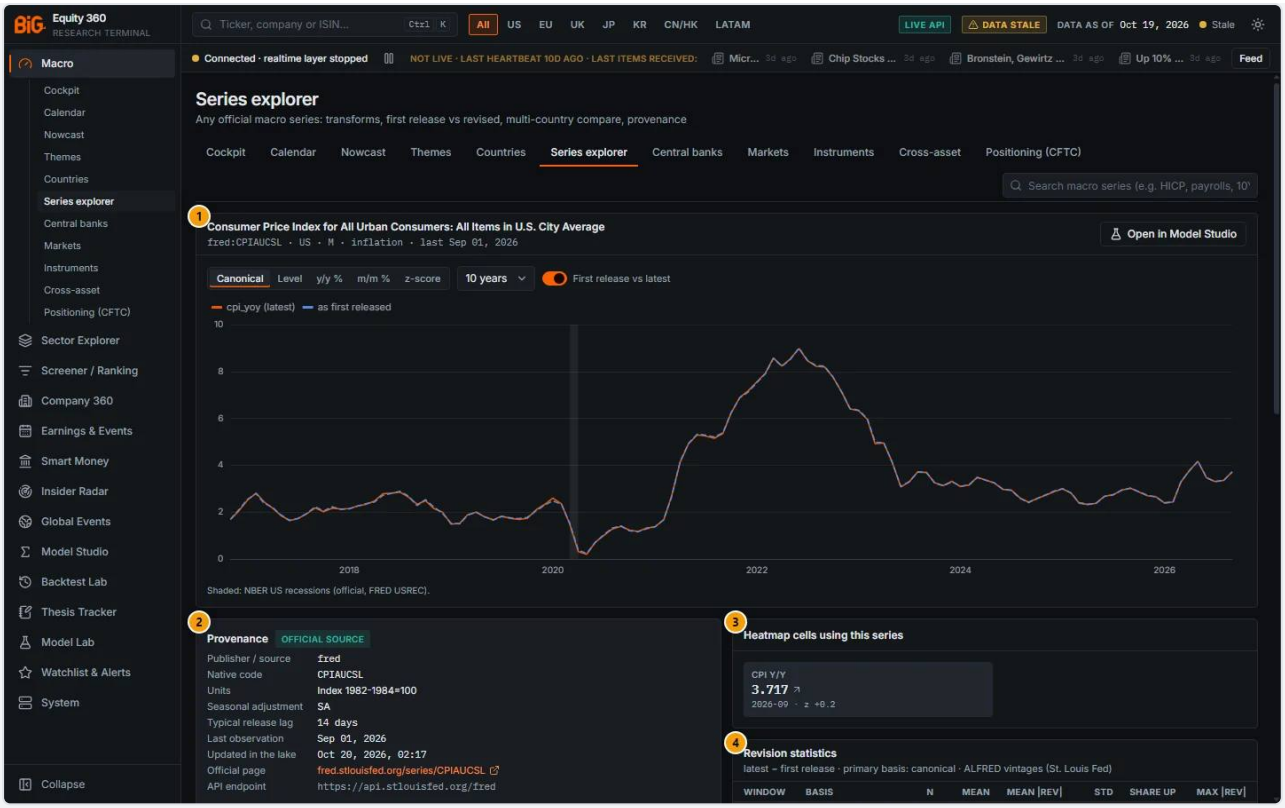


*Figure 2A.23: US CPI with *First release vs latest** on. The dashed "as first released" line is plotted under "cpi_yoy (latest)"; on the canonical y/y basis the two are almost indistinguishable.*

#	Panel	What it shows
1	Series title	Chart with the latest line and the "as first released" overlay; the switch is on
2	Provenance	As in Figure 2A.21
3	Heatmap cells using this series	US CPI y/y 3.717, z +0.2
4	Revision statistics	Latest – first release by window (5y, 10y, all) on the canonical (primary) and native bases; ALFRED vintages
5	Compare countries	As in Figure 2A.22
6	Release history	As in Figure 2A.21

Figure 2A.24: Revision statistics for US CPI (ALFRED vintages). On the canonical y/y basis the 10-year mean absolute revision is 0.0228 pp (max 0.117); on the native index basis it is 0.189 index points (max 0.72). 198 periods with an index rebase and 7 with an estimated first-release date are excluded.

Controls

Control	What it does
Search box	At least 2 characters; up to 12 matches (title, id, frequency, last date); click to open
Suggested series	One-click shortcuts (search page only)
Transform	**Canonical** (the cell measure, e.g. % y/y for CPI) · **Level** (as published) · **y/y %** · **m/m %** · **z-score** (vs the trailing 10-year window, ≥ 24 points)
Range	5 / 10 (default) / 20 years / All
First release vs latest	Overlays the "as first released" line (dashed) and shows **Revision statistics**. If the series has no vintages: "No vintage history (ALFRED / first-seen) for this series." If the vintage pair is on another basis, a note tells you which transform to choose
Open in Model Studio	Creates a new Model Studio time-series study (method ARIMA) with this series as the target `y`. Frequency is set from the series (Q/A → Q, W → W, else M), the start 20 years back, and a y/y transform is pre-applied when the series' default view is y/y. It opens the Studio builder; nothing runs until you run it
Country chips (Compare countries)	Up to 4 countries; a fifth drops the oldest. Default: the series' country plus US, EA, GB, JP
Official page link (Provenance)	Opens the publisher's page for the series

Methodology

- **Transforms.** y/y = % change against 12 months earlier; m/m = % change against the previous period; z = `(x − μ) / σ` over the trailing 10 years (≥ 24 points); canonical = the macro cell measure from `macro_cell_history`.
- **Vintages.** For US series with ALFRED history (CPI, GDP, payrolls, IP, retail and others), the first-release value of each period comes from ALFRED real-time rows. Elsewhere it comes from first-seen values recorded when the ingest detected a revision. The overlay is aligned by period.
- **Revision statistics.** Revision = latest − first release, per period, for windows 5y, 10y and all, on the canonical and native bases (the primary basis is highlighted). The table reports n, mean, mean |rev|, std, share revised up and max |rev|. Periods whose index was rebased between first release and latest, and periods with an estimated first-release date, are excluded and counted under the table.
- **Release history** lists, per period, the first-release date, the latest vintage date and the number of vintages.

Data sources and refresh

The macro catalogue (`contracts/macro_catalog.yaml`): all official series in the lake, refreshed nightly by the source ingests (FRED/ALFRED, ECB, Eurostat, ONS, BoE, BoJ, BCB, Banxico, BIS, OECD, IMF, World Bank and national offices). The Provenance box shows the time of the last update in the lake. The catalogue search is warmed at API start-up, so the first search is fast.

Validation status

Not applicable: official data. Revisions are measured, not modelled.

Correct usage and worked example

Use **First release vs latest** before relying on any historical relationship involving revised data. A signal built on today's revised history can look better than it was in real time.

Worked example: US CPI (`fred:CPIAUCSL` , Figure 2A.21). Canonical view, 10 years: y/y inflation peaked near 9% in 2022 and stood at 3.717% for September 2026 (last observation 1 Sep 2026, updated in the lake 20 Oct 2026 02:17). Provenance: publisher FRED, native code CPIAUCSL, units Index 1982–1984 = 100, seasonally adjusted (SA), typical release lag 14 days. The cell panel shows the cockpit cell CPI y/y 3.717, z +0.2. Switch on **First**

release vs latest (Figure 2A.23). Over the last 10 years (109 periods), the canonical y/y value moved by 0.0228 pp on average between first release and latest, at most 0.117 pp, and 42% of revisions were upward (Figure 2A.24). For CPI y/y, revisions are therefore negligible. The native index level is revised more (mean |rev| 0.189, max 0.72), mainly because the seasonally adjusted series gets new seasonal factors each year. Then click **Open in Model Studio** to start an ARIMA study on CPI y/y.

Limitations and pitfalls

- Vintage depth differs: ALFRED gives full US history, while first-seen revisions elsewhere only start when the app began ingesting.
- The z-score transform uses the full revised history; it is not point-in-time.
- Compare countries relies on the series having a cell indicator. Series outside the 23 indicators have no Compare panel.
- Series from scraped sources are not in the macro search (official sources only).

Related mnemonics and BQL fields

- None. Model Studio: 9. Model Studio.

Central banks overview

Route: `/macro/cb` · **Menu:** Macro › Central banks · **Mnemonics:** `WIRP` , `GC` · **Refresh:** nightly (`cb_rates` , `cb_curves_implied` , `cb_stance` , `cb_tone` , `cb_projections` , `cb_balance_sheets`)

Purpose

This page monitors 17 central banks: the Fed, ECB, BoE, BoJ, SNB, BoC, RBA, Riksbank, Norges Bank, BCB (Brazil), Banxico, BCCh (Chile), BanRep (Colombia), BCRP (Peru), RBI, BoK and PBoC (LPR). For each it shows the official policy rate and corridor, the last move, the rate-cycle phase, the next official meeting, **what is priced in** (an approximation from official curves), the real policy rate, the gap to a Taylor rule, and the latest communication tone. It answers "who is hiking, who is cutting, and how much more does the curve imply?".

Screenshot

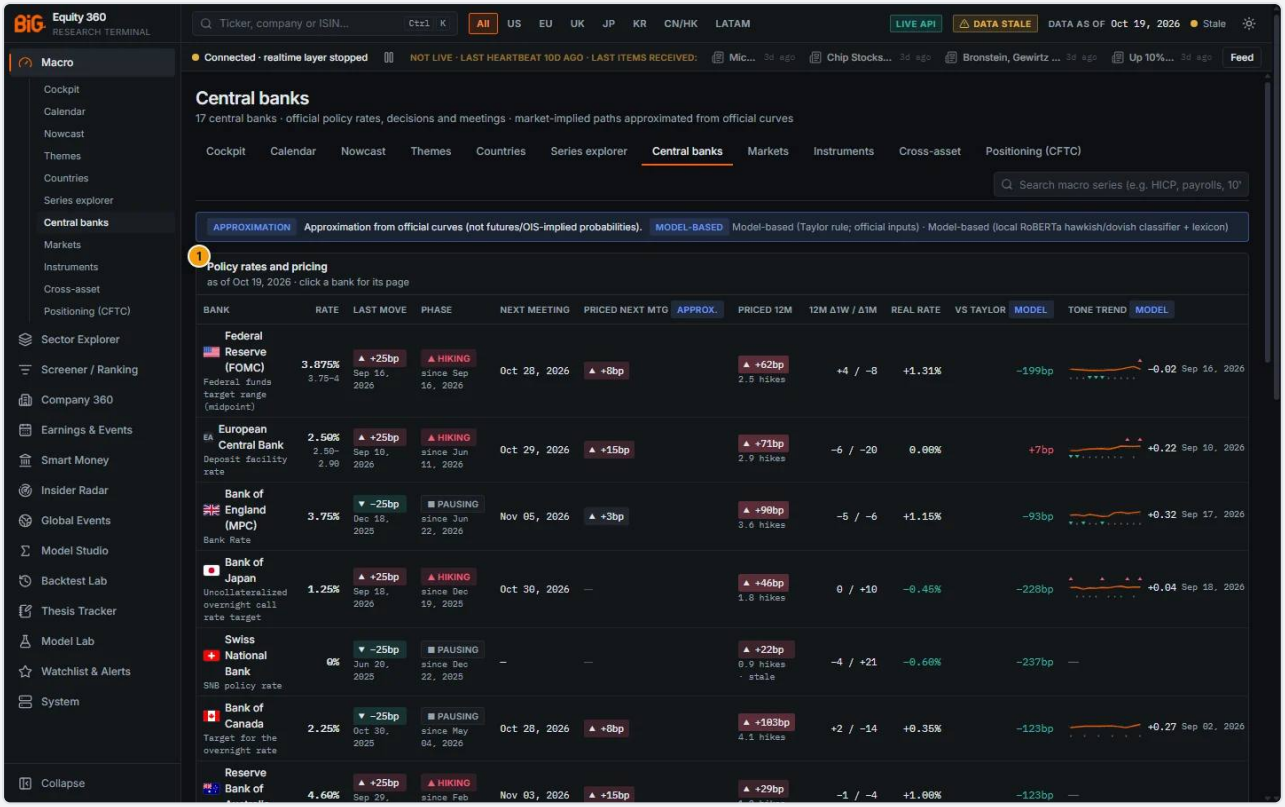


Figure 2A.25: Central banks (`/macro/cb`), as of 19 Oct 2026.

#	Panel	What it shows
1	Policy rates and pricing	One row per bank: Bank (name, rate name), Rate (with corridor), Last move, Phase, Next meeting, Priced next mtg (approx.), Priced 12m, 12m Δ1w / Δ1m, Real rate, vs Taylor (model), Tone trend (model)
2	Priced over the next 12 months	Bar chart of bp priced over 12 months per bank with an official curve; rose = hikes, teal = cuts

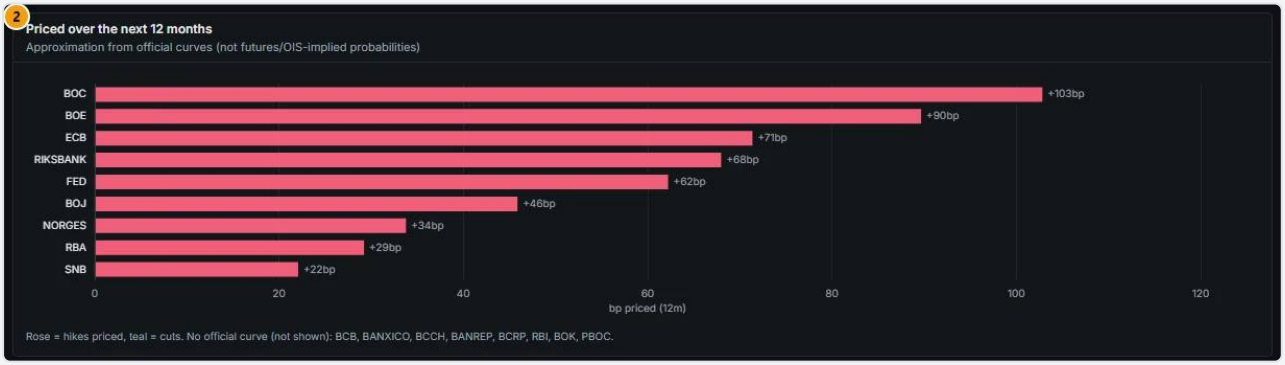


Figure 2A.26: Priced over the next 12 months (below the fold). Banks without an official curve are listed under the chart.

The strip above the table repeats the labels: **APPROXIMATION** "Approximation from official curves (not futures/OIS-implied probabilities)." and **MODEL-BASED** "Model-based (Taylor rule; official inputs) · Model-based (local RoBERTa hawkish/dovish classifier + lexicon)".

Controls

Control	What it does
Bank name / row	Opens the bank page `/macro/cb/:bank`

Control	What it does
Priced next mtg (hover)	Text such as "+8bp · 0.3 hikes of 25bp"
Tone trend (hover)	Latest document type and date; "−1 dovish … +1 hawkish (model-based)"
Bar (hover / click)	Priced 12m, implied rate, policy rate and curve source; click opens the bank page

Reading the chips. ▲ / ▼ / ■ show the direction (hike, cut, unchanged). Rose means tightening and teal easing; the tint deepens at 1.5 or more 25 bp moves. The Phase badge shows ▲ hiking (rose), ▼ cutting (teal) or ■ pausing (grey). A negative real rate and a negative "vs Taylor" are teal (accommodative); a positive "vs Taylor" is rose. "no official curve" or "—" means the bank has no official short-end curve.

Methodology

POLICY RATES, DECISIONS, RATE CYCLE

- **Policy rate** is the bank's official rate, e.g. the Fed's target-range midpoint, the ECB deposit facility rate, the Selic target or the PBoC 1-year LPR. Each is cross-checked against the BIS policy-rate database: a 0 bp difference on all banks at build time (`docs/MACRO_V3_PLAN.md` W3.1). The **corridor** is shown as lower–upper, or as the deposit rate.
- **Decisions** are derived from official rate changes and official meeting dates, with vote splits where published (Fed statements, BoE MPC votes since 1997).
- **Rate cycle** is rule-based: hiking or cutting when the most recent rate change within the trailing **183 days** was a hike or a cut; otherwise pausing, including long holds.

WHAT IS PRICED (IMPLIED PATH, METHOD `IMPLIED_V1`)

For every bank with an official curve and every curve date:

1. **Build the curve.** Bootstrap discount factors from official yields (bills as simple or bond-equivalent yields, coupon bonds as par) with log-linear interpolation, which gives piecewise-flat forwards. Alternatively use a closed-form forward curve (ECB Svensson parameters, BoE OIS instantaneous forwards).
2. **Anchor to the policy rate.** `basis = average forward over the stub [0, next decision) − current policy rate` . If the next decision is less than 10 days away, or there is no meeting calendar, the first month is used instead (the meta flags this).
3. **Implied rate per meeting** `k` , effective the day after the decision: `implied_k = average forward over [e_k, e_{k+1}) − basis` .
4. **Horizons** 3/6/12/24 months: `average forward over the last month before the horizon − basis` .
5. **Priced** = `(implied − policy rate) × 100` bp. **Moves priced** = bp / 25. **This step** = the increment from the previous meeting. **Δ1w / Δ1m** = change of the same point against 1 week and 1 month earlier.

The basis absorbs instrument spreads (bills or government bonds against the overnight rate, overnight rate against the policy rate) and is **assumed constant** along the path.

Bank	Official curve	Resolves meetings
Fed	US Treasury daily par curve incl. bills + NY Fed EFFR; NY Fed ACM term premia for the adjusted path	yes
BoE	BoE UK OIS (SONIA) instantaneous forwards, 1–60m (closest to a true OIS path)	yes
ECB	ECB euro-area AAA government curve, Svensson parameters (govvies, not OIS)	yes
BoC	Bank of Canada T-bills 1M–1Y + benchmark bonds 2Y/3Y/5Y	yes
RBA	RBA bank bills 1M/3M/6M + CGS 2Y/3Y/5Y (RBA OIS series discontinued Dec 2022)	yes
Riksbank	Swedish T-bills 3M/6M + government bonds 2Y/5Y	yes

Bank	Official curve	Resolves meetings
Norges Bank	T-bills 3M/6M/12M + government bonds 3Y/5Y	3–24m horizons
BoJ	MoF JGB benchmark par yields 1Y–10Y (no bills)	12/24m only
SNB	SNB Confederation spot yields 1Y–10Y (monthly)	12/24m only
BCB, Banxico, BCCh, BanRep, BCRP, RBI, BoK, PBoC	no official machine-readable short-end curve	not available

STANCE

- **Real rate** = policy rate − core inflation y/y (or headline when there is no core), from official statistics.
- **vs Taylor** = policy rate − the Taylor rule with official r*, where an r* is published (US, Canada, euro area); otherwise policy rate − Taylor (1993). See Central bank page for the rules.

TONE

Available for **Fed, ECB, BoE, BoJ and BoC** only; the other banks show "—". The tone trend sparkline plots the last 12 scored documents on a fixed −1…+1 scale, with ▲ hike markers at the top, ▼ cut markers at the bottom and dots for holds. The number is the latest document's score. The method is described under Central bank page.

LABELS RETURNED BY THE API

`GET /api/v1/macro/v3/central-banks` returns these `labels` (verbatim, 19 Oct 2026). The page and the bank pages display them:

Key	Label
`implied_path`	Approximation from official curves (not futures/OIS-implied probabilities)
`implied_path_tp_adjusted`	Term-premium-adjusted with NY Fed ACM term premia (model-based, official NY Fed estimates)
`taylor_rule`	Model-based (Taylor rule; official inputs)
`expected_inflation`	Model-based (Cleveland Fed model for the US; our AR(1) elsewhere)
`tone`	Model-based (local RoBERTa hawkish/dovish classifier + lexicon)
`rate_cycle`	Rule-based: most recent policy-rate change within 183 days decides hiking/cutting, else pausing
`decisions`	Derived from official rate changes and official meeting dates

Data sources and refresh

Job	Writes	Sources
`cb_rates`	rates, decisions, meetings, rate cycle	FRED, ECB, BoE, BoC, RBA, Riksbank, Norges Bank, BCB, BanRep, PBoC LPR; BIS WS_CBPOL cross-check
`cb_curves_implied`	implied paths (~2 years daily history), curves, meta	the curves in the table above
`cb_stance`	real rates, Taylor rules, r*	official CPI cells, CBO potential (FRED GDPPOT), OECD Economic Outlook gaps, NY Fed LW/HLW r*, Cleveland Fed expected inflation
`cb_tone`	tone	official statements and minutes (Fed, ECB, BoE, BoJ, BoC)
`cb_projections`	official projections	Fed SEP (FRED/ALFRED), ECB/Eurosystem staff projections (ECB MPD)
`cb_balance_sheets`	balance sheets	Fed H.4.1, ECB ILM, BoE IADB, BoJ accounts

All run in the nightly DAG. Next meetings come from the official meeting calendars (shared with the Release calendar).

Validation status

- Policy rates: cross-checked against the BIS policy-rate database (0 bp difference at build).
- Implied paths: **approximation**, labelled on every view. They have not been validated against OIS- or futures-implied paths; no gate exists.
- Taylor rules, expected inflation and tone: **model-based**, labelled. They are benchmarks and text measures, not forecasts; no gate exists.
- Rate cycle: rule-based, labelled.

Correct usage and worked example

Read **Priced 12m** together with **Phase** and **vs Taylor**. A bank that is hiking, with more hikes priced and a policy rate well below its Taylor benchmark, looks consistent. A bank pausing with a lot priced deserves a look at its curve and caveats.

Worked example (Fed row, 19 Oct 2026): rate **3.875%** (range 3.75–4.00). Last move **+25 bp** on 16 Sep 2026; phase **hiking** since that date; next meeting **28 Oct 2026**. **+8 bp** priced for that meeting (0.3 hikes). Priced 12m **+62 bp** (2.5 hikes of 25 bp), changed **+4 / −8 bp** over 1 week / 1 month. Real rate **+1.31%** (against core). vs Taylor **−199 bp**. Tone **−0.02** (statement of 16 Sep 2026).

Across banks, the 12-month bars show tightening priced at every bank with a curve, from BoC (+103 bp) and BoE (+90 bp) down to SNB (+22 bp, stale curve). BCB, Banxico, BCCh, BanRep, BCRP, RBI, BoK and PBoC have no official curve.

Limitations and pitfalls

- **Not market probabilities.** Government-bond and bill curves carry term, credit, collateral and convenience premia. The constant-basis assumption fails around debt-ceiling episodes, quarter ends and when the market prices a move inside the stub. For the US, the term-premium-adjusted path reduces, but does not remove, the term-premium bias (see the Fed example below).
- The ECB, Riksbank, Norges Bank, BoC and BoJ paths use government curves, not OIS. The RBA path uses bank bills (bank credit premium). BoJ and SNB resolve only 12/24-month horizons.
- A **stale** curve (SNB, monthly publication) is flagged "· stale" in Priced 12m.
- "Real rate" uses realised inflation. The bank page also shows a real rate against model-based expected inflation.
- Tone covers 5 banks. A score is a property of the text, not a policy signal.

Related mnemonics and BQL fields

- `WIRP` opens this page. `WIRP <country or bank>` opens a bank page (e.g. `WIRP US` → Fed, `WIRP EU` or `WIRP DE` → ECB, `WIRP UK` → BoE, `WIRP BR` → BCB). An unknown argument opens the overview with a note.
- `GC` opens this page ("policy and yield curves per bank").
- The rate-cycle phase feeds the regime-performance tables in 2B.

Central bank page

Route: `/macro/cb/:bank` with `fed` , `ecb` , `boe` , `boj` , `snb` , `boc` , `rba` , `riksbank` , `norges` , `bcb` , `banxico` , `bcch` , `banrep` , `bcrp` , `rbi` , `bok` , `pboc` · **Menu:** Macro › Central banks › (bank) · **Mnemonics:** `WIRP <cc or bank>` · **Refresh:** nightly

Purpose

The bank page holds everything about one central bank: the official rate and corridor, rate cycle, next meeting, the implied policy path from its official curve (with a term-premium-adjusted variant for the US), the curve itself (forwards or spot), official projections against the market, the decision history with votes, stance (real rates, r*, Taylor rules), balance sheet and QT pace, and the hawkish/dovish tone of its statements. The page ends with the full method and caveats text.

Screenshot

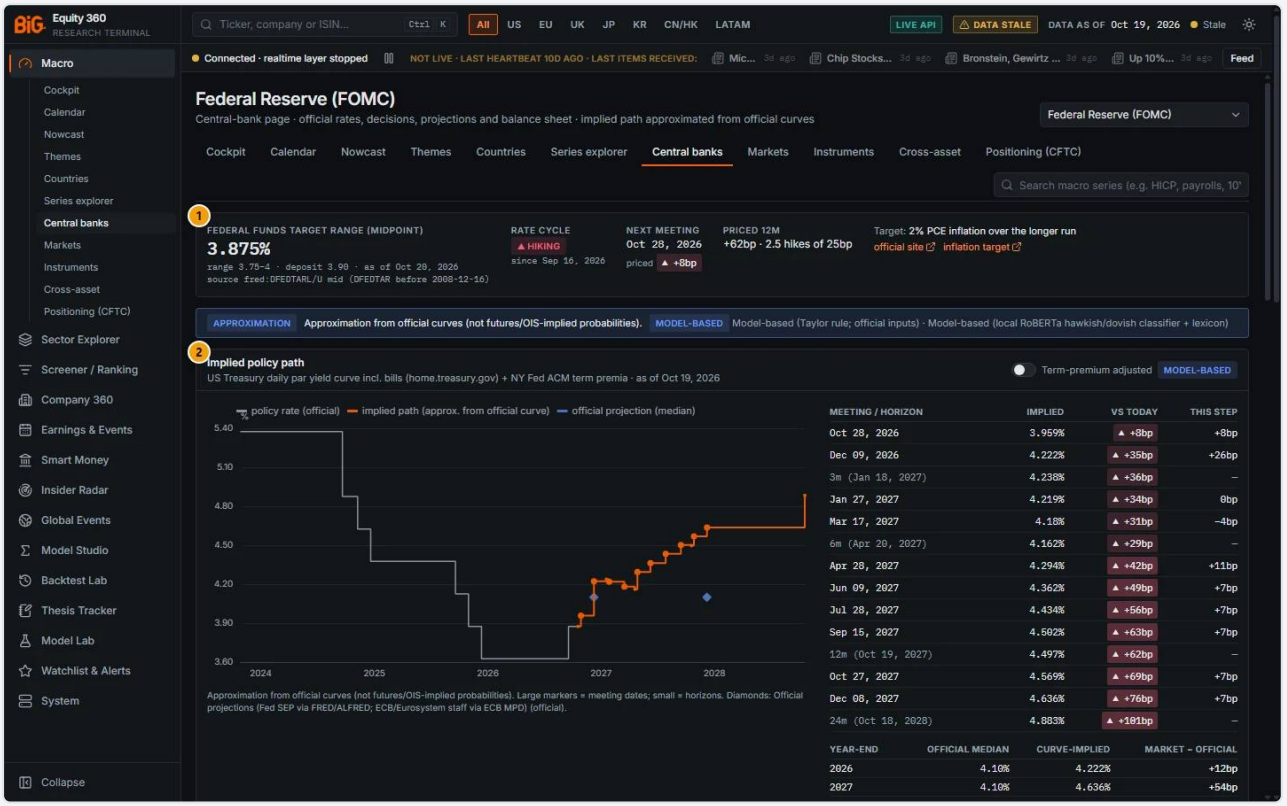


Figure 2A.27: Federal Reserve (`/macro/cb/fed`), as of 19 Oct 2026.

#	Panel	What it shows
1	Federal funds target range (midpoint)	Header: rate, range, deposit (IORB), as-of, source; rate cycle; next meeting with priced change; priced 12m; inflation target with links to the official site and the target statement
2	Implied policy path	Step chart of the official policy rate (3 years) and the implied path (large markers = meetings, small = horizons), official projections as diamonds; table by meeting/horizon; official median vs curve-implied table; "How the priced rate has moved"
3	Official curve used	Forward or spot (zero) curves of the source, with snapshots latest / 1w / 1m / 3m / 6m / 1y
4	Decision history	Decision date, change, rate after, vote, basis (official meeting / unscheduled)
5	Stance	Real rates, r*, output gap, policy − Taylor; charts of real rates against r* and of the policy rate against three Taylor rules
6	Balance sheet	Components (4 charted), latest, 13-week change, annualised pace, 1-year change, source
7	Communication tone	Tone of each scored document over time with decision markers; list of documents with score, vote and link
8	Method and caveats	Implied-path method text, caveats, and every label

Figure 2A.28: Official curve used: US Treasury forwards, latest (orange) against snapshots back to 1 year.

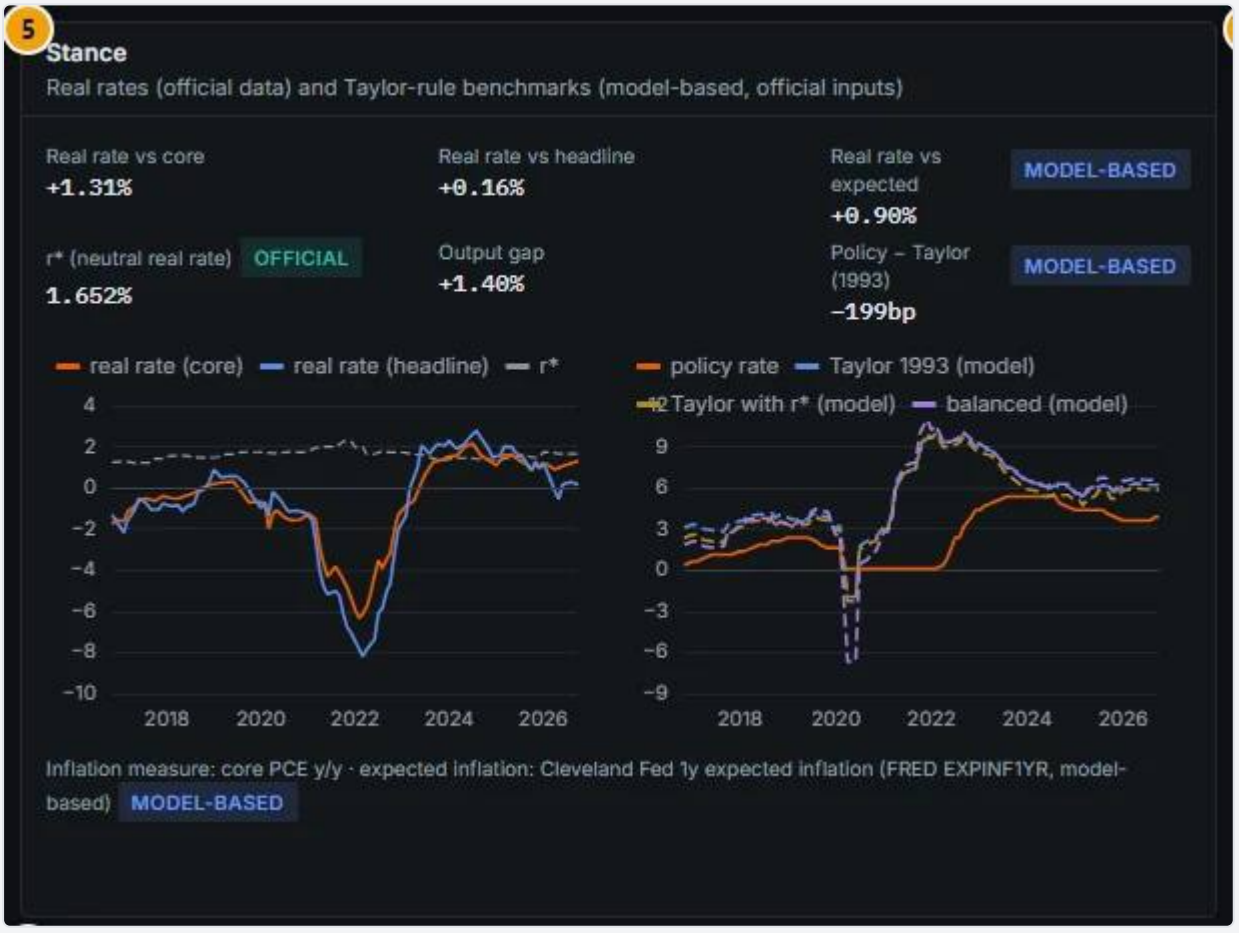


Figure 2A.29: Fed stance: real rate +1.31% (core), +0.16% (headline), +0.90% (expected, model-based); r 1.652% (official, NY Fed LW); output gap +1.40%; policy − Taylor −199 bp.*

Figure 2A.30: Fed communication tone since 2021, with hikes (rose ▲ at the top), cuts (teal ▼ at the bottom) and holds (grey).

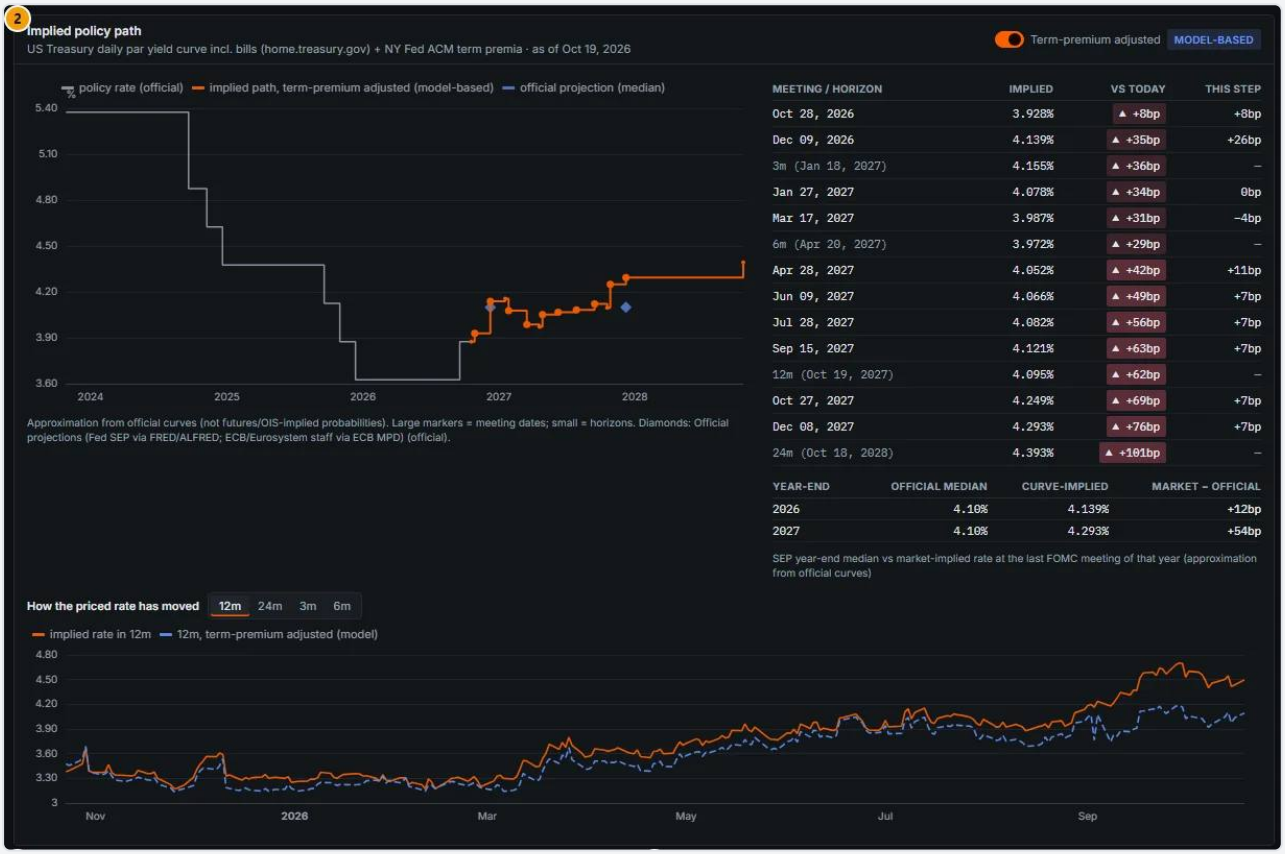


*Figure 2A.31: Fed implied policy path with **Term-premium adjusted** on (badge 2 of the Fed page). The line becomes "implied path, term-premium adjusted (model-based)": 4.095% at 12m and 4.393% at 24m. In "How the priced rate has moved" the dashed adjusted 12m history sits below the raw one, and the gap has widened since mid-2026.*

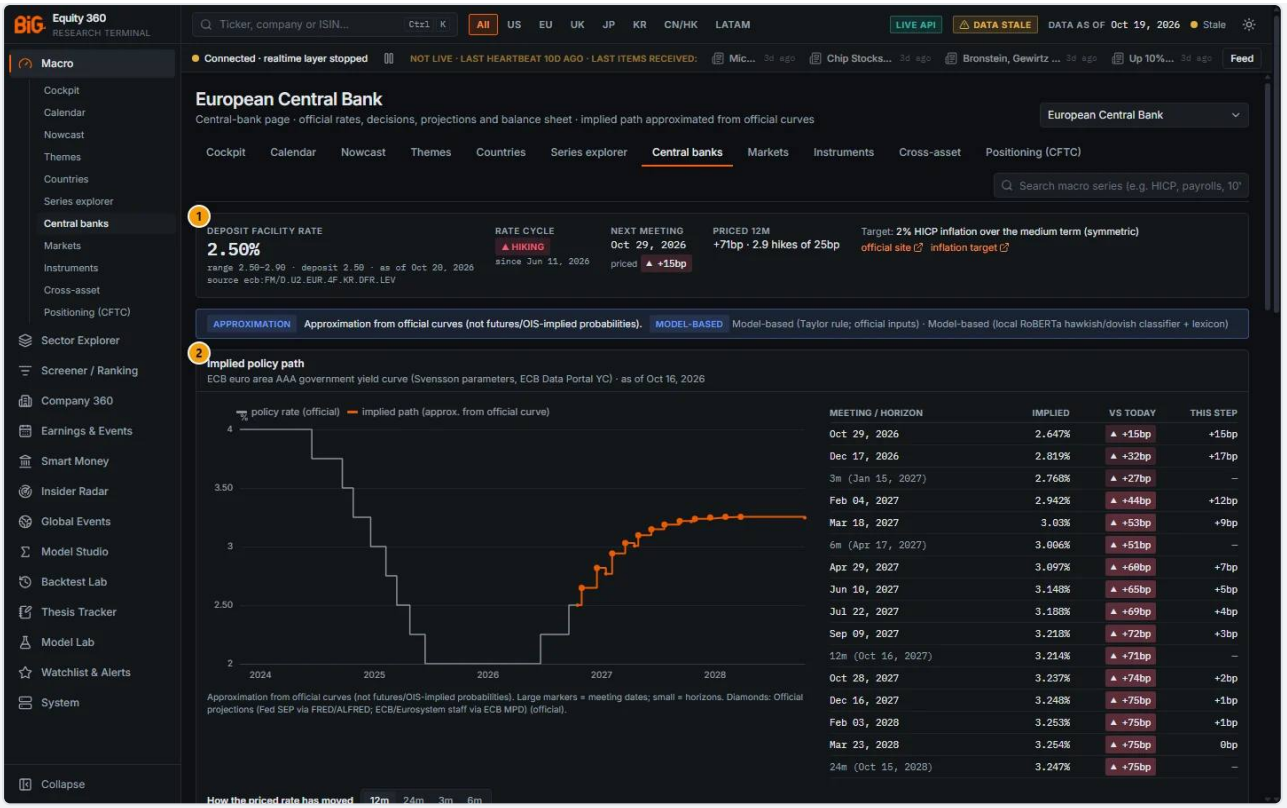


Figure 2A.32: European Central Bank (`/macro/cb/ecb`): implied path from the ECB AAA Svensson curve, as of 16 Oct 2026.

#	Panel	What it shows
1	Deposit facility rate	2.50% (range 2.50–2.90), hiking since 11 Jun 2026; next meeting 29 Oct 2026 (+15 bp priced); priced 12m +71 bp (2.9 hikes); target "2% HICP inflation over the medium term (symmetric)"
2	Implied policy path	Path to 24 months; horizon segmented control 12m / 24m / 3m / 6m for the history chart; no term-premium switch (ACM is US only)
3	Official curve used	ECB AAA Svensson forwards / spot
4	Decision history	ECB decisions
5	Stance	Real rates, HLW r* (euro area), OECD output gap, Taylor rules
6	Balance sheet	Eurosystem (ECB ILM)
7	Communication tone	Monetary policy decision press releases
8	Method and caveats	Svensson method, govvie-curve caveat, labels

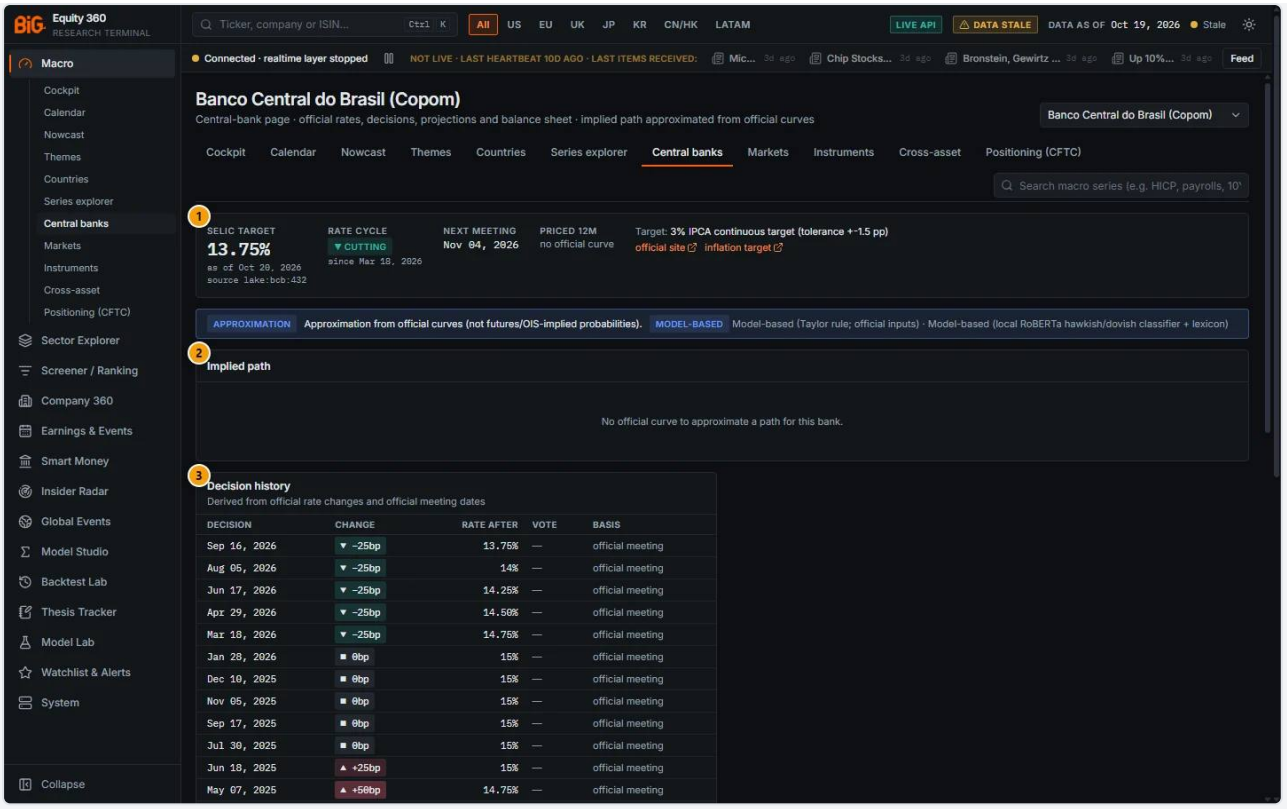


Figure 2A.33: Banco Central do Brasil (`/macro/cb/bcb`): a bank without an official curve.

#	Panel	What it shows
1	Selic target	13.75%, cutting since 18 Mar 2026; next meeting 4 Nov 2026; Priced 12m "no official curve"; target "3% IPCA continuous target (tolerance ±1.5 pp)"
2	Implied path	"No official curve to approximate a path for this bank."
3	Decision history	Copom decisions from the official Selic series and meeting dates (no votes published)
4	Stance	Real rates only (vs core +13.31%, vs headline +9.17%, vs expected +9.28%). No output gap and no r*, so no Taylor-rule values ("—")
5	Balance sheet	"No official balance-sheet series for this bank."
6	Communication tone	"No official statements scored for this bank."
7	Method and caveats	Labels

Controls

Control	What it does
Central bank (top right)	Switches bank
official site / inflation target links	Open the bank's official pages
Term-premium adjusted switch	**Fed only.** Replaces the implied path, the table values, the curve-implied projection comparison and the history chart with the ACM-adjusted values (badged model-based)
Implied path chart (hover)	Date, every line's value, change vs today ("+35bp · 1.4 hikes of 25bp") and 1w/1m change
Horizon (How the priced rate has moved)	3m / 6m / 12m / 24m: history of the curve-implied rate at that fixed horizon (dashed TP-adjusted line when the switch is on)
Forwards / Spot (zero)	Curve kind in the Official curve used panel
Snapshot slider and labels	Highlights one snapshot (1y, 6m, 3m, 1m, 1w, latest); the latest stays dashed for comparison

Control	What it does
Show all N decisions / Show recent only	Expands the decision table beyond 12 rows
Vote cell (hover)	The source of the vote (official statement URL)
Tone point (hover)	Date, document type, tone (model), hawkish/dovish sentence shares, vote
Document link icon	Opens the scored official document

Methodology

IMPLIED PATH AND TERM-PREMIUM ADJUSTMENT

The path follows `implied_v1` (see Central banks overview). For the Fed, the method text reads: "UST par curve bootstrapped to zero/forward rates (bills ≤ 6M as simple 365-day yields, 1Y as bond-equivalent zero, 2–10Y semiannual par); implied fed funds midpoint = average forward per inter-meeting period minus the stub basis; term-premium-adjusted path subtracts NY Fed ACM forward term premia (model-based, official NY Fed estimates)."

Term-premium-adjusted path (US only). For each window $[t_1, t_2]$ the app subtracts the ACM forward term premium over that window, net of the term premium over the stub:

$$\text{implied_tp} = \text{implied} - (\text{TPfwd}[t_1, t_2] - \text{TPfwd}[\text{stub}])$$

It uses the latest ACM date on or before the curve date (ACM is published with a lag).

Forwards vs spot (zero). The forward curve is the instantaneous or short-window forward rate implied for each future date, which is what the path is built from. The spot (zero) curve is the yield to each maturity. Snapshots are kept for the latest date and 1 week, 1 month, 3, 6 and 12 months earlier.

Official projections vs market. The Fed SEP median (FRED/ALFRED vintages since 2015) and ECB/Eurosystem staff projections (ECB MPD) are compared with the curve-implied rate at the last meeting of each year. The table's label reads "SEP year-end median vs market-implied rate at the last FOMC meeting of that year (approximation from official curves)". BoE MPR projections are not machine-readable, so the BoE has none.

STANCE AND TAYLOR RULES (METHOD `STANCE_V1`, MONTHLY)

π is the bank's target measure (US core PCE y/y; other banks core CPI, else headline), π^* the official inflation target, and `gap` the output gap (US: CBO, $100 \times (\text{GDPC1} / \text{GDPPOT} - 1)$; others: OECD Economic Outlook, where current and next year are OECD projections).

$$\begin{aligned}
&\text{Taylor 1993} && = 2 + \pi + 0.5\,(\pi - \pi^*) + 0.5\,\text{gap} \\
&\text{Taylor with } r^* && = r^* + \pi + 0.5\,(\pi - \pi^*) + 0.5\,\text{gap} && \text{(official } r^* \text{ where published)} \\
&\text{Balanced} && = r^*|2 + \pi + 0.5\,(\pi - \pi^*) + 1.0\,\text{gap} && (r^* \text{ if available, else } 2)
\end{aligned}$$

- **r*** comes from official NY Fed estimates: Laubach–Williams (US, one-sided) and Holston–Laubach–Williams (Canada, euro area). Other banks have no machine-readable r*.
- **Expected inflation** for the US is the Cleveland Fed 1-year expected inflation (FRED EXPINF1YR, model-based by the Cleveland Fed). Elsewhere it is our AR(1) forecast of headline y/y 12 months ahead, fitted on the trailing 120 months (model-based, `ar1_12m`).
- **Real rate vs core / headline / expected** = policy rate − the respective inflation measure.
- **Policy − Taylor** uses the r* rule when an official r* exists, otherwise Taylor 1993 (see the pitfall below).

BALANCE SHEET

Official weekly or monthly statements: Fed H.4.1 (FRED), ECB ILM, BoE IADB, BoJ accounts. **QT pace** = change against about 13 weeks earlier × 365/91. Negative means run-off and is shown in rose.

COMMUNICATION TONE (METHOD `TONE_V1`)

- **Documents.** One official document per past policy meeting since 2020: FOMC statements (with dissents parsed), ECB monetary policy decision press releases, the BoE Monetary Policy Summary and minutes, BoJ Statements on Monetary Policy, and BoC rate announcements.
- **Model.** Sentences of 40–600 characters (at most 120 per document) are classified by a local CPU model, `mrince/CBRT-RoBERTa-HawkishDovish-Classifier` (MIT licence; RoBERTa-base fine-tuned on central-bank texts; labels neutral, hawkish, dovish).
- **Score.** `tone = mean(P(hawkish) − P(dovish))` over the sentences, in [−1, +1]. The hawkish and dovish shares are the shares of sentences whose top class is hawkish or dovish.
- **Lexicon.** A transparent lexicon score (per sentence `(h − d)/(h + d)`, averaged over sentences with a match) is stored alongside and used if the model cannot be loaded.
- In the code's words, the scores "are model-based text measures, not policy signals".

Data sources and refresh

As listed under Central banks overview. The implied-path history is daily over about 2 years. Tone scores only new documents each night (sentence cache).

Validation status

Output	Status in the app
Policy rate, decisions	official; BIS cross-check 0 bp
Implied path	approximation from official curves (not futures/OIS-implied probabilities); not validated against OIS/futures
TP-adjusted path	model-based (NY Fed ACM, official estimates)
Taylor rules	model-based (official inputs); benchmarks, not forecasts
Expected inflation	model-based (Cleveland Fed for the US; our AR(1) elsewhere)
r*	official (NY Fed LW/HLW), badged official
Tone	model-based; not validated as a predictor of decisions
Rate cycle	rule-based

Correct usage and worked example

Compare the raw and term-premium-adjusted paths: the gap between them shows how much of the "priced" move could be term premium rather than expected policy. Then check the curve snapshots, to see whether pricing moved recently or has been stable.

Worked example: Fed, 19 Oct 2026 (Figure 2A.27, API values).

- **Anchor.** Policy rate 3.875%; stub basis +19.3 bp (anchor: stub to the next meeting).
- **Path.** 28 Oct 2026: 3.959% (+8 bp). 9 Dec 2026: 4.222% (+35 bp; this step +26 bp). 12m horizon (19 Oct 2027): 4.497% (**+62 bp, 2.5 hikes**). 24m: 4.883% (+101 bp).
- **Term-premium adjusted** (Figure 2A.31). The 12m point falls to **4.095%** (+22 bp), so about two-thirds of the raw 12-month pricing sits in the ACM term premium.
- **Projections against the curve.** SEP median 4.10% for both 2026 and 2027. Curve-implied 4.222% at the December 2026 meeting (**+12 bp** over SEP) and 4.636% at the December 2027 meeting (**+54 bp**).
- **Taylor rules.** Core PCE 3.008%, target 2%, CBO gap +1.4%, r* 1.652% (NY Fed LW):

    - Taylor 1993 = 2 + 3.008 + 0.504 + 0.70 = **6.21%**
    - Taylor with r* = **5.86%**
    - Balanced = **6.56%**
    - Policy − Taylor (r*) = 3.875 − 5.864 = **−199 bp**: on these rules policy is accommodative.
- **Tone.** The 16 Sep 2026 statement (vote 12–0, a hike) scored −0.02. Earlier 2026 statements scored +0.28 (29 Jul, vote 9–3) and +0.15 (17 Jun).

Limitations and pitfalls

- The Stance panel labels its key figure "Policy − Taylor (1993)", but the value is policy − **Taylor with r*** whenever an official r* exists (Fed, BoC, ECB). In the example it is −199 bp against the r* rule; against Taylor 1993 it would be −234 bp. For other banks it is Taylor 1993.
- With **Term-premium adjusted** on, only the **Implied** column, the chart line and **Curve-implied** switch to adjusted values. The **vs today** chips, **this step**, **Market − official** and the header's **Priced 12m** stay on the raw path. In Figure 2A.31, the 12m row reads 4.095% next to "+62bp", although 4.095% is only +22 bp above today's 3.875%, and 2026 shows "+12bp" although 4.139% is +4 bp above the SEP's 4.10%. Compute adjusted changes from the Implied column yourself.
- The "Diamonds: Official projections ..." note appears under the path chart whenever the bank has projections, even when none can be plotted. For the ECB on 19 Oct 2026 the projections are available but the comparison with the market is empty, so no diamonds are drawn (Figure 2A.32).
- When a bank has no official r*, the r* figure shows "—%" but can still carry the green **official** badge (e.g. BCB). Read "—" as "not published".
- Taylor-rule levels depend heavily on the output gap. OECD gaps for the current and next year are projections and are revised.
- BoJ and SNB resolve only 12/24m horizons, so their **Priced next mtg** column is empty.
- Tone history starts in 2020 and covers 5 banks. Statement wording is sticky, so tone moves slowly and can lag decisions (Figure 2A.30).
- Vote splits are parsed from official texts. "1 against" style entries show the number of dissents when the full split is not stated.

Related mnemonics and BQL fields

- `WIRP US` (Fed), `WIRP EU` or `WIRP DE` (ECB), `WIRP UK` (BoE), `WIRP JP` (BoJ), `WIRP BR` (BCB), and the bank ids themselves (`WIRP FED` , `WIRP BOC` , ...).
- No BQL fields expose central-bank data. The rate-cycle phase and macro factors feed 2B. Markets and cross-asset.

2B. Markets and cross-asset

This chapter covers the market-facing half of the Macro menu: the morning **Markets overview**, the **Instruments** browser and instrument pages (FX, futures, indices, ETFs), the four **macro → markets** tools (Exposures, Regime performance, Scenario builder, Episode replay), the **Cross-asset dashboard** and **Positioning (CFTC)**.

All of these pages share the macro sub-navigation (Cockpit · Calendar · Nowcast · Themes · Countries · Series explorer · Central banks · Markets · Instruments · Cross-asset · Positioning (CFTC)) and the macro series search box on the right; both are described in 2A. Macro. The data date shown in the top bar (19 Oct 2026 in every screenshot of this chapter) is the snapshot date; each panel also carries its own dates, and these often differ by region and source.

Page or tab	Route	Section
Markets overview	`/macro/markets`	Markets overview
Instruments browser	`/macro/markets?tab=instruments`	Instruments browser
Instrument page	`/instruments/:id`	Instrument page
Exposures	`/macro/markets?tab=exposures`	Exposures
Regime performance	`/macro/markets?tab=regimes`	Regime performance
Scenario builder	`/macro/markets?tab=scenarios`	Scenario builder
Episode replay	`/macro/markets?tab=episodes`	Episode replay
Cross-asset dashboard	`/macro/cross-asset`	Cross-asset dashboard
Positioning overview	`/macro/positioning`	Positioning (CFTC)
CFTC market page	`/macro/positioning/:code`	CFTC market page

> **Note:** Yahoo Finance market data (futures, indices, ETFs, futures curves, the instruments browser) is **not licensed for commercial use** (docs/OPERATIONS_ROADMAP.md, data licensing review). Treat those series as indicative until a licensed vendor feed replaces them. Official series (FRED / Fed H.15, ECB, central banks, CFTC) are labelled as such on each panel.

Markets overview

Route: `/macro/markets` (default tab Overview; `/market` redirects here) · **Menu:** Macro › Markets › Overview · **Mnemonics:** `WEI` · **Refresh:** prices and macro in the nightly DAG (weekly profile by default: Saturday 08:00 Lisbon; daily profile Tue–Sat 01:30), cross-asset signals nightly, CFTC weekly

Purpose

The Overview is the portfolio manager's morning view. On one screen it answers: what moved (index board, movers), how broad the move was (breadth, sector heatmap), whether risk is on or off (model-based risk gauge, cross-asset tape), what rates, credit, policy pricing and the dollar are doing, and, crucially, **how fresh each region's data is**. It was rebuilt after the 19 Oct 2026 functional audit (docs/AUDIT_2026-10-19.md §4), which found that the earlier page ignored most of its own payload and mixed dates without saying so.

The region selector in the top bar (All · US · EU · UK · JP · KR · CN/HK · LATAM) drives the whole page.

Screenshot

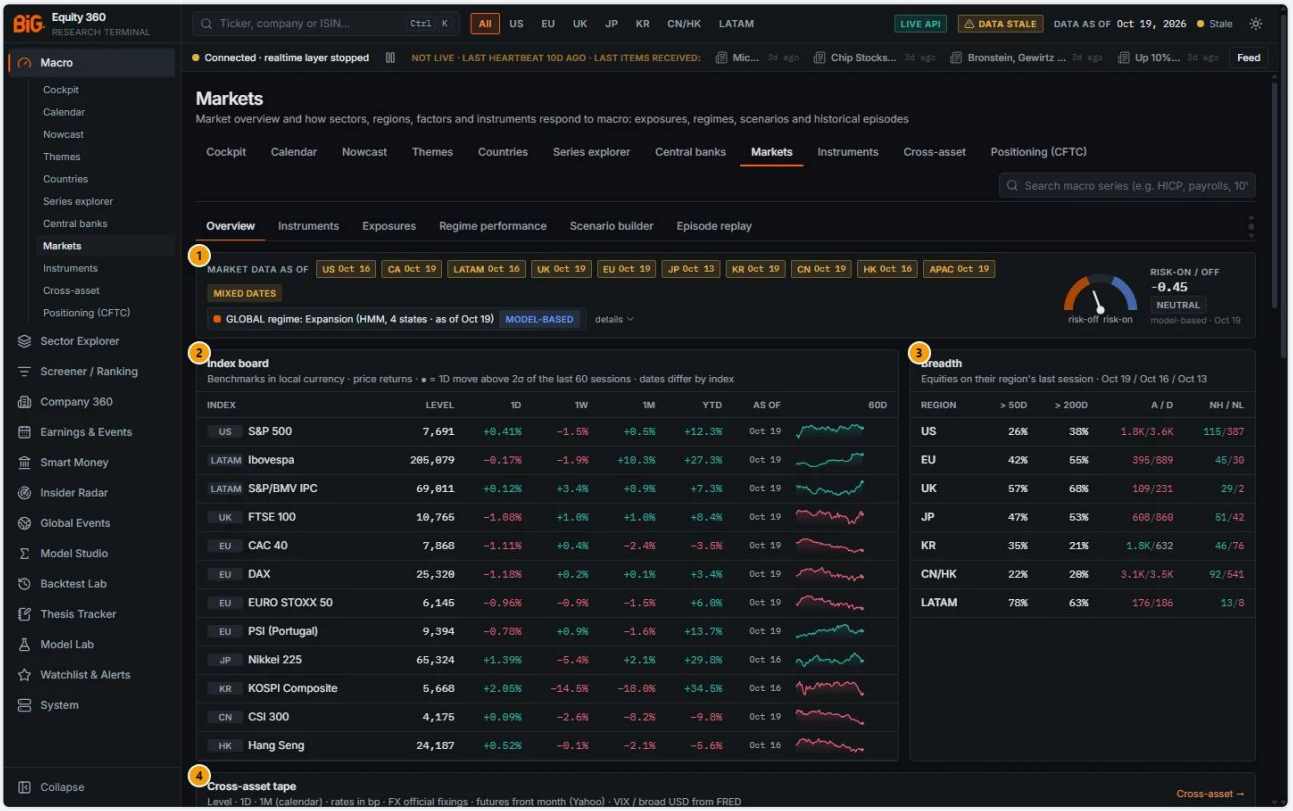


Figure 2B.1: Markets overview (`/macro/markets`), region All, data as of 19 Oct 2026.

#	Panel	What it shows
1	Data dates, regime and risk appetite	One chip per region with its last equity session ("Market data as of"), a "mixed dates" badge, the macro regime chip (HMM, model-based) and the risk-on / risk-off gauge (model-based)
2	Index board	Benchmark indices in local currency: level, 1D, 1W, 1M, YTD price returns, the date of each index's last bar and a 60-session sparkline
3	Breadth	Per region: share of names above their 50-day and 200-day averages, advancers / decliners, new 52-week highs / lows
4	Cross-asset tape	Level, 1D and 1M change for rates, FX, commodities, volatility, crypto and credit
5	Sector performance	Sector × region heatmap of cap-weighted returns (or treemap), total or relative to the region's benchmark
6	Movers (1D)	Top liquid gainers and losers of the day after a server-side liquidity screen
7	Rates	A sovereign yield curve on a numeric tenor axis, current vs 1M and 1Y ago, with the policy rate as a dotted marker, plus a one-year US 2s10s sparkline
8	Credit spreads	ICE BofA option-adjusted spreads and the Moody's Baa yield spread: level, 1D and calendar 1M change, empirical percentile, one-year spark
9	Policy rate vs priced 12m	Each central bank's policy rate, next meeting and the change priced over 12 months, labelled as an approximation
10	Macro headlines	Up to six headline macro tiles per region (three with region All), as rates of change with their observation dates

Panels 4 to 10 sit below the fold. Close-ups follow.

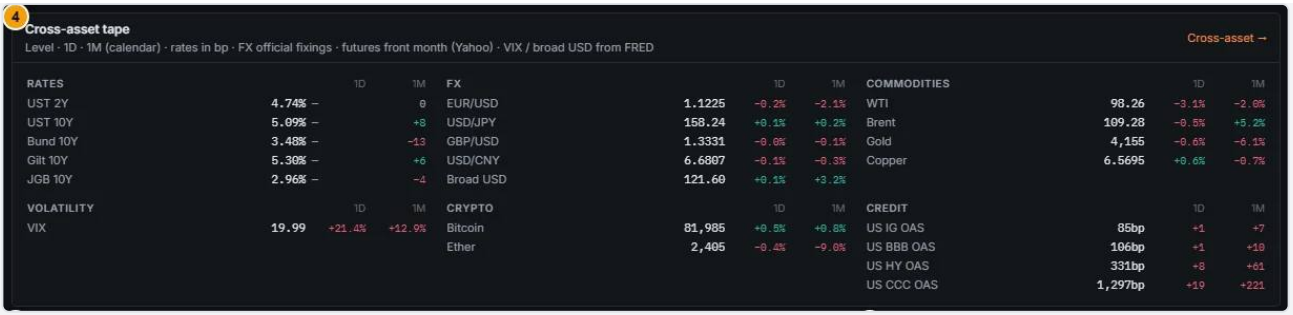


Figure 2B.2: Cross-asset tape. Rates in per cent with changes in bp; FX, commodities and crypto in price terms with % changes; credit in bp.

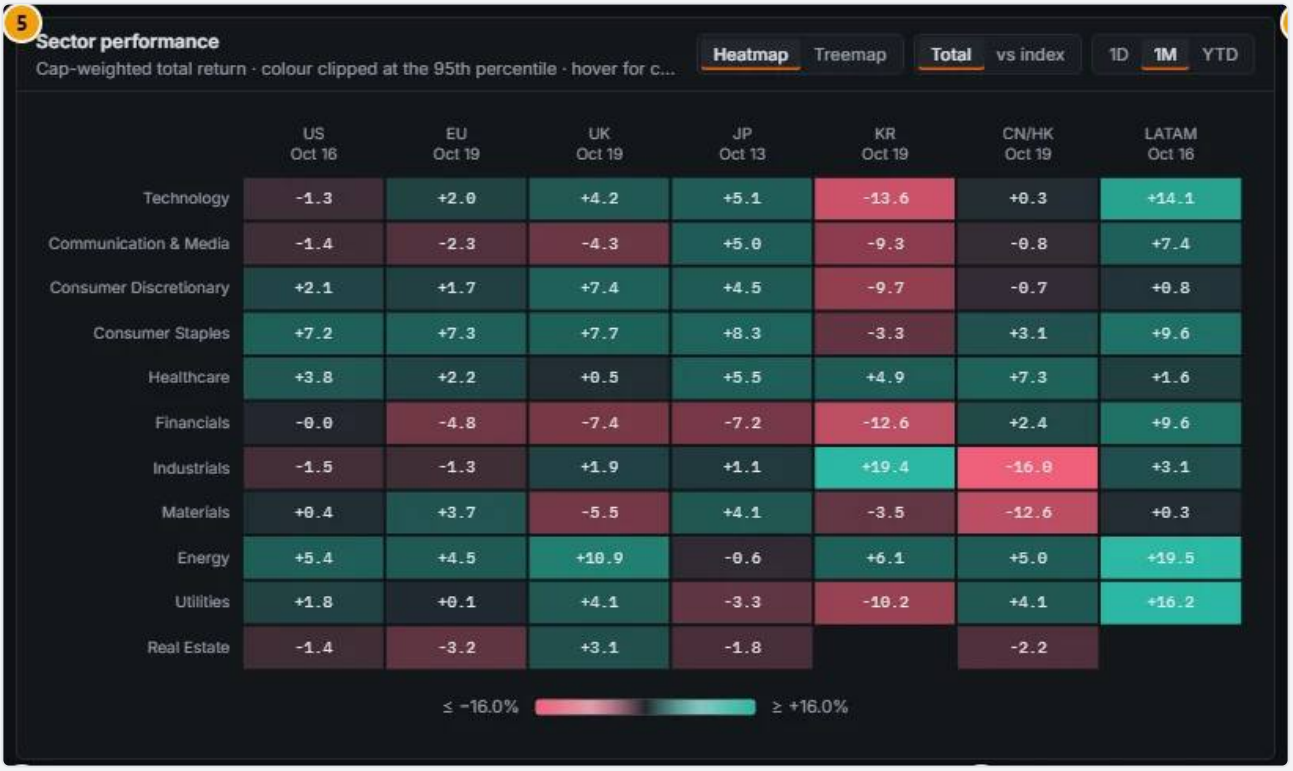


Figure 2B.3: Sector performance, heatmap view, Total basis, 1M horizon. Each column header carries the region's data date.

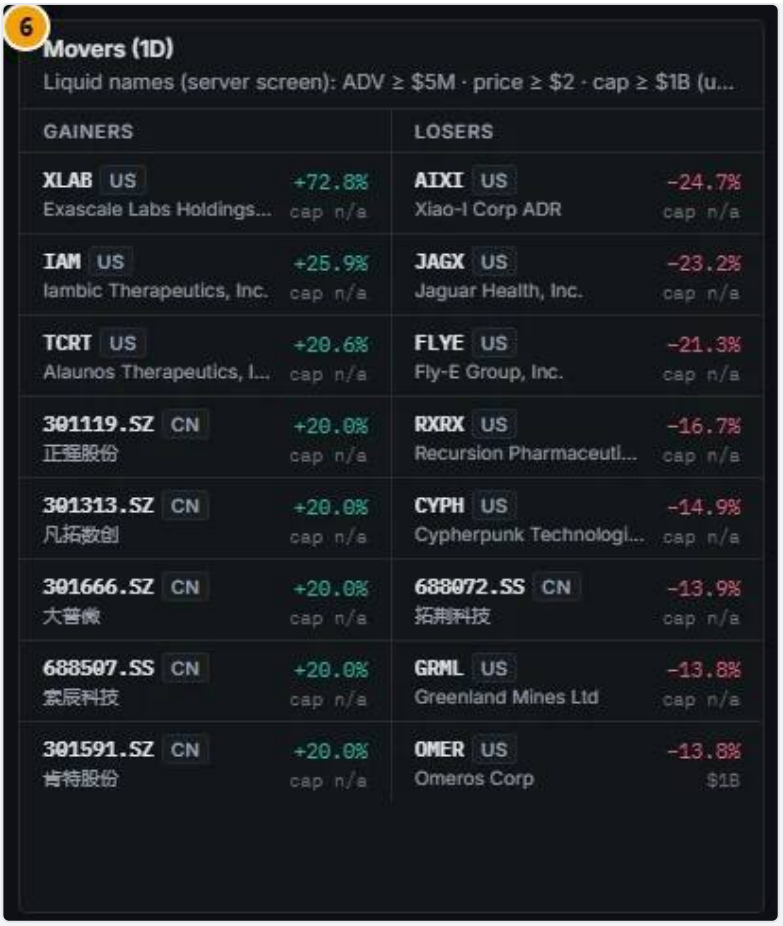


Figure 2B.4: Movers (1D) after the server-side liquidity screen.

Figure 2B.5: Rates panel, US Treasury curve on the log tenor axis, with the dotted policy-rate line at 3.88%.

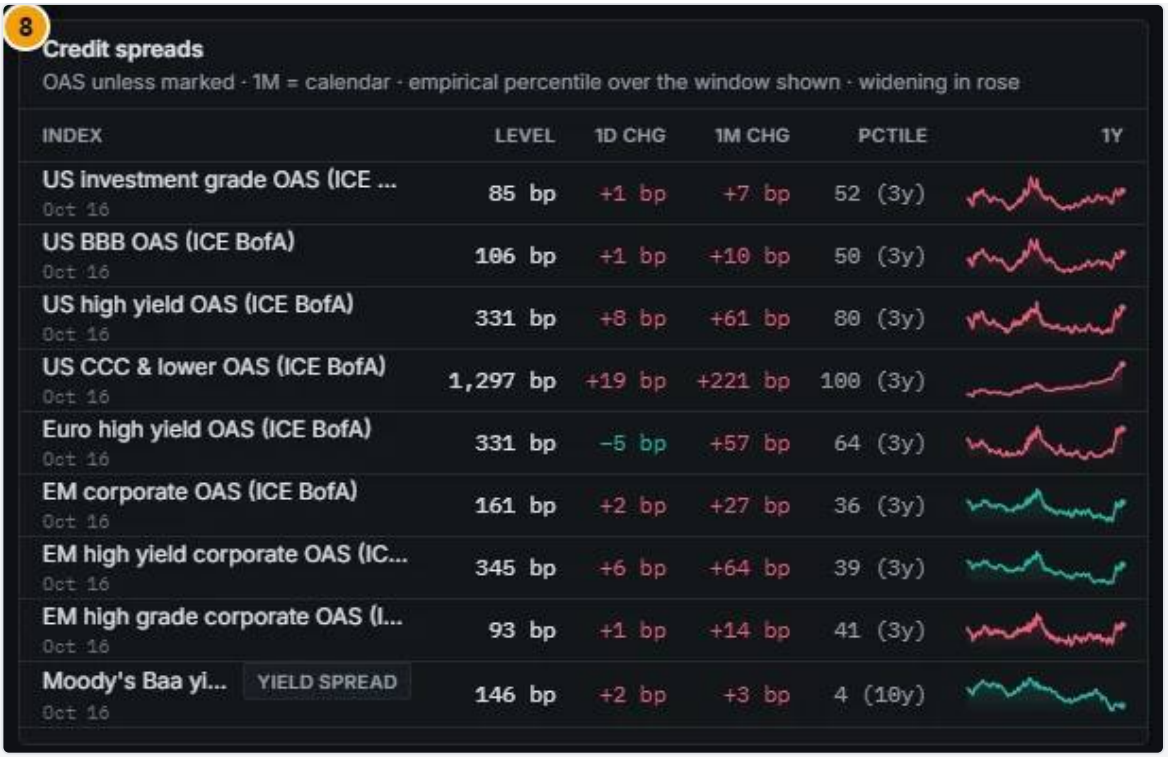


Figure 2B.6: Credit spreads. The Baa row carries a "yield spread" badge because it is not option-adjusted.

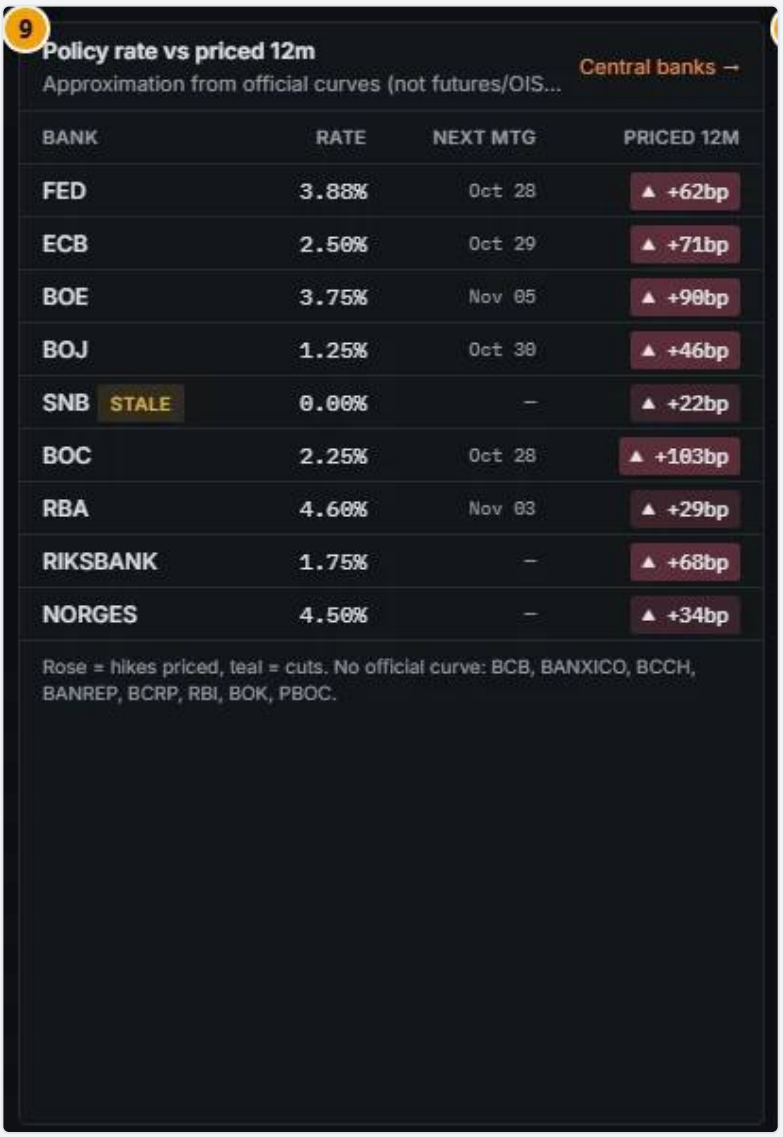


Figure 2B.7: Policy rate vs priced 12m. Rose chips = hikes priced, teal = cuts.

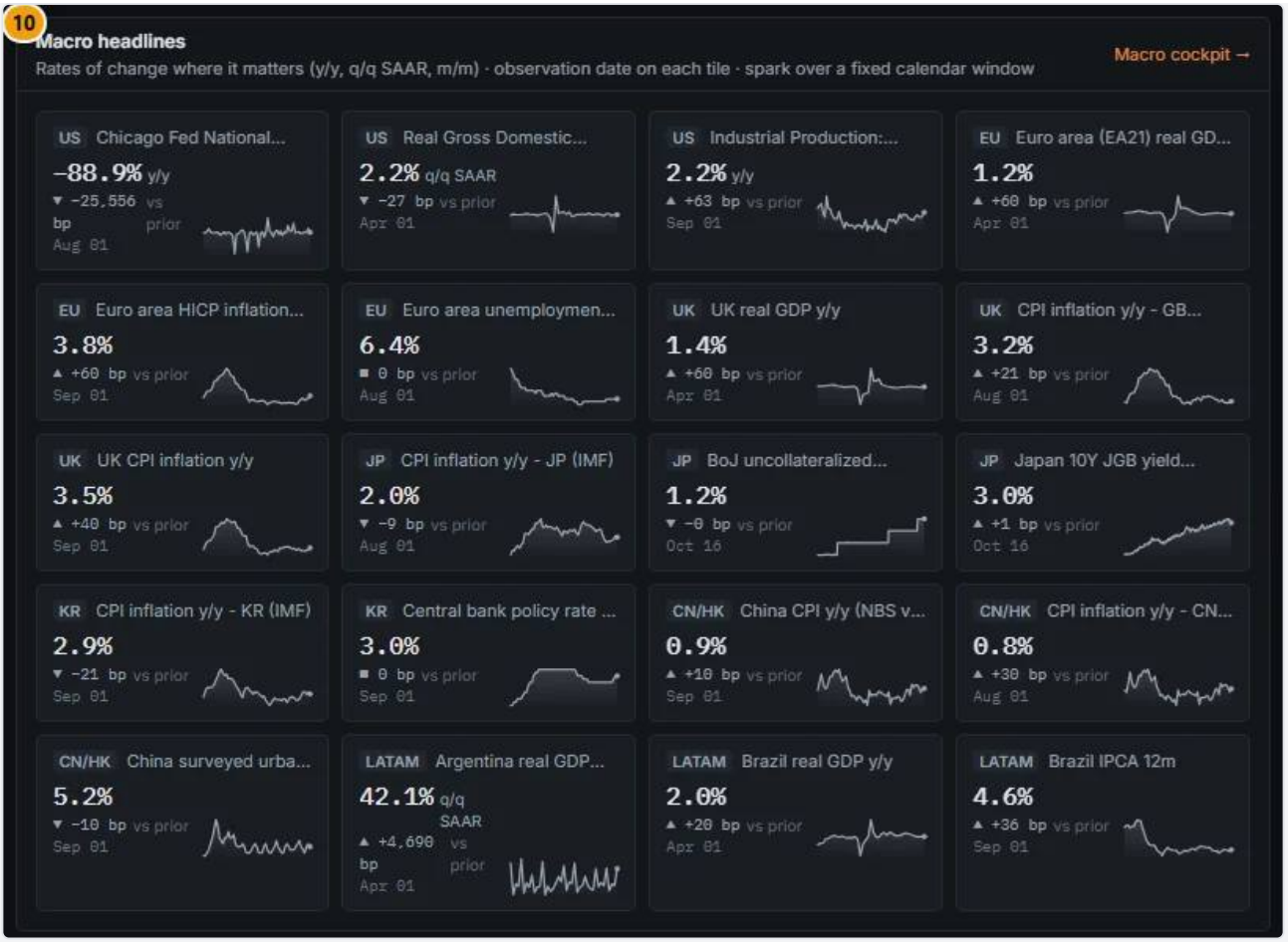


Figure 2B.8: Macro headlines, three tiles per region with region All.

Controls

Control	What it does
Region selector (top bar)	Reloads the page for one region or All; also sets the default yield curve (All and US: US Treasury; EU: Germany; UK: Gilt; JP: JGB; KR: KTB; CN/HK: China; LATAM: Brazil; other: Australia) and the number of macro tiles per region (3 for All, 6 otherwise)
Region date chip (hover)	Tooltip: the region's last session, how many sessions it is behind the latest region, how many weekdays before today, and the share of the region's names whose last bar is older
"mixed dates" badge (hover)	Appears when regions are on different sessions: cross-region 1D figures then compare different days
Regime chip	Opens the Macro cockpit (`/macro`)
details ▾ (next to the regime chip)	Popover with the HMM's posterior probability per state, the monthly regime history (hatched months have no snapshot), the model inputs (level and z vs the model fit) and a link to the cockpit
Risk-on / off gauge	Opens the Cross-asset dashboard (`/macro/cross-asset`)
● on an index row (hover)	"1D move larger than two standard deviations of the last 60 daily moves"
Index name (hover)	Full name and Yahoo symbol (for example `^GSPC`)
A / D cell (hover)	Exact advancer and decliner counts
Cross-asset →	Opens the Cross-asset dashboard
Tape row (hover)	Source, as-of date, cockpit trend signal and realised-vol z (when known) and the definition "1D = vs previous observation · 1M = vs the value 30 calendar days earlier"
Tape row (click)	FX, commodity and crypto rows open the Cross-asset dashboard; VIX and Broad USD open the series explorer (`/macro/series/fred:VIXCLS` , `/macro/series/fred:DTWEXBGS`)
View: Heatmap / Treemap	Heatmap = sector × region grid; Treemap = one box per sector, area = market cap, colour = cap-weighted return

Control	What it does
Basis: Total / vs index	Heatmap only. "vs index" = sector return minus the region's benchmark index over the same horizon
Horizon: 1D / 1M / YTD	Return horizon of the sector panel (default 1M)
Heatmap cell (hover)	Return, constituents (and how many traded on the session for 1D), market cap, data date per region; in "vs index" mode also the raw sector return, the index return and a warning when the index bar date differs
Mover row	Opens Company 360 for that name; hover shows close, bar date, market cap and volume z
Liquid / All	Shown only when the API does not screen movers itself (older APIs): Liquid keeps names with a known market cap ≥ $1bn and US price ≥ $2
Yield curve (select)	Any curve with at least two market tenors; curves with only two are flagged "(2 tenors)"
Tenor axis: log / linear	Log spreads the short end; linear shows true distances
Policy chip (hover)	"Policy / overnight rate: drawn as a dotted reference line, not as a curve tenor"
As of chip	Turns amber when tenors carry different dates; hover shows the oldest and newest
Central banks → / bank code	Opens the Central banks page or one bank's page (`/macro/cb/:bank`)
Priced 12m chip (hover)	Text of what is priced and the curve date
Macro tile	Opens the series in the series explorer; hover shows the full name, transform, source and observation date
Macro cockpit →	Opens `/macro`

Methodology

DATA DATES PER REGION

For each region the API computes the **modal last-bar date** of its equities that traded in the last 21 days (`as_of`), and the share of names whose last bar is older (`stale_share`). A chip is drawn amber ("stale") when it is more than **2 weekdays** before today's date on your computer, or when more than **25%** of the region's names have an older bar. All 1D aggregates (sector 1D, movers) only use names that are on their region's latest session.

INDEX BOARD

- Level, 1D, 1M and YTD come from the price statistics of each benchmark index (price returns, local currency).
- 1W is computed in the browser from the 60-session spark: last close vs the last close at least 7 calendar days earlier.
- The ● flag is set when $|1D| > 2\sigma$, where σ is the standard deviation of the daily returns inside the 60-session spark (at least 11 returns).
- Rows are ordered by region (US, CA, LATAM, UK, EU, JP, KR, CN, HK…) then by name.

BREADTH

Over the region's equities with a bar in the last five days: share above the 50-day and 200-day moving averages, number of names up and down on their last session, and the number of new 52-week highs and lows. The bar next to a share turns green at ≥ 60% and red at ≤ 40%.

CROSS-ASSET TAPE

Group	Rows	Source and definition
Rates	UST 2Y, UST 10Y, Bund 10Y, Gilt 10Y, JGB 10Y	Official curves of the overview payload; level in %, 1M change in bp (value on or before 30 calendar days earlier). No 1D for rates.
FX	EUR/USD, USD/JPY, GBP/USD, USD/CNY, Broad USD	Pairs from official fixings via the cross-asset asset detail; Broad USD = Fed broad dollar index (FRED DTWEXBGS)
Commodities	WTI, Brent, Gold, Copper	Yahoo continuous front month

Group	Rows	Source and definition
Volatility	VIX	FRED VIXCLS (official copy of the CBOE close)
Crypto	Bitcoin, Ether	CME front-month futures (Yahoo)
Credit	First four OAS series (US IG, US BBB, US HY, US CCC)	ICE BofA OAS via FRED, in bp

For prices, `1D = last / previous observation − 1` and `1M = last / value on or before (last date − 30 days) − 1`. For yields and spreads the same changes are differences in bp. Colours are inverted for VIX and credit (a rise is shown as adverse).

SECTOR PERFORMANCE

```
sector return = Σ (rᵢ · capᵢ) / Σ capᵢ     over names with a return and a market cap > 0
                (simple mean when no name in the cell has a market cap)
vs index      = sector return − return of the region benchmark over the same horizon
```

Benchmarks for "vs index": US `^GSPC`; EU `^STOXX50E` (else `^GDAXI`); UK `^FTSE`; JP `^N225`; KR `^KS11`; CN/HK `^HSI` (else `000300.SS`); LATAM `^BVSP`. The colour scale is **winsorised**: it is clipped at the 95th percentile of absolute values (with a floor of 0.5% for 1D and 2% for 1M / YTD), so a single outlier no longer washes out the grid; cell labels keep the true value. The treemap drops sectors with no known market cap and lists them under the chart.

MOVERS

The API screens movers server-side (API v1.30 and later): a **known** 20-day average daily value ≥ $5m, a USD price ≥ $2 and a market cap ≥ $1bn; a name with an unknown market cap (common for Japanese, Chinese and Korean listings that lack share counts) qualifies only with ADV ≥ $20m (4 × the minimum). An unknown ADV never passes. The panel shows the top eight each side.

RATES

The curve is drawn on a **numeric** tenor axis (years; log or linear). The O/N point of each curve is the policy or overnight rate (for the US, the effective fed funds rate EFFR, else SOFR; for euro-area countries the ECB deposit rate) and is drawn as a dotted horizontal line, not as a tenor. "1M ago" and "1Y ago" use the last value on or before 30 and 365 days before each tenor's latest date. 2s10s and 5s30s are spreads of the current curve in bp; "10Y 1M chg" is the 10-year change over one month. The sparkline under the chart is FRED T10Y2Y (10-year minus 2-year Treasury, official).

CREDIT SPREADS

Nine series: US IG, US BBB, US HY, US CCC & lower, Euro HY, EM corporate, EM HY corporate, EM high grade corporate (all ICE BofA OAS via FRED) and Moody's Baa yield minus the 10-year Treasury (labelled "yield spread, not OAS"). For each:

```
1D chg  = last − previous observation
1M chg  = last − value on or before (last date − 30 days)
pctile  = empirical percentile of the latest level within the last 10 years
          (or the available window, shown in brackets: "3y" for the ICE BofA series on FRED)
          = (number below + ½ · number equal) / n, needs ≥ 20 observations
spark   = fixed one-year calendar window
```

POLICY RATE VS PRICED 12M

Taken from the central-bank implied paths of 2A. Macro. The panel's subtitle is the app's own label: **"Approximation from official curves (not futures/OIS-implied probabilities)"**. Banks without an official curve are listed at the bottom ("No official curve: ..."); a "stale" badge marks banks whose curve is out of date.

MACRO HEADLINES

Tiles show rates of change rather than index levels: inflation, growth, housing and trade series as y/y; quarterly growth series as q/q SAAR ($(X_t / X_{t-1})^4 - 1$); monthly labour counts as the m/m change; rates and series already in per cent

stay as levels. The change under the value is vs the previous print (in bp for rates). The spark covers a fixed calendar window: 1 year for daily and weekly series, 5 years for monthly, 10 years for quarterly.

REGIME CHIP AND RISK GAUGE

- **Regime chip.** The latest state of the Gaussian HMM regime model for the selected region's scope. If the region has no model, the chip says "No {region} model · GLOBAL regime (fallback)" with a "fallback" badge; if nothing is modelled it says "No regime model for {region}". The popover note reads: "Posterior probabilities of the model's own states (an HMM is often over-confident)."
- **Risk gauge.** The cross-asset risk composite (see Cross-asset dashboard); the gauge spans −2 to +2 with the neutral band from −0.5 to +0.5.

Data sources and refresh

Panel	Source	Cadence
Index board	Yahoo index bars in the price lake	Nightly price ingestion
Breadth, sectors, movers	Equity price lake (Polygon / yfinance) and the serving snapshot	Nightly
Tape: FX	Official fixings (ECB > FRED H.10 > central bank)	Nightly
Tape: commodities, crypto	Yahoo continuous front month	Nightly
Tape: VIX, Broad USD; credit spreads; 2s10s	FRED (official)	Nightly; H.15 data is known the next day
Yield curves	Official curves (Fed H.15, ECB, Bundesbank, BoE, MoF Japan, BoK...)	Nightly; several countries publish monthly, so tenors can carry different dates
Policy pricing	Central-bank implied paths (macro W3)	Nightly
Macro headlines	Official statistical agencies	Nightly
Regime chip	`macro_regimes` HMM	Weekly fit (`macro_regime_fit`), latest state nightly
Risk gauge	`cross_asset_signals` job	Nightly

Validation status

The Overview is a descriptive page built on official and market data; it has **no validation gate**. Two model-based items appear on it:

- the **regime chip** carries a "model-based" badge; it is not formally validated, and the audit (docs/AUDIT_2026-10-19.md, F07 / F08) recorded a 99.99% "Expansion" posterior from this over-confident HMM;
- the **risk gauge** is labelled "model-based"; docs/CROSS_ASSET_DEFINITIONS.md §3 lists only sanity checks (see the Cross-asset section).

The policy-pricing panel is labelled as an **approximation**.

Correct usage and worked example

Read the page top-down, and always read the date chips first.

Worked example (region All, 19 Oct 2026, Figure 2B.1):

1. **Dates.** US equities are on **16 Oct** while CA, UK, EU and CN are on 19 Oct, and JP on 13 Oct; the "mixed dates" badge is on. The S&P 500 row of the index board shows "Oct 19" because the index bar is newer than the bulk of US equities, so the US breadth and US sector columns describe 16 Oct, not 19 Oct.
2. **Risk.** The gauge reads **−0.45, Neutral** (model-based, 19 Oct). It sits just inside the neutral band; the Cross-asset dashboard shows that all eight components are present and the composite fell 0.88 over one month.
3. **What moved.** KOSPI is +2.05% on the day but −14.5% on the week and −18.0% on the month; the S&P 500 is +0.41% on the day and +12.3% YTD.
4. **Credit.** US HY OAS is 331bp, +8bp on the day and +61bp on the month, at the 80th percentile of its (3-year) window: spreads are widening but are not historically extreme over that short window. The Baa yield spread (146bp) sits at

the 4th percentile of 10 years.

5. **Rates and policy.** On the US Treasury curve (as of 16 Oct) 2s10s is +35bp and 5s30s +54bp; the policy marker is at 3.88%. The policy panel shows +62bp priced for the Fed over 12 months, an approximation from official curves.

Limitations and pitfalls

- **"Today" is your computer's clock.** When you open the page days after the snapshot (as in the screenshot, captured after 19 Oct), every chip turns amber even if the snapshot is complete.
- **Cross-region 1D is not one day.** With mixed dates, comparing "1D" across regions compares different sessions.
- **Price returns only.** Index board figures exclude dividends.
- **Limit moves.** Chinese A-shares appear at +20.0% / −20.0% (daily limit) in the movers list; many Asian names show "cap n/a" because share counts are missing.
- **Short credit history.** The ICE BofA series on FRED keep only about 3 years, so their percentile describes a 3-year window, not 10 years.
- **Macro tiles inherit the transform rules.** A diffusion-type index (such as the Chicago Fed National Activity Index, shown as −88.9% y/y in Figure 2B.8) or a volatile quarterly series (Argentina real GDP, 42.1% q/q SAAR) can produce a misleading rate of change; open the series explorer to check the raw level.
- **VIX on the tape is FRED VIXCLS**, which can lag the Yahoo `^VIX` level shown on the instrument page by a session.
- **No USD view.** The index board is in local currency only.

Related mnemonics and BQL fields

- Mnemonics: `WEI` (this page), `GC` (central-bank curves), `WIRP` (implied policy rates), `ECO`, `ECST` (see 14. BQL console and mnemonics).
- BQL fields with the same per-security inputs: `chg_pct_1d`, `chg_pct_1w`, `chg_pct_1m`, `chg_pct_ytd`, `above_50d`, `above_200d`, `adv_usd_20d`.
- Related: Cross-asset dashboard, 2A. Macro (cockpit, central banks, series explorer), 3. Sector Explorer.

Instruments browser

Route: `/macro/markets?tab=instruments` (also `/instruments` and `/macro/markets/instruments`, which redirect) · **Menu:** Macro › Instruments, or Macro › Markets › Instruments · **Mnemonics:** `FXIP` (alias `FX`) · **Refresh:** nightly price ingestion

Purpose

A one-page browser of every non-equity market series in the app: FX pairs, continuous futures, cash indices (equity, volatility, yields, dollar and commodity indices) and ETFs. Use it to check a price, compare short-term moves within a group and jump to the instrument page.

Screenshot

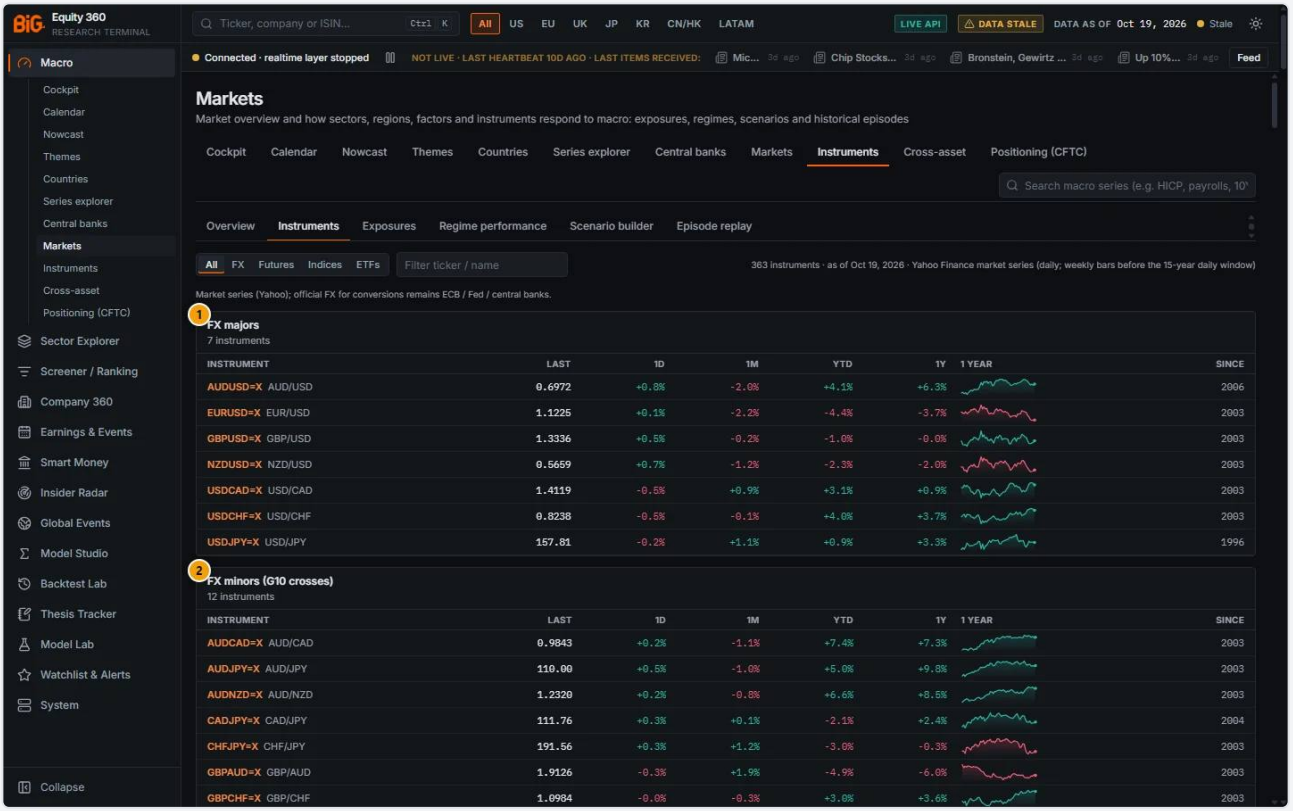


Figure 2B.9: Instruments browser, type All, 363 instruments as of 19 Oct 2026.

#	Panel	What it shows
1	FX majors	7 pairs vs USD
2	FX minors (G10 crosses)	12 crosses
3	EUR crosses	23 pairs
4	FX emerging markets	35 pairs
5	Energy	7 futures (Brent, WTI, heating oil, Henry Hub, E-mini crude, RBOB, Dutch TTF)
6	Metals	9 futures
7	Grains & oilseeds	8 futures
8	Softs	6 futures
9	Livestock & dairy	4 futures
10	Treasury futures	4 futures (2-, 5-, 10-year notes and the bond)
11	Equity index futures	4 futures
12	Currency futures	3 futures
13	Crypto futures	2 CME futures (Bitcoin, Ether)
14	Volatility	13 indices (VIX family, VVIX, SKEW, MOVE, OVX, GVZ, India VIX, ASX VIX...)
15	Equity indices	40 cash indices
16	US Treasury yields	4 yield indices (13-week, 5-, 10-, 30-year)
17	Dollar & commodity indices	2 (US Dollar Index, S&P GSCI)
18	ETFs: US equity & factors	34 ETFs
19	ETFs: sectors & themes	36 ETFs
20	ETFs: international & countries	56 ETFs
21	ETFs: bonds	21 ETFs

#	Panel	What it shows
22	ETFs: commodities	26 ETFs
23	ETFs: currencies	4 ETFs
24	ETFs: crypto	2 ETFs
25	ETFs: volatility	1 ETF

Every group panel has the same columns; the Energy group is shown as an example.

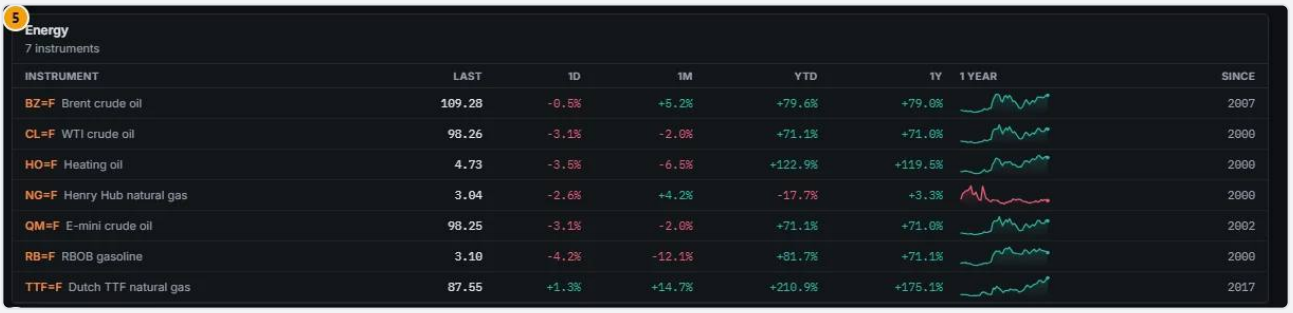


Figure 2B.10: Energy group. The "Since" column is the first year of stored history.

Controls

Control	What it does
All / FX / Futures / Indices / ETFs	Filters the groups by instrument type
Filter ticker / name	Free-text filter across all groups; "No instruments match." when nothing is left
Ticker (orange link)	Opens the Instrument page (`/instruments/<id>` , e.g. `/instruments/IDX%3AEURUSD%3DX`)

Methodology

Column	Definition
Last	Last close, with decimals adapted to the price level
1D, 1M, YTD, 1Y	Price changes from the price statistics (1Y = 12-month return)
1 year	Sparkline of one year of weekly closes
Since	Year of the first stored bar

Names drop the "(Yahoo market rate)" and "(continuous front month)" suffixes in the table; the instrument page shows them in full.

Data sources and refresh

The header line states the source: "Yahoo Finance market series (daily; weekly bars before the 15-year daily window)". The note under it applies to FX: **"Market series (Yahoo); official FX for conversions remains ECB / Fed / central banks."** In other words, these FX rows are market rates for display; every currency conversion in the app uses official fixings. Prices refresh with the nightly price ingestion (weekly profile: Saturday).

Validation status

Not applicable: market data, no model.

Correct usage and worked example

Use the type filter, then sort by eye within a group. **Example (19 Oct 2026, Figure 2B.10):** in Energy, Dutch TTF natural gas is +210.9% YTD and heating oil +122.9%, while Henry Hub is −17.7%; WTI (`CL=F`) is 98.26, −3.1% on the day and +71.1% YTD. Click `CL=F` to open its instrument page with the futures curve and carry (Figures 2B.15 to 2B.17).

Limitations and pitfalls

- **Continuous front months include roll gaps.** A `=F` series jumps when the front contract rolls, so its returns behave like spot-price changes, not futures excess returns.
- **Yield indices are levels.** The US Treasury yield rows are yields in per cent (for example `^TNX` 5.136 on 19 Oct); their "returns" are relative changes of a yield, not investable returns. The "(x10)" in their names does not match the stored values, which are already in per cent.
- **Dates differ.** Some volatility indices (OVX, GVZ, VXN, India VIX, ASX VIX) last printed on 16 Oct in this snapshot; the row does not show a date, the instrument page does.
- **Licensing.** Yahoo data is not licensed for commercial use.

Related mnemonics and BQL fields

- `FXIP` / `FX` opens this tab (all types; pick FX with the filter). A security reference with a yellow key opens the instrument page directly: `EURUSD Curncy DES`, `SX5E Index GP`, `CL1 Comdty GP` (instrument references are resolved by `GET /bql/resolve`, API v1.37).
- Related: Instrument page, Cross-asset dashboard.

Instrument page

Route: `/instruments/:id` (for example `/instruments/IDX:EURUSD=X`, `/instruments/IDX:CL=F`, `/instruments/ETF:SPY`) · **Menu:** from the Instruments browser, the Cross-asset dashboard or a CFTC market page · **Mnemonics:** `DES`, `GP`, `GIP`, `TECH` with an Index / Curncy / Comdty reference; `THES` starts a cross-asset thesis · **Refresh:** nightly

Purpose

The detail page of one FX pair, future, index or ETF: full price history, return and risk statistics, correlations with related markets, and, for instruments linked to a cross-asset asset, the carry and futures-curve panels. It is the instrument equivalent of Company 360.

Screenshot

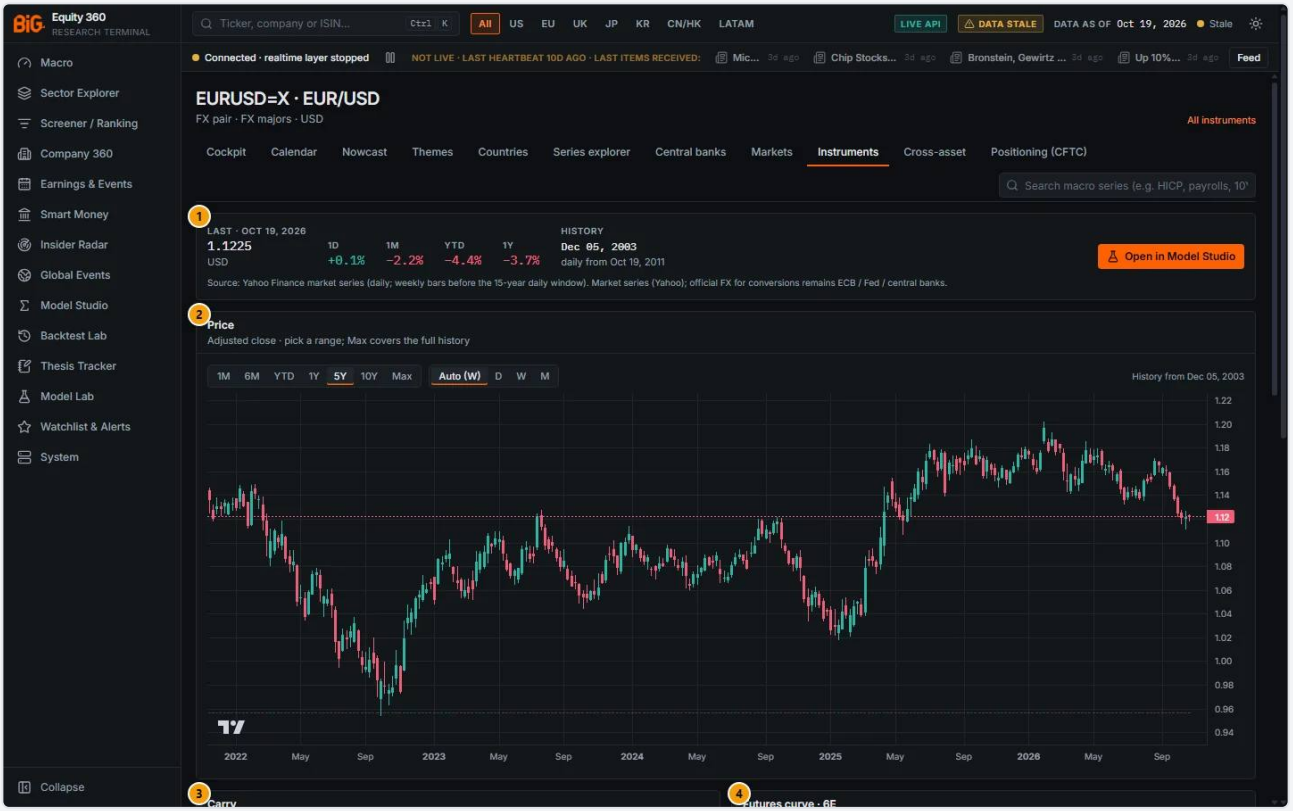


Figure 2B.11: Instrument page for EUR/USD (`/instruments/IDX:EURUSD=X`), 5Y range, weekly bars.

#	Panel	What it shows
1	LAST · OCT 19, 2026	Last price and currency, 1D / 1M / YTD / 1Y changes, history start ("daily from ..."), the source line, and the action buttons
2	Price	Candlestick chart of adjusted closes with range and bar-interval selectors
3	Carry	Cross-asset carry, roll yield and curve label with their history (only for linked instruments)
4	Futures curve · 6E	Term structure of the linked futures root now vs ~1m and ~1y ago (only when a curve is stored)
5	Returns and risk	Annualised return, volatility and maximum drawdown over 1Y, 5Y, 10Y and Max
6	Correlations	Correlation of weekly log returns with up to ten related instruments over 3 and 10 years
7	Rolling 1-year volatility	52-week volatility of weekly log returns, annualised

Figure 2B.12: Carry panel for EUR/USD: policy-rate differential (ECB minus Fed) and, below, the roll yield of the 6E future.

WINDOW	ANN. RETURN	ANN. VOL	MAX DRAWDOWN	FROM
1Y	−3.4%	+5.4%	−6.3%	Oct 24, 2025
5Y	−0.7%	+7.5%	−16.6%	Oct 22, 2021
10Y	+0.3%	+7.3%	−22.1%	Oct 21, 2016
Max	−0.4%	+8.8%	−38.9%	Dec 05, 2003

Figure 2B.13: Returns and risk for EUR/USD.

Figure 2B.14: Correlations for EUR/USD, 3 and 10 years.

For a commodity future the same page shows the futures roll carry and the crude-oil curve:

Figure 2B.15: Instrument page for WTI crude oil (`/instruments/IDX:CL=F`). The "CFTC positioning" button appears because a CFTC market is linked.

Badges 1 to 7 have the same meaning as in Figure 2B.11.

Figure 2B.16: Carry panel for WTI: futures roll yield ln(F1/F2) annualised, front vs next and 12-month (seasonality-neutral).

Figure 2B.17: Futures curve · CL, 16 Oct 2026 (backwardation), vs ~1m and ~1y ago.

An index without a linked cross-asset asset has no Carry or Futures curve panel, so the badges are renumbered:

Figure 2B.18: Instrument page for the CBOE Volatility Index (`/instruments/IDX:^VIX`), 5Y range. The "CFTC positioning" button links to the VIX futures market.

#	Panel	What it shows
1	LAST · OCT 19, 2026	Last level 18.44, 1D −7.8%, 1M +24.5%, YTD +23.3%, 1Y +1.2%, history from 5 Jan 1990, daily from 19 Oct 2011
2	Price	Candlestick chart of the index level
3	Returns and risk	Statistics with the "Level series" note (below)
4	Correlations	Correlations with related volatility indices and the anchors
5	Rolling 1-year volatility	52-week volatility of the weekly log changes of the level

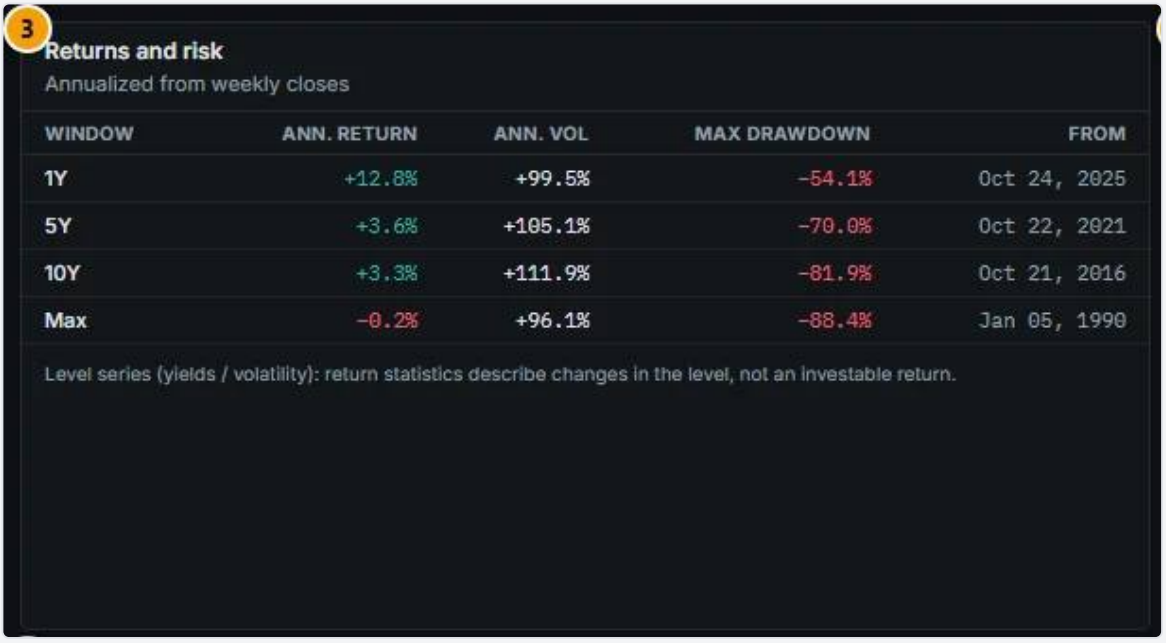


Figure 2B.19: Returns and risk for the VIX: "Level series (yields / volatility): return statistics describe changes in the level, not an investable return."

Controls

Control	What it does
All instruments	Back to the Instruments browser
CFTC positioning	Shown when a CFTC market is linked to the instrument (futures contract, mini / micro, or a labelled price proxy); opens `/macro/positioning/:code`
Open in Model Studio	Opens Model Studio with a draft study: weekly log returns of this instrument, time series from 2000, ARIMA (see 9. Model Studio)
Range: 1M / 6M / YTD / 1Y / 5Y / 10Y / Max	Chart range (default 5Y); Max covers the full history
Interval: Auto / D / W / M	Bar interval; Auto picks one for the range (weekly for 5Y). A badge "weekly (daily capped at 10y)" appears when daily bars are not available for the whole range
Cross-asset cockpit (Carry panel)	Opens the Cross-asset dashboard with this asset selected
Curve viewer (Futures curve panel)	Opens the Cross-asset dashboard with this futures root in the curve viewer
Correlation ticker	Opens that instrument's page; hover a value for the number of weekly returns

Methodology

- **Returns and risk** (weekly closes): `ann. return = (C_end / C_start)^(1 / years) − 1` ; `ann. vol = sd(weekly log returns) × √52` ; `max drawdown = min(C / running peak − 1)` . A window is reported only if history starts within 30 days of its nominal start, otherwise "not enough history".
- **Correlations:** Pearson correlation of weekly log returns over the last 3 and 10 years (at least 20 common weeks). The related list is up to five instruments of the same group plus the anchors S&P 500, US Dollar Index, US 10-year yield, gold, WTI, VIX and EUR/USD (ten at most).
- **Rolling 1-year volatility:** standard deviation of the last 52 weekly log returns × √52.
- **Carry and curve:** from the cross-asset signals (definitions in the Cross-asset dashboard section). The "Curve" stat is the **model-based** curve-shape label.

Data sources and refresh

Prices are Yahoo market series, daily for the last 15 years and weekly before ("daily from 19 Oct 2011" in Figure 2B.11). FX carry uses official central-bank policy rates; futures curves are Yahoo contract-month data with approximate expiry dates; positioning on the carry panel is CFTC (official), point-in-time on the release date. All refresh nightly; CFTC weekly.

Validation status

Not applicable to prices and statistics. The curve-shape label and carry series are model-based transformations (see Cross-asset).

Correct usage and worked example

EUR/USD, 19 Oct 2026 (Figures 2B.11 to 2B.14): the pair is 1.1225, −2.2% over the month and −4.4% YTD. Over one year it returned −3.4% annualised with 5.4% volatility and a −6.3% maximum drawdown; since December 2003 it returned −0.4% a year with a −38.9% maximum drawdown. Its strongest correlations are with the US Dollar Index (−0.81 over 3 years) and GBP/USD (+0.82). The carry is **−1.4% p.a.**: the "Policy-rate differential base - quote (official central-bank rates)", i.e. the ECB rate (2.50%) minus the Fed rate (3.88%), so a long EUR/USD position pays carry.

WTI, same date (Figures 2B.15 to 2B.17): the curve dated 16 Oct is in backwardation, from about 101 at the front to about 83 twelve months out, while one year ago only far-dated contracts near 58 are stored.

VIX, same date (Figures 2B.18, 2B.19): the table shows a 1Y "return" of +12.8% with 99.5% "volatility" and a −54.1% maximum drawdown. These describe changes in the index level, which cannot be held as an investment, so do not compare them with an asset's return statistics.

Limitations and pitfalls

- **Latest carry can be blank.** When the futures curve is dated earlier than the price (WTI: curve 16 Oct, price 19 Oct), the Carry, Roll yield and Curve stats show "—" for the latest row while the history is intact.
- **Short curve history.** Yahoo serves only contracts that are still trading, so the "~1y ago" curve holds far-dated contracts only and its front is missing (panel note).
- **Level series.** For yield and volatility indices the page notes: "Level series (yields / volatility): return statistics describe changes in the level, not an investable return."
- **Roll gaps** in continuous futures distort drawdowns and returns.
- **The FX price is a Yahoo market rate**, the carry uses official policy rates and the cross-asset signals use official fixings, so the levels can differ slightly.

Related mnemonics and BQL fields

- `EURUSD Curncy DES` or `GP`, `SX5E Index GP`, `CL1 Comdty GP`; `THES` with an instrument reference opens a new thesis with subject type cross-asset (12. Thesis Tracker).
- Related: Cross-asset dashboard, CFTC market page, 9. Model Studio.

Exposures

Route: `/macro/markets?tab=exposures` · **Menu:** Macro › Markets › Exposures · **Mnemonics:** none · **Refresh:** weekly (`macro_exposures` job, Saturday under every profile's weekly DAG)

Purpose

Shows how sectors, subsectors, regions, factor portfolios, ETFs, indices and futures have responded to official macro factors: rates, the curve, breakevens, credit spreads, the dollar, oil and the part of the equity market not explained by macro. Analysts use it to see, for example, which sectors are most rate-sensitive today, and portfolio managers to check the macro tilt implied by a sector allocation. The same betas drive the Scenario builder and Episode replay.

Screenshot

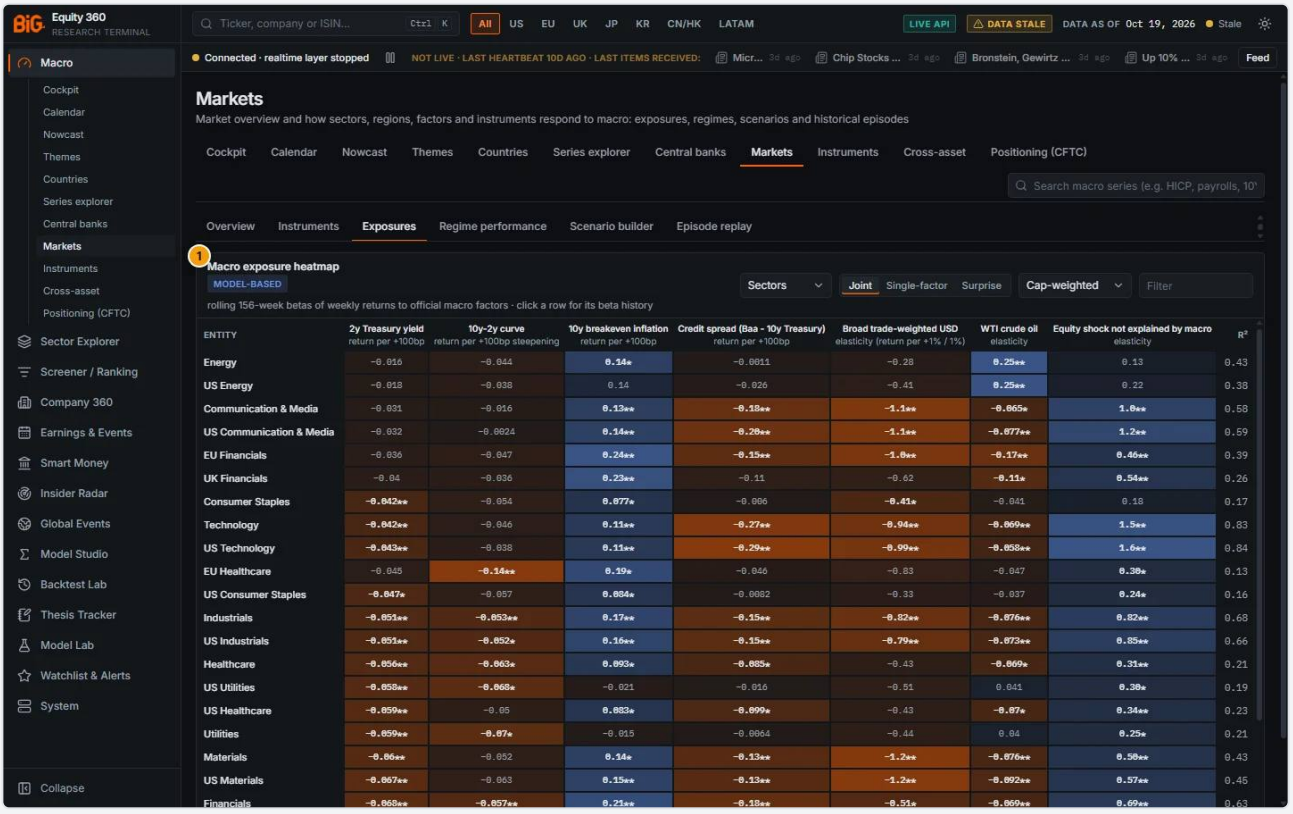


Figure 2B.20: Macro exposure heatmap, Sectors, Joint model, Cap-weighted, as of 19 Oct 2026.

#	Panel	What it shows
1	Macro exposure heatmap	One row per entity, one column per factor: beta (with significance stars), colour by sign and size, and R^2 of the regression. "MODEL-BASED rolling 156-week betas of weekly returns to official macro factors · click a row for its beta history"

After clicking a row, the beta history opens below the heatmap:

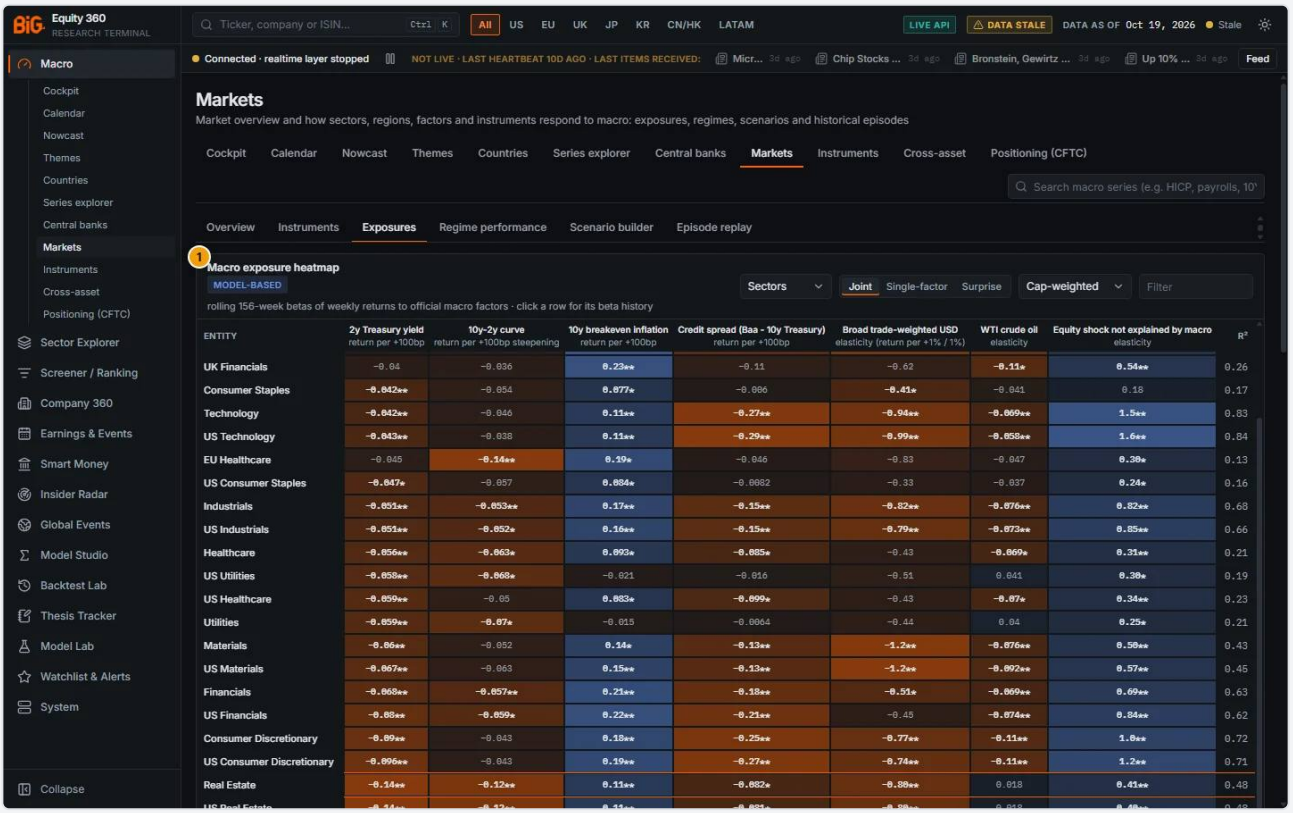


Figure 2B.21: Exposures tab after clicking the Real Estate row.

#	Panel	What it shows
1	Macro exposure heatmap	As in Figure 2B.20; the selected row is outlined
2	Beta history: Real Estate	One chart per factor: point-in-time month-end betas with 90% bands (±1.645 SE), from 2011 to 2026

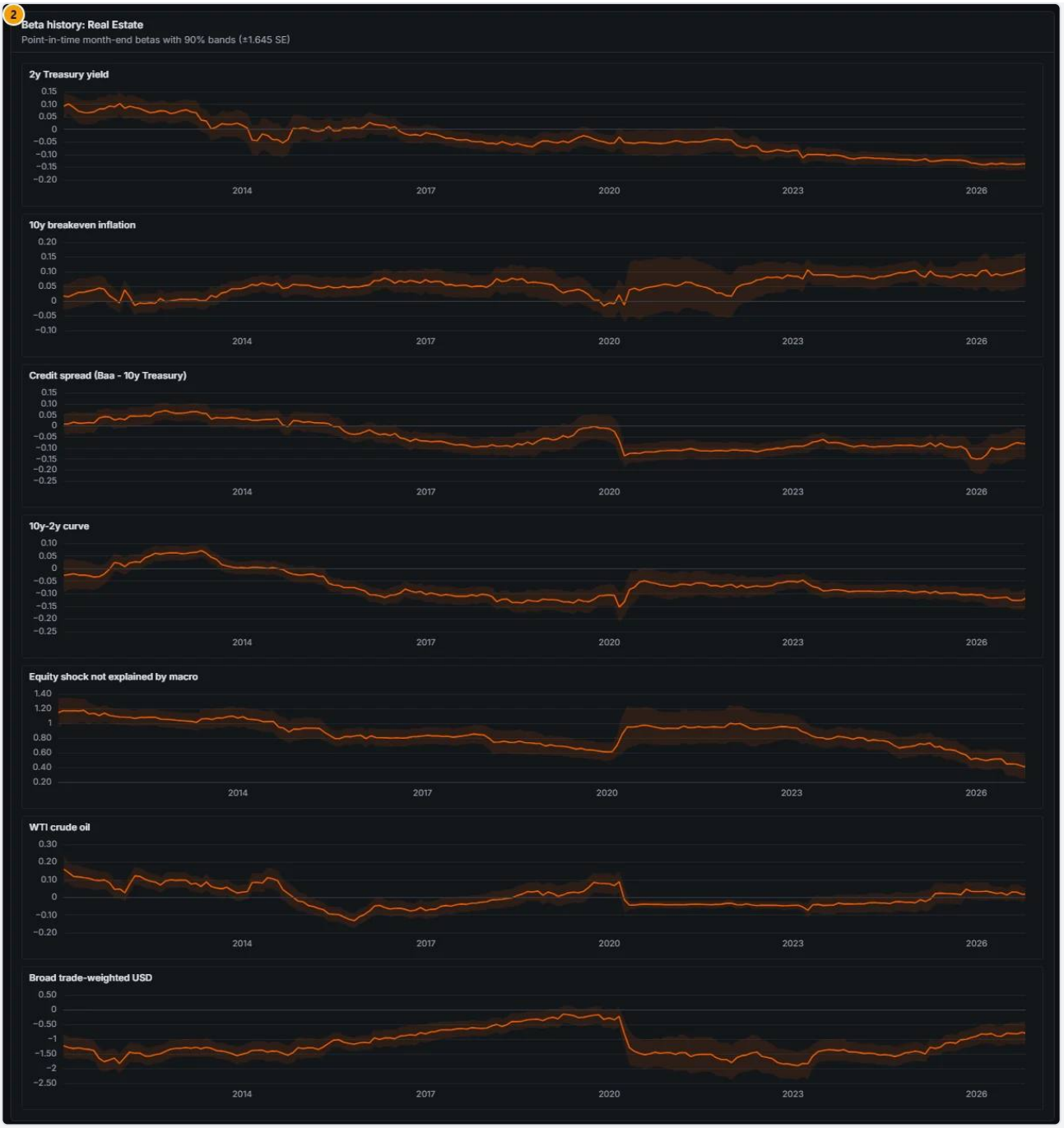


Figure 2B.22: Beta history: Real Estate. The 2-year yield beta drifted from about +0.10 in 2011 to about −0.14 in 2026; the equity-residual beta fell from about 1.2 to about 0.4.

Controls

Control	What it does
Entities	Sectors (default), Subsectors, Regions, Factors, ETFs, Indices, Futures
Model: Joint / Single-factor / Surprise	Which regression is shown (see Methodology)
Weighting: Cap-weighted / Equal-weighted	For sectors, subsectors and regions only
Filter	Free-text filter on name, ticker or id
Column header (hover)	Factor label, unit ("return per +100bp", "elasticity...") and source series
Cell (hover)	β to three decimals, t-statistic and "(1% sig.)", "(5% sig.)" or "(not significant)"
Row (click)	Opens "Beta history: {name}" below: point-in-time month-end betas with 90% bands (±1.645 SE)

The footnote under the table states the method label, the as-of date, the frequency (weekly) and how many rows are shown out of the total.

Methodology

The app's method label: **"Model-based: rolling weekly OLS betas (156 weeks, >= 52), Newey-West HAC(4) standard errors"**.

FACTORS (ALL OFFICIAL EXCEPT THE S&P 500)

Column	Series	Unit of the beta
2y Treasury yield	FRED DGS2 (Fed H.15), weekly change in pp	return per +100bp
10y-2y curve	FRED T10Y2Y, change in pp	return per +100bp steepening
10y breakeven inflation	FRED T10YIE, change in pp	return per +100bp
Credit spread (Baa - 10y Treasury)	FRED BAA10Y (Moody's), change in pp	return per +100bp
Broad trade-weighted USD	FRED DTWEXBGS (Federal Reserve), % change	elasticity (return per +1% / 1%)
WTI crude oil	FRED DCOILWTICO (EIA), % change	elasticity
Equity shock not explained by macro	S&P 500 orthogonalised on the macro factors	elasticity

The credit factor uses the Moody's Baa spread rather than an ICE BofA OAS because the OAS series on FRED keep only about three years of history (code note in `app/macro/markets/factors.py`).

THE THREE MODELS

```
Joint:          r_t = a + b'f_t + b_u · u_t + e_t
                f = (Δ2y, Δcurve, Δbreakeven10y, Δcredit, USD, oil)
                u = S&P 500 return − [1 f]γ   (S&P 500 orthogonalised on f in the same window)
Single-factor:  r_t = a + b · f_k,t + e_t, one regression per factor
                (Δ2y, Δ10y, Δcurve, Δreal 10y, Δbreakeven, Δcredit, USD, oil, equity = CAPM beta)
Surprise:       r_t = a + b's_t + b_u · u_s,t + e_t
                s = changes of the growth / inflation / labour surprise indices of the entity's home economy
                (model-based expectations), per +100 index points; u_s = equity orthogonalised on s
```

Because the equity term is orthogonalised, the macro betas of the joint model are **total effects** (they include the macro factor's effect through the equity market). In the joint model the 2-year beta holds the curve fixed, so it is the response to a **parallel** shift; the curve beta is the response to a steepening with the 2-year unchanged.

ESTIMATION

- Returns: local-currency weekly total returns from the price lake; cap-weighted portfolios use lagged market caps (2010 onwards), equal-weighted portfolios equal weights.
- Window: the trailing **156 weeks** ending at each month end (and the run date); at least **52** weekly returns; missing weeks are masked per entity.
- Standard errors: Newey–West HAC with 4 lags and a small-sample factor n / (n − k).
- Significance in the grid: `**` = $|t| \geq 2.58$ (1%), `*` = $|t| \geq 1.96$ (5%); bold = significant, faint = not significant. Blue = positive beta, orange = negative; intensity is scaled within each column.
- Shrinkage (securities only; the heatmap shows unshrunk aggregates): empirical Bayes toward the sector median, `w = τ² / (τ² + se²)`, `β = m + w (b_raw − m)`.
- Beta history: month-end point-in-time betas from 2004 for aggregates (24 months for single securities), band `β ± 1.645 · SE`.

Data sources and refresh

Factors come from FRED (H.15 yields, TIPS, breakevens, Baa spread, Fed broad dollar, EIA WTI) using the latest vintage per observation date; the equity factor is the S&P 500 from the price lake; surprise indices are the app's model-based

surprise indices (2A. Macro). The `macro_factors` job runs nightly; `macro_exposures` runs weekly (about 5 to 10 minutes) and also writes the episode replay.

Validation status

Not formally validated (no gate in the validation documents). The app labels the panel **MODEL-BASED**. The only diagnostic shipped is the Episode replay, which applies today's betas to past episodes and compares with realised returns.

Correct usage and worked example

Use the joint model for a multi-factor view, the single-factor model when you want the plain sensitivity to one variable, and the surprise model to see how an entity reacts to data surprises. Look at significance before size, and at R^2 before trusting any beta.

Example (Sectors, Joint, Cap-weighted, 19 Oct 2026):

Sector	2y yield	Credit	USD	Oil	Equity residual	R^2
Real Estate	−0.136 (t −9.3)	−0.082	−0.80	+0.02	+0.41	0.48
Technology	−0.042 (t −3.0)	−0.27 (t −9.9)	−0.94	−0.069	+1.55	0.83
Energy	−0.016 (t −0.6)	−0.001	−0.28	+0.25 (t 4.6)	+0.13	0.43

Reading: the model associates a 100bp parallel **rise** in yields with a return of about −13.6% for Real Estate (strongly significant), against about −4.2% for Technology; Energy's dominant exposure is oil (+0.25 elasticity), and Technology's is the equity residual (1.55) and credit.

Limitations and pitfalls

- **Betas describe the last three years.** They are not structural and drift; check the beta history before relying on one. Real Estate's 2-year yield beta was positive until about 2014 and has only been strongly negative since 2022 (Figure 2B.22).
- **Weekly betas scaled to 100bp** are linear extrapolations; large moves rarely happen in one week.
- **US factors for every region.** The macro factors are US series (Treasuries, Fed dollar index); for a Korean sector they measure sensitivity to US conditions, in local-currency returns.
- **Multicollinearity.** In the joint model, yields, breakevens and credit move together; individual coefficients can flip sign while their sum is stable.
- **Not a forecast** and not a trading signal.

Related mnemonics and BQL fields

- BQL (per security, model-based, snapshot only): `macro.beta_rates` ("Beta to the 10y yield (+100bp)"), `macro.beta_usd`, `macro.beta_oil`, `macro.beta_credit`, `macro.beta_growth`, `macro.rates_down_100bp` ("Modelled impact if rates fall 100bp").
- Per company: the Macro tab of Company 360 (5C); screener macro columns (4. Screener).
- Related: Scenario builder, Episode replay.

Regime performance

Route: `/macro/markets?tab=regimes` · **Menu:** Macro › Markets › Regime performance · **Mnemonics:** none · **Refresh:** nightly (`regime_performance` job, after `macro_factors`)

Purpose

Shows how sectors, subsectors, regions, factor portfolios, ETFs and indices have performed **relative to their benchmark** in each macro regime: phases of the business cycle, HMM regimes, central-bank rate cycles and US inflation buckets, with the current regime highlighted. It answers "what has historically done well in a regime like today's?", with sample sizes and t-statistics so you can judge how much history stands behind each number.

Screenshot

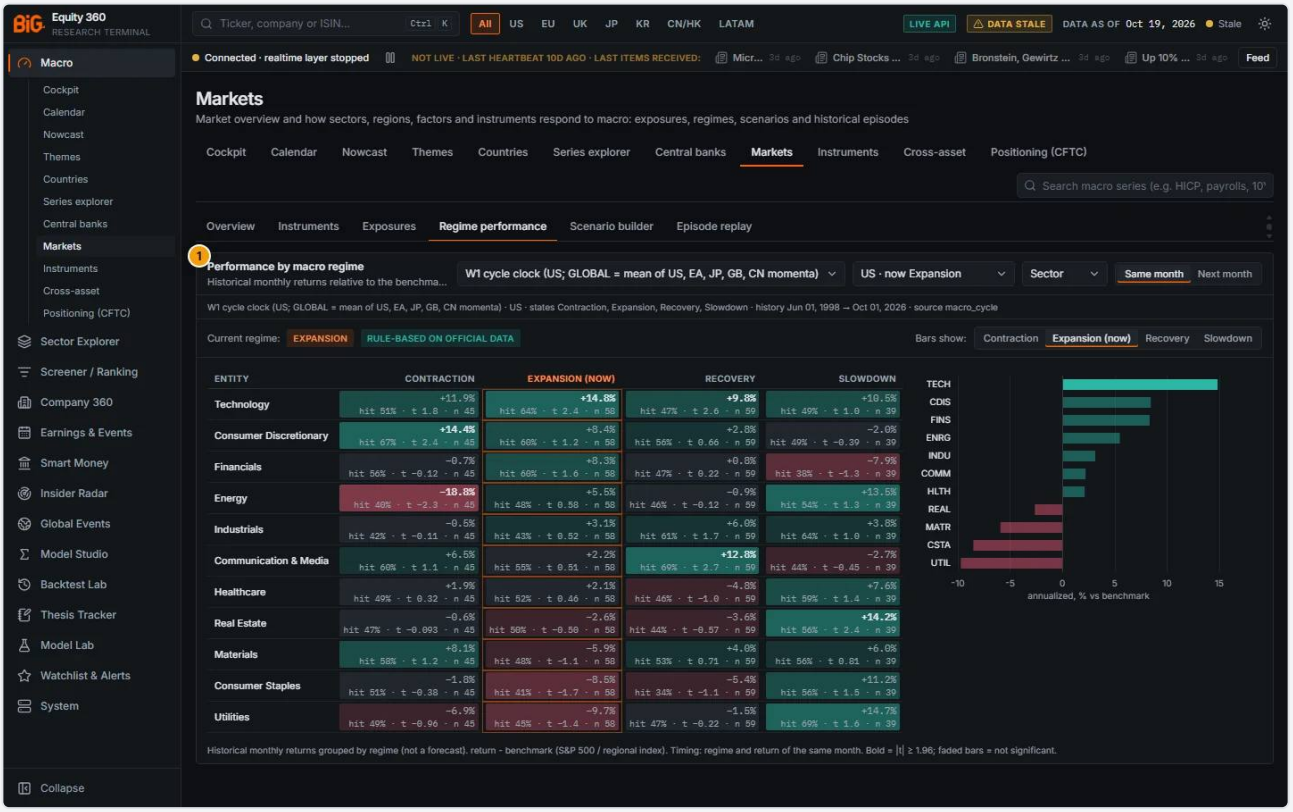


Figure 2B.23: Performance by macro regime: W1 cycle clock, US scope (now Expansion), Sector, Same month.

#	Panel	What it shows
1	Performance by macro regime	Scope line (states, history, source), current-regime badge, a table of annualised relative returns per regime (with hit rate, t and n) and a bar chart for the regime picked in "Bars show"

Controls

Control	What it does
Regime type	W1 cycle clock (US; GLOBAL = mean of US, EA, JP, GB, CN momenta) · Gaussian HMM regimes (model-based) · hiking / pausing / cutting (macro-w3 cb_rate_cycle) · US CPI y/y buckets
Scope	US or GLOBAL for the cycle and HMM; a central bank for the rate cycle (Fed, ECB, BoE, BoJ and PBoC have statistics; other banks show "(no performance data)"); US for inflation. Each option shows the current state ("US · now Expansion")
Entities	Sector, Subsector, Region, Factor, Etf, Index
Timing: Same month / Next month	Same month = regime and return of the same month; Next month = regime known at month end, return of the following month
Bars show	Which regime the bar chart ranks (the current one is marked "(now)")
Cell (hover)	Hit rate, t and n
Bar (hover)	Annualised mean, hit rate, t-stat and months for the chosen regime

Methodology

Method label: **"Historical monthly returns grouped by regime (not a forecast)"**.

```
relative return (month m) = r(m) − benchmark(m)
    benchmark = S&P 500 for global aggregates and instruments, the regional index for region aggregates
annualised     = 12 × mean of the monthly relative returns in the regime
hit rate       = share of those months with a relative return > 0
t              = mean / (sd / √n),  n = number of months (cells with n < 3 are dropped)
```

The app requests the "relative" metric (an "excess" metric over the 3-month T-bill exists in the API but is not offered in the UI). Factor long-short portfolios are self-financing, so they have no relative version. Rows are sorted by the mean in the current regime; the table shows the top 60 rows and the chart the top 25. Bold = $|t| \geq 1.96$; faded bars are not significant.

Regime type	States	Label in the app
Cycle clock (W1)	Expansion, Recovery, Slowdown, Contraction	"rule-based on official data"
HMM regimes	Expansion, Recovery, Slowdown, Contraction	"model-based regimes"
Rate cycle	Hiking, Pausing, Cutting	"rule-based on official data"
Inflation regime (US)	CPI < 2%, CPI 2–3%, CPI 3–5%, CPI > 5%	"rule-based on official data"

Timing note from the method: "next_month = regime of the previous month (publication lags make it only approximately investable)".

Data sources and refresh

Monthly returns of the aggregate portfolios come from the price lake (from 1995); regime states come from the macro cycle (from June 1998), the HMM model (from February 2006), the central-bank rate cycles (macro W3) and US CPI (FRED CPIAUCSL). The job runs nightly.

Validation status

Not formally validated. These are historical averages, labelled "not a forecast". There is no correction for multiple testing across the many entity × regime cells, so some $|t| \geq 1.96$ cells are expected by chance alone.

Correct usage and worked example

Prefer the Next month timing when you want something closer to an investable rule, prefer cells with large n, and treat the table as a prior, not a signal.

Example (cycle clock, US, Sector, Same month; Figure 2B.23): the US cycle is in Expansion. In Expansion months since 1998, Technology beat the S&P 500 by **+14.8% annualised** (hit rate 64%, t 2.4, n 58), while Utilities lagged by −9.7% (t −1.4, not significant) and Consumer Staples by −8.5% (t −1.7). Energy is the clearest Contraction loser (−18.8%, t −2.3, n 45).

Limitations and pitfalls

- **Same-month timing is not tradable**: the regime is only known after the month's data are published.
- **Few regime episodes.** n counts months, not independent episodes; 58 Expansion months may come from only a handful of cycles.
- **Regimes are revised**, and the HMM is model-based and over-confident.
- **Benchmark mismatch.** Global sectors are measured against the S&P 500.

Related mnemonics and BQL fields

- Related: 2A. Macro (cycle clock, regime model, central-bank rate cycles), Company 360 Macro tab (5C), which shows the regime table of the company's sector.

Scenario builder

Route: `/macro/markets?tab=scenarios` · **Menu:** Macro › Markets › Scenario builder · **Mnemonics:** none · **Refresh:** betas weekly; runs are computed on request

Purpose

A stress tester: set instantaneous macro shocks (rates, breakevens, real yields, credit, dollar, oil, equity, data surprises) and see the model-implied impact on universes, sectors, regions, style factors, ETFs and indices, and individual securities (winners and losers, or your watchlist), each with a 90% band. Ten presets cover the usual cases; you can save your own scenarios.

Screenshot

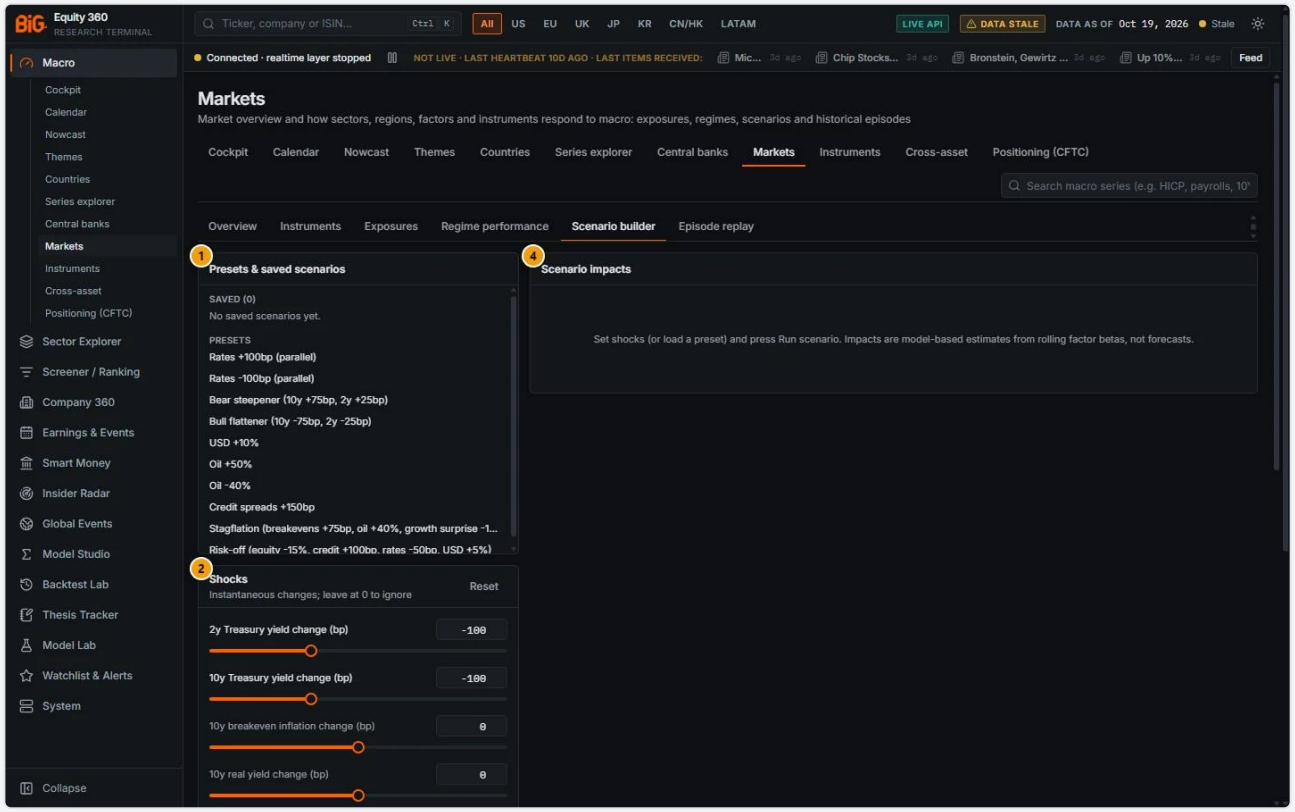


Figure 2B.24: Scenario builder before a run (default shocks: 2y and 10y −100bp).

#	Panel	What it shows
1	Presets & saved scenarios	Your saved scenarios (with date and a delete icon) and the ten read-only presets
2	Shocks	One slider and number box per shock field; "Instantaneous changes; leave at 0 to ignore"
3	Run settings	Propagation mode, universe and filters, watchlist, scenario name, Run / Save buttons
4	Scenario impacts	Results after a run; before it, the note "Impacts are model-based estimates from rolling factor betas, not forecasts."

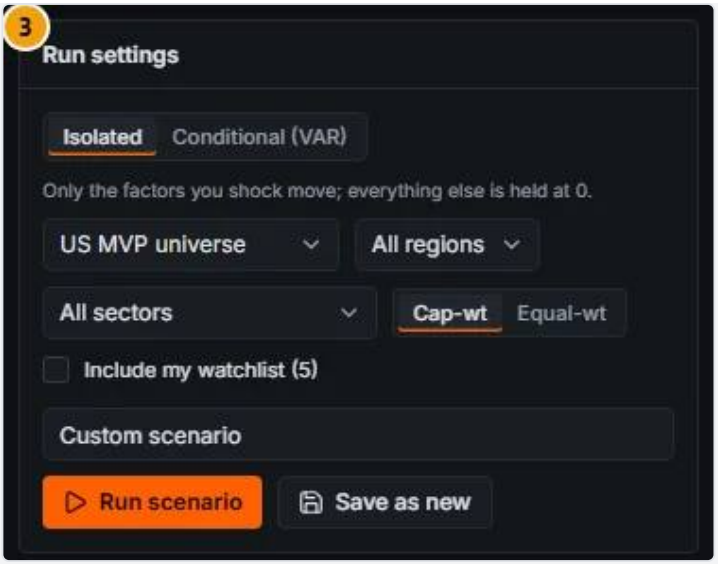


Figure 2B.25: Run settings.

After loading the preset "Rates -100bp (parallel)" and pressing **Run scenario**:

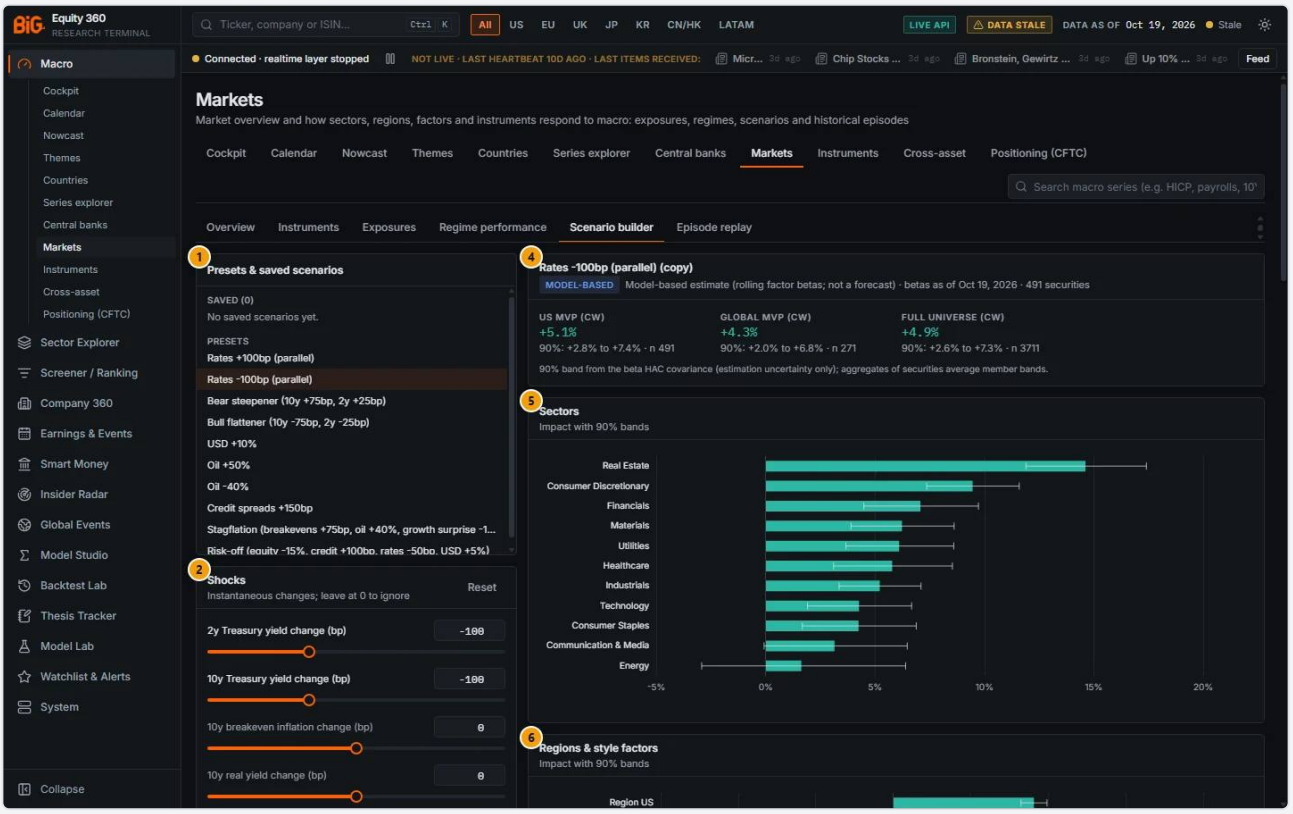


Figure 2B.26: Scenario builder after running the preset "Rates -100bp (parallel)" (Isolated, US MVP universe, cap-weighted), betas as of 19 Oct 2026.

#	Panel	What it shows
1	Presets & saved scenarios	The loaded preset is highlighted
2	Shocks	2y and 10y at −100bp, everything else 0
3	Run settings	As in Figure 2B.25
4	Rates -100bp (parallel) (copy)	Result header with the model-based label, the betas date, the number of securities, and one stat per universe: impact, 90% band and n
5	Sectors	Impact per sector with 90% whiskers
6	Regions & style factors	Impact per region and per factor long-short portfolio with 90% whiskers
7	Winners & losers	The 20 securities with the largest and smallest model-implied moves, with sector, region and 90% band

#	Panel	What it shows
8	Instruments	Top and bottom 10 ETFs and indices
9	Factor vector	The shocked (first-round) and total factor moves with their units

Figure 2B.27: Regions & style factors. Several bands include zero.

7

Winners & losers

Securities with the largest model-implied moves

WINNERS				
CBRE	CBRE Group, Inc.	Real Estate · US	+21.4%	[+15.8%, +27.3%]
PLD	Prologis, Inc.	Real Estate · US	+19.4%	[+14.5%, +24.4%]
CCI	Crown Castle Inc.	Real Estate · US	+17.6%	[+11.4%, +24.3%]
PSA	Public Storage	Real Estate · US	+17.4%	[+13.2%, +21.7%]
EXR	Extra Space Storage Inc.	Real Estate · US	+17.2%	[+11.6%, +23.1%]
ESS	Essex Property Trust, Inc.	Real Estate · US	+17.2%	[+13.4%, +21.0%]
SPG	Simon Property Group, Inc.	Real Estate · US	+16.6%	[+11.8%, +21.6%]
SBAC	SBA Communications Corporation	Real Estate · US	+16.1%	[+8.9%, +23.7%]
AMT	American Tower Corporation	Real Estate · US	+14.4%	[+9.1%, +19.9%]
IRM	Iron Mountain Incorporated	Real Estate · US	+12.9%	[+7.7%, +18.3%]
VMRK	Vivmark Residential	Real Estate · US	+12.7%	[+9.5%, +15.9%]
VICI	VICI Properties Inc.	Real Estate · US	+12.5%	[+8.7%, +16.4%]
RKT	Rocket Companies, Inc.	Financials · US	+12.4%	[+7.8%, +17.3%]
O	Realty Income Corporation	Real Estate · US	+11.2%	[+8.2%, +14.2%]
BAM	Brookfield Asset Management Ltd.	Financials · US	+10.7%	[+7.1%, +14.4%]
VTR	Ventas, Inc.	Real Estate · US	+10.4%	[+7.1%, +13.7%]
SPGI	S&P Global Inc.	Financials · US	+10.3%	[+6.9%, +13.8%]
WELL	Welltower Inc.	Real Estate · US	+10.2%	[+6.9%, +13.6%]
KKR	KKR & Co. Inc.	Financials · US	+10.1%	[+6.3%, +14.1%]
BX	Blackstone Inc.	Financials · US	+10.0%	[+5.9%, +14.2%]
LOSERS				
IBKR	Interactive Brokers Group, Inc.	Financials · US	+0.2%	[−3.4%, +3.9%]
LPLA	LPL Financial Holdings Inc.	Financials · US	+0.6%	[−3.1%, +4.4%]
CME	CME Group Inc.	Financials · US	+0.9%	[−1.9%, +3.9%]
PGR	The Progressive Corporation	Financials · US	+2.0%	[−1.6%, +5.7%]
IMO	Imperial Oil Limited	Energy · US	+2.0%	[+0.8%, +3.3%]
FANG	Diamondback Energy, Inc.	Energy · US	+2.1%	[+0.8%, +3.4%]
XOM	ExxonMobil Holdings Corporation	Energy · US	+2.1%	[+0.8%, +3.4%]
LNG	Cheniere Energy, Inc.	Energy · US	+2.1%	[+0.8%, +3.4%]
COP	ConocoPhillips	Energy · US	+2.1%	[+0.8%, +3.4%]
EOG	EOG Resources, Inc.	Energy · US	+2.1%	[+0.8%, +3.4%]
OXY	Occidental Petroleum Corporation	Energy · US	+2.1%	[+0.8%, +3.4%]
VG	Venture Global, Inc.	Energy · US	+2.1%	[+0.8%, +3.4%]
HAL	Halliburton Company	Energy · US	+2.1%	[+0.8%, +3.4%]
DVN	Devon Energy Corporation	Energy · US	+2.1%	[+0.8%, +3.4%]
EQT	EQT Corporation	Energy · US	+2.1%	[+0.8%, +3.4%]
BKR	Baker Hughes Company	Energy · US	+2.1%	[+0.9%, +3.4%]
WMB	The Williams Companies, Inc.	Energy · US	+2.1%	[+0.9%, +3.4%]
VLO	Valero Energy Corporation	Energy · US	+2.2%	[+0.9%, +3.4%]
EXE	Expand Energy Corporation	Energy · US	+2.2%	[+0.9%, +3.4%]
CQP	Cheniere Energy Partners, L.P.	Energy · US	+2.2%	[+0.9%, +3.5%]

Figure 2B.28: Winners & losers. In a pure rate cut, even the "losers" are mostly positive.

8

Instruments

ETFs, indices and futures

TOP				
ETHA	iShares Ethereum Trust	US	+29.4%	[+13.2%, +47.8%]
ITB	iShares US Home Construction	US	+22.0%	[+18.1%, +26.0%]
XHB	SPDR S&P Homebuilders	US	+18.5%	[+15.0%, +22.2%]
KRE	SPDR S&P Regional Banking	US	+18.4%	[+12.5%, +24.7%]
ARKK	ARK Innovation	US	+18.0%	[+11.8%, +24.5%]
KBE	SPDR S&P Bank	US	+17.0%	[+11.9%, +22.3%]
XBI	SPDR S&P Biotech	US	+16.3%	[+10.7%, +22.1%]
TAN	Invesco Solar	US	+15.6%	[+5.8%, +26.4%]
TLT	iShares 20+ Year Treasury Bond	US	+15.6%	[+14.5%, +16.6%]
VNQ	Vanguard Real Estate	US	+15.5%	[+12.8%, +18.2%]
BOTTOM				
^MERV	MERVAL	LATAM	−12.4%	[−21.3%, −2.5%]
UNG	United States Natural Gas Fund	US	−7.9%	[−18.9%, +4.5%]
ASHR	Xtrackers Harvest CSI 300 China A	US	−4.0%	[−9.4%, +1.7%]
WEAT	Teucrium Wheat	US	−3.7%	[−7.7%, +0.4%]
CORN	Teucrium Corn	US	−3.4%	[−6.1%, −0.5%]
DBE	Invesco DB Energy	US	−2.9%	[−6.6%, +1.0%]
DBO	Invesco DB Oil	US	−2.6%	[−7.3%, +2.4%]
BNO	United States Brent Oil Fund	US	−1.9%	[−5.5%, +1.9%]
PDBC	Invesco Optimum Yield Diversified Commodity	US	−1.8%	[−3.8%, +0.2%]
DJP	iPath Bloomberg Commodity ETN	US	−1.8%	[−3.7%, +0.3%]

Figure 2B.29: Instruments, top and bottom 10.

9

Factor vector

Isolated shock: no propagation

FACTOR	SHOCKED	TOTAL	UNIT
d2y	−1.000	−1.000	pp
dcurve	0.000	0.000	pp
dbe10	0.000	0.000	pp
dcredit	0.000	0.000	pp
usd	0.000	0.000	log
oil	0.000	0.000	log
equity_resid	0.000	0.000	log
d10y	−1.000	−1.000	pp
dreal10	−1.000	−1.000	pp
equity	+0.053	+0.053	log

Figure 2B.30: Factor vector, "Isolated shock: no propagation". The rows below the joint basis show the implied 10y, real 10y and S&P 500 moves.

Controls

Control	What it does
Saved scenario name	Loads it (shocks, mode, horizon, name)
Delete icon	Deletes that saved scenario
Preset name	Loads the preset; the name becomes "{preset} (copy)" so saving creates a new scenario
Shock sliders and boxes	2y Treasury yield change (bp, ±300) · 10y Treasury yield change (bp, ±300) · 10y breakeven inflation change (bp, ±200) · 10y real yield change (bp, ±200) · Baa-Treasury credit spread change (bp, −200 to +400) · Broad USD change (%, ±20) · WTI oil change (%, −60 to +100) · S&P 500 change (%, −40 to +30) · Growth / Inflation / Labor surprise change (points, ±200). The box accepts values outside the slider range; % shocks must be above −100
Reset	Sets every shock to 0
Propagation: Isolated / Conditional (VAR)	Isolated: "Only the factors you shock move; everything else is held at 0." Conditional: "Unshocked factors respond through a monthly VAR on the factor history (model-based propagation)."

Control	What it does
VAR horizon	1m, 3m, 6m or 12m (Conditional only)
Universe	US MVP universe, Global MVP universe, Full universe (for winners and losers)
Region, Sector, Subsector	Filters for winners and losers (Subsector appears once a sector is chosen)
Weighting: Cap-wt / Equal-wt	Weighting of the sector and region bars
Include my watchlist (n)	Adds a table of your watchlist's securities with their impacts
Scenario name	Name used when saving
Run scenario	Runs the scenario (disabled when every shock is 0)
Save as new / Update saved	Saves a new scenario, or updates the loaded saved scenario
Detach	Unlinks the loaded saved scenario so the next save creates a copy
Ticker in Winners & losers	Opens Company 360
Bar (hover)	Impact and 90% band of that sector, region or factor

Saved scenarios are stored locally in the app's state database; presets cannot be changed or deleted.

Methodology

Label returned with every result: **"Model-based estimate (rolling factor betas; not a forecast)"**.

FROM SHOCKS TO FACTOR MOVES (JOINT BASIS)

The joint basis is `z = (Δ2y, Δcurve, Δbreakeven, Δcredit, USD, oil, equity residual)`, with yields and spreads in percentage points and USD, oil and equity in log terms (`log(1 + pct)`).

- **Isolated** (default): unspecified factors are 0. A single 2y or 10y shock is applied as a parallel shift (curve unchanged). A 10y shock without a breakeven shock leaves breakevens unchanged, so real yields move one-for-one. A real-yield shock without a 10y shock sets `10y = real + breakeven`. An S&P 500 shock sets only the equity residual `u = equity − γ′f` (γ = the S&P 500's loading on the macro block), so an equity shock on top of macro shocks counts only the part not already explained by them.
- **Conditional (VAR)**: unspecified factors are first set to their Gaussian conditional mean given the specified ones, `E[z | A z = s] = Σ A′ (A Σ A′)⁻¹ s`, with Σ the weekly covariance of the factors over the last 10 years ("what usually moves with it"). Then a monthly VAR on the factor changes since 2006 (lag order by BIC, at most 3; Model Studio's `var_fit`) propagates the shock: the cumulative response over the chosen horizon, $\Sigma_{h=0..H} \Psi_h s_0$, replaces the first-round vector. The Factor vector table shows both ("Shocked" and "Total").

FROM FACTOR MOVES TO IMPACTS

```
impact     = exp(b′z) − 1
90% band   = exp(b′z ± 1.645 · √(z′ C z)) − 1      C = HAC covariance of the betas (shrunk for securities)
```

Surprise shocks use the surprise model's betas and are **added** to the macro impact (the two channels can overlap). Universe and watchlist aggregates average their members' impacts and bands. The result's band note reads: "90% band from the beta HAC covariance (estimation uncertainty only); aggregates of securities average member bands".

PRESETS

Rates +100bp (parallel) · Rates −100bp (parallel) · Bear steepener (10y +75bp, 2y +25bp) · Bull flattener (10y −75bp, 2y −25bp) · USD +10% · Oil +50% · Oil −40% · Credit spreads +150bp · Stagflation (breakevens +75bp, oil +40%, 10y +50bp, growth surprise −100) · Risk-off (equity −15%, credit +100bp, rates −50bp, USD +5%). All presets run in Isolated mode.

Data sources and refresh

The betas are the latest joint and surprise fits of the weekly `macro_exposures` job ("betas as of" in the result header); the VAR and the conditional covariance use the `macro_factors` history. Runs are computed on demand and not stored unless you save the scenario.

Validation status

Not formally validated. Every result is badged **model-based** with the label above. The bands cover **estimation error only**, not idiosyncratic noise, so realised outcomes fall outside them far more often than 10% of the time (see the error column of the Episode replay).

Correct usage and worked example

Start from a preset, run it Isolated, then rerun in Conditional (VAR) mode to see how much of the answer depends on co-movement assumptions. Compare sectors through their bands, not their point estimates.

Example (preset "Rates -100bp (parallel)", Isolated, US MVP universe, cap-weighted; Figure 2B.26): both yields fall 100bp, so in the joint basis only `Δ2y = −1.00` is non-zero. The Factor vector (Figure 2B.30) also shows the implied 10y and real 10y at −1.000 and an S&P 500 move of +0.053 in log terms, the part explained by the macro block.

- **Universes:** US MVP **+5.1%** (90% band +2.8% to +7.4%, n 491), Global MVP +4.3% (+2.0% to +6.8%, n 271), Full universe +4.9% (+2.6% to +7.3%, n 3,711).
- **Sectors:** Real Estate leads at about +14.6% (band about +12% to +17%), followed by Consumer Discretionary (about +9%) and Financials (about +7%); Energy is last at under +2%, with a band that includes zero.
- **Regions and styles** (Figure 2B.27): Region US about +5%; the Momentum and Quality long-short portfolios lose (about −4.5% and −3%), and most factor and non-US region bands include zero.
- **Securities** (Figure 2B.28): the winners are US REITs and asset managers (CBRE +21.4%, band +15.8% to +27.3%; Prologis +19.4%); the "losers" are US energy names at about +2.1% and some brokers near zero.
- **Instruments** (Figure 2B.29): ETHA (Ether ETF) +29.4% with a band of +13.2% to +47.8%, homebuilders (ITB +22.0%), TLT +15.6%; at the bottom the MERVAL index −12.4% and commodity ETFs.

You can check the Real Estate figure by hand from the Exposures tab: its 2y beta is −0.136 (SE ≈ 0.015), so `exp(0.136) − 1 ≈ +14.6%` and `exp(0.136 ± 1.645 × 0.015) − 1 ≈ +11.9% to +17.3%`.

Limitations and pitfalls

- **Instantaneous and linear.** Weekly betas are scaled to whatever shock you set; a −40% oil shock is far outside the data the betas were estimated on.
- **Bands are too narrow** for decision making (estimation error only).
- **Isolated mode is unrealistic** for large shocks: a 100bp rate move rarely comes without credit, dollar and equity moves. Use Conditional (VAR) for a co-movement-aware view and compare.
- **The "Instruments" panel** (subtitle "ETFs, indices and futures") lists ETFs and indices only.
- **Wide bands on short histories.** Instruments with short price histories (such as ETHA) get very wide bands; read the band before the point estimate.
- **Watchlist securities without betas** (new listings, short histories) are omitted ("None of the watchlist securities has macro betas.").

Related mnemonics and BQL fields

- BQL: `macro.rates_down_100bp` ("Modelled impact if rates fall 100bp", model-based, snapshot only).
- Related: Exposures, Episode replay, Company 360 Macro tab (5C), 13. Watchlists and alerts.

Episode replay

Route: `/macro/markets?tab=episodes` · **Menu:** Macro › Markets › Episode replay · **Mnemonics:** none · **Refresh:** weekly (written by the `macro_exposures` job)

Purpose

Replays six historical macro episodes: it takes the official factor moves over each window, applies **today's** joint betas, and compares the model-implied returns with what sectors, regions, factors, instruments and securities actually returned. It shows how well (or badly) the exposure model would have explained a known episode, which is the best available guide to how much to trust a scenario.

Screenshot

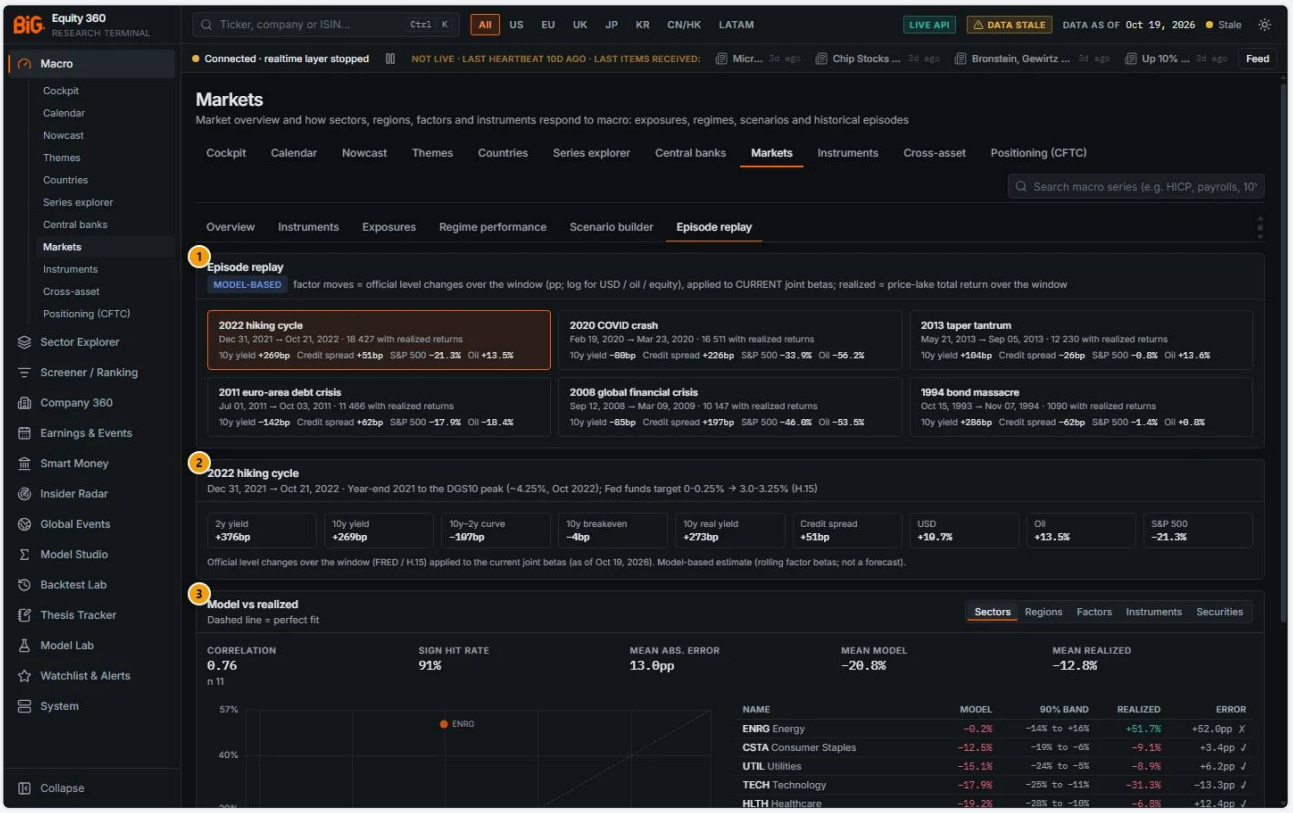


Figure 2B.31: Episode replay, 2022 hiking cycle selected, Sectors block.

#	Panel	What it shows
1	Episode replay	Six episode cards (newest first): window, number of entities with realised returns, and the 10y yield, credit spread, S&P 500 and oil moves
2	2022 hiking cycle	The selected episode's window and rationale, and the nine factor moves (2y, 10y, curve, breakeven, real yield, credit, USD, oil, S&P 500)
3	Model vs realized	Fit statistics, a scatter of model-implied vs realised returns (dashed line = perfect fit) and a table with model, 90% band, realised and error per row

Controls

Control	What it does
Episode card	Selects the episode (default: 2022 hiking cycle)
Block: Sectors / Regions / Factors / Instruments / Securities	Which entities are compared
Scatter point (hover)	Model and realised return

Control	What it does
Error column	Realised minus model in pp, with ✓ when the signs agree and ✗ when they do not

Methodology

Method line: **"factor moves = official level changes over the window (pp; log for USD / oil / equity), applied to CURRENT joint betas; realized = price-lake total return over the window"**.

Episode	Window	Rationale (as shown in the app)
1994 bond massacre	15 Oct 1993 → 7 Nov 1994	DGS10 trough (~5.2%) to peak (~8.0%); Fed funds 3.0% → 4.75% (H.15)
2008 global financial crisis	12 Sep 2008 → 9 Mar 2009	Friday before the Lehman bankruptcy to the equity trough
2011 euro-area debt crisis	1 Jul 2011 → 3 Oct 2011	Euro-area sovereign stress and the US downgrade
2013 taper tantrum	21 May 2013 → 5 Sep 2013	Day before Bernanke's 22-May testimony to the DGS10 peak
2020 COVID crash	19 Feb 2020 → 23 Mar 2020	S&P 500 peak to the Fed's open-ended QE announcement
2022 hiking cycle	31 Dec 2021 → 21 Oct 2022	Year-end 2021 to the DGS10 peak; Fed funds 0–0.25% → 3.0–3.25%

Factor moves come from FRED (DGS2, DGS10, T10YIE, DFII10, BAA10Y, DTWEXBGS, DCOILWTICO) and the S&P 500 from the price lake. Factors without data for a window (for example the broad dollar index before 2006, breakevens and TIPS before 2003) are listed under the moves and treated as 0. Statistics per block:

```
Correlation    = corr(model, realised) across the rows of the block
Sign hit rate  = share of rows where model and realised have the same sign
Mean abs. error = mean |realised − model|, in pp
```

For Securities, statistics cover all securities with betas and realised returns; the chart and table show the 20 best and 20 worst by model.

Data sources and refresh

Official FRED series for the factor moves, the price lake for realised total returns, and the current joint betas of the weekly exposures job ("as of 19 Oct 2026" in Figure 2B.31).

Validation status

Not a formal validation, but the closest thing the macro → markets tools have to one. The panel is badged **MODEL-BASED**. For the 2022 hiking cycle (Sectors, 11 rows) the app shows: **correlation 0.76, sign hit rate 91%, mean absolute error 13.0pp, mean model −20.8% vs mean realised −12.8%** (Figure 2B.31).

Correct usage and worked example

Use it before trusting a scenario: pick the episode most like your scenario and look at the error column.

Example (2022 hiking cycle): the window saw 2y +376bp, 10y +269bp, curve −107bp, breakeven −4bp, real yield +273bp, credit +51bp, USD +10.7%, oil +13.5% and S&P 500 −21.3%. The model ranks sectors reasonably (correlation 0.76) but overstates the average loss by 8pp, and it misses the biggest story: **Energy**, model −0.2% vs realised **+51.7%** (error +52.0pp, ✗). The oil factor only captures the point-to-point oil move (+13.5%), not the spike inside the window, and today's betas are not 2022's betas.

Limitations and pitfalls

- **Current betas on past episodes.** Betas estimated on the last three years are applied to 1994 or 2008; sector compositions and sensitivities were different then.
- **Point-to-point moves** ignore what happened inside the window.
- **Coverage shrinks with age**: only 1,090 entities have realised returns for 1994 against 18,427 for 2022.

- **Missing factors are set to 0**, which biases older episodes.
- Survivorship: realised returns exist only for securities with prices over the window.

Related mnemonics and BQL fields

- Related: Scenario builder, Exposures, Company 360 Macro tab (5C), which shows the episode replay for one company.

Cross-asset dashboard

Route: `/macro/cross-asset` (URL parameters `class`, `asset`, `root`) · **Menu:** Macro › Cross-asset · **Mnemonics:** `CMDTY` · **Refresh:** nightly (`futures_curve` then `cross_asset_signals`); CFTC positioning weekly

Purpose

A cockpit of 73 assets across commodities, FX, equity indices, US Treasuries, US credit and crypto futures, scored on trend, momentum, carry, value, curve slope, realised and implied volatility and CFTC positioning, with a futures-curve viewer, the correlation regime ("is everything moving together?") and the model-based risk-on / risk-off composite. Use it for cross-asset allocation views, to spot crowded or stretched markets, and as a launch point for backtests, theses and Model Studio studies.

Screenshot

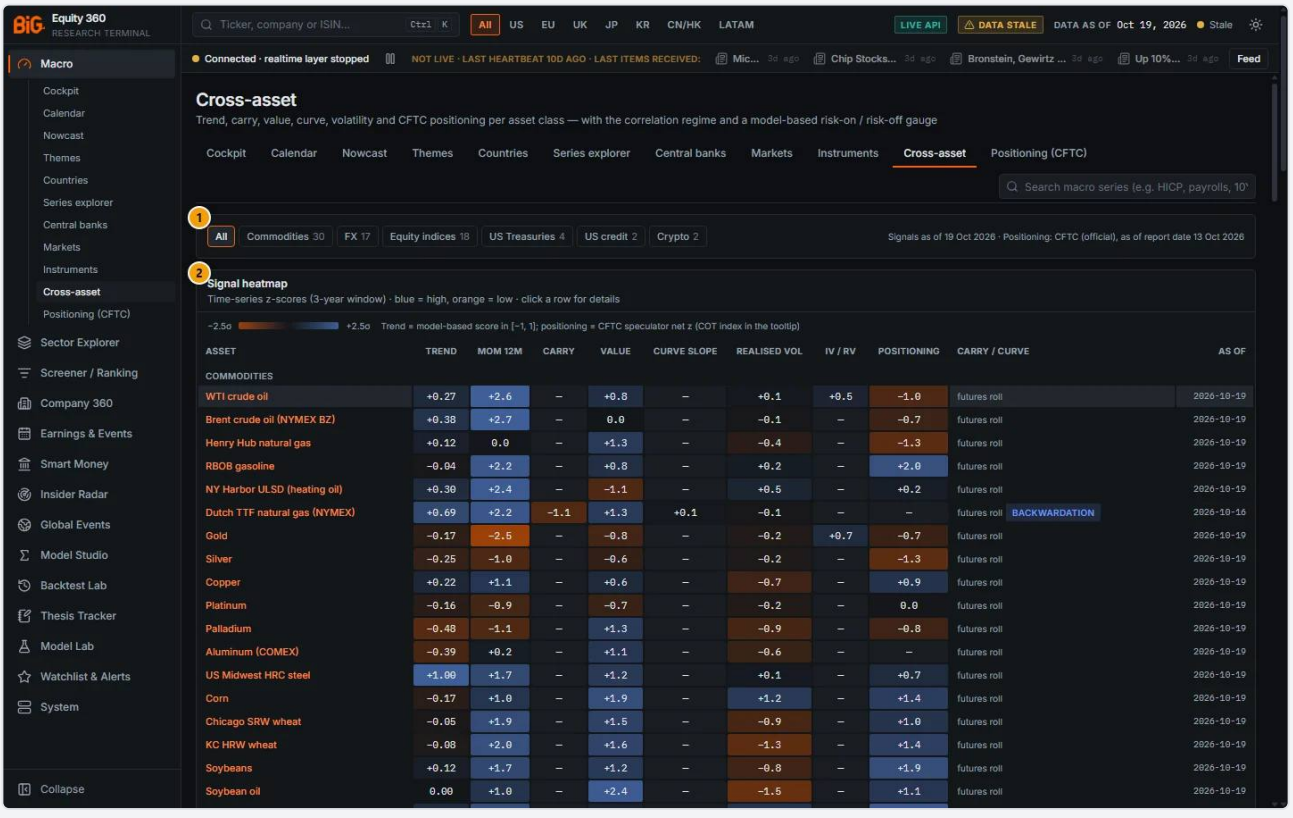


Figure 2B.32: Cross-asset dashboard, All classes, signals as of 19 Oct 2026, positioning report of 13 Oct 2026.

#	Panel	What it shows
1	Asset-class filter	All · Commodities 30 · FX 17 · Equity indices 18 · US Treasuries 4 · US credit 2 · Crypto 2, and the signal and positioning dates
2	Signal heatmap	One row per asset: Trend, Mom 12m, Carry, Value, Curve slope, Realised vol, IV / RV, Positioning, carry method / curve shape, as-of date

#	Panel	What it shows
3	Risk-on / risk-off	Gauge, composite value, regime badge, number of components, 1m change and the eight component z-scores
4	WTI crude oil (selected asset)	Key stats, methods, signal history and the action buttons
5	Futures curve viewer	Term structure now vs ~1m and ~1y ago, slope and roll-yield history, contract coverage
6	Correlation regime	First-factor share of a fixed 12-asset basket over time and the correlation matrix
7	Risk composite history	The composite since 2001 with the risk-on / risk-off zones and its eight components

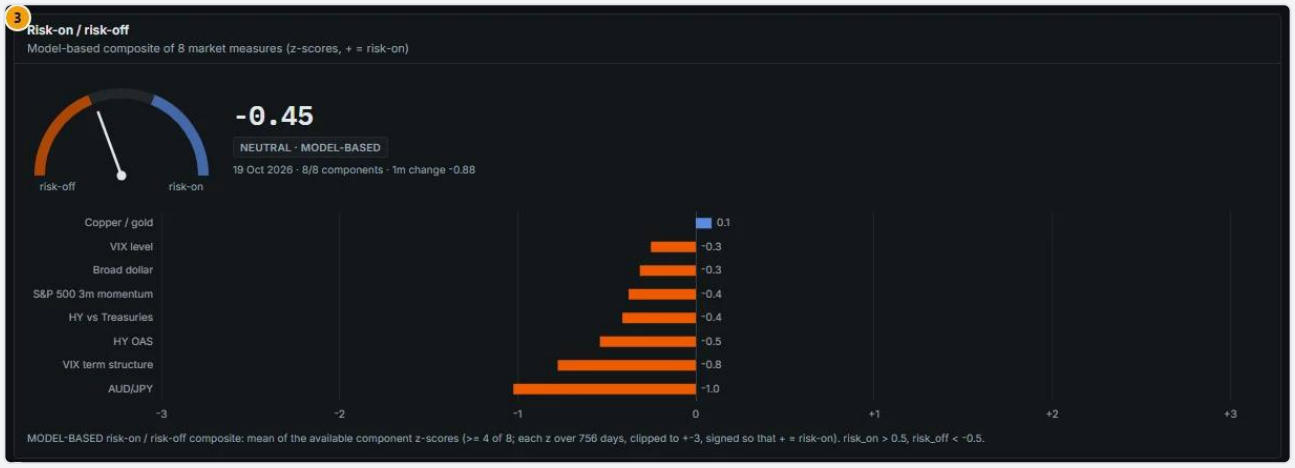


Figure 2B.33: Risk-on / risk-off, −0.45 Neutral, 8/8 components, 1m change −0.88.

Figure 2B.34: Selected asset (WTI crude oil), 5Y history: trend signal, futures-roll carry and CFTC COT index 3y with its extreme zones.

Figure 2B.35: Futures curve viewer, CL.

Figure 2B.36: Correlation regime, 120-day window, matrix "Now".

Figure 2B.37: Risk composite history with the four-component charts below.

With an asset-class button pressed, the heatmap keeps only that class; the other panels keep the badge numbers of Figure 2B.32, and badge 4 shows the class's first asset.

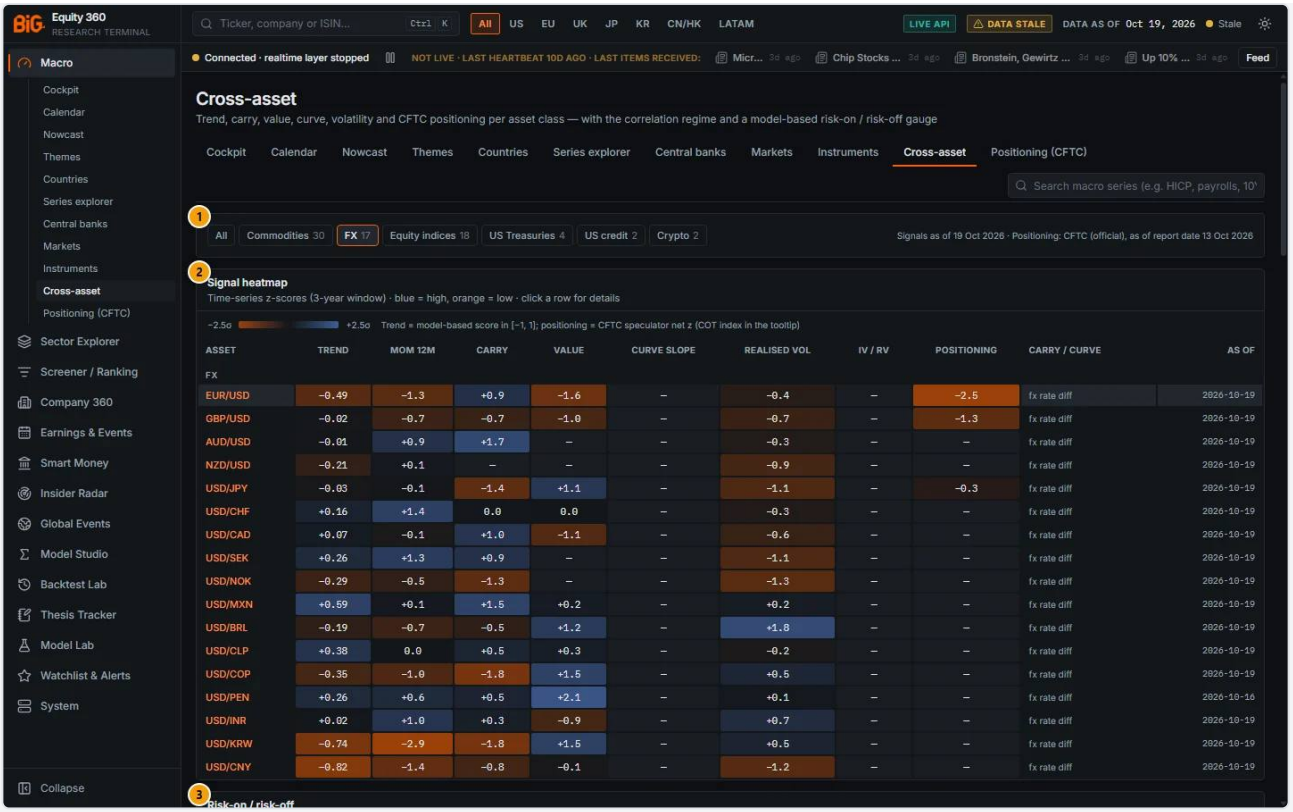


Figure 2B.38: Cross-asset dashboard filtered on FX (`/macro/cross-asset?class=fx`). Curve slope and IV / RV are empty for FX pairs.

Figure 2B.39: Selected asset EUR/USD (official fixings): fx rate diff carry, real-appreciation value method and TFF leveraged-funds positioning.

Controls

Control	What it does
Asset-class buttons	Filter the heatmap (`?class=`); counts show the number of assets per class
Signal column header (hover)	Definition of the signal, with "(model-based)" where applicable
Asset name (click)	Selects the asset (`?asset=`) for panel 4
Heatmap cell (hover)	Raw value, z-score, and the carry or value method; for Positioning: COT index 3y, z and the report / release dates, or "no CFTC market"
As of (amber)	"Latest row older than 10 days"

Control	What it does
History range: 1Y / 5Y / 10Y / Max	Range of the selected asset's history
Instrument page	Opens the linked instrument page
CFTC market	Opens the CFTC market page of the linked contract
Curve {root}	Loads that root into the curve viewer
Signal to backtest	Trend, Carry, Value or CFTC COT index
Backtest this signal	Opens the Backtest Lab pre-filled with a cross-sectional backtest of that signal across the asset's class (see Methodology)
Create thesis	Starts a thesis with this asset as the subject (12. Thesis Tracker)
Open in Model Studio	Opens a draft study: weekly OLS of the asset's log return on its trend signal, carry and COT index, each lagged one week, from 2008
Futures root (select)	Any root with a stored curve
Contract badge (hover)	Symbol, status (served, missing, unchecked, expired), first and last bar, approximate expiry
Window: 60d / 120d	Correlation window (default 120d)
Matrix date: Now / ~1m ago / ~1y ago	Which correlation matrix is shown

Methodology

All definitions below are quoted from docs/CROSS_ASSET_DEFINITIONS.md. "Everything labelled **MODEL-BASED** below is our own transformation, not a published statistic." Windows count rows (trading days of each asset); all returns are log returns.

HEATMAP COLUMNS

Column	Definition	Colour
Trend	**MODEL-BASED**, in [−1, 1]: mean over h ∈ {21, 63, 252} of `clip(mom_h / (σ63 · √h), −2, 2) / 2` , σ63 = 63-day std of daily log returns; needs at least 2 horizons	the score itself
Mom 12m	$\ln(P_t / P_{t-252})$	3-year z-score
Carry	headline carry, annualised fraction; method per asset class (table below)	3-year z-score
Value	value proxy (table below), labelled model-based	3-year z-score
Curve slope	OLS slope of ln F on years to expiry, contracts with 5 to 400 days to expiry (at least 3); positive = contango	z-score
Realised vol	std of daily log returns over 21 rows × √252	3-year z-score
IV / RV	vol-index close / 100 divided by realised vol (SPX ^VIX, NDX ^VXN, DJI ^VXD, CL ^OVX, GC ^GVZ, NIFTY ^INDIAVIX, AS51 ^AXVI; UST10Y: MOVE in bp over realised bp vol of the 10y yield)	3-year z-score
Positioning	CFTC speculator net position: COT index 3y in the tooltip, colour = z 3y	z of net over 156 reports

All z-scores are time-series z-scores, `z = (x − rolling mean) / rolling std` over 756 rows (minimum 126). The colour scale runs from −2.5σ (orange) to +2.5σ (blue).

CARRY BY ASSET CLASS

Method	Assets	Definition
futures_roll	commodities, crypto	roll yield `ln(F1 / F2) · 365 / (T2 − T1)` , nearest two contracts with ≥ 5 days to expiry, front within 200 days; positive = backwardation
fx_rate_diff	FX	(policy rate base − policy rate quote) / 100, official rates; no NZD, so NZDUSD carry is null
bond_carry_rolldown	UST 2/5/10/30y	`(y_T − y_3m)/100 + D_mod(T, y_T) · (y_T − y_{T−1})/100` , with `D_mod = (1/y)(1 − (1 + y/2)^(−2T))`

Method	Assets	Definition
equity_div_funding	equity indices	dividend yield of the tracking ETF − local policy rate / 100; HSI carry is null (no HKD bank)
credit_oas	US HY, US IG	OAS / 100; expected default losses are **not** deducted

The carry panel also stores `carry_12m`, the front vs the contract about one year further out (300 to 430 days), which is seasonality-neutral for natural gas, heating oil, grains and livestock.

VALUE BY ASSET CLASS

Method	Assets	Definition
reversal_5y	commodities, equity indices	`ln(mean P over rows t−1386 … t−1134) − ln(P_t)` : the price 4.5 to 5.5 years ago vs now
fx_real_5y	FX	minus the 5-year real appreciation of the base currency (official CPI, known 46 days after the period start); null for AUD, SEK, NOK, NZD
real_yield	UST 5Y / 10Y	TIPS real yield DFII5 / DFII10 (per cent)
oas_level	credit	OAS (per cent)

CURVE VIEWER

- Curve shape (**MODEL-BASED** label): contango if slope > +1%/yr, backwardation if < −1%/yr, otherwise flat.
- Roll yield front vs next: `ln(F1/F2)` annualised, + = backwardation.
- Snapshots: last stored curve on or before latest − 30 days (~1m) and latest − 365 days (~1y).
- Expiry dates are **approximate**: each root's exchange last-trading-day rule on the NYSE holiday calendar (for example "NYMEX CL: 3 business days before the 25th of the month before delivery").
- "Missing months are recorded gaps, never interpolated."

CORRELATION REGIME

The basket is fixed: SPX, SX5E, NKY, EEM, CL, GC, HG, UST10Y, EURUSD, USDJPY, AUDUSD, US_HY (N = 12; complete from August 2007), daily log returns on the union calendar (prices forward-filled up to 3 days, dates with ≥ 75% of members, windows with ≥ 80% complete rows).

```
first-factor share (w) = largest eigenvalue / N of the w-day correlation matrix   (w = 60, 120)
                         = 1/N (8%) with no common factor, 1 when everything moves together
z (120d, 3y)           = time-series z of the 120-day share over 756 days (min 252)
avg corr               = mean off-diagonal correlation (now, ~1m ago, ~1y ago)
```

RISK-ON / RISK-OFF COMPOSITE

Label: **"MODEL-BASED risk-on / risk-off composite: mean of the available component z-scores (>= 4 of 8; each z over 756 days, clipped to +-3, signed so that + = risk-on). risk_on > 0.5, risk_off < -0.5."**

Component (label in the app)	Measure	Sign
S&P 500 3m momentum	S&P 500 63-day log return	+
VIX level	VIX close	−
VIX term structure	ln(VIX / VIX3M); inversion = stress	−
HY vs Treasuries	63-day change of ln(HYG / IEF), total return	+
HY OAS	ICE BofA US HY OAS (FRED, official; history from 2023-10)	−
Copper / gold	63-day change of ln(copper / gold) front months	+
AUD/JPY	63-day AUD/JPY log return (official fixings)	+
Broad dollar	63-day change of the Fed broad dollar index (official)	−

BACKTEST THIS SIGNAL

The Backtest Lab opens with a draft (you review and run it there): universe = the assets of the selected asset's class; signal = the chosen field; long-short terciles with inverse-volatility weights; monthly rebalancing (weekly for the CFTC COT index, which is traded **contrarian**, i.e. low COT index = long); benchmark = equal-weighted universe. The draft carries the notes "spot-like prices: futures front months include roll gaps; trend / value / z metrics are model-based". See 11. Backtest Lab.

Data sources and refresh

From the page footnote: futures curves and futures / index / ETF prices are Yahoo market data (expiry dates approximate); FX from official fixings (ECB, Fed H.10, central banks); US Treasuries are a synthetic total-return index from official H.15 CMT yields; OAS and the broad dollar from FRED; positioning from the CFTC Commitments of Traders (official), point-in-time on the release date.

Positioning uses the disaggregated futures-only report (managed money) for commodities and the TFF futures-only report (leveraged funds) for financial futures; each row takes the latest report **released** on or before its date, and reports older than 28 days are dropped. USD/JPY uses the inverted yen future (6J). The nightly `futures_curve` job stores the next listed contracts of 43 roots; `cross_asset_signals` runs after it (about 75 seconds).

Validation status

No validation gate. The page states: "Trend signal, curve shape, value proxies, z-scores and the risk composite are model-based transformations, not published statistics." For the risk composite, docs/CROSS_ASSET_DEFINITIONS.md §3 records sanity checks only: **2008-10-15 gave −2.8 (risk_off), 2020-03-20 gave −2.5 (risk_off) and 2017-06-15 gave +0.3 (neutral)**. No out-of-sample test of the trend, carry, value or positioning signals is documented; use "Backtest this signal" (with the Backtest Lab's robustness checks) before acting on any of them.

Correct usage and worked example

Read across a row (is the asset trending, carrying, cheap, stretched, crowded?), then down a column (which assets rank highest on one signal), then check the regime panels for the backdrop.

Example (19 Oct 2026):

1. **WTI crude oil** (Figures 2B.32, 2B.34): trend **+0.27** (model-based), 12-month momentum +53.6% (z +2.6, stretched), value z +0.8 (5-year reversal), realised vol 39.0% (z +0.1), IV / RV z +0.5, and positioning z **−1.0** with a COT index 3y of 27: managed money is not crowded long despite the rally. The curve is in backwardation (shape "backwardation", 83% of the 230 stored days in backwardation; Figure 2B.35).
2. **Risk backdrop** (Figure 2B.33): composite **−0.45, Neutral**, with AUD/JPY (−1.0) and the VIX term structure (−0.8) the largest drags and only copper / gold positive (+0.1).
3. **FX** (Figures 2B.38, 2B.39): EUR/USD has a trend of **−0.49**, 12-month momentum −3.5% (z −1.3), carry −1.4% p.a. (z +0.9), value z −1.6 and positioning z **−2.5**: leveraged funds are near the short end of their three-year range (COT index close to 0 at the end of the chart). Value is "—" for AUD/USD, NZD/USD, USD/SEK and USD/NOK because their CPI history is too short (definitions doc).
4. **Correlation regime** (Figure 2B.36): the 120-day first-factor share is **41%** (floor 8%), **z +1.5σ** above its 3-year average, and the average correlation is 0.18 now vs 0.07 a year ago: assets are moving together more than usual, so diversification across this basket is weaker.

Limitations and pitfalls

- **Front-month prices are unadjusted across rolls**, so momentum, trend and value on futures behave like spot-price changes rather than futures excess returns (definitions doc, "Note on front-month prices").
- **Short curve history.** Yahoo serves only live contracts: CL deferred months from 2017, ES from 2021, Treasuries and livestock only a few months. Curve slope and roll-yield z-scores are therefore missing or short for many roots.
- **Blank latest carry.** When the stored curve is older than the price row (WTI on 19 Oct), the Carry and Curve slope cells are "—".
- **Positioning caveats**: Treasury leveraged-fund shorts are dominated by the cash-futures basis trade, and CME BTC / ETH leveraged funds are structurally short (basis), so their COT index is "a positioning-vs-own-history measure, not a directional view".

- **Asynchronous closes** across Asia, Europe and the US bias cross-region correlations down.
- **HY OAS history** starts only in 2023-10, so that component's z uses a short window.
- In the Risk composite history panel, the second chart is captioned "copper / gold, FX, dollar" but also contains the HY OAS component.

Related mnemonics and BQL fields

- Mnemonics: `CMDTY` (this page), `FXIP`, `COT`.
- Model Studio source `cross_asset` (field `<asset_id>:<metric>`, point-in-time on the knowledge date); Backtest Lab signal sources `cross_asset` and `cftc`.
- Related: Instrument page, Positioning (CFTC), Markets overview (risk gauge and tape), 9. Model Studio, 11. Backtest Lab, 12. Thesis Tracker.

Positioning (CFTC)

Route: `/macro/positioning` (URL parameters `report`, `class`) · **Menu:** Macro › Positioning (CFTC) · **Mnemonics:** `COT` · **Refresh:** weekly, Friday 19:00 ET (`cftc_weekly` DAG: `cftc_cot` then `cftc_signals`) under every schedule profile

Purpose

Shows who is long and who is short across US futures markets, from the CFTC Commitments of Traders: where each trader category's net position sits within its one- and three-year range (COT index), how unusual it is (z-score), which markets are at extremes or crowded, and where positioning and price disagree. Traders and allocators use it as a sentiment and crowding check.

Screenshot

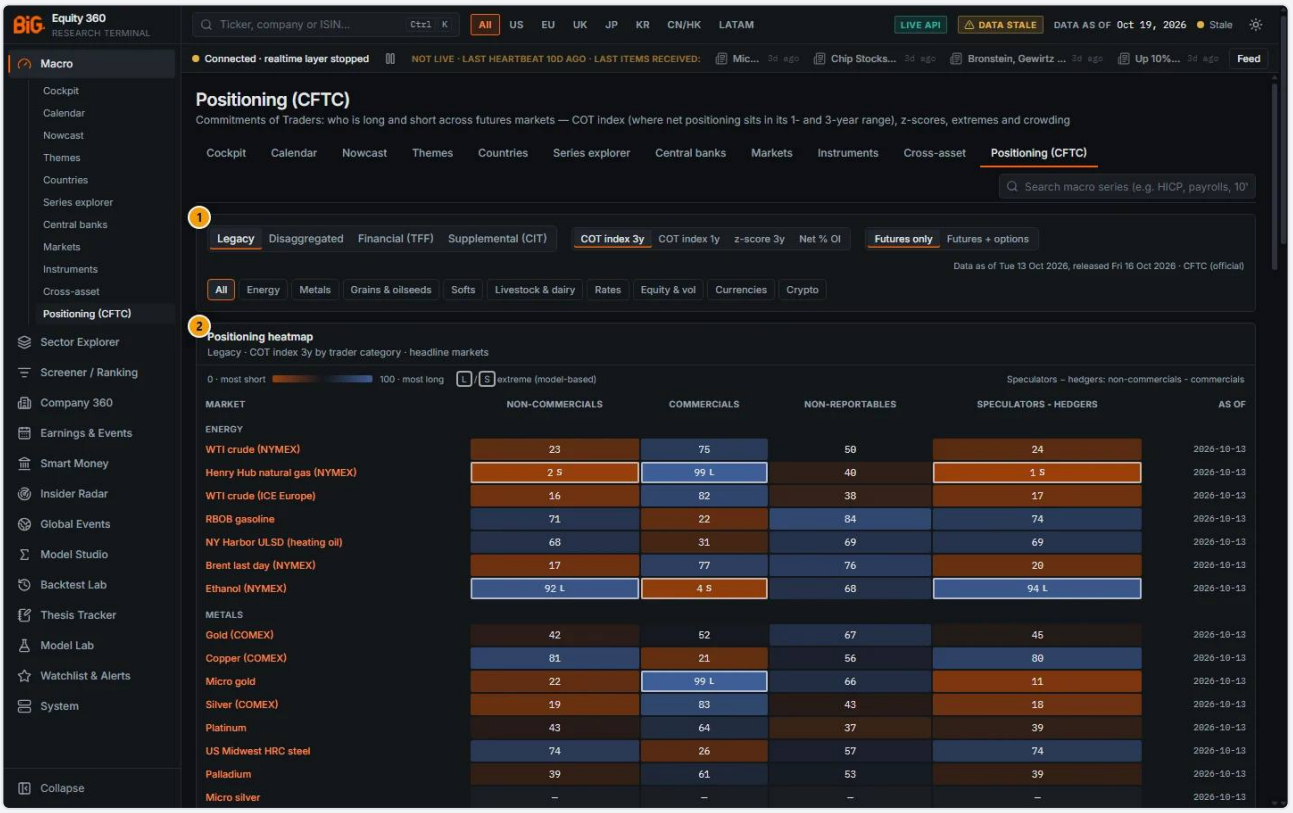


Figure 2B.40: Positioning (CFTC), Legacy report, COT index 3y, futures only, all classes; report of Tuesday 13 Oct 2026, released Friday 16 Oct 2026.

#	Panel	What it shows
1	Report, metric and filter bar	Report type, metric, futures only vs futures + options, the data date line and the asset-class buttons
2	Positioning heatmap	Headline markets by asset class × trader category, coloured by the chosen metric, with L / S marks on model-based extremes
3	Crowded longs	Markets whose headline speculators have a COT index 3y ≥ 90
4	Crowded shorts	Markets with a COT index 3y ≤ 10
5	Positioning vs price divergences	Markets where the 13-week change in net positioning and the price return have opposite signs, each with an absolute z of at least 1

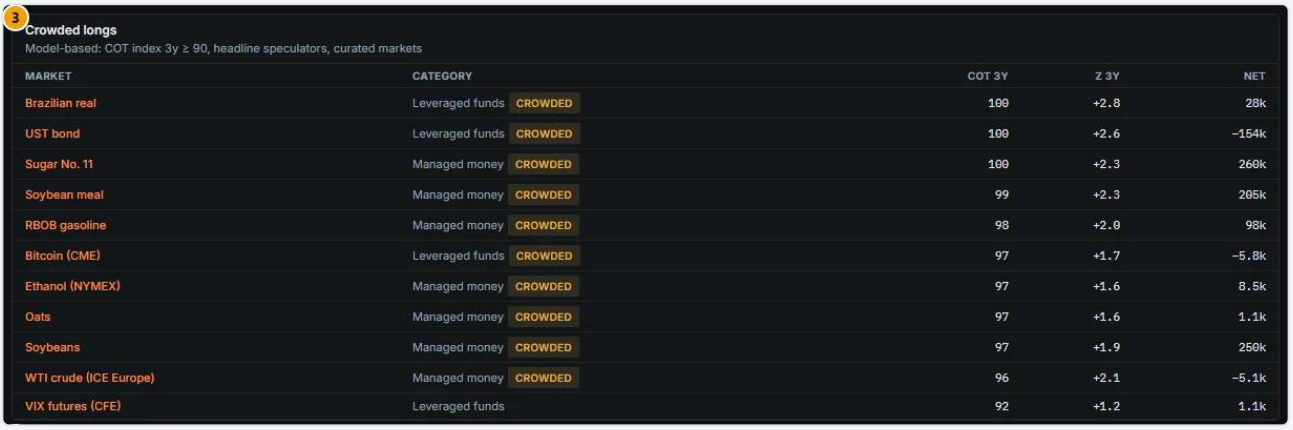


Figure 2B.41: Crowded longs. A "crowded" badge adds $|z\ 3y| \geq 1.5$ to the extreme.

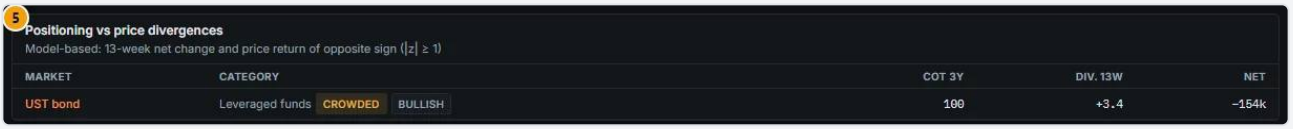


Figure 2B.42: Positioning vs price divergences.

The same page with the Disaggregated report selected:

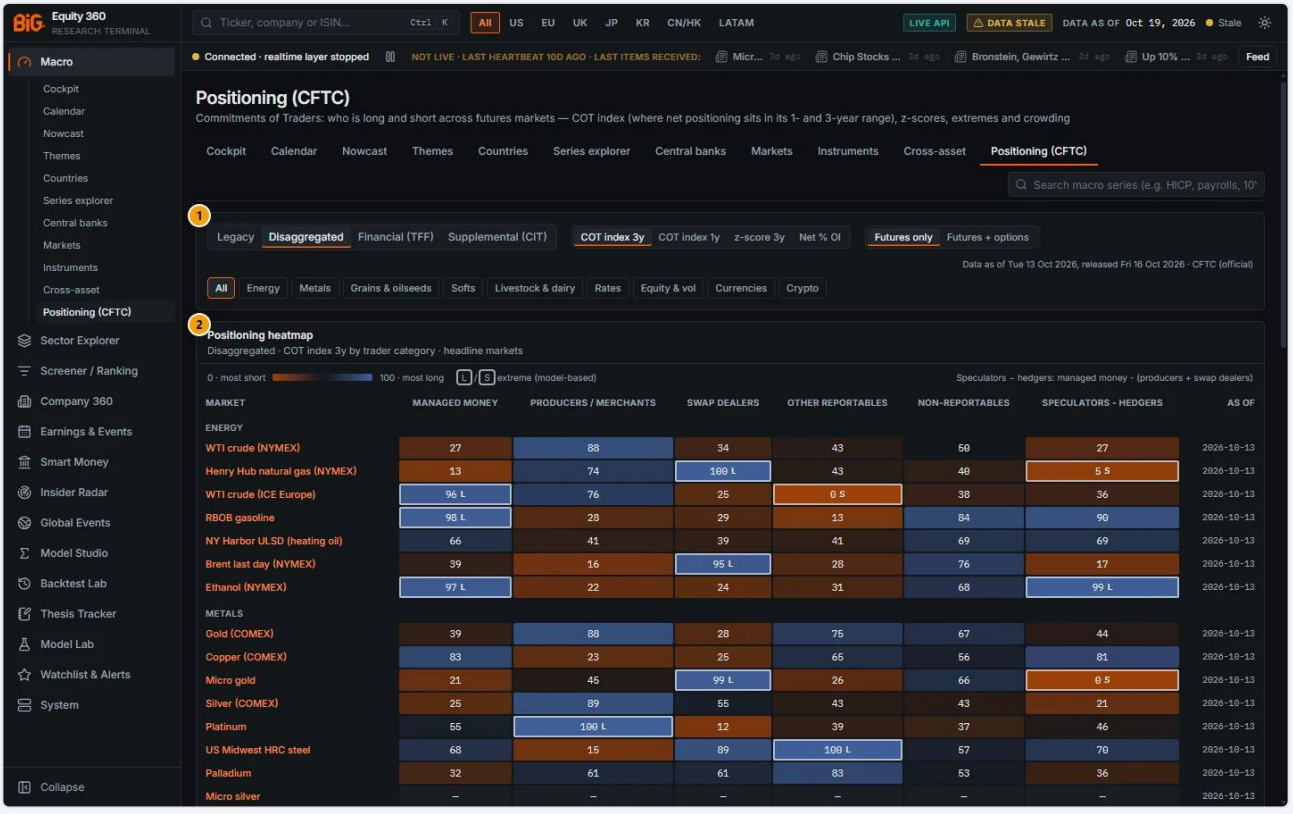


Figure 2B.43: Positioning (CFTC), Disaggregated report (`/macro/positioning?report=disagg`): managed money, producers / merchants, swap dealers, other reportables, non-reportables and speculators − hedgers.

#	Panel	What it shows
1	Report, metric and filter bar	Disaggregated selected; same metric, futures-only and class controls
2	Positioning heatmap	Physical-commodity markets by disaggregated category; the speculators − hedgers column is managed money − (producers + swap dealers)
3	Crowded longs	Unchanged: the lists always use each market's headline category, whatever report is selected
4	Crowded shorts	Unchanged, as above
5	Positioning vs price divergences	Unchanged, as above

Controls

Control	What it does
Report type	Legacy ("All markets: commercials vs non-commercials") · Disaggregated ("Physical commodities: producers, swap dealers, managed money") · Financial (TFF) ("Financial futures: dealers, asset managers, leveraged funds") · Supplemental (CIT) ("13 agricultural markets: commodity index traders")
Metric	COT index 3y · COT index 1y · z-score 3y · Net % OI
Futures only / Futures + options	Futures-only or combined report (CIT exists only combined)
Asset-class buttons	All · Energy · Metals · Grains & oilseeds · Softs · Livestock & dairy · Rates · Equity & vol · Currencies · Crypto
Market name	Opens the CFTC market page with the same report type
Heatmap cell (hover)	Metric value, "long / short extreme (model-based)" and "crowded" when set

Methodology

The page's own definitions: "Positions as of Tuesday, published Friday 15:30 ET. COT index = position of the net (long − short) within its trailing 52- or 156-week range (0 = most short, 100 = most long)." In detail (`app/analytics/cftc_signals.py` ; windows count reports, roughly weeks):

```
net            = long − short                          (spreading positions excluded)
net % OI       = net / open interest
COT index (w)  = (net − min) / (max − min) over the last w reports
                 w = 52 (1y, min 26) or 156 (3y, min 78); shown 0–100
z-score (w)    = (net − rolling mean) / rolling std over w reports
Speculators − hedgers = speculator net − hedger net:
                 Legacy: non-commercials − commercials
                 Disaggregated: managed money − (producers + swap dealers)
                 TFF: leveraged funds − dealers
                 CIT: index traders − commercials
```

MODEL-BASED heuristics (labelled in the app, "not CFTC data"):

Flag	Rule
Extreme (L / S)	COT index 3y ≥ 0.9 (long) or ≤ 0.1 (short)
Crowded	extreme and abs(z 3y) ≥ 1.5
Divergence	z of the 13-week net change and z of the 13-week log price return (156-report windows); "bullish" when positioning z ≥ +1 and price z ≤ −1, "bearish" when the reverse; "Div. 13w" = z(net change) − z(return)

The heatmap shows **headline (curated, active) markets**; the crowded and divergence lists use the **headline speculator category** of each market's primary report (managed money for physical commodities, leveraged funds for financials,

non-commercials otherwise), futures-only unless "Futures + options" is selected. Legend colours: COT index from 0 (most short, orange) to 100 (most long, blue); z from −2.5σ to +2.5σ; Net % OI from −40% to +40%.

Data sources and refresh

CFTC Commitments of Traders via the CFTC Public Reporting Environment (official). Reports: Legacy (futures-only from 1986), Disaggregated and TFF (from June 2006), each futures-only and combined, and the Supplemental CIT report (from 2006). **Point in time:** the knowledge date is the observed publication time where available (rows loaded after the September 2022 bulk load), otherwise the scheduled release, report Tuesday + 3 days at 15:30 ET, moved to the next business day after a Wednesday–Friday federal holiday; the 2013-10 and 2018-12 to 2019-01 government shutdowns are treated conservatively (estimated release dates). The data line says "(scheduled)" when the release date is estimated. The weekly job runs on Friday at 19:00 ET.

Validation status

COT data are official. The extremes, crowding and divergence flags are **model-based heuristics** and **not validated** as signals; the per-market event study (next section) is "indicative only".

Correct usage and worked example

Use the COT index to locate positioning within its own history and the z-score to judge how unusual it is; read crowding as a risk flag, not as a timing signal.

Example (report of 13 Oct 2026): on the Disaggregated heatmap (Figure 2B.43), WTI crude (NYMEX) managed money sits at **27** and producers / merchants at **88**, the same figures as on the WTI market page; for Henry Hub, swap dealers are at a long extreme (100 L) while speculators − hedgers are at a short extreme (5 S). The Crowded longs list (Figure 2B.41) is headed by the Brazilian real, the UST bond and Sugar No. 11, all at a COT index 3y of 100. The UST bond also appears in the divergences list as **bullish** (Div. 13w +3.4): leveraged funds added to positions while the price fell. Note that its net position is **−154k contracts**: a "long extreme" means the net is at the long end of its own three-year range, not that the category is net long, and for Treasuries leveraged-fund shorts reflect the basis trade.

Limitations and pitfalls

- **Three-day lag.** Positions are as of Tuesday and published on Friday; by then the market may have moved.
- **Category meaning differs by report.** "Non-commercials" (Legacy) and "Managed money" (Disaggregated) are not the same population; compare like with like.
- **Structural positions.** Treasury and crypto leveraged-fund shorts are basis trades; asset managers are structural real-money longs.
- **Ranges move.** A COT index of 100 can coexist with a smaller position than a year ago if the 156-week range has shifted.
- **The crowded and divergence lists ignore the report selector**: they always use each market's headline category.
- **CIT** covers 13 agricultural markets and is published combined only, so the futures-only toggle has no effect on it.

Related mnemonics and BQL fields

- Mnemonic: `COT` .
- Model Studio source `cftc` (field `market:report_type:category:metric`); Backtest Lab signal source `cftc` and the event trigger "CFTC extreme".
- Related: CFTC market page, Cross-asset dashboard (positioning column), 7. Smart Money for equity positioning.

CFTC market page

Route: `/macro/positioning/:code` (for example `/macro/positioning/067651` , WTI crude on NYMEX; optional `?report=`) · **Menu:** from the Positioning heatmap, an instrument page or the Cross-asset dashboard · **Mnemonics:** none

(open via `COT`) · **Refresh:** weekly

Purpose

The full Commitments of Traders history of one futures market: the latest positions of every trader category, net positions over time against the price, the COT index history with its extreme zones, open interest and concentration, trader counts, past extreme episodes and an event study of what prices did after extremes.

Screenshot

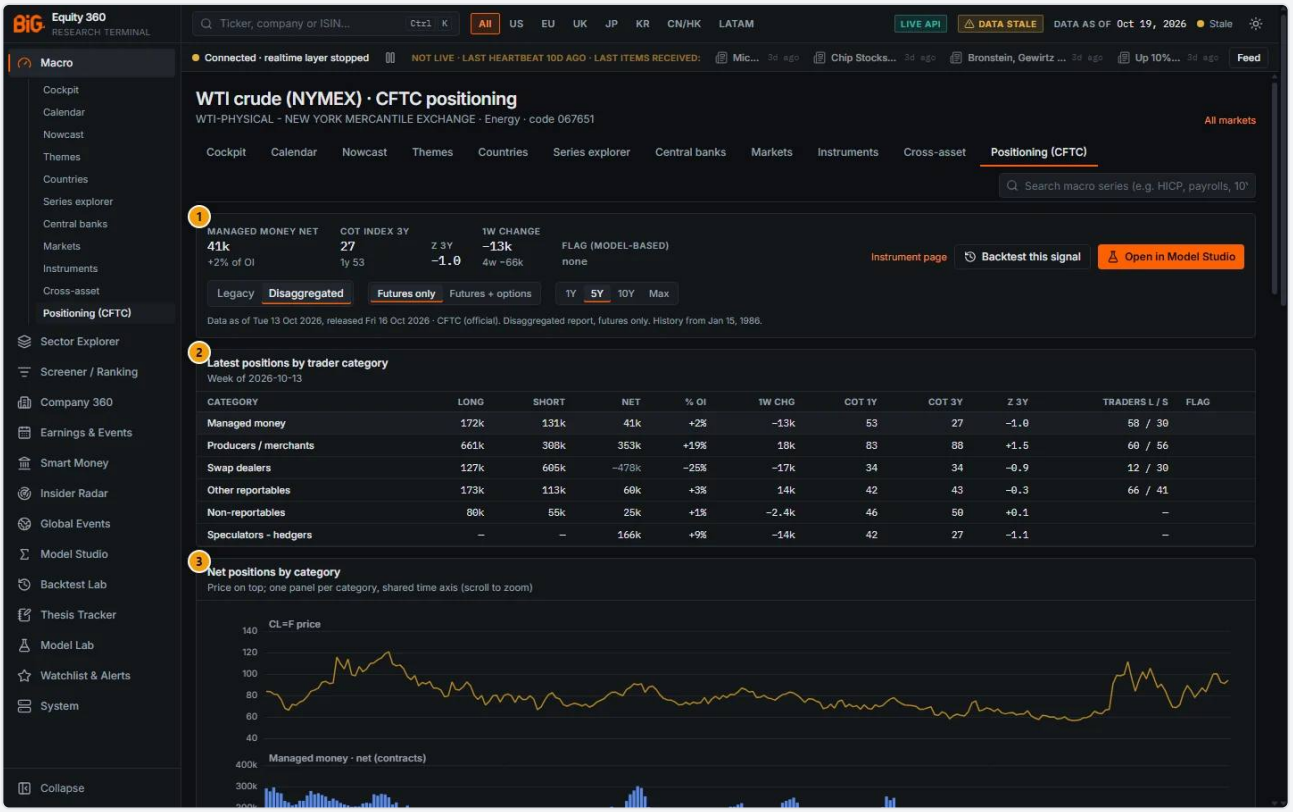


Figure 2B.44: WTI crude (NYMEX) · CFTC positioning, code 067651, Disaggregated, futures only, 5Y range.

#	Panel	What it shows
1	MANAGED MONEY NET	Headline category net (and % of OI), COT index 3y (and 1y), z 3y, 1-week and 4-week change, the model-based flag, buttons, report and range selectors, and the data line
2	Latest positions by trader category	Long, short, net, % OI, 1w change, COT 1y and 3y, z 3y, trader counts and flags for each category
3	Net positions by category	The mapped price on top, then one signed-bar panel per category, on a shared time axis
4	COT index history (3y window)	Headline and speculators − hedgers COT index with shaded model-based extreme zones
5	Open interest	Open interest and the net concentration of the four largest traders, long and short
6	Number of traders	Headline category, long vs short trader counts
7	Extreme episodes	Spells with the headline COT index 3y ≥ 90 or ≤ 10 in the selected range
8	After extremes: forward returns	Model-based event study of forward returns after extremes, full history, futures-only report

Figure 2B.45: COT index history (3y window), managed money and speculators – hedgers.

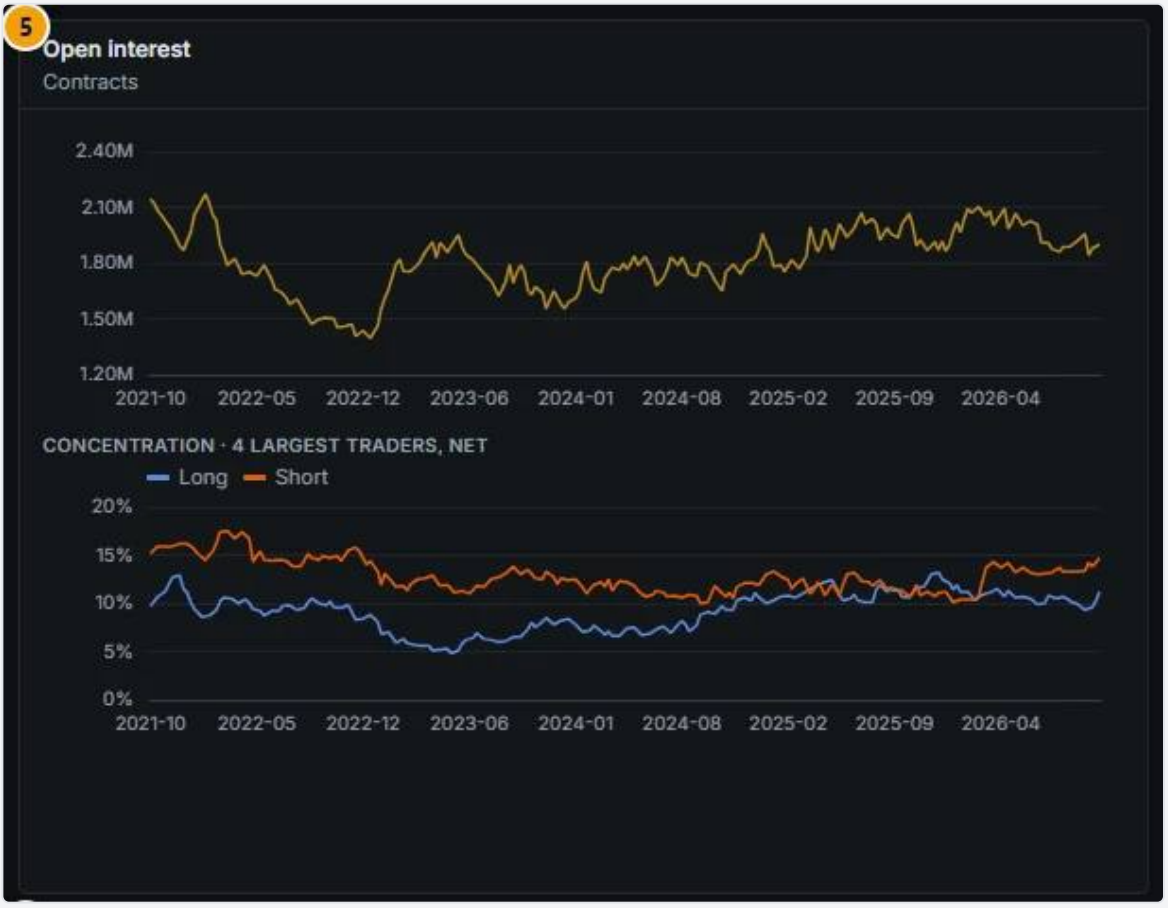


Figure 2B.46: Open interest and concentration of the four largest traders.

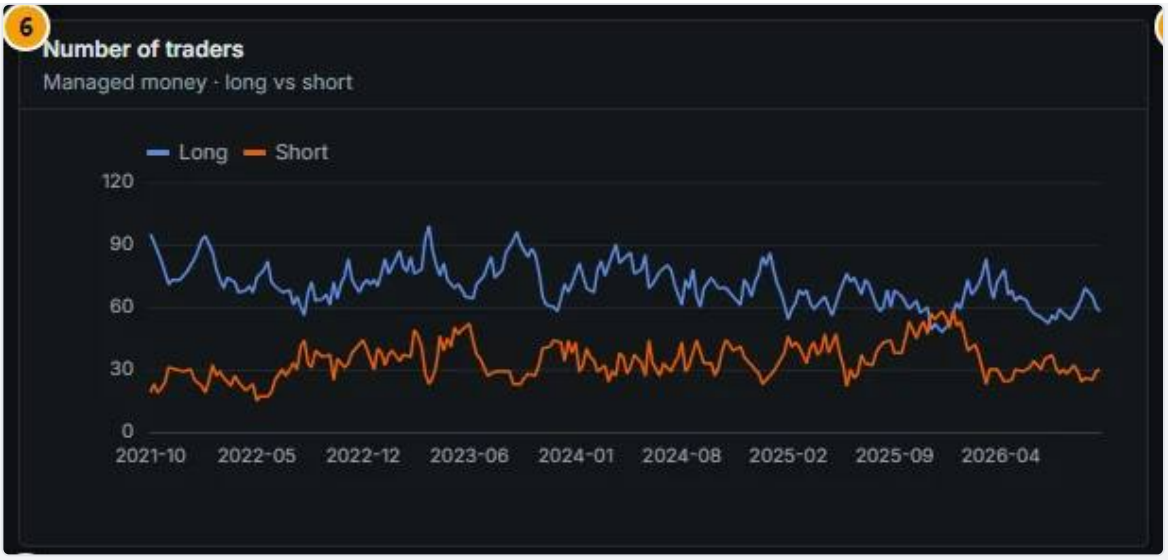


Figure 2B.47: Number of managed-money traders, long vs short.

7

Extreme episodes

Model-based: headline COT index 3y ≥ 90 or ≤ 10 (in range)

SIDE	FROM	TO	WEEKS	PEAK COT
SHORT	2025-11-10	2025-12-02	4	0
SHORT	2025-08-12	2025-10-28	12	0
SHORT	2025-04-08	2025-04-08	1	1
SHORT	2025-02-25	2025-02-25	1	9
SHORT	2024-02-06	2024-02-06	1	9
SHORT	2024-01-16	2024-01-16	1	8
SHORT	2024-01-02	2024-01-02	1	6

Figure 2B.48: Extreme episodes in the 5Y range.

8

After extremes: forward returns

Model-based event study (full history, futures-only report): log return from the first close after the release; indicative only — small samples, overlapping windows

CATEGORY	EVENT	HORIZON	EVENTS	MEAN FWD	ALL WEEKS	EXCESS	HIT RATE	T
Managed money	long extreme	4w	10	+0.7%	+0.3%	+0.4%	60%	0.3
Managed money	long extreme	13w	10	+0.7%	+0.4%	+0.3%	50%	0.1
Managed money	long extreme	26w	10	+0.0%	+0.7%	-0.7%	50%	-0.1
Managed money	short extreme	4w	12	-2.8%	+0.3%	-2.3%	50%	-0.5
Managed money	short extreme	13w	12	-2.3%	+0.4%	-2.7%	58%	-0.2
Managed money	short extreme	26w	12	-8.4%	+0.7%	-9.1%	50%	-0.8
Producers / merchants	long extreme	4w	15	+3.2%	+0.3%	+2.9%	73%	1.6
Producers / merchants	long extreme	13w	15	-3.5%	+0.4%	-3.9%	60%	-0.7
Producers / merchants	long extreme	26w	15	-6.4%	+0.7%	-7.1%	40%	-0.9
Producers / merchants	short extreme	4w	8	+1.1%	+0.3%	+0.8%	50%	0.3
Producers / merchants	short extreme	13w	8	-0.3%	+0.4%	-0.7%	50%	-0.1
Producers / merchants	short extreme	26w	8	+5.1%	+0.7%	+4.5%	50%	0.5

Prices: CL=F. Model-based, descriptive only.

Figure 2B.49: After extremes: forward returns, WTI (prices: CL=F).

Controls

Control	What it does
All markets	Back to the Positioning overview
Instrument page	Opens the linked instrument page (`/instruments/IDX:CL=F`)

Control	What it does
Backtest this signal	Opens the Backtest Lab with an event-driven draft: contrarian trades on COT-index extremes (enter when COT index 3y crosses ≥ 0.9 or ≤ 0.1, hold 20 days, at most 10 positions, shorts allowed, entry one day after the signal), on this commodity's root or on all assets of the class that have a CFTC join
Open in Model Studio	Draft study: weekly OLS of the mapped instrument's log return on the headline COT index 3y, the weekly net change and the speculators − hedgers z 3y, each lagged one week, from 2008
Report type	The report families available for this market (WTI: Legacy and Disaggregated)
Futures only / Futures + options	Hidden for CIT
Range: 1Y / 5Y / 10Y / Max	History range of the charts and the Extreme episodes table (default 5Y)
Charts	Scroll to zoom the shared time axis; hover for values

Methodology

Definitions are the same as on the Positioning (CFTC) page (the page footer repeats them: "COT index = (net - min) / (max - min) over the trailing 52 / 156 weekly reports; extremes (COT index 3y >= 0.9 or <= 0.1), crowding (extreme and |z 3y| >= 1.5), divergence vs price and the event study are model-based heuristics, not CFTC data.").

Event study (model-based, full history, futures-only report; categories: Legacy non-commercials and commercials, Disaggregated managed money and producers / merchants, TFF leveraged funds, asset managers and dealers):

```
event         = the COT index 3y crosses into ≥ 0.9 (long extreme) or ≤ 0.1 (short extreme),
                at least 13 reports after the previous event of the same side
entry         = first close strictly after the release date (point in time)
forward return = log return from the entry close to the first close ≥ entry + 7·h days, h = 4, 13, 26 weeks
All weeks     = mean forward return over every report (unconditional)
Excess        = Mean fwd − All weeks
t             = Excess / (sd of event returns / √n)
```

Prices are the adjusted closes of the mapped futures instrument, or a labelled proxy when the contract has no futures series ("Price overlay uses … as a proxy").

Data sources and refresh

CFTC (official), weekly. The data line under the selectors reads, for WTI, "Data as of Tue 13 Oct 2026, released Fri 16 Oct 2026 · CFTC (official). Disaggregated report, futures only. History from Jan 15, 1986."

Validation status

The event study is **model-based, descriptive only** and "indicative only — small samples, overlapping windows". For WTI managed money, no horizon reaches |t| ≥ 1 (Figure 2B.49): there is **no evidence** in this market that managed-money extremes predict prices.

Correct usage and worked example

WTI crude, report of 13 Oct 2026 (Figure 2B.44): managed money is net long **41k** contracts (+2% of open interest: 172k long vs 131k short), with a COT index 3y of **27** (1y: 53) and z 3y **−1.0**; it cut 13k contracts in the week and 66k over four weeks; no model-based flag. Producers / merchants are net long 353k with a COT index 3y of 88 and swap dealers net short 478k. So speculators have been reducing longs into a +71% YTD rally, and they are in the lower third of their three-year range.

The event study (Figure 2B.49) puts this in context: after managed-money **short** extremes (12 events, 2008 to 2025), WTI's mean 26-week return was −8.4% against +0.7% in all weeks (excess −9.1%, t −0.8, hit rate 50%); after long extremes (10 events) the excess was between −0.7% and +0.4%. Neither is statistically meaningful.

Limitations and pitfalls

- **Small samples**: 8 to 15 events per row, with overlapping 13- and 26-week windows.
- **Proxy prices** for some contracts (for example a cash index or ETF) do not match the futures' returns.
- **Report families differ by market**: WTI has no TFF report; financial futures have no Disaggregated report.
- **Combined (futures + options) positions** are delta-adjusted by the CFTC and are not directly comparable with futures-only numbers.
- **Concentration** ("4 largest traders, net") is "Not published for this report." for some report types.

Related mnemonics and BQL fields

- Mnemonic `COT` (overview); the instrument page's "CFTC positioning" button leads here.
- Model Studio source `cftc` ; Backtest Lab event trigger "CFTC extreme" (11. Backtest Lab).
- Related: Positioning (CFTC), Cross-asset dashboard, Instrument page.

3. Sector Explorer

The Sector Explorer is the top-down entry point for fundamental work: pick a sector, drill into a subsector, and see how the companies in it are spread on the KPIs that matter for that business model. It also lists the best and worst composite scores in the selection and the model-based peer-group anomaly panel.

Sector Explorer

Route: `/sectors` , `/sectors?sector=<sector id>` , `/sectors?sector=<sector id>&sub=<subsector id>` · **Menu:** Sector Explorer · **Mnemonics:** `PEERS` (with no security loaded) · **Refresh:** nightly analytics (`ratios_kpis` , `scoring`); default weekly profile = Saturday refresh, daily profile Tue-Sat

Purpose

The page answers three questions for a sector or subsector: how large it is and how it has moved, where a given company sits on the sector's own KPIs (net interest margin for banks, FFO for REITs, Rule of 40 for software and so on), and which names score best and worst on the composite. Analysts use it to build a peer view before opening individual companies; portfolio managers use it to see the dispersion inside a sector before allocating within it.

The KPI set changes with the business model. A bank is not judged on gross margin and a REIT is not judged on ROIC; each subsector belongs to one KPI set (28 sets in all, listed under Methodology).

Screenshot

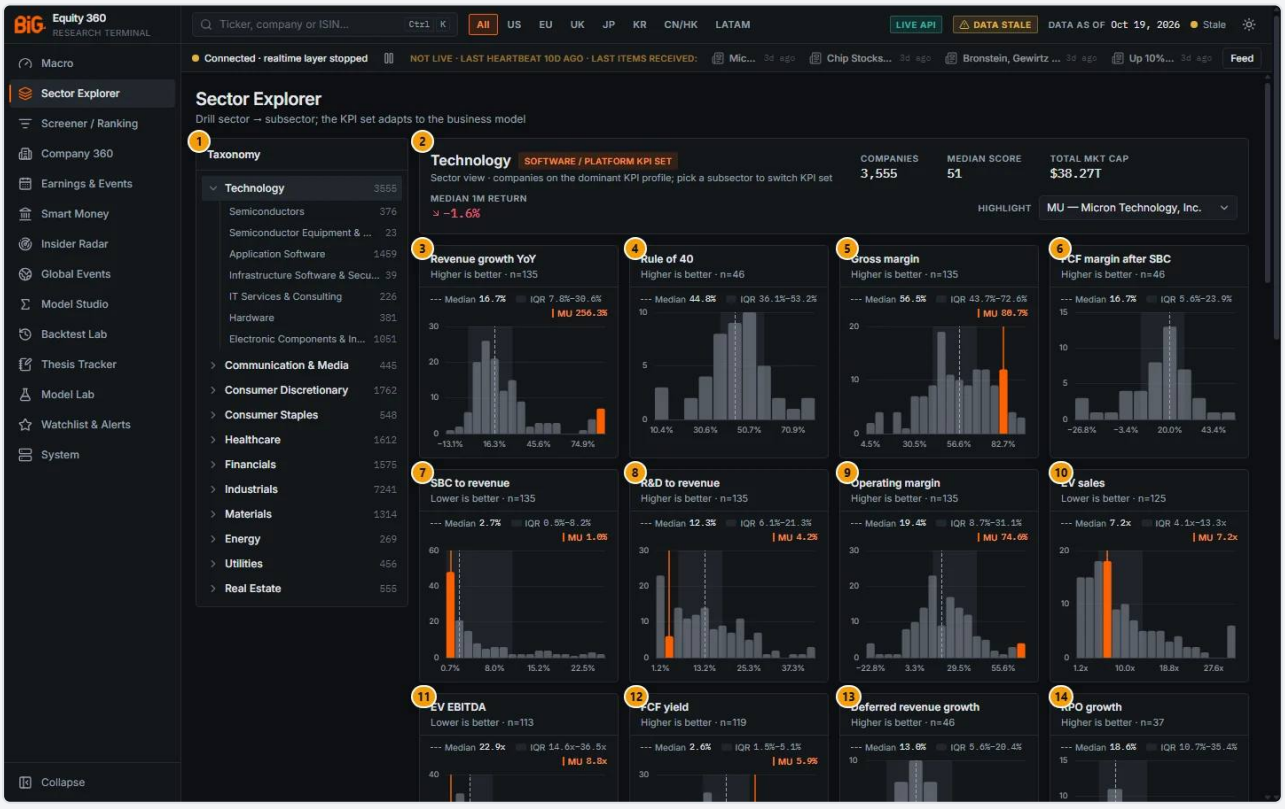


Figure 3.1: Sector Explorer (`/sectors` , Technology sector view, region All, highlight MU), data as of 19 Oct 2026.

#	Panel	What it shows
1	Taxonomy	The 11 sectors with company counts for the current region; the open sector lists its subsectors with their counts
2	Technology (header)	Selection name, KPI-set badge, Companies, Median score, Total mkt cap, Median 1M return, Highlight selector
3	Revenue growth YoY	Histogram of the KPI across the selection (higher is better, n = 135)
4	Rule of 40	Revenue growth + FCF margin, software KPI (n = 46)
5	Gross margin	Generic metric, headline for software and semis (n = 135)
6	FCF margin after SBC	(FCF − stock-based compensation) / revenue, software KPI (n = 46)
7	SBC to revenue	Stock-based compensation / revenue (lower is better, n = 135)
8	R&D to revenue	R&D intensity (n = 135)
9	Operating margin	Generic metric (n = 135)
10	EV sales	Valuation multiple (lower is better, n = 125)
11	EV EBITDA	Valuation multiple (lower is better, n = 113)
12	FCF yield	FCF / market cap (n = 119)
13	Deferred revenue growth	Software KPI (n = 46)
14	RPO growth	Remaining performance obligations growth, software KPI (n = 37)
15	Opex leverage	Revenue growth minus growth of SG&A + R&D, software KPI (n = 47)
16	DIO	Days inventory outstanding, semis / hardware headline (lower is better, n = 98)
17	ROIC	Return on invested capital (n = 118)
18	FCF margin	Free cash flow / revenue (n = 134)
19	P/E TTM	Trailing P/E (lower is better, n = 112)
20	R&D intensity (3y avg)	Semis KPI (n = 39)
21	Inventory days change	DIO change vs a year earlier, semis / hardware KPI (lower is better, n = 78)
22	Capex intensity	Capex / revenue, semis KPI; a target metric (see pitfalls) (n = 39)
23	Inventory turns	Hardware headline (n = 98)
24	DSO	Days sales outstanding, services headline (lower is better, n = 134)
25	Revenue per employee (USD)	Services KPI; empty state "No distribution" because no company has a value (n = 0)
26	Leaders	Ten highest composite scores in the selection
27	Laggards	Ten lowest composite scores in the selection
28	Peer-group joint anomaly in this selection	Model-based anomaly percentile of the peer groups in the selection (descriptive, not a trading signal)

Panels 15 to 28 are below the fold. The three tables at the bottom of the page:

Figure 3.2: Leaders (callout 26): MU 81, SK hynix 75, PLTR 74 ... TSMC 68. Score = stored balanced composite.

Figure 3.3: Laggards (callout 27): MSTR 14, LONGi 15, CoreWeave 23 ...

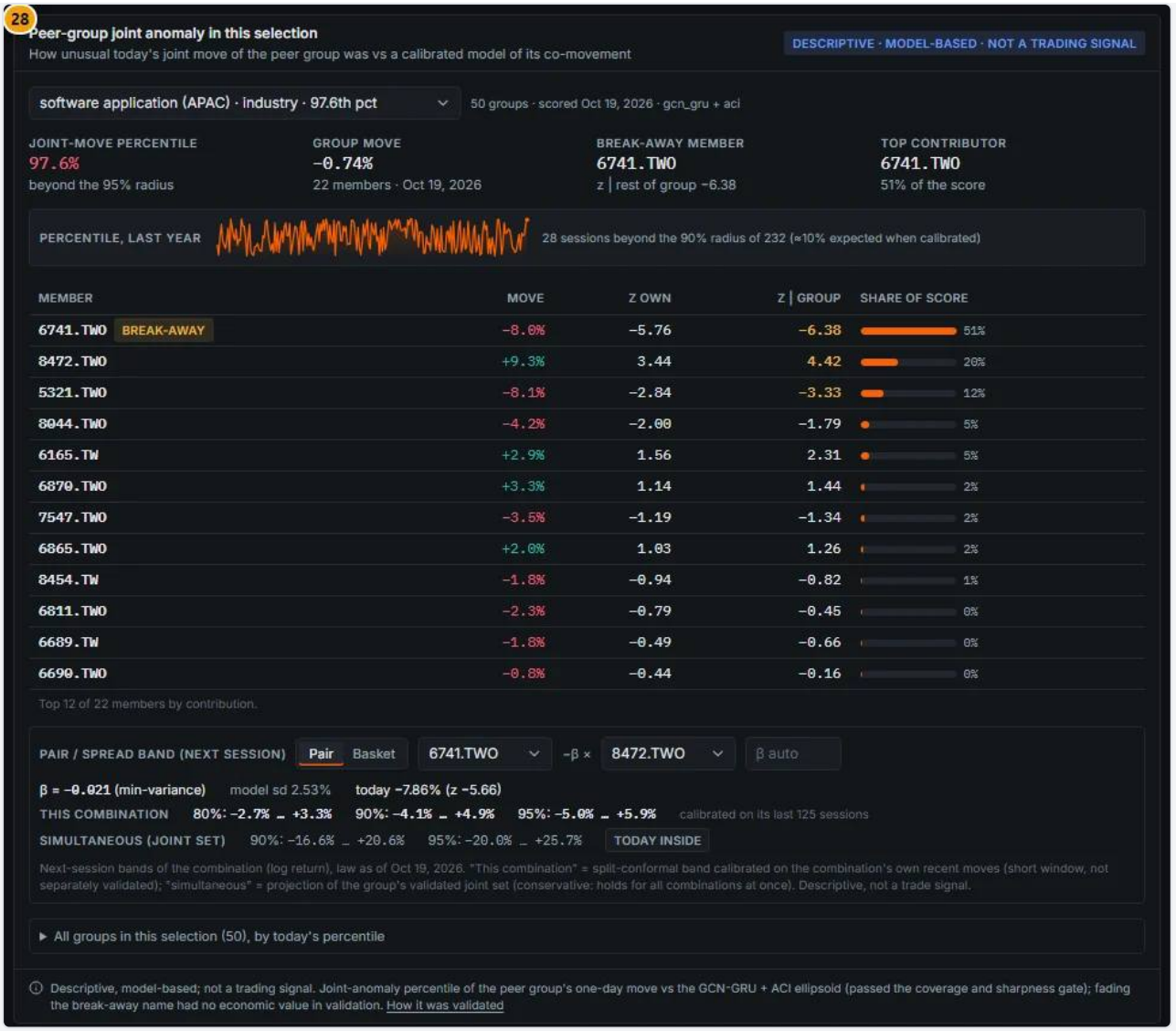


Figure 3.4: Peer-group joint anomaly in this selection (callout 28), labelled "Descriptive · model-based · not a trading signal".

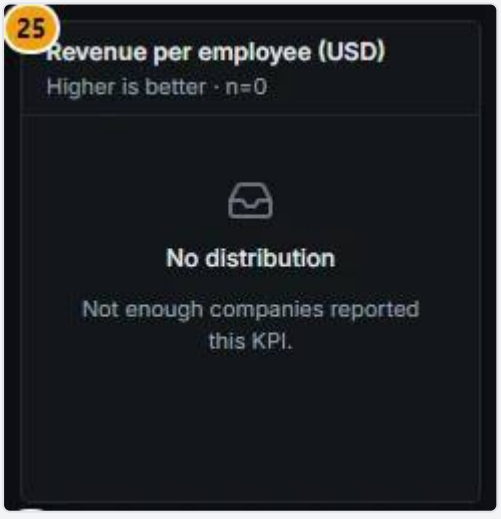


Figure 3.5: A KPI card with no data (callout 25): "No distribution · Not enough companies reported this KPI."

SUBSECTOR VIEWS

Clicking a subsector switches the cards to its KPI set. Three examples follow: a bank set, a semiconductor set and a REIT set.

Figure 3.6: Global & Diversified Banks (`/sectors?sector=FINS&sub=banks_global` , region All, highlight UCG.MI), data as of 19 Oct 2026.

#	Panel	What it shows
1	Taxonomy	Financials expanded; Global & Diversified Banks (76) selected
2	Global & Diversified Banks (header)	"Bank KPI set" badge; 76 companies, median score 52, total cap $9.63T, median 1M return −3.2%; highlight UCG.MI (UniCredit)
3	Return on tangible equity	Median 13.2%, IQR 10.4%-17.4%, UCG.MI 17.5% (n = 57)
4	ROE	Median 12.0%, UCG.MI 16.8% (n = 57)
5	Net interest margin	Median 2.5%, IQR 1.3%-3.9% (n = 42); no value for UCG.MI
6	Efficiency ratio	Lower is better; median 57.9% (n = 29)
7	Cost of risk	Lower is better; median 0.4%, UCG.MI 0.1% (n = 28)
8	NPL ratio	Lower is better; median 0.7% (n = 11)
9	Loans / deposits	Scored on distance from 85% although labelled "Higher is better"; median 80.8%, UCG.MI 97.2% (n = 44)
10	CET1 ratio	Median 12.9% (n = 5, anecdotal)
11	P TBV	Price / tangible book, lower is better; median 1.7x, UCG.MI 1.7x (n = 39)
12	P/E TTM	Median 11.8x (n = 37)
13	Dividend yield	Median 1.8%, UCG.MI 0.6% (n = 39)
14	Allowance / NPL	NPL coverage; median 1.8x (n = 11)
15	Fee income share	Non-interest income / total revenue (n = 45)
16	Pre-provision profit / assets	PPOP / average assets (n = 29)
17	Equity / assets	Leverage measure (n = 57)
18	Loan growth	Scored on distance from 7% although labelled "Higher is better" (n = 44)
19	Deposit growth	Year-on-year deposit growth (n = 45)

#	Panel	What it shows
20	Leaders	UniCredit 70, China Merchants Bank 69, PNC 69 … Intesa Sanpaolo 63
21	Laggards	Lowest composites, led by Citigroup and Lloyds
22	Peer-group joint anomaly in this selection	As in Figure 3.4, for the bank peer groups

Figure 3.7: Leaders of the global banks (callout 20). JPMorgan is eighth with 65.

Figure 3.8: Semiconductors (`/sectors?sector=TECH&sub=semiconductors` , region All, highlight MU), data as of 19 Oct 2026.

#	Panel	What it shows
1	Taxonomy	Technology expanded; Semiconductors (376) selected
2	Semiconductors (header)	"Semiconductor KPI set" badge; 376 companies, median score 54, total cap $17.13T, median 1M return −5.2%; highlight MU
3	Revenue growth YoY	Median 30.6%, MU 256.3% (n = 26)
4	Gross margin	Median 58.5%, IQR 54.4%-66.7%, MU 80.7% (n = 26)
5	Operating margin	Median 26.0%, MU 74.6% (n = 26)
6	R&D to revenue	Median 17.5%, MU 4.2% (n = 26)
7	DIO	Lower is better; median 134 days, MU 133 days (n = 26)
8	ROIC	Median 7.2%, MU 106.2% (n = 22), an outlier (see pitfalls)
9	FCF margin	Median 23.9%, MU 44.3% (n = 26)
10	P/E TTM	Lower is better; median 45.4x, MU 11.7x (n = 22)
11	EV EBITDA	Lower is better; median 26.7x, MU 8.8x (n = 20)
12	R&D intensity (3y avg)	Semis KPI; median 16.9%, MU 9.4% (n = 26)
13	Inventory days change	Semis KPI, lower is better; median +3 days, MU −7 days (n = 26)
14	Capex intensity	Semis KPI scored on distance from 10% although labelled "Higher is better"; median 6.3%, MU 23.1% (n = 26)
15	Leaders	MU 81, SK hynix 75, Samsung Electronics 70, TSMC 68, NVIDIA 67 ...
16	Laggards	Ten lowest composites in the subsector
17	Peer-group joint anomaly in this selection	Semiconductor peer groups

Figure 3.9: Leaders of the Semiconductors subsector (callout 15).

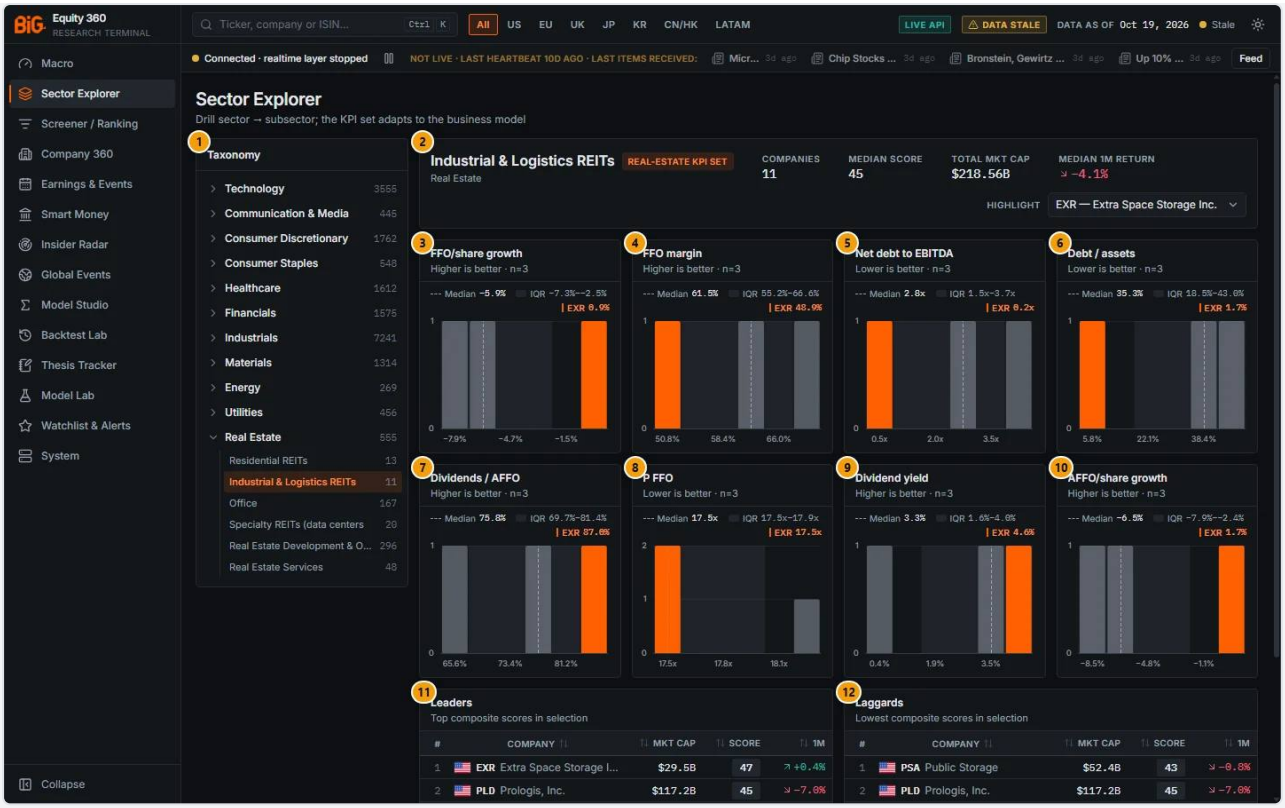


Figure 3.10: Industrial & Logistics REITs (`/sectors?sector=REAL&sub=reit_industrial` , region All, highlight EXR), data as of 19 Oct 2026.

#	Panel	What it shows
1	Taxonomy	Real Estate expanded; Industrial & Logistics REITs (11) selected
2	Industrial & Logistics REITs (header)	"Real-estate KPI set" badge; 11 companies, median score 45, total cap $218.56B, median 1M return −4.1%; highlight EXR
3	FFO/share growth	Median −5.9%, EXR 0.9% (n = 3)
4	FFO margin	Median 61.5%, EXR 48.9% (n = 3)
5	Net debt to EBITDA	Lower is better; median 2.8x, EXR 0.2x (n = 3)
6	Debt / assets	Lower is better; median 35.3%, EXR 1.7% (n = 3)
7	Dividends / AFFO	Payout scored on distance from 75% although labelled "Higher is better"; median 75.8%, EXR 87.6% (n = 3)
8	P FFO	Price / FFO, lower is better; median 17.5x (n = 3)
9	Dividend yield	Median 3.3%, EXR 4.6% (n = 3)
10	AFFO/share growth	Median −6.5% (n = 3)
11	Leaders	Only three scored names: EXR 47, PLD 45, PSA 43
12	Laggards	The same three names in reverse order
13	Peer-group joint anomaly in this selection	REIT peer groups (below the fold)

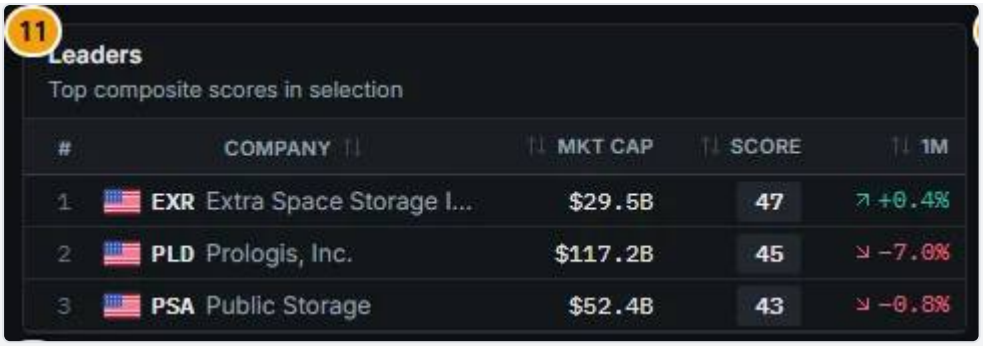


Figure 3.11: With only three scored companies, Leaders and Laggards list the same names (callout 11).

Figure 3.10 shows how thin some subsectors are: 11 companies, but every REIT card has n = 3. The bars are then individual companies, not a distribution; compare the values one by one.

Controls

Control	What it does
Region chips (top bar: All, US, EU, UK, JP, KR, CN/HK, LATAM)	Restrict every number on the page (counts, KPIs, leaders, laggards) to listings of that region. The choice is global and also applies to the Screener
Sector row (Taxonomy)	Opens the sector view and expands its subsectors. The URL becomes `/sectors?sector=<id>`
Subsector row (Taxonomy)	Opens the subsector view; the KPI set switches to that subsector's business model. The URL becomes `/sectors?sector=<id>&sub=<id>`
Company count (grey number)	Active securities in that node for the current region
KPI-set badge (header)	Shows which KPI family is displayed. Hover: "KPI definitions switch with the business model (banks use NIM/CET1, software uses NRR/Rule of 40, REITs use FFO/LTV ...)"
Highlight	Picks one company to mark in every histogram (orange bar and solid line, value printed top right of each card). Options: the leaders and laggards, then the tickers of the first KPI. Default: the top leader. "No highlight" clears it
KPI card hover	Hover a bar: value range of the bin, number of companies and up to six tickers. Hover the card body: KPI name and n
Leaders / Laggards row	Click a row to open that company in Company 360
Peer-group panel	Group selector, Pair / Basket band builder, "All groups in this selection" list and the "How it was validated" link (see Peer-group joint anomaly for the full description)
Retry (error state)	Re-requests the sector after "Could not load data"

The sector and subsector ids for direct links:

Sector	`sector=` id	Companies (All regions)
Technology	`TECH`	3,555
Communication & Media	`COMM`	445
Consumer Discretionary	`CDIS`	1,762
Consumer Staples	`CSTA`	548
Healthcare	`HLTH`	1,612
Financials	`FINS`	1,575
Industrials	`INDU`	7,241
Materials	`MATR`	1,314
Energy	`ENRG`	269
Utilities	`UTIL`	456

Sector	`sector=` id	Companies (All regions)
Real Estate	`REAL`	555

Subsector ids are the lower-case ids in the KPI-set table below (for example `banks_global`, `semiconductors`, `reit_industrial`). Opening `/sectors` with no parameter shows the first sector of the taxonomy (Technology).

Methodology

TAXONOMY

Every issuer is mapped to one of 11 sectors and 79 subsectors (`contracts/sector_taxonomy.yaml`). Vendor industry labels are translated into this taxonomy; one issuer has one subsector, whichever listing you look at. Each subsector belongs to exactly one **KPI set**, and the KPI set decides three things: which sector-specific KPIs are computed for the company, which "most relevant" metrics are shown to compare it with peers, and which generic metrics are switched off (a bank has no gross margin, ROIC or inventory turns).

KPI SETS

The 28 KPI sets (`contracts/sector_kpis.yaml` , `kpi_sets`). "Own KPIs" are computed only for companies in that set; "Headline peer metrics" are the cards shown first, in order of relevance.

KPI set	Subsectors	Own KPIs	Headline peer metrics
software_internet	software_application, software_infrastructure, internet_platforms, video_games, payments_fintech	Rule of 40, FCF margin after SBC, deferred revenue growth, RPO growth, opex leverage	revenue growth, Rule of 40, gross margin, FCF margin after SBC, SBC / revenue, R&D / revenue, operating margin, EV/sales, EV/EBITDA, FCF yield
semis	semiconductors, semi_equipment	R&D intensity (3y avg), inventory days change, capex intensity	revenue growth, gross margin, operating margin, R&D / revenue, DIO, ROIC, FCF margin, P/E, EV/EBITDA
hardware	hardware_networking, electronic_components	inventory days change	revenue growth, gross margin, operating margin, ROIC, inventory turns, FCF margin, P/E, EV/EBITDA
services	it_services, business_services, consumer_services, health_providers_services, insurance_brokers, real_estate_services	revenue per employee (USD)	revenue growth, operating margin, ROIC, FCF margin, DSO, revenue per employee, P/E, EV/EBITDA
telecom	telecom_operators	dividend FCF coverage	EBITDA margin, capex / revenue, FCF margin, net debt incl. leases / EBITDA, dividend yield, EV/EBITDA, FCF yield
media	media_entertainment, advertising_publishing	none	revenue growth, operating margin, FCF margin, net debt / EBITDA, EV/EBITDA, FCF yield
autos_durables	autos, auto_parts, household_durables_leisure	R&D + capex / revenue	revenue growth, operating margin, ROIC, inventory turns, FCF margin, P/E, EV/EBITDA
homebuilders	homebuilders	inventory / equity	revenue growth, gross margin, ROE, debt / equity, BVPS CAGR 3y, P/B, P/E

KPI set	Subsectors	Own KPIs	Headline peer metrics
retail	ecommerce, specialty_retail, staples_retail	inventory turns change, GMROI	revenue growth, gross margin, operating margin, inventory turns, GMROI, ROIC, cash conversion cycle, P/E, EV/EBITDA
consumer_brands	apparel_luxury, beverages, packaged_food, tobacco, household_personal_products	pricing power (gross margin vs 5y mean)	revenue growth, gross margin, operating margin, ROIC, FCF margin, dividend yield, P/E, EV/EBITDA
restaurants_leisure	restaurants, hotels_travel_leisure	none	revenue growth, operating margin, EBITDA margin, ROIC, net debt incl. leases / EBITDA, EV/EBITDA, FCF yield
pharma_biotech	pharma, biotech	cash runway (years), R&D share of opex	revenue growth, gross margin, operating margin, R&D / revenue, cash runway, FCF margin, P/E, EV/sales
medtech	medtech, life_science_tools	none	revenue growth, gross margin, operating margin, R&D / revenue, ROIC, P/E, EV/EBITDA
managed_care	managed_care	medical cost ratio	revenue growth, medical cost ratio, operating margin, ROE, P/E
banks	banks_global, banks_regional	NIM, efficiency ratio, loans / deposits, NPL ratio, allowance / NPL, cost of risk, RoTE, fee income share, PPOP / assets, equity / assets, CET1, loan growth, deposit growth	RoTE, ROE, NIM, efficiency ratio, cost of risk, NPL ratio, loans / deposits, CET1, P/TBV, P/E, dividend yield
consumer_finance	consumer_specialty_finance	NIM, cost of risk, assets / equity	ROE, NIM, cost of risk, assets / equity, revenue growth, P/B, P/E
insurance_pc	insurance_pc_reinsurance	loss ratio, expense ratio, combined ratio, investment yield, premium growth, reserves / equity	combined ratio, loss ratio, ROE, premium growth, BVPS CAGR 3y, P/B, P/E, dividend yield
insurance_life	insurance_life_health	benefits ratio, investment yield, assets / equity, premium growth	ROE, BVPS CAGR 3y, investment yield, assets / equity, premium growth, P/B, P/E, dividend yield
asset_light_fin	asset_managers, capital_markets_exchanges	opex / revenue	revenue growth, operating margin, ROE, FCF margin, P/E, dividend yield
holdings	diversified_holdings	none	BVPS CAGR 3y, ROE, debt / equity, P/B, P/E
industrials	aerospace_defense, industrial_machinery, electrical_equipment, construction_building, industrial_conglomerates	backlog (RPO) growth, backlog / revenue	revenue growth, operating margin, ROIC, FCF conversion, backlog growth, net debt / EBITDA, P/E, EV/EBITDA
distributors	trading_distributors, health_distribution	gross profit / op. working capital	revenue growth, gross margin, operating margin, ROIC, cash conversion cycle, P/E, EV/EBITDA

KPI set	Subsectors	Own KPIs	Headline peer metrics
transport	airlines, logistics_freight, rail_marine_infra	none	revenue growth, operating margin, EBITDA margin, ROIC, net debt incl. leases / EBITDA, EV/EBITDA, FCF yield
materials	the seven Materials subsectors, agri_products, coal_uranium	capex / CFO (reinvestment), ROIC 5y mean	revenue growth, EBITDA margin, ROIC, net debt / EBITDA, FCF margin, EV/EBITDA, P/B, dividend yield
energy	integrated_oil_gas, exploration_production, refining_marketing, midstream, oilfield_services	capex / CFO, ROIC 5y mean, (dividends + buybacks) / CFO	EBITDA margin, ROIC, net debt / EBITDA, FCF yield, shareholder yield, EV/EBITDA, dividend yield
utilities	electric, gas, water, multi-utilities, renewable_ipp	FFO / debt (rating-agency style), PP&E growth (rate-base proxy)	ROE, net debt / EBITDA, FFO / debt, PP&E growth, dividend yield, P/E, EV/EBITDA
reits	reit_residential, reit_industrial, reit_office_retail, reit_specialty	FFO margin, FFO/share growth, AFFO/share growth, dividends / AFFO, debt / assets	FFO/share growth, FFO margin, net debt / EBITDA, debt / assets, dividends / AFFO, P/FFO, dividend yield
real_estate_dev	real_estate_development	inventory / equity, net debt / equity	ROE, net gearing, gross margin, BVPS CAGR 3y, P/B, P/E, dividend yield

The KPI-set badge on the header groups these sets into ten labels: Software / platform (software_internet), Semiconductor (semis), Bank (banks, consumer_finance), Insurance (insurance_pc, insurance_life), Pharma (pharma_biotech), Retail (retail), Utility / telecom (telecom, utilities), Energy (energy), Real-estate (reits, real_estate_dev). Every other set shows the generic label "Industrial KPI set", even where the set is hardware, services or media.

SELECTED KPI DEFINITIONS

All KPIs are evaluated per issuer on TTM rows (FY where no TTM exists), point-in-time (each value as known on its filing date). `avg(x)` is the mean of the balance at period end and one year earlier; `growth_yoy(x) = x / x₋₁ᵧ − 1`.

KPI	Formula	Better
Rule of 40	`revenue growth YoY + FCF margin`	higher
FCF margin after SBC	`(FCF − SBC) / revenue`	higher
Opex leverage	`revenue growth − growth(SG&A + R&D)`	higher
Inventory days change	`DIO − DIO₋₁ᵧ`	lower
Capex intensity (semis)	`capex / revenue`	closest to 10%
Pricing power	`gross margin − 5y mean gross margin`	higher
Cash runway (years)	`cash and ST investments / (−FCF)`, clipped 0-20 (20 when FCF ≥ 0)	higher
Net interest margin	`NII / avg(net loans + securities + cash)`	higher
Efficiency ratio	`non-interest expense / (NII + non-interest income)`	lower
Cost of risk	`credit-loss provisions / avg(gross loans)`	lower
RoTE	`net income to common / avg(tangible book)`	higher
Loans / deposits	`net loans / deposits`	closest to 85%
Loan growth	`growth_yoy(net loans)`	closest to 7%

KPI	Formula	Better
Combined ratio	`(losses and LAE + underwriting expenses) / premiums earned`	lower
Medical cost ratio	`losses (or cost of revenue) / premiums (or revenue)`	lower
FFO / debt (utilities)	`(CFO − change in working capital) / total debt`	higher
FFO margin (REITs)	`FFO / revenue`	higher
Dividends / AFFO	`dividends paid / AFFO`	closest to 75%
Capex / CFO (reinvestment)	`capex / CFO`	closest to 50%
ROIC 5y mean	`mean of ROIC over the last 5 fiscal years`	higher

"Closest to" marks a band metric: it is scored on the distance from the target, `−|x − target|`, not on its level.

WHICH CARDS APPEAR

- **Subsector view:** the subsector's KPI set: first its headline peer metrics, then its own KPIs not already listed.
- **Sector view:** the cards of every KPI set present in the sector, one set after another in the contract order, with duplicates removed, **capped at 24 cards**. Each company only has values for the KPIs of its own set, so `n` varies from card to card (in Technology, Rule of 40 has n = 46 because only software and internet companies carry it).

DISTRIBUTION CHART

Each card is a histogram of the KPI over the companies in the selection (`components/charts/Distribution.tsx`):

- 6 to 18 bins (about 1.6 × √n).
- The x-axis is limited to the 10th and 90th percentiles plus 60% of their spread on each side, so a few extreme values do not flatten the chart; values beyond fall into the edge bins.
- Shaded band = interquartile range (25th to 75th percentile); dashed line = median. The legend prints the median and IQR.
- The highlighted company's bin is orange, with a solid orange line at its exact value.
- Quartiles and the median are computed over every company in the selection; the bars use the 500 largest companies by market cap (all of them in practice, since n is far below 500 on every card in Figure 3.1).
- If the API returns quantiles only, the card shows a quantile strip instead (p10, IQR, median, p90, "quantiles from API"). If no company has a value, it shows "No distribution".

HEADER STATISTICS

Statistic	Definition
Companies	Number of active securities in the selection and region
Median score	Median of the stored **balanced** composite (see Scores and factors)
Total mkt cap	Sum of USD market capitalisations
Median 1M return	Labelled "median", but computed as the **market-cap-weighted average** one-month price change (simple average where caps are missing)

HOW SECTOR SCORES ARE FORMED

The Sector Explorer does not compute a separate "sector score". The numbers come from the company scores:

- **Leaders / Laggards**: the ten highest and ten lowest stored balanced composites in the selection (companies without a composite are excluded), with market cap and one-month return.
- **Median score**: the median of the same composite.
- Factor scores are percentiles **within sector × peer region**, so a typical sector median is close to 50 by construction (the live sector medians on 19 Oct 2026 range from 47.9 for Energy to 53.9 for Communication

& Media). The composite is therefore a ranking *inside* a sector, not a judgement *between* sectors. The method is in Scores and factors.

PEER-GROUP JOINT ANOMALY PANEL

The bottom panel lists the conformal peer groups that fall inside the selection (here 50 groups, "scored Oct 19, 2026 · gcn_gru + aci") and shows for the selected group how unusual today's joint move was against a calibrated model of the group's co-movement: joint-move percentile, group move, the break-away member and the top contributor, the last year of percentiles, a member table (move, z own, z given the group, share of score) and next-session pair / basket bands. It is the same component as on the company Peers tab; see Peer-group joint anomaly. The panel is passed a region only for US, EU, UK, JP, KR and LATAM; for All and CN/HK it covers every region.

Data sources and refresh

- **Fundamentals:** official filings where available (SEC EDGAR XBRL, ESEF, EDINET / TDnet and the other national systems) with yfinance as fallback, normalised to canonical line items, point-in-time by filing date. See Data and coverage.
- **KPIs:** job `ratios_kpis` (nightly DAG) evaluates the generic metrics and the KPI sets on the point-in-time panels; table `sector_kpis`.
- **Prices, market cap, returns:** Polygon and yfinance, nightly.
- **Composite scores:** job `scoring` (nightly DAG), then the serving database rebuild.
- **Peer-group anomaly:** job `conformal_peer_groups` (nightly), refitted weekly.
- **Cadence:** with the default **weekly** schedule profile the nightly DAG runs inside the Saturday 08:00 Lisbon refresh; with the **daily** profile it runs Tue-Sat 01:30. The top-bar "Data as of" date applies.

Validation status

- **KPIs and distributions:** deterministic accounting ratios and descriptive statistics, not models. Not formally validated beyond the data-quality checks; their reliability is that of the underlying filings.
- **Composite (Leaders, Laggards, Median score):** no validation gate; see Scores and factors.
- **Peer-group joint anomaly (Conformal B, GCN-GRU + ACI):** PASS as a calibrated descriptive measure: coverage 90.4% / 95.2% at the 90% / 95% targets, worst regime Covid 85.3% at 90%, about 10% sharper than an EWMA baseline. Break-away mean reversion FAILED as a trading signal (negative Sharpe). The app labels it "Descriptive · model-based · not a trading signal" (`docs/CONFORMAL_VALIDATION.md`, Verdict and section B).

Correct usage and worked example

Work from the subsector, not the sector. The subsector view uses one KPI set, so every card describes the same kind of company, and the peer groups behind the scores are built at subsector level too.

Example 1: Micron in Technology (Figure 3.1). With Technology selected and MU highlighted:

1. Gross margin (callout 5): MU sits at 80.7%, far right of the sector median of 56.5% (IQR 43.7%-72.6%).
2. SBC to revenue (callout 7): 1.0% against a median of 2.7%, in the best part of a "lower is better" distribution.
3. R&D to revenue (callout 8): 4.2% against a median of 12.3%, low for Technology.
4. Rule of 40 (callout 4) carries no orange marker: MU is in the semis set, which does not compute Rule of 40.
5. Leaders (Figure 3.2): MU has the top balanced composite in the sector, 81.

For a like-for-like view open **Semiconductors** (Figure 3.8): the badge switches to "Semiconductor KPI set", and the cards become revenue growth, gross margin, operating margin, R&D / revenue, DIO, ROIC, FCF margin, P/E, EV/EBITDA, followed by R&D intensity (3y), inventory days change and capex intensity. Each card has n = 20 to 26 of the subsector's 376 securities. Against chip peers MU's gross margin (80.7% vs a median of 58.5%) is still

top of the range, its P/E of 11.7x is far below the 45.4x median, and its R&D intensity (9.4% vs 16.9%) is low: a memory maker set against a peer group that includes design houses.

Example 2: global banks (Figure 3.6, live API 19 Oct 2026). `/sectors?sector=FINS&sub=banks_global` covers 76 securities ($9.63T, cap-weighted 1M return −3.2%, median composite 51.7, shown as 52). The KPI medians are RoTE 13.2% (IQR 10.4%-17.4%, n = 57), NIM 2.50% (n = 42), efficiency ratio 57.9% (n = 29), cost of risk 0.41% (n = 28), loans / deposits 80.8% (n = 44), P/TBV 1.66x (n = 39). With UniCredit highlighted: RoTE 17.5% and ROE 16.8% sit in the upper quartile, cost of risk 0.1% is in the best bin, loans / deposits 97.2% is above the 85% target, and P/TBV 1.7x is at the median. UniCredit leads the composite with 70 (Figure 3.7), ahead of China Merchants Bank (69.2) and PNC (68.8); Citigroup (23.1) and Lloyds (29.3) are the two lowest. CET1 has n = 5 and NPL ratio n = 11: read those two cards as anecdotal.

Limitations and pitfalls

- **Sector view of mixed sectors.** The 24-card cap and the fixed set order mean that a sector view can be dominated by one minority set. In **Financials** (Figure 3.12) the sector view starts with the payments / fintech software metrics (Rule of 40 with n = 5, SBC / revenue ...), the badge reads "Software / platform KPI set" although Payments & Fintech has only 5 companies, the bank KPIs only start at card 21, and the insurance KPIs (combined ratio, loss ratio) are not shown at all. In **Real Estate** the services metrics come before the REIT ones. Always drill into the subsector for banks, insurers and REITs.
- **Coverage.** Distributions only include companies with fundamentals: in Technology (all regions) 135 of 3,555 securities have revenue growth. About 4,900 of 24,600 issuers have fundamentals (audit F25), so the full universe is mostly price-only outside the US, Japan and Europe. A small n (CET1 n = 5 for global banks) is not a peer distribution.
- **Band metrics show the wrong direction label.** Loans / deposits, capex intensity, loan growth, dividends / AFFO and capex / CFO are scored on distance from a target, but their card subtitle reads "Higher is better" (for example Capex intensity, callout 22).
- **"Median 1M return"** is a market-cap-weighted average, not a median; a few mega caps dominate it.
- **Region is global.** The region chip in the top bar also filters this page; a sector that looks thin may simply be filtered to one region.
- **Composite is sector-relative.** Do not compare sector median scores to rank sectors (see How sector scores are formed).
- **Unknown ids.** A hand-typed URL with an invalid `sector=` id shows "Could not load data" with a Retry button; use the ids in the table above or click the sector in the taxonomy.
- **Truncated subsector names.** Five subsector names are currently shortened in the app (for example "Hotels" for Hotels, Travel, Cruise & Casinos, "Apparel" for Apparel, Footwear & Luxury, "Business" for Business, Environmental & Professional Services, "Office" for Office, Retail & Diversified REITs, and "Specialty REITs (data centers" with the bracket left open). The membership is correct.
- **Outliers.** Accounting ratios can be implausible for a few names (very high ROIC from a small invested-capital base, for example); the chart clamps the axis but the value printed for the highlighted company is the raw one.
- **Peer-group panel.** Descriptive only. Fading the break-away member had no economic value in validation.

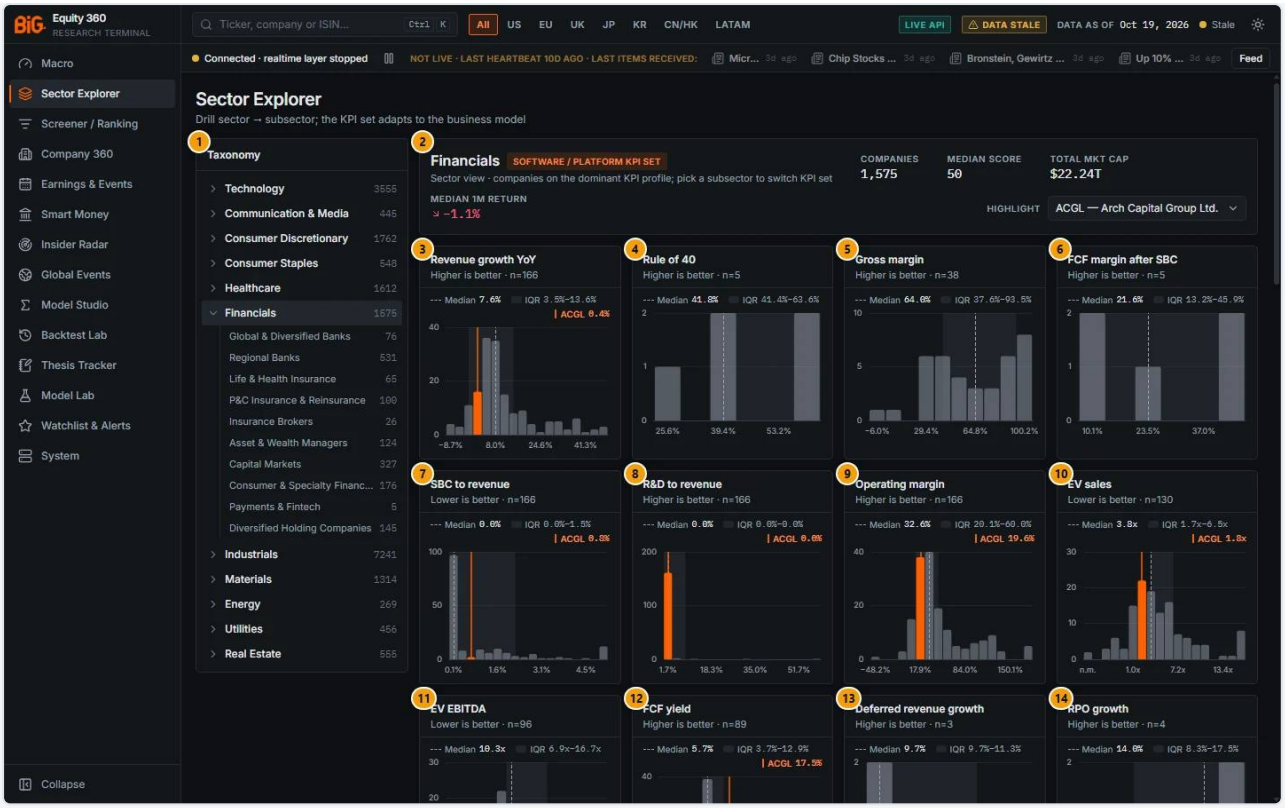


Figure 3.12: Financials sector view (`/sectors?sector=FINS` , region All, highlight ACGL), data as of 19 Oct 2026. The first cards are software / fintech metrics with tiny n; drill into a subsector instead.

#	Panel	What it shows
1	Taxonomy	Financials expanded with its ten subsectors (Global & Diversified Banks 76, Regional Banks 531 … Payments & Fintech 5, Diversified Holding Companies 145)
2	Financials (header)	"Software / platform KPI set" badge; 1,575 companies, median score 50, total cap $22.24T, median 1M return −1.1%; highlight ACGL (Arch Capital)
3	Revenue growth YoY	Median 7.6%, ACGL 0.4% (n = 166)
4	Rule of 40	Software KPI, n = 5
5	Gross margin	n = 38 (most financials have no gross margin)
6	FCF margin after SBC	Software KPI, n = 5
7	SBC to revenue	Median 0.0% (n = 166)
8	R&D to revenue	Median 0.0% (n = 166): uninformative for financials
9	Operating margin	Median 32.6%, ACGL 19.6% (n = 166)
10	EV sales	Median 3.8x, ACGL 1.8x (n = 130)
11	EV EBITDA	Median 10.3x (n = 96)
12	FCF yield	Median 5.7%, ACGL 17.5% (n = 89)
13	Deferred revenue growth	Software KPI, n = 3
14	RPO growth	Software KPI, n = 4
15	Opex leverage	Software KPI, n = 5
16	ROIC	n = 158
17	FCF margin	n = 147
18	DSO	Services headline, n = 87
19	Revenue per employee (USD)	Services KPI, n = 0 ("No distribution")

#	Panel	What it shows
20	P/E TTM	n = 130
21	Return on tangible equity	First bank KPI: median 13.3%, IQR 11.2%-17.9% (n = 82)
22	ROE	n = 168
23	Net interest margin	n = 66
24	Efficiency ratio	Lower is better, n = 47
25	Cost of risk	Lower is better, n = 48
26	NPL ratio	Lower is better, n = 21 (the 24-card cap stops here)
27	Leaders	Ten highest composites in Financials
28	Laggards	Ten lowest composites in Financials
29	Peer-group joint anomaly in this selection	Financial peer groups

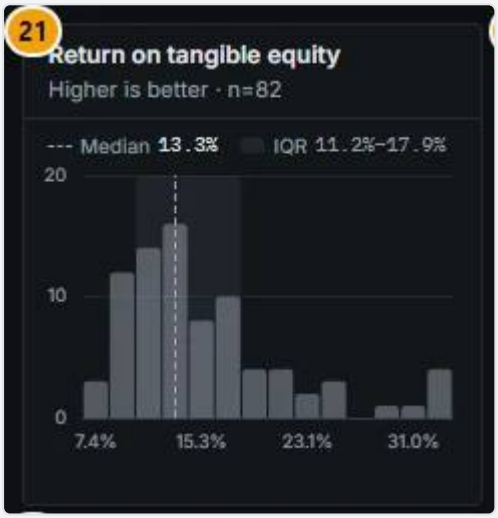


Figure 3.13: The first bank KPI in the Financials sector view arrives at card 21 (callout 21).

Related mnemonics and BQL fields

- **Mnemonics:** `PEERS` with no security loaded opens the Sector Explorer; with a security it opens that company's Peers tab, as does `RV` . See BQL and mnemonics.
- **BQL fields:** every sector KPI is a field `kpi.<kpi_id>` (for example `kpi.rule_of_40` , `kpi.nim` , `kpi.rote` , `kpi.combined_ratio` , `kpi.ffo_margin`); generic ratios such as `roic` , `gross_margin` ; scores `composite` , `grade` , `f.<factor>` ; universe functions `sector('Technology')` and `subsector(...)` in `for(...)` .
- **API:** `GET /api/v1/sectors` (tree with counts and medians), `GET /api/v1/sectors/{sector_id}?subsector_id=` (summary, KPI distributions, leaders, laggards, factor heatmap, revenue share), `GET /api/v1/sectors/{sector_id}/scatter?x=&y=` . The factor heatmap, revenue share and scatter are served but not displayed on this page.
- **Related sections:** Scores and factors, Screener, Company 360 Peers, Data and coverage.

4. Screener and Ranking

The Screener ranks the equity universe on a composite of ten factor families and lets you re-weight the factors and watch the ranking change as you drag. This chapter describes the page and, in Scores and factors, the method behind every composite and factor score in the app (Screener, Sector Explorer, Company 360, BQL).

Screener

Route: `/screener` · **Menu:** Screener / Ranking · **Mnemonics:** `EQS` · **Refresh:** nightly `scoring` job (default weekly profile = Saturday refresh; daily profile Tue-Sat); weights re-rank instantly in the browser

Purpose

The Screener is the top-down ranking: it loads the 1,000 best-scoring stocks of the region, shows the composite score with its ten factor percentiles, and lets you filter by sector, subsector, country, size, moat class, minimum score and model-based macro sensitivity. The factor-weight sliders express a view ("more value, less momentum") and re-rank the list immediately. From here you can open a company, add it to the watchlist, start a thesis, export the list, backtest the screen point-in-time or open it as a BQL query.

Screenshot

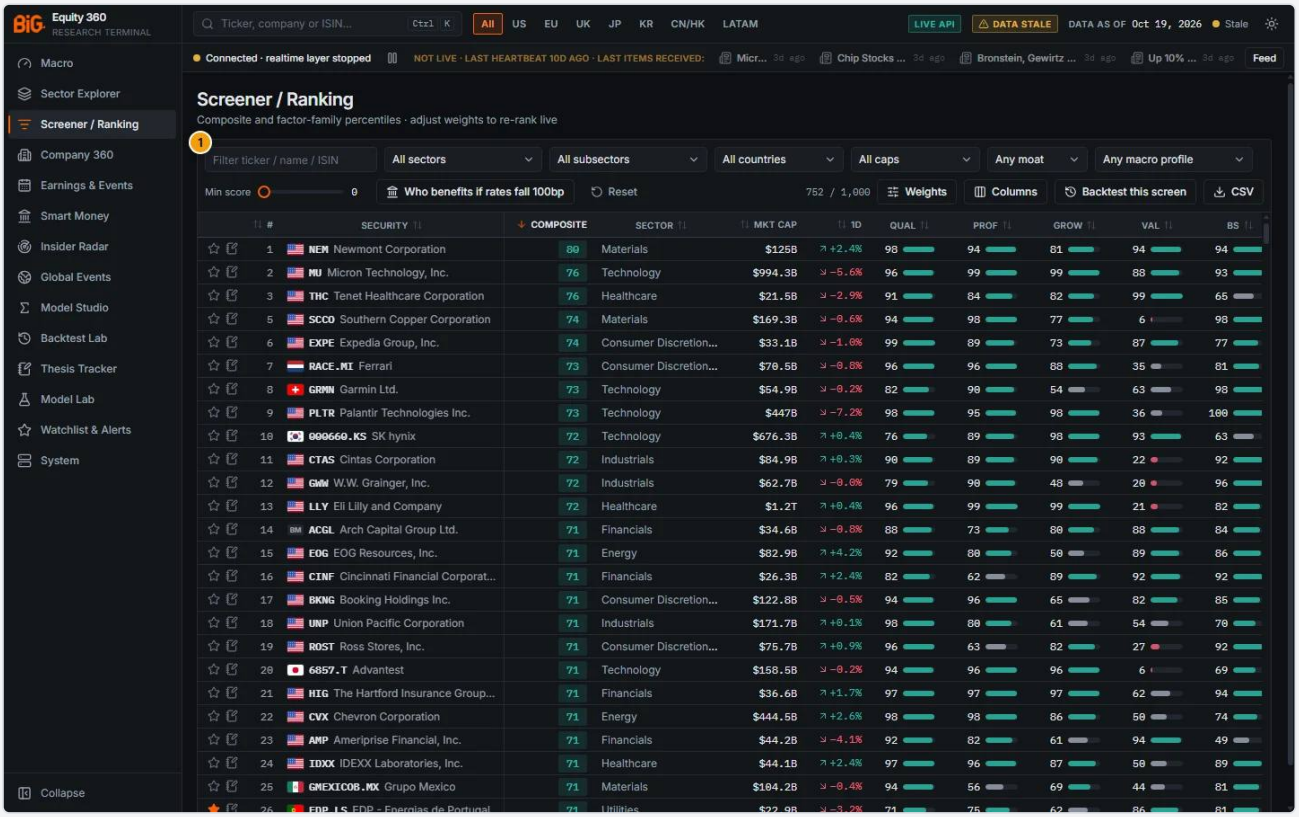


Figure 4.1: Screener (`/screener` , region All, no filters, default weights), data as of 19 Oct 2026.

#	Panel	What it shows
1	Screener / Ranking	The whole working area: filter bar, action buttons and the ranked table

The single panel has three regions:

- **Filter bar, first row:** the text filter ("Filter ticker / name / ISIN"), then All sectors, All subsectors, All countries, All caps, Any moat and Any macro profile.
- **Filter bar, second row:** the Min score slider (at 0), the "Who benefits if rates fall 100bp" preset, Reset, the counter "752 / 1,000" (rows shown / rows loaded), and the Weights, Columns, Backtest this screen and CSV buttons. The current build

also has an "Open as BQL" button between Backtest this screen and CSV; it was added after Figure 4.1 was captured (see Controls and Figure 4.6).

- **Table:** star (watchlist) and notebook (create thesis) icons, live rank, security with flag and ticker, Composite (sorted descending), Sector, Mkt cap, 1D, then the factor percentiles QUAL, PROF, GROW, VAL, BS; the remaining columns (MOM, INS, SENT, MAC, MOAT, 1M, 12M, P/E, EV/EBIT, FCF yld, ROIC, Rev g, E[xs] 3M, Moat) scroll to the right. Factor bars are teal at 67 and above, rose at 33 and below, grey in between; the Composite pill uses the same thresholds.

In Figure 4.1 Newmont leads with 80, followed by Micron and Tenet (76). Rank 4 is missing from the sequence; Limitations and pitfalls explains why.

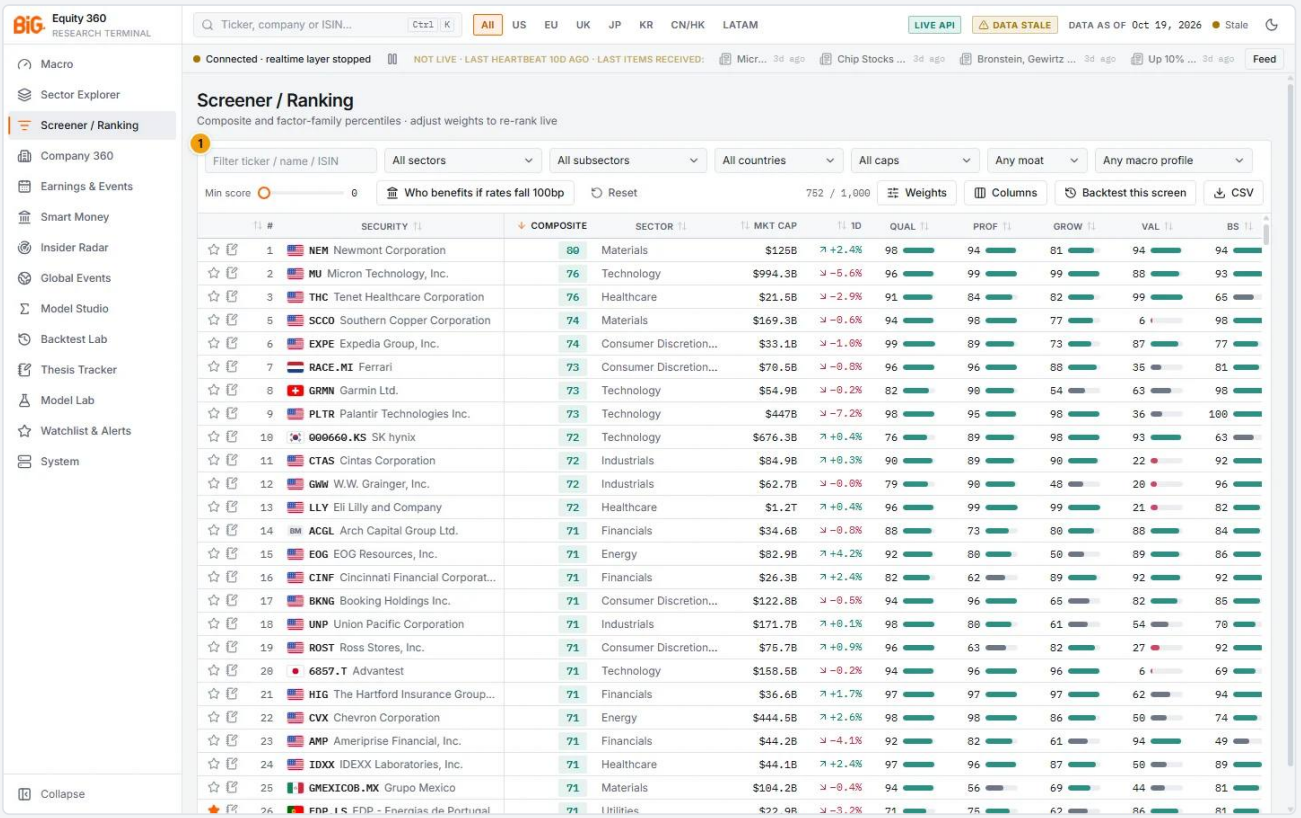


Figure 4.2: The same view in the light theme.

#	Panel	What it shows
1	Screener / Ranking	Same layout as Figure 4.1

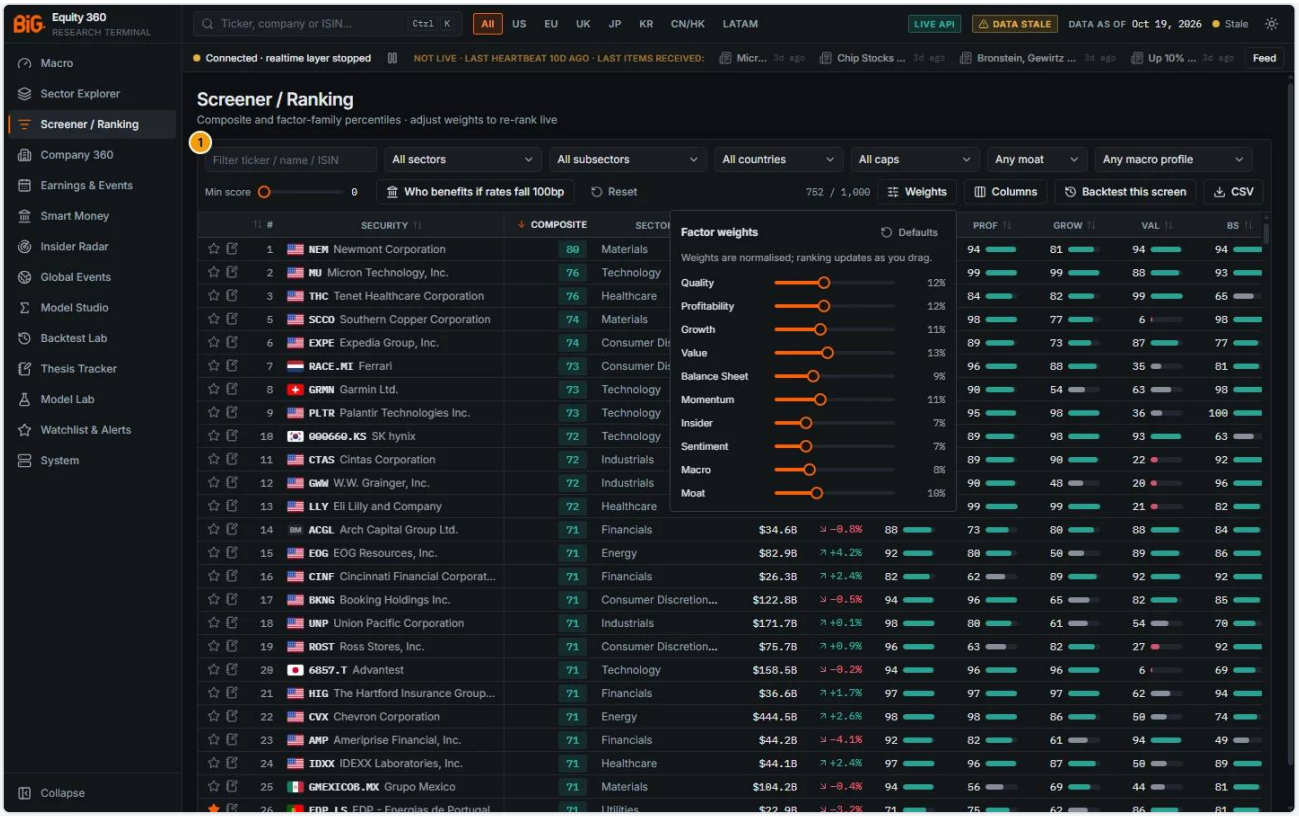


Figure 4.3: The Weights popover (click Weights). Ten sliders, Quality to Moat, at the defaults; the right-hand column shows each weight's share of the total (Quality 12%, Value 13%, Balance Sheet 9%, Insider and Sentiment 7% ...). Defaults is at the top right.

#	Panel	What it shows
1	Screener / Ranking	The working area with the "Factor weights" popover open over the table: "Weights are normalised; ranking updates as you drag."

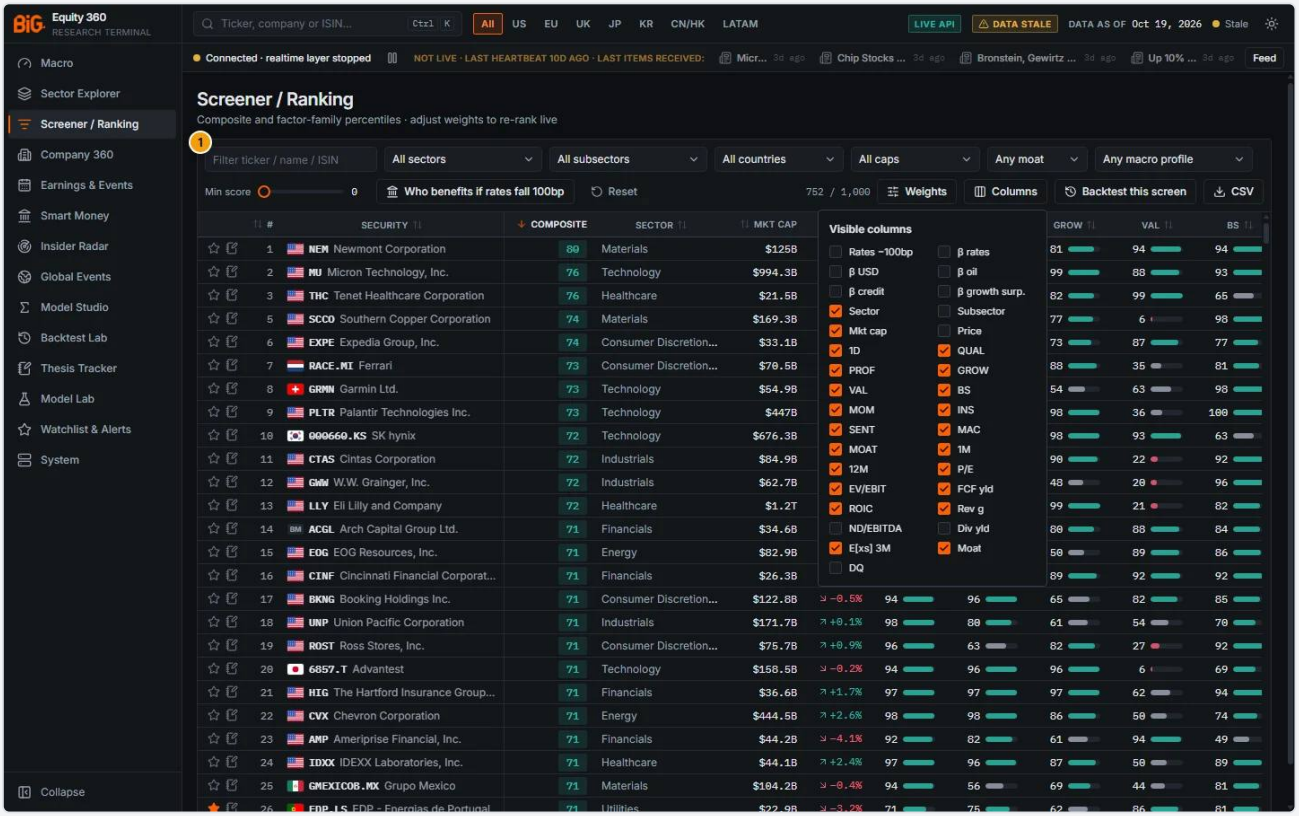


Figure 4.4: The Columns chooser (click Columns), in its default state. Unticked: Rates −100bp, β rates, β USD, β oil, β credit, β growth surp., Subsector, Price, ND/EBITDA, Div yld and DQ. Everything else is ticked.

#	Panel	What it shows
1	Screener / Ranking	The "Visible columns" list in two columns, one checkbox per optional column

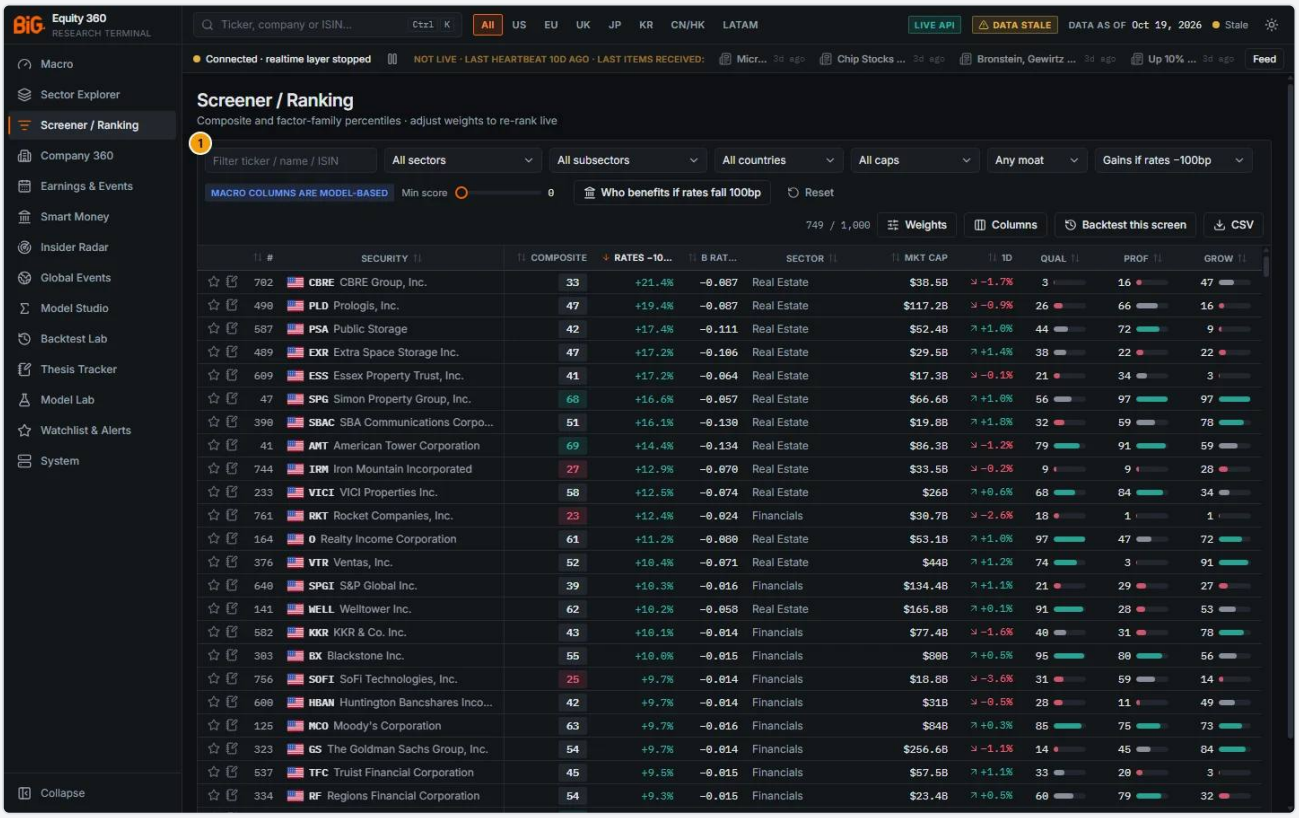


Figure 4.5: The "Who benefits if rates fall 100bp" preset. The macro filter reads "Gains if rates −100bp", the "Macro columns are model-based" badge appears next to Min score, and the table is sorted on the Rates −100bp column (header truncated to "RATES −10...") next to β rates.

#	Panel	What it shows
1	Screener / Ranking	Counter 749 / 1,000. The top of the list is real estate (CBRE +21.4%, Prologis +19.4%, Public Storage +17.4%, Extra Space +17.2% ...) followed by financials (Rocket, S&P Global, KKR, Blackstone ...). The # column keeps the live composite rank (CBRE is 702nd)

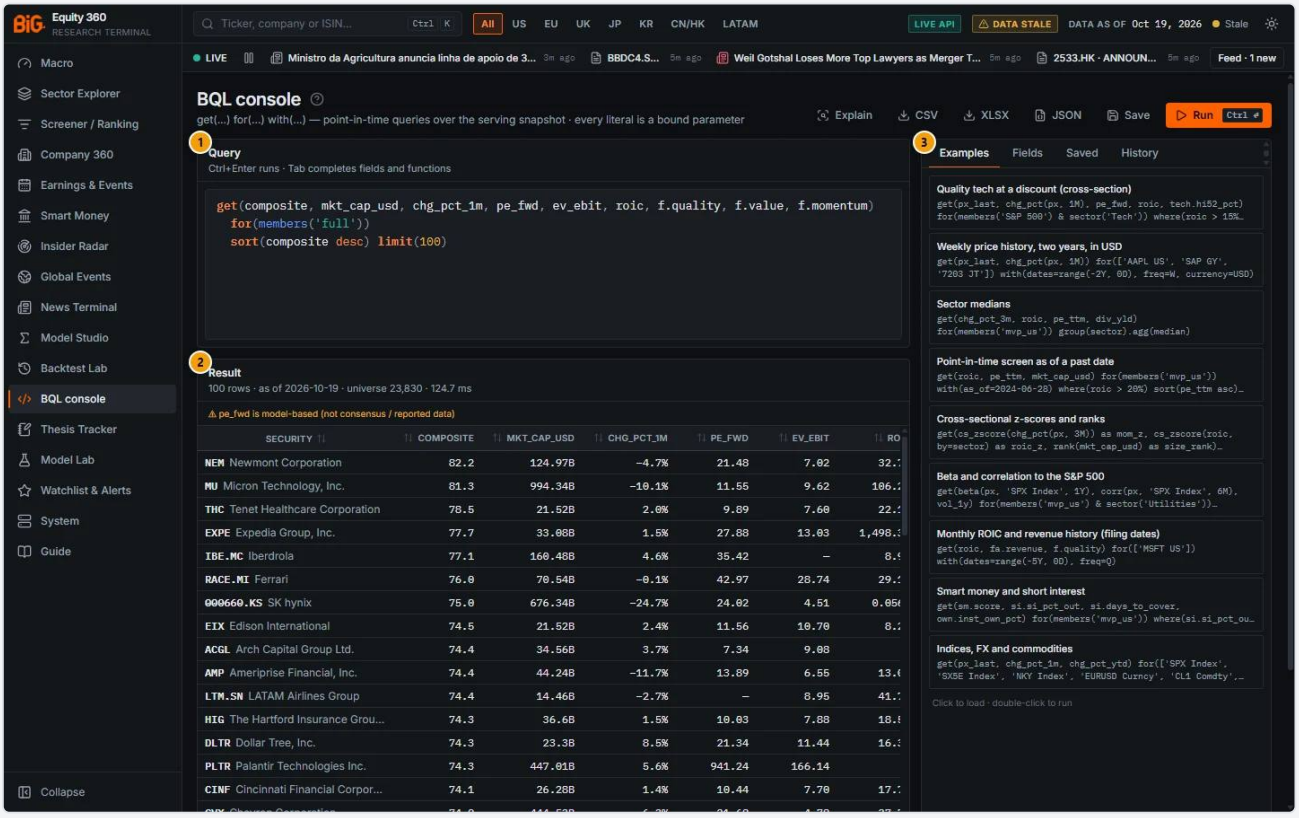


Figure 4.6: "Open as BQL" from the Screener with default filters (region All, no filters, sorted on Composite): the BQL console (`/bql`) with the generated query already run, data as of 19 Oct 2026.

#	Panel	What it shows
1	Query	The generated query: `get(composite, mkt_cap_usd, chg_pct_1m, pe_fwd, ev_ebit, roic, f.quality, f.value, f.momentum) for(members('full')) sort(composite desc) limit(100)` . With no filters set there is no `where(...)` and no `region(...)`
2	Result	"100 rows · as of 2026-10-19 · universe 23,830 · 124.7 ms", the warning "pe_fwd is model-based (not consensus / reported data)", and the rows: Newmont 82.2, Micron 81.3, Tenet 78.5, Expedia 77.7, Iberdrola 77.1 ...
3	Examples / Fields / Saved / History	The console's side panel with example queries (click to load, double-click to run)

The result ranks on the stored balanced composite, so the scores differ from the Screener's live composite with default weights (Micron 81.3 here, 76 in Figure 4.1) and the order changes: Iberdrola, fifth here, is not among the first rows of Figure 4.1, either because the default weights rank it lower or because a missing factor score hides the row (see Limitations and pitfalls). The console toolbar (Explain, CSV, XLSX, JSON, Save, Run) is described in BQL console.

Controls

FILTERS

All filters work in the browser on the 1,000 loaded rows and combine with AND. The region chips in the top bar (All, US, EU, UK, JP, KR, CN/HK, LATAM) decide which 1,000 rows are loaded.

Control	What it does
Filter ticker / name / ISIN	Keeps rows whose ticker or name contains the text, or whose ISIN starts with it
All sectors	One sector (the list holds only sectors present in the loaded rows). Changing it clears the subsector
All subsectors	One subsector; the list follows the chosen sector
All countries	Country of listing
All caps	USD market-cap bucket: Mega ≥ $200bn, Large $10-200bn, Mid $2-10bn, Small $0.3-2bn, Micro < $0.3bn
Any moat	Wide, Narrow or None (None also contains "insufficient data")
Any macro profile (model-based)	Gains if rates −100bp, Loses if rates −100bp, Rate beta < 0, Rate beta > 0, USD beta > 0, Oil beta > 0, Has macro betas. Names without a macro estimate are excluded by every option
Min score	Slider 0-90 in steps of 5: keeps rows whose live composite (with your current weights) is at least this value
Counter `752 / 1,000`	Rows passing the filters / rows loaded

FACTOR WEIGHTS

Control	What it does
Weights	Opens the factor-weight popover: "Weights are normalised; ranking updates as you drag."
Slider per factor (Quality ... Moat)	Integer weight 0-30. The number on the right is that factor's share of the total in %
Defaults	Restores the default weights: Quality 12, Profitability 12, Growth 11, Value 13, Balance Sheet 9, Momentum 11, Insider 7, Sentiment 7, Macro 8, Moat 10

Weights are saved in the browser (local storage) and used the next time the page loads. A factor-column header tooltip shows the factor name and its current weight ("Quality percentile (weight 12)").

COLUMNS

Control	What it does
Columns	Check-box list of every optional column. Hidden by default: Subsector, Price, ND/EBITDA, Div yld, DQ and the six macro columns. The star, #, Security and Composite columns are always shown and pinned on the left
Column header	Click to sort; click again to reverse. Composite descending is the default sort

The columns available:

Column	Content
#	Live rank under the current weights (over all 1,000 loaded rows, before filters)
Composite	Weighted composite of the ten factor percentiles with your weights
Rates −100bp	Model-based impact of a parallel 100bp fall in rates (2y and 10y), current betas
β rates, β USD, β oil, β credit, β growth surp.	Model-based sensitivities: return per +100bp in the 10y yield; elasticity to the broad USD; elasticity to WTI; return per +100bp in the credit spread; per +100 points of the growth surprise index
Sector, Subsector	Taxonomy (see Sector Explorer)
Mkt cap, Price, 1D	USD market cap, last price in trading currency, one-day change
QUAL, PROF, GROW, VAL, BS, MOM, INS, SENT, MAC, MOAT	Factor scores 0-100 with bar
1M, 12M	Total return over one and twelve months
P/E	P/E NTM from the base-case forecast (model-based, not consensus)
EV/EBIT, FCF yld, ROIC, Rev g, ND/EBITDA, Div yld	Latest ratios (Rev g = revenue growth LTM)
E[xs] 3M	Model expected excess return over 63 trading days (ML model)
Moat	Wide / Narrow / None badge
DQ	Data quality: High (≥ 0.8), Medium (0.5-0.8 or unknown), Low

ACTIONS

Control	What it does
Who benefits if rates fall 100bp	Resets the filters, sets the macro filter to "Gains if rates −100bp", shows the Rates −100bp and β rates columns and sorts by Rates −100bp, best first. Tooltip: "Model-based: shows the impact of a 100bp parallel fall in rates, keeps names that gain, sorted best first"
"macro columns are model-based" badge	Appears whenever a macro filter is active or a macro column is visible
Reset	Clears every filter and the sort (weights and columns are kept)
Backtest this screen	Sends the filters, sort and weights to the Backtest Lab, which re-evaluates the screen point-in-time at each monthly rebalance and opens a draft backtest. Fields without point-in-time history are reported (the text filter is ignored, P/E NTM becomes P/E TTM, the subsector is dropped). See Backtest Lab
Open as BQL	Opens the BQL console with the screen written as a `get / for / where / sort` query and runs it. A note appears when Min score is set: "min score uses the stored balanced composite (the Screener re-weights live)"
CSV	Downloads `big360-screener-YYYY-MM-DD.csv` with the filtered rows

"Open as BQL" builds, for Technology, Large caps, Wide moat and Min score 60:

```
get(composite, mkt_cap_usd, chg_pct_1m, pe_fwd, ev_ebit, roic, f.quality, f.value, f.momentum)
  for(members('full') & sector('Technology'))
  where(mkt_cap_usd >= 10e9 and mkt_cap_usd < 200e9 and moat == 'wide' and composite >= 60)
  sort(composite desc) limit(100)
```

A region chip other than All adds `region(...)` to `for(...)`. The text filter, the macro filter and your factor weights are not carried over; the query ranks on the stored balanced composite. The console is described in BQL and mnemonics.

The CSV holds: rank, id, ticker, name, isin, country, sector, subsector, market_cap_usd, composite (one decimal), the ten factor scores, return_1m, return_12m, pe_ntm, ev_ebit, fcf_yield, roic, revenue_growth, moat, data_quality. Returns and ratios are decimal fractions (0.05 = 5%). The header contains `moat` twice: the first is the Moat factor score, the second the moat class. Rows follow the live rank, not the column you sorted on, and the macro columns are not exported.

ROW ACTIONS

Control	What it does
Row click	Opens the company in Company 360
Star	Adds or removes the company from the watchlist (synced to the server). See Watchlists and alerts
Notebook icon	"Create thesis (thesis tracker)": opens a new thesis on the company, recording the origin "screener rank N". See Thesis Tracker

There is no saved-screen feature on this page. To keep a screen, use Open as BQL and save the query in the console as a screen (`members('screen:NAME')`).

Methodology

What happens when the page loads and when you move a slider:

1. **Load.** The browser sends `POST /api/v1/screener/query` with the region, the current weights and `limit 1000` . The server computes the composite with those weights for every equity in the `full` universe, sorts descending and returns the top 1,000 with their ten factor scores and the extra columns.
2. **Re-rank.** Every slider move recomputes, for each loaded row, `Composite = Σ` $w_i \cdot F_i$ `/ Σ` w_i over the ten factor scores F_i , and renumbers the live rank. No request is sent; the set of 1,000 rows stays the one loaded.
3. **Filter.** Text, sector, subsector, country, cap bucket, moat, min score and macro filter are applied to the re-ranked rows.

The factor scores, the server composite and their definitions are in Scores and factors.

The macro columns come from the weekly macro-exposure model (job `macro_exposures`): rolling regressions of weekly local-currency returns on weekly changes of official macro factors over the trailing 156 weeks (minimum 52), with Newey-West errors and empirical-Bayes shrinkage of each stock towards its sector median.

- **β rates** (and USD, oil, credit) are univariate betas.
- **β growth surp.** comes from a regression on the home economy's model-based surprise indices.
- **Rates −100bp** uses the joint model (2y, curve, breakevens, credit, USD, oil and the equity residual): impact $= \exp(-\beta_{2y}) - 1$ for a 100bp fall in the 2-year yield with the curve slope unchanged, that is a parallel fall in the 2y and 10y.

The method is described in Markets and cross-asset.

> **Model-based:** the macro columns and filters, P/E NTM and E[xs] 3M are model outputs. The app marks the macro columns with the "macro columns are model-based" badge and column tooltips beginning "Model-based".

Data sources and refresh

- **Factor scores and composite:** job `scoring` in the nightly DAG, after ratios / KPIs, valuation, insider signals, sentiment, the macro regime nowcast and the forecast. Inputs: official filings (yfinance fallback), Polygon / yfinance prices, Form 4 and other insider filings, news and GDELT sentiment, the weekly moat job.
- **Macro betas:** weekly DAG (`macro_exposures`).
- **E[xs] 3M:** nightly ML inference with the production model.
- **P/E NTM:** nightly forecast job (base case).
- **Cadence:** with the default weekly profile everything above runs in the Saturday 08:00 Lisbon refresh; the daily profile runs the nightly DAG Tue-Sat 01:30. The score date can trail the "Data as of" date by a session or two (Micron's scores on 19 Oct 2026 were dated 16 Oct).

Validation status

- **Composite and factor scores:** a descriptive, peer-relative ranking. They have **no validation gate** as a return forecast in the validation documents, and the app does not label them as validated.
- **Macro columns:** labelled model-based in the app ("macro columns are model-based" badge, column tooltips). Not formally validated as forecasts.
- **P/E NTM:** model-based base-case forecast, not consensus.
- **E[xs] 3M (ML model):** out-of-sample rank-IC about 0.035 (XGBoost): treat it as one input, not proven alpha. Backtests before 2026-10-10 use today's company list (survivorship bias) (`docs/USER_GUIDE.md` §3).
- **Momentum:** the 12-1 momentum measure that carries 35% of the Momentum factor did not pass when tested on its own in the Technicals v3 gate (IC 20d +0.011, t 1.6; FAIL) (`docs/TECHNICALS_V3_VALIDATION.md` , Verdict #1). The scoring factor itself was not separately tested.

Correct usage and worked example

Start from a question, set the filters, then adjust the weights. Read the factor bars before the composite: two names on 72 can have opposite profiles. Open the company (row click) to see each factor's coverage and the metric-level percentiles on the Overview tab before acting.

Worked example: why Micron is 81 in the Sector Explorer and 76 here. Micron's factor scores on 19 Oct 2026 (live API) are Quality 95.5, Profitability 98.5, Growth 98.5, Value 87.6, Balance sheet 92.6, Momentum 87.6, Insider 14.2, Sentiment 18.9, Macro 59.7, Moat 58.8.

- With the **balanced profile** (the stored composite used by the Sector Explorer, Company 360 and BQL): `0.18·95.5 + 0.12·98.5 + 0.12·98.5 + 0.15·87.6 + 0.08·92.6 + 0.12·87.6 + 0.05·14.2 + 0.05·18.9 + 0.05·59.7 + 0.08·58.8 = 81.3`, grade A, global rank 2.
- With the **Screener default weights** (12/12/11/13/9/11/7/7/8/10): `(12·95.5 + 12·98.5 + 11·98.5 + 13·87.6 + 9·92.6 + 11·87.6 + 7·14.2 + 7·18.9 + 8·59.7 + 10·58.8) / 100 = 76.5`, shown as 76 in Figure 4.1.

The Screener default weights give Insider, Sentiment, Macro and Moat more weight than the balanced profile, which costs Micron about five points. Press Defaults to get back to this view. To reproduce the stored composite in the Screener, set the sliders to 18 / 12 / 12 / 15 / 8 / 12 / 5 / 5 / 5 / 8.

Worked example: a defensive screen. Set All caps = Large, Any moat = Wide, raise Balance Sheet and Quality to 25, set Momentum and Sentiment to 0, and Min score to 60. The list now ranks large wide-moat companies mainly on quality and balance-sheet strength. Check the counter: if only a handful of rows remain, lower Min score rather than trusting a short list.

Limitations and pitfalls

- **Only the top 1,000 are loaded.** The server picks them with the weights in force when the page loaded. Re-weighting re-ranks those 1,000; a stock that would enter the top under the new weights appears only after a reload (the new weights are remembered). Narrow the region chip to load a different 1,000.
- **Rows with a missing factor score are hidden.** In the current build, a company with no score for one or more factors (for example no Growth score because it has fewer than two years of fundamentals) cannot be re-weighted in the browser and is left out of the table, although the server ranked it. This is why the counter reads 752 / 1,000 with no filter set, and why the rank sequence skips numbers (rank 4 in Figure 4.1).
- **Two composites.** The Screener composite uses your weights; the Sector Explorer, Company 360, BQL `composite`, `grade` and Open as BQL use the stored balanced profile. Expect different numbers (see the worked example).
- **Grades are not shown here.** The A-E grade belongs to the stored composite. See it on the API (`/companies/{id}/scores`) or in BQL (`grade`).
- **Sector-relative scores.** Factor percentiles are computed within sector × peer region, so a top-ranked bank and a top-ranked software company are each top of their own sector, not equally attractive in absolute terms.
- **Moat "None" includes "insufficient data".** Young companies and those with fewer than seven years of history cannot be Wide.
- **Coverage outside US / JP / EU is thin.** Many securities in the `full` universe are price-only (audit F25: about 4,900 of 24,600 issuers have fundamentals); they cannot reach the top 1,000 without fundamental factors.
- **Model-based columns.** Macro betas, P/E NTM and E[xs] 3M are model outputs; use them as context, not targets. In Figure 4.5 the "Gains if rates −100bp" filter keeps 749 of the 752 displayed names: in this model almost every stock gains from a parallel fall in rates, so the information is in the size and ordering of the impact (real estate and financials first), not in its sign.
- **Open as BQL** sorts on the stored composite and drops the text and macro filters. Sorting the table on BS (Balance Sheet) before Open as BQL produces a query sorted on the composite.
- **Backtest this screen** is a point-in-time re-run of the rules, not of today's list; it can hold different names than you see.

Related mnemonics and BQL fields

- **Mnemonics:** `EQS` opens the Screener; `BQL` opens the console. See BQL and mnemonics.
- **BQL fields:** `composite` (balanced, 0-100), `grade`, `rank_global`, `c.<profile>` / `grade.<profile>` / `rank_sector.<profile>` for each profile (balanced, quality_value, growth, momentum, defensive, ml_blend), `f.<factor>` with `.rank_sector` and `.chg_1m`, `moat`, `moat_score`, `dq_score`, `insider_signal`, `pe_fwd`, `ml.exp_excess_3m`, `ml.rank_pct_3m`, `macro.beta_rates`, `macro.beta_usd`, `macro.beta_oil`, `macro.beta_credit`, `macro.beta_growth`, `macro.rates_down_100bp`.
- **API:** `GET /api/v1/screener/fields`, `POST /api/v1/screener/query`, `POST /api/v1/screener/export`, `POST /api/v1/backtest/from-screen`.
- **Related sections:** Sector Explorer, Company 360 Overview, Backtest Lab, Watchlists and alerts, Thesis Tracker.

Scores and factors

Applies to: Screener, Sector Explorer (Leaders, Laggards, Median score), Company 360 header and Overview (factor profile, score breakdown), BQL `composite`, `grade`, `f.*` · **Refresh:** nightly `scoring` job; monthly history for charts and BQL time series

Purpose

Every company in the scored universe gets ten factor scores from 0 to 100 and a composite score from 0 to 100 with a grade from A to E. The scores answer one question: *how does this company compare with its closest peers, and with its own past, on each dimension?* They are relative and descriptive. This section is the reference for how they are built (`contracts/scoring.yaml`, `backend/app/analytics/scoring.py`).

Methodology

OVERVIEW

```
metric value (point-in-time)
  → metric score 0-100  = blend of peer percentile, own-history score and (valuation only) size-adjusted score
  → factor score 0-100  = weighted mean of the factor's metric scores, re-percentiled within sector × peer region
  → composite 0-100     = weighted mean of the factor scores (profile or your weights)
  → grade A-E           = quintile of the composite over the scored universe
```

A score of 100 is always best: metrics where lower is better (P/E, net debt / EBITDA, cost of risk ...) are flipped before ranking, and band metrics (loans / deposits, capex intensity ...) are scored on `−|x − target|`.

INPUTS AND POINT-IN-TIME RULE

For an as-of date D, every input must have a knowledge date ≤ D.

- **Fundamentals:** the latest TTM row (FY, or half-year for semi-annual reporters, if no TTM) with period end within 400 days.
- **Valuation:** the last valuation row within 10 days of D.
- **Market metrics:** computed from adjusted closes and need at least 126 sessions of history.

The scored set of metrics for a company depends on its KPI set: the factor members (with the KPI-set overrides below), the KPI set's quality weights, the accounting block and the KPI set's headline metrics, minus the generic metrics the KPI set excludes (see KPI sets).

1. PEER PERCENTILE

Each metric is ranked against peers by average rank: `percentile = (rank − 0.5) / n × 100`. The peer group is the first of these levels with at least **8** companies that have a valid value (the company included):

Level	Peer group
1	subsector × peer region × size
2	subsector × peer region
3	subsector, global
4	sector × peer region
5	sector, global (used with as few as 3 if nothing else qualifies)

- **Peer regions:** NA (US, Canada), EUROPE (UK, EU), JP, ASIA_EX_JP (Korea, other Asia-Pacific), GREATER_CHINA (China, Hong Kong), LATAM, GLOBAL (other).
- **Size buckets:** large ≥ $10bn, mid $2-10bn, small < $2bn. These are the scoring buckets, not the Screener's five cap buckets.
- **Momentum, sentiment and macro metrics:** compared across the whole peer region (falling back to global), not within subsectors.
- **Negative or missing multiples:** a P/E, EV/EBITDA, EV/sales, P/B, P/TBV or P/FFO that is ≤ 0 (negative earnings) gets the worst score, 5, instead of being dropped. The same applies to net debt / EBITDA when EBITDA is not positive.

2. OWN-HISTORY SCORE

For fundamental and valuation metrics, the current value is compared with the company's own history:

```
z = (x − μ_hist) / max(σ_hist, 0.05·|μ_hist|),   clipped to ±3
self score = Φ(z) × 100
```

The history covers up to 10 years of quarter-end TTM rows (quarter-end valuation samples for multiples). It needs at least 8 observations spanning at least 5 years; otherwise there is no self score.

3. SIZE-ADJUSTED VALUATION SCORE

For valuation metrics only (earnings yield, FCF yield, EV/EBITDA, EV/sales, P/E, P/B, P/TBV, P/FFO, dividend yield, shareholder yield), a cross-sectional regression within the peer group (at least 20 observations, else wider) asks whether the stock is cheap *for its size, quality and growth*:

```
log(multiple) = a + b·log(market cap) + c·Quality score + d·revenue CAGR 3y + residual
```

Yields are regressed without the log, and values are winsorised at 1% / 99%. The residual is direction-adjusted (cheaper than predicted = better) and turned into a percentile.

4. METRIC SCORE

Metric type	Blend
Fundamental (ratios, sector KPIs)	70% peer + 30% own history
Valuation	40% peer + 25% own history + 35% size-adjusted
Market, insider, sentiment, macro	100% peer (scores already on 0-100 pass through)

Missing components are dropped and the remaining weights renormalised (no history means 100% peer).

5. FACTOR SCORES

Factor	Members (default weights)	Overrides by KPI set	No data
Quality	80% the KPI set's quality weights (e.g. software: Rule of 40 25%, FCF margin after SBC 20%, gross margin 15% ...; banks: RoTE 25%, efficiency 20%, NIM 15% ...) + 20% accounting block (Piotroski F 50%, Sloan accruals 25%, Beneish M 25%)	via the KPI set	null
Profitability	ROIC 25, gross profitability 20, operating margin 20, FCF margin 20, ROE 15	banks: RoTE 40, ROE 30, PPOP / assets 30; insurers, REITs, utilities, consumer finance have their own	null
Growth	revenue growth 20, revenue CAGR 3y 20, EPS growth 15, EPS CAGR 3y 15, FCF / share CAGR 3y 15, gross-profit growth 10, revenue acceleration 5	banks: BVPS CAGR 30, EPS CAGR 30, revenue 20, deposits 20; insurers: premiums and BVPS; REITs: FFO / AFFO per share	null
Value	FCF yield 25, earnings yield 20, EV/EBITDA 20, EV/sales 10, P/B 10, shareholder yield 10, reverse-DCF implied vs historical growth 5	banks: earnings yield 35, P/TBV 35, dividend yield 30; REITs: P/FFO 50 ...; holdings: P/B 60	null
Balance sheet	net debt / EBITDA 25, interest coverage 20, Altman Z'' 20, cash / assets 15, current ratio 10, liabilities / assets 10	banks: equity / assets 40, CET1 30, loans / deposits 30; insurers, REITs, consumer finance have their own	null
Momentum	12-1 month return 35, 6-1 month 25, proximity to 52-week high 15, price vs 200-day average 15, 1-year volatility (lower better) 10	none	null
Insider	rules-based insider signal 0-100 (passes through unchanged)	none	50
Sentiment	news sentiment EWM 30d 40, "Lazy Prices" filing similarity 40, GDELT tone EWM 7d 20	none	50
Macro sensitivity	macro regime fit 70 (Σ current regime probability × the stock's return beta in that regime), absolute rate beta (lower better) 30	none	50
Moat	moat score 0-100 (passes through unchanged)	none	null

How a factor score is built:

- **Raw score:** the weighted mean of the available metric scores, with weights renormalised.
- **Coverage:** the weight share of members that have a real (non-imputed) value. Below **30%** the factor is null, except Insider, Sentiment and Macro, which are set to **50 (neutral) with coverage 0**.
- **Re-percentiling:** the raw score is re-ranked within **sector × peer region** (falling back to peer region, then global, when fewer than 8 companies), so factor scores are percentiles. Insider and Moat are already on a 0-100 scale and are not re-ranked.
- **Valuation confidence:** the reverse-DCF Value member is weighted by the forecast's valuation confidence and is dropped when confidence is below 40.
- **Smart money:** an optional smart-money member of the Insider factor exists but is switched off.

6. COMPOSITE, PROFILES AND GRADES

The stored composite is the weighted mean of the usable factor scores (coverage ≥ 30%, or a neutral 50), with weights renormalised over the factors present. A company needs at least **5** non-null factors, including at least **2** of Quality, Profitability, Value and Growth; otherwise its composite is null. The equivalent form shown on Company 360, "Composite = 50 + Σ weight × (score − 50)", gives the same number.

Profile	Quality	Profit.	Growth	Value	Bal. sheet	Momentum	Insider	Sentiment	Macro	Moat
balanced (default)	18	12	12	15	8	12	5	5	5	8
quality_value	25	15	5	25	10	5	-	-	-	15
growth	15	15	30	5	-	20	-	5	-	10
momentum	10	-	15	-	-	40	10	15	10	-
defensive	25	10	-	15	25	-	-	-	10	15
Screener defaults	12	12	11	13	9	11	7	7	8	10

- **ml_blend** profile: `0.5 × balanced + 0.5 × the production 3-month ML model's rank percentile`. It is available in BQL (`c.ml_blend`) and the API, not on the Screener.
- **Custom weights** (Screener, `factor_weights` in the API): the weights are normalised to sum to 1, and the composite is the weighted mean over the non-null factors. It is null when the factors present carry less than half of the requested weight.
- **Grade:** quintile of the composite over the scored universe. A = top 20%, B = next 20%, down to E = bottom 20%. With custom weights the API grades over the requested universe before filters.
- **Ranks:** `rank_global` , and `rank_sector` within sector × peer region.

WHAT 50 MEANS

50 is the middle of the peer distribution. It is also the value given to Insider, Sentiment and Macro when there is **no data** (no insider filings in the region, no news coverage, no macro betas), with coverage 0. A 50 on those three factors is therefore "neutral or unknown", not "average evidence"; check the coverage on Company 360. Quality, Profitability, Growth, Value, Balance sheet, Momentum and Moat are never filled with 50: when they lack data they are null and drop out of the composite.

Data sources and refresh

- **Job:** `scoring` (nightly DAG). It writes `peer_scores` (every metric with its peer group key, level, count, median, peer percentile, self z, size-adjusted score and metric score), `factor_scores` (score, coverage, rank in sector, one-month change) and `composite_scores` (per profile: score, grade, ranks, coverage, data-quality score). In history mode it writes month-end scores for the last 10 years, which feed the ML models and the BQL history.
- **Upstream jobs:** `ratios_kpis` , `valuation_daily` , `insider_signals` , `sentiment_daily` , `macro_regime_nowcast` and `forecast` (nightly); `moat` and `macro_betas` (weekly); `composite_ml_blend` after ML inference.
- **Data-quality score:** the mean data-quality score of the fundamental inputs; the Screener's DQ column buckets it (High ≥ 0.8, Low < 0.5).
- **Sources and coverage:** see Data and coverage.

Validation status

- **Composite and factors:** not formally validated. There is no gate for the composite or the factor scores as predictors of returns in the validation documents. They are a transparent, rules-based peer ranking; the app shows them without a "model-based" label because every step is a deterministic rule.
- **Inputs that are models:**

    - the reverse-DCF member of Value (forecast, confidence-weighted);
    - Macro regime fit and rate beta (model-based regime probabilities and betas);
    - Sentiment (FinBERT / XLM-R news sentiment, GDELT tone).
- **Insider factor:** uses the rules-based insider signal only. The insider XGBoost model showed no out-of-sample edge and is not used (`docs/USER_GUIDE.md` §6).
- **ml_blend:** the ML model behind it has an out-of-sample rank-IC of about 0.035 (XGBoost): one input, not proven alpha.
- **Moat thresholds:** calibrated on live data (2026-10-11) so that Wide stays a minority: Wide needs a score ≥ 75, ROIC > WACC in ≥ 90% of years and ≥ 7 years of history; Narrow needs a score ≥ 55 and ≥ 60% persistence. This is a classification rule, not a validated forecast.
- **12-1 momentum:** see the Screener validation status: the stand-alone 12-1 momentum test failed the Technicals v3 gate.

Correct usage and worked example

Use the scores to find where to look, then read the evidence: open the company, check coverage, the peer group used for each metric and the self-history score. A high composite built on a 50 Insider / Sentiment / Macro with zero coverage is a fundamentals-and-price ranking only.

Worked example 1: Micron's gross margin and EV/EBITDA (live API, scores as of 16 Oct 2026). Micron is in the semis KPI set, peer group `subsector_region_size: semiconductors | NA | large`.

- **Gross margin** 80.7%: peer percentile 97.4 (top of 19 peers, peer median 58.3%), own-history z = 2.41 so self score 99.2. Metric score `0.7 × 97.4 + 0.3 × 99.2 = 97.9`.
- **EV/EBITDA** 8.8x against a peer median of 29.7x: peer percentile 96.9 (16 peers), self score 58.0, size-adjusted score 92.5. Metric score `0.40 × 96.9 + 0.25 × 58.0 + 0.35 × 92.5 = 85.6`.
- **R&D intensity (3y)** 9.4% against a peer median of 20.5%: metric score 9.9, which pulls Quality down. Quality still ends at 95.5 after re-percentiling within Technology × NA.

Worked example 2: peer widening for JPMorgan (live API). JPM's peer group for most bank metrics is `subsector_region_size` (global banks × NA × large, 8-10 peers). For **NIM** fewer than 8 of those peers report the inputs, so the level widens to `subsector_global` (14 peers). JPM's NIM of 5.5% is at the 96.4th peer percentile, self score 92.2, metric score `0.7 × 96.4 + 0.3 × 92.2 = 95.1`. For **P/TBV** 2.81x the three components are 25.0 (peer), 1.7 (own history: the most expensive in its 10-year range) and 32.1 (size-adjusted), giving `0.40 × 25.0 + 0.25 × 1.7 + 0.35 × 32.1 = 21.7`. JPM's composite is 65.1 (grade A, global rank 104), with Profitability 98.1 and Growth 90.1 but Value 14.9 and Balance sheet 13.6.

Limitations and pitfalls

- **Relative, not absolute.** Factor scores are percentiles within sector × peer region; a 90 Value in a richly valued sector can still be an expensive stock. The composite does not rank sectors against each other.
- **Small peer groups.** With 8 peers, one place in the ranking moves the percentile by about 12 points. Look at the peer count on Company 360.
- **Missing data is not penalised for most factors.** Null fundamental factors drop out and the remaining weights are renormalised; a company with only five factors can outrank one with ten.
- **Neutral 50.** For Insider, Sentiment and Macro, no data scores 50. In regions without insider filings (or before the news history starts, about August 2026) these factors carry no information.
- **History.** Factor history is monthly; growth and own-history components need at least two fiscal years and five years respectively. Survivorship: universe membership history starts on 2026-10-10.
- **Accounting outliers.** A ratio computed on a tiny denominator (ROIC above 100%, for example) still gets a top percentile; the own-history z is capped at ±3 but the peer rank is not.
- **Do not read the composite as an expected return.** It has no validation gate as a forecast (see Validation status).

Related mnemonics and BQL fields

- **BQL:** `composite`, `grade`, `rank_global`, `c.<profile>`, `grade.<profile>`, `rank_sector.<profile>`, `f.quality` … `f.moat` (aliases `quality_score` …), `f.<factor>.rank_sector`, `f.<factor>.chg_1m`, `moat_score`, `insider_signal`, `dq_score`, `kpi.<kpi_id>`. Scores have monthly point-in-time history; the `grade` snapshot does not.
- **API:** `GET /api/v1/companies/{security_id}/scores` (composite, factors with coverage and rank in sector, every metric with its peer group, peer count, peer percentile, self z, size-adjusted and metric score, monthly history, accounting flags).
- **Mnemonics:** `DES` (Company 360 overview with the factor profile and score breakdown), `EQS`, `PEERS`.
- **Related sections:** Company 360 Overview, Moat, Sector Explorer, Methodology appendix.

5A. Company 360: fundamentals

Company 360 is the single-company page of BiG Equity 360. This chapter covers how to open it, the header that sits above every tab, and the eight fundamental tabs: Overview, Financials, Forecast, Peers (with the peer-group joint anomaly panel), Moat, Insiders, Ownership and News & Sentiment. The Technicals tab is covered in 5B. Macro, Earnings, Event risk, Network & Geo and Model are covered in 5C.

All worked examples use the snapshot with data as of 19 Oct 2026 (built 22 Oct 2026). Valuation outputs carry the forecast job date, 16 Oct 2026.

Company 360 at a glance

Route: `/company/:id` · **Menu:** Company 360 · **Mnemonics:** `DES` (and the tab mnemonics listed per section) · **Refresh:** with each data refresh (weekly profile by default, daily optional)

Purpose

Company 360 brings everything the app knows about one listed company onto one page: scores, statements, model-based valuation, peers, competitive position, insider and institutional activity, news and, on the other tabs, technicals, macro exposure, earnings, events and the ML ranking. Analysts use it to prepare a view on a name. Portfolio managers use it to check a name surfaced by the Screener, a watchlist alert or a thesis.

Screenshot

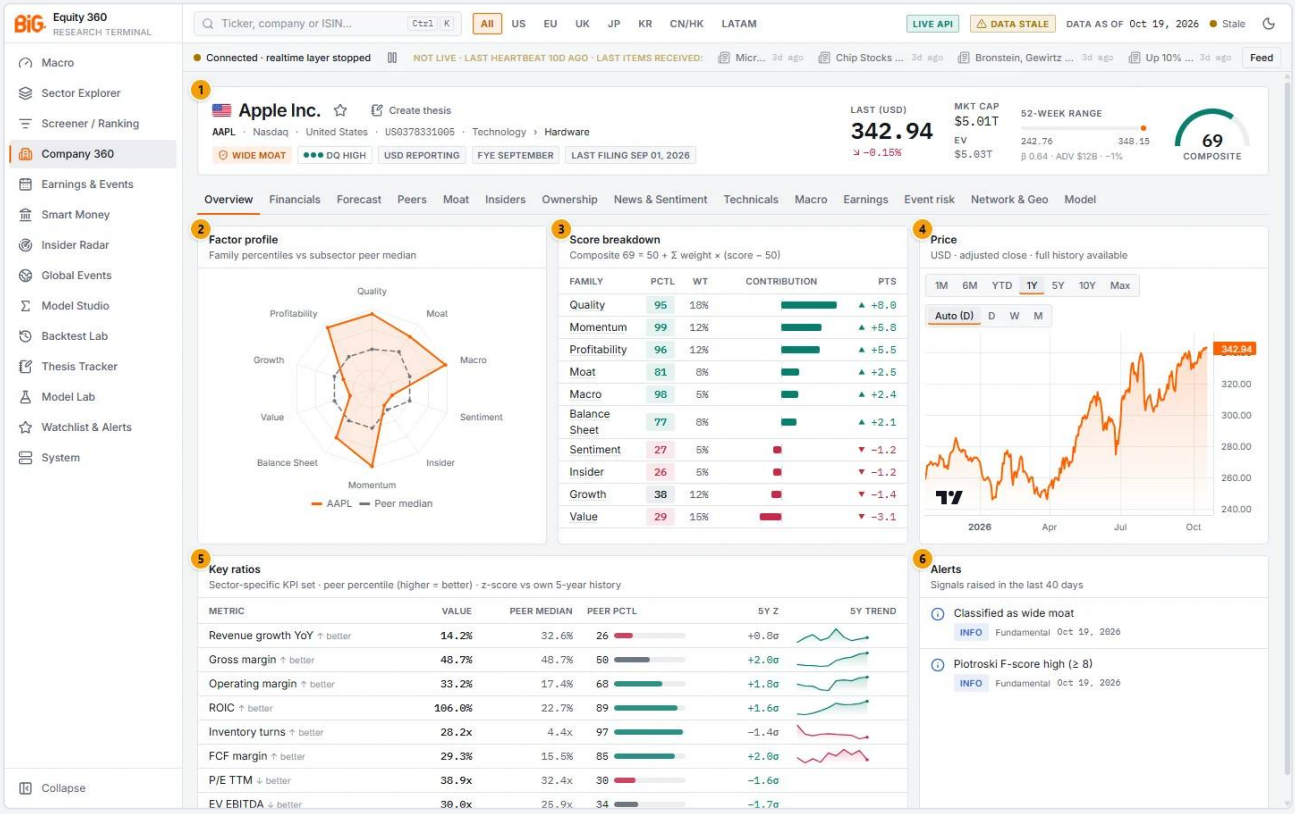


Figure 5A.1: Company 360 for Apple (`/company/AAPL`), light theme, data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	Name, identifiers, badges, price, size, 52-week range and composite gauge (see Company header)
2	Factor profile	Radar of the ten factor scores against the peer median

#	Panel	What it shows
3	Score breakdown	How each factor moves the composite away from 50
4	Price	Adjusted close with range and bar-interval controls
5	Key ratios	The sector-specific KPI set with peer percentile and self-history z-score
6	Alerts	Current flags raised for the company

The page follows the app theme (sun / moon toggle in the top bar). The dark theme is used in the rest of this chapter.

Controls

Control	What it does
Opening the page	Search with **Ctrl+K** (ticker, name or ISIN), type a mnemonic such as `AAPL US Equity DES`, or click any company row in the Screener, Sector Explorer, peer tables, watchlists or alert feeds
`:id` in the URL	Accepts a ticker (`AAPL`, `ASML.AS`, `7203.T`, `PETR4.SA`) or an internal security id. `/company` without an id opens Microsoft (`/company/MSFT`). Index, FX and futures ids are redirected to the instrument page (`/instruments/:id`)
Tab strip	14 tabs: Overview, Financials, Forecast, Peers, Moat, Insiders, Ownership, News & Sentiment, Technicals, Macro, Earnings, Event risk, Network & Geo, Model. On narrow screens the strip wraps onto a second row
`?tab=` deep link	Each tab has a URL id: `overview`, `financials`, `forecast`, `peers`, `moat`, `insiders`, `ownership`, `sentiment`, `technicals`, `macro`, `earnings`, `events`, `network`, `model`. Clicking a tab updates the URL (without adding a browser history entry), so the address can be shared or bookmarked. An unknown or missing value opens Overview
Retry	If the overview request fails, an error panel with a retry button replaces the page
`?` icon	Next to Create thesis in the header: opens this guide at the section of the current tab (tooltip "Guide: how this page works"). Typing `HELP` in the command bar opens the in-app guide (`/help`)

Methodology

The page is a front end over the company endpoints of the API (`/api/v1/companies/{security_id}/overview`, `/financials`, `/forecast`, `/peers`, `/moat`, `/insiders`, `/ownership`, `/sentiment` and others). The ticker in the URL is resolved to the internal security id first. Each tab loads its own data only when it is opened, so a slow tab never blocks the header or the Overview.

ADR detection. A listing is treated as an ADR when its name contains "ADR" or "ADS", or when it trades on a US venue (NYSE, Nasdaq, NYSE Arca, NYSE American, Cboe BZX, OTC Markets), the issuer is not American and no composite score exists. The app then looks up the home listing of the same issuer and shows the ADR banner described under Company header.

Data sources and refresh

Every number on the page comes from the serving snapshot built at the end of the refresh pipeline. With the default **weekly** schedule profile the nightly, weekly and (when due) monthly job chains run on Saturday morning (Lisbon time). With the **daily** profile the nightly chain runs Tuesday to Saturday at 01:30 and the weekly chain on Saturday. See Data and coverage for sources and point-in-time rules. The data date is shown in the top bar ("Data as of").

Validation status

Not applicable to the page as a whole. Each tab section below states the validation status of its models.

Correct usage and worked example

Open Apple from the command bar: type `AAPL` and press Enter, or `AAPL US Equity FA` to land directly on the Financials tab (`/company/AAPL?tab=financials`). To send a colleague the valuation, copy the address while on the Forecast tab: `/company/AAPL?tab=forecast`.

Limitations and pitfalls

- Coverage differs by market. US names have the deepest fundamentals, insider and ownership data. Several tabs show an empty or "pending" state for names outside the main coverage, which means "no data", not "nothing happened".
- Tab data are as of the snapshot, not live. The realtime layer (new-filing badge, live news) is separate and may be stopped (top-bar status).

Related mnemonics and BQL fields

- `DES` opens Overview. `FA` and `DVD` open Financials, `EE` Forecast, `RV` and `PEERS` Peers, `HDS`, `SI` and `SMRT` Ownership, `CN` the News Terminal (or News & Sentiment when the News Terminal is not installed), `GP` and `TECH` Technicals, `ERN` Earnings, `EVTS` Event risk, `NET` Network & Geo, `THES` a new thesis. There is no mnemonic for Moat or Insiders. Full list: BQL console and mnemonics.
- In the BQL console, most header and tab values are queryable, for example `get(composite, grade, moat, fair_value, dcf_upside) for(['AAPL'])`.

Company header

Route: `/company/:id` (every tab) · **Menu:** Company 360 · **Mnemonics:** `DES` · **Refresh:** with each data refresh; new-filing badge polled every 2 minutes

Purpose

The header is the summary card that stays on top of all 14 tabs. It identifies the security, gives its size, price position and composite score at a glance, and holds the two actions you take most often from a company page: add it to the watchlist and open a thesis.

Screenshot

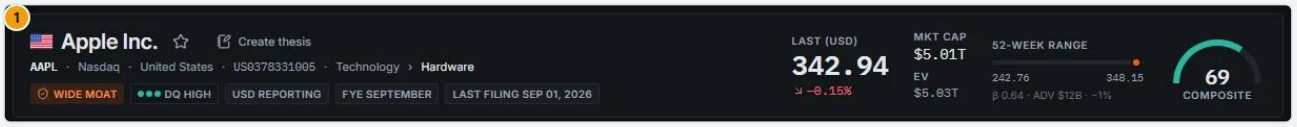


Figure 5A.2: Company header for Apple, data as of 19 Oct 2026.

The header has a left block (identity and badges) and a right block (market data and composite).

Element	What it shows
Flag, name	Country flag and issuer name
Watchlist star	Adds or removes the company from your watchlist (filled = on the list)
Create thesis	Opens the Thesis Tracker with this company pre-filled as the subject
`?` icon	Opens the user guide at the section of the tab you are on
Ticker · exchange · country · ISIN · sector › subsector	Identity line. The sector name links to the Sector Explorer for that sector
ADR banner	Only for ADRs (see Figure 5A.3)
Moat badge	Wide moat, Narrow moat or No moat (see Moat)
DQ badge	Data-quality bucket with three dots: DQ high, DQ med, DQ low; hover for the definition and any notes
`XXX reporting`	Reporting currency of the financial statements
`FYE <month>`	Fiscal year-end month
`Last filing <date>`	Date of the most recent filing ingested for the issuer

Element	What it shows
New filing badge	Appears only when the realtime layer has seen new filings for the company; hover for the five latest, click to open the latest
Last (currency)	Last close in the trading currency, with the one-day change
Mkt cap, EV	Market capitalisation and enterprise value, in USD
52-week range	Low and high with a marker at the current price; underneath: β (1-year beta), `ADV` (average daily traded value, USD, 20 days) and the distance from the 52-week high
Composite gauge	Composite score 0-100 (balanced profile); green from 67, red at or below 33

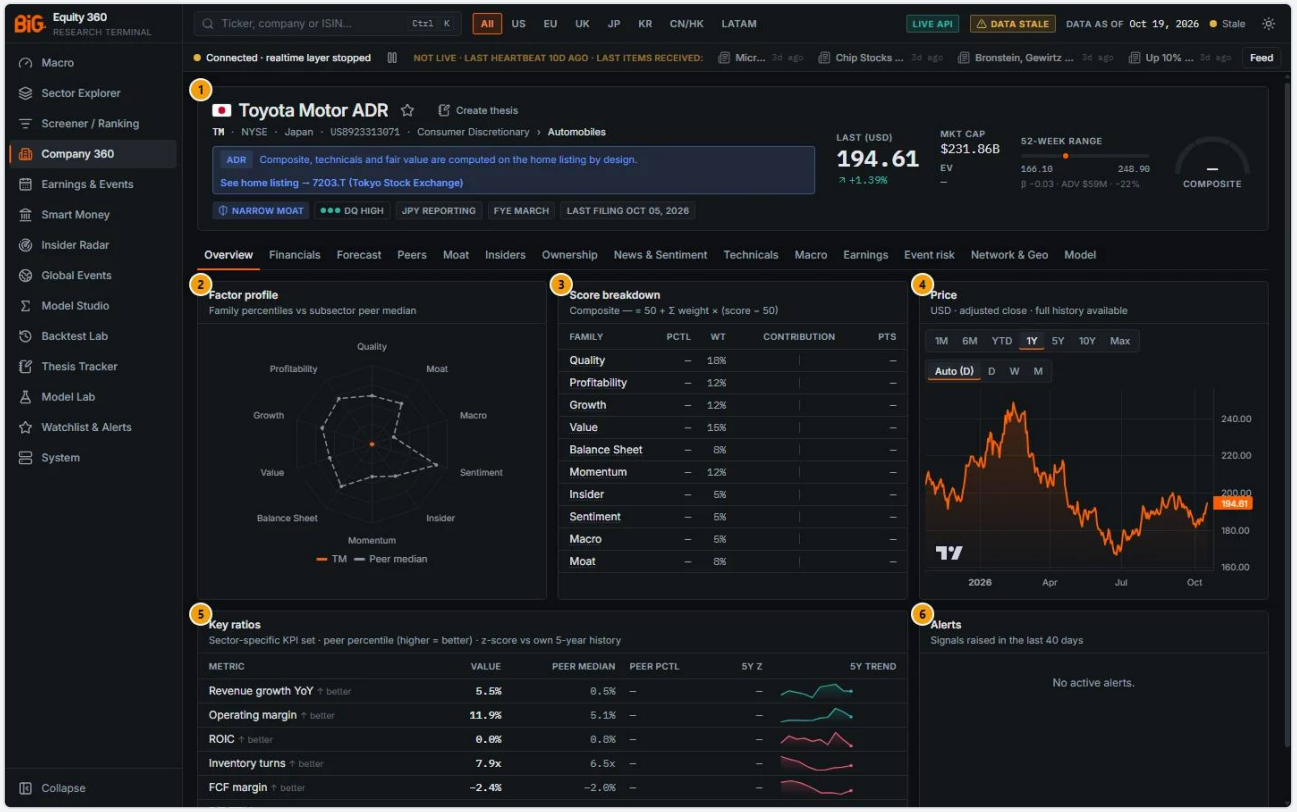


Figure 5A.3: Toyota Motor ADR (`/company/TM`), data as of 19 Oct 2026. The banner links to the home listing 7203.T.

#	Panel	What it shows
1	Company header	ADR banner: "Composite, technicals and fair value are computed on the home listing by design", with the link "See home listing → 7203.T (Tokyo Stock Exchange)"
2	Factor profile	Empty radar: ADRs have no factor scores
3	Score breakdown	All families show "—" and the composite is "—"
4	Price	The ADR's own USD price history
5	Key ratios	Ratios of the issuer (shared with the home listing); peer percentiles and z-scores are not computed for the ADR
6	Alerts	"No active alerts"

Controls

Control	What it does
Watchlist star	Toggles membership of your watchlist. The list is stored in this browser (it is not shared with other users) and drives the watchlist filters and alert rules (see Watchlists and alerts)
Create thesis	Opens `/theses/new` with subject type "security", the company id and origin "company" pre-filled (see Thesis Tracker)
`?` icon (next to Create thesis)	Opens the in-app guide (`/help`) at the section for the current tab, for example this chapter's Forecast section from `?tab=forecast` . Hover: "Guide: how this page works"

Control	What it does
Sector link	Opens the Sector Explorer on the company's sector
DQ badge hover	Shows "Data quality: high — full statements, point-in-time, reconciled" (or the medium / low text) and notes such as "stale fundamentals"
New filing badge	Hover lists up to five recent filings with time; click opens the latest filing in a new browser tab
See home listing →	On ADR pages, opens the home listing's Company 360
52-week strip hover	The small grey line has the tooltip "Beta · average daily traded value · distance from 52-week high"

Methodology

- **Market data.** Last close, one-day change, 52-week high and low, 20-day average dollar volume, market cap and enterprise value come from the serving snapshot (`quote` block of the overview endpoint). Market cap and EV are in USD at the snapshot FX rate.
- **Beta.** The header `β` is the **1-year beta of daily log returns against the region benchmark**. It is a different number from the beta in the cost of capital (3-year weekly regression, Blume-adjusted and shrunk to the subsector, see Forecast). Apple shows β 0.64 in the header and 1.12 in the cost of capital.
- **Composite gauge.** The composite of the default ("balanced") factor profile, 0-100 (see Overview). The A-E grade (quintiles of the global scored universe) is not drawn on the gauge; it is shown in the Screener and available as the BQL field `grade` .
- **DQ badge.** The composite's `dq_score` (mean data-quality score of the fundamental inputs, 0-1) bucketed as high (≥ 0.8), medium (≥ 0.5) or low (< 0.5). A missing score shows as medium.
- **Moat badge.** The latest moat class. A company classified "insufficient data" is shown with the **No moat** badge in the header; the Moat tab states "insufficient history to classify a moat".
- **Fiscal year-end** comes from the issuer reference data (month only).

Data sources and refresh

Prices from Polygon (US) and yfinance or national sources (other markets), FX from the FX job, reference data from the identity jobs, filings from SEC EDGAR and the national filing systems. The header refreshes with each snapshot. The new-filing badge is the only live element: it polls the realtime filings feed every two minutes and listens to the streamed feed while the realtime layer runs.

Validation status

Not applicable (reported data). The composite is a ranking, not a validated return forecast; see Overview.

Correct usage and worked example

Apple on 19 Oct 2026: last close 342.94 USD (−0.15%), market cap $5.01T, EV $5.03T, 52-week range 242.76-348.15 with the marker near the top (1% below the high), β 0.64, ADV $12B, composite 69. Badges: wide moat, DQ high, USD reporting, FYE September, last filing 1 Sep 2026.

Toyota is a cross-listed example. The Tokyo line 7203.T has a composite of 57 (Figure 5A.8). The NYSE ADR TM shows "—" because composite, technicals and fair value are computed on the home listing by design: use the banner link to jump to 7203.T before drawing conclusions.

Figure 5A.4: ADR header for Toyota Motor ADR (`/company/TM`): the info banner, the "See home listing → 7203.T (Tokyo Stock Exchange)" link, and an empty composite gauge and EV.

The market-data values in an ADR header belong to the US line (price 194.61 USD, 52-week range 166.10-248.90, β −0.03, ADV $59M), while the moat, DQ, reporting-currency and fiscal-year badges are the issuer's.

Limitations and pitfalls

- **Do not compare header beta with WACC beta.** They use different windows, frequencies and adjustments.
- Market cap and EV are in USD even when the price is in another currency.
- On ADR pages the factor, score and fair-value panels are intentionally empty. If the home listing is not in the universe, the banner says "Home listing not found in the universe".
- The watchlist lives in the browser profile. Clearing site data or using another machine starts with an empty list.

Related mnemonics and BQL fields

`DES` , `THES` . BQL: `px_last` , `chg_pct_1d` , `market_cap_usd` , `beta_1y` , `composite` , `grade` , `moat` , `dq_score` .

Overview

Route: `/company/:id?tab=overview` · **Menu:** Company 360 › Overview · **Mnemonics:** `DES` · **Refresh:** with each data refresh

Purpose

The Overview answers "where does this company stand against its peers and its own history?". It shows the ten factor scores against the peer median, how they add up to the composite, the price, the sector-specific key ratios and any current flags. It is the first stop after the Screener and the place to see why a name ranks where it does.

Screenshot

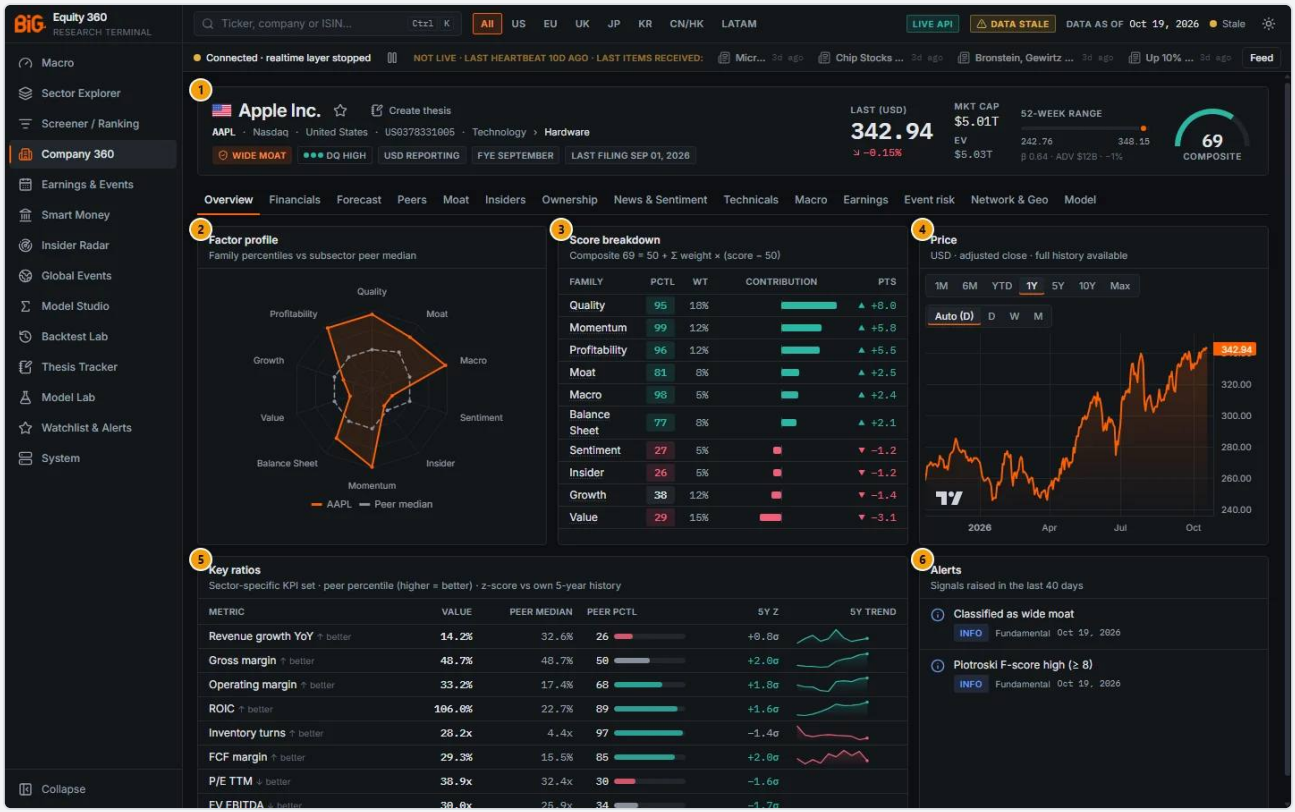


Figure 5A.5: Overview tab for Apple (`/company/AAPL?tab=overview`), data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	See Company header
2	Factor profile	Radar of the ten factor percentiles (orange) against the peer median (dashed)

#	Panel	What it shows
3	Score breakdown	"Composite 69 = 50 + Σ weight × (score − 50)": percentile, weight and points contributed by each factor family, sorted by contribution
4	Price	Adjusted close as an area chart, default range 1Y, "full history available"
5	Key ratios	Sector-specific KPI set: value, peer median, peer percentile bar, 5y z and trend sparkline
6	Alerts	"Signals raised in the last 40 days": current flags with severity, category and date

Controls

Control	What it does
Radar hover	Shows the company's and the peer median's value for each factor
Family name hover (Score breakdown)	Dotted-underlined names show a tooltip with the key-ratio metric scores that belong to that factor
Price range	`1M` , `6M` , `YTD` , `1Y` (default), `5Y` , `10Y` , `Max`
Bar interval	`Auto (D)` and `D` , `W` , `M` . Auto uses daily bars up to 1Y, weekly for 5Y and 10Y, monthly for Max. If daily bars are capped (10 years), a "weekly (daily capped at 10y)" badge appears. The right-hand note gives the first available date ("History from …")
Key ratios row hover	Highlights the row; the `↑ better` / `↓ better` hint after each label gives the metric's direction
Beneish breakdown	When a Beneish alert is raised, an expandable "Breakdown" line under the alert shows the eight indices, their contributions, the growth-neutral M-score, red flags and corroboration

Methodology

FACTOR SCORES AND THE COMPOSITE

Every metric is scored in up to three ways and blended into a 0-100 metric score (contract `contracts/scoring.yaml`):

1. **Against peers.** Percentile within the peer group. Peer groups are tried in order, and the first with at least 8 companies with a value wins: subsector × peer region × size bucket, subsector × peer region, subsector (global), sector × peer region, sector (global). Peer regions are North America, Europe, Japan, Asia ex-Japan, Greater China, Latin America and Global.
2. **Against its own history.** A z-score of the current value against the company's own history (10-year window, at least 5 years and 8 observations), direction-adjusted so that a positive z is always "better than usual": `z = d × (x − μ_hist) / σ_hist` , with `d = +1` when higher is better and `−1` when lower is better, clipped at ±3 and turned into a score with the normal CDF.
3. **Against market cap** (valuation metrics only). Within the peer group (at least 20 names), `log(multiple)` is regressed on `log(market cap)` , the quality factor score and 3-year revenue growth; the residual percentile ("cheap versus what its size and quality justify") is the third component.

The blend weights are peer 0.70 / self 0.30 for fundamental ratios and peer 0.40 / self 0.25 / market-cap 0.35 for valuation metrics; market, insider, sentiment and macro metrics use the peer percentile only. Missing components are renormalised away.

Metric scores are averaged into ten **factor families**: Quality, Profitability, Growth, Value, Balance sheet, Momentum, Insider, Sentiment, Macro sensitivity and Moat. Each family score is then re-percentiled within peer region × sector (at least 8 names, else the peer region), so a family score is itself a percentile. A family with less than 30% coverage is null for fundamental families and set to 50 (neutral) for the others.

The **composite** is the weighted mean of the family scores with coverage ≥ 0.3, weights renormalised:

```
composite = Σ_f w_f × score_f  /  Σ_f w_f        (equivalently 50 + Σ_f w_f × (score_f − 50) when Σ w_f = 1)
balanced weights: Quality 18%, Value 15%, Profitability 12%, Growth 12%, Momentum 12%,
                  Balance sheet 8%, Moat 8%, Insider 5%, Sentiment 5%, Macro 5%
```

A composite needs at least five non-null families including at least two of Quality, Profitability, Value and Growth. Grades A-E are quintiles of the global scored universe.

PANELS

- **Factor profile.** The orange polygon is the company's family scores. The dashed polygon is the median family score of companies in the same **sector** and region (at least 8 names, else the global sector). Because family scores are percentiles within region × sector, this median sits close to 50 for most families; it departs from 50 mainly for Insider and Moat, where many companies share the same value.
- **Score breakdown.** Each row shows the family percentile (`Pctl`), the default-profile weight (`Wt`), a bar and the points it adds or removes: `points = weight × (score − 50)` . The rows are sorted from the largest positive to the largest negative contribution, and the title restates the composite. The composite itself is the API value; small rounding differences against the sum of the rows are normal.
- **Price.** Adjusted close (splits and dividends) in the trading currency.
- **Key ratios.** The KPI set of the company's business model from `contracts/sector_kpis.yaml` (for example inventory turns for hardware, net interest margin and efficiency ratio for banks, R&D to revenue for semiconductor equipment). Columns: value; peer median of the peer group used by scoring; peer percentile (higher = better, already direction-adjusted); `5y z` , the direction-adjusted self-history z-score described above (positive = better than the company's own history); and a sparkline of the annual history.
- **Alerts.** Current flags computed for the snapshot date, each shown with the snapshot date. The flags are: insider cluster buy; CEO or CFO open-market buy; Altman Z″ below 1.1 ("distress zone", critical); Beneish screen (model-based; "Beneish screen" warning when the recalibrated, growth-neutral M-score is above −1.78 with corroborating red flags, "Beneish watch" info when only the textbook M-score is above −1.78; not applied to financials and REITs); classified as wide moat; Piotroski F-score ≥ 8; fundamentals stale (last period end more than 400 days before the price date); new 52-week high or low; ML ranking model in the top decile (3M); data-quality score below 0.6.

Data sources and refresh

Fundamentals from SEC XBRL, EDINET / TDnet, ESEF and national filing systems, with yfinance as a fallback where no official filing is available; prices from Polygon and yfinance or national exchanges. The `scoring` job runs in the nightly chain; the moat input is weekly. The ratio history comes from `/companies/{id}/ratios` (fiscal years, up to 10).

Validation status

- **Composite and factor scores:** a transparent, rules-based ranking. They are not a model trained on returns and have **not been validated as a return forecast**. No "model-based" label is attached because there is no estimated model; treat them as a structured description.
- **Beneish screen:** labelled **model-based** in the alert ("A statistical screen, not an accounting finding; high sales growth alone lifts the textbook score"). The 19 Oct 2026 audit (finding F15) found the textbook rule flagged 4.6% of names, mostly hyper-growth; the screen was then recalibrated to the growth-neutral version and downgraded to a warning.
- **ML top decile flag:** labelled model-based; the ML model's out-of-sample rank-IC is about 0.035 (see Model).

Correct usage and worked example

Apple (Figure 5A.5). Composite 69, grade A, rank 43 globally and 4th in its sector. The breakdown shows the score is carried by Quality (95th percentile, +8.0 points), Momentum (99th, +5.8) and Profitability (96th, +5.5), offset by Value (29th, −3.1), Growth (38th, −1.4), Insider (26th, −1.2) and Sentiment (27th, −1.2). Key ratios are compared within "hardware networking, North America, large caps" (17 peers): revenue growth 14.2% against a peer median of 32.6% (26th percentile) but +0.8σ against Apple's own history; ROIC 106.0% against 22.7% (89th percentile, +1.6σ); P/E TTM 38.9x against 32.4x (30th percentile, −1.6σ, i.e. more expensive than Apple's own history). Alerts: "Classified as wide moat" and "Piotroski F-score high (≥ 8)".

Read it as: a high-quality, expensive large cap whose rank rests on quality and momentum rather than value.

Figure 5A.6: Key ratios for JPMorgan Chase (`/company/JPM?tab=overview`), bank KPI set, data as of 19 Oct 2026.

JPMorgan (bank KPI set). The KPI set changes with the business model: return on tangible equity 20.4% (85th percentile), ROE 17.8% (95th), net interest margin 5.5% (96th), efficiency ratio 51.4% against a peer median of 59.5% (lower is better; 94th percentile), cost of risk 0.9% (15th). NPL ratio and CET1 show "—" because JPMorgan's values are not in the data. Composite 65: Quality, Profitability, Growth and Momentum lift it; Value (15th percentile, −5.3 points) and Balance sheet (14th, −2.9) hold it back.

Figure 5A.7: Score breakdown for Petrobras (`/company/PETR4.SA?tab=overview`), data as of 19 Oct 2026.

Petrobras. Composite 53. Momentum (99th, +5.9) and Growth (90th, +4.8) are offset by Quality at the 6th percentile (−8.0 points). Several key ratios (EBITDA margin, net debt to EBITDA, FCF yield, EV/EBITDA) show "—" for the company: check the Financials tab before relying on the Value and Balance-sheet scores.

Figure 5A.8: Score breakdown for Toyota Motor on its home listing (`/company/7203.T?tab=overview`), data as of 19 Oct 2026.

Toyota (7203.T). Composite 57 with Growth (81st), Profitability (74th) and Quality (61st) positive and Momentum (24th, −3.2) and Macro (18th, −1.6) negative.

The two European examples follow the same layout: ASML (`/company/ASML.AS` , composite 68, wide moat) and SAP (`/company/SAP.DE` , composite 57, narrow moat).

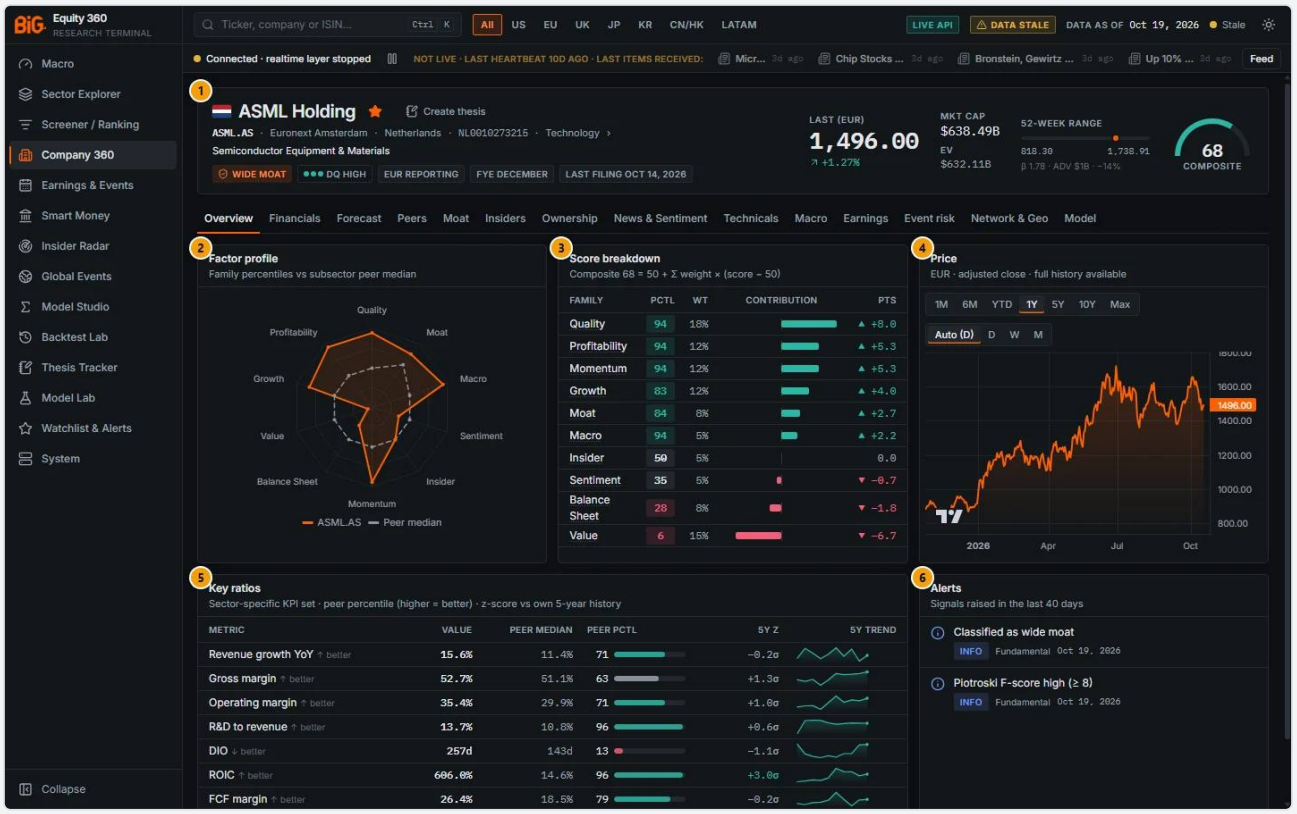


Figure 5A.9: Overview for ASML Holding on Euronext Amsterdam (`/company/ASML.AS?tab=overview`), data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	EUR price, wide moat, EUR reporting, FYE December

#	Panel	What it shows
2	Factor profile	ASML.AS against the sector peer median
3	Score breakdown	Composite 68: Quality, Profitability and Momentum at the 94th percentile; Insider 50 (neutral); Value at the 6th percentile (−6.7 points)
4	Price	EUR adjusted close
5	Key ratios	Semiconductor-equipment KPI set, scored against global subsector peers (fewer than 8 European peers)
6	Alerts	"Classified as wide moat" and "Piotroski F-score high (≥ 8)"

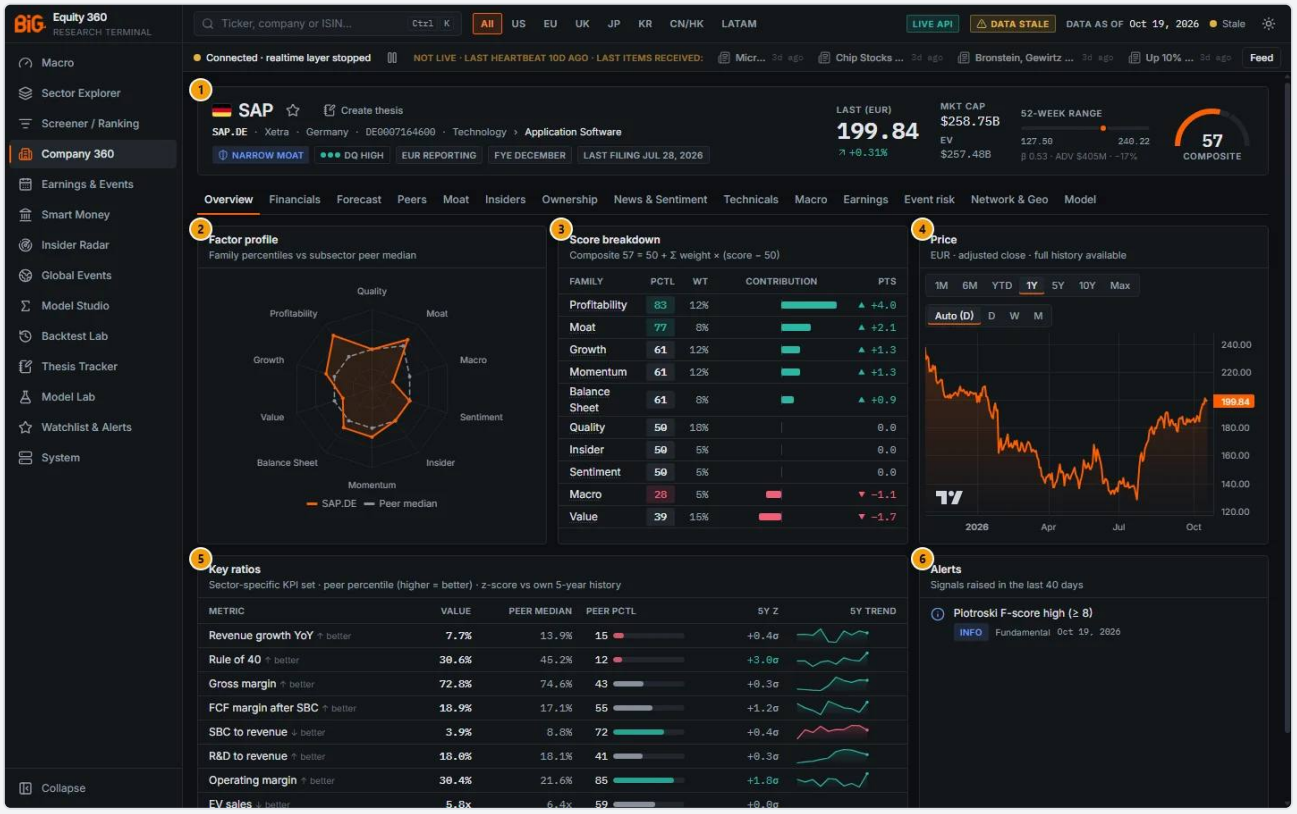


Figure 5A.10: Overview for SAP on Xetra (`/company/SAP.DE?tab=overview`), data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	EUR price, narrow moat, EUR reporting
2	Factor profile	SAP.DE against the sector peer median
3	Score breakdown	Composite 57: Profitability (83rd percentile) and Moat (77th) add most; Macro (28th) and Value (39th) subtract
4	Price	EUR adjusted close
5	Key ratios	Software KPI set
6	Alerts	"Piotroski F-score high (≥ 8)"

Limitations and pitfalls

- **Percentiles are relative.** A 95th percentile Quality score in a weak sector is not the same as in a strong one. Check the peer group level: when a subsector × region group is too small, scoring falls back to wider groups (ASML's key ratios use global semiconductor-equipment peers).
- **The radar's dashed line is a sector median**, not a subsector median as its subtitle says, and it is close to 50 by construction for most families.
- **"5y z" uses up to 10 years of history** (5 years minimum), and its sign is direction-adjusted. A negative z on P/E means "more expensive than usual", not "lower P/E".
- **Score breakdown tooltips list only the key-ratio metrics** of each family, not every metric that feeds it.
- **Alerts are current-state flags.** They all carry the snapshot date; the "last 40 days" in the subtitle is not a look-back window over historical alerts.

- Implausible values usually mean a data problem, not a business fact (for example an ROIC of several hundred percent, or 0.0% for a large industrial). Check the Financials tab and the DQ badge before acting on them.
- A composite of "—" means the minimum-coverage rule failed (or the page is an ADR).

Related mnemonics and BQL fields

`DES`, `EQS` (Screener). BQL: `composite`, `grade`, `grade.<profile>`, `f.quality`, `f.profitability`, `f.growth`, `f.value`, `f.balance_sheet`, `f.momentum`, `f.insider`, `f.sentiment`, `f.macro`, `f.moat` (each also `.rank_sector`, `.chg_1m`), `dq_score`, `roic`, `gross_margin`, `pe_ttm`, `ev_ebitda`. See Screener and Ranking for profiles, custom weights and the full factor definitions.

Financials

Route: `/company/:id?tab=financials` · **Menu:** Company 360 › Financials · **Mnemonics:** `FA`, `DVD` · **Refresh:** nightly chain (filings and normalisation); weekly full rebuild

Purpose

The Financials tab shows the standardised income statement, balance sheet and cash-flow statement as filed, annual or quarterly, in the reporting currency or in USD, as values, common-size percentages or growth rates. Use it to check the numbers behind a ratio, to see the trend of a line and to export the statements.

Screenshot

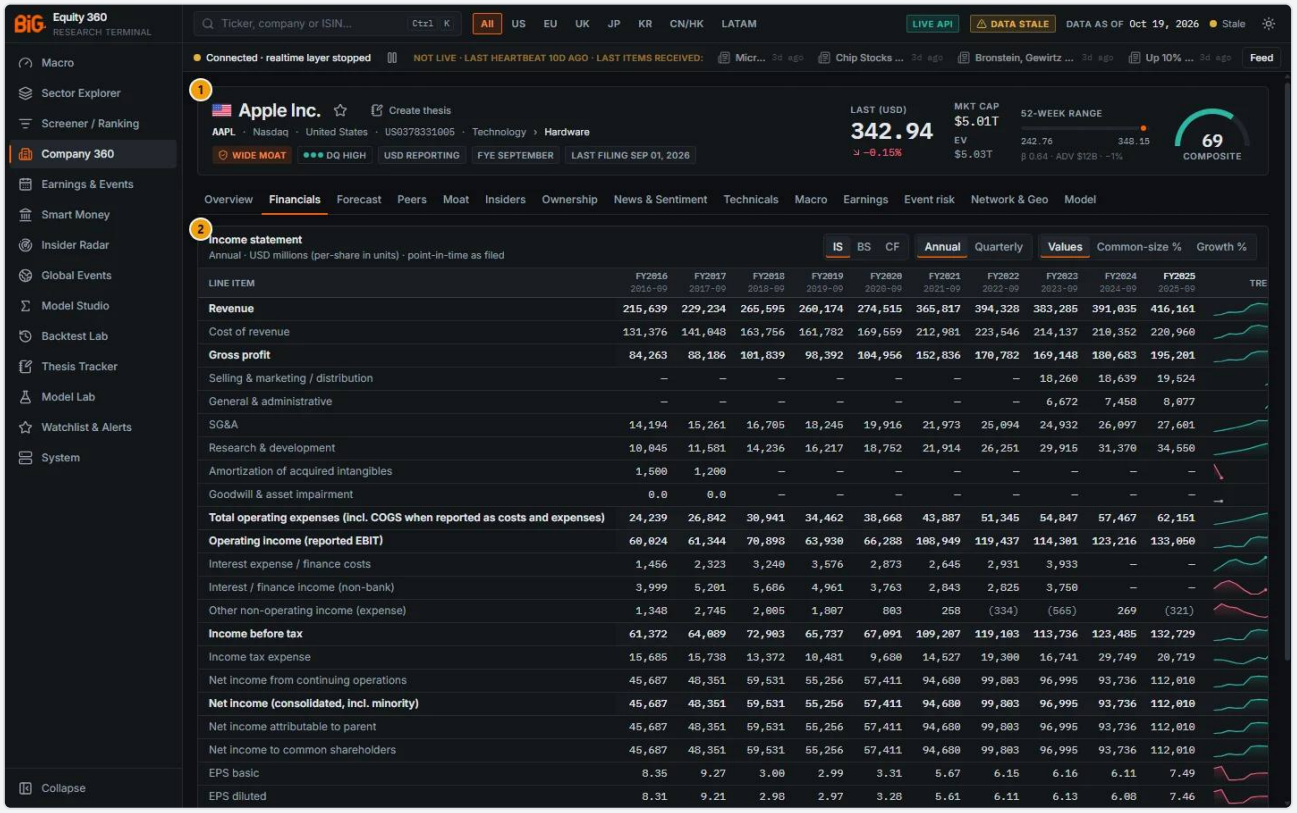


Figure 5A.11: Financials tab for Apple (`/company/AAPL?tab=financials`), income statement, annual, data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	See Company header

#	Panel	What it shows
2	Income statement	"Annual · USD millions (per-share in units) · point-in-time as filed": up to ten fiscal years (FY2016-FY2025 for Apple), with the period-end month under each year and a trend sparkline per line

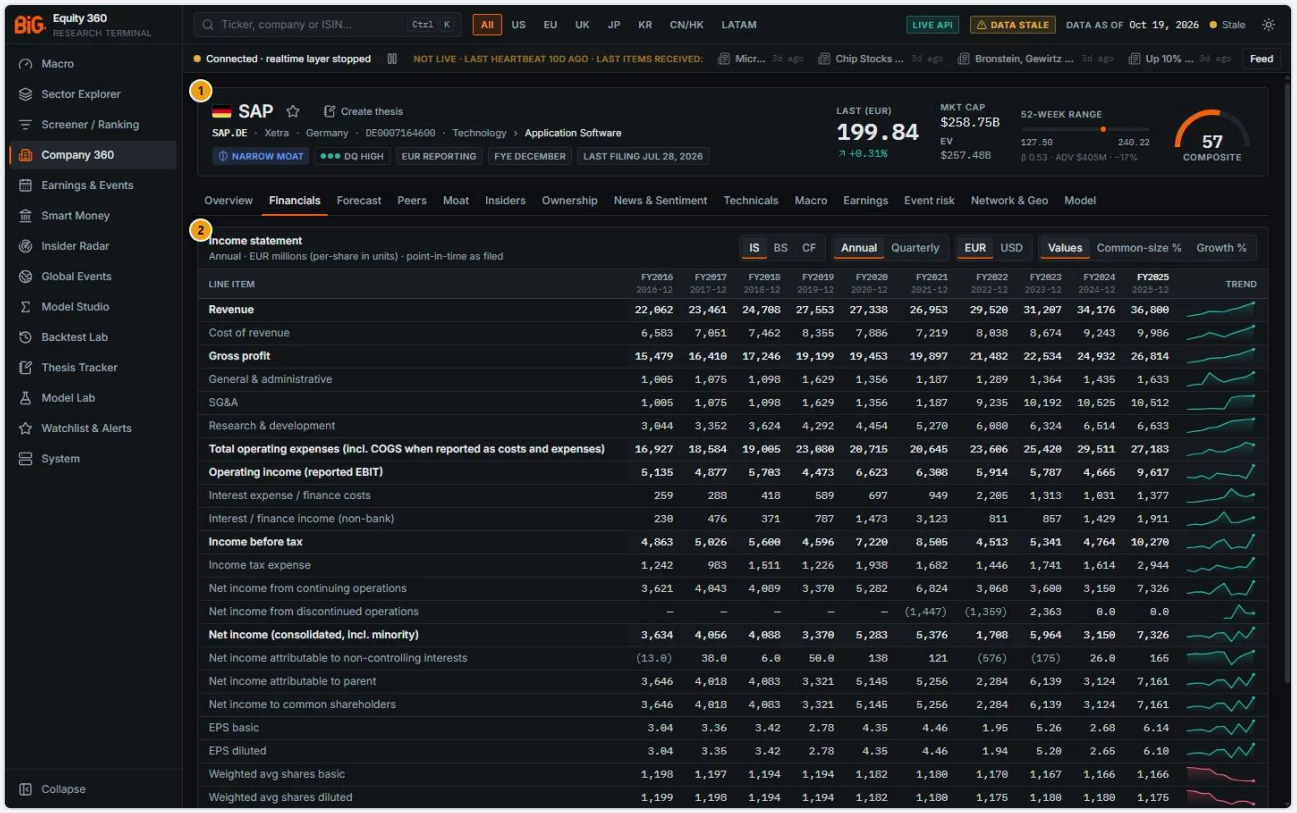


Figure 5A.12: Financials tab for SAP (`/company/SAP.DE?tab=financials`), EUR, with the EUR / USD switch, data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	EUR reporting
2	Income statement	Annual, EUR millions; the `EUR` / `USD` segmented control appears because the reporting currency is not USD

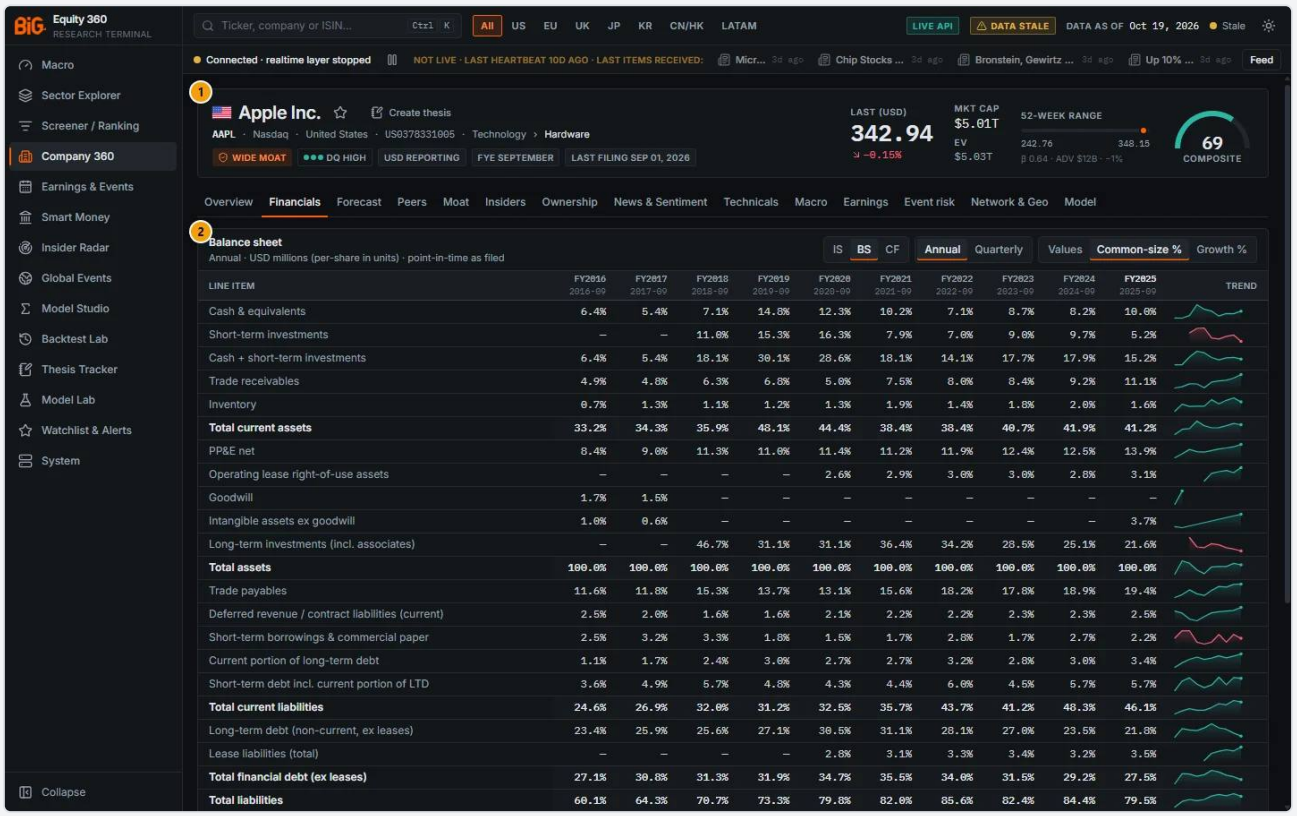


Figure 5A.13: Balance sheet for Apple in the Common-size % view (`/company/AAPL?tab=financials` , `BS` then `Common-size %`), data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	See Company header
2	Balance sheet	Each line as a share of total assets (total assets = 100.0% in every year): for FY2025 cash and short-term investments 15.2%, total current assets 41.2%, long-term investments 21.6%, total liabilities 79.5%

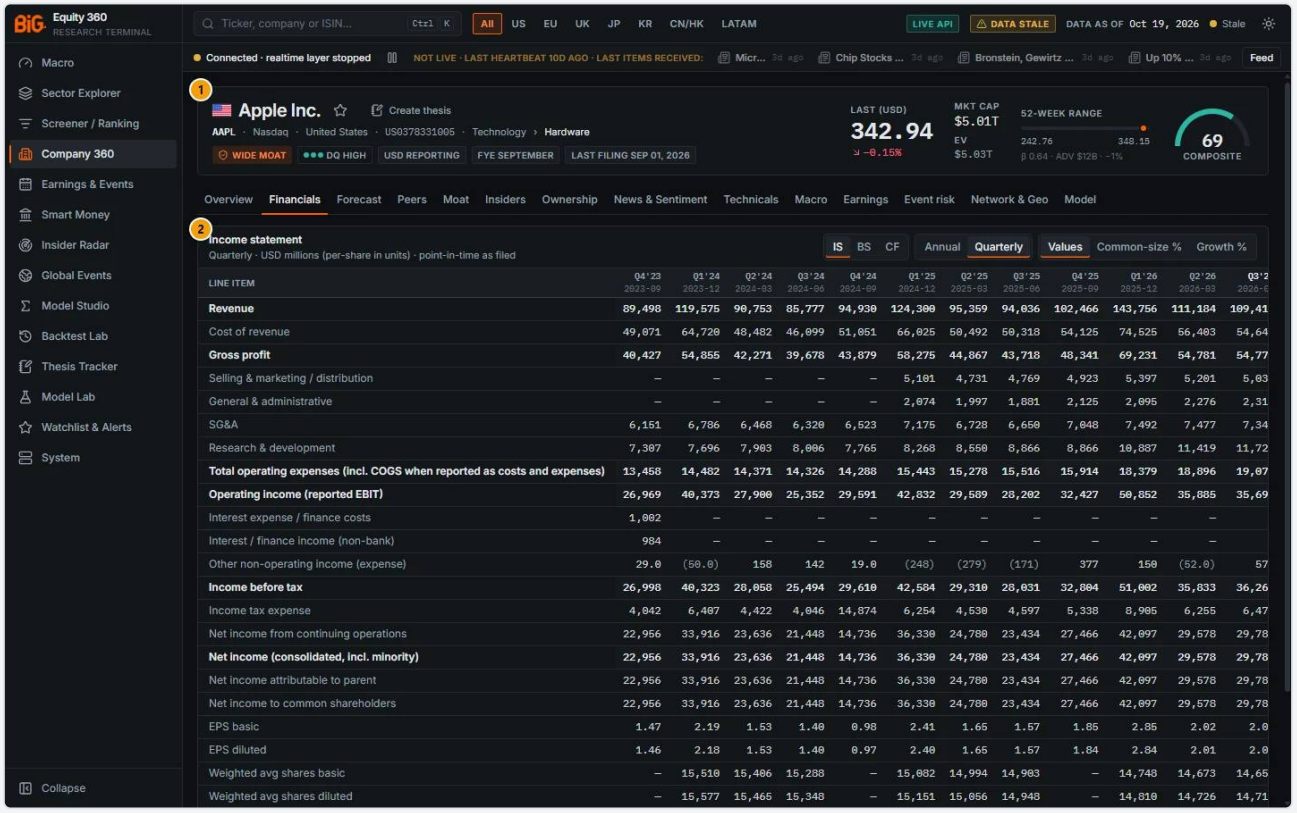


Figure 5A.14: Income statement for Apple, Quarterly view, data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	See Company header
2	Income statement	Twelve fiscal quarters (Q4'23 to Q3'26), labelled by fiscal quarter with the calendar period-end month underneath (Apple's fiscal Q1 ends in December)

Controls

Control	What it does
`IS` / `BS` / `CF`	Income statement, balance sheet, cash-flow statement (hover shows the full name)
`Annual` / `Quarterly`	Fiscal years (up to 10) or fiscal quarters (up to 12)
Reporting currency / `USD`	Shown only for non-USD reporters. USD converts each period at that period's FX rate ("Converted at period-end FX")
`Values` / `Common-size %` / `Growth %`	Amounts; each currency line as a share of revenue (IS and CF) or of total assets (BS); or year-on-year growth (quarterly: same quarter of the prior year)
Trend sparkline	Per-line history over the displayed periods
`CSV` (bottom right)	Downloads the current view as `<id>-<statement>-<period>-<view>.csv`
Horizontal scroll	The line-item column stays pinned on the left

Methodology

- **Standardised line items.** Reported items are mapped to the canonical line-item dictionary (`contracts/canonical_line_items.yaml`). Lines with no value in any displayed period are hidden. Totals (revenue, gross profit, operating income, EBIT, EBITDA, pre-tax income, net income, totals, CFO, CFI, CFF, FCF) are shown in bold.
- **Point-in-time storage.** Every value is stored with its knowledge date (the filing date). Restatements are kept as new versions rather than overwriting the original. The tab shows, for each period, **the latest version known at the snapshot date**, so a restated year shows the restated number. The API can also return statements "as known on" an earlier date (`as_of` parameter), but the tab has no control for it; backtests and the BQL console use the point-in-time versions.
- **Units.** Amounts are in millions of the reporting currency (or USD); per-share lines (EPS, DPS) in units; share counts in millions. Negative values are shown in parentheses.
- **Common-size and growth** are computed in the browser from the displayed values: `line / revenue` (or `/ total assets`) for common-size, `x_t / |x_{t-k}| - x_{t-k} / |x_{t-k}|` for growth with `k = 1` year or 4 quarters (this form keeps the sign meaningful when the base is negative).
- **USD view.** Uses the per-period FX implied by the API's USD values, i.e. each period at its own period-end rate, so USD growth rates mix business growth and currency moves.

Data sources and refresh

US: SEC XBRL (10-K, 10-Q, 20-F, 40-F). Japan: EDINET and TDnet. Europe: ESEF via filings.xbrl.org and national systems (FCA, AFM, BaFin and others); Hong Kong: HKEX; Brazil: CVM; China, Korea and Taiwan per the coverage notes in Data and coverage. Where no official filing is available, yfinance fills the gap and the DQ badge reflects it. Filings are ingested and normalised in the nightly chain; the weekly chain runs a full rebuild.

Validation status

Not a model. Data quality is checked by the data-quality jobs (DQ badge, System › Data Health). See System.

Correct usage and worked example

Apple, annual income statement (Figure 5A.11): revenue grew from 215,639 (FY2016, USD millions) to 416,161 (FY2025); operating income (reported EBIT) 133,050 in FY2025; diluted EPS 7.46. To see margins, switch to `Common-size %` : each line becomes a share of revenue. On the balance sheet (Figure 5A.13) the same view expresses every line as a share of total assets: Apple's total liabilities were 79.5% of assets in FY2025 against 60.1% in FY2016. The Quarterly view (Figure 5A.14) shows the seasonality of the December quarter (fiscal Q1): revenue 143,756 in Q1'26 against 102,466 in Q4'25. Note the gaps in the quarterly data: the interest lines are empty after Q4'23, and weighted share counts are missing in the fiscal Q4 columns, which come from the annual report rather than a 10-Q. To check whether a high ROIC on the Overview is genuine, open `BS` and look at total equity and debt over several years.

SAP (Figure 5A.12) reports in EUR. Switch to `USD` only when comparing with US peers in absolute size; keep EUR for trend analysis.

Limitations and pitfalls

- **Semi-annual reporters.** Many European, UK and Australian companies publish half-year accounts only. The tab offers Annual and Quarterly; for half-year reporters the Quarterly view may be sparse or empty.
- **"Point-in-time as filed" means the latest version known**, not the first-reported number. For first-reported values use the BQL console or the Backtest Lab.
- **Mapping differences.** Lines such as "SG&A" versus "selling and marketing" depend on how the company tags its filing. A dash can mean "not reported separately". Apple, for example, shows separate selling and G&A lines only from FY2023.
- Older periods sourced from a provider rather than an official filing may have estimated knowledge dates.
- Growth in the USD view includes currency effects.

Related mnemonics and BQL fields

`FA` , `DVD` (dividends: DPS, payout and yield rows). BQL: the `fa.*` statement fields (for example `fa.revenue` , `fa.short_term_debt`) and the ratio fields; use `with(currency=USD)` for conversion. See BQL console.

Forecast

Route: `/company/:id?tab=forecast` · **Menu:** Company 360 › Forecast · **Mnemonics:** `EE` · **Refresh:** nightly chain (`forecast` job)

Purpose

The Forecast tab gives a **model-based** five-year projection of the company's statements with bear, base and bull cases, and the resulting fair value, 12-month target, cost of capital, sensitivity grid and reverse DCF. It is not a consensus estimate: the app has no consensus data. Use it to understand what the current price implies and how sensitive a value is to its assumptions, not as a price target.

> **Model-based:** every number on this tab comes from the app's own projection and valuation models. The `EE` mnemonic is labelled "MODEL-BASED, not consensus".

Screenshot

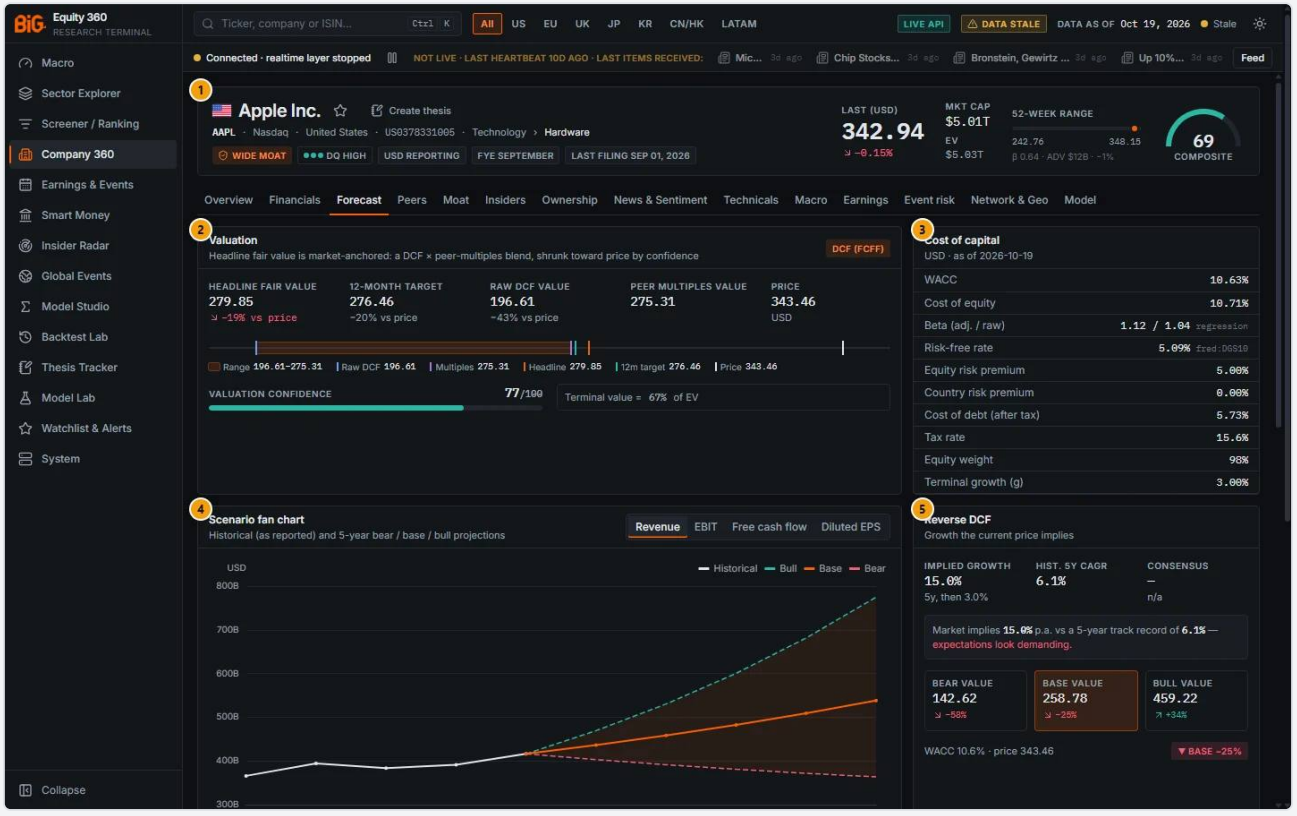
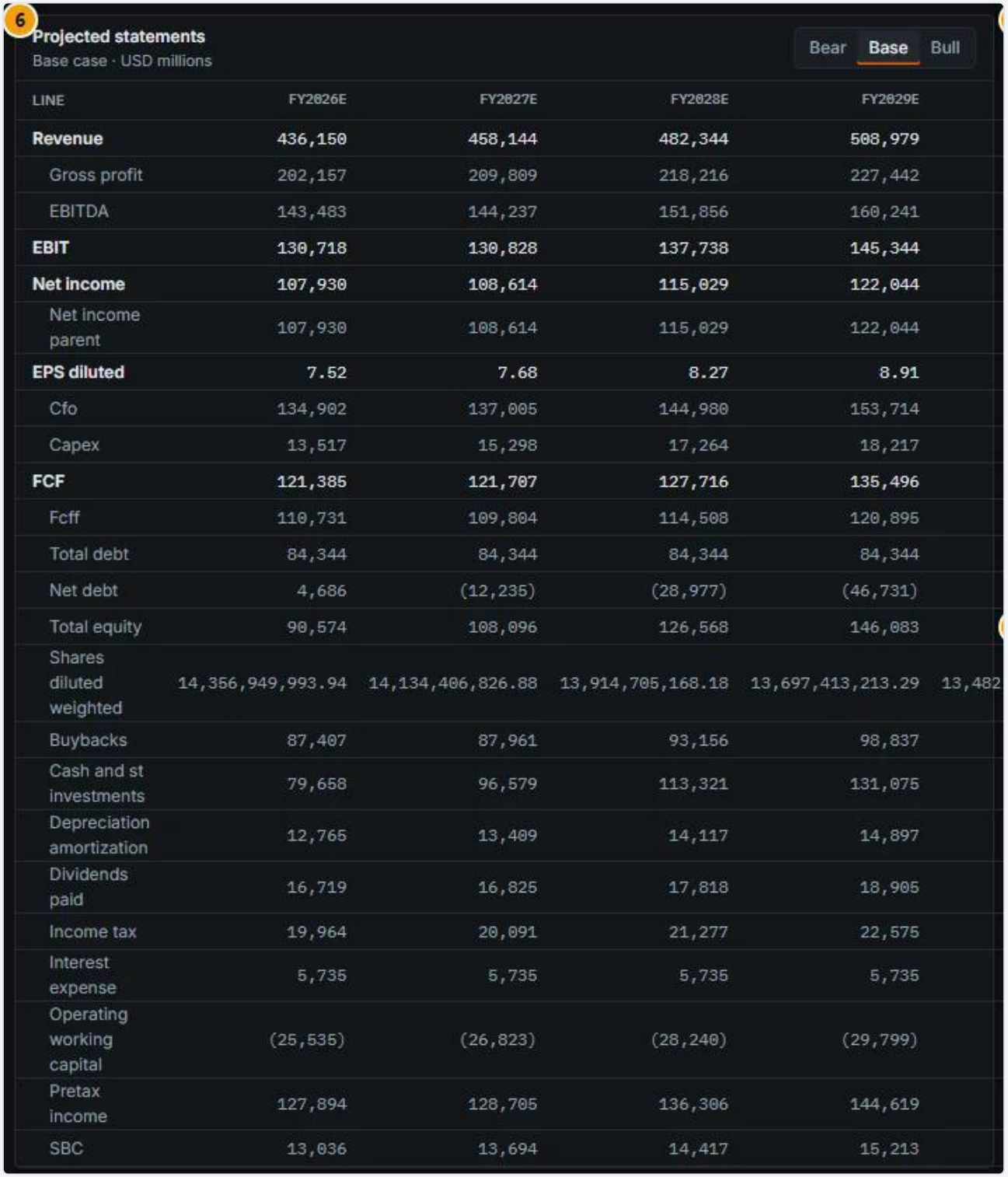


Figure 5A.15: Forecast tab for Apple (`/company/AAPL?tab=forecast`), valuation as of 16 Oct 2026, data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	See Company header
2	Valuation	"Headline fair value is market-anchored: a DCF × peer-multiples blend, shrunk toward price by confidence": headline fair value, 12-month target, raw DCF value, peer multiples value, price, the fair-value range bar, valuation confidence 0-100 with reasons, terminal-value share and flags. A badge names the model: DCF (FCFF) or Excess-return model

#	Panel	What it shows
3	Cost of capital	WACC, cost of equity, beta (adjusted / raw, with source), risk-free rate (with series), equity and country risk premia, after-tax cost of debt, tax rate, equity weight, terminal growth
4	Scenario fan chart	Historical values (as reported) and the 5-year bear / base / bull paths with the shaded bear-bull band
5	Reverse DCF	"Growth the current price implies": implied growth, historical 5-year CAGR, consensus (always n/a), a verdict line and the bear / base / bull values
6	Projected statements	Projected lines for the chosen scenario, in millions (below the fold)
7	Assumptions	Driver inputs by scenario against the 5-year history (below the fold)
8	DCF sensitivity	Fair value per share on a WACC × terminal-growth grid (below the fold)



6

Projected statements

Base case · USD millions

Bear | Base | Bull

LINE	FY2026E	FY2027E	FY2028E	FY2029E	
Revenue	**436,150**	**458,144**	**482,344**	**508,979**	
Gross profit	202,157	209,809	218,216	227,442	
EBITDA	143,483	144,237	151,856	160,241	
EBIT	**130,718**	**130,828**	**137,738**	**145,344**	
Net income	**107,930**	**108,614**	**115,029**	**122,044**	
Net income parent	107,930	108,614	115,029	122,044	
EPS diluted	**7.52**	**7.68**	**8.27**	**8.91**	
Cfo	134,902	137,005	144,980	153,714	
Capex	13,517	15,298	17,264	18,217	
FCF	**121,385**	**121,707**	**127,716**	**135,496**	
Fcff	110,731	109,804	114,508	120,895	
Total debt	84,344	84,344	84,344	84,344	
Net debt	4,686	(12,235)	(28,977)	(46,731)	
Total equity	90,574	108,096	126,568	146,083	
Shares diluted weighted	14,356,949,993.94	14,134,406,826.88	13,914,705,168.18	13,697,413,213.29	13,482
Buybacks	87,407	87,961	93,156	98,837	
Cash and st investments	79,658	96,579	113,321	131,075	
Depreciation amortization	12,765	13,409	14,117	14,897	
Dividends paid	16,719	16,825	17,818	18,905	
Income tax	19,964	20,091	21,277	22,575	
Interest expense	5,735	5,735	5,735	5,735	
Operating working capital	(25,535)	(26,823)	(28,240)	(29,799)	
Pretax income	127,894	128,705	136,306	144,619	
SBC	13,036	13,694	14,417	15,213	

Figure 5A.16: Projected statements, Apple, base case (panel 6).

Figure 5A.17: Assumptions by scenario, Apple (panel 7). The 5Y column is empty when the driver history is not published.

Figure 5A.18: DCF sensitivity grid, Apple (panel 8). Rows are WACC, columns terminal growth; colour is the gap to the price.

Controls

Control	What it does
Model badge hover	"Free cash flow to the firm, discounted at WACC" or "Banks / insurers: residual-income (excess return on equity) model"
Range bar hover	Each marker shows its label and value: Raw DCF, Multiples, Headline, 12m target, Price; the shaded block is the fair-value range (low-high)

Control	What it does
Fan chart metric	`Revenue` , `EBIT` , `Free cash flow` , `Diluted EPS` . For banks the first three become `Net income` , `Residual income` and `Book value`
Fan chart hover	Values per year for Historical, Bull, Base, Bear
Projected statements scenario	`Bear` / `Base` / `Bull`
Sensitivity hover	"WACC x% · g y%", fair value and the gap to price

Methodology

The projection and valuation are computed by the nightly `forecast` job (`backend/app/analytics/forecast.py`), point-in-time at the as-of date: only fundamentals filed by that date, prices up to that date and rates known by then.

NON-FINANCIAL COMPANIES: LINKED THREE-STATEMENT PROJECTION AND FCFF DCF

Five annual years after the last reported fiscal year:

- **Revenue growth** starts at g_1 = 0.5 × Theil-Sen trend of log revenue (up to 7 years) + 0.5 × last-3-year CAGR (clipped to −25%…+50%) and fades linearly to g_5 = 0.5 × subsector median 5-year CAGR + 0.5 × terminal growth .
- **Margins** (EBITA, i.e. EBIT plus amortisation of acquired intangibles) mean-revert at no more than 2 pp a year: normally toward `0.9 × (0.6 × own 5-year median + 0.4 × last) + 0.1 × subsector median` ; for cyclical subsectors (semiconductors, semi equipment, mining, oil and gas, autos, airlines and others) or volatile margins, from `0.5 × last + 0.5 × own 10-year median` toward the 10-year median; for loss-makers toward 0.8 × the median margin of profitable peers, at up to 4 pp a year.
- **Other drivers**: D&A, capex, stock-based compensation and operating working capital as a percentage of revenue (own 3-year medians, subsector fallback); normalised tax rate; interest on average debt and interest income on average cash; payout from the 3-year median. Capex reaches a growth-consistent level within three years. The balance sheet balances every year, with cash as the plug and a revolver when cash would fall below 2% of revenue.
- **Free cash flow to the firm**: `FCFF = NOPAT + D&A − capex − ΔNWC` (stock-based compensation is treated as a real cost), discounted at WACC over fractional years from the as-of date to each fiscal year-end.
- **Fade and terminal value**: a further five-year fade stage (growth to the terminal rate, margin to the long-run target, reinvestment = Δrevenue / sales-to-capital), then `TV = NOPAT × (1 − g / RONIC) / (WACC − g)` with `RONIC = max(WACC, (ROIC + WACC) / 2)` . The terminal-value share of EV is shown and flagged above 85%.
- **Equity value** = EV + cash − debt − minorities − preferred, per diluted share, converted to the trading currency (and divided by the ADR ratio for ADRs).
- **Scenarios**: bear and bull shift the growth path by ∓1.28 × 0.6 × σ(year-on-year growth) and the margin target by ∓1.28 × σ(EBIT margin), using the company's own 10-year history. Probabilities 25% / 50% / 25%.

BANKS AND INSURERS: EXCESS-RETURN MODEL

Banks, insurers, consumer-finance companies, holdings and levered financials use a residual-income model on book value:

$$\text{value} = BV_0 + \sum_t (ROE_t - k_e) \times BV_{t-1} / (1 + k_e)^t + RI_6 / (k_e - g)$$

ROE fades to `0.6 × own 5-year median + 0.4 × subsector median` at up to 2 pp a year; the payout comes from history. The cost of capital is the cost of equity only ("financial (equity-only)" in the panel subtitle).

HEADLINE FAIR VALUE

1. **Two legs.** The model value per share (DCF or excess return) and a **peer-multiples value**: the geometric mean of the values implied by peer EV/EBITA, P/E on normalised earnings and, for loss-makers, EV/Sales (banks: P/E and P/B). Only 50% of the gap between the price and the multiples value is assumed to close.
2. **Blend in log space**: `l = w × ln(DCF / P) + (1 − w) × 0.5 × ln(Multiples / P)` , with the DCF weight `w` between 0.2 and 0.5 depending on model stability.
3. **Shrink toward the price** by a credibility factor `k = 0.4 + 0.6 × confidence / 100` : `FV_scenario = P × exp(k × l)` . Bear and bull are scaled from the base by the ratio of their DCF values (bounded 0.25×-4×).

4. **Headline** = the probability-weighted value of the three scenarios, `0.25 × bear + 0.5 × base + 0.25 × bull`. The upside is `headline / price − 1`.
5. **12-month target** = base fair value × (1 + k_e) − expected dividends per share.
6. **Range** = the lower and higher of the raw DCF value and the multiples value. The raw legs are always shown separately.

VALUATION CONFIDENCE (0-100)

```
confidence = 100 × data quality × model stability × agreement between DCF and multiples
```

- Data quality starts from the fundamentals' DQ score and is cut for share-count or currency problems, short history (fewer than 5 years), an old fiscal year (more than 15 months) and a stale price.
- Model stability is cut when the terminal value exceeds 75-95% of EV, FCFF is negative in several years, margins are volatile, the company is a loss-maker or deep loss-maker, the subsector is cyclical, the discount rate hit its floor or first-year growth exceeds 30%.
- Agreement falls as `|ln(DCF / Multiples)|` grows (reason "dcf multiples disagree" beyond a factor of 2).
- Confidence is capped at 35 when the upside exceeds +100% or is below −90%.

Each cut adds a reason to the list under the meter (for example "model: cyclical normalized margin"). **Below 40 the fair value is indicative only.** In scoring, valuation inputs derived from the DCF are ignored when confidence is below 40 and weighted by confidence / 100 above it; the Value factor then relies on the multiples (contract `forecast_confidence`). The meter is green from 60, amber from 40, red below 40.

COST OF CAPITAL

From `backend/app/analytics/cost_of_capital.py`, point-in-time:

- **Risk-free rate**: the 10-year government yield of the reporting currency (for example FRED DGS10 for USD, the ECB AAA curve for EUR, MoF JGB for JPY). Where no series is available a dated fallback table is used (flag `rf fallback`). For emerging-market local yields the sovereign default spread is removed to avoid double counting with the country premium.
- **Beta**: weekly log returns against the local benchmark over 3 years (at least 52 weeks), Blume-adjusted `0.67 × β + 0.33`, clipped to 0.4-2.0, then shrunk 50% toward the subsector median. Shown as "adj. / raw".
- **Equity risk premium**: the panel shows the contract ERP of **5%** plus a country risk premium from a country table (in the examples: 0% for the US and the Netherlands, 0.7% for Japan, 3.5% for Brazil).
- **Cost of debt**: interest expense / average debt, floored at the risk-free rate plus a synthetic-rating spread and clipped to rf + 0.5%...rf + 8%, after tax. Tax: 3-year median effective rate, clipped to 0-40%.
- **WACC**: market-value equity weight, book debt; clipped to 5%...15% (upper bound widened for high-rate currencies). Cost of equity is floored at `max(rf + 0.6 × (ERP + CRP), 5%)`.
- **Terminal growth**: `min(long-run nominal GDP proxy − 0.5%, rf, 3% + EM excess)`, clipped to 0.5%...7%.

Market-calibrated ERP in the valuation. The valuation itself does **not** use the 5% ERP. With the default `erp_mode=implied`, the forecast job solves, for each region bucket (North America, Europe, Japan, China / Hong Kong, other) and model family (FCFF or excess return), the ERP between 1% and 10% at which the **median base-case upside of the bucket is zero** (at least 12 valued names; otherwise the region bucket, then a pooled value). The discount rates behind the fair value, the sensitivity grid and the scenario valuations therefore differ from the WACC shown in the Cost of capital panel. To value with the fixed 5% instead, run the forecast job with `--param erp_mode=fixed`.

SENSITIVITY GRID AND REVERSE DCF

- **Sensitivity**: the base-case value recomputed for WACC −2...+2 pp (1 pp steps) and terminal growth −1...+1 pp (0.5 pp steps) around the valuation's own WACC and g, then **scaled to the blended headline base value** so the centre cell equals the base fair value. Cells where WACC ≤ g + 0.5 pp are empty. Colour is the gap to the price (teal above, rose below).
- **Reverse DCF** (non-financials): the constant 5-year revenue CAGR that, at base-case margins, makes the DCF equal to the market EV (root search), then the terminal growth. It is compared with the historical 5-year revenue CAGR: more than 2 pp above reads "expectations look demanding", more than 2 pp below "expectations look undemanding". For banks the reverse calculation solves for an implied steady ROE instead, which the panel does not display.

Data sources and refresh

Inputs: standardised fundamentals (see Financials), prices, FX, government yields from official sources (FRED, ECB, MoF Japan, BoE, BoC, RBA, SNB, ChinaBond and others), subsector statistics. The `forecast` job runs in the nightly chain and also writes the cost-of-capital table used by the moat job. Issuers are skipped when the last fiscal year is more than two years old, fewer than two usable years exist, or price or share count is missing ("No forecast scenarios published for this company yet").

Validation status

Not formally validated. There is no backtest of the fair value or the 12-month target against realised returns. The headline is by construction anchored to the market price and calibrated so that the median company in each region is close to fair value; it is a relative, model-based valuation. Confidence is an internal quality score, not a calibrated probability. The tab states "market-anchored" in the Valuation subtitle and the `EE` mnemonic is labelled model-based, not consensus.

Correct usage and worked example

Apple (Figure 5A.15, valuation of 16 Oct 2026, price 343.46 USD).

- Raw DCF value 196.61 (−43% versus price); peer multiples value 275.31; range 196.61-275.31.
- Scenario values after blending and shrinkage: bear 142.62, base 258.78, bull 459.22. Headline fair value `0.25 × 142.62 + 0.5 × 258.78 + 0.25 × 459.22 = 279.85` (−19% versus price). 12-month target 276.46 (−20%).
- Valuation confidence 77/100 with no reasons listed; terminal value 67% of EV (below the 85% flag).
- Cost of capital panel: WACC 10.63%, cost of equity 10.71%, beta 1.12 adjusted / 1.04 raw (regression), risk-free 5.09% (FRED DGS10), ERP 5.00%. The sensitivity grid, however, is centred on WACC 7.3% and g 3.0% (centre cell 258.78 = the base value): the North American implied ERP on that date was much lower than 5% (the valuation's cost of equity of 7.28% implies about 1.9%).
- Reverse DCF: the price implies 15.0% revenue growth a year for five years, then 3.0%, against a 5-year track record of 6.1%: "expectations look demanding".

Reading: the market price requires growth well above Apple's history; the model's value sits 19% below the price with good confidence. This is a statement about what the price implies under the model, not a sell recommendation.

Figure 5A.19: Valuation panel for JPMorgan Chase (`/company/JPM?tab=forecast`), excess-return model, valuation as of 16 Oct 2026.

JPMorgan (bank model). The badge reads "Excess-return model" and the raw value is labelled "Raw model value": 243.65 (−28%); peer multiples (P/E and P/B) 235.62; headline 274.34 (−19% versus 339.72); 12-month target 284.25; confidence 85. The Cost of capital panel is "financial (equity-only)": WACC = cost of equity = 9.91%. The fan chart offers Net income, Residual income, Book value and Diluted EPS. The Reverse DCF shows "—" for implied growth because the bank model solves for an implied ROE that is not displayed.

Figure 5A.20: Cost of capital for JPMorgan Chase, equity-only.

Low-confidence examples. Both ASML and Petrobras have a valuation confidence below 40, so the meter turns red and reads "Below 40 — scoring down-weights this valuation", with the reasons listed underneath.

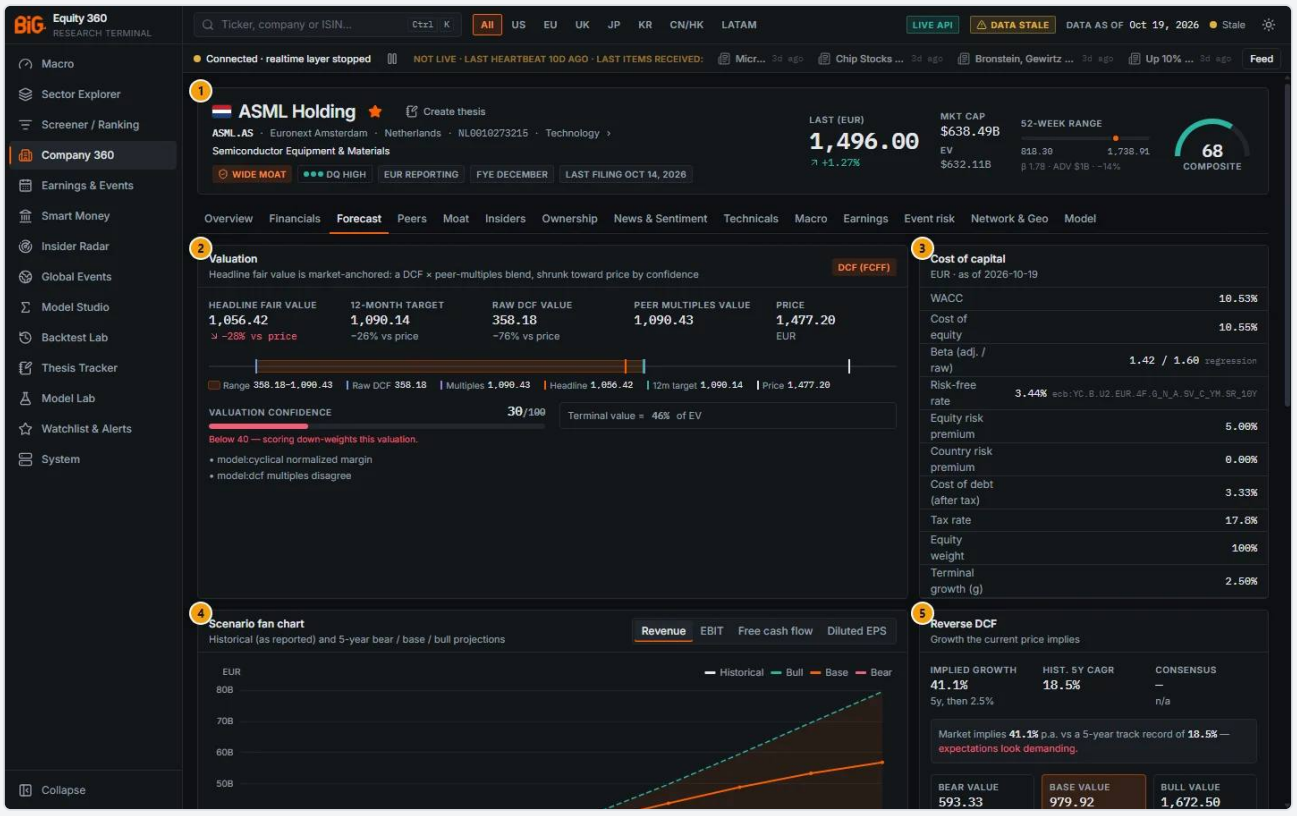


Figure 5A.21: Forecast tab for ASML (`/company/ASML.AS?tab=forecast`), valuation as of 16 Oct 2026.

#	Panel	What it shows
1	Company header	ASML Holding, EUR
2	Valuation	Headline 1,056.42 EUR (−28% vs 1,477.20); 12-month target 1,090.14; raw DCF 358.18 (−76%); peer multiples 1,090.43; confidence 30/100 with reasons "model:cyclical normalized margin" and "model:dcf multiples disagree"; terminal value 46% of EV
3	Cost of capital	WACC 10.53%, beta 1.42 adj. / 1.60 raw, risk-free 3.44% (ECB AAA 10-year), ERP 5.00%, CRP 0.00%, terminal growth 2.50%
4	Scenario fan chart	Revenue paths; the bull case diverges strongly from the base

#	Panel	What it shows
5	Reverse DCF	Implied growth 41.1% a year for 5 years, then 2.5%, against a 5-year record of 18.5%: "expectations look demanding"; bear 593.33, base 979.92, bull 1,672.50
6	Projected statements	Base case, EUR millions (below the fold)
7	Assumptions	Drivers by scenario (below the fold)
8	DCF sensitivity	WACC × terminal-growth grid (below the fold)

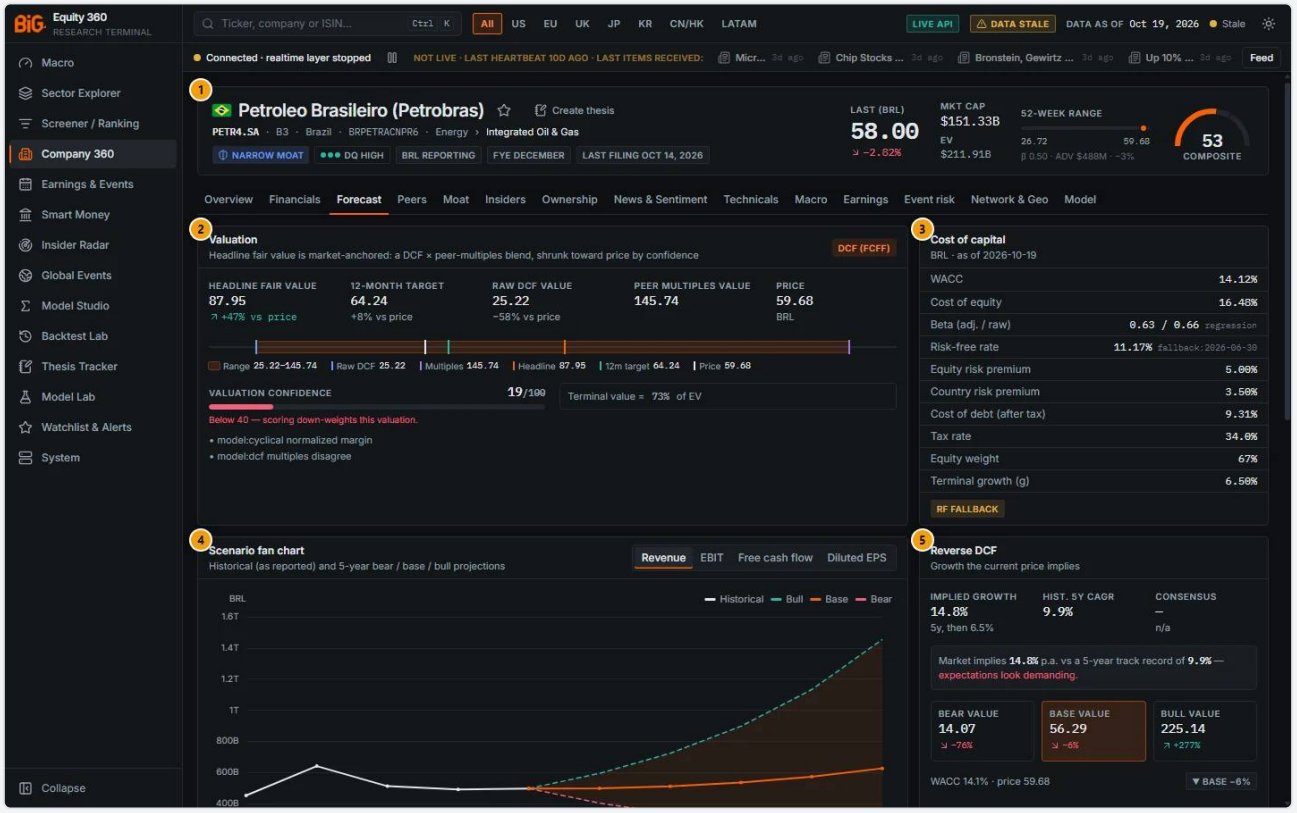


Figure 5A.22: Forecast tab for Petrobras (`/company/PETR4.SA?tab=forecast`), valuation as of 16 Oct 2026.

#	Panel	What it shows
1	Company header	Petrobras, BRL
2	Valuation	Headline 87.95 BRL (+47% vs 59.68); 12-month target 64.24 (+8%); raw DCF 25.22 (−58%); peer multiples 145.74; range 25.22-145.74; confidence 19/100 with the same two reasons; terminal value 73% of EV
3	Cost of capital	WACC 14.12%, cost of equity 16.48%, risk-free 11.17% from the dated fallback table ("fallback:2026-06-30", `RF FALLBACK` flag), ERP 5.00%, CRP 3.50%, terminal growth 6.50%
4	Scenario fan chart	Revenue: a wide bear-bull band
5	Reverse DCF	Implied growth 14.8% against a 5-year record of 9.9%: "expectations look demanding"; bear 14.07 (−76%), base 56.29 (−6%), bull 225.14 (+277%)
6	Projected statements	Base case, BRL millions (below the fold)
7	Assumptions	Drivers by scenario (below the fold)
8	DCF sensitivity	WACC × terminal-growth grid (below the fold)

Read them together. For ASML the raw DCF (358) and the multiples value (1,090) are a factor of three apart: the cyclical normalisation of margins and a high discount rate depress the DCF while peers trade on high multiples. The low confidence pulls the headline toward the multiples and the price. For Petrobras the headline of +47% looks attractive, but it is driven by a bull case at +277% while the base value is −6%. With confidence 19, a fallback risk-free rate and 73% of the value in the terminal value, the fair value is indicative only and is excluded from scoring. Note also the skew effect for Toyota (7203.T): base value 3,107 JPY (+0.4%) but bull 9,278, so the probability-weighted headline is 4,067 (+31%).

Limitations and pitfalls

- **The headline is a probability-weighted scenario value**, so a very wide bull case can lift it well above the base value (Toyota). Look at the bear / base / bull boxes, not only the headline.
- **The Cost of capital panel shows the contract cost of capital (ERP 5%)**, while the valuation discounts at the market-implied ERP. The Reverse DCF footer ("WACC x%") also shows the panel WACC. Use the sensitivity grid's centre row for the WACC actually used.
- **Market anchoring.** The implied ERP makes the median stock in each region fairly valued by design and the credibility shrink pulls every value toward the price. Upside is relative to the region, not an absolute statement about market level.
- **Confidence below 40**: indicative only, and excluded from scoring. The meter text says "scoring down-weights this valuation"; in fact, below 40 the DCF-derived input is dropped entirely.
- **Not consensus.** "Consensus" in the Reverse DCF is always n/a.
- **Projection mechanics are generic.** Company guidance, M&A, buyback plans and regulatory changes are not modelled. Cyclical companies are valued on normalised margins.
- Terminal value above 85% of EV means the value rests mostly on the terminal assumption.
- The Reverse DCF verdict line compares implied with historical growth even when the implied growth is missing (banks); ignore the verdict there.

Related mnemonics and BQL fields

`EE` . BQL: `fair_value` (probability-weighted DCF × multiples fair value, model-based), `dcf_upside` , `implied_cagr` (reverse-DCF implied 5-year revenue CAGR), `coc.wacc` , `coc.cost_of_equity` , `coc.beta` , `coc.rf` , `coc.erp` , `coc.crp` , `coc.terminal_growth` , `coc.flags` . Related: Earnings for model-based revenue and EPS expectations.

Peers

Route: `/company/:id?tab=peers` · **Menu:** Company 360 › Peers · **Mnemonics:** `RV` , `PEERS` · **Refresh:** nightly chain (scoring); peer-group anomaly nightly, model weekly

Purpose

The Peers tab compares the company with the companies most like it on the metrics that matter for its business model. It ranks the company on each metric, plots it against peers on any two metrics, compares composite scores by size and, at the bottom, shows how unusual the latest joint move of its peer group was (see Peer-group joint anomaly).

Screenshot

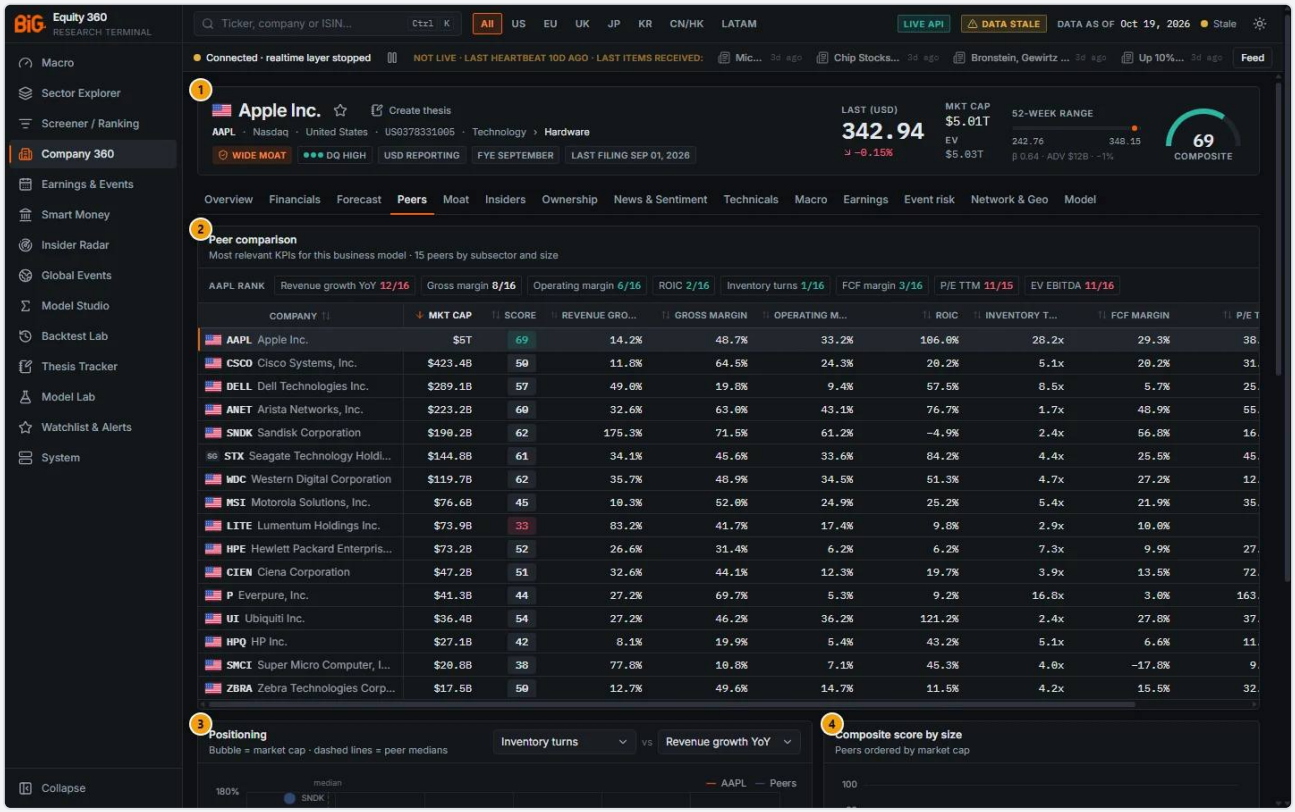


Figure 5A.23: Peers tab for Apple (`/company/AAPL?tab=peers`), data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	See Company header
2	Peer comparison	"Most relevant KPIs for this business model · 15 peers by subsector and size": a rank strip (`AAPL rank`) and a sortable table of market cap, composite score and each KPI
3	Positioning	Bubble chart of two chosen metrics; bubble size = market cap, dashed lines = peer medians
4	Composite score by size	Composite of each peer as bars, ordered by market cap; the company in orange
5	Peer-group joint anomaly (this company's group)	See Peer-group joint anomaly (below the fold)

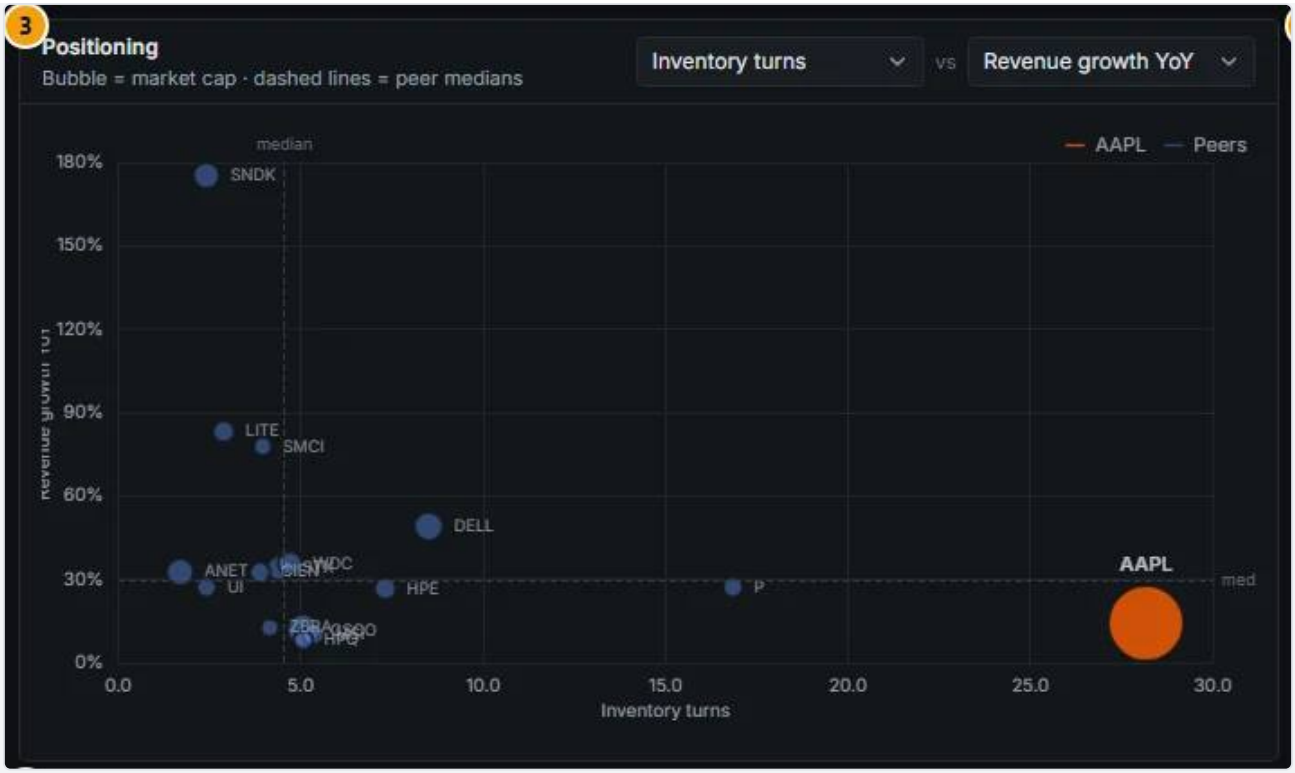


Figure 5A.24: Positioning, Apple: inventory turns against revenue growth (panel 3).

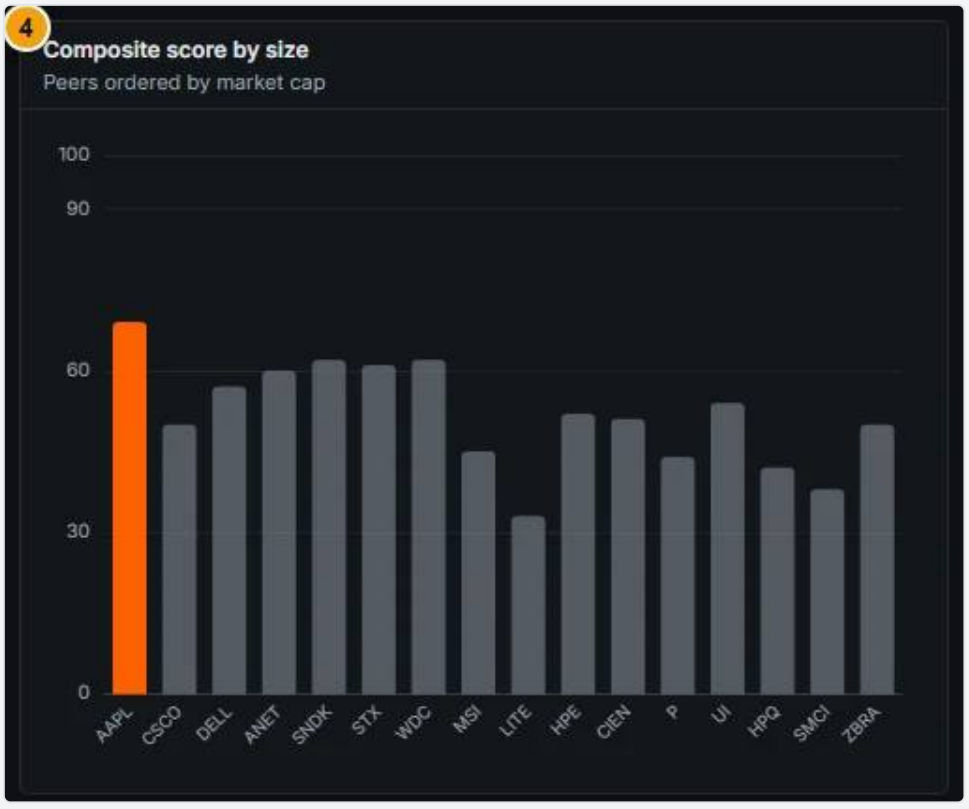


Figure 5A.25: Composite score by size, Apple's peers (panel 4).

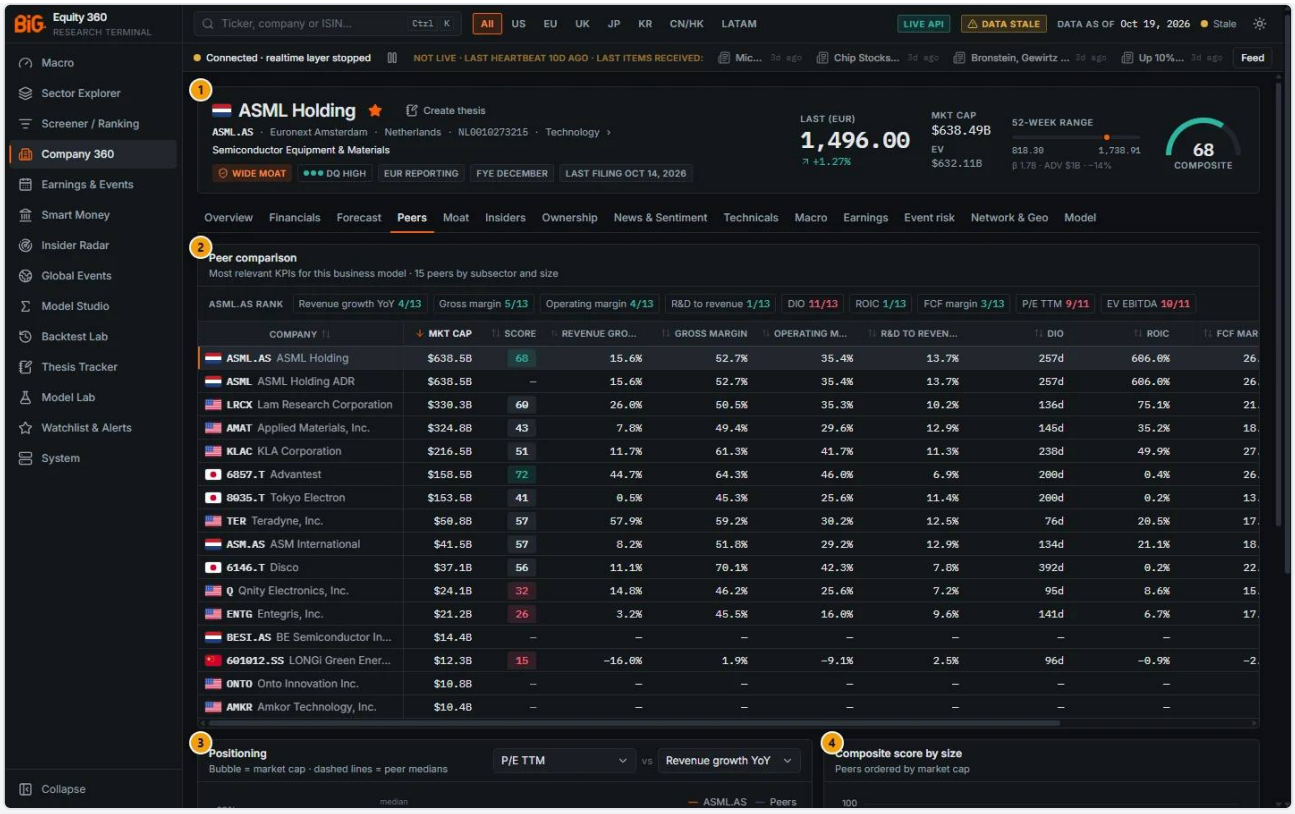


Figure 5A.26: Peers tab for ASML (`/company/ASML.AS?tab=peers`), global semiconductor-equipment peers, data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	ASML Holding, Euronext Amsterdam
2	Peer comparison	Semiconductor-equipment KPIs (revenue growth, gross and operating margin, R&D to revenue, DIO, ROIC, FCF margin, P/E, EV/EBITDA) against global peers
3	Positioning	P/E TTM against revenue growth
4	Composite score by size	Peers by market cap
5	Peer-group joint anomaly	ASML's peer groups (below the fold)

Controls

Control	What it does
Peer group (select, panel 2)	Appears only when a Model Studio peer set contains the company: "Subsector peers (default)" or "Peer set: (name)". With a peer set, the comparison uses the company's cluster in that set (or the whole set)
Rank strip	`rank/of` per metric, green in the top third and red in the bottom third, direction-aware
Table header click	Sorts by that column (default: market cap, descending); empty values sort last. Header hover gives the metric's formula and direction
Table row click	Opens that peer's Company 360 (the company's own row is highlighted and not clickable)
`X metric` / `Y metric`	Choose the two scatter axes from the KPI list (default: the first multiple against the first "higher is better" percentage)
Scatter and bar hover	Ticker, both metric values, market cap or composite

Methodology

- **Peer group.** The level used by scoring for this company (the most common `peer_group_level` of its metric scores): subsector × peer region × size bucket, then subsector × peer region, subsector (global), sector × region, sector (global), the first with at least 8 companies. The table shows the company plus the 15 largest peers by market cap;

the peer count in the API covers the whole group. With a Model Studio peer set (`peer_set_id`), the group is the company's cluster in that set.

- **Metrics.** The `peer_metrics` list of the company's KPI set (`contracts/sector_kpis.yaml`). Defaults: revenue growth, gross margin, operating margin, ROIC, FCF margin, P/E, EV/EBITDA, FCF yield.
- **Rank.** `1 + number of displayed peers strictly better` on the metric (direction-aware), out of the peers with a value.
- **Scatter medians.** Medians of the plotted points.

Data sources and refresh

The serving snapshot's latest metric values (from the `valuation_daily` and ratio jobs) and composite scores (`scoring`). Model Studio peer sets are user-defined (see Model Studio).

Validation status

Not a model; descriptive comparison. For the joint-anomaly panel see Peer-group joint anomaly.

Correct usage and worked example

Apple is compared within "hardware networking, North America, large caps" (19 companies in the group, 15 shown). The rank strip reads: revenue growth 12/16, gross margin 8/16, operating margin 6/16, ROIC 2/16, inventory turns 1/16, FCF margin 3/16, P/E TTM 11/15, EV/EBITDA 11/16. So Apple is among the best on capital efficiency and cash conversion, mid-pack on margins, and expensive and slower-growing than the group. In the composite-by-size chart, Apple (69) is among the highest scores of the large peers.

For ASML the subsector-region group is too small, so the comparison is global (23 semiconductor-equipment companies): Lam Research, Applied Materials, KLA, Advantest, Tokyo Electron and others.

Limitations and pitfalls

- The subtitle always reads "by subsector and size", even when scoring had to widen the group (ASML: global subsector). The `level` is in the API response (`peer_group_key`), not on screen.
- **Cross-listings can appear as peers.** ASML's own ADR (ASML) appears in its peer table with identical fundamentals. Ignore such rows when ranking.
- Peers without data show "—" and are excluded from the rank denominator, so `of` differs by metric.
- The table is capped at the 15 largest peers; small peers may be missing from view but are in the medians.
- Sanity-check outliers: an implausible ratio (for example an ROIC of several hundred percent, or near zero for a large profitable company) usually reflects a data or unit issue.

Related mnemonics and BQL fields

`RV` , `PEERS` (with a security: this tab; without: the Sector Explorer). BQL: the KPI fields (`roic` , `gross_margin` , `pe_ttm` , `ev_ebitda` , `kpi.*`), `composite` . See Sector Explorer.

Peer-group joint anomaly

Route: `/company/:id?tab=peers` (bottom panel) · **Menu:** Company 360 › Peers · **Mnemonics:** `RV` , `PEERS` · **Refresh:** nightly scoring (`conformal_peer_groups`); model refit weekly (`conformal_peer_groups_fit`); group definitions rebuilt every 90 days

Purpose

This panel answers "was the latest one-day move of this company's peer group unusual, given how these stocks normally move together?", and if so, which members drove it and which one broke away from the rest. It also gives

calibrated next-session bands for a pair spread or basket of group members. It is a **descriptive, model-based** measure. It is **not a trading signal**: the badge on the panel says "descriptive · model-based · not a trading signal".

Screenshot

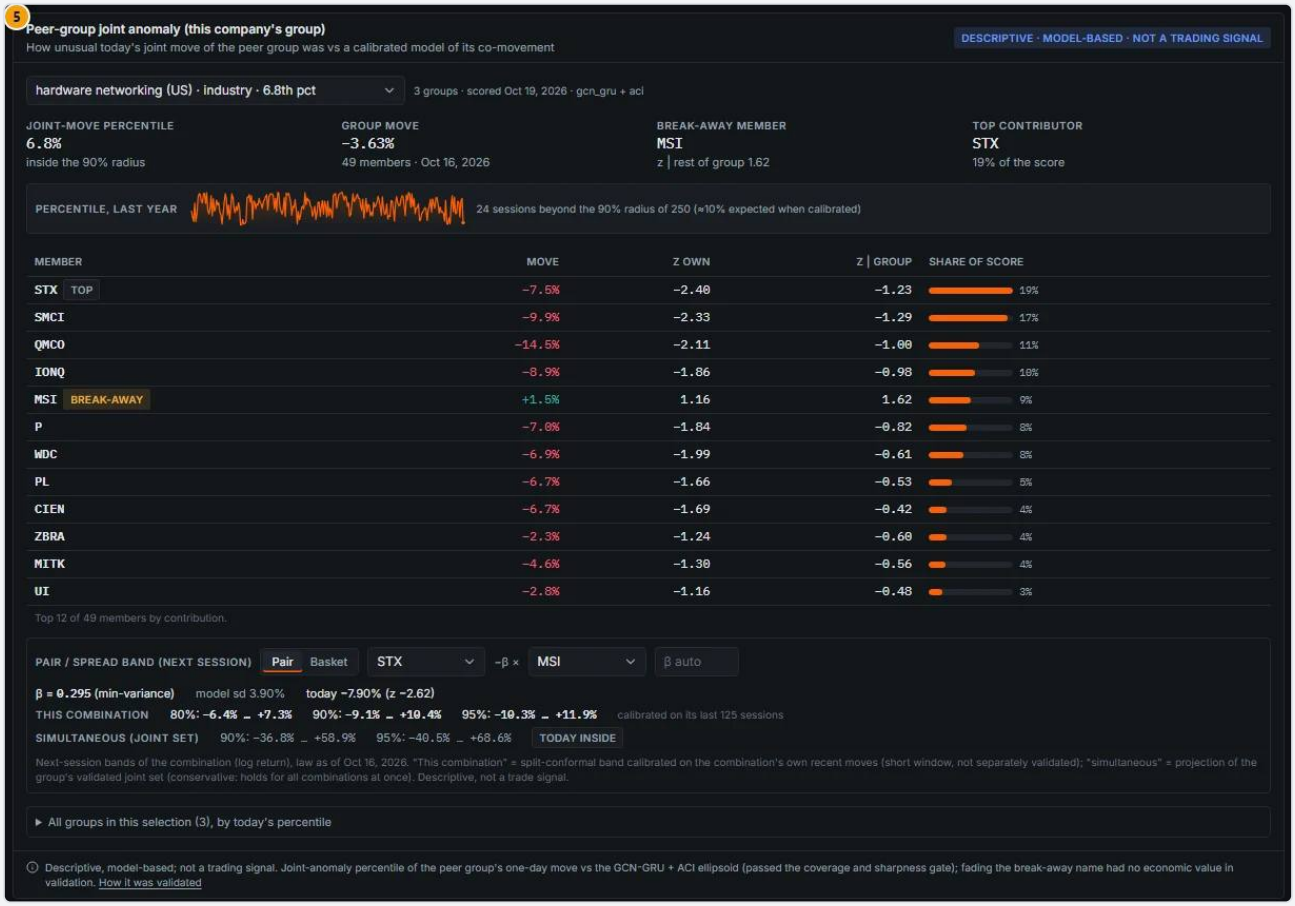


Figure 5A.27: Peer-group joint anomaly for Apple's industry group "hardware networking (US)", scored 16 Oct 2026 (snapshot 19 Oct 2026).

Element	What it shows
Peer group select	The groups that contain the company, each with kind (industry or supply chain) and today's percentile; the industry group is selected first
Status line	Number of groups, scoring date, filter and radius ("gcn_gru + aci")
Joint-move percentile	Calibrated percentile of today's joint-move score, with "inside the 90% radius", "beyond the 90% radius" or "beyond the 95% radius"
Group move	Average one-day move of the members, number of members and date
Break-away member	The member whose move was least explained by the others, with its `z | rest of group`
Top contributor	The member with the largest share of the joint score
Percentile, last year	Sparkline of the daily percentile and the count of sessions beyond the 90% radius ("≈10% expected when calibrated")
Member table	Top 12 members by contribution: move, `z own`, `z | group`, share of score; badges `top` and `break-away`
Pair / spread band (next session)	Calibrated band for `long − β × short` or a weighted basket
All groups in this selection	Expandable list of every group containing the company, by today's percentile
Footer note	The model-based label and the link "How it was validated"

Controls

Control	What it does
Peer group	Switches the detail to another group containing the company
All groups (expandable)	Lists the groups; click one to select it
Member ticker	Opens that member's page
Break-away / Top contributor ticker	Opens that member's page
`Pair / Basket`	Band tool mode
Long leg / Short leg	Pair members (defaults: the first member and the break-away member)
`β auto`	Hedge ratio; blank = minimum-variance β under the predicted covariance
Basket weights	Free text `TICKER:weight`, comma-separated (for example `AAA:0.5, BBB:0.5, CCC:-1`)
How it was validated	Opens the Conformal validation page (see System)

Methodology

From `backend/app/conformal/` and the contract description of `GET /conformal/peer-groups` and `GET /conformal/pair-band`.

- **Groups.** Each group holds 20-50 liquid names of one listing region, either an **industry** group (the most liquid names of one subsector) or a **supply-chain** group (names linked around a company in the disclosed supply-chain network). A company belongs to its industry group and to any supply-chain groups it is part of.
- **Model.** For each group a graph-native learned filter (**GCN-GRU**, shared across groups) predicts, at each close, the next session's mean vector μ and covariance Σ of the members' log returns (low rank plus diagonal). The fallback filter is an EWMA low-rank model.
- **Score.** The squared Mahalanobis distance of the realised one-day returns from that predicted law: $R = (r - \mu)^\top \Sigma^{-1} (r - \mu)$.
- **Calibration.** The score is compared with the last 500 realised scores of the same group (rolling window). The **joint-move percentile** is the rank of today's score in that window. The **90% and 95% radii** are conformal quantiles adjusted online by **ACI** (adaptive conformal inference, $\gamma = 0.005$): when recent sets miss too often the radius widens, when they over-cover it narrows. "Beyond the 90% radius" means today's score exceeded the 90% radius.
- **Attribution.** $\text{contrib}_i = r_i \times (\Sigma^{-1} r)_i$ splits the score exactly across members ("share of score"; negative = the member offset the others). $z\ \text{own} = r_i / \sqrt{\Sigma_{ii}}$ is the member's own standardised move. $z \mid \text{group} = (\Sigma^{-1} r)_i / \sqrt{(\Sigma^{-1})_{ii}}$ is the move **given all the other members**, i.e. how far it broke away from what the rest of the group implied. The **break-away member** has the largest $|z \mid \text{group}|$; the **top contributor** the largest positive contribution. $|z \mid \text{group}| \geq 2.5$ is highlighted.
- **Pair / spread band.** For weights a (pair: $+1$ and $-\beta$), the "simultaneous (joint set)" band is $a'\mu \pm \sqrt{(q \times a'\Sigma a)}$ with the group's 90% / 95% radius q. It is the projection of the validated joint set, so it is **conservative**: it holds for all combinations at once. "This combination" is a split-conformal band calibrated on the combination's own standardised moves over about 125 sessions (80 / 90 / 95%); it is narrower but was **not separately validated**. "Today" shows the realised combination move and its z against the band emitted at the previous close, and whether it was inside the 90% / 95% joint band. All bands are in log-return units for the next session.

Data sources and refresh

Daily adjusted closes of the members; supply-chain links from the network data (from 2023, point-in-time). Scores are produced nightly (`conformal_peer_groups`), the GCN-GRU is refitted weekly on a block ending 520 sessions before the as-of date (`conformal_peer_groups_fit`), and group definitions are rebuilt every 90 days. Realised coverage is audited nightly (`conformal_coverage`) and shown on System › Data Health ("Conformal coverage (model-based sets)") and the Conformal validation page.

Validation status

From `docs/CONFORMAL_VALIDATION.md`, section B and the Verdict (walk-forward, 2,955 group-years, 707,115 group-days, test years 2010-2026):

Criterion	Result
Coverage within ±1.5 pp	**PASS**: 90.4% at the 90% target, 95.2% at 95% (GCN-GRU + ACI)
No regime more than 5 pp under target	**PASS**, borderline: worst is Covid (15 Feb-30 Jun 2020) at 85.3% at 90% (−4.7 pp); 2022 90.9%; VIX > 25 90.1%
Sharper than baselines at matched coverage	**PASS**: w_proj −10.1% versus EWMA, −9.1% versus static Ledoit-Wolf, −12.4% versus a PCA factor model, −4.7% versus GARCH-DCC
Economic value: break-away mean reversion	**FAIL (null)**: fading the break-away member (group beyond its 90% radius and `abs(z | group) ≥ 2.5` or 3, hold 5 days, long-short, costs on) had Sharpe −0.09 / −0.15; versus a naive fade the difference was +0.3 to +2.1% a year with t ≤ 1.17, not significant
Gate	**PASS as a calibrated descriptive measure** (anomaly percentile, pair / basket bands). **Not a trading signal.**

Per-member coverage of the projected bands is 99.9% in validation: the simultaneous pair bands are very conservative. The app shows the panel with the label "Descriptive, model-based; not a trading signal" and the API label "Coverage is marginal over history (all securities and days of the pool), not a statement about any single day".

> **Model-based:** the percentile and the bands are calibrated over history: about 10% of sessions fall beyond the 90% radius across all groups and days. That is not "a 10% chance today" for this group.

Correct usage and worked example

Apple, industry group "hardware networking (US)", scored 16 Oct 2026 (Figure 5A.27). Joint-move percentile 6.8%, inside the 90% radius: although the group fell on the day (average move −3.63% over 49 members), the move was entirely ordinary given how these stocks co-move: a broad, correlated decline. The top contributor was STX (19% of the score, −7.5%, `z own` −2.40). The break-away member was MSI (+1.5% while the group fell, `z | group` +1.62, below the 2.5 highlight). Over the last year, 24 of 250 sessions were beyond the 90% radius (9.6%, close to the 10% expected when calibrated). Apple belongs to two other groups: the supply-chain group "links around AAPL (US)" (27th percentile) and "links around ADSK (US)" (69th percentile).

Pair band. STX − β × MSI, β = 0.295 (minimum variance): model sd 3.90%; today −7.90% (z −2.62). "This combination" 90% band −9.1% … +10.4%; the simultaneous 90% band −36.8% … +58.9%, and today is inside. Read: the STX-MSI spread moved about 2.6 standard deviations, within the combination's own 90% band and far inside the conservative joint band.

Good practice. Use the percentile to ask "is something group-specific happening?" (for example a sector-wide shock or a single member's news). A high percentile with a single dominant contributor points to stock-specific news; a high percentile with spread-out contributions points to a group-wide re-pricing. Then check News & Sentiment and Event risk.

Limitations and pitfalls

- **Not a trading signal.** Fading the break-away name had no economic value after costs in validation.
- **Calibrated over history**, marginal over time and groups; per-regime coverage is reported, not guaranteed. In a sudden crash (Covid) coverage fell to 85.3%.
- The "this combination" band is short-window and not separately validated; the simultaneous band is conservative.
- Group membership and the universe are current (survivorship); supply-chain edges only exist from 2023.
- The scoring date can lag the snapshot date by a session or two (16 Oct versus 19 Oct here).
- Companies that are not in a scored group see "This selection is not part of a scored peer group".
- The member table highlights the company's own row and defaults the pair's long leg to it only when the page was opened with the internal security id; opened by ticker, the defaults fall back to the first member.

Related mnemonics and BQL fields

`RV` , `PEERS` . BQL: the `cf.*` conformal fields (for the single-stock calibrated range, see Technicals). Validation page and coverage audit: System. Other peer-group views (by sector or region) are in the Sector Explorer.

Moat

Route: `/company/:id?tab=moat` · **Menu:** Company 360 › Moat · **Mnemonics:** none · **Refresh:** weekly chain (`moat` job)

Purpose

The Moat tab classifies the company's durable competitive advantage as **Wide**, **Narrow**, **None** or **Insufficient data**, and shows the evidence behind the rating: returns on capital against the cost of capital over up to ten years, margin level and stability, margin premium over peers, share of subsector revenue, intangibles intensity, revenue stability and customer lock-in. Use it to judge whether high returns are likely to persist.

Screenshot

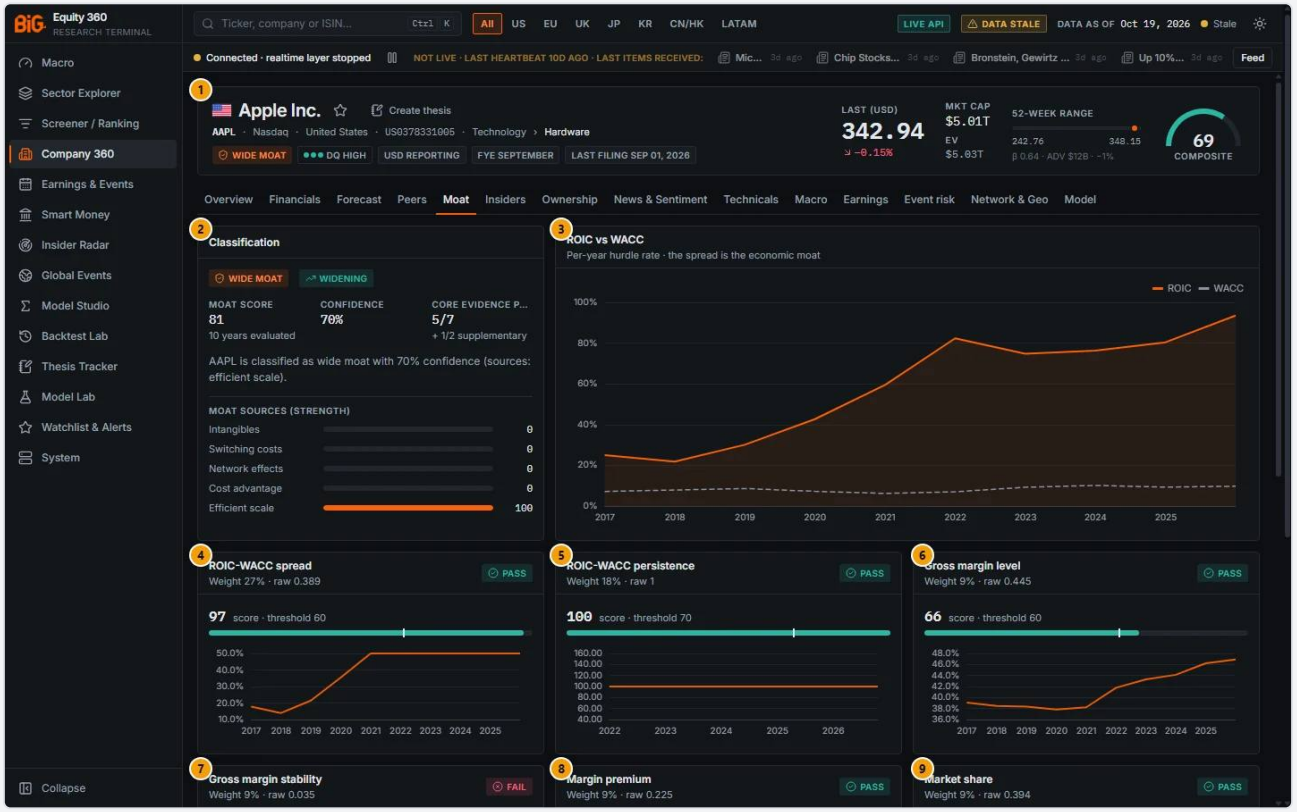


Figure 5A.28: Moat tab for Apple (`/company/AAPL?tab=moat`), moat as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	See Company header
2	Classification	Moat badge, trend (widening / stable / narrowing), moat score, confidence, core evidence passed, summary sentence and moat sources with strength
3	ROIC vs WACC	Yearly ROIC against the per-year WACC (or hurdle)
4	ROIC-WACC spread	Component card: score, threshold 60, pass / fail, raw value, yearly history (weight 27%)
5	ROIC-WACC persistence	Share of years with ROIC above WACC (threshold 70, weight 18%)
6	Gross margin level	(threshold 60, weight 9%)
7	Gross margin stability	(threshold 60, weight 9%)
8	Margin premium	Operating margin above the subsector median (threshold 60, weight 9%)
9	Market share	Share of subsector revenue (threshold 60, weight 9%)
10	Intangibles intensity	R&D, intangible amortisation and capitalised intangibles over revenue (threshold 60, weight 9%; below the fold)
11	Revenue stability	Supplementary (threshold 60, weight 5%; below the fold)

#	Panel	What it shows
12	Customer lock-in	Supplementary (threshold 60, weight 4%; below the fold)
13	Market share	Time series of the share of subsector revenue (below the fold)

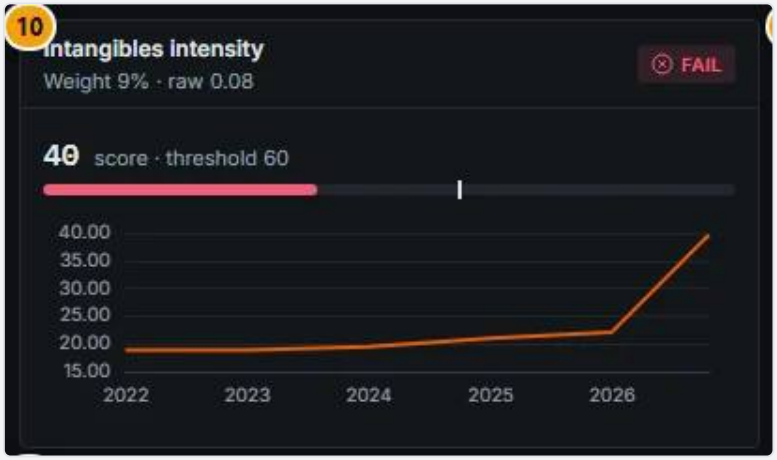


Figure 5A.29: Intangibles intensity, Apple (panel 10): score 40, below the threshold of 60.

Figure 5A.30: Revenue stability, supplementary component, Apple (panel 11).

Figure 5A.31: Customer lock-in, supplementary component, Apple (panel 12).

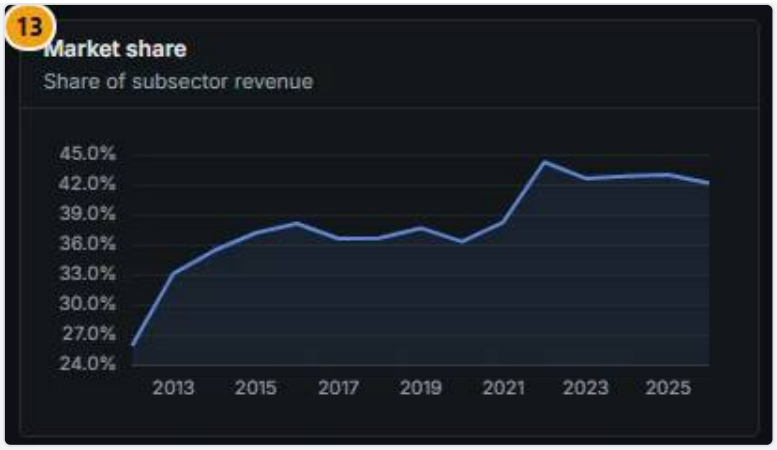


Figure 5A.32: Share of subsector revenue over time, Apple (panel 13).

Controls

Control	What it does
Chart hover	Values per year (ROIC, WACC, hurdle; component raw values)
`financial model` badge hover	"Financial-company variant: deposit franchise / underwriting economics replace ROIC-based components"
`supplementary` badge hover	"Supplementary component — lower weight, shown for context"
Threshold tick	The vertical tick on each score bar marks the pass threshold

Methodology

From `backend/app/analytics/moat.py` and `contracts/scoring.yaml` (`moat`), on the point-in-time fiscal-year panel of the last 10 years.

Component	Weight	Score (0-100)	Pass at
ROIC-WACC spread	0.30	Mean of `ROIC − WACC` : 0.5 × percentile within sector + 0.5 × absolute map (−5% → 0, +15% → 100). `ROIC = EBIT × (1 − t) / average invested capital` ; per-year WACC = today's WACC shifted by the change in the currency's 10-year yield	60
ROIC-WACC persistence	0.20	`100 × share of years with ROIC > WACC`	70
Gross margin level	0.10	5-year mean gross margin: 0.5 × subsector percentile + 0.5 × absolute anchor (60% margin → 100)	60
Gross margin stability	0.10	−stdev of gross margin: 0.5 × subsector percentile + 0.5 × absolute anchor (0 → 100, 8 pp → 0)	60
Margin premium	0.10	Mean of `EBIT margin − subsector median that year` , global percentile	60
Market share	0.10	Revenue (USD) / total subsector revenue of the app's universe: 0.5 × subsector percentile + 0.5 × `min(100, 100 × share × n_peers / 3)`	60
Intangibles intensity	0.10	`(R&D + intangible amortisation + capitalised intangibles) / revenue` , 3-year mean, sector percentile	60
Revenue stability (supplementary)	0.06	−stdev of year-on-year revenue growth, global percentile (switching-cost proxy)	60
Customer lock-in (supplementary)	0.04	Deferred revenue / revenue, 3-year mean, global percentile (switching-cost proxy; null if not reported)	60

The card shows each weight after renormalisation (27%, 18%, 9% … 5%, 4%). Percentile groups fall back from subsector to sector to all companies when a group has fewer than 5 members.

```
moat score = weight-renormalised mean of the available component scores
Wide         : score ≥ 75  and persistence ≥ 0.90  and ≥ 7 years
Narrow       : score ≥ 55  and persistence ≥ 0.60
None         : otherwise
Insufficient : fewer than 5 years, or component coverage < 0.6
confidence   = 0.4 × min(years / 10, 1) + 0.3 × coverage + 0.3 × (1 − normalised stdev of the yearly spreads)
```

The thresholds were raised on live data on 11 Oct 2026 (from 70 / 0.8 and 50 / 0.6) so that Wide stays a minority of the large-cap universe. **Commodity producers** (mining, energy and commodity processors) are **capped at Narrow**: a Wide result is downgraded with the reason "commodity cyclical", because a commodity boom alone should not earn a wide moat.

Financial-company variant (banks, insurers, consumer finance, holdings, levered broker-dealers; `financial model` badge): ROIC − WACC is replaced by ROE − cost of equity, the gross-margin components use the pre-tax margin and the margin premium uses the net margin. The card titles stay the same ("ROIC-WACC spread", "Gross margin level"), and the chart title becomes "ROIC vs hurdle" when a per-year hurdle differs from WACC.

Sources of moat (Intangibles, Switching costs, Network effects, Cost advantage, Efficient scale) are mapped from the component evidence with a 0-100 strength; for example scale and market share map to "Efficient scale", gross-margin stability to "Switching costs".

Trend compares the moat score now with up to four records earlier: more than +3 = widening, less than −3 = narrowing, otherwise stable. **Core evidence passed** counts passed core components (and, in small type, the supplementary ones).

Data sources and refresh

Standardised fundamentals (10 years, point-in-time), cost of capital (shared with the forecast job), subsector statistics from the app's universe. The `moat` job runs in the weekly chain; the 19 Oct 2026 audit noted the moat table was then dated 9 Oct because the weekly chain had not run (finding F34). The moat score is also the input of the Moat factor in scoring.

Validation status

Not formally validated against future returns or future profitability. The classification is a transparent rules-based assessment with thresholds calibrated on the cross-section of live data (to keep Wide a minority), not on outcomes. No model-based label is shown.

Correct usage and worked example

Apple (Figure 5A.28). Wide moat, score 81 (81.48), 10 years evaluated, confidence 70%, trend widening, 5 of 7 core components passed plus 1 of 2 supplementary. ROIC-WACC spread score 97 (mean spread +38.9 pp), persistence 100 (ROIC above WACC every year), gross margin level 66, margin premium 93, market share 100 (39% of the subsector revenue in the universe). Failed: gross margin stability (51) and intangibles intensity (40, 8% of revenue). Source: efficient scale.

Figure 5A.33: Moat classification for JPMorgan Chase (`/company/JPM?tab=moat`), financial-company variant.

JPMorgan (bank). Narrow moat with the `financial model` badge, confidence 91%, 5 of 7 core components passed. The score is displayed as 75 but is 74.65, just below the Wide bar of 75, so the class is Narrow even though persistence is 100%. Sources: efficient scale (100) and switching costs (68). Intangibles intensity has no score for the bank (shown as Fail).

SAP and Petrobras (API). SAP has a score of 76.7, above the Wide bar, but ROE persistence is 0.80 (ROIC above WACC in 8 of 10 years), below the 0.90 Wide requirement: Narrow. Petrobras is Narrow with a score of 64.6, persistence 0.71 and 7 years of history; as a commodity producer it could not be Wide in any case.

Use the cards to see *why* a rating is what it is: a Wide rating carried by scale and market share is different from one carried by margin stability and lock-in.

Limitations and pitfalls

- **Market share is the share of the subsector revenue within the app's universe**, not true market share.
- The ten-year window is shorter for many non-US issuers; fewer than 5 years gives "Insufficient data", which the header shows as **No moat**.
- Scores near a threshold are fragile (JPMorgan 74.65 against 75). Read the confidence and the components, not only the badge.
- A missing component shows as "Fail" on its card although it is simply not scored.
- The 5Y history of each component shows the **raw** metric, while pass / fail applies to the 0-100 score.
- The weekly cadence means the rating can lag a new annual report by up to a week (or more if the weekly chain does not run).

Related mnemonics and BQL fields

No mnemonic. BQL: `moat` (wide / narrow / none / insufficient_data), `moat_score` , `f.moat` , `roic` , `roic_mean_5y` , `coc.wacc` .

Insiders

Route: `/company/:id?tab=insiders` · **Menu:** Company 360 › Insiders · **Mnemonics:** none (cross-company view: Insider Radar) · **Refresh:** nightly chain (`sec_form4` , national filings, `insider_signals`); models weekly

Purpose

The Insiders tab shows what the company's directors and officers have been doing with their own shares: the full transaction ledger from Form 4 (or the national equivalent), open-market buys and sells on the price chart, cluster buying, an anomaly score per transaction and a rules-based insider score. Open-market purchases by several insiders, by the CEO or CFO, or after a drawdown are the classic signal of interest.

Screenshot

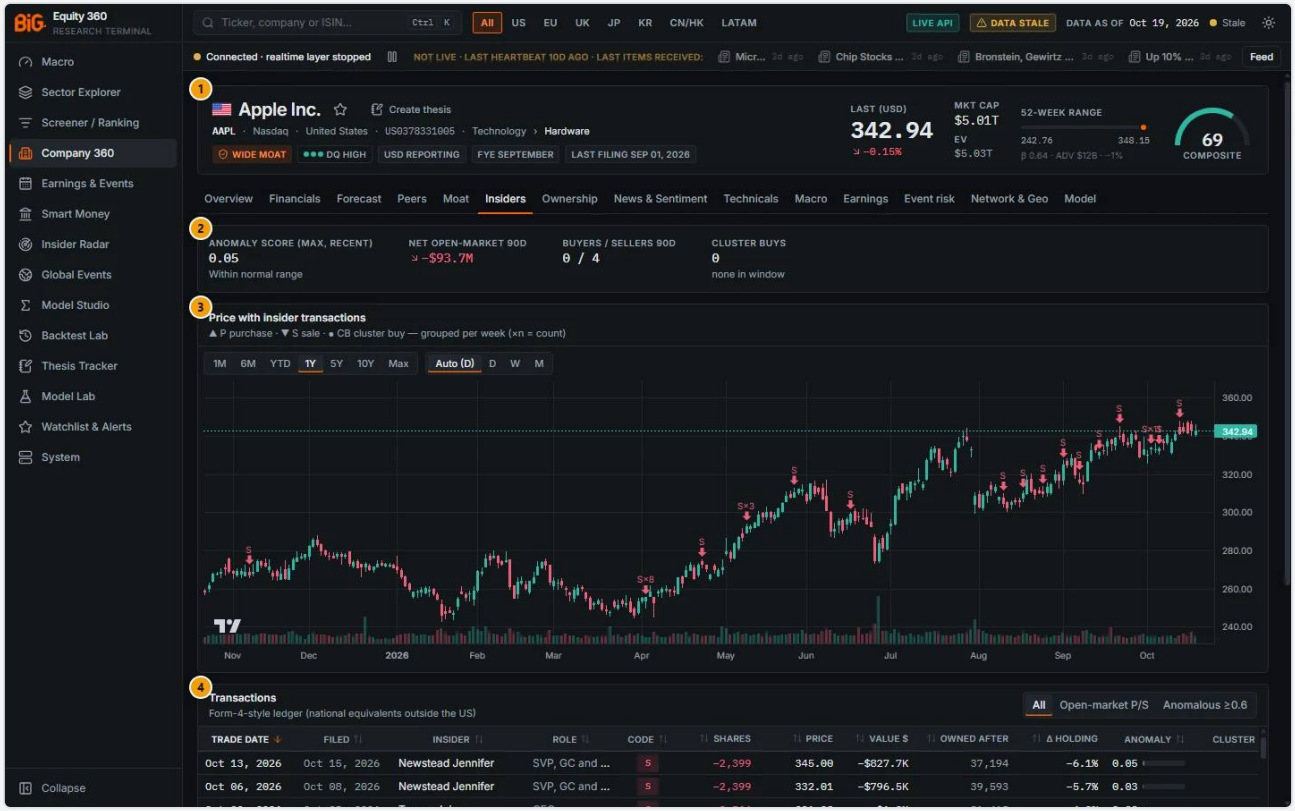


Figure 5A.34: Insiders tab for Apple (`/company/AAPL?tab=insiders`), data as of 19 Oct 2026 (signal as of 16 Oct 2026).

#	Panel	What it shows
1	Company header	See Company header
2	Summary stats	Anomaly score (max, recent) with "Within normal range" or "Unusual activity"; net open-market value over 90 days; buyers / sellers over 90 days; cluster buys
3	Price with insider transactions	Candles with markers: ▲ P purchase, ▼ S sale, ● CB cluster buy, grouped per week (`×n` = count)
4	Transactions	"Form-4-style ledger (national equivalents outside the US)": trade date, filed date, insider, role, code, shares, price, value, owned after, Δ holding, anomaly bar, cluster

Controls

Control	What it does
Price range / Bar interval	As on the Overview price chart; default 1Y daily candles. Markers are drawn on the daily view only
Marker hover	Kind and count for that week
`All` / `Open-market P/S` / `Anomalous ≥0.6`	Filters the ledger: every transaction type; open-market purchases and sales only; transactions with an anomaly score of at least 0.6
Column header click	Sorts the ledger (default: trade date, newest first)

Methodology

From `backend/app/analytics/insider.py` and `insider_models.py` .

- **Point in time.** A transaction is usable from its **filing date** (knowledge date), never from the trade date. Amended filings (4/A) replace the original from the amendment's filing date.
- **Ledger.** The tab loads two years of transactions of **all** types: purchases (P), sales (S), grants and awards (A), option exercises and conversions (M), tax withholding (F) and gifts (G). Sales, withholding and gifts are shown as negative share counts and values.
- **Signals use open-market trades only.** The stats, chart markers, cluster logic and scores use only **open-market buys (P) and sells (S)**; grants, exercises, withholding, gifts and derivative-table rows are excluded.
- **Per-transaction features.** USD value, role weight (CEO 3.0, Chair 2.75, CFO 2.5, President 2.25, COO 2.0, other officers 1.5, directors 1.25, 10% owners 0.75), 10b5-1 plan flag, change in holding, number of other insiders trading in the same direction within 30 days, first buy by this insider in two years, size against the insider's own past trades, the Cohen-Malloy-Pomorski **routine** flag (traded in the same calendar month in each of the three prior years), pre-filing price drawdown and run-up, volatility, liquidity and filing lag.
- **Anomaly score (0-1).** An **IsolationForest** on those features; the score is the empirical CDF of the training anomaly scores, so 1 = the most unusual transaction seen. It is unsupervised: it measures how strange a transaction is, not whether it predicts returns. The summary shows the maximum over the latest filing day in the last 90 days; "Unusual activity" from 0.75.
- **Clusters.** Open-market buys of one issuer are chained while consecutive trade dates are at most 30 days apart; a chain with at least 2 distinct insiders gets a cluster id (● CB markers, "Cluster buys" count). The **cluster-buy flag** in the insider signal (and the Overview alert) is stricter: at least 3 distinct buyers within 30 days.
- **Rules-based insider score (0-100).** Over the last 90 days each open-market trade contributes a weight that decays with a 45-day half-life:
  - buy: `role weight × (1 + 0.5 × cluster + 0.4 × first buy in 2 years + 0.4 × drawdown worse than −20%) × 0.5 if 10b5-1 × 0.6 if routine × ln(1 + value / $10k)`;
  - sell: `−role weight × 0.35 × (0.2 if 10b5-1 or routine) × (1 + 0.3 if the stock rose more than 30% in 6 months) × ln(1 + value / $10k)`;
  - `score = 50 + 50 × tanh(c / 2.5)` with `c = sign(raw) × ln(1 + |raw| / 5)`. 50 = no activity; one CEO buy of about $1M scores about 74, a large multi-insider cluster about 94, heavy selling about 10. This score is the Insider factor of the composite.
- **XGBoost model.** A gradient-boosted model of P(3-month excess return > 0) after insider filing events is trained weekly with purged, embargoed walk-forward validation. It is blended into the score (30%) only if it passes its out-of-sample promotion gate. It has not (see Validation status), so **only the rules-based score is used**.

Data sources and refresh

SEC Form 4 (US, "full" coverage) and national regulator filings: UK (FCA), Germany (BaFin), the Netherlands (AFM), Korea (OpenDART, when its key is set), Hong Kong (HKEX) and Brazil (CVM, monthly). The API labels coverage as full for the US and partial for Korea, the EU and the UK. Form 4 is ingested nightly; `insider_signals` runs nightly; the IsolationForest and XGBoost are refitted weekly (`insider_models_fit`).

Validation status

- **Rules-based insider score:** a transparent rule, not trained on returns; not formally validated as a stand-alone signal. Its rank-IC is computed as the benchmark in the model's walk-forward report.
- **Insider XGBoost: no out-of-sample edge.** The current model (version 20261020) reports, over 91 out-of-sample months and 170,131 events: 3-month AUC 0.493, pooled rank-IC −0.013, monthly rank-IC t-statistic −1.17; the promotion gate was **not passed** (`gate_passed = false`). Only the rules-based score is used in the app.
- **Anomaly score:** unsupervised descriptive measure; not validated as predictive.

Correct usage and worked example

Apple (Figure 5A.34, signal as of 16 Oct 2026). Anomaly score 0.05, "Within normal range"; net open-market value over 90 days −$93.7M; 0 buyers, 4 sellers; no cluster buys. The chart shows only ▼ S markers (sales, several weeks with `S×n`). In the two-year ledger there are 248 transactions: 73 sales, 106 option exercises, 37 grants, 25 tax-withholding rows and 7 gifts, and **no** open-market purchases. The rules-based score is 26, the Insider factor value on the Overview.

Read: routine executive selling, much of it linked to vesting and plans, with nothing unusual. Selling is a weak signal; buying, especially by several insiders or after a fall, is the informative event. Use the `Open-market P/S` filter to cut out

grants and exercises, and `Anomalous ≥0.6` to find the transactions worth reading in the filing.

Limitations and pitfalls

- **The default ledger includes grants, exercises and withholding.** They are not signals; filter to `Open-market P/S`.
- The anomaly score measures unusualness, not direction or predictive value.
- Non-US coverage is partial and lags (national registers publish with different delays; CVM monthly).
- The 19 Oct 2026 audit found Form 4 data stale by about a week at that time; check System › Data Health for freshness.
- Insider XGBoost has no out-of-sample edge, so do not read `xgb_prob_up` (available in the API) as a forecast.
- Planned (10b5-1) and routine trades are down-weighted but still appear in the ledger.

Related mnemonics and BQL fields

No company-tab mnemonic; the cross-company view is the Insider Radar. BQL: `insider_signal`, `f.insider`, `ins.anomaly_score`, `ins.anomaly`, `ins.cluster_buy`, `ins.cluster_buy_flag`, `sm.insider_signal`.

Ownership

Route: `/company/:id?tab=ownership` · **Menu:** Company 360 › Ownership · **Mnemonics:** `HDS`, `SI`, `SMRT` · **Refresh:** 13F weekly; short interest weekly; EU / UK shorts and 13D / 13G nightly; smart-money score nightly

Purpose

The Ownership tab shows who owns and who is shorting the company: institutional holders from SEC 13F filings with their latest changes, the ownership trend, short interest, insider transactions, EU / UK disclosed net short positions, 13D / 13G (activist and passive 5% holders) filings and a model-based smart-money score. Use it to judge crowding, flows and short pressure. Definitions follow `docs/SMART_MONEY.md`.

Screenshot

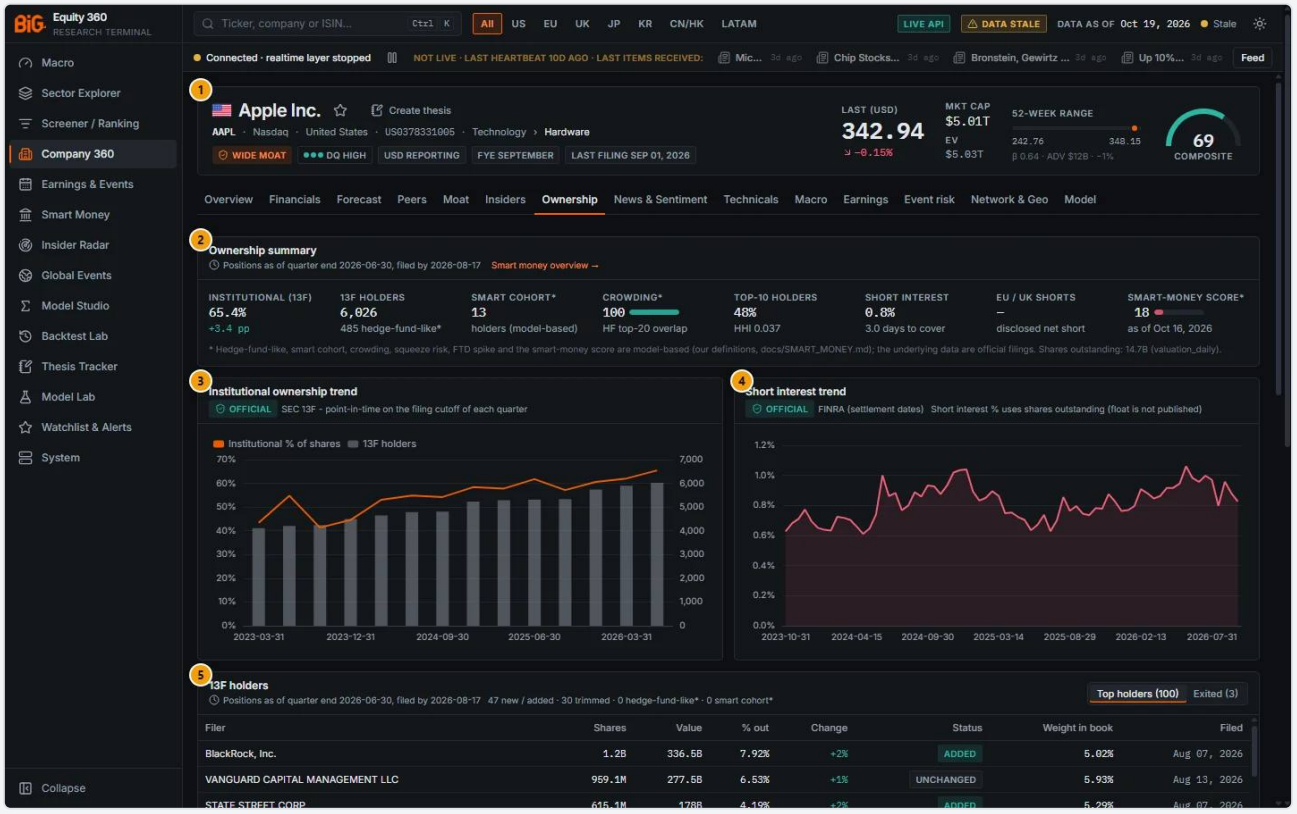


Figure 5A.35: Ownership tab for Apple (`/company/AAPL?tab=ownership`), 13F positions as of quarter end 30 Jun 2026, filed by 17 Aug 2026.

#	Panel	What it shows
1	Company header	See Company header
2	Ownership summary	Institutional (13F) % and change; 13F holders and hedge-fund-like*; smart cohort*; crowding*; top-10 holders and HHI; short interest and days to cover (with squeeze risk* when flagged); EU / UK shorts; smart-money score* and date. Link "Smart money overview →"
3	Institutional ownership trend	OFFICIAL, SEC 13F: institutional % of shares (line) and number of 13F holders (bars) per quarter, point-in-time on each quarter's filing cutoff
4	Short interest trend	OFFICIAL, FINRA (settlement dates): short interest as % of shares outstanding
5	13F holders	Top holders: filer, shares, value, % out, change, status, weight in the filer's book, filed date; badges HF-like* and smart*
6	Insider timeline	OFFICIAL, Form 4 / national regulator filings: open-market buys and sells, 2 years (below the fold)
7	EU / UK net short positions	OFFICIAL, regulator registers (≥ 0.5% of capital) (below the fold)
8	13D / 13G filings	OFFICIAL, SEC (below the fold)
9	Smart-money score history	MODEL-BASED percentile within the universe (below the fold)

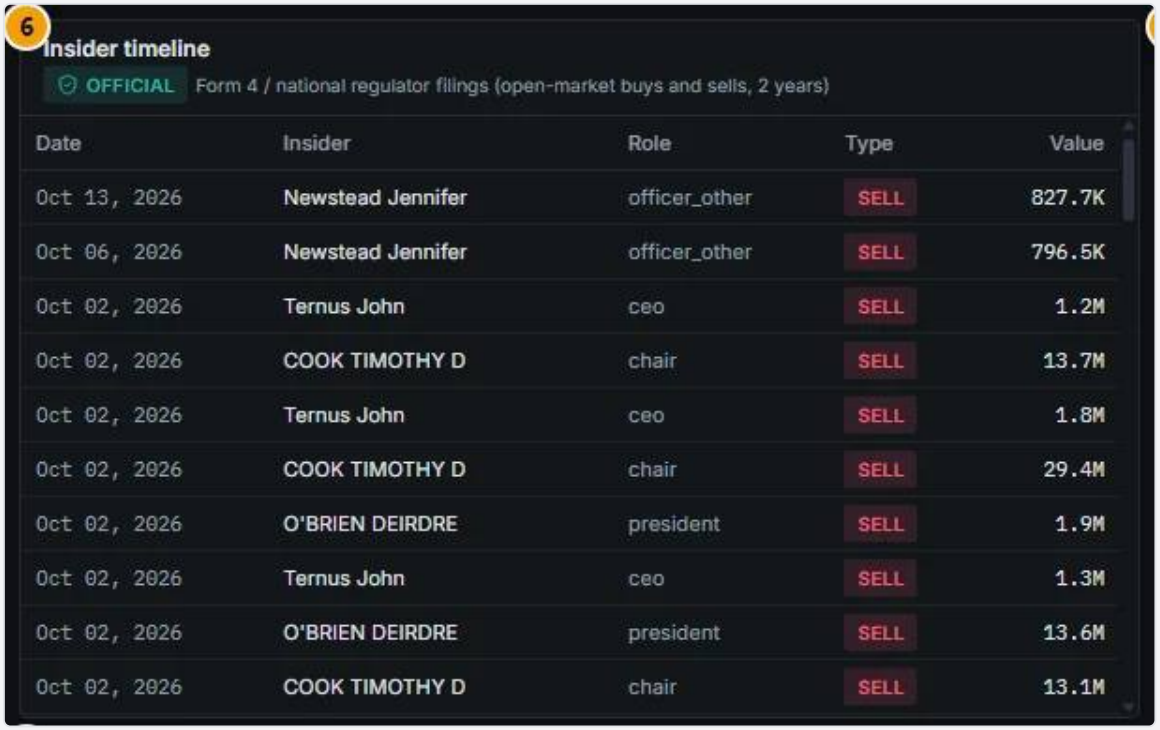


Figure 5A.36: Insider timeline, Apple (panel 6).

Figure 5A.37: EU / UK net short positions, Apple (panel 7): empty, as expected for a US issuer.

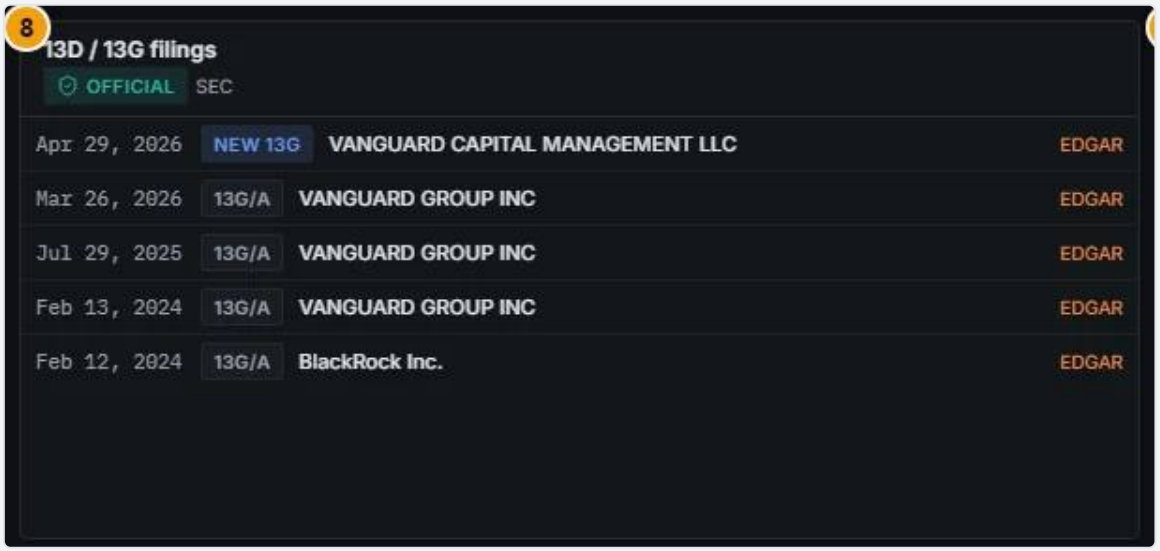


Figure 5A.38: 13D / 13G filings, Apple (panel 8).

Figure 5A.39: Smart-money score history, Apple (panel 9).

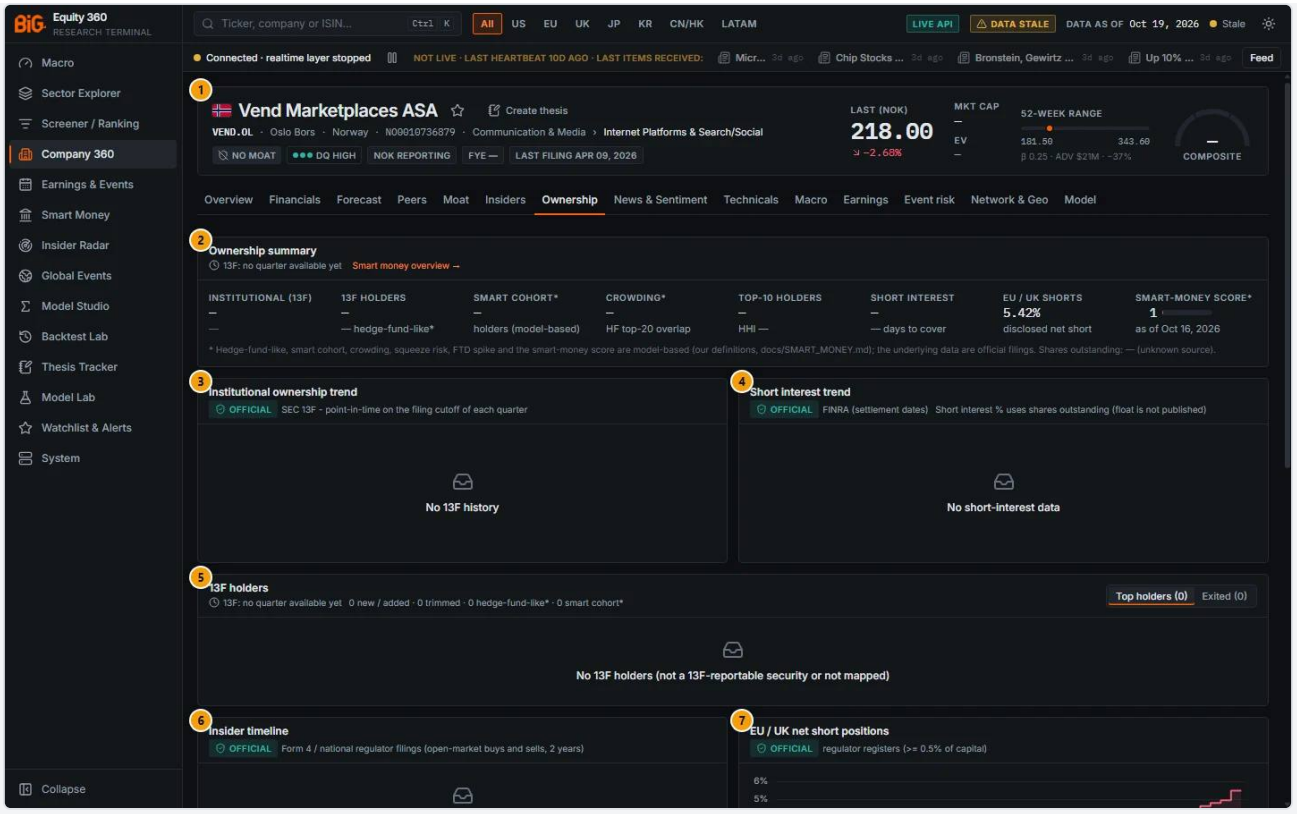


Figure 5A.40: Ownership tab for Vend Marketplaces (`/company/VEND.OL?tab=ownership`), a Norwegian issuer, data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	Vend Marketplaces ASA, Oslo Børs, NOK
2	Ownership summary	"13F: no quarter available yet"; all 13F and FINRA fields empty; EU / UK shorts 5.42% disclosed net short; smart-money score at the 1st percentile as of 16 Oct 2026
3	Institutional ownership trend	"No 13F history"
4	Short interest trend	"No short-interest data" (FINRA covers US-traded securities only)
5	13F holders	"No 13F holders (not a 13F-reportable security or not mapped)"
6	Insider timeline	No transactions in the window
7	EU / UK net short positions	Step chart of the disclosed net short since January 2026 and the current register (Figure 5A.41)
8	13D / 13G filings	Below the fold (empty for a non-US issuer)
9	Smart-money score history	Below the fold

Figure 5A.41: EU / UK net short positions for Vend Marketplaces (panel 7): the Finanstilsynet aggregate (5.42%) and the active named holders at or above 0.5%.

Controls

Control	What it does
Smart money overview →	Opens `/smart-money` (see Smart Money)
`Top holders (n)` / `Exited (n)`	13F holders of the latest quarter, or last quarter's top holders that exited
Badge hover (HF-like, *smart*)	The model-based definition
`EDGAR` (13D / 13G)	Opens the filing on SEC EDGAR
Chart hover	Quarter or settlement-date values
Sources footer	The official sources used, with their lags

Methodology

- **13F point in time.** A quarter's snapshot uses only filings made by the cutoff: the filing deadline (quarter end + 45 days, next SEC business day) + 1 business day. The header states "Positions as of quarter end Q, filed by D". Restatements filed after the deadline drop out (conservative). Previous-quarter share counts are restated for splits before changes are computed.
- **Holder status**: new; added (> +1%); trimmed (< −1%); unchanged; exited. `Top-10 holders` is the share of the 13F total held by the ten largest; `HHI` is the sum of squared holder shares.
- **Hedge-fund-like filers*** (model-based): 10 to 1,000 positions, AUM ≥ $50M, ≥ 60% of the book in single stocks, and quarterly turnover ≥ 15% or option value ≥ 5% or (top-10 weight ≥ 50% and turnover ≥ 5%).
- **Crowding*** (model-based): number of hedge-fund-like filers holding the stock among their top-20 positions, as a percentile within the quarter.
- **Smart cohort*** (model-based): hedge-fund-like filers whose reported books beat the average 13F filer over the past four quarters (top decile, positive excess), selected point-in-time.
- **Short interest**: FINRA short interest / **shares outstanding** (float is not published; documented proxy), days to cover. **Squeeze risk*** (model-based): short interest ≥ 15% of shares, days to cover ≥ 5, 20-day return ≥ +10%, and shares ≤ 150M or market cap ≤ $2B.

- **Fails-to-deliver**: SEC FTD data feed the model-based **FTD spike** flag (largest daily fail ≥ 0.5% of shares outstanding, or z ≥ 3 against the previous 12 half-month files with at least 100k shares). FTD spikes are listed on the Smart Money overview; this tab lists FTD only among the sources.
- **EU / UK disclosed short %**: the regulator aggregate where published (UK FCA since 10 Jul 2026, Norway), otherwise the sum of active holder positions ≥ 0.5% (latest notification per holder, within two years); the maximum across regulators.
- **Activism**: new 13D, 13D amendment, 13G → 13D switch (the filer's previous filing on the company within three years was a 13G), new or amended 13G.
- **Smart-money score*** (model-based): z-scores within the date (winsorised at 1/99%) of 13F flow and breadth, smart-cohort flow and the insider signal, minus short-interest level and change, minus EU short level and 30-day change; grouped (institutional, smart, insider, short interest, EU short), averaged over available groups and shown as a 0-100 percentile. It is **not** part of the production composite (optional input, disabled by default).

Data sources and refresh

All underlying data are official: SEC Form 13F data sets (quarterly, from 2023 Q1), FINRA equity short interest (twice monthly, published about 7 business days after settlement), SEC fails-to-deliver (half-monthly), SEC Schedules 13D / 13G (EDGAR indices), Form 4 and national insider filings, and the short-position registers of the FCA (UK), AMF (France), CNMV (Spain), AFM (Netherlands), CBI (Ireland), FSMA (Belgium), FI (Sweden) and Finanstilsynet (Norway). Italy (CONSOB) and Germany (BaFin, Bundesanzeiger) are not fetched (bot challenge; robots.txt). Jobs: `sec_13f` and `smart_money_13f` weekly, `finra_short_interest` weekly, `eu_short_positions` and `activist_filings` nightly, `smart_money_signals` nightly.

Validation status

- 13F, FINRA, FTD, 13D / 13G and EU / UK registers: official data, labelled **OFFICIAL**.
- Hedge-fund-like, smart cohort, crowding, squeeze risk, FTD spike and the smart-money score: **model-based** (asterisked; "our definitions, docs/SMART_MONEY.md").
- **Smart-money score sanity check** (`docs/SMART_MONEY.md` §4, 2023-06-30 to 2026-10-16, 6,145 securities, point-in-time, non-overlapping): rank-IC of the composite −0.004 at 21 days (t −0.67) and −0.005 at 63 days (t −0.56). **None of the groups is statistically significant**; the composite has no predictive power in this short sample. Treat it as a descriptive screen, not an alpha signal. The weights were not tuned on this result.

Correct usage and worked example

Apple (Figure 5A.35). Institutional ownership 65.4% of shares (+3.4 pp on the quarter), 6,026 13F holders of which 485 hedge-fund-like, 13 in the smart cohort; crowding at the 100th percentile (Apple is in the top 20 of more hedge-fund-like books than almost any stock); top-10 holders 48% (HHI 0.037). Short interest 0.8% of shares outstanding, 3.0 days to cover, no squeeze flag; no EU / UK disclosures. Smart-money score 18th percentile as of 16 Oct 2026. Holders: 47 new or added, 30 trimmed; BlackRock holds 7.92% of shares outstanding (added).

Read: widely and increasingly owned, very crowded among active managers, lightly shorted. Crowding is a risk factor in a de-risking episode rather than a signal of direction.

Vend Marketplaces (Figures 5A.40 and 5A.41). The opposite case: a Norwegian issuer with no 13F or FINRA data, where the short side is visible only through the regulator register. Norway's Finanstilsynet publishes an aggregate, so the summary shows 5.42% disclosed net short (the aggregate row, −0.06 pp on 16 Oct 2026). The step chart shows the disclosed short rising from about 0.5% in January 2026. Named holders at or above 0.5% include Kintbury Capital (2.35%, +0.06 pp), PDT Partners (0.90%), JPMorgan Asset Management (0.60%), Capital Fund Management (0.55%) and two new positions (Pertento Partners 0.52%, Eleva Capital 0.50%). In the Change column an increase in a short is shown in red and a decrease in green. A steady build-up of disclosed shorts across several funds, together with a smart-money score at the 1st percentile, is worth investigating; it describes positioning and is not a forecast.

Limitations and pitfalls

- **13F lag.** Positions are as of quarter end and filed up to 45 days later; they are long positions only (no shorts, no non-13F managers).
- **CUSIP mapping gaps.** The 19 Oct 2026 audit (F29) found about 40% of 2025-26 13F rows without a mapped security; holder counts can be understated, notably for ADRs and smaller names.

- Short interest % uses shares outstanding, not float, so it understates short pressure for low-float stocks.
- EU / UK coverage excludes Italy and Germany; UK holder names are aggregated since 10 Jul 2026.
- 13F is a US filing regime: for non-US home listings, the holders shown are US managers' positions reported under 13F, typically via ADRs or US-listed lines.
- The smart-money score has no demonstrated predictive power.

Related mnemonics and BQL fields

`HDS` (holders), `SI` (short interest), `SMRT` (smart money). BQL: `own.inst_own_pct`, `own.inst_own_chg`, `own.n_holders`, `own.n_hf_holders`, `own.crowding_pct`, `own.n_smart_holders`, `own.smart_flow_pct_out`, `shrt.short_interest_shares`, `shrt.short_pct_float`, `sm.eu_short_pct`, `sm.score_pct`, `sm.g_inst`, `sm.g_short_interest`. See Smart Money.

News & Sentiment

Route: `/company/:id?tab=sentiment` · **Menu:** Company 360 › News & Sentiment · **Mnemonics:** `CN` (when the News Terminal is not installed) · **Refresh:** nightly chain (`nlp_sentiment`, `sentiment_daily`, `lazy_prices`, `gdelt_company_daily`); live news hourly

Purpose

The News & Sentiment tab measures the tone of news about the company over time, lists the latest items with their sentiment and relevance, plots GDELT global events linked to the company, and reports the "Lazy Prices" text-change measure of its periodic filings. Use it to see whether news flow is turning, and whether a price move is accompanied by news.

Screenshot

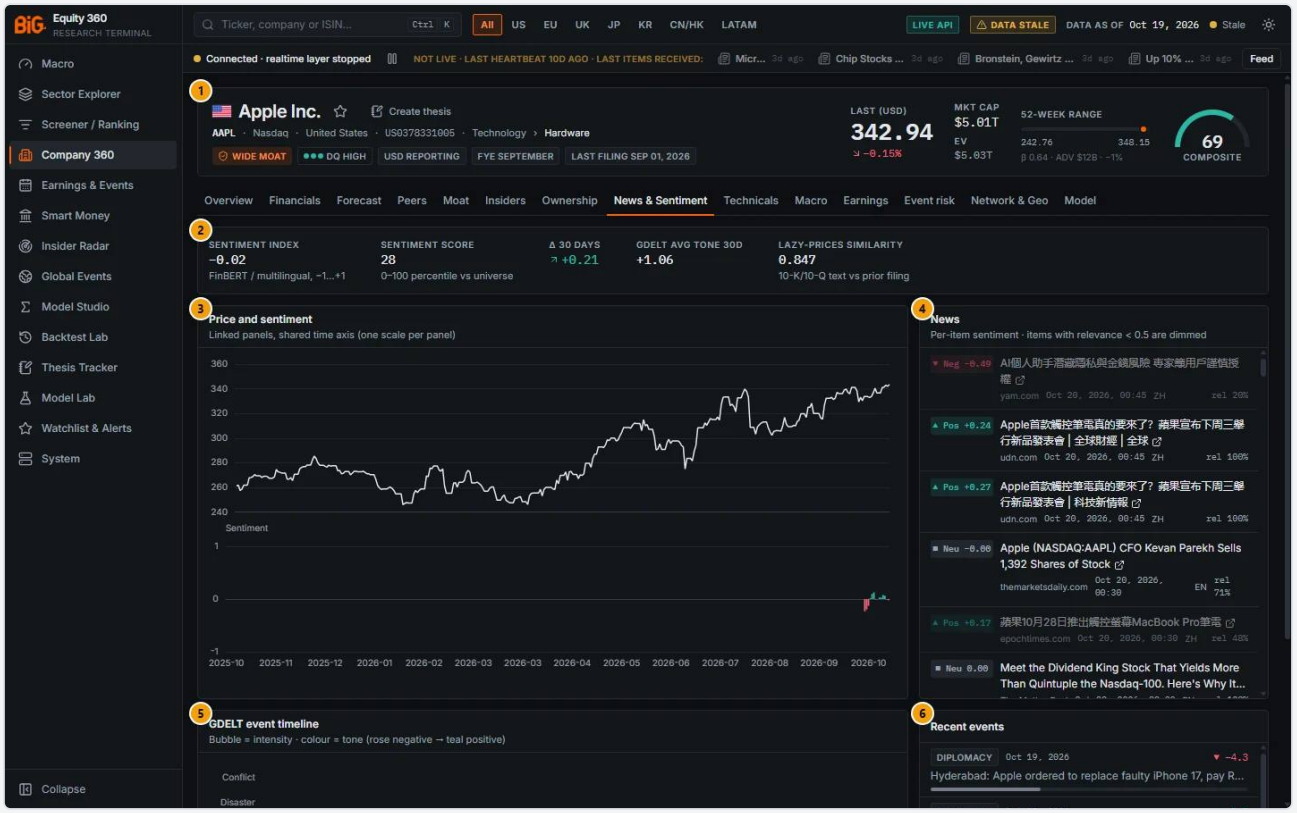


Figure 5A.42: News & Sentiment tab for Apple (`/company/AAPL?tab=sentiment`), data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	See Company header
2	Summary stats	Sentiment index (FinBERT / multilingual, −1…+1); sentiment score (0-100 percentile vs universe); Δ 30 days; GDELT average tone 30 days; Lazy-prices similarity (10-K/10-Q text vs prior filing)
3	Price and sentiment	Linked panels on a shared time axis: close (top) and the daily sentiment index as bars (bottom)
4	News	"Per-item sentiment · items with relevance < 0.5 are dimmed": chip, headline with link, source, time, language, relevance
5	GDELT event timeline	Events by category over time; bubble = intensity, colour = tone (rose negative → teal positive)
6	Recent events	Category, date, tone and an intensity bar for each linked GDELT event

Controls

Control	What it does
Linked chart hover	Close and sentiment index for the date, on both panels
Headline	Opens the article at the source (new tab)
Dimmed item hover	"Low issuer-link confidence (relevance < 0.5)"
Timeline bubble hover	Headline, date, average tone, intensity
Scroll	The News and Recent events lists scroll inside their panels

Methodology

From `backend/app/nlp/` (`sentiment.py` , `aggregate.py` , `lazy_prices.py` , `gdelt_link.py`).

- **Item sentiment.** English text is scored with **FinBERT** (`ProsusAI/finbert`); other languages with **XLM-R** (`cardiffnlp/twitter-xlm-roberta-base-sentiment`). Language comes from the source metadata or a script / stop-word detector. Score = `P(positive) − P(negative)` in −1…+1. The chip shows Pos above +0.15, Neg below −0.15, otherwise Neu.
- **Relevance (0-1)** of each item to the company: ticker or company name in the title 1.0; provider ticker tag 0.85; name in the snippet 0.7; organisation mention only (GDELT GKG) 0.5 × the organisation-link confidence. Ambiguous names (Apple, Meta, Visa, Shell, Target …) need corroborating context in the title. Items linked to many companies are down-weighted (× √(3/n) when n > 3).
- **Daily index.** Each item's score blends the model (0.6), the provider's sentiment where available (0.25) and GDELT tone / 5 (0.15), then is **de-meaned per source** by that source's trailing 30-day universe mean (sources have different tone levels). Per company and day: the **relevance-weighted mean, ignoring items with relevance below 0.5** (they still count for attention, and are dimmed in the list). The **sentiment index** is the 7-day exponentially weighted mean on irregular time: days without news carry no information and only decay the weight; the index becomes empty when it is stale. Days when a source's universe-wide volume falls below 50% of its 30-day median are flagged and do not distort the attention measure.
- **Δ 30 days**: latest index minus the value about 22 data days earlier.
- **Sentiment score (0-100)**: per-date cross-sectional percentiles of the 30-day sentiment EWM (weight 0.40), the 7-day GDELT tone EWM (0.20) and the Lazy Prices percentile (0.40), renormalised over what is available; 50 = no data. It feeds the Sentiment factor of the composite.
- **GDELT.** GDELT organisation mentions are linked to issuers (exact alias, fuzzy match ≥ 92, context rules for ambiguous names; links below 0.5 confidence excluded). The timeline shows linked GDELT events of the last 200 days by category (Policy, Economic, Legal, Corporate, Diplomacy, Protest, Disaster, Conflict). "GDELT avg tone 30d" is the mean daily average tone over the last 30 days.
- **Lazy Prices** (Cohen, Malloy and Nguyen 2020). Each periodic filing is compared with the prior year's equivalent (10-K / 20-F / 40-F with the previous annual report; 10-Q with the same quarter a year earlier), section by section (full text, risk factors, MD&A, legal, business). Measures: TF-IDF cosine similarity, Jaccard, sentence edit ratio, share of new sentences, Loughran-McDonald negative and uncertainty word deltas. A filing is flagged **large change** when the risk-factor (or full) section has a similarity z-score ≤ −2 against filings of the prior 365 days or at least 35% new or rewritten sentences. The stat on the tab shows the cosine similarity of the most recent compared filing (1 =

unchanged); low similarity (a low universe percentile) means an unusually large change, which in the original study preceded weaker returns.

Data sources and refresh

News: Polygon news (US), RSS feeds, GDELT GKG headlines and the realtime news layer. GDELT events and GKG are streamed daily. Filing texts: SEC EDGAR (US only for Lazy Prices). Jobs in the nightly chain: `nlp_sentiment` (scoring new items), `gdelt_company_daily`, `lazy_prices`, `sentiment_daily`. The live news layer folds new items hourly.

Validation status

Not formally validated as a return signal in the app. FinBERT and XLM-R are published pre-trained classifiers used as is; the daily index, relevance weights and the 0-100 score are transparent rules. The tab shows no model-based badge. The Lazy Prices construction follows the published study, but its predictive value has not been re-tested here.

Correct usage and worked example

Apple (Figure 5A.42). Sentiment index −0.02 (neutral), Δ 30 days +0.21, sentiment score 28 (28th percentile against the universe), GDELT average tone +1.06. Lazy-prices similarity 0.847: the July 2026 10-Q compared with the same quarter a year earlier, at about the 11th percentile of similarity in the universe, i.e. a larger-than-usual rewrite. The news list mixes English and Chinese items; a Chinese headline with relevance 20% is dimmed and does not count in the index.

Read: neutral tone, a below-average sentiment score, nothing to act on alone. A combination of a falling index, rising attention and a large Lazy Prices change would deserve a read of the filing and the news.

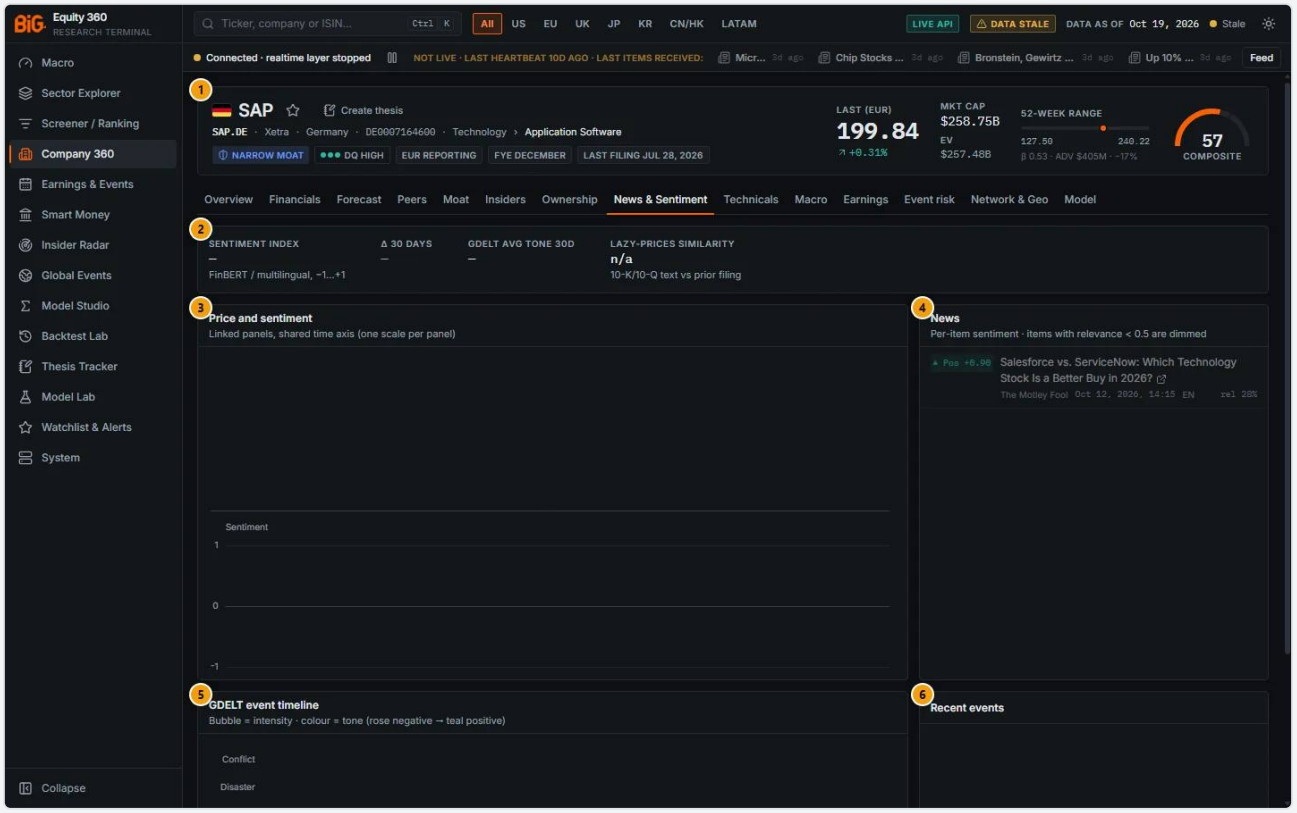


Figure 5A.43: News & Sentiment for SAP (`/company/SAP.DE?tab=sentiment`), a non-US issuer, data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	SAP, Xetra, EUR
2	Summary stats	Sentiment index "—", Δ 30 days "—", GDELT average tone "—", Lazy-prices similarity "n/a"; the sentiment score tile is hidden because no score exists
3	Price and sentiment	Empty: no sentiment history in the window
4	News	One item (The Motley Fool, 12 Oct 2026, English) with relevance 28%, dimmed

#	Panel	What it shows
5	GDELT event timeline	No linked events
6	Recent events	Empty

SAP (Figure 5A.43) shows the coverage gap for a large European issuer. The only linked news item has low relevance (it is about Salesforce and ServiceNow) and is excluded from the index, so the index and Δ 30 days are empty. Lazy Prices is n/a because SAP files no 10-K or 10-Q. SAP's Sentiment factor sits at the neutral 50, which reflects missing data, not neutral news.

Limitations and pitfalls

- **Short history.** News sentiment history starts around August 2026 and GDELT company history in October 2026. In this snapshot Apple's index has non-empty values only from 3 Oct 2026, which is why the bars sit at the right edge of the chart.
- Coverage is uneven by language and market; small and non-US names may have few items, and the index then goes stale (empty) rather than reading 0.
- Relevance is a heuristic; ambiguous company names can still mislink.
- Lazy Prices uses SEC filings only (US issuers). The stat shows the latest compared section, which is not necessarily the risk-factor section; the large-change flag itself is in BQL (`sent.lazy_prices_large_change`).
- The GDELT DOC API is degraded; the raw event files are the main GDELT source (see Global Events).
- News timestamps can be later than the snapshot date (the live layer keeps adding items).

Related mnemonics and BQL fields

`CN` (News Terminal filtered on the ticker when installed, else this tab), `TOP` . BQL: `sentiment` , `f.sentiment` , `news_count_30d` , `sent.news_count` , `sent.news_count_z` , `sent.gdelt_tone_ewm_7d` , `sent.lazy_prices_sim` , `sent.lazy_prices_pct` , `sent.lazy_prices_large_change` . See Global Events and News Terminal.

5B. Company 360: Technicals

This chapter covers the **Technicals** tab of Company 360: the price chart with the calibrated range fan, the context chips, confluence zones and their levels, touch probabilities, the trade setup card with its first-passage and context-adjusted (meta) probabilities, the technical score, the evidence-based technical factors, the breakout panel and the indicators. It is the most model-heavy tab in the app. Read the Technicals overview first: it explains the experimental toggle, how to read a probability that is "calibrated over history", and which outputs passed or failed their validation gate.

Technicals

Route: `/company/:id?tab=technicals` · **Menu:** Company 360 › Technicals tab · **Mnemonics:** `TECH` , `GP` , `GIP` · **Refresh:** nightly jobs in the weekly profile (default) or daily profile; model refits weekly / monthly

Purpose

The Technicals tab answers four practical questions about one listed security: where price has historically reacted (levels and confluence zones), how far price typically travels over the next 1 to 20 sessions (the calibrated range), what a level-based trade setup has historically delivered (first-passage probabilities, expected R), and what the documented, evidence-based technical effects say about the stock (52-week-high proximity, MAX, idiosyncratic volatility, volume-confirmed breakouts). It is used by analysts to time entries and size risk around a fundamental view, and by portfolio managers to check whether a planned stop or target is realistic.

Every probability on this tab is **model-based** and **calibrated over history**: it is the frequency with which similar situations produced the outcome in a walk-forward test, not a statement about this stock today. Outputs that failed their validation gate are hidden behind the **experimental toggle**.

Screenshot

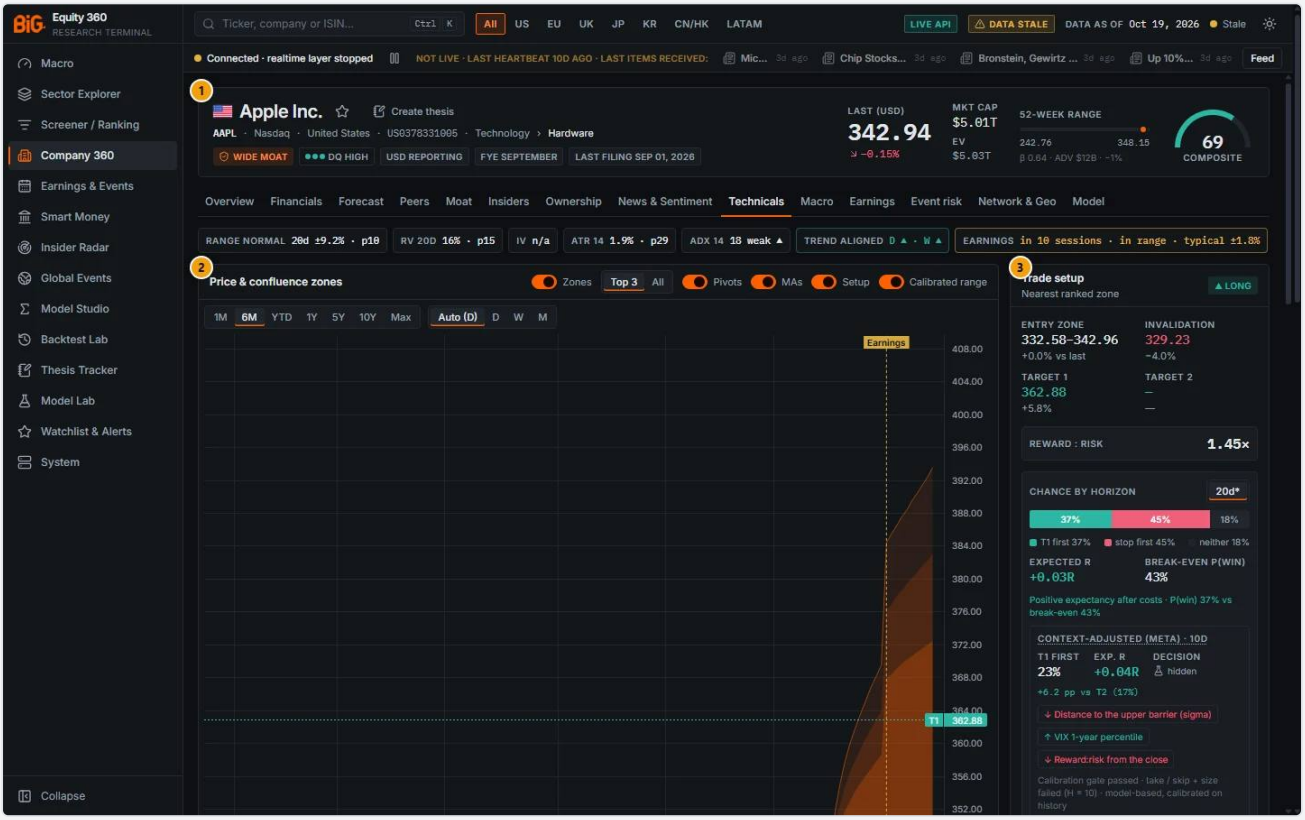


Figure 5B.1: Technicals tab for Apple (`/company/AAPL?tab=technicals`), experimental toggle off (the default), data as of 19 Oct 2026.

#	Panel	What it shows
1	Apple Inc.	The Company 360 header (price, market cap, 52-week range, composite score). See 5A. Company 360: header.
2	Price & confluence zones	Candlestick chart with confluence zones, pivots, moving averages, the trade setup lines and the calibrated range fan to 20 sessions; range strip underneath. See Price chart and calibrated range.

#	Panel	What it shows
3	Trade setup	Bias, entry zone, invalidation, targets, reward : risk, the two-barrier "Chance by horizon" bar, expected R, break-even P(win) and the "Context-adjusted (meta)" row. See Trade setup and Context-adjusted (meta) row.
4	Technical score	The heuristic 0-100 technical score with percentile sub-scores; "No out-of-sample edge" note. See Technical score.
5	Technical factors (evidence-based)	Percentiles of the factors that passed their gate (52-week-high proximity, MAX, idiosyncratic volatility), each with its "rank IC" badge. See Technical factors (evidence-based).
6	Breakout	State of the latest 20 / 55-session breakout and the historical evidence on volume confirmation. See Breakout.
7	Indicators	RSI 14, ATR 14, SMA 50, SMA 200 and the trend label. See Indicators.
8	Confluence zones	Table of zones: rank, type, range, distance, strength, touch probability, hold-given-touch column, contributors, touches. See Confluence zones and Touch probabilities.
9	Level types · backtested hit rates	Best level key per level type for this stock, with shrunk hit rate, average reaction and number of events. See Level types and level-type reliability.
10	Level-type reliability	Bounce hit rate and edge versus a random-level control per level type. See Level types and level-type reliability.

The row of chips directly under the tab bar (no badge) is described in Context chips.

WHAT THE EXPERIMENTAL TOGGLE REVEALS

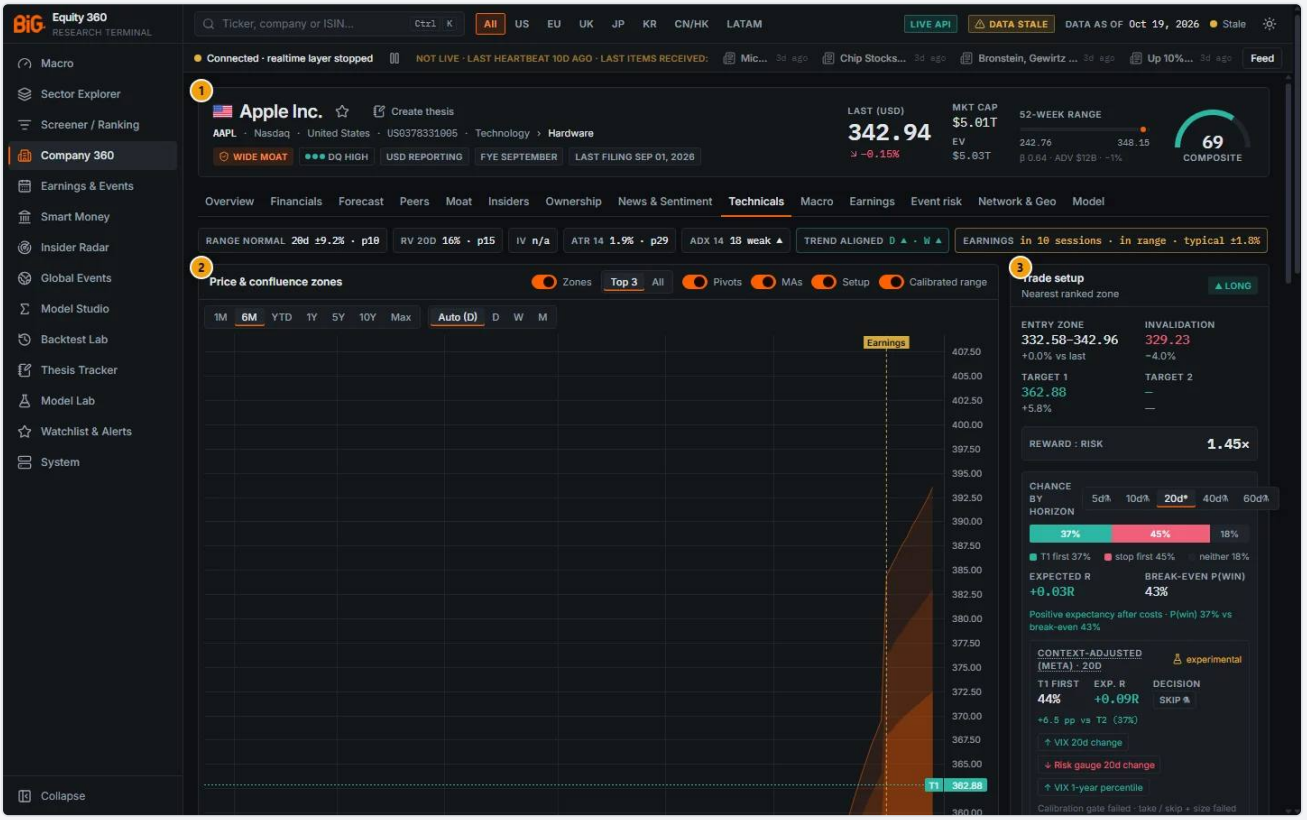


Figure 5B.2: The same page with "show experimental (failed validation)" switched on. Callout numbers are the same as in Figure 5B.1.

Panel	Default (toggle off)	With the toggle on
Trade setup, Chance by horizon	Only the 20-session horizon (the one that passed). A note says "4 experimental horizons hidden."	Horizons 5, 10, 20, 40 and 60 sessions; failed ones carry a flask icon and the line "Experimental horizon: failed the validation gate."
Trade setup, Context-adjusted (meta)	The 10-session row (passed); Decision shows "hidden"; no suggested size.	The meta row follows the horizon on screen (20 or 40 sessions, labelled "experimental"); Decision shows take / skip with a flask icon; a suggested size appears for "take".
Confluence zones, hold-given-touch column	A flask icon in place of the probability, plus the validated edge badge ("no reliable edge" or "±x.x pp vs random").	The calibrated P(hold given touch) in per cent, with a flask icon, next to the same edge badge.
Technical score	Percentile sub-scores and the "No out-of-sample edge" note.	Adds the score bucket's historical outcomes at 5 / 20 / 60 sessions, each badged "no out-of-sample edge".
Technical factors	hi52, MAX, IVOL only; "4 factors without a validated edge hidden."	Adds 12-1 momentum, residual momentum, 1-month reversal and industry momentum, each with a flask icon and "no reliable edge".
Breakout	State chip, confirmation flag and score; the line "P(follow-through) is experimental (failed validation) — hidden."	Adds "P(follow-through, 20d)" with a flask icon, the geometry baseline and the base rate.

> **Note:** The toggle is stored per browser (local storage), not per user account, and is shared by every Technicals card and by the probability tables in the Thesis Tracker. If you switch it on once, it stays on for every security until you switch it off. All screenshots in this chapter show the default view (toggle off) unless the caption says otherwise.

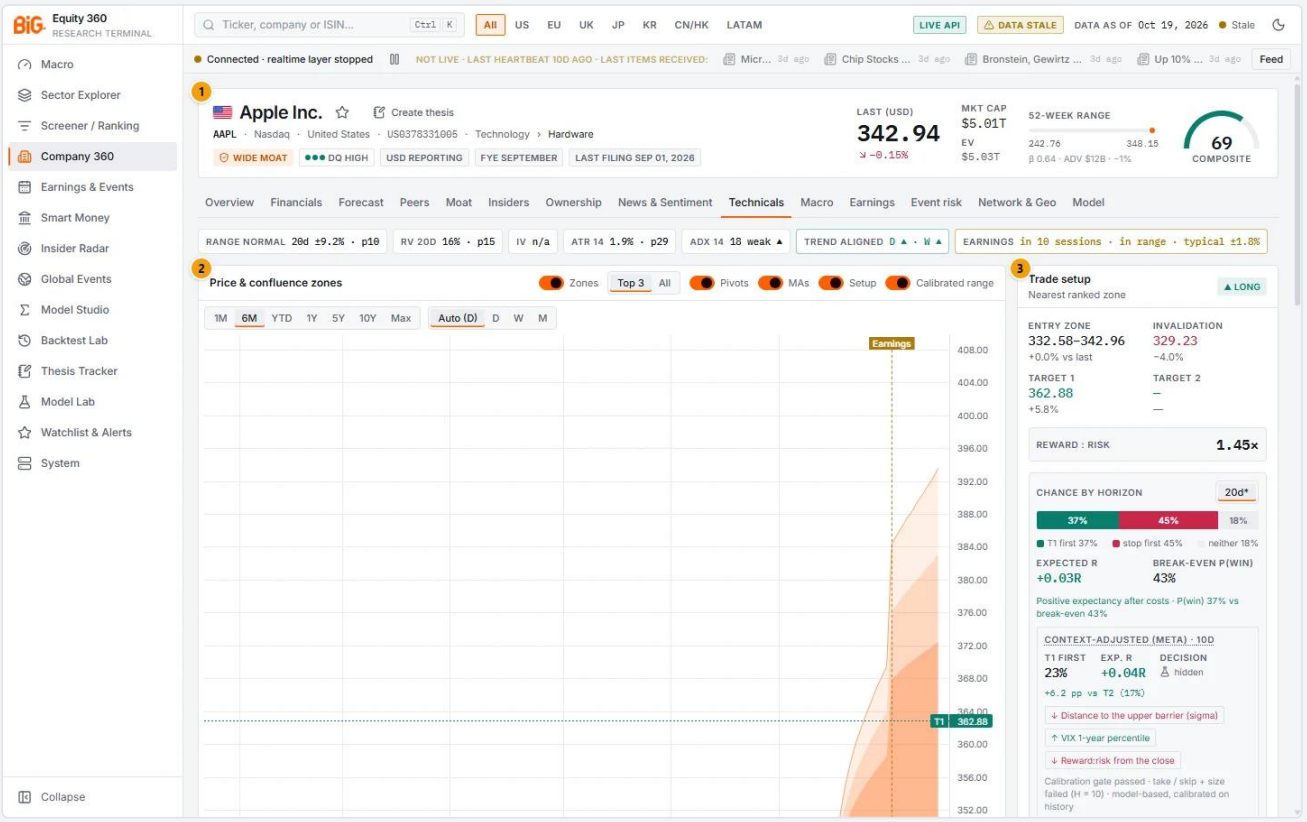


Figure 5B.3: Light theme (`/company/AAPL?tab=technicals`), default view. Every panel, callout number and colour role is the same as in Figure 5B.1.

Controls

Control	What it does
show experimental (failed validation)	Switch with a flask icon. Reveals outputs that failed their validation gate (table above). It sits inside the "calibrated on history" popovers of the Trade setup and Technical score cards and inside the info (ⓘ) popover of the Technical factors card. Off by default; persisted per browser.
calibrated on history (button)	Opens a popover with the model's plain-language definition, one line per horizon or output with its gate verdict ("pass" or "experimental") and summary (ECE, worst regime, Brier against baselines), the experimental switch, and the validation document name.
Chart switches and range strip	Zones, Top 3 / All, Pivots, MAs, Setup, Calibrated range, price range, bar interval, range level and badge horizon. See Price chart and calibrated range.
Touch horizon (1d / 5d / 10d / 20d)	One setting shared by the chart's zone badges and the Confluence zones table.
Chance by horizon (5d / 10d / 20d / 40d / 60d)	Selects the horizon of the two-barrier probabilities on the Trade setup card. An asterisk marks the setup's own horizon; a flask marks an experimental horizon.
Row hover (Confluence zones)	Previews that zone on the chart.
Hover tooltips	Every dotted-underlined label (Touch, Hold given touch, Context-adjusted (meta), factor names, context chips, probability segments) has a tooltip with its definition.
Create thesis (header)	Opens the Thesis Tracker editor for this security; the thesis target / stop then get the same calibrated probabilities. See 12. Thesis Tracker.

There is no export button on this tab. The same numbers are available through the API endpoints and BQL fields listed in each section.

Methodology

The tab combines four model families. Each is described in detail in its own section; this subsection gives the common concepts.

CALIBRATED OVER HISTORY VERSUS CHANCE TODAY

A probability such as "T1 first 37%" means: in the walk-forward test, among historical setups with the same standardised geometry and in the same pool (region, volatility regime, earnings window), the target was reached first about 37% of the time, and the model's 37% buckets realised close to 37% on average. It is a **marginal** statement over many securities and days. It is **not** "a 37% chance for Apple today": the app has no way to verify a single-day, single-stock probability, and conditions specific to today (news, a market shock) are not in the pool. This is why every popover says "calibrated on history" and the range popover says "Calibrated to 90% over history (2010–2026 walk-forward), not "90% chance today"".

CONFORMAL CALIBRATION (THE RANGE FAN AND TOUCH PROBABILITIES)

Conformal prediction turns any volatility forecast into a band with a guaranteed long-run hit rate, using the model's own past errors instead of a Gaussian assumption.

```
V(h)      = h · s²ₜ  (+ Jₜ if an earnings day falls in (t, t+h])      EWMA λ = 0.94, earnings days excluded
score     = y²ₕ / V(h)            yₕ = log(C₍ₜ₊ₕ₎ / Cₜ), on past origins of the same pool
q̂(α)      = the ⌈(n+1)(1 − α)⌉-th smallest of the last 500 realised scores
band      = Cₜ · exp(± √(q̂ · V(h)))     α = 0.20 / 0.10 / 0.05 for the 80 / 90 / 95% bands
```

Scores are pooled per region × sector (separate pool for origins whose window contains an earnings day), so $\hat{q}$ is estimated on 15,000 or more scores per pool. The filter decides the shape (how wide relative to recent volatility), the conformal quantile decides the size. The band has **no direction call**: its centre is the last close.

FIRST-PASSAGE PROBABILITIES (THE TRADE SETUP)

A first-passage probability asks which of two price levels is touched first. For a long setup: does the high reach T1 before the low reaches the invalidation, within H sessions? The answer depends on the path, not only on where price ends, so it cannot be read off a terminal-price band. The app estimates it empirically: for every daily origin in the gate universe it records, for a 20 × 20 grid of barrier distances, whether the upper or lower barrier was hit first or neither. Distances are standardised by σ_H = √V(H), so one surface serves every security. A same-bar touch of both barriers counts as a loss (intrabar order is unknown).

META-LABELLING (THE CONTEXT-ADJUSTED ROW)

Meta-labelling (Lopez de Prado, *Advances in Financial Machine Learning*, ch. 3-4) separates two jobs. The **primary model** proposes a trade (here: the level-based setup with direction, entry, T1 and invalidation). A **secondary (meta) model** then estimates the probability that this proposal succeeds, given context the primary model ignores. Outcomes are labelled with the **triple barrier**: win = upper barrier (T1) first, loss = lower barrier (invalidation) first, and a vertical barrier at H sessions. The meta model starts from the T2 first-passage probability (as a fixed offset) and adjusts it for market regime, VIX, peer break-aways, insider and short-interest flow, earnings proximity, squeeze, ADX and trend alignment.

PURGED WALK-FORWARD

Every model on this tab was tested **walk-forward**: for each test period, the model is fitted only on data available before that period, with a **purge / embargo** gap at least as long as the outcome horizon, so that no training label overlaps the test period. Examples: the T2 surface for test year Y uses origin quarters before Q(Y) − 1; the zone model uses training origins before the block start − 425 days and calibrates on [start − 365 days, start − 60 days); the meta model embargoes 1.5 × H sessions (28 / 49 / 91 calendar days at H = 10 / 20 / 40). Leakage tests (causal invariance, shifted-feature peeks, label permutation) were run on every model.

ECE, BRIER SCORE AND LOG-LOSS

- **ECE** (expected calibration error): sort the predictions into 15 quantile bins, compare the mean prediction with the realised frequency in each bin, and average the absolute gaps weighted by bin size. Reported in percentage points (pp). ECE 1.47 pp means predictions were on average about 1.5 pp away from what happened. The gates require ECE ≤ 2.5 pp.
- **Worst regime gap**: the same comparison in calibration-in-the-large within stress windows (Covid 2020 = origins 15 Feb to 30 Jun 2020, the 2022 rates shock, VIX > 25, earnings windows, each region, volatility regime and side). The gates require every regime with n ≥ 2,000 to be within 5 pp.
- **Brier score**: the mean squared difference between predicted probabilities and outcomes (0 is perfect; lower is better). **Log-loss** penalises confident mistakes more heavily. A model must beat simple baselines on both, for example the driftless Brownian formula or the naive "R:R only" rule.

A model can be well calibrated (low ECE) and still useless (no better than the baseline), or useful on average but wrong in a crash. That is why each gate has all three parts.

HOW A GATE DECIDES WHAT YOU SEE

Gate result	How the app shows it
PASS	Shown by default; the popover says "pass".
FAIL	Hidden unless the experimental toggle is on; shown with a flask icon and the word "experimental".

Gate result	How the app shows it
Not validated (region or universe outside the test)	Treated as experimental (meta row) or flagged "region not validated" (factors).

Verdicts are re-read from the gate files at every weekly fit, so a re-run validation updates the display without a code change.

Data sources and refresh

- **Prices:** split- and dividend-adjusted daily bars (open, high, low, close, volume) from the app's price store (Polygon and yfinance). All calculations use the security's own trading sessions.
- **Earnings dates:** the earnings calendar (company announcement, provider estimate or model projection, labelled) and past releases for the earnings jump term and the typical move.
- **Market context (meta model):** FRED VIX, the cross-asset risk gauge, macro cycle states (lagged two months), conformal peer groups, Form 4 insider flow, FINRA short interest and the smart-money composite.
- **Jobs (nightly DAG):** `tech_levels` → `confluence_zones` (levels, zones, setup, technical score), `conformal_envelopes` (range fan and touch curves), `technicals_probabilities` (T2 / T3 / T4 serving), `tech_meta_score`, `tech_factors`, `tech_breakouts`. **Weekly DAG:** `level_type_backtest`, `technicals_probabilities_fit`, `tech_meta_fit`. **Monthly DAG:** `tech_factor_eval` (factor monitor and P(follow-through) refit). The context chips are computed when the page is opened.
- **Schedule:** with the default weekly profile the nightly, weekly and (when due) monthly DAGs run back to back on Saturday 08:00 Lisbon; the daily profile runs the nightly DAG every night. See 1. Data and coverage.
- **As-of dates:** the zones, setup and probabilities carry the as-of date of the last nightly run (for example 16 Oct 2026), which can be one or more sessions older than the last candle on the chart. The range strip states its origin ("from the Oct 16 close").

Validation status

Output	Gate	Key numbers	How the app shows it	Source
Calibrated range fan (marginal bands, h = 1..20)	Validated by-product	Per-horizon coverage 79.6-80.0 / 89.6-90.0 / 94.6-94.9%; about 40% narrower than 2×ATR (which covers 98%)	Fan and range strip, labelled "Calibrated range"	CONFORMAL_VALIDATION A.2, A.5
Joint 1/5/10/20 envelope (ellipsoid)	FAIL	Covid Feb-Jun 2020: 77-78% at 90%; w_proj +35% vs a calibrated ATR box	Not on this tab ("failed validation and is not shown")	CONFORMAL_VALIDATION Verdict
Next bar (H, L, C)	FAIL (sharpness)	w_proj +43% vs calibrated ATR box	Not on this tab	same
Touch probabilities	Validated by-product	Predicted = realised within 0.2 pp	Touch column and chart badges	CONFORMAL_VALIDATION A.4
T2 P(win / loss), H = 20	PASS	ECE 1.13 / 1.47; worst regime Covid 2020 win -4.6 / loss +4.0; Brier 0.3741 vs 0.3759 / 0.3834	Trade setup bar, default	TECHNICALS_V2 Verdict
T2, H = 5 / 10	FAIL	Covid 2020 loss +5.6 / +5.1 pp	Experimental	same
T2, H = 40 / 60	FAIL	Not better than the Brownian baseline (H = 60 also ECE 2.68 / 2.73)	Experimental	same
Expected R filter (economic check)	Null result	+1.0%/yr at H = 20, t = 1.33	Expected R shown as a description only	same
T3 P(hold given touch)	FAIL	Covid 2020 -8.7 pp	Experimental	same
T3 per-zone edge vs random	PASS	Bucket predicted vs realised edge within 1.9 pp	Edge badge, default	same
T4 score → outcomes	FAIL	Rank IC -0.003 / -0.012 / -0.014 (n.s.)	"No out-of-sample edge"; bucket stats experimental	same
Meta P(win), H = 10	PASS	ECE 0.50 / 0.78; Brier meta − T2 -0.00040 [-0.00071, -0.00006]	Context-adjusted row, default	TECHNICALS_V3 Meta-labelling
Meta P(win), H = 20 / 40	FAIL	Not better than T2 / the context-free model	Experimental	same
Meta take / skip + size	FAIL (economic check)	US meta-filtered − all -4.3%/yr (t -2.55) at H = 20	Hidden unless experimental	same
Factors hi52, MAX, IVOL	PASS	20d IC +0.023 / -0.038 / -0.044; not profitable stand-alone after costs	Shown by default, "rank IC" badge	TECHNICALS_V3 #1
Factors mom, rmom, strev, indmom	FAIL	See the factor table	Experimental	same
Factor composite	FAIL	G6 vs v2 t 1.66; G7 economics	Not served	same
P(follow-through), H = 10 / 20	FAIL	Covid 2020 +18.3 / +17.2 pp	Experimental	TECHNICALS_V3 #5

Output	Gate	Key numbers	How the app shows it	Source
Volume-confirmation edge	PASS	Geometry-adjusted +1.6 pp [+1.2, +2.0]; no tradable edge	State chip, flag and score shown	same
Level-type ranking	Out-of-sample caveat	Top-ranked level types beat random levels by about 1.5 pp in bounce rate	"use as an entry filter, not a standalone strategy"	Level-type reliability panel note

Correct usage and worked example

Use the tab in this order: (1) read the context chips (is the range compressed, is earnings inside the window?); (2) look at the fan and the range strip to see what a normal 1-20 session move is; (3) check where the confluence zones sit relative to the fan and their touch probabilities; (4) read the trade setup's 20-session first-passage bar, expected R and break-even; (5) treat the technical score as a description, and the factor percentiles as rank context. Never treat any single number as a forecast.

Worked example: Apple (AAPL), as of the 16 Oct 2026 close (343.46 USD), default view.

1. Context: "Range normal 20d ±9.2% · p10", "RV 20d 16% · p15", "Trend aligned D ▲ · W ▲", "Earnings in 10 sessions · in range · typical ±1.8%". The range is narrow for Apple (only about 10% of the last two years were narrower), and the next release falls inside the 20-session window, so the fan widens at the earnings marker.
2. Range strip at 90%: 1d 337.11–349.93 (±1.9%), 5d 329.45–358.06 (±4.2%), 10d 324.11–363.97 (±5.8%), 20d 307.98–383.03 (±10.9%).
3. Setup: long, entry zone 332.58–342.96, invalidation 329.23, Target 1 362.88, reward : risk 1.45× from the limit entry.
4. Chance by horizon, 20d (the setup's own horizon): T1 first 37%, stop first 45%, neither 18%; expected R +0.03R after costs; break-even P(win) 43%.
5. Context-adjusted (meta) 10d: T1 first 23%, +6.2 pp vs T2 (17%), Exp. R +0.04R; decision hidden.
6. Technical score 88 (p100), with the note that it has no out-of-sample edge; factors: 52-week-high proximity p100, MAX p98, idiosyncratic vol p98, all "rank IC" badges.

Reading: Apple is near its 52-week high in a quiet, aligned uptrend; the nearest support zone is right under the price, and historically similar 20-session setups reached T1 first in roughly 37% of cases, enough for a slightly positive expectancy only because 18% of paths resolve neither way (see Trade setup). None of this says Apple will rise.

Limitations and pitfalls

- **Survivorship and current classification.** All validation universes use current index membership and current sectors (documented in every validation report).
- **Gate universe.** The probability models were validated on about 650 large and liquid names (mvp US 490 + EU 90 + UK 27 + JP 40). They are served for every security with a setup; outside US / EU / UK / JP and for ETFs or indices the meta row is always experimental and the factors carry "region not validated".
- **Daily data only.** Barrier events use the day's high and low, so the order of touches within a day is unknown (both on the same bar counts as a loss); there is no intraday or options data.
- **Regime breaks.** Every model's worst case is the Feb-Jun 2020 crash: sudden shocks arrive after the forecast origin. Calibration over history does not protect a single position in a crash.
- **The toggle persists.** If numbers look different from a colleague's screen, check the experimental switch first.

Related mnemonics and BQL fields

- Command bar: `TECH` (Technicals), `GP` (price graph, opens this tab for equities), `GIP` (daily data only: opens the price graph). On an instrument (index, FX, commodity) these open the instrument page instead. `HELP` with no argument opens this user guide (`/help`); `HELP TECH` describes the mnemonic. See 14. BQL console and mnemonics.
- BQL groups: `setup.*` (trade setup and score), `zone.*` (top zone), `cf.*` (conformal range), `tp.*` (calibrated score), `tech.*` (evidence-based factors), `bo.*` (breakout). There are no BQL fields for the T2 first-passage or meta probabilities; use the API endpoints `GET /api/v1/technicals/{security_id}/setup-probabilities` and `/meta-setup` .
- Related: Peers tab (conformal peer-group panel), 11. Backtest Lab (experimental triggers `meta_setups` , calibrated-band breakout, source `tech_factors`), 13. Watchlists and alerts (zone touch, price outside calibrated range), 15. System (Conformal validation page), 16. Methodology.

Price chart and calibrated range

Route: `/company/:id?tab=technicals` · **Panel:** Price & confluence zones · **Mnemonics:** `GP` , `TECH` · **Refresh:** prices nightly; fan nightly (`conformal_envelopes`)

Purpose

The chart shows the price history with the technical structure the other panels refer to (zones, swing pivots, moving averages, the setup's entry, invalidation and targets), and extends the time axis 20 sessions into the future to draw the **calibrated range fan**: the 80 / 90 / 95% bands for every horizon from 1 to 20 sessions. It lets you see at a glance whether a target or stop lies inside a normal move or far outside it.

Screenshot

2
Price & confluence zones
Zones
Top 3
All
Pivots
MAs
Setup
Calibrated range
1M
6M
YTD
1Y
5Y
10Y
Max
Auto (D)
D
W
M
Earnings
T1
362.88
Entry
342.96
342.94
invalid.
329.23
touch 20d: 93%
touch 20d: 24%
S1
S2
S3
HH
HL
HL
408.00
404.00
400.00
396.00
392.00
388.00
384.00
380.00
376.00
372.00
368.00
364.00
360.00
356.00
352.00
348.00
340.00
336.00
332.00
328.00
324.00
320.00
316.00
312.00
308.00
304.00
300.00
296.00
292.00
288.00
284.00
280.00
276.00
272.00
268.00
264.00
260.00

Figure 5B.4: Price & confluence zones (callout 2), Apple, 6M daily view. Shaded fan = calibrated range 95 / 90 / 80% to 20 sessions; dashed vertical = expected earnings date (2 Nov 2026); green bands S1-S3 = the three strongest support zones; badges "touch 20d: 93%" and "touch 20d: 24%" at the right edge; price lines Entry, Invalid. and T1.

Controls

Control	What it does
Zones (switch)	Shows or hides the confluence zone bands (green = support, red = resistance; opacity grows with strength; labels S1, R1 … by rank).
Top 3 / All	Top 3 (default) draws the three strongest zones; All draws every zone. Hovering a row of the Confluence zones table previews any zone.
Pivots (switch)	Swing-pivot markers. Market-structure pivots are labelled HH / HL (green) or LH / LL (red); other swing highs and lows are grey dots.
MAs (switch)	SMA 50 and SMA 200 lines (computed on the loaded daily bars).
Setup (switch)	Horizontal price lines Entry, Invalid. (dashed red), T1 and T2 (dashed green) with axis labels.
Calibrated range (switch)	Shows the fan, the earnings marker and the zone badges. Disabled, with the tooltip "Calibrated range not available for this security yet", when no envelope exists.
Price range: 1M, 6M, YTD, 1Y, 5Y, 10Y, Max	Time window; 6M by default.
Bar interval: Auto, D, W, M	Auto = daily up to 2 years, weekly up to 10 years, monthly for Max. Overlays (zones, pivots, setup, fan) are drawn on the daily view only; on weekly / monthly bars the header says "zones, levels, setup and the calibrated range shown on the daily view only". A "weekly (daily capped at 10y)" badge appears when the server returns weekly bars.
History label (right of the selectors)	"History from" a date, plus "weekly bars before" a date when part of the range predates the daily table.
Crosshair in the future area	Tooltip "by Nov 6: 80% 318–371 · 90% 312–378 · 95% 306–385" (the band at that date, all three levels).
Range strip: 80% / 90% / 95%	Level shown in the strip below the chart (90% default). Each cell is h = 1d, 5d, 10d, 20d with low–high and ± half-width in per cent; hover a cell for the target date.
Zone badges: 1d / 5d / 10d / 20d	Horizon of the touch badges (shared with the zones table).
About (popover)	Calibration details: headline "Calibrated to 90% over history (2010–2026 walk-forward), not "90% chance today"", the calibration pool (region and sector, for example US and TECH), walk-forward and last-year coverage at 1 / 5 / 10 / 20 sessions, the earnings-pool warning, the fan note and a "Validation report" link to System › Conformal validation (`/system/conformal`).
Volume histogram	Daily volume at the bottom of the pane, coloured by up / down day.

> Screenshot pending: the future-area crosshair tooltip and the "About" calibration popover (`/company/AAPL?tab=technicals` , hover the fan near the 20th future session, then click About in the range strip).

Methodology

Zones on the chart are the confluence zones of the Confluence zones table. **Pivots** come from the levels engine (fractal swings and the ZigZag(3 ATR) market-structure pivots). **Moving averages** are simple 50- and 200-session averages of the closes on screen.

The fan. For each future session h = 1..20 the app draws the marginal conformal band described in the Technicals methodology:

```
half-width(h, α) = √(q̂ₕ(α) · V(h))          in log-price units
lower / upper    = C₀ · exp(∓ half-width)       C₀ = the envelope's origin close
V(h)             = h · s²  (+ J when an expected earnings day falls inside (0, h])
```

- **Per-horizon calibration.** Each horizon has its own pooled split-conformal radius $\hat{q}_h$ (rolling window of 500 realised scores, outcome lag h, region × sector pool, separate earnings-event pool). An older snapshot without per-h radii falls back to interpolating between the 1 / 5 / 10 / 20-session radii (the About popover says so).
- **Nested, non-narrowing envelope.** For display, the half-widths are made non-decreasing in h (running maximum) and strictly nested (80 inside 90 inside 95). Both adjustments can only widen a band; when they changed anything the About popover adds "Drawn as a non-narrowing envelope where a longer horizon came out slightly narrower (only ever wider)".
- **Marginal, not joint.** Each date's band is calibrated alone: about 90% of h-session closes ended inside the h-session 90% band. It is **not** a path band: the chance that the whole path stays inside the 90% fan for 20 sessions is lower than 90%. The joint (simultaneous) 1/5/10/20 envelope that would answer that question failed validation and is not shown.
- **No direction.** The centre (dashed) is the last close. The fan says how far, not which way.
- **Earnings.** When the expected earnings date falls inside the window, the variance includes a jump term (mean excess squared reaction-day return over the last 8 events, shrunk) and the radius comes from the event pool, so the fan steps out at the marker.
- **Future dates** are real exchange sessions from the exchange calendar (holidays skipped).

Zone badges are drawn for zones that lie at least partly inside the 95% band at its widest horizon, at the zone's mid-price within the fan, with the touch probability at the selected horizon ("in zone" when the close is inside).

Data sources and refresh

Prices: adjusted daily bars (the chart loads up to about two years for the default view; longer ranges are fetched on demand). Fan: `conformal_envelopes` (nightly, full universe, EWMA filter λ 0.94, plain radius), served by `GET /api/v1/conformal/envelope/{security_id}` with the `fan` and `fan_info` fields. Earnings date: the earnings calendar (`fan_info.earnings` carries the date source, for example `provider_estimate`). Coverage audit: the nightly rolling audit (`conformal_coverage`) and the walk-forward rows shown in the About popover.

Validation status

Validated (by-product of the conformal gate). From `docs/CONFORMAL_VALIDATION.md` :

- **A.2 (marginal bands, target 90%):** coverage at h = 1 / 5 / 10 / 20 was 90.0 / 89.9 / 89.8 / 89.6% with widths 0.0605 / 0.1386 / 0.1979 / 0.2818 (log units), against the heuristic 2×ATR band at 98.4 / 98.1 / 98.2 / 98.2% with widths 0.1035 / 0.2314 / 0.3272 / 0.4628: "about 40% narrower than the heuristic 2×ATR band, which over-covers at 98%". They are **not sharper than a conformally calibrated ATR band**: "the gain comes from calibration, not from the filter".
- **A.5 (fan, every h = 1..20):** "Per-horizon calibration is on target at every h: 79.6-80.0 / 89.6-90.0 / 94.6-94.9%". In the Covid window the 90% band was 2-4 pp under (for example 86.0% at h = 10); with VIX > 25 slightly over at long horizons (92.3% at h = 20); the earnings pool within 0.6 pp. Worst calendar year at h = 20: 86.1% (2017).
- **Not shown:** the joint envelope (FAIL: Covid Feb-Jun 2020 77-78% at 90%; w_proj +35% vs a calibrated ATR box) and the next-bar set (FAIL on sharpness). Both stay "experimental" in Model Studio and the Backtest Lab.
- **Economic checks:** "Exiting the calibrated band is not a better breakout trigger than a naive N-day high" (FAIL). Calibrated band stops beat ATR stops (band 95% minus 3×ATR +6.2%/yr, t +3.56) but for the momentum strategy tested "any stop hurts, and no stop is best" (band 90% minus no stop -2.2%/yr, t -2.64).

Correct usage and worked example

Use the fan to judge **distance**, not direction: a target outside the 95% band at the setup's horizon is historically rare; a stop inside the 80% band is likely to be touched by normal noise. Compare the band with the earnings marker: a window that contains earnings is wider by construction.

Example (Apple, origin close 343.46 on 16 Oct 2026, pool US / TECH, event pool because earnings on 2 Nov fall at h = 11). The 90% range strip reads 20d 307.98–383.03 (±10.9%). The setup's T1 (362.88, +5.7% from that close) lies inside the 10-session 90% band (324.11–363.97) and well inside the 20-session band; the invalidation (329.23, -4.1%) lies just below the 5-session 90% band (329.45–358.06) and inside the 10-session band. In words: both barriers are within normal 1-4 week moves, which is why the 20-session first-passage bar shows only 18% "neither". The About popover shows the walk-forward 90% coverage at 20 sessions and the last year's realised coverage for US names (90.0% for 1-session bands, for example).

Limitations and pitfalls

- Coverage is **marginal over time and securities.** In a sudden crash the bands under-cover (Covid 2020: 2-4 pp under at 90%); per-regime coverage is reported, not guaranteed.
- The fan starts from the envelope's origin close. When the envelope is older than the last candle (the API flags `stale`), the fan origin is the earlier close stated in the strip.
- Do not read the 95% fan as "price will stay inside": that is a joint statement the model does not make.
- Weekly / monthly views hide all overlays.

Related mnemonics and BQL fields

`GP`, `TECH`. BQL: `cf.qm_{1,5,10,20}_{80,90,95}` (calibrated radii), `cf.v_{1,5,10,20}` (variance), `cf.fan_v`, `cf.fan_qm_80`, `cf.fan_qm_90`, `cf.fan_qm_95` (fan arrays), `cf.band_5d_90` (half-width of the 90% 5-session band as a fraction of the close), `cf.touch_20d`, `cf.earn_in_window`, `cf.pool_key`. Alert rule `price_outside_calibrated_range` in 13. Watchlists and alerts; coverage audit in 15. System.

Context chips

Route: `/company/:id?tab=technicals` · **Panel:** chip row above the chart · **Mnemonics:** `TECH` · **Refresh:** computed when the page loads (from the latest snapshot)

Purpose

The chips put today's volatility and trend into the stock's own historical context: is the range compressed (squeeze) or stretched (expansion), how strong is the trend, do the daily and weekly trends agree, and is an earnings release inside the next 20 sessions? They are descriptive context for reading the fan and the setup, not signals.

Screenshot

The chips appear in Figure 5B.1 (Apple: "Range normal 20d ±9.2% · p10", "RV 20d 16% · p15", "IV n/a", "ATR 14 1.9% · p29", "ADX 14 18 weak ▲", "Trend aligned D ▲ · W ▲", "Earnings in 10 sessions · in range · typical ±1.8%").

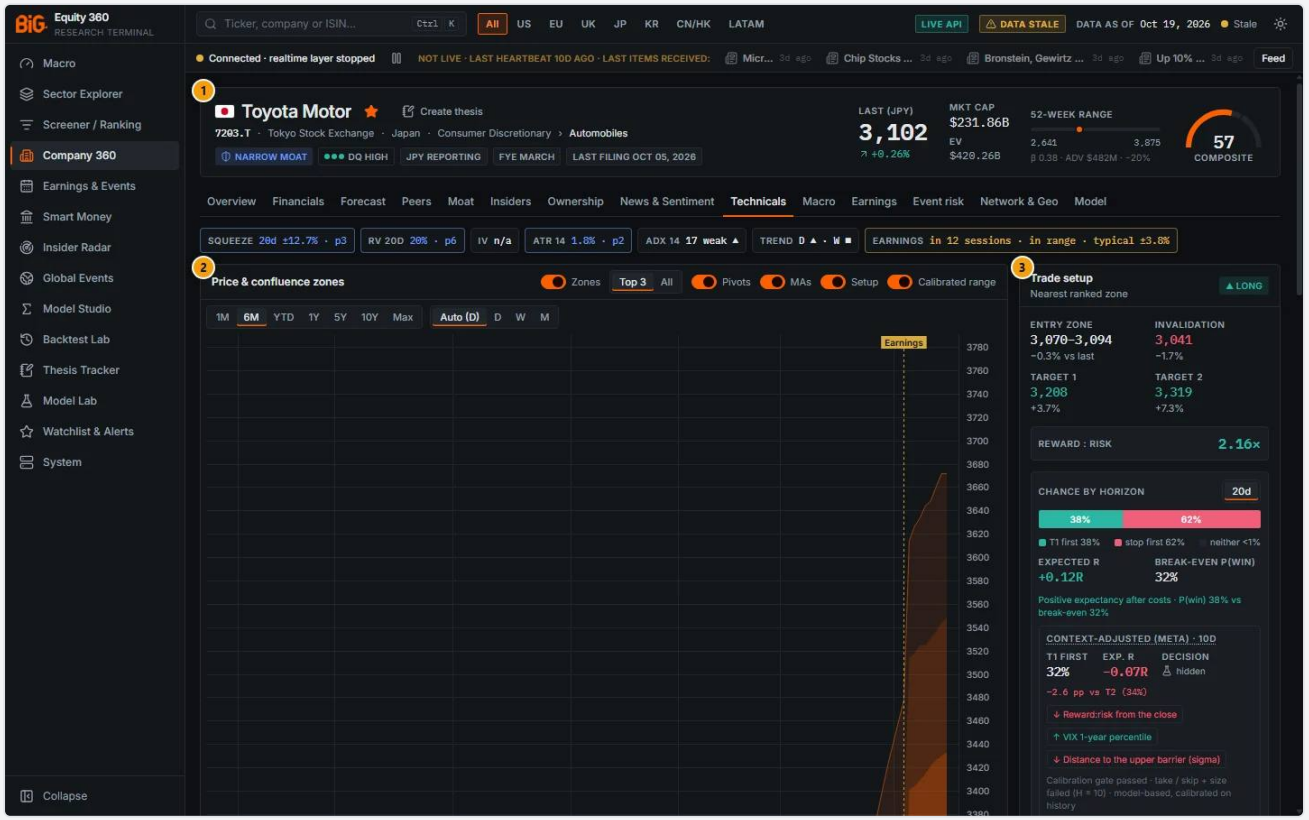


Figure 5B.5: Toyota Motor (`/company/7203.T?tab=technicals`), default view. The chip row shows "Squeeze 20d ±12.7% · p3" and "ATR 14 1.8% · p2" (blue), "Trend D ▲ · W ■" (weekly flat) and "Earnings in 12 sessions · in range · typical ±3.8%" (amber). Callouts as in Figure 5B.1 (1 header, 2 Price & confluence zones, 3 Trade setup; 4-10 are below the fold).

Controls

Chip	Value shown	Colour	Tooltip
Squeeze / Expansion / Range normal	20-session ± half-width of the calibrated 90% range and its percentile (for example p10)	Blue = squeeze, amber = expansion	Definition, plus "Today: pNN of the last 504 sessions (n = ...)". Ends "Descriptive: a squeeze is not a breakout signal."
RV 20d	Annualised realised volatility and percentile	Blue / amber as above	Definition

Chip	Value shown	Colour	Tooltip
IV	"n/a"	Neutral	"Implied volatility: n/a - there is no per-stock option-implied volatility source (for the US only VIX-type index volatility exists)."
ATR 14	ATR as % of price and percentile	Blue / amber	Definition
ADX 14	ADX value, state (weak / emerging / trending / strong) and direction arrow	Green / red when trending or strong	Definition plus "+DI x / −DI y". "Strength only - it says nothing about what happens next."
Trend aligned / Trend mixed / Trend	Daily and weekly arrows (▲ up, ▼ down, ■ flat)	Green aligned up, red aligned down, amber mixed	Definition
Earnings	"in N sessions", "· in range" when inside the 20-session window, "· typical ±x%"	Amber when inside the window	Date source and method; "The calibrated range includes the earnings-event pool."

Chips are buttons for keyboard focus; hovering or focusing shows the tooltip. There are no other controls.

Methodology

Computed from the security's adjusted daily bars (about 3.5 years), over a lookback of 504 sessions (about two years). A percentile is the share of the lookback whose value is at or below today's (today included).

Chip	Definition
Range width	Width of today's calibrated 20-session 90% band, $2 \cdot \sqrt{(q_{90,20} \cdot V(20))}$, compared with the same band on each of the last 504 sessions, using today's calibrated radius (so the percentile compares the volatility term). Squeeze ≤ 10th percentile, expansion ≥ 90th, otherwise normal. Model-based.
RV 20d	Standard deviation of the last 20 daily log returns × √252.
ATR 14	Wilder ATR(14) / close.
ADX 14	Wilder ADX(14) with +DI / −DI. Below 20 weak, 20-25 emerging, 25 or more trending, 40 or more strong. Direction = up when +DI > −DI.
Trend	Daily = close vs SMA 50 and the SMA 50's 10-session slope; weekly = last weekly close vs the 30-week SMA and its 4-week slope. Aligned when both point the same way, mixed when they disagree, otherwise flat.
Earnings	Next expected date (company, provider or model-projected, labelled); "in range" when within 20 sessions (the fan then uses the earnings-event pool). Typical move = median absolute cumulative abnormal return over [-1, +1] sessions around the stock's past releases (market model).

Data sources and refresh

`GET /api/v1/technicals/{security_id}/context`, computed at request time from the serving snapshot's bars, the calibrated radius (`cf_envelopes`) and the earnings tables. Refreshes whenever the snapshot is rebuilt (nightly DAG).

Validation status

Not formally validated (descriptive statistics, no gate). The range-width chip uses the validated calibrated radius but the squeeze / expansion classification itself has no predictive gate, and the tooltip says so: "a squeeze is not a breakout signal". The trend and ADX chips are rule-based.

Correct usage and worked example

Use the chips to set expectations for the fan and the setup: in a squeeze the bands are narrow and stops placed by eye tend to sit too close; with earnings "in range" the 20-session numbers include the event pool.

Example (Toyota, Figure 5B.5): "Squeeze 20d ±12.7% · p3" means today's calibrated 20-session 90% range is narrower than 97% of the last two years for Toyota, while "Earnings in 12 sessions · in range · typical ±3.8%" warns that the window includes a release whose median abnormal move was 3.8%. Apple (Figure 5B.1) shows "Range normal 20d ±9.2% · p10": the percentile rounds to 10 but is just above the 10th-percentile threshold (0.1012), so the state is normal.

Limitations and pitfalls

- Percentiles are relative to the stock's **own** last two years, not to peers. A "normal" range for a volatile stock can be wider than an "expansion" for a utility.
- The chip's 20-session width is a context measure computed on the latest bars; it need not equal the strip's 20d band (Apple: chip ±9.2%, strip ±10.9% at 90%).
- The trend chip (daily / weekly alignment) and the setup's "trend" badge use different rules and can disagree (JPMorgan: chip "Trend mixed", setup badge "Trend sideways").
- There is no per-stock implied volatility source.

Related mnemonics and BQL fields

`TECH` . BQL: `cf.earn_in_window` , `cf.earn_days_ahead` , `cf.sigma_d` ; the chip values themselves are not BQL fields (use the API). Earnings dates and reactions: 5C. Earnings, 6. Earnings & Events.

Confluence zones

Route: `/company/:id?tab=technicals` · **Panel:** Confluence zones (callout 8) · **Mnemonics:** `TECH` · **Refresh:** nightly (`tech_levels` , `confluence_zones` , `technicals_probabilities`); zone model refit weekly

Purpose

A confluence zone is a price band where several independent technical levels coincide (for example a prior weekly low, a Camarilla pivot, a Fibonacci retracement and an anchored VWAP). The table lists every zone with its strength, the chance of touching it within the selected horizon, and the validated flag that says whether zones like it have historically held more or less often than random price bands.

Screenshot

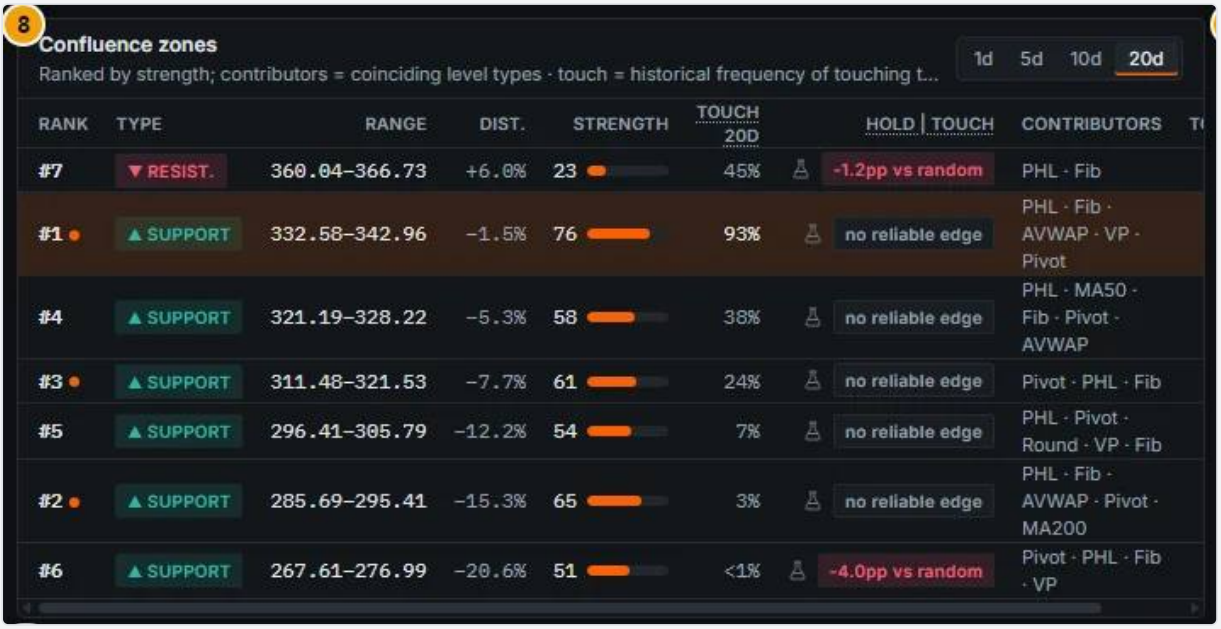


Figure 5B.6: Confluence zones (callout 8), Apple, default view, touch horizon 20d. The highlighted row (#1) is the setup's zone; the orange dots mark the three strongest zones drawn on the chart. The hold-given-touch column shows a flask icon (experimental, hidden) and the validated edge badge.

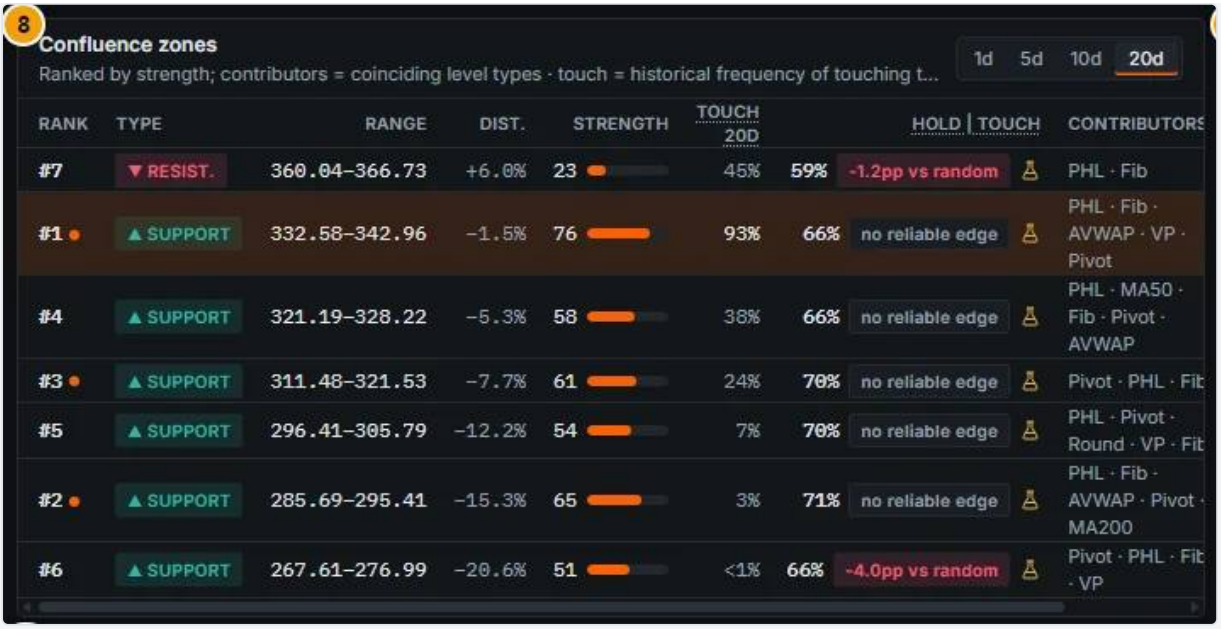


Figure 5B.7: The same table with the experimental toggle on: the calibrated P(hold given touch) appears (66%, 59%, 70% ...) with a flask icon.

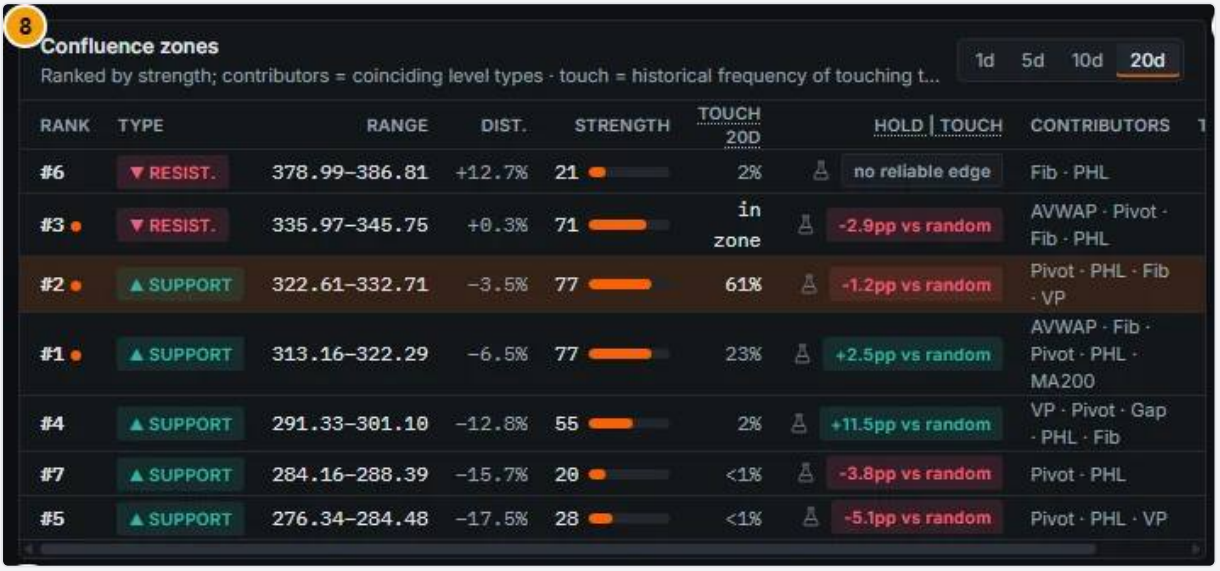


Figure 5B.8: JPMorgan (`/company/JPM?tab=technicals`), default view (hold probabilities hidden behind the flask icon; edge badges visible). Zones with a significant edge show "+2.5pp vs random", "+11.5pp vs random" (better) or "-2.9pp vs random", "-5.1pp vs random" (worse); #3 resistance shows "in zone" in the touch column.

Controls

Column / control	What it shows
Touch horizon (1d / 5d / 10d / 20d)	Horizon of the Touch column and of the chart badges.
Rank	Strength rank (#1 = strongest). An orange dot marks the three strongest zones (drawn on the chart by default).
Type	▲ Support (below the price) or ▼ Resist. (above).
Range	Zone low–high in the trading currency.
Dist.	Distance of the zone mid-point from the last price, in per cent.
Strength	0-100 with a bar (see Methodology).
Touch Nd	Calibrated chance of touching the zone's near edge within N sessions; "in zone" when the close is inside; "<1%" beyond the calibrated tail. Tooltip: "Historical frequency of touching the zone's near edge within N trading day(s) for similar setups ... Calibrated within 0.2 pp over 2011–2026; model-based, not a forecast of direction." See Touch probabilities.
Hold given touch	Experimental P(hold given touch) (toggle on) and the validated edge badge: "+x.xpp vs random" (green), "-x.xpp vs random" (red) or "no reliable edge" (grey). Cell tooltip: P(hold given touch), the random-control probability, the edge with its walk-forward CI, P(touch) and P(touch and hold), and "EXPERIMENTAL (failed the validation gate)" where relevant.

Column / control	What it shows
Contributors	Level families present: PHL (prior period high / low, opens, period pivots, Camarilla, opening range, 52-week and all-time extremes), Fib (Fibonacci retracements and Fibonacci pivots), Pivot (swing pivots, market structure), AVWAP, VP (volume profile), Round, MA50, MA200, Gap.
Touches	Sum of past touches of the member levels.
Row hover	Previews the zone on the chart (also zones outside the top 3).
Row highlight	The zone the trade setup is built on (accent background).

The column header of the hold column reads "Hold" and "touch" separated by a vertical bar (P of hold given touch). Rows are listed by price, highest first (resistance at the top, support below); the Rank column gives the strength rank.

Methodology

LEVEL FAMILIES

Levels are generated point-in-time from daily bars (a level becomes usable only after the information that defines it is complete). Families:

Family	Levels	Timeframes
Period OHLC	Prior period open, high, low, close, mid; current period open	D, W, M, Q, Y (current open W and above)
Opening range	High / low of the first bars of the period	W, M, Q, Y
Floor pivots	P, R1-R3, S1-S3 of the prior period	W, M, Q, Y
Camarilla	H3-H5, L3-L5	W, M, Q, Y
Fibonacci pivots	R1-R3, S1-S3	W, M, Q, Y
Swing pivots	Fractal swing highs / lows, k = 2, 3, 5, 10 bars (daily), k = 2 (weekly)	D, W
Market structure	ZigZag (3 ATR) pivots labelled HH / HL / LH / LL; break-of-structure (BOS) and change-of-character (CHoCH) levels	—
Fibonacci	23.6 / 38.2 / 50 / 61.8 / 78.6% retracements, 127.2 / 161.8% extensions of the last ZigZag leg	—
Anchored VWAP	Anchored at YTD and QTD start, last swing high / low, last earnings filing, 52-week high / low	—
Volume profile	Point of control, value-area high / low, high-volume nodes over 6 months and 1 year	—
Extremes	52-week high / low, all-time high	—
Round numbers	Major and minor psychological grid	—
Moving averages	SMA 50, SMA 200	—
Gaps	Near edge of an unfilled price gap (≥ 0.25 ATR)	—

LEVEL WEIGHT

```
weight = reliability × recency × touches × timeframe × volume-node strength × role-flip
reliability = (type_rank_score / 0.5)²       from the level-type backtest; neutral 0.5 → 1.0
recency     = 0.5^(age_days / 180)            swing, structure, Fibonacci and gap levels only
touches     = 1 + 0.1 × min(touches, 5)
timeframe   = D 0.55 · W 0.8 · M 1.0 · Q 1.15 · Y 1.3
volume node = 0.5 + 0.5 × node_strength       volume-profile levels only
role flip   = 0.7 if the level was closed through (it now acts from the other side)
```

Levels at the identical price are merged (weight = max + 0.25 × (sum − max)), so one pivot seen by several swing orders is not counted six times. Level types that test clearly worse than random levels for this stock (type rank score < 0.4) are left out.

ZONES AND STRENGTH

1. Each level's distance from the close is expressed in ATR units, `x = (price − close) / ATR14`, for $|x| \le 12$.
2. A weighted Gaussian kernel density (bandwidth 0.35 ATR) is computed over x.
3. Zones are density peaks of at least 15% of the highest peak, bounded where the density falls to 50% of the peak, at the valley to a neighbouring peak, or at ±0.75 ATR from the peak.
4. Evidence `S = weight_sum × (1 + 0.2 × (distinct families − 1))`; **strength = 100 × S / (S + 20)** (S = 20 was the median zone of the first live run, giving strength 50).

Strength is a weight-of-evidence score. It is **not** a probability and it was not validated as a predictor of holding; the validated output is the edge badge below.

P(HOLD GIVEN TOUCH) AND THE EDGE VERSUS RANDOM (T3)

- **Touch** = the near edge is reached within 20 sessions. **Break** = a close beyond the far edge by at least 0.5σ. **Hold** = a reversal of at least 1.5σ beyond the near edge before any break, within 10 sessions of the touch (σ = the EWMA daily volatility).
- **P(hold given touch):** a HistGradientBoosting classifier with isotonic calibration, purged walk-forward (2-year test blocks 2010-2026). Inputs: the 14 level-family flags, member counts, kernel density, width and distance in σ, approach speed (5 and 20 sessions), trend and structure, RSI, distances to SMA 50 / 200, nearest volume node, prior touches and hold rate of the same band, volatility regime, VIX > 25, earnings within 20 sessions, region and side. The old strength is not an input.
- **Random control:** for every real zone, a price band of the same width shifted by ±U(0.75, 2.5) ATR, labelled by the same rule; the same pipeline trained on these bands gives P_control. **Edge = P(hold) − P_control.**
- **Per-zone flag:** the zone is placed in a walk-forward quintile of predicted edge; the flag is shown as significant only when that quintile's realised edge had a 95% confidence interval excluding zero. Quintile 5 → "holds more often than random" (green badge); quintiles 1-2 → "holds less often than random" (red); quintiles 3-4 → "no reliable edge".
- **P(touch and hold)** = P(touch within H) × P(hold given touch), in the cell tooltip.

Data sources and refresh

Levels and zones: `tech_levels` and `confluence_zones` (nightly, every security with at least 60 bars; latest plus month-end history kept), served by `GET /api/v1/companies/{security_id}/technicals`. Zone hold model: `GET /api/v1/technicals/{security_id}/zones?horizon=N` (nightly scoring by `technicals_probabilities`, weekly refit by `technicals_probabilities_fit` on the gate universe). Touch: `GET /api/v1/conformal/touch`.

Validation status

From `docs/TECHNICALS_V2_VALIDATION.md`:

- **T3 P(hold given touch): FAIL → experimental.** ECE 1.45 pp; better than climatology and the old strength (Brier 0.2410 vs 0.2470 / 0.2470), but the worst regime was **Covid 2020: -8.7 pp** (2020 full year -4.6): "zones broke far more often than predicted". The old strength "carries no information beyond the base rate: Brier identical to climatology".
- **T3 edge vs random (per-zone flag): PASS → production.** "Bucket predicted vs realised edge within 1.9 pp"; level features beat the random-control model (Brier -0.00045 [-0.00066, -0.00025]).
- **The honest headline:** "on average, touched confluence zones hold **less** often than random price bands with the same width, approach, trend and regime (-0.63 pp vs the control model, CI [-1.26, -0.01]; raw rate 55.5% vs 56.9%)". By predicted-edge quintile, the realised edge was:

Quintile	Predicted edge	Realised edge	95% CI	Shown as
1	-5.6 pp	-4.4 pp	[-5.2, -3.6]	holds less often than random
2	-1.8	-1.3	[-1.9, -0.7]	holds less often than random
3	-0.4	-0.2	[-1.0, +0.5]	no reliable edge
4	+0.6	+0.7	[-0.2, +1.6]	no reliable edge
5	+4.0	+2.1	[+1.4, +2.7]	holds more often than random

Correct usage and worked example

Use the edge badge as a **filter**: prefer zones flagged "holds more often than random", be wary of zones flagged "holds less often", and treat "no reliable edge" as an ordinary price band. Combine with the touch probability: a zone with a high edge that is unlikely to be touched within your horizon is irrelevant to the trade.

Example (Apple, 20d, as of 16 Oct 2026). Zone #1 (support 332.58–342.96, strength 76, 25 member levels from 9 families: PHL · Fib · AVWAP · VP · Pivot) has Touch 20d 93% and "no reliable edge" (experimental P(hold given touch) 66% vs control 65.6%: edge +0.4 pp, not significant). Zone #6 (support 267.61–276.99) is flagged "-4.0pp vs random": zones in its quintile held less often than random bands, but its touch probability is "<1%". Zone #7 (resistance 360.04–366.73, whose centre 362.88 is the setup's T1) shows "-1.2pp vs random" and Touch 20d 45%: historically, resistance like it tended to give way slightly more often than a random band, which is mildly favourable for a long whose target sits there.

Limitations and pitfalls

- The badge number is the model's **point estimate for this zone**; the significance comes from its quintile. Large numbers such as "+11.5pp vs random" (JPMorgan zone #4) should be read against the validation: the top quintile realised **+2.1 pp** on average, with predicted +4.0 pp.
- High strength does not mean the zone will hold; strength was never validated as a predictor.

- In the walk-forward history the level-type reliability weights were neutral (no point-in-time history exists), while live zones use them; zone sets therefore differ slightly from the tested ones (documented deviation).
- The zone snapshot is as of the last nightly run; if the price has moved since, a zone may already contain the price even though the table shows a touch percentage.
- Rows are ordered by price, not by rank.

Related mnemonics and BQL fields

`TECH` . BQL: `zone.side` , `zone.center` , `zone.distance_atr` , `zone.reward_risk` (top zone); `setup.support_low` , `setup.support_high` , `setup.resistance_low` , `setup.resistance_high` , `setup.dist_support_atr` , `setup.dist_resistance_atr` . Alert rule "zone touch" in 13. Watchlists and alerts.

Touch probabilities

Route: `/company/:id?tab=technicals` · **Panel:** Touch column of Confluence zones; zone badges on the chart · **Mnemonics:** `TECH` · **Refresh:** nightly (`conformal_envelopes` touch curves)

Purpose

"Will price get to this level within the next h sessions?" is a path question. The touch probability answers it with a calibrated historical frequency, for every confluence zone at 1, 5, 10 and 20 sessions, and is reused by the Thesis Tracker for targets and stops.

Screenshot

See the Touch column in Figures 5B.6 to 5B.8 and the "touch 20d: 93%" / "touch 20d: 24%" badges in Figure 5B.4. The horizon selector is shared by both.

Controls

Control	What it does
1d / 5d / 10d / 20d (zones table header and range strip)	Horizon of the Touch column and the chart badges (20d default).
Touch column header tooltip	Definition and validation sentence (see Confluence zones).

Methodology

For each pool (region × sector, normal or earnings-event origins) and horizon, the app stores the distribution of the **standardised running maximum** over its calibration origins:

```
M_up(h) = max over k ≤ h of log(H_(i+k) / C_i) / √V(h)     (M_dn with lows for levels below)
u       = |log(level / close)| / √V(h)
P(touch within h) = share of calibration origins with M ≥ u
```

This is an empirical (conformal-style) probability of a path event, accurate to about 1/(n+1) under exchangeability. It is **not** a Gaussian probability. Zones use their near edge (support: the zone high; resistance: the zone low); a close inside the zone is shown as "in zone" (probability 1). Values at or below 0.5% are displayed "<1%" (beyond the calibrated tail). Horizons above 20 sessions (Thesis Tracker only) reuse the 20-session curve with the variance scaled by h and are flagged "extrapolated", not validated.

Data sources and refresh

`GET /api/v1/conformal/touch?security_id=…&horizon=20` (automatic mode: the confluence zones plus the nearest still-valid technical levels), from the nightly touch curves of `conformal_envelopes` .

Validation status

Validated (conformal gate, §A.4). US mvp, 2011-2026, curves calibrated on year Y−1 for test year Y. Predicted / realised P(touch):

h	u = 0.5	u = 1.0	u = 1.5	u = 2.0	u = 2.5	u = 3.0
1	0.556 / 0.557	0.284 / 0.286	0.132 / 0.133	0.060 / 0.061	0.028 / 0.028	0.014 / 0.014
5	0.583 / 0.584	0.297 / 0.297	0.137 / 0.137	0.061 / 0.061	0.028 / 0.028	0.015 / 0.014

h	u = 0.5	u = 1.0	u = 1.5	u = 2.0	u = 2.5	u = 3.0
20	0.601 / 0.600	0.311 / 0.311	0.142 / 0.142	0.063 / 0.063	0.029 / 0.029	0.015 / 0.015

"Calibration is within 0.2 pp. For comparison, a Gaussian reflection-principle formula gives P(u = 3) = 0.0027, 5× too low." Real prices have fat tails: a level three standard deviations away was touched about 1.4-1.5% of the time, not 0.27%.

Correct usage and worked example

Read the touch probability together with the horizon of your plan. **Example (Apple, 16 Oct 2026 close 343.46):** zone #1's near edge (342.96, 0.15% below the close) has a touch probability of 75% within 1 session, 88% within 5, 91% within 10 and 93% within 20; zone #3's near edge (321.53) has 24% within 20 sessions; zone #2 (295.41, u ≈ 2.6 at 20 sessions) has 3%. So a buy order resting at zone #3 would, historically, be filled within a month in roughly one case in four.

Limitations and pitfalls

- The probability is about touching, not holding or breaking (see Confluence zones for hold).
- Validation is on US names; other regions use the same method on their own pools.
- It is marginal over the pool: a stock with an imminent catalyst may behave differently from its pool.

Related mnemonics and BQL fields

`TECH` . BQL: `cf.touch_20d` (calibrated 20-session move to the 90% band edge). Thesis targets and stops: 12. Thesis Tracker.

Level types and level-type reliability

Route: `/company/:id?tab=technicals` · **Panels:** Level types · backtested hit rates (callout 9), Level-type reliability (callout 10) · **Mnemonics:** `TECH` · **Refresh:** weekly (`level_type_backtest`)

Purpose

These two tables answer "which kinds of level does this stock actually respect?" They show, per level type, how often a touch of that type was followed by a bounce, and how that compares with touching a random price. The results feed the reliability weight of each level in the confluence zones.

Screenshot

Figure 5B.9: Level types · backtested hit rates (callout 9), Apple. One row per type with its best level key (for example "Swing pivots · swing_high_k10@D"), shrunk hit rate, average reaction and number of events.

Figure 5B.10: Level-type reliability (callout 10), Apple. Edge versus random from +1.4 pp (Prior period H/L) down to +0.1 pp (Volume profile), with the out-of-sample note underneath.

Controls

Column	What it shows
Level type (callout 9)	Type label · the best-ranked level key of that type for this stock (key = type@timeframe, for example `prev_low@D`, `fibp_s1@W`, `vp_poc_6m@NA`).
Hit rate	Shrunk bounce rate, with a bar scaled to the highest row.
Avg reaction	Mean favourable excursion after a touch, in ATR converted to per cent of the price.
n	Number of touch events of that key.
Level type (callout 10)	One row per type, all keys of the type pooled.
Hit rate	Event-weighted shrunk bounce rate of the type.
Edge vs random	Hit rate minus the random-level control, in pp (▲ green positive, ▼ red negative). Rows sorted by edge.
Events	Total events.
Note	"Out-of-sample: top-ranked level types beat random levels by ~1.5pp bounce rate — use as an entry filter, not a standalone strategy."

Methodology

Event study (per stock, point-in-time). A bar is a touch event of a level when it opens more than 0.25 ATR away from the level and its range then comes within 0.25 ATR of it (support test if the open was above, resistance test if below). One test is counted once until it resolves. An event is used only when its whole outcome window is in the past.

Outcome (10 bars, ATR = ATR14 of the previous bar).

- **Hit / bounce:** a favourable excursion of at least 1.0 ATR before an adverse close more than 0.5 ATR beyond the level (a bar that does both counts as a break: conservative).
- **Break:** the adverse close comes first.
- **Reaction:** maximum favourable / adverse excursion in ATR.

Random-level control. Every level is replayed with its price shifted by ±U(0.75, 2.5) ATR (a new shift every 20 bars for moving levels); same event and outcome logic. The control's hit rate is what touching any price yields for this stock; **edge = level hit rate − control hit rate**.

Shrinkage. Rates are shrunk hierarchically (universe → sector × region pool → ticker) with a beta-binomial prior of 2,000-5,000 pseudo-events, because per-ticker deviations carry little information: `(hits + m · p_parent) / (n + m)`. The reliability score `logistic(edge_shrunk / 0.02)` is the `type_rank_score` used in the level weight.

The reliability table aggregates this stock's level keys by type, weighting by events.

Data sources and refresh

`level_type_backtest` (weekly DAG, 10-year lookback) → `level_type_stats`; served in `GET /api/v1/companies/{security_id}/technicals` (`level_stats`, `type_reliability`).

Validation status

Out-of-sample caveat, not a gated model. The panel's note and the original user guide state the result: tested out of sample, **the top-ranked level types beat randomly placed levels by about 1.5 pp in bounce rate**. Use them as an **entry filter, not a stand-alone strategy**. The Technicals v2 plan records level-type bounce edges against random levels of only "+0.3 to +1.4 pp". The prior strength

of the shrinkage was chosen by internal validation (scores from events resolved before 2017 ranked the 2017-2020 out-of-sample edge best at about 3,000 pseudo-events; the 2021+ hold-out confirmed it).

Correct usage and worked example

Use the tables to decide which contributors of a zone deserve attention. **Example (Apple):** prior period highs / lows have the largest edge (+1.4 pp on 5,746 events), Fibonacci +1.1 pp, while volume-profile levels (+0.1 pp) and anchored VWAPs (+0.2 pp) are barely different from random prices. A zone made mostly of prior-period and Fibonacci levels is therefore slightly better supported than one made of VWAP and volume nodes, but the difference is a percentage point or so of bounce rate.

Limitations and pitfalls

- Edges are small (about 1-1.5 pp at best). A 60% hit rate looks impressive until you see that touching a random price gives about 58%.
- The "Hit rate" bar in callout 9 is scaled to the best row, which visually exaggerates small differences.
- The reliability table pools this stock's keys with heavy shrinkage toward its sector × region pool; it is not a cross-sectional study.
- Hit and break are defined in ATR on daily bars; intraday behaviour is not captured.

Related mnemonics and BQL fields

`TECH` . No dedicated BQL fields for the level-type statistics (API only).

Trade setup

Route: `/company/:id?tab=technicals` · **Panel:** Trade setup (callout 3) · **Mnemonics:** `TECH` · **Refresh:** setup nightly (`confluence_zones`); probabilities nightly (`technicals_probabilities`), surface refit weekly

Purpose

The Trade setup card turns the nearest relevant zone into a concrete plan (bias, entry zone, invalidation, targets, reward : risk) and then says, from history, how such plans tended to resolve: how often T1 was reached before the invalidation within H sessions, how often the stop came first, how often neither happened, and the resulting expected R after costs. It is the main decision-support card of the tab.

Screenshot

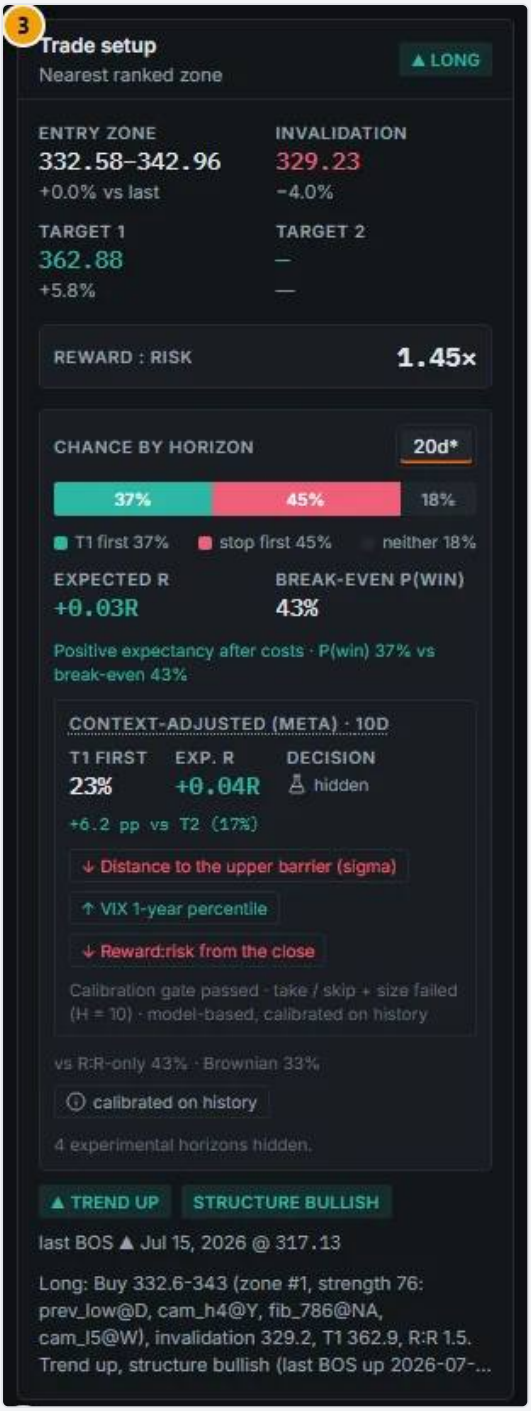


Figure 5B.11: Trade setup (callout 3), Apple, default view. Chance by horizon shows only 20d (the production horizon and the setup's own horizon); "4 experimental horizons hidden."*

Figure 5B.12: The same card with the experimental toggle on: horizons 5d, 10d, 40d and 60d appear with flask icons, and the meta row switches to 20 sessions (experimental).

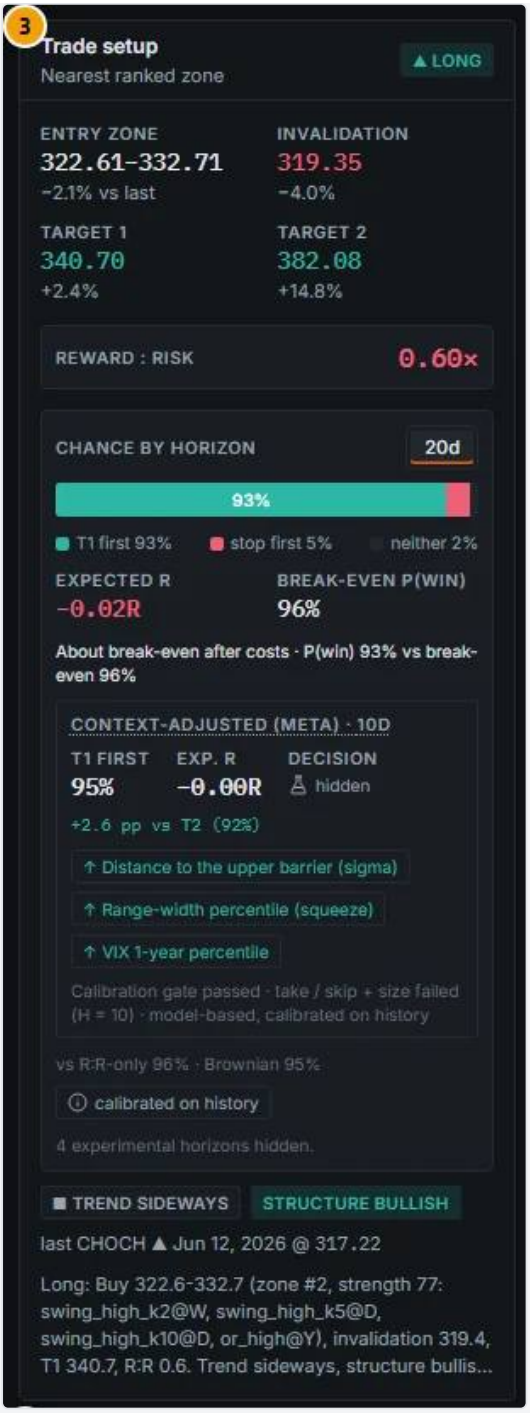


Figure 5B.13: JPMorgan, default view. The price (339.72) is above the entry zone (322.61–332.71) and T1 (340.70) is only 0.3% above the close: at 20 sessions T1 first 93%, stop first 5%, neither 2%, but the break-even P(win) is 96% and expected R -0.02R ("About break-even after costs"); R:R from the limit entry 0.60× in red. The meta row (10d) reads 95%, +2.6 pp vs T2 (92%).

Controls

Element	What it shows / does
Bias badge	▲ LONG, ▼ SHORT or ■ NEUTRAL.
Entry zone / Entry	Zone low–high (or a single entry price) and the distance to the last price.
Invalidation	The stop level (red) and its distance from the entry.
Target 1 / Target 2	Target prices (green) and distance from the entry; "—" when there is no second target.
Reward : risk	From the limit entry: green ≥ 2, red < 1.
Chance by horizon	Segmented control 5d / 10d / 20d / 40d / 60d (only 20d without the toggle). An asterisk = the setup's own horizon; a flask = experimental.
Probability bar	Green "T1 first", red "stop first", grey "neither"; hover a segment for "T1 first: 37.2% (within 20 sessions)".
Expected R	After costs; green > 0, red < 0. Hover: "After costs (0.01R round trip). Marked to market incl. the time exit: +0.07R".

Element	What it shows / does
Break-even P(win)	Hover: the R:R from the close and the break-even including costs.
Expectancy line	"Positive / Negative expectancy after costs" or "About break-even after costs", with "P(win) x% vs break-even y%".
Context-adjusted (meta) row	See Context-adjusted (meta) row.
Baselines line	"vs R:R-only x% · Brownian y%" (hover for the definition).
calibrated on history	Popover: definitions of the setup probabilities, expected R and break-even; pool line ("Pool US · vol low · earnings window · n = 141,388 historical origins · R:R from the close 1.30× (card: from the entry)"; "(all-region fallback)" or "distances beyond the calibrated grid (clamped)" when applicable); gate lines per horizon; the experimental switch.
Trend / structure badges	"▲ TREND UP", "STRUCTURE BULLISH", and "last BOS ▲ (date) @ (price)" (break of structure or change of character).
Note	One-line plan, for example "Long: Buy 332.6-343 (zone #1, strength 76: prev_low@D, cam_h4@Y, fib_786@NA, cam_l5@W), invalidation 329.2, T1 362.9, R:R 1.5. Trend up, structure bullish (last BOS up 2026-07-15)."

When no horizon is visible (all failed and the toggle is off) the card shows "Setup probabilities are experimental (failed validation) — enable in the popover".

Methodology

THE SETUP

```
trend score   = share of {close > SMA200, SMA50 > SMA200, close > SMA50, SMA50 20-session slope > 0}
                → up ≥ 75%, down ≤ 25%, else sideways
structure     = ZigZag(3 ATR) market structure: bullish 100, bearish 0, range 50
direction     = 0.55 × trend score + 0.45 × structure → long ≥ 60, short ≤ 35, else neutral
```

For a long (or neutral) bias the card uses the nearest strong support zone (strength ≥ 40; otherwise the strongest support): entry = min(zone high, close), entry zone = zone low to entry, **invalidation = zone low − 0.5 ATR**, **T1 / T2 = centres of the next resistance zones above** with strength ≥ 20 (if none, T1 = entry + 3 ATR, noted "no overhead zone: 3 ATR projection"). Shorts mirror this. `R:R = (T1 − entry) / (entry − invalidation)`.

TWO-BARRIER FIRST-PASSAGE PROBABILITIES (T2)

- **Barriers from the close.** The probabilities assume a market entry at the last close, so a = log(T1 / close) / σ_H and b = log(close / invalidation) / σ_H, with σ_H = √V(H) from the conformal EWMA filter (including the earnings jump).
- **Outcomes within H sessions:** win = the high reaches T1 first; loss = the low reaches the invalidation first, or both on the same bar; neither = no barrier by H.
- **Estimator (grid, production):** for every daily origin of the gate universe (3.2 million origins × 5 horizons), the outcome at each (a, b) of a 20 × 20 log grid (0.1 to 8 σ_H), counted per region × volatility regime × earnings window; shrinkage pool → region → all regions (500 pseudo-origins); monotone smoothing; bilinear interpolation. Volatility regime = σ_d over its median of the previous 252 sessions: low < 0.85 ≤ mid ≤ 1.2 < high.
- **Setup's own horizon:** the first H in 5 / 10 / 20 / 40 / 60 at which T1 is within one σ_H of the close (|log(T1 / close)| ≤ σ_H), else 60. Marked with an asterisk.

EXPECTED R AND BREAK-EVEN

```
R:R(close)    = (T1 − close) / (close − invalidation)                    (long; mirrored for a short)
Expected R    = P(win) × R:R(close) − P(loss) × 1 − round-trip cost (in R)
E[R] (MTM)    = Expected R + P(neither) × mean R of the paths that hit neither by H
Break-even    = 1 / (1 + R:R(close))       a win-or-lose trade with this R:R breaks even
```

Costs follow the Backtest Lab cost model (commission 1 bp, ADV-bucket half spread, square-root impact on a 200k USD position, borrow for shorts). Baselines: **R:R only** = 1 / (1 + R:R) (ignores the horizon and "neither"); **Brownian** = the driftless two-barrier formula.

Data sources and refresh

Setup: `tech_setups` (nightly). Probabilities: `GET /api/v1/technicals/{security_id}/setup-probabilities` (nightly `technicals_probabilities` for every security with a setup; surface `tech_prob_surfaces` refitted weekly on the gate universe). With `target` and `stop` parameters the same surface is evaluated at query time for custom barriers (Thesis Tracker).

Validation status

From `docs/TECHNICALS_V2_VALIDATION.md` (gate: ECE ≤ 2.5 pp, worst regime ≤ 5 pp, better than both baselines on Brier and log-loss, leakage tests pass):

Output	ECE win / loss (pp)	Worst regime	Better than baselines	Gate	Shown as
T2, H = 5	0.67 / 0.89	Covid 2020: loss +5.6	yes (Brier 0.4575 vs Brownian 0.4650 / R:R-only 0.6494)	FAIL	experimental
T2, H = 10	0.91 / 1.03	Covid 2020: loss +5.1	yes (0.4253 vs 0.4297 / 0.4801)	FAIL	experimental
T2, H = 20	1.13 / 1.47	Covid 2020: win -4.6 / loss +4.0	yes (0.3741 vs 0.3759 / 0.3834; log-loss 0.617 vs 0.648 / 1.073)	**PASS**	production
T2, H = 40	1.96 / 2.14	Covid 2020 3.9	no (Brier 0.3485 vs Brownian 0.3482)	FAIL	experimental
T2, H = 60	2.68 / 2.73	Covid 2020 2.8	no (0.3453 vs 0.3429 / 0.3430)	FAIL	experimental

- Why H = 5 / 10 fail: "the Feb-Jun 2020 crash window had 5.1-5.6 pp more stop-outs than predicted". Why 40 / 60 fail: "the grid loses its edge over the Brownian formula".
- Baselines: the naive R:R rule "is badly miscalibrated at short horizons (ECE 15.8 pp at H = 5: it ignores 'neither by H')"; the driftless Brownian formula "is 1.2-2.7 pp too pessimistic on P(win) at H ≤ 10".
- **Expected R** "is well ordered and roughly calibrated" (H = 20 deciles: predicted -0.33 … +0.085 R, realised -0.32 … +0.030 R); "the top decile is optimistic by ~0.05 R".
- **Economic check: null result.** "Filtering a simple setup-following strategy by expected R > 0 did **not** significantly improve it over taking every setup (+1.0%/yr at H = 20, t = 1.33; -0.6%/yr at H = 10, t = -0.75)." It beat the naive "R:R ≥ 1.5" rule (+5.6 / +5.8%/yr, t = 3.4 / 3.2), which was itself the worst variant. PBO 0.07; DSR about 0.99 for every configuration (does not discriminate).

Correct usage and worked example

Use the **20-session** bar (production) to check whether a planned trade is realistic, and expected R to compare setups, not to select them mechanically (the filter added nothing significant after costs). Always compare the setup's horizon with your holding period.

Example (Apple, 16 Oct 2026 close 343.46; setup long, entry 342.96, invalidation 329.23, T1 362.88).

1. Card R:R from the limit entry: (362.88 − 342.96) / (342.96 − 329.23) = **1.45×**. The probabilities use the close: R:R(close) = (362.88 − 343.46) / (343.46 − 329.23) = **1.30×** (shown in the popover).
2. Setup's own horizon: T1 is 0.93 σ_{20} away (within one σ at 20 sessions, not at 10), so **20d*** is selected.
3. Pool US · vol low · earnings window (earnings on 2 Nov is inside 20 sessions) · n = 141,388 historical origins: **T1 first 37%, stop first 45%, neither 18%**.
4. Expected R = 0.372 × 1.30 − 0.446 − 0.010 = **+0.03R** (marked-to-market +0.07R). Break-even = 1 / (1 + 1.30) = **43%**. Baselines: R:R-only 43%, Brownian 33%.

The line "Positive expectancy after costs · P(win) 37% vs break-even 43%" can look contradictory. It is not: the break-even refers to a trade that must end as a win or a loss, while here 18% of paths end neither way and contribute nothing to expected R. Among resolved paths, T1 came first 0.372 / (0.372 + 0.446) = 45% of the time, just above the 43% break-even.

Limitations and pitfalls

- **Entry assumption.** The probabilities assume entry at the close. If the price is above the entry zone (JPMorgan, Figure 5B.13), the card's R:R from the limit entry (0.60×) and the probability's R:R from the close differ widely; the break-even (96%) then tells you that buying at market is a poor plan even though "T1 first 93%" (expected R -0.02R).
- Only H = 20 passed. Experimental horizons are shown with a flask and should not be quoted.
- The engine's backtest exits are close-based while the probability model uses intraday highs and lows ("touched", not closed through), so realised P&L and the model's win / loss events are not identical.
- The thesis tracker's conformal touch frequency of a target and the T2 "target first" probability are calibrated separately and can differ by a few pp (seen: 18% touch vs 20% target-first for one 20-session example).
- A missing T2 ("—") or a 3 ATR target means there is no overhead zone; the plan is then a projection.

Related mnemonics and BQL fields

`TECH` . BQL: `setup.bias` , `setup.entry` , `setup.entry_low` , `setup.entry_high` , `setup.invalidation` , `setup.target1` , `setup.target2` , `setup.reward_risk` , `setup.trend` , `setup.structure_state` , `setup.last_break_kind` , `setup.last_break_date` , `setup.note` . The T2 probabilities are API-only (`/technicals/{security_id}/setup-probabilities`). Thesis target / stop probabilities: 12. Thesis Tracker.

Context-adjusted (meta) row

Route: `/company/:id?tab=technicals` · **Panel:** inside Trade setup (callout 3) · **Mnemonics:** `TECH` · **Refresh:** scoring nightly (`tech_meta_score`), refit weekly (`tech_meta_fit`)

Purpose

The meta row re-estimates the chance that the setup wins (T1 before the invalidation within H sessions) by adding market and stock context to the T2 probability. Its validated value is a **better-calibrated P(win)** at 10 sessions; it does not demonstrate that the context adds an edge, and its take / skip decision failed its economic check.

Screenshot

The meta row is the boxed block inside the Trade setup card in Figures 5B.11 (default: "CONTEXT-ADJUSTED (META) · 10D", T1 first 23%, Exp. R +0.04R, Decision "hidden", "+6.2 pp vs T2 (17%)", three driver chips, footer "Calibration gate passed · take / skip + size failed (H = 10) · model-based, calibrated on history") and 5B.12 (toggle on: "· 20D", "experimental", T1 first 44%, Exp. R +0.09R, Decision "SKIP" with a flask icon).

Controls

Element	What it shows
Label "Context-adjusted (meta) · Nd"	Hover: "Context-adjusted (meta-labelling): the chance of T1 before the stop re-estimated from the T2 probability plus the context at the setup close (…). Model-based, calibrated on the 2013–2026 purged walk-forward; not a statement about this trade."
"experimental" (amber, flask)	The horizon failed its gate or the security is outside the validated universe.
T1 first	Meta P(win).
Exp. R	Meta expected R (hover: "Context-adjusted P(T1) x R:R - P(stop) - costs + the marked-to-market time exit").
Decision	"hidden" (flask) by default; with the toggle, "take" or "skip" with a flask.
"+x.x pp vs T2 (y%)"	Difference from the T2 probability at the **same** horizon (hover shows the T2 value).
Suggested size	Only for "take" with the toggle on: "x.xx% of equity at risk · y× the median taken setup" (quarter Kelly, capped at 2% at risk per 1R).
Driver chips (up to 3)	↑ raises / ↓ lowers the chance, for example "↓ Distance to the upper barrier (sigma)", "↑ VIX 1-year percentile". Hover: the feature value and its contribution in logit units versus T2.
Footer	Gate line: "Calibration gate passed / failed · take / skip + size passed / failed (H = N)".

When the meta horizon is hidden the card says "Context-adjusted (meta) Nd is experimental (failed the validation gate | outside the validated universe) — turn on “show experimental” to see it."

Which horizon is shown. The meta row follows the Chance-by-horizon selection when that horizon exists (10 / 20 / 40) and is visible; otherwise it falls back to the first production horizon (10 sessions). By default, therefore, the bar above shows **20** sessions while the meta row shows **10**.

Methodology

- **Primary model:** the v2 setup generator (direction, entry, T1, invalidation) with its T2 probabilities.
- **Labels (triple barrier):** win = T1 before the invalidation within H sessions (same-bar touch = loss), loss = invalidation first, vertical barrier at H (H = 10, 20, 40).
- **Features (point-in-time, 89 with extras):** T2 base (logit P(win), logit P(loss), P(neither), E[R], costs in R); geometry in σ units; setup sub-scores; technical context (squeeze percentile, ADX, ±DI, trend alignment, ATR / RV percentiles, 5 / 20-session returns); market (VIX level, 1-year percentile and 20-day change, cross-asset risk gauge and its 20-day change, macro cycle states lagged two months); peer-group break-aways; smart money (insider net buying 90 days, short interest, smart-money composite); earnings proximity; region / sector; the hi52 / MAX / IVOL and breakout features. News sentiment and event exposure are **not** used (history too short).
- **Learners:** L2-regularised logistic regression or XGBoost (chosen in-sample), both with the T2 logit as an **offset** so that "no usable context" reproduces T2; calibration identity / Platt / isotonic chosen on a held-out year; sample weights = average uniqueness (overlapping labels).
- **Purged walk-forward** per test year 2013-2026 (embargo 28 / 49 / 91 days at H = 10 / 20 / 40).
- **Decision and size:** take when meta expected R exceeds a threshold chosen on the training history; size = quarter Kelly `0.25 × E[R] / E[R²]` , capped at 2% of equity at risk.
- **Drivers:** the three largest feature contributions beyond the T2 offset (logit units).

Data sources and refresh

`GET /api/v1/technicals/{security_id}/meta-setup?horizon=10|20|40` and the `meta` block of `/setup-probabilities`; nightly `tech_meta_score` (every T2 setup, about 20.8k securities), weekly `tech_meta_fit`. Rows outside the validated domain (ETFs, indices, regions other than US / EU / UK / JP) are always experimental.

Validation status

From `docs/TECHNICALS_V3_VALIDATION.md`, section Meta-labelling (gate: ECE ≤ 2.5 pp; regimes within 5 pp; Brier and log-loss below T2 alone and the context-free model, with the month-block CI of Brier(meta) − Brier(T2) below 0; permutation and leakage tests; decisions additionally need a significant economic gain):

Output	ECE win / loss	Worst regime	Brier meta / T2 / context-free	Brier meta − T2 [95% CI]	Brier meta − context-free [95% CI]	Gate
Meta P(win), H = 10	0.50 / 0.78	Covid 2020 win -4.1 / loss +2.5 (T2 alone: loss +5.1)	0.16777 / 0.16817 / 0.16799	-0.00040 [-0.00071, -0.00006]	-0.00022 [-0.00050, +0.00013]	**PASS**
Meta P(win), H = 20	0.60 / 0.85	Covid 2020 win -4.7 / loss +2.3	0.16891 / 0.16900 / 0.16888	-0.00009 [-0.00047, +0.00033]	+0.00003 [-0.00032, +0.00037]	**FAIL** (not better than T2 / context-free)
Meta P(win), H = 40	0.50 / 0.50	Covid 2020 3.8	0.16951 / 0.17023 / 0.16928	-0.00072 [-0.00125, -0.00023]	+0.00023 [-0.00022, +0.00067]	**FAIL** (the context-free model is better)
Take / skip + size, H = 10 / 20 / 40						**FAIL** (economic check)

- **What the pass means.** "The H = 10 PASS therefore means 'a better-calibrated P(win) than T2', not 'context has a demonstrated edge'." Meta minus context-free is not significant at any horizon. With three horizons tested, "the H = 10 pass is modest evidence: its Bonferroni (98.3%) interval is [-0.00081, -0.000002], excluding 0 only barely."
- **Stress regimes.** The meta model fixes T2's calibration in stress regimes (H = 10: Covid loss gap +5.1 → +2.5 pp; 2022 +3.7 → +1.7; VIX > 25 +3.5 → +0.5).
- **Economic check: null result.** Meta-filtered minus all setups: US -4.3%/yr (t -2.55) at H = 20, -5.3% (t -2.62) at 10, -1.9% (t -1.71) at 40; meta-sized minus all US -4.2% (t -2.41). Taken setups did realise more per trade (+0.008 R vs -0.058 R for skipped ones at H = 20), but only 12-19% of setups are taken, and "in a long-only equity book over 2013-2026 lower exposure costs more than the better per-trade selection earns". PBO 0.35 over 62 configurations. "Null economic result: take / skip and the suggested size fail and are shown only as experimental."

Correct usage and worked example

Read the meta row as a second, better-calibrated opinion on P(win) at 10 sessions, and the driver chips as an explanation of why it differs from T2. Do not use the decision or the size.

Example (Apple, default view). Meta 10d: T1 first **23%** against T2 10d **17%** (+6.2 pp), Exp. R **+0.04R** (T2 10d: -0.04R). Drivers: "↓ Distance to the upper barrier (sigma)" (contribution -0.20 logit), "↑ VIX 1-year percentile" (value 0.82, +0.12), "↓ Reward:risk from the close" (1.30, -0.07). Footer: "Calibration gate passed · take / skip + size failed (H = 10)". Note that the comparison is with the **10-session** T2 (17%), not with the 37% at 20 sessions shown in the bar above. With the toggle on, the 20-session meta row reads 44% vs T2 37%, but that horizon failed its gate.

Contrast (Toyota, Figure 5B.5): meta 10d 32% vs T2 10d 34% (-2.6 pp) and Exp. R -0.07R, while the 20-session bar above shows T1 first 38% and expected R +0.12R: at 10 sessions the context lowered the estimate. Again, compare like with like (the meta row against T2 at 10 sessions).

Limitations and pitfalls

- The default meta horizon (10) differs from the default T2 horizon (20); always read the "vs T2 (y%)" figure.
- Context history is short for insider (from 2015-16) and short interest / smart money (from 2023); macro states use today's vintages (lagged two months), not a true real-time history.
- Survivorship (current membership) as in v2.
- Outside US / EU / UK / JP, and for ETFs and indices, the row is experimental.

Related mnemonics and BQL fields

`TECH`. No BQL fields (API only). The Thesis Tracker's "Target before stop" panel shows the same meta table for thesis barriers (12. Thesis Tracker); the Backtest Lab has an experimental event trigger `meta_setups` (all / take / sized) (11. Backtest Lab).

Technical score

Route: `/company/:id?tab=technicals` · **Panel:** Technical score (callout 4) · **Mnemonics:** `TECH` · **Refresh:** score nightly; percentiles nightly; bucket table refit weekly

Purpose

The technical score summarises trend, market structure, location relative to the zones, reward : risk and momentum into one 0-100 number. The card shows it with percentile sub-scores so you can see where the stock sits among its region × sector peers, and it says plainly that the score has **no out-of-sample edge**.

Screenshot

Figure 5B.14: Technical score (callout 4), Apple, default view: score 88 (p100), percentile bars per sub-score with the raw sub-score in grey, and the "No out-of-sample edge" note.

Figure 5B.15: Toggle on: the score's decile (10) and its historical outcomes at 5 / 20 / 60 sessions, each badged "no out-of-sample edge".

Controls

Element	What it shows
Score pill	Heuristic technical score 0-100.
pNN under the pill	Percentile of the score within region × sector (hover: "Percentile of the composite within region × sector").
Sub-score rows	Trend, Structure, Location, Reward:risk, Momentum: bar = percentile within region × sector (pNN), grey number = raw sub-score (hover: "raw x (saturating 0–100) · percentile ...").

Element	What it shows
Subtitle	"Percentiles within region × sector (n = 549)" (number of peers). Without calibration data: "Heuristic composite (trend, structure, location, reward:risk, momentum)" and raw sub-scores only.
calibrated on history	Popover: definitions, "No out-of-sample edge" label, the walk-forward rank IC per horizon, gate lines and the experimental switch.
Bucket outcomes (toggle on)	Per horizon: "5d · decile 10", median excess return, "beat region" hit rate with CI, n, and the badge "no out-of-sample edge"; footer "Historically this score bucket → outcomes above (2003–2026, gate universe); not a forecast."
No out-of-sample edge note (toggle off)	"No out-of-sample edge: in the 2010–2026 walk-forward this score's rank IC and decile spread were not significant, so its historical bucket stats are experimental (enable in "calibrated on history")."

Methodology

```
technical score = 25% trend + 20% structure + 35% location + 10% reward:risk + 10% momentum
trend       = 100 × share of {close > SMA200, SMA50 > SMA200, close > SMA50, SMA50 slope > 0}
structure   = 100 bullish / 50 range / 0 bearish (ZigZag 3 ATR)
location    = 50 + 50 × (a − b),  a = support strength × e^(−d_support/2),  b = resistance strength × e^(−d_resistance/2)
              (strengths as fractions, distances in ATR from the zone edge)
reward:risk = 100 × min(R:R / 4, 1)  (0 for a short bias)
momentum    = 100 − 4 × max(0, RSI − 65) − 2 × max(0, 40 − RSI), clipped to 0..100
```

The raw sub-scores saturate (structure is 0 or 100 on every origin; trend has five possible values), so the card shows each as a **percentile within region × sector** on the same date (mid-rank for ties). The bucket statistics map the score's decile to the next 5 / 20 / 60-session excess log return versus the region's equal-weight universe (median, IQR, hit rate, n, month-block bootstrap CIs).

Data sources and refresh

Score: `tech_setups` (nightly). Percentiles and buckets: `GET /api/v1/technicals/{security_id}/score-calibration` (nightly `technicals_probabilities`; decile table refitted weekly).

Validation status

FAIL → experimental; the card always says "no out-of-sample edge". From `docs/TECHNICALS_V2_VALIDATION.md` (T4):

Horizon	Rank IC (OOS)	t (non-overlapping)	Spearman (decile, mean)	Top − bottom decile	Edge
5 d	-0.0031	-0.67	-0.94	-0.09% [-0.22, +0.03]	none
20 d	-0.0117	-0.91	-0.84	-0.35% [-0.72, +0.09]	none
60 d	-0.0139	-1.00	-0.89	-0.59% [-1.32, +0.12]	none

"The composite technical score has **no out-of-sample edge**; if anything it is weakly mean-reverting." The percentile versions "fix the presentation but not the predictive content". The bucket mapping is accurate (hit-rate error 1.1-1.8 pp), "so the statistics are honest descriptions of history - but they are no better than the unconditional distribution". The evidence-based factor composite meant to replace it also failed its gate (see Technical factors (evidence-based)), so the card keeps the heuristic score.

Correct usage and worked example

Use the score and its percentiles as a **description** of the chart (for example, to screen for names in an uptrend near support), not as a forecast. **Example (Apple):** score 88, p100; Trend p91 (raw 100), Structure p79 (raw 100), Location p98 (raw 83), Reward:risk p94 (raw 36), Momentum p77 (raw 100). With the toggle on, decile 10 historically delivered a median excess return of -0.04% at 5 sessions, -0.05% at 20 and -0.07% at 60, beating the region 49-50% of the time (n about 56,000): no better than a coin.

Limitations and pitfalls

- A high score is not a buy signal; historically the top decile slightly lagged.
- Raw sub-scores of 100 are common (saturation); compare percentiles.
- The score is long-side attractiveness; for a short bias the reward:risk component is 0.

Related mnemonics and BQL fields

`TECH`. BQL: `setup.technical_score`, `setup.score_trend`, `setup.score_structure`, `setup.score_location`, `setup.score_rr`, `setup.score_momentum`; percentiles `tp.trend_pct`, `tp.structure_pct`, `tp.location_pct`, `tp.rr_pct`, `tp.momentum_pct`, `tp.technical_pct`, `tp.technical_score_pct`; bucket fields `tp.h5_*`, `tp.h20_*`, `tp.h60_*` (each with `_gate`). See also 4. Screener and Ranking.

Technical factors (evidence-based)

Route: `/company/:id?tab=technicals` · **Panel:** Technical factors (evidence-based) (callout 5) · **Mnemonics:** `TECH` · **Refresh:** nightly (`tech_factors`); validation monitor monthly (`tech_factor_eval`)

Purpose

Instead of heuristic chart scores, this card shows the technical effects documented in the academic literature, each as a percentile against sector peers and each with **its own out-of-sample evidence on this app's data**. Three passed their gate and are shown by default as "rank edge" context; none is a profitable stand-alone strategy after costs.

Screenshot

Figure 5B.16: Technical factors (callout 5), Apple, default view: 52-week-high proximity p100 ("rank IC +0.023"), MAX (lottery) p98 ("rank IC +0.038"), Idiosyncratic vol p98 ("rank IC +0.044"); the economics note; "4 factors without a validated edge hidden."

Figure 5B.17: Toggle on: 12-1 momentum p69, Residual momentum p69, 1-month reversal p30 (orange: unfavourable side), Industry momentum p72, each with a flask and "no reliable edge".

Figure 5B.18: ASML (`/company/ASML.AS?tab=technicals` , Euronext Amsterdam), default view: percentiles against 87 sector peers in the EU region: 52-week-high proximity p62, MAX p64, idiosyncratic vol p66.

Controls

Element	What it shows
Factor name (dotted)	Hover: full name, the direction the literature reports ("near the 52-week high = historically higher returns"), the out-of-sample IC at 5 / 20 / 60 sessions with t-statistics and Benjamini-Hochberg p-values, the stock's region IC, the gate verdict, the economics note and the academic reference.
Badge	"rank IC +0.0xx" (passed; descriptive edge, not tradable after costs) or "no reliable edge" (failed, with a flask).
Bar and pNN	Percentile on the **historically favourable side** (green when ≥ 50, orange below). For negative-sign factors (MAX, IVOL, reversal) this is 100 minus the raw percentile; the bar tooltip shows both.
Subtitle	"Percentile vs sector peers (n 455)" (or "vs region peers" for thin sectors).
ⓘ button	Popover: definitions, number of cells tested and the multiple-testing method, "region not validated" warning outside US / EU / UK / JP, a liquidity warning below 1M USD ADV, the experimental switch and the validation document.
Notes	"Rank IC = the factor sorts stocks' next-month returns (median) out-of-sample, but a stand-alone quintile portfolio did not make money after costs — use as context, not as a strategy." and "Bar = percentile on the historically favourable side. Historical evidence (2005–2026 walk-forward), not a forecast."

Methodology

Factor	Definition	Sign	Reference
52-week-high proximity (hi52)	close / max(close over 252 sessions)	+	George & Hwang (2004)
12-1 momentum (mom)	log C[t−21] − log C[t−252]	+	Jegadeesh & Titman (1993)
Residual momentum (rmom)	Sum of residuals of months t−12..t−2 from a rolling 36-month regression on region market and sector returns, divided by their standard deviation	+	Blitz, Huij & Martens (2011)
1-month reversal (strev)	log C[t] − log C[t−21]	−	Jegadeesh (1990), Lehmann (1990)
MAX (maxret)	Largest daily simple return over the last 21 sessions	−	Bali, Cakici & Whitelaw (2011)
Idiosyncratic volatility (ivol)	Annualised standard deviation of residuals of a 21-session regression on market and sector returns	−	Ang, Hodrick, Xing & Zhang (2006)
Industry momentum (indmom)	Region × sector equal-weight log return over 126 sessions (ranked within the region)	+	Moskowitz & Grinblatt (1999)

Percentile = rank 0-100 of the raw value among **liquid** names (20-session ADV ≥ 1M USD, close ≥ 1 USD) of the same region × sector on the same session. **Rank IC** = Spearman correlation between the percentile and the next 20 sessions' sector-neutral log return (from the next close), averaged over weekly dates 2005-2026, pooled over US / EU / UK / JP, with Newey-West t-statistics. Definitions and signs were fixed in advance from the literature, so the whole sample is out-of-sample for each factor.

Data sources and refresh

`tech_factors` (nightly, every price region; US / EU / UK / JP validated, other regions computed and flagged "region not validated"); `GET /api/v1/technicals/{security_id}/factors`. Monthly `tech_factor_eval` writes the validation rows and a trailing-156-week monitor IC per factor and region.

Validation status

From `docs/TECHNICALS_V3_VALIDATION.md`, Verdict #1. Pre-registered gate: G1 pooled 20-session IC sign-aligned with Holm p < 0.05; G2 monotone deciles ≥ 0.7 and spread CI > 0; G3 aligned in ≥ 3 of 4 regions and ≥ 60% of years; G4 leakage checks; G5 no regime significantly wrong-signed.

Factor	IC 20d (t)	Holm p	Monotone / spread	Economics (US L/S net Sharpe; long-only vs EW)	Gate
hi52	+0.023 (2.9)	0.017	0.82 / +1.3%	-0.27; -1.5%/yr (t -0.8)	**PASS**
mom	+0.011 (1.6)	0.30	0.36	-0.17; -0.1%/yr	FAIL (G1, G2)
rmom	+0.004 (1.1)	0.56	0.84	-0.28; +0.1%/yr	FAIL (G1)
strev	-0.015 (-3.6)	0.002	0.35	-0.66; -4.7%/yr (t -2.5)	FAIL (G2: not monotone)
MAX	-0.038 (-6.4)	9e-10	1.00 / +2.1%	-0.35; -2.0%/yr (t -1.1)	**PASS**
IVOL	-0.044 (-7.5)	6e-13	1.00 / +2.4%	-0.28; -0.7%/yr (t -0.4)	**PASS**
indmom	+0.003 (0.6)	0.56	0.71 / CI incl. 0	n/a	FAIL (G1-G3)
Composite (comp_wf)	+0.035 (5.3)	single test p < 1e-6	1.00 / +1.8%	-0.46; -1.9%/yr (t -1.1)	**FAIL** (G6 vs v2: t 1.66; G7 economics) → not served

- "**Shown by default: hi52, MAX, IVOL** (labelled "rank edge", see the economics). Everything else is experimental."
- "No factor and no composite is profitable as a stand-alone strategy after costs." The IC is a median / rank effect on log returns; simple-return decile spreads are much smaller (hi52 +0.08%, MAX +0.23%, IVOL +0.25% per 20 sessions).
- Effects are strongest in small and micro names and weakest in large caps (hi52 large-cap IC +0.011, t 1.2: not significant). MAX and IVOL "carry the edge in calm markets … and disappear when VIX > 25".
- **Regions:** only US, EU, UK and JP were validated. Elsewhere the values are shown with "This region was not part of the validation (US / EU / UK / JP): values shown, edge not validated."

Correct usage and worked example

Use the percentiles as cross-sectional context: "historically, stocks in this percentile had higher / lower next-month returns than their sector peers, in median". **Example (Apple, US Technology, 455 liquid peers):** 52-week-high proximity p100 (at its high), MAX aligned p98 (raw percentile 1.65: one of the least "lottery-like" names), IVOL aligned p98 (raw 2.31: very low stock-specific volatility). All three sit on the historically favourable side, but for a mega-cap the evidence is weakest (large-cap effects are small or not significant). With the toggle on, 1-month reversal shows p30 (orange): Apple's strong last month is, under the reversal effect, the unfavourable side, but that factor failed its gate.

Limitations and pitfalls

- "Rank IC" is not a return forecast and not a trading signal; the quintile portfolios lost money after costs.
- Large caps carry the weakest evidence.
- Size buckets in the validation use dollar volume, not market cap (no point-in-time shares outside the core universe).
- Survivorship: listing coverage favours current names.

Related mnemonics and BQL fields

`TECH`. BQL: `tech.hi52`, `tech.mom`, `tech.rmom`, `tech.strev`, `tech.maxret`, `tech.ivol`, `tech.indmom` (raw), the matching `_pct` fields (percentiles), `tech.composite` / `tech.composite_pct` (null: the composite failed), `tech.adv20_usd`, `tech.liquid`, `tech.size_bucket`, `tech.n_peers`. The Backtest Lab accepts `tech_factors` as a source (11. Backtest Lab).

Breakout

Route: `/company/:id?tab=technicals` · **Panel:** Breakout (callout 6) · **Mnemonics:** `TECH` · **Refresh:** nightly (`tech_breakouts`); P(follow-through) refit monthly

Purpose

The Breakout panel reports whether the stock has made a fresh 20- or 55-session high or low close in the last 20 sessions, whether that breakout was volume-confirmed, and what has happened since (active, followed through, failed). It also states the historical evidence: volume-confirmed breakouts follow through more often, but not enough to trade after costs.

Screenshot

Figure 5B.19: Breakout (callout 6), Apple: "No breakout in the last 20 sessions." and the evidence line.

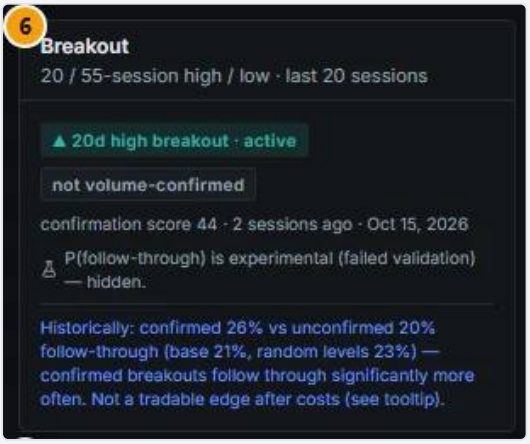


Figure 5B.20: Toyota, default view: "▲ 20d high breakout · active", "not volume-confirmed", confirmation score 44, 2 sessions ago (15 Oct 2026); "P(follow-through) is experimental (failed validation) — hidden."

Figure 5B.21: JPMorgan, default view: "▼ 20d low breakout · active", "volume-confirmed", score 54, 18 sessions ago (22 Sep 2026); the probability is hidden.

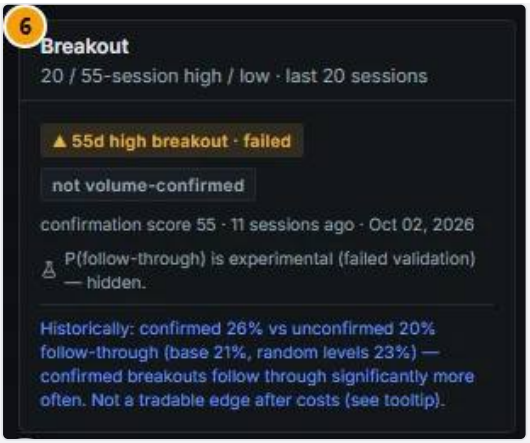


Figure 5B.22: ASML, default view: "▲ 55d high breakout · failed" (closed back inside the level), score 55, 11 sessions ago.

Controls

Element	What it shows
State chip	"▲ 20d high breakout · active / followed through / failed" (or ▼ ... low); hover: definitions of breakout and confirmation.
Confirmation chip	"volume-confirmed" (green) or "not volume-confirmed".
Sub-line	"confirmation score NN · N sessions ago · (date)".
P(follow-through, 20d)	Toggle on only (flask): calibrated probability, with "vs geometry baseline x% · base rate y%". Toggle off: "P(follow-through) is experimental (failed validation) — hidden."
Evidence line	"Historically: confirmed 26% vs unconfirmed 20% follow-through (base 21%, random levels 23%) — confirmed breakouts follow through significantly more often. Not a tradable edge after costs (see tooltip)." Hover for the economic check.

Methodology

- **Breakout (lookback N = 20 or 55):** the close exceeds the highest high of the previous N sessions while the previous close did not, with no breakout of the same N in the previous 10 sessions; level L = that prior high (mirrored for lows).
- **Follow-through (H = 20):** the high reaches `C · exp(σ · √20)` (one 20-session σ beyond the breakout close, σ = EWMA daily volatility) **before** any close back inside the level. Failed = closed back inside first (a bar with both = fail). Active = neither yet.
- **Volume confirmation (pre-registered):** volume ≥ 1.5 × its 50-session average **and** the close in the top quarter of the day's range (bottom quarter for a down breakout).
- **Confirmation score 0-100:** fixed mean of four bounded components: relative volume (log scale), close location in the range, Chaikin money flow (20), OBV slope (20).
- **P(follow-through):** logistic regression with isotonic calibration, purged walk-forward (2-year test blocks 2008-2026), on the breakout-day return, distance beyond the level, trend, relative volume, OBV, gap, volatility ratio and other features. **Geometry baseline:** distance beyond the level, volatility regime, lookback, direction and region only. **Base rate:** all breakouts of that lookback.

Data sources and refresh

`tech_breakouts` / `tech_breakout_state` (nightly; liquid names; events of the last 80 sessions, panel shows the latest of the last 20); `GET /api/v1/technicals/{security_id}/breakout`. The model is refitted monthly by `tech_factor_eval`.

Validation status

From `docs/TECHNICALS_V3_VALIDATION.md`, Verdict #5 (liquid US / EU / UK / JP, 2005-2026, 1.67 million events):

Output	H	ECE	Worst regime	Beats climatology and geometry	Gate
Calibrated P(follow-through)	10	1.23	**Covid 2020 +18.3 pp** (2020 +6.8)	yes: Brier 0.1810 vs 0.1867 / 0.1823	**FAIL** → experimental
Calibrated P(follow-through)	20	1.20	**Covid 2020 +17.2 pp** (2020 +6.5)	yes: Brier 0.1603 vs 0.1651 / 0.1610	**FAIL** → experimental

Effect (H = 20)	Estimate	95% CI
Base rate (all breakouts)	21.1%	[20.1, 22.0]
Confirmed / unconfirmed	26.2% / 19.6%	
Raw difference	+6.6 pp	[+6.1, +7.2]
Geometry-adjusted difference	**+1.6 pp**	[+1.2, +2.0]

Effect (H = 20)	Estimate	95% CI
Confirmed – geometry-matched random control	**+3.5 pp**	[+2.7, +4.6]
Unconfirmed – its random control	+0.6 pp	[-0.1, +1.4]
Volume-confirmation edge		**PASS**
Economic check (US up breakouts, costs on)	confirmed Sharpe -0.35 vs all -0.61; confirmed – random subset of the same size -1.0%/yr (t -0.6); PBO 0.12	**no tradable edge**

"Most of the raw +6.6 pp is geometry (confirmed breakouts close much further beyond the level ...), but +1.6 pp remains within geometry strata." "Breakout trading is negative after costs (fails back inside ~70% of the time)." "The state chip and the confirmation flag / score (deterministic) are shown; the probability is hidden unless the experimental toggle is on."

Correct usage and worked example

Use the panel as context for a level: a fresh, volume-confirmed breakout through a resistance zone is historically a little more likely to keep going than an unconfirmed one, but most breakouts fail back inside the level. **Example (Toyota, 15 Oct 2026):** a 20-session high breakout, still active two sessions later, **not** volume-confirmed (score 44); by default the probability is hidden; with the toggle on, the experimental model gave 17%, slightly below its geometry baseline (19%) and the base rate (20%). **Example (JPMorgan):** a volume-confirmed 20-session **low** breakout on 22 Sep, still active after 18 sessions (experimental P(follow-through) 29% vs geometry 28%, base 22%).

Limitations and pitfalls

- About 70% of breakouts close back inside the level; "active" does not mean "working".
- The probability failed in the 2020 rebound (+17-18 pp): breakouts then followed through far more often than predicted.
- Only liquid names have breakout events.
- Backtest exits are close-based; the label uses intraday highs and lows.

Related mnemonics and BQL fields

`TECH` . BQL: `bo.state` , `bo.direction` , `bo.lookback` , `bo.event_date` , `bo.sessions_since` , `bo.level` , `bo.target_20` , `bo.confirmed` , `bo.confirm_score` , `bo.relvol50` , `bo.clv` , `bo.p_follow_20` , `bo.p_geo_20` , `bo.base_rate_20` , `bo.gate` .

Indicators

Route: `/company/:id?tab=technicals` · **Panel:** Indicators (callout 7) · **Mnemonics:** `TECH` · **Refresh:** nightly (`confluence_zones`)

Purpose

A compact panel of the standard indicators used by the setup and the score: RSI, ATR, the 50- and 200-day moving averages and the trend label.

Screenshot

Figure 5B.23: Indicators (callout 7), Apple: RSI 14 62.1 (neutral), ATR 14 6.70 (2.0% of price), SMA 50 325.69 (+5.3% vs last), SMA 200 292.14 (+17.4% vs last), Trend ▲ Up.

Controls

Element	What it shows
RSI 14	Wilder RSI(14); "overbought" above 70, "oversold" below 30, otherwise "neutral".
ATR 14	Wilder ATR(14) in price units and as a share of the price.
SMA 50 / SMA 200	Moving averages and the last price's distance from them.
Trend	▲ Up / ▼ Down / ■ Sideways from the setup's trend score.

Methodology

RSI and ATR use Wilder smoothing over 14 sessions. SMA 50 / 200 are simple averages of the last 50 / 200 closes. Trend = up when at least 75% of {close > SMA200, SMA50 > SMA200, close > SMA50, positive 20-session SMA50 slope} hold, down when 25% or fewer, otherwise sideways.

Data sources and refresh

`tech_setups` (nightly), via `GET /api/v1/companies/{security_id}/technicals` (`indicators`).

Validation status

Not formally validated (standard indicator definitions, no predictive claim).

Correct usage and worked example

Use ATR to size stops in volatility units and to read zone distances (the zones table measures distance in per cent, the zone engine in ATR). **Example (Apple):** ATR 6.70 = 2.0% of the price; the invalidation 329.23 sits 0.5 ATR below zone #1's low, and T1 is about 3 ATR above the close.

Limitations and pitfalls

The "overbought / oversold" labels are conventional thresholds, not tested signals. The trend label here (SMA rules) can differ from the trend chip (daily / weekly alignment).

Related mnemonics and BQL fields

`TECH` . BQL: `setup.rsi14` , `setup.atr14` , `setup.sma50` , `setup.sma200` , `setup.trend` , `setup.structure_state` .

5C. Company 360: macro, earnings, events, network and model

This chapter covers the last five tabs of Company 360: **Macro**, **Earnings**, **Event risk**, **Network & Geo** and **Model**. Each one adds an outside-in view of the company: how it moves with the macro factors, how it reacts to its own results, which news and geopolitical shocks it is exposed to, who it trades with and where it earns its revenue, and what the cross-sectional ML ranker says about it.

Almost everything on these tabs is **model-based**: our own estimates from public data, not analyst consensus and not forecasts. Every panel says so with a badge, and this chapter repeats the label and the validation status for each output. The company header at the top of every tab (callout 1 in the screenshots) is described in 5A. Company 360.

Examples use Apple (AAPL), Toyota (7203.T), SAP (SAP.DE) and Petrobras (PETR4.SA). Data is as of 19 October 2026 (snapshot built 22 October 2026), unless a panel shows an earlier "as of" date.

Macro

Route: `/company/:id?tab=macro` · **Menu:** Company 360 › Macro · **Mnemonics:** none for this tab (open the company with `DES` , then the tab) · **Refresh:** betas weekly (`macro_exposures`), regime table nightly (`regime_performance`)

Purpose

The Macro tab shows how sensitive the stock is to rates, the yield curve, inflation breakevens, credit spreads, the dollar, oil and the equity market, compared with its sector. It turns those sensitivities into the estimated impact of ten preset shocks (and your saved scenarios), replays six historical episodes against today's betas, and shows how the company's sector has done in each macro regime. Use it to understand what kind of macro bet you take when you own the name, and to check a thesis against a rates or oil scenario. It is a description of past co-movement, labelled "Model-based estimate (rolling factor betas; not a forecast)".

Screenshot

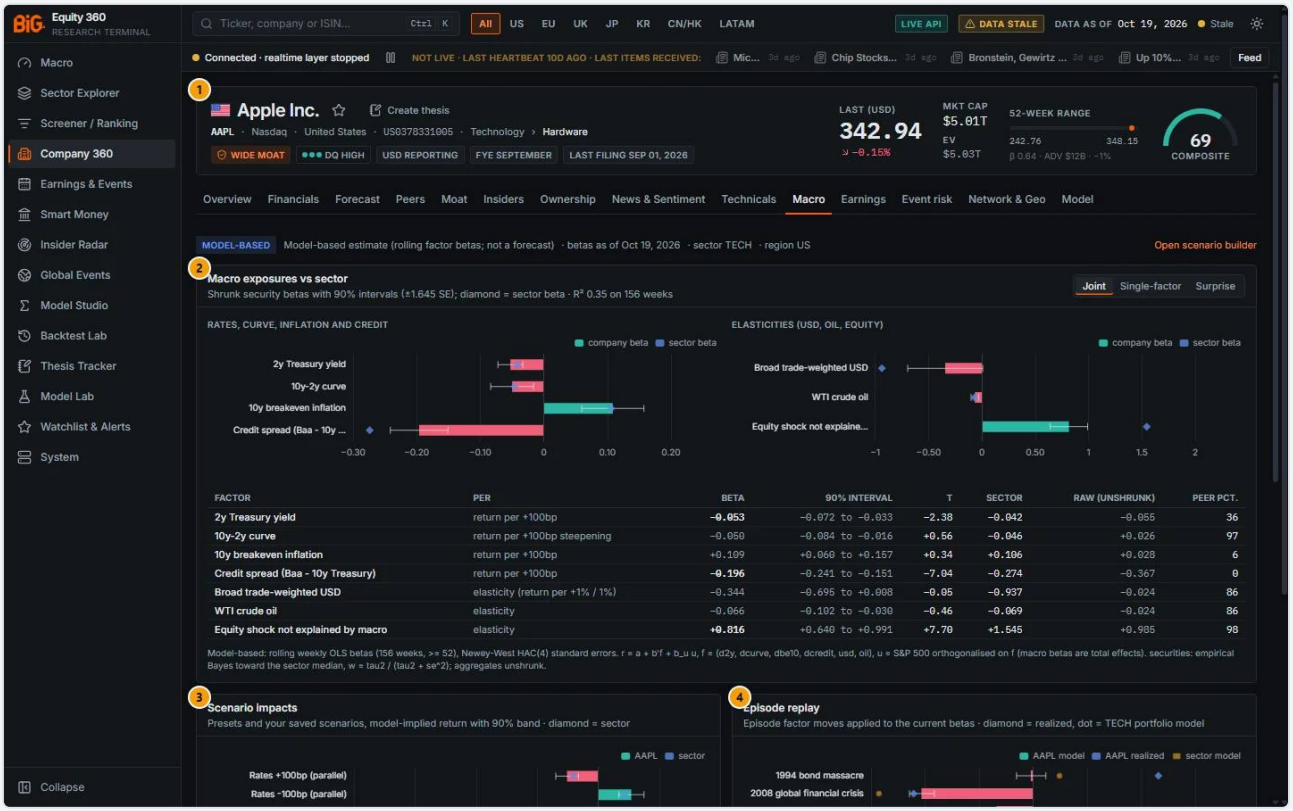


Figure 5C.1: Macro tab for Apple (`/company/AAPL?tab=macro`), betas as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	Name, listing, badges, last price and composite score (see 5A)
2	Macro exposures vs sector	Shrunk betas with 90% intervals, the sector beta (diamond), fit R^2, and the full beta table
3	Scenario impacts	Model-implied return for each preset and saved scenario, with a 90% band and the sector impact
4	Episode replay	Today's betas applied to six historical episodes, against the realised return of the stock and its sector
5	Sector regime performance	Mean monthly return of the sector relative to its benchmark in each macro regime (below the fold, Figure 5C.2)

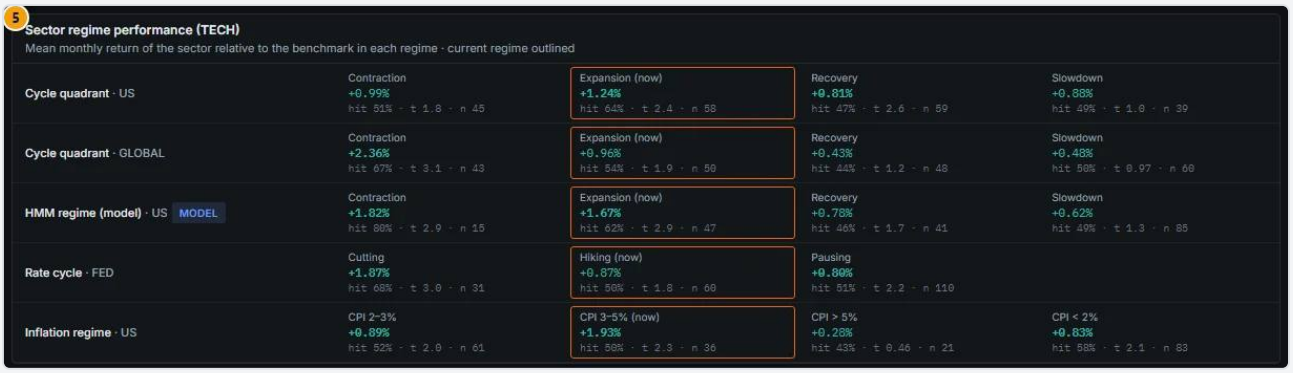


Figure 5C.2: Sector regime performance for Technology (TECH). The current regime of each row is outlined and marked "(now)".

Controls

Control	What it does
Model segmented control (Joint / Single-factor / Surprise)	Switches the exposure model behind the bars and the table (see Methodology). Default: Joint

Control	What it does
Open scenario builder link (top right)	Opens the Scenario builder at `/macro/markets?tab=scenarios`, where you define and save your own shocks
Hover on a bar	Shows the company beta, its 90% interval and the sector beta
Hover on an episode name	Shows the episode window (start → end)
Hover on a model value in the episode table	Shows the 90% band of the model-implied return (and the sector weighting for the sector column)

The tab has no export of its own. The same numbers are available through the macro-to-markets pages and BQL (see the last subsection).

Methodology

FACTORS

All factor inputs come from official series (via FRED unless stated):

Factor	Label in the app	Unit of the beta	Source
`d2y`	2y Treasury yield	return per +100bp	Fed H.15 (DGS2)
`d10y`	10y Treasury yield	return per +100bp	Fed H.15 (DGS10)
`dcurve`	10y-2y curve	return per +100bp steepening	T10Y2Y
`dreal10`	10y real yield (TIPS)	return per +100bp	Fed H.15 (DFII10)
`dbe10`	10y breakeven inflation	return per +100bp	T10YIE
`dcredit`	Credit spread (Baa - 10y Treasury)	return per +100bp	Moody's via FRED (BAA10Y)
`usd`	Broad trade-weighted USD	elasticity (return per +1% / 1%)	Federal Reserve (DTWEXBGS)
`oil`	WTI crude oil	elasticity	EIA (DCOILWTICO)
`equity`	S&P 500 (market beta)	elasticity	price lake (^GSPC)
`equity_resid`	Equity shock not explained by macro	elasticity	S&P 500 orthogonalised on the macro factors
`surp_growth`, `surp_inflation`, `surp_labor`	Growth / Inflation / Labor surprise index	return per +100 index points	our surprise indices (model-based expectations)

A rate beta of −0.05 means about −5% return for a +100bp move. An elasticity of −0.34 means about −0.34% for a +1% move in the dollar.

THE THREE EXPOSURE MODELS

All three are rolling weekly OLS regressions of the stock's local-currency weekly total return on weekly factor changes, over the last **156 weeks** (at least 52), with Newey-West HAC(4) standard errors. They are re-estimated at each month end.

- **Joint** (default): `r = a + b'f + b_u·u + e`, with `f = (Δ2y, Δcurve, Δbe10, Δcredit, USD, oil)` and `u` = the S&P 500 return orthogonalised on `f` in the same window. Because `u` only carries the equity move that the macro factors do not explain, the macro betas are **total effects**.
- **Single-factor** (`univariate`): one regression per factor. The `equity` row is the ordinary CAPM beta to the S&P 500.
- **Surprise**: `r = a + b's + b_u·u_s + e`, where `s` = changes in the growth, inflation and labour surprise indices of the company's **home economy** (model-based expectations, see 2A. Macro) and `u_s` = the equity return orthogonalised on `s`.

SHRINKAGE, INTERVALS AND THE TABLE COLUMNS

Raw security betas are noisy, so they are shrunk toward the sector by empirical Bayes:

$$\beta = m + w \cdot (\beta_{raw} - m) \qquad w = \tau^2 / (\tau^2 + se_{raw}^2) \qquad se = \sqrt{w} \cdot se_{raw}$$

m is the median raw beta of the sector on the same date (the all-security median when the sector has fewer than 5 names) and τ^2 is the cross-sectional variance of the true betas (robust variance of the raw betas minus the mean squared standard error, floored at 5% of the variance). Sector, region and other aggregates are not shrunk.

Column	Meaning
Factor, Per	Factor label and unit of the beta
Beta	Shrunk beta. **Bold** when the absolute t is at least 1.96
90% interval	$\beta \pm 1.645 \cdot se$ (shrunk standard error)
t	t-statistic of the **raw** estimate, β_{raw} / se_{raw}
Sector	Cap-weighted sector portfolio beta (the diamond on the chart)
Raw (unshrunk)	β_{raw} before shrinkage
Peer pct.	Percentile of the shrunk beta among all securities of the same sector (0-100)

The panel subtitle gives the R^2 and the number of weeks of the selected model; the footnote under the table repeats the method text served by the API.

SCENARIO IMPACTS

Each preset is run in **isolated** mode: factors you do not shock are set to zero, and a 2y and 10y shock of the same size is a parallel shift (the curve is unchanged). Percent shocks enter in logs. With z the factor shock vector and C the shrunk HAC covariance of the stock's betas:

$$\text{impact} = \exp(b'z) - 1 \qquad 90\%\ \text{band} = \exp(b'z \pm 1.645 \cdot \sqrt{(z'Cz)}) - 1$$

Surprise shocks (in the Stagflation preset) use the Surprise model and are added to the macro impact; the two channels can overlap. The band covers **estimation error only**, not the stock's idiosyncratic noise. The ten presets are: Rates ±100bp (parallel), Bear steepener (10y +75bp, 2y +25bp), Bull flattener (10y −75bp, 2y −25bp), USD +10%, Oil +50%, Oil −40%, Credit spreads +150bp, Stagflation (breakevens +75bp, oil +40%, growth surprise −100) and Risk-off (equity −15%, credit +100bp, rates −50bp, USD +5%). Scenarios you saved in the Scenario builder appear below them, marked "(saved)". On this tab the optional VAR second-round propagation of the Scenario builder is switched off.

EPISODE REPLAY

For each episode the factor moves are the official level changes over the window (log changes for USD, oil and equity). The model value is today's betas × those moves; "Realized" is the stock's actual return over the window, or "not listed" when its prices do not cover it. The sector columns show the same for the sector portfolio (cap-weighted where available, else equal-weighted).

Episode	Window
1994 bond massacre	15 Oct 1993 → 7 Nov 1994
2008 global financial crisis	12 Sep 2008 → 9 Mar 2009
2011 euro-area debt crisis	1 Jul 2011 → 3 Oct 2011
2013 taper tantrum	21 May 2013 → 5 Sep 2013
2020 COVID crash	19 Feb 2020 → 23 Mar 2020
2022 hiking cycle	31 Dec 2021 → 21 Oct 2022

SECTOR REGIME PERFORMANCE

Historical monthly returns of the company's sector (cap-weighted) **minus the benchmark** (S&P 500 for global aggregates), grouped by the regime of the same month (contemporaneous timing). Each cell shows the mean monthly

relative return, the hit rate (share of months above zero), the t-statistic `t = mean / (sd / √n)` and the number of months `n`. The mean is bold when |t| ≥ 1.96.

Row	Regime definition
Cycle quadrant · US	Cycle clock quadrant of the US (expansion, slowdown, contraction, recovery)
Cycle quadrant · GLOBAL	Same, on the mean of the US, euro area, Japan, UK and China momenta
HMM regime (model) · US	Gaussian hidden Markov model regimes (model-based, badge "model")
Rate cycle · FED	Hiking / pausing / cutting
Inflation regime · US	US CPI y/y buckets (< 2%, 2-3%, 3-5%, > 5%)

Data sources and refresh

- **Factors:** FRED (Fed H.15, Federal Reserve, EIA, Moody's), the S&P 500 from the price lake, and the surprise indices built from official releases. Job `macro_factors` (nightly).
- **Stock returns:** local-currency weekly total returns from the price lake (Polygon and yfinance).
- **Betas and episode replay:** job `macro_exposures`, **weekly** (about 20,000 securities plus sector, region and other aggregates). The header line shows "betas as of" the date of the latest run.
- **Regime table:** job `regime_performance`, nightly, monthly history from 1995.
- **Scenarios:** presets are built in; your saved scenarios live in the app state database.
- Under the default weekly schedule profile, all of these run in the Saturday refresh.

Validation status

Not formally validated. There is no pass / fail gate for the macro exposures, scenario impacts or episode replay. Every output carries the badge **MODEL-BASED** and the label "Model-based estimate (rolling factor betas; not a forecast)". The regime table is labelled "Historical monthly returns grouped by regime (not a forecast)", and its HMM row carries a "model" badge.

The uncertainty bands cover estimation error only. The episode replay is the best available check of how well betas explain big moves, and it often shows large gaps (see the worked example). The aggregate model-versus-realised statistics are on the Episode replay page of 2B. Markets and cross-asset.

Correct usage and worked example

Use the tab to answer "what macro risk am I taking?", not "what will the stock do?". Read the **Joint** model first; switch to **Single-factor** for the familiar CAPM beta and simple sensitivities, and to **Surprise** to see whether the stock reacts to data surprises in its home economy. Treat a beta as meaningful only when it is bold (|t| ≥ 1.96 on the raw estimate) and its interval does not straddle zero.

Worked example: Apple (AAPL), betas as of 19 Oct 2026 (Figure 5C.1). The Joint model explains R^2 0.35 of Apple's weekly returns over 156 weeks.

1. The strongest macro sensitivity is to **credit spreads**: β −0.196 per +100bp (90% −0.241 to −0.151), t −7.04, against a sector beta of −0.274. Apple is less credit-sensitive than the average technology stock.
2. The **equity shock not explained by macro** has elasticity +0.816 (t +7.70; sector +1.545). Apple carries less pure equity-market risk than its sector, which is consistent with its low market beta in the header (β 0.64).
3. **Credit spreads +150bp** therefore gives `exp(−0.196 × 1.5) − 1 = −25.5%` (band −30.4% to −20.2%), against −33.7% for the sector. **Rates +100bp (parallel)** gives −5.1% (band −7.0% to −3.3%).
4. In the episode replay, the 2008 crisis is well explained (model −41.3%, realised −44.2%), but the 1994 bond massacre is not (model −0.8%, realised +46.4%): over that window Apple's return was driven by company news, not by macro.
5. In the regime table, Technology is currently in **Expansion (now)** on the US cycle quadrant, with a mean monthly relative return of +1.24% (hit 64%, t 2.4, n 58).

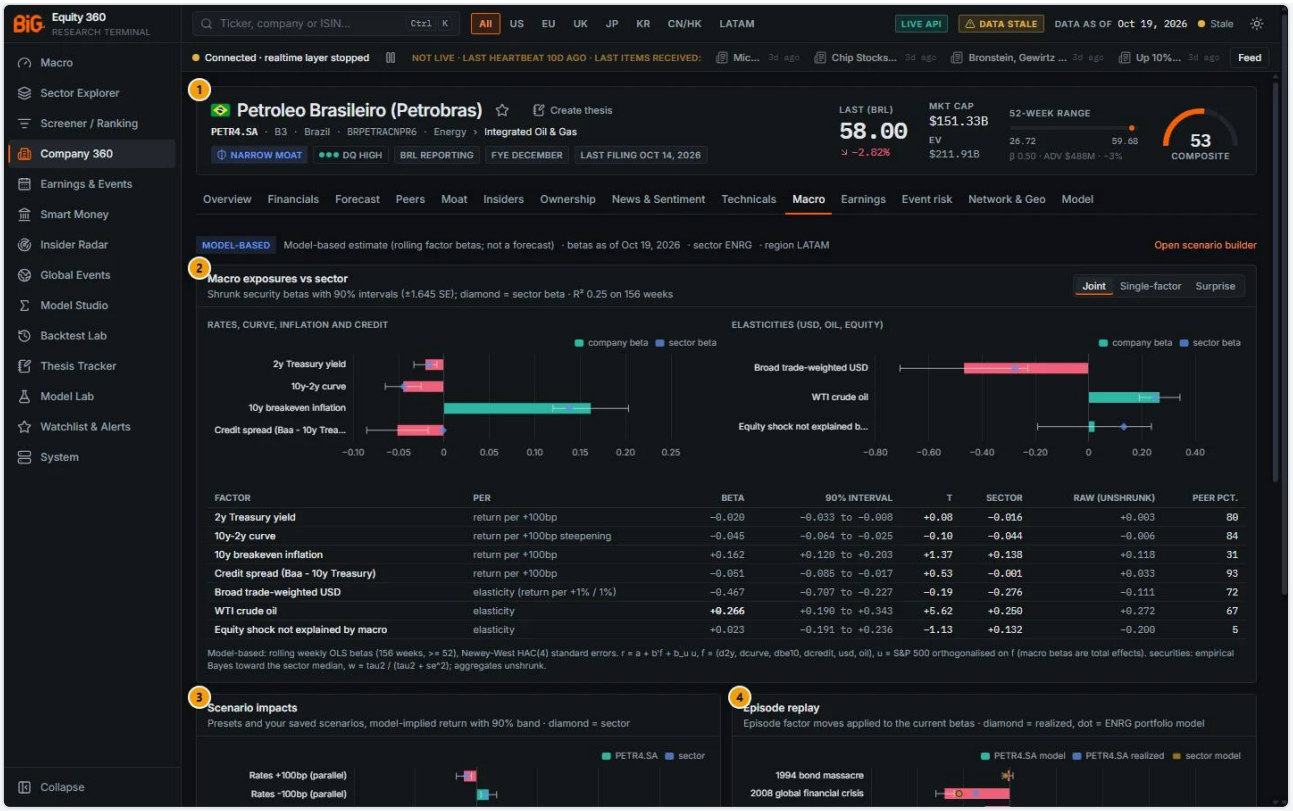


Figure 5C.3: Macro tab for Petrobras (`/company/PETR4.SA?tab=macro`), betas as of 19 Oct 2026. The callouts are the same as in Figure 5C.1.

Second example: Petrobras (PETR4.SA). The factors are US factors and the returns are in BRL. The Joint R^2 is 0.25. The dominant exposure is **WTI crude oil**: elasticity +0.266 (90% +0.190 to +0.343, t +5.62). **Oil +50%** gives +11.4% (band +8.0% to +14.9%) and **Oil −40%** gives −12.7%. The dollar elasticity is −0.467, but its raw t is only −0.19, so most of that number is the sector prior, not evidence from Petrobras. In the 2022 hiking cycle the model expected −7.0% while the stock returned +98.5%: oil, the real and Brazilian politics dominate, and US macro betas cannot capture them. The Surprise model uses Brazil's surprise indices and explains almost nothing (R^2 0.03).

Limitations and pitfalls

- **Not a forecast.** Betas describe the last three years of co-movement. They change with the business mix, and the replay of old episodes applies today's betas to yesterday's shocks.
- **Bold depends on the raw t, the bar on the shrunk beta.** A large shrunk beta with a small raw t (Petrobras USD above) mostly reflects the sector median. The two can even have opposite signs when the raw estimate is near zero.
- **Bands are too narrow for a single stock.** They exclude idiosyncratic risk; a realised move far outside the band is not a model failure in itself.
- **US-centric factor set.** The Joint and Single-factor models use US rates, US credit, the Fed broad dollar and WTI for every security, including non-US stocks. Local rates and local currency effects are only captured to the extent that they co-move with these factors.
- **The regime table is for the sector, with US / global / Fed regimes**, whatever the company's country. It uses contemporaneous regimes, which are only known after the month, so it is not directly investable.
- **Weekly refresh.** Betas can be up to a week old (more if a weekly run is missed; the header shows the date).
- **Minimum history.** A security with fewer than 52 weeks of prices shows "No macro exposures".

Related mnemonics and BQL fields

- No command-bar function opens this tab directly. `DES` opens the company; `WEI` opens the Markets overview.
- BQL (snapshot only, model-based): `macro.beta_rates` (Single-factor 10y beta), `macro.beta_usd` , `macro.beta_oil` , `macro.beta_credit` (Single-factor), `macro.beta_growth` (Surprise model, growth index) and `macro.rates_down_100bp` (modelled impact of rates −100bp, from the Joint 2y beta).

- Related sections: macro exposures heatmap, regime performance, Scenario builder and Episode replay in 2B. Markets and cross-asset; surprise indices and the cycle clock in 2A. Macro.

Earnings

Route: `/company/:id?tab=earnings` · **Menu:** Company 360 › Earnings · **Mnemonics:** `ERN` (also `EE` for the model-based forecast tab) · **Refresh:** nightly jobs `earnings_*` (weekly under the default profile)

Purpose

The Earnings tab is the company's earnings file: when it reports next and how sure we are of that date, how the stock has reacted to its past releases, our **model-based** expectation of the next EPS and revenue figure, and every past release with its official time, surprise and price reaction. It also lists dividends, splits and other corporate events. Use it before a release to size the usual move, and after a release to see whether the reaction was unusual. There are **no analyst consensus estimates** in the app; every expectation is a time-series model on the company's own history.

Screenshot

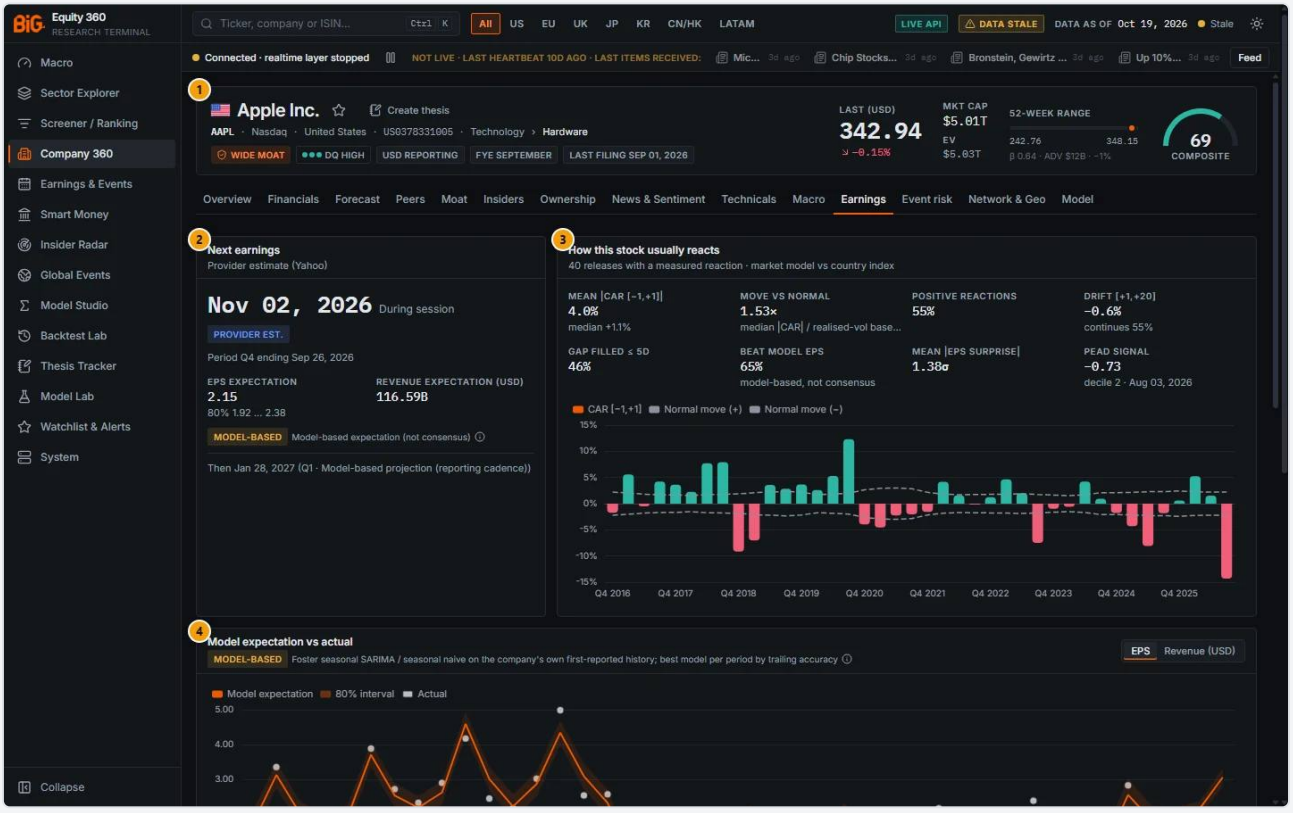


Figure 5C.4: Earnings tab for Apple (`/company/AAPL?tab=earnings`), data as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	See 5A
2	Next earnings	Next date, session, date-source badge, period, model-based EPS and revenue expectation, later dates
3	How this stock usually reacts	Eight reaction statistics and a bar chart of CAR [−1,+1] per release against the normal move
4	Model expectation vs actual	Model expectation with its 80% interval and the reported actuals, for EPS or revenue
5	Release history	One row per release: time, source, expectation, actual, surprises, CARs, move ratio, gap (Figure 5C.5)
6	Corporate events	Dividends, splits, 8-K items, proxies and exchange disclosures (Figure 5C.6)

Figure 5C.5: Release history for Apple. "OFFICIAL" = SEC 8-K Item 2.02 acceptance time; "doc" opens the filing.

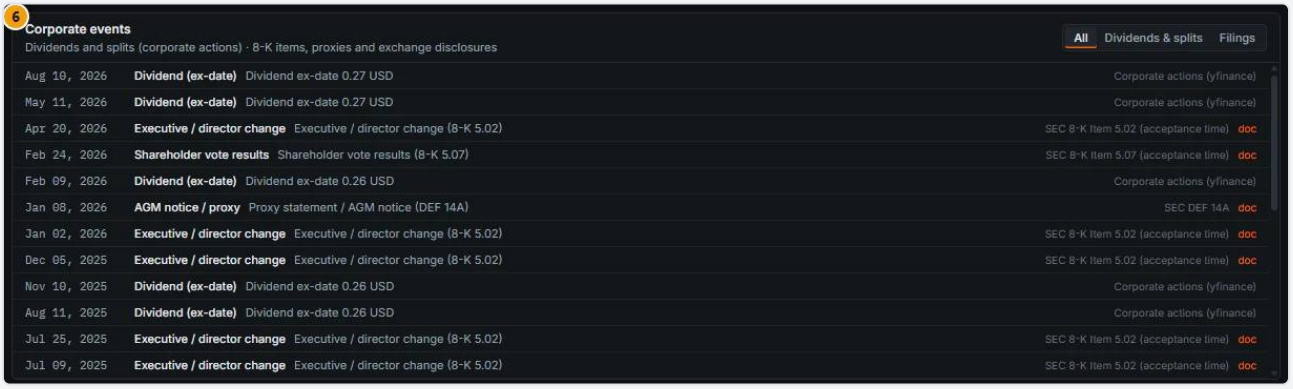


Figure 5C.6: Corporate events for Apple, with the source of each row on the right.

Controls

Control	What it does
Date-source badge (hover)	Explains what the date is based on; for a projection, also the plausible window
Item segmented control (EPS / Revenue) on panel 4	Switches the expectation chart between diluted EPS and revenue (reporting currency)
Hover on the reaction chart	Period, release date, CAR [−1,+1], normal move (±) and drift [+1,+20]
Hover on the expectation chart	Period, model expectation, 80% interval, actual, surprise (or "pending") and the model used
Hover on the **Model-based** tag	States that expectations come from time-series models on the company's own reported history and are not analyst consensus
`*` after a period (hover)	"Period inferred from the calendar / cadence (no filing matched)"
Source badge in the history (hover)	The full source label, e.g. "SEC 8-K Item 2.02 (acceptance time)"
doc links	Open the filing or exchange disclosure in a new browser tab
Event kind segmented control (All / Dividends & splits / Filings) on panel 6	Filters the corporate events list (up to 120 rows)
"awaiting release" badge	Shown when the next date has passed but no release has been recorded yet

Methodology

NEXT DATE AND ITS SOURCE

Every date carries a source badge, always visible:

Badge	Meaning
Announced	Company-announced date from the exchange schedule (for example the JPX schedule for Tokyo)
Provider est.	Provider estimate (Yahoo)
Projection	Model-based projection from the company's reporting cadence, with a plausible window
Official	Past release at the official time (SEC 8-K Item 2.02 acceptance time, TDnet kessan tanshin, HKEX results announcement)
Filing	Past periodic report filing (10-Q, 10-K, 20-F, 40-F or national equivalent), which may lag the first public release
Provider	Past release dated by the provider (Yahoo)

Projection. The next period end follows the company's cadence (52/53-week reporters use the same period a year earlier + 364 days). The expected date is the same-period release a year earlier + 364 days (which keeps the weekday), or the period end plus the median lag of the last four releases. The window is ± max(3, 1.5 × MAD of the lags) days, capped at 21. A company-announced date overrides the projection; otherwise a Yahoo date mapped to the same period does.

Past releases. A dated release is assigned to the latest known fiscal period end P with 4 ≤ release − P ≤ 100 days (130 outside the US); otherwise the period is extrapolated and flagged `*`. For each period the release is the earliest official release, else the provider date, else the periodic report filing itself.

PRICE REACTIONS

The reaction uses a **market model** against the company's country index (for example the S&P 500 for Apple, the Nikkei 225 for Toyota; ACWI as fallback), estimated by OLS on trading days [−250, −30] before the release (at least 60 observations):

```
AR_t   = r_t − (α + β · r_m,t)
CAR    = Σ AR_t over the window
BHAR   = Π(1 + r_t) − Π(1 + α + β · r_m,t)
normal move = σ_resid · √3        move ratio = |CAR[−1,+1]| / normal move
```

Day 0 is the first trading day on or after the reaction date (the next session for an after-close release). Windows: [−1,+1] (the announcement reaction), [0,+5] and [+1,+20] (post-announcement drift). Returns are split- and dividend-adjusted; a daily move above 50% is treated as a data error. There is no option-implied move in the app: the normal move is the stock's usual 3-day abnormal volatility.

Gap = open on day 0 ÷ previous close − 1. A gap of at least half a daily σ is "filled" if the price trades back to the previous close within six sessions (day 0 to +5), else "held".

REACTION STATISTICS (PANEL 3)

Statistic	Definition
Mean abs. CAR [−1,+1]	Mean absolute announcement reaction; "median" = median signed CAR
Move vs normal	Median of the absolute CAR [−1,+1] divided by the normal move, across releases
Positive reactions	Share of releases with CAR [−1,+1] > 0
Drift [+1,+20]	Mean CAR [+1,+20]; "continues" = share where the drift has the same sign as the announcement reaction
Gap filled ≤ 5d	Share of qualifying gaps that were filled
Beat model EPS	Share of releases with a positive model-based EPS surprise (labelled "model-based, not consensus")
Mean abs. EPS surprise	Mean absolute EPS surprise in σ units (z)
PEAD signal	Latest post-earnings drift score, its decile and signal date (model-based)

MODEL-BASED EXPECTATIONS AND SURPRISES

Expectations for quarterly (or half-yearly) **revenue** and **diluted EPS** come from the company's own first-reported history, point in time: for a release on date R the model is fitted on data known at R − 1 day, with EPS put on the share

basis of that date.

- **Foster seasonal model:** SARIMA(1,0,0)×(0,1,0)$_s$ with drift, the classic time-series earnings expectation of the SUE / PEAD literature, on EPS levels and on log revenue. Needs at least two seasons plus four observations.
- **Seasonal naive:** the same period a year earlier (EPS plus the mean seasonal change of the last eight periods; revenue times trailing seasonal growth).
- **Forecast path** (upcoming periods only): the base case of the Forecast tab for the fiscal year × the period's average share of the year.
- **Primary model:** the one with the lower mean absolute percentage error over the company's previous (up to) eight released periods, if at least four exist; else the Foster model; else seasonal naive.

The interval is the **80% interval** of the h-step forecast (for seasonal naive, from the company's own past errors). When the structured actual arrives (10-Q / 10-K XBRL or the national filing):

```
surprise % = (actual − expectation) / |expectation|        surprise z = (actual − expectation) / σ
```

The **PEAD signal** combines the standardised EPS surprise (SUE: surprise ÷ standard deviation of the previous eight surprises, at least four), the revenue surprise z and CAR [−1,+1] ÷ normal move, each capped at ±3, and averages them. It is ranked into deciles within region × signal month (at least 20 names). The signal date waits until the actual is public, so for US companies it often follows the 10-Q, not the press release.

Data sources and refresh

- **Official release times:** SEC EDGAR submissions (8-K Item 2.02 acceptance time; 10-Q / 10-K / 20-F / 40-F as periodic reports; other 8-K items and DEF 14A as corporate events), TDnet kessan tanshin (Japan) and HKEX results announcements (Hong Kong).
- **Dates:** JPX schedule (company-announced), Yahoo (provider dates and estimates).
- **Actuals:** `fundamentals_canonical` (XBRL and national filing systems).
- **Corporate actions:** dividends and splits from yfinance, including declared future ex-dates; buybacks and forecast revisions from TDnet; HKEX announcements.
- **Jobs (nightly DAG):** `earnings_dates_fetch`, `earnings_events`, `earnings_reactions`, `earnings_expectations`, `earnings_signals`. Under the default weekly profile they run in the Saturday refresh. Reaction history starts in 2012; expectations are back-filled from 2016.

Validation status

Not formally validated. There is no pass / fail gate for the expectations, the reactions or the PEAD signal.

- Expectations are labelled **Model-based** "Model-based expectation (not consensus)". The API note reads: "Expectations are model-based (Foster seasonal SARIMA / seasonal naive on the company's own first-reported quarterly series, plus the analytics forecast path), NOT analyst consensus."
- Projected dates are labelled "Model-based projection (reporting cadence)" and provider dates "Provider estimate (Yahoo)".
- The PEAD signal is labelled model-based. Its forward outcomes are stored only as **labels** for the Backtest Lab, never as inputs; no out-of-sample result is recorded in the validation documents.
- The 19 Oct 2026 audit confirmed that earnings source labels are handled point-in-time correctly.

Correct usage and worked example

Before a release, read the **date badge** first (an Announced date is firm; a Projection is a window), then **Move vs normal** and **Mean |CAR|** to size the usual reaction. After a release, compare the move ratio of the latest row with the stock's median. Treat "beat" and "miss" as relative to a mechanical time-series model, not to what the market expected.

Worked example: Apple (AAPL), 19 Oct 2026 (Figure 5C.4).

1. **Next earnings:** Nov 02, 2026, "During session", badge **Provider est.** (Yahoo), fiscal Q4 ending 26 Sep 2026. Model-based EPS expectation 2.15 (80% 1.92 … 2.38) and revenue USD 116.59B. The following date, 28 Jan 2027, is a **Projection** with a plausible window of 25-31 Jan 2027.

2. **Usual reaction (40 releases):** mean |CAR [−1,+1]| 4.0% (median +1.1%), move vs normal 1.53×, positive 55%, drift [+1,+20] −0.6% (continues 55%), gaps filled 46%, beat model EPS 65%, mean |EPS surprise| 1.38σ.
3. **Latest release:** fiscal Q3 2026 (period ending 27 Jun 2026), released on 30 Jul 2026 after the close, at the SEC 8-K Item 2.02 acceptance time (badge **Official**). EPS was 2.02 against a seasonal-naive expectation of 1.80 (+12.3%, z +0.72), and revenue USD 109.4B against 107.1B (+2.2%). Yet CAR [−1,+1] was **−14.3%**, 6.4× the normal move of about 2.2%, with an opening gap of −8.6% that held. A "beat" against our model said nothing about the market's expectations.
4. **PEAD signal:** −0.73, decile 2, signal date 3 Aug 2026 (the large negative announcement reaction outweighs the positive surprises).

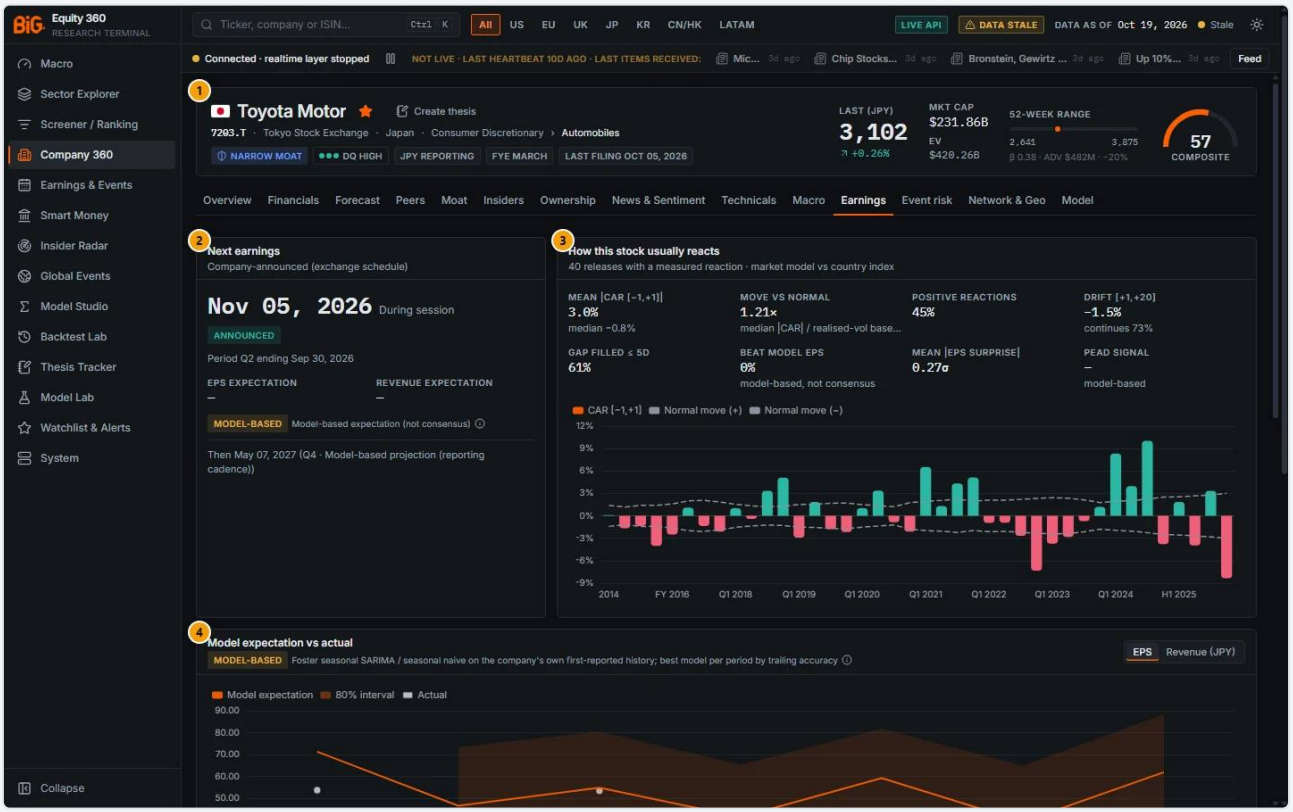


Figure 5C.7: Earnings tab for Toyota (`/company/7203.T?tab=earnings`), data as of 19 Oct 2026. The callouts are the same as in Figure 5C.4.

Second example: Toyota (7203.T). The next date is Nov 05, 2026 with the badge **Announced** (JPX schedule), for fiscal Q2 ending 30 Sep 2026. There is no EPS or revenue expectation for it (dashes): the company's quarterly series is too short or too stale for the models. Over 40 releases measured against the Nikkei 225, the mean |CAR| is 3.0%, the move vs normal 1.21× and the drift continues 73% of the time. "Beat model EPS 0%" rests on only three measured surprises (2022-2023), so ignore it. In this snapshot all of Toyota's past releases are Yahoo provider dates, so the history shows no release time and "Time n/a" for most sessions.

Limitations and pitfalls

- **No consensus.** Beat / miss is against a time-series model on the company's own history. A company can "beat" our model and still disappoint the market (Apple, Q3 2026).
- **Surprises arrive late.** The surprise appears when the structured actual lands (XBRL 10-Q / 10-K or the national filing), which can be days or weeks after the press release.
- **Thin non-US coverage.** Official release times exist only where the filing system provides them (SEC, TDnet, HKEX) and were ingested; elsewhere the date is the provider's or the periodic filing's. Expectations need enough quarterly or half-yearly history; many non-US issuers have none (Toyota).
- **Provider dates can be wrong.** A Yahoo estimate is labelled as such; check the company's investor-relations page before acting on it. "Projection" dates are only a window.
- **Times are shown in your local time zone.** Release timestamps are displayed in the browser's time zone, so an after-close US release appears after midnight in Lisbon.

- **EPS history is on the share basis of each forecast date.** The EPS chart is not restated across splits; Apple's 2020 four-for-one split shows as a level drop.
- **Small samples.** Reaction statistics over a few releases, and PEAD deciles in thin months, are noisy.
- **No option-implied move.** "Normal" means the stock's own realised abnormal volatility.

Related mnemonics and BQL fields

- `ERN` opens this tab; `EE` opens the model-based Forecast tab; `EVTS` without a security opens the Earnings & Events calendar.
- BQL: `earn.*` (latest release: `earn.release_date`, `earn.date_source`, `earn.car_m1_p1`, `earn.car_0_p5`, `earn.car_p1_p20`, `earn.bhar_*`, `earn.move_ratio`, `earn.gap_pct`, `earn.gap_filled`, `earn.eps_expected`, `earn.eps_lo80`, `earn.eps_hi80`, `earn.eps_actual`, `earn.eps_surprise_pct`, `earn.eps_surprise_z`, `earn.rev_*`), `eps_surprise`, `eps_surprise_z`, `rev_surprise`, and `pead.*` (`pead.sue`, `pead.ear_z`, `pead.pead_score`, `pead.decile`, `pead.signal_date`; `pead.outcome_car_p2_p20` / `_p60` are labels for back-tests).
- Alerts: "earnings upcoming" and "earnings surprise" (|model surprise z| above a threshold) in 13. Watchlists and alerts.
- Related sections: 6. Earnings & Events calendar and Earnings season; 5A. Forecast.

Event risk

Route: `/company/:id?tab=events` · **Menu:** Company 360 › Event risk · **Mnemonics:** `EVTS` (with a security) · **Refresh:** nightly (`event_exposure`, `event_escalation`, GDELT jobs); event-study library weekly

Purpose

The Event risk tab measures how exposed the company is to news and geopolitical events in the countries where it earns its revenue, and what similar shocks have done to prices in the past. It also shows the GDELT news coverage labelled to the company itself, with its tone and the most salient stories. Use it to spot names whose revenue base is under unusual news pressure, and to read the news flow on the company in one place. All exposure figures are **model-based**.

Screenshot

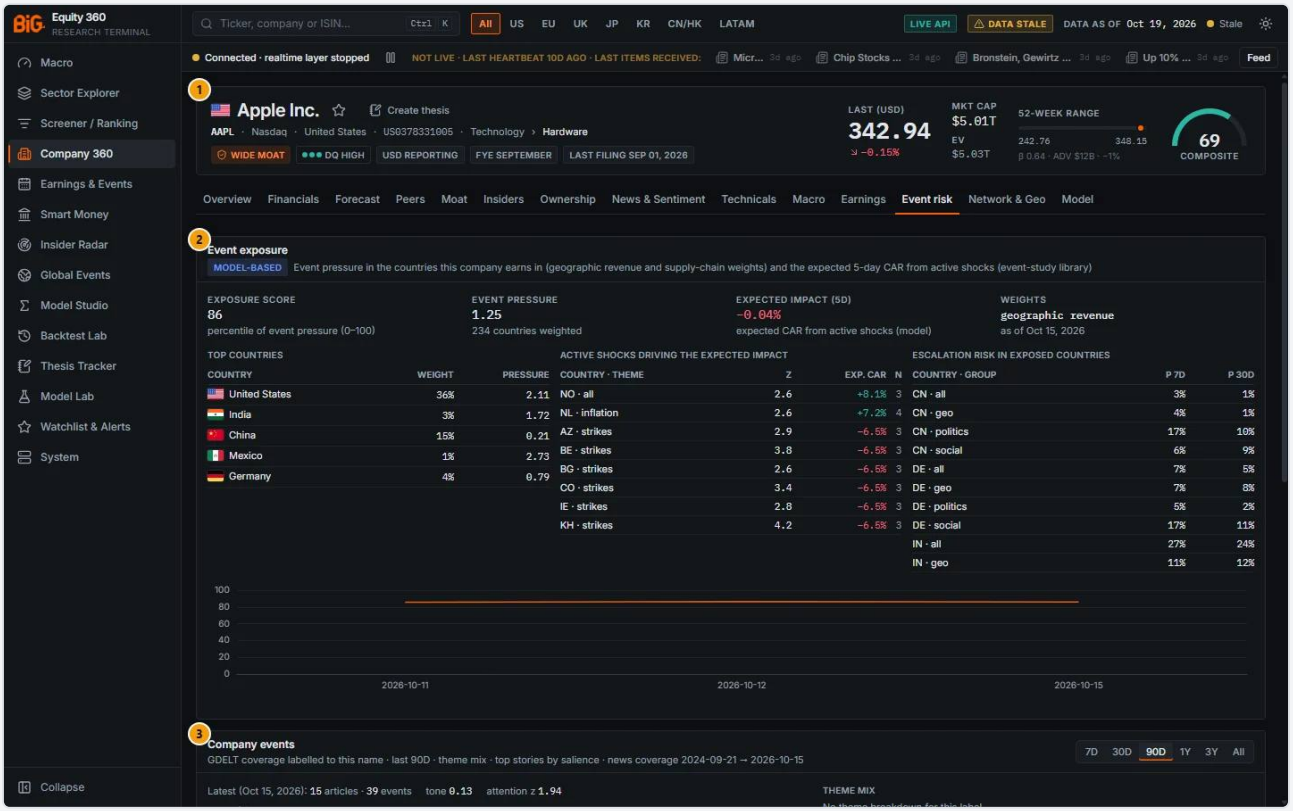


Figure 5C.8: Event risk tab for Apple (`/company/AAPL?tab=events`), exposure as of 15 Oct 2026.

#	Panel	What it shows
1	Company header	See 5A
2	Event exposure	Exposure score, event pressure, expected 5-day impact, weights; top countries, active shocks, escalation risk; exposure score history
3	Company events	GDELT coverage of the company: articles and tone, theme mix, recent shocks and top stories (Figure 5C.9)

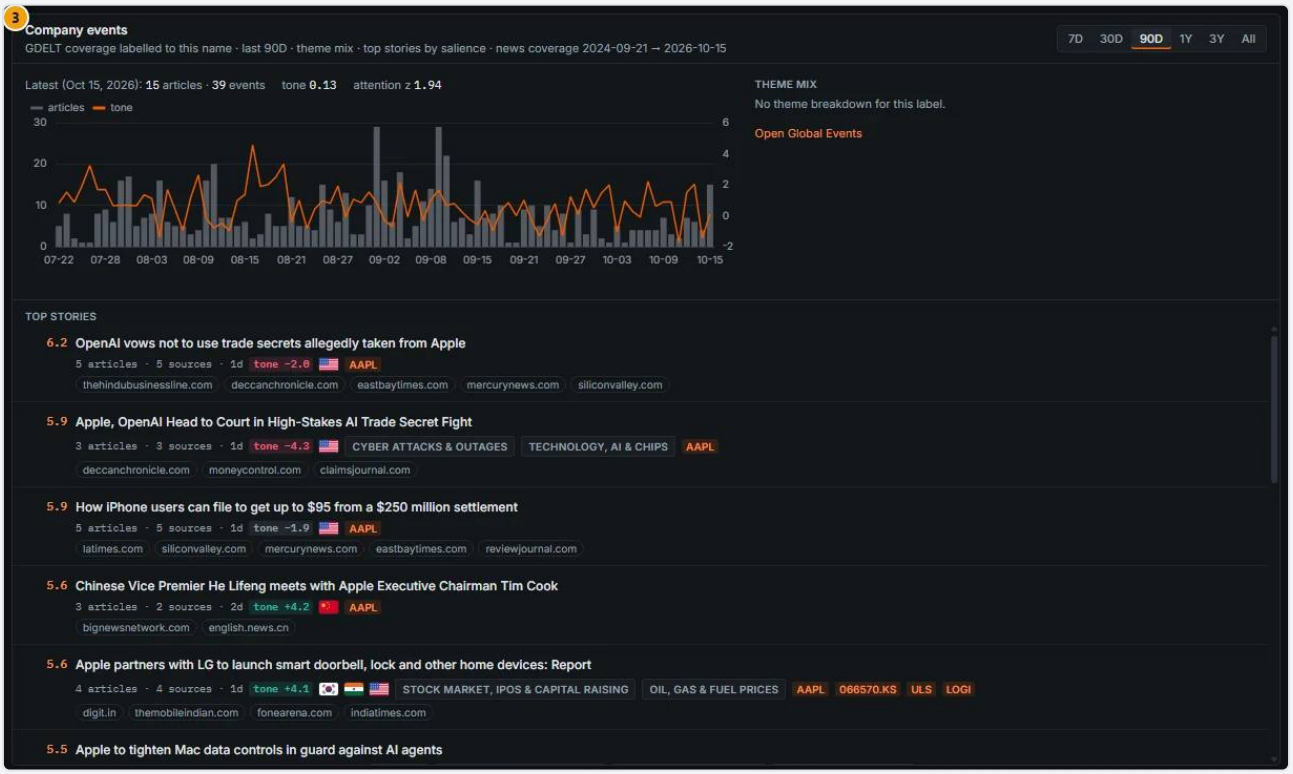


Figure 5C.9: Company events for Apple: daily articles (bars) and tone (line) over 90 days, and the top stories by salience.

Controls

Control	What it does
Events history range (7D / 30D / 90D / 1Y / 3Y / All)	Sets the window of the coverage chart and theme mix; ranges over a year are shown weekly. Default 90D
Hover on the coverage chart	Daily articles and tone
Click a story	Opens the story drawer with its sources, labels and links
Open Global Events link	Opens the Global Events page (`/events`)
Hover on the exposure history	Exposure score by date

Methodology

COUNTRY WEIGHTS

The company's exposure to each country comes from its **geographic revenue** (the issuer-country exposure weights of the linkage layer; regional segments are spread over member countries by GDP). Without a disclosure, 100% goes to the home country. If the company has disclosed customers, their home countries are added with weight 0.25 × the customer's share (capped at 1), and the weights are re-normalised. The **Weights** tile names the source: "geographic revenue", "home country only", or either "+ supply chain".

EVENT PRESSURE AND EXPOSURE SCORE

For each country, from the daily GDELT aggregates (all themes) over the last 7 days:

```
pressure_c   = mean(max(attention_z, 0)) + mean(max(−tone_z, 0))
pressure     = Σ_c w_c · pressure_c
exposure score = percentile of the company's pressure across the universe (0-100)
```

`attention_z` is the article / event count against its own history and `tone_z` the average tone against its history, so pressure rises with unusual attention and unusually negative tone. "Top countries" lists the five countries with the largest `w_c × pressure_c`.

EXPECTED IMPACT (5D)

An **active shock** is a country (× theme) day in the last 7 days with attention z ≥ 2.5 and at least 20 items, not within 10 days of the previous shock of the same kind. For each active shock the app looks up the **event-study library**: the mean CAR [0,+5] after past shocks of the same kind, trying theme × country × sector, then theme × country, then theme × sector, then theme alone (and the issuer, country-index or sector-ETF level, in that order). The library CAR is shrunk toward zero by `n / (n + 10)` for a cell with `n` past events; concurrent shocks in one country are averaged, not added; and the result is weighted by the country weight:

```
expected impact = Σ_c w_c · mean_shocks( CAR̄_cell · n / (n + 10) )
```

The "Active shocks" table shows each shock's attention z, the **unshrunk** library CAR ("Exp. CAR") and the number of past events `n` in the library cell.

ESCALATION RISK

For each country × theme group (all, geo, politics, social, ...) a bivariate discrete-time **Hawkes process** (the country and its neighbours, with a day-of-week background) is fitted by EM. **P 7d / P 30d** is the Monte Carlo probability that the event count over the next 7 / 30 days exceeds the 90th percentile of the country's own historical 7 / 30-day counts. An **explosive** badge appears when the branching ratio (expected offspring per event) is above 1. The table shows the first ten rows for the exposed countries.

COMPANY EVENTS PANEL

GDELT articles and events labelled to the issuer (organisation matching at ingest, link confidence ≥ 0.5): daily article counts, mean tone, the latest attention z, the theme mix (top eight themes by share), recent attention or tone shocks, and the top stories. Near-duplicate articles are clustered into **stories**; the salience score shown before each story is

```
salience = ln(1 + articles) + ln(1 + sources) + 0.5 · ln(1 + events) + |tone| / 5 + issuer bonus (0.5 or 1)
```

The top-stories list shows up to 15 stories linked to the company in the **last 30 days**, whatever the range selected above it.

Data sources and refresh

- **GDELT 2.0** Events, Mentions and GKG, streamed and labelled at ingest (security, sector, country, about 40 market themes); news stories also merge Polygon and RSS items. Jobs `gdelt_*`, `events_agg` and `story_cluster` (nightly).
- **Exposure:** job `event_exposure` (nightly, universe `full`, 7-day window). **Escalation:** `event_escalation` (nightly, 2,000 Monte Carlo paths). **Event-study library:** `event_study_library` (weekly, market-model CARs for windows [0,+1], [0,+5], [−1,+5], [0,+10]).
- **Weights:** geographic revenue and customer links from the linkage layer (see Network & Geo).
- **History:** the panel subtitle shows the news coverage actually loaded (2024-09-21 → 2026-10-15 in Figure 5C.9). A deep GDELT backfill is extending history back toward 2015, newest first.

Validation status

Not formally validated as a predictor. The exposure panel carries the badge **MODEL-BASED** ("Event pressure in the countries this company earns in … and the expected 5-day CAR from active shocks (event-study library)"), and the expected impact is described as "expected CAR from active shocks (model)".

What has been checked, per the plan documents:

- **Labels:** a 60-row spot check of the GDELT pilot found security links correct 82% of the time (49/60), country 84%, theme 68% and sector about 68% (`EVENTS_V2_PLAN.md`, E5).
- **Change-point detector** (used for shocks elsewhere): calibrated to about 0.13 false alarms per 200 days with 97% detection of 1.5 sd shifts (`EVENTS_MODELS_REALTIME_PLAN.md`, C2).
- **Event-study library:** each cell has a standard error and p-value, but many cells have only a handful of events; the shrinkage `n / (n + 10)` is there for that reason.
- **Hawkes escalation probabilities:** no calibration result is recorded.

Correct usage and worked example

Use the exposure score to rank names by **current** news pressure on their revenue base, then read the top countries and active shocks to see why. Treat the expected impact as a size check: if it is a few basis points, the shocks are either far from the company's revenue or historically weak. Use escalation probabilities to see which exposed countries could get worse, not as a forecast for the stock.

Worked example: Apple (AAPL), exposure as of 15 Oct 2026 (Figure 5C.8).

1. **Exposure score 86**: Apple's event pressure (1.25) is higher than that of 86% of the universe. Weights come from **geographic revenue** spread over 234 countries.
2. **Top countries:** the United States (weight 36%, pressure 2.11) drives most of it; India (3%, 1.72) and Mexico (1%, 2.73) are under high pressure but carry little weight; China (15%) is calm (0.21).
3. **Active shocks:** the largest library CARs are Norway, all themes (+8.1%, n 3), the Netherlands, inflation (+7.2%, n 4) and strikes in several countries (−6.5%, n 3). The identical −6.5% on six "strikes" rows shows that the lookup fell back to the theme-level cell. With tiny weights in these countries and shrinkage of 3/13, the **expected impact (5d) is −0.04%**: no meaningful event-driven expectation.
4. **Escalation:** India, all themes, has P 7d 27% and P 30d 24%; China, politics, 17% and 10%.

5. **Company events:** on 15 Oct 2026, 15 articles and 39 events, tone 0.13, attention z 1.94. The top story ("OpenAI vows not to use trade secrets allegedly taken from Apple", salience 6.2, 5 articles from 5 sources, tone −2.0) is company news rather than a country shock.

Limitations and pitfalls

- **Media attention is not economic impact.** Pressure measures unusual coverage and negative tone in GDELT, which is weighted toward English-language and large-country media.
- **Exposure is only as good as the revenue geography.** Without a geographic disclosure the company is 100% home country; regional segments are spread by GDP, which can put small weights on many countries.
- **Library cells are small.** A CAR from three past events is anecdotal, even after shrinkage. Look at `n` before quoting "Exp. CAR".
- **The library mixes levels.** The fallback can use country-index or sector-ETF reactions, not the company's own.
- **Label errors.** About one in five issuer links in the pilot spot check was wrong; read the stories before drawing conclusions.
- **History is short for some uses.** Coverage starts where the loaded GDELT history starts (see the panel subtitle); exposure history on this tab only has the dates the job has run.

Related mnemonics and BQL fields

- `EVTS` with a security opens this tab; without a security it opens the Earnings & Events calendar. `TOP` and `N` open top news.
- No BQL field exposes the company event exposure in this snapshot (`sent.*` covers news sentiment).
- Related sections: 8. Global Events (escalation, GPR, contagion, event studies); news sentiment in 5A. News & Sentiment; the country weights in Network & Geo.

Network & Geo

Route: `/company/:id?tab=network` · **Menu:** Company 360 › Network & Geo · **Mnemonics:** `NET` · **Refresh:** network nightly (`supply_chain_network`); links and geography weekly / monthly

Purpose

The Network & Geo tab maps the company's disclosed **customers and suppliers**, its **statistical neighbours** (stocks that move with it), and where it **earns its revenue**, with country risk, event and macro overlays. A shock simulator propagates a revenue shock through the supply chain. Use it to find hidden concentration (a single large customer, one dominant country), to see second-order country exposure through customers, and to trace who is hit if a company or country is shocked. Disclosed links come from filings by rules (no LLM); look-through exposure, country risk and shock propagation are **model-based**.

Screenshot

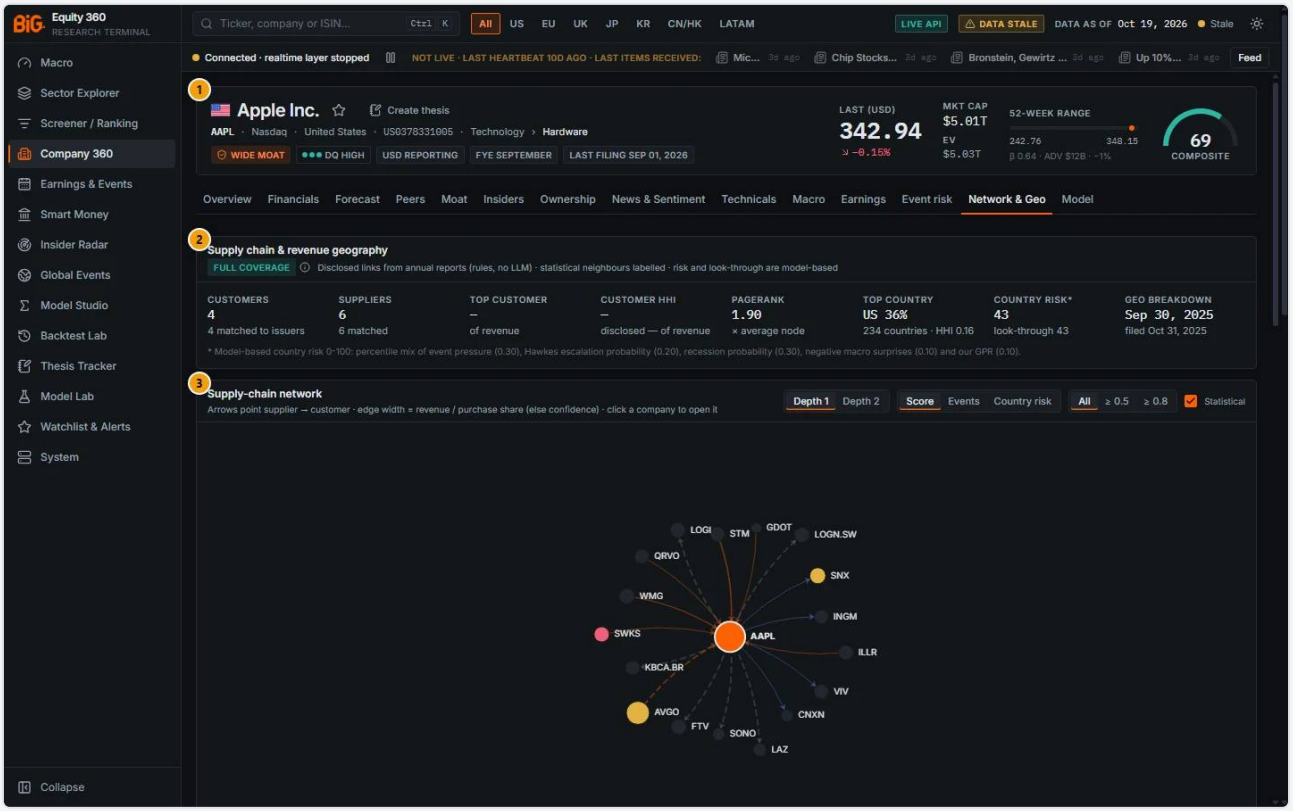


Figure 5C.10: Network & Geo tab for Apple (`/company/AAPL?tab=network`), network as of 19 Oct 2026.

#	Panel	What it shows
1	Company header	See 5A
2	Supply chain & revenue geography	Coverage badge and eight summary tiles (customers, suppliers, concentration, centrality, top country, country risk, geography date)
3	Supply-chain network	Force-directed graph of disclosed links and statistical neighbours
4	Relationships & provenance	Customer, supplier or statistical rows with share or correlation, confidence and source (Figure 5C.11)
5	Revenue geography	World map of revenue exposure or of a country overlay (Figure 5C.12)
6	Exposure by country	Direct, second-order and look-through shares with risk, events, escalation, hotspot, recession and surprise overlays (Figure 5C.13)
7	Shock scenario	Revenue-shock propagation over the supply chain, with an optional macro overlay (Figures 5C.14 and 5C.15)

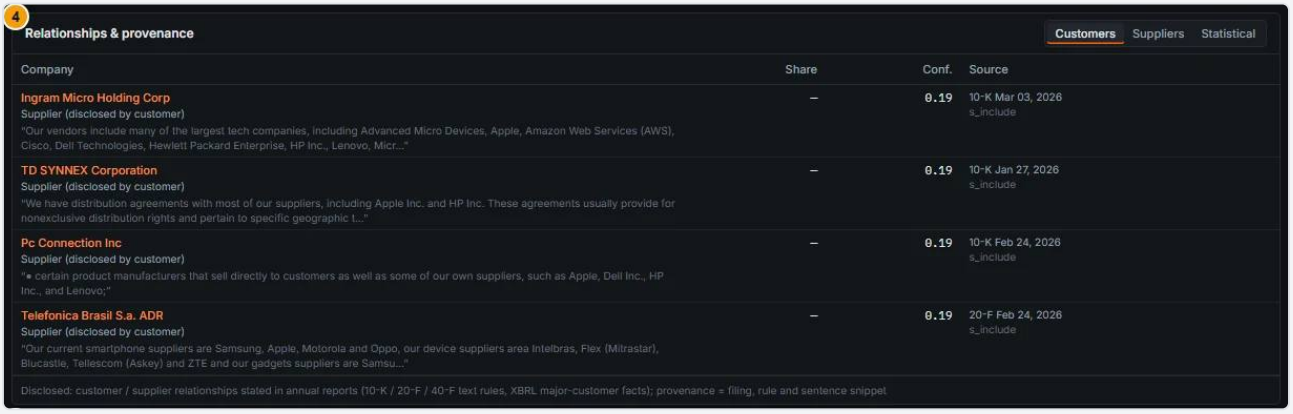


Figure 5C.11: Relationships & provenance for Apple, Customers view: each row shows the rule, the filing and the matched sentence.

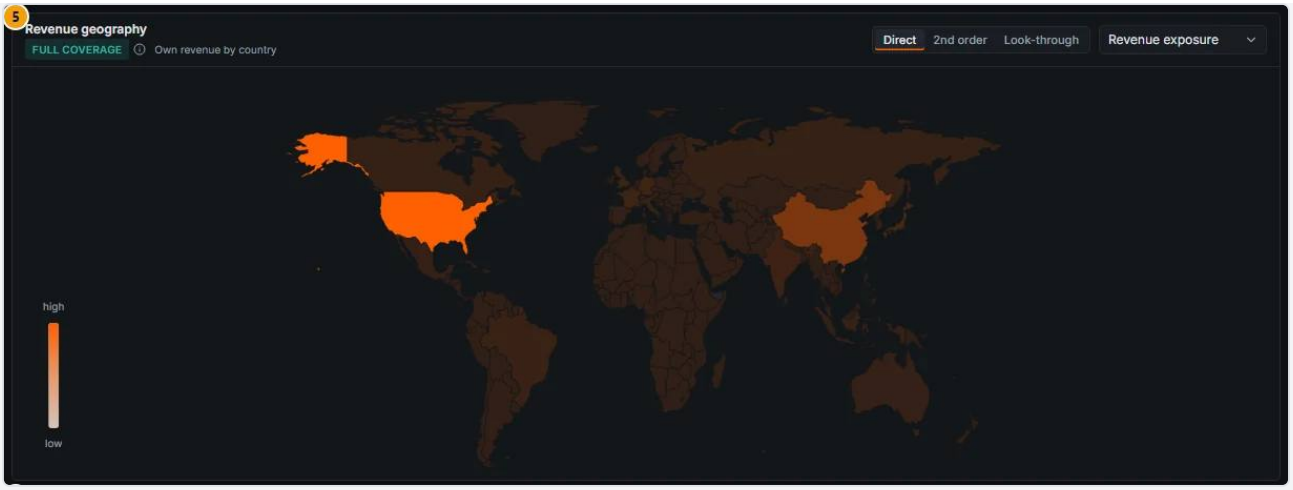
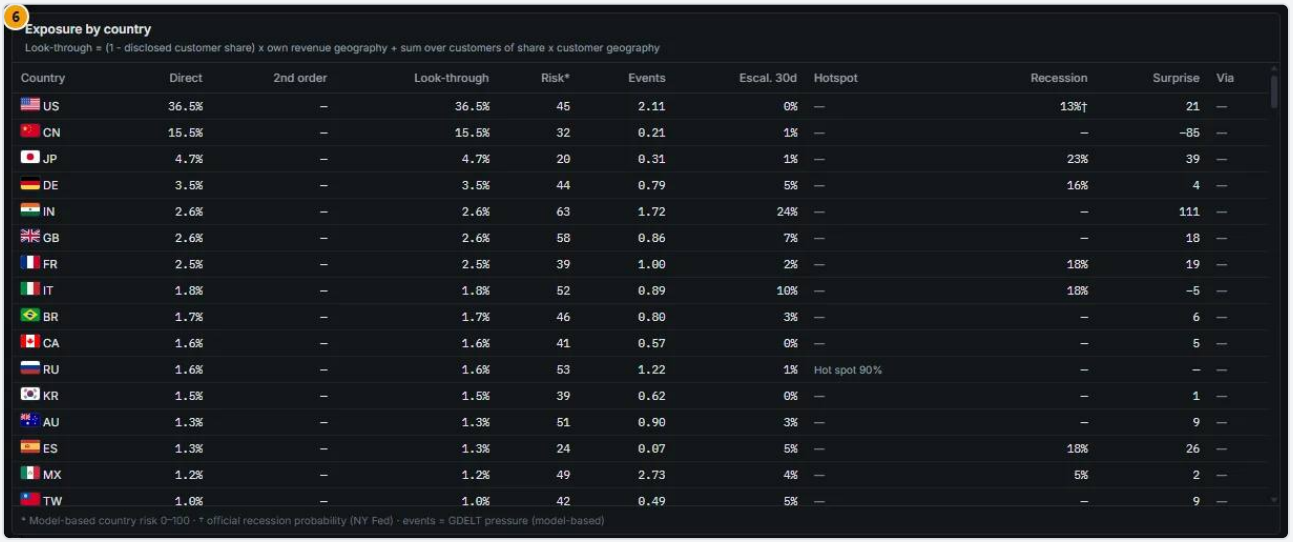


Figure 5C.12: Revenue geography for Apple, Direct basis, Revenue exposure layer.



Country	Direct	2nd order	Look-through	Risk*	Events	Escal. 30d	Hotspot	Recession	Surprise	Via
US	36.5%	–	36.5%	45	2.11	0%	—	13%†	21	—
CN	15.5%	–	15.5%	32	0.21	1%	—	–	-85	—
JP	4.7%	–	4.7%	20	0.31	1%	—	23%	39	—
DE	3.5%	–	3.5%	44	0.79	5%	—	16%	4	—
IN	2.6%	–	2.6%	63	1.72	24%	—	–	111	—
GB	2.6%	–	2.6%	58	0.86	7%	—	–	18	—
FR	2.5%	–	2.5%	39	1.00	2%	—	18%	19	—
IT	1.8%	–	1.8%	52	0.89	10%	—	18%	-5	—
BR	1.7%	–	1.7%	46	0.80	3%	—	–	6	—
CA	1.6%	–	1.6%	41	0.57	0%	—	–	5	—
RU	1.6%	–	1.6%	53	1.22	1%	Hot spot 90%	–	–	—
KR	1.5%	–	1.5%	39	0.62	0%	—	–	1	—
AU	1.3%	–	1.3%	51	0.90	3%	—	–	9	—
ES	1.3%	–	1.3%	24	0.07	5%	—	18%	26	—
MX	1.2%	–	1.2%	49	2.73	4%	—	5%	2	—
TW	1.0%	–	1.0%	42	0.49	5%	—	–	9	—

Figure 5C.13: Exposure by country for Apple. † = official recession probability (NY Fed).

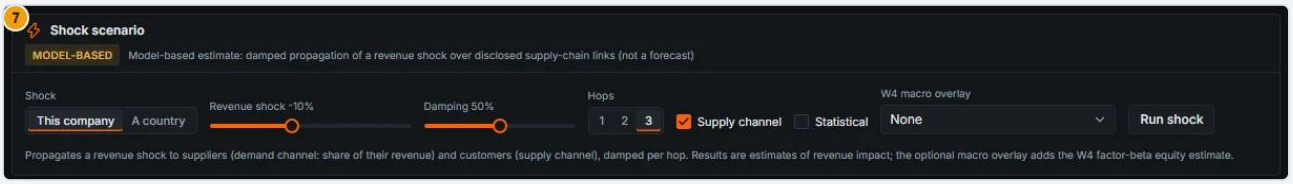
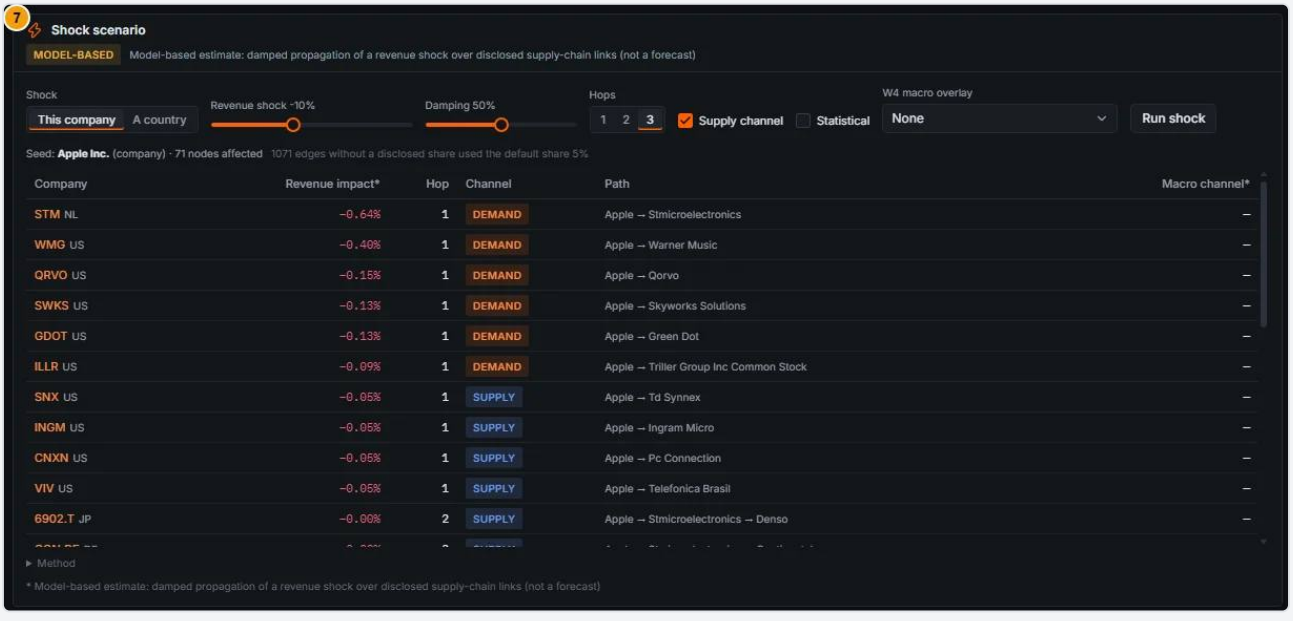


Figure 5C.14: Shock scenario controls before a run.



Company	Revenue impact*	Hop	Channel	Path	Macro channel*
STM NL	-0.64%	1	DEMAND	Apple → Stmicroelectronics	–
WMG US	-0.40%	1	DEMAND	Apple → Warner Music	–
QRVO US	-0.15%	1	DEMAND	Apple → Qorvo	–
SWKS US	-0.13%	1	DEMAND	Apple → Skyworks Solutions	–
GDOT US	-0.13%	1	DEMAND	Apple → Green Dot	–
ILLR US	-0.09%	1	DEMAND	Apple → Triller Group Inc Common Stock	–
SNX US	-0.05%	1	SUPPLY	Apple → Td Synnex	–
INGM US	-0.05%	1	SUPPLY	Apple → Ingram Micro	–
CNXN US	-0.05%	1	SUPPLY	Apple → Pc Connection	–
VIV US	-0.05%	1	SUPPLY	Apple → Telefonica Brasil	–
6902.T JP	-0.00%	2	SUPPLY	Apple → Stmicroelectronics → Denso	–

*Figure 5C.15: Shock scenario for Apple after *Run shock** with the defaults (This company, −10%, damping 50%, 3 hops, supply channel on): seed line, result table and the Method fold-out.*

Controls

Control	What it does
Coverage badge (hover)	Full / Partial / Sparse coverage, with the note and how the geography was obtained
Depth (Depth 1 / Depth 2)	Shows direct links only, or also the links of the company's counterparties
Colour nodes by (Score / Events / Country risk)	Colours nodes by composite score (0-100), event pressure or model-based country risk
Minimum confidence (All / ≥ 0.5 / ≥ 0.8)	Hides links below the confidence threshold
Statistical neighbours checkbox	Shows or hides the return co-movement layer (dashed). On by default
Graph: drag, scroll, hover, click	Drag nodes, zoom and pan; hover a node or edge for details (an edge shows its share, ρ, confidence, filing and sentence); click a listed company to open its Company 360
Legend items (hover)	Explain each edge type
Relation (Customers / Suppliers / Statistical)	Switches the relationship table; click a company name to open it
Exposure basis (Direct / 2nd order / Look-through)	Changes the map and sorts the country table
Map layer select	Revenue exposure, Country risk, Event pressure, Escalation 30d or Recession prob.
Hover on a map country	Exposure and every overlay component
Hover on "Via" or "Recession" cells	The customers behind second-order exposure; the recession method
Shock (This company / A country), **Country** select	Chooses the seed of the shock (a country from the company's exposure list)
Revenue shock slider (−50% to +50%, step 5)	Size of the shock to the seed's revenue. Default −10%
Damping slider (0-100%, step 10)	Damping per hop. Default 50%
Hops (1 / 2 / 3)	Number of propagation rounds. Default 3
Supply channel checkbox	Also propagates to customers (on by default; the demand channel is always on)
Statistical checkbox	Adds a one-hop co-movement estimate for statistical neighbours
Home-country-only checkbox (country shocks)	Also seeds companies without a geographic disclosure, at 100% home country
W4 macro overlay select	Adds the macro-beta equity estimate of a preset scenario for each affected company
Run shock button	Runs the propagation on demand; results appear below with a **Method** fold-out

Methodology

The definitions below follow `docs/SUPPLY_CHAIN.md`, which states: "Data sources are company filings and our own models. Everything model-based is labelled as such; statistical neighbours are never shown as disclosed relationships."

DISCLOSED LINKS (RULES ONLY, NO LLM)

Customer and supplier relationships come from annual reports, point-in-time by filing date: the XBRL `MajorCustomers` axis (SEC financial statement data sets), and customer / supplier sentences and customer tables in 10-K, 20-F and 40-F text. Each extraction rule has a prior precision, for example:

Rule	Example	Prior
`c_sales_to_pct`	"Sales to Walmart Inc. accounted for 21% of net sales"	0.95

Rule	Example	Prior
`c_name_pct`	"Samsung Electronics Co., Ltd. accounted for 11% of total net sales"	0.90
`c_largest`	"Our largest customer is PACCAR Inc."	0.75
`c_include`	"Our customers include Ford, General Motors and Stellantis"	0.55
`s_purchases_pct`	"Purchases from McKesson accounted for 40% of total purchases"	0.90
`s_sole`	"TSMC is our sole foundry"	0.80
`s_include`	"Our suppliers include BASF SE, Dow Inc."	0.55

Guards reject competition sentences, combined shares ("A and B accounted for 25%"), names after a geographic preposition and generic names; weak-context matches are kept only when the name resolves exactly to one of our issuers. **Confidence** = rule prior × name-match confidence (0 for an unmatched name, drawn as a grey diamond). A supplier named by its customer becomes the edge supplier → customer ("Supplier (disclosed by customer)"); when both sides disclose the pair the type is "Disclosed by both sides". Each edge keeps its filing, rule and the matched sentence (up to 200 characters). Edges older than two years are flagged **stale** and not used for weights.

Edge type (legend)	Meaning
Customer (disclosed by supplier)	The supplier names this customer in its annual report
Supplier (disclosed by customer)	The customer names this supplier in its annual report (reverse edge)
Disclosed by both sides	Named by the supplier and by the customer
Statistical neighbour (dashed)	Weekly return co-movement, not a disclosed relationship

STATISTICAL NEIGHBOURS

Weekly log returns over 104 weeks (at least 52 common weeks), top ten per issuer: Pearson correlation ≥ 0.5 (`return_corr`), or the partial correlation after removing the region's equal-weight market return, ≥ 0.3. In the docs' words: "These are co-movements, not relationships."

REVENUE GEOGRAPHY

`company_geo_revenue` combines tagged geographic segments in SEC XBRL (US filers and 20-F / 40-F filers), IFRS geographic-axis facts from ESEF and other national filings, and rule-based parsing of ESEF segment-note tables (FY2022 onwards; confidence 0.8 when the table total reconciles within 3%, else 0.7). Regions and rest-of-world are spread over member countries by GDP. **Without a breakdown, 100% goes to the home country (flagged "home").** Japanese, Korean and Chinese filers are covered only where they file with the SEC.

EXPOSURE BASES

With `s_c` the disclosed revenue share of customer `c` :

```
Direct          = latest annual breakdown known at the as-of date
2nd order       = Σ_c s_c · direct_c            (sums to the disclosed customer share)
Look-through    = (1 − Σ_c s_c) · Direct + 2nd order        (sums to 1)
```

Only resolved customers with a disclosed share count for second-order exposure. Sector exposure via customers is computed the same way and shown in an extra panel when more than one sector results.

SUMMARY METRICS

Tile	Definition
Customers / Suppliers	Number of disclosed counterparties, and how many are matched to listed issuers
Top customer	Largest disclosed customer share of revenue
Customer HHI	`Σ s_c²` over disclosed shares: a **lower bound**, because the undisclosed remainder is treated as diversified
PageRank	PageRank on the supplier → customer graph (α 0.85, weighted by confidence), × the average node

Tile	Definition
Top country	Largest country share, with the number of countries and the geographic HHI
Country risk*	Revenue-weighted model-based country risk, on direct and look-through weights
Geo breakdown	Period end of the geographic breakdown and its filing date, or "home only"

COUNTRY RISK (MODEL-BASED)

Each component is a percentile across countries, combined with these weights: event pressure 0.30, Hawkes P(escalation 30d) 0.20, recession probability 0.30 (NY Fed official where available, else our yield-curve probit, else HMM; the euro-area value for members without one), negative macro surprises 0.10 and our geopolitical-risk index 0.10. Missing components are re-weighted, and the score is shrunk toward 50 by the covered weight: `50 + (raw − 50) × covered`. The "Hotspot" column shows Getis-Ord Gi* classes (hot or cold spot at 90 / 95 / 99%) from the Global Events models.

SHOCK PROPAGATION (MODEL-BASED)

The shock is a fractional **revenue** change (−0.10 = −10%). Transmission from node j to node i:

- **Demand channel (upstream):** supplier j of customer c moves by `s_rev(j, c) × x_c`, where `s_rev` is the share of j's revenue that comes from c.
- **Supply channel (downstream):** customer c of supplier j moves by `γ × s_pur(j, c) × x_j`, with γ = 0.5 (partly substituted or buffered).
- Links without a disclosed share use a default share of 0.05 (flagged "assumed"); link confidence multiplies the transmission.
- **Hops:** `x(h) = δ^(h−1) · M^h · x(0)`, damping δ (default 0.5), total `x(0) + Σ x(h)` for up to three hops; each node is clipped to |seed|, so there is no amplification.
- **Country shock** of size m on country K: each company's seed is m × its direct revenue share in K (only companies with a geographic disclosure, unless Home-country-only is ticked).
- **Statistical channel:** one hop, `ρ × δ × seed`, reported separately "Statistical co-movement (not causal)".
- **Macro channel (W4 overlay):** the equity-return estimate of the chosen preset from each company's macro betas, shown next to the revenue estimate and never added to it (the units differ).

The result table lists Company, Revenue impact*, Hop, Channel (demand / supply), Path (the strongest predecessor at the first hop reached) and Macro channel*.

Data sources and refresh

- **Filings:** SEC 10-K / 20-F / 40-F text and XBRL (EDGAR, financial statement data sets), ESEF and other national filings. Jobs `linkage_text_units` (weekly), `linkage_links` and `linkage_georev` (monthly), `linkage_esef_geo` (monthly).
- **Prices:** weekly returns for statistical neighbours, job `supply_chain_neighbours` (weekly).
- **Overlays:** GDELT event pressure and Hawkes escalation, recession probabilities (NY Fed, our models), surprise indices, our GPR index.
- **Network, metrics, exposure and country risk:** job `supply_chain_network` (nightly). Under the default weekly profile, it runs in the Saturday refresh.
- **Shock scenario:** computed on demand when you click **Run shock**; nothing is saved.

Validation status

Not formally validated. There is no gate for link extraction, look-through exposure, country risk or shock propagation.

- The panel subtitle reads "Disclosed links from annual reports (rules, no LLM) · statistical neighbours labelled · risk and look-through are model-based".
- Country risk is labelled "Model-based country risk 0-100" with its weights; the shock panel carries the badge **MODEL-BASED** and the label "Model-based estimate: damped propagation of a revenue shock over disclosed supply-chain links (not a forecast)".
- The rule priors (0.55-0.95) are prior precisions assigned per rule, not measured accuracy.

- Statistical neighbours are labelled "NOT disclosed relationships".

Correct usage and worked example

Start with the **coverage badge**. With Full coverage, read the concentration tiles and the relationship table (open the sentence snippets: they are the evidence). With Sparse coverage, treat the tab as a home-country view plus a co-movement list. Use **Look-through** to see where your revenue really ends up when big customers are disclosed with shares, and the **shock** to rank second-round victims, not to forecast revenue.

Worked example: Apple (AAPL), 19 Oct 2026 (Figures 5C.10 to 5C.13).

1. **Coverage: Full.** Geography from tagged XBRL segment facts for the fiscal year ending 30 Sep 2025, filed 31 Oct 2025. Top country US 36% of 234 countries (HHI 0.16); country risk 43 on both bases; PageRank 1.90× the average node.
2. **Suppliers 6** (companies that name Apple as a customer). STMicroelectronics' 20-F states that Apple was **17.7% of its revenue** (rule `c_name_pct`, confidence 0.36) and Warner Music Group's 10-K gives 11%. Qorvo, Skyworks and Green Dot call Apple their largest customer. Broadcom's 10-K for fiscal 2023 gave 20% (rule `c_sales_to_pct`, confidence 0.38), but that filing is more than two years old, so the edge is **stale** (dashed) and excluded from weights and from the shock.
3. **Customers 4** (companies that name Apple among their suppliers): Ingram Micro, TD SYNNEX, PC Connection and Telefônica Brasil, all by the list rule `s_include` (confidence 0.19). None discloses a share, so the "Top customer" and "Customer HHI" tiles are empty and the **2nd-order column is empty: look-through equals direct**.
4. **Exposure by country:** US 36.5% (risk 45, recession 13%† from the NY Fed), China 15.5% (risk 32, surprise index −85), Japan 4.7%, Germany 3.5%, India 2.6% (risk 63, escalation 24% in 30 days).
5. **Statistical neighbours:** Logitech (ρ 0.56), Sonos (0.53), Fortive (0.51) and others: co-movement, not trade.
6. **Shock (Figure 5C.15):** with the defaults (Apple's revenue −10%, damping 50%, 3 hops), the seed line reads "Apple Inc. (company) · 71 nodes affected", with the note that 1,071 edges without a disclosed share used the default share of 5%. The largest hits are first-hop suppliers on the **demand** channel: STMicroelectronics −0.64% (−10% × 0.177 share × 0.36 confidence) and Warner Music −0.40%. Qorvo (−0.15%), Skyworks and Green Dot (−0.13% each) and Triller (−0.09%) have no disclosed share, so they use the assumed 5%. The four distributors (TD SYNNEX, Ingram Micro, PC Connection, Telefônica Brasil) are hit on the **supply** channel at −0.05% each (γ 0.5 × 5% × 0.19). Second-hop effects, such as Denso via STMicroelectronics, round to −0.00%. Broadcom does not appear because its edge is stale. The Macro channel column shows "—" because no W4 overlay was selected.

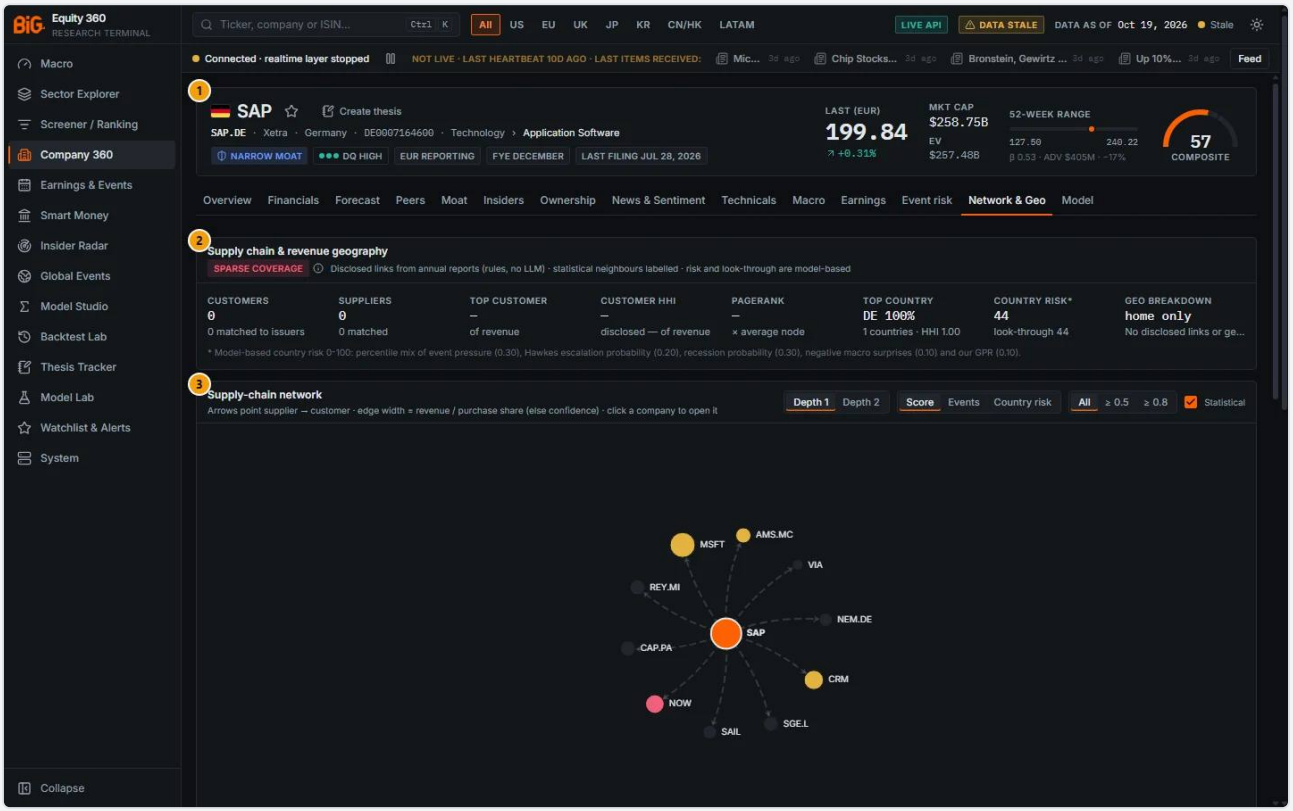


Figure 5C.16: Network & Geo tab for SAP (`/company/SAP.DE?tab=network`). The callouts are the same as in Figure 5C.10.

Second example: SAP (SAP.DE). Coverage is **Sparse**: "No disclosed links or geographic revenue: home country and statistical neighbours only". The tiles show 0 customers, 0 suppliers, Germany 100% (geo breakdown "home only") and country risk 44. The graph contains only the ten statistical neighbours, led by Nemetschek (ρ 0.68), Sage Group (0.63) and Microsoft (0.63). SAP's worldwide revenue base is not reflected here, so its event exposure and country risk are those of Germany alone.

Limitations and pitfalls

- **Coverage is uneven.** Disclosure-based links are mostly US 10-K / 20-F filers; European geography needs tagged ESEF facts or a parsable segment table (FY2022+). Many large non-US companies (SAP, Petrobras) fall back to home country 100%.
- **Disclosed shares are rare.** Only "10% customer" style disclosures give shares; list mentions ("our customers include …") have no share and low confidence. Second-order and look-through exposure only move when shares exist.
- **HHI is a lower bound** and the "Top customer" tile only covers disclosed customers.
- **Statistical neighbours are not suppliers or customers.** A correlation of 0.6 can come from a common factor (Apple and Lazard, for instance).
- **Name matching errors.** Unmatched counterparties (diamonds) cannot be opened; a fuzzy match can be wrong. Read the snippet.
- **Shock results are first-order revenue arithmetic,** not price forecasts. Default shares (0.05) for links without a share can dominate small results.
- **Staleness.** Links more than two years old are dashed ("stale") and excluded from weights; the geography date is the latest annual report known.

Related mnemonics and BQL fields

- `NET` with a security opens this tab; without a security it opens contagion & spillovers in Global Events.
- No BQL field exposes the supply-chain metrics in this snapshot.
- Related sections: watchlist exposure map in 13. Watchlists and alerts; escalation, hotspots and GPR in 8. Global Events; the W4 macro scenarios in 2B. Markets and cross-asset; the event exposure that uses the same weights in Event risk.

Model

Route: `/company/:id?tab=model` · **Menu:** Company 360 › Model · **Mnemonics:** none for this tab (`MDL` opens Model Lab) · **Refresh:** predictions nightly (`ml_predict`); retraining weekly (`ml_train_xgb` , `ml_ensemble`)

Purpose

The Model tab shows where the cross-sectional machine-learning ranker places the stock: its predicted percentile of 3-month (63-day) excess return, the expected excess return of that percentile, a heuristic confidence, and the features that pushed the score up or down (SHAP). Use it as **one input** among the scores, valuation and technicals. The production model's out-of-sample rank-IC is about **0.035**: a real but small edge on average across hundreds of names, and not proven alpha for any single stock.

Screenshot

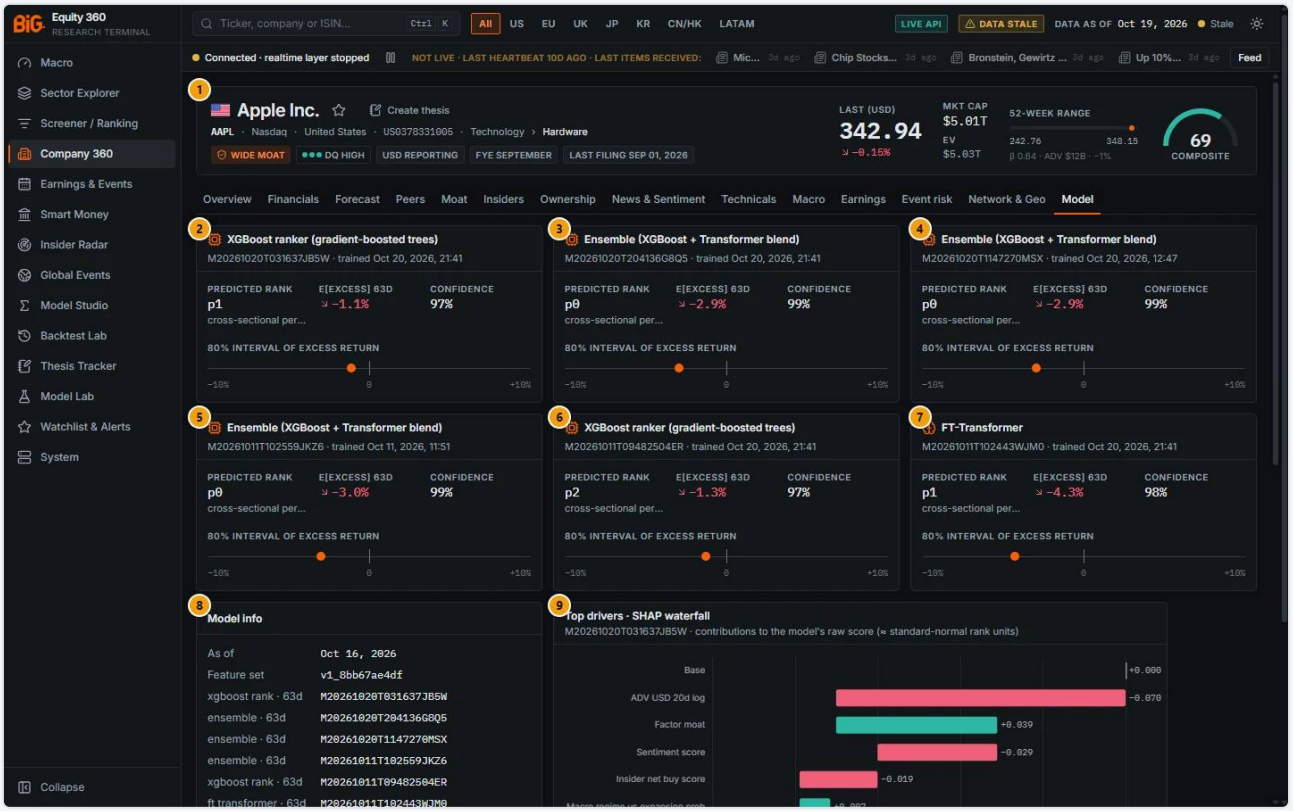


Figure 5C.17: Model tab for Apple (`/company/AAPL?tab=model`), predictions as of 16 Oct 2026.

#	Panel	What it shows
1	Company header	See 5A
2	XGBoost ranker (gradient-boosted trees), run M20261020T031637JB5W	The **production** model: predicted rank, E[excess] 63d, confidence, 80% interval bar
3	Ensemble (XGBoost + Transformer blend), run M20261020T204136G8Q5	Comparison model, not production (see Validation status)
4	Ensemble, run M20261020T1147270MSX	Earlier ensemble run, comparison only
5	Ensemble, run M20261011T102559JKZ6	Earlier ensemble run, comparison only
6	XGBoost ranker, run M20261011T09482504ER	Previous XGBoost run, comparison only
7	FT-Transformer, run M20261011T102443WJM0	Demoted Transformer, comparison only
8	Model info	Prediction date, feature set, run id per model, target, validation scheme, agreement badges
9	Top drivers · SHAP waterfall	Top ten feature contributions of the production model, in model-score units
10	Predicted rank history	Weekly predicted percentile of the production model (below the fold, Figure 5C.18)

Figure 5C.18: Predicted rank history of the production model for Apple (weekly percentile).

Controls

Control	What it does
Hover on a SHAP bar	Feature, SHAP contribution, feature value and feature family; on the last bar, the sum of the top contributions
Hover on the rank-history chart	Date and predicted percentile
Header ADR banner, "See home listing"	On an ADR page (for example TM), links to the home listing (7203.T), where scores, technicals and the model live

The tab has no inputs of its own. Model Lab (10. Model Lab) has the run-level metrics, folds and feature importance.

Methodology

TARGET AND FEATURES

- **Label:** the forward return from the close of the first trading day after the prediction date to h trading days later on the stock's own calendar (21 / 63 / 126 days for 1M / 3M / 6M), **minus the equal-weighted mean forward return of its sector and region** in the training universe. The model learns the cross-sectional rank of that excess return on each date.
- **Features:** point-in-time factor scores (quality, value, moat, profitability, ...), price, volatility and liquidity features, insider and sentiment scores, and macro regime probabilities, on a weekly and month-end grid (feature set `v1_8bb67ae4df` in Figure 5C.17).
- **Universe:** trained on the `mvp_us` and `mvp_global` universes.

The Model info panel describes the target as "63-day excess return vs regional benchmark, cross-sectionally ranked"; the training code defines it against the sector-and-region mean, as above.

MODELS

- **XGBoost ranker** (production for 1M, 3M and 6M): gradient-boosted trees on the rank label, with TreeSHAP explanations.

- **FT-Transformer:** a feature-tokenised Transformer. Demoted (see Validation status).
- **Ensemble:** the average of the XGBoost and Transformer percentiles.

Validation scheme: expanding walk-forward folds (24-month test blocks) with a **purge** equal to the label horizon plus a 10-day **embargo** between training and test, and point-in-time features. A label-permutation leakage test is required before any non-XGBoost model may become production.

WHAT EACH CARD SHOWS

The tab shows the **3M (63-day)** horizon: the production model first, then the other runs that have predictions for the stock.

Item	Definition
Predicted rank	Cross-sectional percentile of the model score on the prediction date (p0 = lowest predicted excess return)
E[excess] 63d	Mean out-of-sample excess return of the prediction's decile for that run
Confidence	`clip(t_rankIC / 3, 0, 1) × abs(2 · rank − 1)` : the run's rank-IC t-statistic (capped) times how far the rank is from the median. A heuristic, **not a probability**
80% interval of excess return	10th to 90th percentile of the decile's out-of-sample returns, drawn on a −10% to +10% axis; the dot is E[excess]
Agreement badge	"high" when the first two cards are within 15 percentile points; the gap badge shows the difference

SHAP WATERFALL

Contributions come from TreeSHAP on the production XGBoost run and are in **model-score units** ("contributions to the model's raw score (≈ standard-normal rank units)"): positive values push the stock up the ranking. The chart starts from a base of 0 and adds the ten largest contributions; the last bar ("Score") is **the sum of those ten**, not the model's full score.

RANK HISTORY

The chart plots the weekly predicted percentile of the **production** model only, under its family name (XGBoost here). The Transformer line stays empty when the production model is XGBoost.

Data sources and refresh

- **Features:** job `ml_features` (nightly) from the serving snapshot (scores, prices, insider, sentiment, macro regimes).
- **Predictions:** job `ml_predict` (nightly) re-scores the latest feature date with the latest completed run of each model type, so several runs appear side by side.
- **Training:** `ml_labels` , `ml_train_xgb` and `ml_ensemble` weekly; `ml_train_transformer` is manual. The cards show each run's training time.
- Under the default weekly profile, these run in the Saturday refresh. The prediction date (16 Oct 2026 for AAPL) can trail the data date.

Validation status

> **Model-based:** out-of-sample rank-IC about 0.035 (XGBoost). Treat the ML score as one input, not proven alpha.

- **Production XGBoost runs of 20 Oct 2026** (Model Lab, walk-forward out-of-sample): 3M rank-IC mean 0.0349 (t 12.2, hit rate 68%), 1M 0.0352 (t 11.7); 6M 0.021. The 3M long-short decile spread is 12.0% annualised (Sharpe 1.18), before trading costs. These are averages across the universe; an IC of 0.035 explains very little of any one stock's return.
- **Transformer: failed.** "The 3M FT-Transformer FAILS the label-permutation test (IC +0.06–0.09 on shuffled labels …). Demoted. XGBoost (passes) is production for 1M/3M/6M" (`docs/PHASE3_TODO.md`). No Transformer or ensemble run is recorded as passing the test, so none can be promoted.
- **Ensemble and Transformer metrics are not credible.** Their Model Lab rank-ICs (0.10-0.14, hit rates above 90%) are consistent with the leakage found in the Transformer, which the ensemble averages in. Their cards are shown for comparison only.

- **Survivorship bias.** Before 2026-10-10 the training labels and back-tests use **today's company list** applied retroactively, and names that delisted before ingestion are missing. This flatters the historical metrics.

Correct usage and worked example

Read the **first card** (production XGBoost) and ignore the confidence of the others. Use the predicted rank together with the composite score and valuation: a low rank is a reason to look for what the model sees (in the SHAP drivers), not a sell signal. Check the rank history: a percentile that jumps week to week carries little information.

Worked example: Apple (AAPL), predictions as of 16 Oct 2026 (Figure 5C.17).

1. **Production XGBoost (3M):** predicted rank **p1** (1.4th percentile), E[excess] 63d **−1.1%**, confidence 97%. The 97% only says that the rank is extreme and the model's IC is statistically significant; it is not a 97% chance of underperforming.
2. **Interval:** the bar shows only the dot. In this snapshot the decile's interval has no width, so no interval is available; this is not certainty.
3. **SHAP drivers:** ADV USD 20d log −0.070 (Apple's very high liquidity pushes it down the ranking), Factor moat +0.039, Sentiment score −0.029, Insider net buy score −0.019, then smaller macro-regime, profitability, volatility and momentum terms. The ten contributions sum to −0.083.
4. **Rank history:** the 3M percentile was p63 on 2 Oct, p0.4 on 9 Oct and p1 on 16 Oct 2026. Moves this large in two weeks are a reason for caution.
5. The other cards (ensembles at p0, FT-Transformer at p1, E[excess] −2.9% to −4.3%, "confidence" 98-99%) agree in direction but rest on models that failed or never passed the leakage test.

For an ADR such as Toyota's TM, the header banner reads "Composite, technicals and fair value are computed on the home listing by design" with a link to 7203.T; use the home listing's Model tab, where the model has predictions (Toyota 7203.T: production rank p25, E[excess] −0.8% on 16 Oct 2026).

Limitations and pitfalls

- **Small edge.** A rank-IC of 0.035 is useful in a diversified portfolio of many names, not for a single-stock call.
- **Survivorship.** Back-tests before 2026-10-10 are survivorship-biased and look better than reality.
- **Only the 3M horizon is shown.** 1M and 6M predictions exist (BQL fields below, Model Lab) but are not on this tab.
- **Confidence is a heuristic.** It rises with the extremity of the rank; it is not calibrated.
- **E[excess] is a decile average** from the run's own out-of-sample history, not a stock-specific forecast.
- **SHAP is in score units**, not in percent of return, and the waterfall's last bar is the sum of the top ten contributions only.
- **Feature values are not shown** ("—" in the tooltip) in this snapshot.
- **ADRs.** ADR pages do not carry their own model output; follow the header link to the home listing.
- **Coverage.** Securities outside the training universes may have no prediction.

Related mnemonics and BQL fields

- `MDL` opens Model Lab (model registry and monitoring).
- BQL (production model, snapshot only, model-based): `ml.rank_pct_1m`, `ml.rank_pct_3m`, `ml.rank_pct_6m`, `ml.decile_*` (10 = best), `ml.score_*` and `ml.exp_excess_*` (OOS decile mean).
- Alerts: the `model_decile` rule type in 13. Watchlists and alerts.
- Related sections: 10. Model Lab, 9. Model Studio, 11. Backtest Lab, the composite score in 5A. Overview, and the model gate summary in 16. Methodology.

6. Earnings and events calendar

This chapter covers the two pages under **Earnings & Events** in the menu: the calendar of earnings releases and corporate events, and the earnings-season dashboard. Both pages use the same data. Release dates and times come from official filings where they exist, and every date shows its source. Revenue and EPS expectations are **model-based, not analyst consensus**: the app has no consensus feed.

For one company's earnings history (past reactions, expectation versus actual, the next date and the post-earnings drift signal), use the Company 360 Earnings tab. Every company link on these two pages opens that tab.

Earnings & Events calendar

Route: `/earnings` · **Menu:** Earnings & Events › Calendar · **Mnemonics:** `EVTS` (with no security loaded; with a security loaded, `EVTS` opens its Event risk tab) · **Refresh:** nightly jobs `earnings_dates_fetch`, `earnings_events`, `earnings_reactions`, `earnings_expectations`, `earnings_signals`, run under the active schedule profile (weekly by default, daily as an option)

Purpose

The calendar shows what is reported or happening, when, and where each date comes from. It lists earnings releases (upcoming and already released), dividends, splits and other corporate events (SEC 8-K items, AGMs, buybacks, guidance revisions) in a day, week or month grid. Analysts use it to plan the reporting week, to check whether a holding reports before a decision date, and to see at a glance which releases surprised our model and how the price reacted.

Screenshot

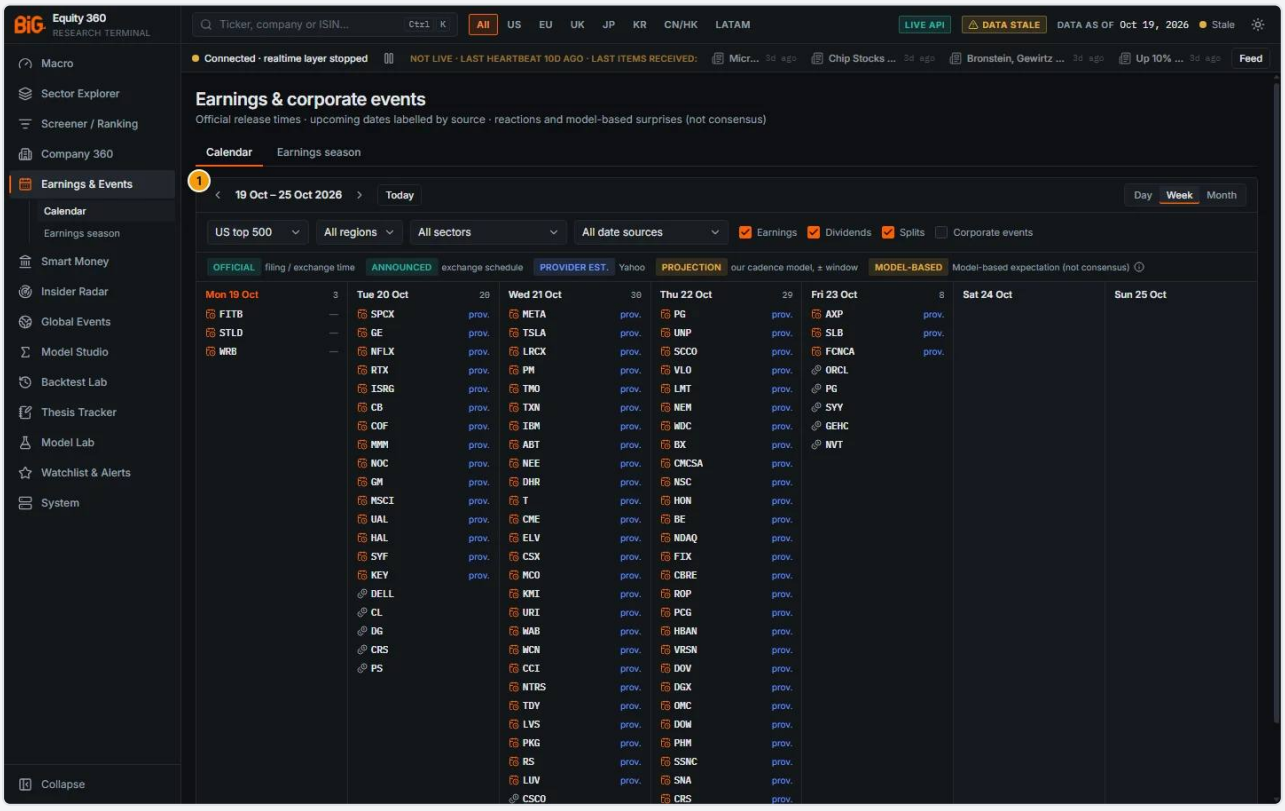


Figure 6.1: Earnings & corporate events calendar (`/earnings`), week view of 19–25 Oct 2026, US top 500, data as of 19 Oct 2026.

#	Panel	What it shows
1	Calendar (19 Oct – 25 Oct 2026)	The whole calendar panel: date navigation and view switch, the filter bar, the date-source legend and the grid of events (here the week view).

What the figure shows:

- **Mon 19 Oct** is highlighted because it is the data as-of date. FITB, STLD and WRB had released, and they show a dash because no model surprise exists yet.
- From Tue 20 Oct onwards, US releases carry a blue **prov.** tag: the date is a Yahoo provider estimate.
- The items with a coin icon (DELL, CL, DG and others) are dividend events.
- Saturday and Sunday are shaded.

Day view. The day view replaces the grid with a table of every event that day.

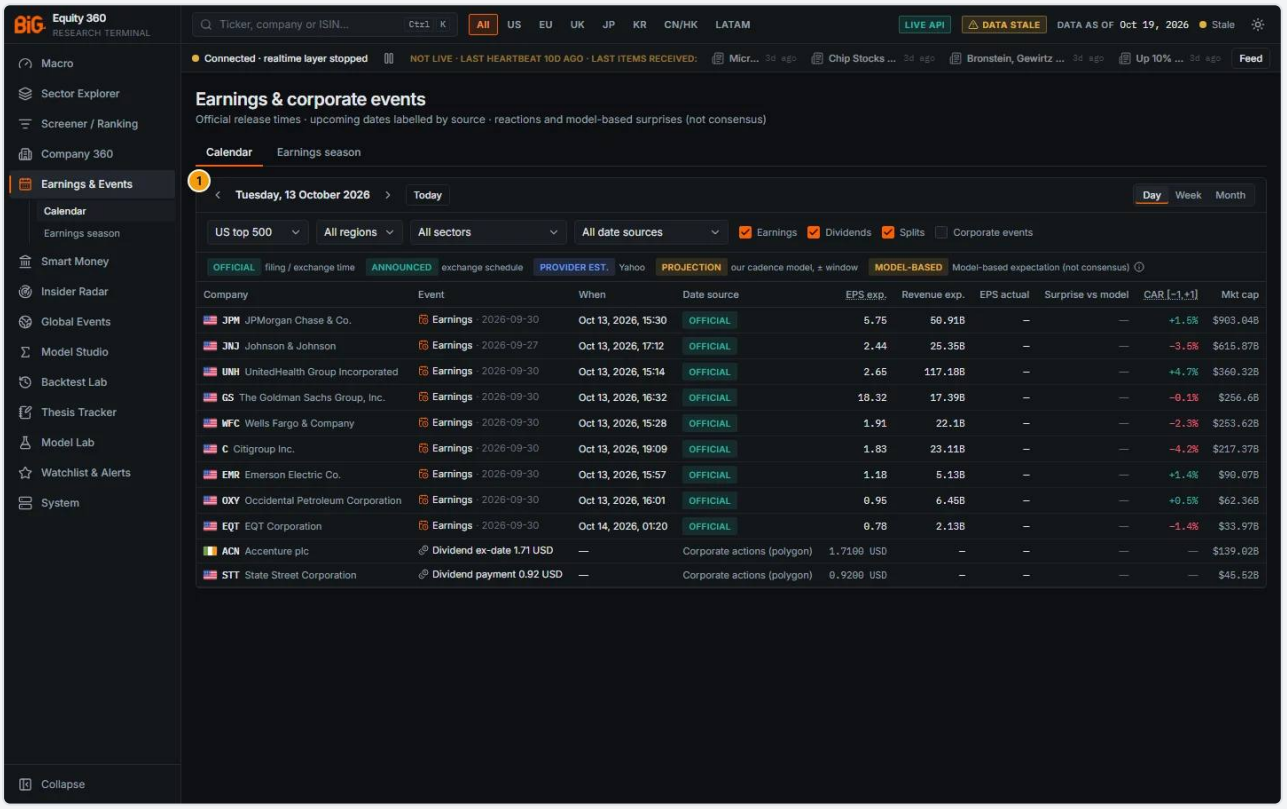


Figure 6.2: Day view (`/earnings?view=day&date=2026-10-13`), Tuesday 13 Oct 2026, US top 500, data as of 19 Oct 2026.

#	Panel	What it shows
1	Calendar (Tuesday, 13 October 2026)	The day table: nine bank and industrial releases with **OFFICIAL** badges (SEC 8-K Item 2.02 acceptance times), their model-based EPS and revenue expectations and CAR [−1,+1], followed by two dividend rows (Accenture ex-date 1.71 USD, State Street payment 0.92 USD). EPS actual and Surprise vs model are still empty: the 10-Q actuals had not arrived.

In the table, the dividend rows show the dividend amount in the EPS exp. column and the source "Corporate actions (polygon)" as their date source. Timestamps in the **When** column are shown in your computer's time zone. In the figure, JPM's 8-K acceptance at 14:30 UTC appears as 15:30.

Month view. The month view shows one cell per day with event counts.

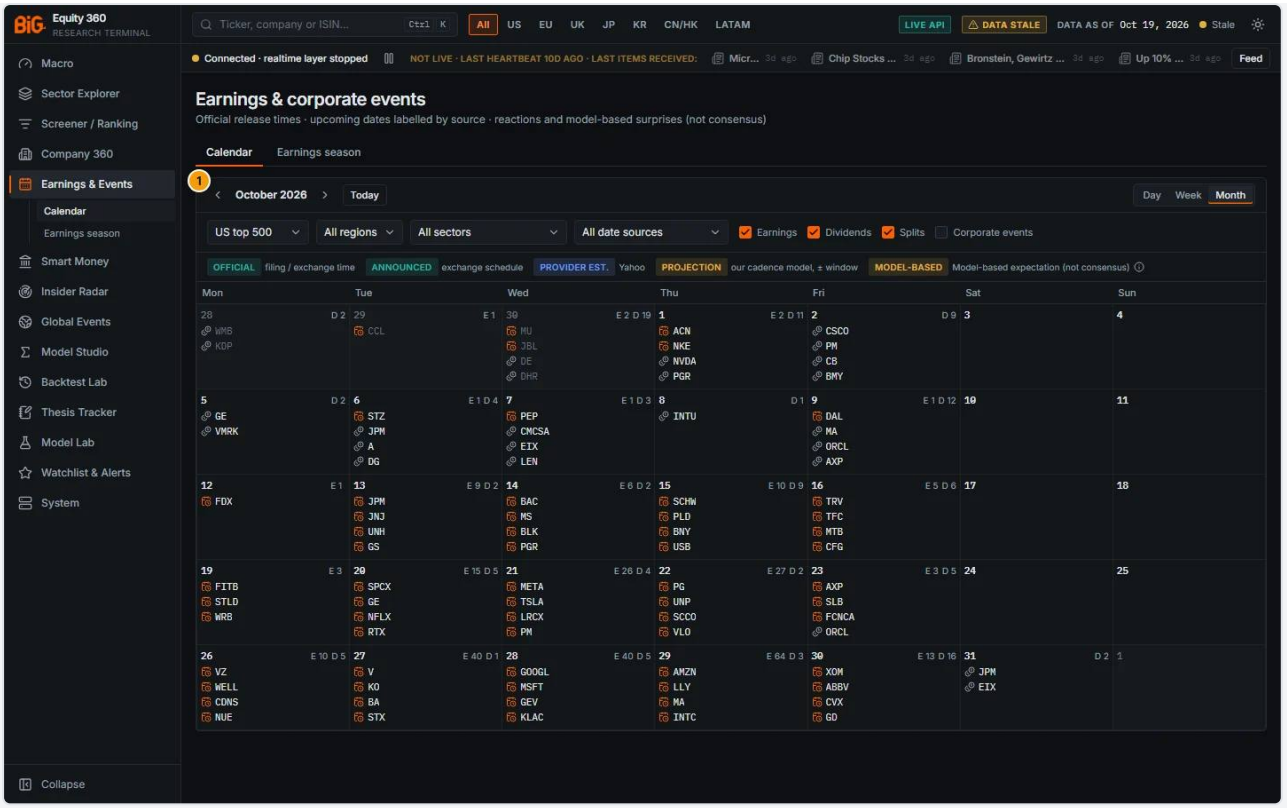


Figure 6.3: Month view (`/earnings?view=month&date=2026-10-01`), October 2026, US top 500.

#	Panel	What it shows
1	Calendar (October 2026)	Month grid, Monday to Sunday. Each cell shows the day number, the counts **E** (earnings) and **D** (dividends) and the first four tickers. Days outside the month (28–30 Sep, 1 Nov) are dimmed. The heaviest days are Wed 28 Oct (E 40, D 5) and Thu 29 Oct (E 64, D 3).

Corporate events. Ticking the Corporate events checkbox adds SEC 8-K items and other corporate events to the grid.

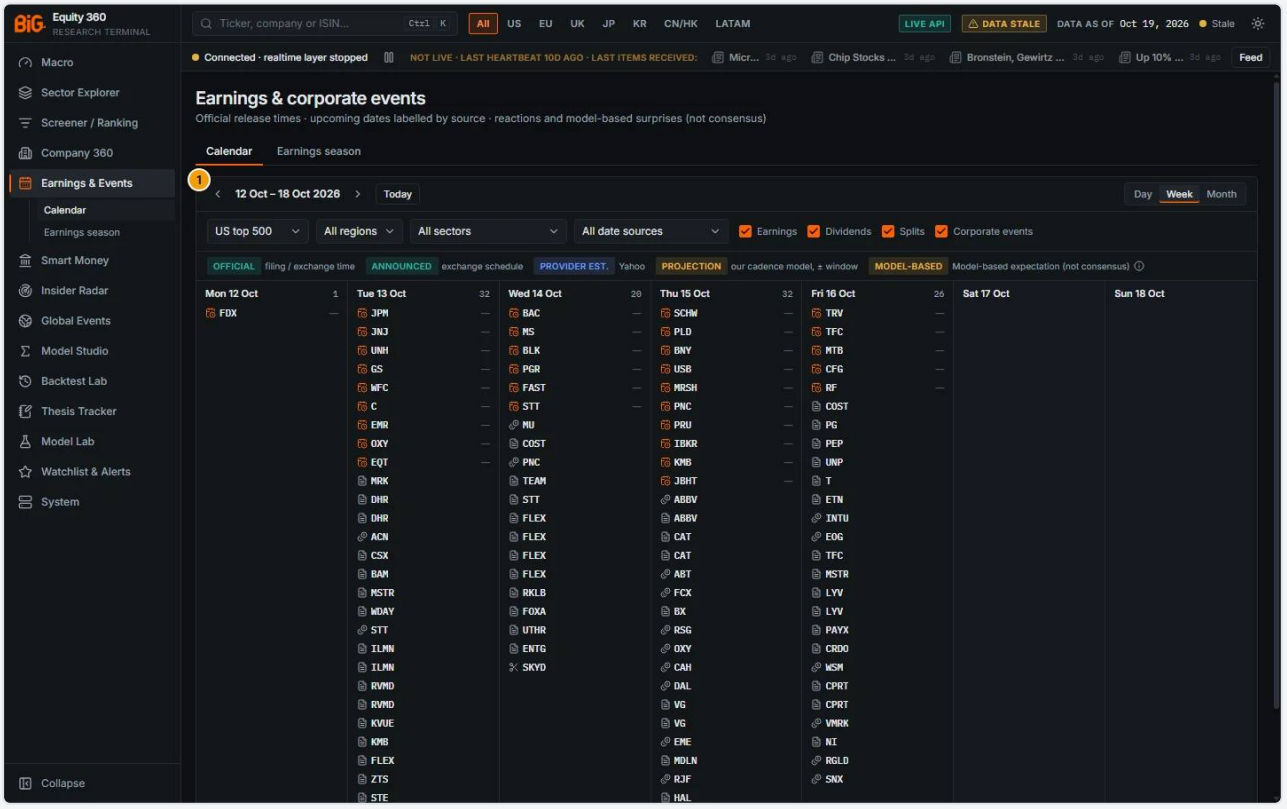


Figure 6.4: Week of 12–18 Oct 2026 with all four event types ticked (`/earnings?types=earnings,dividend,split,corporate&date=2026-10-12`), US top 500.

#	Panel	What it shows
1	Calendar (12 Oct – 18 Oct 2026)	Earnings (calendar icon, listed first) followed by corporate events with a document icon. Examples are 8-K items such as MRK and DHR (5.02 executive change, 8.01 other material event). Dividends have a coin icon; the SKYD 2:1 split on Wed 14 Oct has a scissors icon. Day counts rise to 32 on Tue 13 and Thu 15 Oct. A company can appear several times when it filed several 8-K items.

Controls

Every control writes its value into the page URL (`view` , `date` , `universe` , `region` , `sector` , `src` , `types`), so a filtered view can be bookmarked or sent to a colleague.

Control	What it does
Section tabs **Calendar** / **Earnings season**	Switch between this page and the Earnings season dashboard.
‹ / **›** (Previous / Next)	Move back or forward by one day (skipping weekends), one week or one month, depending on the view.
Today	Returns to the current date (the browser's date, not the data as-of date).
Day / Week / Month	Calendar view. Week is the default.
Universe select	US top 500 (default), Global MVP, Europe broad, Full universe, or **My watchlist** (the companies starred on Company 360). With an empty watchlist the page shows "Your watchlist is empty".
Region select	All regions, US, EU, UK, JP, KR, CN, HK, LATAM, APAC, CA.
Sector select	All sectors or one of the 11 sectors (Consumer Discretionary ... Utilities).
Date source select	All date sources; Official (released); Company-announced; Provider (Yahoo); Model-based projection. Applies to earnings items only.
Earnings / Dividends / Splits / Corporate events checkboxes	Event types shown. The first three are ticked by default. Corporate events (8-K items, AGMs, buybacks and others) is off by default. Unticking all four leaves Earnings on.
Legend row	Explains the date-source badges: **OFFICIAL** filing / exchange time, **ANNOUNCED** exchange schedule, **PROVIDER EST.** Yahoo, **PROJECTION** our cadence model, ± window, and the **MODEL-BASED** tag (hover: how the expectations are built).
Day header (week view)	Shows the number of events that day. Click it to open the day view. Each day lists up to 30 events (earnings first, then by market cap); **+N more** opens the day view.
Event chip (week view)	Hover for the company name, event, session, date source with its window, the EPS model expectation with its 80% interval, the surprise versus the model, and the dividend amount. Click the ticker to open Company 360 › Earnings.
Chip tags	**prov.** (blue) = provider-estimated date; **proj.** (amber) = model-based projected date; a signed % on released items = EPS surprise versus the model; a dash = released, no surprise yet.
Month cell	Day number, counts **E n** (earnings) and **D n** (dividends), the first four tickers. Click a cell to open the day view.
Day table	Columns Company, Event, When, Date source, EPS exp., Revenue exp., EPS actual, Surprise vs model, CAR [−1,+1], Mkt cap. Hover **EPS exp.** for "Model-based expectation (not consensus), 80% interval" and a value for its interval. Hover **CAR [−1,+1]** for "Cumulative abnormal return over days -1..+1 (market model vs country index)".
Date-source badge (day table)	Hover for the full label and, for projections, the plausible window (for example "Plausible window 2026-10-09 → 2026-10-29"). An amber **awaiting** badge means the projected or provider date has passed but the release is not in the data yet.
Truncation banner	"Showing the first N of M events — narrow the filters." appears when the API caps the list (3,000 items per request).

Date-source badges. Earnings dates carry one of six badges:

Badge	Meaning (tooltip text)	Used for
Official	Official earnings release (filing / exchange disclosure time)	Released: SEC 8-K Item 2.02 acceptance time, TDnet kessan tanshin, HKEX results announcement
Filing	Official periodic report filing (may lag the first public release)	Released, when only the 10-Q / 10-K / 20-F or national report is known
Provider	Provider date (Yahoo)	Released, dated by Yahoo only
Announced	Company-announced (exchange schedule)	Upcoming, from the JPX schedule
Provider est.	Provider estimate (Yahoo)	Upcoming, Yahoo date
Projection	Model-based projection (reporting cadence)	Upcoming, our own estimate with a ± window

The **When** column shows the exact timestamp when one exists (official release time), otherwise the session: Before open, During session, After close or Time n/a.

Corporate events. The calendar groups non-earnings items as follows:

Group (checkbox)	Event types	Source
Dividends	Dividend ex-date, dividend payment, special dividend, stock dividend (amount and currency shown)	Corporate actions feed (labelled, e.g. "Corporate actions (polygon)")
Splits	Split (ratio, e.g. "Split 2.0:1"), spin-off, rights issue	Corporate actions feed
Corporate events	8-K Item 1.01 material definitive agreement; 2.01 acquisition / disposition completed; 2.03 direct financial obligation; 3.02 unregistered equity sale; 5.02 executive / director change; 5.07 shareholder vote results; 8.01 other material event; DEF 14A proxy / AGM notice	SEC EDGAR, labelled "SEC 8-K Item x.xx (acceptance time)"
Corporate events	Share buyback, guidance revision	TDnet disclosures (Japan)
Corporate events	AGM results, share buyback, board meeting, general meeting	HKEXnews announcements (Hong Kong)

Methodology

Release dates and periods. The job `earnings_events` builds one row per issuer and fiscal period:

- For each period, the release is the earliest official release (SEC 8-K Item 2.02 acceptance time, TDnet tanshin, HKEX results). If there is none, it is the provider date. If there is neither, it is the periodic report filing.
- A dated release is assigned to the latest known fiscal period end P with $4 \le D - P \le 100$ days (130 days outside the US). Otherwise P is extrapolated from the issuer's reporting cadence and flagged as inferred.
- The session is classified against the listing's local regular trading hours. It can be pre-market, intraday or post-market, or unknown when no time exists.

Upcoming dates: precedence. For the next periods, the date with the highest precedence wins:

1. A company-announced date (JPX schedule) is labelled **Announced**.
2. Otherwise a Yahoo date mapped to the same period is labelled **Provider est.**
3. Otherwise our **model-based projection** from the reporting cadence:

```
expected date = same-period release a year earlier + 364 days   (keeps the weekday)
                or period end + median lag of the last 4 releases (if no prior-year release,
                or if the year-earlier rule is > 30 days off the median lag)
window        = ± max(3, 1.5 × MAD of the last 8 lags) days, capped at 21 days
```

Model-based expectations (job `earnings_expectations`). These are point-in-time forecasts of quarterly (or half-yearly) diluted EPS and revenue, built only from the company's own first-reported history:

- `sarima`: Foster (1977) seasonal model SARIMA(1,0,0)×(0,1,0)$_s$ with drift, estimated by conditional least squares on seasonal differences. The 80% and 95% intervals come from the forecast variance. Revenue is modelled in logs when all values are positive.
- `seasonal_naive`: the same period a year earlier plus the mean seasonal change (EPS), or times the trailing seasonal growth (revenue). σ comes from the issuer's own past errors.
- `forecast_path` (upcoming periods only): the analytics forecast for the fiscal year multiplied by the period's average share of the year over the last 3 years.
- **Primary model**: the one with the lower mean absolute % error over the issuer's previous ≤ 8 released targets (at least 4 needed). Otherwise `sarima`, then `seasonal_naive`.
- **Point in time**: for a release on date *R*, the model is fitted at origin *R* − 1 day on values known by then. EPS is put on the share basis at the origin (splits applied). Q4 is derived as FY − 9M when needed and flagged.

Surprise (shown when the structured actual arrives):

```
surprise      = actual − expectation
surprise %    = (actual − expectation) / |expectation|
surprise z    = (actual − expectation) / σ
```

Price reaction (job `earnings_reactions`). The method is a market-model event study, the same one Model Studio uses:

- $r_i = a + b \cdot r_m + e$, estimated by OLS on trading days [−250, −30] around day 0 (at least 60 observations). r_m is the issuer's country index (ACWI as fallback).
- Day 0 is the first trading day on or after the reaction date (the next session for after-close releases).
- `CAR [−1,+1]` is the sum of abnormal returns over days −1 to +1. The Company 360 tab also shows [0,+5], [+1,+20] and BHAR.

Data sources and refresh

Data	Source	Notes
US release times	SEC EDGAR submissions: 8-K Item 2.02 acceptance timestamps; 10-Q / 10-K / 20-F / 40-F filings	Nightly re-request of companies that filed in the last 4 days
Japan	TDnet kessan tanshin (release time); JPX announcement schedule (company-announced upcoming dates)	
Hong Kong	HKEX results announcements (release time); HKEXnews announcements (corporate events)	
Other regions	Periodic report filings (EDINET, DART, CVM, ESEF via filings.xbrl.org, FCA NSM, CNINFO / exchange)	Report filing dates may lag the first public release
Upcoming provider dates	Yahoo earnings dates (dates only, no estimates)	Fetched for US top 500, Global MVP and Europe broad; re-fetched after 7 days or near a known date
Dividends, splits	Corporate actions feed, including declared future ex-dates	
Expectations, surprises	Our model on fundamentals (XBRL for the US, national filings elsewhere)	Model-based, not consensus
Reactions	Daily prices (adjusted close) and country indices	

All five earnings jobs run in the nightly DAG. Under the default **weekly** profile the nightly DAG runs on Saturday at 08:00 (Lisbon), followed by the weekly DAG. Under the **daily** profile it runs Tuesday to Saturday at 01:30 (Lisbon).

Validation status

Not formally validated. No accuracy gate exists for the date projections, the expectation models or the reaction statistics.

- The app labels upcoming dates by source. The expectations and surprises carry the badge **MODEL-BASED** with the text "Model-based expectation (not consensus)".
- The tag's tooltip reads: "Expectations come from time-series models on the company's own reported history (Foster seasonal SARIMA, seasonal naive, analytics forecast path). They are not analyst consensus."
- The primary model is chosen per issuer by its own trailing error, as described in Methodology. This is a model-selection rule, not a validation of the forecast against consensus or against outcomes.

Correct usage and worked example

Use the calendar to answer three questions:

- When does a name report?
- How reliable is that date?
- Did the last release beat *our model*, and how did the price react?

Treat **Projection** dates as a window, not a date. Treat a model surprise as "different from the company's own seasonal pattern", not as "beat the Street".

Worked example: JPMorgan, week of 12–18 Oct 2026 (US top 500), data as of 19 Oct 2026.

1. Click ‹ once from the 19–25 Oct week, then click the **Tue 13 Oct** header to open the day view (Figure 6.2).
2. JPM's row shows **Official**: the source is "SEC 8-K Item 2.02 (acceptance time)", timestamp 13 Oct 2026 14:30:38 UTC (shown as 15:30 in a Lisbon browser), session *During session*.
3. **EPS exp.** shows 5.75. Hovering gives the 80% interval 4.45 … 7.04. This is our seasonal model's expectation for the quarter ending 30 Sep 2026, not consensus.
4. **EPS actual** and **Surprise vs model** show a dash. The 8-K press release is in, but the structured EPS actual arrives with the 10-Q XBRL, which can be days later. The surprise appears then.
5. **CAR [−1,+1]** is +1.5%: over the three sessions around the release, JPM did 1.5 percentage points better than its market model against the US index predicted.
6. Click JPM to open Company 360 › Earnings and compare this reaction with JPM's usual move (the "move vs normal" ratio).

Projection example. In Global MVP, Contemporary Amperex (300750.SZ) shows **Projection** on 19 Oct 2026. Hovering the badge shows "Plausible window 2026-10-09 → 2026-10-29": the cadence model expects the Q3 release somewhere in that 21-day window. Disco (6146.T) on 22 Oct 2026 shows **Announced** because JPX published the company's schedule.

Limitations and pitfalls

- **No consensus.** "Surprise vs model" is not comparable with the beat / miss figures in the press. A company can beat consensus and miss our model, or the reverse.
- **US dates are provider estimates.** There is no company-announced source for the US, so upcoming US dates show **Provider est.** (Yahoo) even when the company has confirmed the date (audit item F42).
- **Acceptance time is not press-release time.** The 8-K acceptance time can come later than the press release. A bank that publishes before the open can appear as *During session* (JPM above: 14:30 UTC).
- **Surprises lag the release.** The EPS and revenue actuals come from structured filings (10-Q / 10-K XBRL or the national filing). Until they arrive, a released item shows a dash.
- **Overdue projections.** An **awaiting** badge (or the tooltip "Projected date passed - release not in the data yet") means our expected date has passed. Either the release was late, or our data has not captured it yet.
- **Coverage.** Yahoo dates are fetched only for US top 500, Global MVP and Europe broad. Other names rely on projections. Outside the US, Japan and Hong Kong, past releases are often dated by the periodic report filing

(badge **Filing**), which can lag the first announcement.

- **Buybacks.** Share buybacks appear as their own event type only for Japan (TDnet) and Hong Kong (HKEX). In the US they appear only inside 8-K Item 8.01 "Other material event", or not at all.
- **Data quality.** At the 19 Oct 2026 audit the Yahoo date fetch had timed out and some provider rows carried a sentinel date (audit items F14 and F26). Check the freshness chip before relying on far-future dates.
- **Browser date. Today** uses your computer's date, which can differ from the data as-of date shown in the top bar.

Related mnemonics and BQL fields

- Mnemonics: `EVTS` (calendar; with a security loaded it opens Company 360 › Event risk); `ERN` (Company 360 › Earnings: history, reactions, model surprises); `EE` (Company 360 › Forecast, model-based estimates).
- BQL fields (latest release per security; expectations model-based, not consensus):
  - Release: `earn.release_date`, `earn.session`, `earn.date_source`.
  - Expectation and actual: `earn.eps_expected`, `earn.eps_lo80`, `earn.eps_hi80`, `earn.eps_actual`.
  - Surprises: `earn.eps_surprise_pct`, `earn.eps_surprise_z`, `earn.rev_surprise_pct`.
  - Reaction: `earn.car_m1_p1`, `earn.car_0_p5`, `earn.car_p1_p20`, `earn.move_ratio`, `earn.gap_filled`, `earn.drift_continuation`.
  - Post-earnings drift: `pead.sue`, `pead.pead_score`, `pead.decile`.
- Alerts: "earnings upcoming" (watchlist names reporting within N days, source label in the message) and "earnings surprise" (|model surprise z| above a threshold). See Watchlists and alerts.
- Related sections: Company 360 Earnings, Company 360 Event risk, Earnings season, BQL and mnemonics.

Earnings season

Route: `/earnings/season` · **Menu:** Earnings & Events › Earnings season · **Mnemonics:** none (open from the menu or the Calendar / Earnings season tabs) · **Refresh:** nightly job `earnings_signals`, after `earnings_reactions` and `earnings_expectations`

Purpose

The earnings-season dashboard summarises a reporting season in one screen:

- how many companies have reported;
- what share beat **our model-based expectation** for EPS and revenue;
- whether prices reacted positively.

Results are broken down by region and sector and compared with earlier seasons. Use it to judge whether a season is going better or worse than usual, and which sectors or regions stand out. The beats are measured against a model, not against consensus.

Screenshot

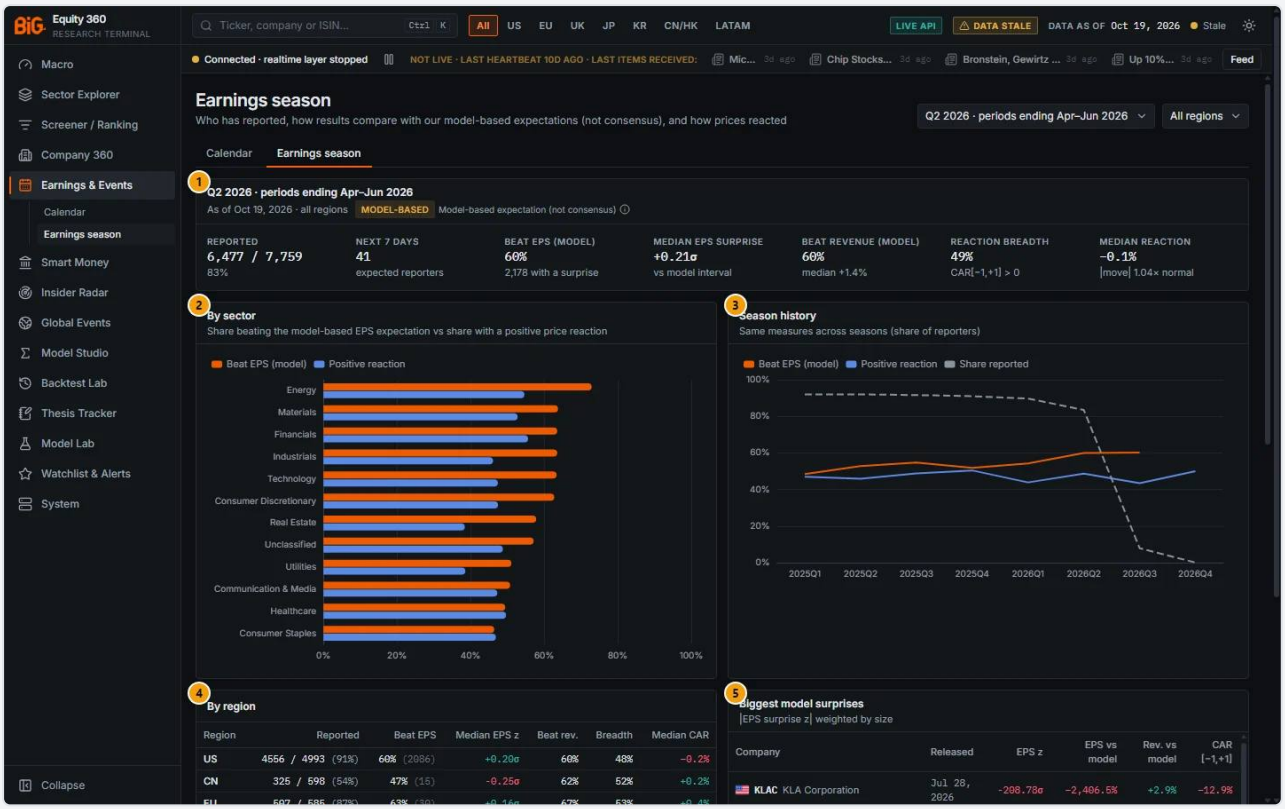


Figure 6.5: Earnings season (`/earnings/season`), season Q2 2026 (periods ending Apr–Jun 2026), all regions, data as of 19 Oct 2026.

#	Panel	What it shows
1	Season summary (Q2 2026 · periods ending Apr–Jun 2026)	Seven headline numbers: Reported, Next 7 days, Beat EPS (model), Median EPS surprise, Beat revenue (model), Reaction breadth, Median reaction. Tagged **MODEL-BASED** "Model-based expectation (not consensus)".
2	By sector	Bars per sector: share beating the model-based EPS expectation (orange) and share with a positive price reaction (blue). Sectors are sorted by beat rate.
3	Season history	Lines across the last eight seasons: Beat EPS (model), Positive reaction, Share reported (dashed).
4	By region	Table per region: Reported, Beat EPS (with the number of surprises), Median EPS z, Beat rev., Breadth, Median CAR.
5	Biggest model surprises	Up to 30 releases ranked by the absolute EPS surprise z weighted by size: Company, Released, EPS z, EPS vs model, Rev. vs model, CAR [−1,+1].
6	Reporting in the next 7 days	Companies expected to report in the next seven days, each with its date-source badge (Projection, Provider est., Announced).

Panel 6 is below the fold in the main image:

Figure 6.6: "Reporting in the next 7 days", 19–20 Oct 2026. Chinese, Mexican and smaller US names carry *Projection**; most US names carry ****Provider est.***

Controls

Control	What it does
Season select	Seasons available in the data, labelled e.g. "Q2 2026 · periods ending Apr–Jun 2026". The default is the most recent season with at least 25% reported.
Region select	All regions, US, EU, UK, JP, KR, CN, HK, LATAM, APAC. Changes the summary, the sector bars, the history lines, the biggest surprises and the next-7-days list. The **By region** table always shows every region.
Section tabs	**Calendar** / **Earnings season**.
MODEL-BASED tag	Hover: how the expectations are built (not consensus).
Chart tooltips	Hover a sector bar for Beat EPS (model), Positive reaction and Reported n / N. Hover the history chart for the three shares per season.
Company links	Open Company 360 › Earnings.
Date-source badges	Hover for the full label and, for projections, the window.

When the earnings jobs have not produced season aggregates yet, the page shows a pending state ("The earnings jobs have not produced season aggregates yet.").

Methodology

Season = the calendar quarter of the fiscal **period end**. For example, 2026Q3 is periods ending Jul–Sep 2026, whatever the company's fiscal labels. The job computes the current season, the quarter in progress and the previous seasons, per region × sector with "all" roll-ups.

Measure	Definition
Expected (N)	Issuers that reported this season, or reported the same quarter a year earlier, or have a calendar row for the season
Reported (n)	Releases of the season with release date ≤ as-of date; share reported = n / N
Next 7 days	Expected issuers with a calendar date in the next seven days
Beat EPS (model)	Share of reporters with a primary EPS surprise where actual > model expectation; "with a surprise" = how many have one
Median EPS surprise	Median of `z = (actual − expectation) / σ` (shown in σ, "vs model interval")

Measure	Definition
Beat revenue (model)	Same for revenue; sub-line = median revenue surprise %
Reaction breadth	Share of reporters with `CAR[−1,+1] > 0`
Median reaction	Median `CAR[−1,+1]` ; the "move" sub-line = median of abs(CAR[−1,+1]) / normal 3-day move
Normal move	Realised-volatility baseline = residual σ of the market model × √3 (there is no option-implied move)
Biggest surprises rank	`abs(EPS surprise z) × ln(1 + market cap / $100M)`

Data sources and refresh

The page uses the same sources as the calendar: official release times and filings, Yahoo and JPX dates, our model-based expectations, and market-model reactions. The aggregates are rebuilt nightly by `earnings_signals` , which also computes the post-earnings drift (PEAD) signal used in the Backtest Lab.

Validation status

Not formally validated. The beat rates and surprises are descriptive statistics against a **model-based** expectation, labelled in the app with the **MODEL-BASED** tag "Model-based expectation (not consensus)". No study in the project documents tests whether the season aggregates predict anything.

Correct usage and worked example

Compare a season with its own history (panel 3) and across regions. Do not compare the beat rate with consensus beat rates published elsewhere. Read beat rates together with their counts. A rate based on a handful of surprises (shown in brackets in the region table) means little.

Worked example: Q2 2026 (periods ending Apr–Jun 2026), all regions, as of 19 Oct 2026.

1. **Reported:** 6,477 of 7,759 expected issuers (83%).
2. **Beat EPS (model):** 60%, but only 2,178 reporters have an EPS surprise. The others have no structured actual or no usable model.
3. **Median EPS surprise:** +0.21σ. **Beat revenue (model):** 60%, median revenue surprise +1.4%.
4. **Reaction breadth** is 49% and the **median reaction** is −0.1%, with a typical move of 1.04× normal. Results ran slightly ahead of the model, but prices did not reward them on balance.
5. **By region:** the US accounts for 4,556 of 4,993 reported (91%), a 60% beat rate on 2,086 surprises and 48% breadth. Japan shows a 100% beat rate, but on only 3 surprises; disregard it.
6. **Season history:** the Q2 2026 beat rate (60%) is the highest of the eight seasons shown (48–60%), while breadth stays in its usual 44–50% range.

Limitations and pitfalls

- **Model, not consensus.** A 60% "beat" means 60% of reporters came in above *our* seasonal time-series expectation. It says nothing about analyst expectations.
- **Small denominators.** Outside the US the number of reporters with a surprise is small: 3 in Japan, 15 in China, 30 in the EU for Q2 2026. Rates there are noisy. Korea has none.
- **Extreme z-scores.** The ranking of biggest surprises is driven by z = surprise / σ. A very small σ, an expectation near zero, a derived Q4 (FY − 9M) or a share-basis change can produce extreme values, for example KLA at −208.78σ (−2,406.5%) in Q2 2026. Check such rows on Company 360 › Earnings before drawing conclusions.
- **The current season is incomplete.** Early in a season the share reported is low; the default season is therefore the latest one with at least 25% reported. In the figure, Q3 2026 stood at about 8% on 19 Oct 2026.

- **Season = calendar quarter of the period end.** A company with a June fiscal year-end reports its fiscal Q4 in the "Q2" season.
- **Unclassified sector.** Issuers without a sector are grouped as "Unclassified" (1,108 reporters in Q2 2026).

Related mnemonics and BQL fields

- Mnemonics: none opens this page directly. `EVTS` opens the calendar and `ERN` a company's earnings history.
- BQL fields: per-security release fields `earn.*` and the drift signal `pead.*` (see the calendar). Season aggregates are not exposed as BQL fields.
- Related sections: Earnings & Events calendar, Company 360 Earnings, Backtest Lab (the PEAD signal), Methodology appendix.

7. Smart Money and Insider Radar

This chapter covers the **Smart Money** pages (overview, EU / UK short register, activism, sectors) and the **Insider Radar**. They show who is buying, selling, shorting and crowding, using official disclosures: SEC Form 13F, FINRA short interest, SEC fails-to-deliver, SEC Schedules 13D / 13G, the EU / UK net short-position registers, CFTC Commitments of Traders, and insider filings (SEC Form 4 and national equivalents).

The underlying data are official. Several classifications built on top of them are **our own definitions and are labelled model-based** in the app: hedge-fund-like, smart cohort, crowding, squeeze risk, FTD spike and the combined smart-money score. The definitions below are quoted from `docs/SMART_MONEY.md` and the API labels.

For a single company, the same data appear on the Company 360 Ownership tab (13F holders, short-interest trend, EU shorts, activism, score history) and the Insiders tab. Company links on the Smart Money pages open the Ownership tab; rows in the Insider Radar open the Insiders tab.

> **Model-based:** the app's footnote reads "Hedge-fund-like, smart cohort, crowding, squeeze risk, FTD spike and the smart-money score are model-based (our definitions, docs/SMART_MONEY.md); the underlying data are official filings".

Smart Money overview

Route: `/smart-money` · **Menu:** Smart Money › Overview · **Mnemonics:** `SMRT`, `HDS`, `SI` (with no security loaded; with a security loaded they open Company 360 › Ownership) · **Refresh:** nightly `smart_money_signals`, `eu_short_positions`, `activist_filings`, `insider_signals`; weekly `sec_13f`, `smart_money_13f`, `finra_short_interest`; CFTC weekly

Purpose

The overview is a one-page screen of institutional and short-side activity across the coverage universe:

- the largest 13F accumulations and distributions;
- what the best-performing hedge-fund-like managers bought;
- the most crowded and the most shorted names, and possible short squeezes;
- fails-to-deliver spikes and futures positioning;
- the combined smart-money score, the latest activist 13D filings and the latest EU / UK short positions.

Use it to generate ideas and to check positioning around a name. It is a **descriptive** screen. The combined score has shown no predictive power so far (see Validation status).

Screenshot

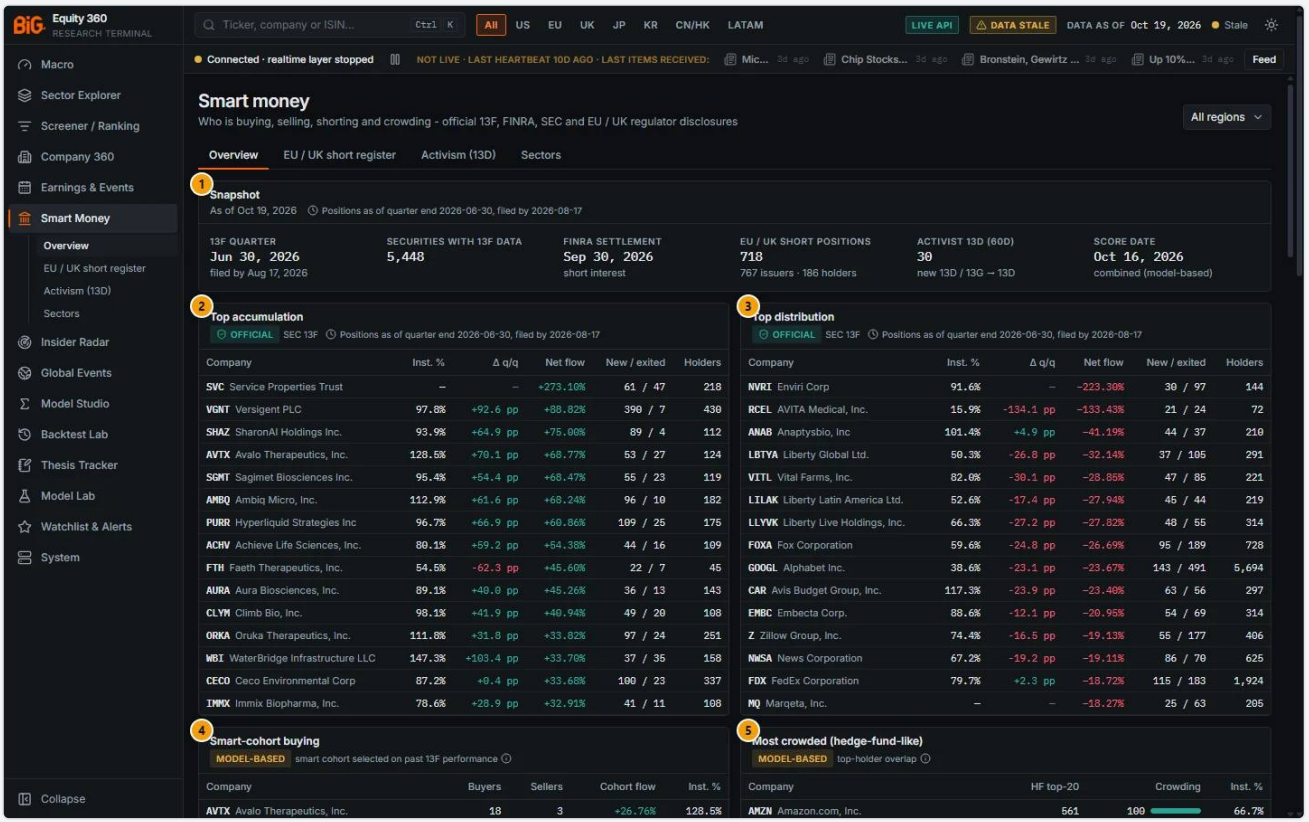


Figure 7.1: Smart money overview (`/smart-money`), all regions, data as of 19 Oct 2026. 13F positions as of quarter end 30 Jun 2026, filed by 17 Aug 2026.

#	Panel	What it shows
1	Snapshot	As-of date and the 13F lag note. Six figures: 13F quarter (filed by), securities with 13F data, FINRA settlement date, EU / UK short positions (issuers, holders), activist 13D in the last 60 days, score date.
2	Top accumulation	**OFFICIAL** SEC 13F. The 15 largest net 13F inflows as % of shares outstanding: Inst. %, Δ q/q, Net flow, New / exited holders, Holders.
3	Top distribution	Same columns, the 15 largest net 13F outflows.
4	Smart-cohort buying	**MODEL-BASED** "smart cohort selected on past 13F performance". Buyers, Sellers, Cohort flow, Inst. %.
5	Most crowded (hedge-fund-like)	**MODEL-BASED** "top-holder overlap". HF top-20 count, Crowding percentile bar, Inst. %.
6	Most shorted	**OFFICIAL** FINRA short interest; "Short interest % uses shares outstanding (float is not published)". SI % out, Days to cover, Δ SI, z 1y.
7	Squeeze watch	**MODEL-BASED** "high SI + rising price + low float". SI % out, DTC, 20d ret., Risk bar, **SQUEEZE RISK** flag.
8	Short-interest risers	**OFFICIAL** FINRA. Largest rises in short interest since the previous settlement.
9	Fails-to-deliver spikes	**MODEL-BASED** SEC FTD data. FTD % out, SI % out.
10	Futures positioning (CFTC)	**OFFICIAL** CFTC COT. Leveraged funds (TFF) in six equity-index futures: Net % OI, COT 3y index, report date; link **all markets** →.
11	Combined score - top	**MODEL-BASED** "13F flows · smart cohort · insiders · short interest · EU shorts". Score, Pctl, SI % out, EU short.
12	Combined score - bottom	**MODEL-BASED** "lowest scores". Same columns.
13	Activist feed	**OFFICIAL** SEC 13D / 13G. New 13D and 13G → 13D switches in the last 60 days: date, event badge, filer → subject, **EDGAR** link; **full feed** →.
14	EU / UK short-seller register	**OFFICIAL** FCA · AMF · CNMV · AFM · CBI · FSMA · FI · Finanstilsynet. Latest named net short positions: Holder, Issuer (regulator), Net short, Change, Date; **register** →.

Panels 6 to 14 are below the fold. The close-ups below show the ones that need explanation.

6

Most shorted

OFFICIAL FINRA short interest Short interest % uses shares outstanding (float is not published)

Company	SI % out	Days to cover	Δ SI	z 1y
WOLF Wolfspeed, Inc.	50.7%	11.4	+3%	+1.4
SOUN Soundhound Ai, Inc.	36.2%	9.7	−4%	+1.0
SHOE Shoe Station Group Inc	34.7%	7.2	+8%	–
TRIP TripAdvisor, Inc.	32.6%	8.7	+7%	+2.5
CAPR Capricor Therapeutics, Inc.	32.4%	4.5	−4%	+1.2
NTLA Intellia Therapeutics, Inc.	32.2%	9.5	−3%	+0.4
FCEL Fuelcell Energy Inc	30.6%	4.6	−2%	+2.7
GRPN Groupon, Inc.	29.6%	8.6	−8%	+0.3
UPST Upstart Holdings, Inc.	29.4%	6.2	+3%	+1.4
RUN Sunrun Inc.	29.4%	8.9	−1%	+2.3
CSIQ Canadian Solar Inc.	29.3%	9.3	+8%	+2.3
ASST Strive, Inc.	29.1%	2.6	−8%	−0.5
RH Rh	28.0%	7.1	+6%	+0.9
NVAX Novavax Inc	27.9%	8.1	−8%	−0.7
KSS KOHLS Corp	27.4%	8.0	−7%	+0.8

Figure 7.2: Most shorted (FINRA settlement 30 Sep 2026). Wolfspeed (WOLF) leads at 50.7% of shares outstanding with 11.4 days to cover.

7

Squeeze watch

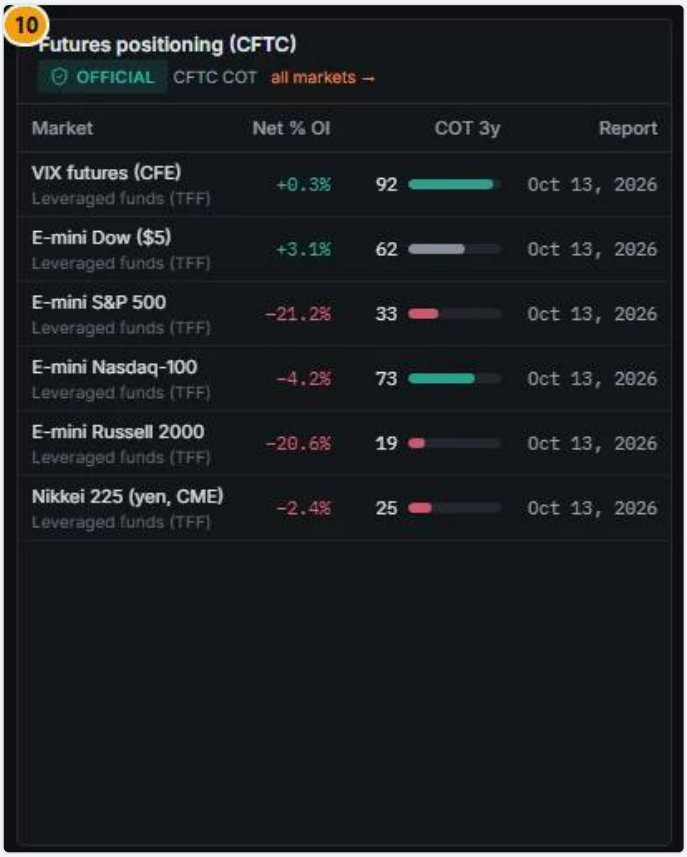
MODEL-BASED high SI + rising price + low float ⓘ

Company	SI % out	DTC	20d ret.	Risk	Flag
COHU Cohu Inc	19.0%	9.1	+49%	89	SQUEEZE RISK
MNRO Monro, Inc.	25.1%	6.7	+13%	89	SQUEEZE RISK
IBTA Ibotta, Inc.	13.8%	15.5	+5%	91	–
LGND Ligand Pharmaceuticals Inc	13.4%	8.7	+9%	91	–
CLFD Clearfield, Inc.	9.6%	9.2	+8%	90	–
GRPN Groupon, Inc.	29.6%	8.6	+5%	88	–
WMK Weis Markets Inc	11.1%	9.7	+4%	88	–
WOLF Wolfspeed, Inc.	50.7%	11.4	+6%	88	–
GCO Genesco Inc	11.7%	7.6	+2%	88	–
EBF Ennis, Inc.	8.3%	10.2	+7%	86	–
TMDX TransMedics Group, Inc.	24.7%	13.1	−1%	86	–
USPH U S Physical Therapy Inc	8.7%	6.9	+10%	86	–
FTK Flotek Industries Inc	10.6%	7.4	+25%	86	–
SIG Signet Jewelers Ltd	14.5%	6.3	+25%	86	–
CHH Choice Hotels International Inc	14.7%	11.1	+2%	86	–

Figure 7.3: Squeeze watch. Only COHU and MNRO meet all four conditions of the squeeze-risk flag; the others are ranked by the risk score alone.

Company	FTD % out	SI % out
SECZ Securitize Corp.	3,055.22%	–
WYFI WhiteFiber, Inc.	11.36%	12.3%
DFNS T3 Defense Inc.	5.63%	36.4%
ONDS Ondas Inc.	3.67%	41.6%
SHAZ SharonAI Holdings Inc.	2.35%	12.8%
PLAB Photronics Inc	2.29%	7.0%
POET Poet Technologies Inc.	1.28%	17.3%
FIGR Figure Technology Solutions, Inc.	1.28%	7.2%
FWDI Forward Industries, Inc.	1.19%	8.7%
KHC The Kraft Heinz Company	1.18%	7.8%
PURR Hyperliquid Strategies Inc	1.16%	12.3%
CSWC Capital Southwest Corp	1.02%	5.6%
TJGC TJGC GROUP Ltd	0.95%	6.8%
FET Forum Energy Technologies, Inc.	0.90%	5.4%
CCOI Cogent Communications Holdings, ...	0.84%	11.7%

Figure 7.4: Fails-to-deliver spikes. The first row (3,055.22% of shares outstanding) is a share-count error, not a real fail (see Limitations).



Market	Net % OI	COT 3y	Report
VIX futures (CFE) Leveraged funds (TFF)	+0.3%	92	Oct 13, 2026
E-mini Dow ($5) Leveraged funds (TFF)	+3.1%	62	Oct 13, 2026
E-mini S&P 500 Leveraged funds (TFF)	-21.2%	33	Oct 13, 2026
E-mini Nasdaq-100 Leveraged funds (TFF)	-4.2%	73	Oct 13, 2026
E-mini Russell 2000 Leveraged funds (TFF)	-20.6%	19	Oct 13, 2026
Nikkei 225 (yen, CME) Leveraged funds (TFF)	-2.4%	25	Oct 13, 2026

Figure 7.5: Futures positioning (CFTC), leveraged funds, report of 13 Oct 2026.

11

Combined score - top

MODEL-BASED 13F flows · smart cohort · insiders · short interest · EU shorts

Company	Score	Pctl	SI % out	EU short
CENT Central Garden & Pet Co	+1.89	95	0.6%	–
BBDC4.SA Banco Bradesco	+1.33	94	0.1%	–
BRVE Braveheart Bio, Inc.	+1.28	93	4.6%	–
MFP Midera Food Processing, Inc.	+1.27	93	4.1%	–
CLBK Columbia Financial, Inc.	+1.27	93	8.2%	–
TMS Teamshares Inc	+1.24	93	0.2%	–
ODTX Odyssey Therapeutics, Inc.	+1.23	93	4.6%	–
AVBC Avidia Bancorp, Inc.	+1.21	93	2.4%	–
KLRA Kailera Therapeutics, Inc.	+1.21	93	6.7%	–
NMM Navios Maritime Partners L.P.	+1.18	93	1.7%	–
TSN Tyson Foods, Inc.	+1.15	92	2.5%	–
FCBM First Carolina Financial Services, Inc.	+1.12	92	0.6%	–
LWAY Lifeway Foods, Inc.	+1.09	92	4.5%	–
WELL Welltower Inc.	+1.08	92	2.5%	–
DPC DPC Holdings PLC	+1.07	92	3.9%	–

Figure 7.6: Combined score - top (score date 16 Oct 2026).

13

Activist feed

OFFICIAL SEC 13D / 13G full feed →

Oct 16, 2026	NEW 13D	**ELI LILLY & Co → TRXB** TRex Bio, Inc.	EDGAR
Oct 16, 2026	NEW 13D	**Pani Corp. → AIOK** AI OKTO Corp.	EDGAR
Oct 16, 2026	13G → 13D	**Qin Zhen → LHAI** Linkhome Holdings Inc.	EDGAR
Oct 16, 2026	NEW 13D	**SV Biotech Crossover Opportunities Fund LP → TRXB** TRex Bio, Inc.	EDGAR
Oct 15, 2026	13G → 13D	**Third Point LLC → SRTA** Strata Critical Medical, Inc.	EDGAR
Oct 15, 2026	NEW 13D	**Ruen Hua Dyeing & Weaving Co., Ltd. → GGR** Gogoro Inc.	EDGAR
Oct 15, 2026	NEW 13D	**Liontrust Investment Partners LLP → MXCT** Maxcyte, Inc.	EDGAR
Oct 15, 2026	NEW 13D	**Fairmount Funds Management LLC → TCRT** Alaunos Therapeutics, Inc.	EDGAR
Oct 14, 2026	NEW 13D	**Saba Capital Management, L.P. → TCPC** BlackRock TCP Capital Corp.	EDGAR
Oct 14, 2026	NEW 13D	**Prime Ocean Holdings Ltd → TROO** Troops, Inc. /Cayman Islands/	EDGAR
Oct 14, 2026	NEW 13D	**Mariselle a.s. → MDA** MDA Space Ltd.	EDGAR
Oct 14, 2026	NEW 13D	**Keller Eric L. → RTSN** Retension Pharmaceuticals, Inc.	EDGAR
Oct 14, 2026	NEW 13D	**Powell Anderson Capital Partners LLC → FSP** Franklin Street Properties...	EDGAR
Oct 14, 2026	NEW 13D	**Seto Wai Yue → TDIC** Dreamland Ltd	EDGAR
Oct 13, 2026	NEW 13D	**Sun Valley Investments AG →** Canagold Resources Ltd.	EDGAR

Figure 7.7: Activist feed, new 13D and 13G → 13D switches in the 60 days to 19 Oct 2026.

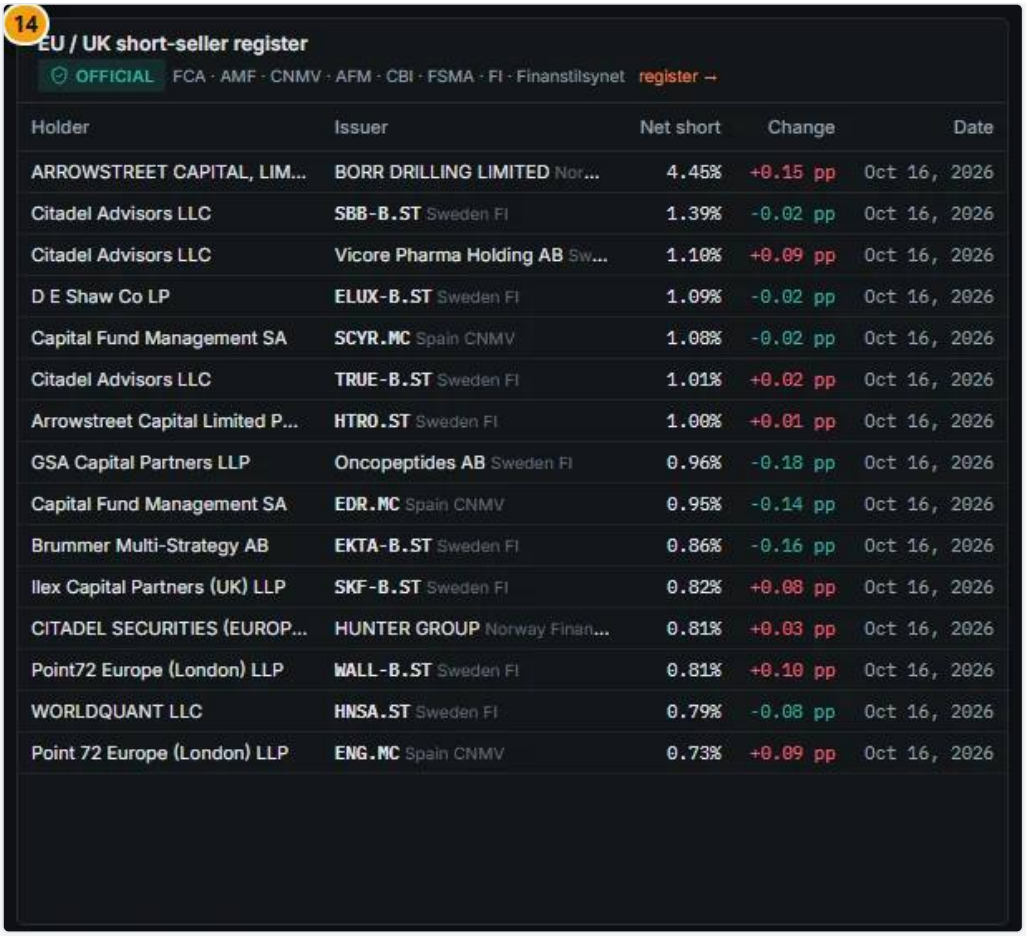


Figure 7.8: Latest named EU / UK net short positions on the overview. An increase in a short position is shown in red.

Controls

Control	What it does
Region select	All regions, US, EU, UK, JP, KR, CN, HK, LATAM, APAC. Filters the security lists (panels 2–9, 11 and 12) and is kept in the URL (`?region=`). The CFTC, activist and EU register panels are not filtered.
Section tabs	**Overview**, **EU / UK short register**, **Activism (13D)**, **Sectors**.
Lag note (clock icon)	"Positions as of quarter end Q, filed by D". Hover: "13F filings report quarter-end long positions and are due 45 days after the quarter. We use only filings made by the cutoff (point-in-time)."
MODEL-BASED tags	Hover for the exact definition (hedge-fund-like, smart cohort, crowding, squeeze, FTD spike, score; quoted in Methodology).
OFFICIAL tags	Name the official source of the panel.
Column headers with a tooltip	Inst. % ("13F long shares / shares outstanding"), Δ q/q (percentage points), Net flow ("13F net share flow / shares outstanding"), Buyers / Sellers (smart-cohort filers that opened or added / trimmed or exited), HF top-20, SI % out, Δ SI, Risk, FTD % out, Score, EU short.
Company links	Open Company 360 › Ownership.
EDGAR links	Open the filing on sec.gov in a new tab.
full feed → / **register →**	Open Activism (13D) and the EU / UK short register.
all markets → and market names	Open Positioning (CFTC) and the market's page.
Sources footer	Each official source with a hover note, e.g. FINRA "Twice-monthly settlement dates, published about 7 business days later".

Sign colours never stand alone: every change carries its sign in the text. For short interest and EU shorts the colours are inverted (a rise is red).

Methodology

Lists. The security lists keep listings with a market cap of at least $300M (the fails-to-deliver list has no market-cap filter).

- The flow lists (accumulation, distribution, smart-cohort buying) need at least 20 13F holders in both quarters and an institutional ownership of at most 150%.
- Accumulation and distribution are sorted by `flow_pct_out` . Smart-cohort buying is sorted by buyers minus sellers, then cohort flow. Most crowded is sorted by `hf_top20_count` .

13F point-in-time (quoted from `docs/SMART_MONEY.md`):

> "A quarter's snapshot uses only rows filed on or before the cutoff = filing deadline (period end + 45 days, next SEC business day; Presidents' Day aware) + 1 business day (EDGAR dates after-hours filings next day) or the run date while filings still arrive (`is_complete = false`); `knowledge_date = cutoff`. The UI shows "positions as of quarter end Q, filed by D"."

Example: quarter end 30 Jun 2026 + 45 days = Fri 14 Aug 2026, plus one business day gives the cutoff Mon 17 Aug 2026 shown in the figure. Previous-quarter share counts are re-expressed on the current share basis (split-adjusted) before changes are computed.

Measure	Definition
Inst. %	13F long shares / shares outstanding
Holder changes	Among filers that filed both quarters: new, added (> +1%), trimmed (< −1%), unchanged, exited
Net flow	`net_flow_shares` = sum of share changes; shown as % of shares outstanding
Breadth	`(new + added − trimmed − exited) / positions compared`
Hedge-fund-like (model-based)	"10-1,000 positions, AUM >= $50M, >= 60% single stocks, and quarterly turnover >= 15% or options >= 5% of value or (top-10 weight >= 50% and turnover >= 5%)". Turnover = min(buys, sells) / mean AUM, with buys / sells = abs(share change) × price, so price moves are not turnover. Index-like books (> 1,000 positions) never qualify.
Crowding (model-based)	"number of hedge-fund-like filers holding the stock among their top-20 positions (percentile)": `hf_top20_count`, `crowding_pct` = its percentile within the quarter
Smart cohort (model-based)	"hedge-fund-like filers whose reported books beat the average 13F filer over the past 4 quarters (top decile, positive excess), selected point-in-time". Each book is held for the next quarter (value weights, coverage ≥ 50% of AUM); excess = return minus the equal-weight mean of all filers; `smart_score` = mean excess of books q−4 … q−1 (at least 3). Cohort flows = cohort members' share changes q−1 → q.
SI % out	Short interest / shares **outstanding** ("float is not published"); null if unknown or > 150%
Days to cover	FINRA days to cover
Δ SI	`si_chg_pct`: change versus the previous settlement
z 1y	`si_z_1y`: z-score of shares short versus the previous 24 settlements (at least 8)
Squeeze risk flag (model-based)	"short interest >= 15% of shares outstanding, days to cover >= 5, 20-day return >= +10%, shares <= 150M or market cap <= $2B"
Risk (squeeze score)	Mean percentile (within the settlement date) of SI %, days to cover, 20-day return and small share count; shown 0–100
FTD spike (model-based)	"daily fails >= 0.5% of shares outstanding, or z >= 3 vs the previous 12 files with >= 100k shares". FTD % out = largest daily fail / shares outstanding
Activism	new 13D, 13D amendment, 13G → 13D switch (the filer's previous filing on the subject within 3 years was a 13G), new / amended 13G
EU / UK disclosed short %	Regulator aggregate when published (FCA since 2026-07, Norway), else the sum of active holder positions (latest notification per holder ≥ 0.5%, within 2 years); maximum across regulators

Combined smart-money score (model-based; `docs/SMART_MONEY.md` §2):

- Inputs, all known on the score date, are winsorised at 1% / 99% and z-scored within the date: 13F flow and breadth, smart-cohort flow, the insider signal, short-interest level and change, and EU short level and 30-day change.
- Each input is signed so that higher = more bullish: short interest and EU shorts enter negatively.
- The inputs are averaged into five groups: inst, smart, insider, short_interest, eu_short.

```
score     = mean of the available group z-scores
score_pct = percentile of score within the date (0..100)
```

The top and bottom lists require at least two groups. The score is "forward-label-free". It is an optional input to the scoring Insider family behind `contracts/scoring.yaml optional_inputs.smart_money.enabled` (default **false**; production scores unchanged).

Futures positioning. For six equity-index futures (VIX, E-mini Dow, E-mini S&P 500, E-mini Nasdaq-100, E-mini Russell 2000, Nikkei 225 yen), the panel shows the CFTC TFF leveraged-funds net position as % of open interest and its 3-year COT index. See Positioning (CFTC) for the method.

Data sources and refresh

Source	Cadence and lag	History in the app
SEC Form 13F data sets	Quarterly; "Quarter-end long positions of managers with > $100M; filed up to 45 days after the quarter". Job `sec_13f` weekly, analytics `smart_money_13f` weekly	2023Q1 … 2026Q2 (14 quarters)
FINRA equity short interest	"Twice-monthly settlement dates, published about 7 business days later". Job `finra_short_interest` weekly	2023-01 … 2026-09

Source	Cadence and lag	History in the app
SEC fails-to-deliver	Half-month files	2022-09 …
SEC Schedules 13D / 13G	EDGAR full and daily index; job `activist_filings` nightly	2022-01 …
EU / UK net short positions	Regulator files; job `eu_short_positions` nightly	see EU / UK short register
Insider signal	Job `insider_signals` nightly (see Insider Radar)	
CFTC COT	Weekly, Tuesday positions, released Friday	

The combined score is computed by `smart_money_signals` (nightly; scores dated weekly on Fridays plus the run date). Under the default weekly profile these jobs run on Saturday morning (Lisbon), and the CFTC job runs on Friday evening. On 19 Oct 2026 the latest 13F quarter was 30 Jun 2026 (cutoff 17 Aug 2026), the latest FINRA settlement was 30 Sep 2026 and the score date was 16 Oct 2026.

Validation status

The combined score was checked once for predictive power. The result is negative. From `docs/SMART_MONEY.md` §4 (2026-10-19, report `data/logs/reports/smart_money_ic_20261016.json`): Spearman rank IC against forward returns, point-in-time and purged, non-overlapping signal dates, 6,145 securities, 2023-06-30 … 2026-10-16:

Column	IC 21d (t, n dates)	IC 63d (t, n dates)
score (composite)	−0.004 (−0.67, 33)	−0.005 (−0.56, 11)
g_inst (13F flow + breadth)	−0.015 (−1.30, 31)	−0.023 (−1.15, 10)
g_smart (smart-cohort flow)	−0.002 (−0.23, 27)	−0.019 (−1.16, 9)
g_insider	+0.014 (+0.79, 32)	+0.024 (+1.09, 11)
g_short_interest (low SI = bullish)	+0.016 (+1.39, 32)	+0.015 (+0.67, 10)
g_eu_short (low disclosed short = bullish)	+0.003 (+0.18, 31)	+0.011 (+0.42, 9)

> "None is statistically significant; the composite has no predictive power in this short sample (the 13F history starts 2023Q1 and the smart cohort needs 4 quarters of track record). The short-interest and insider groups have the expected sign; lagged 13F flows are slightly contrarian (consistent with crowding / the 45-day lag). Treat the score as a descriptive screen, not an alpha signal; the backtesting lab can re-test it with longer history. The weights were NOT tuned on this result (no data snooping)."

In the app the score, smart cohort, crowding, squeeze and FTD-spike panels carry the **MODEL-BASED** tag. The 13F, FINRA, 13D / 13G, EU register and CFTC panels carry **OFFICIAL**: they show reported data, not a model.

Correct usage and worked example

Use the overview to find names worth a closer look, then open Company 360 › Ownership. Always read 13F numbers with their lag: on 19 Oct 2026 they describe positions at 30 Jun 2026, more than three months earlier. Read short interest with its settlement date.

Worked example: squeeze watch, FINRA settlement 30 Sep 2026.

1. Scroll to **Squeeze watch** (Figure 7.3). Cohu (COHU) shows SI 19.0% of shares outstanding, DTC 9.1, 20-day return +49%, risk 89 and the **SQUEEZE RISK** badge. It meets all four conditions: SI ≥ 15%, DTC ≥ 5, return ≥ +10% and the low-float test.
2. Ibotta (IBTA) has a higher risk score (91) but no badge. Its SI (13.8%) is below 15% and its 20-day return (+5%) is below +10%. The risk score ranks names; only the flag applies the rule.
3. Wolfspeed (WOLF) has the highest SI in the universe (50.7%, DTC 11.4, Figure 7.2), but its 20-day return is only +6%, so it is not flagged.
4. The FINRA data are as of 30 Sep 2026 and were published about 7 business days later. The 15 Oct settlement was not yet in the snapshot on 19 Oct.
5. Open COHU (click the ticker) to see the short-interest trend and holders on the Ownership tab before acting.

Crowding example. In **Most crowded**, Amazon (AMZN) is held among the top-20 positions of 561 hedge-fund-like filers (crowding percentile 100). This describes how concentrated the hedge-fund-like ownership is. It does not mean the stock will fall.

Limitations and pitfalls

- **13F lag and scope.** The 45-day lag means positions can be stale by up to 4.5 months. 13F covers long US-listed equity and options positions of managers with more than $100M only: no shorts, no non-US listings, no cash.
- **Implausible ratios.** Ratios to shares outstanding depend on the share count.
  - A stale or wrong count produces values above 100% (Inst. % 128.5% for AVTX) or absurd flows (SVC +273% net flow).
  - In the FTD list, Securitize (SECZ) shows 3,055.22%.

    - Treat such rows as data errors.
- **Spin-offs and IPOs.** New listings show huge "accumulation" because every holder is new (VGNT: 390 new holders). The 20-holder filter removes only the smallest cases.
- **Short interest uses shares outstanding**, not free float, so SI % understates the short share of the float. FINRA symbols are not all mapped to securities: about 4,100 securities have a short-interest row. `docs/PHASE3_TODO.md` records low symbol-resolution coverage.
- **EU coverage gaps.** Germany (BaFin) and Italy (CONSOB) are not ingested, so German and Italian shorts are missing. Since 10 Jul 2026 the UK publishes aggregates only (no holder names).
- **Activist 13D (60d) count.** The snapshot figure counts the items returned to the feed, which is capped at 30. With more activity, the true 60-day count is higher. Use the Activism page for counts.
- **Model-based labels.** Hedge-fund-like and smart cohort are transparent rules on 13F books, not knowledge of a manager's strategy. The smart cohort needs four quarters of history, and the 13F history starts in 2023Q1.
- **The score is not alpha.** See Validation status.

Related mnemonics and BQL fields

- Mnemonics: `SMRT` (smart money), `HDS` (holders), `SI` (short interest); all open this page without a security and Company 360 › Ownership with one.
- BQL fields:
    - 13F: `own.inst_own_pct`, `own.inst_own_chg`, `own.flow_pct_out`, `own.breadth`, `own.n_new`, `own.n_exited`, `own.hhi`, `own.hf_top20_count`, `own.crowding_pct`, `own.smart_buyers`, `own.smart_sellers`, `own.smart_flow_pct_out`.
    - Short interest and FTD: `si.si_pct_out`, `si.days_to_cover`, `si.si_chg_pct`, `si.si_z_1y`, `si.squeeze_score`, `si.squeeze_flag`, `si.ftd_pct_out`, `si.ftd_spike`.
    - Combined score: `sm.score`, `sm.score_pct`, `sm.g_inst`, `sm.g_smart`, `sm.g_insider`, `sm.g_short_interest`, `sm.g_eu_short`, `sm.eu_short_pct`.
- Alerts: "activist 13D", "short-interest spike", "EU short position" (see Watchlists and alerts).
- Related sections: Company 360 Ownership, Positioning (CFTC), Backtest Lab, BQL and mnemonics.

EU / UK short register

Route: `/smart-money/eu-shorts` · **Menu:** Smart Money › EU / UK short register · **Mnemonics:** none (tab of Smart Money; `SMRT` opens the overview) · **Refresh:** nightly job `eu_short_positions`

Purpose

The register lists the net short positions at or above 0.5% of issued capital that the European and UK regulators publish. It names the short seller where the regulator does. Use it to see who is short a European stock, how a holder's positions have changed, and which funds run the most disclosed shorts.

Screenshot

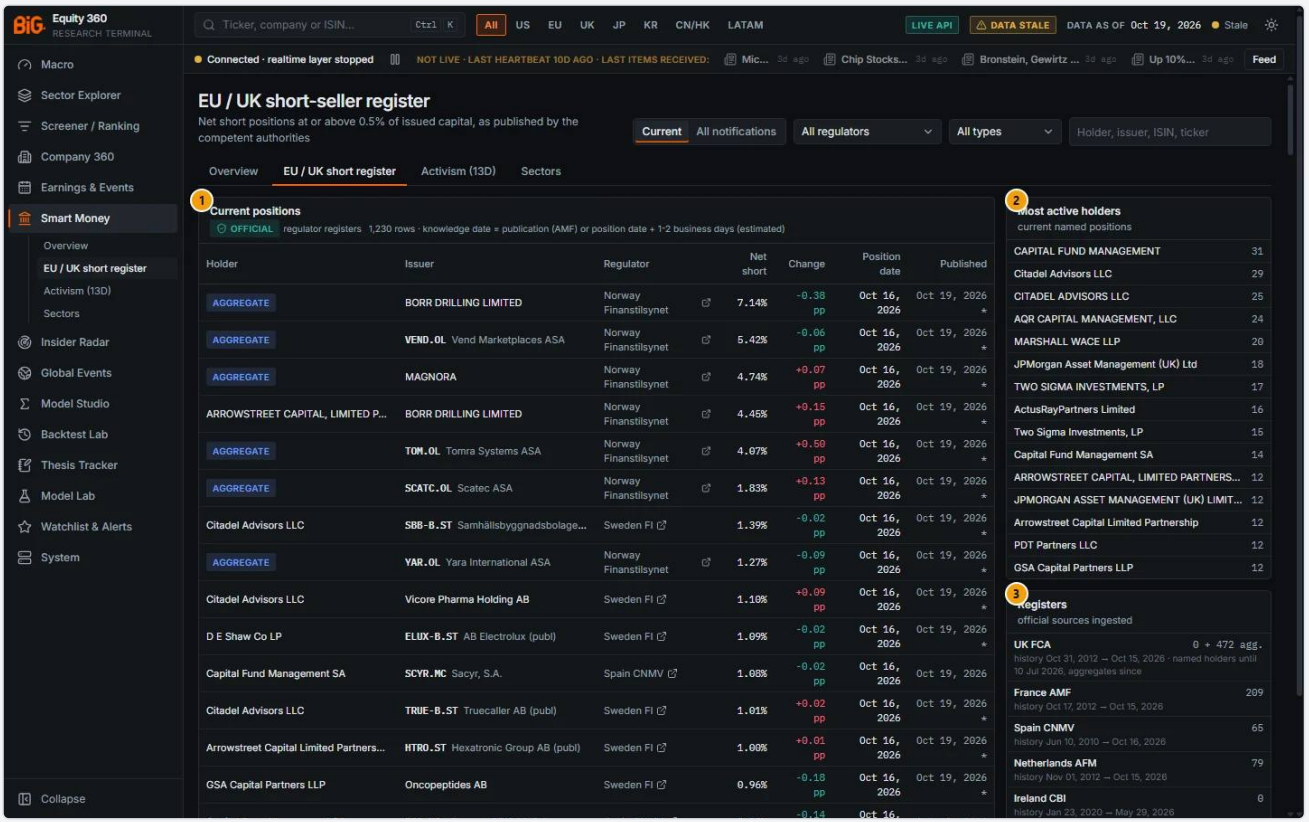


Figure 7.9: EU / UK short-seller register (`/smart-money/eu-shorts`), Current mode, all regulators, data as of 19 Oct 2026.

#	Panel	What it shows
1	Current positions	**OFFICIAL** regulator registers; "1,230 rows · knowledge date = publication (AMF) or position date + 1-2 business days (estimated)". Holder (or **AGGREGATE** badge), Issuer, Regulator (link to the source), Net short, Change, Position date, Published.
2	Most active holders	"current named positions": the 15 holders with the most current named positions, with their count.
3	Registers	"official sources ingested": per regulator the number of current named positions (+ aggregates) and the history range. The UK FCA line notes "named holders until 10 Jul 2026, aggregates since".
4	Not ingested	"and why": registers that could not be ingested (below the fold, Figure 7.10).

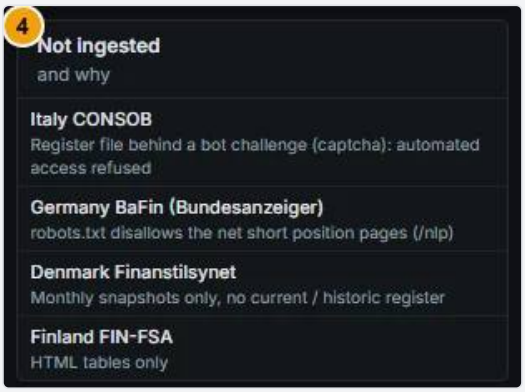


Figure 7.10: Registers not ingested and the reasons: Italy CONSOB, Germany BaFin (Bundesanzeiger), Denmark Finanstilsynet, Finland FIN-FSA.

All notifications, one holder. The figure below shows the register in All notifications mode, filtered on one holder.

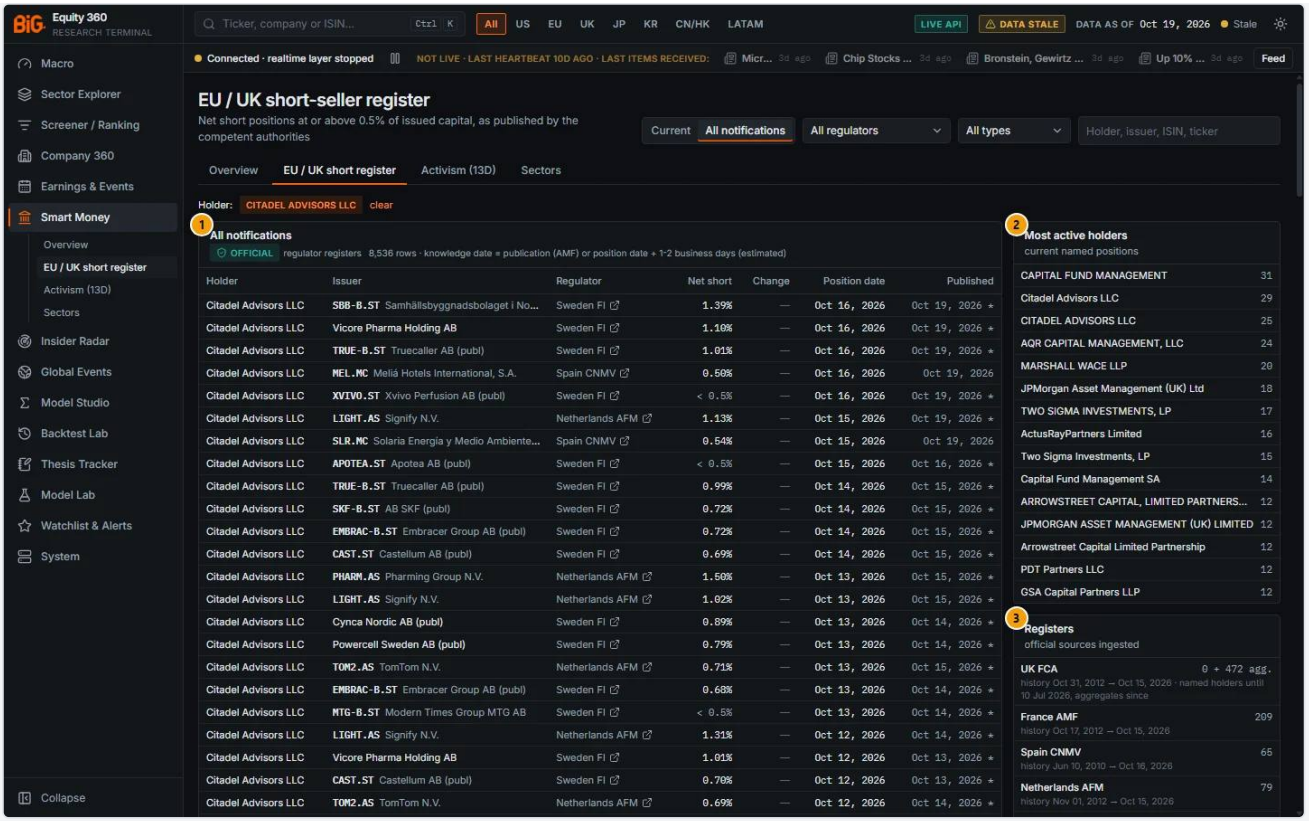


Figure 7.11: All notifications filtered on the holder Citadel Advisors LLC (`/smart-money/eu-shorts?mode=history&holder=Citadel%20Advisors%20LLC`), data as of 19 Oct 2026.

#	Panel	What it shows
1	All notifications	The holder's full notification history across regulators, newest first: 8,536 rows (Sweden FI, Spain CNMV, Netherlands AFM and others). Rows showing "< 0.5%" are closing notifications. The Change column is empty in this mode.
2	Most active holders	Unchanged by the filter: it always ranks current named positions.
3	Registers	Unchanged by the filter: the ingested registers and their history ranges.
4	Not ingested	Below the fold, as in Figure 7.10.

The **Holder:** chip above the table shows the active holder filter; **clear** removes it. The chip displays the name in capitals, but the filter is the exact name as filed ("Citadel Advisors LLC"). Notifications filed under other spellings, such as "CITADEL ADVISORS LLC", are not included.

Controls

Control	What it does
Current / **All notifications**	Register mode. *Current*: each holder's latest notification per issuer that is still ≥ 0.5% and less than two years old, plus the latest regulator aggregates. *All notifications*: every notification ever published (about 240,000 rows), without the Change column.
Regulator select	All regulators, UK FCA, France AMF, Spain CNMV, Netherlands AFM, Ireland CBI, Belgium FSMA, Sweden FI, Norway Finanstilsynet.
Type select	All types, **Named holders** (individual positions), **Aggregates** (regulator totals per issuer, no names).
Search box (Holder, issuer, ISIN, ticker)	Press Enter to apply. Holder and issuer match on a substring; ISIN and ticker must match exactly (e.g. `SBB-B.ST` , not `SBB`).
Holder name (table or Most active holders)	Click to filter the register on that exact holder. A **Holder:** chip appears with **clear**.
Issuer link	Opens Company 360 › Ownership when the ISIN maps to a covered security; otherwise the issuer name is plain text (hover for the ISIN).
Regulator link	Opens the regulator's source file or page in a new tab.
AGGREGATE badge	Hover: "Regulator-published aggregate net short position of the issuer (no holder names)".
Net short **< 0.5%**	The holder's position fell below the publication threshold (the closing notification).
Change	Current mode only: change versus the holder's previous notification in percentage points; **new** when there is no earlier notification. A rise is red (more short).
Published (with an asterisk)	Publication date. The asterisk marks an estimated publication date (position date + 1–2 business days).

Control	What it does
← previous / next →	Pages of 200 rows.

Filters are kept in the URL (`mode` , `regulator` , `type` , `q` , `holder` , `page`).

Methodology

The register shows the regulator files as published, with these processing rules (from `docs/SMART_MONEY.md` §1 and §2):

- **Threshold.** Net short positions at or above 0.5% of issued capital are published by the regulators. A notification of "<0,5" (Sweden) or below 0.5% closes a position.
- **Current register.** For each regulator, holder and issuer, the latest notification is kept if it is still ≥ 0.5% and its position date is within two years; regulator aggregates are added.
  - A regulator whose last named notification is more than 30 days older than its aggregates is treated as aggregate-only.
  - This is the UK case: "since 10 Jul 2026 the FCA publishes **aggregated** net short positions per issuer only (no holder names). The frozen named-holder history stays searchable ("All notifications"); the current register shows FCA aggregates only".
- **Knowledge dates** (point-in-time). Position date + 2 business days (estimated) for FCA, CNMV, AFM, CBI and FSMA; + 1 business day for Sweden FI and Norway; the observed publication date for AMF. All are capped by the file publication date and the download date, and never before the position date.
- **Disclosed short % per issuer** (used in the smart-money score and the sector table): the regulator aggregate when the regulator publishes one, else the sum of active named positions; maximum across regulators.

Data sources and refresh

Regulator	Format	History	Current named / aggregates (19 Oct 2026)
UK FCA	xlsx (individual, frozen 10 Jul 2026) + aggregate CSVs	named 2012-11 … 2026-07-10; aggregates since 2026-07	0 / 472
France AMF	CSV via data.gouv.fr (holder LEI, publication dates)	since 2012-11	209 / 0
Spain CNMV	xls (Current, Series, Previous sheets)	since 2010	65 / 0
Netherlands AFM	CSV (current + historic)	since 2012	79 / 0
Ireland CBI	xlsx (Current / Historical)	since 2020 in the app; last update 2026-05	0 / 0
Belgium FSMA	CSV, current only (history accumulates nightly)	since 2023-06	14 / 0
Sweden FI	ODS (current + historic)	since 2010	262 / 0
Norway Finanstilsynet	Open JSON API (aggregates and active holders)	last 2 years	89 / 40

Not ingested. The reasons are listed on the page and in `docs/SMART_MONEY.md` :

- Italy CONSOB: the register file is behind a bot challenge (captcha), so automated access is refused.
- Germany BaFin: positions are published only in the Bundesanzeiger, whose robots.txt disallows the net short-position pages (`/nlp`).
- Denmark: monthly snapshots only, with no current or historic register.
- Finland: HTML tables only.

The job `eu_short_positions` downloads the full files every night, one request per file, respecting each site's robots.txt and crawl delay.

Validation status

Not applicable: official data. The register reproduces regulator disclosures and carries the **OFFICIAL** tag. Only the knowledge dates are estimated (marked with an asterisk). The EU short inputs to the smart-money score had no significant rank IC (g_eu_short IC 21d +0.003, t +0.18; see Smart Money overview).

Correct usage and worked example

Use **Current** to see who is short now and **All notifications** to follow a holder's history. Search by exact ticker or ISIN to see every disclosed short in one issuer.

Worked example: SBB (Samhällsbyggnadsbolaget i Norden), as of 19 Oct 2026.

1. Type `SBB-B.ST` in the search box and press Enter. Eight current named positions appear, all reported to Sweden FI.
2. The largest is Samlyn Capital LLC at 3.18% (position date 25 Sep 2026, down from 3.25%). It is followed by Brooklands Fund Management at 2.69% (badge **new**, first notified 1 Oct 2026) and Arrowstreet Capital at 1.40%.

3. Citadel Advisors LLC holds 1.39% (16 Oct 2026, change −0.02 pp from 1.41%). The **Published** date 19 Oct 2026 carries an asterisk: estimated as position date + 1 business day, rounded to the next file.
4. Together the named holders disclose about 12.0% of issued capital. Positions below 0.5% are not public, so the true total short interest is higher.
5. Click **Citadel Advisors LLC** to filter on the holder and see its other positions. Then switch to **All notifications** to see its full history in SBB.

Limitations and pitfalls

- **Only positions ≥ 0.5% are public.** The disclosed total is a lower bound on the short interest.
- **No name normalisation across regulators.** The same fund can appear under several spellings ("Citadel Advisors LLC" and "CITADEL ADVISORS LLC"; "CAPITAL FUND MANAGEMENT" and "Capital Fund Management SA"). The Most active holders list counts them separately.
- **UK names stop on 10 Jul 2026.** After that date the UK shows aggregates per issuer only. Named UK history remains searchable in All notifications.
- **Germany and Italy are missing.** DAX and FTSE MIB names show no disclosed shorts in the app even when positions exist.
- **Stale "current" positions.** A position stays current until the holder files a change or it is two years old. An old notification (for example January 2026) may already have been reduced below the threshold without a record in the file.
- **Unmapped issuers.** Many issuers (small caps, non-covered listings) do not map to a security and have no Company 360 link.
- **Estimated publication dates** (asterisk) are approximations for point-in-time use.

Related mnemonics and BQL fields

- Mnemonics: `SI` opens Smart Money (no security) or Company 360 › Ownership (with a security), which shows the company's EU / UK positions.
- BQL fields: `sm.eu_short_pct`, `sm.eu_short_chg_30d`, `sm.eu_short_holders`, `sm.z_eu_level`, `sm.z_eu_chg`, `sm.g_eu_short`.
- Alerts: "EU short position" (new or increased position published), see Watchlists and alerts.
- Related sections: Smart Money overview, Company 360 Ownership.

Activism (13D)

Route: `/smart-money/activism` · **Menu:** Smart Money › Activism (13D) · **Mnemonics:** none (tab of Smart Money) · **Refresh:** nightly job `activist_filings`, events built by `smart_money_signals`

Purpose

The activism feed lists SEC Schedule 13D and 13G filings: beneficial owners crossing 5% of a US-registered company. A 13D signals active or control intent, while a 13G is passive. The most informative events are a **new 13D** and a **13G → 13D switch**, where a holder that was passive declares active intent. Use the page to follow activist campaigns and to see which filers are most active.

Screenshot

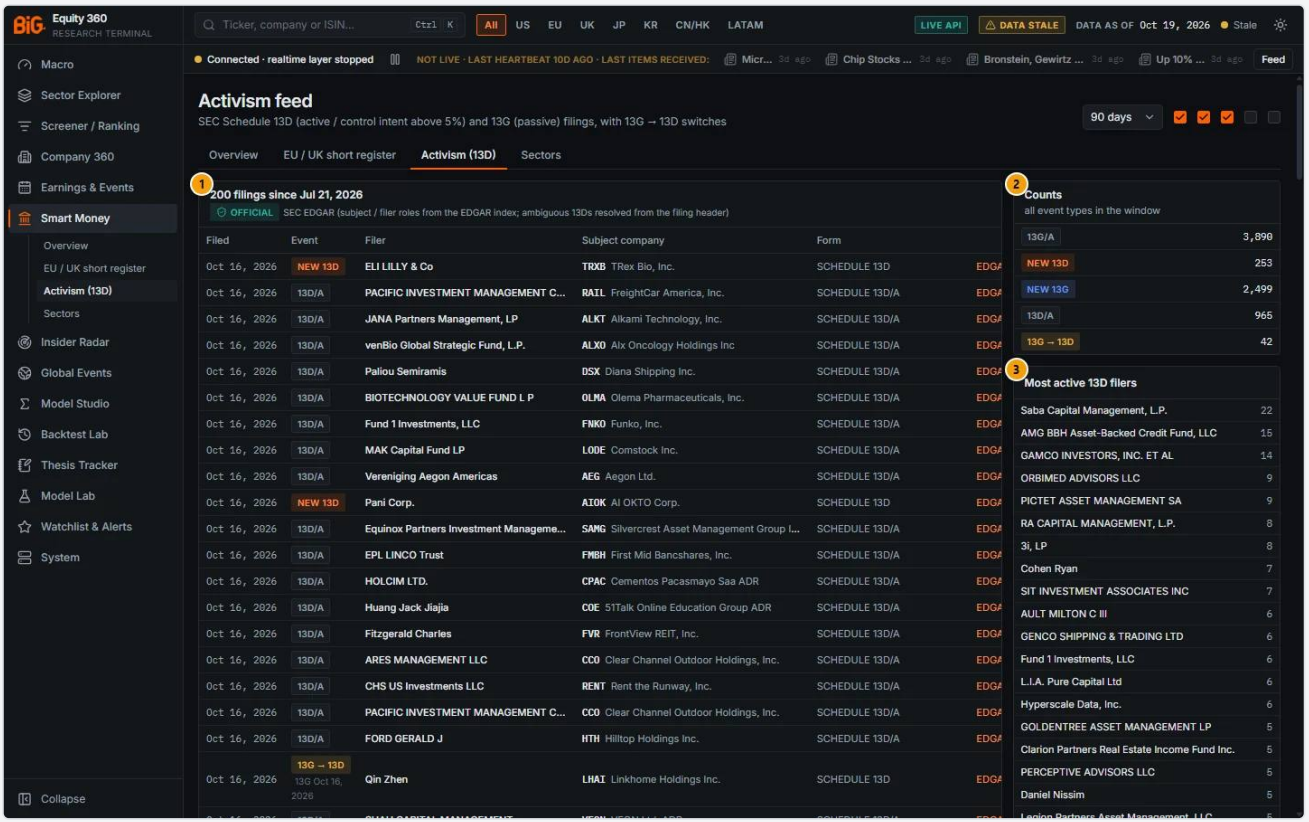


Figure 7.12: Activism feed (`/smart-money/activism`), 90-day window, default event types, data as of 19 Oct 2026.

#	Panel	What it shows
1	200 filings since Jul 21, 2026	**OFFICIAL** "SEC EDGAR (subject / filer roles from the EDGAR index; ambiguous 13Ds resolved from the filing header)". Filed, Event badge, Filer, Subject company, Form, **EDGAR** link.
2	Counts	"all event types in the window": counts per event type (13G/A 3,890; New 13D 253; New 13G 2,499; 13D/A 965; 13G → 13D 42).
3	Most active 13D filers	Filers with the most 13D filings in the window (e.g. Saba Capital Management 22).

Controls

Control	What it does
Window select	30 days, 90 days (default), 180 days, 1 year, 3 years.
Event-type checkboxes (top right)	Five unlabelled checkboxes, in this order: **New 13D**, **13G → 13D**, **13D/A**, **New 13G**, **13G/A**. The first three are ticked by default. Screen readers announce the names; the boxes have no visible text. Unticking all restores the default.
Event badges	New 13D (orange), 13G → 13D (amber), 13D/A, New 13G (blue), 13G/A. A switch row also shows the date of the filer's previous 13G (e.g. "13G Oct 16, 2026").
+n after a filer	The filing has n further joint filers.
inferred badge (Form column)	Hover: "Subject / filer roles inferred (no ticker match and no filing-header check)".
Subject company link	Opens Company 360 › Ownership when the subject is a covered security.
EDGAR	Opens the filing on sec.gov.

The window and the ticked types are kept in the URL (`days` , `types`).

Methodology

The event types are defined as follows (quoted from `docs/SMART_MONEY.md`):

> "**Activism**: new 13D, 13D amendment, 13G → 13D switch (filer's previous filing on the subject within 3 years was a 13G), new / amended 13G."

Subject and filer are taken from the EDGAR master index (quarterly full index and daily index files). When the index does not tell which party is the subject, the job reads the filing header; up to 400 headers are fetched per run. The list is sorted by filing date, newest first, and shows at most 200 filings. The Counts and Most active 13D filers panels cover the whole window.

Data sources and refresh

SEC EDGAR Schedules 13D and 13G, history since January 2022 (backfilled from the quarterly full index). The job `activist_filings` runs nightly after the EDGAR daily index; activist events are rebuilt by `smart_money_signals`.

Validation status

Not applicable: official filings. The feed reproduces SEC filings with the **OFFICIAL** tag. Roles resolved by name only are marked **inferred**. No study in the project measures returns after 13D events.

Correct usage and worked example

Tick only **New 13D** and **13G → 13D** to see fresh activist intent; add **13D/A** to follow campaigns already under way. Read the filing on EDGAR for the stated purpose (Item 4 of the 13D) before drawing conclusions.

Worked example: Third Point and Strata Critical Medical, 15 Oct 2026.

1. Untick **13D/A** (the third checkbox) so only new 13Ds and switches remain.
2. The row of 15 Oct 2026 shows **13G → 13D**: Third Point LLC → SRTA (Strata Critical Medical, Inc.), Form SCHEDULE 13D. The filer's previous filing on the company was a 13G on 14 Nov 2025, so a holder that had been passive for eleven months now declares active intent.
3. Click **EDGAR** to read the 13D. Click **SRTA** to see the 13F holders and short interest on Company 360 › Ownership.

Limitations and pitfalls

- **Panel title count is capped.** The title ("200 filings since ...") counts the rows returned, at most 200. Use the **Counts** panel for the true numbers (in the figure, 965 13D amendments alone).
- **Unlabelled checkboxes.** The event-type checkboxes show no text. Use the order given in Controls.
- **Same-day switches.** A 13G and a 13D filed on the same day by the same filer appear as a switch (Qin Zhen → LHAI on 16 Oct 2026, previous 13G the same day). Check the filing history on EDGAR.
- **US filings only.** Schedules 13D / 13G exist only for SEC-registered companies. European large-holder notifications are not included.
- **Filers are not normalised.** Individuals and funds appear as filed, so related entities count separately in Most active 13D filers.

Related mnemonics and BQL fields

- Mnemonics: `SMRT` (overview, with the activist feed of the last 60 days). With a security loaded, `HDS` / `SMRT` open Company 360 › Ownership, which lists the company's 13D / 13G events.
- BQL fields: none for individual filings.
- Alerts: "activist 13D" (new 13D or 13G → 13D switch on a watched name; amendments optional), see Watchlists and alerts.
- Related sections: Smart Money overview, Company 360 Ownership.

Smart-money sectors

Route: `/smart-money/sectors` · **Menu:** Smart Money › Sectors · **Mnemonics:** none (tab of Smart Money) · **Refresh:** with the serving snapshot after `smart_money_13f` (weekly), `finra_short_interest` (weekly), `eu_short_positions` and `smart_money_signals` (nightly)

Purpose

The sector page aggregates the smart-money measures by sector, optionally within one region:

- institutional ownership and its quarterly change;
- 13F breadth and smart-cohort activity;
- crowding, short interest and squeeze flags;
- disclosed EU / UK shorts and the average combined score.

Use it to see where institutions are adding, where short sellers are concentrated, and which sectors are crowded.

Screenshot

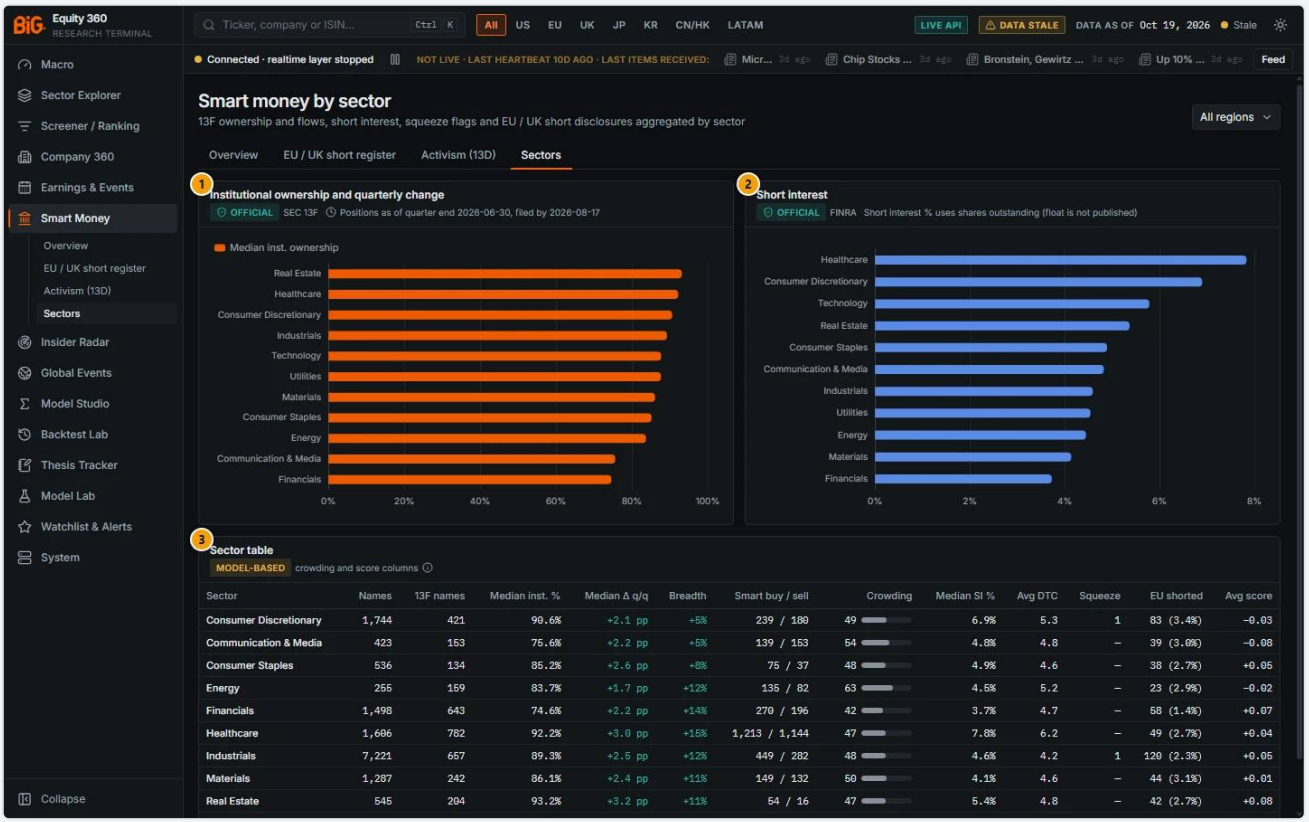


Figure 7.13: Smart money by sector (`/smart-money/sectors`), all regions, data as of 19 Oct 2026; 13F positions as of quarter end 30 Jun 2026, filed by 17 Aug 2026.

#	Panel	What it shows
1	Institutional ownership and quarterly change	**OFFICIAL** SEC 13F with the lag note. Bars of median institutional ownership per sector (hover adds the median change q/q and the number of 13F names).
2	Short interest	**OFFICIAL** FINRA; "Short interest % uses shares outstanding (float is not published)". Bars of median SI % per sector (hover adds average days to cover and squeeze flags).
3	Sector table	**MODEL-BASED** "crowding and score columns". Sector, Names, 13F names, Median inst. %, Median Δ q/q, Breadth, Smart buy / sell, Crowding, Median SI %, Avg DTC, Squeeze, EU shorted, Avg score.

Controls

Control	What it does
Region select	All regions, or one of APAC, CA, CN, EU, HK, JP, KR, LATAM, UK, US. Kept in the URL (`?region=`).
Section tabs	Overview, EU / UK short register, Activism (13D), Sectors.
Chart tooltips	Hover a bar for the sector's figures.
Lag note and **MODEL-BASED** tag	Hover for the 13F lag and the crowding / score definitions.

Methodology

All securities of the sector (and region) are counted in **Names**; the other columns use the securities that have the data.

Column	Definition
13F names	Securities with a 13F ownership row in the latest quarter
Median inst. % / Median Δ q/q	Median institutional ownership and its change (percentage points)
Breadth	Mean of `(new + added − trimmed − exited) / positions compared`
Smart buy / sell	Sum over the sector's securities of smart-cohort filers buying / selling (filer-security pairs)
Crowding	Mean crowding percentile (model-based)
Median SI % / Avg DTC	FINRA short interest (settlements of the last 45 days)
Squeeze	Number of securities with the squeeze-risk flag (model-based)
EU shorted	Securities with a disclosed EU / UK short (count and average %)

Column	Definition
Avg score	Mean combined smart-money score (scores of the last 14 days; model-based)

Definitions of each measure are in Smart Money overview › Methodology.

Data sources and refresh

The page uses the same sources as the overview: SEC 13F (weekly), FINRA short interest (weekly), EU / UK registers and the combined score (nightly).

Validation status

Not formally validated: descriptive aggregates. The crowding and score columns are labelled **MODEL-BASED** in the app. The combined score has no significant predictive power (overview validation).

Correct usage and worked example

Compare sectors with each other in the same snapshot, and read the 13F columns as of the quarter end shown in the lag note.

Worked example: Healthcare, all regions, 13F quarter 30 Jun 2026.

1. Healthcare has 1,606 names, of which 782 have 13F data.
2. Median institutional ownership is 92.2%, up a median +3.0 pp on the quarter, with breadth +15%: more holders opened or added than trimmed or exited.
3. Smart-cohort filers made 1,213 buys against 1,144 sells, roughly balanced.
4. Healthcare has the highest median short interest of all sectors (7.8% of shares outstanding, average 6.2 days to cover) and the tallest bar in the Short interest chart.
5. 49 Healthcare securities have a disclosed EU / UK short (average 2.7%). The average combined score is +0.04, close to neutral.

Limitations and pitfalls

- **Coverage is mostly US.** 13F and FINRA cover US-listed securities: about 4,000 of 19,000 names have 13F or short-interest data. Non-US sectors rely mostly on EU shorts and the score.
- **Medians of ratios to shares outstanding** inherit the share-count errors described in the overview pitfalls.
- **Smart buy / sell counts are pairs**, not distinct filers or names.
- **Sector "Unclassified"** collects securities without a sector.

Related mnemonics and BQL fields

- Mnemonics: none opens this tab directly; `SMRT` opens the overview.
- BQL fields: the per-security fields `own.*`, `si.*`, `sm.*` listed under the overview, which can be grouped by sector in the BQL console.
- Related sections: Sector Explorer, Smart Money overview.

Insider Radar

Route: `/insiders` · **Menu:** Insider Radar · **Mnemonics:** none (type "insider" in the command palette to find the page) · **Refresh:** nightly insider ingestion jobs and `insider_signals`; weekly `insider_models_fit`

Purpose

The Insider Radar lists unusual insider dealings across markets: open-market purchases and sales by directors and officers, cluster buys (several insiders buying within weeks), and CEO / CFO purchases. Each trade has an anomaly score that shows how unusual it is. Use it to find insider activity worth a closer look and open the company's Insiders tab for the full history.

Screenshot

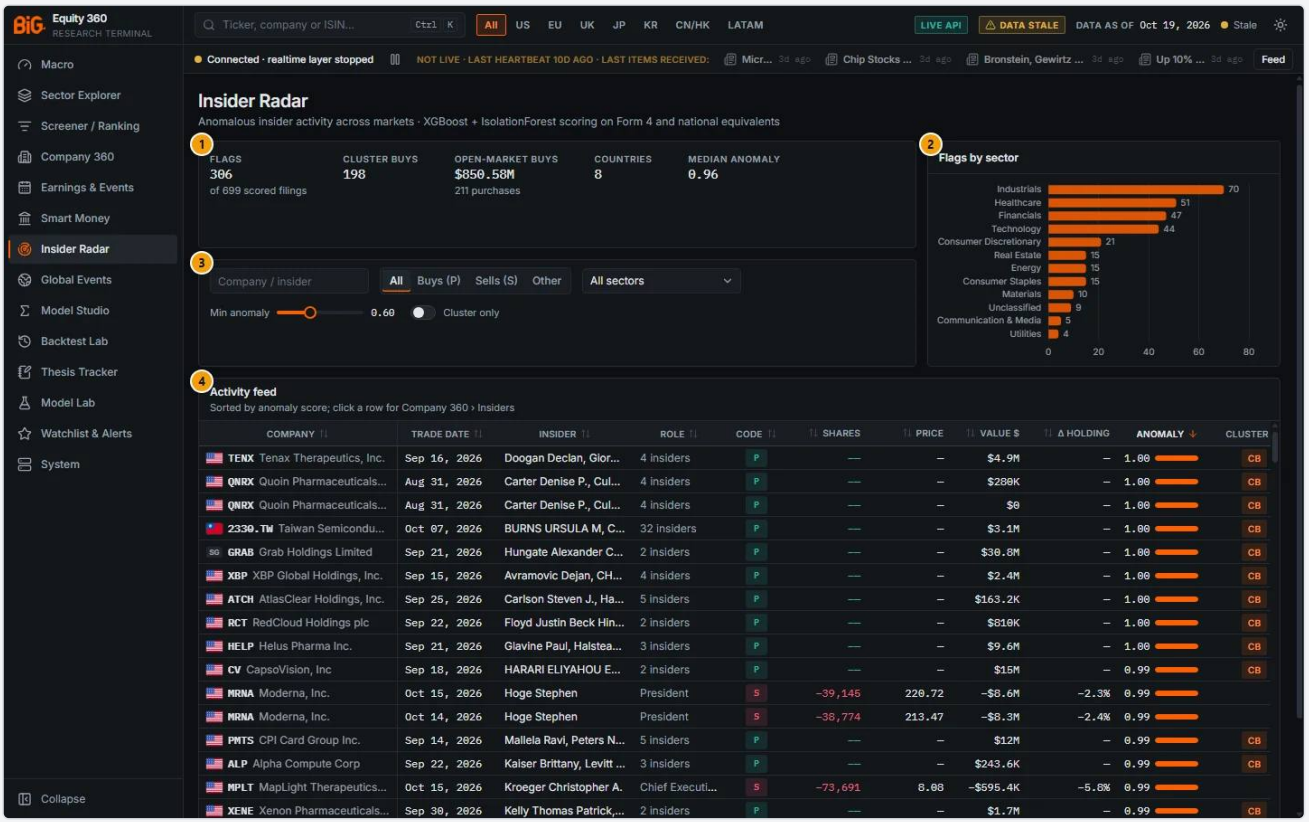


Figure 7.14: Insider Radar (`/insiders`), all regions, minimum anomaly 0.60, data as of 19 Oct 2026.

#	Panel	What it shows
1	Flags (summary)	**Flags** 306 of 699 scored filings; **Cluster buys** 198; **Open-market buys** $850.58M in 211 purchases; **Countries** 8; **Median anomaly** 0.96. All figures follow the filters.
2	Flags by sector	Bar chart: number of flagged rows per sector (Industrials 70, Healthcare 51, Financials 47 ...).
3	Filters	Company / insider search, transaction-code switch, sector select, minimum-anomaly slider, cluster-only switch.
4	Activity feed	"Sorted by anomaly score; click a row for Company 360 › Insiders". Company, Trade date, Insider, Role, Code, Shares, Price, Value $, Δ holding, Anomaly, Cluster, Mkt cap, Why flagged.

Controls

Control	What it does
Top-bar region buttons (All, US, EU, UK, JP, KR, CN/HK, LATAM)	The page follows the global region filter in the top bar.
Company / insider search	Filters by ticker, company name or insider name as you type.
All / Buys (P) / Sells (S) / Other	Transaction code. *Other* = grants (A), option exercises (M), tax withholding (F), gifts (G).
Sector select	All sectors or one sector.
Min anomaly slider	0.40 to 0.95 in steps of 0.05; default 0.60. Rows below the threshold are hidden.
Cluster only switch	Show only rows that are part of a cluster buy.
Column headers	Click to sort (the feed starts sorted by Anomaly, descending). The Company column stays pinned when scrolling sideways.
Code badge	Hover for the meaning, e.g. "P — Open-market purchase", "S — Open-market sale".
Anomaly cell	Score 0–1 with a bar (accent ≥ 0.75, amber ≥ 0.5). Hover for the features behind it: role weight, insiders same direction (30d), cluster size, first buy in 2 years, routine trader, size vs own history, drawdown before (1y), return before (6m), value (USD).
CB badge	Part of a cluster buy; hover for the cluster id.
Why flagged	The reason for the row: "Cluster buy: n insiders date → date ($ value)", "C-suite open-market purchase", "Insider model P(outperform) x%" (see Validation status) or the transaction type and value.
Row click	Opens Company 360 › Insiders for the company.

Methodology

What the page loads. The radar combines three API calls:

- The 500 most recent insider filings of the last 90 days worth at least $50,000, of any transaction type.
- Cluster buys of the last 90 days. A cluster is a run of open-market purchases of one security where consecutive buys are at most 30 days apart and at least 2 distinct insiders buy. A cluster without a transaction in the 500 filings appears as one summary row: role "n insiders", source "Cluster", value = cluster total.
- The 200 strongest bullish insider signals.

Open-market trades only for scoring. The signal pipeline uses open-market buys (code P) and sells (code S) only: "grants/exercises/tax withholding/gifts (A/M/F/G/...) and derivative-table rows are excluded". The feed itself also lists other codes (filter **Other**). Those rows have no per-trade anomaly score, and the radar shows the security's insider signal instead (see Limitations).

Point in time. "an insider transaction is usable from its filing date (knowledge_date), never from the trade date". Amended filings (4/A) replace the original from the amendment's date.

Anomaly score (per transaction). An IsolationForest is fitted weekly on per-transaction features:

- direction (buy or sell), log USD value, value relative to average daily volume, size versus the insider's own past trades;
- role weight (CEO > Chair > CFO > President > COO > other officers > directors > 10% owners);
- the 10b5-1 plan flag, the change in the insider's holding, the filing lag;
- the number of distinct insiders trading in the same direction within 30 days, first buy in two years;
- the Cohen-Malloy-Pomorski "routine" flag (same calendar month in each of the 3 prior years);
- returns over 1, 3 and 6 months, the 1-year drawdown and the 3-month volatility before the filing.

The score is the empirical CDF of the training anomaly scores, 0..1, with 1 = most anomalous. It is unsupervised: it measures how *unusual* a trade is, not whether it predicts returns.

Rules-based insider score (0–100, 50 = no activity). This score feeds the Insider factor and the smart-money score; BQL exposes it as `ins.signal_score`. For each issuer, the open-market trades filed in the last 90 days are weighted with a 45-day half-life:

```
buy weight  = role weight × (1 + 0.5·[≥ 3 insiders buying within 30 d] + 0.4·[first buy in 2 y] + 0.4·[1-y drawdown < −20%])
              × 0.5 if under a 10b5-1 plan × 0.6 if routine × ln(1 + value / $10k)
sell weight = role weight × 0.35 × (0.2 if 10b5-1 or routine) × (1 + 0.3·[6-m return > 30%]) × ln(1 + value / $10k)
raw         = Σ decayed buy weights − Σ decayed sell weights
score       = 50 + 50 · tanh( sign(raw) · ln(1 + |raw| / 5) / 2.5 )
```

The issuer signal's **cluster buy** flag needs at least 3 distinct buyers within 30 days. The **CB** badge in the feed marks any cluster of at least 2 insiders. The **C-suite buy** flag is an open-market purchase by the CEO or CFO.

Data sources and refresh

Market	Source	Notes
United States	SEC Forms 3 / 4 / 5 (+ amendments)	Job `sec_form4`, nightly
United Kingdom	FCA National Storage Mechanism: director / PDMR dealing notices (RNS "DSH", MAR art. 19)	Job `fca_nsm`, nightly; parsed from notice text
Germany	BaFin directors' dealings database (MAR art. 19)	Job `eu_pdmr_bafin`, nightly (archived nightly because the portal keeps a limited window)
Netherlands	AFM register "Transacties leidinggevenden MAR 19"	Job `eu_pdmr_afm`, nightly
Hong Kong	HKEX Disclosure of Interests (director / substantial-shareholder dealings)	Job `hkex_filings`, nightly
Brazil	CVM VLMO monthly consolidated insider positions / trades	Monthly data
Korea	FSS OpenDART executive and major-holder filings	Needs the `OPENDART_APIKEY` key; skipped without it
France	AMF BDIF directors' dealings	Metadata only (who / when); no transactions yet

Signals are recomputed nightly by `insider_signals`; the IsolationForest and the XGBoost model are refitted weekly by `insider_models_fit`.

Validation status

> **Model-based:** "Insider XGBoost model: shows no out-of-sample edge, so only the rules-based insider score is used." (`docs/USER_GUIDE.md` §6)

The supervised XGBoost model predicts the 3-month excess return versus the sector ETF after an insider filing day. It is trained walk-forward with purging and an embargo. Its promotion gate is "OOS monthly rank-IC Newey-West t > 2 and AUC > 0.52"; when the gate fails, "the model probability is reported but NOT blended into signal_score".

The current fit (version v20261020T034351, events 2016-01 … 2026-07, 91 test months) **fails** the gate:

Out-of-sample, 3 months	AUC	Monthly rank IC (mean)	NW t-stat
XGBoost, all events	0.493	−0.015	−1.17
XGBoost, buy events	0.492	+0.030	+1.75
Rules-based score (benchmark), all events	n/a	+0.010	+1.38

The rules-based score is used because it is transparent, not because it passed a gate: its own benchmark IC is not significant either. The IsolationForest anomaly score is unsupervised and has no validation gate. It ranks how unusual a trade is.

> **Caution:** the page subtitle still reads "XGBoost + IsolationForest scoring", and **Why flagged** can show "Insider model P(outperform) x%". That probability comes from the XGBoost model that failed its gate. Treat it as uninformative.

Correct usage and worked example

Start from **Buys (P)** with **Cluster only** on: open-market purchases by several insiders are the classic signal of the insider literature. Then check the role (CEO / CFO), the size against the insider's own history, and whether the trade was under a 10b5-1 plan. Treat sells with caution: most are planned or for diversification.

Worked example: Tenax Therapeutics (TENX), as of 19 Oct 2026.

1. The first row of the feed (Figure 7.14) is TENX: trade date 16 Sep 2026, Insider "Doogan Declan, Gior…", Role "4 insiders", Code **P**, Value $4.9M, Anomaly 1.00, **CB**.
2. **Why flagged** reads "Cluster buy: 4 insiders 2026-08-20 → 2026-09-16 ($4.9M)". Four insiders (Declan Doogan, Christopher Thomas Giordano, Thomas R. Staab II, Daniel Schneeberger) bought in the open market within four weeks.
3. Clicking the row opens Company 360 › Insiders, with the anomaly score, 90-day net buying, buyers and sellers, cluster buys and the trades on the price chart. The issuer signal has both the cluster-buy and the C-suite-buy flags set, with 4 buyers and 0 sellers in 90 days. Its rules-based insider score (BQL `ins.signal_score`) is 82.6 out of 100, where 50 = no activity.
4. For contrast, Moderna (MRNA) also scores 0.99: two sales of about 39,000 shares each by the President (−$8.6M and −$8.3M, Δ holding −2.3%). The anomaly tooltip shows a size 3.7× the insider's usual trade and a 6-month return before the sale of +306%. The filing is marked as a 10b5-1 plan sale. The trade is unusual, but planned selling after a strong run carries little information.

Limitations and pitfalls

- **The feed covers a short window.** The radar loads the 500 most recent filings worth at least $50,000. On 19 Oct 2026 these covered only filings of 15–16 Oct 2026, almost all US. Older dealings appear only as cluster summary rows. Use Company 360 › Insiders, or a single region in the top bar, for a longer history.
- **Anomaly ≠ signal.** The anomaly score ranks how unusual a trade is against all trades, and large trades are unusual by construction: the median flagged row scores 0.96. The minimum-anomaly slider therefore filters little.
- **Rows without a per-trade score.** Grants, option exercises, tax withholding and gifts, and cluster summary rows, show the security-level score instead: the issuer's anomaly score or its rules-based signal / 100. Rows with no score at all are never hidden by the slider.
- **Summary counts include cluster rows.** "Open-market buys" adds the totals of cluster summary rows, so it is not a count of single filings.
- **Coverage.** Korea needs an OpenDART key and France has no transactions. The UK depends on parsing free-text notices. Brazil is monthly and consolidated. Values in local currency are converted to USD; a missing price gives "$0" or "—".
- **10b5-1 and routine trades** are down-weighted in the rules-based score but can still score high on anomaly.
- **"Insider model P(outperform)"** in Why flagged comes from the XGBoost model that failed its gate (see Validation status).

Related mnemonics and BQL fields

- Mnemonics: none opens the radar; open it from the menu or the command palette. For one company use Company 360 › Insiders (`HDS` opens Ownership, which includes the insider timeline).
- BQL fields:
  - Activity: `ins.net_buy_value_usd_90d`, `ins.net_buy_mcap_bps_90d`, `ins.n_buyers_90d`, `ins.n_sellers_90d`.
  - Flags: `ins.cluster_buy`, `ins.c_suite_buy`, `ins.cluster_buy_flag`, `ins.c_suite_buy_flag`.
  - Opportunistic value: `ins.opportunistic_buy_value_usd_90d`.
  - Scores: `ins.anomaly_score`, `ins.anomaly`, `ins.signal_score`, `ins.xgb_prob_up` (failed its gate, not used in the score).
- Alerts: "insider buy" (see Watchlists and alerts).

- Related sections: Company 360 Insiders, Screener › Insider factor, Smart Money overview (the insider group of the combined score), Methodology appendix.

8. Global events and news

This chapter covers two pages that track what is happening in the world and in the news:

- **Global Events** (`/events`) turns the GDELT event and article stream into labelled, compact coverage by country, sector, company and market theme. It adds model-based layers for escalation risk, a geopolitical-risk index, contagion between markets and an event-study library.
- **News Terminal** (`/news` , `/news/top`) is a keyboard-first live headline terminal that groups articles into stories and offers watchlist and story-velocity alerts.

> **Note:** Every page header has a **?** button next to the title that opens the matching section of this guide.

Global Events

Route: `/events` (tabs `?tab=overview` , `risk` , `contagion` , `studies`) · **Menu:** Global Events · **Mnemonics:** `NET` (no security: Contagion & spillovers tab) · **Refresh:** nightly (labelled coverage, shocks, hotspots, change points, escalation, GPR, connectedness); weekly (spillovers, impulse responses, event-study library)

Purpose

Global Events answers three questions: *where is the world's news attention right now, and on which themes*; *is a situation escalating or spreading to other markets*; and *how have markets historically reacted to similar shocks*. It is used by macro and country analysts for situational awareness, by sector analysts to see which themes are hitting their names, and by anyone preparing an event study in Model Studio. Every number on the page is either a count from labelled news coverage (descriptive) or the output of a statistical model, which carries a **MODEL-BASED** badge.

Screenshot

The page opens on the Overview tab. Figure 8.1 in Overview shows the full page; the page-level elements are:

- **Header:** the title, the subtitle "Labelled GDELT coverage (41 themes) and clustered news stories by country, sector and company · model-based risk, contagion and event-study layers", and the line **News coverage: 2024-09-21 → 2026-10-15 · 738 days with labelled coverage (backfill in progress adds older history)**. This line tells you how much history the page can use; it grows as the historical backfill continues.
- **Tabs:** Overview, Escalation & GPR, Contagion & spillovers, Event studies. The selected tab is kept in the URL (`?tab=risk`), so a link opens the same tab.

Controls

Control	What it does
? (next to the title)	Opens this section of the user guide.
Tabs	Switch between Overview, Escalation & GPR, Contagion & spillovers and Event studies. The URL parameter `tab` follows the selection.
`?country=XX` in the URL	Pre-selects a country (ISO2) for the Overview story list, as when you click a country on the map.

The controls of each panel are described in the tab sections.

Methodology

Labelling, then forgetting. Every 15-minute GDELT 2.0 file (Events, Mentions and GKG articles) is streamed into memory, parsed and **kept only if it gets at least one label**; the raw files are never written to disk. The labels are:

- **Securities / issuers:** organisation names and the page title are matched to issuers through the app's alias table (23.7k aliases for 19.1k issuers), with context and block-list rules. Only links with confidence ≥ 0.5 are kept.
- **Country:** ISO2 from the action geography, the actors and the article locations (GDELT FIPS codes converted to ISO2), or **GLOBAL** when no country applies.
- **Sector:** from the linked issuers, plus a theme → sector map (for example oil and gas → Energy).
- **Market theme:** 41 market-relevant themes in 11 groups, stored as a bitmask:

Group	Themes
macro	Inflation & prices; Rates & central banks; Fiscal policy & taxes; Sovereign debt & IMF; Currencies & FX; Growth, recession & outlook; Jobs & labour market
geo	Trade & tariffs; Sanctions & embargoes; Armed conflict & military; Terrorism; Geopolitical tension & diplomacy
politics	Elections & politics; Coups & regime change
social	Strikes & labour action; Protests & civil unrest
corporate	M&A & deals; Earnings & guidance; Bankruptcy & default; Litigation & legal; Fraud, corruption & scandals; Product recalls & safety
markets	Stock market, IPOs & capital raising; Banking & financial stress; Crypto & digital assets
policy	Regulation & antitrust; Nationalization & state intervention; Foreign investment & industrial policy
tech	Cyber attacks & outages; Technology, AI & chips
risk	Natural disasters & extreme weather; Climate & energy transition; Pandemics & epidemics; Supply chain & shipping
sector	Healthcare & pharma; Housing & real estate; Aviation & transport disruption
commodities	Oil, gas & fuel prices; Power & utilities; Metals & mining; Agriculture & food

Storage tiers. Daily aggregates per country, sector and issuer × theme (counts, tone, Goldstein scale, mentions, attention z) are kept forever. Labelled event and article rows are kept in full for 180 days, then only the salient ones. **Stories** compress the news: near-duplicate headlines are clustered (MinHash, Jaccard ≥ 0.6 within a 72-hour linking window) into one row per story with its canonical headline, first and last seen times, article and source counts, mean tone, labels and up to three representative URLs. Sentiment (FinBERT for English, XLM-R for other languages) is computed once per story.

Attention z and tone z. For each label × theme × day:

```
attention_z = (ln(1 + events + articles) − mean₉₀) / sd₉₀
tone_z      = (tone − mean₉₀) / sd₉₀
```

where $mean_{90}$ and sd_{90} are taken over the label's previous 90 daily observations (at least 10 required; the current day is excluded). A **shock** is a day with attention z ≥ 2 or $|\text{tone z}| \geq 2$.

Data sources and refresh

- **GDELT 2.0** Events, Mentions and GKG (open data, data.gdeltproject.org), pulled nightly by `gdelt_stream`; the weekly profile catches up to 10 days at a time.
- **Other news** (Polygon, RSS, GDELT DOC titles) through `news_ingest`, merged into the same story layer by `story_cluster`; SEC 8-K and exchange announcements as corporate news by `news_filings`.
- **Model jobs:** nightly `event_hotspots`, `event_changepoints`, `event_escalation`, `event_gpr`, `event_connectedness`, `event_exposure`; weekly `event_spillovers`, `event_irf`, `event_study_library`.
- **Linkage data** for spatial weights: Natural Earth borders and distances, IMF and UN Comtrade trade, OECD ICIO input-output, BIS banking claims, SEC geographic revenue segments.

- **Coverage:** labelled history from 2024-09-21 at the time of writing. The deep backfill continues newest-first towards 2015-02-19 (the start of GDELT 2.0), so older history appears over time without any action from you.

Validation status

The labelling pipeline was spot-checked on the 90-day pilot (EVENTS_V2_PLAN E5): **security links 82% correct (49 of 60), country 84%, theme 68%, sector about 68%**. This is a precision spot check, not a statistical gate. The validation of each model layer is given in its tab section.

Correct usage and worked example

Start on Overview to see where attention is concentrated, open the stories behind it, then move to Escalation & GPR or Contagion & spillovers only if you need a model view of persistence or spread. Use Event studies to put a number on how markets reacted to similar shocks in the past, and send the dates to Model Studio when you want a proper study on your own entity. A walk-through for each tab follows in its section.

Limitations and pitfalls

- Coverage measures **media attention**, not the importance of an event. Large, English-language and US news systems dominate GDELT.
- One label in three (themes) and one in six (security links) is wrong in the spot check, so treat a single story's labels as indicative.
- The z-scores need 90 days of history. Near the start of the coverage window, or for thinly covered labels, they are missing ("n/a").
- **Licensing:** GDELT is open. Headlines and URLs are kept for research display only; article text is never stored.

Related mnemonics and BQL fields

- Command bar: `NET` with no security opens the Contagion & spillovers tab; with a company it opens Company 360 › Network & Geo. See Mnemonic reference.
- BQL: `attention_z` (GDELT media attention z of a security, snapshot only), `sent.news_count_z`, `sent.gdelt_tone_ewm_7d`, `sent.sentiment_ewm_7d` (see Field dictionary).
- Company and country event panels: Company 360 › Event risk and the country pages in Macro › Countries.

Overview

Route: `/events?tab=overview` · **Menu:** Global Events › Overview · **Refresh:** nightly

Purpose

The Overview tab is the situational-awareness view: a world map of event intensity and tone by country with a time slider, a feed of attention and tone shocks with model-based change points, the top stories ranked by salience, and a country × theme heatmap. Clicking a country on the map or the heatmap filters the story list, so you can go from "Russia is lit up" to the headlines behind it in one click.

Screenshot

Figure 8.1: Global Events, Overview tab (`/events?tab=overview`), window 13-19 Oct 2026, data as of 19 Oct 2026.

#	Panel	What it shows
1	World map	GDELT coverage by country for the selected window: intensity, attention z, tone or conflict share, with an optional hotspot overlay and a time slider of daily article counts.
2	Shocks feed	Labels with attention z or tone breaks ≥ 2 in the last 14 days, then model-based change points of the last 30 days.
3	What's moving	Top stories by salience in the window, with filters for country, sector, ticker, theme, tone and minimum sources.
4	Country × theme heatmap	Maximum daily attention z per country and theme in the window (or relative attention when z is not available).

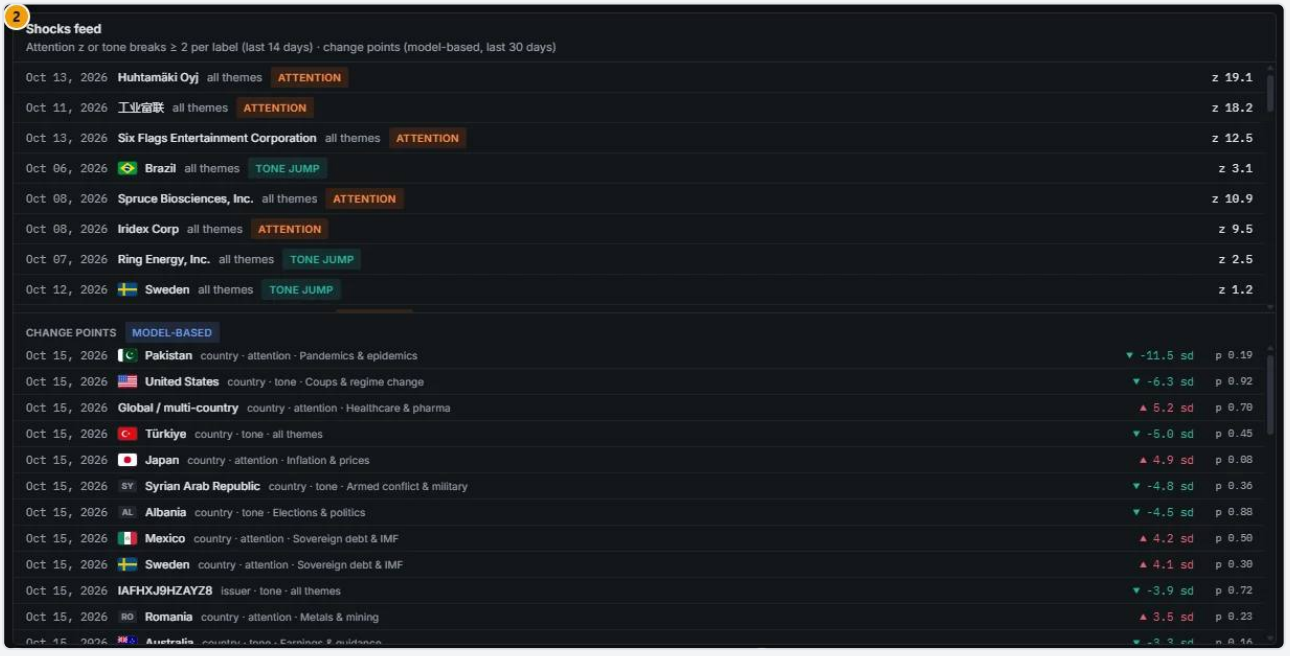


Figure 8.2: Shocks feed (top) and change points (bottom).

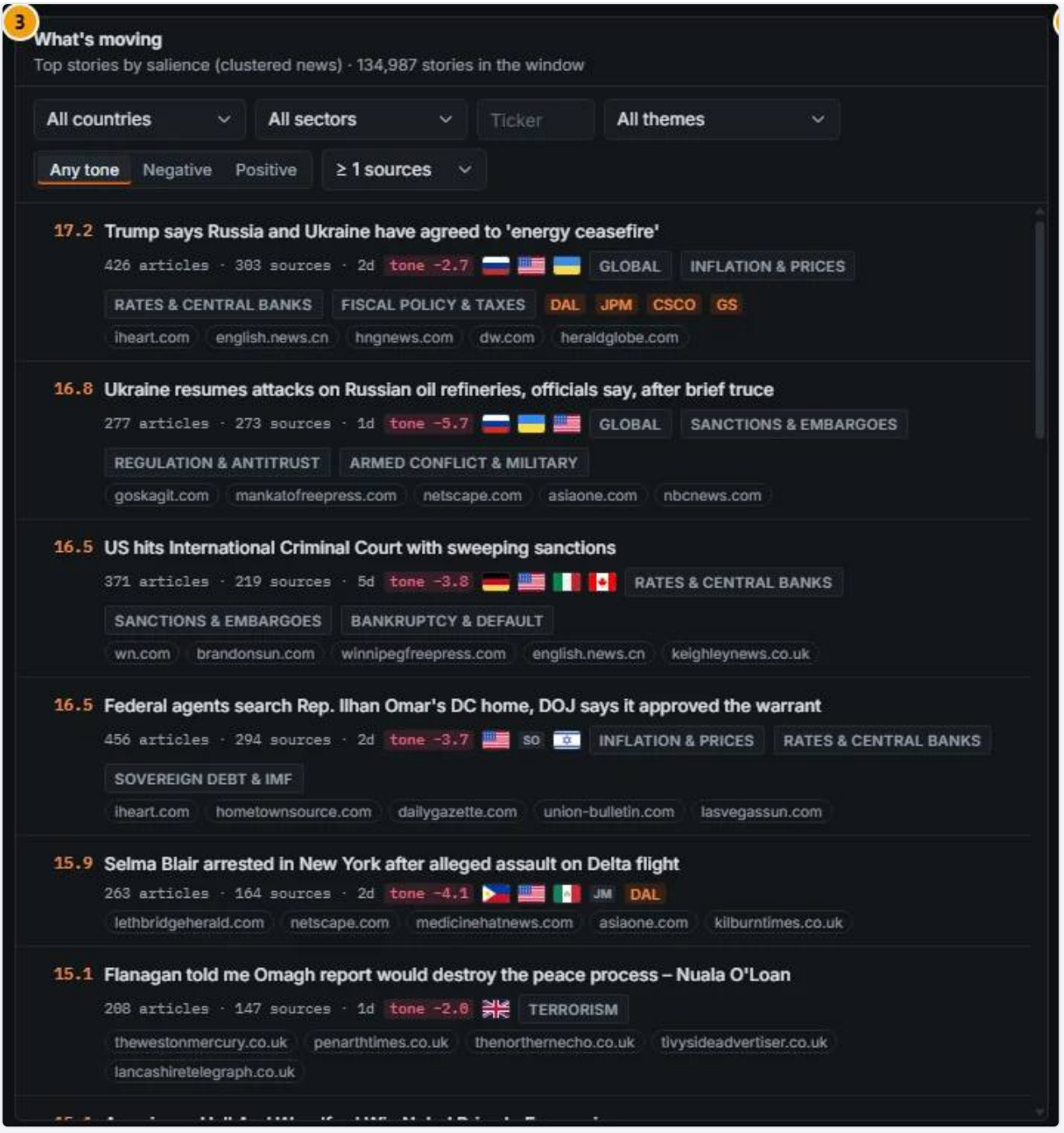


Figure 8.3: What's moving, 7-day window; 134,987 stories in the window.

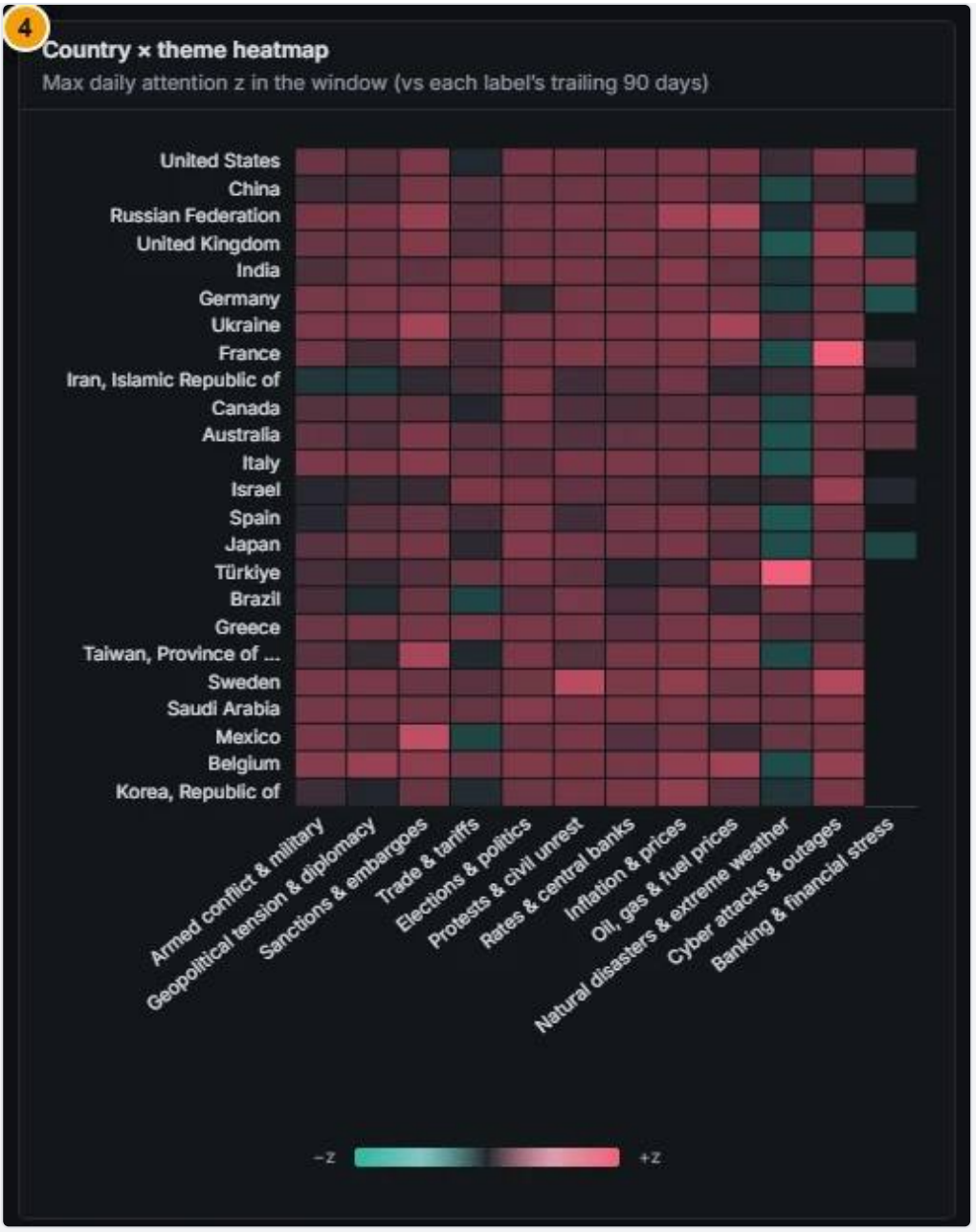


Figure 8.4: Country × theme heatmap (maximum daily attention z vs each label's trailing 90 days).

Controls

WORLD MAP (PANEL 1)

Control	What it does
History range `7D` `30D` `90D` `1Y` `3Y` `All`	Sets the map window (ending on the snapshot's as-of date) and the span of the time slider (at least 90 days). Ranges above one year use weekly bars. `All` starts at the first covered date (maximum 2,600 days).
Themes (multi-select, "All themes")	Restricts the map, the story list and the heatmap to up to 12 themes, grouped by taxonomy group. "All themes" clears the selection. With one theme selected, the story list and hotspot overlay use it; with two or more, the heatmap shows them as columns.
Map metric `Intensity` `Attention z` `Tone` `Conflict share`	Chooses what colours the countries (see Methodology).
Hotspots switch (MODEL badge)	Overlays Getis-Ord Gi* hot and cold spots (fill colour, 90 / 95 / 99%) and emerging-hotspot markers. A caption gives the theme, measure, week, global Moran's I with its p-value and the spatial weights used.
Click a country	Filters What's moving to that country; click it again to clear. Drag to pan and scroll to zoom (1× to 8×).

Control	What it does
Hover a country	Tooltip with intensity, maximum attention z, tone, conflict share and events / articles; with the overlay on, also the Gi* class and z, the LISA quadrant and p-value, and the emerging-hotspot class.
Time slider (two thumbs)	Selects a custom window inside the slider span; the bar chart above it highlights the selected days. Releasing a thumb applies the window to the map, the stories and the heatmap. The text on the right shows the dates and the number of days.

When attention z is selected but not yet available for the window, a "no data" badge explains that the z baseline needs 90 days of history.

SHOCKS FEED (PANEL 2)

The upper list shows up to 50 shocks of the last 14 days: date, flag, label name (country, sector or company), theme, a kind badge (**ATTENTION**, **TONE DROP**, **TONE JUMP**) and a z value. The lower list (**CHANGE POINTS**, MODEL-BASED) shows up to 40 change points of the last 30 days: date, label, label type, stream (attention or tone), theme, the shift in standard deviations of the previous regime (▲ up, ▼ down, ■ flat) and the change-point probability `p`. Both lists scroll.

WHAT'S MOVING (PANEL 3)

Control	What it does
Country	All countries, GLOBAL, or one country. Kept in sync with the map click.
Sector	All sectors or one of the 11 sectors.
Ticker	A security id or ticker; only stories linked to it are shown.
Theme	All themes (or "Map theme" when one theme is selected on the map) or one theme.
Tone `Any tone` `Negative` `Positive`	Negative = mean tone ≤ −2; Positive = mean tone ≥ +2.
Minimum sources `≥ 1` `3` `10` `25` `50`	Hides stories carried by fewer distinct sources (applied in the browser to the 60 stories fetched).
Story headline	Opens the **story drawer** (below).
Ticker chips	Open the linked company's Company 360 on the Event risk tab. Hover for the name and link relevance.

Each row shows the salience score (left), the headline, articles · sources · days, a tone chip (rose ≤ −2, teal ≥ +2, grey otherwise), up to four country flags, three theme badges, up to four linked tickers and up to five source domains. The list shows the top 25 rows.

STORY DRAWER

Clicking a headline opens a side drawer with: the article, source and GDELT event counts; the tone chip and, when available, the story sentiment marked "(model)"; first and last seen times; an **Articles per hour** chart; **Linked securities** with their relevance (each opens Company 360 › Event risk); **Sources** (domains with article counts); **Languages**; **Labels** (country codes link to the country macro page, plus sector and theme badges); and **Representative articles** (up to three external links). Close it with the × button or Esc.

COUNTRY × THEME HEATMAP (PANEL 4)

The rows are the 24 countries with the most articles in the window (GLOBAL excluded); the columns are the themes selected on the map when two or more are selected, otherwise 12 default themes (armed conflict, geopolitical tension, sanctions, trade, elections, protests, rates and central banks, inflation, oil and gas, natural disasters, cyber, banking stress). Hover a cell for the value and the article count; click a cell to filter the story list to that country. A **z pending** badge appears when the heatmap falls back to relative attention.

COMPANIES AFFECTED AND LABEL PANELS

Stories carry their linked securities, so the ticker chips in What's moving and the drawer's Linked securities list are the page's "companies affected" view. For a single company or country, the same data is presented as a **label event panel**: Company 360 › Event risk (title "Company events") and each country macro page (title "Events") show a coverage and tone chart, the theme mix, recent shocks and the top 15 stories for that label, with its own range selector (`7D` to `All` , default 90D) and an **Open Global Events** link. See Company 360 › Event risk.

Methodology

Map metrics (country rows of the daily aggregates over the window, all themes or the selected ones):

```
intensity      = ln(1 + events + articles) / max over countries of ln(1 + events + articles)   (0-100%)
attention z    = maximum daily attention_z in the window
tone           = mean GDELT tone (article tone, else event tone), weighted by items
conflict share = events with CAMEO quad class 3-4 (verbal or material conflict) / all events
```

Hotspot overlay (MODEL). Weekly, per theme and measure (attention, tone or intensity; the conflict-share metric uses attention), the job `event_hotspots` computes on the country map with queen-contiguity weights (islands linked to their nearest neighbour):

- global **Moran's I** and local **LISA** with 499 permutations (quadrants HH / LL / HL / LH kept at $p < 0.05$);
- **Getis-Ord Gi*** z-scores, classed as hot or cold spots at 90%, 95% and 99% ($p < 0.10, 0.05, 0.01$);
- ESRI-style **emerging-hotspot** classes over the weekly history (new, consecutive, intensifying, persistent, diminishing, sporadic, oscillating, historical), using a Mann-Kendall trend test at $p < 0.05$. Markers on the map use one symbol per class, rose for hot and blue for cold.

Shocks feed. Label-days in the last 14 days with attention z ≥ 2 or |tone z| ≥ 2 and at least 20 items (5 for a company). The kind is *attention* when attention z ≥ 2 and at least as large as |tone z|, otherwise *tone drop* or *tone jump*. Sorted by the larger of attention z and |tone z|.

Change points (MODEL-BASED). Bayesian online change-point detection (Adams and MacKay 2007) on the daily attention and tone streams of each country, sector and the 300 most-covered issuers, per theme, over the last 365 days. The model is Gaussian with unknown mean and variance (Normal-Gamma prior) and a constant hazard of 1/100 (an expected regime length of 100 days). A change point is reported with its probability, the shift in standard deviations of the previous regime and its direction (|shift| < 0.5 sd is "flat").

Salience of a story (model-free):

```
salience = ln(1 + articles) + ln(1 + sources) + 0.5 · ln(1 + GDELT events) + |tone| / 5 + link bonus
link bonus = 1.0 if a security link has relevance ≥ 80, 0.5 for any weaker link, else 0
```

Heatmap. Each cell is the maximum daily attention z of that country and theme in the window. When no z exists for the window, the cell is the **relative attention** $\log_2(\text{(country's theme articles / all theme articles)} / \text{(country's articles / all articles)})$, so positive means the country is over-represented in that theme.

Data sources and refresh

Daily aggregates and stories from the nightly GDELT and news jobs (see Global Events); hotspots and change points nightly. The default 7D window ends on the snapshot's as-of date (19 Oct 2026 on the screenshot, window 13-19 Oct), which can be a few days after the last covered GDELT day (15 Oct 2026 in the coverage line), so the last days of a window may be partial.

Validation status

- Map, shocks, stories, salience and heatmap are **descriptive** (counts and z-scores); no validation gate applies. Label precision: see the spot check under Global Events.

- **Change points** (MODEL-BASED): calibrated on the detector's own tests at about **0.13 false alarms per 200 days and 97% detection of 1.5 sd shifts** (EVENTS_MODELS_REALTIME_PLAN C2). Not validated as a predictor of market moves.
- **Hotspots** (MODEL): permutation inference re-implemented to match the reference `esda` library; a statistical description of spatial clustering, not a forecast. No gate.

Correct usage and worked example

Use the map to find where attention is unusual, not where it is large: the United States always has the highest intensity, so switch to **Attention z** to see what is abnormal for each country.

Worked example (7D window, 13-19 Oct 2026). On **Intensity** the United States is at 100% (158,069 events and 90,865 articles). Switching to **Attention z** shows the Russian Federation at 1.66 against 0.93 for the United States, with a tone of −2.86 and a conflict share of 37%. Clicking Russia filters What's moving; the top story overall that week is "Trump says Russia and Ukraine have agreed to 'energy ceasefire'" with salience 17.2 (426 articles, 303 sources, 2 days, tone −2.7). Its drawer lists the linked countries (RU, US, UA, GLOBAL) and the representative articles. The same story also carries the tickers DAL, JPM, CSCO and GS: these come from the issuer linker and should be checked against the articles before you treat them as affected companies.

Limitations and pitfalls

- **Intensity is relative to the largest country in the window**, so it changes when you change the window or the themes; compare countries within one view only.
- In the shocks feed, the number on the right is the **attention z when it exists, even on a tone row**; the tone z is shown only when attention z is missing.
- Change-point arrows are coloured by direction only (▲ rose, ▼ teal). A ▼ on a *tone* stream is a deterioration even though it is teal.
- Small companies produce very large attention z from a handful of articles (for example z 19.1 for Huhtamäki on 13 Oct 2026 from 4 events and 4 articles). The feed requires at least 5 items for a company, which is still few.
- Theme labels are broad: a single story can carry 10 or more themes (the ceasefire story above has 19), so theme filters and heatmap cells over-count multi-topic stories.
- Ticker links are the weakest label in the spot check after themes; a story linking JPM or GS to a ceasefire headline is usually a mention, not an exposure.
- The minimum-sources filter is applied to the 60 stories fetched, so it can return fewer than 25 rows.

Related mnemonics and BQL fields

- `EVTS AAPL` (or `AAPL EVTS`) opens the company's Event risk tab with its label panel; see Mnemonic reference.
- BQL: `attention_z`, `sent.news_count_z`, `sent.gdelt_tone_ewm_7d`.

Escalation & GPR

Route: `/events?tab=risk` · **Menu:** Global Events › Escalation & GPR · **Refresh:** nightly (`event_escalation`, `event_gpr`; published GPR downloaded weekly)

Purpose

This tab holds two model-based risk views. **Escalation probabilities** ask whether the recent burst of events in a country and theme group is likely to continue above its usual level in the next 7 or 30 days. The **geopolitical risk index (GPR)** gives a daily, country-level index of geopolitical coverage, compared with the published Caldara-Iacoviello index. Both are for monitoring and for ranking where to look; neither is a trading signal.

Screenshot

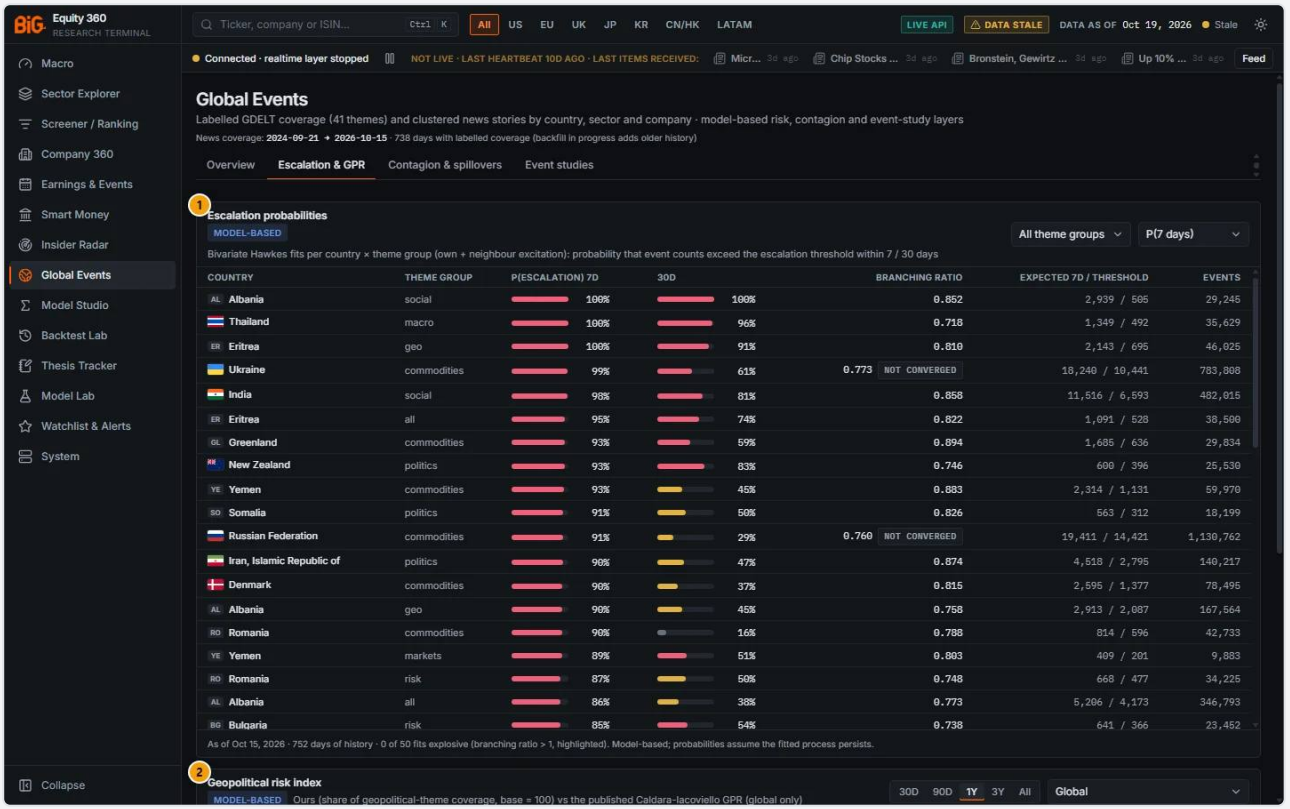


Figure 8.5: Escalation & GPR tab (`/events?tab=risk`), as of 15 Oct 2026.

#	Panel	What it shows
1	Escalation probabilities	Bivariate Hawkes fits per country × theme group: P(escalation) within 7 and 30 days, branching ratio, expected 7-day count vs threshold, total events.
2	Geopolitical risk index	Our daily GPR (daily and 7-day average) against the published Caldara-Iacoviello GPR, with correlations and the countries with the highest 7-day GPR.

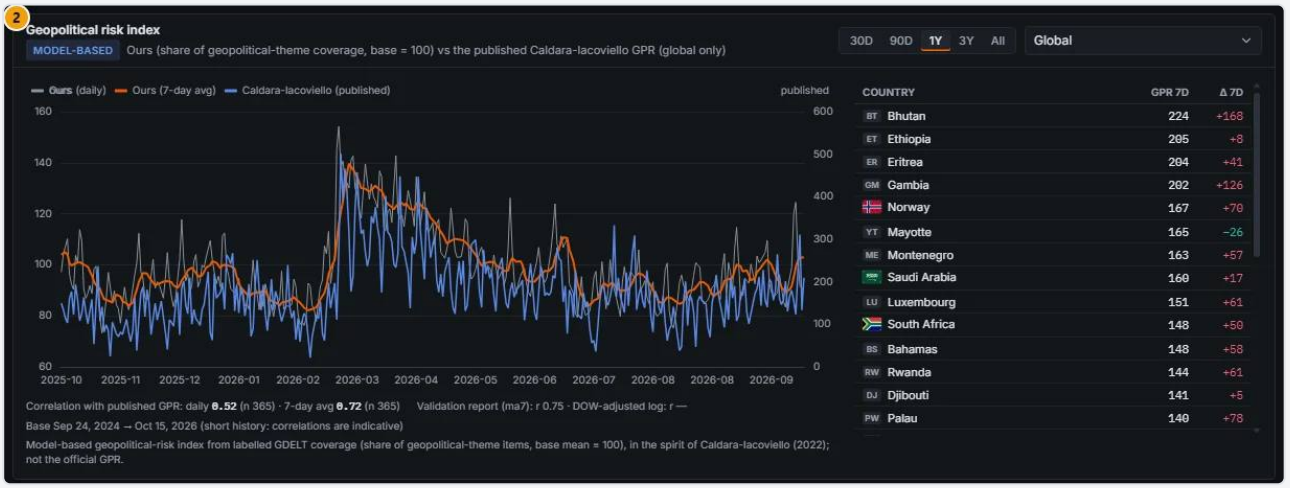


Figure 8.6: Geopolitical risk index, Global, 1Y range.

Controls

Control	What it does
Theme group (Escalation)	All theme groups (also lists the fits on all themes together, group "all"), or one group: commodities, corporate, geo, macro, markets, policy, politics, risk, sector, social, tech.
Sort (Escalation)	`P(7 days)` (default), `P(30 days)` , `Branching ratio` , `Expected 7d` . The table shows the top 50 fits with at least 100 events.

Control	What it does
explosive badge	Shown when the branching ratio is above 1; hover: "Raw branching ratio > 1: the self-exciting fit is explosive (non-stationary), so the probabilities are not reliable." The row is highlighted.
not converged badge	The estimation did not converge; read the row with extra caution.
GPR history range `30D` `90D` `1Y` `3Y` `All`	Chart window (default 1Y).
Country (GPR)	Global (default) or a country from the top-30 list. Clicking a row of the country table does the same.
Chart legend	Toggle *Ours (daily)*, *Ours (7-day avg)* and *Caldara-Iacoviello (published)*.

The footer of the escalation table gives the as-of date, the days of history and the number of explosive fits. Under the GPR chart, the panel shows the correlation of our index with the published one for the visible window (daily and 7-day average, with n), the latest validation-report figures, the base period and the full model label.

Methodology

ESCALATION (BIVARIATE HAWKES PROCESS)

For each country and theme group, daily event counts are modelled as a self-exciting process with a second series for the country's neighbours:

```
λ_k(t) = μ_k · s_k(day of week) + Σ_j α_kj · Σ_{l≥1} g(l) · N_j(t − l)
g(l)   = (1 − e^(−β)) · e^(−β(l − 1))
```

- `μ` is the background rate with a day-of-week factor (the weekly news cycle).
- `α_kj` is the expected number of type-k events triggered by one type-j event (own country and neighbours). The **branching ratio** is the spectral radius of the α matrix: below 1 the process is stable; above 1 it is explosive.
- Estimation by EM on the latent branching structure, with β chosen by profile likelihood on a grid.
- **Escalation probability:** 2,000 Monte-Carlo continuations of the fitted process from today's state, with negative-binomial over-dispersion. The **threshold** is the 90th percentile of the country's own historical rolling 7-day (30-day) counts. P(escalation 7d) is the share of paths whose next-7-day count exceeds the threshold; likewise for 30 days. **Expected 7d** is the mean path.

GEOPOLITICAL RISK INDEX

```
share_t = items with geopolitical themes / all labelled items (events + articles of the country-day)
GPR_t   = 100 · share_t / mean(share over the base window)
```

- Geopolitical themes: *threats* (geopolitical tension, sanctions, trade and tariffs) and *acts* (armed conflict, terrorism, coups and regime change); both sub-indices are also computed.
- Base window: calendar 2021 when covered; until then, all days covered so far (currently 24 Sep 2024 → 15 Oct 2026).
- The table shows the 7-day average (GPR 7D) and its change over 7 days (Δ 7D).
- The published Caldara-Iacoviello daily (GPRD) and monthly country (GPRC) indices are downloaded weekly as the reference.

Data sources and refresh

Labelled GDELT daily aggregates (country × theme); neighbours from the Natural Earth contiguity weights. Published GPR from matteoiacoviello.com (cached, refreshed weekly). Both models rerun nightly after `events_agg`.

Validation status

- **Escalation probabilities: not formally validated.** There is no out-of-sample calibration test of the 7- and 30-day probabilities. The app labels the panel **MODEL-BASED** and states in its footer: "Model-based; probabilities assume the fitted process persists." Explosive fits are flagged and should be disregarded.
- **GPR: model-based, compared with the published index; no pass / fail gate.** The panel's label reads: "Model-based geopolitical-risk index from labelled GDELT coverage (share of geopolitical-theme items, base mean = 100), in the spirit of Caldara-Iacoviello (2022); not the official GPR." The validation report of 20 Oct 2026 (window 24 Sep 2024 → 15 Oct 2026, 735 days) gives:

Comparison	Result
Daily global level vs GPRD	Pearson 0.323, Spearman 0.188 (n 735)
7-day average vs GPRD	Pearson 0.747, Spearman 0.562 (n 729)
Threats / acts sub-indices	Pearson 0.320 / 0.376
Monthly cross-section of 44 countries vs GPRC	Spearman 0.83 to 0.92 each month (23 months)
Monthly time series per country vs GPRC	median Pearson 0.56; from −0.03 (Hong Kong) to 0.90 (Denmark)

In short: our index ranks countries much like the published one and tracks its weekly swings reasonably, but day-to-day values are noisy and some countries (Japan 0.18, China 0.21, Hong Kong −0.03) track poorly. The panel itself warns "short history: correlations are indicative".

Correct usage and worked example

Read the escalation table as a ranked watch-list of *where coverage is running hot relative to its own history*, then check the stories on the Overview tab. Compare **Expected 7d** with the **threshold** before reading the probability.

Worked example (as of 15 Oct 2026, theme group geo). Eritrea shows P(escalation 7d) 99.7% (displayed 100%), P(30d) 90.7%, branching ratio 0.810, expected 7-day count 2,143 against a threshold of 695, from 46,025 events. The fitted process expects about three times its own 90th-percentile weekly count, so almost every simulated path crosses the threshold: the model says the recent burst is large and self-sustaining, not that a conflict will escalate. Probabilities near 100% with expected counts far above the threshold (Albania social: 2,939 vs 505 on the screenshot) usually reflect a recent spike in coverage of a small country.

For GPR, select **Global** and **1Y**: on 15 Oct 2026 our 7-day average is 102.9 (daily 90.8), and over the visible year the correlation with the published index is 0.52 daily and 0.72 for the 7-day average (n 365). The marked jump in early March 2026 appears in both series.

Limitations and pitfalls

- Escalation probabilities are **not calibrated over history** and must not be read as "X% chance of escalation". They describe the fitted counting process of news coverage.
- The threshold is relative to each country's own history, so small countries with low baselines dominate the top of the table.
- Fits marked *not converged* (for example Ukraine and Russia commodities on the screenshot) or *explosive* are unreliable.
- GPR is a share of coverage: a quiet news day elsewhere raises it; translation and source coverage changes in GDELT move it.
- The country table is led by small countries with volatile shares (Bhutan 224 with +168 in a week); use the 7-day average and a longer range before drawing conclusions.
- The published index is global only on a daily basis; country lines have no daily reference.

Related mnemonics and BQL fields

No mnemonic opens this tab directly; type `Global Events` in the command bar or use the menu. There are no BQL fields for escalation or GPR.

Contagion & spillovers

Route: `/events?tab=contagion` · **Menu:** Global Events › Contagion & spillovers · **Mnemonics:** `NET` (no security) · **Refresh:** connectedness nightly; spillovers and impulse responses weekly

Purpose

This tab measures how shocks travel between markets. The **contagion network** shows which equity markets or sectors transmit volatility and returns to others. **Spatial spillovers** estimate whether news attention in neighbouring countries moves a country's market beyond the effect of its own news. **Impulse responses** trace the response of returns or volatility to a news-attention shock over 10 days. It is used by cross-asset and country strategists to judge whether a local event is likely to stay local.

Screenshot

Figure 8.7: Contagion & spillovers tab (`/events?tab=contagion`), countries · returns, 16 Oct 2026.

#	Panel	What it shows
1	Contagion network	Diebold-Yilmaz connectedness network: net transmitters (rose) and receivers (blue), arrows for the strongest net pairwise spillovers, total connectedness history and a To / From / Net table.
2	Spatial spillovers	OLS / SLX / SAR / SEM / SDM panel models of event attention on country responses: direct, indirect (spillover) and total effects, diagnostics, and top transmitters and receivers.
3	Impulse responses	Response of returns or volatility to a 1-sd event-attention shock: own and spatial components with 95% bands.

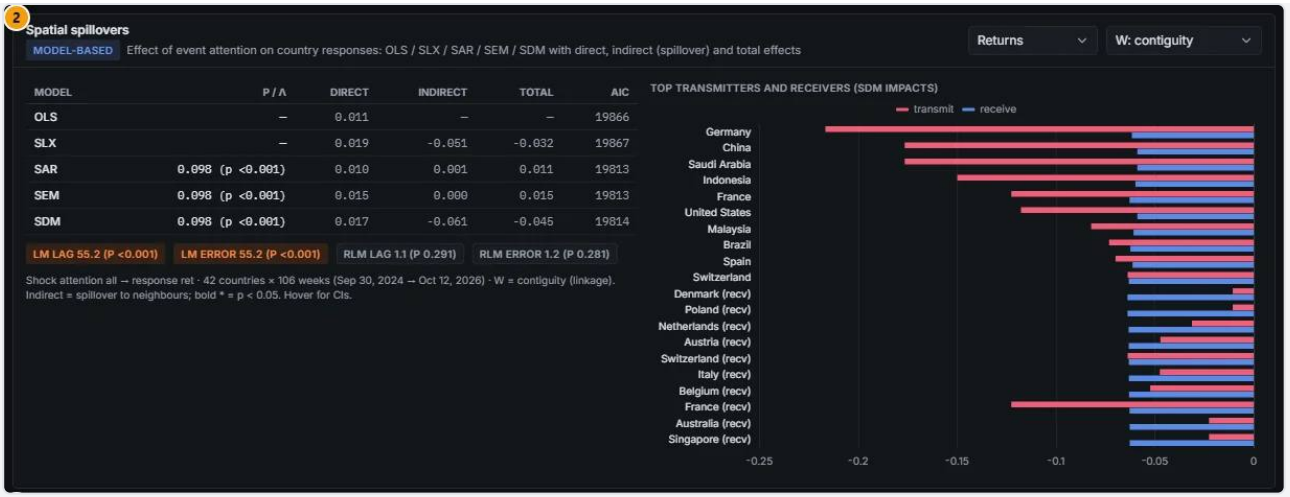


Figure 8.8: Spatial spillovers, response returns, W contiguity.

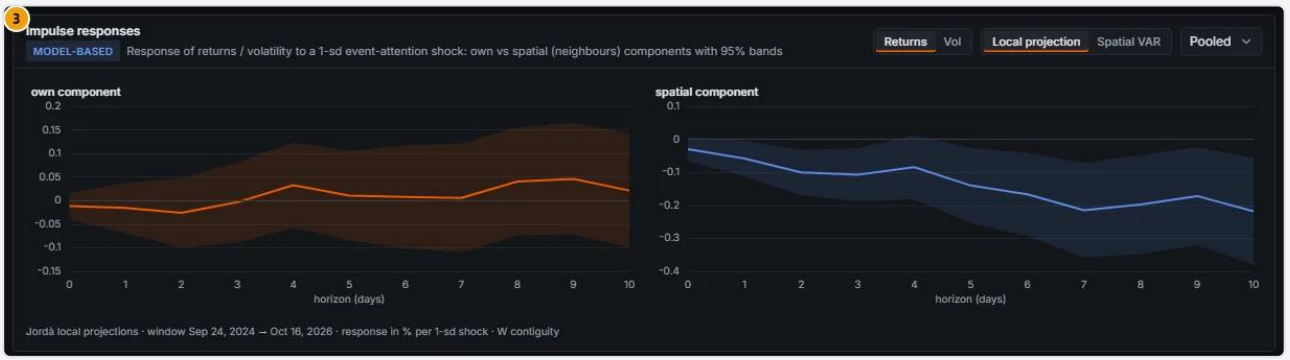


Figure 8.9: Impulse responses, returns, local projection, pooled.

Controls

Control	What it does
Universe `Countries · returns` `Countries · vol` `Sectors · returns` `Sectors · vol`	Network data set: 42 country equity indices or 11 US sector ETFs, on daily returns or on range-based volatility.
Network	Drag nodes to rearrange, scroll to zoom; hover a node for to / from / net, hover an arrow for the net pairwise spillover in percentage points. Node size is proportional to the absolute net value.
Response (spillovers)	Returns, Volatility or Macro surprise.
Weights (spillovers)	`W: contiguity` or `W: inverse distance` (the spatial weight matrix).
Effect cells	Bold with * when p < 0.05; hover for the confidence interval and p-value.
Response `Returns` `Vol` (impulse responses)	Which response to show.
Model `Local projection` `Spatial VAR`	Panel local projections or a pooled spatial VAR(1).
Scope	Pooled (all countries) or one of the eight largest markets (spatial VAR).

Methodology

Connectedness (Diebold-Yilmaz). A rolling VAR(1) with ridge shrinkage (0.05) on a 200-day window, moved forward 5 days at a time from 2012, with a 10-day horizon generalised forecast-error variance decomposition. For each market:

```
to others   = share of other markets' forecast-error variance due to this market's shocks
from others = share of this market's variance due to other markets
net         = to − from      (> 0: transmitter, < 0: receiver)
total       = average off-diagonal share (%), shown as "Total connectedness"
```

Volatility uses the Parkinson high-low range estimator. Arrows are the strongest net pairwise spillovers.

Spatial spillovers. A country × week panel (daily when the data allow) with two-way fixed effects. The shock is event attention (all themes); the response is the country index return, its range volatility or the change in its macro surprise index. Models:

- **OLS-FE** (no spatial term), **SLX** (adds neighbours' shock W·x), **SAR** (spatial lag of the response, ρ), **SEM** (spatial error, λ), **SDM** (SAR plus W·x), estimated by maximum likelihood.
- Direct, indirect (spillover to neighbours) and total effects by the LeSage-Pace decomposition, with Monte-Carlo inference (500 draws).
- Lagrange-multiplier diagnostics: LM lag, LM error and their robust versions.
- The bar chart gives each country's transmit and receive spillover from the SDM impacts.

Impulse responses. Jordà panel local projections of returns or volatility on the own shock and the spatially lagged shock, with Driscoll-Kraay standard errors, for horizons 0 to 10 days; or a pooled spatial VAR(1) with simulation bands (300 draws). Responses are in % per 1-sd attention shock.

Data sources and refresh

Daily index and sector ETF prices from the price history (country indices from 2012); weekly country attention from the labelled GDELT aggregates; spatial weights (contiguity, inverse distance) from the linkage data (Natural Earth). Connectedness is updated nightly after prices; spillovers and impulse responses weekly.

Validation status

Not formally validated; MODEL-BASED. These are in-sample econometric estimates with standard inference (p-values, confidence bands, LM tests); the panel badges read MODEL-BASED. The estimators were checked to reproduce reference libraries (spreg ML lag for the spatial models, esda for Moran / LISA), which is a correctness check, not evidence of predictive value. No out-of-sample test of spillover forecasts exists.

Correct usage and worked example

Use connectedness to judge the *regime* (high total connectedness means diversification across markets is weak) and to see who leads; use spillovers and impulse responses to ask whether news attention spreads.

Worked example (countries · returns, 16 Oct 2026). Total connectedness is 84.67% (window 200 days, horizon 10, VAR(1), 42 nodes), close to the top of its range since 2023 apart from a brief fall to about 72% around the start of 2026. The largest net transmitters are South Africa (+59.1: to 150.4, from 91.2), Poland (+52.7) and Italy (+49.1); the largest receivers are Singapore (−54.6), Malaysia (−50.9) and Taiwan (−50.2). In the spillover panel (returns, contiguity, 42 countries × 106 weeks), the spatial parameter is 0.098 ($p < 0.001$) in SAR, SEM and SDM; the direct effect of attention is small (0.010 to 0.019) and the SDM indirect effect is −0.061. LM lag and LM error are both 55.2 ($p < 0.001$) but the robust versions are not significant (1.1 and 1.2), so the data cannot tell a spatial-lag model from a spatial-error model. The impulse responses show an own component close to zero at every horizon, and a spatial component drifting to about −0.2% by days 7 to 10 with a band mostly below zero.

Limitations and pitfalls

- Connectedness in returns is high almost everywhere in a global sell-off; a high total is a description of co-movement, not causality.
- Node colours depend on the sign of net, so a market near zero can flip between transmitter and receiver from one window to the next.

- Spillover models are estimated on about two years of attention data; effects are small and sensitive to the weight matrix. Compare contiguity with inverse distance before relying on a sign.
- Responses are for country indices and US sector ETFs only, not individual companies.
- Asian markets close before European news arrives, so daily return linkages partly reflect time zones.

Related mnemonics and BQL fields

- `NET` with no security opens this tab; `AAPL NET` opens Company 360 › Network & Geo. See Mnemonic reference.
- The same spatial methods are available in Model Studio (spatial lag, spatial error, spatial Durbin, Moran / LISA, connectedness, local projection): see Model Studio.

Event studies

Route: `/events?tab=studies` · **Menu:** Global Events › Event studies · **Refresh:** weekly (`event_study_library`)

Purpose

The event-study library records how markets moved after past labelled news shocks, by theme × country × sector, at three levels (country index, sector ETF, individual issuers). It provides an "expected reaction" reference for a new shock of the same kind. The second panel sends the shock dates of any country, sector or company to Model Studio as a ready-made event study, so you can run your own analysis on your own entity.

Screenshot

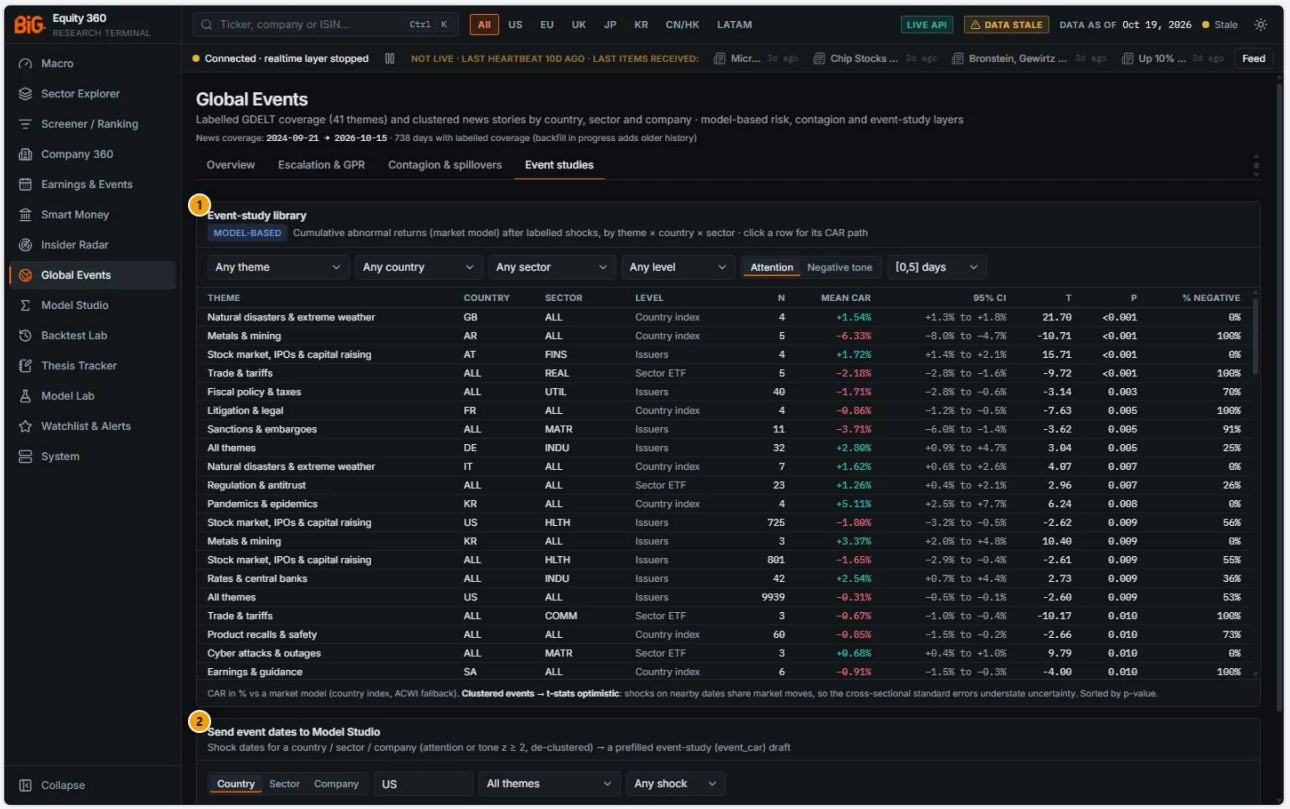


Figure 8.10: Event studies tab (`/events?tab=studies`), attention shocks, window [0,5].

#	Panel	What it shows
1	Event-study library	Cells of mean cumulative abnormal return (CAR) after shocks, with n, 95% CI, t, p and the share of negative CARs, sorted by p-value. Click a row for its CAR path.

#	Panel	What it shows
2	Send event dates to Model Studio	The de-clustered shock dates of a chosen country, sector or company, with a **Send to Model Studio** button that opens a prefilled event study.

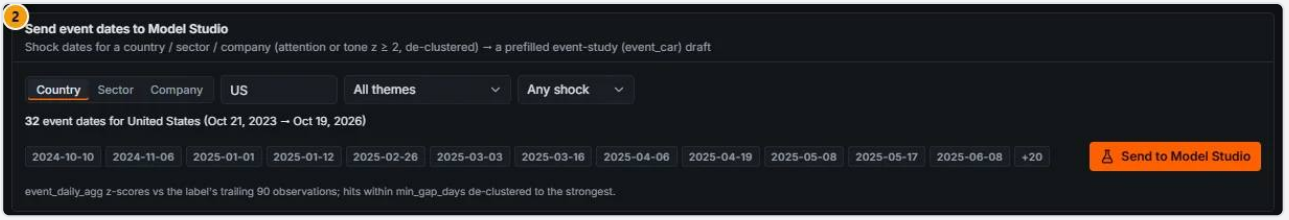


Figure 8.11: Send event dates to Model Studio, United States, all themes, any shock.

Controls

EVENT-STUDY LIBRARY

Control	What it does
Theme	Any theme, "All-theme shocks" (shocks on total coverage) or one theme.
Country	Any country, "Pooled (ALL)" or one country.
Sector	Any sector, "Pooled (ALL)" or one sector.
Level	Any level, Country index, Sector ETF or Issuers.
Shock kind `Attention` `Negative tone`	Which shock definition the cells use.
Window `[-1,5]` `[0,1]` `[0,5]` `[0,10]` days	The CAR window in trading days around the shock.
Click a row	Opens a **CAR path** panel below the table for that cell.

The table shows up to 100 cells with at least 3 events. The **CAR path** panel joins the stored windows [0,1], [0,5] and [0,10] into a line with a 95% band, shows [−1,5] as a separate diamond at day 5, labels each point with its number of events, and gives the shock dates covered and the benchmark.

SEND EVENT DATES TO MODEL STUDIO

Control	What it does
Label type `Country` `Sector` `Company`	Switches the entity type (defaults: US, TECH, AAPL).
Entity	ISO2 code, sector, or security id / ticker.
Theme	All themes or one theme.
Shock kind	Any shock, Attention, Tone drop, Tone jump.
Date badges	The first 12 event dates; "+N" for the rest.
Send to Model Studio	Opens Model Studio's builder with a new event-study draft (disabled when there are no dates).

Methodology

Library. For each labelled shock (attention or negative-tone z ≥ 2 on the label's daily aggregate), the abnormal return is measured with a market model and cumulated over the window:

- **Country index** level: country × theme shocks, country index against ACWI.
- **Sector ETF** level: sector × theme shocks, the US sector SPDR ETF against SPY.
- **Issuers** level: issuer shocks, the primary security against its country index (ACWI fallback); the theme is the issuer's dominant article theme that day, plus "all".

Per cell: number of events, mean and median CAR (%), standard error, 95% CI, t-statistic, p-value, mean buy-and-hold abnormal return and the share of negative CARs. The footer warns: "**Clustered events → t-stats optimistic**: shocks on nearby dates share market moves, so the cross-sectional standard errors understate uncertainty."

Event dates for Model Studio. Days on which the label's attention z (or tone z) reaches 2 against its trailing 90 observations, over the last 1,095 days; hits within 5 days of each other are de-clustered to the strongest, up to 200 dates. The app's caption reads: "event_daily_agg z-scores vs the label's trailing 90 observations; hits within min_gap_days de-clustered to the strongest."

The draft sent to Model Studio is an `event_car` study named "Event study · ‹label› · ‹theme› (‹kind›)": daily panel, market model, estimation window [−250, −30], event window [−5, +10], daily total return as the dependent variable, entity level country, sector or security, universe mvp_global primary listings, starting on 1 January two years before the first event.

Data sources and refresh

Shocks from the labelled GDELT daily aggregates; returns from the price history (country indices, US sector SPDRs, ACWI and SPY, issuer primary listings). The library is rebuilt weekly; event dates are read live from the serving snapshot.

Validation status

Not formally validated; MODEL-BASED. The cells are historical in-sample averages with conventional t-tests. Two facts limit their weight:

- Of the first 2,000 cells for attention shocks at [0,5], **1,468 have fewer than 10 events**, and only **115 have p < 0.05**, about what chance alone would produce at a 5% threshold across that many cells.
- Shocks cluster in time, so the t-statistics are optimistic, as the panel states.

The library is also the source of the News Terminal's "impact estimate", which is labelled "Model-based: historical average reaction of similar labelled events (event-study library), not a forecast".

Correct usage and worked example

Filter to the theme, country or sector you care about, prefer cells with many events, check that the CI excludes zero *and* that the share of negative CARs agrees with the sign, then open the CAR path.

Worked example. The top cell for attention shocks at [0,5] is Natural disasters & extreme weather · GB · Country index: n = 4, mean CAR +1.54%, 95% CI +1.3% to +1.8%, t 21.70, p < 0.001, 0% negative. Four events are far too few, and the very tight interval comes from four similar values, not from robustness; treat it as an anecdote. A more informative cell is All themes · US · Issuers: n = 9,939, mean CAR −0.31% (CI −0.5% to −0.1%, p 0.009, 53% negative), a small average drift after attention shocks to US companies.

Sending dates. With Country, US, All themes, Any shock, the panel lists **32 event dates** between 21 Oct 2023 and 19 Oct 2026, the first being 2024-10-10. **Send to Model Studio** opens the builder with "Event study · United States · all themes (any)", start 2022-01-01, ready to run. See Model Studio.

Limitations and pitfalls

- Sorting by p-value across thousands of cells guarantees impressive-looking rows at the top; this is multiple testing, not discovery.
- History starts in late 2024 (coverage from 21 Sep 2024 and a 90-day z baseline), so most cells have few events. The library deepens as the GDELT backfill adds earlier years.
- Sector ETF cells use US sector ETFs even for non-US shocks.
- The CAR path joins three stored windows; it is not a daily path.
- When there are no dates, the Send panel explains that the z baseline needs 90 days of history.

Related mnemonics and BQL fields

- Model Studio method `event_car` : see Model Studio.
- Post-earnings event studies are covered separately in Earnings & Events and the BQL fields `earn.car_0_p5` , `pead.ear` .

News Terminal

Route: `/news` (Live) and `/news/top` (Top stories; also `/news?view=top`) · **Menu:** News Terminal › Live / Top stories · **Mnemonics:** `N` , `TOP` , `CN` (with a ticker) · **Refresh:** live over server-sent events (target under 60 s from publication), hourly fold into the intraday store

Purpose

The News Terminal is a fast, keyboard-first headline screen for the working day. It merges official sources (central banks, statistics offices, regulators, exchanges), publisher RSS headlines, Polygon news and GDELT's 15-minute web articles into one live list, groups near-duplicates into stories, colours them by model-based sentiment and lets you filter by ticker, watchlist, country, sector, theme, source tier and language. The detail pane shows the story's timeline, the price move of linked securities since first publication, related filings and GDELT events, and a model-based impact estimate. Alerts can notify you of watchlist news or a sudden burst of articles on a story.

Screenshot

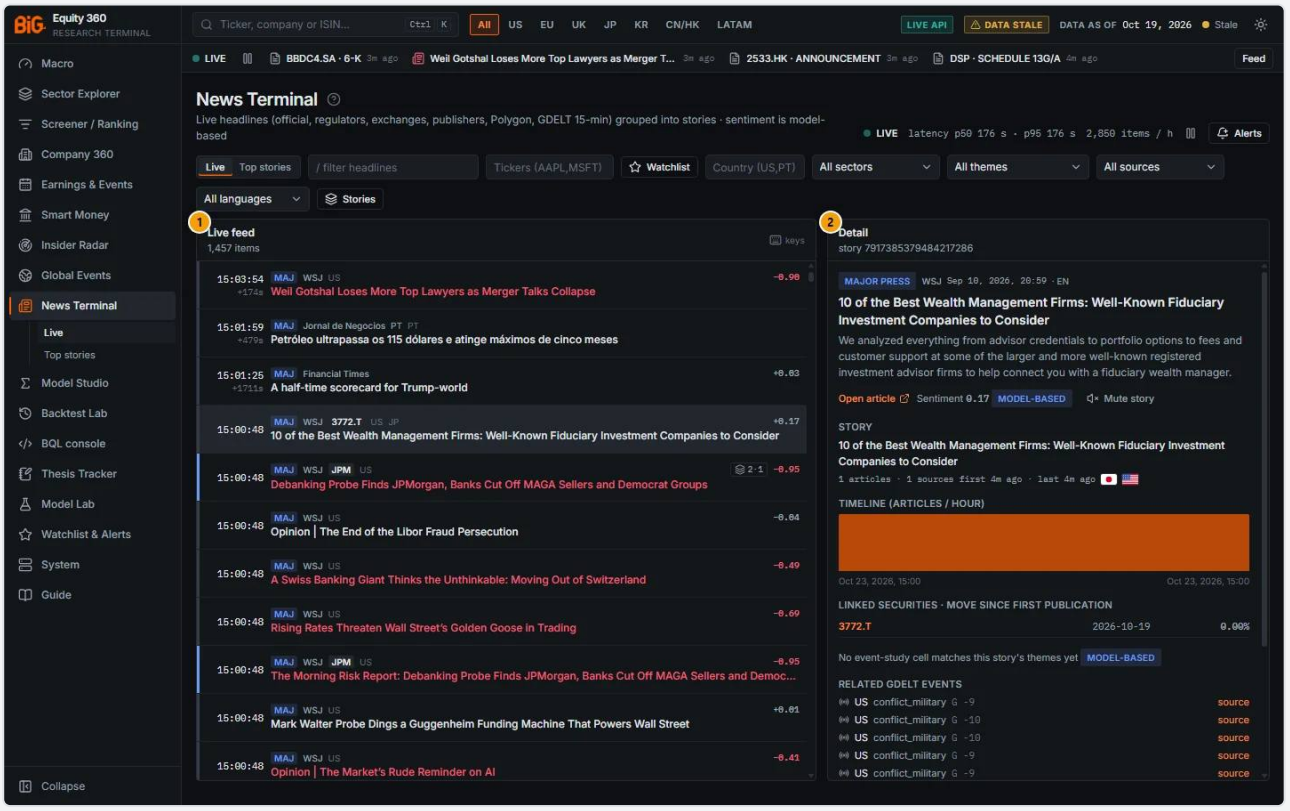


Figure 8.12: News Terminal, Live view (`/news`), 23 Oct 2026 around 15:00, the fourth item selected with `j` .

#	Panel	What it shows
1	Live feed	The live list (1,457 items loaded): time first seen and latency, tier badge, publisher, tickers, countries, story size, sentiment and headline coloured by sentiment.
2	Detail	The selected item (WSJ, major press) and its story: timeline, linked securities with the move since first publication, impact estimate (here "No event-study cell matches this story's themes yet") and related GDELT events.

Figure 8.13: Detail pane: item, story, timeline, linked security 3772.T, impact estimate and related GDELT events.

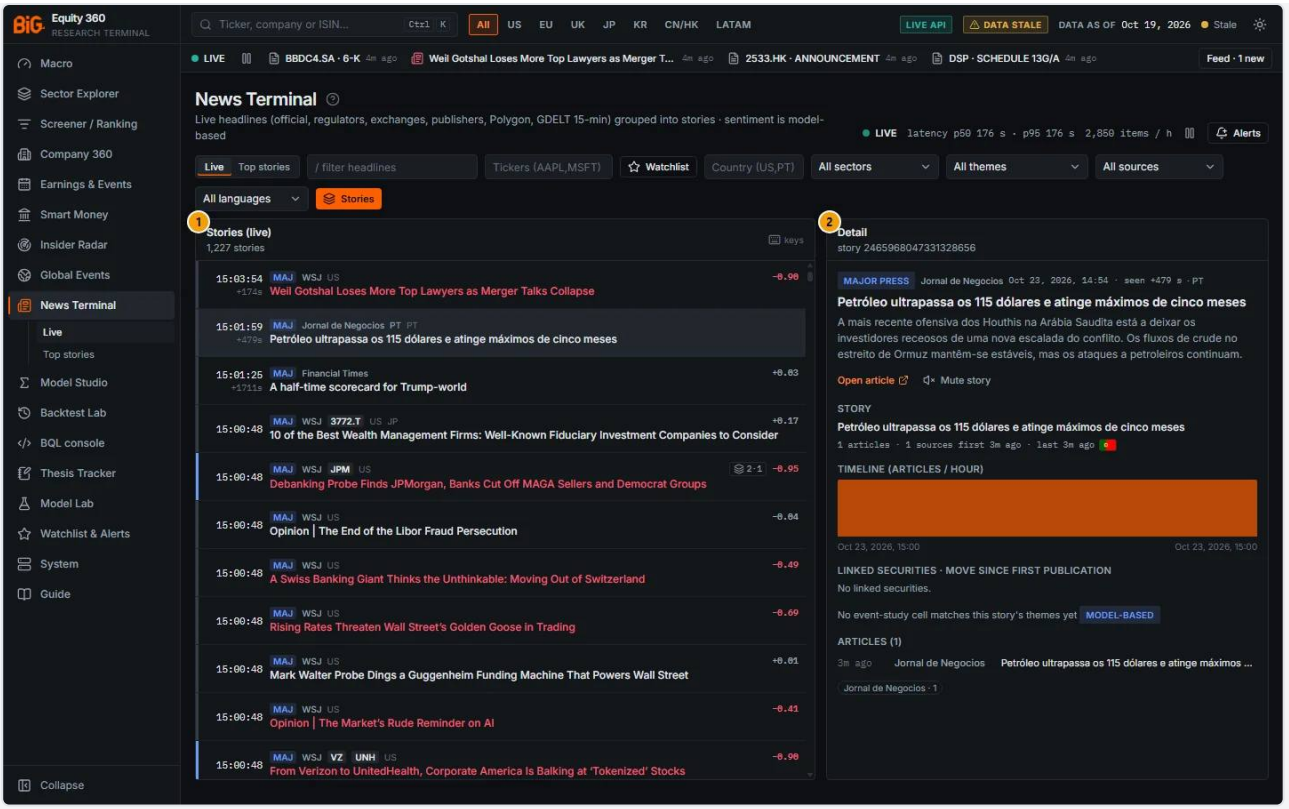


*Figure 8.14: Live view grouped by story (**Stories** on, 1,227 stories) with a Portuguese-language story open after `Enter`.*

#	Panel	What it shows
1	Stories (live)	One row per story, headed by its newest item; the `n·sources` chip shows the story size.
2	Detail	The selected story (Jornal de Negócios, seen +479 s after publication): snippet, story counts, timeline, linked securities ("No linked securities"), impact estimate and the member articles.

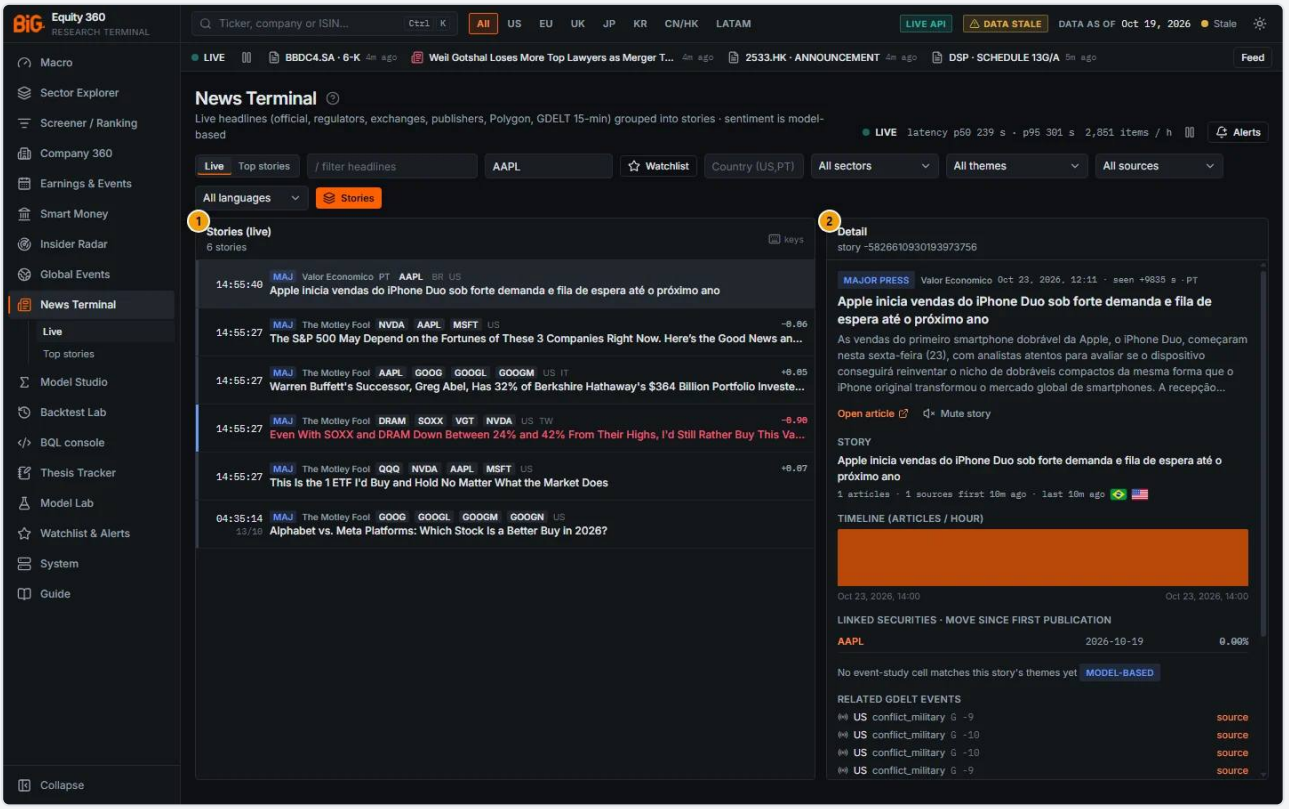


Figure 8.15: Live view filtered on the ticker AAPL (`/news?ticker=AAPL`) and grouped by story: 6 stories.

#	Panel	What it shows
1	Stories (live)	The six stories linked to AAPL, with all their tickers (NVDA, MSFT, GOOG, …).
2	Detail	The Valor Econômico story on the iPhone Duo: AAPL linked, base date 2026-10-19, move 0.00%.

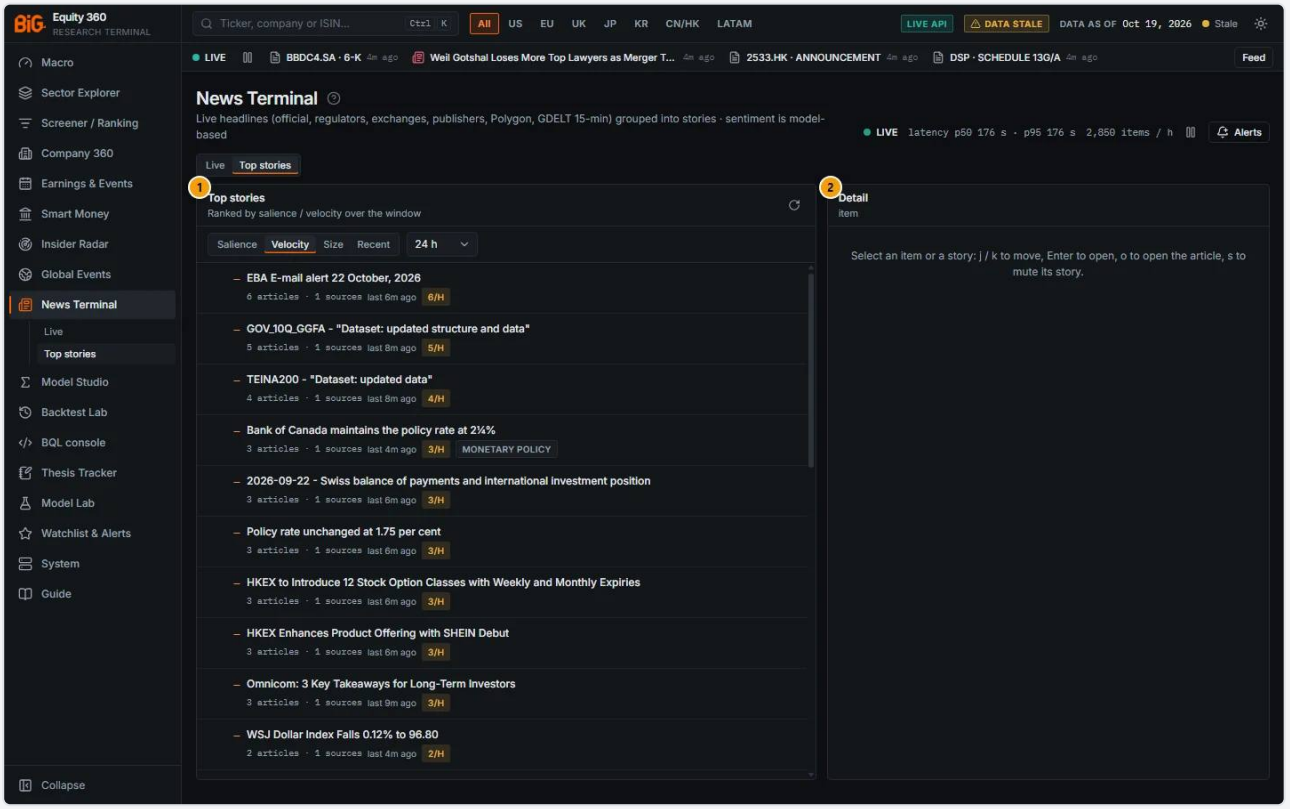


Figure 8.16: Top stories view (`/news/top`), Sort = Velocity, window 24 h; nothing selected yet.

#	Panel	What it shows
1	Top stories	Stories ranked by velocity: articles · sources, last article time and the velocity badge (6/H, 5/H, …).
2	Detail	Empty until a story is picked: "Select an item or a story: j / k to move, Enter to open, o to open the article, s to mute its story."

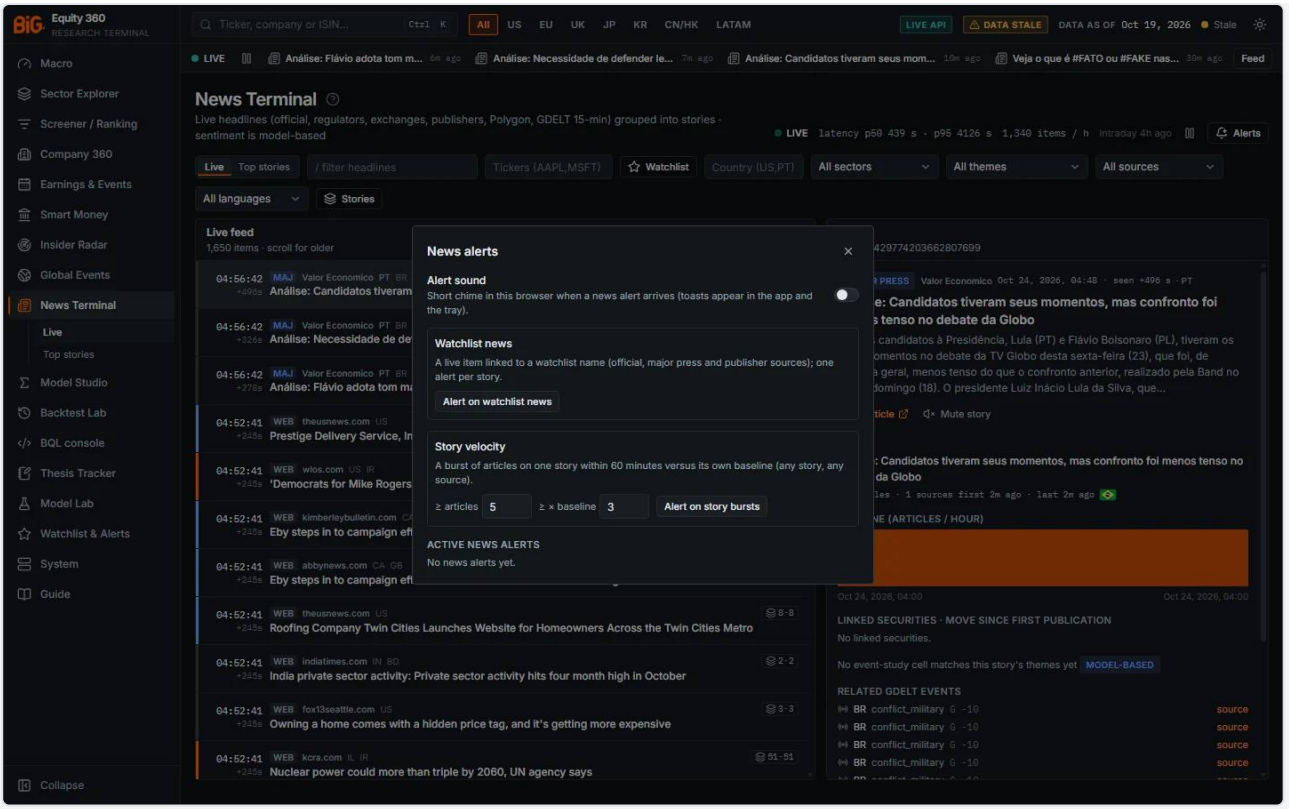


*Figure 8.17: News alerts dialog (**Alerts** in the header), 24 Oct 2026: alert sound, Watchlist news, Story velocity (defaults 5 articles, 3× baseline) and "No news alerts yet". The dialog covers the panel badges.*

The layout has three areas:

Area	What it shows
Page header	Title, subtitle "Live headlines (official, regulators, exchanges, publishers, Polygon, GDELT 15-min) grouped into stories · sentiment is model-based", the live state, publication-to-bus latency (p50 / p95), items per hour, the intraday store age, a pause button and **Alerts**.
List panel (left)	**Live feed** (or **Stories (live)** when grouped), or **Top stories** in the Top view.
Detail panel (right)	The selected item and its story.

Controls

HEADER

Control	What it does
? (next to the title)	Opens this section of the user guide.
Live state	**LIVE** (green, pulsing), **Connected · service stopped** (the stream is open but the realtime service is not running), **Connecting...**, **Reconnecting...**, **Disconnected · retrying**, **Paused** or **Off**.
latency p50 / p95	Hover: "Publish → bus latency (news.item first seen - source publication time) over recent items".
items / h	Items received in the last hour, all sources.
intraday ‹age›	When the hourly intraday store was last built.
Pause / Resume	Pauses the live stream; on resume, missed items are replayed from the last event id.
Alerts	Opens the News alerts dialog.

FILTERS (LIVE VIEW)

Control	What it does
View `Live` `Top stories`	Switches between `/news` and `/news/top`.
"/ filter headlines"	Free-text filter on headlines (sent to the server).
Tickers (AAPL,MSFT)	Comma- or space-separated tickers; applied on Enter or when you leave the box.
Watchlist	Shows only items linked to names on your watchlist (matched by security id or ticker). Shortcut `w`.
Country (US,PT)	ISO2 codes, applied when you leave the box.
Sector, Theme	One sector or one of the 41 themes.
Source tier	All sources; Official; Official + major; No GDELT web; GDELT web only.
Language	All languages, or EN, PT, ES, FR, DE, IT, JA, ZH, KO, RU, AR.
Stories	Groups items by story (one row per story, the newest item as head). Shortcut `g`.
"N muted · unmute"	Appears when stories are muted; restores them.
"N new"	Count of unread arrivals; click to mark all as read.
keys	Hover for the shortcut summary: "j / k move · / filter · w watchlist · o open · s mute story · g group · Enter detail".

Filters can be set from the URL: `ticker` (or `tickers`), `country`, `sector`, `theme`, `tier`, `lang`, `q`, `watchlist=1` and `view=top`. The command `AAPL US CN` opens `/news?ticker=AAPL`.

KEYBOARD SHORTCUTS

These work when the focus is not in a text box (they are ignored with Ctrl, Alt or Cmd held):

Key	Action
`j` or `↓`	Select the next item.
`k` or `↑`	Select the previous item.
`/`	Focus the headline filter.
`Esc`	Leave a text box (returns the keys to the list).
`w`	Toggle watchlist-only.
`o`	Open the selected item's article in a new tab.
`s`	Mute the selected item's story (hidden until you unmute).
`g`	Toggle grouping by story.
`Enter`	Move the focus to the detail pane and scroll it into view.

Selecting an item marks it as read.

LIST ROWS

Each row is 54 px high and shows:

- a **salience bar** at the left edge (none, grey, blue or orange for increasing salience);
- the **time first seen** (local, 24-hour clock), with the date when not today or, for recent items, the latency `+Ns` since publication;
- the **tier badge**: OFF (official), MAJ (major press), PUB (publisher), WEB (GDELT web);
- publisher, language when not English, up to four tickers and two countries;
- the **story size** `n·sources` (articles · distinct sources) when the story has more than one article;
- the **sentiment** score (hover: "Sentiment ... (model-based: ‹model›)");

- an **unread dot**; new arrivals flash for 3 seconds;
- the headline, in bold when unread and coloured teal / rose when the sentiment is positive / negative.

The list is virtualised: only the visible rows are drawn, and older rows load as you scroll (up to 6,000 from the server; the live buffer keeps 8,000).

TOP STORIES VIEW

Control	What it does
Sort `Salience` `Velocity` `Size` `Recent`	Salience (score, then articles); Velocity (articles in the last hour, then the ratio to the story's baseline); Size (articles, then sources); Recent (last article time).
Window `6 h` `24 h` `3 d` `7 d`	Stories with an article in the window (up to 150 listed, refreshed every minute).
Refresh (circular arrow)	Clears the selected story.
Story row	Salience, headline, articles · sources, "last ‹time› ago", a velocity badge `N/h`, up to three tickers and two themes. Click to show it in the detail pane.

DETAIL PANE

With nothing selected, the pane reads: "Select an item or a story: j / k to move, Enter to open, o to open the article, s to mute its story." For a selected item it shows:

- **Item:** tier badge, publisher, publication time, "seen +N s" (hover: first seen by the live path minus source publication time), language, headline (coloured by sentiment), the ≤ 300-character snippet, **Open article**, the sentiment with a *model-based* badge, and **Mute story**.
- **Story:** headline; articles and sources (live counts); first and last seen; a velocity badge `N/h · R×` (hover: "articles in the last 60 min vs the story's own baseline"); flags, themes and the story sentiment.
- **Timeline (articles / hour):** the last 48 hours of the story.
- **Linked securities · move since first publication:** each linked security with the close before first publication (base date) and the move to the latest close; the ticker opens Company 360.
- **Impact estimate** (model-based): the mean CAR over [0,5] days of the closest event-study cell, its CI, n and p, with the matched theme, country, sector and level and the label "Model-based: historical average reaction of similar labelled events (event-study library), not a forecast". When nothing matches: "No event-study cell matches this story's themes yet".
- **Related filings** (form type, title, age) and **Related GDELT events** (country, themes, Goldstein scale, source link), when any.
- **Articles (N):** up to 60 member articles, newest first, with age, publisher and link, then the top source domains.

NEWS ALERTS

The **Alerts** button opens a dialog with:

Control	What it does
Alert sound	A short chime in this browser when a news alert arrives (toasts appear in the app and the tray). Turning it on plays the chime once.
Alert on watchlist news	Creates a `watchlist_news` rule on your server watchlist: a live item from official, major press or publisher sources (tiers 1-3) linked to a watchlist name; one alert per story.
≥ articles, ≥ × baseline, **Alert on story bursts**	Creates a `story_velocity` rule (defaults 5 articles in 60 minutes, 3× the story's baseline, at least 2 sources) for any story from any source.
Active news alerts	Lists the news rules (on / off, description) with a delete button.

Rules are evaluated live by the realtime service, not by the nightly alert job. Other alert types are in Watchlists and alerts.

Methodology

Pipeline per item (the news desk consumer):

1. **Normalise:** URL hash, headline, a snippet of at most 300 characters, publisher, language, publication time and *first seen* time.
2. **Dedupe:** the same URL seen in the window is not published again; near-duplicate headlines are grouped, not dropped.
3. **Link:** provider tickers when present, otherwise the headline linker (the Global Events alias table with context rules).
4. **Label:** countries, sectors, the 41 themes and the source tier.
5. **Sentiment:** FinBERT for English headlines (batched and cached); other languages are scored with XLM-R in the hourly job.
6. **Story attach:** MinHash-LSH (Jaccard ≥ 0.6 on cleaned headline tokens) against stories of the last 72 hours; the story id is the same one the nightly lake uses.
7. **Publish** `news.item` and, for stories with at least two members or an official source, `news.story` on the realtime bus; the page receives them over the SSE stream `/api/v1/live/stream` (topics `news.item`, `news.story`).

GDELT web articles join stories but appear as items only when they are issuer-linked, strongly toned (|tone| ≥ 4) or part of a multi-source story, so the list is not flooded with about 100,000 web articles a day.

Source tiers: 1 Official (central banks, statistics offices, regulators, exchanges), 2 Major press (wire and major financial press, including Polygon), 3 Publisher, 4 GDELT web.

Live salience (list bar and Top stories):

```
salience = ln(1 + articles) + ln(1 + sources) + |sentiment| + min(1, |tone| / 10)
           + tier bonus (1.0 official, 0.6 major press, 0.3 publisher, 0 web) + 0.8 if issuer-linked
bar: grey ≥ 1.8, blue ≥ 3, orange ≥ 4.5
```

Velocity: articles in the last 60 minutes, and its ratio to the story's own baseline rate (floored at 0.5 per hour).

Sentiment colour: a headline is teal above +0.2 and rose below −0.2 (FinBERT score; GDELT tone ÷ 10 when there is no score).

Move since publication: last close ÷ the last close *before* the story's first publication − 1, from the serving snapshot.

Hourly compaction (N3). Every hour at :10 UTC the `hourly` job folds the hot store into partial-day lake files (stories, members, labels), hourly event and sentiment aggregates (`event_hourly_agg`, `sentiment_hourly`) and an intraday DuckDB for the last 7 days that the API hot-swaps without a snapshot rebuild; a daily finalise step re-clusters the day and applies retention. The page reads three layers merged by key, newest first: the realtime bus (last 72 hours), the intraday store (last 7 days) and the nightly snapshot (older stories).

Data sources and refresh

- **Curated RSS / Atom feeds** (`config/news_feeds.yaml`), each verified for terms, robots.txt and RSS intent, polled every 60-120 s (statistics offices every 5 minutes) with conditional requests and per-host politeness. Only the headline, a ≤ 300-character snippet, URL, publisher, time and language are stored, never article text. Verified feeds at the time of writing:
  - central banks (tier 1): Federal Reserve, ECB, Bank of England, Bank of Japan, Bank of Canada, RBA, SNB, Riksbank, Norges Bank, Banco Central do Brasil;
  - statistics offices (tier 1): BLS, Eurostat, ONS, Statistics Canada;
  - regulators (tier 1): SEC, CFTC, FCA, ESMA, EBA; exchanges (tier 1): JPX, HKEX;
  - major financial press (tier 2): Bloomberg, Financial Times, WSJ, MarketWatch, Nikkei Asia, SCMP; European and LatAm business press (tier 2): Handelsblatt, Expansión, ECO, Jornal de Negócios, Valor

Econômico;
    - other publishers (tier 3): BBC, The Guardian.
- **Rejected feeds** are kept in the file with the reason: for example Reuters (public RSS discontinued in 2020), AP (no public RSS), CNBC (robots.txt disallows), Les Echos (503 / 403), Banco de Portugal (403), BEA and Destatis (robots.txt), INSEE and INE Portugal (errors at verification).
- **Polygon** news (plan permitting) and **GDELT** 15-minute files, polled every minute.
- **Targets:** end-to-end p95 under 60 s for RSS and under 3 minutes for GDELT; first paint under 1 s and live updates rendered in under 100 ms (checked by the QA smoke suite). At the time of the screenshots the header showed latency p50 176 s / p95 176 s with 2,850 items per hour (23 Oct, 15:00) and p50 439 s / p95 4,126 s with 1,340 items per hour (24 Oct, 04:56), so the 60-second target was not met on that machine.
- The realtime service runs as the fourth child of the app's supervisor (RAM cap 1.5 GB, restarted on crash), with a tray toggle **Live news on/off**. Its health (latency per source, items per hour, last poll, failures) is on Data Health › Live news.

Validation status

- **Sentiment:** FinBERT / XLM-R scores are **model-based** and labelled so on every row and in the detail pane; they are not validated as a predictor of returns in this app.
- **Impact estimate: model-based**, labelled "historical average reaction of similar labelled events (event-study library), not a forecast"; see Event studies for the library's limits.
- **Linking and story grouping:** the same alias table and clustering rule as Global Events (spot check: security links 82%, theme 68%). The live headline linker has no separate published spot check.
- **Latency and rendering budgets** are performance targets, not model validation.

Correct usage and worked example

Keep the terminal open on a second screen with **Official + major** and your **Watchlist** filter, use `j` / `k` to scan and `Enter` to read the story, and mute recurring stories with `s`. Use **Top stories › Velocity** to catch stories that are accelerating.

Worked example (23 Oct 2026, Figure 8.15). Type `AAPL US CN` in the command bar (Ctrl+K): the terminal opens as `/news?ticker=AAPL`, filtered on Apple. Press `g` to group by story: six stories are linked to AAPL, most of them Motley Fool articles that mention several tickers. Press `j` to select the first story, "Apple inicia vendas do iPhone Duo sob forte demanda e fila de espera até o próximo ano" (Valor Econômico, published 12:11, seen +9,835 s, that is almost three hours later), and `Enter` to read it. It has 1 article from 1 source, so it is not yet confirmed by other outlets. The linked-securities row shows AAPL with base date 2026-10-19 and a move of 0.00%: the snapshot's last close is 19 Oct, before the article, so no reaction can be measured yet. No event-study cell matches its themes, so there is no impact estimate. When a cell does match, check its n and p: an impact cell with n < 10 is an anecdote. Then open **Alerts › Alert on story bursts** with the defaults (5 articles in 60 minutes, 3× baseline): a new story needs at least 5 articles within the hour before it fires, because the baseline is floored at 0.5 per hour.

Limitations and pitfalls

- **Licensing:** RSS headlines, GDELT and Common Crawl are suitable for personal research. Use inside the bank needs licensed news (Dow Jones / Factiva, LSEG, Bloomberg), as noted in `docs/it/DATA_LICENSING.md`. Article pages are never scraped and full text is never stored.
- Several major wires (Reuters, AP, CNBC) have no usable public RSS, so first reports often arrive via GDELT web articles or later publisher feeds.
- MarketWatch publication dates were found unreliable at verification; latency figures for undated or estimated entries are excluded.
- Non-English headlines get sentiment only after the hourly job (XLM-R), so they appear uncoloured for up to an hour.
- The move since publication uses daily closes from the snapshot, not intraday prices, and starts from the close *before* the first article.

- Muted stories are stored in this browser only (last 500), as are the read marks of the current session.
- Story groups can merge unrelated headlines that share most words, or split one event into several stories when wording differs.
- **Move since publication is 0.00% until the snapshot has a close after the article.** The snapshot is rebuilt by the weekly or daily refresh, so for today's news the move is usually blank or zero (Figure 8.15).
- Old feed entries that a publisher re-lists appear as new: in Figure 8.12 a WSJ article dated 10 Sep 2026 was first seen on 23 Oct. Check the publication date in the detail pane.
- Ticker links come from the headline linker and can be wrong: the same WSJ headline about "Wealth Management Firms" is linked to 3772.T (a Japanese company whose name contains "Wealth Management").
- **Related GDELT events** are matched by the story's countries and themes only; for a story without themes they are simply recent events in the same country (the US armed-conflict events in Figure 8.13 are unrelated to the headline).
- Top stories by **Velocity** are often single-source official feeds (statistics datasets, EBA e-mails in Figure 8.16). The story-velocity alert requires at least 2 sources, so these do not fire it.

Related mnemonics and BQL fields

- `N` opens the Live view, `TOP` the Top stories view, `<ticker> CN` the Live view filtered on the ticker (on a two-token input, `CN` is the Canadian market code: type `RY CN CN` or `AAPL US CN`). See Mnemonic reference.
- BQL: `sent.sentiment_ewm_7d`, `sent.news_count`, `sent.news_count_z` (daily sentiment and attention per security).
- Company-level news history: Company 360 › News & Sentiment.

9. Model Studio

Model Studio is the app's econometrics and machine-learning workbench. You build a point-in-time dataset from any table in the data lake (prices, valuation, factor scores, fundamentals, ratios, macro series, CFTC positioning, cross-asset signals, insider, sentiment, smart-money and technical signals), pick one of 37 estimation methods, run it, and read the coefficients, tests, diagnostics, forecasts or clusters in a set of result tabs. Every run is pinned to a data snapshot, a specification hash and a random seed, can be saved as a study, compared with other runs, re-run on a schedule with drift alerts, and exported as a standalone Python script.

Model Studio is a research tool. It does not produce validated signals by itself: a regression coefficient or a high in-sample R^2 in Model Studio is a finding to be tested, not an edge. The chapter explains each part of the page, the methods and their formulas, and the guardrails that stop the most common mistakes (look-ahead, too few observations, collinearity, overlapping cross-validation folds).

> **Note:** screenshots that show saved studies and completed runs (figure captions mention the "test copy") were captured from a test copy of the app seeded with example content on 23 Oct 2026, on the same data snapshot. The red **WORKER OFF** chip in their top bar comes from that test copy, not from your installation.

The nightly ML ranking models (XGBoost, Transformer) are monitored in the **ML monitor** tab, which is the same view as the Model Lab page. Strategy simulation with costs and overfitting controls lives in the Backtest Lab.

Model Studio

Route: `/studio` · **Menu:** Model Studio · **Mnemonics:** none (open from the sidebar or the command palette) · **Refresh:** on demand (your runs); scheduled studies re-run weekly or monthly via the `studio_scheduled` job

Purpose

Model Studio answers ad-hoc quantitative questions without writing code: "how sensitive are bank returns to the 10-year yield, with lags?", "do our factor scores carry a premium in the cross-section?", "what does a VAR say about the response of industrial production to a rates shock?", "which companies cluster together on profitability and valuation?". Analysts use it to test a hypothesis before writing it into a thesis; the owner uses it to prototype models before they become nightly jobs. Results can be saved, compared, scheduled, exported to Python, turned into a thesis, and (for panel ML runs) sent to the Backtest Lab as a signal.

Screenshot

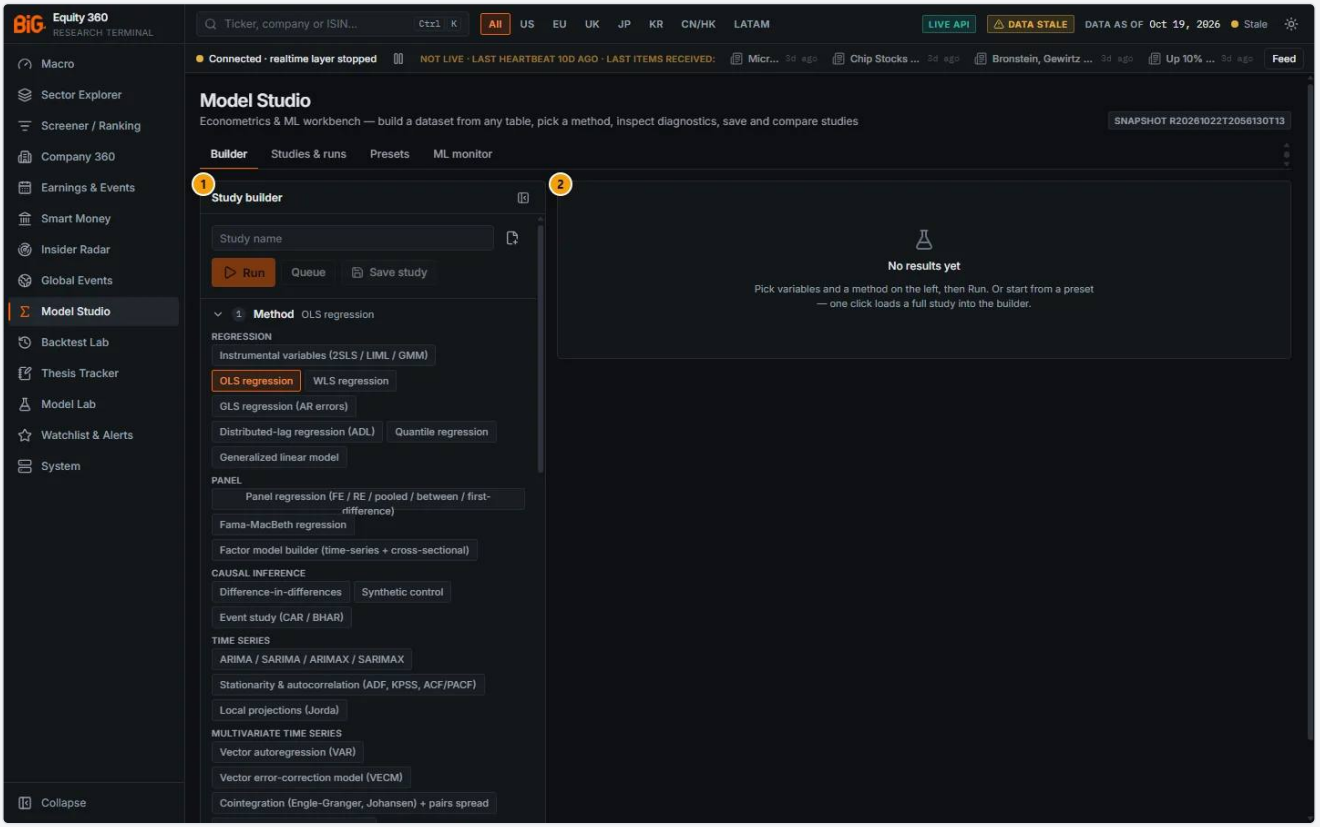


Figure 9.1: Model Studio (`/studio`), Builder tab with an empty draft, snapshot R20261022T2056130T13, data as of 19 Oct 2026.

#	Panel	What it shows
1	Study builder	The left-hand builder: study name, **Run** / **Queue** / **Save study** buttons, then five numbered sections (1 Method, 2 Dataset, 3 Variables, 4 Parameters, 5 Preview). The figure shows section 1 open with the method families.
2	Results pane	The results of the selected run. With no run selected it shows "No results yet: Pick variables and a method on the left, then Run. Or start from a preset — one click loads a full study into the builder."

Controls

Control	What it does
Tabs **Builder** / **Studies & runs** / **Presets** / **ML monitor**	Switch between the builder with its results pane, the list of saved studies and runs (Studies & runs), the preset library (Presets) and the nightly ML monitor (ML monitor). The last tab you used is remembered in the browser.
SNAPSHOT badge (top right)	The serving snapshot every run is pinned to (hover: "Data snapshot every run is pinned to"). Two runs on the same spec, snapshot and seed give the same numbers.
Hide builder (icon at the top right of the builder panel)	Collapses the builder so the results use the full width. **Show builder** brings it back.
Toasts	"Run complete", "Run queued on the worker" (with the time estimate), "Run failed", "Run rejected", "Run finished: ‹name›" for queued runs that complete while you are elsewhere in the app.

The builder draft, the selected run, the runs being watched and the runs ticked for comparison are stored in the browser (local storage key `big360.studio`), so they survive a page reload.

Methodology

A study has three parts:

1. **Dataset specification**: shape (time series, panel or cross-section), frequency (D/W/M/Q), dates, entity filter, entity level and a list of variables, each with a source, a field, aggregations and a chain of transforms. The dataset engine builds it point-in-time from the Parquet lake (see Builder, Methodology).
2. **Method and parameters**: one of the 37 registered methods (Methods catalogue) with its parameter form, a seed (default 42) and a timeout (default 600 s).
3. **Run**: the engine estimates the cost. Runs estimated under **2 seconds** execute inline in the API and come back finished ("sync"); longer runs are queued on the background worker ("async", job `studio_run`) and polled. Every run is stored under `data/modeling/runs/<run_id>/` with its spec, status, result and a reproducibility block (spec hash, snapshot id, seed, library versions, timings).

Data sources and refresh

Model Studio reads the same lake as the rest of the app, read-only, at the current serving snapshot. Nothing is downloaded when you run a study. A source is only as fresh as the nightly / weekly job that fills it (see Refresh schedule). Built datasets are cached under `data/modeling/cache/` by a hash of the spec, the snapshot id and the engine version, so a repeated run on the same snapshot reuses the cached dataset ("cached" in the preview and in the Data tab).

Validation status

Model Studio is a tool, not a model, so it has no validation gate of its own. The statistical tests it reports (Breusch–Pagan, Ljung–Box, Hausman, GRS, Diebold–Mariano and so on) are standard textbook tests; they are labelled pass / fail or reject / do-not-reject by fixed rules, not by a validated model. One method carries a validation verdict: the filtered conformal ellipsoid is **experimental** in Model Studio (see Methods catalogue, Validation status). The ML models shown in the ML monitor tab have their own status, described in Model Lab.

Correct usage and worked example

Start from a preset, inspect it, then change one thing at a time.

1. Open **Presets** and click **Load & run** on "GARCH: S&P 500 volatility and 1-day VaR".
2. The study runs on the worker or inline (the live run history shows this preset finishing in 769 ms on 6,734 daily observations, AIC 17,927).
3. Read the **Summary** tab, then the **Volatility** tab (conditional volatility and the VaR backtest), then **Diagnostics** (Ljung–Box on squared standardised residuals, Kupiec test).
4. Change one parameter in **4 · Parameters** (for example `vol` from `gjr` to `egarch`), run again, then tick both runs in **Studies & runs** and press **Compare**.

Limitations and pitfalls

- A significant coefficient is not a trading signal. Most studies are in-sample fits; only the out-of-sample evaluation blocks (rolling-origin, purged cross-validation, leaderboard) say anything about forecasting ability, and even those are subject to multiple testing if you try many variants.
- The universe history starts on 2026-10-10 (survivorship): panels before that date use today's member list. See Survivorship.
- Some sources are short: news sentiment starts in October 2025, technical setups are stored for the latest date only, smart-money signals start in mid-2023. The variable picker shows each variable's first and last date.
- Bayesian (PyMC) and deep sequence models (LSTM / TFT) listed in the original plan were not built.

Related mnemonics and BQL fields

No command-bar mnemonic opens Model Studio. Related: BQL console (expressions usable as a Studio variable source, see BQL query variable source), Backtest Lab, Model Lab, Purged and embargoed cross-validation, Multiple testing.

Builder

Route: `/studio` (tab **Builder**) · **Menu:** Model Studio › Builder · **Mnemonics:** none · **Refresh:** on demand

Purpose

The builder turns a question into a study specification: which method, on which data shape, over which dates and entities, with which variables and transforms, and with which parameters. It validates the draft as you type and lists any problem that blocks a run.

Screenshot

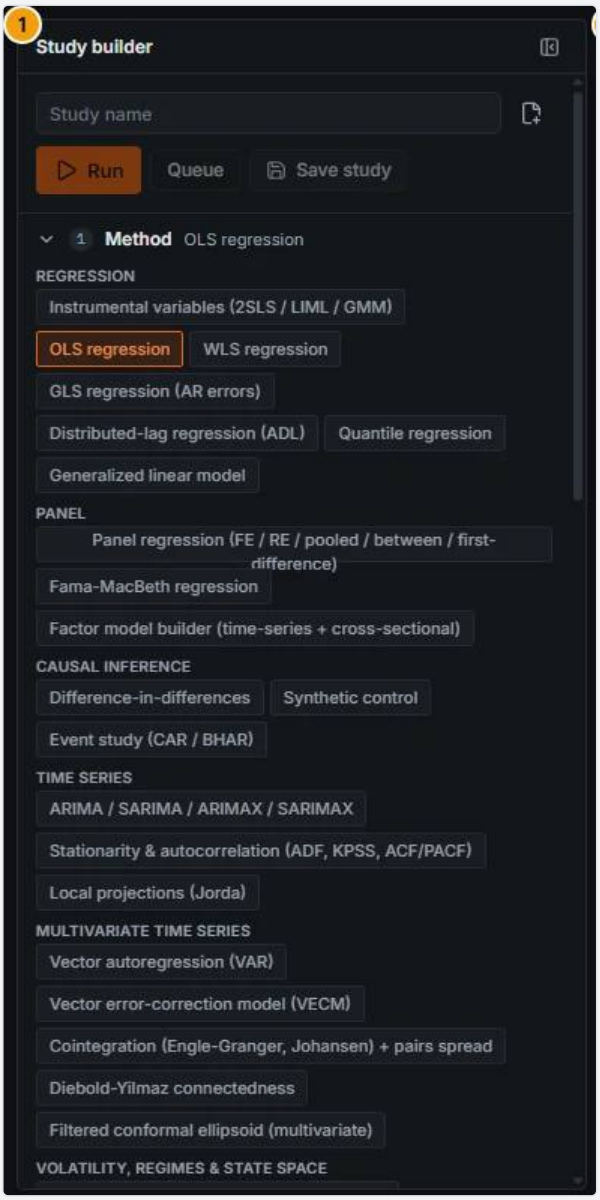


Figure 9.2: Study builder close-up (callout 1 of Figure 9.1): run bar and section 1 · Method with the method families.

> Screenshot pending: builder with a loaded preset showing sections 2 · Dataset, 3 · Variables (two variable cards with a lag of t−1 and a transform chip) and 4 · Parameters open (route `/studio`, Presets › "Sector returns vs lagged rates" › Load).

Controls

RUN BAR

Control	What it does
Study name	Free text; used as the run name, the saved study name and the export file name.
New empty study (icon)	Clears the draft (method OLS, time series, monthly, start 2005-01-01, seed 42, timeout 600 s).
Run (or **Run leaderboard** for the leaderboard method)	Submits the study in "auto" mode: inline if the estimate is under 2 s, otherwise queued on the worker. Disabled while the draft has problems.

Control	What it does
Queue	Hover: "Queue on the background worker even if it would be quick". Use it for runs you want to leave running.
Save study / **Save** / **Save as new**	Saves the draft as a study (name defaults to "Untitled study"). Once a draft is opened from a saved study, **Save** updates it and **Save as new** creates a copy; a **saved study** badge appears.
Problems list (amber bullets)	Lists every blocking problem, for example "Add at least one variable", "OLS regression needs exactly one target (y)", "Variable names must be unique", "IV needs at least as many instruments as endogenous regressors", "x: a regressor uses future data (lead / forward transform) — look-ahead", "Set “Treatment” in Parameters".

1 · METHOD

Control	What it does
Method chips, grouped by family	Regression, Panel, Causal inference, Time series, Multivariate time series, Volatility, regimes & state space, Machine learning, Clustering & factors, Spatial. Hover a chip for the method description. Picking a method keeps your variables and re-maps their roles (for example, unsupervised methods turn y and x into features).
Description and badges	The method description, the data shapes it accepts (the current one highlighted) and **needs y** or **unsupervised**.

2 · DATASET

Control	What it does
Data shape	**Time series** (one row per date: entities pooled or macro series), **Panel** (entity × date), **Cross-section** (one row per entity at a date).
Start / **End**	Date window. End empty = latest data.
Frequency	Daily, Weekly (Fridays), Monthly (month ends), Quarterly (quarter ends).
Entity level	Security, Sector (aggregated), Region (aggregated), Country (aggregated). Aggregated levels average (or sum) the securities of each group per date.
Macro alignment	**Auto**, **Point-in-time (as published)** or **Reference period**. See Methodology.
Entities box	Universe (Any, `mvp_us`, `mvp_global`, `full`, `watch`, `eu_broad`), **Primary listings only** switch (on by default), **Sectors** and **Regions** toggles, **Securities** search (ticker or name; chips with a remove cross). Greyed out with "(macro-only study: not used)" when every variable is a macro series in a time-series study.

3 · VARIABLES

Control	What it does
Add variable / **Close catalog**	Opens the variable picker.
Picker search	"Search variables (e.g. 10y yield, roic, hicp)". Matches titles and field ids.
Source table select	All sources, or one source with its number of variables (see Data sources). Choosing **BQL query** adds an expression row (see BQL query variable source).
Frequency select	Any frequency, daily, weekly, monthly, quarterly, annual.
Variable row	Country flag (macro), title, `source:field`, native frequency, units, first → last date, number of observations, number of entities, and a **PIT** badge when the source is knowledge-dated (sampled as published).
Add buttons on each row	**+ y**, **+ x**, and for IV methods **+ endog** / **+ instrument**; for unsupervised methods **+ add**. Adding a second y turns it into an x. A toast confirms "Added ‹field› as ‹role›".
Load more	Shows 40 more variables.

Each added variable becomes a **variable card**:

Control	What it does
Name field	The column name in the dataset (letters, digits, underscore; red border when invalid). Spaces become underscores.
Role select	Target (y), Regressor (x), Weights, Feature (or "System variable" for VAR-type methods), Endogenous (IV), Instrument (IV), Not used. The card border is orange for y and blue for IV roles.
Move up / Move down / Remove	Reorder or delete. Order matters for VAR ordering (Cholesky) and for methods that read "the first x".
`source:field — title` line	Where the variable comes from.
lag spinner (– / +)	Sets the net lag: `t` = same period, `t-1` = one period back, `t+1` = one period ahead (amber, "lead").
Transforms chips and **+ transform** select	The transform chain applied after resampling. Click a chip to edit its parameters (k, window, min periods, quantiles, threshold, factor, scope); the cross removes it.
Amber note "Look-ahead: a regressor uses future values. Only targets may lead."	Shown when an x uses `lead`, `fwd_sum` or `fwd_compound`. The run is blocked.
Aggregation & entity pooling (expander; dot when customised)	**time agg.** (default, last, first, mean, sum, min, max, median, count), **pool entities** (none, mean, median, sum, count, min, max), **Before resampling** transform chain (applied per entity on the native frequency), the variable's own entity filter (with **clear**), and for `factor_ls` variables **long-short quantiles** (2–10, "top minus bottom, sorted on the previous period's score over the entity filter").

TRANSFORMS

Transform	Definition (as the catalogue documents it)
lag	`x[t-k]` (k periods back, per entity)
lead	`x[t+k]`. LOOK-AHEAD: allowed only as a target
diff	`x[t] - x[t-k]`

Transform	Definition (as the catalogue documents it)
log	`ln(x)` , null for x ≤ 0
log diff	`ln(x[t]) − ln(x[t−k])`
% change	`x[t]/x[t−k] − 1`
rolling mean / std / sum / min / max	trailing statistic over `window` periods
rolling z	`(x − trailing mean) / trailing std` over `window`
drawdown	`x / trailing max(x, window) − 1` (use on a price level)
winsorize	clip to [lower, upper] quantiles; scope cross-section (per date, panel default), whole sample or per entity
clip	clip to [lower, upper] in absolute values
standardize	z-score; scope cross-section, whole sample or per entity
cs-rank	cross-sectional percentile rank 0..1 per date
cs-z	cross-sectional z-score per date
cs-demean	minus the cross-sectional mean per date
fwd sum (target)	sum of x over t+1..t+k. LOOK-AHEAD (targets only)
fwd compound (target)	`∏(1+x) − 1` over t+1..t+k (forward return from periodic returns). LOOK-AHEAD (targets only)
binarize	1 if x > threshold else 0
scale / abs / square / negate	`x·factor` , `|x|` , `x²` , `−x`

4 · PARAMETERS

The form is generated from the method's parameter schema. Required inputs come first and are marked `*` ; a dot marks a value changed from its default; hover a dotted-underlined label for its description. Only changed values are sent. **Defaults** resets the form. Under the form, **Seed** and **Timeout (s)** (minimum 5) apply to every method.

Special editors:

Editor	Used by	What it does
Treatment editor	Difference-in-differences	Segmented **Entities** (treated tickers), **Attribute** (Sector, Region or Country ISO-2 with toggles), **Variable** (a 0/1 column of the dataset) or **Events** (explicit entity / date pairs for staggered adoption). Entities and Attribute modes take one **Treatment date**.
Events editor	Event study (CAR / BHAR)	Rows of entity + date. "Set the date first, then add entities (Enter)."
Order tuples	ARIMA	`order` as p, d, q and `seasonal_order` as P, D, Q, s.
Lists	Quantile regression (`quantiles` , e.g. 0.1, 0.5, 0.9), GARCH (`var_levels`), distributed lag (`lag_overrides` as `var=4, other=2`)	Comma-separated values.
Variable pickers	Parameters such as `panel_variable` , `pivot_var` , `offset` , `market` , `event_column` , `factor_scores`	Chips or a select of your dataset's variable names.
Nested object	`cv` of the ML methods	Folds, scheme (auto, expanding, purged_kfold, kfold), purge, embargo.
Candidates editor	Model leaderboard	Checkboxes for Historical mean (benchmark), OLS, Logit, Ridge, Lasso, ElasticNet (alpha by inner CV), Random forest, Gradient boosting (HistGradientBoosting), XGBoost and ARIMA(X) (time series only); each ticked candidate gets an editable name. Footer: "N selected · every candidate is scored on the same cross-validation splits; Diebold–Mariano tests compare each with the best."

5 · PREVIEW

Control	What it does
Preview dataset / **Refresh preview**	Builds the dataset only (no model) and shows rows, entities, periods, date range and build time ("cached" when reused). "Builder changed since this preview" warns when the preview is stale. "Add a variable first" when empty.
Dataset warnings (amber)	Coverage and alignment notes from the engine (for example stale values dropped, empty variables).
Coverage / **Summary** / **Rows (n)**	Per-variable coverage bar (green ≥ 80%, amber ≥ 50%, red below) with first / last date and a "future" tag for look-ahead columns; summary statistics (n, mean, std, min, median, max); the first 50 rows.

Methodology

POINT-IN-TIME DATASET CONSTRUCTION

The dataset engine (`backend/app/modeling/_core.py` , embedded verbatim in exported scripts) builds the frame in this order:

1. **Grid.** Period ends at the dataset frequency: D = trading days in the data (or Monday–Friday), W = Fridays, M = month ends, Q = quarter ends. Only buckets up to the data end are kept, so there is no partial last period.
2. **Entities.** The entity filter is resolved to securities (universe, sectors, regions, tickers, primary listings). With `pit_universe` on (the default), an (entity, date) row is kept only while the entity is a member of the universe, but see the survivorship caveat below.
3. **Event-dated sources** (prices, valuation, factor and composite scores, sentiment, insider, technicals, smart money, technical factors) are aggregated into each bucket (default `last` ; `sum` for volumes) and forward-filled per entity for at most D 5 / W 2 / M 1 / Q 1 buckets.
4. **Knowledge-dated sources** (ratios, fundamentals, macro in point-in-time alignment, CFTC, cross-asset signals) are sampled as-of: the value at grid date D is the latest period whose `knowledge_date ≤ D` . A restated or revised number only appears from the date it became known.

5. **Macro alignment. Point-in-time** samples macro series as published (with their release lag). **Reference period** indexes them by observation period at the latest vintage, which is what a pure macro time-series model wants (for example an ARIMA on CPI). **Auto** uses reference period when every variable is a macro series and point-in-time otherwise. Mixing equities with reference-period macro data would leak information that was not yet published, which is why Auto switches.
6. **Transforms.** `pre_transforms` run per entity before cross-entity aggregation; `transforms` run after. Time-series transforms run on the complete entity × grid frame; cross-sectional transforms (cs-rank, cs-z, cs-demean, cross-section winsorize / standardize) only use universe members at each date.
7. **Entity pooling.** In a time-series dataset built from securities, each variable is pooled across entities per date (`entity_agg` , default mean). Sector / region / country entity levels aggregate the same way.

Limits: 1 to 60 variables, 3,000 entities and 5,000,000 rows by default.

GUARDRAILS

Before fitting, the engine rejects the study (HTTP 422, shown as "Guardrail: ‹code›") when:

Code	Rule
`lookahead`	an x, weight or interaction uses `lead` , `fwd_sum` or `fwd_compound`
`too_few_obs`	fewer than k + 2 complete observations for k parameters
`singular`	the design matrix is rank-deficient (lists the collinear columns)
`bad_shape` , `missing_y` , `missing_x` , `too_many_x` , `weights_unsupported`	the method does not accept this data shape or number of variables
`unknown_column` , `dataset` , `invalid_spec`	a referenced column does not exist or the specification is invalid

Non-blocking warnings appear above the result tabs: `few obs per param` (observations per parameter), `near singular` (condition number above 30 on standardised regressors), `high vif` (VIF above 10), `nonstationary` (ADF / KPSS suggest unit roots in y and x: spurious-regression risk), `oos worse than naive` , and method-specific ones.

Data sources and refresh

The live catalogue (`GET /api/v1/studio/catalog/datasets` , snapshot R20261022T2056130T13) lists 15 sources:

Source	Table	Native frequency	PIT rule	Coverage (first → last)
Daily prices & returns	`prices_daily`	D	event-dated	1927-12-30 → 2026-10-20, 24,943 securities
Daily valuation multiples	`valuation_daily`	D	event-dated (price × PIT fundamentals)	2010-01-01 → 2026-10-16
Factor scores (0-100)	`factor_scores`	M	event-dated	2016-10-31 → 2026-10-16
Composite scores by profile	`composite_scores`	M	event-dated	2016-10-31 → 2026-10-16
News sentiment (daily)	`sentiment_daily`	D	event-dated	2025-10-09 → 2026-10-16
Insider signals	`insider_signals`	D	event-dated	2017-01-31 → 2026-10-16
Technical setups	`tech_setups`	D	latest date only	2026-10-16 only
Financial ratios	`ratios`	Q	knowledge-dated (TTM)	1993-12-31 → 2026-09-30
Fundamentals line items	`fundamentals_wide`	Q	knowledge-dated (TTM, USD)	1993-12-31 → 2026-09-30
Long-short factor returns (`factor_ls`)	`factor_scores`	M	built from the previous period's score	2016-10-31 → 2026-10-16
Macro series (vintage-aware)	`macro_observations`	M	knowledge-dated	1854-12-01 → 2026-10-22, 1,062 series
CFTC positioning	`cftc_signals`	W	knowledge-dated	1986-01-15 → 2026-10-13
Cross-asset signals	`cross_asset_signals`	D	knowledge-dated; trend / curve shape / value / z model-based	2000-01-01 → 2026-10-20
Smart-money signals	`smart_money_signals`	W	event-dated; score model-based	2023-06-30 → 2026-10-16
Technical factors	`tech_factors`	D	event-dated	2004-01-01 → 2026-10-19

Benchmarks, ETFs, indices, futures and FX pairs (363 instruments, for example `IDX:^GSPC` , `ETF:SPY`) are usable as `prices` entities through the securities search. The BQL source is described in BQL query variable source.

Validation status

Not formally validated: the builder is an input tool. The look-ahead guardrail and the point-in-time sampling are covered by the engine's tests (`tests/modeling`).

Correct usage and worked example

Building "financials' monthly returns on rate changes" by hand:

1. Method **Distributed-lag regression (ADL)**; Dataset: Time series, Monthly, start 2005-01-01, universe `mvp_us` , sector Financials.
2. Add `prices:ret` as **y**; set **pool entities** to mean (the default) so the series is the equal-weighted sector return.

3. Add the US 10-year Treasury yield from the macro source as **x**, with transform **diff** (k = 1). Leave the lag at `t` : the method adds lags 0..3 itself (parameter `lags`).
4. Open **5 · Preview**: check that both columns cover the whole window and that no column is tagged "future".
5. Set `cov_type` to HAC in Parameters and **Run**. The preset "Sector returns vs lagged rates" is this study ready-made.

Limitations and pitfalls

- **Macro alignment matters.** For a regression of equity returns on macro data use point-in-time (or Auto, which picks it). Reference-period alignment with equity returns uses data before it was published.
- **Lags are per period of the dataset frequency.** `t-1` on a monthly dataset is one month.
- **Forward-fill limits.** A quarterly fundamental sampled on a daily grid is carried forward as-of, but an event-dated value is only carried 5 days on a daily grid; sparse event sources then show gaps.
- **Survivorship.** Universe membership history starts on 2026-10-10, so earlier dates use today's members.
- The missing-value policy (`ffill_limit` , `fill` , `drop`), dummies and interactions are part of the dataset specification but are not exposed in the builder; they can be set in presets and appear in the exported JSON.

Related mnemonics and BQL fields

Point-in-time rules, Point-in-time data, Field dictionary.

Methods catalogue

Route: `/studio` (Builder › 1 · Method) · **API:** `GET /api/v1/studio/catalog/methods` · **Refresh:** fixed in code (`backend/app/modeling/methods/*`)

Purpose

The catalogue lists every estimation method registered in the backend, with its accepted data shapes, its outputs and its diagnostics. This section is the reference for what each method estimates, its key parameters and its main assumptions.

Controls

Methods are chosen in section **1 · Method** of the builder. Parameters are set in **4 · Parameters** (see Builder).

Methodology

CATALOGUE

Defaults are in brackets. "ts" = time series, "pn" = panel, "cs" = cross-section.

Method (title in the app)	Family	What it estimates	Key parameters	Diagnostics	Assumptions / pitfalls
OLS regression	Regression	Linear coefficients, fit statistics; rolling-origin OOS for ts	`cov_type` [HC1]: nonrobust, HC0–HC3, HAC, cluster; `hac_maxlags` ; `cluster_by` ; `rolling_eval`	Breusch–Pagan, White, Durbin–Watson, Ljung–Box, Jarque–Bera, VIF, condition number, ADF/KPSS screen	Use HAC on overlapping or autocorrelated data; spurious regression on non-stationary levels
WLS regression	Regression	Weighted least squares	as OLS; a variable with role Weights	as OLS	Weights must be positive (e.g. inverse variance)
GLS regression (AR errors)	Regression	Feasible GLS with AR(p) errors (GLSAR, iterated)	`ar_order` [1], `max_iter`	as OLS	Time series only
Distributed-lag regression (ADL)	Regression	Lag profile, long-run multiplier (sum of lags) with SE, joint F per lag block	`lags` [3], `lag_overrides` , `y_lags` [0], `cov_type` [HAC]	as OLS + lag-block F	Collinear lags: read the joint F and the multiplier, not single lags
Quantile regression	Regression	Conditional quantiles (Koenker–Bassett), coefficient process	`quantiles` [0.1, 0.25, 0.5, 0.75, 0.9]	pseudo R^2 (Koenker–Machado), OLS comparison	Crossing quantiles possible in small samples
Generalized linear model	Regression	Gaussian, binomial (logit / probit), Poisson, gamma, negative binomial; exp(coef), AME	`family` [gaussian], `link` , `offset` , `cov_type` [HC1]	deviance, Pearson χ^2, overdispersion, AUC / Brier / log-loss (binomial), VIF	Poisson with overdispersion understates SEs: check dispersion, use negative binomial
Instrumental variables (2SLS / LIML / GMM)	Regression	IV coefficients for endogenous regressors	`estimator` [2sls], `cov_type` [robust]; roles endog / instrument	first-stage F / partial R^2, Sargan / Hansen J, Wu–Hausman	Weak instruments (first-stage F < 10) bias toward OLS
Panel regression (FE / RE / pooled / between / first-difference)	Panel	Within / between / overall R^2, effects	`estimator` [fe], `entity_effects` [on], `time_effects` , `cov_type` [clustered]: kernel = Driscoll–Kraay; `hausman` [on]	Hausman FE vs RE, F poolability, VIF	Time-invariant regressors drop out under FE
Fama-MacBeth regression	Panel	Risk premia λ (time means of cross-sectional slopes), NW SEs	`nw_lags` [|4(T/100)^(2/9)|], `min_obs`	Newey–West t, λ autocorrelation, share of positive λ	Errors-in-variables if betas are estimated; needs many dates

Method (title in the app)	Family	What it estimates	Key parameters	Diagnostics	Assumptions / pitfalls
Factor model builder	Panel	Long-short factor returns, alphas / betas of test assets, GRS, second-pass premia	`factor_scores`, `quantiles` [5], `include_market` [on], `test_assets` [portfolios]	GRS F-test, alpha t, factor Sharpe	Factors sorted on last period's score; equal weight
Difference-in-differences	Causal	ATT: 2×2, TWFE, event study (leads / lags), staggered (Callaway–Sant'Anna)	`design` [event_study], `treatment`, `leads` [6], `lags` [12], `control` [never], `bootstrap` [199]	joint pre-trend Wald test, treated / control counts	Parallel trends; TWFE is biased with staggered, heterogeneous effects: use `staggered`
Synthetic control	Causal	Treated minus synthetic path, donor weights	`treated`, `treatment_date`, `donors`, `placebo` [on], `max_donors` [60]	pre-period RMSPE, placebo rank p-value	Poor pre-fit invalidates the effect
Event study (CAR / BHAR)	Causal	AAR, CAAR path, per-event CAR and BHAR	`events` or `event_column`, `model` [market], `est_window` [−250, −30], `event_window` [−5, 5]	cross-sectional t, Patell Z	Clustered event dates make t-statistics overconfident
ARIMA / SARIMA / ARIMAX / SARIMAX	Time series	ARMA coefficients, forecast with intervals, OOS vs baselines	`order` [1,0,0], `seasonal_order`, `auto`, `criterion` [aic], `forecast_steps` [12], `exog_future` [last], `eval_folds` [24]	Ljung–Box, Jarque–Bera, heteroskedasticity, ADF / KPSS of y, residual ACF	ARIMAX forecasts need future regressor values
Stationarity & autocorrelation	Time series	ADF and KPSS in levels and differences, ACF / PACF, suggested d	`regression` [c], `max_diff` [1], `nlags` [24]	ADF, KPSS	Low power near unit root; read both tests
Local projections (Jorda)	Time series	Impulse responses by direct regression at each horizon	`horizon` [12], `y_lags` [2], `cumulative` [on], `fe` [entity], `spatial`	IRF bands (Driscoll–Kraay)	The first x is the shock; needs an identified shock
Vector autoregression (VAR)	Multivariate	Coefficients, IRFs, FEVD, Granger matrix, forecast	`maxlags` [8], `ic` [bic], `irf_periods` [24], `orthogonalized` [on]	stability roots, Portmanteau, normality	Cholesky IRFs depend on variable order
Vector error-correction model (VECM)	Multivariate	Johansen rank, β (cointegrating vectors), α (loadings), forecast	`k_ar_diff` [1], `det_order` [0], `rank`	Johansen trace / max-eigenvalue	Series must be I(1)
Cointegration + pairs spread	Multivariate	Engle–Granger and Johansen tests, hedge ratio, spread z-scores, half-life	`z_window` [60], `entry_z` [2.0]	Engle–Granger, Johansen trace, ADF on spread	In-sample hedge ratio; cointegration breaks
Diebold-Yilmaz connectedness	Multivariate	Total, to / from / net and pairwise connectedness (generalised FEVD)	`lags` [1], `horizon` [10], `window`, `step` [5]	none	Sensitive to lag and horizon choice
Filtered conformal ellipsoid (multivariate)	Multivariate	Calibrated joint 1−α set for the next observation, anomaly scores, projection bands	`filter` [ewma], `alpha` [0.1], `window` [250], `aci_gamma` [0.005]	binomial coverage test, per-coordinate projection coverage	**Experimental.** Coverage is marginal over time, not a statement about one date
GARCH volatility (GARCH / EGARCH / GJR)	Volatility	Conditional volatility, persistence, half-life, VaR, volatility forecast	`vol` [garch], `p` / `q` [1/1], `dist` [t], `horizon` [10], `var_levels` [0.01, 0.05]	Ljung–Box on squared std. residuals, Kupiec POF, Jarque–Bera	y only (no x); returns scaled by 100
Markov-switching regression / AR	Volatility	Regime means / variances, smoothed probabilities, transition matrix, durations	`k_regimes` [2], `model` [regression], `switching_variance` [on], `search_reps` [10]	expected durations, classification certainty	Local optima; labels of regimes are arbitrary
State space: unobserved components	Volatility	Smoothed level / trend / seasonal with bands, forecast	`level` [local level], `seasonal`, `forecast_steps` [12]	Ljung–Box, Jarque–Bera, heteroskedasticity	Component choice drives the result
Time-varying beta (Kalman filter)	Volatility	Random-walk α and β paths with bands vs static OLS	`add_constant` [on], `alpha` [0.05]; 1–5 x	Ljung–Box on one-step errors, share of β variation	Smoothed paths use the whole sample (not real-time)
Regularized regression (Lasso / Ridge / ElasticNet)	Machine learning	Penalised coefficients, coefficient paths, CV curve, OOS vs OLS	`model` [elasticnet], `l1_ratio` [0.5], `n_alphas` [50], `selection` [min] or 1se, `cv`	OOS R^2 vs OLS, selected features, CV folds	Standardised features; selection is unstable with correlated x
Tree ensemble (random forest / gradient boosting)	Machine learning	OOS metrics, impurity and permutation importance, SHAP, partial dependence	`model` [hgb]: rf, hgb, xgb; `task` [auto]; `n_estimators` [300]; `learning_rate` [0.05]; `min_samples_leaf` [20]; `cv`; `shap` [on]	OOS AUC / RMSE / rank IC, train vs OOS gap	Importance is not causality; use expanding CV for a backtestable signal
Model leaderboard	Machine learning	Ranking of candidates on identical folds, DM tests vs the best	`candidates`, `task` [auto], `metric` (rmse, mae, auc, logloss), `cv`	Diebold–Mariano	Many candidates = multiple testing
K-means clustering	Clustering & factors	Clusters, centroids, profiles, PCA view; k by silhouette	`k` or `k_min` .. `k_max` [2..10], `standardize` [on], `layout` [rows] or pivot, `impute` [drop]	silhouette, Calinski–Harabasz, Davies–Bouldin	Assumes round, similar-size clusters
Hierarchical (agglomerative) clustering	Clustering & factors	Dendrogram, cut by k, distance or best silhouette	`linkage` [ward], `metric` [euclidean], `n_clusters`, `distance_threshold`, `k_max` [10]	cophenetic correlation, silhouette	Ward needs euclidean distances
Principal component analysis	Clustering & factors	Explained variance, loadings, scores, score time series	`n_components`, `standardize` [on]	components for 80% / 90% variance	Signs of components are arbitrary
Gaussian mixture clustering	Clustering & factors	Soft clusters, k by BIC	`k_min` .. `k_max` [1..8], `covariance_type` [full]	BIC, mean max membership probability, silhouette	Full covariance needs many rows per cluster
Density clustering (HDBSCAN / DBSCAN)	Clustering & factors	Clusters of any shape plus noise (cluster −1)	`algorithm` [hdbscan], `min_cluster_size` [10], `min_samples` [5], `eps` [0.5]	noise share, silhouette (non-noise)	Many points can end up as noise

Method (title in the app)	Family	What it estimates	Key parameters	Diagnostics	Assumptions / pitfalls
Spatial lag model (SAR)	Spatial	ρ, β, direct / indirect / total effects across countries	`w_kind` [contiguity]: distance, kNN, trade, input-output, banking; `k` [5]; `fe` [entity]; `effects_draws` [500]	LM lag / error (robust), Moran's I of residuals	Country panels; W choice drives spillovers
Spatial error model (SEM)	Spatial	λ (spatially correlated errors), β	as SAR	as SAR	as SAR
Spatial Durbin model (SDM)	Spatial	ρ, β, θ (spatially lagged x), effects with simulation inference	as SAR	as SAR	as SAR
Spatial autocorrelation (Moran's I, LISA, Getis-Ord Gi*)	Spatial	Global Moran's I, LISA clusters HH / LL / HL / LH, hot / cold spots	`period` [latest], `permutations` [999]	permutation inference	y only; descriptive

REGRESSION FAMILY

```
OLS            β̂ = (X'X)⁻¹ X'y
HC1            V̂ = n/(n−k) · (X'X)⁻¹ X' diag(ê²) X (X'X)⁻¹
Newey–West     Ŝ = Γ̂₀ + Σ_{l=1..L} (1 − l/(L+1)) (Γ̂_l + Γ̂_l'),   L = ⌊4 (n/100)^(2/9)⌋ by default
Cluster        V̂ = (X'X)⁻¹ [Σ_g X_g' ê_g ê_g' X_g] (X'X)⁻¹      (entity, date or both)
ADL            y_t = α + Σ_{j=0..L} β_j x_{t−j} + Σ_{k=1..p} ϕ_k y_{t−k} + ε_t
               long-run multiplier = Σ β_j,  SE = √(1' V̂_β 1)
Quantile       β̂(τ) = argmin Σ ρ_τ(y_i − x_i'β),  ρ_τ(u) = u (τ − 1{u < 0})
GLM            g(E[y|x]) = x'β;  logit P = 1/(1 + e^(−x'β));  Poisson E[y] = exp(x'β + offset)
               exp(β) = odds ratio (logit) or incidence-rate ratio (Poisson);  AME = mean_i ∂E[y_i]/∂x
2SLS           β̂ = (X' P_Z X)⁻¹ X' P_Z y,  P_Z = Z (Z'Z)⁻¹ Z'
```

The coefficient table shows estimate (with stars: *** p < 0.01, ** p < 0.05, * p < 0.1, unadjusted), standard error, t or z, p-value, a Benjamini–Hochberg adjusted p-value when there are several tests, the 1−α interval, and where relevant exp(coef), the average marginal effect and VIF (amber above 10).

PANEL FAMILY

```
Fixed effects  y_it = α_i + λ_t + x_it'β + ε_it        (within transformation)
Hausman        H = (b_FE − b_RE)' [V_FE − V_RE]⁻¹ (b_FE − b_RE) ~ χ²_k
Fama–MacBeth   at each date t:  r_{i,t+1} = γ_0t + γ_t' z_it + e_it
               λ̂ = (1/T) Σ_t γ̂_t,   t(λ̂) = λ̂ / (σ̂_NW(γ̂) / √T)
GRS            GRS = (T/N) · (T−N−L)/(T−L−1) · α̂' Σ̂⁻¹ α̂ / (1 + μ̄' Ω̂⁻¹ μ̄) ~ F(N, T−N−L)
```

Driscoll–Kraay standard errors (`cov_type = kernel`) are robust to both cross-sectional and serial correlation, which is the usual situation in return panels.

CAUSAL FAMILY

```
2×2 DiD        ATT = (Ȳ_T,post − Ȳ_T,pre) − (Ȳ_C,post − Ȳ_C,pre)
Event study    y_it = α_i + λ_t + Σ_{k ≠ −1} δ_k 1{t − g_i = k} + ε_it     (reference period −1)
               pre-trend test: H0 δ_k = 0 for all k < −1 (joint Wald)
Callaway–Sant'Anna  ATT(g,t) = E[Y_t − Y_{g−1} | G = g] − E[Y_t − Y_{g−1} | control],
               control = never treated or not yet treated; aggregated overall and by event time; entity bootstrap
Synthetic control   w* = argmin ‖Y₁,pre − Y₀,pre w‖  s.t. w ≥ 0, Σ w = 1;   gap_t = Y₁t − Σ_j w_j Y_jt
               p-value = rank of post/pre RMSPE ratio among in-space placebos
CAR            AR_it = R_it − (α̂_i + β̂_i R_mt)   (market model on [−250, −30])
               CAR_i = Σ_{t ∈ window} AR_it;  CAAR = mean_i CAR_i;  BHAR = Π(1+R_it) − Π(1+R_mt)
```

TIME-SERIES AND MULTIVARIATE FAMILY

```
SARIMAX        ϕ(L) Φ(L^s) (1−L)^d (1−L^s)^D y_t = c + β'x_t + θ(L) Θ(L^s) ε_t
               order search: minimum AIC / BIC / AICc over a bounded grid (max_models 64)
OOS evaluation rolling origin, horizon h, folds n; RMSE relative to naive (no change), random walk + drift,
               seasonal naive and historical mean; "RMSE vs naive" < 1 = beats the no-change forecast
VAR(p)         Y_t = c + Σ_{j=1..p} A_j Y_{t−j} + u_t;  p by AIC / BIC;  stable if all roots lie outside the unit circle
VECM           ΔY_t = α β' Y_{t−1} + Σ_j Γ_j ΔY_{t−j} + u_t;  rank r by Johansen trace / max-eigenvalue
Pairs          y_t = a + b x_t + e_t;  spread s_t = y_t − b̂ x_t;  z_t = (s_t − mean) / sd  (full sample and rolling)
               half-life = −ln 2 / ln ϕ̂  from  s_t = c + ϕ s_{t−1} + η_t
Local projection y_{t+h} − y_{t−1} = α_h + β_h shock_t + controls + ε_{t+h};  IRF = {β_h}
Connectedness  total = 100 · Σ_{i≠j} θ̃_ij(H) / N   (generalised FEVD, rows normalised)
```

VOLATILITY, REGIMES AND STATE SPACE

```
GARCH(1,1)     σ²_t = ω + α ε²_{t−1} + β σ²_{t−1};   persistence α + β;  half-life = ln 0.5 / ln(α + β)
GJR            σ²_t = ω + (α + γ 1{ε_{t−1} < 0}) ε²_{t−1} + β σ²_{t−1}
VaR            VaR_q = μ + σ̂_{t+1} F⁻¹(q),  backtested in-sample with Kupiec's proportion-of-failures test
Markov switching y_t = μ_{S_t} + x_t'β_{S_t} + ε_t,  ε_t ~ N(0, σ²_{S_t}),  P(S_t = j | S_{t−1} = i) = p_ij
               expected duration of regime i = 1 / (1 − p_ii);  smoothed probabilities P(S_t | all data)
Kalman beta    y_t = α_t + β_t x_t + ε_t,   β_t = β_{t−1} + η_t   (variances by MLE, Kalman smoother)
UCM            y_t = μ_t + γ_t + ε_t,   μ_t = μ_{t−1} + ν_{t−1} + η_t
```

MACHINE LEARNING

```
ElasticNet       β̂ = argmin (1/2n) ‖y − Xβ‖² + α [ρ ‖β‖₁ + (1−ρ)/2 ‖β‖²₂]     (ρ = l1_ratio; Lasso ρ = 1, Ridge ρ = 0)
                 α chosen by CV: minimum MSE ("min") or the largest α within one SE of it ("1se")
SHAP             f(x) = E[f] + Σ_j ϕ_j(x);  importance = mean_i |ϕ_j(x_i)|
Permutation imp. increase of the held-out loss when feature j is shuffled
Diebold–Mariano  d_t = L(e_A,t) − L(e_B,t);  DM = d̄ / √(LRV̂(d)/T)
```

Cross-validation for regularised regression, tree ensembles and the leaderboard is time-aware (`backend/app/modeling/_ml.py`):

- `expanding` (default for time series): walk-forward on blocks of unique dates; the training set is every date before the test block minus `purge` periods.
- `purged_kfold` (default for panels): contiguous date blocks as test sets; the training set excludes `purge` periods before and `embargo` periods after each test block (López de Prado).
- `kfold` (default for cross-sections): shuffled K-fold on rows.
- `purge` defaults to the target's forward horizon (the k of a `lead` / `fwd_*` transform), at least 1 period. Default 5 folds.

Only `expanding` gives predictions that never used later data for training, which is why the **Backtest this signal** button warns when another scheme was used. See Purged and embargoed cross-validation.

CLUSTERING AND FACTORS

```
K-means          minimise Σ_c Σ_{x ∈ c} ‖x − μ_c‖²;  k by best silhouette s = (b − a) / max(a, b)
Hierarchical     linkage ward / average / complete / single on euclidean, 1 − correlation or cosine distance
GMM              k by BIC = −2 ln L + p ln n;  soft membership P(c | x)
HDBSCAN          density clusters on mutual-reachability distance; points in no dense region = noise (−1)
PCA              eigen-decomposition of the (standardised) covariance; share_k = λ_k / Σ λ
```

`layout = rows` clusters each row (entity) on its features; `layout = pivot` turns one variable's dates into features, for example clustering sectors by their return series with correlation distance.

SPATIAL

```
SAR   y = ρ W y + X β + ε            SEM   y = X β + u,  u = λ W u + e            SDM   y = ρ W y + X β + W X θ + ε
Effects  S(W) = (I − ρ W)⁻¹ (β I + θ W);  direct = mean of the diagonal, total = mean row sum, indirect = total − direct
Moran's I = (N / S₀) · z'W z / z'z
```

W is built from the linkage tables by `w_kind` (contiguity, distance, k nearest neighbours, trade, input-output, banking) as of `w_as_of` .

FILTERED CONFORMAL ELLIPSOID

```
score     s_t = (y_t − μ̂_t)' Σ̂_t⁻¹ (y_t − μ̂_t)       (μ̂, Σ̂ from a frozen filter: EWMA, static shrinkage, or EWMA vol + static correlation)
radius    q_t = the ⌈(1−α)(n+1)⌉-th smallest of the last `window` realised scores
ACI       α_{t+1} = α_t + γ (α − 1{y_t outside the set})
band      μ̂_j ± √(q_t Σ̂_jj)   (projection of the ellipsoid on coordinate j)
```

Validation status

> **Experimental:** the filtered conformal ellipsoid. `docs/CONFORMAL_VALIDATION.md` (Verdict) states that the joint envelope "A stays out of the main UI. It remains 'experimental' in Model Studio and the Backtest Lab": application A failed its gate (Covid February–June 2020 coverage 77–78% at a 90% target; projection width +35% against a calibrated ATR box). The Model Studio method applies the same construction to any multivariate selection and inherits that status. The builder does not show an experimental badge on the chip; the result carries a "model based" note: "coverage is marginal over time ('calibrated to X% over the test period'), never a statement about one date".

Every other method is a standard estimator from statsmodels, linearmodels, arch, scikit-learn, xgboost, shap or scipy and is **not formally validated** on our data as a model; the verdicts in the Tests tables are fixed-rule readings of textbook tests.

Correct usage and worked example

Pick the method by question, not by habit:

Question	Method
Does x move y, controlling for z?	OLS (HAC errors for time series, clustered for panels)
Does it work across many stocks?	Panel FE (Driscoll–Kraay) or Fama–MacBeth
What happened after an event?	Event study (CAR) or DiD
What will the series do next?	ARIMA / SARIMAX, compare with the baselines in the Out-of-sample block
How do shocks propagate?	VAR (IRF / FEVD) or local projections
Which variables predict out of sample?	Regularised regression, tree ensemble, leaderboard
Which names belong together?	K-means, hierarchical, GMM, HDBSCAN

Limitations and pitfalls

- Clustering, PCA and Markov switching labels (cluster 0, PC1, regime 1) are arbitrary; read the profiles.
- Smoothed state-space and Markov probabilities use the whole sample; they are not what you would have seen in real time.
- The leaderboard and the coefficient table's many p-values are subject to multiple testing; use the adjusted p-values.

Related mnemonics and BQL fields

Conformal prediction, Event studies and abnormal returns, Gate status of every model.

Presets

Route: `/studio` (tab **Presets**) · **API:** `GET /api/v1/studio/presets` · **Refresh:** fixed (`backend/app/modeling/presets/*.json`)

Purpose

Presets are complete, ready-to-run studies covering every method family on real data. They are the fastest way to see what a method produces and a template to adapt.

Screenshot

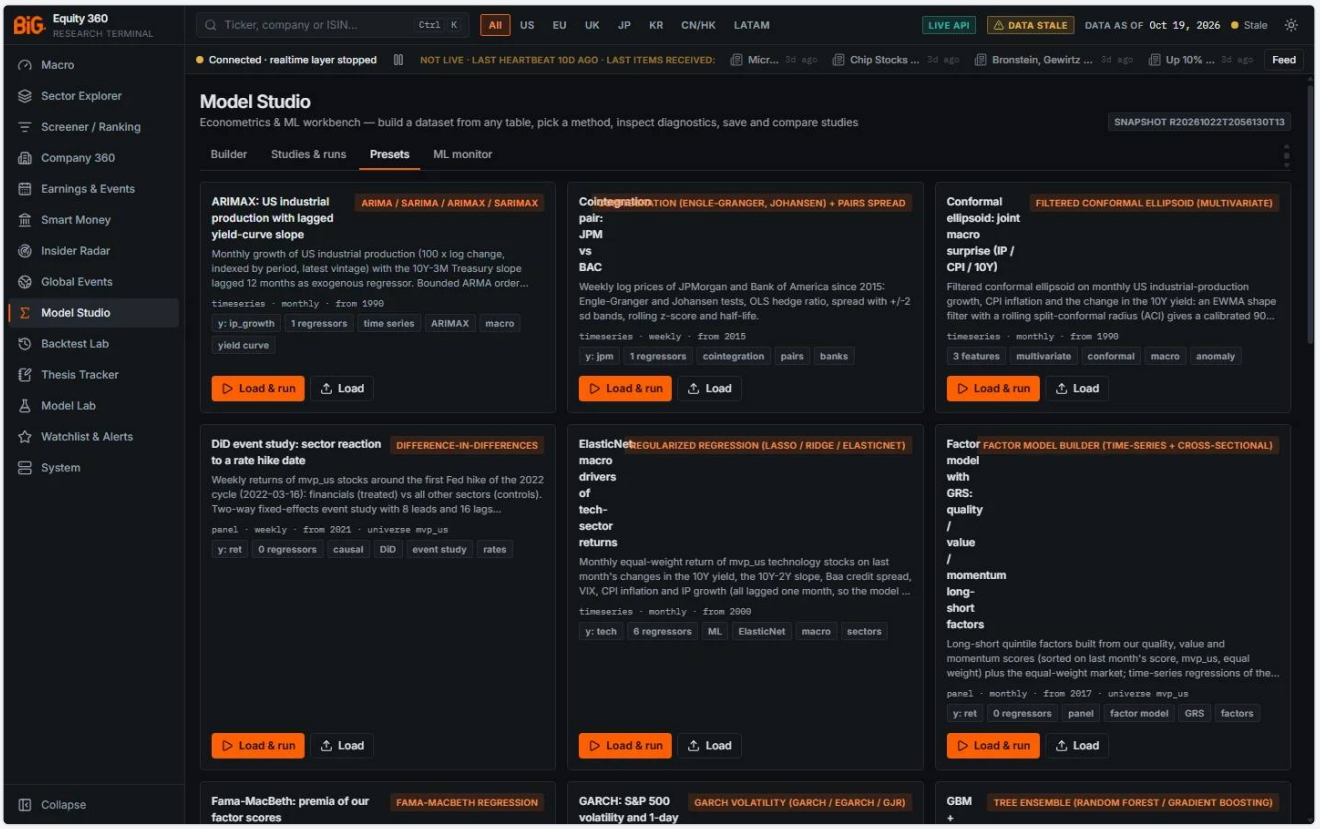


Figure 9.3: Presets tab (`/studio`), first six of 22 preset cards.

Each card shows the preset title, the method badge, a three-line description, the dataset summary (shape · frequency · start year · universe), the target `y:` , the number of regressors or features and up to four tags.

Controls

Control	What it does
Load & run	Loads the study into the builder and runs it immediately.
Load	Loads the study into the builder only (toast "Preset loaded into the builder"). Change it, then run or save it.

Methodology

The live API returns 22 presets:

Preset	Method	What it does
ARIMAX: US industrial production with lagged yield-curve slope	ARIMA	Monthly US IP growth with the 10Y–3M slope lagged 12 months as exogenous regressor; ARMA order searched by BIC; 12-month forecast using known future regressor values.

Preset	Method	What it does
Cointegration pair: JPM vs BAC	Cointegration	Weekly log prices since 2015: Engle–Granger and Johansen tests, hedge ratio, ±2 sd spread bands, rolling z-score (52 weeks), half-life.
Conformal ellipsoid: joint macro surprise (IP / CPI / 10Y)	Conformal ellipsoid (**experimental**)	Calibrated 90% joint set for next month's US IP growth, CPI inflation and 10Y change; months outside it are joint macro anomalies.
DiD event study: sector reaction to a rate hike date	Difference-in-differences	Weekly `mvp_us` returns around 2022-03-16: financials treated vs other sectors; TWFE event study, 8 leads, 16 lags, pre-trend test.
ElasticNet: macro drivers of tech-sector returns	Regularized regression	Monthly tech-sector return on last month's changes in 10Y, 10Y–2Y, Baa spread, VIX, CPI, IP; penalty by expanding-window CV.
Factor model with GRS: quality / value / momentum long-short factors	Factor model builder	Quintile long-short factors from our scores plus the market; time-series regressions of 15 score portfolios; GRS test; second-pass premia.
Fama-MacBeth: premia of our factor scores	Fama-MacBeth	Monthly `mvp_us` cross-sections: next-month return on standardised quality, value, momentum, growth and profitability scores.
GARCH: S&P 500 volatility and 1-day VaR	GARCH	Daily `IDX:^GSPC` returns since 2000: GJR-GARCH(1,1) with Student-t errors, 10-day volatility band, 1-day VaR at 1% / 5% with Kupiec backtest.
GBM + SHAP: 3M outperformance from factor scores	Tree ensemble	HistGradientBoosting classifier of "next 3-month return beats the cross-sectional mean"; purged K-fold (3-month purge); SHAP and PDP.
GMM regimes of US macro states	Gaussian mixture	Monthly US macro state since 1990 (IP, CPI, 10Y–3M, Baa, unemployment change); states chosen by BIC with soft probabilities.
HDBSCAN of global companies	Density clustering	`mvp_global` companies on profitability, growth, leverage and valuation ratios; dense peer clusters and outliers as noise.
Hierarchical clustering of sectors by return correlation	Hierarchical	Equal-weighted monthly sector returns since 2012; average linkage on 1 − correlation; dendrogram and best-silhouette cut.
Kalman time-varying beta: AAPL vs S&P 500	Time-varying beta	Weekly AAPL return on `IDX:^GSPC` since 2005 with random-walk alpha and beta vs the static OLS beta.
K-means peer discovery: MVP companies on profitability/growth/valuation ratios	K-means	Latest PIT TTM ratios of `mvp_us` , winsorised and standardised; k by silhouette over 3..10.
Leaderboard: OLS / ElasticNet / GBM / ARIMAX for US IP growth	Leaderboard	One-month-ahead IP growth from its own lags, the lagged slope and Baa spread; five candidates on identical expanding folds, ranked by OOS RMSE with Diebold–Mariano tests.
Logit: probability of 3M outperformance from factor scores	GLM (binomial)	Monthly `mvp_us` panel; y = 1 if the next-3-month return beats the cross-sectional mean; date-clustered errors.
Markov switching: US market regimes	Markov switching	Monthly S&P 500 return since 1960 with two regimes in mean and variance; smoothed probabilities, transition matrix, durations.
PCA of global yield curves	PCA	Monthly changes of the US curve (3M–30Y) and nine other 10Y yields since 1992; PC1 is the global level factor.
Poisson: insider cluster-buy counts vs drawdown	GLM (Poisson)	Monthly count of `mvp_us` companies with an insider cluster buy on last month's average drawdown and 3-month return; HAC errors; dispersion check.
SARIMA: EA HICP monthly	ARIMA	Euro-area HICP inflation (ECB) with a seasonal ARIMA (s = 12), order by AIC, 24-month fan, rolling-origin evaluation.
Sector returns vs lagged rates (distributed lag OLS, HAC)	Distributed lag	Monthly US financials return on current and three lagged changes in the 10Y yield and 10Y–2Y slope; joint F-tests, long-run multipliers.
VAR: rates / inflation / IP impulse responses	VAR	Monthly US IP growth, CPI inflation and 10Y change since 1990; lag by BIC; Cholesky IRFs over 24 months (ordering IP → inflation → rates).

The research-recipe library with about 40 presets grouped by question, a recipe search and "Save as my preset", planned in `docs/ROADMAP_V3_PLAN.md` section S, is not built yet: the app serves the 22 presets above.

Data sources and refresh

Presets reference sources and series ids, not data; they run on the current snapshot.

Validation status

Not formally validated. Presets demonstrate methods; their results are not validated findings. The conformal ellipsoid preset is experimental (see Methods catalogue).

Correct usage and worked example

Load "PCA of global yield curves" with **Load & run**. In the PCA tab, the explained-variance table states how many components explain at least 80% of the variance; the loadings heatmap shows PC1 loading with the same sign on every yield (a global level factor). The live run history shows this study on 416 monthly observations.

Limitations and pitfalls

- Presets with `mvp_us` panels start in 2016–2017 because factor-score history starts on 2016-10-31.
- Presets whose description says "latest vintage" (ARIMAX, SARIMA) use reference-period macro alignment: fine for a macro forecast, not for an equity strategy.

Related mnemonics and BQL fields

Methods catalogue, Results.

Studies & runs

Route: `/studio` (tab **Studies & runs**) · **Refresh:** live list; scheduled studies re-run weekly or monthly (job `studio_scheduled`, weekly DAG)

Purpose

This tab keeps your saved studies, the history of every run (yours and scheduled ones), the comparison of runs, and the peer sets saved from clustering runs.

Screenshot

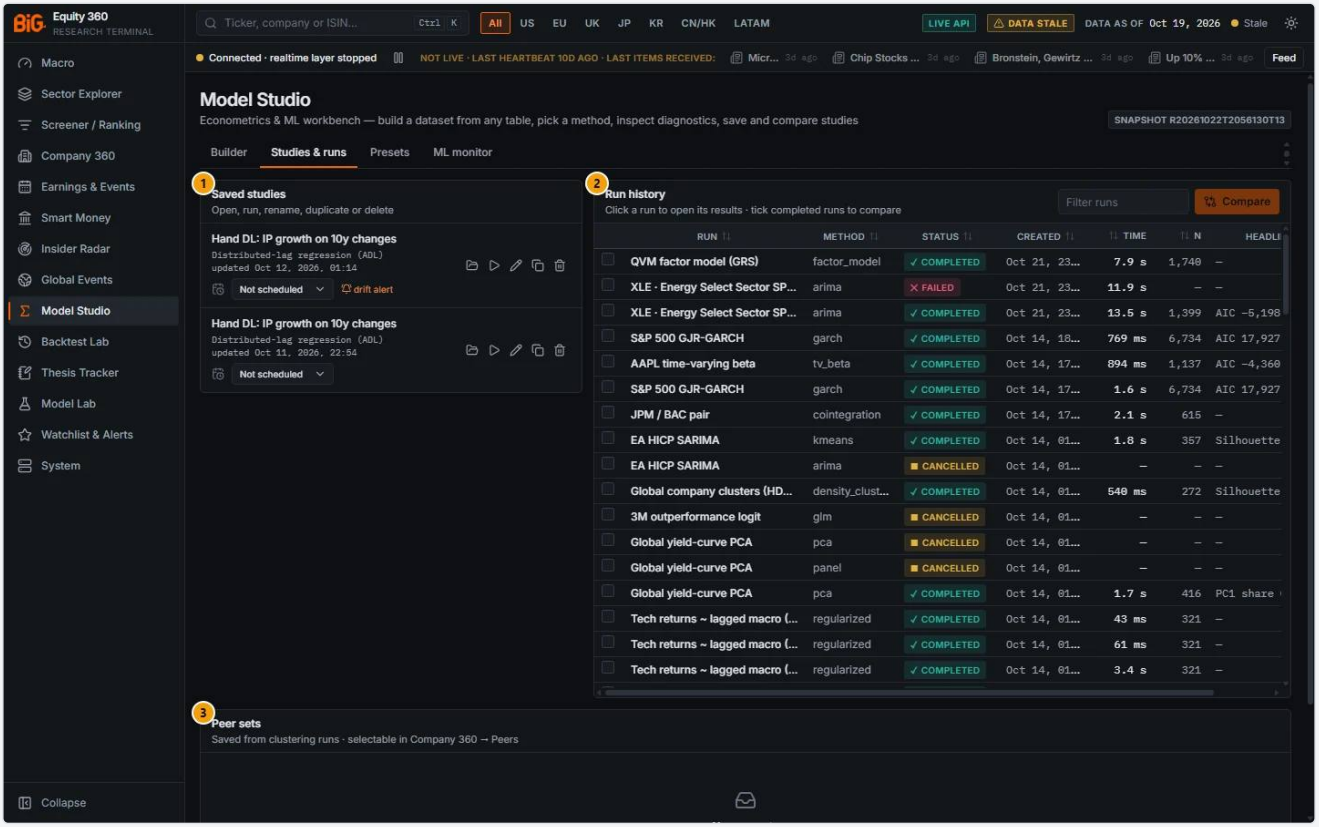


Figure 9.4: Studies & runs (`/studio`): two saved studies, the run history and the empty peer-set panel.

#	Panel	What it shows
1	Saved studies	"Open, run, rename, duplicate or delete". Each row: name, method, last update, "from preset", last run id, and the schedule row. The first study shows a **drift alert** link.
2	Run history	"Click a run to open its results · tick completed runs to compare". Columns: checkbox, Run, Method, Status (completed, failed, cancelled, queued, running), Created, Time, n, Headline (the first available of adjusted R^2, R^2, pseudo R^2, AUC, silhouette, PC1 share, OOS relative RMSE, AIC).
3	Peer sets	"Saved from clustering runs · selectable in Company 360 → Peers". Empty: "No peer sets. Run a security-level clustering (k-means, GMM, HDBSCAN…) and use “Save as peer set”."

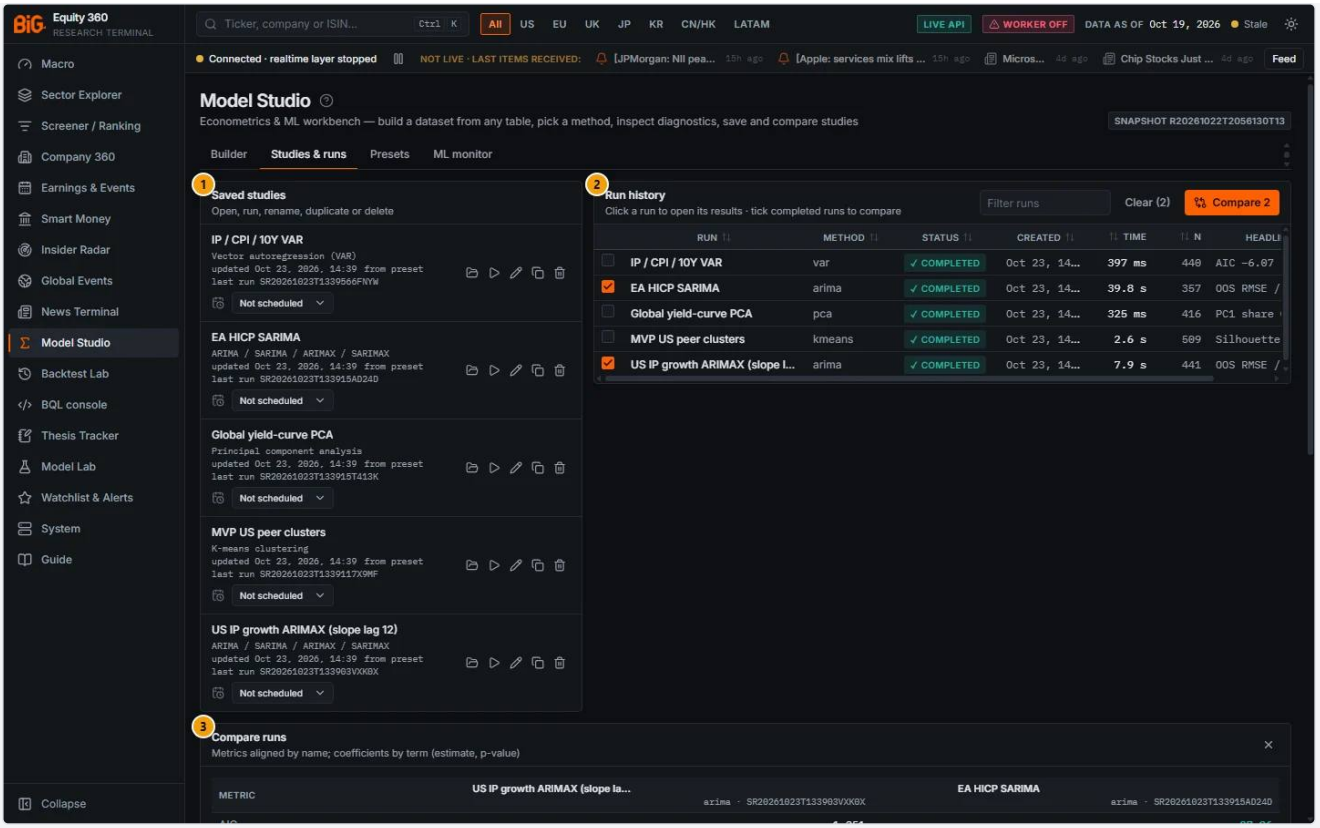


*Figure 9.5: Studies & runs on the test copy: five saved studies loaded from presets, the run history with two ARIMA runs ticked and the *Compare runs** panel open.*

#	Panel	What it shows
1	Saved studies	Five studies created from presets ("from preset"), each with its last run id and the schedule select (here "Not scheduled").
2	Run history	Five completed runs; "US IP growth ARIMAX (slope lag 12)" and "EA HICP SARIMA" are ticked, so the header shows **Clear (2)** and **Compare 2**. Headline column: AIC, OOS RMSE, PC1 share, silhouette.
3	Compare runs	"Metrics aligned by name; coefficients by term (estimate, p-value)": one column per run with its method and run id.
4	Peer sets	Below the fold.

Controls

Control	What it does
Study name (link) / **Open in builder** (folder icon)	Loads the study and its last run into the builder.
Run now (play icon)	Opens the study and runs it ("auto" mode).
Rename (pencil)	Inline rename with **Save** / cancel.
Duplicate	Creates "‹name› (copy)".
Delete (bin) → **Delete?**	Two-step delete.
Schedule select	**Not scheduled**, **Re-run weekly**, **Re-run monthly**. Scheduling creates a `studio_drift` alert rule; "Schedule removed" deletes it.
drift alert at N % worse	The relative degradation that triggers the alert (default 20%).
"last scheduled run" / **drift alert** link	When the study last re-ran; the link opens Watchlist & Alerts.
Filter runs	Filters the history by name, method or run id.
Row click	Opens that run in the Builder tab's results pane.
Checkboxes, **Clear (n)**, **Compare n**	Tick up to six completed runs; **Compare** needs at least two.
Peer set **run** / delete	Opens the source clustering run or deletes the peer set.

Methodology

Run modes. "auto" runs inline when the estimated cost is under 2 seconds, otherwise it is queued on the worker. **Queue** forces the worker. Queued runs show "Queued on the worker" or "Running" with the elapsed time, the estimate, a progress bar capped at 95% and **Cancel run**. A run that stops sending heartbeats for 180 s beyond its timeout is marked failed ("run stopped without a result (worker died or was killed)").

Run watcher. An app-wide watcher polls every queued run every 2 seconds, wherever you are in the app, and raises a toast when it finishes ("Run finished: ‹name›", with "check the warnings" when there are non-info warnings), fails or is cancelled. If no run is open in the results pane, the finished run opens there.

Compare (`POST /studio/compare`) aligns metrics by name and coefficients by term. In the metrics table the best value per row is green (lower is better for AIC, BIC, RMSE, MAE, MAPE, Brier, log-loss, deviance, inertia, Davies–Bouldin; counts such as n_obs or k are not ranked). The coefficient table shows each estimate with its p-value, bold when p < 0.05. The footnote warns: "Only meaningful when runs share the same target and sample."

Scheduled re-runs and drift (`backend/app/modeling/scheduled.py`). A weekly study is due 6 days after its last scheduled run, a monthly one after 27 days. After the re-run, the job compares the result with the previous completed run of the study:

- **Metric drift:** relative change `(new − old) / |old|` worse than the threshold in the bad direction. Higher-is-better metrics include R^2, adjusted R^2, within R^2, pseudo R^2, AUC, OOS R^2, silhouette, OOS rank IC, accuracy; lower-is-better include OOS RMSE / MAE, in-sample RMSE, Brier, OOS log-loss, OOS relative RMSE.
- **Coefficient shift:** `|β_new − β_old| > 3 × max(SE_old, SE_new)` for a term present in both runs.

When either fires, a `studio_drift` event is appended to the study's alert ("Study '‹name›' drifted: ...").

Data sources and refresh

Studies and run metadata live in `app_state` (`studies` , `study_runs`); run artefacts under `data/modeling/runs/` . The list shows the 100 most recent runs.

Validation status

Not applicable (workflow).

Correct usage and worked example

In Figure 9.4 the run history contains two runs of "S&P 500 GJR-GARCH" with the same n (6,734) and AIC (17,927): same spec, same snapshot, same numbers. Tick both and **Compare** to confirm that every metric matches; a difference would mean the spec or snapshot changed (check the spec hash in the Data & reproducibility tab).

Limitations and pitfalls

- Compare only what is comparable: same target, same sample. Comparing R^2 across different targets is meaningless.
- A drift alert on a scheduled study is a prompt to look, not a verdict: new data can legitimately change a coefficient.

Related mnemonics and BQL fields

Watchlists and alerts, Peers (peer sets).

Results

Route: `/studio` (Builder tab, results pane) · **API:** `GET /api/v1/studio/runs/{run_id}`

Purpose

The results pane shows one run: its header with actions, the warnings, and a set of tabs that depends on what the method produced.

Screenshot

All result figures come from the test copy (runs of 23 Oct 2026 on snapshot R20261022T2056130T13). In each figure callout 1 is the study builder with the study loaded and callout 2 the results pane.

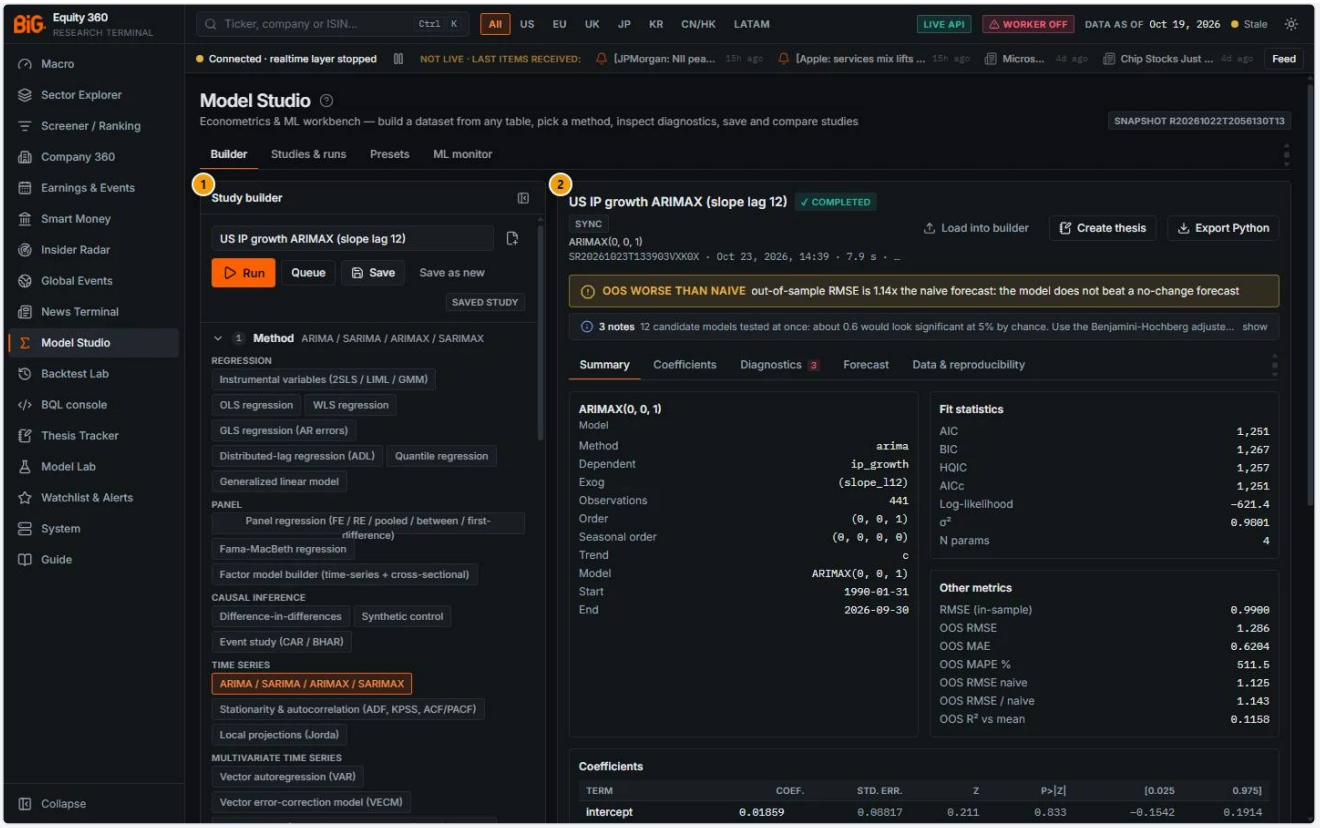


Figure 9.6: "US IP growth ARIMAX (slope lag 12)", Summary tab (test copy).

#	Panel	What it shows
1	Study builder	The saved study (badge **SAVED STUDY**, buttons **Save** / **Save as new**).
2	Results pane	Header (✓ COMPLETED, **SYNC**, ARIMAX(0, 0, 1), run id, 7.9 s), the amber warning "OOS WORSE THAN NAIVE out-of-sample RMSE is 1.14x the naive forecast: the model does not beat a no-change forecast", "3 notes" (the first: "12 candidate models tested at once: about 0.6 would look significant at 5% by chance. Use the Benjamini-Hochberg adjusted..."), the tabs and the Summary: Model card (dependent `ip_growth`, exog `slope_l12`, 441 observations, order (0, 0, 1), 1990-01-31 → 2026-09-30), Fit statistics (AIC 1,251, BIC 1,267, σ^2 0.9801) and Other metrics (OOS RMSE 1.286 vs naive 1.125, OOS RMSE / naive 1.143, OOS R^2 vs mean 0.1158).

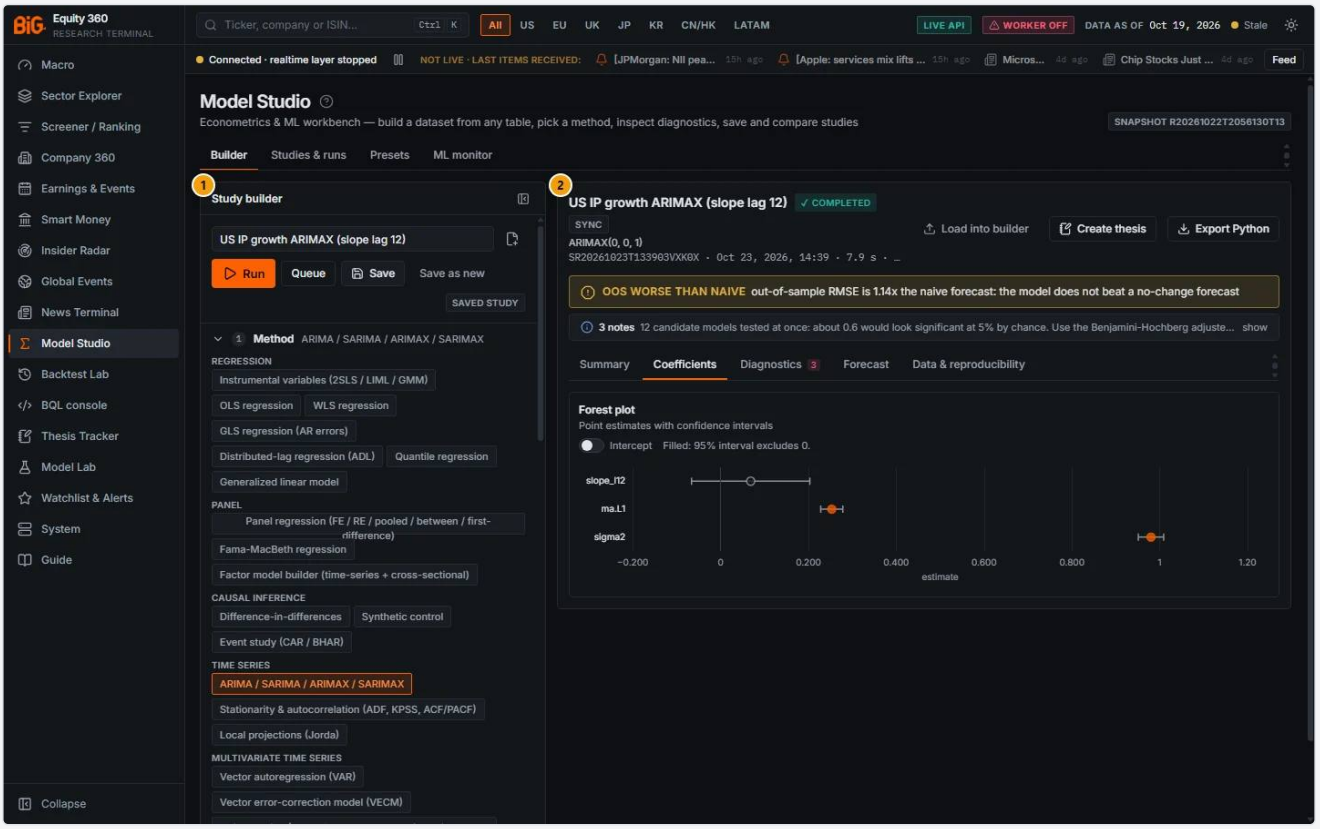


Figure 9.7: Coefficients tab: forest plot of `slope_l12`, `ma.L1` and `sigma2` (filled marker = 95% interval excludes 0; the Intercept switch adds the constant).

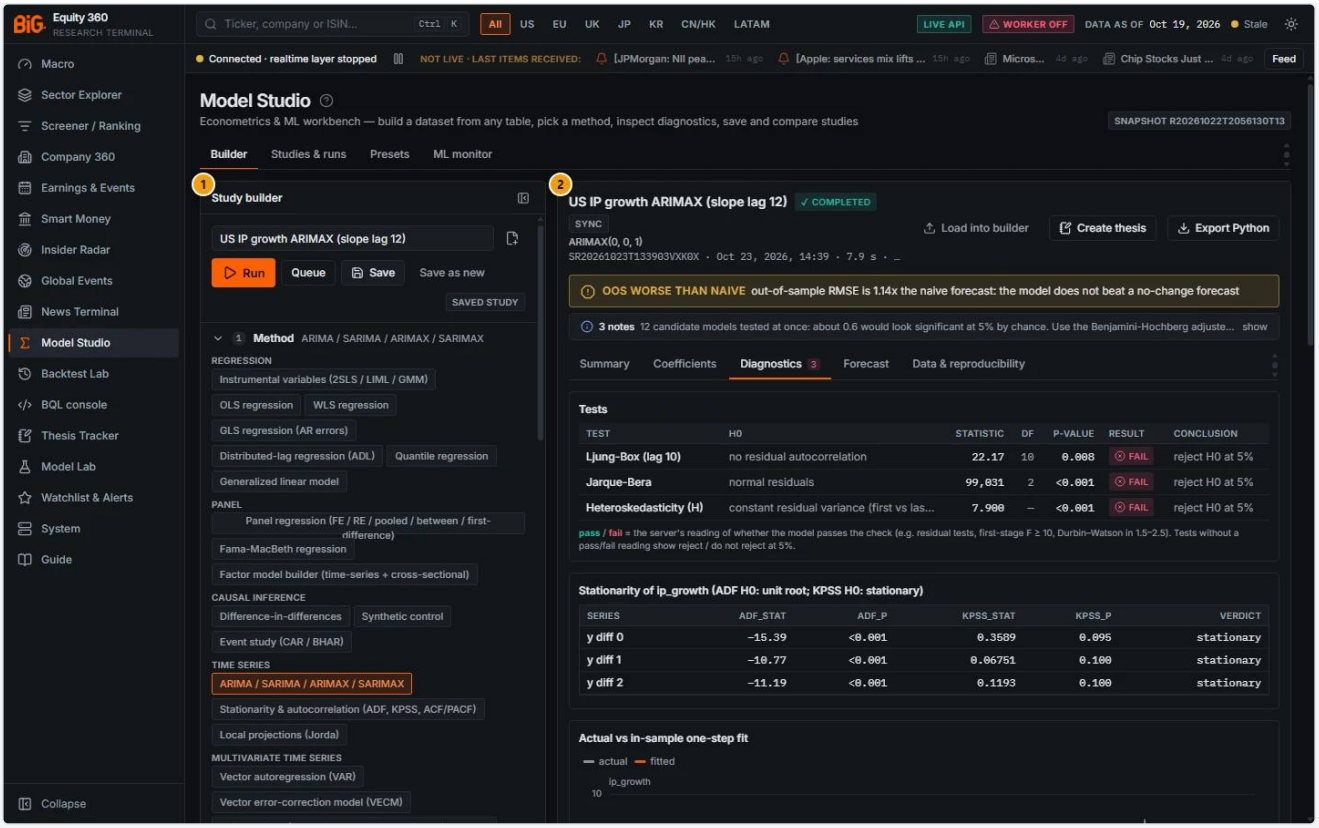


Figure 9.8: Diagnostics tab (badge 3 = three failed tests): Ljung–Box (lag 10) p 0.008, Jarque–Bera p < 0.001 and heteroskedasticity (H) p < 0.001 all FAIL; the stationarity table finds `ip_growth` stationary in levels (ADF −15.39, KPSS 0.3589).

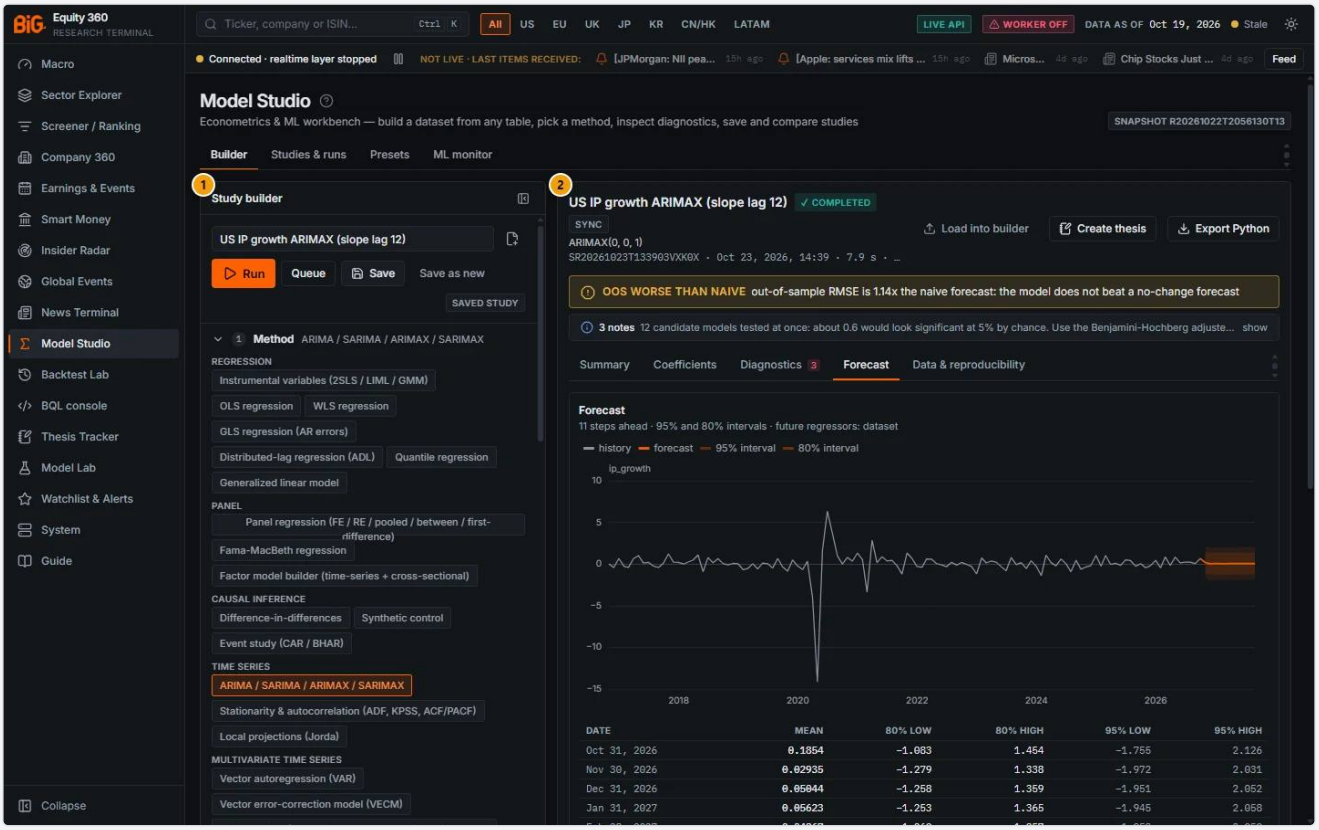


Figure 9.9: Forecast tab: "11 steps ahead · 95% and 80% intervals · future regressors: dataset", fan chart and the forecast table (Oct 2026 mean 0.1854, 80% −1.083 to 1.454).

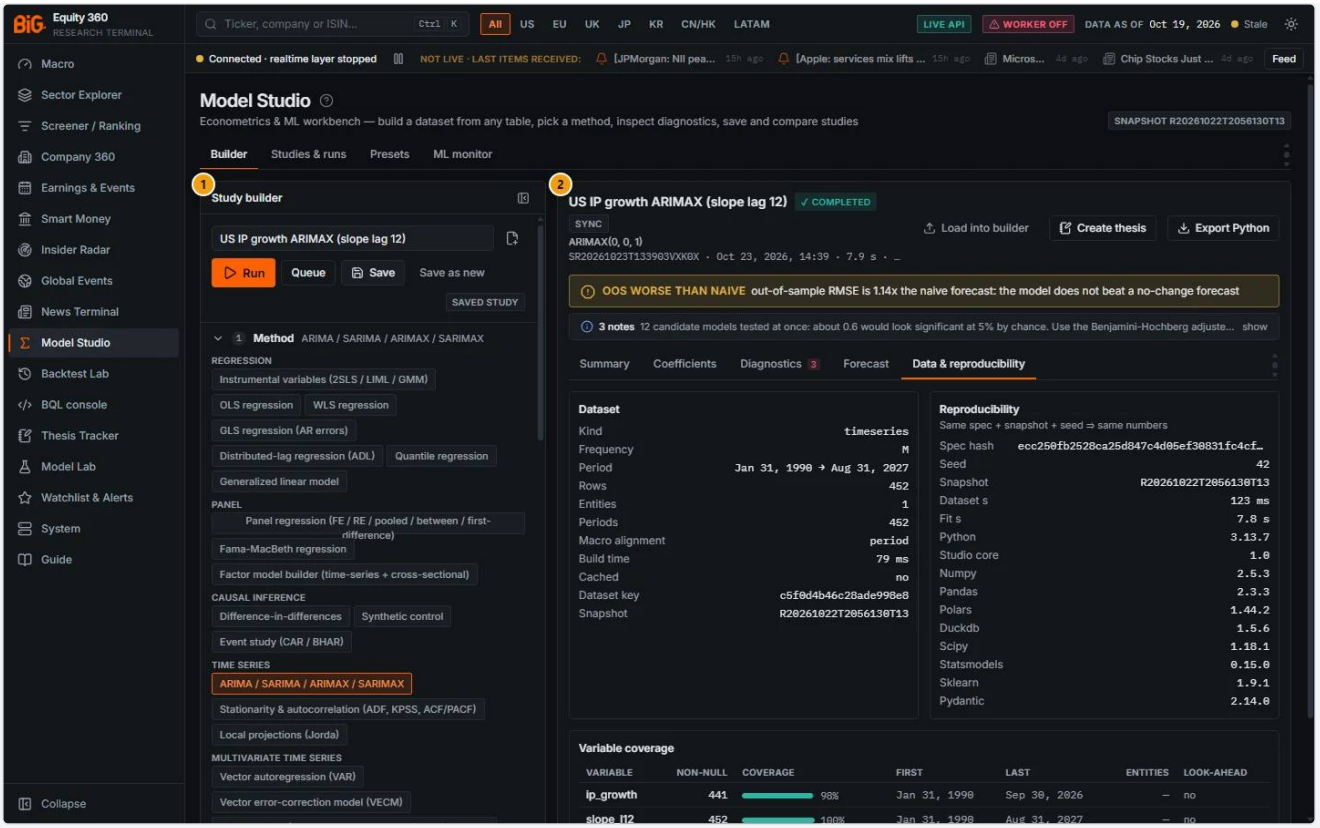


Figure 9.10: Data & reproducibility: dataset (time series, monthly, 452 rows, macro alignment `period`), reproducibility (spec hash, seed 42, snapshot, timings, library versions) and variable coverage (`slope_l12` runs to Aug 2027 because a 12-month lag of a known series is known a year ahead).

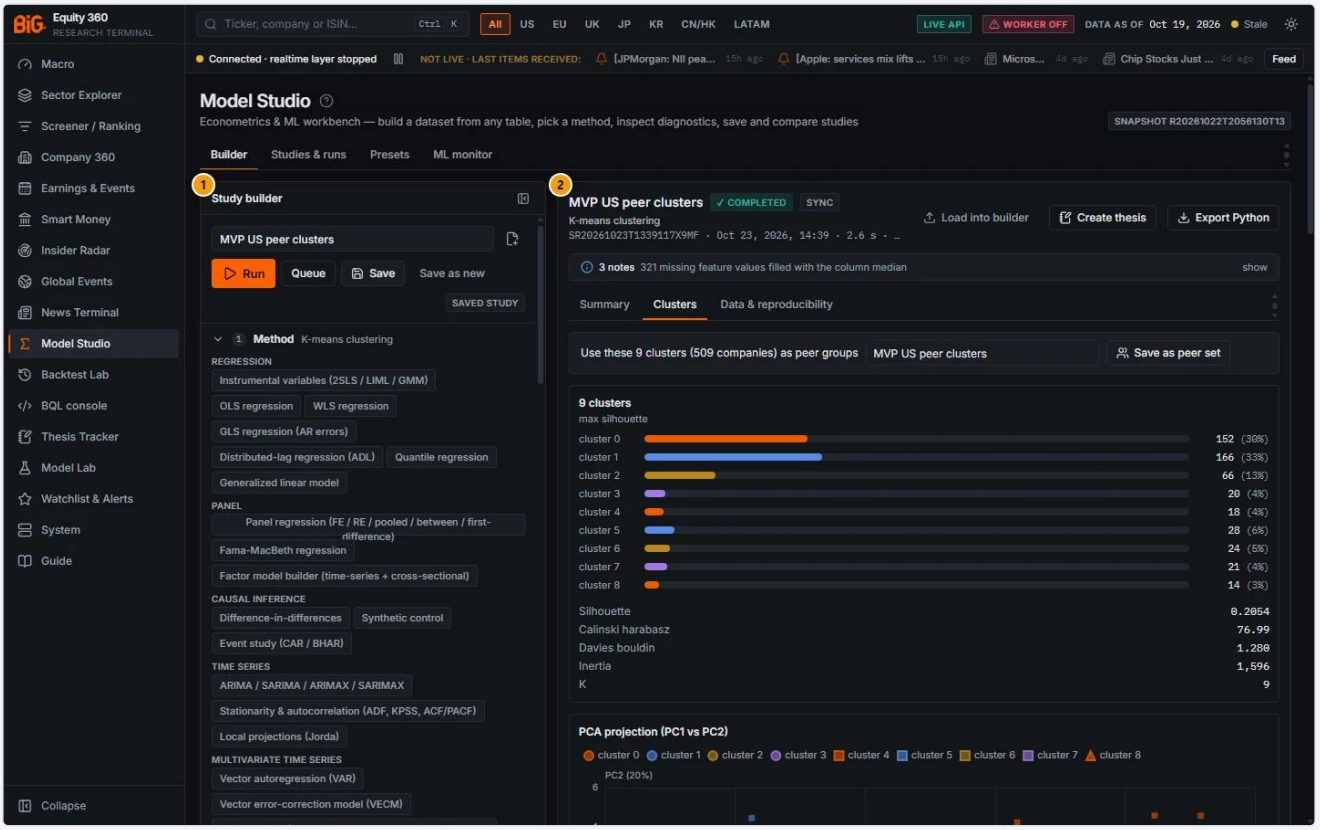


*Figure 9.11: "MVP US peer clusters" (k-means), Clusters tab: the "Use these 9 clusters (509 companies) as peer groups" bar with *Save as peer set*, cluster sizes chosen by maximum silhouette (silhouette 0.2054, Calinski–Harabasz 76.99, Davies–Bouldin 1.280), the PCA projection below. The note "321 missing feature values filled with the column median" comes from `impute = median`.*

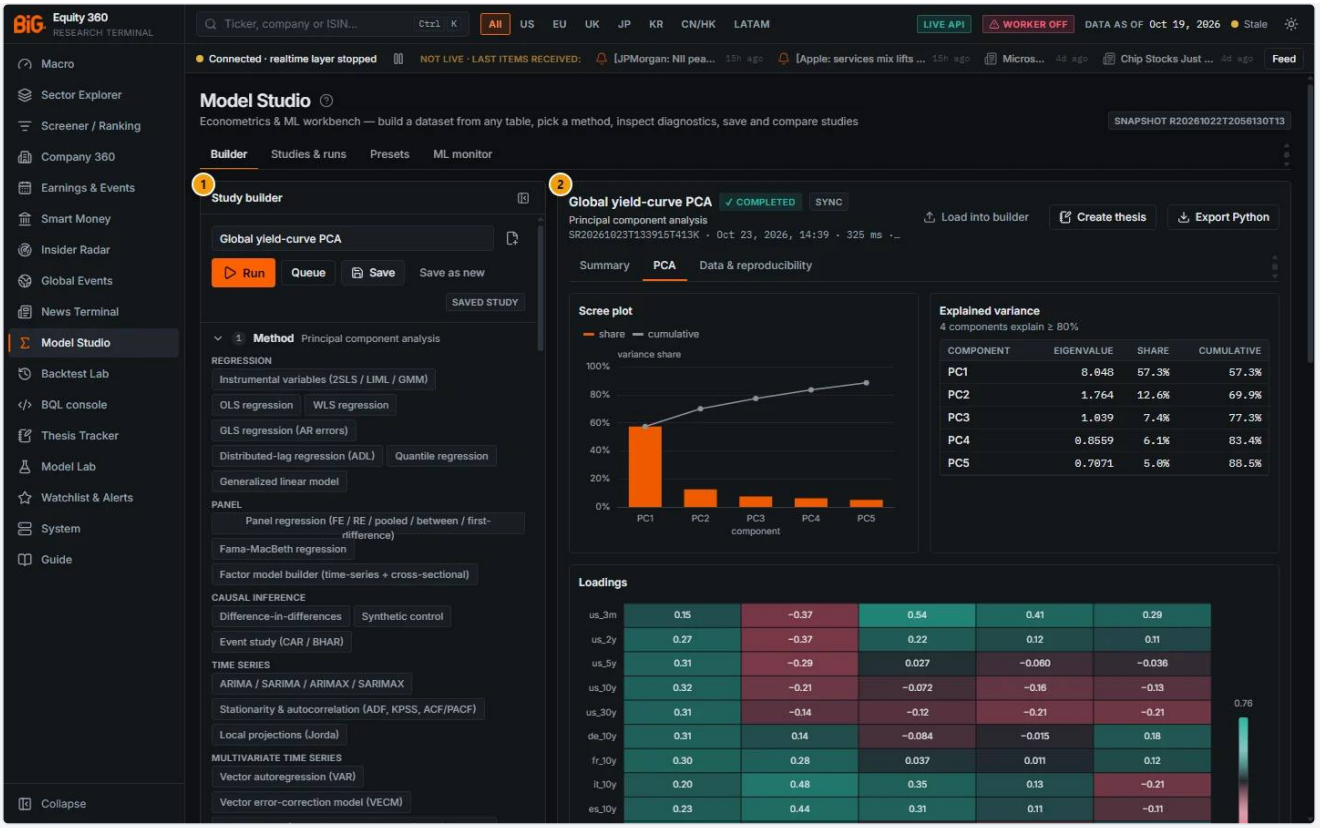


Figure 9.12: "Global yield-curve PCA", PCA tab: scree plot, explained variance (PC1 57.3%, four components explain ≥ 80%) and the loadings heatmap (PC1 loads positively on every yield).

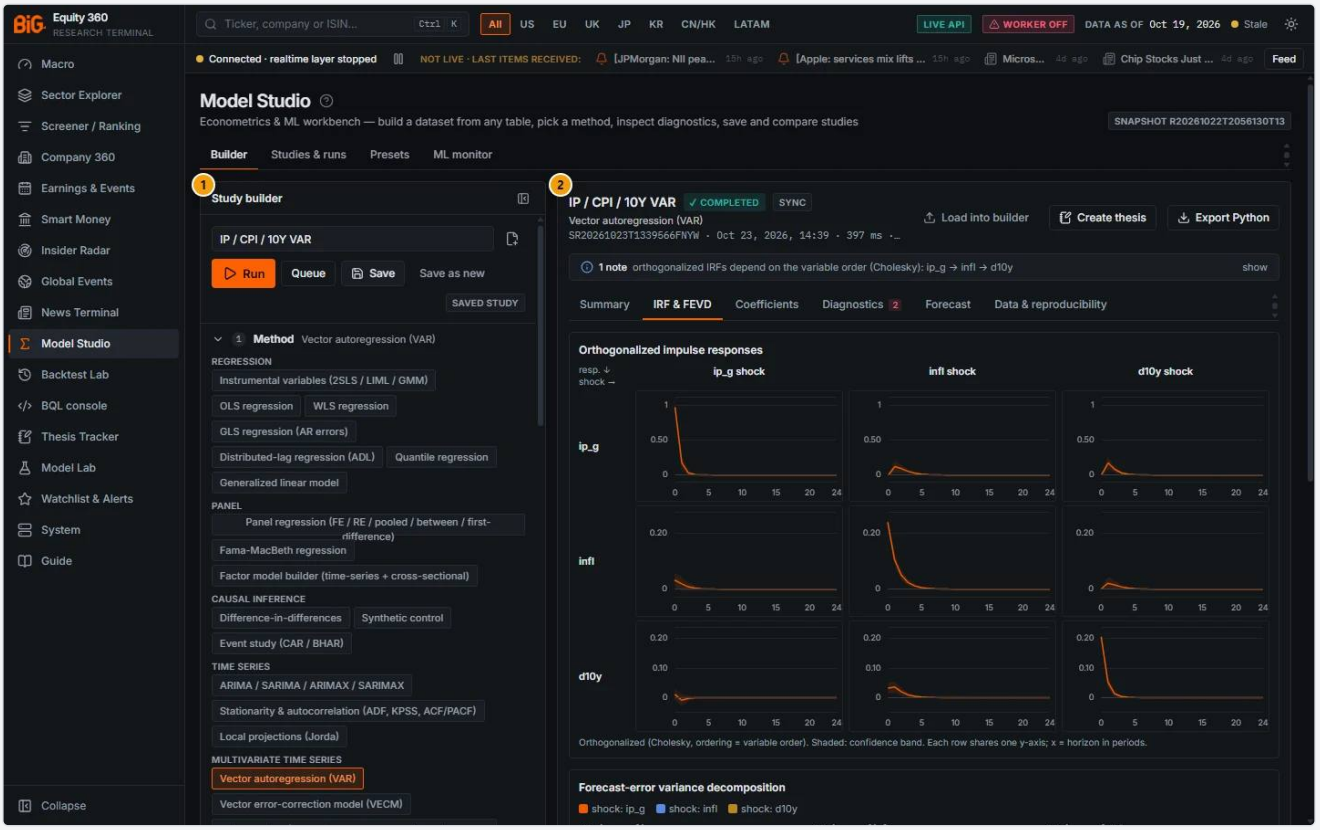


Figure 9.13: "IP / CPI / 10Y VAR", IRF & FEVD tab: 3 × 3 grid of orthogonalised impulse responses (rows = responses, columns = shocks, 24 periods) with the note "orthogonalized IRFs depend on the variable order (Cholesky): ip_g → infl → d10y"; the FEVD starts below.

Controls

RUN HEADER

Control	What it does
Title, status badge, mode badge	Run name, status (✓ completed, ✕ failed, ■ cancelled, … queued, ◔ running) and **sync** or **async**.

Control	What it does
Sub-lines	Method title; run id · finish time · duration · n · first 10 characters of the spec hash.
Load into builder	Hover: "Replace the builder contents with this run's study".
Backtest this signal	Only for completed **panel** runs of the tree ensemble or regularised regression: opens the Backtest Lab with this run's out-of-fold predictions as the signal (source `studio_run`). Hover: "Use walk-forward CV (scheme expanding) - other schemes are flagged look-ahead."
Create thesis	Opens the thesis editor linked to this run (Thesis Tracker).
Export Python	Downloads `<study_name>_<run_id>.py` (Export to Python).
Pending view	"Queued on the worker" / "Running", elapsed time, estimate, progress bar, "Long runs execute on the background worker. You can keep editing, switch tabs or leave the page — you will get a notification when it finishes.", **Cancel run**.
Failed view	"Run failed — ‹code›" and the error message.
Guardrail banner	"Guardrail: ‹code›" with the HTTP status, or "Study rejected"; **Dismiss**.

WARNINGS

Serious warnings (severity warn or error) appear as red or amber banners with the warning code in capitals and the message. Informational notes and dataset notes are folded into "N notes · show / hide".

RESULT TABS

Only the tabs with content appear. Their order is fixed:

Tab	When it appears	What it shows
Summary	always	**Model** card (method, dependent variable, sample, covariance type and other specification items), **Fit statistics**, **Other metrics**, the coefficient table, **Key tests** (specification, identification, inference and cointegration tests such as Hausman, first-stage F, Johansen, Patell, with their conclusion), **Tests at a glance** (up to 16 verdict badges with p-values and a **Diagnostics** → link) and, for ARIMA, the order-search table.
Method tab (label varies)	methods with special outputs	**Treatment effect** (DiD: pre-trend badge "no evidence against parallel trends" or "pre-treatment effects are not zero: parallel trends is doubtful", ATT, 2×2 means, event path, group-time ATT heatmap), **Event study** (CAAR path, AAR, events), **Synthetic control** (treated vs synthetic, gaps, donor weights bar), **IRF & FEVD** (VAR: IRF grid, FEVD, lag selection, Granger matrix), **Cointegration** (VECM: Johansen, β, α, error-correction term), **Spread** (pairs: spread with z bands, z-score, legs), **Volatility** (GARCH: conditional volatility, VaR backtest), **Regimes** (Markov: regime cards with expected duration and share of time, probability chart, transition matrix), **Components** (state space), **Beta paths** (Kalman), **Premia** (Fama–MacBeth λ series), **Regularization path** (coefficient paths, CV curve, selected features as ▲ / ▼ badges with the standardised size, unselected struck through), **Explainability** (tree ensembles: feature importance with permutation vs impurity, SHAP beeswarm, partial dependence, CV folds), **Factors** (factor model: cumulative factor returns, beta heatmap, alphas, the GRS line "F … · p … · conclusion", second-pass premia), **Leaderboard** (ranked table with the best highlighted, cumulative loss difference, CV folds). Other methods with path plots get a tab named **Paths**.
Coefficients	coefficients or coefficient plots exist	Forest plot (point estimates with intervals; not for quantile regression), lag-profile charts and the lag-block table ("Long-run multiplier = sum of the lag coefficients; F tests all lags of a variable jointly."), the quantile coefficient process, pseudo R^2 by quantile.
Diagnostics	tests or diagnostic plots exist; a red count badge shows failed tests	Full **Tests** table grouped by test group (Test, H0, Statistic, df, p-value, Result, Conclusion), stationarity and first-stage tables, and the plots: fitted vs actual, residuals vs fitted, QQ, residual ACF, residual histogram, residuals over time, ACF of y, ROC and calibration (binomial).
Forecast (or **Out-of-sample**)	forecast or OOS evaluation exists	Forecast fan with 80% and 1−α bands (variable switch for VAR / VECM), table of the first 12 steps; for GARCH a "Volatility forecast (sigma)" with the next-period VaR. **Out-of-sample evaluation** card: scheme, initial train, horizon, folds, purge; "RMSE vs naive" (green below 1, "beats the no-change forecast"), "OOS R^2 vs historical mean", a metrics table per forecaster (Model, OLS (unpenalized), Naive (no change), Random walk + drift, Historical mean, Seasonal naive) with RMSE, MAE, MAPE %, OOS R^2, correlation, AUC, log-loss, Brier, accuracy, "RMSE vs model" and a **best** badge, and the OOS chart.
Clusters	clustering methods	"Use these k clusters (n companies) as peer groups" bar with a name and **Save as peer set** (security-level clusterings only); cluster sizes (and noise count); soft-clustering note with the average membership probability; PCA scatter coloured by cluster (faded markers = uncertain members for GMM); dendrogram with **use this cut** for hierarchical runs (copies `distance_threshold` into the builder); BIC, elbow and silhouette curves; correlation and profile heatmaps; **Cluster profiles** (most distinctive features as mean z-scores ▲ / ▼ and dominant sectors); **Members** table (cluster, member with a Company 360 link, sector, distance to centre, membership probability) with a filter and a cluster select.
PCA	PCA	Scree plot, explained-variance table ("N components explain ≥ 80%"), loadings heatmap, biplot, scores scatter, score time series, loadings table.
Other output	outputs not routed elsewhere	Remaining plots and tables.
Data & reproducibility	always	**Dataset** card (kind, frequency, period, rows, entities, periods, macro alignment, build time, cached, dataset key, snapshot), **Reproducibility** card ("Same spec + snapshot + seed ⇒ same numbers": spec hash, seed, snapshot, timings, library versions), **Variable coverage** (non-null count, coverage bar, first / last date, entities, "uses future" badge) and **Study spec (JSON)** ("Show the exact request body").

Methodology

Test verdicts. "pass / fail = the server's reading of whether the model passes the check (e.g. residual tests, first-stage F ≥ 10, Durbin–Watson in 1.5–2.5). Tests without a pass/fail reading show reject / do not reject at 5%." The badges read **pass**, **fail**, **reject H0**, **do not reject** and **info**.

Coefficient significance. Stars use unadjusted p-values; bold marks significance at the run's α (default 0.05). Where several coefficients are tested, the "p adj." column gives Benjamini–Hochberg adjusted p-values, which control the false discovery rate.

Peer sets. Saving a security-level clustering as a peer set (`POST /studio/runs/{run_id}/peer-set`) stores its membership; Company 360 › Peers can then compare a company with its cluster instead of its sub-sector.

Data sources and refresh

The result is read from `data/modeling/runs/<run_id>/result.json` ; it does not change after the run.

Validation status

Not applicable; see the method's entry in the Methods catalogue.

Correct usage and worked example

Figures 9.6 to 9.10 show this reading on the ARIMAX preset: the model's out-of-sample RMSE is 1.14 times that of the no-change forecast and all three residual tests fail, so despite a sensible lagged-slope coefficient the study gives no usable forecast. For a time-series regression, read in this order: Summary (sign and size of the coefficients, covariance type), Diagnostics (fail count: residual autocorrelation means the standard errors are wrong unless HAC was used), Out-of-sample ("RMSE vs naive" above 1 means the model does not beat a no-change forecast, which also raises the `oos worse than naive` warning), then Data & reproducibility (coverage and look-ahead flags).

Limitations and pitfalls

- In-sample fit statistics (R^2, AIC) reward complexity; prefer the out-of-sample blocks.
- SHAP and permutation importance explain the fitted model, not the economy.
- The **Save as peer set** bar only appears for security-level clusterings with sector information.

Related mnemonics and BQL fields

Calibration metrics, Rank IC and factor evaluation.

Export to Python

Route: `/studio` (run header › **Export Python**) · **API:** `GET /api/v1/studio/runs/{run_id}/export.py`

Purpose

The export turns a completed run into a standalone script that rebuilds the same dataset and refits the same model outside the app, for audit, peer review or further work in a notebook.

Controls

Control	What it does
Export Python (completed runs only)	Downloads `<study_name>_<run_id>.py` ; toast "Python script downloaded". Runs that are not completed return HTTP 409 "only completed runs export".

Methodology

The script (`backend/app/modeling/export.py`):

- embeds verbatim the dataset engine (`_core.py`), the point-in-time as-of helper and the method cores (`_algos.py` , `_ml.py` , `_conformal.py` , `_spatial.py`);
- has a header with the run id, title, method, creation time, snapshot id, spec hash and seed;
- rebuilds the dataset read-only from the Parquet lake under `--data-dir` (default: the run's data directory), or loads the exact dataset saved with the run with `--from-artifact` ;
- refits with plain statsmodels / scikit-learn calls, with the orders or k chosen by the run's search fixed in the script;
- prints a summary and a final `RESULT_JSON {...}` line with the key estimates, so a reproduction can be checked mechanically.

Requirements: duckdb, polars, numpy, pandas, scipy, statsmodels, scikit-learn.

Validation status

The plan records that the scripts reproduce the presets exactly (`docs/MODEL_STUDIO_PLAN.md` , A3).

Limitations and pitfalls

- Variables from the **BQL query** source cannot be rebuilt by the script: the BQL engine needs the backend and the serving snapshot. Use `--from-artifact` for such runs.
- The script reads the lake as it is when you run it; to reproduce the original numbers exactly use the same snapshot or `--from-artifact` .

Related mnemonics and BQL fields

Backtest Lab export.

ML monitor

Route: `/studio` (tab **ML monitor**) · **Menu:** Model Studio › ML monitor (same content as Model Lab, `/models`) · **Refresh:** weekly ML retraining

Purpose

The ML monitor tab embeds the nightly ML monitor so that you can check the production ranking model while working in Model Studio. It shows exactly the same panels as the Model Lab page, under the line "Nightly ML monitor — walk-forward evaluation of the XGBoost baseline and the multi-factor Transformer."

Screenshot

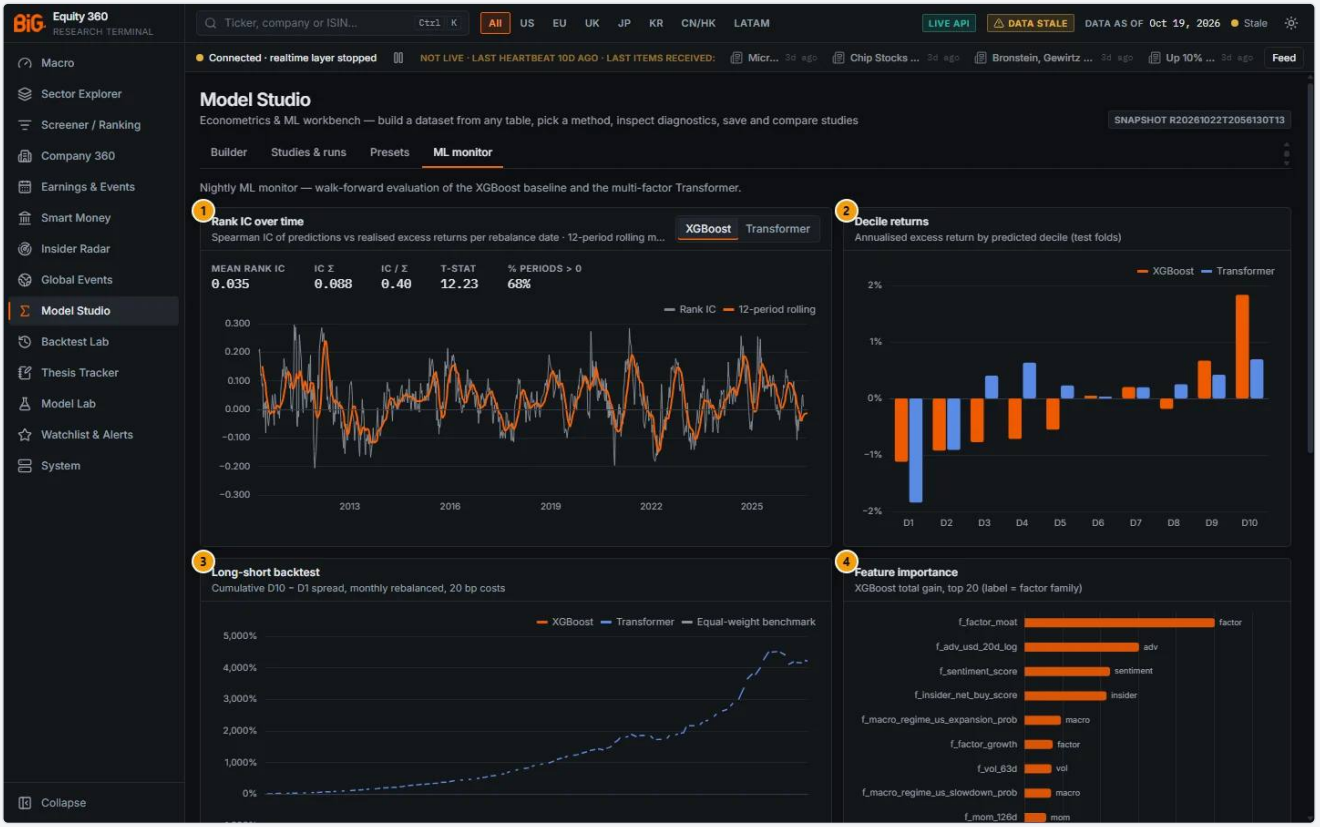


Figure 9.14: ML monitor tab (`/studio`), XGBoost selected, data as of 19 Oct 2026.

#	Panel	What it shows
1	Rank IC over time	Spearman IC per rebalance date with a 12-period rolling mean and the summary statistics.
2	Decile returns	Excess return by predicted decile for XGBoost and the Transformer.
3	Long-short backtest	Cumulative D10 − D1 spread with the equal-weight benchmark.
4	Feature importance	XGBoost total gain, top 20, labelled by factor family.
5	Walk-forward folds	Train / test windows per fold with rank IC and Sharpe.
6	Training runs	Every ranking-model run; `prod` marks the promoted model.

Controls

As in Model Lab.

Methodology

See Model Lab.

Validation status

> **Caution:** only XGBoost is production. Its out-of-sample rank IC is about 0.035: treat the ML score as one input, not proven alpha. The Transformer and the ensemble are comparison models; the 3M FT-Transformer failed the label-permutation leakage test (`docs/PHASE3_TODO.md`). Details in Model Lab › Validation status.

Related mnemonics and BQL fields

`MDL` opens Model Lab. See Model Lab.

Open in Model Studio entry points

Route: several pages → `/studio` · **Refresh:** not applicable

Purpose

Other pages can open Model Studio with a draft already filled in, so that a series or a set of event dates you are looking at becomes a study in one click. The draft replaces the current builder content, clears the selected run and opens the Builder tab; nothing runs until you press **Run**.

Controls

Page	Button	Draft it creates
Macro › Series explorer and the macro series drawer	**Open in Model Studio** (hover: "New Model Studio study with this series as the target")	Time series, the series' own frequency, start 20 years back, macro alignment **Reference period**, method **ARIMA**, the series as y (with a 12 / 4 / 52-period % change × 100 when the series is shown year on year).
Global Events › Event studies, panel "Send event dates to Model Studio"	**Send to Model Studio**	An **Event study (CAR / BHAR)** draft over the shock dates of a country, sector or company (attention or tone z ≥ 2, de-clustered), filtered by theme and shock kind (any, attention, tone drop, tone jump). Disabled with "No event dates for this label".
Instrument page	**Open in Model Studio**	Weekly log returns of the instrument since 2000, **ARIMA**.
Cross-asset dashboard	**Open in Model Studio**	Weekly **OLS** of the asset's log return on its lagged cross-asset signals (source `cross_asset` : trend, carry, positioning where available), since 2008.
CFTC market page	**Open in Model Studio**	Weekly **OLS** of the mapped instrument's log return on CFTC positioning (source `cftc` , regressors lagged one week), since 2008.

Correct usage and worked example

From the Global Events event-studies page, choose Country `US` , theme "All themes", shock kind "Any shock" and press **Send to Model Studio**. The builder opens with an event-study draft whose events are the listed dates. Before running, check the event list in Parameters and the estimation window (default −250 to −30 days); the panel notes that the z baseline needs 90 days of history (data from 2026-07-14), so few dates exist before mid-October 2026 unless the 2020 backfill has landed.

Limitations and pitfalls

- The draft overwrites the current builder content without asking. Save your study first.
- Drafts are starting points: check the alignment, lags and dates before drawing conclusions.

Related mnemonics and BQL fields

Backtest entry points (the reverse direction: "Backtest this signal").

BQL query variable source

Route: `/studio` (Builder › 3 · Variables › Add variable › Source table **BQL query**) · **Roadmap item:** v3 B1

Purpose

The BQL source turns any BQL-lite expression into a dataset column, evaluated point-in-time at every date of the dataset. It brings the query language of the BQL console (cross-sectional z-scores, ranks, changes over windows, dictionary fields) into Model Studio without building each transform by hand.

Controls

Control	What it does
Source table **BQL query**	Lists the dictionary fields that have point-in-time history (field = the BQL mnemonic) and shows an expression row.
Expression row	"BQL expression — evaluated point-in-time at every dataset date on the serving snapshot (cross-sectional functions run over the dataset's securities)". Default `cs_zscore(chg_pct(px, 3M))` ; placeholder `cs_zscore(roic) - cs_zscore(pe_ttm)` . Add buttons **+ y** / **+ x** (or **+ add**).

Methodology

For each grid date D the expression sees only data known at D. Cross-sectional functions such as `cs_zscore` or `rank` run over the dataset's entities at each date. Security-level panels and cross-sections use the dataset's entity filter (an `options.universe` BQL universe overrides it); time-series datasets aggregate the panel across entities per date (`entity_agg` , default mean). Non-BQL variables are built by the dataset engine and joined on (entity, date).

Data sources and refresh

The serving snapshot (not the raw lake).

Validation status

Not formally validated.

> **Note:** on the live snapshot used for this guide (R20261022T2056130T13) the BQL source does not appear in the source list: the catalogue lists 15 sources without "BQL query". It is offered only when the BQL field dictionary loads (roadmap v3 B1 "go live" step: rebuild the app and restart the API). If you do not see it, use the BQL console to explore and build the equivalent transforms in the builder.

Limitations and pitfalls

- BQL variables are not reproducible by the exported script (see Export to Python).
- Expressions are evaluated on the serving snapshot, which holds a shorter history for some fields than the lake.

Related mnemonics and BQL fields

`BQL` opens the console. BQL language reference, Field dictionary, BQL elsewhere in the app.

10. Model Lab

Model Lab monitors the machine-learning models that rank stocks by expected excess return: the production XGBoost model and the comparison models (the multi-factor FT-Transformer and the rank-average ensemble). It shows how well each model's walk-forward, out-of-sample predictions ranked stocks over time, how returns line up by predicted decile, the fold-by-fold results, the features that drive the XGBoost model, and the registry of training runs with the production flag.

The predictions feed the ML score on Company 360 (Model tab), the Screener's ML fields and the Backtest Lab's `ml` signal source. Read this chapter before relying on any of them: the measured out-of-sample edge is small, and only XGBoost is production.

The same page is embedded in Model Studio as the ML monitor tab.

Model Lab

Route: `/models` · **Menu:** Model Lab · **Mnemonics:** `MDL` · **Refresh:** weekly (jobs `ml_labels`, `ml_train_xgb`, `ml_ensemble`); the Transformer is retrained by hand only (`ml_train_transformer`, manual since 2026-10-20)

Purpose

Model Lab is the governance view of the ML ranking models. Portfolio managers use it to judge how much weight the ML score deserves (answer: little, see Validation status); the owner uses it to check each weekly retraining (did the rank IC hold up, did a fold turn negative, which run is in production). It also shows why the Transformer is not in production even though its headline numbers look better.

Screenshot

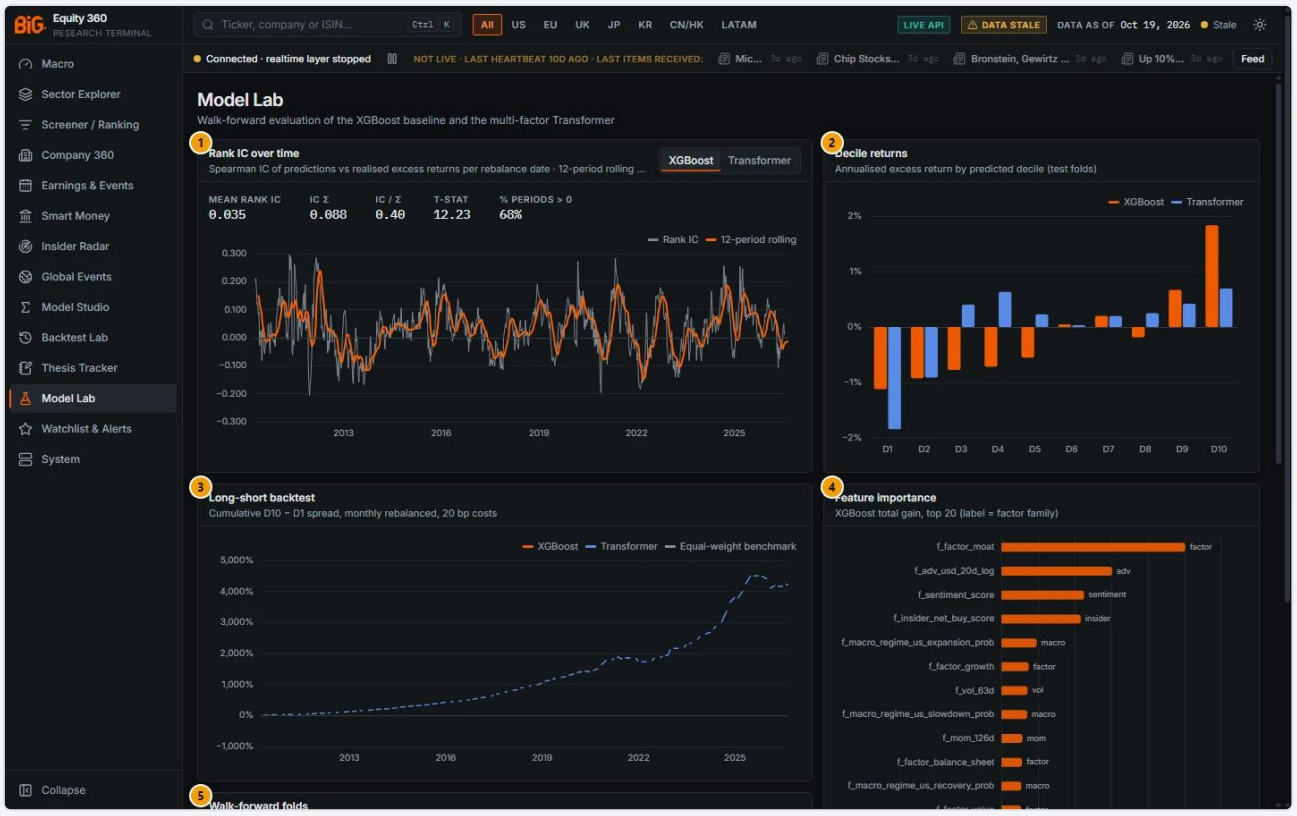


Figure 10.1: Model Lab (`/models`), XGBoost selected in the Rank IC panel, data as of 19 Oct 2026.

#	Panel	What it shows
1	Rank IC over time	"Spearman IC of predictions vs realised excess returns per rebalance date · 12-period rolling mean". Statistic tiles (here for XGBoost): MEAN RANK IC 0.035, IC σ 0.088, IC / σ 0.40, T-STAT 12.23, % PERIODS > 0 68%. Grey line = rank IC per date, orange = 12-period rolling mean.
2	Decile returns	"Annualised excess return by predicted decile (test folds)": one bar per decile D1…D10 for XGBoost (orange) and the Transformer (blue).
3	Long-short backtest	"Cumulative D10 − D1 spread, monthly rebalanced, 20 bp costs": XGBoost, Transformer and the equal-weight benchmark (dashed).
4	Feature importance	"XGBoost total gain, top 20 (label = factor family)". Here `f_factor_moat` leads, then `f_adv_usd_20d_log`, `f_sentiment_score`, `f_insider_net_buy_score` and the US macro-regime probabilities.
5	Walk-forward folds	"Expanding train window (grey) → 12-month test (orange), purged & embargoed", one row per fold with its rank IC and Sharpe. Below the fold in the main image.
6	Training runs	"Most recent first · 'prod' = promoted to scoring". Below the fold in the main image.

Figure 10.2: Walk-forward folds (callout 5) of the XGBoost 3M production run: nine folds, rank IC from −0.031 to 0.083; the last fold is a partial window (2026-04-15 → 2026-10-09).

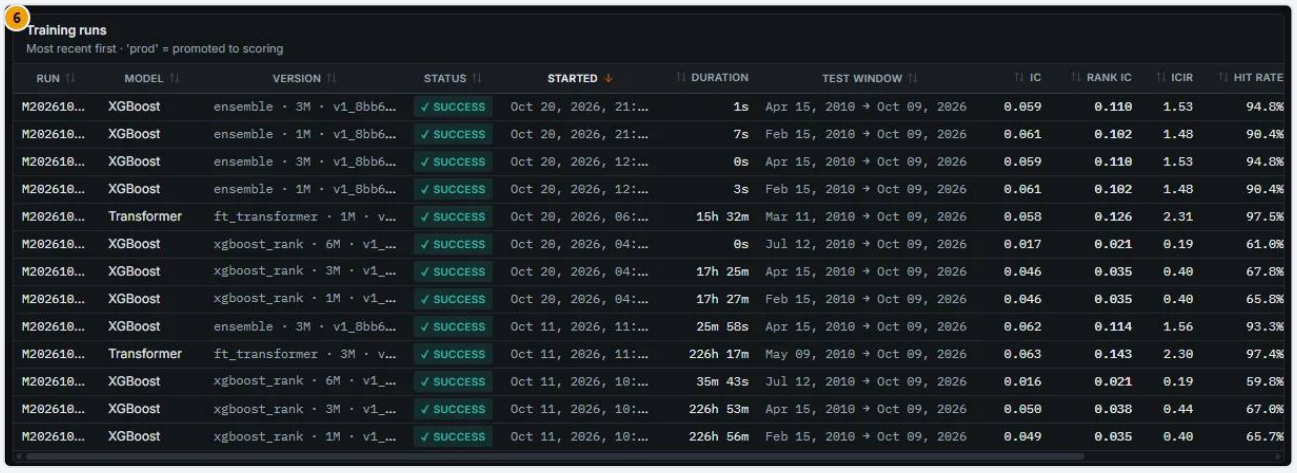


Figure 10.3: Training runs (callout 6): ensemble, FT-Transformer and XGBoost runs of 11 and 20 Oct 2026 with test windows, IC, rank IC, ICIR and hit rate.

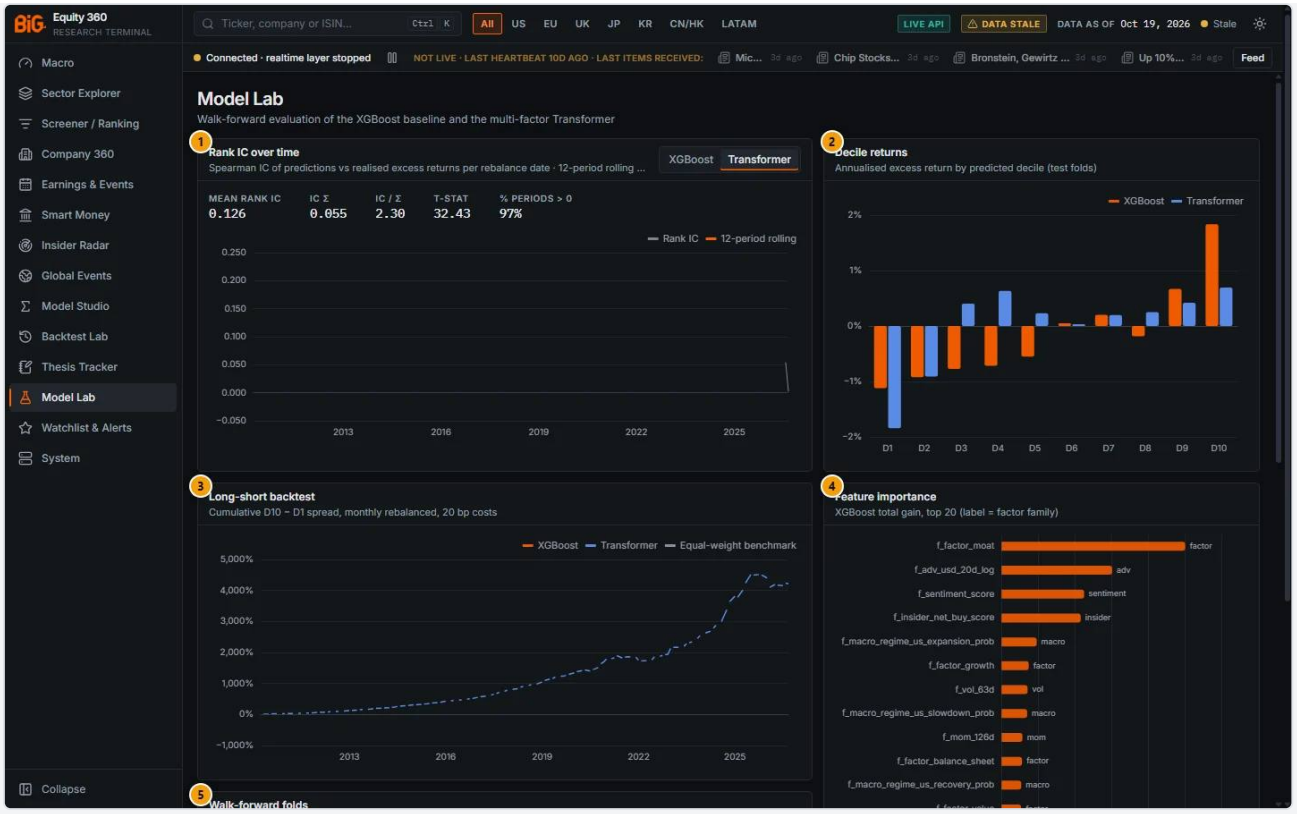


*Figure 10.4: The same page with *Transformer** selected: MEAN RANK IC 0.126, IC σ 0.055, IC / σ 2.30, T-STAT 32.43, % PERIODS > 0 97%. These figures come from a model that is not production; see Validation status.*

Controls

Control	What it does
XGBoost / **Transformer** (Rank IC panel)	Switches the Rank IC chart and its five statistic tiles between the two model families. The other panels always show both models (decile returns, long-short) or XGBoost only (feature importance, folds).
Chart hover	Rank IC and rolling mean per date; decile values; cumulative spread per date; feature name with its family and gain share.
Fold bar hover	"Train ‹start› → ‹end› · Test ‹start› → ‹end›".
Training runs table	Sortable columns: Run, Model, Version, Status, Started (default sort, newest first), Duration, Test window, IC, Rank IC, ICIR, Hit rate, D10−D1 ann., Features. A **prod** badge marks the promoted run.

There is no promote or retrain button on the page. Promotion is done by the weekly production gate or by hand through `POST /api/v1/models/runs/{id}/promote`; retraining through the scheduled jobs or `POST /api/v1/models/train`.

Methodology

WHICH RUNS THE CHARTS SHOW

The page reads the training-run registry (`GET /api/v1/models/runs`) and, for the charts, the run detail (`GET /api/v1/models/runs/{id}`) of one run per family:

- **XGBoost family** (model types `xgboost_rank` and `ensemble`): the production run if one exists, otherwise the most recent completed run. On 19 Oct 2026 this is the **3M** XGBoost production run `M20261020T031637JB5W`.
- **Transformer family** (model types `ft_transformer`, `cs_attention`): the production run if one exists, otherwise the most recent completed run. No Transformer is production, so the charts show the most recent run, the **1M** FT-Transformer `M20261020T050950HN87` of 20 Oct 2026.

The two families can therefore be on different horizons (3M vs 1M here). In the Training runs table, rows with Version `ensemble · …` are the rank-average ensemble even though the Model column reads XGBoost.

LABELS AND FEATURES

- **Rebalance grid:** the last trading day of every ISO week plus every month end (S&P 500 calendar). XGBoost trains on the full weekly + month-end grid; the Transformer on month ends only.
- **Label:** forward return from close(D+1) to close(D+1+h), h = 21 / 63 / 126 trading days for 1M / 3M / 6M, on the security's own calendar. `fwd_excess_sector` subtracts the equal-weighted mean forward return of the training universe in the same (date, horizon, sector, region) bucket; `rank_label` is its cross-sectional percentile. Labels are written to their own table so a feature pipeline can never read them.
- **Features:** 31 live features from `feature_registry.yaml`, cross-sectionally rank-Gaussianised: price momentum (21d, 63d, 126d, 12m, 12m−1m), 5-day reversal, realised volatility (21d, 63d), liquidity (log 20-day dollar volume), US macro-regime HMM probabilities (expansion, slowdown, contraction, recovery, stagflation, recession), rolling macro betas (market, 10Y, 2s10s, credit spread, USD, oil), our fundamental factor scores (value, quality, growth, profitability, balance sheet, moat), technical confluence, insider and sentiment scores. Fundamental, insider and sentiment features are lagged one day.
- **Universe:** `mvp_us + mvp_global`.

WALK-FORWARD CROSS-VALIDATION

Expanding window: 5 years of initial training history, then out-of-sample test folds of **24 months** stepped by 24 months (nine folds from 2010 to October 2026; the last one partial). Between training and test there is a **purge** equal to the label horizon (about 1.4 × h calendar days) plus a 10-day **embargo**, so no training label window overlaps the test period. Hyperparameters are fixed modest defaults (XGBoost: 300 trees, depth 4, learning rate 0.05, subsample 0.8), not searched across folds, to avoid fitting them to walk-forward noise. A static guard (`assert_no_label_leakage`) checks on every training call that the feature columns are exactly the registry's live features and that none has a label-like name.

METRICS

```
IC_t       = Pearson corr(score_i,t , fwd_excess_sector_i,t)        over names at date t (≥ 5 names)
Rank IC_t  = Spearman corr(score_i,t , fwd_excess_sector_i,t)
Mean rank IC (tile)  = mean_t Rank IC_t
IC σ                 = sd_t Rank IC_t
IC / σ  (ICIR)       = mean / sd
t-stat (tile)        = mean / sd · √n   (n = number of dates); the run registry stores a Newey–West t instead
% periods > 0 (Hit rate) = share of dates with Rank IC_t > 0
```

- **Decile returns:** at each date the model's own scores are cut into deciles; the bar is the mean `fwd_excess_sector` of each decile over all test dates, i.e. the average sector-relative excess return over the label horizon.
- **Long-short series:** the D10 − D1 difference of those decile means per date. Because the label windows of consecutive weekly dates overlap, the stored series keeps only dates at least one horizon apart (non-overlapping subsample) and compounds them geometrically: `cum_t = ∏(1 + LS_s) − 1`. **D10−D1 ann.** in the runs table = mean LS × periods per year (12 / 4 / 2 for 1M / 3M / 6M); the long-short Sharpe = mean / sd × √(periods per year). The stored evaluation does not deduct trading costs.
- **Feature importance:** XGBoost total gain per feature as a share of the total, top 20.
- **Turnover:** average week-over-week share of top-decile names that changed.

ENSEMBLE AND PRODUCTION GATE

The `ml_ensemble` job blends the latest XGBoost and Transformer predictions by averaging their rank percentiles on common (month-end) dates and evaluates the blend the same way. The production gate picks, per horizon, the model type with the highest OOS rank IC, but a non-XGBoost model (Transformer or ensemble) is eligible only when its run notes record a passed label-permutation leakage check (`[LEAKAGE_CHECK] passed`): train on labels shuffled within each date and require |rank IC| < 0.01 on the real test fold. XGBoost is always eligible. Production runs are consumed by the nightly scoring and by `ml_predictions`.

Data sources and refresh

Item	Source / job	Cadence
Labels	`ml_labels` from `prices_daily`	weekly
Features	`ml_features` from prices, factor scores, insider, sentiment, macro regimes, macro betas, technicals	with the nightly / weekly pipeline

Item	Source / job	Cadence
XGBoost training (1M, 3M, 6M)	`ml_train_xgb`	weekly
FT-Transformer training	`ml_train_transformer`	manual only since 2026-10-20 ("DEMOTED (failed label-permutation leakage test)")
Ensemble and gate	`ml_ensemble`	weekly
Run registry, evaluation series	`model_runs` , `model_eval_series`	written by the jobs

The test windows in Figure 10.3 run from February–July 2010 to 9 Oct 2026.

Validation status

> **Caution:** the production model is **XGBoost** (`xgboost_rank` , production for 1M, 3M and 6M). Its out-of-sample rank IC is about **0.035** (3M production run: mean rank IC 0.035, ICIR 0.40, hit rate 68%; 1M run: 0.035; 6M run: 0.021). Treat the ML score as **one input, not proven alpha**. Backtests and evaluation dates before **2026-10-10** use today's company list (survivorship bias), which flatters them.

> **Caution:** the **Transformer and the ensemble are comparison models, not production.** `docs/PHASE3_TODO.md` records: "ML leakage audit: the 3M FT-Transformer FAILS the label-permutation test (IC +0.06–0.09 on shuffled labels; `row + lag1 + lag2` in FTTransformerCS.row_repr). Demoted. XGBoost (passes) is production for 1M/3M/6M." Trained on shuffled labels, the 3M Transformer still reached an out-of-sample rank IC of +0.06 to +0.09 against the true labels, which a model without a defect cannot do. The architecture fix ("gate or concat the lag path", then a new permutation test) is still open. The Backtest Lab's `ml` source states the same: "options.model_type=ft_transformer is DEMOTED (failed the label-permutation test)".

What this means on the page:

- The Transformer's tiles (mean rank IC 0.126, t-stat 32.43, 97% of periods positive) and its long-short line are **not evidence of skill**. Numbers this high on monthly stock ranking are implausible and were the reason for the audit; the run shown (1M, 20 Oct) uses the same architecture as the 3M run that failed, and no passed leakage check is recorded for it.
- The **ensemble** rows (rank IC 0.102–0.114) blend XGBoost with Transformer predictions (the 3M ensemble of 20 Oct uses the 3M Transformer run that failed the test), so they inherit the same problem.
- Only the XGBoost numbers describe the model that feeds the app.

See also Gate status of every model and Rank IC and factor evaluation.

Correct usage and worked example

Reading the XGBoost 3M production run (Figure 10.1 and 10.2):

1. **Size of the edge.** Mean rank IC 0.035 with IC σ 0.088: the average cross-sectional correlation between score and subsequent sector-relative return is small and noisy; in about a third of the weeks it is negative (68% positive).
2. **Stability.** The folds range from −0.031 (2012–2014) to 0.083 (2024–2026); 2022–2024 is close to zero (0.011, long-short Sharpe −0.20). An edge that disappears for two years is not something to size a position on.
3. **Monotonicity.** The decile bars rise from about −1.1% (D1) to about +1.8% (D10) per 3-month horizon, but not monotonically (D8 is below D7). Most of the spread sits in the extreme deciles.
4. **Drivers.** The top features are the moat score, liquidity and sentiment, not a single dominant signal.

Use the ML score as a tie-breaker inside a fundamental process, alongside the factor scores, not as a stand-alone ranking. The partial last fold (Sharpe 9.00 over six months) is too short to mean anything.

Limitations and pitfalls

- **Survivorship.** Universe membership history starts on 2026-10-10. Names that left the universe earlier, and delisted names not in the price history (most of the non-US universe comes from yfinance, which drops delisted tickers), are

missing; this biases the evaluation upward.

- **Short histories for some features.** The technical confluence feature comes from a latest-snapshot table and is effectively null in historical training rows; sentiment starts in late 2025.
- **Overlapping labels.** Weekly rebalances with 3-month labels make consecutive rank ICs strongly autocorrelated; the naive t-stat in the tile overstates significance (the run registry stores a Newey–West t).
- **Different horizons on one chart.** The two families may be shown on different horizons (3M XGBoost vs 1M Transformer on 19 Oct 2026).
- **Panel labels.** The long-short series is compounded over non-overlapping horizon-length windows (quarterly for 3M) without cost deduction in the stored evaluation, and the decile bars are mean returns per label horizon. Read the subtitles together with this section.
- **Display gaps.** When the two families have different date grids, the chart lines can break into isolated points (the Transformer rank-IC chart in Figure 10.4 is almost empty; the XGBoost long-short line is hard to see in Figure 10.1).
- **Insider XGBoost** (`insider_xgb`) runs also exist in the registry but are not shown here; that model shows no out-of-sample edge, so only the rules-based insider score is used in the app.

Related mnemonics and BQL fields

- `MDL` (Model registry) opens this page.
- Screener and BQL fields: `ml_rank_pct_1m` , `ml_decile_1m` , `ml_expected_excess_1m` and the 3M / 6M equivalents (see the Field dictionary).
- Company 360 › Model, Backtest Lab (preset "ML 3M top decile (walk-forward OOS)"), Purged and embargoed cross-validation.

11. Backtest Lab

The Backtest Lab tests whether a signal or a screen would have made money, point-in-time and net of realistic costs, and how much of the result could be luck. You define a strategy as universe → signal → construction → costs → period, run it, and read performance, risk, attribution, trades and a robustness block with the deflated Sharpe ratio, the probability of backtest overfitting, a label-permutation test and a stale-signal check. Every strategy keeps a multiple-testing counter, every run keeps a mandatory out-of-sample holdout, and every run can be exported as a standalone Python script.

> **Note:** screenshots with saved strategies and completed runs (captions mention the "test copy") were captured from a test copy of the app seeded with example content on 23 Oct 2026, on the same data snapshot. The red **WORKER OFF** chip in their top bar comes from that test copy.

A backtest here is evidence about the past under stated assumptions. Read the survivorship banner at the top of the page before trusting any number before 2024–2026, and read the Robustness tab before trusting any Sharpe ratio.

Backtest Lab

Route: `/backtest` · **Menu:** Backtest Lab · **Mnemonics:** `PORT` (alias `BT`) · **Refresh:** on demand; paper strategies re-run weekly (job `backtest_scheduled`)

Purpose

Analysts use the Backtest Lab to check a screen ("Backtest this screen"), a factor or ML signal, a cross-asset or CFTC signal, or an event rule (insider cluster buys, earnings surprises) before building a thesis around it. Portfolio managers use it to see the cost drag, turnover and drawdowns of an idea, and to run a strategy weekly on paper with drift alerts. The robustness block is there to tell you when a good-looking backtest is likely noise.

Screenshot

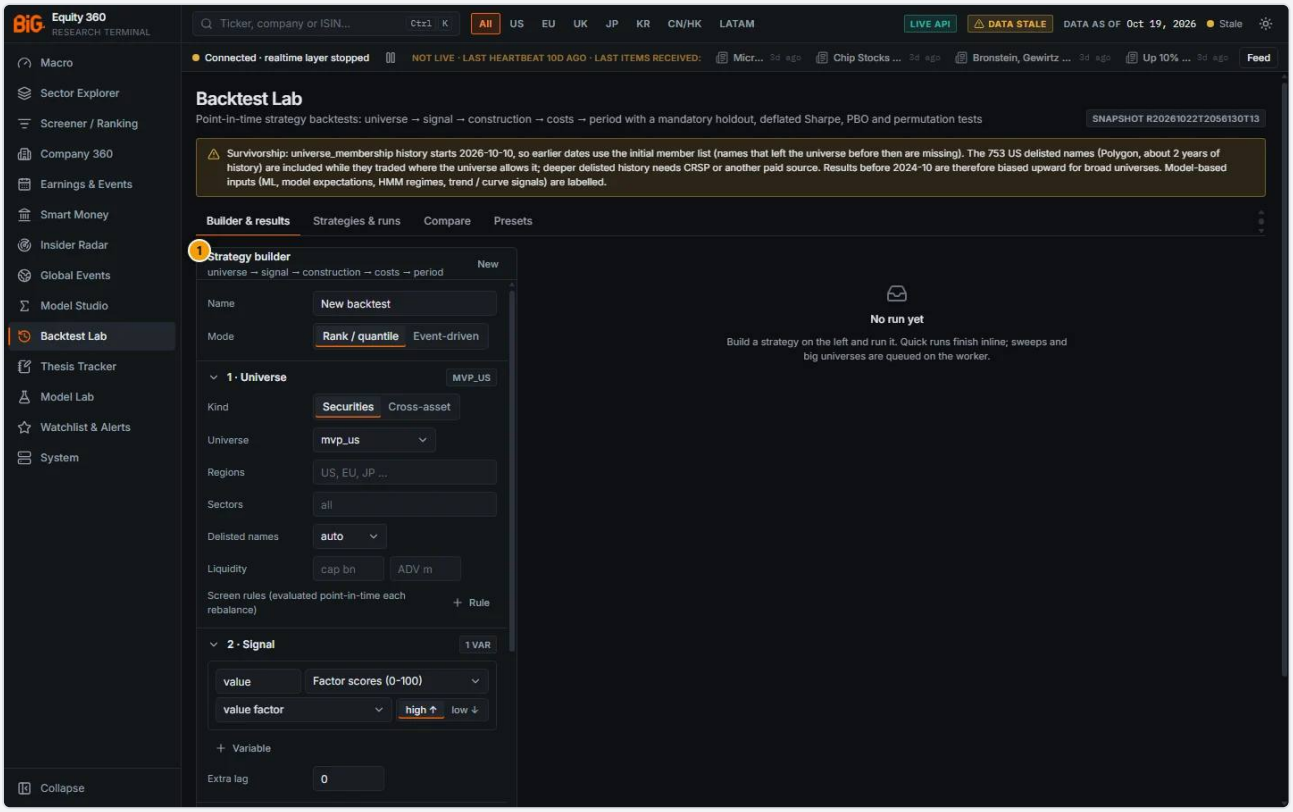


Figure 11.1: Backtest Lab (`/backtest`), Builder & results tab with the default strategy and no run, snapshot R20261022T2056130T13, data as of 19 Oct 2026.

#	Panel	What it shows
1	Strategy builder	"universe → signal → construction → costs → period": name, mode, then the numbered sections (1 · Universe, 2 · Signal, 3 · Construction, 4 · Costs, 5 · Period & holdout, 6 · Robustness) and the run buttons at the bottom. The default draft is a long-only value strategy on `mvp_us` .

The empty results pane reads "No run yet: Build a strategy on the left and run it. Quick runs finish inline; sweeps and big universes are queued on the worker."

Controls

Control	What it does
Survivorship banner (amber)	"Survivorship: universe_membership history starts 2026-10-10, so earlier dates use the initial member list (names that left the universe before then are missing). The 753 US delisted names (Polygon, about 2 years of history) are included while they traded where the universe allows it; deeper delisted history needs CRSP or another paid source. Results before 2024-10 are therefore biased upward for broad universes. Model-based inputs (ML, model expectations, HMM regimes, trend / curve signals) are labelled."
SNAPSHOT badge	The snapshot the catalogue was measured on.
Tabs **Builder & results** / **Strategies & runs** / **Compare** / **Presets**	See Strategy builder, Results, Strategies & runs, Compare, Presets.

The draft, its origin, the open strategy, the selected run, the runs ticked for comparison and the active tab are kept in the browser (key `big360.backtest.v1`).

Methodology

Timing convention (no look-ahead). The signal row dated t uses only information known at the close of t (the panel builder guarantees it). Trades execute at the close of the `execution_lag_days` -th trading day strictly after t (default 1, the same convention as the ML labels: close(D+1)); the new weights earn returns from the following day. Positions drift

with prices between rebalances. Returns are simple daily returns in the base currency: USD by default (adjusted close × official `usd_per_unit`), or local currency. A delisted name is closed at its last price.

Run engine. A run estimated under **4 seconds** (cached panel, small robustness settings) executes inline; longer runs (sweeps, big universes) are queued on the worker (job `backtest_run`). Each run is stored under `data/backtest/runs/<run_id>/` with its spec, plan, status, result, spec hash and snapshot id. Point-in-time panels are cached under `data/cache/backtest/` (bounded, least recently used).

Holdout. Every run splits its period into an in-sample window and a mandatory out-of-sample holdout: from **Holdout start** if set, else the last 30% of the period (`oos_fraction`, 10–60%). The holdout is never used for parameter selection; the deflated Sharpe is computed on the in-sample window.

Data sources and refresh

Signals, screens and prices come from the Parquet lake and the serving snapshot, read-only, as known at each rebalance date. The catalogue (`GET /api/v1/backtest/catalog`) measures every source's point-in-time coverage on the current snapshot (see Strategy builder, Data sources). Benchmarks: SPY (total return), ACWI, VEA, EEM, IWM, QQQ, VGK, EWJ, the S&P 500 price index, universe equal weight or none. Risk-free rate: the 3-month T-bill (`macro_factors` cash) or zero.

Validation status

The engine has correctness tests (look-ahead, permutation ≈ 0, toy portfolios, costs, known DSR / PBO values; `docs/ROADMAP_V2_PLAN.md` item 4.8) and the exported scripts reproduce runs. The engine is not a validated model: the Robustness tab is how you judge each strategy. Several signals carry their own validation status; see Experimental signals and Strategy builder, Validation status.

Correct usage and worked example

1. Open **Presets** and click **Open** on "US value long-short (quintiles, monthly)", then **Run backtest**.
2. Read the Performance tab, then go straight to **Robustness**: the verdict badge and the DSR tell you whether the Sharpe ratio is distinguishable from luck.
3. If you change parameters, save the strategy first (**Save**) so that every variant counts toward its multiple-testing counter, and prefer a **parameter sweep** over repeated manual runs.
4. Look at the holdout column last, and as few times as possible.

The worked example of the Results section uses this preset.

Limitations and pitfalls

- **Survivorship.** Before 2026-10-10 the universe is today's member list; only 753 US delisted names with about two years of history are added. Results before 2024-10 are biased upward for broad universes.
- **Short histories.** Factor scores start in October 2016, news sentiment in October 2025, technical setups and event exposure have only their latest dates (not backtestable).
- **Execution.** Trades fill at the close with modelled costs; there is no intraday fill model, no market-impact feedback on prices and no capacity limit beyond the impact term.
- **Licensing.** Several price sources (yfinance and others) are not licensed for bank production use; see Licensing.

Related mnemonics and BQL fields

`PORT` / `BT` open the page. Related: Screener (Backtest this screen), Model Studio, Multiple testing, Survivorship.

Strategy builder

Route: `/backtest` (tab **Builder & results**) · **API:** `GET /api/v1/backtest/catalog`, `POST /api/v1/backtest/validate`, `POST /api/v1/backtest/runs`

Purpose

The builder defines a strategy in five or six collapsible sections. In **Rank / quantile** mode it builds portfolios from a cross-sectional ranking on a calendar; in **Event-driven** mode it enters positions on events and exits on time, stop or target.

Screenshot

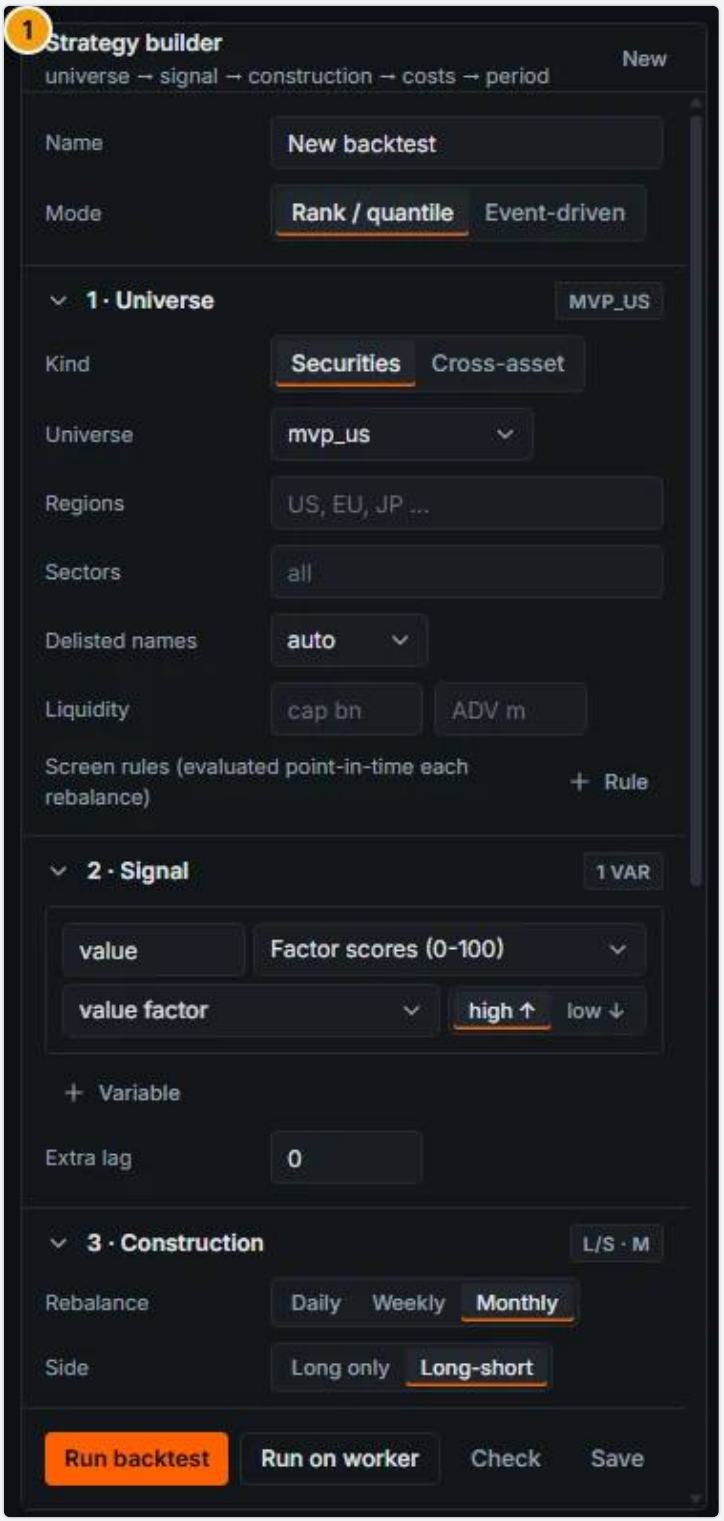


Figure 11.2: Strategy builder (callout 1 of Figure 11.1): name, mode, 1 · Universe and 2 · Signal.

> Screenshot pending: builder scrolled to 3 · Construction, 4 · Costs (expanded), 5 · Period & holdout and 6 · Robustness with a two-parameter sweep (route `/backtest`).

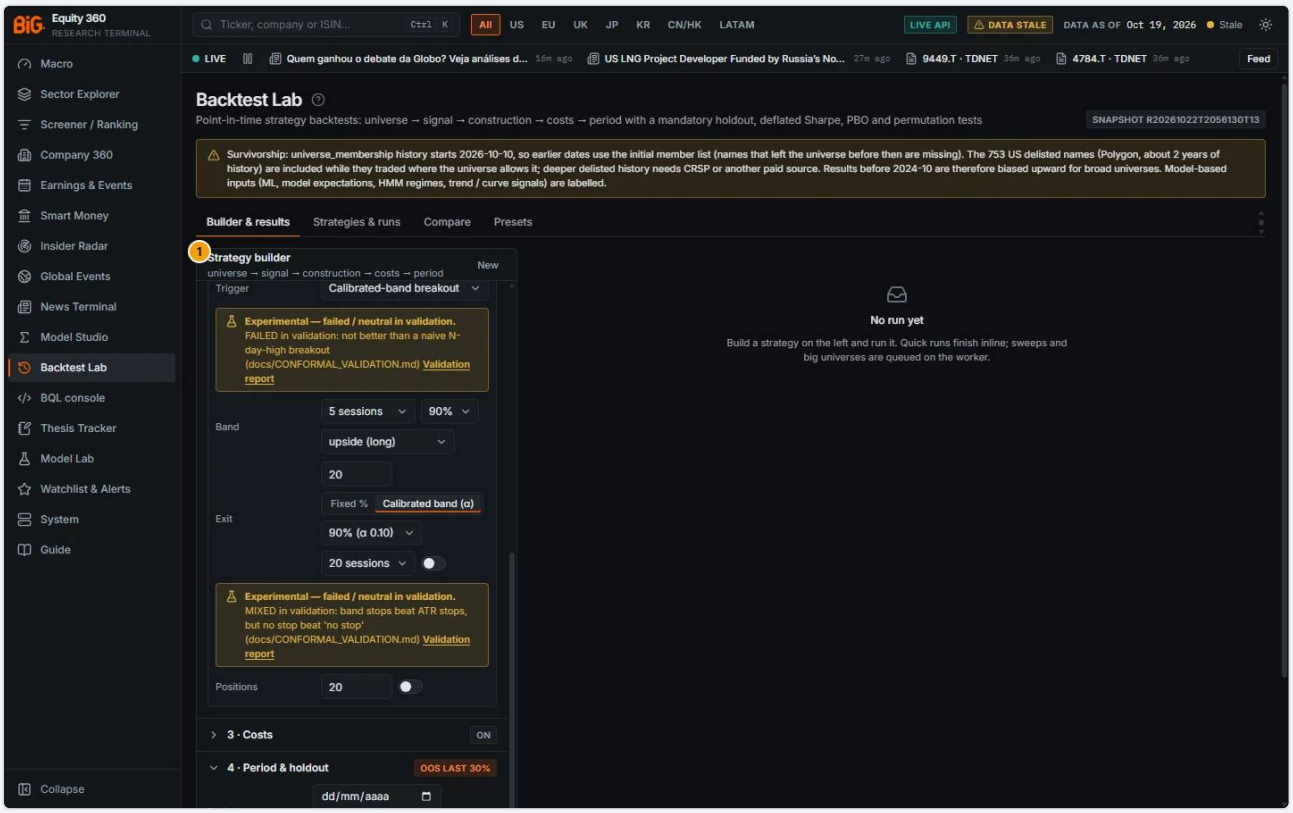


*Figure 11.3: Strategy builder (callout 1) in Event-driven mode with the trigger "Calibrated-band breakout" (band 5 sessions, 90%, upside (long), hold 20) and the exit *Calibrated band (α)** (90%, 20 sessions): both carry the amber "Experimental — failed / neutral in validation." note with a **Validation report** link.*

Controls

HEADER AND RUN BUTTONS

Control	What it does
New	Resets to the default spec.
Origin notes	When the draft came from an entry point, its notes are listed under the title (for example "signal = out-of-fold predictions of run SR... (CV scheme expanding)" or "dropped (no point-in-time history): ...").
Name	Strategy name.
Mode	**Rank / quantile** ("quantile portfolios rebalanced on a calendar") or **Event-driven** ("enter on an event, exit on time / stop / target").
Run backtest	Runs in "auto" mode (inline under 4 s, else on the worker).
Run on worker	Hover: "Queue on the worker (long sweeps / big universes)".
Check	Hover: "Check the spec and estimate the run time". Toast "Spec OK · sync / async (≈ N s)" with the benchmark, the number of sweep configurations, whether delisted names are included, and warnings.
Save	Hover: "Save as a strategy (multiple-testing counter, paper schedule)". Opens a name field; afterwards runs of this draft count toward the strategy's counter (toast "runs now count toward its multiple-testing counter").

1 · UNIVERSE

Control	What it does
Kind	**Securities** or **Cross-asset** (commodity, FX, equity-index, rates, credit and crypto assets of the cross-asset registry). Switching resets the signal and benchmark.

Control	What it does
Universe	`mvp_us` , `mvp_global` , `full` , `eu_broad` or "none (filters only)". Hover: "Point-in-time membership (history starts 2026-10-10; earlier dates use the initial list)".
Regions / **Sectors**	Comma-separated lists (e.g. `US, EU, JP` ; sector ids).
Delisted names	auto, include, exclude. Hover: "Inactive US names (Polygon, ~2 years) while they traded". Auto = on for `full` and region-only universes, off for the curated `mvp_*` lists.
Liquidity	Minimum market cap (USD bn) and minimum 20-day ADV (USD m), both point-in-time.
Screen rules	**+ Rule**: field, operator (≥, >, ≤, <, =, ≠, in, has value) and value, "evaluated point-in-time each rebalance". 61 screen fields (factor and composite scores, valuation multiples, price changes and momentum, volatility, beta, ADV, insider, sentiment, ML rank / decile / expected excess, revenue growth, net debt / EBITDA, sector, region, country, size bucket) plus any ratio metric id.
Cross-asset: **Asset classes** / **Assets**	Class checkboxes and a comma-separated asset list (CL, GC, SPX, EURUSD ...).

2 · SIGNAL (AND EVENT)

Control	What it does
Variable card	Name, **Signal source**, **Signal field** (labelled "(model)" when model-based), direction **high ↑** / **low ↓**, and a **Weight** when there are several variables. For `studio_run` the field is a Studio run id (`SR…`); for `bql` an expression.
+ Variable	Up to 12 variables.
Combine	**z blend** (weighted sum of cross-sectional z-scores), **rank blend** (weighted sum of percentile ranks) or **formula** (formula-engine syntax over the variable names, e.g. `0.5*value + 0.5*mom - abs(vol)` ; panel functions such as lag or growth are not allowed).
Extra lag	"Lag the signal by n rebalance periods (on top of the t+1 execution)", 0–24.

In Event-driven mode a box below the signal holds the event settings:

Control	What it does
Trigger	Event rule (see the trigger table). Experimental triggers show an amber note.
Trigger parameters	Surprise field and threshold; COT thresholds and contrarian / with-the-crowd mode; macro-surprise threshold z, country, theme and target instruments; maximum distance in ATR; signal threshold; band horizon, level and direction; break-away |z| and signal source; meta-setup horizon, mode, source and direction.
Exit	**Hold days**; stop / target mode **Fixed %** (stop loss %, take profit %) or **Calibrated band (α)** (band level 80 / 90 / 95%, band horizon 1 / 5 / 10 / 20 sessions, "Also take profit at the opposite band edge"). Hover: "Stop / target: fixed fractions, or the EXPERIMENTAL calibrated band (stop at the band edge emitted on the event date)".
Positions	**Max positions** and **Allow shorts**.

3 · CONSTRUCTION (RANK MODE)

Control	What it does
Rebalance	Daily, Weekly, Monthly.
Side	Long only, or Long-short (top minus bottom).
Selection	Quantiles (2–20; the top quantile is held long, the bottom short), Top N, or All (signal-weighted).
Weighting	Equal, Market cap, Signal, Inverse volatility, Risk parity (ERC). Volatility uses a 63-day lookback.
Neutralize	None, Sector (Asset class for cross-asset), Country.
Limits	Max weight per name (%) and max one-way turnover per rebalance (%).
Holding	Holding periods K ("Overlapping tranches: each rebalance re-forms 1/K of the book (Jegadeesh-Titman)") and the execution lag in days (default 1).
Regime filter / **Invest in**	"Go to cash (or half) when the macro regime known at t is not in the selected states (model-based states labelled)". Regimes offered: cycle quadrant (US, GLOBAL), HMM regime (US, GLOBAL; model-based), inflation regime (US), rate cycle (Fed, ECB). The builder applies the cash action.

4 · COSTS (3 · COSTS IN EVENT MODE)

Control	What it does
Costs switch	Apply trading costs (on by default; the badge turns amber "off").
Commission bp	Per unit traded, default 1 bp.
Spread	**ADV buckets** (hover: "Half-spread by 20-day ADV bucket (≥1bn 1bp · ≥100m 3bp · ≥20m 8bp · ≥5m 20bp · else 50bp) or fixed") or **Fixed** (default 5 bp).
Impact	"Square-root impact: coef × daily vol × √(trade / ADV), with the capital below": coefficient (default 0.1) and capital (default USD 10 m).
Borrow bp/yr	General collateral (default 50) and hard to borrow (default 300, applied when ADV < USD 5 m).
FX	Base currency **USD (official FX)** or **Local currency**, and the FX conversion cost on non-USD trades (default 2 bp).

5 · PERIOD & HOLDOUT (4 · IN EVENT MODE)

Control	What it does
Period	Start and end dates.
Holdout start	"Mandatory out-of-sample holdout: never used for parameter selection". Empty = the last **Holdout fraction %** (default 30%). The section badge shows "OOS ‹date›" or "OOS last 30%".
Benchmark	auto (SPY for US, ACWI for global, assets equal weight for cross-asset) or one of the listed benchmarks.
Risk-free	3m T-bill or zero.

6 · ROBUSTNESS (5 · IN EVENT MODE)

Control	What it does
Permutations	Label-permutation runs, 0–200 (default 20). Hover: "Label permutation test: the signal shuffled across names within each date (should give ~0 Sharpe)".
Stale-signal check	On by default.
Factor attribution	On by default.
Parameter sweep (walk-forward)	**+ Parameter** (up to 4): `construction.quantiles` , `top_n` , `weighting` , `holding_periods` , `rebalance` , `side` , `neutralize` , `max_weight` , `signal.lag_periods` , `event.hold_days` , `event.stop_loss` , `event.take_profit` , `costs.commission_bps` , each with comma-separated values. At most 64 combinations. Hover: "Every combination is simulated; selection uses the in-sample window only (walk-forward), PBO via CSCV, all configurations count as trials for the deflated Sharpe". The badge shows the number of configurations.
WF / CSCV	With a sweep: walk-forward folds (2–12, default 4) and CSCV blocks (4–16, default 10).

Methodology

SIGNAL SOURCES

Source (title in the app)	Universe	Point-in-time rule (catalogue text)	Coverage on the live snapshot
Factor scores (0-100)	securities	monthly point-in-time history (scoring job, knowledge-dated inputs)	2016-10-31 → 2026-10-16
Composite scores by profile	securities	monthly point-in-time history	2016-10-31 → 2026-10-16
ML predictions (XGBoost, walk-forward OOS)	securities	expanding walk-forward out-of-sample predictions (purged, embargoed); production model xgboost_rank; ft_transformer is DEMOTED (failed the label-permutation test). Model-based	2010-02-19 → 2026-10-16
Technical score and components	securities	stored for the latest as-of date only	latest date only, **not backtestable**
Technical factors (evidence-based)	securities	daily, bars ≤ t; percentiles within region × sector among liquid names; gate per factor in TECHNICALS_V3_VALIDATION	2004-01-01 → 2026-10-19
Insider signals	securities	Form 4 filing dates (knowledge-dated)	2017-01-31 → 2026-10-16

Source (title in the app)	Universe	Point-in-time rule (catalogue text)	Coverage on the live snapshot
News sentiment / attention	securities	daily, FinBERT / GDELT tone (model-based)	2025-10-09 → 2026-10-16
Geopolitical / event exposure	securities	snapshot as-of dates (model-based)	three dates, **not backtestable**
Earnings PEAD / model surprise	securities	known on signal date; surprises vs model-based expectations (SARIMA), not consensus	2016-01-07 → 2026-10-13
Valuation multiples (daily)	securities	price × PIT fundamentals	2010-01-01 → 2026-10-16
Financial ratios (point-in-time)	securities	as of knowledge date (filing date); any ratio metric id	1994-04-30 → 2026-10-20
Price-derived	both	daily adjusted prints on or before the rebalance date	1927-12-30 → 2026-10-20
Model Studio run (out-of-fold predictions)	securities	field = Studio run id of a panel tree_ensemble / regularized run; use expanding CV, other schemes are flagged LOOK-AHEAD. Model-based	per run
Cross-asset signals	assets	as of knowledge date; trend / curve shape / value / z metrics model-based	2000-01-01 → 2026-10-20
CFTC positioning via cross-asset join	assets	official CFTC COT, latest report released on or before t (stale > 28 days → null)	2000-01-01 → 2026-10-20
BQL expression	securities	evaluated point-in-time per rebalance (BQL-lite)	not in the live catalogue (see below)

Signal transforms use the Model Studio transform set; look-ahead transforms (`lead` , `fwd_*`) are rejected in backtest signals. The builder does not expose a transform editor; transforms come from presets or entry points.

> **Note:** the **BQL expression** signal source (roadmap v3 B1) is implemented but not offered by the live catalogue of 19 Oct 2026 (snapshot R20261022T2056130T13), which lists the 15 sources above it. It appears after the B1 go-live step (rebuild the app and restart the API). Then: choose source "BQL expression (cross-sectional, point-in-time per rebalance)" and type an expression such as `cs_zscore(chg_pct(px, 3M)) - cs_zscore(pe_ttm)` ; see the BQL language reference.

EVENT TRIGGERS

Trigger	Universe	Rule	Label
Insider cluster buy	securities	first insider-signals snapshot with a cluster buy per name; long	
Earnings model surprise	securities	|field| ≥ threshold (default 2.0) on the signal date, field SUE, PEAD score, announcement-return z or revenue z; direction = sign (shorts need **Allow shorts**)	model-based (not consensus)
CFTC positioning extreme	assets	3-year COT index crossing above high (0.9) / below low (0.1); contrarian fades the crowd	
Macro surprise	both	|signed z| vs model-based expectations ≥ threshold (1.5): trade the target instruments	model-based
Confluence-zone touch	securities	price inside / within x ATR (0.25) of the top-ranked zone: long at support, short at resistance	model-based; latest date only, not backtestable
Signal threshold crossing	both	the combined signal crossing the threshold at a rebalance date	
Calibrated-band breakout (experimental)	securities	the h-session move leaves the marginal conformal band emitted h sessions earlier	**experimental**, failed validation
Peer break-away fade (experimental)	securities	fade the member with |z_cond| ≥ z on days the peer group's joint score exceeded its 90% radius	**experimental**, failed validation

Trigger	Universe	Rule	Label
Meta-labelled setups (Technicals v3)	securities	the v2 weekly trade setups (target T1, stop at the invalidation): all, only those the meta-labelling model takes, or taken and sized by its quarter-Kelly fraction	**experimental** in the catalogue; see Experimental signals

Events enter at the close after `entry_lag_days` (default 1) and exit on the first of: hold days, stop, target. Each position gets an equal share of the book up to **Max positions**.

PORTFOLIO CONSTRUCTION

At each rebalance date the eligible names (universe member, screen rules passed, liquidity limits met, signal not null, at least 5 names) are ranked by the combined signal. Quantile selection holds the top quantile long and, for long-short, the bottom quantile short; gross exposure defaults to 1 long-only and 2 long-short (100/100). Neutralisation ranks within each sector or country. Weighting schemes: equal; market cap; signal; inverse volatility $w_i \propto 1/\sigma_i$; risk parity (equal risk contribution) on the 63-day covariance. Weights are capped at **Max weight**, and the one-way turnover per rebalance at **Max turnover**. With K holding periods, each rebalance replaces 1/K of the book (overlapping tranches).

COST MODEL

For a trade of Δw_i (fraction of NAV) in name i:

```
cost_i   = |Δw_i| · (commission + half_spread(ADV_i) + 1{non-base currency} · fx) / 10⁴
         + |Δw_i| · impact_coef · σ_i,daily · √( |Δw_i| · capital · NAV / ADV_i )
borrow_i = w_i,short · (ADV_i < 5 m ? htb_bps : borrow_bps) / 10⁴ / 252        (per day held short)
half_spread(ADV) = 1 bp (≥ 1 bn) · 3 bp (≥ 100 m) · 8 bp (≥ 20 m) · 20 bp (≥ 5 m) · 50 bp (otherwise)
```

ADV is the 20-day average dollar volume; σ is the daily volatility. The equity curve shows the strategy net of costs and a dashed "Gross of costs" line.

METRICS

```
Sharpe   = mean(r − r_f) / sd(r − r_f) · √252           (daily, excess of the 3m T-bill)
Sortino  = mean(r − r_f) / downside sd · √252
CAGR     = NAV_end^(1/years) − 1;   Calmar = CAGR / |max drawdown|
Beta, alpha (ann.)  from the regression of strategy on benchmark returns
Information ratio = annualised excess return over the benchmark / tracking error
Hit rate (days) = share of days with a positive return;  Positive months = share of months > 0
```

Metrics are shown for the full period, the in-sample window and the holdout.

Data sources and refresh

See the source and trigger tables; coverage is measured on the current snapshot. Screen fields without point-in-time history (DCF fair value, forecasts, zones, KPIs, data quality, short interest) are dropped from backtested screens with a warning.

Validation status

Signals carry the validation status of the model behind them:

- **ML predictions:** XGBoost OOS rank IC about 0.035, one input, not proven alpha; the Transformer is demoted (failed the label-permutation test). See Model Lab.
- **Technical factors:** hi52, MAX and IVOL passed as a "rank edge" (descriptive; not profitable stand-alone after costs); momentum, residual momentum, reversal and industry momentum failed and are experimental; the composite failed (`docs/TECHNICALS_V3_VALIDATION.md` #1).
- **Earnings surprises, macro surprises, HMM regimes, sentiment, cross-asset trend / value / z metrics:** model-based.
- **Conformal triggers and band stops, meta setups:** experimental (Experimental signals).

Correct usage and worked example

Building "quality plus momentum, long-only, sector-neutral" by hand: universe `mvp_us`; signal variables `quality` (Factor scores) and `mom` (Price-derived `mom_12_1`), Combine **z blend**; construction Monthly, Long only, Quintiles 5, Equal, Neutralize Sector, Max weight 5%; costs on; holdout last 30%; Robustness: permutations 20, a sweep on `construction.quantiles` = 3, 5, 10. Press **Check** first: the toast reports whether the run will be sync or async and its estimated time. This is the "Quality + momentum" preset plus a sweep.

Limitations and pitfalls

- Fields marked "(model)" are model outputs; a backtest of a model-based signal tests the model too.
- A screen built in the Screener may lose fields that have no history ("dropped (no point-in-time history): ..." in the origin notes).
- Event triggers on latest-only tables (confluence touch) have no history to backtest.

Related mnemonics and BQL fields

Screener, Scores and factors, Technical factors (evidence-based), Positioning (CFTC).

Results

Route: `/backtest` (Builder & results, right-hand pane) · **API:** `GET /api/v1/backtest/runs/{run_id}`, `/holdings`, `/trades`

Purpose

The results pane shows one completed run in five tabs: Performance, Risk, Attribution (or Events for event-driven runs), Trades & holdings and Robustness.

Screenshot

The figures show the "US value long-short (quintiles, monthly)" preset run on the test copy (run `BR20261023T133814A3986`, spec hash A1D0FE78, 2016-11-01 → 2026-10-09, holdout from 2023-10-16).

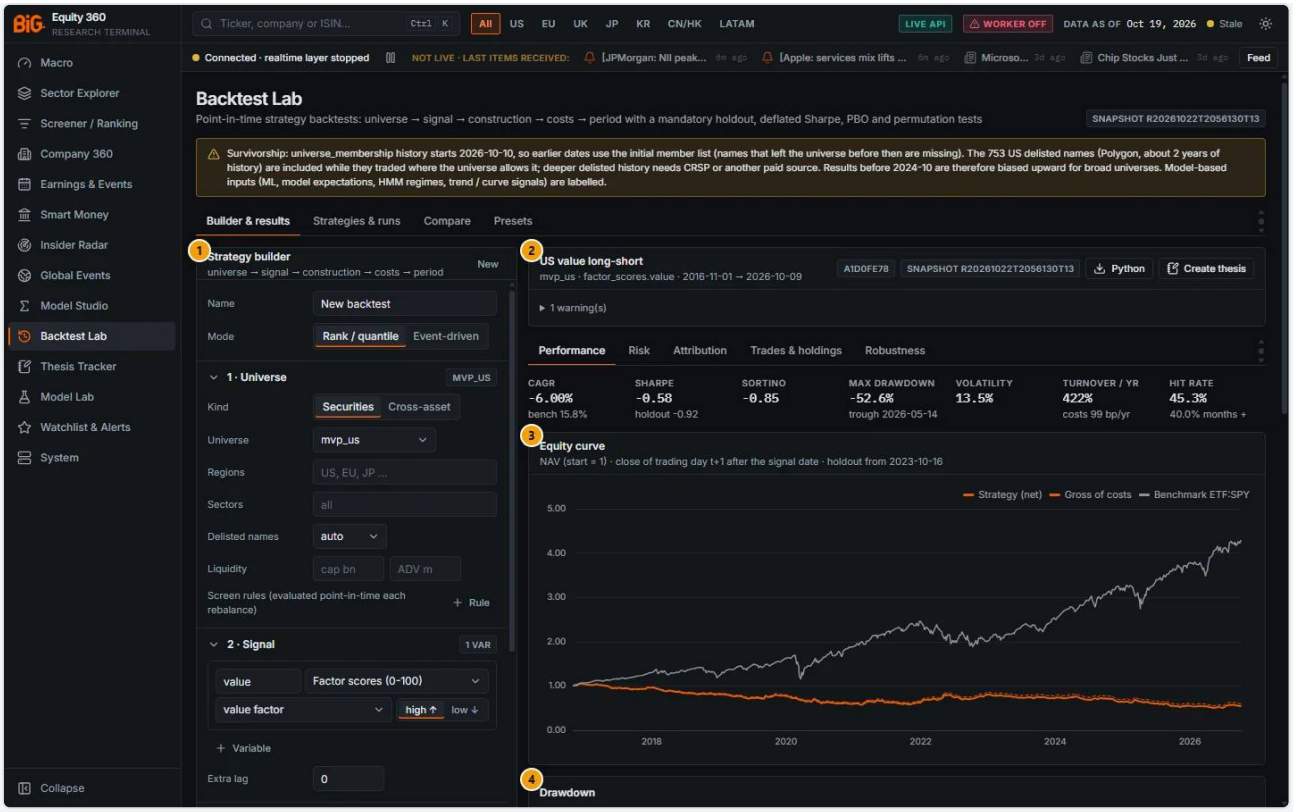


Figure 11.4: Performance tab of the US value long-short run.

#	Panel	What it shows
1	Strategy builder	The draft (left).
2	Run header	"US value long-short · mvp_us · factor_scores.value · 2016-11-01 → 2026-10-09", spec hash and snapshot badges, **Python** and **Create thesis** buttons, "1 warning(s)".
3	Equity curve	"NAV (start = 1) · close of trading day t+1 after the signal date · holdout from 2023-10-16": strategy net, gross of costs (dashed), benchmark ETF:SPY.
4	Drawdown	Strategy and benchmark drawdowns.
5	Metrics	"net of costs · Sharpe / Sortino on excess of the 3m T-bill": Full, In-sample, Holdout (OOS). Below the fold.
6	Monthly returns	Year × month grid of net returns (%). Below the fold.

Figure 11.5: Metrics table (callout 5): full period, in-sample and holdout.

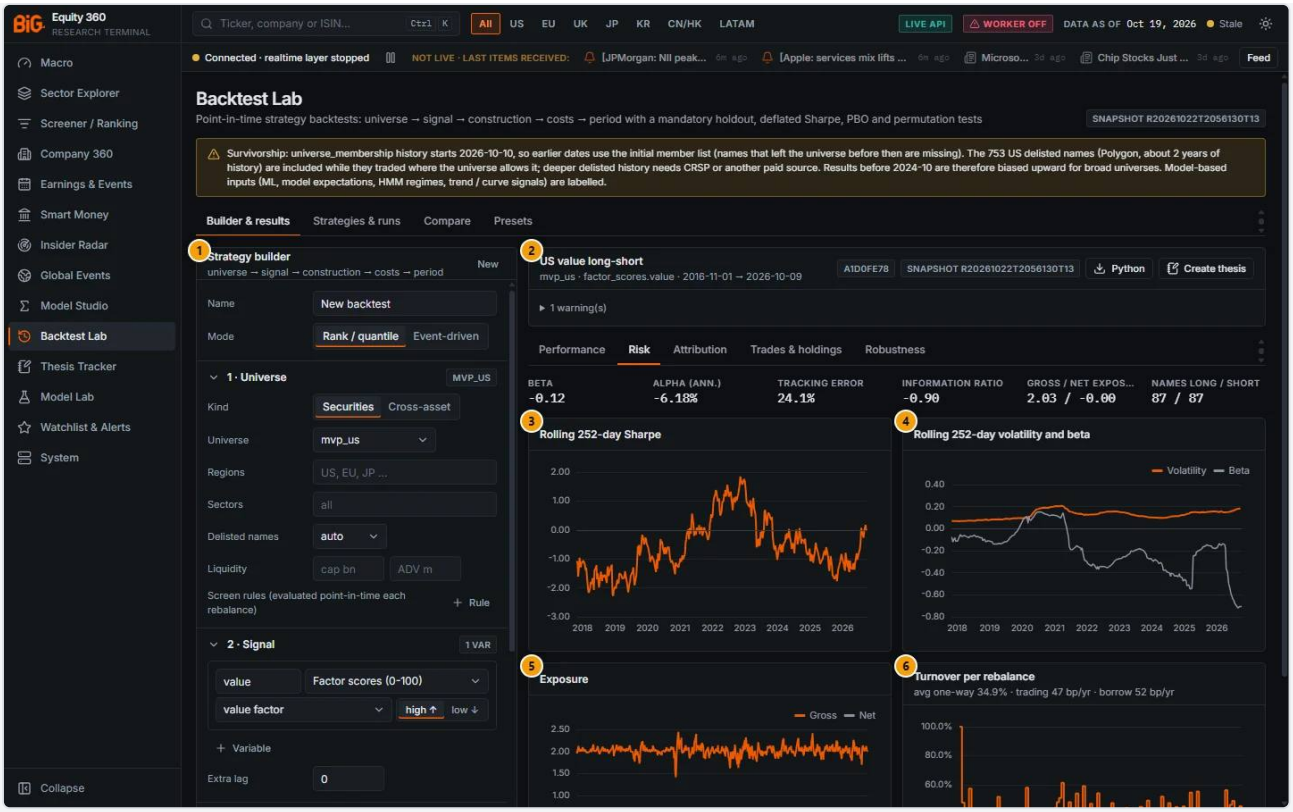


Figure 11.6: Risk tab.

#	Panel	What it shows
1	Strategy builder	The draft.
2	Run header	As in Figure 11.4.
3	Rolling 252-day Sharpe	One-year rolling Sharpe ratio.
4	Rolling 252-day volatility and beta	Rolling volatility and beta to the benchmark.
5	Exposure	Gross and net exposure over time.
6	Turnover per rebalance	One-way turnover per rebalance; subtitle "avg one-way 34.9% · trading 47 bp/yr · borrow 52 bp/yr".

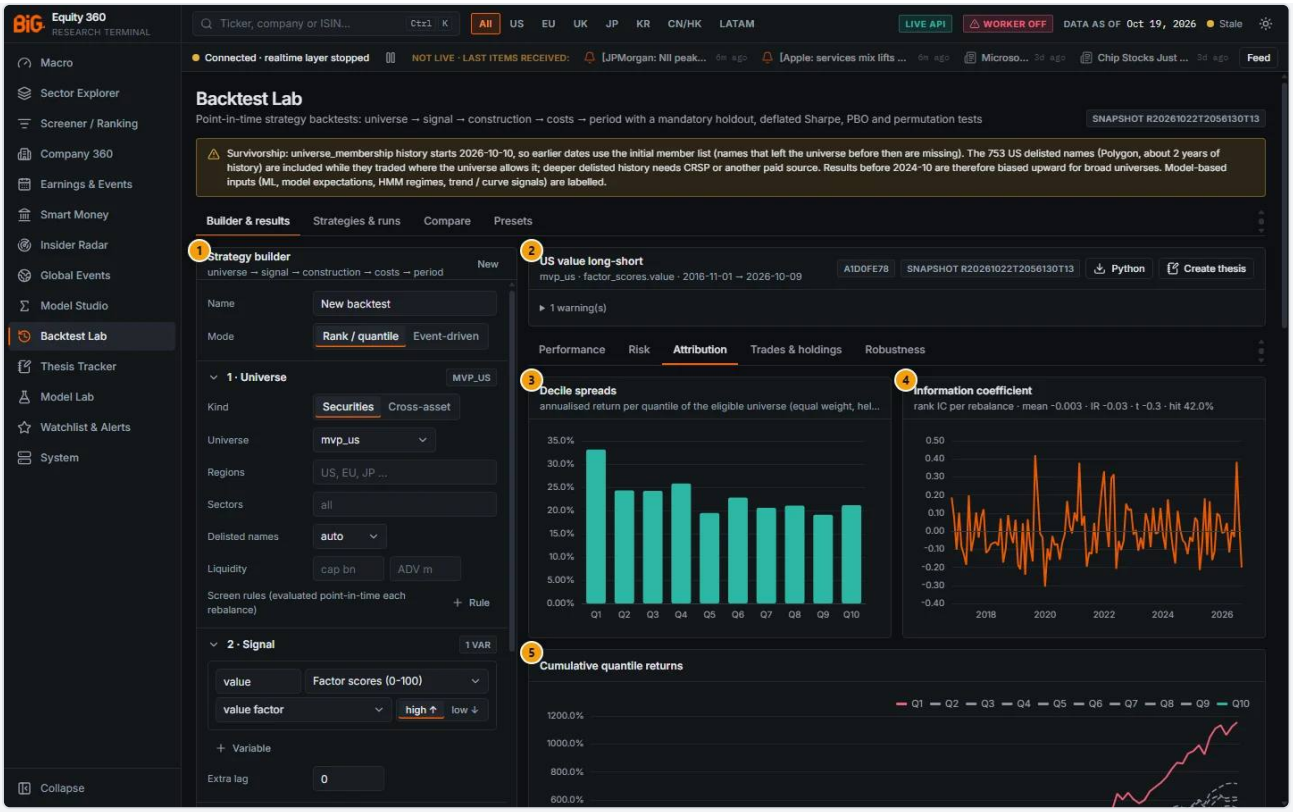
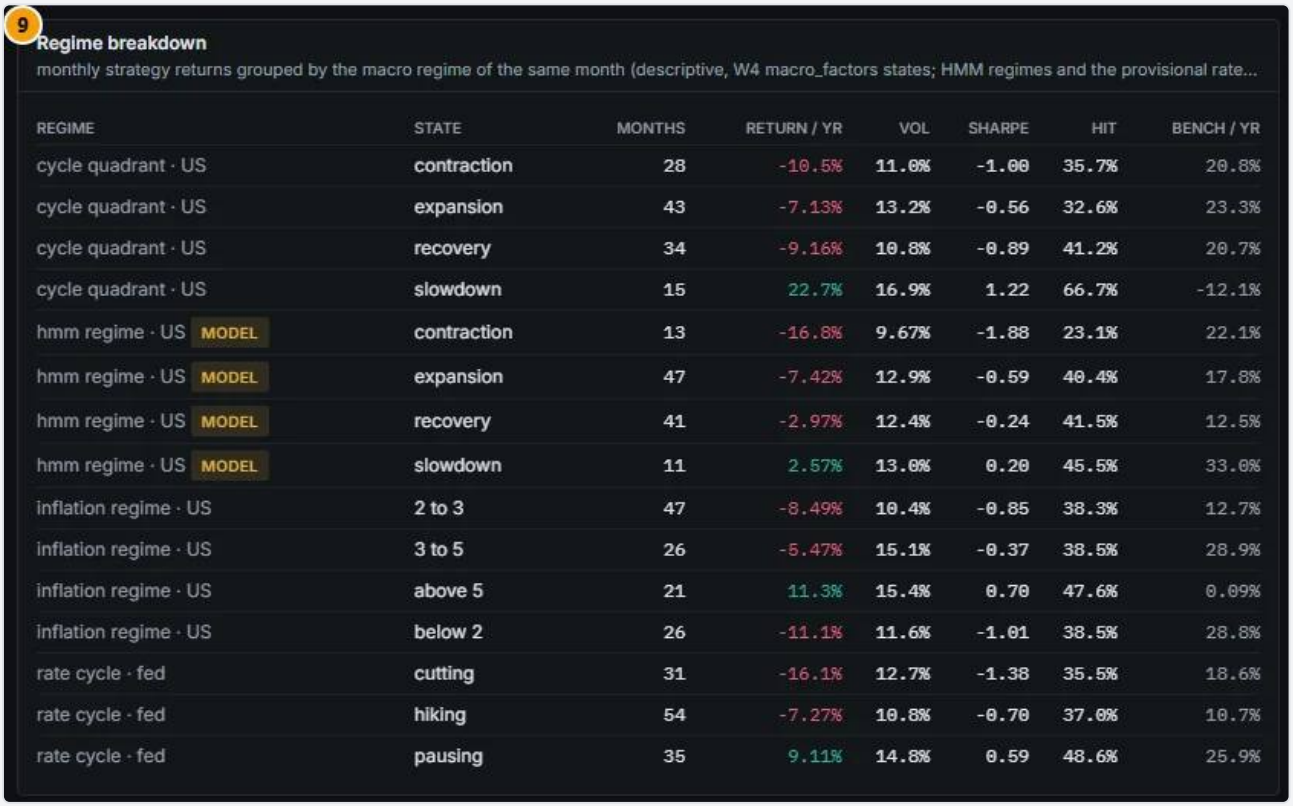


Figure 11.7: Attribution tab.

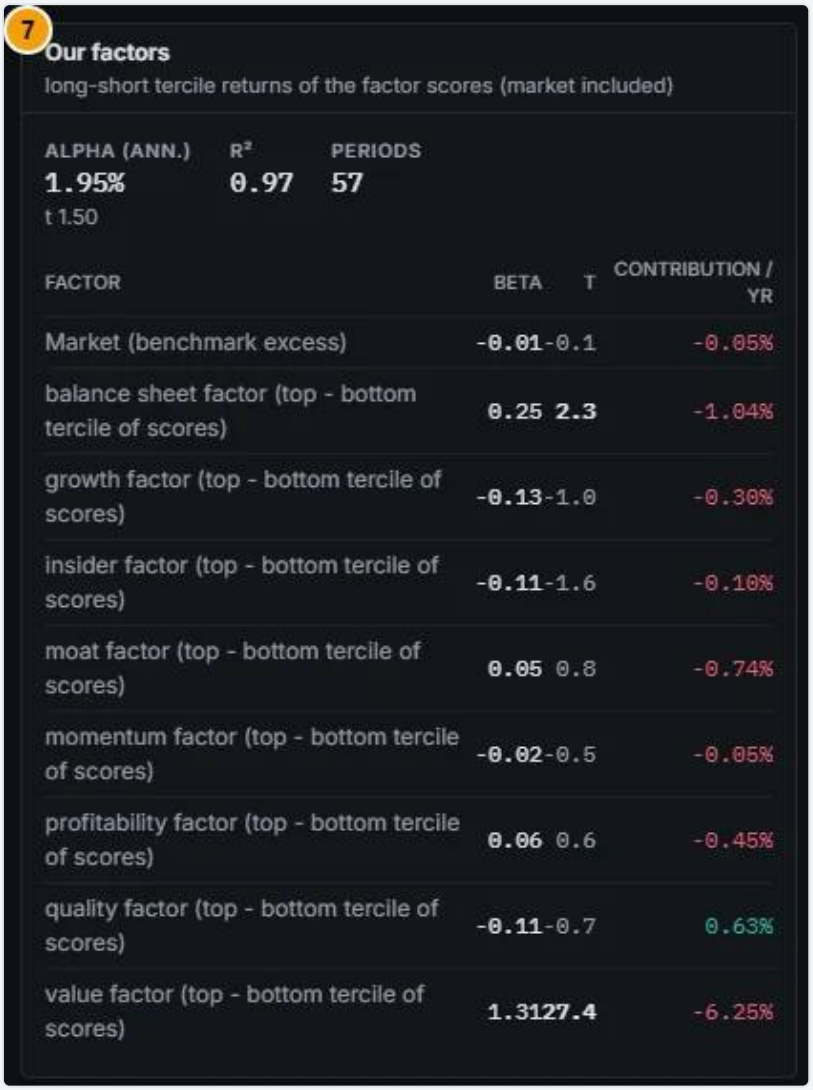
#	Panel	What it shows
1	Strategy builder	The draft.
2	Run header	As in Figure 11.4.
3	Decile spreads	Annualised return per signal quantile of the eligible universe (equal weight, held to the next rebalance), with the top − bottom spread and its t.
4	Information coefficient	Rank IC per rebalance: "mean -0.003 · IR -0.03 · t -0.3 · hit 42.0%".
5	Cumulative quantile returns	Q1…Q10 cumulative returns (Q10 and Q1 solid, the others dashed).
6	Style attribution	Market / size / value / momentum proxies (below the fold, Figure 11.8).
7	Our factors	Long-short tercile returns of our factor scores plus the market (below the fold).
8	Average book exposures	Weighted cross-sectional z-score of the holdings (long − short) per factor (below the fold).
9	Regime breakdown	Monthly returns grouped by macro regime (below the fold, Figure 11.9).

Figure 11.8: Style attribution (callout 6): value beta 1.24 (t 42.4), alpha −0.57% a year (t −0.70), R^2 0.95 over 119 periods.



REGIME	STATE	MONTHS	RETURN / YR	VOL	SHARPE	HIT	BENCH / YR
cycle quadrant · US	contraction	28	-10.5%	11.0%	-1.00	35.7%	20.8%
cycle quadrant · US	expansion	43	-7.13%	13.2%	-0.56	32.6%	23.3%
cycle quadrant · US	recovery	34	-9.16%	10.8%	-0.89	41.2%	20.7%
cycle quadrant · US	slowdown	15	22.7%	16.9%	1.22	66.7%	-12.1%
hmm regime · US MODEL	contraction	13	-16.8%	9.67%	-1.88	23.1%	22.1%
hmm regime · US MODEL	expansion	47	-7.42%	12.9%	-0.59	40.4%	17.8%
hmm regime · US MODEL	recovery	41	-2.97%	12.4%	-0.24	41.5%	12.5%
hmm regime · US MODEL	slowdown	11	2.57%	13.0%	0.20	45.5%	33.0%
inflation regime · US	2 to 3	47	-8.49%	10.4%	-0.85	38.3%	12.7%
inflation regime · US	3 to 5	26	-5.47%	15.1%	-0.37	38.5%	28.9%
inflation regime · US	above 5	21	11.3%	15.4%	0.70	47.6%	0.09%
inflation regime · US	below 2	26	-11.1%	11.6%	-1.01	38.5%	28.8%
rate cycle · fed	cutting	31	-16.1%	12.7%	-1.38	35.5%	18.6%
rate cycle · fed	hiking	54	-7.27%	10.8%	-0.70	37.0%	10.7%
rate cycle · fed	pausing	35	9.11%	14.8%	0.59	48.6%	25.9%

Figure 11.9: Regime breakdown (callout 9): cycle quadrant, HMM regime (labelled MODEL), inflation regime and Fed rate cycle.



7

Our factors

long-short tercile returns of the factor scores (market included)

ALPHA (ANN.)	R^2	PERIODS
1.95%	0.97	57
t 1.50		

FACTOR	BETA	T	CONTRIBUTION / YR
Market (benchmark excess)	-0.01	-0.1	-0.05%
balance sheet factor (top - bottom tercile of scores)	0.25	2.3	-1.04%
growth factor (top - bottom tercile of scores)	-0.13	-1.0	-0.30%
insider factor (top - bottom tercile of scores)	-0.11	-1.6	-0.10%
moat factor (top - bottom tercile of scores)	0.05	0.8	-0.74%
momentum factor (top - bottom tercile of scores)	-0.02	-0.5	-0.05%
profitability factor (top - bottom tercile of scores)	0.06	0.6	-0.45%
quality factor (top - bottom tercile of scores)	-0.11	-0.7	0.63%
value factor (top - bottom tercile of scores)	1.31	27.4	-6.25%

Figure 11.10: Our factors (callout 7).

Figure 11.11: Average book exposures (callout 8).

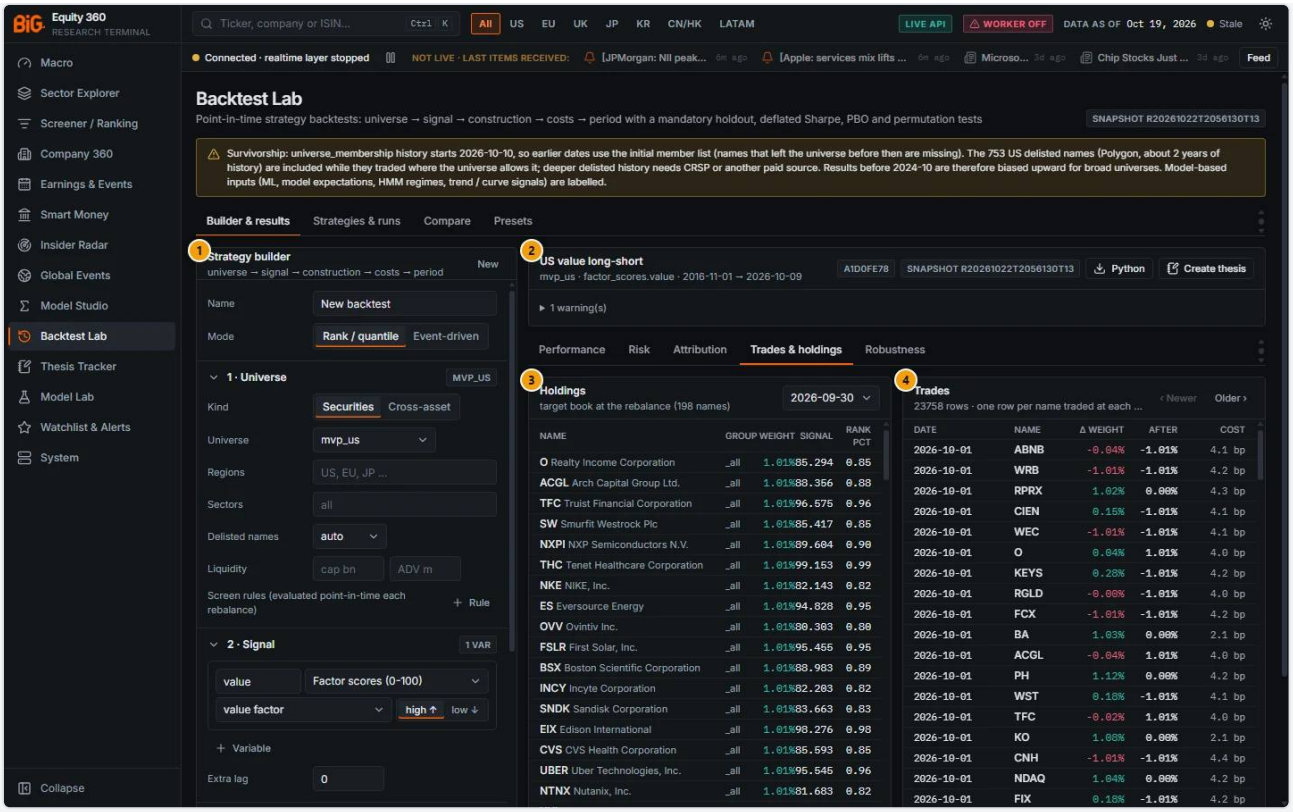


Figure 11.12: Trades & holdings tab.

#	Panel	What it shows
1	Strategy builder	The draft.
2	Run header	As in Figure 11.4.
3	Holdings	"target book at the rebalance (198 names)" for the selected date (2026-09-30): name, group, weight, signal, rank percentile.
4	Trades	"23758 rows · one row per name traded at each rebalance": date, name, Δ weight, weight after, cost in bp.

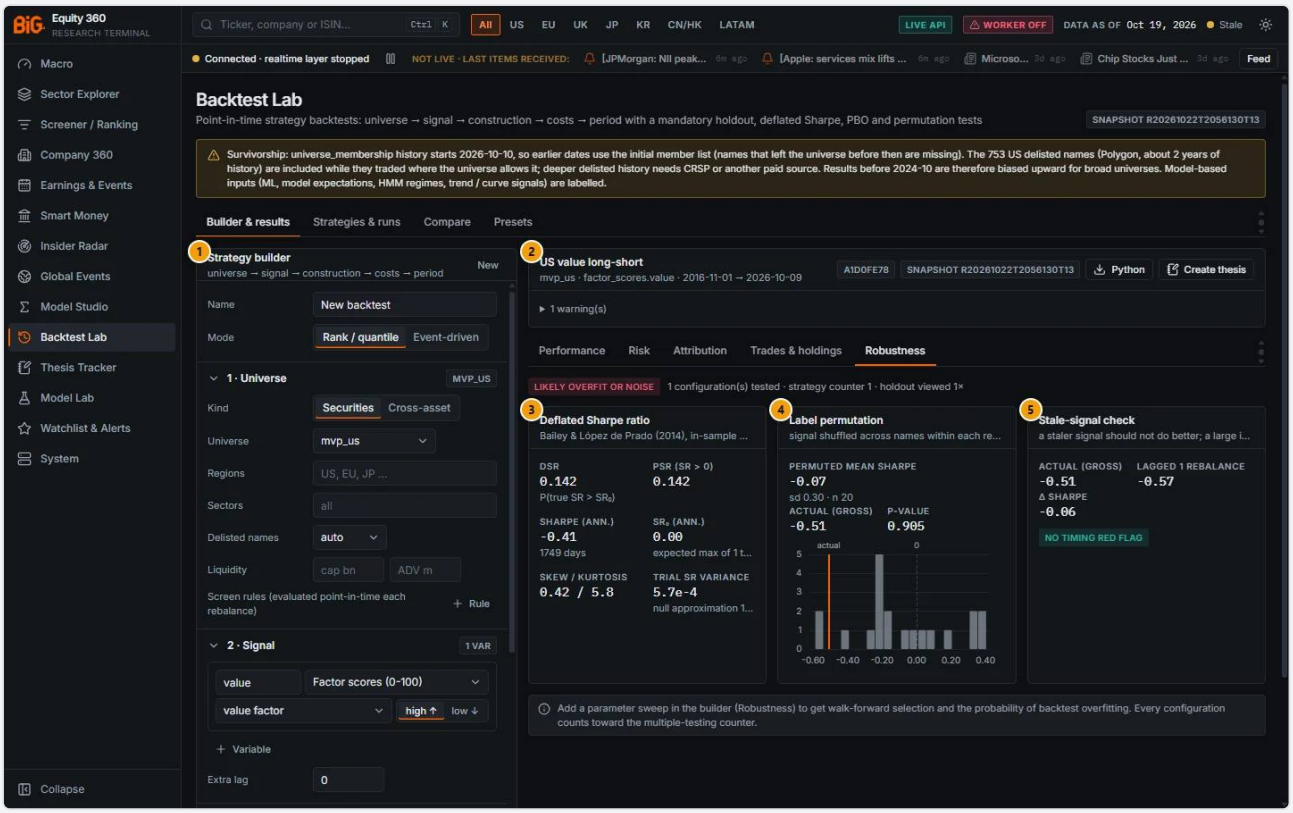


Figure 11.13: Robustness tab with the verdict "LIKELY OVERFIT OR NOISE".

#	Panel	What it shows
1	Strategy builder	The draft.
2	Run header	As in Figure 11.4.
3	Deflated Sharpe ratio	DSR, PSR, annualised Sharpe and number of days, SR_0 (expected maximum of the trials), skew / kurtosis, trial Sharpe variance and its source.
4	Label permutation	Permuted mean Sharpe, actual gross Sharpe, p-value and the histogram of permuted Sharpes with the actual marked.
5	Stale-signal check	Actual vs one-rebalance-lagged Sharpe and the flag.

> Screenshot pending: Robustness tab of a run with a parameter sweep, showing the Parameter sweep table, PBO (CSCV) histogram, Walk-forward NAV with fold lines and the Holdout of the in-sample choice (route `/backtest`).

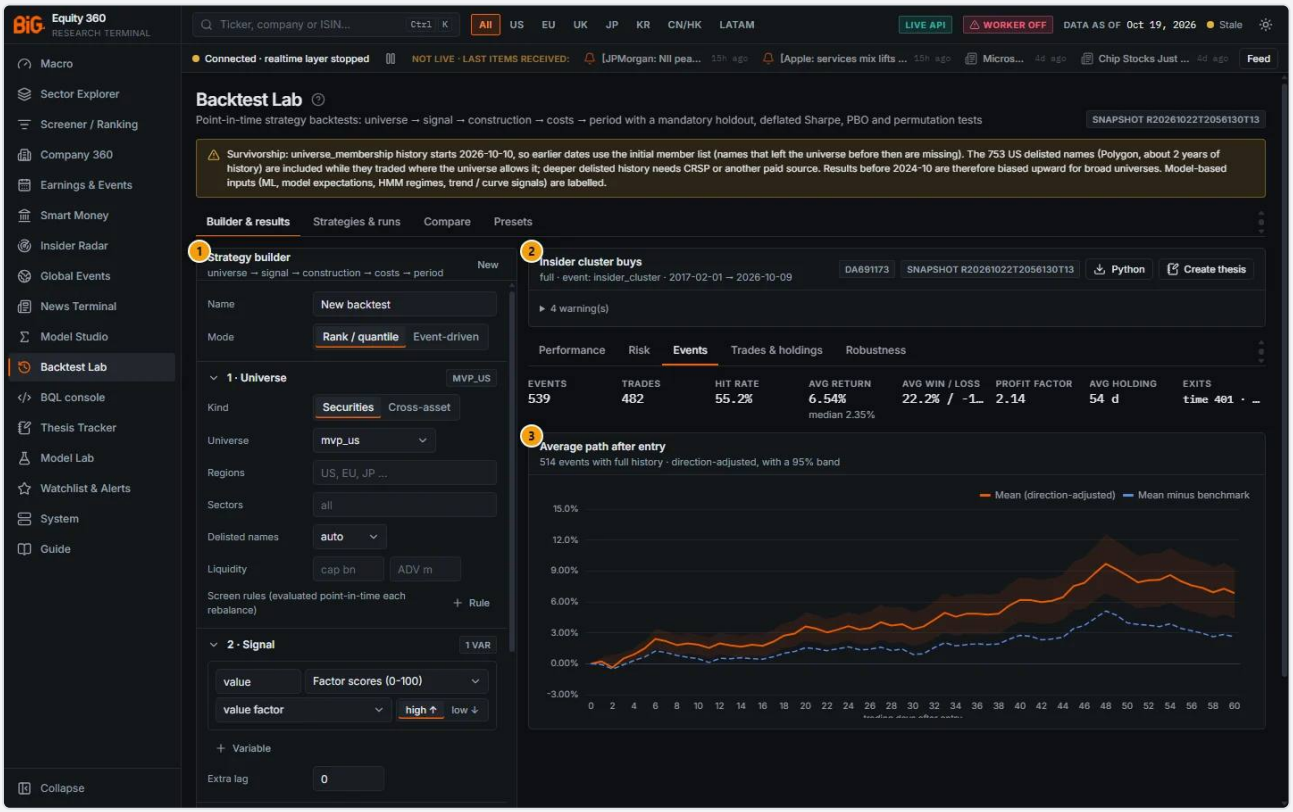


Figure 11.14: Events tab of the "Insider cluster buys (60-day hold)" preset run (test copy, `full` universe, 2017-02-01 → 2026-10-09).

#	Panel	What it shows
1	Strategy builder	The draft.
2	Run header	"Insider cluster buys · full · event: insider_cluster · 2017-02-01 → 2026-10-09", spec hash DA691173, "4 warning(s)". Tiles: 539 events, 482 trades, hit rate 55.2%, average return 6.54% (median 2.35%), average win 22.2%, profit factor 2.14, average holding 54 days, exits mostly by time.
3	Average path after entry	"514 events with full history · direction-adjusted, with a 95% band": the mean path (orange) peaks near +10% around day 48; the mean minus benchmark (dashed blue) stays near +3 to +5%.

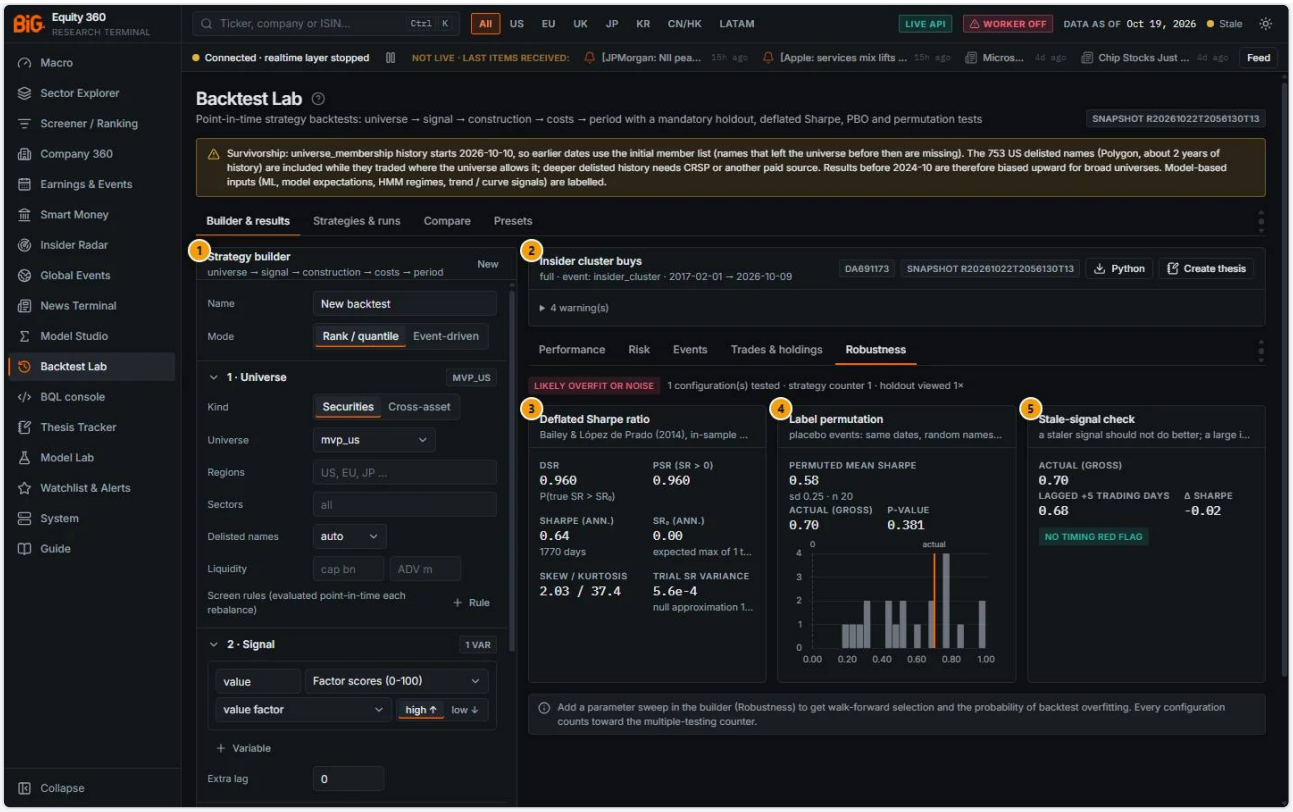


Figure 11.15: Robustness tab of the same run: DSR 0.960 but placebo events give a mean Sharpe of 0.58 against an actual 0.70 (p 0.381), so the badge reads "LIKELY OVERFIT OR NOISE".

#	Panel	What it shows
1	Strategy builder	The draft.
2	Run header	As in the previous figure.
3	Deflated Sharpe ratio	DSR 0.960, PSR 0.960, Sharpe 0.64 over 1,770 in-sample days, SR_0 0.00 (one trial), skew 2.03, kurtosis 37.4.
4	Label permutation	"placebo events: same dates, random names from the universe (gross of costs)": permuted mean 0.58 (sd 0.25, n 20), actual 0.70, p-value 0.381.
5	Stale-signal check	Actual 0.70, lagged +5 trading days 0.68, Δ −0.02: no timing red flag.

Controls

Control	What it does
Run header badges	Spec hash (first 8 characters) and snapshot.
Python	"Export a standalone Python script that reproduces this run" (Export to Python).
Create thesis	"Create a thesis linked to this backtest run (pick the subject in the editor)".
Experimental note	"Experimental — failed / neutral in validation. Conformal triggers / band stops were not promoted after validation: ..." with a **Validation report** link, when the run used them.
Notes	The survivorship caveat for cross-asset runs and "Model-based inputs: ..." when model-based sources were used.
N warning(s)	Expandable list of run warnings (dropped screen rules, look-ahead in fitting, short history, ignored regime filter, experimental inputs).
Tabs	**Performance**, **Risk**, **Attribution** (rank mode) or **Events** (event mode), **Trades & holdings**, **Robustness**.
Holdings **Rebalance date**	Select any rebalance date (latest first).
Trades **‹ Newer** / **Older ›**	Pages of 200 rows.

PERFORMANCE TAB

Tiles: CAGR (with the benchmark's), Sharpe (with the holdout's), Sortino, Max drawdown (with the trough date), Volatility, Turnover / yr (with the cost drag in bp a year), Hit rate (with the share of positive months). Then the equity curve, the drawdown chart, the **Metrics** table (CAGR, total return, volatility, Sharpe, Sortino, max drawdown, longest drawdown in days, Calmar, hit rate, positive months, beta, alpha, information ratio, tracking error, excess CAGR vs benchmark, benchmark CAGR and Sharpe, skew, kurtosis, period; hover a metric for its definition) and the **Monthly returns** grid (green / red cells, bold yearly total).

RISK TAB

Tiles: beta, alpha (annualised), tracking error, information ratio, gross / net exposure, average names long / short. Charts: rolling 252-day Sharpe, rolling volatility and beta, gross and net exposure, one-way turnover per rebalance (with trading and borrow cost in bp a year).

ATTRIBUTION TAB (RANK MODE)

Decile (or quantile) spreads, the information coefficient per rebalance, cumulative quantile returns, **Style attribution**, **Our factors** (alpha, R^2, periods and a table of factor, beta, t, contribution per year; $|t| \geq 2$ in bold), **Average book exposures** and the **Regime breakdown** table (regime, state, months, return per year, volatility, Sharpe, hit rate, benchmark return; model-based regimes carry a **model** badge).

EVENTS TAB (EVENT MODE)

Tiles: events, trades, hit rate, average and median return, average win / loss, profit factor, average holding days, exit reasons (time, stop, target, open). Chart: "Average path after entry": mean direction-adjusted cumulative return over the days after entry with a 95% band and the excess path.

TRADES & HOLDINGS TAB

Holdings at any rebalance and the trade list. Event runs list round trips (name, entry, exit, direction, return, exit reason).

ROBUSTNESS TAB

See Robustness statistics.

Methodology

- **Quantile spreads and IC.** At each rebalance the eligible universe is cut into quantiles of the signal (10 by default); each quantile is held equal-weighted to the next rebalance. The tab shows the annualised return per quantile, the top − bottom spread with its t-statistic, and the rank IC per rebalance (Spearman correlation of signal and next-period return) with mean, IR, t and hit rate.
- **Attribution.** "Time-series regression of strategy excess returns per rebalance period on long-short factor returns built from the same point-in-time panel (equal-weight terciles); Newey-West t-stats". Style factors: market (benchmark excess), size (small minus big on PIT market cap), value (value factor score), momentum (12−1 price momentum). "Our factors": top − bottom tercile returns of each factor score, plus the market. Contribution per year = beta × mean factor return, annualised.
- **Regime breakdown.** "Monthly strategy returns grouped by the macro regime of the same month (descriptive, W4 macro_factors states; HMM regimes and the provisional rate-cycle rule are model-based)". It is descriptive: the regime of a month is known only at the end of it.

Data sources and refresh

The result is stored with the run and does not change; holdings and trades are paged from the run directory.

Validation status

Not applicable (reporting).

Correct usage and worked example

The US value long-short preset (Figures 11.4–11.13):

1. **Performance.** CAGR −6.00% a year against +15.8% for SPY; Sharpe −0.58 (holdout −0.92); maximum drawdown −52.6% (trough 2026-05-14); turnover 422% a year with a cost drag of 99 bp a year.
2. **Risk.** Beta −0.12, gross / net exposure 2.03 / −0.00 (a dollar-neutral 100/100 book), 87 names long and 87 short on average; trading costs 47 bp and borrow 52 bp a year.
3. **Attribution.** The style regression explains the strategy almost entirely by the value factor (beta 1.24, t 42.4, R^2 0.95, alpha −0.57% a year with t −0.70): this is a pure value bet, and value lost money over 2016–2026 in US large caps. The rank IC is −0.003 (t −0.3). The regime table shows the only positive states are the US cycle "slowdown" (+22.7% a year over 15 months) and inflation above 5% (+11.3%).
4. **Robustness.** DSR 0.142, permutation p-value 0.905, verdict "likely overfit or noise". The stale-signal check shows no timing problem (−0.51 vs −0.57 lagged).

Conclusion: the backtest gives no evidence of a value premium in `mvp_us` over this period, and the holdout is worse than the in-sample window.

Limitations and pitfalls

- The monthly-returns grid and the metrics use the net series; the gross line shows how much costs matter.
- The regime breakdown is descriptive and the HMM states are model-based.
- A run whose NAV falls below zero, or whose cost drag runs into thousands of basis points, is invalid. The test-copy "ML 3M top decile" run in Figures 11.16–11.17 shows such values (CAGR −100.0%, maximum drawdown −42,983,631%): do not interpret it, and report runs like it.

Related mnemonics and BQL fields

Rank IC and factor evaluation, Regime performance.

Robustness statistics

Route: `/backtest` (results › **Robustness**)

Purpose

The Robustness tab answers "could this result be luck or a mistake?" with five tools: the deflated Sharpe ratio, a label-permutation test, a stale-signal check and, when a parameter sweep is defined, walk-forward selection and the probability of backtest overfitting. A verdict badge summarises them, next to the number of configurations tested, the strategy's multiple-testing counter and the number of holdout views.

Controls

Control	What it does
Verdict badge	**survives the overfitting checks** (green), **inconclusive: treat with caution** (amber), **likely overfit or noise** (red) or **not enough information**.
Counter line	"N configuration(s) tested · strategy counter N · holdout viewed N×".
Panels	Deflated Sharpe ratio; Label permutation (histogram); Stale-signal check (badge "no timing red flag" or "staler signal does better: check timing"); with a sweep: Parameter sweep table (IS Sharpe, IS CAGR, OOS Sharpe; the in-sample best highlighted), Probability of backtest overfitting (CSCV) with the logit histogram, Walk-forward NAV with one line per fold, Holdout of the in-sample choice. Without a sweep: "Add a parameter sweep in the builder (Robustness) to get walk-forward selection and the probability of backtest overfitting. Every configuration counts toward the multiple-testing counter."

Methodology

PROBABILISTIC AND DEFLATED SHARPE RATIO

On the **in-sample** window, with the per-period Sharpe ratio $\hat{SR}$, n observations, skewness γ_3 and kurtosis γ_4 (not excess):

$$PSR(SR^*) = \Phi\left((\hat{SR} - SR^*) \cdot \sqrt{n-1} \,/\, \sqrt{1 - \gamma_3 \hat{SR} + \frac{\gamma_4 - 1}{4} \hat{SR}^2} \right)$$

$$SR_0 = \sqrt{V[SR]} \cdot \left((1-\gamma)\, \Phi^{-1}\left(1 - \frac{1}{N}\right) + \gamma\, \Phi^{-1}\left(1 - \frac{1}{N e}\right) \right), \quad \gamma = 0.5772 \text{ (Euler–Mascheroni)}$$

$$DSR = PSR(SR_0)$$ PSR shown in the tab = PSR(0)

N = configurations in this run's sweep (at least 1) + the trials recorded for the same saved strategy under other spec hashes. V[SR] = the variance of the recorded trial Sharpe ratios (sweep and previous trials) when there are at least two; otherwise the null approximation 1/T ("null approximation 1/T (fewer than 2 recorded trials)"). DSR is the probability that the true Sharpe exceeds what the best of N unskilled trials would show (Bailey & López de Prado 2014).

LABEL PERMUTATION

The signal is shuffled across names within each rebalance date and the strategy re-simulated gross of costs (default 20 times). The permuted mean should be about 0 and the actual Sharpe in the right tail. `p = (1 + #{permuted Sharpe ≥ actual}) / (1 + n)`. Event mode uses placebo events: the same dates with random names.

STALE-SIGNAL CHECK

The signal is lagged one more rebalance (event mode: entry 5 more days later) and re-simulated gross of costs. If the staler signal does better by more than two standard errors of an annualised Sharpe (`2 · √(252 / T)`), the result is flagged: a signal that improves when older usually has a timing error.

PARAMETER SWEEP, WALK-FORWARD AND HOLDOUT

Every configuration of the grid (at most 64) is simulated net of costs. On the in-sample window:

- **Walk-forward:** the window is split into `wf_folds + 1` equal segments; for each segment f ≥ 1 the configuration with the best Sharpe on the data before it is chosen and its returns on the segment kept; the stitched returns form the walk-forward NAV. Needs an in-sample window longer than 40 × folds days.
- **Holdout of the in-sample choice:** the configuration with the best in-sample Sharpe, measured on the holdout.

PROBABILITY OF BACKTEST OVERFITTING (CSCV)

The in-sample returns of the N configurations are split into S blocks (even, default 10). For each of the C(S, S/2) ways of choosing half the blocks as in-sample:

$$n^* = \text{configuration with the best in-sample Sharpe}$$

$$\omega = \text{rank of } n^* \text{ among the N configurations on the other half} \,/\, (N+1)$$

$$\lambda = \ln\left(\frac{\omega}{1-\omega}\right)$$

$$PBO = \text{share of combinations with } \lambda \le 0 \quad \text{(the in-sample winner ranks at or below the median out of sample)}$$

The panel also shows P(OOS loss) (share of combinations where the winner loses money out of sample) and the IS → OOS slope (regression of the winner's out-of-sample Sharpe on its in-sample Sharpe; near or below zero means in-sample performance does not carry over). Needs an in-sample window longer than 20 × blocks days (Bailey, Borwein, López de Prado and Zhu 2017).

VERDICT THRESHOLDS

```
survives the overfitting checks : DSR ≥ 0.95 and PBO < 0.2 and permutation p ≤ 0.05   (each when available)
likely overfit or noise         : DSR < 0.5 or PBO > 0.5 or permutation p > 0.2
inconclusive                    : otherwise
```

MULTIPLE-TESTING COUNTER

A saved strategy records one trial entry per distinct spec hash, with n = the size of its sweep grid and its trial Sharpe ratios (up to 64 per entry). The **strategy counter** (Trials column) is the sum of n over all entries; it feeds the N of the deflated Sharpe of the next runs. **OOS views** counts the completed runs that computed holdout metrics; it turns amber above 5 ("each look at the holdout erodes its independence"). Unsaved drafts have no counter: their DSR uses only the current sweep.

PAIRED COMPARISONS

The Compare tab shows runs side by side but does not run a paired test. The validation documents compare strategies by **paired daily-return differences** (annualised difference with its t-statistic), for example in `docs/CONFORMAL_VALIDATION.md` : calibrated band stops minus 2×ATR stops +6.7% a year (t +2.69), band 90% minus no stop −2.2% (t −2.64); and in `docs/TECHNICALS_V3_VALIDATION.md` : meta-filtered minus all setups, US H = 20, −4.3% (t −2.55). To make that comparison yourself, export both runs to Python and difference their daily return series.

Validation status

The DSR and PBO implementations are tested against known values (`docs/ROADMAP_V2_PLAN.md` 4.8). In the validation studies the DSR "does not discriminate" when the trial Sharpe variance is small (about 0.99 for every breakout configuration in `docs/CONFORMAL_VALIDATION.md` ; 0.79–0.96 for the meta configurations): a high DSR alone is not enough.

Correct usage and worked example

In Figure 11.13 the US value long-short run shows DSR 0.142 = PSR 0.142 (one trial, so SR_0 = 0.00), an in-sample Sharpe of −0.41 over 1,749 days, a trial Sharpe variance 5.7e−4 from the null approximation, a permuted mean Sharpe of −0.07 (sd 0.30, n 20) against an actual gross Sharpe of −0.51 (p 0.905), and no timing flag. The red badge comes from DSR < 0.5 and p > 0.2. The insider cluster-buy run (Figure 11.15) shows the opposite case: a DSR of 0.960 looks reassuring, but placebo events on the same dates with random names earn a Sharpe of 0.58 against the actual 0.70 (p 0.381), so most of the return comes from when the events happened (market exposure), not from which names were bought; the badge is red because p > 0.2. For the "Quality + momentum" preset the Runs table shows DSR 0.98 with one trial; before believing it, add a sweep (each configuration raises N) and check PBO.

Limitations and pitfalls

- With one trial and no recorded history, the DSR equals the PSR against zero: it is not deflated for the variants you tried by hand in unsaved drafts. Save the strategy.
- Permutations are gross of costs and keep the dates, so they test the cross-sectional information of the signal, not market timing.
- Looking at the holdout repeatedly and then changing the strategy turns the holdout into in-sample data; the OOS-views counter records this, it cannot undo it.

Related mnemonics and BQL fields

Multiple testing, Multiple testing when screening.

Experimental signals

Route: `/backtest` (builder › Event trigger / Exit)

Purpose

Three conformal and meta-labelling tools are offered for research only. The builder and the results mark them with the amber note "Experimental — failed / neutral in validation." and a **Validation report** link to the Conformal validation page.

Controls

Tool	Where	Label in the app (catalogue text)
Calibrated-band breakout (experimental)	Event trigger	"FAILED in validation: not better than a naive N-day-high breakout (docs/CONFORMAL_VALIDATION.md)"
Peer break-away fade (experimental)	Event trigger	"FAILED in validation: negative Sharpe after costs, null vs a naive fade (docs/CONFORMAL_VALIDATION.md)"
Calibrated band (α) stop / target mode	Exit	"MIXED in validation: band stops beat ATR stops, but no stop beat 'no stop' (docs/CONFORMAL_VALIDATION.md)"; the builder note reads "Band stops beat ATR stops, but no stop beat 'no stop' in validation."
Meta-labelled setups (Technicals v3)	Event trigger	"see docs/TECHNICALS_V3_VALIDATION.md (Meta-labelling): the run notes carry the current gate verdict". Flagged experimental in the catalogue.

Run warnings repeat the status: "stop mode calibrated band: EXPERIMENTAL - band stops beat ATR stops but no stop beat 'no stop' ..." and "event trigger ‹id›: EXPERIMENTAL - failed in validation ...".

Methodology

- **Calibrated-band breakout:** enters when the h-session move leaves the marginal conformal band emitted h sessions earlier (walk-forward EWMA band on the panel prices); long, or both directions with shorts.
- **Peer break-away fade:** on days a peer group's joint score exceeds its 90% radius, fades the member with $|z_{cond}| \geq z$ (default 2.5), using the GCN-GRU walk-forward signals 2010–2026 or the live table.
- **Calibrated band stop:** per trade, the stop sits at the lower (long) edge of the calibrated band of the chosen level and horizon emitted on the event date; optionally the target at the opposite edge.
- **Meta setups:** the v2 weekly trade setups (target T1, stop at the invalidation); **all setups**, **meta: take only** (setups the meta-labelling model takes) or **meta: take + Kelly size** (taken and sized by a quarter-Kelly fraction); horizon 10 / 20 / 40 sessions; source walk-forward 2013–2026 or the live table.

Validation status

Experimental: results of the validation studies, Backtest Lab engine with costs on.

- **Breakout on exit of the calibrated band: FAIL.** Not better than a naive N-day-high breakout: band h = 5 minus 20-day high −1.4% a year (t −0.66); band h = 20 minus 20-day high −5.6% (t −2.32); joint band −11.3% (t −3.23); PBO 0.54 (`docs/CONFORMAL_VALIDATION.md` , Economic value (i)).
- **Band stops: MIXED.** Better than ATR stops (band 90% minus 2×ATR +6.7% a year, t +2.69), but no stop beat "no stop" (band 90% minus no stop −2.2%, t −2.64) on 12−1 momentum (same document, (ii)).
- **Peer break-away mean reversion: FAIL (null).** Sharpe −0.09 ($z \geq 2.5$) and −0.15 ($z \geq 3$); break-away minus naive fade +0.3 to +2.1% a year, $t \leq 1.17$; PBO 0.47 (same document, (iii)). The peer-group ellipsoid itself passed as a calibrated descriptive measure, not as a trading signal.
- **Meta-labelling:** P(win) at H = 10 PASSED (a better-calibrated P(win) than T2; context edge not demonstrated); H = 20 and 40 FAILED; **take / skip and the suggested size FAILED the economic check**: meta-filtered minus all setups, US H = 20, −4.3% a year (t −2.55); meta-sized −4.2% (t −2.41); PBO 0.35 over 62 configurations (`docs/TECHNICALS_V3_VALIDATION.md` , Meta-labelling). The only positive comparison was long + short (+8.0% a year, t 2.89) because the model mostly skips short setups in a rising market.
- **T2 E[R] > 0 setup filter:** null economic result after costs (`docs/TECHNICALS_V2_VALIDATION.md` , Economic check).

Use these triggers to reproduce or extend the validation work, not to build a strategy.

Limitations and pitfalls

- The conformal joint envelope (application A) failed its gate and is not in the main UI; the marginal single-horizon bands used by the breakout trigger are validated as calibrated bands, not as a breakout signal.
- A good-looking run of an experimental trigger is still subject to the multiple-testing counter and the robustness checks.

Related mnemonics and BQL fields

Conformal prediction, Meta-labelling, Context-adjusted (meta) row, Peer-group joint anomaly.

Strategies & runs

Route: `/backtest` (tab **Strategies & runs**) · **API:** `GET /api/v1/backtest/strategies` , `GET /api/v1/backtest/runs`

Purpose

Saved strategies carry the multiple-testing counter, the holdout-view counter and the weekly paper schedule; the runs table lists every run with its headline statistics and lets you pick runs to compare.

Screenshot

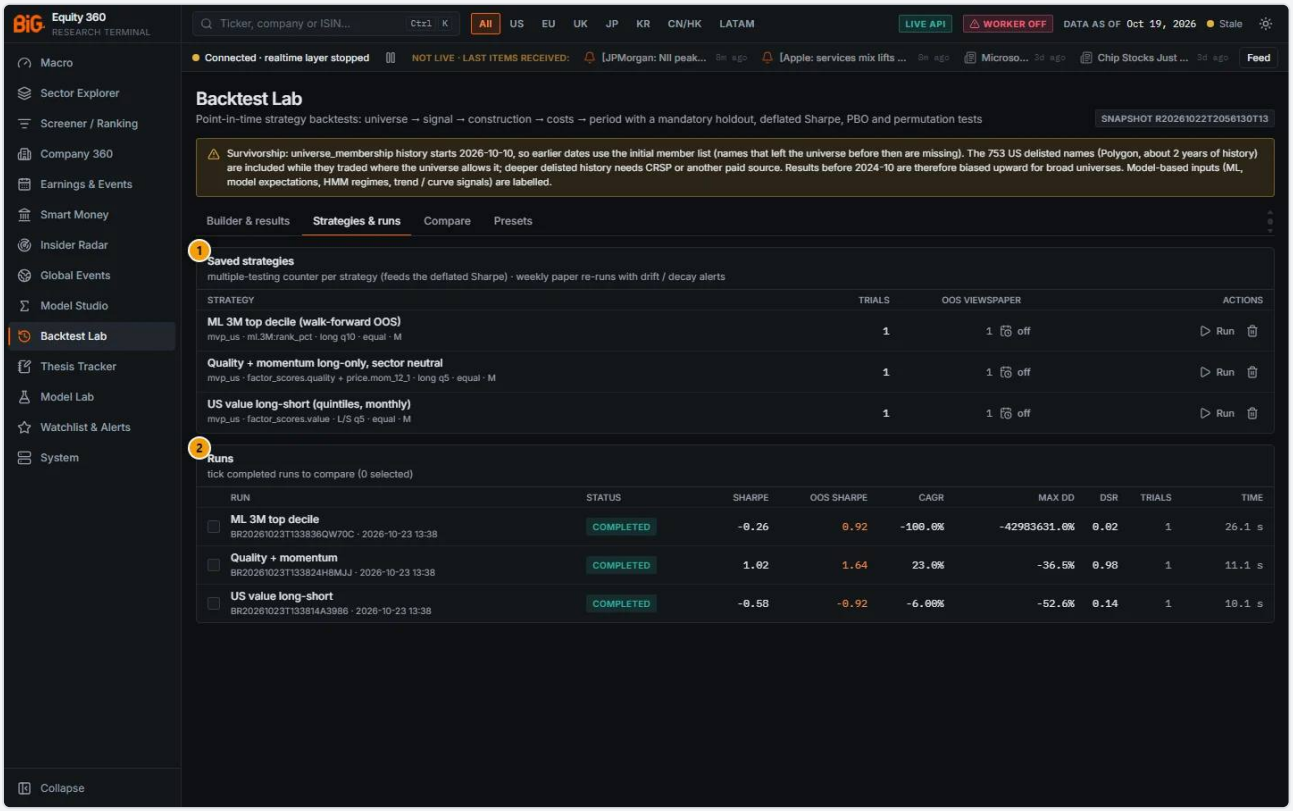


Figure 11.16: Strategies & runs (`/backtest`) on the test copy: three saved strategies and their runs.

#	Panel	What it shows
1	Saved strategies	"multiple-testing counter per strategy (feeds the deflated Sharpe) · weekly paper re-runs with drift / decay alerts". Columns: Strategy (name and spec summary), Trials, OOS views, Paper, Actions.
2	Runs	"tick completed runs to compare (N selected)". Columns: checkbox, Run (name, run id, time, **paper** badge for scheduled runs), Status, Sharpe, OOS Sharpe, CAGR, Max DD, DSR, Trials, Time.

> **Caution:** the "ML 3M top decile" row in Figure 11.16 shows impossible values (CAGR −100.0%, Max DD −42,983,631%): that run is invalid. The two other rows are normal: "Quality + momentum" Sharpe 1.02, OOS Sharpe 1.64, CAGR 23.0%, Max DD −36.5%, DSR 0.98; "US value long-short" Sharpe −0.58, OOS −0.92, CAGR −6.00%, Max DD −52.6%, DSR 0.14.

Controls

Control	What it does
Strategy name	Opens the strategy in the builder with its last run.
Trials / **OOS views**	The multiple-testing and holdout-view counters (amber above 5 views).
Paper button (**off** / **weekly**)	Hover: "Re-run weekly on the latest data and alert on drift / decay". Toast "Weekly paper re-runs on (drift alerts in Watchlist & Alerts)" or "Paper schedule off".
Run	Runs the saved strategy now (it counts as a trial).
Delete (bin)	Deletes the strategy.
Run name	Opens the run in the Builder & results tab.
Checkboxes, **Compare N**	Tick up to six completed runs, then compare.

Methodology

See Robustness statistics (counters) and Paper strategies and drift alerts.

Data sources and refresh

`app_state` tables `backtest_strategies` and `backtest_runs` ; run artefacts under `data/backtest/runs/` .

Validation status

Not applicable.

Limitations and pitfalls

- Deleting a strategy deletes its counters; re-creating it starts from zero trials, which flatters the next DSR.
- Runs of unsaved drafts do not increase any counter.

Related mnemonics and BQL fields

Watchlists and alerts.

Compare

Route: `/backtest` (tab **Compare**) · **API:** `POST /api/v1/backtest/compare`

Purpose

Compare puts the equity curves and every metric of up to six completed runs side by side.

Screenshot

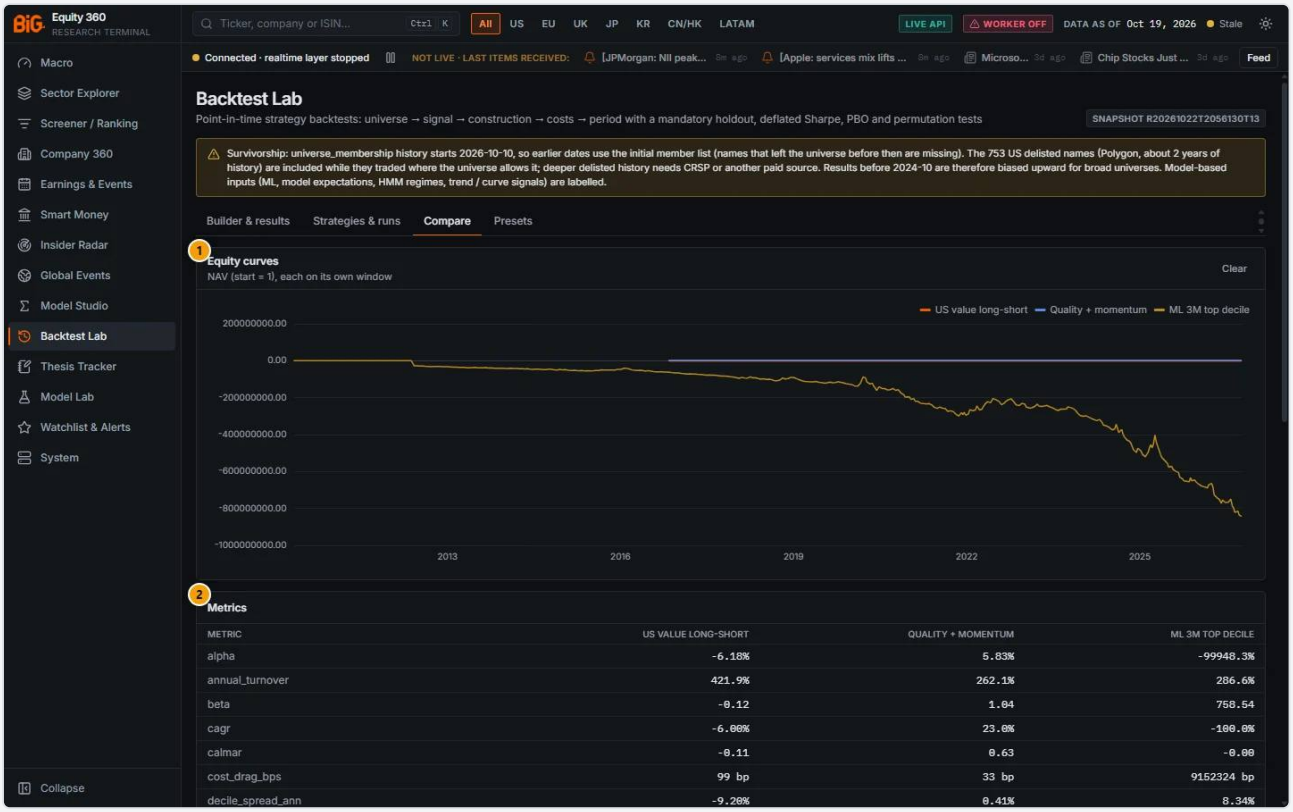


Figure 11.17: Compare (`/backtest`) of three runs on the test copy.

#	Panel	What it shows
1	Equity curves	"NAV (start = 1), each on its own window", one line per run; **Clear** removes the selection.
2	Metrics	One row per metric (alpha, annual turnover, beta, CAGR, Calmar, cost drag, decile spread ...), one column per run.

> **Caution:** the y-axis of Figure 11.17 is distorted by the invalid "ML 3M top decile" run (NAV down to about −1 billion), which flattens the two valid curves. Compare valid runs only.

With nothing ticked the tab shows "Nothing to compare: Tick completed runs in Strategies & runs."

Controls

Control	What it does
Clear	Empties the comparison.
Chart hover	NAV of each run per date.

Methodology

Metrics are the flat metric set of each run (percentages for CAGR, volatility, drawdown, returns, hit rates, alpha, spreads and turnover; basis points for `*_bps` metrics). Each curve is plotted on its own window, so runs over different periods are not aligned. No paired test is run (see Robustness statistics, Paired comparisons).

Limitations and pitfalls

- Runs on different universes, periods or cost settings are not comparable on Sharpe or CAGR alone.

Related mnemonics and BQL fields

Strategies & runs.

Presets

Route: `/backtest` (tab **Presets**) · **API:** `GET /api/v1/backtest/catalog` (`presets`)

Purpose

Nine ready-made strategies covering the main signal families. **Open** loads one into the builder (origin "preset · ‹title›"); nothing runs until you press **Run backtest**.

Screenshot

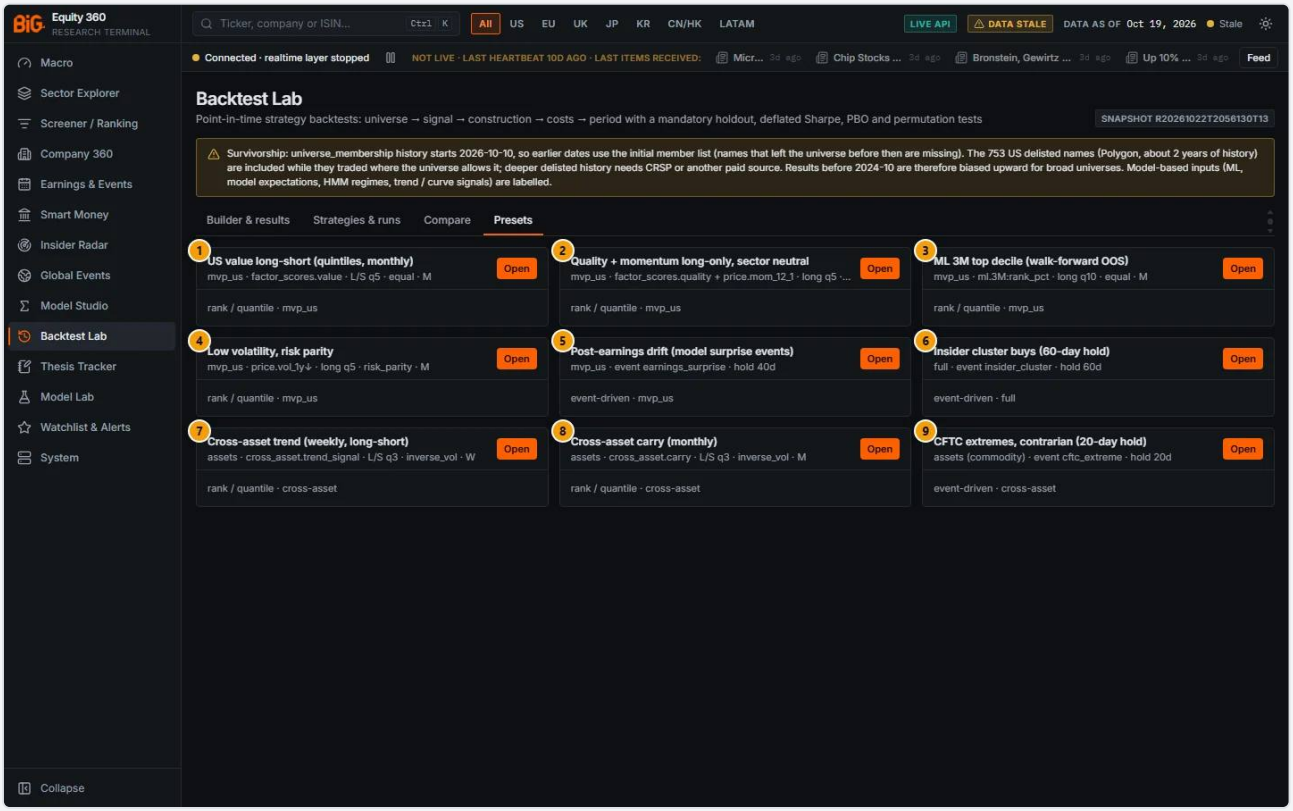


Figure 11.18: Backtest presets (`/backtest`).

#	Preset	Specification (card subtitle)	One line
1	US value long-short (quintiles, monthly)	mvp_us · factor_scores.value · L/S q5 · equal · M	Top minus bottom quintile of the value factor score, monthly.
2	Quality + momentum long-only, sector neutral	mvp_us · factor_scores.quality + price.mom_12_1 · long q5 · equal · M	z blend of quality and 12–1 momentum, top quintile, sector-neutral, 5% weight cap.
3	ML 3M top decile (walk-forward OOS)	mvp_us · ml.3M:rank_pct · long q10 · equal · M	Top decile of the 3M ML rank (walk-forward OOS predictions), three overlapping monthly tranches; model-based.
4	Low volatility, risk parity	mvp_us · price.vol_1y↓ · long q5 · risk_parity · M	Lowest-volatility quintile with equal-risk-contribution weights.
5	Post-earnings drift (model surprise events)	mvp_us · event earnings_surprise · hold 40d	Enter on |SUE| ≥ 2 vs model-based expectations (not consensus), hold 40 days, up to 40 positions, shorts allowed.
6	Insider cluster buys (60-day hold)	full · event insider_cluster · hold 60d	Long on insider cluster buys, hold 60 days, 20% stop, up to 30 positions.
7	Cross-asset trend (weekly, long-short)	assets · cross_asset.trend_signal · L/S q3 · inverse_vol · W	Long top tercile, short bottom tercile of the model-based trend signal, inverse-volatility weights, weekly.

#	Preset	Specification (card subtitle)	One line
8	Cross-asset carry (monthly)	assets · cross_asset.carry · L/S q3 · inverse_vol · M	Tercile long-short on carry, inverse-volatility weights, monthly.
9	CFTC extremes, contrarian (20-day hold)	assets (commodity) · event cftc_extreme · hold 20d	Fade 3-year COT-index extremes (above 0.9 / below 0.1) in commodities, hold 20 days, up to 10 positions, shorts allowed.

Validation status

Presets are examples, not validated strategies. The ML preset uses the XGBoost walk-forward predictions (OOS rank IC about 0.035, see Model Lab).

Limitations and pitfalls

- Each **Open** + run of a preset is a new unsaved draft: save it to track trials.

Related mnemonics and BQL fields

Strategy builder.

Paper strategies and drift alerts

Route: `/backtest` (Strategies & runs › **Paper**) and Watchlist & Alerts · **Refresh:** weekly (job `backtest_scheduled`)

Purpose

A saved strategy switched to **weekly** is re-run every week on the latest data as a paper portfolio. The job compares each new run with the previous completed run and raises a `backtest_drift` alert when the strategy decays.

Methodology

A strategy is due 6 days after its last scheduled run. Its scheduled runs carry the **paper** badge and count as trials. Drift rules (`backend/app/backtest/scheduled.py` , default threshold 30%):

- **Metric drift:** full-period Sharpe or CAGR falls by more than the threshold (relative), or the maximum drawdown deepens by more than the threshold.
- **Live decay:** over the days added since the previous run, `z = (ln(1 + R_new) − n · ln(1 + μ_d)) / (σ_d · √n)` with μ_d and σ_d the previous run's daily mean (from its CAGR) and volatility; z < −2 fires ("live decay: … over N new days (z … vs the backtest)").
- **Drawdown breach:** the current drawdown is deeper than the previous run's worst drawdown.
- **Holdings drift** (informational): one-way turnover between the two runs' latest books.

Switching the schedule on creates a `backtest_drift` alert rule (parameters: strategy id, name, schedule, threshold); switching it off deletes the rule. Events appear in Watchlist & Alerts as "Paper strategy '<name>': …".

Controls

The **Paper** button in Strategies & runs. The drift threshold is not editable on this page (default 30%).

Validation status

Not applicable (monitoring).

Limitations and pitfalls

- The weekly re-run uses the whole history again; with a revised snapshot the backtest itself can change, not only the new days.
- A paper portfolio has no fills, slippage or capacity beyond the cost model.

Related mnemonics and BQL fields

Watchlists and alerts, Studies & runs (the Model Studio equivalent, `studio_drift`).

Backtest entry points

Route: several pages → `/backtest` · **API:** `POST /api/v1/backtest/from-screen`

Purpose

Other pages open the builder with a draft and an origin note, so you can test what you are looking at.

Controls

Page	Button	Draft it creates
Screener	**Backtest this screen** (hover: "Backtest this screen: the filters and ranking re-evaluated point-in-time at each monthly rebalance (Backtest Lab)")	The screener's filters become point-in-time screen rules and its ranking the signal, monthly rebalance. Origin "from the Screener (rules evaluated point-in-time)"; fields without history are listed as "dropped (no point-in-time history): ..."; sub-sector filters are dropped.
Model Studio run header	**Backtest this signal** (panel tree-ensemble or regularised-regression runs)	Source `studio_run` with the run's out-of-fold predictions, long-short quintiles, the run's universe. Notes: "signal = out-of-fold predictions of run SR... (CV scheme ...)" and, for non-expanding schemes, "non walk-forward CV trains on future data: the backtest will flag LOOK-AHEAD IN FITTING".
Cross-asset dashboard	**Signal to backtest** (Trend, Carry, Value, CFTC COT index) + **Backtest this signal**	Long-short of the chosen signal across the asset's class, inverse-volatility weights. Notes: "spot-like prices: futures front months include roll gaps; trend / value / z metrics are model-based".
CFTC market page	**Backtest this signal** (hover: "event-driven strategy on COT-index extremes")	Event strategy fading COT-index extremes for the mapped asset or asset class ("contrarian: fade COT-index extremes (official CFTC positioning)").

Correct usage and worked example

In the Screener, filter `mvp_us` on quality ≥ 70 and sort by value, then press **Backtest this screen**. Read the origin notes before running: any filter listed as dropped is not in the backtest, so the backtest tests a different screen from the one on screen.

Limitations and pitfalls

- A Studio signal trained with purged K-fold or K-fold uses later data to predict earlier dates; the backtest flags it as look-ahead in fitting. Only `expanding` CV gives a point-in-time signal.

Related mnemonics and BQL fields

Open in Model Studio entry points.

Export to Python

Route: `/backtest` (results header › **Python**) · **API:** `GET /api/v1/backtest/runs/{run_id}/export.py`

Purpose

The export produces `backtest_<run_id>.py` , a standalone script that rebuilds the point-in-time panel from the lake and re-simulates the strategy, for audit and independent review.

Methodology

The script (`backend/app/backtest/export.py`) embeds verbatim the point-in-time as-of helper, the Model Studio dataset engine, the formula engine, the backtest engine core (`_engine.py`) and the panel builder (`_panel.py`), plus the run's normalised spec and panel plan. Its header records the run id, name, creation time, snapshot, spec hash, plan key and seed. It reads the lake read-only under `--data-dir` (default: the run's data directory), re-simulates the strategy, prints the headline metrics and ends with a `RESULT_JSON {...}` line so the reproduction can be checked mechanically. Requirements: duckdb, polars, numpy. Toast: "Script exported — standalone: rebuilds the panel from the lake and re-simulates".

Validation status

Export reproduction is part of the backtest test suite (`docs/ROADMAP_V2_PLAN.md` 4.8).

Limitations and pitfalls

- Signals from the BQL source and Studio runs depend on the backend (BQL engine, Studio run artefacts) and are not rebuilt from the lake alone.
- Re-running on a later lake gives the numbers of that data; reproduce on the same snapshot for identical results.

Related mnemonics and BQL fields

Model Studio export.

12. Thesis Tracker

The Thesis Tracker is the app's decision log. A thesis records what you believe about a security, a sector, a country, a macro theme or a cross-asset market, how you will know you are wrong, and what you plan to do about price (entry band, target, stop). The app then holds you to it: it generates alert rules from the thesis, re-evaluates an **automatic status** every night, measures performance against a benchmark and a sector benchmark, schedules reviews, keeps a dated journal and, when you close the thesis, records the outcome and a post-mortem template. Closed theses feed hit-rate statistics.

The tracker has three pages: the **board** (`/theses`), the **thesis page** (`/theses/:id`) and the **editor** (`/theses/new` , `/theses/:id/edit`).

> **Note:** The screenshots in this chapter come from a separate test instance seeded with example theses (Apple long, ASML idea, JPMorgan short, Toyota closed, a US country thesis and Petrobras). Prices are from the snapshot with data as of 19 Oct 2026; the theses were created on 23 Oct 2026. The examples illustrate the mechanics; they are not research views.

> **Model-based:** The automatic status, the KPI medians, the event-shock z-scores, the scenario impact and every probability on the thesis page are model-based. The touch and target-before-stop panels show frequencies **calibrated over history**, not the probability that the thesis is right.

Thesis board

Route: `/theses` · **Menu:** Thesis Tracker · **Mnemonics:** `THES` (global) · **Refresh:** on every change; automatic status nightly (`thesis_eval`)

Purpose

The board is a kanban of every thesis in three columns, **Idea**, **Active** and **Closed**, with a statistics strip on top. It answers "what am I tracking, what is at risk, which reviews are due, and how have my closed theses done?". Portfolio managers use it as the weekly agenda; analysts use it to move ideas into active positions and to close them with an outcome.

Screenshot

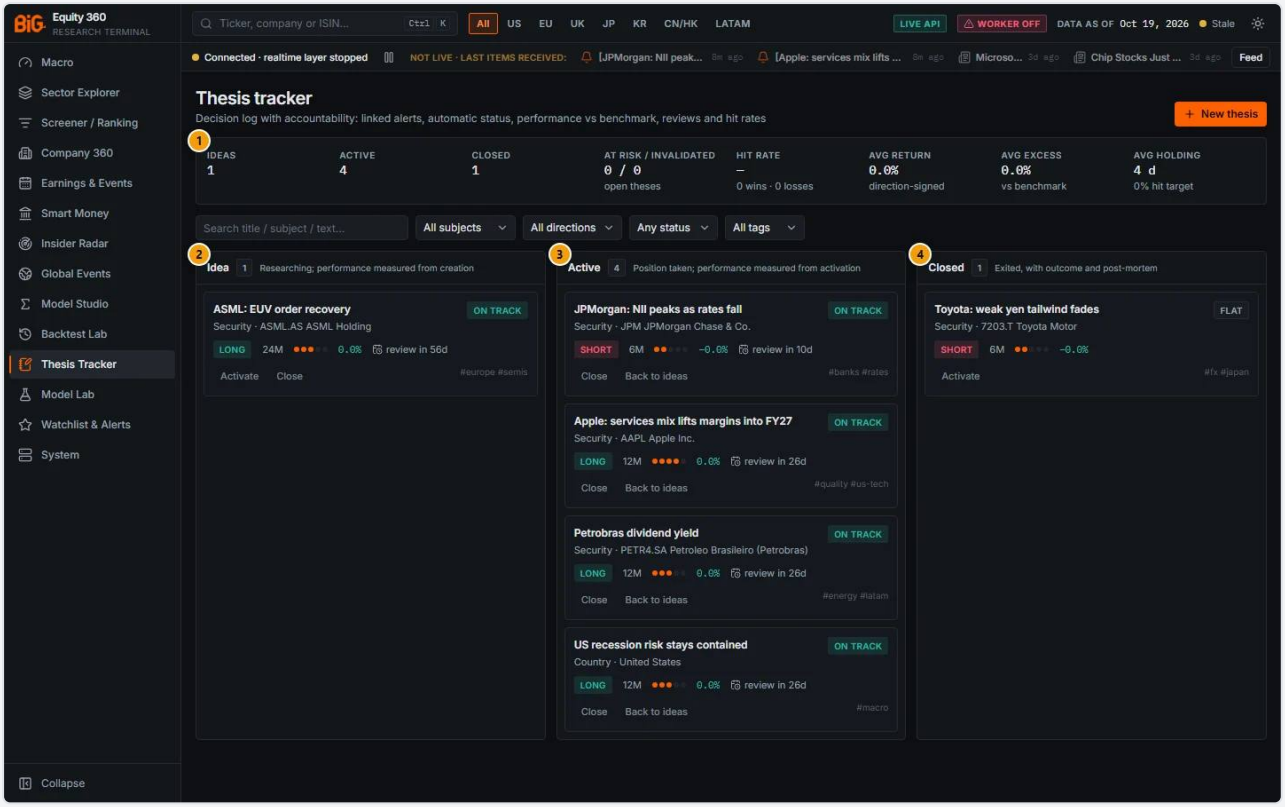


Figure 12.1: Thesis board (`/theses`), test instance, data as of 19 Oct 2026.

#	Panel	What it shows
1	Statistics strip	Ideas 1, Active 4, Closed 1; at risk / invalidated 0 / 0 among open theses; hit rate "—" (no win or loss yet); average return and excess 0.0%; average holding 4 days, 0% hit target.
2	Idea column	"Researching; performance measured from creation". One card: ASML EUV order recovery (long, 24M, conviction 3, review in 56 days).
3	Active column	"Position taken; performance measured from activation". JPMorgan (short), Apple, Petrobras (long, security) and the US recession thesis (country).
4	Closed column	"Exited, with outcome and post-mortem". Toyota (short, 6M) closed **FLAT**.

Controls

Control	What it does
New thesis	Opens the empty editor (`/theses/new`).
Search title / subject / text...	Free-text filter over the title, subject label, subject id and thesis text.
All subjects	Filter by subject type: Security, Sector, Country, Macro theme, Cross-asset.
All directions	Long, Short or Neutral.
Any status	Automatic status: Invalidated, At risk, On track, Unknown.
All tags	Appears once any thesis has tags; filters on one tag (`#quality`).
Clear filters	Appears when a filter is set; resets all of them. Filters live in the URL, so a filtered board can be bookmarked.
Card title	Opens the thesis page.
Drag a card to another column	Moves the thesis (same rules as the buttons below). A move that is not allowed shows a toast, for example "A closed thesis cannot move to idea".

Control	What it does
Activate / **Close** / **Back to ideas** (card buttons)	Idea → Active or Closed; Active → Closed or Idea; Closed → Active. **Close** opens the close dialog.
Close dialog: **Reason**	Target hit, Stopped out, Thesis invalidated, Time stop / horizon over, Thesis changed, Other (default).
Close dialog: **Outcome**	"From the return" (default), Win, Loss or Flat.
Close dialog: **Close thesis** / **Cancel**	Closes the thesis. The dialog explains: "The exit price, the performance since inception and a post-mortem template are recorded; linked alert rules are disabled."

Each card shows the title, subject type and label, the automatic status badge (open theses) or the outcome badge (closed theses), the direction badge, the horizon, five conviction dots, the direction-signed return since inception (closes), the review countdown ("review in 26d" or, in amber, "review 3d overdue"), the most serious status reason (an invalid or risk message) and the tags. Inside a column, open theses are sorted riskiest first (invalidated, at risk, on track, unknown), then most recently updated; closed theses by close date.

When there are no theses the board shows "No theses yet" with a **New thesis** button and the reminder that theses can be created from Company 360, the Screener, Model Studio, the Cross-asset dashboard or the Backtest Lab (see Thesis editor).

Methodology

Statistics strip. Counts are over all theses; "At risk / invalidated" counts open theses by their stored automatic status. The performance figures use **closed theses only**:

Figure	Definition
Hit rate	`wins / (wins + losses)` . Flat outcomes are excluded from both, so one flat close leaves the hit rate at "—".
Wins · losses	Outcomes chosen in the close dialog or, with "From the return", derived from the direction-signed return: win if `> +0.5%` , loss if `< −0.5%` , flat in between.
Avg return	Mean direction-signed thesis return at close (see Thesis page, Methodology).
Avg excess	Mean excess return versus the benchmark at close.
Avg holding	Mean days from inception to close; the sub-line is the share of closed theses whose close reason is "Target hit".

The return of a closed thesis is taken from the performance stored in its close snapshot; if that is missing, from `sign × (exit price / inception price − 1)` on unadjusted closes.

Hit rates by group. `GET /api/v1/theses/stats` also returns the same statistics by direction, by conviction, by subject type and by close reason, plus the stopped-out rate (close reason "Stopped out" or "Thesis invalidated"). The board shows only the overall row; the breakdown is available from the API only.

Card return. Direction-signed (`+` when the thesis is right): `sign × (last close / inception price − 1)` , with sign `−1` for shorts and `+1` for longs and neutral theses. On a closed thesis the exit price replaces the last close. Both prices are unadjusted closes, so the card can differ from the Performance panel, which uses adjusted closes (see Thesis page).

Moves and what they record.

Move	Effect
Created as idea	Inception date = the data as-of date of the snapshot, inception price = its close. A "creation" snapshot and a journal entry are written.
Idea → Active	Inception is **reset** to the current as-of date and close ("performance measured from activation"); an "activation" snapshot is taken.
Active → Idea	Status change only; inception is kept.

Move	Effect
→ Closed	Exit price = latest close; close reason recorded (default Other); outcome = your choice or derived from the return; a "close" snapshot with the performance summary; a post-mortem template written into the thesis if none exists; next review cleared; every linked alert rule disabled.
Closed → Active	Exit price, close reason and outcome cleared; inception kept; linked rules re-enabled; the review cadence restarts from today.

Every move is logged in the journal as a "status change" entry.

Data sources and refresh

Theses, journal, reviews and snapshots live in the local application state (`app_state`), not in the market-data snapshot. Prices, scores and KPIs come from the current serving snapshot. The stored automatic status is refreshed by the nightly `thesis_eval` job (after `alerts_eval`), on every save and by **Evaluate** on the thesis page. Under the default weekly schedule profile the nightly DAG runs inside the Saturday 08:00 (Lisbon) refresh; under the daily profile, Tuesday to Saturday at 01:30 (see Refresh schedule).

Validation status

Not formally validated: the hit-rate statistics are a descriptive record of your own decisions, and the automatic status is a set of rules (see Thesis page). The status badge carries the tooltip "Automatic status (model-based rules on the latest data)".

Correct usage and worked example

Use the board as the weekly agenda: filter **Any status › At risk**, then work through overdue reviews (amber "review … overdue"). Move an idea to **Active** only when you have taken the position, because activation resets the inception price used for performance.

Example (Figure 12.1). The Toyota short was closed on 23 Oct 2026 with reason "Thesis changed" and outcome derived from the return. The direction-signed return over four days was −0.0%, inside the ±0.5% flat band, so the outcome is **FLAT**. Because flat outcomes count neither as a win nor as a loss, the hit rate shows "—" with "0 wins · 0 losses", while Avg holding shows 4 days and "0% hit target" (the close reason was not "Target hit").

Limitations and pitfalls

- The statistics are only as honest as the closes: choosing "Win" in the dialog overrides the return-based outcome. Prefer "From the return" and explain exceptions in the post-mortem.
- Hit rates on a handful of closed theses are noise. The by-group breakdown is API-only.
- Moving a thesis Active → Idea → Active resets the inception on the second activation, which can flatter or hurt the recorded performance. Use **Back to ideas** only for genuine mistakes.
- The board is per installation: theses are stored in this PC's application state, not shared between users.

Related mnemonics and BQL fields

`THES` without a security opens the board; with a security (`AAPL US THES`) or a cross-asset id it opens the editor pre-filled (see 14. BQL console and mnemonics). No BQL fields expose theses. API: `GET /api/v1/theses` , `GET /api/v1/theses/stats` , `PATCH /api/v1/theses/{id}` (moves).

Thesis page

Route: `/theses/:id` · **Menu:** Thesis Tracker › card · **Mnemonics:** `THES` (board) · **Refresh:** live status on each view; stored status nightly (`thesis_eval`), on save and on **Evaluate**

Purpose

The thesis page is the single place to follow one thesis: price against the plan, automatic status with its reasons, the historical chance of touching the target or stop, KPIs, performance against the benchmarks, the thesis text, catalysts, invalidation conditions, the journal, reviews and snapshots, linked evidence, the linked macro scenario and the alert rules generated from the thesis. It is also where you close a thesis, write the post-mortem and export the record.

Screenshot

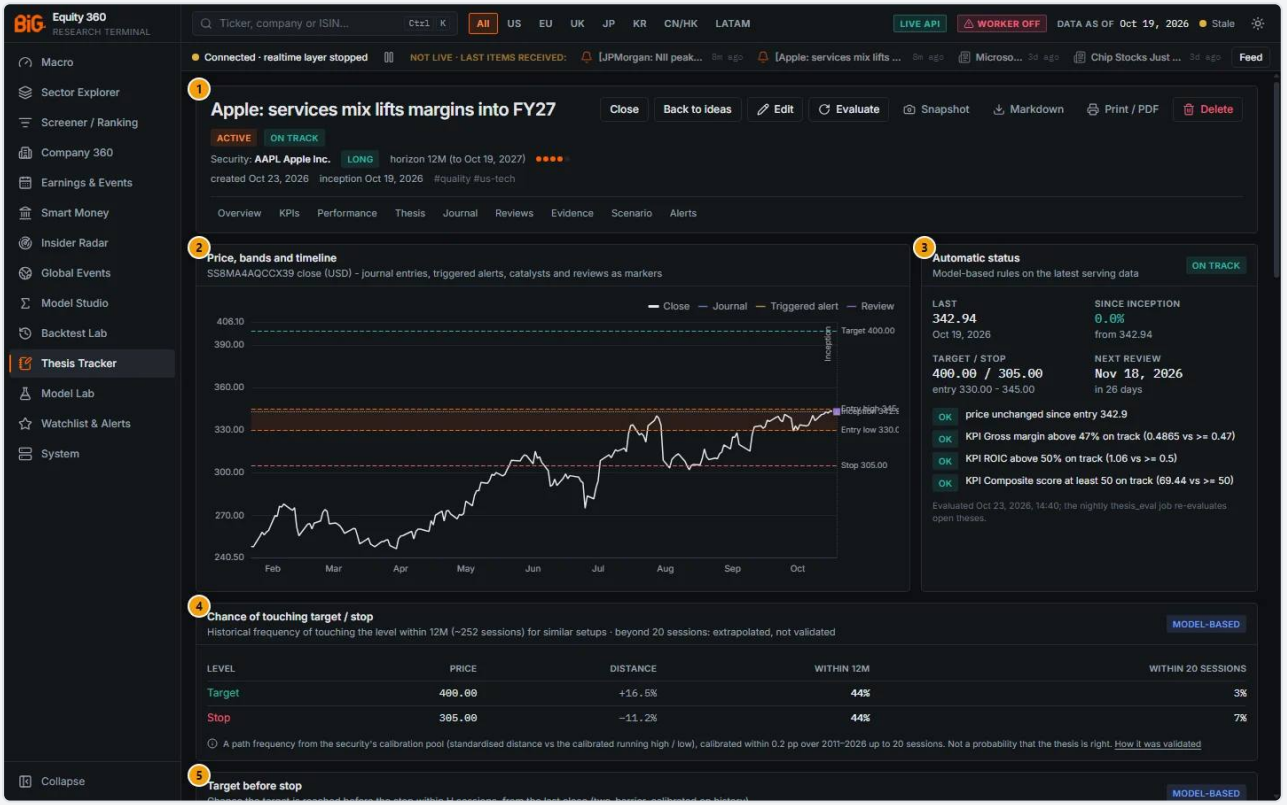


Figure 12.2: Thesis page for "Apple: services mix lifts margins into FY27" (`/theses/:id`), test instance, data as of 19 Oct 2026. Panels 6 to 16 are below the fold and shown as close-ups further down.

#	Panel	What it shows
1	Summary header	Title, ACTIVE and ON TRACK badges, subject (AAPL Apple Inc.), LONG, horizon 12M (to 19 Oct 2027), conviction 4/5, created 23 Oct 2026, inception 19 Oct 2026, tags; action buttons; section links.
2	Price, bands and timeline	AAPL close (USD) with target 400, entry band 330-345, stop 305, the inception line and the journal / alert / review markers.
3	Automatic status	Last 342.94 (19 Oct 2026), since inception 0.0%, target / stop 400.00 / 305.00, next review 18 Nov 2026 (in 26 days), and the reasons (all OK).
4	Chance of touching target / stop	Touch frequency within the horizon (12M, ~252 sessions) and within 20 sessions.
5	Target before stop	Two-barrier probabilities (T2) at H = 20 and the context-adjusted (meta) row at H = 10.
6	KPI tracker	Each KPI's live value, healthy condition, role and state.

#	Panel	What it shows
7	Performance since inception	Thesis return, excess versus benchmark and sector, max drawdown, rebased chart.
8	Thesis	The thesis text (Markdown) and key drivers; post-mortem when closed.
9	Catalysts	Dated catalysts with kind and an "in N d" badge within two weeks.
10	Invalidation conditions	Each condition with its kind and level or date.
11	Links	Benchmark, sector benchmark, linked Studio / Backtest ids and the origin.
12	Journal	New-note form and the dated entries.
13	Reviews & accountability	Review form, review list and snapshot table.
14	Linked evidence	Filings, news stories, event shocks, earnings and macro surprises.
15	Scenario impact (W4)	Model-based impact of the linked macro scenario on the subject.
16	Linked alerts	The 19 rules generated from the thesis and recent triggered alerts.

Controls

HEADER

Control	What it does
Status badge (idea / active / closed)	Board column.
Automatic status badge	Stored status from the last evaluation (On track, At risk, Invalidated, Unknown). On a closed thesis the outcome badge (win / loss / flat) replaces it.
Subject link	Opens Company 360 (security), the Sector Explorer (sector), the country page (country), the theme page for growth / inflation / labour themes (macro theme) or the Cross-asset dashboard.
`?`	Opens this chapter in the in-app Help.
Activate / **Close** / **Back to ideas**	Same moves as on the board. **Close** opens the close dialog (reason, outcome).
Edit	Opens the editor (`/theses/:id/edit`).
Evaluate	"Run the nightly evaluator now (stores the status, fires due alerts)". A toast reports the status and the number of alert events fired.
Snapshot	"Capture a snapshot (price, benchmarks, score, KPIs)": adds a *manual* row to the snapshot table.
Markdown	Downloads the thesis as a `.md` file (see Export).
Print / PDF	"Print / save as PDF (light theme)": switches to the light theme, opens the browser print dialog and restores the theme afterwards. Buttons, forms and section links are hidden in print.
Delete	Asks "Delete ... with its journal, reviews, snapshots and linked alert rules?" and returns to the board.
Section links	Overview, KPIs, Performance, Thesis, Journal, Reviews, Evidence, Scenario, Alerts: jump within the page.

PRICE, BANDS AND TIMELINE

Figure 12.3: Price, bands and timeline for the Apple thesis.

The chart plots the **unadjusted** daily close of the price instrument (the unit of the bands) from about nine months before creation to today (weekly bars when the span exceeds three years; for a closed thesis, up to 30 days after the close). Dashed horizontal lines mark the target (green), entry high and entry low (accent, with the band shaded) and the stop (red); a dotted line marks the inception price, and dotted vertical lines mark inception and close. Markers: circle = journal entry, triangle = triggered alert, diamond = catalyst, square = review; several markers of one kind on one day are stacked into one symbol, and the tooltip lists them. Future catalysts are listed in the Catalysts panel, not plotted. The legend shows only the marker kinds present. The y-axis always includes every band, so a distant target is never clipped.

Subjects without a price instrument (a macro theme without a proxy) show "No price instrument" and a hint to set a proxy such as `ETF:TIP` in the editor; without one there are no bands, no price alerts and no performance.

AUTOMATIC STATUS

The panel re-evaluates the thesis **live** on the current snapshot every time you open the page (the badge in the panel header), while the header badge shows the **stored** status of the last evaluation; the two can differ until the next evaluation. It shows Last (close and date), Since inception (direction-signed, from the inception price), Target / stop (with the entry band) and Next review (with "in N days" or "N days overdue"), then the list of reasons, each with a level badge: **ok** (green), **info** (blue), **risk** (amber) or **invalid** (red). The footer gives the time of the last stored evaluation: "the nightly thesis_eval job re-evaluates open theses".

On a closed thesis this panel is replaced by **Outcome** ("Recorded at close (performance from the close snapshot)"): close reason and date, exit / inception price, thesis return with days held, return versus the benchmark, and the status at close with its non-ok reasons.

CHANCE OF TOUCHING TARGET / STOP

Figure 12.4: Chance of touching target / stop (Apple thesis).

Shown for security subjects that are not closed. Columns: Level (Target, Stop), Price, Distance from the last close, **Within** followed by the horizon (e.g. **Within 12M**) and **Within 20 sessions** (the validated horizon). The

subtitle gives the horizon in sessions (1M = 21, 3M = 63, 6M = 126, 12M and longer = 252) and, beyond 20 sessions, "extrapolated, not validated". The badge reads **MODEL-BASED** and the note under the table states: "Not a probability that the thesis is right." Without a target or stop the panel asks you to set one; when the security has no calibration it says "Touch calibration is not available for this security yet."

TARGET BEFORE STOP

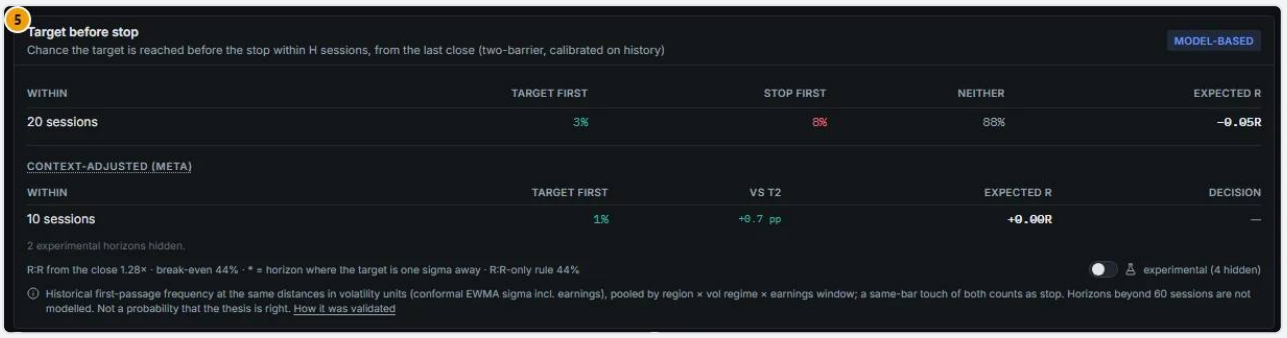


Figure 12.5: Target before stop (Apple thesis), experimental outputs hidden.

Shown for security subjects that are not closed and have both a target and a stop on opposite sides of the last close (otherwise: "Target and stop must lie on opposite sides of the last close for this view."). The table gives, per horizon H in sessions, **Target first**, **Stop first**, **Neither** and **Expected R** (`P(target) × R:R − P(stop) − costs`). By default only H = 20 is shown; an asterisk marks the horizon where the target is one sigma away.

Below it, the **Context-adjusted (meta)** table (hover the dotted title for its definition) gives Target first, the change versus T2 in percentage points, Expected R and the take / skip Decision. By default only H = 10 is shown and Decision shows "—"; "2 experimental horizons hidden" refers to H = 20 and 40.

The footer shows the R:R from the close, the break-even probability, the R:R-only rule and the switch **Show experimental outputs** ("experimental (4 hidden)"), shared with the Technicals tab. Turning it on adds T2 at 5, 10, 40 and 60 sessions, meta at 20 and 40, and the take / skip decision, each marked experimental.

KPI TRACKER

Figure 12.6: KPI tracker (Apple thesis).

One row per KPI: label, **Source** (field id, formula, "cross-asset" or "macro cell" followed by the reference, plus the basis: *security*, *median of members*, *cross-asset latest*, *macro cell (latest)*), **Value**, **Healthy if** (the condition, e.g. "≥ 0.470"), **Role** (confirm, invalidate, watch) and **State**:

State	Meaning
OK (green)	The healthy condition holds.
Breached (amber)	Condition fails on a *confirm* KPI: the status turns at risk.
Invalidates (red)	Condition fails on an *invalidate* KPI: the status turns invalidated.
Watch (blue)	Condition fails on a *watch* KPI: information only.

State	Meaning
No data	No value in the snapshot.
Error	The KPI could not be computed; hover for the message (e.g. unknown field).

With no KPIs the panel says "Add KPIs in the editor: each becomes a linked alert and feeds the status."

PERFORMANCE SINCE INCEPTION

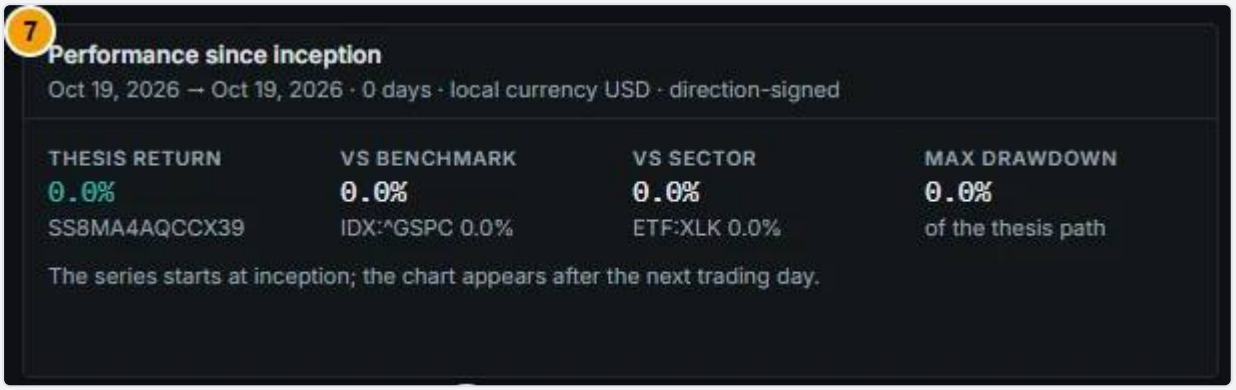


Figure 12.7: Performance since inception (Apple thesis, created on the as-of date, so 0 days).

The subtitle gives the period, the number of days, the currency basis ("local currency USD" or "in USD (official FX)") and "direction-signed". Stats: **Thesis return** (price instrument id below), **vs benchmark** (benchmark id and its own return), **vs sector** (sector benchmark id and return, or "none"), **Max drawdown** "of the thesis path". The chart rebases the subject (solid), the benchmark (dashed) and the sector benchmark to 100 at inception; it appears from the first trading day after inception ("The series starts at inception; the chart appears after the next trading day.").

THESIS, CATALYSTS, INVALIDATION CONDITIONS AND LINKS

Figure 12.8: Thesis text and key drivers.

The **Thesis** panel renders the thesis text (Markdown subset) and the key drivers. On a closed thesis it adds **Post-mortem** with an **Edit** button (Markdown text area, **Save** / **Cancel**).

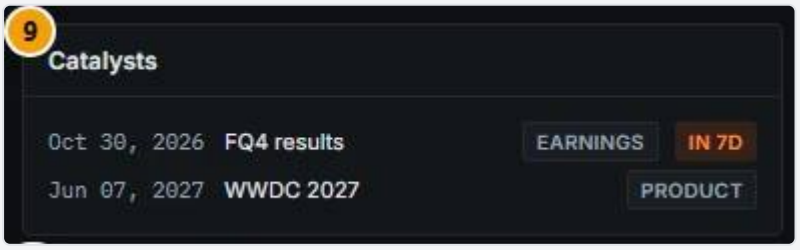


Figure 12.9: Catalysts, sorted by date; undated ones last.

Each catalyst shows its date ("undated" if none), label and kind badge; an accent badge "in N d" appears when the date is within 14 days.

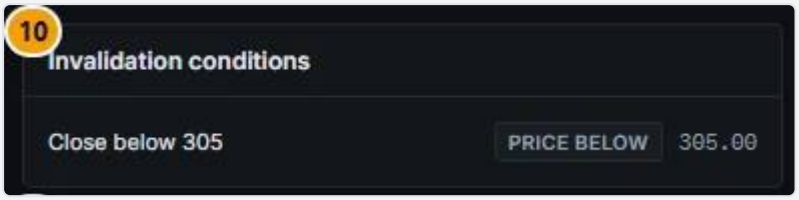


Figure 12.10: Invalidation conditions.

Each condition shows its label, kind badge (price below, price above, kpi, date, manual) and level or date. With none, the panel reminds you that the stop still invalidates.

Figure 12.11: Links.

Links lists the benchmark and sector benchmark ids, links to the Studio study or run and the Backtest strategy or run when set, and the origin ("Created from company (AAPL)").

JOURNAL

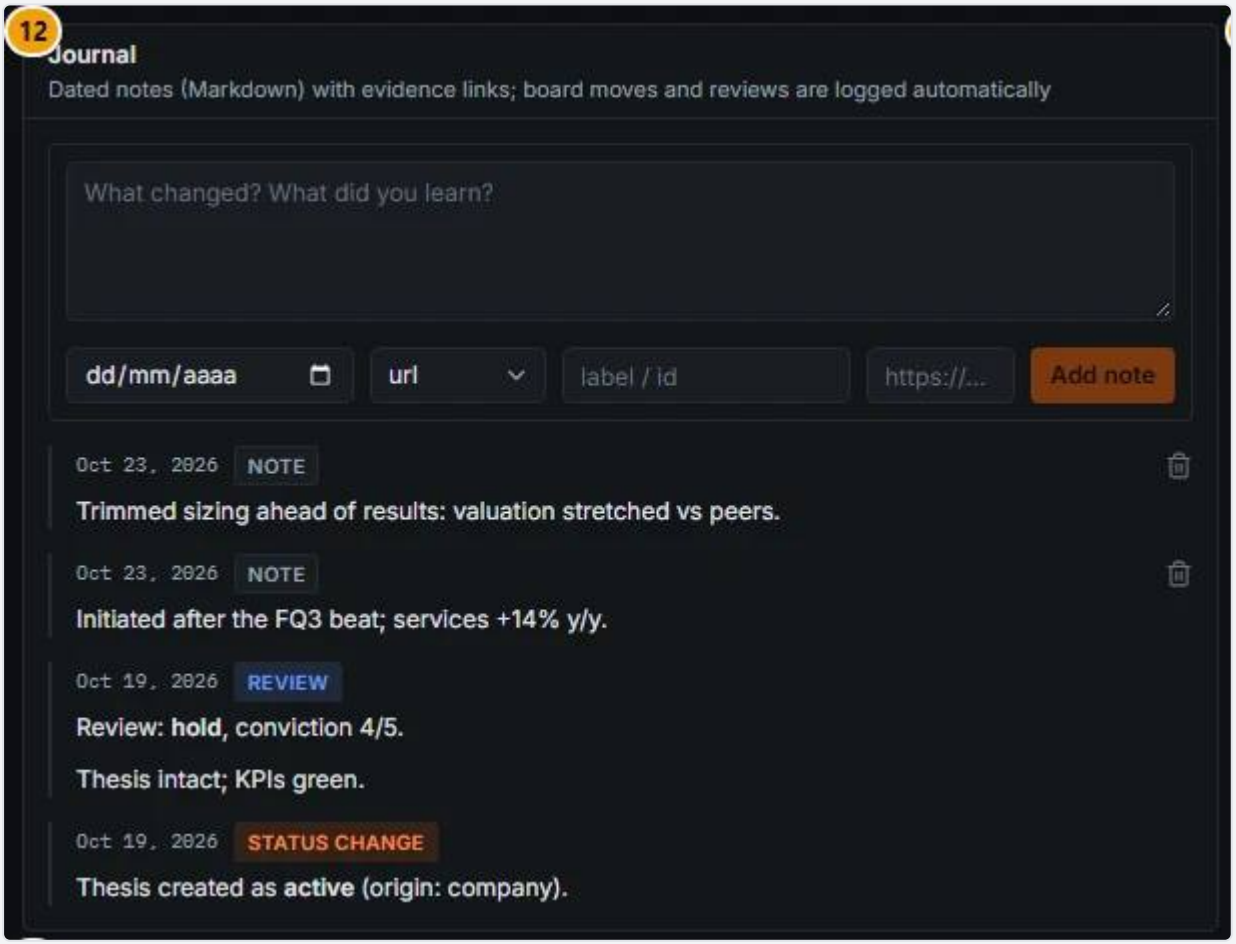


Figure 12.12: Journal ("Dated notes (Markdown) with evidence links; board moves and reviews are logged automatically").

Control	What it does
Note text area	"What changed? What did you learn?" (Markdown).
Date	Entry date (default today).
Link kind	url, filing, story, event, alert, earnings, study, backtest.
label / id, https://...	Optional evidence link: a label or id, and a URL.
Add note	Saves the entry (enabled when the text is not empty).
Bin icon	Deletes a *note* (asks for confirmation). Automatic entries (status change, review) cannot be deleted here.

Entries are newest first, with a kind badge: NOTE, REVIEW, STATUS CHANGE (also catalyst, alert, post-mortem when written through the API). Links with an `http(s)` URL open in a new tab.

REVIEWS AND SNAPSHOTS

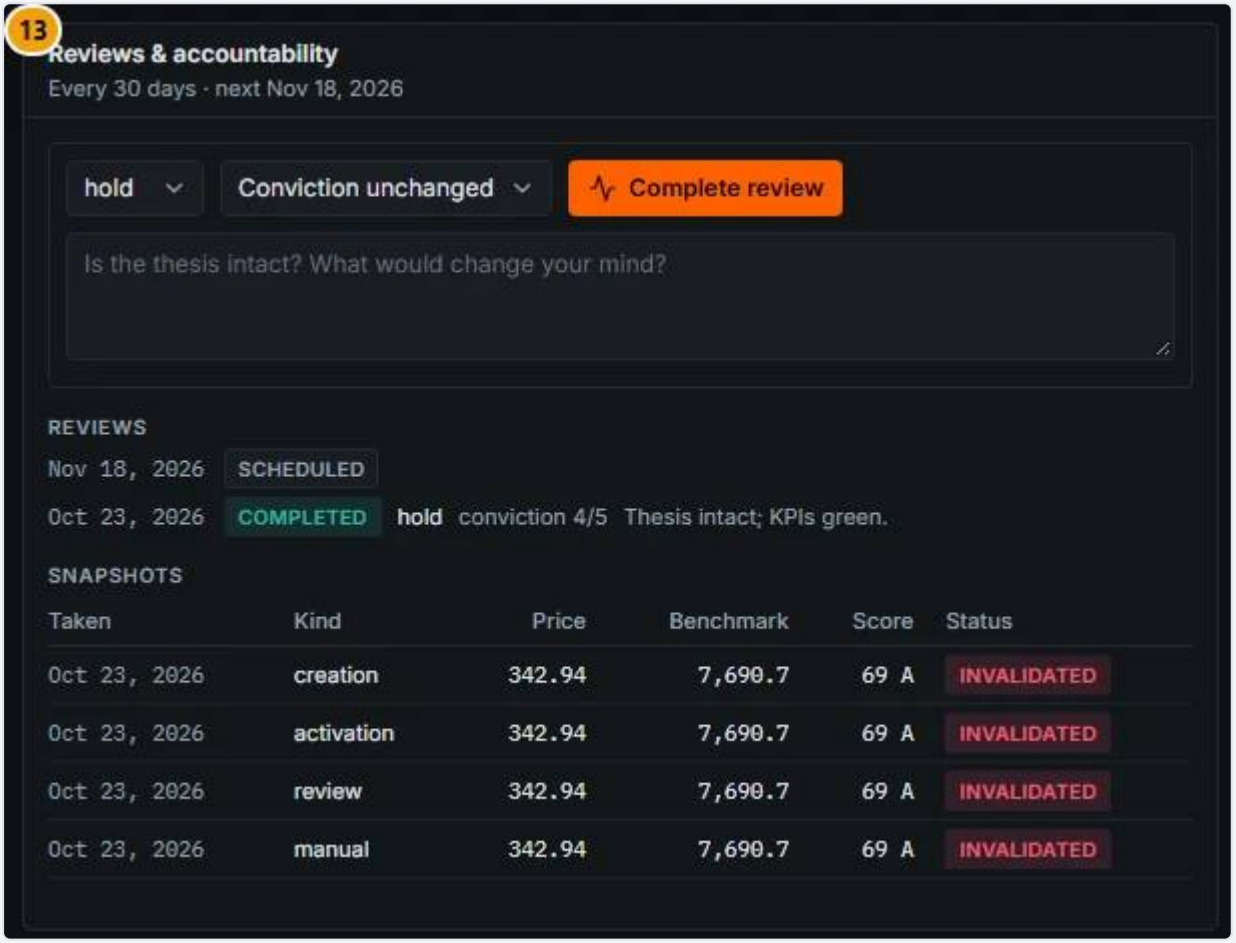


Figure 12.13: Reviews & accountability ("Every 30 days · next Nov 18, 2026").

Control	What it does
Decision	hold (default), add, trim, close, none. Recorded only; choosing "close" does not close the thesis.
Conviction after	"Conviction unchanged" or 1/5 to 5/5; a value updates the thesis conviction.
Notes	"Is the thesis intact? What would change your mind?" (Markdown).
Complete review	Completes the scheduled review (or records an unscheduled one), captures a *review* snapshot, writes a journal entry and schedules the next review `cadence` days after the snapshot's as-of date.

The form is hidden on closed theses. **Reviews** lists scheduled and completed reviews (date, status badge, decision, conviction, notes). **Snapshots** is the accountability table: date taken, kind (*creation*, *activation*, *review*, *manual*, *close*), price, benchmark level, composite score with grade, and the automatic status **at the time of the**

snapshot. A snapshot is never rewritten, so its status can differ from today's (in Figure 12.13 the four snapshots were taken while an earlier version of the composite-score KPI was breached).

LINKED EVIDENCE

Figure 12.14: Linked evidence ("Filings, news stories, event shocks (GDELT attention / tone z, model-based), earnings and macro surprises (model-based expectations, not consensus)").

Four columns over the thesis life, and at least the last 365 days:

Column	Content
Filings	Up to 15 filings of the issuer (date, form type, link). Security subjects only.
News stories	Up to 12 story clusters linked to the issuer (date, title, link). Security subjects only.
Event shocks (model-based)	Days when the GDELT attention z ≥ 2.5 or the absolute tone z ≥ 2.5 for the issuer, country or sector, with the item count. When there are none the column says so.
Earnings & macro	Upcoming earnings dates with their source label (accent), past releases with the model-based EPS surprise z and CAR[−1,+1], and, for country and macro-theme subjects, macro surprises with an absolute z of 1.5 or more.

SCENARIO IMPACT (W4)

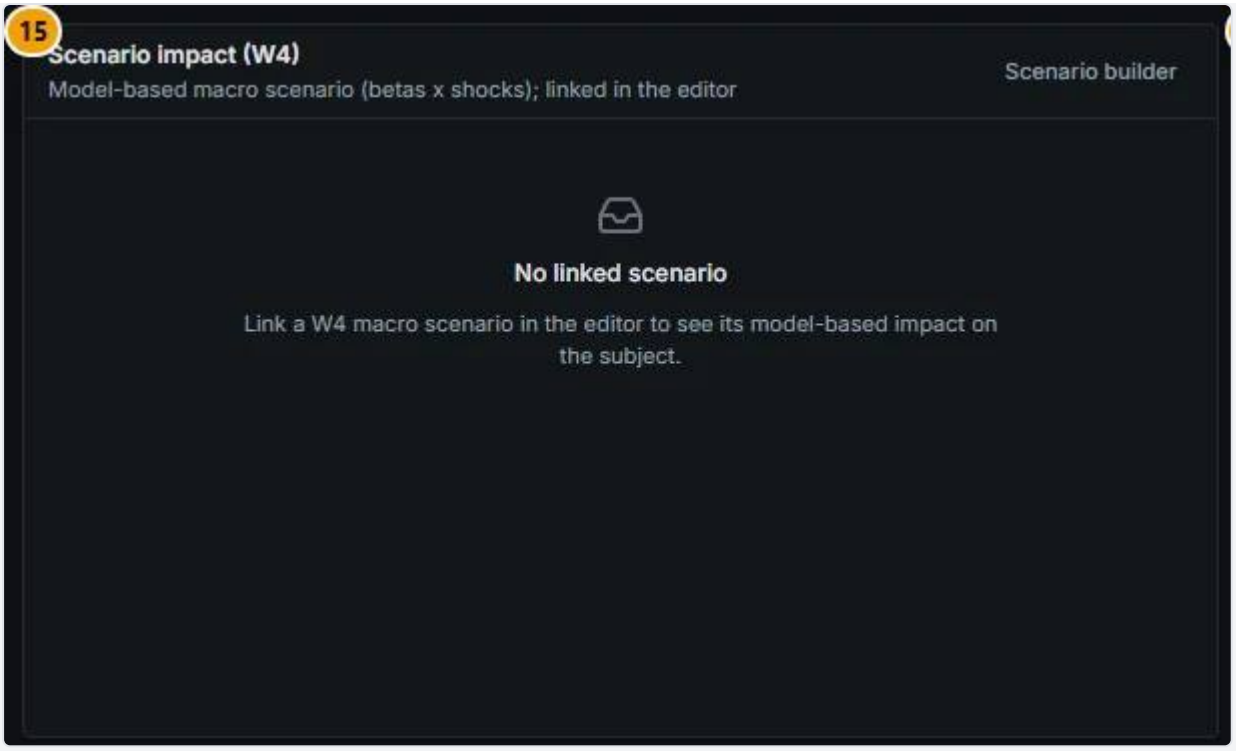


Figure 12.15: Scenario impact (W4), no scenario linked.

When a macro scenario is linked in the editor, the panel runs it and shows the model-based impact on the subject with its 90% band, the universe aggregates and the model label. For a country subject the closest region is used as a proxy (for example "Region EU (proxy for DE)"); for a sector, the sector row; for macro and cross-asset subjects, the price instrument when it is among the top or bottom instruments. **Scenario builder** opens `/macro/markets?tab=scenarios` (see Scenario builder).

LINKED ALERTS

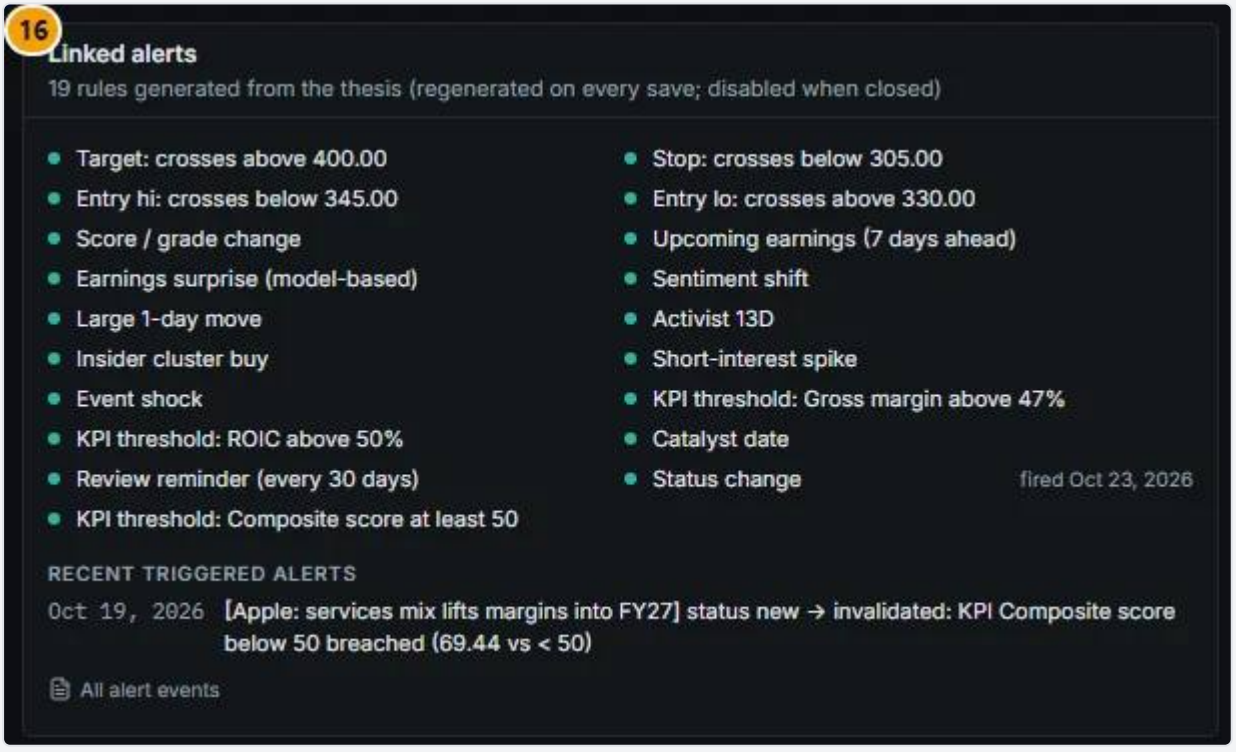


Figure 12.16: Linked alerts ("19 rules generated from the thesis (regenerated on every save; disabled when closed)").

Each rule shows a green dot when enabled, a readable label (for example "Target: crosses above 400.00", "KPI threshold: ROIC above 50%", "Review reminder (every 30 days)") and "fired" with the date when it has fired.

Recent triggered alerts lists up to 20 events with their data date; empty: "None yet. Price and data rules are evaluated nightly (alerts_eval, thesis_eval)." **All alert events** opens the Watchlist & Alerts page. Rule definitions are in Alert rule reference.

EXPORT AND PRINT

Markdown downloads a `.md` file named after the title with: title line (direction, subject, horizon, conviction, status, automatic status), dates and thesis id, the thesis text, key drivers, price plan (instrument, entry band, target, stop, inception and last price), catalysts, KPIs with their current value and state, invalidation conditions, automatic-status reasons, performance, linked analysis ids, linked alert rules, triggered alerts (up to 50), reviews, the snapshot table, the full journal and the post-mortem. It ends with "Exported from BiG Equity 360. Automatic status, KPI medians and z-scores are model-based; prices from the serving snapshot." PDF output is the browser's print-to-PDF of the page in the light theme; there is no server-side PDF.

Methodology

AUTOMATIC STATUS

`thesis_eval` (and every save, **Evaluate** and each view of the status panel) evaluates the thesis on the latest serving snapshot. The reference price is the inception price, else the middle of the entry band. Direction sign: `s = +1` long, `−1` short; price rules are skipped for neutral theses. Each rule adds a reason:

Level	Code	Rule
invalid	stop_hit	`s × (price − stop) ≤ 0` : the close is at or beyond the stop.
invalid	invalidation	An invalidation condition is met: price below / above its level, the linked KPI breached, or a manual condition marked triggered.
invalid	kpi_breach	An *invalidate* KPI fails its healthy condition.
risk	near_stop	Price is within a third of the distance from the reference price to the stop.
risk	adverse_move	No stop set and the price has moved 10% or more against the thesis.
risk	kpi_breach	A *confirm* KPI fails its healthy condition.
risk	deadline	A deadline condition's date has passed and it is not marked triggered ("confirm or mark it").
risk	score_change	The balanced composite score fell 15 points or more since the activation (else creation) snapshot; for a short, rose 15 or more.
risk	horizon_passed	Active thesis past its horizon date ("review the time stop").
info	target_reached	Price at or beyond the target ("consider closing or raising the target").
info	price_against	Price against the thesis since entry (less than the adverse-move threshold, or with a stop set).
info	in_entry_band	An idea whose price is inside the entry band.
info	kpi_breach	A *watch* KPI fails.
info	kpi_error / kpi_missing	A KPI cannot be computed or has no data.
info	review_overdue	More than 7 days past the next review date.
ok	price_in_favour / price_unchanged	Price in favour since entry, or unchanged (within ±0.05%).
ok	kpi_ok	A KPI meets its healthy condition.

The status is **Invalidated** if any reason is invalid; else **At risk** if any is risk; else **Unknown** if there is no price and no KPI value ("no price instrument and no KPI values to evaluate"); else **On track**. Info reasons never change the status.

Horizon dates: inception plus 30 (1M), 91 (3M), 182 (6M), 365 (12M), 730 (24M) or 1,095 days (36M+).

KPI VALUES

A KPI threshold is the **healthy** condition (e.g. `gross_margin ≥ 0.47`); a breach is that condition failing. Operators: `>`, `≥`, `<`, `≤`, `between` (inclusive). Values come from the serving snapshot:

Kind	Value
Metric	A numeric screener field id (`GET /api/v1/screener/fields`), plus `composite` (balanced composite score) and `rank`. Security subject: the security's value; sector or country subject: the **median** across primary listings of the sector or country.
Formula	An expression of the formula engine over screener field ids (e.g. `roic - 0.08`), evaluated on the same values (medians for sector / country).
Cross-asset	`<asset_id>:<signal>` (or `<signal>` for the thesis's own asset) from the cross-asset latest table.
Macro cell	`<CC>:<indicator>` from the latest macro heatmap cell on or before the as-of date.

PERFORMANCE

```
start = inception date (as-of date at creation, or at activation from idea)
end   = close date for a closed thesis, else the snapshot as-of date
r_X   = adjusted close at end / adjusted close at start − 1     (X = subject, benchmark, sector)
thesis return       = s × r_subject              (s = −1 short, +1 long or neutral)
excess vs benchmark = s × (r_subject − r_benchmark)
excess vs sector    = s × (r_subject − r_sector)
max drawdown        = worst peak-to-trough of the path 1 + s × (P_t / P_start − 1)
```

Each series is anchored on the last bar on or before the start date. If the subject, benchmark and sector benchmark trade in one currency the comparison is in that currency (basis *local*); otherwise all three are converted to USD with the official FX rate of each day (basis *usd*, "in USD (official FX)"). Prices are adjusted closes (splits and dividends) from the serving snapshot.

Default benchmarks: a security uses its region's benchmark index (`IDX:^GSPC` for the US and Canada, `IDX:^STOXX50E` for the EU, `IDX:^FTSE` for the UK, `IDX:^N225` for Japan, `IDX:^BVSP` for Latin America) and the US sector ETF of its sector (`ETF:XLK` for technology; none for sectors without one); a country uses its main index as the price instrument and a world ETF (`ETF:ACWI`, else `ETF:VT`, else the S&P 500) as the benchmark; a sector uses its sector ETF as the price instrument; a cross-asset subject uses its instrument. All can be overridden in the editor.

TOUCH AND TARGET-BEFORE-STOP PANELS

- **Touch** (Conformal technicals): the share of calibration origins in the security's pool whose standardised running high (or low) reached the level's standardised distance within h sessions. The same curves as the Touch probabilities on the Technicals tab, evaluated at the thesis target and stop. Beyond 20 sessions the 20-session curve is reused with the variance scaled by h: an extrapolation, labelled as such.
- **Target before stop** (Technicals v2 T2): two-barrier first-passage frequencies at the same distances in volatility units (conformal EWMA sigma including earnings), pooled by region × vol regime × earnings window; a same-bar touch of both counts as stop. The same surface as the Trade setup card, computed at query time at the thesis barriers, from the last close. Horizons beyond 60 sessions are not modelled.
- **Context-adjusted (meta)** (Technicals v3 #2): a secondary model that re-estimates the T2 probability from the market and stock context; see Context-adjusted (meta) row and Meta-labelling.

EVALUATION AND LINKED EVENTS

`thesis_eval` runs every open thesis, stores the status and reasons, and appends `thesis_*` alert events: a KPI that newly breaches, the status newly turning at risk or invalidated, a review due, a dated non-earnings catalyst within 3 days, a CFTC positioning extreme (cross-asset subjects) and an events-v2 shock. Events are

idempotent per rule, target and data date, so running **Evaluate** and the nightly job on the same snapshot does not duplicate them. The other linked rules (prices, scores, earnings, smart money, macro) are evaluated by `alerts_eval`. Details in Alert rule reference.

Data sources and refresh

Element	Source	Refresh
Prices, benchmarks, scores, KPI metrics	Serving snapshot (Polygon / yfinance prices, official FX, screener fields)	Nightly DAG (weekly profile: Saturday)
Automatic status (stored)	`thesis_eval`	Nightly after `alerts_eval`, on save, on **Evaluate**
Automatic status (panel)	Live evaluation	Each page view
Touch frequencies	`conformal_envelopes` touch curves (`GET /api/v1/conformal/touch`)	Nightly
Target before stop, meta	T2 surface (weekly refit) and meta model (`GET /api/v1/technicals/{id}/setup-probabilities` with the thesis target, stop and direction)	Query time on the weekly-refitted models
Evidence	SEC / company filings, events-v2 stories and GDELT aggregates, earnings events and calendar, macro surprises	Nightly; GDELT up to 10 days per weekly run
Scenario	W4 macro exposures	Weekly
Journal, reviews, snapshots, rules	Local application state	Immediate

Validation status

Output	Status	Source
Touch frequency, h ≤ 20 sessions	**Validated**: "predicted equals realised within 0.2 pp at every distance and horizon" (US mvp, 2011-2026). The panel note says "calibrated within 0.2 pp over 2011–2026 up to 20 sessions".	`docs/CONFORMAL_VALIDATION.md` §A.4
Touch frequency, h > 20 (thesis horizons 1M+)	**Extrapolated, not validated** (labelled in the subtitle).	same, and the panel
T2 target-before-stop, H = 20	**PASS → production** (ECE 1.13 / 1.47 pp; worst regime Covid 2020 win −4.6 / loss +4.0 pp; better than both baselines).	`docs/TECHNICALS_V2_VALIDATION.md` Verdict
T2, H = 5, 10, 40, 60	**FAIL → experimental**, hidden by default (H = 5 / 10: "the Feb-Jun 2020 crash window had 5.1-5.6 pp more stop-outs than predicted"; H = 40 / 60 lose their edge over the Brownian formula).	same
Expected R	Well ordered and roughly calibrated; the economic check (filtering setups by E[R] > 0) is a **null result**.	same
Meta P(target first), H = 10	**PASS** (default row): "a better-calibrated P(win) than T2, not 'context has a demonstrated edge'".	`docs/TECHNICALS_V3_VALIDATION.md`, Meta-labelling
Meta, H = 20 and 40	**FAIL → experimental**.	same
Meta take / skip and size	**FAIL** (economic check) → hidden unless experimental is on.	same
Automatic status, KPI states, hit rates	Not formally validated: deterministic rules on snapshot data, labelled "model-based rules".	code (`app.theses.evaluate`)
Event shocks, earnings and macro surprises	Model-based z-scores and model-based expectations (not consensus).	panel subtitles
Scenario impact	Model-based (betas × shocks).	panel subtitle

Correct usage and worked example

Read the page top-down once a week: status reasons first, then KPIs, then the probabilities, then the journal. Treat the probabilities as a sanity check on the plan, not as a forecast: if the stop is touched far more often than the target within the validated 20-session window, the plan is asymmetric against you in volatility terms whatever your view.

Example 1, Apple long (Figure 12.2 and 12.5). Last close 342.94 on 19 Oct 2026; target 400 (+16.5%), stop 305 (−11.2%), entry band 330-345. The touch panel gives 3% (target) and 7% (stop) **within 20 sessions** (validated) and 44% for each **within 12M** (extrapolated, not validated). T2 at H = 20 gives target first 3%, stop first 8%, neither 88%, Expected R −0.05R, with R:R from the close 1.28× and a break-even of 44%. The context-adjusted row at H = 10 gives 1% (+0.7 pp versus T2). Reading: over the next month the most likely outcome by far is that neither level is reached, and a stop-out is somewhat more frequent than a target hit, so the setup has no short-term edge after costs. The thesis is a 12-month fundamental view; the right action is to rely on the KPIs (gross margin 0.487 ≥ 0.47, ROIC 1.06 ≥ 0.5, composite 69.4 ≥ 50, all OK) and the review cadence, not to expect the price plan to resolve soon.

Example 2, JPMorgan short. Last 339.72; target 300 (−11.7%), stop 365 (+7.4%). Within 20 sessions the touch panel gives target 2% and stop 11%; T2 gives target first 3%, stop first 17%, neither 79%, Expected R −0.12R (R:R 1.73×, break-even 37%). The stop is close in volatility units, so the plan is exposed to a squeeze. The touch (2%) and target-first (3%) figures for the target differ slightly because they are calibrated separately on different pools, a known limitation ("seen: 18% touch vs 20% target-first for one 20-session example").

Example 3, closing. For Toyota (short, 7203.T, JPY) the benchmark is the Nikkei 225 and the sector benchmark a US ETF, so performance is computed in USD with official FX. Closed after four days with reason "Thesis changed": exit 3,102.00 / inception 3,102.00, thesis return −0.0%, outcome FLAT, status at close ON TRACK; the post-mortem records "The BoJ path changed; closed flat."

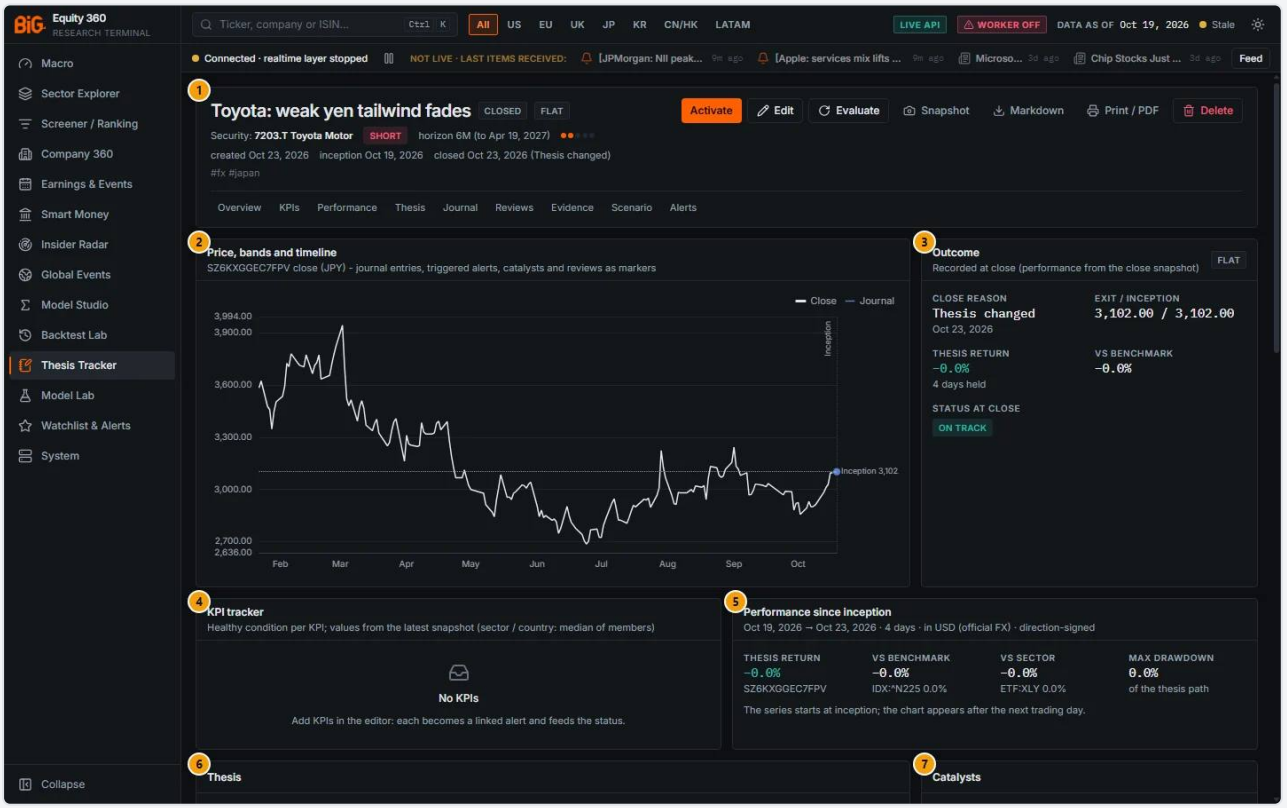


Figure 12.17: A closed thesis (`/theses/:id` , Toyota short), test instance, data as of 19 Oct 2026.

#	Panel	What it shows
1	Summary header	CLOSED and FLAT badges, SHORT, horizon 6M, closed 23 Oct 2026 (Thesis changed); **Activate** replaces Close / Back to ideas.

#	Panel	What it shows
2	Price, bands and timeline	7203.T close (JPY) with the inception line at 3,102 and a journal marker.
3	Outcome	Close reason, exit / inception 3,102.00 / 3,102.00, thesis return −0.0% (4 days held), vs benchmark −0.0%, status at close ON TRACK.
4	KPI tracker	"No KPIs".
5	Performance since inception	19 → 23 Oct 2026, 4 days, in USD (official FX); benchmark IDX:^N225, sector ETF:XLY.
6	Thesis	Thesis text and the post-mortem with **Edit**.
7	Catalysts	Catalysts of the thesis.
8	Invalidation conditions	As for an open thesis.
9	Links	Benchmarks and origin.
10	Journal	Notes and the status changes.
11	Reviews & accountability	"Every 30 days · next —": no review form and no next review once closed; the snapshot table includes the *close* snapshot.
12	Linked evidence	As for an open thesis.
13	Scenario impact (W4)	As for an open thesis.
14	Linked alerts	10 rules, all disabled (grey dots) while closed.

The touch and target-before-stop panels are not shown on a closed thesis.

Limitations and pitfalls

- **Not a probability that the thesis is right.** Touch and target-before-stop frequencies describe price paths of similar setups; they know nothing about your fundamental view.
- **Thesis horizons exceed the validated window.** Only the 20-session column and T2 at H = 20 (and meta at H = 10) are validated; the "Within 12M" touch figure is an extrapolation.
- **Two estimators, two numbers.** The touch frequency of the target and the T2 "target first" are calibrated separately and can differ by a few points.
- **Unadjusted versus adjusted.** The chart, the bands, the card return and the status rules use unadjusted closes; the Performance panel uses adjusted closes. After a dividend or split the two can differ; after a split the bands must be edited by hand.
- **Sector and country KPIs are medians** of primary listings, not index-weighted values. Coverage outside the US is thinner, so a median can rest on few names.
- **Snapshot statuses are historical.** They record the status at capture and are never recomputed.
- **Editing a rule-generating field deletes that rule's history.** When a target, KPI or other parameter changes, the old rule and its fired events are deleted and a new rule is created; unchanged rules keep their history.
- **Survivorship and coverage** caveats of the underlying models apply (current mvp membership in the technicals validation; EU / JP earnings dates only from about 2014-2015).
- The scenario panel for a country uses a regional proxy, not the country itself.

Related mnemonics and BQL fields

`THES` (board, or editor pre-filled from a security or cross-asset context). No BQL fields expose thesis data; the touch and setup figures have Technicals counterparts (`cf.touch_20d` , see Touch probabilities). API: `GET /api/v1/theses/{id}` and its `/status` , `/performance` , `/timeline` , `/evidence` , `/journal` , `/reviews` , `/export.md` sub-paths; `POST .../evaluate` , `.../snapshots` , `.../reviews` , `.../journal` . Related: Trade setup, First-passage probabilities, Conformal prediction, Checklist for an investment decision.

Thesis editor

Route: `/theses/new` , `/theses/:id/edit` · **Menu:** Thesis Tracker › New thesis, or **Edit** on a thesis · **Mnemonics:** `THES` with a security or cross-asset id · **Refresh:** saving regenerates the linked rules and re-evaluates the status immediately

Purpose

The editor captures the whole plan in one form: subject, view, thesis text, KPIs, catalysts, invalidation conditions, price plan, benchmarks, links to other analysis, review cadence and tags. "Saving regenerates the linked alert rules and re-evaluates the automatic status."

Screenshot

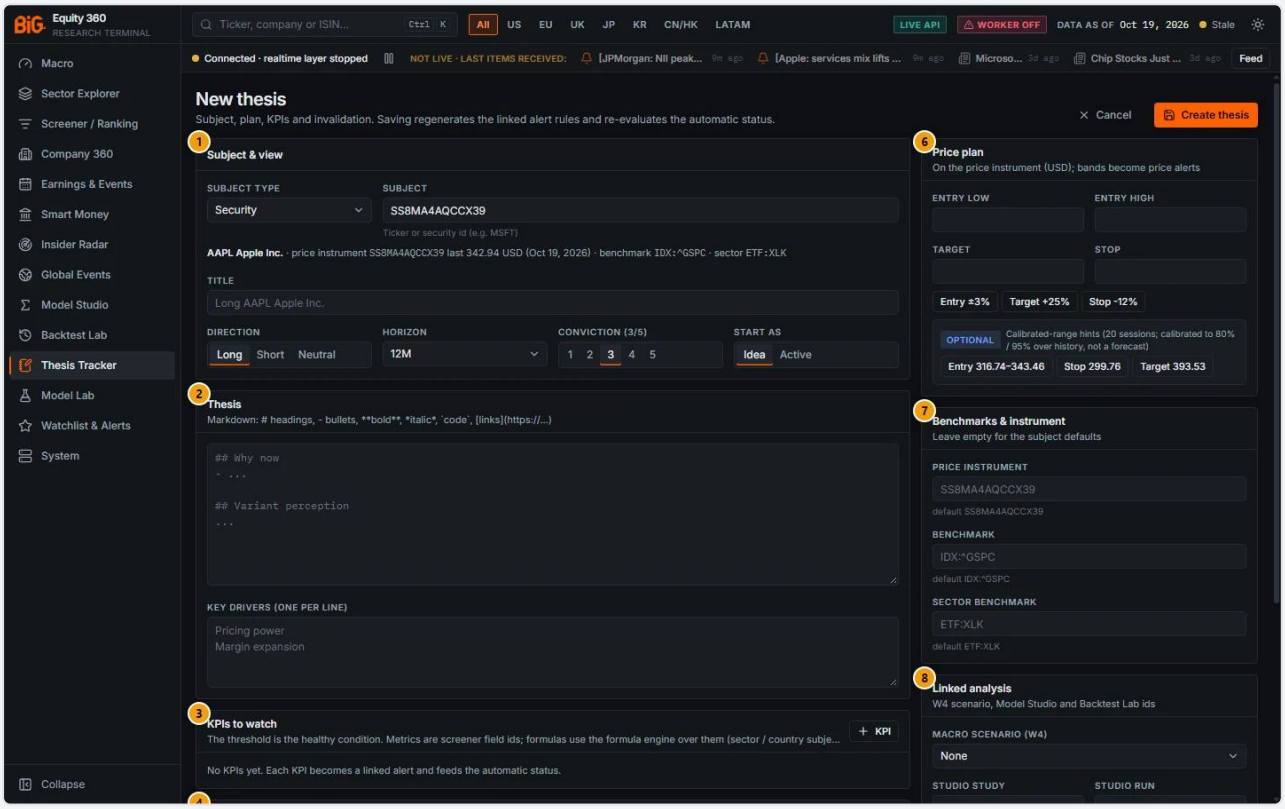


Figure 12.18: New thesis (`/theses/new?subject_type=security&subject_id=...`), opened from Company 360 for Apple, test instance, data as of 19 Oct 2026.

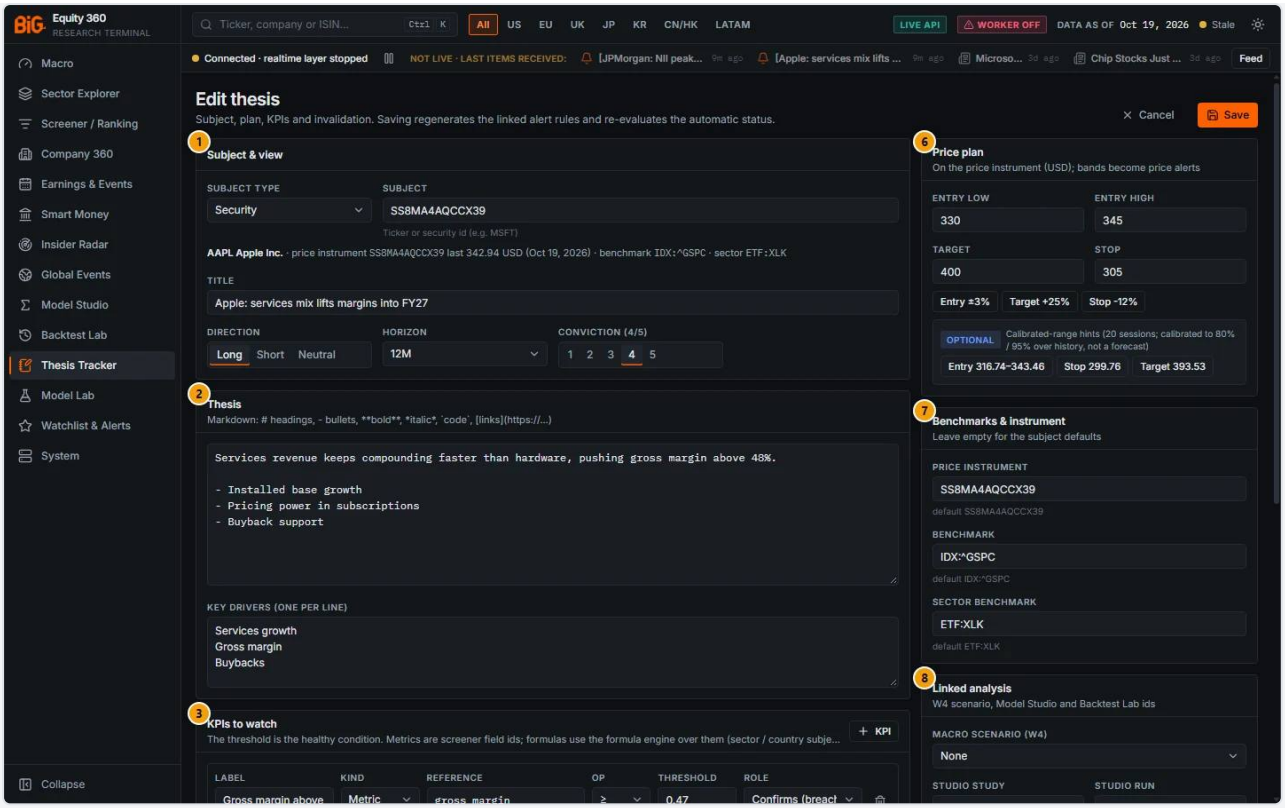


Figure 12.19: Edit thesis (`/theses/:id/edit`) for the Apple thesis.

The badges are the same in both figures:

#	Panel	What it shows
1	Subject & view	Subject type and subject, the resolved subject line, title, direction, horizon, conviction and (new thesis only) Start as.
2	Thesis	Thesis text (Markdown) and key drivers.
3	KPIs to watch	One row per KPI (label, kind, reference or formula, operator, threshold, role).
4	Catalysts	Suggested earnings dates and the catalyst rows.
5	Invalidation conditions	One row per condition.
6	Price plan	Entry low / high, target, stop, quick-fill buttons and the calibrated-range hints.
7	Benchmarks & instrument	Price instrument, benchmark, sector benchmark (defaults shown).
8	Linked analysis	Macro scenario (W4), Studio study / run, Backtest strategy / run.
9	Housekeeping	Review cadence and tags.

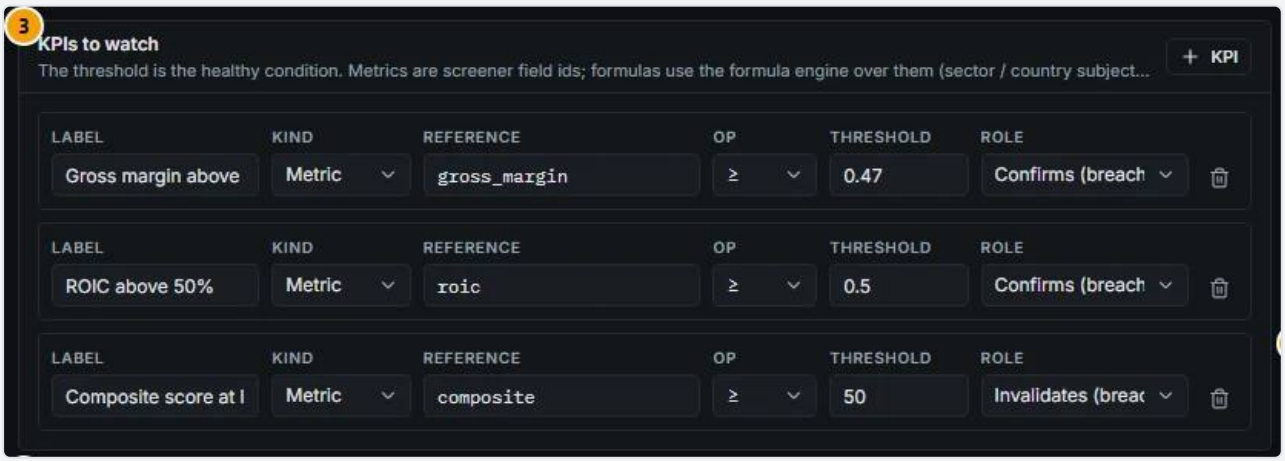


Figure 12.20: KPIs to watch (Apple thesis).

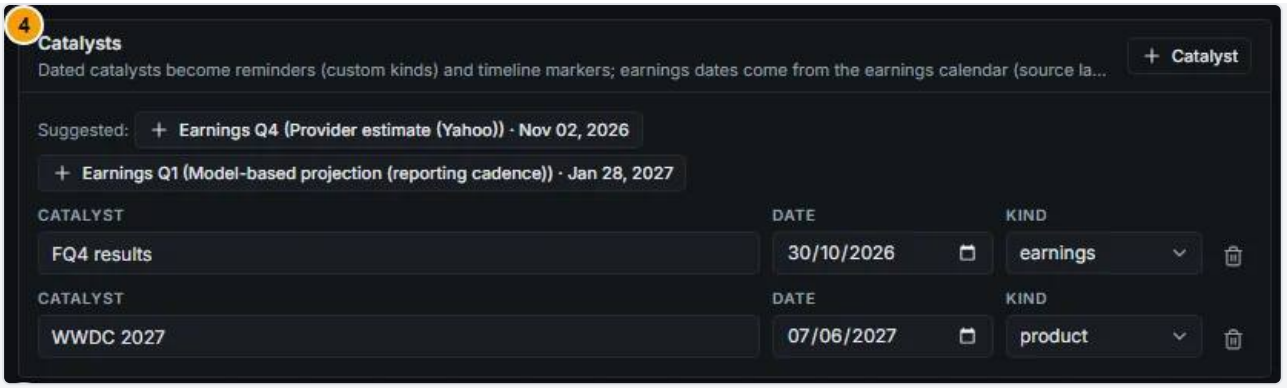


Figure 12.21: Catalysts, with suggestions from the earnings calendar.

Figure 12.22: Invalidation conditions.

Figure 12.23: Housekeeping.

Controls

HEADER

Control	What it does
Cancel	Goes back without saving.
Create thesis / Save	Validates and saves. Create: toast "Thesis created · N linked alert rules generated." and opens the thesis page. Save: "Linked alert rules regenerated and status re-evaluated." Validation errors are listed in a red box at the top.

SUBJECT & VIEW

Field	What it does
Subject type	**Security**, **Sector**, **Country**, **Macro theme**, **Cross-asset**. Fixed after creation.
Subject	Security: "Ticker or security id (e.g. MSFT)". Sector: "Sector id (e.g. TECH)". Country: "ISO2 country (e.g. DE)". Macro theme: "Theme, optionally with a country: 'inflation' or 'US:inflation'". Cross-asset: "Cross-asset id (e.g. CL, GC, EURUSD, UST10Y)". Fixed after creation.
Resolved subject line	As you type, the app resolves the subject and shows its label, the price instrument with the last price and date, the default benchmark and sector benchmark, or "Unknown subject - check the id." or "no price instrument (set a proxy below for bands and performance)".

Field	What it does
Title	Up to 160 characters. On a new thesis an empty title defaults to "Long", "Short" or "Watch" followed by the subject label (shown as the placeholder).
Direction	**Long**, **Short**, **Neutral** (neutral: no price-based status rules; target above the stop like a long for the price alerts).
Horizon	1M, 3M, 6M, 12M (default), 24M, 36M+. Sets the horizon date and the touch-panel horizon.
Conviction	1 to 5 (default 3).
Start as	New thesis only: **Idea** (default) or **Active**.

THESIS

Field	What it does
Thesis text	"Markdown: # headings, - bullets, **bold**, *italic*, `code`, links". Also block quotes, numbered lists, code blocks and horizontal rules. Rendered by a safe subset renderer (no HTML).
Key drivers	One per line (a leading `-` is removed); shown as a list on the thesis page.

KPIS TO WATCH

+ KPI adds a row; the bin removes it. "The threshold is the healthy condition."

Field	Values
Label	Required (e.g. "ROIC above 50%").
Kind	**Metric** (screener field id; the reference box suggests field ids), **Formula** (formula-engine expression over field ids, e.g. `roic - 0.08`), **Cross-asset** (`<asset>:<signal>`, e.g. `CL:carry`), **Macro cell** (`<CC>:<indicator>`, e.g. `US:cpi_yoy`). New rows default to Cross-asset for cross-asset subjects, Macro cell for macro themes, Metric otherwise.
Reference / Formula	The field id, expression or reference. Special metric ids: `composite`, `rank`.
Op	`≥` (default), `>`, `≤`, `<`, `between` (adds a second box for the upper bound).
Threshold	A number (comma or point decimals).
Role	**Confirms (breach → at risk)** (default), **Invalidates (breach → invalidated)**, **Watch only**.

The server checks every KPI on save (unknown field ids, invalid expressions, a missing upper bound for `between`) and rejects the save with the reason.

CATALYSTS

+ Catalyst adds a row: label, date (optional) and kind (**earnings**, **macro**, **corporate**, **product**, **regulatory**, **custom**). For security subjects, **Suggested:** buttons add the next earnings dates from the earnings calendar with their source label, for example "Earnings Q4 (Provider estimate (Yahoo)) · Nov 02, 2026" or "Earnings Q1 (Model-based projection (reporting cadence))". Dated catalysts that are not earnings become reminders (one linked rule, firing 3 days ahead); all dated catalysts are timeline markers. Earnings reminders come from the separate upcoming-earnings rule.

INVALIDATION CONDITIONS

"Any condition met turns the automatic status to invalidated (deadlines passed: at risk)." **+ Condition** adds a row: label (e.g. "Loses the key contract") and kind:

Kind	Third field	Met when
Price below	Level	Last close < level.
Price above	Level	Last close > level.

Kind	Third field	Met when
KPI breached	KPI (choose one of this thesis's KPIs)	That KPI fails its healthy condition, whatever its role.
Deadline	Date	Not "met": once the date has passed without the condition being marked, the status turns **at risk** ("confirm or mark it").
Manual	Triggered: No / Yes	You set Yes.

With no conditions, the stop still invalidates.

PRICE PLAN

"On the price instrument (USD); bands become price alerts." (the currency of the instrument)

Control	What it does
Entry low, Entry high, Target, Stop	Prices in the instrument's trading currency (unadjusted). Checks: entry low ≤ entry high; a long needs target > stop, a short target < stop; all > 0.
Entry ±3%	Entry band 3% either side of the last close.
Target +25%	Last close × 1.25 (× 0.75 for a short).
Stop -12%	Last close × 0.88 (× 1.12 for a short).
Calibrated-range hints (**OPTIONAL**)	Security subjects: "Calibrated-range hints (20 sessions; calibrated to 80% / 95% over history, not a forecast)". **Entry** fills the band between the 80% band edge and the last close; **Stop** sets the 95% edge on the adverse side; **Target** the opposite 95% edge. For a long: entry from the lower 80% edge to the close, stop at the lower 95% edge, target at the upper 95% edge; mirrored for a short.

BENCHMARKS & INSTRUMENT

"Leave empty for the subject defaults." **Price instrument** (default shown; e.g. `ETF:TIP` for an inflation theme; must have prices in the snapshot), **Benchmark** (default per subject, e.g. `IDX:^GSPC`), **Sector benchmark** (security subjects: the US sector ETF, or "no US sector ETF for this sector").

LINKED ANALYSIS

Field	What it does
Macro scenario (W4)	**None**, a **Preset: ...** scenario or a saved scenario; drives the Scenario impact panel.
Studio study, Studio run	Model Studio ids (`ST...` , run id); shown as links on the thesis page.
Backtest strategy, Backtest run	Backtest Lab ids (`BS...` , run id); shown as links.

HOUSEKEEPING

Field	What it does
Review every (days)	"Review reminders fire as alerts; empty = none." Default 30 on a new thesis; 1 to 730. The next review is set this many days after the current as-of date.
Tags (comma separated)	Stored lower-case, de-duplicated; used by the board's tag filter.

On a closed thesis the editor notes: "Closed thesis: the close reason, outcome and post-mortem are edited on the thesis page."

ENTRY POINTS ("CREATE THESIS")

Where	Control	Pre-filled
Company 360 header	**Create thesis** next to the watchlist star (see Company header)	Security, origin "company".

Where	Control	Pre-filled
Screener	Notebook icon in the "Watchlist · create thesis" column	Security, origin "screener" with the rank.
Cross-asset dashboard	**Create thesis** on the asset view	Cross-asset subject, origin "cross_asset".
Model Studio	**Create thesis** on a completed run (see Results)	Studio run and study ids, title "Thesis from ..."; subject type Macro theme with an empty subject: pick the subject.
Backtest Lab	**Create thesis** in the run results header (see Results)	Backtest run and strategy ids, title "Thesis from backtest ..."; subject type Macro theme, empty subject: pick the subject.
Command bar	`THES` with a security or cross-asset id	Subject type and id.

The origin is shown in the thesis page's Links panel ("Created from company (AAPL)").

Methodology

Linked alert rules. Every save rebuilds the rule set from the thesis; unchanged rules are kept (with their history), changed ones replaced, and all are disabled while the thesis is closed.

Subject	Rules generated (cooldown in days)
Any with a price instrument	`price_cross` on target, stop, entry high and entry low, each when set (1). Long / neutral: target "crosses above", stop "crosses below"; short: the reverse. Entry high "crosses below", entry low "crosses above".
Security	`score_change` (balanced composite, ±10 points over 7 days, 5); `earnings_upcoming` (7 days ahead, 1); `earnings_surprise` (absolute z ≥ 2 on EPS or revenue, model-based, 1); `sentiment_shift` (absolute z ≥ 2.5, 5); `price_change_pct_1d` (±7%, 1); `activist_13d` (1); `insider_buy` (cluster buy ≥ $250,000, 5); `short_interest_spike` (US only, +25%, 5); `eu_short_position` (EU / UK only, ≥ 0.5%, increases ≥ 0.1 pp, 5); `thesis_event_shock` on the issuer (z ≥ 3, 3).
Country	`macro_surprise` (country, absolute z ≥ 2, tier ≤ 2, 5); `regime_change` (country, any kind, level 0.5, 5); `thesis_event_shock` on the country (3).
Macro theme	`macro_surprise` (theme if growth / inflation / labour, country if given, 5); `regime_change` (country if given, 5); `thesis_event_shock` on the country when a country is given (3).
Sector	`thesis_event_shock` on the sector (3).
Cross-asset	`thesis_positioning` (CFTC COT index over 3 years ≤ 10% or ≥ 90%, 5) when the asset has a futures root.
All	One `thesis_kpi` per KPI (5); `thesis_catalyst` when a dated, non-earnings, not-done catalyst exists (3 days ahead, 1); `thesis_review` when a cadence is set (1); `thesis_status` (1).

The Apple thesis (four price levels, three KPIs, one product catalyst, a 30-day cadence) therefore has 19 rules; the US country thesis (no target or stop, no KPIs) has 5.

Subject resolution. A security is matched by security id or ticker (the primary listing with the largest market capitalisation wins); a sector must exist in the taxonomy; a country must be a two-letter code; a macro theme accepts any theme name (only growth, inflation and labor are used as theme filters for macro alerts); a cross-asset id must exist in the cross-asset asset list.

Data sources and refresh

Subject defaults, last prices and suggested catalysts come from `GET /api/v1/theses/subject-defaults` on the serving snapshot; the metric list from `GET /api/v1/screener/fields`; scenarios from the macro scenario list; the calibrated-range hints from the conformal envelope (`GET /api/v1/conformal/envelope/{id}`, nightly). Saving writes to the local application state at once.

Validation status

The editor computes nothing itself. The calibrated-range hints use the marginal conformal bands, which validated as a by-product of the conformal gate ("on target per horizon (89.6-90.0%), about 40% narrower than the conventional 2×ATR band"; `docs/CONFORMAL_VALIDATION.md` Verdict). They are labelled "optional" and "not a forecast", and the bands carry no direction. The **Entry ±3%**, **Target +25%** and **Stop -12%** buttons are plain arithmetic, not calibrated.

Correct usage and worked example

Write the KPI thresholds as the conditions that must stay true for the thesis to hold, and give each a role deliberately: *invalidate* for the one or two facts that would end the thesis, *confirm* for supporting evidence, *watch* for context. Add at least one invalidation condition besides the stop, and a deadline for catalysts that must happen by a date.

Example (Figures 12.18 to 12.23). Starting from **Create thesis** in the Apple header, the subject is pre-filled and resolved: "AAPL Apple Inc. · price instrument … last 342.94 USD (Oct 19, 2026) · benchmark IDX:^GSPC · sector ETF:XLK". The calibrated-range hints offer entry 316.74-343.46, stop 299.76 and target 393.53 (20 sessions, 80% / 95% edges). The analyst chose a rounder plan: entry 330-345, target 400, stop 305, plus an invalidation "Close below 305" (price below 305), KPIs `gross_margin ≥ 0.47` (confirm), `roic ≥ 0.5` (confirm) and `composite ≥ 50` (invalidate), catalysts FQ4 results (30 Oct 2026, earnings) and WWDC 2027 (7 Jun 2027, product), review every 30 days and tags quality, us-tech. Saving generated the 19 rules of Figure 12.16.

Limitations and pitfalls

- Subject type and subject cannot be changed after creation; create a new thesis instead.
- Prices in the plan are unadjusted and in the instrument's currency; for an ADR or a second listing use the instrument you actually trade.
- A Studio or Backtest entry starts with subject type Macro theme and an empty subject: choose the real subject before saving.
- The `GET /api/v1/theses/subject-defaults` response also carries suggested KPIs (for example the composite score), but the current editor offers only the suggested catalysts.
- KPI references are not checked until you save; a typo is rejected by the server with the reason.
- Earnings suggestions carry their source label: a "Model-based projection (reporting cadence)" date is an estimate, not a company announcement.

Related mnemonics and BQL fields

`THES` with a security (`AAPL US THES`) or a cross-asset id opens this editor pre-filled; see 14. BQL console and mnemonics. Metric KPIs use the same field ids as the Screener. API: `POST /api/v1/theses` , `PATCH /api/v1/theses/{id}` , `GET /api/v1/theses/subject-defaults` .

13. Watchlists and alerts

This chapter covers the **Watchlist & Alerts** page (`/watchlist`) and every alert rule type the app can evaluate. Alerts in BiG Equity 360 are rules stored in the local application state and evaluated against data: most of them by the nightly `alerts_eval` job on the latest snapshot, the thesis rules by `thesis_eval` , the Model Studio and Backtest Lab drift rules by their weekly re-run jobs, and the live news rules by the realtime alert consumer. A rule that fires writes an **alert event** (a message with the data date that triggered it), which appears in the live strip and feed, as a toast, on the thesis page for thesis rules, and in the Macro alerts panel for country rules.

> **Note:** The screenshots in this chapter come from a separate test instance with the Default watchlist seeded with seven names and alert rules created through the API. The live watchlist of the app is the owner's and is not shown.

Watchlist & Alerts

Route: `/watchlist` · **Menu:** Watchlist & Alerts · **Mnemonics:** `ALRT` · **Refresh:** watchlist on every change; rules evaluated nightly (`alerts_eval` ; weekly profile: Saturday); live rules by the realtime layer when it runs

Purpose

The page keeps the names you follow and the alert rules on them. It shows the watchlist with live snapshot metrics, lets you add security-level rules (price, scores, insiders, sentiment, zones, earnings, smart money, conformal checks) and country-level macro rules, and maps where the watchlist earns its revenue. Analysts use it as a personal monitor; the rules turn the nightly refresh into a short list of things that changed.

Screenshot

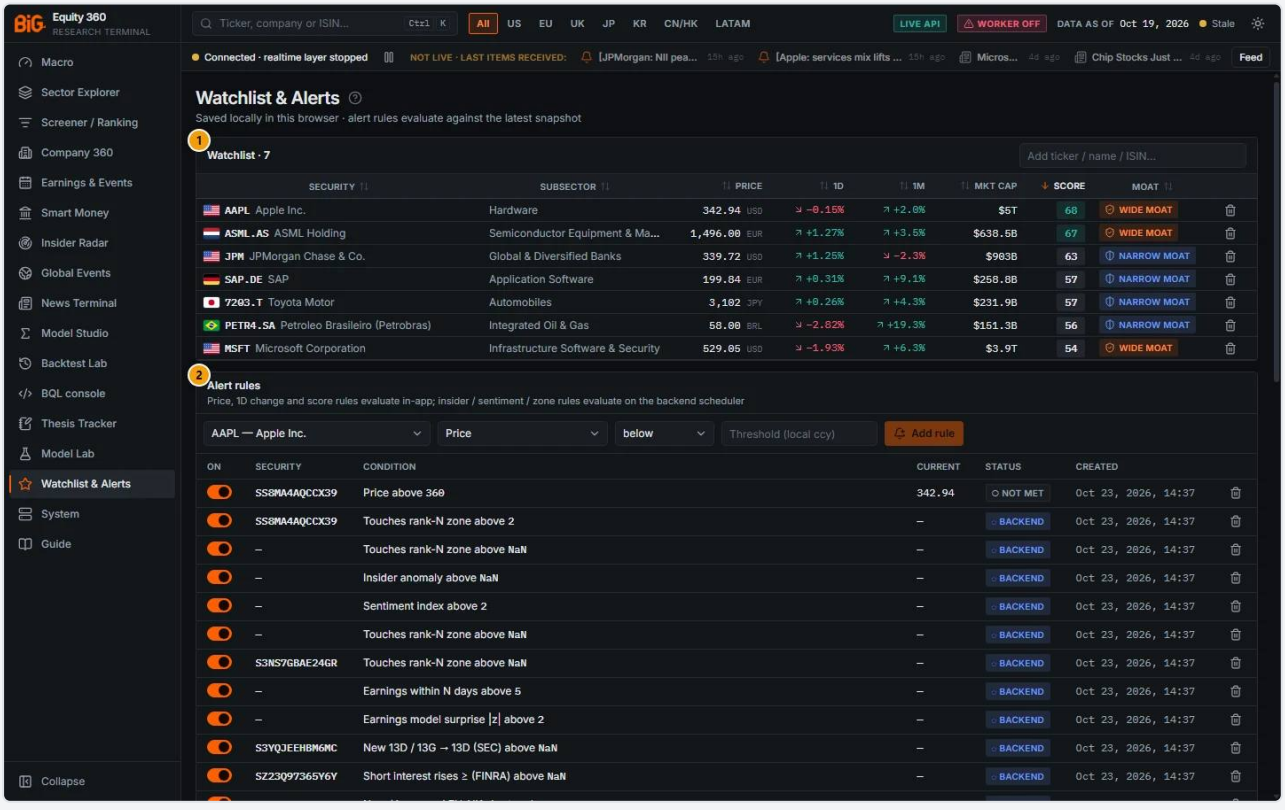


Figure 13.1: Watchlist & Alerts (`/watchlist`), test instance, data as of 19 Oct 2026. Panels 3 and 4 are below the fold and shown further down.

#	Panel	What it shows
1	Watchlist · 7	AAPL, ASML.AS, JPM, SAP.DE, 7203.T, PETR4.SA and MSFT with price in local currency, 1D and 1M change, market cap, composite score (sorted, 68 down to 54) and moat; search box to add names.
2	Alert rules	The rule editor (security, metric, operator, threshold, **Add rule**) and the table of security rules with their Not met / Backend status.
3	Watchlist exposure map	Equal-weight revenue geography of the watchlist with country overlays (direct or look-through).
4	Macro alerts	Country-level rule editor, the macro rule table and recent macro events.

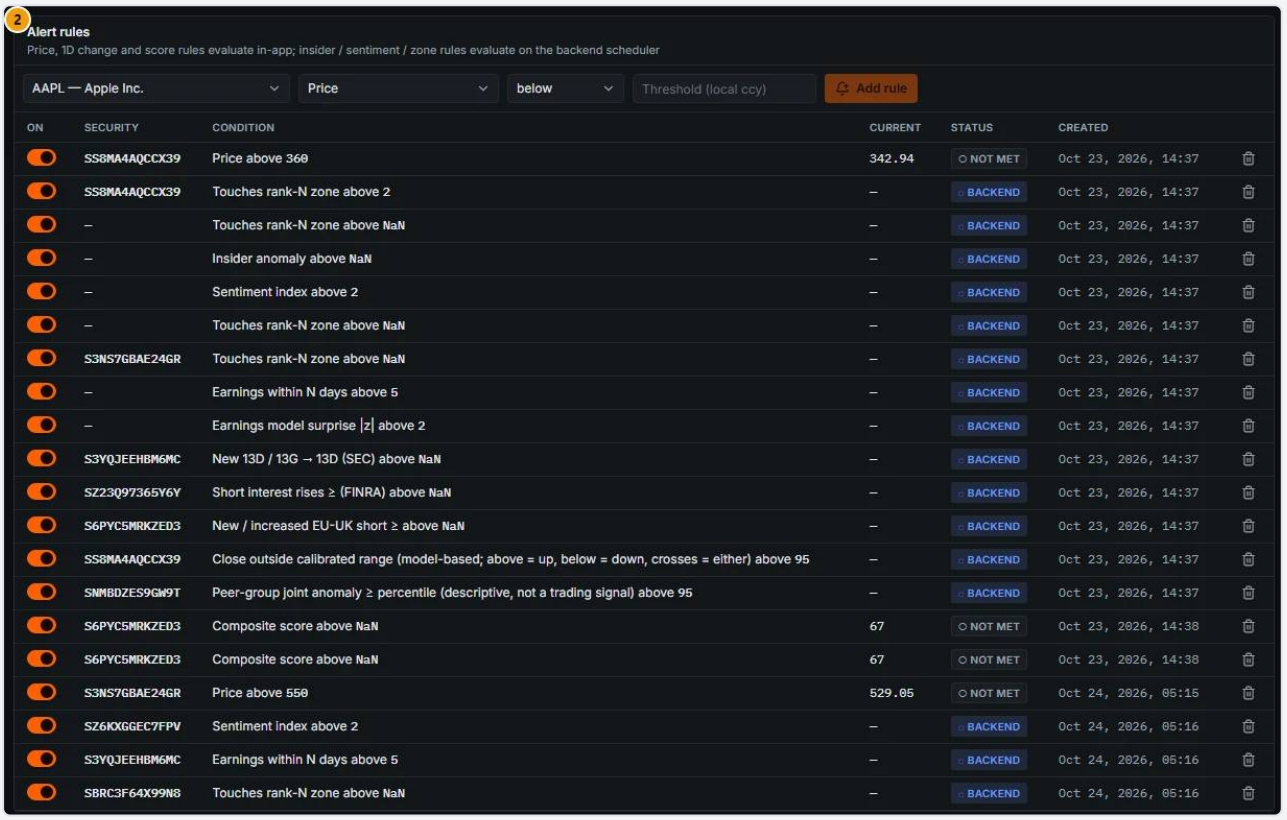


Figure 13.2: Alert rules panel. Rules created through the API show raw security ids or "—" and "above NaN" conditions (see Limitations).

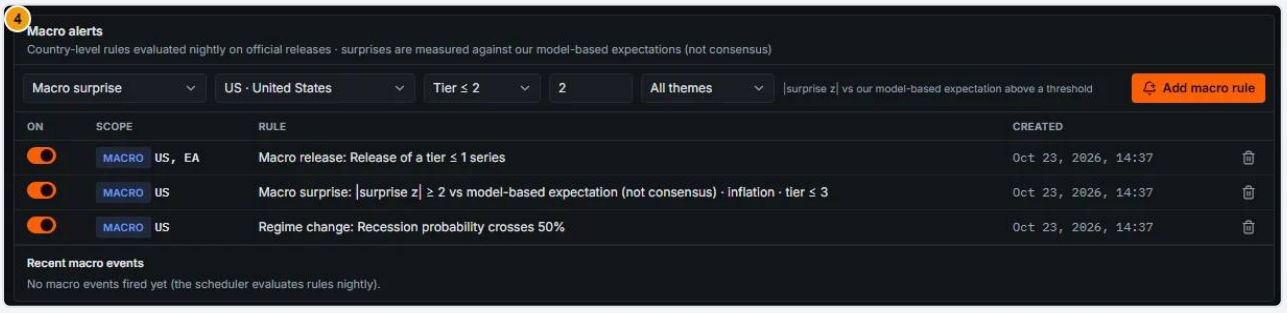


Figure 13.3: Macro alerts panel ("Country-level rules evaluated nightly on official releases · surprises are measured against our model-based expectations (not consensus)") with three rules: Macro release (US, EA; tier ≤ 1), Macro surprise (US; absolute z ≥ 2, inflation, tier ≤ 3) and Regime change (US; recession probability crosses 50%). No macro events had fired yet.

Controls

WATCHLIST

Control	What it does
Add ticker / name / ISIN…	Searches securities; the first 10 matches are listed with flag, ticker and name; a filled star marks names already watched. Click to add.
Star in a Company 360 header	Adds or removes the company (see Company header).
Star in the Screener	First column ("Watchlist · create thesis"): adds or removes the row's security (see Screener).
Table columns	Security (flag, ticker, name), Subsector, Price (local currency), 1D, 1M, Mkt cap (USD), Score (composite), Moat. Sorted by score, highest first; click a header to re-sort.
Row click	Opens Company 360.
Bin	Removes the security from the watchlist.

The page holds **one** watchlist. It is kept in the browser ("Saved locally in this browser") and mirrored to a server watchlist named **Default**, which the News Terminal's watchlist alert and watchlist-targeted rules use. On start-up the app merges the two: names on the server are loaded, and local names are resolved to security ids and pushed up. Further named watchlists can be created through the API (`POST /api/v1/watchlists`), but the page shows only the Default list.

ALERT RULES (SECURITY LEVEL)

Control	What it does
Security	A security from the watchlist (first one by default). Rules can only be added for watched names.
Metric	The rule type; see the table below.
Operator	**above** (`>`), **below** (`<`), **crosses**. Used by Price and the in-app status; mapped to a side for the calibrated-range rule; ignored by the others.
Threshold	Number whose unit is shown in the placeholder (for example "Threshold (local ccy)"). Required, even for rules without a threshold (type any number for the 13D rule).
Add rule	Adds the rule locally and sends it to the server.
On (switch)	Enables or disables the rule (mirrored to the server).
Bin	Deletes the rule.

Metric (label in the page)	Server rule	What your inputs become
Price	`price_cross`	Level = threshold; above or crosses → crosses above, below → crosses below.
1D change %	sent as `price_cross`	See the caution below.
Composite score	`score_change`	See the caution below.
Insider anomaly	`insider_buy`	Minimum value 0, cluster not required; the threshold is stored but not used by the server rule.
Sentiment index	`sentiment_shift`	Threshold used as the absolute z threshold (despite the "−1…1" unit hint).
Touches rank-N zone	`zone_touch`	Threshold = highest zone rank (N).
Earnings within N days	`earnings_upcoming`	Days ahead = N (0 is treated as the default 3).
Earnings model surprise absolute z	`earnings_surprise`	z threshold (model-based).
New 13D / 13G → 13D (SEC)	`activist_13d`	No threshold used.
Short interest rises ≥ (FINRA)	`short_interest_spike`	Threshold % → minimum rise (e.g. 25 → +25%).

Metric (label in the page)	Server rule	What your inputs become
New / increased EU-UK short ≥	`eu_short_position`	Threshold = minimum net short in % of capital (floor 0.1).
Close outside calibrated range	`price_outside_calibrated_range`	Threshold = band level 80, 90 or 95 (else 95); above → up side, below → down side, crosses → either; horizon 1 session.
Peer-group joint anomaly ≥ percentile	`conformal_peer_anomaly`	Threshold as a percentile (95 or 0.95), scope group or member.

> **Caution:** In the current build the page sends the **1D change %** rule as a price-level rule without a level and the **Composite score** rule in "level" form without `mode: level`, and the server rejects both. They are kept only in this browser: they show a Triggered / Not met status in the table but are never evaluated by the scheduler, and they disappear when the app next reloads the rule list from the server. Until this is fixed, use the API for 1-day moves (`price_change_pct_1d`) and score changes (`score_change`).

The table shows **On**, **Security** (Model Studio and Backtest Lab drift rules show a "studio" or "backtest" badge and the study or strategy name, linking to those pages), **Condition**, **Current** (price, 1D change or composite score from the snapshot), **Status** and **Created**. Status is an **in-app** check against the current snapshot for Price, 1D change and Composite score: ● **Triggered** when the value is above / below the threshold (for "crosses", within 1% of it), ○ **Not met** otherwise; every other rule shows ◌ **Backend** (evaluated by the scheduler). Rules generated by a thesis and the live news rules are not listed here: they are managed on the thesis page and in the News Terminal.

MACRO ALERTS

Control	What it does
Macro rule type	**Macro release** ("a linked official series is published"), **Macro surprise** ("absolute surprise z vs our model-based expectation above a threshold"), **Regime change** ("cycle quadrant changes or a recession probability crosses a level").
Country	One country (default US) or **All countries**.
Maximum tier	Release and surprise: **Tier 1**, **Tier ≤ 2** (default), **Any tier** (importance of the series).
absolute z (surprise)	Threshold, default 2.
Theme (surprise)	**All themes**, Growth, Inflation, Labor.
Regime trigger	**Quadrant or recession prob.** (default), **Cycle quadrant change**, **Recession probability**; with a recession trigger, **level** in % (default 50).
Add macro rule	Creates the rule (cooldown 1 day).
Rule table	On switch, scope (macro badge and country), rule description (e.g. "Macro surprise: absolute surprise z ≥ 2 vs model-based expectation (not consensus) · tier ≤ 2"), created, delete.
Recent macro events	The last 50 macro events: time fired, country (links to the country page) and message; unread events in full colour. "No macro events fired yet (the scheduler evaluates rules nightly)."

WATCHLIST EXPOSURE MAP

Figure 13.4: Watchlist exposure map (Direct, Revenue exposure) for the seven names: country risk 43, top country US 29%, 5 / 7 names with geographic revenue.

Control	What it does
Direct / **Look-through**	Exposure basis: the members' own revenue geography, or that plus the exposure carried through their disclosed customers.
Map layer	**Revenue exposure**, **Country risk**, **Event pressure**, **Escalation 30d**, **Recession prob.**
Click a country	Shows its share of watchlist revenue and the contributing names with their weights; click a name to open it.
Stats	**Country risk** with an asterisk (exposure-weighted, 0-100), **Top country** with its share, **Countries** (count, basis).
Coverage badges	One per name, green / amber / red for full / partial / sparse coverage; hover for "home country only".
"N / M with geographic revenue"	How many names disclose a geographic split; the rest use the home country only ("common outside the US"). Also the count with disclosed supply-chain links.
Shared customers / suppliers	Counterparties disclosed by two or more watched names, with the relation and the holders.
Footnote (asterisk)	"Model-based country risk 0-100: percentile mix of event pressure (0.30), Hawkes escalation probability (0.20), recession probability (0.30), negative macro surprises (0.10) and our GPR (0.10)".

FIRED ALERT EVENTS

Events appear in several places:

- the **live strip** and the **Feed** side panel (filter **Alerts**), with a bell icon (see Live strip);
- a toast "Alert ·" followed by the rule type when an event arrives on the live stream (a chime for news alerts if the sound is on);
- the thesis page's **Linked alerts** panel for thesis rules (see Thesis page);
- the **Recent macro events** list for country rules.

Events carry a *read* flag. The page shows unread macro events in full colour, but the current UI has no mark-as-read control; `POST /api/v1/alerts/events/mark-read` with a list of event ids marks them read, and `GET /api/v1/alerts/events?unread_only=true` lists the unread ones.

Methodology

Rule targets. A security rule targets one security; a rule created through the API with a `watchlist_id` targets every member of that watchlist at evaluation time. Macro rules have no security target: they fire per country, and

their events carry the target `macro:<CC>`. Thesis rules without a security fire on `thesis:<id>`; untargeted story-velocity rules on `news:<story id>`.

Noise control.

- **One event per rule, target and data date.** A rule never fires twice for the same security and data date, whichever job evaluates it.
- **Cooldown.** After a rule fires for a security, it does not fire again for that security until `cooldown_days` have passed. Defaults: 5 days for rules created on the page or through the API, 1 day for macro rules created on the page, 0 for the News Terminal's rules; thesis rules have their own (1 to 5 days, see the Thesis editor).
- **Transitions.** Several rules fire on a change of state rather than a level (crossings, entering a zone, a new insider signal, entering the top decile), so a condition that stays true does not repeat.

In-app status versus server rule. The page's Triggered / Not met badge is a *state* check on the current snapshot ("price is below 300"). The server's `price_cross` rule fires on a *crossing* between the two latest closes. A price that has been below 300 for a week shows Triggered but does not fire again.

Exposure map. Each name's direct exposure is its latest annual geographic revenue split known at the as-of date (regions spread over member countries by GDP; no split = 100% home country). Second order sums, over disclosed customers with a revenue share `s_c`, `s_c ×` the customer's direct exposure; look-through = `(1 − Σ s_c) × direct + second order`. The watchlist map is the **equal-weight** average of its members. Country risk is the exposure-weighted average of the model-based country-risk score. Details in Network & Geo.

Data sources and refresh

What	Source and timing
Watchlist metrics	Serving snapshot (screener rows); nightly DAG (weekly profile: Saturday 08:00 Lisbon; daily profile: Tuesday-Saturday 01:30).
Rule evaluation	`alerts_eval` in the nightly DAG, right after the serving snapshot is built; `thesis_eval` after it.
Live evaluation	The realtime alert consumer evaluates `new_filing`, `insider_buy`, `macro_release`, `macro_surprise`, `watchlist_news` and `story_velocity` as items arrive, when the realtime layer runs.
Drift rules	`studio_scheduled` and `backtest_scheduled` jobs (weekly DAG).
Exposure map	Supply-chain jobs (`POST /api/v1/supply-chain/portfolio-exposure`), nightly.
Rules and events	Local application state; changes are immediate.

Validation status

The page itself computes no model. The labels of the underlying data carry over: earnings and macro surprises are measured against **model-based** expectations, not consensus; country risk and the map overlays are model-based; the conformal rules are calibrated over history (see Alert rule reference).

Correct usage and worked example

Keep the watchlist short enough to read the nightly events, put rules on the facts you would act on, and prefer transition rules (crossings, new filings, new 13D) to level rules. Check the **Status** column after adding a rule: ○ **Backend** means the scheduler will evaluate it; a rule that disappears after a reload was rejected by the server.

Example. To be told when Apple's close falls through 300 USD: add AAPL to the watchlist (search box or the star in its header), choose Security AAPL, Metric **Price**, operator **below**, threshold 300, **Add rule**. The server stores `price_cross {level: 300, direction: below}` with a 5-day cooldown. On the snapshot of 19 Oct 2026 (close 342.94) the status shows ○ **Not met**. If a later nightly run sees the close go from above 300 to 300 or below, `alerts_eval` writes "close crossed below 300 (...)" with that data date, and the event shows in the live strip.

For a macro example: **Macro surprise**, United States, Tier ≤ 2, z 2, Inflation. After each nightly run any US inflation release of tier 1 or 2 whose surprise against the app's model-based expectation is 2 standard deviations or more fires one event per country and release date, with the message ending "(not consensus)".

Limitations and pitfalls

- **Two rule types do not reach the server** (1D change %, Composite score; see the caution above).
- **Rules evaluate on the snapshot, not intraday.** Under the weekly profile a price rule is checked once a week, on the two latest sessions only: a level crossed earlier in the week does not fire, and window-based rules (filings, 13D, short interest, earnings and macro surprises) look back one business day only (see Alert rule reference, Limitations).
- **Rules created elsewhere display poorly in the rule table.** The table can only describe rules that carry the page's own display hints. Rules created through the API (and some created by other pages) show the raw security id (`SS8MA4AQCCX39`) instead of the ticker, "—" for watchlist-targeted rules, and "above NaN" where the page cannot map the rule's parameters to its single threshold (for example `insider_buy` , `activist_13d` , `short_interest_spike` , `eu_short_position` , and `score_change` in delta mode). Rule types the page has no label for (`new_filing` , `model_decile` , `price_change_pct_1d`) are shown as "Touches rank-N zone above NaN". In Figure 13.2 these are display artefacts only: the server stores the rules with their real parameters and evaluates them correctly. Check a rule's true parameters with `GET /api/v1/alerts` .
- **One watchlist on the page.** Other server watchlists and watchlist-targeted rules are API-only.
- **News rules are not live yet.** The realtime layer and the News Terminal (roadmap v3 N2-N4) are built but not yet switched on, so `watchlist_news` and `story_velocity` rules do not fire in the current release.
- **Insider and sentiment inputs.** The insider threshold is not used by the server rule, and the sentiment threshold is a z-score, not a sentiment level.
- **Exposure coverage.** Many non-US companies disclose no geographic split; their revenue is assigned to the home country, which overstates home exposure.
- If the watchlist shows 0 names while the exposure badges list tickers, the browser list has not been synchronised with the server (API unreachable at start-up); reload once the API is up.

Related mnemonics and BQL fields

`ALRT` opens this page; see 14. BQL console and mnemonics. No BQL fields expose rules or events. API: `GET/POST /api/v1/watchlists` , `POST /api/v1/watchlists/{id}/items` , `GET/POST /api/v1/alerts` , `PATCH/DELETE /api/v1/alerts/{id}` , `GET /api/v1/alerts/events` , `POST /api/v1/alerts/events/mark-read` . Related: Thesis Tracker, News alerts, Macro cockpit.

Alert rule reference

Route: rules on `/watchlist` , `/theses/:id` , `/news` , Model Studio and Backtest Lab; API `/api/v1/alerts` · **Mnemonics:** `ALRT` · **Refresh:** see each rule

Purpose

This section lists every rule type in the alert contract (`contracts/schemas/alerts.yaml` , version 8): what triggers it, its parameters and defaults, which job evaluates it and when, and where it can be created. Use it when an event message needs explaining, or when creating rules through the API.

Controls

Where each rule type can be created:

Rule type	Watchlist page	Other page	API (`POST /alerts`)	Generated automatically
`price_cross`	Price		yes	thesis price bands
`zone_touch`	Touches rank-N zone		yes	
`score_change`	Composite score (rejected, see caution)		yes	thesis (security)
`price_change_pct_1d`	(1D change % is sent wrongly)		yes	thesis (security)
`insider_buy`	Insider anomaly		yes	thesis (security)
`sentiment_shift`	Sentiment index		yes	thesis (security)
`new_filing`			yes	
`model_decile`			yes	
`earnings_upcoming`	Earnings within N days		yes	thesis (security)
`earnings_surprise`	Earnings model surprise		yes	thesis (security)
`activist_13d`	New 13D / 13G → 13D		yes	thesis (security)
`short_interest_spike`	Short interest rises		yes	thesis (US security)
`eu_short_position`	New / increased EU-UK short		yes	thesis (EU / UK security)
`price_outside_calibrated_range`	Close outside calibrated range		yes	
`conformal_peer_anomaly`	Peer-group joint anomaly		yes	
`macro_release`	Macro alerts panel		yes	
`macro_surprise`	Macro alerts panel		yes	thesis (country, macro theme)
`regime_change`	Macro alerts panel		yes	thesis (country, macro theme)
`watchlist_news`		News Terminal › Alerts	yes	
`story_velocity`		News Terminal › Alerts	yes (no target needed)	
`thesis_kpi` , `thesis_review` , `thesis_status` , `thesis_catalyst` , `thesis_positioning` , `thesis_event_shock`			no	thesis tracker only
`studio_drift`	listed (badge "studio")	Model Studio schedule	no	when a study is scheduled
`backtest_drift`	listed (badge "backtest")	Backtest Lab paper schedule	no	when a strategy is scheduled weekly

Through the API every non-macro rule needs a `security_id` or a `watchlist_id` (except `story_velocity`); `cooldown_days` is 0-365 (default 5). The server fills defaults and rejects invalid parameters with the reason.

Methodology

Each entry gives the trigger, the parameters with defaults, and the evaluator. "Window" means releases or filings whose data became known after the previous business day and up to the as-of date.

PRICE_CROSS

Fires when the close crosses a level between the two latest daily bars: for `above`, previous close below the level and latest at or above it; for `below`, the reverse. Parameters: `level` (required), `direction` above (default) or below, `metric` close (default) or adj_close. Message: "close crossed below 300 (305.1 → 298.7)". Evaluator: `alerts_eval`. Thesis bands carry `band` (target, stop, entry_hi, entry_lo) and use cooldown 1.

ZONE_TOUCH

Fires when the latest bar's high-low range enters a confluence zone that the previous bar did not touch. One event per security (the best-ranked zone entered). Parameters: `zone_rank_max` (default 3), `side` support, resistance or any (default). Message: "price entered support zone #2 [295.4-301.2]". Evaluator: `alerts_eval`. Zones as in Confluence zones.

SCORE_CHANGE

Composite (`profile_id`, default balanced) or factor (`factor_id`) score change. `mode` delta (default): fires when the change over `lookback_days` (default 1) is at least `delta` (default 10) in `direction` up, down or any. `mode` level: fires when the score crosses `level` (up from below, down from above, or either). Message: "composite balanced score 61.0 → 72.5 (+11.5)". Evaluator: `alerts_eval`. Thesis rule: balanced, delta 10, any, lookback 7 days.

PRICE_CHANGE_PCT_1D

Fires when the 1-day change of the adjusted close is at least `pct` (required, a fraction: 0.05 = 5%) in `direction` up, down or any (default). Message: "1-day move -7.4% (threshold 7.0%, any)". Evaluator: `alerts_eval`. Thesis rule: 7%, any.

INSIDER_BUY

Fires when the security's insider signal newly qualifies: 30-day net open-market buying of at least `min_value_usd` (default 0) and above zero, and a cluster buy if `cluster_only` (default false). Message: "insider buying $1,250,000 (30d) — cluster buy". Evaluators: `alerts_eval`, and the realtime consumer on a live Form 4. Uses the rules-based insider signal (see Insider Radar). Thesis rule: ≥ $250,000, cluster only.

SENTIMENT_SHIFT

Fires when the 7-day smoothed news sentiment is at least `threshold_z` (required; default 2.0 if absent) standard deviations from its mean over `window_days` (default 90; at least 10 days of history). Message: "sentiment shift z=+2.71 (positive)". Model-based (sentiment model scores). Evaluator: `alerts_eval`. Thesis rule: 2.5.

WATCHLIST_NEWS

Fires on a live news item linked to a target security (the security, or any member of the target watchlist; linked by security or issuer). Parameters: `tiers` (default [1, 2, 3]: official, major press, publishers; add 4 for GDELT web articles), `min_abs_sentiment` (0-1, default 0), `languages`, `sources`, `max_per_day` (default 10). One event per rule, security and **story** (later articles of the same story do not fire again). Message: "news (negative):" followed by the headline and the publisher. Evaluator: realtime consumer only (no-op in `alerts_eval`). The News Terminal creates it on the Default watchlist with tiers 1-3 and cooldown 0.

> **Note:** `watchlist_news` and `story_velocity` depend on the realtime layer and the News Terminal (roadmap v3 N2-N4), which are built but not yet live. Until they are switched on these rules can be created but do not

> fire.

STORY_VELOCITY

Fires on a burst of articles on one story: at least `min_articles` (default 5) in the last 60 minutes, at least `min_ratio` (default 3.0) times the story's own baseline rate (floored at 0.5 per hour, so a brand-new story needs 5 articles within the hour), and at least `min_sources` (default 2) sources; optional `themes` and `countries` filters. With a security or watchlist target only stories linked to the targets fire; without a target any matching story fires. One event per rule, target and story. Message: "story burst: 9 articles in 60 min (4.5x baseline, 3 sources): ...". Evaluator: realtime consumer only. See News alerts.

NEW_FILING

Fires when the security's issuer has filings of the chosen classes in the window. Parameters: `form_classes` (default annual, quarterly, current_report; allowed also semiannual, insider, holdings, proxy, prospectus, other), optional `lookback_days` . Message: "new filing: 10-Q, 8-K". Evaluators: `alerts_eval` and the realtime consumer. API only.

MODEL_DECILE

Fires when the security's live ML prediction enters decile `min_decile` or higher (from below). Parameters: `horizon` 1M, 3M (default) or 6M; `min_decile` 1-10 (default 9 when created through the API). Message: "model 3M decile 10 (>= 9)". Model-based: the excess-return model is one input, not proven alpha (see Model). Evaluator: `alerts_eval` . API only.

MACRO_RELEASE

Fires when a tracked official release is published in the window. Parameters: `countries` (ISO2 or EA), `series_ids` , `max_importance` 1-3 (default 1). One event per country and release date; target `macro:<CC>` . Message: "US CPI: CPIAUCSL = 3.1 (prior 3.0)". Evaluators: `alerts_eval` and the realtime consumer.

MACRO_SURPRISE

Fires when the absolute surprise z of a release, measured against the app's **model-based expectation (not consensus)**, is at least `threshold_z` (default 2.0). Filters: `countries` , `themes` (growth, inflation, labor), `series_ids` , `max_importance` (default 3). Message: "US CPIAUCSL surprise z=-2.40: actual ... vs model-based expectation ... (not consensus)". Evaluators: `alerts_eval` and the realtime consumer. Thesis rule (country and macro theme): z 2, tier ≤ 2.

REGIME_CHANGE

`kind` cycle_quadrant: fires when a country's cycle-clock quadrant changes. `kind` recession_probability: fires when a recession probability crosses `level` (default 0.5) up or down; `methods` restricts the methods (for example `nyfed_yield_curve_official` , `yield_curve_probit` , `hmm_regime`). `kind` any (default): either. Filter `countries` . Message: "US recession probability (yield_curve_probit) crossed up 50%: 46.0% → 52.3%". The official NY Fed series is official; the probit and HMM methods are model-based (see Macro cockpit). Evaluator: `alerts_eval` .

EARNINGS_UPCOMING

Fires when the security has earnings expected within `days_ahead` (default 3; 0-60) from the as-of date; optional `date_sources` (company_announced, provider_estimate, model_projection). Fires once per expected date. The message carries the source label, and for model projections the date window: "earnings expected 2026-11-02 after the close - Q4 - Provider estimate (Yahoo)". Evaluator: `alerts_eval` . Thesis rule: 7 days ahead.

EARNINGS_SURPRISE

Fires when an actual that became known in the window has an absolute surprise z of at least `threshold_z` (default 2.0) against the **model-based expectation (not consensus)**, for `items` eps_diluted and / or revenue (default both); the largest per security. Message: "EPS surprise z=+2.80 (+6.1%): actual 1.85 vs model-based expectation 1.74 (not consensus)". Evaluator: `alerts_eval` .

ACTIVIST_13D

Fires on a new Schedule 13D filed on the security in the window; also a 13G → 13D switch if `include_switch` (default true) and amendments if `include_amendments` (default false). Source SEC. Message: "new 13D filed by … (SC 13D, SEC)". Evaluator: `alerts_eval` . See Activism (13D).

SHORT_INTEREST_SPIKE

Fires when FINRA short interest published in the window rose by at least `min_chg_pct` (default 0.25 = +25%) versus the previous settlement and is at least `min_si_pct` of shares outstanding (default 0). US securities. Message: "short interest +32% to 41,200,000 shares, 2.8% of shares outstanding (settlement …, FINRA)". Evaluator: `alerts_eval` .

EU_SHORT_POSITION

Fires on a new named net short position of at least `min_pct` (default 0.5% of issued capital) published by an EU or UK regulator in the window, or an increase of at least `min_increase_pp` (default 0.1 pp) when `include_increases` (default true). Message: "new net short position:" followed by the holder, the percentage of the issuer and the position date and regulator. Evaluator: `alerts_eval` . See EU / UK short register.

THESIS_* RULES

Generated by the thesis tracker for each thesis (tagged with its id), regenerated on every save, disabled when the thesis is closed, deleted with it. Evaluated by `thesis_eval` (nightly after `alerts_eval`), on save and by **Evaluate**; no-op in `alerts_eval` .

Rule	Fires when	Default
`thesis_kpi`	A KPI newly breaches its healthy condition (message "KPI … breached: value vs condition").	one per KPI, cooldown 5
`thesis_status`	The automatic status newly turns at risk or invalidated; the message lists the risk / invalid reasons.	cooldown 1
`thesis_review`	The next review date is reached ("review due …"), once per due date.	when a cadence is set
`thesis_catalyst`	A dated, non-earnings, not-done catalyst falls within `days_ahead` of the as-of date.	3 days
`thesis_positioning`	Cross-asset subject with futures: CFTC COT index over 3 years ≤ `lo` (crowded short) or ≥ `hi` (crowded long). Official CFTC data.	lo 0.1, hi 0.9
`thesis_event_shock`	Events-v2 GDELT aggregate for the issuer, country or sector with attention z ≥ `threshold_z` or absolute tone z ≥ `threshold_z` in the last 3 days, on at least 5 items (issuer) or 20 (country / sector). Model-based z.	z 3.0, cooldown 3

The thesis tracker also generates ordinary rules (price bands, score, earnings, smart money, macro); see the Thesis editor for the full list.

BACKTEST_DRIFT

Created when a Backtest Lab strategy is scheduled as a weekly paper strategy (parameters `strategy_id` , `strategy_name` , `schedule` , `drift_threshold` , default 0.3). The weekly `backtest_scheduled` job re-runs it and appends an event on metric drift (Sharpe or CAGR worse, or max drawdown deeper, by more than the threshold, relative), live decay (the new days' returns more than 2 standard deviations below the backtest) or a drawdown breach (current drawdown beyond the previous worst). Removed when the schedule is set to none. See Paper strategies and drift alerts.

STUDIO_DRIFT

Created when a Model Studio study is scheduled weekly or monthly (parameters `study_id`, `study_name`, `schedule`, `drift_threshold`, default 0.2). The weekly `studio_scheduled` job re-runs due studies and appends an event when a headline metric degrades by more than the threshold (relative) or coefficients shift by more than three standard errors. See Studies & runs.

PRICE_OUTSIDE_CALIBRATED_RANGE

Fires when the move over the last `horizon` sessions (1, 5, 10 or 20; default 1), on adjusted closes, ended outside the **marginal** conformal band of `level` (80, 90 or 95%; default 95) emitted at its origin h sessions earlier; `side` up, down or any (default). Message: "close outside the 95% calibrated 1-session range (down: move -6.2% vs +-4.1%; band from ...; model-based)". By construction about `100 - level` % of moves end outside the band, so at 95% roughly one session in twenty fires per security before cooldown. Evaluator: `alerts_eval`.

CONFORMAL_PEER_ANOMALY

Scope `group`: fires when the security's peer group's joint-anomaly percentile is at least `percentile` (default 0.95). Scope `member`: fires when the security's absolute conditional z within its group is at or above its own trailing-year percentile `percentile` (needs 60 days of history). Scope `any` (default): either, strongest kept. Message ends "- descriptive, model-based". Evaluator: `alerts_eval`. See Peer-group joint anomaly.

Data sources and refresh

Evaluator	Rule types	When
`alerts_eval` (nightly DAG)	All except thesis, drift and news rules	After the serving snapshot is built: Saturday 08:00 Lisbon chain under the weekly profile (default); Tuesday-Saturday 01:30 under the daily profile.
`thesis_eval` (nightly DAG)	`thesis_*`	After `alerts_eval`; also on save and **Evaluate**.
`studio_scheduled`, `backtest_scheduled` (weekly DAG)	`studio_drift`, `backtest_drift`	Weekly refresh.
Realtime alert consumer	`new_filing`, `insider_buy`, `macro_release`, `macro_surprise`, `watchlist_news`, `story_velocity`	As items arrive, when the realtime layer runs (rules re-read every 60 s). Not live in the current release.

The nightly and live evaluators share the same idempotency (one event per rule, target and data date) and cooldown, so an event fired live is not fired again at night.

Validation status

Rule	Label and status
`earnings_surprise`, `macro_surprise`, `thesis_kpi` on surprise data	Model-based expectations, not consensus (the messages say so).
`price_outside_calibrated_range`	Marginal single-horizon conformal bands: validated by-product, "on target per horizon (89.6-90.0%)" at 90% (`docs/CONFORMAL_VALIDATION.md` Verdict and §A.2). Calibrated over history; per-regime coverage is lower in crashes (Covid window 82-88% at 90% for the EWMA scale).
`conformal_peer_anomaly`	Conformal B **PASS** as a calibrated descriptive measure (90.4 / 95.2% coverage); **not** a trading signal (break-away mean reversion FAIL).
`regime_change`	NY Fed probability official; probit and HMM methods model-based.
`sentiment_shift`, `thesis_event_shock`, `story_velocity` sentiment	Model-based scores (sentiment model, GDELT z-scores).
`model_decile`	ML excess-return model: out-of-sample rank IC about 0.035; one input, not proven alpha.

Rule	Label and status
`insider_buy`	Rules-based insider score (the insider ML model showed no out-of-sample edge and is not used).
Price, filing, 13D, short-interest, EU short, earnings-date rules	Data rules on official or vendor data; no model.
Drift rules	Re-run comparisons; not formally validated.

Correct usage and worked example

Read an event as "this happened in the data", then open the page that explains it. Use thresholds that fire rarely: at a 95% calibrated range, a 1-session rule on 50 names fires on about two or three names on an average day.

Example: building a smart-money monitor through the API. For a watchlist id `W…`, `POST /api/v1/alerts` with `{"watchlist_id": "W…", "rule_type": "activist_13d", "params": {}}` stores `include_amendments: false, include_switch: true` and cooldown 5; each nightly run then fires one event per member with a new 13D (or 13G → 13D switch) filed since the previous business day, with the filer and form in the message.

Limitations and pitfalls

- **The window is one business day.** `alerts_eval` runs once per refresh on the snapshot's as-of date, and window-based rules (`new_filing`, `activist_13d`, `short_interest_spike`, `eu_short_position`, `earnings_surprise`, `macro_release`, `macro_surprise`) only consider items that became known after the previous business day. Under the weekly profile, items from earlier in the week are therefore not alerted by the nightly job. The live consumer covers filings, insider buys and macro releases / surprises when the realtime layer runs, and `new_filing` accepts a `lookback_days` parameter through the API (for example 7). Use the calendars (Earnings & Events, Release calendar, Smart Money) for completeness.
- A changed thesis parameter replaces the rule and deletes its past events.
- Macro and thesis events have no ticker; their target is `macro:<CC>` or `thesis:<id>`.
- The model-based labels matter: a "surprise" is relative to the app's model, not to market consensus.

Related mnemonics and BQL fields

`ALRT` (this page), `THES` (thesis tracker), `N` (News Terminal, when installed). No BQL fields. Contract: `contracts/schemas/alerts.yaml`, `contracts/schemas/alert_events.yaml`; API tag `alerts`.

14. BQL console and mnemonic reference

This chapter covers the two "power-user" entry points of BiG Equity 360:

- the **BQL console** (`/bql`), where you write short declarative queries (`get(...) for(...) with(...)`) that run point-in-time over the app's data and can be exported, saved and reused elsewhere in the app;
- the **command bar** (Ctrl+K or `/`), which understands Bloomberg-style mnemonics such as `AAPL US Equity DES` , `SX5E Index GP` , `ECO US` or `WIRP EU` .

> **Note:** The **?** button next to each page title, the **Guide** menu item and the command `HELP` open this user guide.

The BQL language is "BQL-lite": it borrows the shape of Bloomberg's query language but is the app's own, smaller language. It is not compatible with Bloomberg BQL and does not access Bloomberg data.

BQL console

Route: `/bql` (deep link `/bql?q=<query>&run=1`) · **Menu:** BQL console · **Mnemonics:** `BQL` , `BQL get(...)` , `<ticker> BQL` · **Refresh:** queries run on demand against the current serving snapshot (rebuilt by the weekly or daily refresh profile)

Purpose

The console answers ad-hoc questions that no page answers directly: "which large European industrials have ROIC above 15% and are within 5% of their 52-week high", "how did these three stocks' P/E evolve weekly over two years in USD", "what was this screen's membership on 28 June 2024". It is used by analysts who want a screen or a panel of numbers quickly, and by anyone who wants to feed a custom expression into Model Studio or the Backtest Lab. Every query is compiled to a safe, parameterised, point-in-time SQL query; it never changes data.

Screenshot

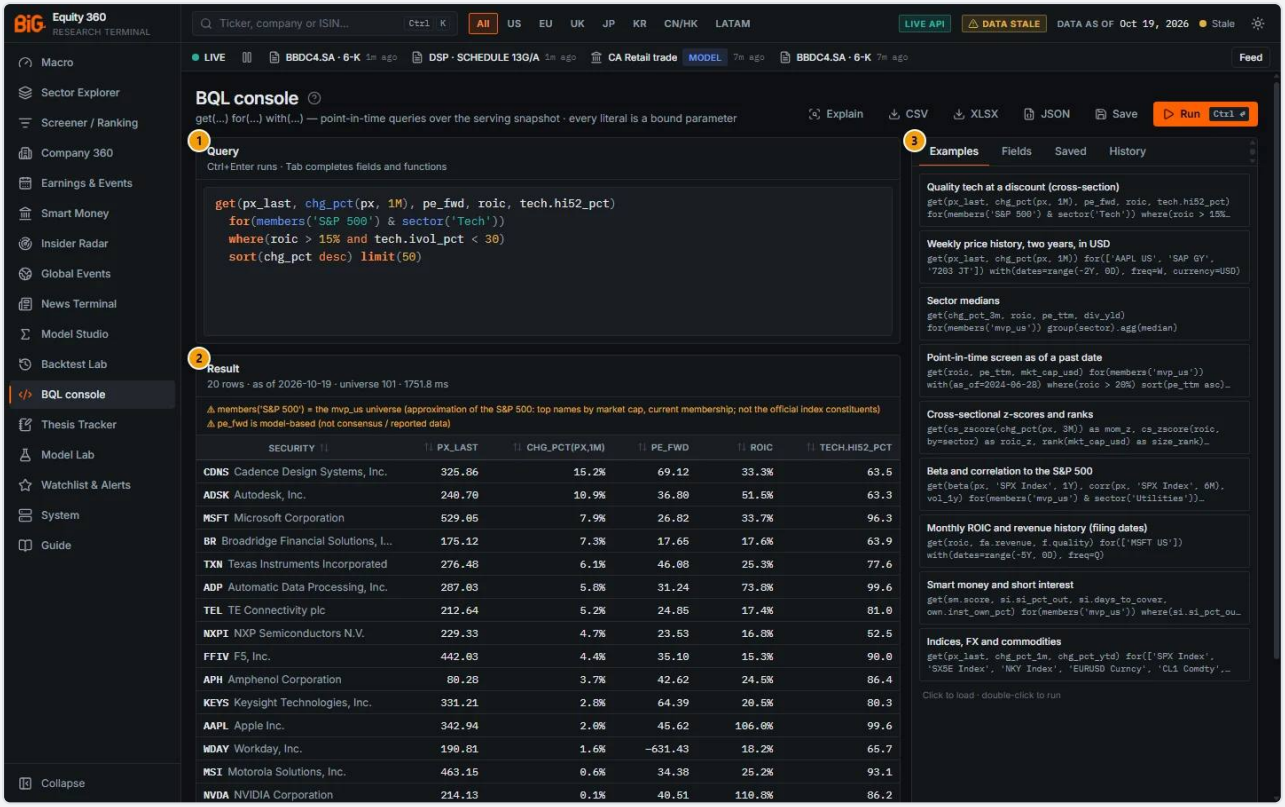


Figure 14.1: BQL console (`/bql`) with the default example "Quality tech at a discount" run; snapshot as of 19 Oct 2026.

#	Panel	What it shows
1	Query	The syntax-highlighted editor with the default query.
2	Result	20 rows · as of 2026-10-19 · universe 101 · 1,751.8 ms; two warnings (the S&P 500 approximation and "pe_fwd is model-based (not consensus / reported data)"), then the result table.
3	Side panel	Tabs Examples, Fields, Saved and History; here the nine examples ("Click to load · double-click to run").

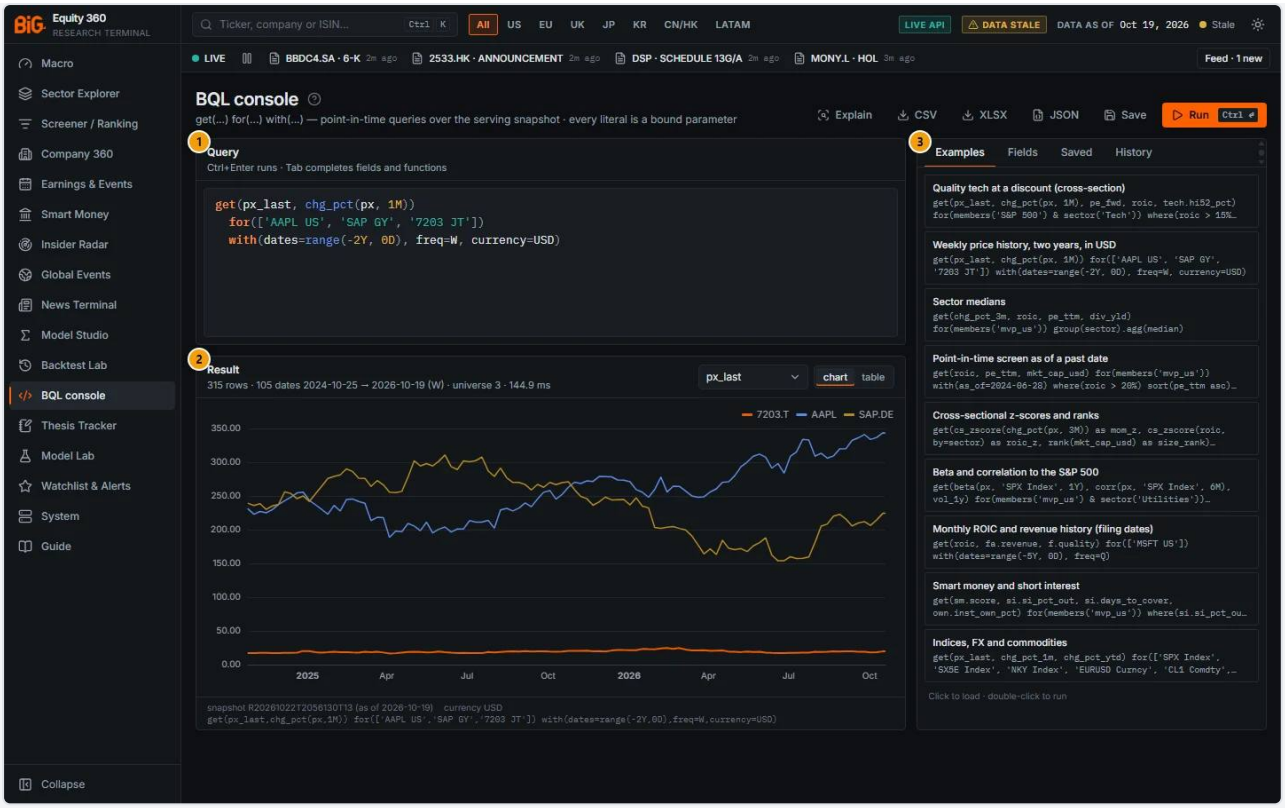


Figure 14.2: The example "Weekly price history, two years, in USD" run: 315 rows, 105 weekly dates 2024-10-25 → 2026-10-19, chart of px_last in USD for 7203.T, AAPL and SAP.DE.

#	Panel	What it shows
1	Query	`get(px_last, chg_pct(px, 1M)) for(['AAPL US', 'SAP GY', '7203 JT']) with(dates=range(-2Y, 0D), freq=W, currency=USD)` .
2	Result	Chart view with the Chart column selector (px_last) and the chart / table switch; the footer shows the snapshot id, its as-of date, the currency and the canonical query.
3	Side panel	Examples tab.

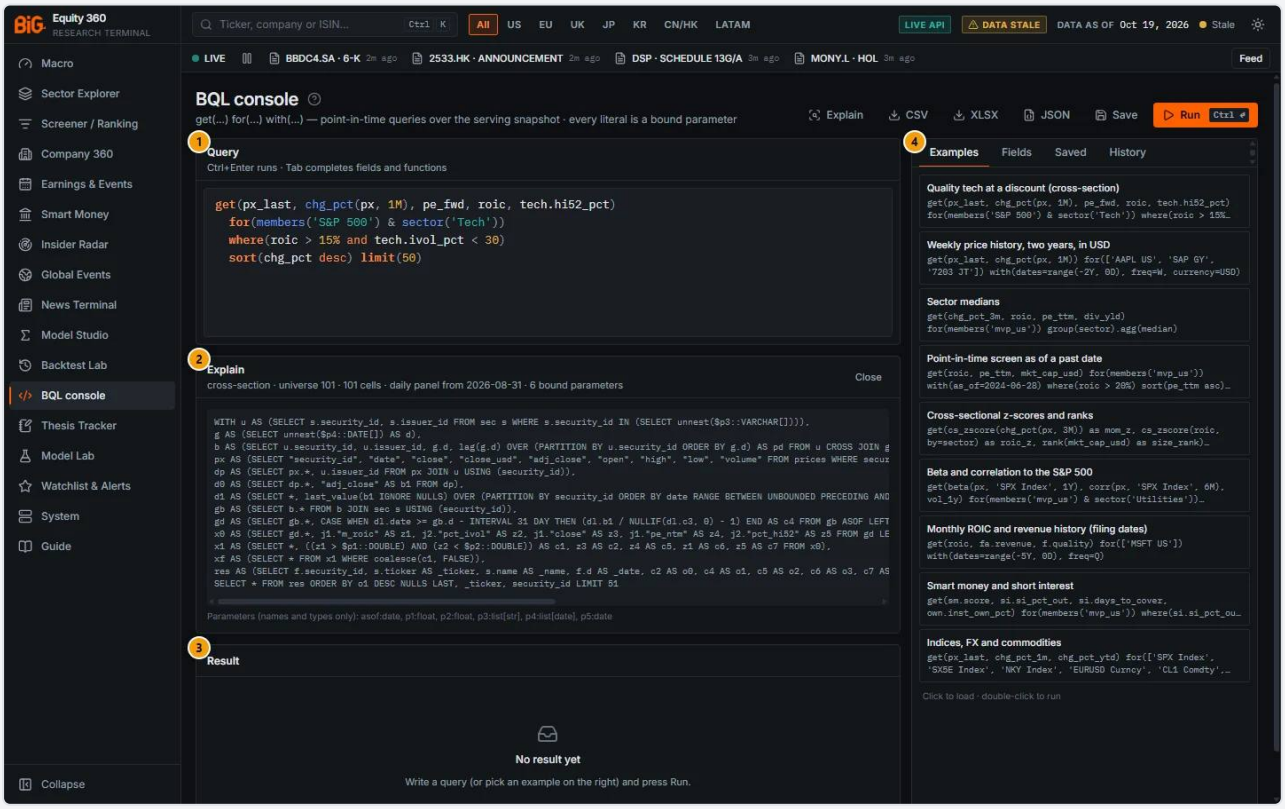


*Figure 14.3: *Explain** on the default query, before running it.*

#	Panel	What it shows
1	Query	The default query.
2	Explain	"cross-section · universe 101 · 101 cells · daily panel from 2026-08-31 · 6 bound parameters", the compiled SQL and "Parameters (names and types only)". **Close** hides it.
3	Result	"No result yet": Explain does not run the query.
4	Side panel	Examples tab.

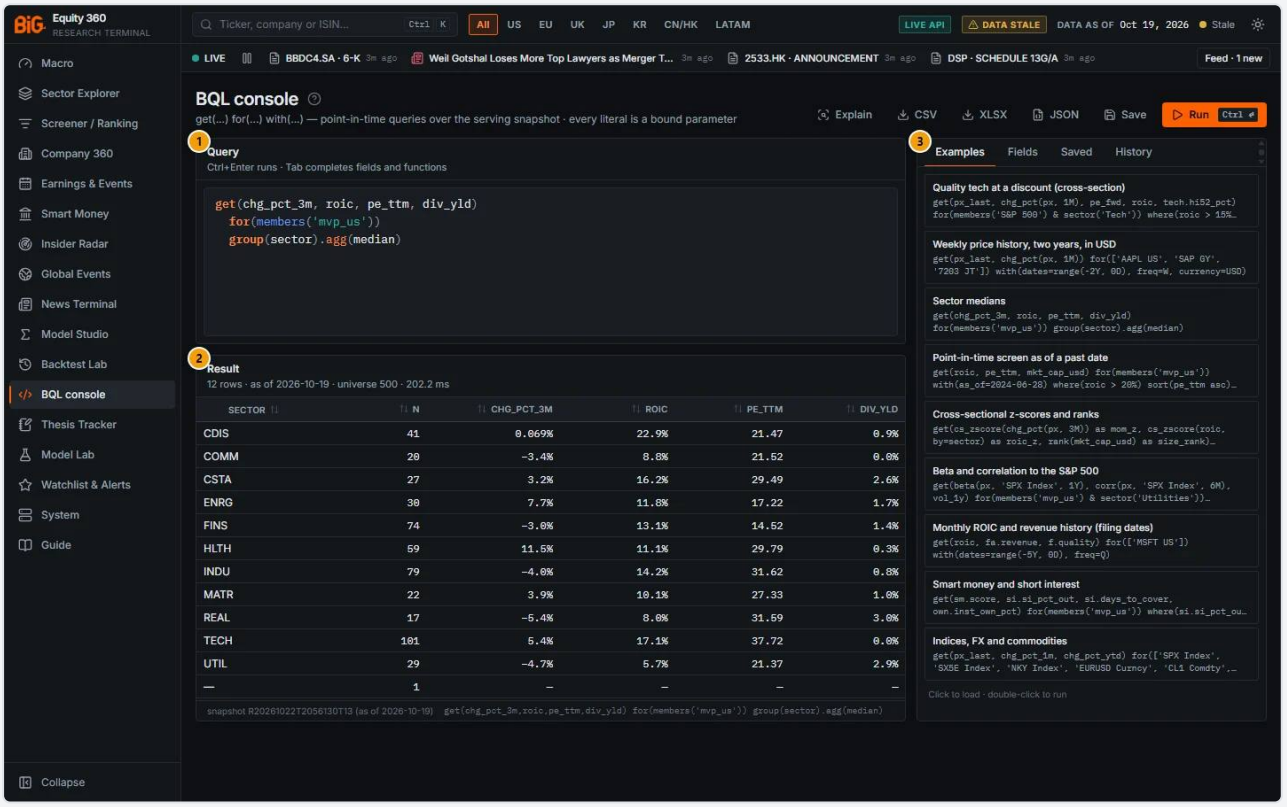


Figure 14.4: The example "Sector medians": `group(sector).agg(median)` over mvp_us, 12 rows, universe 500.

#	Panel	What it shows
1	Query	`get(chg_pct_3m, roic, pe_ttm, div_yld) for(members('mvp_us')) group(sector).agg(median)` .
2	Result	One row per sector with **N** (securities in the group) and the median of each field; the last row "—" (N = 1) collects securities without a sector.
3	Side panel	Examples tab.

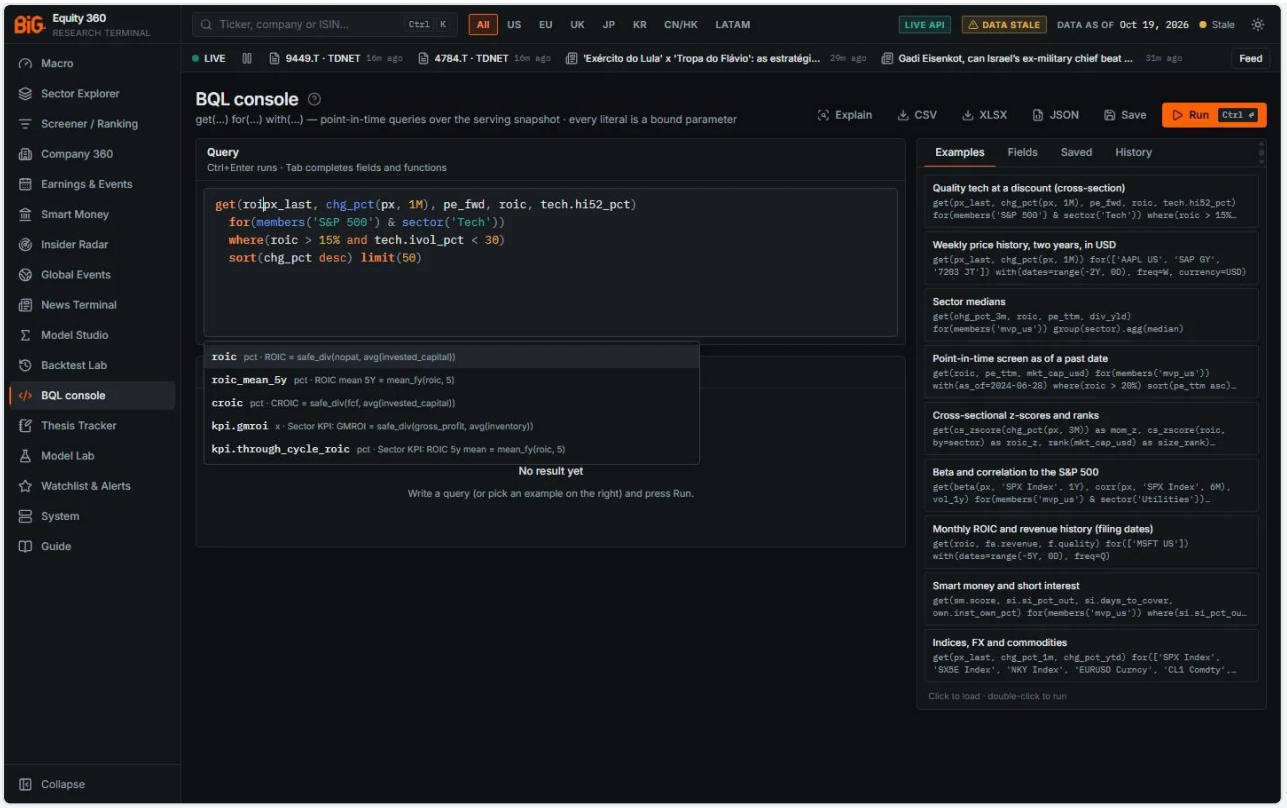


Figure 14.5: Autocomplete after typing "roi" inside `get(` : roic, roic_mean_5y, croic, kpi.gmroi, kpi.through_cycle_roic, each with its unit and definition.

The page has a header with the actions, the **Query** editor and the **Result** panel on the left, and a side panel with four tabs on the right:

Area	What it shows
Header	Title, subtitle "get(...) for(...) with(...) — point-in-time queries over the serving snapshot · every literal is a bound parameter", and the buttons Explain, CSV, XLSX, JSON, Save and **Run** (Ctrl ↵).
Query	The syntax-highlighted editor ("Ctrl+Enter runs · Tab completes fields and functions").
Explain	Appears after **Explain**: the compiled SQL and its parameters.
Result	The result table (or chart for time series), warnings and the footer.
Side panel	Tabs **Examples**, **Fields**, **Saved**, **History**.

Controls

HEADER AND EDITOR

Control	What it does
? (next to the title)	Opens this section of the user guide.
Run / Ctrl+Enter	Runs the query in the editor. A successful run is added to History.
Explain	Compiles the query without running it and shows the mode (cross-section or time series), universe size, number of cells, whether a daily panel is needed (and from which date), the SQL and the parameter names and types (never their values). **Close** hides it.
CSV, **XLSX**, **JSON**	Run the query and download the result as `bql_<as-of>.csv` , `.xlsx` or `.json` . XLSX needs the openpyxl package on the server; otherwise an error suggests CSV or JSON.

Control	What it does
Save	Opens a name box, a Kind switch (`query` or `screen` ; hover on screen: "usable as members('screen:NAME')"), **Save** and **Cancel**. A toast confirms; for a screen it reminds you of `members('screen:NAME')` .
Editor	Colours clauses, functions, fields, strings, numbers / dates / durations and operators; characters outside the grammar are underlined in red. The draft is kept in this browser between visits.
Autocomplete	After two letters, up to nine suggestions (fields from the dictionary with unit and description, then functions). ↑ / ↓ move, **Tab** or **Enter** accepts (a function is inserted with its opening bracket), **Esc** closes.

RESULT PANEL

Element	What it shows
Subtitle	Rows (and "(truncated)"), then either "N dates start → end (freq)" for a time series or "as of ‹date›" for a cross-section, the universe size, the run time in ms and "(cached)" when served from the cache.
Warnings	Lines starting with ⚠: unresolved tickers, index approximations ("members('S&P 500') = the mvp_us universe ..."), model-based fields ("pe_fwd is model-based (not consensus / reported data)"), ignored options, truncation, `as_of` clamped to the snapshot.
View `chart` `table`	For a time series with numeric columns; the chart is the default.
Chart column	Which numeric column to plot when there are several. The chart shows the first 8 series ("first 8 series shown · switch to the table for all rows").
Table	A **Security** column (ticker and name, linking to Company 360 or the instrument page) and one column per output. Hover a header for the field description, unit, currency and "model-based". Click a header to sort.
Footer	The snapshot run id and its as-of date, the result currency, and the canonical query text.

SIDE PANEL

Tab	What it does
Examples	Nine example queries (titles below). Click loads one into the editor; double-click loads and runs it.
Fields	Searches the field dictionary ("Search fields (pe, hi52, roic ...)") with a group filter. Shows "N of 817 fields · 616 with point-in-time history"; each field has its unit, a **PIT** badge when it has history, a **model** badge when model-based, its description and aliases ("aka ..."). Hover for the point-in-time rule. A **Functions** section lists every function with its signature.
Saved	Saved queries with their kind badge (`query` or `screen`); click to load, double-click to run, bin icon to delete. Empty state: "No saved queries. Use Save next to Run; kind “screen” makes a query usable as members('screen:NAME')."
History	The last 30 queries run in this browser, with time; click to load; **Clear history**.

The examples are: *Quality tech at a discount (cross-section)*; *Weekly price history, two years, in USD*; *Sector medians*; *Point-in-time screen as of a past date*; *Cross-sectional z-scores and ranks*; *Beta and correlation to the S&P 500*; *Monthly ROIC and revenue history (filing dates)*; *Smart money and short interest*; *Indices, FX and commodities*.

Figure 14.6: Side panel, Examples tab.

Figure 14.7: Fields tab searching "hi52": 5 of 817 fields, 616 with point-in-time history; PIT badges and aliases; the Functions section below.

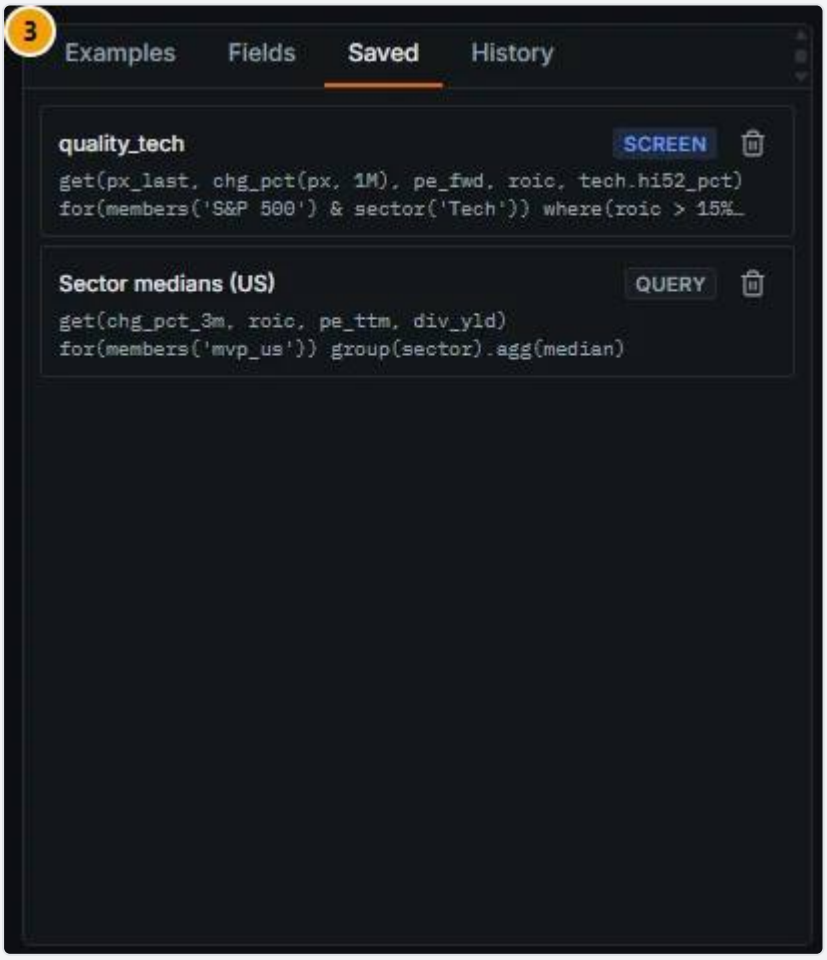


Figure 14.8: Saved tab with a saved screen (quality_tech, SCREEN) and a saved query (Sector medians (US), QUERY).

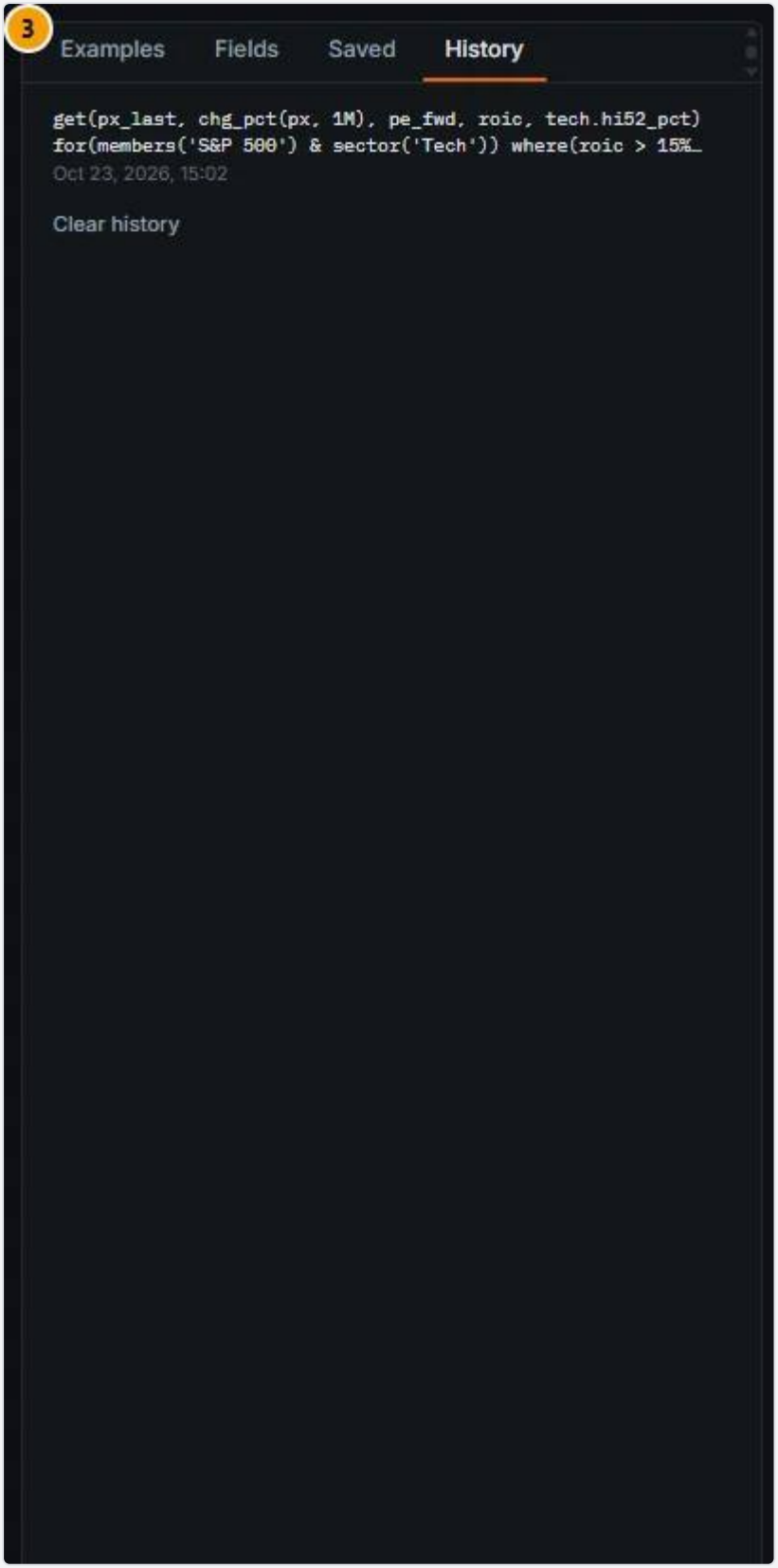


*Figure 14.9: History tab after one run, with its time and *Clear history*.*

Methodology

A query goes through four steps on the server:

1. **Parse.** A hand-written lexer and recursive-descent parser turns the text into a syntax tree. Any character outside the grammar (for example ； or a comment) is rejected.

2. **Resolve.** Field names are looked up in the field dictionary (`contracts/bql_fields.yaml`), tickers through the Bloomberg-style resolver, and the universe is resolved to a list of security ids.
3. **Compile.** The tree becomes one DuckDB SQL statement over the serving snapshot (and, for long histories, the data lake). Each value is evaluated at the right level: static identity, a dated stream sampled as of a date, a daily price panel, or the output grid. Time-series functions run on the daily panel, cross-sectional functions per grid date over the rows that pass `where()` .
4. **Execute** with limits, a timeout and a result cache, then return columns with their unit, currency, description and model-based flag.

The full language is in the BQL language reference and the fields in the Field dictionary.

Data sources and refresh

- The **serving snapshot** (the read-only DuckDB the app reads), as of the snapshot date shown in the result footer. Daily price bars cover 15 years; before that, weekly bars are used.
- **Lake histories** for fields whose history is not in the snapshot (valuation, technical factors, conformal envelopes), and the lake FX history for currency conversion.
- Results reflect whatever the last refresh built: the weekly profile (Saturday 08:00 Lisbon) by default, or the daily profile. See Data and coverage.

Validation status

The BQL engine is a query tool, not a model, so there is no model gate. Its correctness is covered by tests: golden syntax-tree tests for the grammar, a test that every field mnemonic and alias resolves and runs, and a headless-browser check of the console on a test API in both themes (Roadmap v3 B1/B2).

The **fields** keep the validation status of the model that produced them. The console marks them with a **model** badge (Fields tab) and "model-based" (column header tooltip). Notable cases:

Fields	Status (source)
`pe_fwd` , `fair_value` , `dcf_upside` , `implied_cagr`	Model-based forecast and DCF, **not consensus**
`eps_surprise` , `rev_surprise` , `earn.*` , `pead.*`	Surprises against **model-based expectations** (seasonal model on own history), not consensus
`ml.*`	ML excess-return model: out-of-sample rank IC about 0.035; one input, not proven alpha
`tech.hi52_pct` , `tech.maxret_pct` , `tech.ivol_pct`	Technicals v3 factors that **passed** (descriptive rank edge, not profitable stand-alone after costs)
`tech.mom*` , `tech.rmom*` , `tech.strev*` , `tech.indmom*`	Technicals v3 factors that **did not pass** the full gate; experimental (hidden by default) in the app
`tp.*`	Calibrated technical score: **failed** (no out-of-sample edge), experimental; `tp.h5_gate` , `tp.h20_gate` , `tp.h60_gate` give the stored verdict
`cf.qm_*` , `cf.band_*` , `cf.touch_20d`	Marginal conformal bands: validated (on-target coverage per horizon)
`sm.score` , `ins.anomaly`	Model-based scores

See Methodology appendix for the full gate table.

Correct usage and worked example

Start from an example, run it, then change one clause at a time. Use **Explain** before running a large time series to see the number of cells. Keep `where()` conditions on fields with history when you use `as_of` or `dates=` .

Worked example (Figure 14.1, snapshot of 19 Oct 2026). Run the default query "Quality tech at a discount". The universe `members('S&P 500') & sector('Tech')` has 101 securities; after `where(roic > 15% and tech.ivol_pct < 30)` only **20 rows** remain, so `limit(50)` does not bite. The result is sorted by the one-month

change: Cadence Design Systems is first (+15.2%, forward P/E 69.12, ROIC 33.3%, hi52 percentile 63.5). Apple appears with px_last 342.94, +2.0% over the month, forward P/E 45.62, ROIC 106.0% and hi52 percentile 99.6; the ROIC matches the 106.0% in Company 360's Key ratios. Two warnings explain the result: `members('S&P 500')` is the mvp_us approximation, and `pe_fwd` is model-based, not consensus. Workday's forward P/E of −631.43 shows why you should add `pe_fwd > 0` before sorting or filtering on it.

Now add `with(dates=range(-2Y, 0D), freq=M)`: the query becomes a time series with one row per security and month-end, and `pe_fwd` is refused with "pe_fwd has no point-in-time history (snapshot only): it cannot be used with dates=", because the forward P/E is a model output stored only for the snapshot date. Replace it with `pe_ttm`, which has history.

Limitations and pitfalls

- `members('S&P 500')` and `members('STOXX 600')` are **approximations**: the app's mvp_us and eu_broad universes (top names by market cap, current membership), not the official constituents. The result warns about this. Using them with a past `as_of` introduces **survivorship bias**.
- Identity attributes (sector, country, region, subsector) are **current**, not point-in-time, so `sector('Tech')` in a 2019 screen uses today's classification.
- Units: `pct` fields are fractions; write `15%` (= 0.15) in conditions, not `15`. `pctile` fields are 0-100.
- A daily value whose last price bar is more than 31 days old is returned empty (stale listing).
- The console reads the snapshot: intraday prices and live news are not visible.
- Grouped results include an **N** column and a "—" group for securities without a value of the grouping field (Figure 14.4); read medians of small groups with care.
- History is per browser; saved queries live in the app state on the server.

Related mnemonics and BQL fields

- `BQL` opens the console; `BQL get(...) for(...)` runs a query from the command bar; `AAPL BQL` opens it pre-filled. See Mnemonic reference.
- Fields: Field dictionary. Other uses: BQL elsewhere in the app.

BQL language reference

Query structure

A query is a sequence of clauses. `get` and `for` are required; the others are optional and can come in any order:

```
get(px_last, chg_pct(px, 1M), pe_fwd, roic, tech.hi52_pct)
  for(members('S&P 500') & sector('Tech'))
  with(dates=range(-2Y, 0D), freq=W, currency=USD, fill=prev)
  where(roic > 15% and tech.ivol_pct < 30)
  sort(chg_pct desc) limit(50)
```

Clause	Meaning
`get(expr [as name], ...)`	The output columns: fields, expressions and functions, each optionally renamed with `as`.
`for(universe)`	The securities (see Universes).
`with(key=value, ...)`	Options: dates, frequency, currency, fill, as-of date, row cap.
`where(condition)`	Keeps rows that satisfy the condition.

Clause	Meaning
`group(expr, ...).agg(fn, ...)`	Aggregates per group (and per date in a time series).
`sort(expr [asc or desc], ...)`	Orders the rows; ascending unless you add `desc`.
`limit(n)`	Keeps the first n rows (in a time series: the first n securities).

Queries are limited to 8,000 characters. Line breaks and indentation are free.

Universes

Form	Meaning
`members('mvp_us')` or `members('S&P 500')`, `'SPX'`, `'SPX Index'`, `'sp500'`, `'US large caps'`	The app's US large-cap universe (approximates the S&P 500).
`members('mvp_global')` or `'global'`	The app's global universe.
`members('eu_broad')` or `'STOXX 600'`, `'SXXP'`, `'Europe'`	The app's broad European universe (approximates the STOXX Europe 600).
`members('full')` or `'all'`	Every active primary equity in the database.
`members('watch')`	Your watchlist(s).
`members('watchlist:NAME')`, `members('screen:NAME')`, `members('peers:GROUP')`	A named watchlist, a saved query of kind **screen**, or a peer group.
`region('EU')`, `country('DE')`, `sector('Tech')`, `subsector('...')`	Primary equities by region, country, sector or subsector. Sector accepts ids (`TECH`) or names and synonyms (`'Technology'`, `'banks'`, `'reits'`).
`type('etf')`, `exchange('XETR')`, `currency('EUR')`	By security type, listing MIC or trading currency.
`['AAPL US', 'SAP GY', 'SPX Index', 'EURUSD Curncy', 'CL1 Comdty']`	A ticker list (up to 1,000). Unresolved tickers are listed in the warnings.

Combine universes with `&` (intersection), `|` (union) and `-` (difference), with parentheses: `members('eu_broad') & sector('Industrials') - country('GB')`.

Ticker references accepted in lists and in `corr` / `beta`: `'AAPL US'`, `'AAPL US Equity'`, `'SAP GY'`, `'SAP.DE'`, `'7203 JT'`, `'SPX Index'`, `'EURUSD Curncy'`, `'CL1 Comdty'`, an internal id (`'IDX:^GSPC'`), an ISIN, or a plain ticker (`'AAPL'`: the best listing, preferring an exact ticker, the primary listing and the largest). Index names such as SPX, INDU, CCMP, NDX, RTY, VIX, SX5E, SXXP, DAX, CAC, UKX, NKY, HSI, KOSPI, IBEX, FTSEMIB, SMI, AEX, PSI20, OMX, SHCOMP, CSI300, IBOV, MEXBOL, SPTSX, AS51, TWSE, NIFTY, SENSEX, STI, DXY, MOVE and USGG10YR map to the app's index series; commodity generics (CL, CO, NG, HO, XB, GC, SI, HG, PL, PA, C, W, S, KC, SB, CT, CC, LC, LH, ES, NQ, TY, US, EC, BP, JY, BTC, TTF and others) map to continuous front-month futures. Market codes are listed under Command bar and mnemonics.

Options

Option	Values	Effect
`dates=`	`range(start, end)` or a single date; each a date (`2024-06-28`) or a duration relative to `as_of` (`-2Y`, `0D`)	Turns the query into a **time series**: one row per security and date. Without it, a cross-section at `as_of`. Spans up to 30 years.
`freq=`	`D`, `W`, `M`, `Q`	Grid of business days, Fridays, month-ends or quarter-ends (plus the end date). Default `D`. Ignored without `dates=` (with a warning). At most 8,000 dates.
`currency=`	an ISO code (`USD`, `EUR`, ...) or `local`	Converts `price` and `currency` fields with the FX history; `currency_usd` fields are always USD. Default: trading currency.
`fill=`	`prev` (default) or `none`	`prev` carries the last known value forward; `none` keeps only values published since the previous grid date.

Option	Values	Effect
`as_of=`	a date or a duration relative to the snapshot (`-6M`)	The knowledge cut-off: the query sees only what was known on that date. Later than the snapshot is clamped (warning); before 1980 is refused.
`max_rows=`	a positive integer	Row cap (default 5,000, maximum 100,000).

Conditions, grouping, sorting and limits

- `where()` filters the rows **before** any cross-sectional function in `get()` runs, so `cs_zscore(roic)` is computed over the rows that pass the filter.
- In a time series, `sort()` and `limit()` choose the top N **securities** by their value on the last date, and keep all their dates.
- `group(sector).agg(median)` aggregates every numeric output per group (per group and date in a time series). Aggregations: `median` , `mean` (or `avg`), `sum` , `min` , `max` , `count` , `std` . Grouped queries can only sort by an output column.
- Sorting by a name that matches several outputs is refused as ambiguous; rename with `as` .

Functions

TIME-SERIES FUNCTIONS

Windows are either a number of observations (`21` = 21 trading sessions on daily data) or a calendar duration (`1M`).

Function	Meaning
`chg(x, n or P)`	x minus x n observations ago (or at t – P).
`chg_pct(x, n or P)`	x / x[t – n] – 1, as a fraction; `chg_pct(px, 1M)` = one calendar month, `chg_pct(px, 21)` = 21 sessions.
`lag(x, n or P)`	x n observations ago, or at t – P.
`avg std min max sum median count (x, n or P)`	Trailing-window statistic (`mean = avg`). `min` / `max` with two plain expressions are `least` / `greatest` .
`zscore(x, n or P)`	(x – trailing mean) / trailing standard deviation.
`pct_rank(x, n or P)`	Percentile (0-1) of today's x within its trailing window.
`rolling(fn, x, n or P)`	Same as `fn(x, n or P)` for fn in avg, mean, std, min, max, sum, median, zscore.
`resample(x, W or M or Q, how=last)`	Aggregates x within the calendar week, month or quarter so far; `how` = last, first, mean, sum, min, max.
`ytd(x)`	x / last value of the previous calendar year – 1.
`corr(a, b, n or P)`	Rolling correlation of one-period returns of a and b (`returns=false` for levels); b can be a security, e.g. `'SPX Index'` .
`beta(a, b, n or P)`	Rolling OLS slope of a's returns on b's returns (same conventions).

CROSS-SECTIONAL FUNCTIONS

Computed per date over the universe rows that pass `where()` ; each takes an optional `by=field` (for example `by=sector`) to compute within groups.

Function	Meaning
`rank(x)`	Rank, 1 = largest.
`cs_rank(x)`	Percentile 0-1 (ascending).
`cs_zscore(x)`	Cross-sectional z-score.

Function	Meaning
`cs_demean(x)`	x minus the cross-sectional mean.
`cs_mean(x)` , `cs_median(x)`	Cross-sectional mean or median, repeated on every row.

SCALAR FUNCTIONS AND OPERATORS

- Scalar: `abs` , `ln` , `log` , `log10` , `exp` , `sqrt` , `sign` , `floor` , `ceil` , `round(x, n)` , `least(...)` , `greatest(...)` , `coalesce(...)` , `nz(x)` (null → 0), `if(condition, a, b)` , `clip(x, lo, hi)` , `isnull(x)` .
- Arithmetic: `+` , `-` , `*` , `/` , `^` (power), unary minus.
- Comparisons: `==` (or `=`), `!=` (or `<>`), `<` , `<=` , `>` , `>=` , `in [...]` , `not in [...]` .
- Logic: `and` (or `&`), `or` (or `|`), `not` . Inside `for()` , `&` , `|` and `-` are the universe set operators.

Literals

Literal	Examples
Numbers	`15` , `0.15` , `1e9` , `200e9`
Percentages	`15%` (= 0.15)
Durations	`1D` , `2W` , `3M` , `1Q` , `5Y` (negative with a minus: `-2Y`)
Dates	`2026-01-31`
Strings	`'AAPL US'` or `"AAPL US"` ; double a quote to escape it (`'O''Reilly'`)
Booleans and null	`true` , `false` , `null`
Lists	`['AAPL US', 'SAP GY']` , `[1, 2, 3]` (literal values only, up to 1,000)

Point-in-time rules

A grid date D only sees bars dated on or before D and rows whose knowledge date (filing, publication or as-of date) is on or before D:

- **Prices** (`px` , `px_last` , returns, moving averages, volatility): the daily bar of the last session on or before D; weekly bars before the 15-year daily window.
- **Fundamentals** (ratios, `fa.*` , `kpi.*`): the latest period known at D, preferring TTM, otherwise the fiscal year. A restatement of an older period never overwrites a newer period; a restatement of the same period counts from its own filing date.
- **Dated streams** (scores, factors, short interest, 13F, insider, sentiment, earnings, conformal bands): the latest row with its knowledge or as-of date on or before D (for example FINRA short interest from its publication date, 13F holdings from the quarter's filing cut-off).
- **Snapshot-only fields** (no stored history, e.g. `pe_fwd` , `grade` , `beta_1y` , `ml.*` , `macro.*`) are allowed only at the snapshot date. They are refused with `dates=` , with an earlier `as_of` , and inside a time-series function.
- **Identity attributes** (sector, country, region, ticker) are current, not point-in-time.

Limits

Limit	Value
Universe	25,000 securities
Universe × dates	2,000,000 cells
Daily bars read, including look-back	about 40,000,000
Grid dates	8,000; span 30 years

Limit	Value
Ticker list / literal list	1,000 items
Output rows	`max_rows`, default 5,000, maximum 100,000 (more rows are truncated with a warning)
Query length	8,000 characters
Timeout	`BIG360_BQL_TIMEOUT_S`, default 30 s (between 0.5 and 120 s); the statement is interrupted

Caching

Results are cached in memory (the 128 most recent results of up to 50,000 rows), keyed by the snapshot run id, a hash of the normalised query and a hash of the resolved universe. A cached result is marked "(cached)" in the result subtitle. A new snapshot invalidates the cache automatically, and reformatting a query (spaces, line breaks) still hits the cache.

Security

- Identifiers (tables, columns, SQL fragments of computed fields) come only from the field dictionary and the compiler's function table.
- Every literal you type (strings, numbers, dates, ticker lists) is sent as a **bound parameter**, never pasted into SQL. Explain shows parameter names and types only.
- The lexer rejects anything outside the grammar, including `;` and comments.
- Queries run on the read-only serving snapshot.

API

The same engine is available to scripts, Excel and the C# client:

Endpoint	Purpose
`POST /api/v1/bql`	Run a query (`query`, `max_rows`, `use_cache`).
`POST /api/v1/bql/explain`	Compile only: SQL, parameter names, estimates.
`POST /api/v1/bql/export`	Run and download (`format` = csv, xlsx or json).
`GET /api/v1/bql/fields`	Search the field dictionary (`q`, `group`, `limit`, `offset`); also returns the function list.
`GET /api/v1/bql/resolve`	Resolve a Bloomberg-style reference to a security and asset class.
`GET` / `POST` / `PUT` / `DELETE /api/v1/bql/saved`	Saved queries and screens.

Field dictionary

Source: `contracts/bql_fields.yaml`, generated from the data contracts and the serving schema plus a curated alias layer: **817 fields, 146 aliases, 616 with point-in-time history** (86 curated Bloomberg-style names such as `px_last`, `pe_fwd`, `mkt_cap_usd`, `div_yld`).

Families

Family (prefix)	Fields	Examples
Identity (no prefix)	22	`ticker` , `name` , `isin` , `country` , `region` , `sector` , `subsector` , `currency` , `security_type`
Price and return	20	`px` , `px_last` , `px_volume` , `chg_pct_1m` , `chg_pct_ytd` , `mom_12_1`
Technical (`tech.` , `setup.` , `bo.` , `zone.` , plain)	154	`hi52` , `sma200` , `vol_1y` , `tech.hi52_pct` , `setup.rsi14` , `bo.state`
Valuation (`val.` , `coc.` , plain)	62	`mkt_cap_usd` , `pe_ttm` , `pe_fwd` , `ev_ebit` , `fcf_yield` , `coc.wacc`
Fundamental (`fa.` , plain ratios)	165	`roic` , `gross_margin` , `revenue_growth_yoy` , `fa.revenue` , `fa.fcf`
Sector KPI (`kpi.`)	54	`kpi.rule_of_40` , `kpi.rpo_growth` , `kpi.inventory_days_change`
Scores (`c.` , `grade.` , `rank.` , `rank_sector.` , plain)	30	`composite` , `c.defensive` , `grade` , `moat` , `rank_sector.balanced`
Factors (`f.`)	30	`f.quality` , `f.value` , `f.momentum` , `f.quality.rank_sector`
Smart money (`sm.` , `own.` , `si.` , `shrt.`)	95	`sm.score` , `own.inst_own_pct` , `si.si_pct_out` , `si.days_to_cover`
Insider (`ins.` , plain)	19	`insider_signal` , `ins.cluster_buy_flag` , `ins.net_buy_value_usd_90d`
Sentiment (`sent.` , plain)	22	`sentiment` , `sent.sentiment_ewm_7d` , `sent.news_count_z` , `attention_z`
Earnings (`earn.` , `pead.` , plain)	65	`eps_surprise` , `earn.car_0_p5` , `pead.sue`
Conformal (`cf.`)	57	`cf.qm_20_90` , `cf.band_5d_90` , `cf.touch_20d`
Model (`ml.`) and macro exposure (`macro.`)	18	`ml.exp_excess_3m` , `ml.decile_3m` , `macro.beta_rates`
Technical score calibration (`tp.`)	57	`tp.technical_score_pct` , `tp.h20_hit_rate` , `tp.h20_gate`

Units

Unit	Meaning and display
`pct`	Fraction (0.15 = 15%); write `15%` in queries; shown as a percentage.
`pctile`	Percentile points 0-100.
`score`	Score (0-100 for factor and composite scores).
`price` , `currency`	Trading currency; converted with `currency=` .
`currency_usd`	US dollars.
`x` , `ratio` , `z`	Multiple, ratio, z-score (two decimals).
`shares` , `count` , `days` , `rank` , `decile`	Counts and ranks (integers).
`text` , `bool` , `date`	Labels, flags and dates.

Most useful fields

H = point-in-time history (usable with `dates=` and `as_of`); S = snapshot only; M = model-based.

Field	Family	Unit	Hist.	Meaning
`ticker`	identity	text	S	Display ticker (Yahoo-style suffix outside the US)
`name`	identity	text	S	Security name (alias `long_comp_name`)
`country`	identity	text	S	Issuer country, ISO2 (current)

Field	Family	Unit	Hist.	Meaning
`region`	identity	text	S	US, CA, LATAM, UK, EU, JP, KR, CN, HK, APAC, OTHER
`sector`	identity	text	S	Sector id (TECH, HLTH, FINS, ...; alias `gics_sector`)
`currency`	identity	text	S	Trading currency
`px`	price	price	H	Adjusted close (splits and dividends)
`px_last`	price	price	H	Last unadjusted close (alias `close`)
`turnover_usd`	price	currency_usd	H	Close in USD × volume
`chg_pct_1m`	return	pct	H	21-session return (alias `ret_1m`)
`chg_pct_3m`	return	pct	H	63-session return
`chg_pct_12m`	return	pct	H	252-session return (alias `chg_pct_1y`)
`chg_pct_ytd`	return	pct	H	Year-to-date return
`mom_12_1`	return	pct	H	12-1 month momentum: px[t−21] / px[t−252] − 1
`hi52`	technical	price	H	52-week high (alias `high_52week`)
`px_to_hi52`	technical	ratio	H	Adjusted close / 52-week high
`trend_200d`	technical	pct	H	px / 200-session average − 1
`vol_1y`	technical	pct	H	Annualised volatility, 252 sessions
`beta_1y`	technical	x	S	1-year beta vs the region benchmark
`atr14`	technical	price	S	Average true range, 14 sessions
`tech.hi52_pct`	technical	pctile	H	52-week-high factor percentile (passed gate)
`tech.ivol_pct`	technical	pctile	H	Idiosyncratic volatility percentile (passed gate)
`tech.maxret_pct`	technical	pctile	H	MAX (largest daily return) percentile (passed gate)
`setup.rsi14`	technical	number	H	Wilder RSI(14)
`bo.state`	technical	text	S	Status of the latest 20 / 55-day breakout
`mkt_cap_usd`	valuation	currency_usd	H	Market capitalisation (alias `cur_mkt_cap`)
`pe_ttm`	valuation	x	H	Price / trailing EPS
`pe_fwd`	valuation	x	S, M	Price / next-year EPS of the base-case forecast, not consensus
`pb`	valuation	x	H	Price / book
`ev_ebit`	valuation	x	H	EV / EBIT TTM
`fcf_yield`	valuation	pct	H	Free cash flow TTM / market cap
`div_yld`	valuation	pct	H	Dividend yield (alias `eqy_dvd_yld_ind`)
`dcf_upside`	valuation	pct	S, M	Fair value / last close − 1
`roic`	fundamental	pct	H	NOPAT / average invested capital
`gross_margin`	fundamental	pct	H	Gross profit / revenue
`fcf_margin`	fundamental	pct	H	Free cash flow / revenue
`revenue_growth_yoy`	fundamental	pct	H	Revenue growth year on year
`eps_growth_yoy`	fundamental	pct	H	Diluted EPS growth year on year
`net_debt_to_ebitda`	fundamental	x	H	Net debt / EBITDA
`fa.revenue`	fundamental	currency_usd	H	Revenue, TTM (FY otherwise), USD
`kpi.rule_of_40`	sector KPI	pct	H	Revenue growth + FCF margin

Field	Family	Unit	Hist.	Meaning
`composite`	score	score	H	Composite score 0-100, balanced profile
`grade`	score	text	S	Composite grade A-E
`moat`	score	text	S	Moat class: wide, narrow, none, insufficient_data
`f.quality`	factor	score	H	Quality factor 0-100 (sector-relative)
`f.value`	factor	score	H	Value factor 0-100
`f.momentum`	factor	score	H	Momentum factor 0-100
`sm.score`	smart money	score	H, M	Smart-money composite (mean group z)
`own.inst_own_pct`	smart money	pct	H	13F shares held / shares out
`si.si_pct_out`	smart money	pct	H	FINRA short interest / shares out
`si.days_to_cover`	smart money	days	H	FINRA days to cover
`insider_signal`	insider	score	H	Rules-based insider signal score
`ins.cluster_buy_flag`	insider	bool	S	Three or more insiders buying
`sent.sentiment_ewm_7d`	sentiment	number	H	7-day EWM of daily news sentiment
`sent.news_count_z`	sentiment	z	H	News attention z vs own 90 days
`attention_z`	sentiment	z	S	GDELT media attention z
`eps_surprise`	earnings	pct	H, M	EPS surprise vs the model-based expectation
`earn.car_0_p5`	earnings	number	H, M	CAR [0, +5] after the latest release
`pead.sue`	earnings	number	H, M	Standardised unexpected EPS (model-based)
`ml.exp_excess_3m`	model	pct	S, M	ML expected 3-month excess return
`macro.beta_rates`	macro	x	S, M	Beta to +100 bp in the 10-year yield
`cf.band_5d_90`	conformal	pct	H, M	Half-width of the 90% 5-session calibrated band
`cf.touch_20d`	conformal	pct	H, M	20-session move to the 90% band edge

Use the **Fields** tab to search the remaining fields; every field shows its point-in-time rule on hover.

Snapshot-only fields

182 of the 817 fields (about a fifth) have no history and can only be read at the snapshot date. They include most model outputs and current-state labels: `pe_fwd`, `fair_value`, `dcf_upside`, `implied_cagr`, `shares_out`, `grade`, `rank_global`, `moat`, `moat_score`, `beta_1y`, `atr14`, `volume_z`, `new_hi52`, `ml.*`, `macro.*`, `ins.cluster_buy_flag`, `ins.anomaly`, `attention_z`, `news_count_30d`, `setup.trend`, `bo.*` and `zone.*`. Identity attributes (23 fields) also have no history but are always allowed, as current values. A query that uses one of them with `dates=`, an earlier `as_of` or inside a time-series function stops with "‹field› has no point-in-time history (snapshot only): it cannot be used ...".

Command bar and mnemonics

Opens with: Ctrl+K anywhere, or `/` when the focus is not in a text field; the search box in the top bar · **Menu:** none (global) · **Mnemonics:** all functions below · **Refresh:** not applicable

Purpose

The command bar is the fastest way around the app. Analysts used to a Bloomberg terminal can type a security and a function (`SAP GY FA` , `SX5E Index GP`) or a market-wide function (`ECO US` , `WIRP EU`) and press Enter; anyone else can type a company name and pick it from the list. It also runs BQL queries, explains every function with `HELP <code>` and opens this user guide with `HELP` .

Screenshot

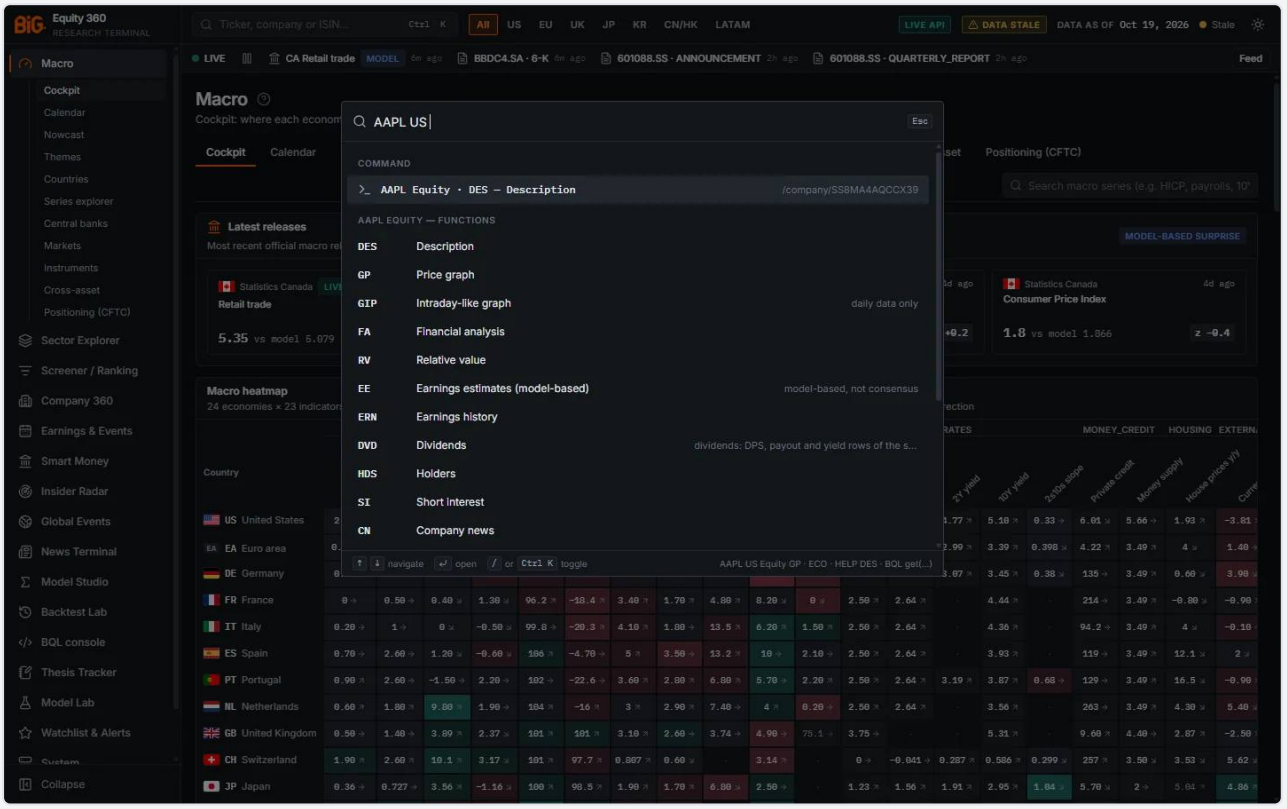


Figure 14.10: `AAPL US` *with a trailing space: the Command item "AAPL Equity · DES — Description" (with its Company 360 path) and the "AAPL EQUITY — FUNCTIONS" list with notes such as "daily data only" and "model-based, not consensus".*

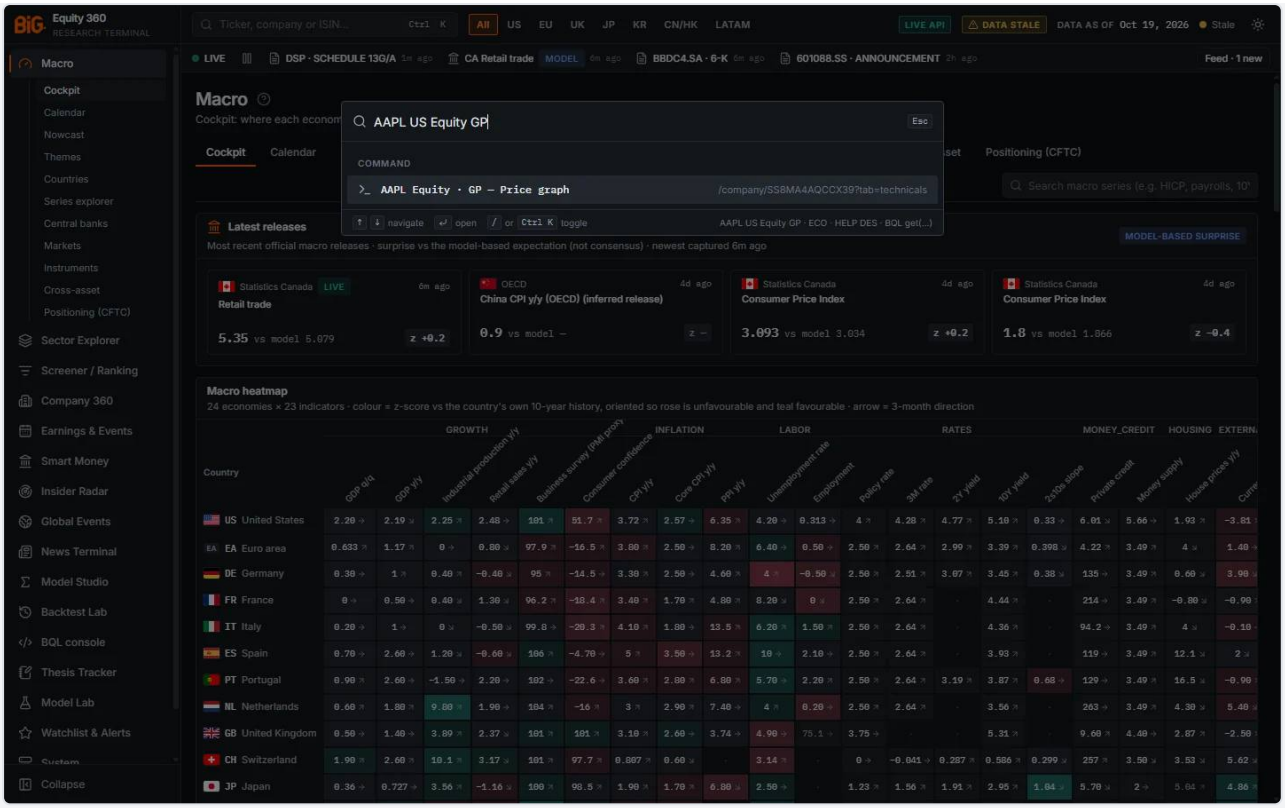


Figure 14.11: `AAPL US Equity GP` resolved to "AAPL Equity · GP — Price graph", destination `/company/<id>?tab=technicals`.

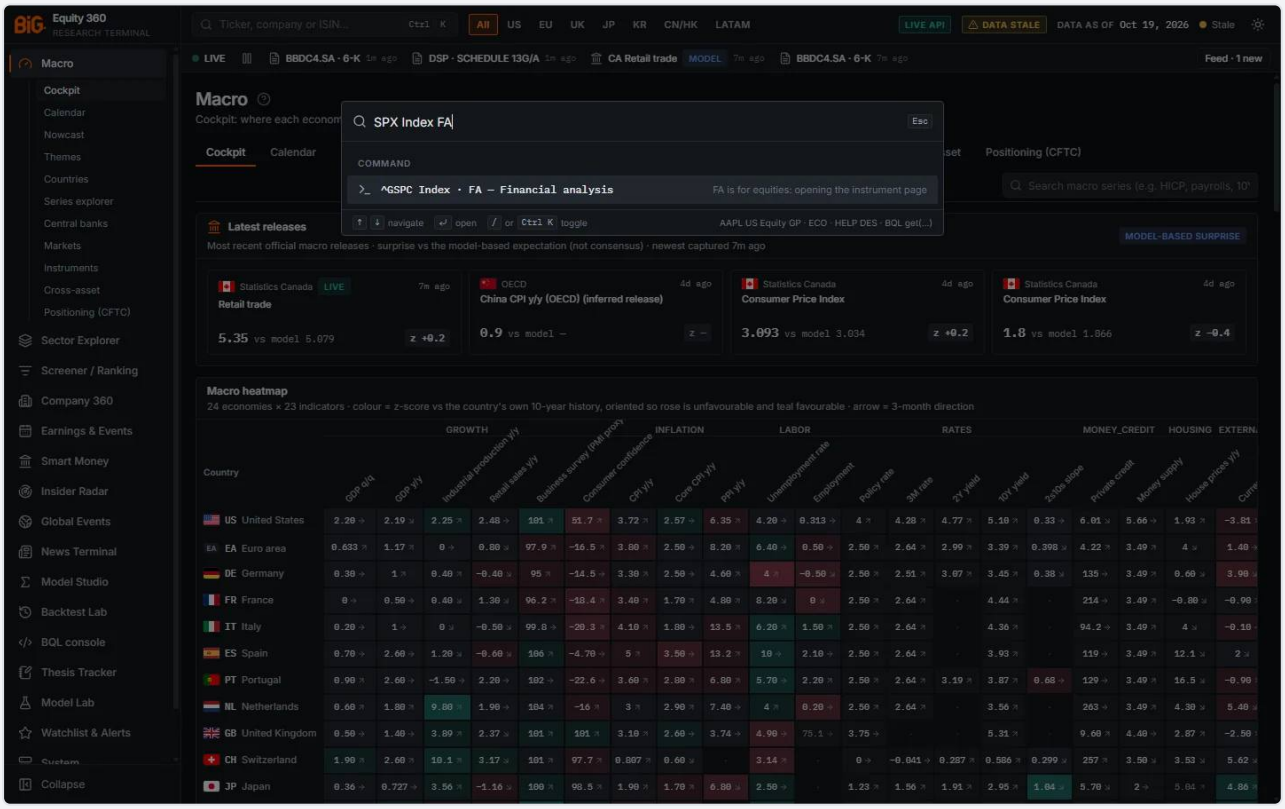


Figure 14.12: `SPX Index FA`: the index resolves to ^GSPC and the note reads "FA is for equities: opening the instrument page".

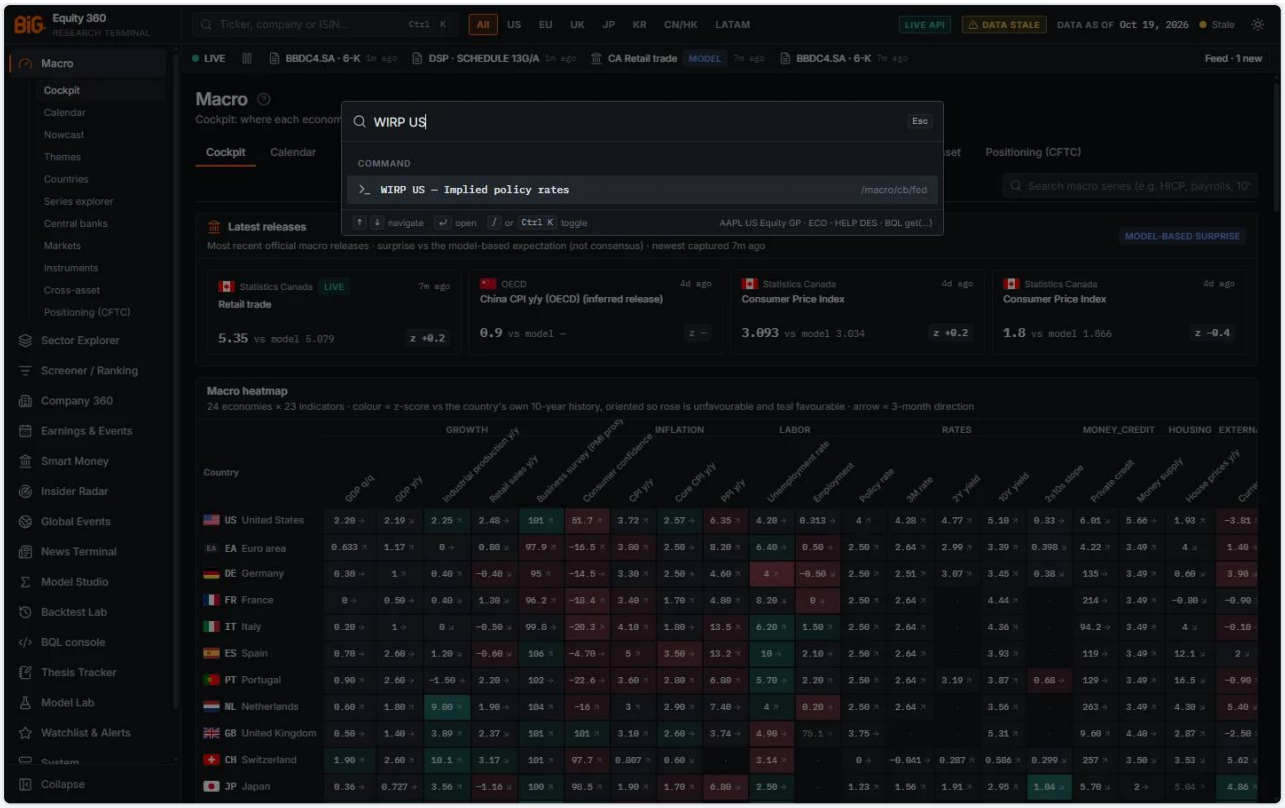


Figure 14.13: `WIRP US` → "WIRP US — Implied policy rates", destination `/macro/cb/fed`.

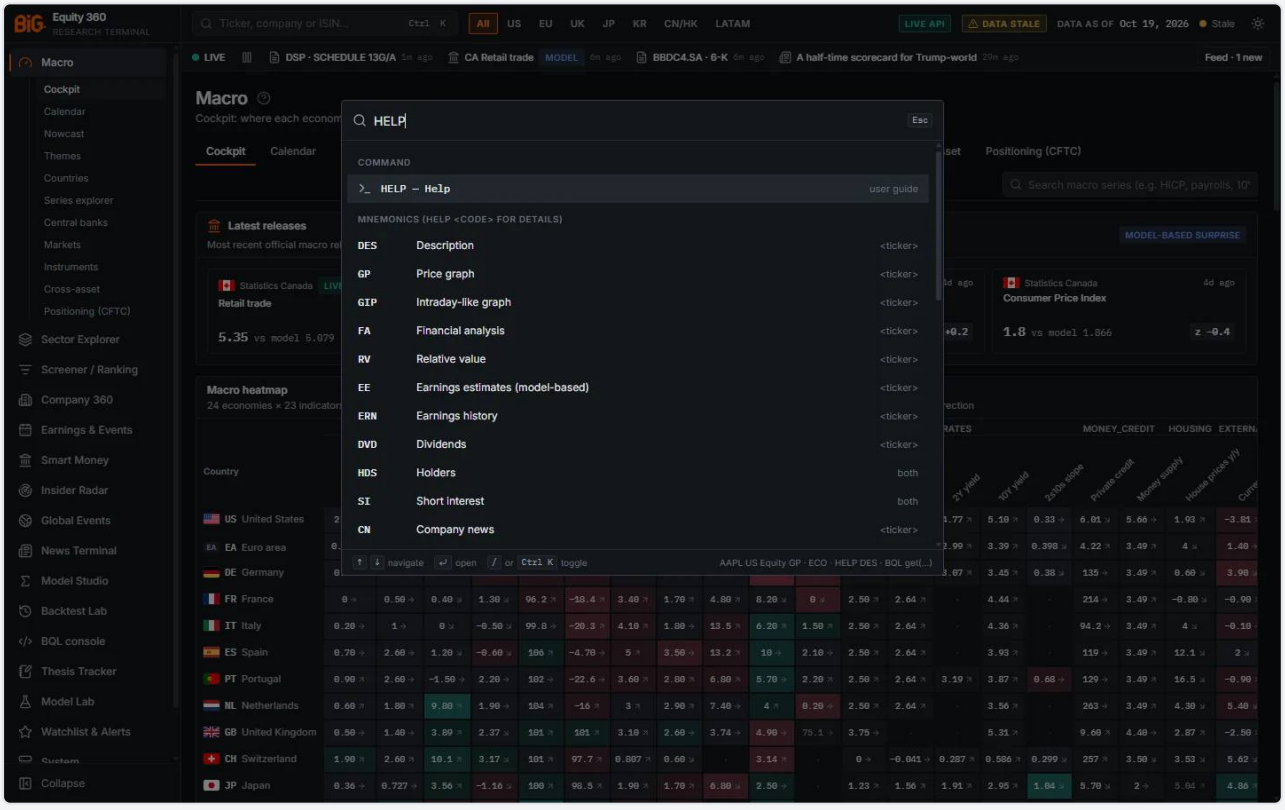


Figure 14.14: `HELP` : the Command item "HELP — Help" (note "user guide") above the list of mnemonics with their scope (`<ticker>` , both, global).

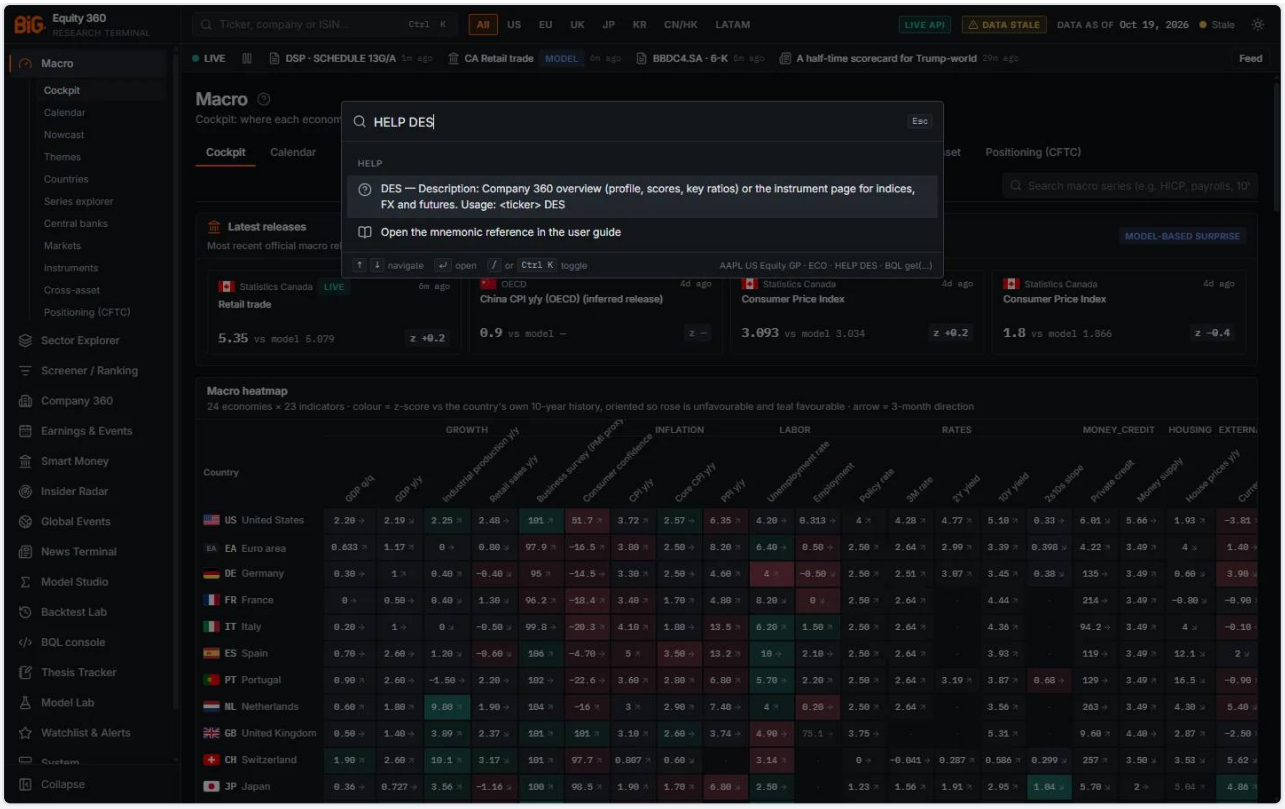


Figure 14.15: `HELP DES` : the one-line description and "Open the mnemonic reference in the user guide".

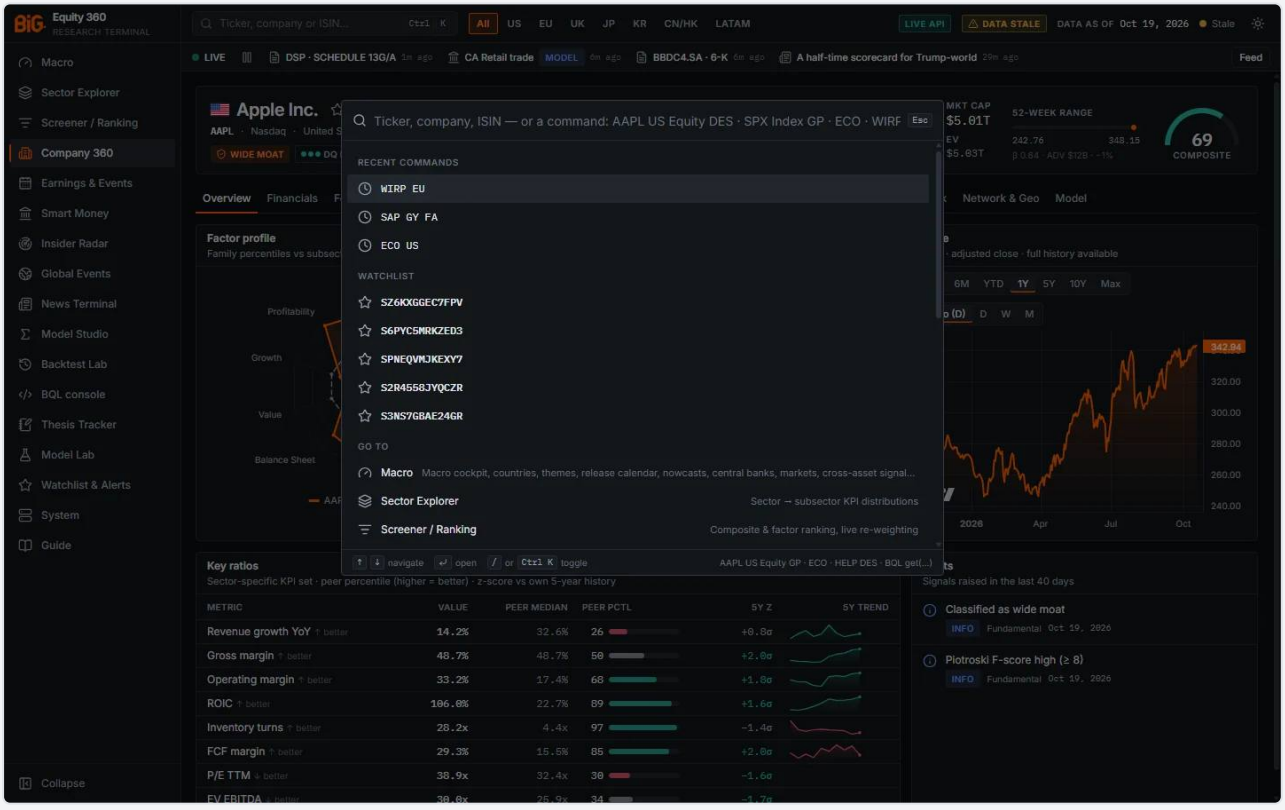


Figure 14.16: Empty input on Company 360 (Apple): Recent commands (WIRP EU, SAP GY FA, ECO US), Watchlist and Go to.

These images have no numbered badges: the command bar is a single dialog over the page.

Controls

Control	What it does
Ctrl+K	Opens or closes the command bar from anywhere.
`/`	Opens it when the focus is not in an input, select or text area.
↑ / ↓	Move through the items.
Enter	Opens the selected item.
Esc	Closes the bar (and clears the input).
Input	Placeholder: "Ticker, company, ISIN — or a command: AAPL US Equity DES · SPX Index GP · ECO · WIRP US · HELP".

The list shows these groups, in order, when they apply:

Group	Content
Command	The parsed command, e.g. "AAPL Equity · GP — Price graph", with its destination path or a note on the right. Disabled (greyed) when it cannot route: "unavailable: no options data", "not applicable", or "type a ticker first: AAPL DES".
Help	For `HELP <code>` : the description and usage (selecting it puts the function in the input), then **Open the mnemonic reference in the user guide**, which opens this chapter at Mnemonic reference (`/help/14-bql-mnemonics#mnemonic-reference`).
Mnemonics (HELP <code> for details)	For `HELP` : below the Command item "HELP — Help" (which opens the user guide), every function with its title and scope (global, `<ticker>` or both). Selecting one puts `HELP <code>` in the input.
<TICKER> <class> — functions	After a security and a space: the functions for its asset class, each with its note.
Functions	When you type 1-5 letters: market-wide functions whose code starts with them.
Securities	The ordinary security search (ticker, name, ISIN, exchange).
Recent commands	With an empty input: your last 6 commands (up to 20 are kept in this browser). Selecting one puts it back in the input.
Watchlist	With an empty input: up to 8 watchlist names.
Go to	Pages whose name or description matches the input.
Actions	Switch to light / dark theme; Open example: Microsoft 360; Command bar help (mnemonics), which types `HELP` ; **Open the user guide** (`/help`).

The footer reminds you of the keys and shows examples: "AAPL US Equity GP · ECO · HELP DES · BQL get(...)".

Methodology

Input forms (case-insensitive):

Form	Example	Result
`<ticker> [<market>] [<yellow key>] <MNEMONIC>`	`AAPL GP` , `AAPL US Equity DES` , `SAP GY FA` , `7203 JT TECH` , `SX5E Index GP` , `EURUSD Curncy GP` , `CL1 Comdty GP`	The function for that security.
`<MNEMONIC> [argument]`	`ECO` , `ECO US` , `WIRP EU` , `ECST DE` , `N` , `TOP`	A market-wide function.
`<MNEMONIC>` alone, security function	`DES` , `GP` , `FA`	Applies to the security of the page you are on (Company 360 or an instrument page); elsewhere it asks for a ticker. Functions that work both ways (`HDS` , `SI` , `SMRT` , `PEERS` , `NET` , `EVTS` , `THES` , `BQL`) also apply to the page's security when you are on one.
A security with a market or yellow key	`AAPL US` , `SPX Index` , `RY CN`	Enter opens `DES` .

Form	Example	Result
A security followed by a space	`AAPL US`	Lists the functions for that asset class.
`BQL ...` or `get(...)`	`BQL get(px_last) for(['AAPL US'])`	Opens the BQL console and runs the query.
`HELP`, `HELP <code>`	`HELP`, `HELP WIRP`	`HELP`: Enter opens the user guide (`/help`), and the function list is shown below. `HELP <code>`: one description, plus a link to the mnemonic reference.
Anything else	`ASML`, `Apple`, an ISIN	Falls back to the security search and page matches.

Yellow keys: `Equity`, `Index`, `Curncy`, `Comdty`. They tell the resolver which kind of instrument you mean; without one, the resolver infers it (six letters such as `EURUSD` are tried as a currency pair, known index names as indices). Generic commodity tickers may contain a space (`C 1 Comdty`).

Market codes (Bloomberg exchange codes):

Region	Codes
United States	`US` (composite), `UN` (NYSE), `UW` `UQ` `UR` (Nasdaq), `UA` (NYSE American), `UP` (NYSE Arca), `UF` (Cboe BZX)
Canada	`CN` (Toronto and TSX Venture), `CT` (Toronto), `CV` (TSX Venture)
UK and Ireland	`LN` (London), `ID` (Dublin)
Germany	`GY` (Xetra), `GR` (Xetra and Frankfurt), `GF` (Frankfurt)
Western Europe	`FP` (Paris), `NA` (Amsterdam), `BB` (Brussels), `PL` (Lisbon), `IM` (Milan), `SM` (Madrid), `SW` `SE` `VX` (SIX Swiss), `AV` (Vienna)
Nordics and Baltics	`SS` (Stockholm), `DC` (Copenhagen), `FH` (Helsinki), `NO` (Oslo), `IR` (Iceland), `ET` (Tallinn), `LH` (Vilnius)
Central and Eastern Europe, Greece, Turkey	`PW` (Warsaw), `CP` (Prague), `HB` (Budapest), `RO` (Bucharest), `GA` (Athens), `TI` (Istanbul)
Japan, Korea	`JP` `JT` (Tokyo), `KS` (Korea Exchange), `KQ` (KOSDAQ)
Greater China	`HK` (Hong Kong), `CH` (Shanghai and Shenzhen), `CG` `C1` (Shanghai), `CS` `C2` (Shenzhen), `TT` (Taiwan, TWSE and TPEx), `TP` (TPEx)
Rest of Asia-Pacific	`AU` `AT` (ASX), `SP` (Singapore), `IN` (NSE and BSE), `IS` (NSE), `IB` (BSE)
Latin America	`BZ` `BS` (B3), `MM` (Mexico), `CI` (Santiago), `CB` (Colombia), `PE` (Lima), `AR` (Buenos Aires)
Middle East and Africa	`IT` (Tel Aviv), `AB` (Saudi Exchange), `SJ` (Johannesburg)

`CN` is both the Canada code and the company-news function. On a two-word input, `CN` is read as the market: `RY CN` is Royal Bank of Canada's Toronto listing. For company news, use three or more words: `AAPL US CN`, `RY CN CN`.

Resolution. The security part is resolved on the server (`GET /api/v1/bql/resolve`) to a security and an asset class (FX → Curncy, futures → Comdty, indices → Index, everything else including ETFs → Equity). Indices, FX pairs, futures and ETFs open the instrument page; equities open Company 360. If nothing matches, the bar shows "Unknown security “‹text›”."

Data sources and refresh

The security search and resolver read the serving snapshot. Recent commands are kept in this browser only (key `big360.cmd.history`, 20 entries); they are lost if browser storage is cleared and are not shared between users.

Validation status

Not applicable (navigation). The parser and router are covered by unit tests (`frontend/src/lib/__tests__/mnemonics.test.ts`) and by the headless-browser check of Roadmap v3 B2.

Correct usage and worked example

Type the security first and the function last, as on a Bloomberg terminal, and use the trailing-space list when you do not remember a code.

Worked example. Press Ctrl+K and type `SAP GY` (with a space). The bar resolves SAP's Xetra listing and lists its functions. Add `FA` and press Enter: Company 360 opens on SAP's Financials tab, and `SAP GY FA` is saved in Recent commands. On that page, press Ctrl+K and type just `RV` : it applies to SAP and opens the Peers tab. Then type `WIRP EU` : the Central banks page for the ECB opens with its implied policy path (an approximation from official curves, see Macro › Central bank page). Finally type `HELP EE` : the bar explains "EE — Earnings estimates (model-based): Three-statement forecast and DCF scenarios — MODEL-BASED, not consensus. Usage: ‹ticker› EE".

Limitations and pitfalls

- A function typed on its own uses the **page's security**; on a page without one (for example the Screener), security-only functions are disabled until you type a ticker.
- `GIP` opens the same daily price graph as `GP` (the app has no intraday data).
- Equity functions on an index, FX pair or future (for example `SPX Index FA`) open the instrument page instead, with a note.
- `OMON` is listed but unavailable: the app has no options data.
- `DES AAPL` (function first) is not a command; it falls back to search. Put the ticker first.
- `N` , `TOP` and `CN` fall back to Global Events or the News & Sentiment tab when the News Terminal is not installed, with a note.

Related mnemonics and BQL fields

- Full list: Mnemonic reference. BQL from the bar: BQL console.
- Keyboard shortcuts of the app shell: Getting started.

Mnemonic reference

Every function in `frontend/src/lib/mnemonics.ts` . "Typed alone" is what happens with no ticker: the page's security on Company 360 or an instrument page, otherwise the market-wide destination.

Security functions

Code	Title	Classes	With a security	Typed alone (no page security)
`DES`	Description	all	Company 360 Overview; instruments: instrument page	needs a ticker
`GP`	Price graph	all	Company 360 › Technicals; instruments: instrument page	needs a ticker
`GIP`	Intraday-like graph	all	as `GP` , note "daily data only"	needs a ticker
`TECH`	Technicals	all	Company 360 › Technicals; instruments: instrument page	needs a ticker

Code	Title	Classes	With a security	Typed alone (no page security)
`FA`	Financial analysis	Equity	Company 360 › Financials	needs a ticker
`RV`	Relative value	Equity	Company 360 › Peers	needs a ticker
`EE`	Earnings estimates (model-based)	Equity	Company 360 › Forecast, note "model-based, not consensus"	needs a ticker
`ERN`	Earnings history	Equity	Company 360 › Earnings	needs a ticker
`DVD`	Dividends	Equity	Company 360 › Financials, note "dividends: DPS, payout and yield rows of the statements"	needs a ticker
`CN`	Company news	Equity	News Terminal `/news?ticker=‹ticker›` ; without it, Company 360 › News & Sentiment	needs a ticker
`HDS`	Holders	Equity	Company 360 › Ownership	Smart Money (`/smart-money`)
`SI`	Short interest	Equity	Company 360 › Ownership	Smart Money
`SMRT`	Smart money	Equity	Company 360 › Ownership	Smart Money
`PEERS`	Peers	Equity	Company 360 › Peers	Sector Explorer (`/sectors`)
`NET`	Network & geo	Equity	Company 360 › Network & Geo	Global Events › Contagion & spillovers
`EVTS`	Events	Equity	Company 360 › Event risk	Earnings & Events (`/earnings`)
`THES`	Thesis	all	New thesis on the security (`/theses/new?subject_type=security` ; instruments: `cross_asset`)	Thesis Tracker (`/theses`)
`BQL`	BQL console	all	`/bql` pre-filled with `get(px_last, chg_pct_1m, pe_ttm, roic) for(['‹security›'])`	BQL console (`/bql`)
`OMON`	Option monitor	all	**unavailable: no options data**	unavailable

For an index, FX pair or future, every equity-only function opens the instrument page with the note "‹code› is for equities: opening the instrument page". After a security and a space, the bar suggests the functions of its class: all of the above for an equity; `DES` , `GP` , `GIP` , `TECH` , `THES` , `BQL` and `OMON` (disabled) for an index, currency or commodity.

Market-wide functions

Code	Title	Opens	Argument
`WEI`	World equity indices	Markets overview (`/macro/markets`)	none
`WIRP`	Implied policy rates	Central banks (`/macro/cb`); with a country or bank, that bank's page with the implied path (approximation)	`US` , `EU` / `EZ` / `EA` (and euro-area members DE, FR, IT, ES, NL), `UK` / `GB` , `JP` , `CH` , `CA` , `AU` , `SE` , `NO` , `BR` , `MX` , `CL` , `CO` , `PE` , `IN` , `KR` , `CN` , or a bank: `FED` , `ECB` , `BOE` , `BOJ` , `SNB` , `BOC` , `RBA` , `RIKSBANK` , `NORGES` , `BCB` , `BANXICO` , `BCCH` , `BANREP` , `BCRP` , `RBI` , `BOK` , `PBOC` ; unknown values open the monitor with a note
`ECO`	Economic calendar	Release calendar (`/macro/calendar`), model-based expectations	optional country: `ECO US`
`ECST`	Economic statistics	Country macro dashboard (`/macro/country/US` by default)	optional ISO2: `ECST DE`
`GC`	Curves	Central banks (`/macro/cb`), note "policy and yield curves per bank"	none
`FXIP` (alias `FX`)	FX	Instruments browser on FX (`/macro/markets?tab=instruments`)	none

Code	Title	Opens	Argument
`CMDTY`	Commodities	Cross-asset dashboard (`/macro/cross-asset`)	none
`COT`	CFTC positioning	Positioning (`/macro/positioning`)	none
`EQS`	Equity screening	Screener (`/screener`)	none
`PORT` (alias `BT`)	Portfolio / backtest	Backtest Lab (`/backtest`)	none
`TOP`	Top news	News Terminal › Top stories (`/news?view=top`); without it, Global Events	none
`N`	News terminal	News Terminal (`/news`); without it, Global Events	none
`ALRT`	Alerts	Watchlist & Alerts (`/watchlist`)	none
`MDL`	Model registry	Model Lab (`/models`)	none
`HELP`	Help	The user guide (`/help` , note "user guide"); the bar also lists every function. Description: "Open the user guide (and list the mnemonics); HELP ‹MNEMONIC› describes one"	optional code: `HELP DES` (one-line description and a link to the mnemonic reference)

HELP

- `HELP` with no argument shows a first Command item, "HELP — Help" with the note "user guide": Enter opens this user guide (`/help`). Below it, the bar lists every function with its title and scope; selecting one shows its description.
- `HELP <code>` shows one line: "‹CODE› — ‹Title›: ‹description›. Usage: ‹form›", where the form is `<ticker> CODE` , `CODE or <ticker> CODE` , or `CODE` . For example: "WIRP — Implied policy rates: Central-bank monitor; with a country / bank (WIRP US, WIRP EU) the bank page with the implied path (approximation). Usage: WIRP".
- Under the description, **Open the mnemonic reference in the user guide** opens this chapter at the mnemonic reference (`/help/14-bql-mnemonics#mnemonic-reference`).
- An unknown code gives "‹CODE›: unknown mnemonic. Type HELP for the list."
- The Actions group (empty input) also has **Open the user guide**, and typing `HE` lists `HELP` among the Functions, which opens the guide as well.

Unavailable functions

`OMON` (option monitor) is shown so that Bloomberg users are not surprised, but it is disabled: "Not available: the app has no options data". There is no intraday data either, so `GIP` is the daily graph.

Examples

Type	Opens
`AAPL GP`	Apple, Company 360 › Technicals
`AAPL US Equity DES`	Apple, Company 360 Overview
`SAP GY FA`	SAP (Xetra), Financials
`7203 JT TECH`	Toyota (Tokyo), Technicals
`ASML NA RV`	ASML (Amsterdam), Peers
`PETR4 BZ ERN`	Petrobras preferred (B3), Earnings

Type	Opens
`RY CN`	Royal Bank of Canada (Toronto), Overview
`AAPL US CN`	News Terminal filtered on AAPL
`SX5E Index GP`	Euro Stoxx 50 instrument page
`EURUSD Curncy GP`	EUR/USD instrument page
`CL1 Comdty GP`	WTI front-month instrument page
`ECO US`	US release calendar
`ECST DE`	Germany macro dashboard
`WIRP EU`	ECB page with the implied path
`COT`	CFTC positioning
`BQL get(chg_pct_ytd) for(['SPX Index', 'SX5E Index'])`	BQL console, query run
`HELP`	This user guide (`/help`)
`HELP SI`	Description of the short-interest function, with a link to the mnemonic reference

BQL elsewhere in the app

BQL expressions can be used in three other places.

Screener: Open as BQL

Route: `/screener` · button **Open as BQL** (hover: "Open this screen in the BQL console (the filters and sort as a get / for / where query)").

The button turns the Screener's current filters and sort into a query and runs it in the console:

- universe `members('full')` narrowed by the selected regions, country, sector and subsector;
- `where()` conditions for the size bucket (mega ≥ $200 bn, large $10-200 bn, mid $2-10 bn, small $0.3-2 bn, micro < $0.3 bn), the moat class and the minimum score;
- fields `composite, mkt_cap_usd, chg_pct_1m, pe_fwd, ev_ebit, roic, f.quality, f.value, f.momentum` plus the sort column, `limit(100)` .

When a minimum score is set, a toast notes that "min score uses the stored balanced composite (the Screener re-weights live)": the Screener's custom factor weights are not carried over. Save the query as a **screen** to reuse it as `members('screen:NAME')` . See Screener.

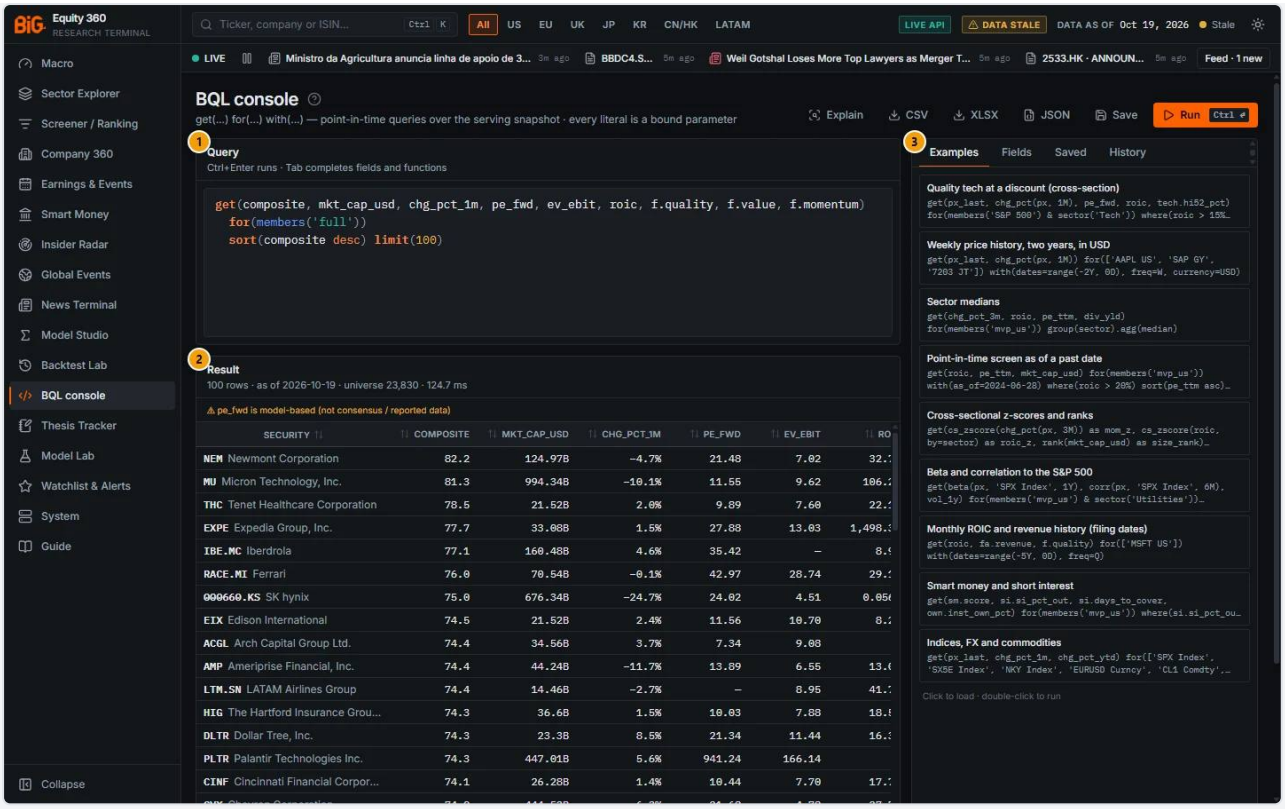


*Figure 14.17: *Open as BQL** from the Screener with no filters: the generated query over `members('full')`, 100 rows, universe 23,830, sorted by composite.*

#	Panel	What it shows
1	Query	`get(composite, mkt_cap_usd, chg_pct_1m, pe_fwd, ev_ebit, roic, f.quality, f.value, f.momentum) for(members('full')) sort(composite desc) limit(100)` .
2	Result	100 rows · as of 2026-10-19 · universe 23,830 · 124.7 ms, with the pe_fwd model-based warning; Newmont (composite 82.2) first.
3	Side panel	Examples tab.

Model Studio: BQL query variable source

In Model Studio's variable picker, the source **BQL query (expression, point-in-time on the serving snapshot)** adds any BQL expression as a dataset column ("BQL expression — evaluated point-in-time at every dataset date on the serving snapshot (cross-sectional functions run over the dataset's securities)"). Type an expression such as `cs_zscore(roic) - cs_zscore(pe_ttm)` and add it with the role you need (y, x, ...). Snapshot-only fields are refused for dated panels, as in the console. See Model Studio.

Backtest Lab: BQL expression signal source

In the Backtest Lab builder, the signal source **BQL expression (cross-sectional, point-in-time per rebalance)** takes an expression such as `cs_zscore(chg_pct(px, 3M)) - cs_zscore(pe_ttm)` (hover: "Evaluated point-in-time at every rebalance date (BQL-lite; see the BQL console for fields and functions)"). The expression is evaluated on each rebalance date with only the data known then, and cross-sectional functions run over the backtest universe. The run notes record how many rebalance dates were evaluated. See Backtest Lab.

15. System, Data Health and QA

The System section is where you check whether the numbers on every other page can be trusted today. It has three pages and one background guard:

Page or component	Route	Answers the question
System overview	`/system`	Is the scheduler running, when did the last refresh finish, how fresh is each source, which jobs failed?
Data Health	`/system/data-health`	What data should the app hold, what is missing or stale beyond its SLA, and how is the background gap filler repairing it?
Conformal validation	`/system/conformal`	Which conformal outputs passed their validation gate, and are they still hitting their target coverage in the nightly audit?
QA guard	panel on `/system/data-health` , CLI `python -m app.qa`	Did the API contract sweep, the golden data checks and the UI smoke suite pass after the last refresh?

All three pages are read-only views except one control: the schedule profile switch on the System page. Nothing on these pages shows an API key value; the Backend panel lists only key **names** with a tick or a cross.

System overview

Route: `/system` · **Menu:** System › Overview · **Mnemonics:** none · **Refresh:** status is read on page load and with the **Refresh** button; the Real-time layer panel polls every 30 s

Purpose

The System page is the operations dashboard of the app. Use it first thing in the morning, or whenever a number looks old, to confirm that the scheduler is on, that the last refresh completed, which data date each region is on, and whether any job failed. It is also where you switch the refresh schedule between the weekly and daily profiles. Portfolio managers need only the top panels (alerts, Scheduler, data as of); the source and job tables are for the app's owner and IT.

Screenshot

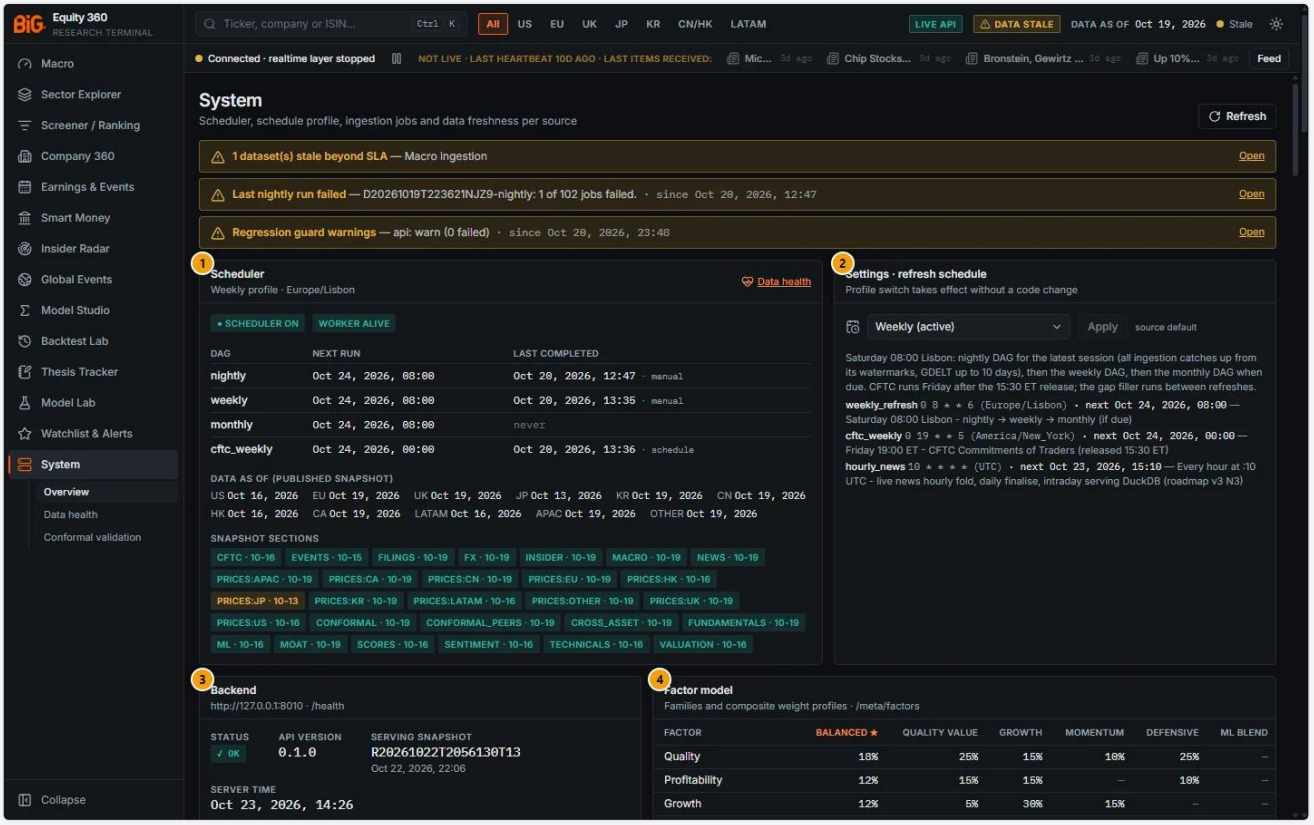


Figure 15.1: System overview (`/system`), status read on 23 Oct 2026 14:26 Lisbon time; serving snapshot R20261022T2056130T13, data as of 19 Oct 2026.

Above the numbered panels, the **ops alert banner** lists red (error) and amber (warn) alerts, each with an **Open** link (see "Alerts" under Methodology). In the figure it shows three amber alerts. Info alerts are not shown in the banner.

#	Panel	What it shows
1	Scheduler	Scheduler on / off, worker alive / down, catch-up due, the next run and last completed run (with its trigger) of each DAG (`nightly` , `weekly` , `monthly` , `cftc_weekly`), the published data date per region and the snapshot section badges. Link **Data health**.
2	Settings · refresh schedule	The schedule profile select (**Weekly**, **Daily**, **Intraday (placeholder)**) with **Apply**, where the active profile came from (`source default`), the profile description and its entries (cron, time zone, next run).
3	Backend	`/health` status, API version, serving snapshot id and build time, server time, API keys configured (names only, tick or cross), universe counts.
4	Factor model	The factor families and the composite weight profiles used by the composite score (default profile starred, here **Balanced**).
5	Real-time layer	Supervisor, bus, queue and heartbeat of the live layer (`GET /live/status`), one row per stream or consumer, and this browser's SSE connection with **Pause** / **Resume**.
6	Summary stats	Sources fresh (of all region rows), Stale, Errors, Job success rate over the last completed runs, Status as of.
7	Preferences & connection	Data mode (Live API or Mock), API base URL, the number and date locale select, and the chart / map / events attributions.
8	Data sources	Freshness of every source table against its expected cadence, **Per dataset** or **Per region**.
9	Job runs	The last 100 job runs (APScheduler ingestion and analytics jobs) with status, start, duration, rows and message; filter **All**, **Failed / partial**, **Running / queued**.

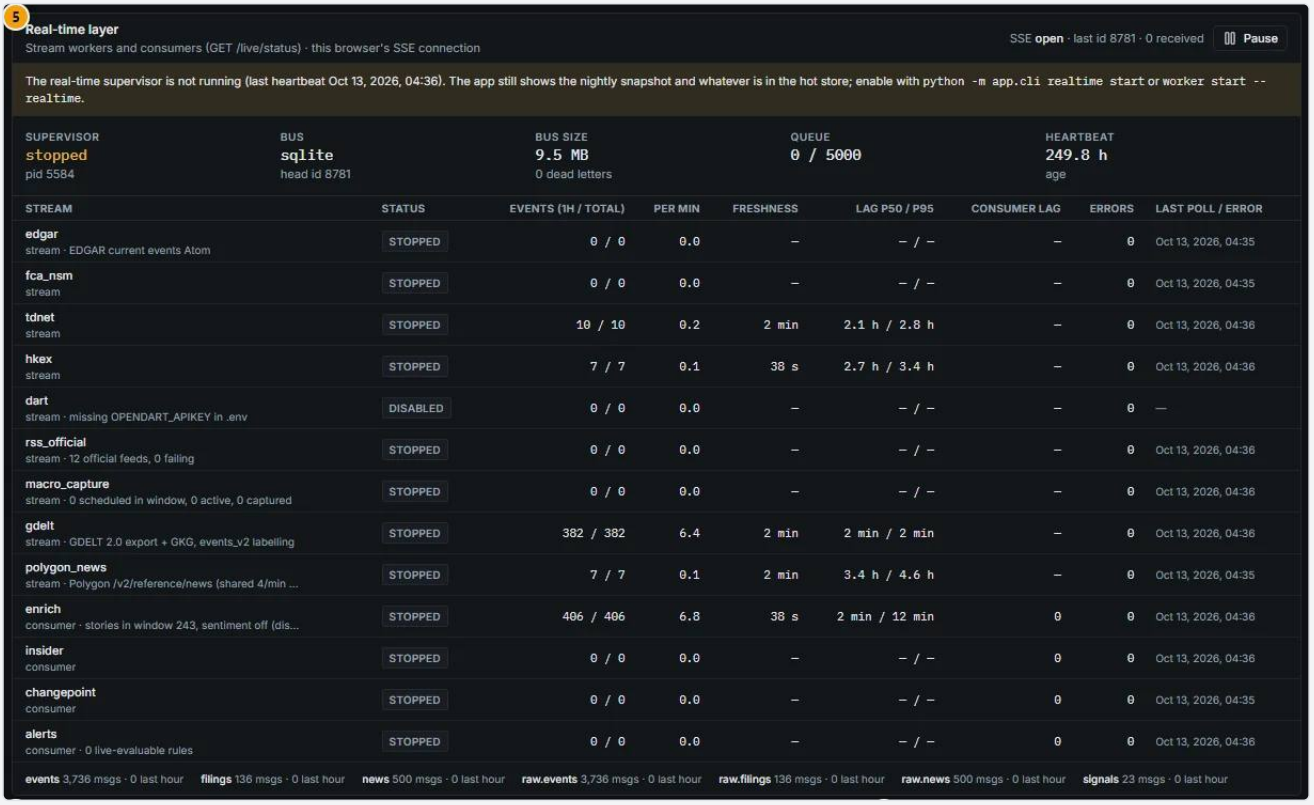
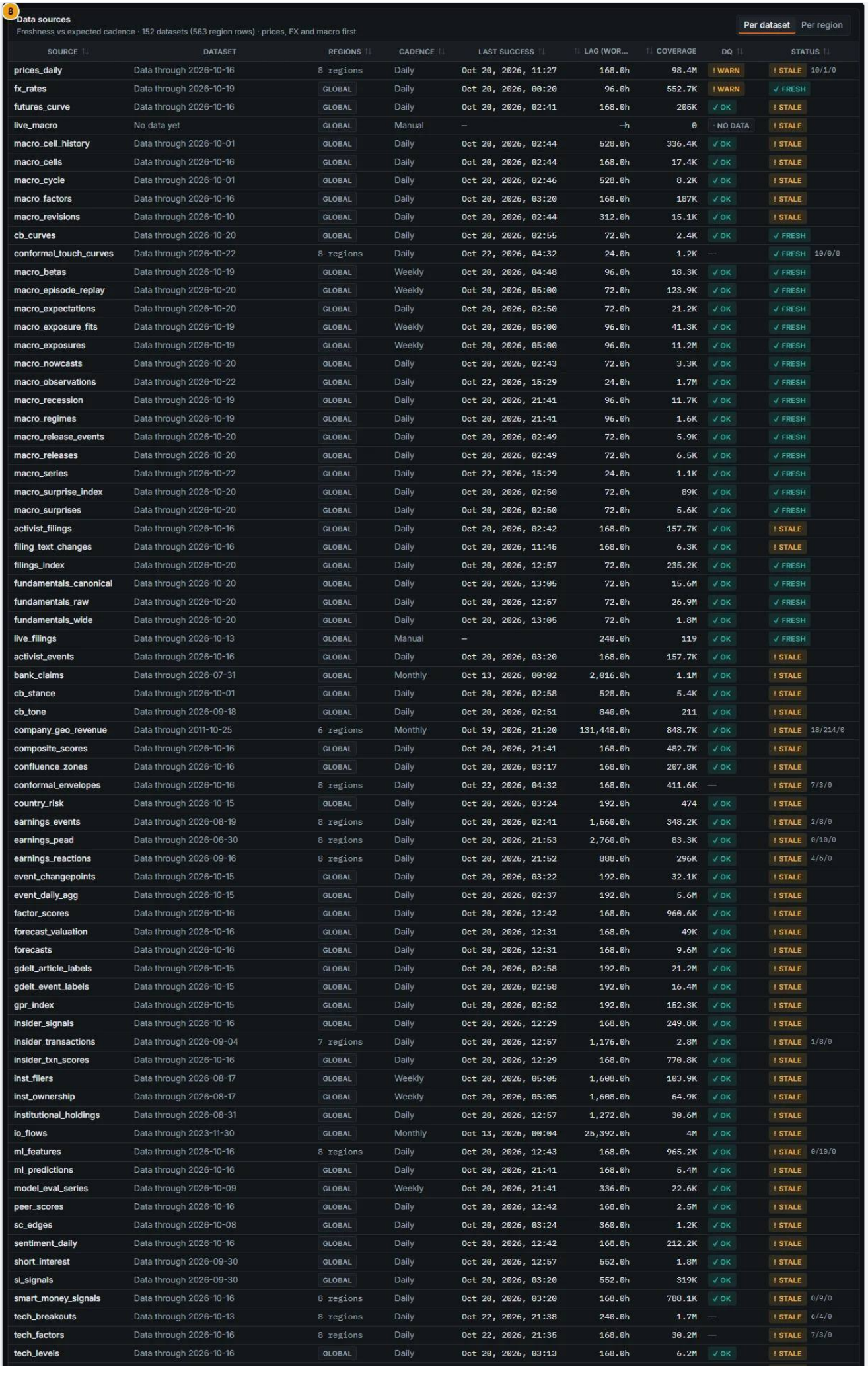


Figure 15.2: Real-time layer (panel 5). Here the real-time supervisor is stopped (last heartbeat 13 Oct 2026); the yellow notice says the app still shows the nightly snapshot and how to start the layer. The `dart` stream is disabled because its key is not configured.

Figure 15.3: Summary stats (panel 6): 221 of 563 source × region rows fresh, 342 stale, 0 errors, job success rate 77% over the last 83 completed runs.

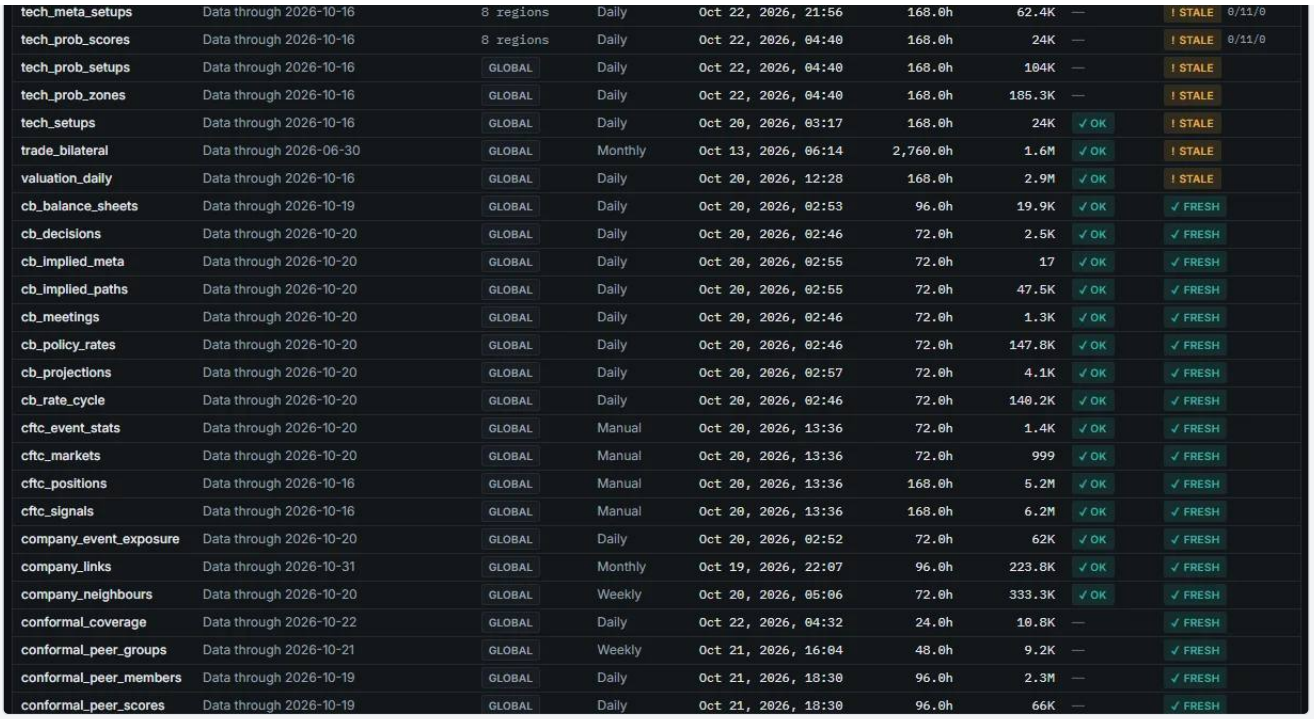


8 Data sources

Freshness vs expected cadence · 152 datasets (563 region rows) · prices, FX and macro first

Per dataset | Per region

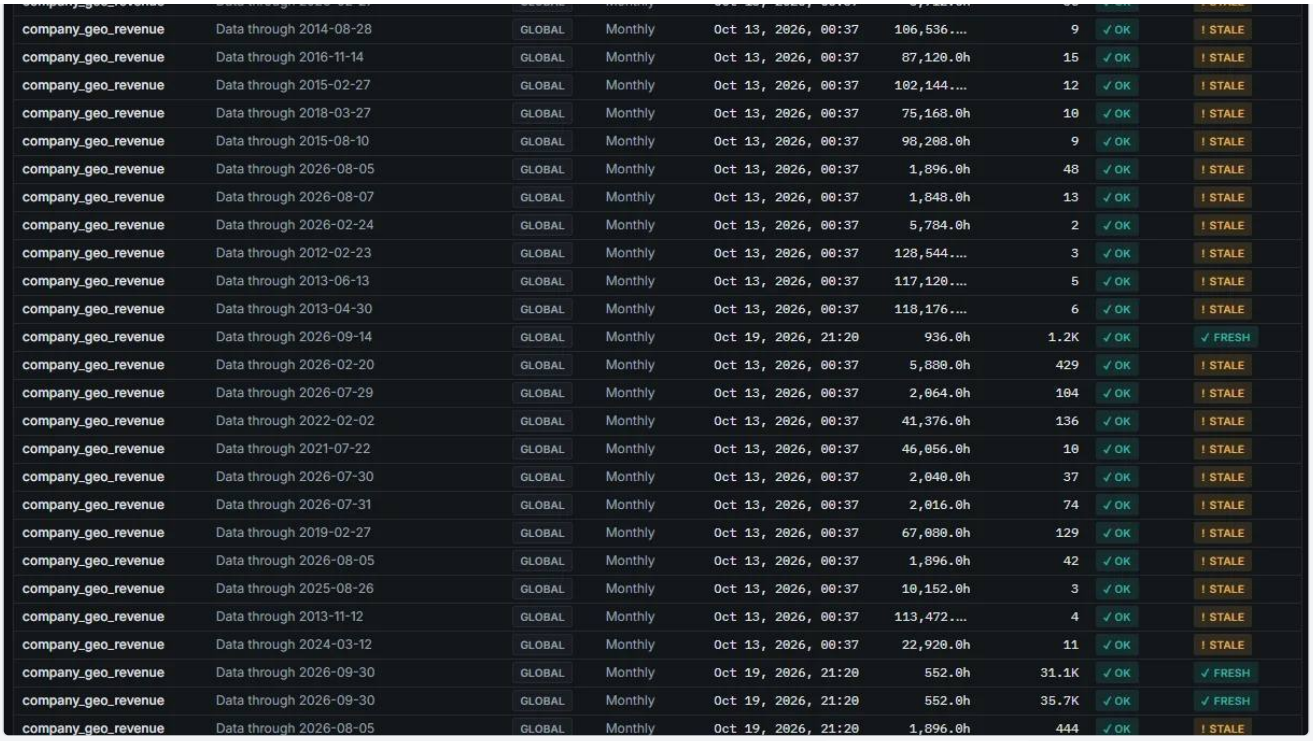
SOURCE	DATASET	REGIONS	CADENCE	LAST SUCCESS	LAG (WOR...	COVERAGE	DQ	STATUS
prices_daily	Data through 2026-10-16	8 regions	Daily	Oct 20, 2026, 11:27	168.0h	98.4M	! WARN	! STALE 10/1/0
fx_rates	Data through 2026-10-19	GLOBAL	Daily	Oct 20, 2026, 00:20	96.0h	552.7K	! WARN	✓ FRESH
futures_curve	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 02:41	168.0h	205K	✓ OK	! STALE
live_macro	No data yet	GLOBAL	Manual	—	—h	0	· NO DATA	! STALE
macro_cell_history	Data through 2026-10-01	GLOBAL	Daily	Oct 20, 2026, 02:44	528.0h	336.4K	✓ OK	! STALE
macro_cells	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 02:44	168.0h	17.4K	✓ OK	! STALE
macro_cycle	Data through 2026-10-01	GLOBAL	Daily	Oct 20, 2026, 02:46	528.0h	8.2K	✓ OK	! STALE
macro_factors	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 03:20	168.0h	187K	✓ OK	! STALE
macro_revisions	Data through 2026-10-10	GLOBAL	Daily	Oct 20, 2026, 02:44	312.0h	15.1K	✓ OK	! STALE
cb_curves	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:55	72.0h	2.4K	✓ OK	✓ FRESH
conformal_touch_curves	Data through 2026-10-22	8 regions	Daily	Oct 22, 2026, 04:32	24.0h	1.2K	—	✓ FRESH 10/0/0
macro_betas	Data through 2026-10-19	GLOBAL	Weekly	Oct 20, 2026, 04:48	96.0h	18.3K	✓ OK	✓ FRESH
macro_episode_replay	Data through 2026-10-20	GLOBAL	Weekly	Oct 20, 2026, 05:00	72.0h	123.9K	✓ OK	✓ FRESH
macro_expectations	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:50	72.0h	21.2K	✓ OK	✓ FRESH
macro_exposure_fits	Data through 2026-10-19	GLOBAL	Weekly	Oct 20, 2026, 05:00	96.0h	41.3K	✓ OK	✓ FRESH
macro_exposures	Data through 2026-10-19	GLOBAL	Weekly	Oct 20, 2026, 05:00	96.0h	11.2M	✓ OK	✓ FRESH
macro_nowcasts	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:43	72.0h	3.3K	✓ OK	✓ FRESH
macro_observations	Data through 2026-10-22	GLOBAL	Daily	Oct 22, 2026, 15:29	24.0h	1.7M	✓ OK	✓ FRESH
macro_recession	Data through 2026-10-19	GLOBAL	Daily	Oct 20, 2026, 21:41	96.0h	11.7K	✓ OK	✓ FRESH
macro_regimes	Data through 2026-10-19	GLOBAL	Daily	Oct 20, 2026, 21:41	96.0h	1.6K	✓ OK	✓ FRESH
macro_release_events	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:49	72.0h	5.9K	✓ OK	✓ FRESH
macro_releases	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:49	72.0h	6.5K	✓ OK	✓ FRESH
macro_series	Data through 2026-10-22	GLOBAL	Daily	Oct 22, 2026, 15:29	24.0h	1.1K	✓ OK	✓ FRESH
macro_surprise_index	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:50	72.0h	89K	✓ OK	✓ FRESH
macro_surprises	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:50	72.0h	5.6K	✓ OK	✓ FRESH
activist_filings	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 02:42	168.0h	157.7K	✓ OK	! STALE
filing_text_changes	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 11:45	168.0h	6.3K	✓ OK	! STALE
filings_index	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 12:57	72.0h	235.2K	✓ OK	✓ FRESH
fundamentals_canonical	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 13:05	72.0h	15.6M	✓ OK	✓ FRESH
fundamentals_raw	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 12:57	72.0h	26.9M	✓ OK	✓ FRESH
fundamentals_wide	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 13:05	72.0h	1.8M	✓ OK	✓ FRESH
live_filings	Data through 2026-10-13	GLOBAL	Manual	—	240.0h	119	✓ OK	✓ FRESH
activist_events	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 03:20	168.0h	157.7K	✓ OK	! STALE
bank_claims	Data through 2026-07-31	GLOBAL	Monthly	Oct 13, 2026, 00:02	2,016.0h	1.1M	✓ OK	! STALE
cb_stance	Data through 2026-10-01	GLOBAL	Daily	Oct 20, 2026, 02:58	528.0h	5.4K	✓ OK	! STALE
cb_tone	Data through 2026-09-18	GLOBAL	Daily	Oct 20, 2026, 02:51	840.0h	211	✓ OK	! STALE
company_geo_revenue	Data through 2011-10-25	6 regions	Monthly	Oct 19, 2026, 21:20	131,448.0h	848.7K	✓ OK	! STALE 18/214/0
composite_scores	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 21:41	168.0h	482.7K	✓ OK	! STALE
confluence_zones	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 03:17	168.0h	207.8K	✓ OK	! STALE
conformal_envelopes	Data through 2026-10-16	8 regions	Daily	Oct 22, 2026, 04:32	168.0h	411.6K	—	! STALE 7/3/0
country_risk	Data through 2026-10-15	GLOBAL	Daily	Oct 20, 2026, 03:24	192.0h	474	✓ OK	! STALE
earnings_events	Data through 2026-08-19	8 regions	Daily	Oct 20, 2026, 02:41	1,560.0h	348.2K	✓ OK	! STALE 2/8/0
earnings_pead	Data through 2026-06-30	8 regions	Daily	Oct 20, 2026, 21:53	2,760.0h	83.3K	✓ OK	! STALE 0/10/0
earnings_reactions	Data through 2026-09-16	8 regions	Daily	Oct 20, 2026, 21:52	888.0h	296K	✓ OK	! STALE 4/6/0
event_changepoints	Data through 2026-10-15	GLOBAL	Daily	Oct 20, 2026, 03:22	192.0h	32.1K	✓ OK	! STALE
event_daily_agg	Data through 2026-10-15	GLOBAL	Daily	Oct 20, 2026, 02:37	192.0h	5.6M	✓ OK	! STALE
factor_scores	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 12:42	168.0h	960.6K	✓ OK	! STALE
forecast_valuation	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 12:31	168.0h	49K	✓ OK	! STALE
forecasts	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 12:31	168.0h	9.6M	✓ OK	! STALE
gdelt_article_labels	Data through 2026-10-15	GLOBAL	Daily	Oct 20, 2026, 02:58	192.0h	21.2M	✓ OK	! STALE
gdelt_event_labels	Data through 2026-10-15	GLOBAL	Daily	Oct 20, 2026, 02:58	192.0h	16.4M	✓ OK	! STALE
gpr_index	Data through 2026-10-15	GLOBAL	Daily	Oct 20, 2026, 02:52	192.0h	152.3K	✓ OK	! STALE
insider_signals	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 12:29	168.0h	249.8K	✓ OK	! STALE
insider_transactions	Data through 2026-09-04	7 regions	Daily	Oct 20, 2026, 12:57	1,176.0h	2.8M	✓ OK	! STALE 1/8/0
insider_txn_scores	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 12:29	168.0h	770.8K	✓ OK	! STALE
inst_filers	Data through 2026-08-17	GLOBAL	Weekly	Oct 20, 2026, 05:05	1,608.0h	103.9K	✓ OK	! STALE
inst_ownership	Data through 2026-08-17	GLOBAL	Weekly	Oct 20, 2026, 05:05	1,608.0h	64.9K	✓ OK	! STALE
institutional_holdings	Data through 2026-08-31	GLOBAL	Daily	Oct 20, 2026, 12:57	1,272.0h	30.6M	✓ OK	! STALE
io_flows	Data through 2023-11-30	GLOBAL	Monthly	Oct 13, 2026, 00:04	25,392.0h	4M	✓ OK	! STALE
ml_features	Data through 2026-10-16	8 regions	Daily	Oct 20, 2026, 12:43	168.0h	965.2K	✓ OK	! STALE 0/10/0
ml_predictions	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 21:41	168.0h	5.4M	✓ OK	! STALE
model_eval_series	Data through 2026-10-09	GLOBAL	Weekly	Oct 20, 2026, 21:41	336.0h	22.6K	✓ OK	! STALE
peer_scores	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 12:42	168.0h	2.5M	✓ OK	! STALE
sc_edges	Data through 2026-10-08	GLOBAL	Daily	Oct 20, 2026, 03:24	360.0h	1.2K	✓ OK	! STALE
sentiment_daily	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 12:42	168.0h	212.2K	✓ OK	! STALE
short_interest	Data through 2026-09-30	GLOBAL	Daily	Oct 20, 2026, 12:57	552.0h	1.8M	✓ OK	! STALE
si_signals	Data through 2026-09-30	GLOBAL	Daily	Oct 20, 2026, 03:20	552.0h	319K	✓ OK	! STALE
smart_money_signals	Data through 2026-10-16	8 regions	Daily	Oct 20, 2026, 03:20	168.0h	788.1K	✓ OK	! STALE 0/9/0
tech_breakouts	Data through 2026-10-13	8 regions	Daily	Oct 22, 2026, 21:38	240.0h	1.7M	—	! STALE 6/4/0
tech_factors	Data through 2026-10-16	8 regions	Daily	Oct 22, 2026, 21:35	168.0h	30.2M	—	! STALE 7/3/0
tech_levels	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 03:13	168.0h	6.2M	✓ OK	! STALE



tech_meta_setups	Data through 2026-10-16	8 regions	Daily	Oct 22, 2026, 21:56	168.0h	62.4K	—	! STALE 0/11/0
tech_prob_scores	Data through 2026-10-16	8 regions	Daily	Oct 22, 2026, 04:40	168.0h	24K	—	! STALE 0/11/0
tech_prob_setups	Data through 2026-10-16	GLOBAL	Daily	Oct 22, 2026, 04:40	168.0h	104K	—	! STALE
tech_prob_zones	Data through 2026-10-16	GLOBAL	Daily	Oct 22, 2026, 04:40	168.0h	185.3K	—	! STALE
tech_setups	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 03:17	168.0h	24K	✓ OK	! STALE
trade_bilateral	Data through 2026-06-30	GLOBAL	Monthly	Oct 13, 2026, 06:14	2,760.0h	1.6M	✓ OK	! STALE
valuation_daily	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 12:28	168.0h	2.9M	✓ OK	! STALE
cb_balance_sheets	Data through 2026-10-19	GLOBAL	Daily	Oct 20, 2026, 02:53	96.0h	19.9K	✓ OK	✓ FRESH
cb_decisions	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:46	72.0h	2.5K	✓ OK	✓ FRESH
cb_implied_meta	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:55	72.0h	17	✓ OK	✓ FRESH
cb_implied_paths	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:55	72.0h	47.5K	✓ OK	✓ FRESH
cb_meetings	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:46	72.0h	1.3K	✓ OK	✓ FRESH
cb_policy_rates	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:46	72.0h	147.8K	✓ OK	✓ FRESH
cb_projections	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:57	72.0h	4.1K	✓ OK	✓ FRESH
cb_rate_cycle	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:46	72.0h	140.2K	✓ OK	✓ FRESH
cftc_event_stats	Data through 2026-10-20	GLOBAL	Manual	Oct 20, 2026, 13:36	72.0h	1.4K	✓ OK	✓ FRESH
cftc_markets	Data through 2026-10-20	GLOBAL	Manual	Oct 20, 2026, 13:36	72.0h	999	✓ OK	✓ FRESH
cftc_positions	Data through 2026-10-16	GLOBAL	Manual	Oct 20, 2026, 13:36	168.0h	5.2M	✓ OK	✓ FRESH
cftc_signals	Data through 2026-10-16	GLOBAL	Manual	Oct 20, 2026, 13:36	168.0h	6.2M	✓ OK	✓ FRESH
company_event_exposure	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:52	72.0h	62K	✓ OK	✓ FRESH
company_links	Data through 2026-10-31	GLOBAL	Monthly	Oct 19, 2026, 22:07	96.0h	223.8K	✓ OK	✓ FRESH
company_neighbours	Data through 2026-10-20	GLOBAL	Weekly	Oct 20, 2026, 05:06	72.0h	333.3K	✓ OK	✓ FRESH
conformal_coverage	Data through 2026-10-22	GLOBAL	Daily	Oct 22, 2026, 04:32	24.0h	10.8K	—	✓ FRESH
conformal_peer_groups	Data through 2026-10-21	GLOBAL	Weekly	Oct 21, 2026, 16:04	48.0h	9.2K	—	✓ FRESH
conformal_peer_members	Data through 2026-10-19	GLOBAL	Daily	Oct 21, 2026, 18:30	96.0h	2.3M	—	✓ FRESH
conformal_peer_scores	Data through 2026-10-19	GLOBAL	Daily	Oct 21, 2026, 18:30	96.0h	66K	—	✓ FRESH

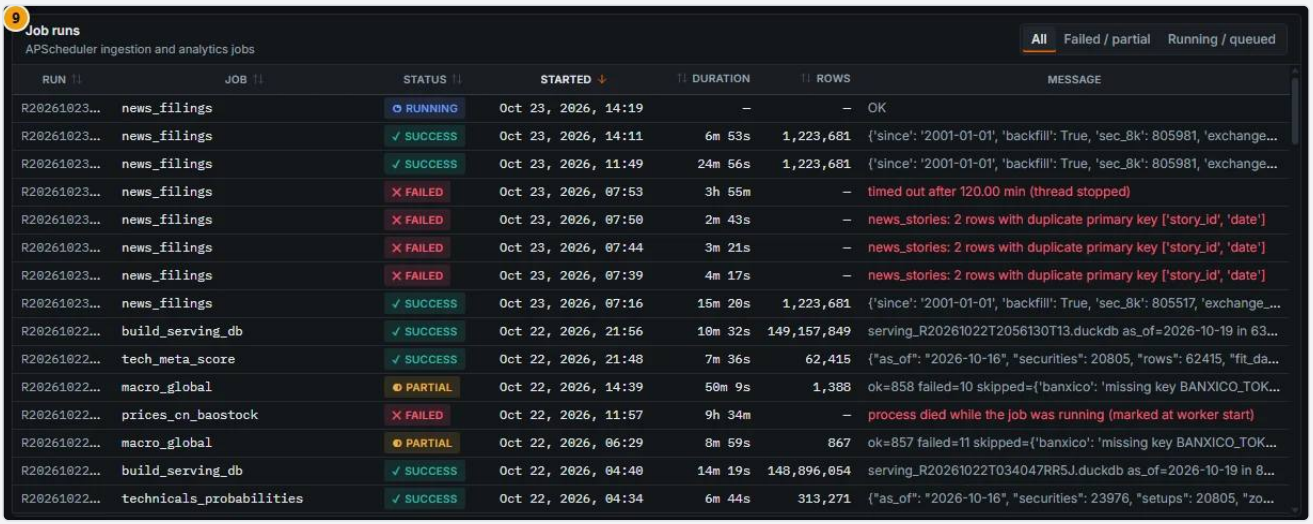
*Figure 15.4: Data sources (panel 8), *Per dataset** view: 152 datasets (563 region rows), prices, FX and macro first. Each row shows source table, "Data through" date, regions, cadence, last success, worst lag, coverage, DQ and status with fresh / stale / error counts for multi-region tables.*

SOURCE	DATASET	REGION	CADENCE	LAST SUCCESS	LAG	COVERAGE	DQ	STATUS
activist_events	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 03:20	168.0h	157.7K	✓ OK	! STALE
activist_filings	Data through 2026-10-16	GLOBAL	Daily	Oct 20, 2026, 02:42	168.0h	157.7K	✓ OK	! STALE
bank_claims	Data through 2026-07-31	GLOBAL	Monthly	Oct 13, 2026, 00:02	2,016.0h	1.1M	✓ OK	! STALE
cb_balance_sheets	Data through 2026-10-19	GLOBAL	Daily	Oct 20, 2026, 02:53	96.0h	19.9K	✓ OK	✓ FRESH
cb_curves	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:55	72.0h	2.4K	✓ OK	✓ FRESH
cb_decisions	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:46	72.0h	2.5K	✓ OK	✓ FRESH
cb_implied_meta	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:55	72.0h	17	✓ OK	✓ FRESH
cb_implied_paths	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:55	72.0h	47.5K	✓ OK	✓ FRESH
cb_meetings	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:46	72.0h	1.3K	✓ OK	✓ FRESH
cb_policy_rates	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:46	72.0h	147.8K	✓ OK	✓ FRESH
cb_projections	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:57	72.0h	4.1K	✓ OK	✓ FRESH
cb_rate_cycle	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:46	72.0h	140.2K	✓ OK	✓ FRESH
cb_stance	Data through 2026-10-01	GLOBAL	Daily	Oct 20, 2026, 02:58	528.0h	5.4K	✓ OK	! STALE
cb_tone	Data through 2026-09-18	GLOBAL	Daily	Oct 20, 2026, 02:51	840.0h	211	✓ OK	! STALE
cftc_event_stats	Data through 2026-10-20	GLOBAL	Manual	Oct 20, 2026, 13:36	72.0h	1.4K	✓ OK	✓ FRESH
cftc_markets	Data through 2026-10-20	GLOBAL	Manual	Oct 20, 2026, 13:36	72.0h	999	✓ OK	✓ FRESH
cftc_positions	Data through 2026-10-16	GLOBAL	Manual	Oct 20, 2026, 13:36	168.0h	5.2M	✓ OK	✓ FRESH
cftc_signals	Data through 2026-10-16	GLOBAL	Manual	Oct 20, 2026, 13:36	168.0h	6.2M	✓ OK	✓ FRESH
company_event_exposure	Data through 2026-10-20	GLOBAL	Daily	Oct 20, 2026, 02:52	72.0h	62K	✓ OK	✓ FRESH
company_geo_revenue	Data through 2026-09-30	GLOBAL	Monthly	Oct 19, 2026, 21:20	552.0h	22.4K	✓ OK	✓ FRESH
company_geo_revenue	Data through 2019-12-20	GLOBAL	Monthly	Oct 13, 2026, 00:37	59,976.0h	10	✓ OK	! STALE
company_geo_revenue	Data through 2026-05-12	GLOBAL	Monthly	Oct 13, 2026, 00:37	3,936.0h	179	✓ OK	! STALE
company_geo_revenue	Data through 2026-03-27	GLOBAL	Monthly	Oct 13, 2026, 00:37	5,040.0h	20	✓ OK	! STALE
company_geo_revenue	Data through 2025-09-30	GLOBAL	Monthly	Oct 13, 2026, 00:37	9,312.0h	96	✓ OK	! STALE
company_geo_revenue	Data through 2021-10-21	GLOBAL	Monthly	Oct 13, 2026, 00:37	43,872.0h	32	✓ OK	! STALE
company_geo_revenue	Data through 2016-09-08	GLOBAL	Monthly	Oct 13, 2026, 00:37	88,728.0h	29	✓ OK	! STALE
company_geo_revenue	Data through 2022-11-10	GLOBAL	Monthly	Oct 13, 2026, 00:37	34,632.0h	29	✓ OK	! STALE
company_geo_revenue	Data through 2026-07-09	GLOBAL	Monthly	Oct 13, 2026, 00:37	2,544.0h	8	✓ OK	! STALE
company_geo_revenue	Data through 2011-10-25	GLOBAL	Monthly	Oct 13, 2026, 00:37	131,448.…	8	✓ OK	! STALE
company_geo_revenue	Data through 2012-02-17	GLOBAL	Monthly	Oct 13, 2026, 00:37	128,688.…	5	✓ OK	! STALE
company_geo_revenue	Data through 2012-06-29	GLOBAL	Monthly	Oct 13, 2026, 00:37	125,496.…	4	✓ OK	! STALE
company_geo_revenue	Data through 2021-08-16	GLOBAL	Monthly	Oct 13, 2026, 00:37	45,456.0h	21	✓ OK	! STALE
company_geo_revenue	Data through 2026-09-30	EU	Monthly	Oct 19, 2026, 21:20	552.0h	604	✓ OK	✓ FRESH
company_geo_revenue	Data through 2026-08-06	GLOBAL	Monthly	Oct 19, 2026, 21:20	1,872.0h	315	✓ OK	! STALE
company_geo_revenue	Data through 2026-08-11	GLOBAL	Monthly	Oct 19, 2026, 21:20	1,752.0h	149	✓ OK	! STALE
company_geo_revenue	Data through 2026-09-01	GLOBAL	Monthly	Oct 13, 2026, 00:37	1,248.0h	169	✓ OK	! STALE
company_geo_revenue	Data through 2022-11-14	GLOBAL	Monthly	Oct 13, 2026, 00:37	34,536.0h	56	✓ OK	! STALE
company_geo_revenue	Data through 2026-08-06	GLOBAL	Monthly	Oct 13, 2026, 00:37	1,872.0h	227	✓ OK	! STALE
company_geo_revenue	Data through 2026-07-29	GLOBAL	Monthly	Oct 13, 2026, 00:37	2,064.0h	188	✓ OK	! STALE
company_geo_revenue	Data through 2021-05-28	GLOBAL	Monthly	Oct 13, 2026, 00:37	47,376.0h	54	✓ OK	! STALE
company_geo_revenue	Data through 2021-03-11	GLOBAL	Monthly	Oct 13, 2026, 00:37	49,248.0h	22	✓ OK	! STALE
company_geo_revenue	Data through 2021-02-25	GLOBAL	Monthly	Oct 13, 2026, 00:37	49,584.0h	37	✓ OK	! STALE
company_geo_revenue	Data through 2026-02-25	GLOBAL	Monthly	Oct 13, 2026, 00:37	5,760.0h	30	✓ OK	! STALE
company_geo_revenue	Data through 2026-08-20	GLOBAL	Monthly	Oct 13, 2026, 00:37	1,536.0h	837	✓ OK	! STALE
company_geo_revenue	Data through 2026-02-27	GLOBAL	Monthly	Oct 13, 2026, 00:37	5,712.0h	140	✓ OK	! STALE
company_geo_revenue	Data through 2019-08-06	GLOBAL	Monthly	Oct 13, 2026, 00:37	63,240.0h	28	✓ OK	! STALE
company_geo_revenue	Data through 2021-03-11	GLOBAL	Monthly	Oct 13, 2026, 00:37	49,248.0h	32	✓ OK	! STALE
company_geo_revenue	Data through 2026-07-31	GLOBAL	Monthly	Oct 13, 2026, 00:37	2,016.0h	296	✓ OK	! STALE
company_geo_revenue	Data through 2020-11-05	GLOBAL	Monthly	Oct 13, 2026, 00:37	52,272.0h	23	✓ OK	! STALE
company_geo_revenue	Data through 2026-07-13	GLOBAL	Monthly	Oct 13, 2026, 00:37	2,448.0h	61	✓ OK	! STALE
company_geo_revenue	Data through 2014-07-30	GLOBAL	Monthly	Oct 13, 2026, 00:37	107,232.…	42	✓ OK	! STALE
company_geo_revenue	Data through 2026-07-30	GLOBAL	Monthly	Oct 13, 2026, 00:37	2,040.0h	63	✓ OK	! STALE
company_geo_revenue	Data through 2017-07-28	GLOBAL	Monthly	Oct 13, 2026, 00:37	80,976.0h	22	✓ OK	! STALE
company_geo_revenue	Data through 2018-03-01	GLOBAL	Monthly	Oct 13, 2026, 00:37	75,792.0h	5	✓ OK	! STALE
company_geo_revenue	Data through 2026-02-12	GLOBAL	Monthly	Oct 13, 2026, 00:37	6,072.0h	33	✓ OK	! STALE
company_geo_revenue	Data through 2026-07-14	GLOBAL	Monthly	Oct 13, 2026, 00:37	2,424.0h	24	✓ OK	! STALE
company_geo_revenue	Data through 2026-08-07	GLOBAL	Monthly	Oct 13, 2026, 00:37	1,848.0h	48	✓ OK	! STALE
company_geo_revenue	Data through 2026-02-24	GLOBAL	Monthly	Oct 13, 2026, 00:37	5,784.0h	2	✓ OK	! STALE
company_geo_revenue	Data through 2012-06-06	GLOBAL	Monthly	Oct 13, 2026, 00:37	126,048.…	3	✓ OK	! STALE
company_geo_revenue	Data through 2013-11-18	GLOBAL	Monthly	Oct 13, 2026, 00:37	113,328.…	3	✓ OK	! STALE
company_geo_revenue	Data through 2023-09-14	GLOBAL	Monthly	Oct 13, 2026, 00:37	27,240.0h	18	✓ OK	! STALE
company_geo_revenue	Data through 2025-04-04	GLOBAL	Monthly	Oct 19, 2026, 21:20	13,608.0h	33	✓ OK	! STALE
company_geo_revenue	Data through 2026-08-13	GLOBAL	Monthly	Oct 19, 2026, 21:20	1,704.0h	571	✓ OK	! STALE
company_geo_revenue	Data through 2020-02-06	GLOBAL	Monthly	Oct 13, 2026, 00:37	58,824.0h	52	✓ OK	! STALE
company_geo_revenue	Data through 2026-07-31	GLOBAL	Monthly	Oct 13, 2026, 00:37	2,016.0h	281	✓ OK	! STALE
company_geo_revenue	Data through 2026-09-28	GLOBAL	Monthly	Oct 13, 2026, 00:37	600.0h	28	✓ OK	✓ FRESH
company_geo_revenue	Data through 2023-03-01	GLOBAL	Monthly	Oct 13, 2026, 00:37	31,968.0h	217	✓ OK	! STALE
company_geo_revenue	Data through 2019-11-26	GLOBAL	Monthly	Oct 13, 2026, 00:37	60,552.0h	18	✓ OK	! STALE
company_geo_revenue	Data through 2023-11-08	GLOBAL	Monthly	Oct 13, 2026, 00:37	25,920.0h	142	✓ OK	! STALE
company_geo_revenue	Data through 2020-02-06	GLOBAL	Monthly	Oct 13, 2026, 00:37	58,824.0h	13	✓ OK	! STALE
company_geo_revenue	Data through 2026-02-27	GLOBAL	Monthly	Oct 13, 2026, 00:37	5,712.0h	36	✓ OK	! STALE



company_geo_revenue	Data through 2014-08-28	GLOBAL	Monthly	Oct 13, 2026, 00:37	106,536.…	9	✓ OK	! STALE
company_geo_revenue	Data through 2016-11-14	GLOBAL	Monthly	Oct 13, 2026, 00:37	87,120.0h	15	✓ OK	! STALE
company_geo_revenue	Data through 2015-02-27	GLOBAL	Monthly	Oct 13, 2026, 00:37	102,144.…	12	✓ OK	! STALE
company_geo_revenue	Data through 2018-03-27	GLOBAL	Monthly	Oct 13, 2026, 00:37	75,168.0h	10	✓ OK	! STALE
company_geo_revenue	Data through 2015-08-10	GLOBAL	Monthly	Oct 13, 2026, 00:37	98,208.0h	9	✓ OK	! STALE
company_geo_revenue	Data through 2026-08-05	GLOBAL	Monthly	Oct 13, 2026, 00:37	1,896.0h	48	✓ OK	! STALE
company_geo_revenue	Data through 2026-08-07	GLOBAL	Monthly	Oct 13, 2026, 00:37	1,848.0h	13	✓ OK	! STALE
company_geo_revenue	Data through 2026-02-24	GLOBAL	Monthly	Oct 13, 2026, 00:37	5,784.0h	2	✓ OK	! STALE
company_geo_revenue	Data through 2012-02-23	GLOBAL	Monthly	Oct 13, 2026, 00:37	128,544.…	3	✓ OK	! STALE
company_geo_revenue	Data through 2013-06-13	GLOBAL	Monthly	Oct 13, 2026, 00:37	117,120.…	5	✓ OK	! STALE
company_geo_revenue	Data through 2013-04-30	GLOBAL	Monthly	Oct 13, 2026, 00:37	118,176.…	6	✓ OK	! STALE
company_geo_revenue	Data through 2026-09-14	GLOBAL	Monthly	Oct 19, 2026, 21:20	936.0h	1.2K	✓ OK	✓ FRESH
company_geo_revenue	Data through 2026-02-20	GLOBAL	Monthly	Oct 13, 2026, 00:37	5,880.0h	429	✓ OK	! STALE
company_geo_revenue	Data through 2026-07-29	GLOBAL	Monthly	Oct 13, 2026, 00:37	2,064.0h	104	✓ OK	! STALE
company_geo_revenue	Data through 2022-02-02	GLOBAL	Monthly	Oct 13, 2026, 00:37	41,376.0h	136	✓ OK	! STALE
company_geo_revenue	Data through 2021-07-22	GLOBAL	Monthly	Oct 13, 2026, 00:37	46,056.0h	10	✓ OK	! STALE
company_geo_revenue	Data through 2026-07-30	GLOBAL	Monthly	Oct 13, 2026, 00:37	2,040.0h	37	✓ OK	! STALE
company_geo_revenue	Data through 2026-07-31	GLOBAL	Monthly	Oct 13, 2026, 00:37	2,016.0h	74	✓ OK	! STALE
company_geo_revenue	Data through 2019-02-27	GLOBAL	Monthly	Oct 13, 2026, 00:37	67,080.0h	129	✓ OK	! STALE
company_geo_revenue	Data through 2026-08-05	GLOBAL	Monthly	Oct 13, 2026, 00:37	1,896.0h	42	✓ OK	! STALE
company_geo_revenue	Data through 2025-08-26	GLOBAL	Monthly	Oct 13, 2026, 00:37	10,152.0h	3	✓ OK	! STALE
company_geo_revenue	Data through 2013-11-12	GLOBAL	Monthly	Oct 13, 2026, 00:37	113,472.…	4	✓ OK	! STALE
company_geo_revenue	Data through 2024-03-12	GLOBAL	Monthly	Oct 13, 2026, 00:37	22,920.0h	11	✓ OK	! STALE
company_geo_revenue	Data through 2026-09-30	GLOBAL	Monthly	Oct 19, 2026, 21:20	552.0h	31.1K	✓ OK	✓ FRESH
company_geo_revenue	Data through 2026-09-30	GLOBAL	Monthly	Oct 19, 2026, 21:20	552.0h	35.7K	✓ OK	✓ FRESH
company_geo_revenue	Data through 2026-08-05	GLOBAL	Monthly	Oct 19, 2026, 21:20	1,896.0h	444	✓ OK	! STALE

*Figure 15.5: Data sources (panel 8) in the *Per region** view: one row per source × region. `company_geo_revenue` alone has dozens of rows (one per region combination, many with data dates years old and therefore "stale"), which is why the Per dataset view is the default.*



RUN	JOB	STATUS	STARTED	DURATION	ROWS	MESSAGE
R20261023…	news_filings	RUNNING	Oct 23, 2026, 14:19	–	–	OK
R20261023…	news_filings	✓ SUCCESS	Oct 23, 2026, 14:11	6m 53s	1,223,681	{'since': '2001-01-01', 'backfill': True, 'sec_8k': 805981, 'exchange…
R20261023…	news_filings	✓ SUCCESS	Oct 23, 2026, 11:49	24m 56s	1,223,681	{'since': '2001-01-01', 'backfill': True, 'sec_8k': 805981, 'exchange…
R20261023…	news_filings	✕ FAILED	Oct 23, 2026, 07:53	3h 55m	–	timed out after 120.00 min (thread stopped)
R20261023…	news_filings	✕ FAILED	Oct 23, 2026, 07:50	2m 43s	–	news_stories: 2 rows with duplicate primary key ['story_id', 'date']
R20261023…	news_filings	✕ FAILED	Oct 23, 2026, 07:44	3m 21s	–	news_stories: 2 rows with duplicate primary key ['story_id', 'date']
R20261023…	news_filings	✕ FAILED	Oct 23, 2026, 07:39	4m 17s	–	news_stories: 2 rows with duplicate primary key ['story_id', 'date']
R20261023…	news_filings	✓ SUCCESS	Oct 23, 2026, 07:16	15m 20s	1,223,681	{'since': '2001-01-01', 'backfill': True, 'sec_8k': 805517, 'exchange_…
R20261022…	build_serving_db	✓ SUCCESS	Oct 22, 2026, 21:56	10m 32s	149,157,849	serving_R20261022T2056130T13.duckdb as_of=2026-10-19 in 63…
R20261022…	tech_meta_score	✓ SUCCESS	Oct 22, 2026, 21:48	7m 36s	62,415	{"as_of": "2026-10-16", "securities": 20805, "rows": 62415, "fit_da…
R20261022…	macro_global	◐ PARTIAL	Oct 22, 2026, 14:39	50m 9s	1,388	ok=858 failed=10 skipped={'banxico': 'missing key BANXICO_TOK…
R20261022…	prices_cn_baostock	✕ FAILED	Oct 22, 2026, 11:57	9h 34m	–	process died while the job was running (marked at worker start)
R20261022…	macro_global	◐ PARTIAL	Oct 22, 2026, 06:29	8m 59s	867	ok=857 failed=11 skipped={'banxico': 'missing key BANXICO_TOK…
R20261022…	build_serving_db	✓ SUCCESS	Oct 22, 2026, 04:40	14m 19s	148,896,054	serving_R20261022T034047RR5J.duckdb as_of=2026-10-19 in 8…
R20261022…	technicals_probabilities	✓ SUCCESS	Oct 22, 2026, 04:34	6m 44s	313,271	{"as_of": "2026-10-16", "securities": 23976, "setups": 20805, "zo…

*Figure 15.6: Job runs (panel 9), filter *All*, newest first. Failed runs show their error in red (here a timeout and a duplicate-key contract violation of `news_filings`).*

Controls

Control	What it does
Refresh (page header)	Re-reads `/system/status` and the job runs. The icon spins while it loads.
Alert **Open** link	Jumps to the page named by the alert (`/system` or `/system/data-health`).
Data health link (Scheduler panel)	Opens Data Health.
Snapshot section badges (hover)	Show the job that feeds the section and how far it is behind the snapshot date. Green = fresh, amber = stale, red = error.
Schedule profile select + **Apply**	Switches the refresh profile (`PUT /api/v1/system/schedule`). **Apply** is disabled until you pick a profile other than the active one. A toast confirms: "The worker applies it within a few seconds" (worker running) or "Applied when the worker starts". A **worker still on …** badge shows until the worker has picked the change up.
SSE **Pause** / **Resume** (Real-time layer)	Pauses or resumes live updates in this browser only. It does not stop the server-side streams.
Number locale select	Sets the number and date format for the whole app in this browser: English (US), English (UK), Português (PT), Deutsch (DE), Français (FR), 日本語 (JP).
Source rows: **Per dataset** / **Per region**	Per dataset shows one row per source table (worst status across its regions; hover the Regions cell for the freshest and stalest region and their lag). Per region shows one row per source × region (Figure 15.5); click a header to sort.

Control	What it does
Job filter: All / Failed / partial / Running / queued	Filters the Job runs table.
Table headers	Click to sort; hover the Dataset cell to see the writing job id; hover the DQ and Coverage headers for their definitions.
TradingView link	Opens the TradingView Lightweight Charts site (attribution).

The tray menu offers the same profile switch (**Schedule ▸ Weekly / Daily**) and **Run refresh now**; see Getting started.

Methodology

SCHEDULER STATE AND PROFILES

The worker maps the active profile in `config/schedule.yaml` to APScheduler triggers:

Profile	When it refreshes	Freshness SLA used by Data Health
Weekly (default)	Saturday 08:00 Lisbon: the nightly DAG for the latest session (ingestion catches up from its watermarks, GDELT up to 10 days), then the weekly DAG, then the monthly DAG when due. CFTC Friday 19:00 ET (after the 15:30 ET release).	9 days
Daily	The `contracts/jobs.yaml` crons: nightly Tue–Sat 01:30, weekly Saturday 06:00, monthly first Sunday 08:00 (Lisbon); CFTC Friday 19:00 ET.	3 days
Intraday (placeholder)	Behaves like Daily today ("placeholder: behaves like daily for now").	as Daily

Every profile also runs `hourly_news` at :10 UTC each hour (live news fold); it never wakes the PC and is never caught up.

The environment variable `BIG360_SCHEDULE_PROFILE` wins over the stored choice; the **source** text next to **Apply** says where the active profile came from (`default` , the stored state, or the environment).

Catch-up rule. If the PC was asleep or off at a scheduled time, an entry is due when its most recent scheduled time has passed and no completed run of its DAG finished after it. The worker checks at start and every 15 minutes; runs older than the profile's `max_catchup_days` are not replayed. A due entry shows as **catch-up due: ...** in the Scheduler panel and as a "Missed scheduled run" alert.

Last completed. The time the DAG last finished, with its trigger (`schedule` , `manual` , `api` , ...). A manual trigger is normal after a hand-run catch-up.

Data as of (published snapshot). The latest market date per region in the serving snapshot. Regions can differ by several sessions (holidays, a region whose prices did not complete).

Snapshot sections. Since the audit fix F16, fresh market data (prices, FX, macro) publishes even when an analytics job failed; the previous analytics outputs stay in the snapshot and their section turns amber. A failure of the data-quality checks in the same run still blocks publishing.

SOURCE FRESHNESS

Each row of **Data sources** is one lake table (and region where the table is regional). Its status is computed by `GET /system/status` :

```
lag = today − last data date          (forward-looking tables: today − last ingest)
fresh  if lag ≤ tolerance of the writing DAG
stale  otherwise, or if the writing job's latest run failed after its last success
missing if the table has no rows
tolerance: nightly 4 days · weekly 10 days · monthly 40 days · manual none
```

DQ is the worst data-quality result of the latest load (`ok` , `info` , `no data` , `warn` , `error`). **Coverage** is the number of entities or series covered.

Per-dataset rows are ordered by criticality for a portfolio manager: prices first, then FX, macro, fundamentals, other datasets, and live streams last; within that, errors and stale rows first.

ALERTS

Alert	Severity	Raised when
Worker not running	error	No worker heartbeat.
Scheduler off	warn, error after 24 h	The worker runs without its scheduler.
Refresh overdue	error	The last completed nightly is older than the profile period + 1 day.
Missed scheduled run	warn	A catch-up is due.
Core data stale beyond SLA	error	Prices US / EU / UK / JP, FX rates or macro releases are stale or missing in Data Health.
N dataset(s) stale beyond SLA	warn	Any other Data Health dataset is stale or missing.
Last nightly run failed	warn	The latest nightly DAG run has status failed (any job failed).

Alert	Severity	Raised when
Low disk space	warn below 25 GB, error below 15 GB	Free space on the data drive.
Regression guard failed / Regression guard warnings	error / warn	The latest QA guard run failed or ended with warnings.
Gap filler not running / paused	info (Data Health only)	No filler heartbeat for 5 minutes, or the resource governor paused it.

JOB SUCCESS RATE

`successful runs / (successful + failed runs)` over the completed runs among the last 100 shown; partial, skipped, queued and running runs are left out.

Data sources and refresh

- `GET /system/status` (scheduler, freshness, DQ, snapshot sections, alerts), `GET /system/jobs/runs?limit=100`, `GET /health`, `GET /system/schedule`, `GET /meta/factors`, `GET /live/status`.
- Job runs come from the runner's `job_runs` table; freshness from the lake tables and the serving snapshot.
- The page reflects the state at load time. It does not auto-refresh except the Real-time layer panel (30 s).

Validation status

Not a model: there is no validation gate. The status logic was reworked after the functional audit of 2026-10-19 (`docs/AUDIT_2026-10-19.md`, findings F01, F03, F13, F14, F16, F22), which found that the worker had been running without its scheduler and that nothing in the UI said so. The scheduler panel, the alerts and the per-dataset grouping are the fixes.

Correct usage and worked example

Read the page top-down: alerts first, then the Scheduler panel, then only the rows of the sources you rely on.

Example (Figure 15.1: 23 Oct 2026, 14:26 Lisbon time; snapshot `R20261022T2056130T13`, built 22 Oct 22:06).

1. The banner shows three amber alerts: "1 dataset(s) stale beyond SLA — Macro ingestion"; "Last nightly run failed — D20261019T223621NJZ9-nightly: 1 of 102 jobs failed"; "Regression guard warnings — api: warn (0 failed)". None is red, so the core datasets (prices US / EU / UK / JP, FX, macro releases) are within their SLA.
2. Scheduler: **SCHEDULER ON**, **WORKER ALIVE**, Weekly profile, Europe/Lisbon. Next `nightly`, `weekly` and `monthly` runs 24 Oct 08:00 (the Saturday chain); next `cftc_weekly` 24 Oct 00:00 Lisbon (Friday 19:00 ET). Last completed nightly 20 Oct 12:47, trigger `manual`; `monthly` has never completed.
3. Data as of: US 16 Oct, EU and UK 19 Oct, **JP 13 Oct** (its snapshot badge `PRICES:JP · 10-13` is amber), HK and LATAM 16 Oct. Before quoting a Japanese price-based number, open Data Health: Prices Japan shows lag 6 sessions against an SLA of 6, so it is still "fresh" by the rule but at its limit.
4. Backend: keys configured `fred`, `polygon`, `edinet`; `opendart`, `krx_openapi_key` and others are not configured, which explains thin Korean coverage elsewhere in the app. Universe counts (from `/system/status`): 24,961 securities, 5,022 with fundamentals, 769 with a composite score.

Limitations and pitfalls

- **Two different freshness rules.** The Data sources table uses fixed tolerances per DAG (4 / 10 / 40 days) whatever the profile, while Data Health uses the profile-adjusted SLA (9 days under Weekly). Under the Weekly profile the System page therefore shows many more "stale" rows than Data Health (Figure 15.3: 342 stale of 563 rows, while Data Health has 28 of 34 datasets fresh). For "is my data usable", trust Data Health.
- "Fresh" means "within tolerance", not "updated today".
- The Job runs table shows only the last 100 runs.
- The header's freshness chip and this page describe the **serving snapshot**. A job that finished after the last snapshot build is not visible on other pages until the next build.
- Changing the profile changes when refreshes run, not what they contain. Switching to Daily on a laptop that sleeps at night produces catch-up runs in the morning.
- The **Report** path in the Regression guard panel (Data Health) is a local file path on the machine that ran the check.

Related mnemonics and BQL fields

- No command-bar function opens the System pages; use the menu or the tray (**Data health**).
- No BQL fields. For scripts, `GET /api/v1/system/status` and `python -m app.cli jobs status` (from `backend\`) give the same information.
- Related: Data and coverage, Troubleshooting and FAQ, Good practice: stale data.

Data Health

Route: `/system/data-health` · **Menu:** System › Data health · **Mnemonics:** none · **Refresh:** on page load and with **Refresh**; the gap detector runs after every nightly build and daily from the gap filler

Purpose

Data Health is the window onto the **Data Ops supervisor** (`docs/OPERATIONS_ROADMAP.md` O1): the app knows what data it should have, finds what is missing or stale, and fills it in the background with the same code as the live updates. Use the page to answer "can I rely on dataset X as of today?", to see why a region or a history is incomplete, and to follow long backfills. It also hosts the regression-guard result, the live-news service health and the conformal coverage audit.

Screenshot

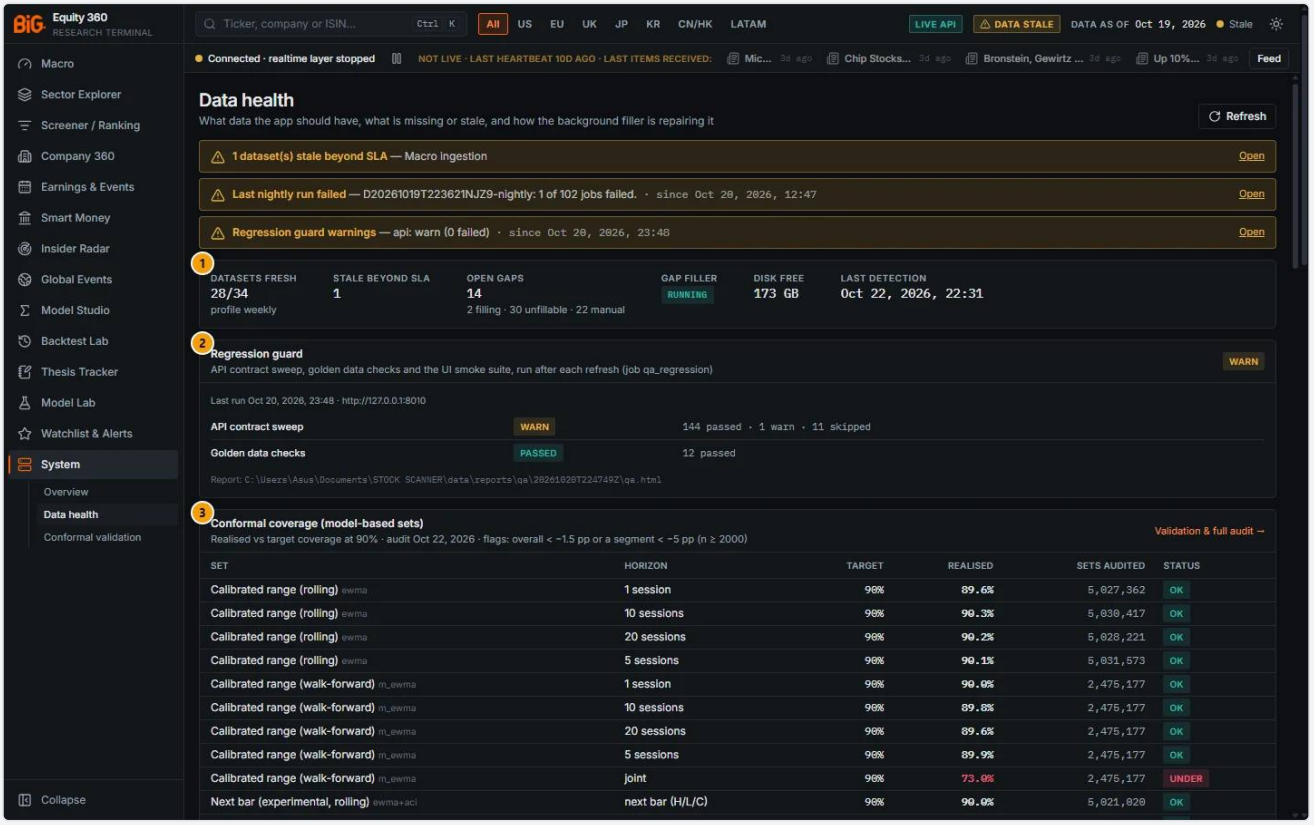


Figure 15.7: Data Health (`/system/data-health`), 23 Oct 2026 14:26 Lisbon time, Weekly profile.

The same **ops alert banner** as on the System page sits above the numbered panels.

#	Panel	What it shows
1	Summary stats	Datasets fresh (with the active profile), Stale beyond SLA, Open gaps (with filling / unfillable / manual counts), Gap filler state and pause reasons, Disk free, Last detection.
2	Regression guard	The latest QA guard run: overall status, one row per suite with counts, the first ten failed checks and the report path.
3	Conformal coverage (model-based sets)	Realised vs target coverage at 90% of the conformal sets (rolling audit and walk-forward rows) and the flagged segments; link **Validation & full audit →**.
4	Freshness heatmap	The day-level datasets × the last 21 days: share of expected entities present each day, status and data as-of; below it the table of every registered dataset.
5	Backfill campaigns	Long resumable backfills run by the gap filler, with progress bars, ETA, rate per hour and last error.
6	Gap filler & resource governor	Filler process state, pause reasons, free RAM, filler RAM, power, network, disk, priority and maximum tasks; running, waiting and recent tasks.
7	Gap queue	Every tracked gap with priority, class, status, period, actual / expected, tries and next try; filter **Active** / **Unfillable** / **Manual** / **All**.
8	Sources	Live-job health per source over 14 days and the filler's rate-budget use in the last hour.
9	SLA breaches & recent failures	Datasets beyond their SLA, failed job runs of the last 7 days, and the info alerts as badges.

> **Note:** The current code also renders a **Live news** panel between the Regression guard and the Conformal coverage panels (real-time news service state and memory, publish-to-bus latency p50 / p95, pipeline latency, GDELT release-to-hot lag, items per hour, last feed poll and failing feeds, hourly fold, intraday store, link **News Terminal**). It is not visible in the build captured for Figure 15.7.

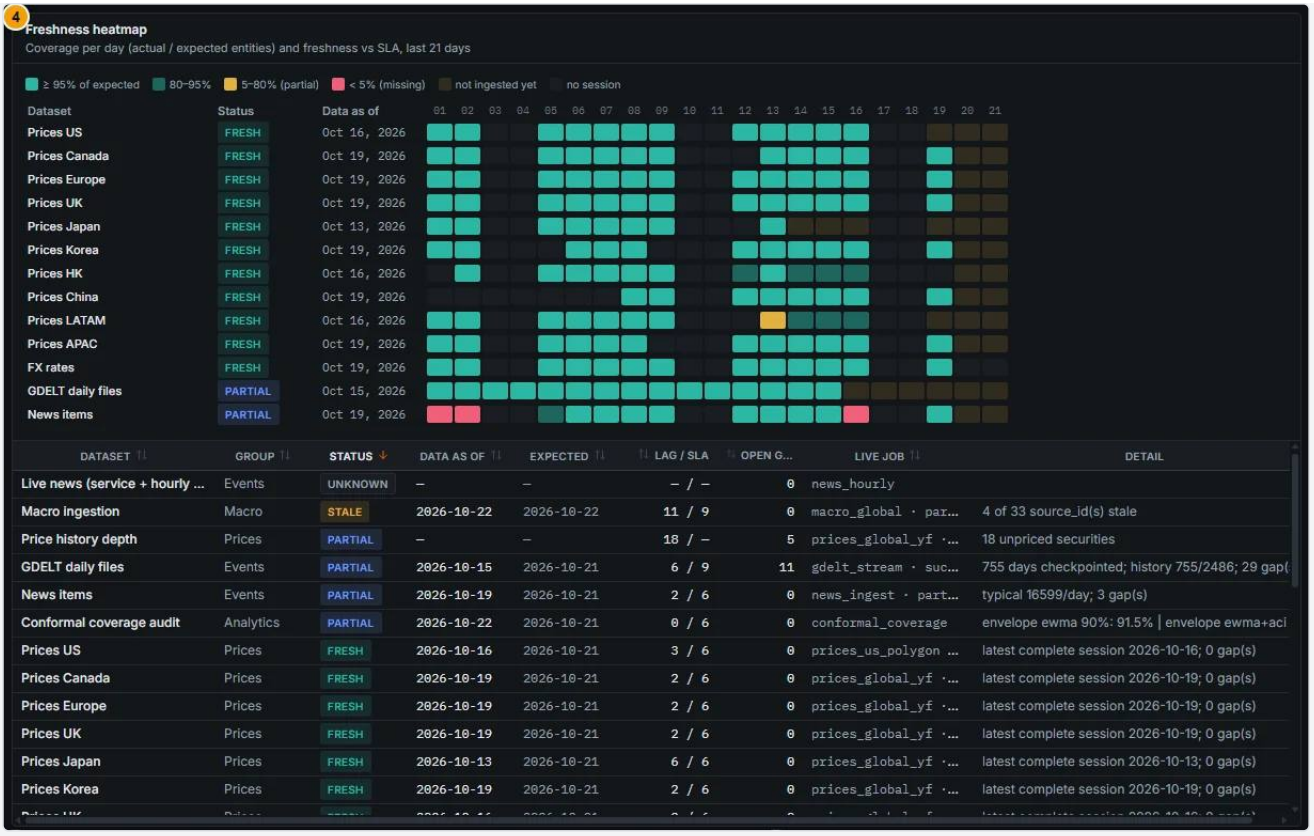


Figure 15.8: Freshness heatmap (panel 4). Prices Japan stops at 13 Oct (later days "not ingested yet"); News items shows red "< 5% (missing)" cells on 1, 2 and 16 Oct. The dataset table below is sorted by status: Macro ingestion is the only stale dataset (lag 11 / SLA 9).

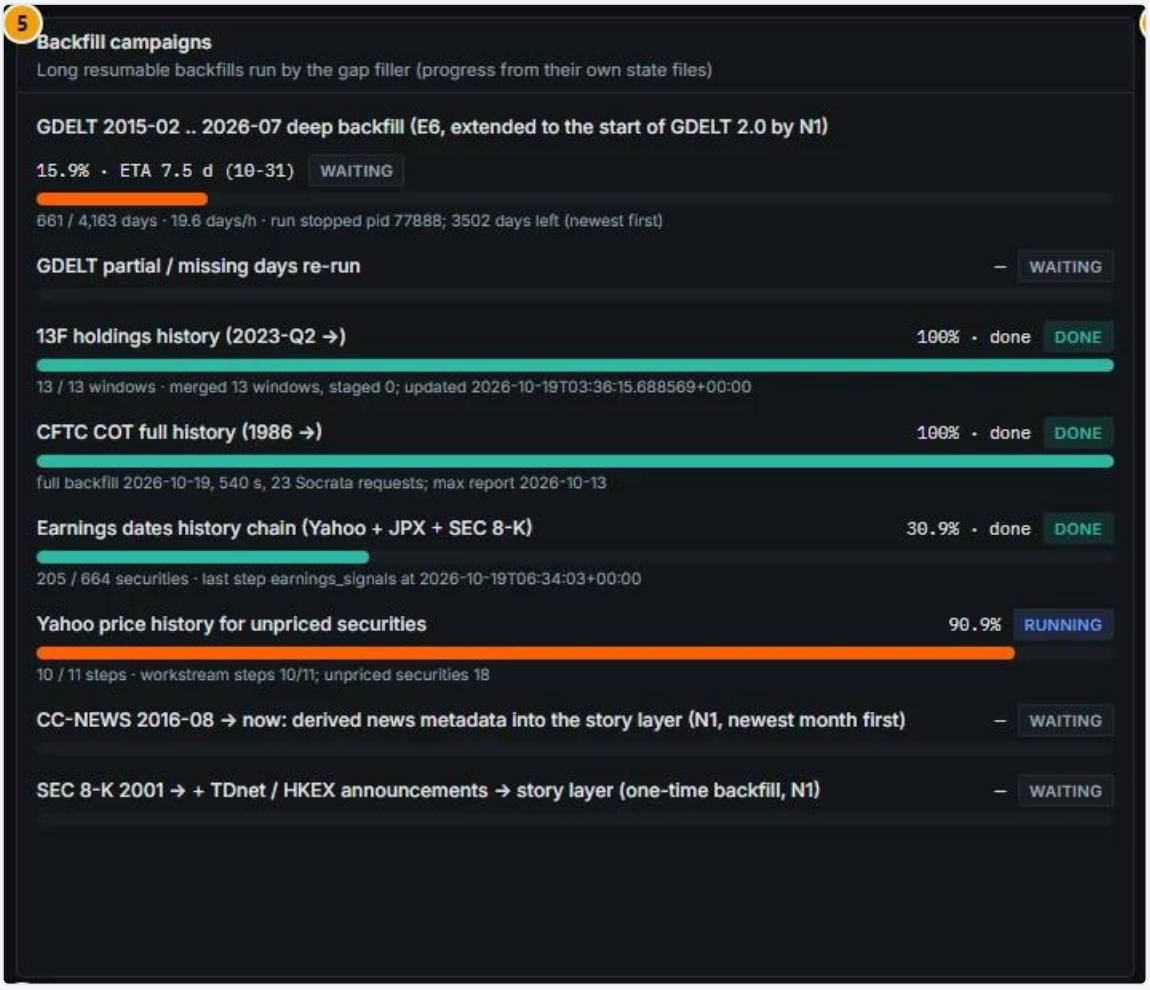


Figure 15.9: Backfill campaigns (panel 5): GDELT deep backfill 15.9% (661 of 4,163 days, 19.6 days per hour, ETA 7.5 days), 13F and CFTC histories done, Yahoo price history for unpriced securities running at 90.9%.

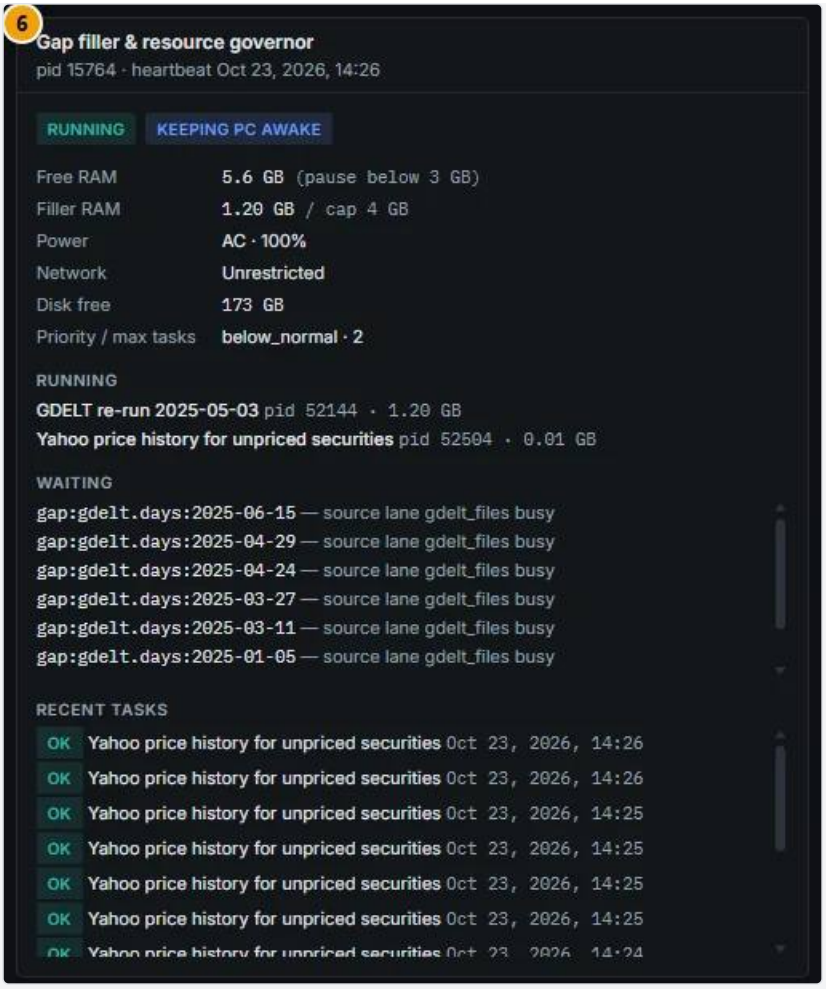


Figure 15.10: Gap filler & resource governor (panel 6): running and keeping the PC awake; 5.6 GB free RAM (pauses below 3 GB), filler 1.20 GB of its 4 GB cap, AC power, unrestricted network, below-normal priority, at most 2 tasks. GDELT gaps wait because the `gdelt_files` source lane is busy.

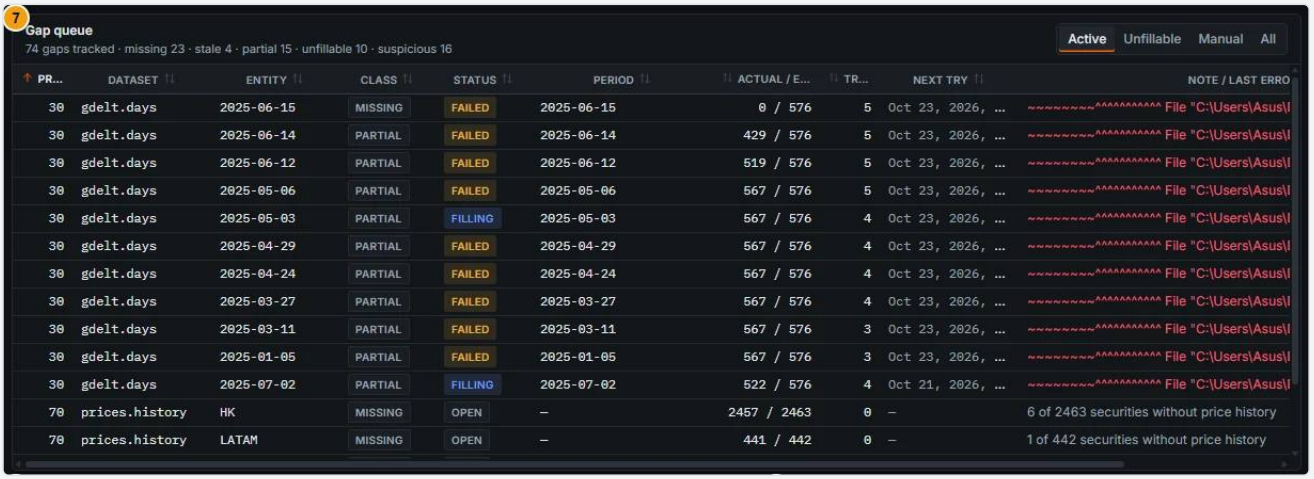


*Figure 15.11: Gap queue (panel 7), filter *Active**: GDELT days with 429–567 of 576 files are partial and being retried; two `prices.history` gaps are open.*

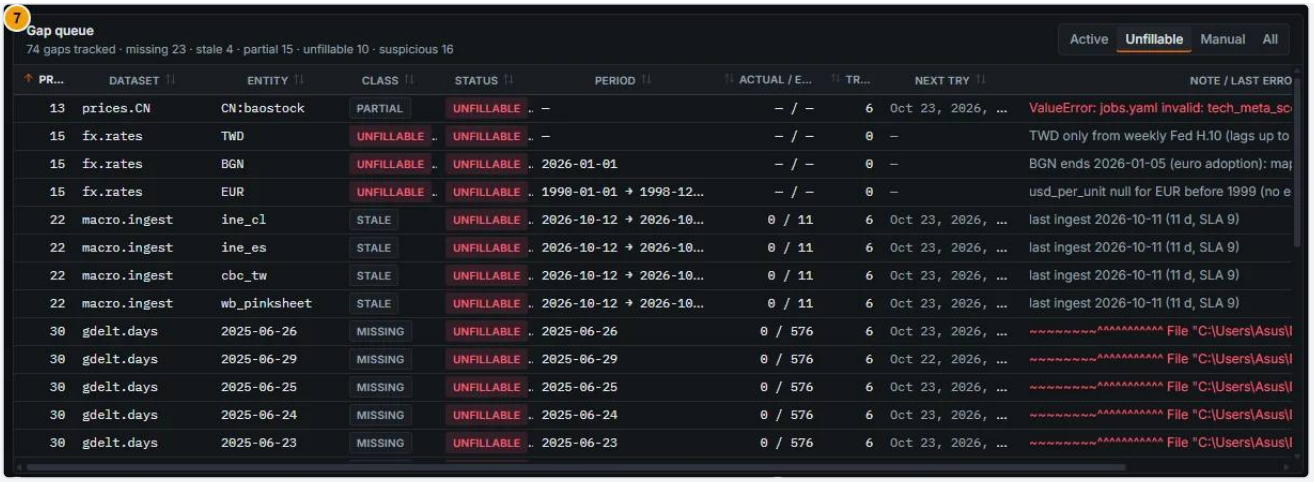


*Figure 15.12: Gap queue with the filter on *Unfillable**: Shanghai names without BaoStock (`CN:baostock`), FX gaps that no official source fills (TWD from the weekly Fed H.10 only, BGN after the euro adoption, EUR before 1999), macro sources stuck beyond their SLA (`ine_cl` , `ine_es` , `cbc_tw` , `wb_pinksheet`), and GDELT days with 0 of 576 files after 6 tries. The note column says why each gap cannot be filled.*

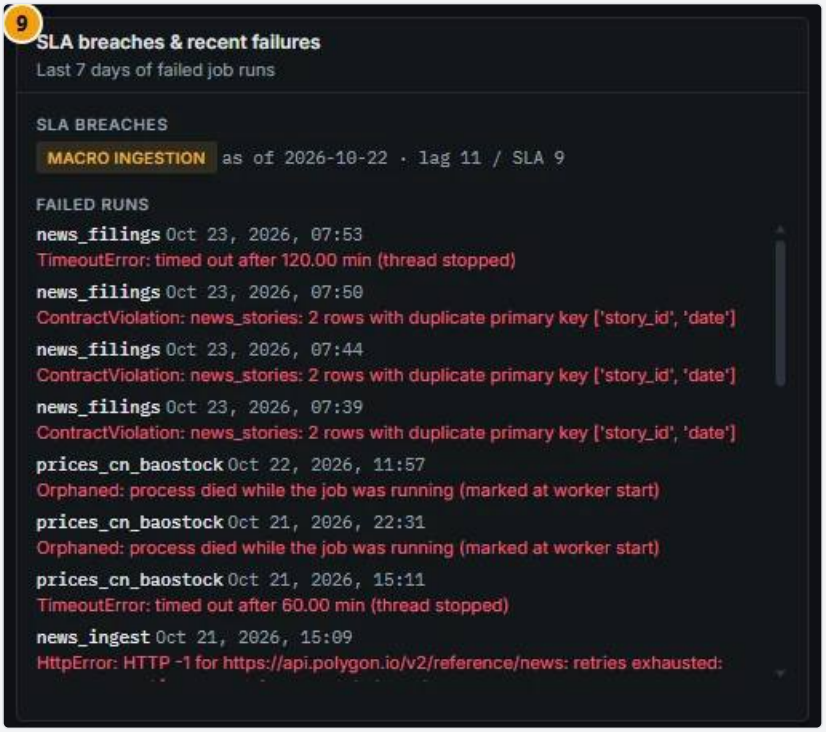


Figure 15.13: SLA breaches & recent failures (panel 9): Macro ingestion as of 22 Oct, lag 11 / SLA 9; failed runs of `news_filings` , `prices_cn_baostock` and `news_ingest` with their error type.

Controls

Control	What it does
Refresh	Re-reads `GET /system/data-health` .
Alert **Open** links	As on the System page.
Heatmap cells (hover or focus)	Tooltip "dataset · date: N% of expected", "no session", or "(after the data as-of: not ingested yet)".
Heatmap legend	≥ 95% of expected · 80–95% · 5–80% (partial) · < 5% (missing) · not ingested yet · no session.
Dataset name (hover)	Shows the dataset's detail line or registry id.
Dataset table headers	Sort by status, data as of, expected, Lag / SLA (hover: "Sessions (or days) behind the expected date vs the profile-adjusted SLA"), open gaps, live job.
Gap filter: **Active** / **Unfillable** / **Manual** / **All**	Active = open, queued, filling and failed gaps; Unfillable = the source has no data; Manual = needs a code or data fix, not a re-fetch; All = every gap including filled ones.
Gap note (hover)	Shows the full last error.
Source last error (hover)	Shows the source note.
Validation & full audit →	Opens Conformal validation.

Pausing the gap filler. There is no pause button on the page. Use the tray menu **Pause / Resume background jobs**, which creates or removes the flag file `data\state\ops\filler.PAUSE`; the filler then reports the reason "manual pause". Scheduled refreshes still run while it is paused. Stopping it entirely uses `data\state\ops\filler.STOP` or the supervisor.

Methodology

COVERAGE REGISTRY AND GAP DETECTOR

Each dataset is declared in `contracts/coverage.yaml` with its expected cadence (exchange calendars for prices, official release calendars for macro, the CFTC Friday schedule, 13F deadlines), the expected set of entities, the freshness SLA, its sources, priority and a backfill function. The job `data_gaps_detect` (after `build_serving_db` in the nightly DAG, and every 24 h from the filler) compares expected with actual:

```
cell(dataset, day) = actual entities / expected entities        (null = no session that day)
lag   = sessions (or days) between the latest complete date and the expected latest date
SLA   = the dataset's SLA, adjusted to the active profile (weekly profile: 9 days)
status: fresh (lag ≤ SLA) · stale (lag > SLA) · partial (fresh but with open partial gaps) · missing · unknown
```

Each gap gets a **class** and a **status**:

Class	Meaning
Missing	Expected data is absent (a whole session or period).
Stale	The dataset lags beyond its SLA.
Partial	Some entities are missing (for example a GDELT day with 512 of 576 files, a session with half the names).
Suspicious	Data exists but failed a check: DQ failure, unit or FX glitch, ticker reuse, and conformal under-coverage.
Unfillable	The source has no data; recorded so that nothing retries it. The note says what would fix it.

Status	Meaning
open → queued → filling	Waiting, scheduled and being repaired by the filler.
failed	The last attempt failed; it is retried with exponential back-off until `max_attempts`, then becomes unfillable.
manual	The filler cannot repair it (needs a code fix, a key or a decision). Known issues from the audit and `docs/PHASE3_TODO.md` are seeded here.
unfillable	No retry.
filled	No longer detected; kept with its resolution time.

GAP FILLER AND RESOURCE GOVERNOR

The filler (`python -m app.ops.filler`, started by the supervisor) is a long-running low-priority process. Every 15 s it reaps finished tasks, asks the governor whether it may run, then starts the most urgent due work that fits, one task per source lane. Defaults from `config/ops.yaml`:

Rule	Default
RAM cap of the filler's process tree	4 GB (above 1.5 × cap the least urgent preemptible task is stopped)
Pause when free RAM is below / resume above	3.0 GB / 3.5 GB
Maximum concurrent tasks	2
CPU priority	below normal
Pause on battery / on a metered network (phone hotspot)	yes / yes
Minimum free disk	15 GB
Yield to refresh DAGs	heavy tasks stop while `nightly`, `weekly` or `monthly` runs

Tasks are resumable, so a pause loses no work. Per-source **rate budgets** (Polygon about 5 per minute, SEC 10 requests per second, GDELT bandwidth, official-API limits) are shown in the Sources panel as "used in the last hour / per hour". The filler keeps the PC awake only while tasks run on AC power, and runs housekeeping every 6 hours (log rotation, two serving snapshots kept, disk guard).

BACKFILL CAMPAIGNS

Campaigns are long backfills folded into the filler (GDELT E6 deep history, GDELT partial-day re-runs, 13F history, CFTC history, earnings-dates chain, Yahoo price history, CC-NEWS metadata, SEC 8-K / TDnet / HKEX announcements). Progress comes from each campaign's own state file; status is running, waiting, backoff or done.

SOURCES

Per source, over 14 days of live-job runs: **failing** after 3 consecutive failed runs, **degraded** after at least one failure in the current streak or when more than a third of runs failed, **ok** otherwise, **idle** without runs.

CONFORMAL COVERAGE FLAGS

The coverage panel and the detector read the nightly audit table `conformal_coverage`. A rolling-audit segment is flagged when realised coverage is **more than 1.5 pp under target overall**, or **more than 5 pp under target in a segment with n ≥ 2,000** (`docs/CONFORMAL_VALIDATION.md`, "Coverage audit"). The detector records such segments as *suspicious* gaps with status *manual*: they are model issues, not data the filler can re-fetch. See Conformal validation for how to read them.

Data sources and refresh

- `GET /system/data-health` (registry status, heatmap, gaps, campaigns, filler, sources, failures, SLA breaches, alerts, QA summary); `GET /conformal/coverage`; `GET /live/news` for the live-news panel.
- State files: `data/state/ops/coverage_status.json` (detector), `data/state/ops/filler.json` (filler), `data/state/ops/qa_regression.json` (QA).
- The heatmap covers the last 21 days. Detection runs nightly after the snapshot build and every 24 h from the filler.

Validation status

Not a model: no validation gate. The coverage registry, detector and filler are operations tooling (O1.1–O1.8 complete; O1.9 "live switch-over and first unattended weekly run observed" was still open in `docs/OPERATIONS_ROADMAP.md` at the time of writing).

Correct usage and worked example

Before using a dataset for a decision, find its row in the dataset table: status, data as of and Lag / SLA. If it is not fresh, check the Gap queue (filter **All**, sort by dataset) for the reason.

Example (Figure 15.7, 23 Oct 2026; profile Weekly, SLA 9 days).

- Summary: 28 of 34 datasets fresh, 1 stale beyond SLA, 14 open gaps (2 filling, 30 unfillable, 22 manual), filler RUNNING, 173 GB free, last detection 22 Oct 22:31.
- **Macro ingestion** is stale: "4 of 33 source_id(s) stale", lag 11 vs SLA 9. It is the only SLA breach and the cause of the amber alert. Macro releases themselves are fresh ("0 of 104 releases in the last 30 days still pending").
- **Prices Japan**: data as of 13 Oct, expected 21 Oct, lag 6 / SLA 6, status fresh. One more missing session would turn it stale.
- **GDELT daily files** are partial: "755 days checkpointed; history 755/2486; 29 gap(s)". Event-based history is still being backfilled (campaign at 15.9%).
- Gap queue, filter **Manual**: for example "SPCX/BNY/FIG" (audit F04: ticker reuse stitched into one price series) and the UK "100x flips" (audit F32) are *suspicious* data that need a fix, not missing data. Filter **Unfillable** (Figure 15.12): "TWD" (FX only from the weekly Fed H.10) and "CN:baostock" (424 Shanghai names without the BaoStock feed).
- Regression guard: WARN — API contract sweep 144 passed, 1 warn, 11 skipped; golden data checks 12 passed.

Limitations and pitfalls

- "Fresh" is relative to the profile SLA. Under the Weekly profile a dataset can be up to 9 days behind and still be fresh.
- Datasets not in `contracts/coverage.yaml` are not tracked here (see the System page's source table for every lake table).
- *Partial* in the heatmap can mean a genuine hole or a late source. A cell after the dataset's data as-of is "not ingested yet", not a hole.
- *Suspicious* gaps with status *manual* (ticker reuse, unit flips, conformal under-coverage) do not block publishing. The affected numbers stay in the app until fixed; check the queue before relying on an outlier.
- The page shows the filler's last heartbeat; if it is older than 5 minutes the filler is treated as not running.
- Error notes in the Gap queue and the Report path of the Regression guard show local file paths of the machine (including the Windows user folder). Crop them out before sharing a screenshot outside your team.

Related mnemonics and BQL fields

- No command-bar function; the tray menu item **Data health** opens this page.
- No BQL fields.
- Related: System overview, Conformal validation, QA guard, Data and coverage, Good practice: coverage gaps by region.

Conformal validation

Route: `/system/conformal` (`/conformal` redirects here) · **Menu:** System › Conformal validation · **Mnemonics:** none · **Refresh:** nightly audit (`conformal_coverage` job); the gate itself changes only when the validation is re-run

Purpose

This page is the audit trail of the conformal sets used on the Technicals chart (calibrated range and fan), in the touch probabilities, and in the peer-group anomaly panels. It shows which applications passed the P4 validation gate of `docs/CONFORMAL_VALIDATION.md` (and are shown in the app, labelled model-based), which failed and stay experimental in Model Studio and the Backtest Lab, and whether the nightly rolling audit

still finds the target coverage. Use it when you cite a "calibrated band" or a touch probability, or when the Data Health panel shows a coverage flag.

Screenshot

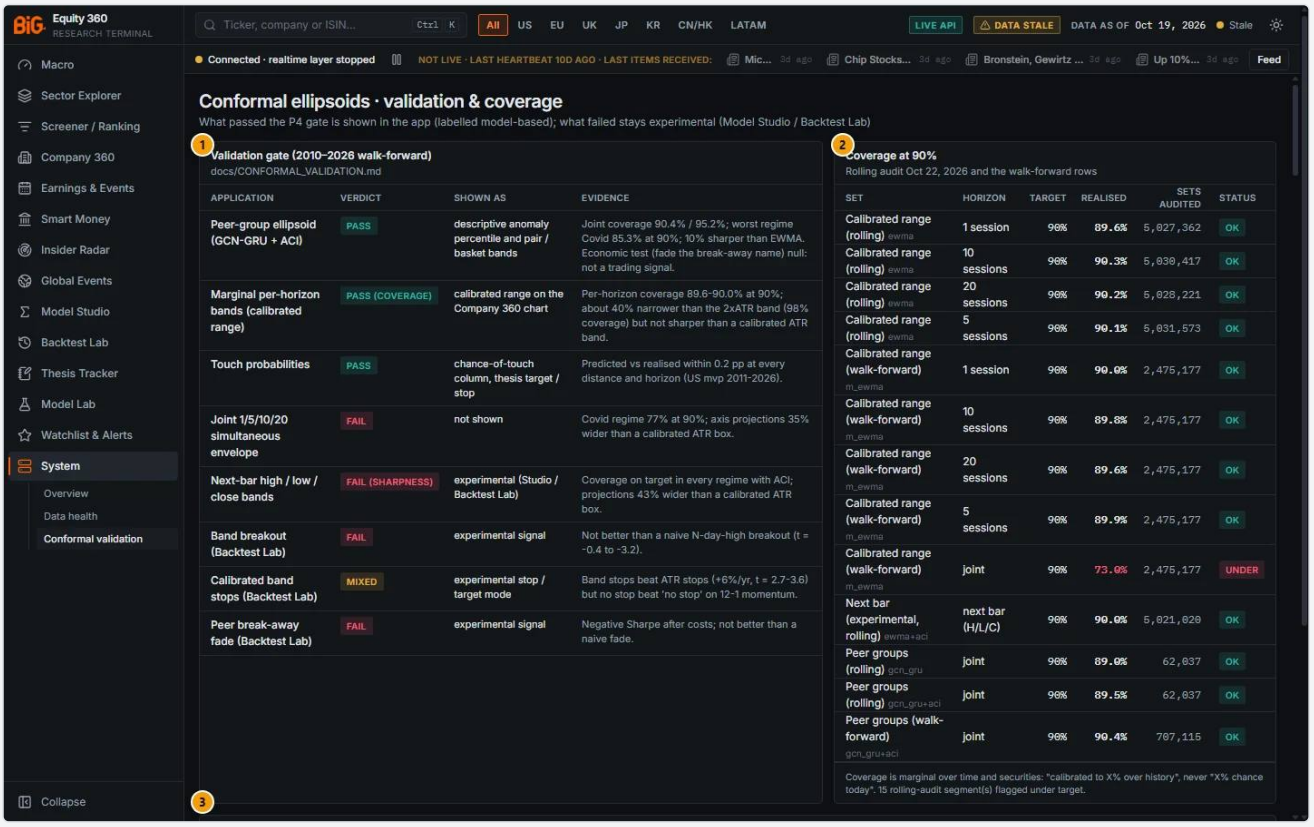


Figure 15.14: Conformal ellipsoids · validation & coverage (`/system/conformal`), rolling audit of 22 Oct 2026.

#	Panel	What it shows
1	Validation gate (2010–2026 walk-forward)	One row per application: verdict badge (PASS, PASS (COVERAGE), MIXED, FAIL (SHARPNESS), FAIL), how it is shown in the app, and the evidence.
2	Coverage at 90%	Headline realised coverage at the 90% level: the rolling audit (latest date) and the walk-forward rows, with status badges and the number of flagged rolling-audit segments.
3	Production configuration	The filters and radii in production (JSON): envelopes `ewma` with a plain radius for the joint and marginal bands and ACI for the next bar, learned head not deployed; peer groups `gcn_gru` with ACI, fallback `ewma_lr` , and the model and group versions.
4	Validation report	The full text of `docs/CONFORMAL_VALIDATION.md` , with its tables rendered as monospace blocks.

Figure 15.15: Production configuration (panel 3).

4 **Validation report**
docs/CONFORMAL_VALIDATION.md

Conformal ellipsoids: P4 validation gate

Plan: `docs/CONFORMAL_TECHNICALS_PLAN.md` . Paper: Limmer, *Filtered Conformal Ellipsoids for Graph-Native Time Series* (arXiv:2606.17014v2). Code: `backend/app/conformal/` . Scripts: `scripts/conformal/` (`wf_envelopes.py` , `wf_peers.py` , `econ.py` , `touch_check.py` , `report.py`). Raw results: `data/reports/conformal/` (`wf_env/` , `wf_peer/` incl. break-away signals, `econ.json` , `econ_returns.parquet` , `touch_reliability.json` , and `summary.md` , the full set of generated tables). The walk-forward coverage rows are also in the lake table `conformal_coverage` (applications `envelope_wf` , `nextbar_wf` , `peer_wf`), next to the nightly rolling audit.

Everything here is **model-based.** Coverage is **marginal over time and securities**: "calibrated to X% over history", never "X% chance today".

Verdict

Application	Coverage ±1.5 pp	No regime > 5 pp under	Sharper than baselines at matched coverage	Economic value	Gate
A, joint 1/5/10/20 envelope (ellipsoid)	PASS (89.6 / 94.5%)	**FAIL** (Covid Feb-Jun 2020: 77-78% at 90%)	**FAIL** on w_proj (+35% vs a calibrated ATR box); PASS on log-volume	(i) breako
A, next bar (H, L, C) (ellipsoid)	PASS (90.0 / 95.0%)	PASS (Covid 91.2%, VIX>25 89.1%)	**FAIL** on w_proj (+43% vs calibrated ATR box); PASS on log-volume	not tested separately	**FAIL
B, peer-group ellipsoid (GCN-GRU + ACI)	PASS (90.4 / 95.2%)	PASS (worst: Covid 85.3% at 90%, i.e. -4.7 pp)	PASS: -10% w_proj vs EWMA, -9% vs static Ledoit-Wolf, -13% vs PCA factor; sharpe

Recommendation:

- **A stays out of the main UI.** It remains "experimental" in Model Studio and the Backtest Lab.
- Two A by-products did validate and are candidates for P3 in place of the heuristic ATR bands:
- The **marginal single-horizon conformal bands** (`qm_*`): on target per horizon (89.6-90.0%), about 40% narrower than the conventional 2×ATR band, which covers 98%. They are not sharper than a conformally calibrated ATR band; the gain comes from calibration, not from the filter.
- The **touch probabilities**: predicted equals realised within 0.2 pp at every distance and horizon (see §A.4). A Gaussian reflection formula is 5× too low in the tail.
- **B passed the statistical gate.** It can be shown as a model-based anomaly percentile and pair/spread band, without a trading-signal framing.

Setup

- **Universe, A:**
- mvp US (485 of the 500 have enough history) plus the mvp_global EU (90), UK (27) and JP (40) names;
- prices 2003 → 2026-10-20; test years 2010 → 2026.
- Survivorship: the current mvp membership is used throughout, which is a known bias. EU/JP earnings dates only exist from about 2014-2015, US from 2010.
- **Universe, B:**
- the 259 current peer groups (version 20261020, see P2);
- per test year, members with ≤ 2% missing / ≤ 20% zero returns over [Y-6, Y], with N ≥ 20. This gives 2,955 group-years and 707,115 group-days.
- **Protocol (Algorithm 1, with refits):**
- Fitted filters are fit on data before Jan 1 of Y-1 and then frozen:
- GARCH / static / factor are fit yearly;
- the GRU / GCN-GRU are fit once per 3-year block on all securities / groups.
- Calibration uses a rolling window of the last 500 realised scores, restricted to origins on or after Jan 1 of Y-1, so only the frozen model's scores are used.
- Calibration is pooled per region × sector for A and per group for B, with separate earnings-event pools for A.
- Outcome lag: 20 sessions for the joint set, 1 for next-bar / B.
- ACI uses γ = 0.005 and restarts each year.
- Warm-up discard: ≥ 50.
- **Regimes:**
- "Covid 2020" = origins from 15 Feb to 30 Jun 2020;
- "2022" = the rates shock year;
- "VIX>25" = FRED VIXCLS > 25 at the origin.
- **Size metrics (paper eq. 3-4):**
- **w_proj** is the mean coordinate-projection length 2·half-width, in log-return units, averaged over coordinates.
- **log-volume** is the normalised log-volume (for boxes: the mean log of the side lengths).
- **"unbounded"** is the share of sets with an infinite radius: ACI at *α ≤ 0, or a Bonferroni quantile index beyond the window. These sets count as covered; their widths are excluded from w*proj.

A. Conformal price envelope (walk-forward)

A.1 JOINT 1/5/10/20-SESSION SET, TARGET 90%

method	radius	joint cov	per-horizon cov h1/h5/h10/h20	worst year	Covid 2020	2022	VIX>25	earnings window	w_proj	log-vol	unbounded
ewma (ellipsoid)	plain	**89.6**	98.7/98.7/98.7/98.8	84.7 (2020)	77.0	89.4	90.2	89.2	0.3134	-1.927	0
ewma	aci	89.7	98.8/98.7/98.7/98.8	87.6 (2020)	78.3	90.2	92.1	89.2	0.3291	-1.913	1.1
gru (learned head)	plain	89.3	98.7/98.6/98.7/98.7	85.0 (2020)	77.3	89.8	90.9	89.2	0.3074	-1.942	0
gru	aci	89.6	98.7/98.7/98.7/98.8	87.8 (2020)	78.6	90.3	92.9	89.4	0.3246	-1.925	1.1
garch	plain	88.7	97.9/97.8/97.7/97.8	81.6 (2020)	72.6	85.4	85.9	88.2	0.3014	-1.955	0
garch	aci	89.4	98.0/97.9/97.9/97.9	85.8 (2018)	76.8	88.3	89.6	88.9	0.3621	-1.915	1.1
static (4×4 Σ per security)	plain	88.5	98.2/98.1/98.2/98.2	76.9 (2020)	54.2	80.5	77.6	87.7	0.3197	-1.842	0
static	aci	90.0	98.4/98.4/98.4/98.4	83.8 (2018)	68.2	87.2	85.3	89.4	0.3281	-1.829	2.9
factor (rank 1 + diag)	plain / aci	88.7 / 90.0	98.6-98.9	77.1 / 83.9	55.6 / 69.4	81.4 / 87.5	78.9 / 86.2	87.9 / 89.4	0.3565 / 0.3665	-1.768 / -1.755	0 / 2.8
atr_cal (box, conformal ATR multiple)	plain	89.7	96.6/96.3/96.3/96.4	85.9 (2020)	77.3	89.5	90.8	89.4	**0.2319**	-1.713	0
atr_cal	aci	89.8	96.6/96.3/96.3/96.4	88.4 (2020)	78.9	90.0	92.3	89.3	0.2461	-1.696	0.7
atr_k2 (heuristic 2×ATR×√h, uncalibrated)	fixed	94.8	98.4/98.1/98.2/98.2	89.6 (2020)	81.7	95.9	95.1	91.2	0.2812	-1.505	0
uni_ewma (per-horizon CP at α/4, Bonferroni)	plain / aci	92.1 / 92.2	97.1-97.5	88.4 / 90.1	80.0 / 82.4	92.0 / 92.5	92.7 / 93.9	92.2	0.2659 / 0.2487	-1.616 / -1.657	0 / 6.8
uni_static	plain / aci	91.8 / 93.0	96.8-97.7	83.7 / 88.7	65.9 / 78.8	87.1 / 91.3	84.8 / 90.9	91.5 / 92.6	0.2662 / 0.2602	-1.544 / -1.562	0 / 6.4

At 95%:

- ewma: 94.5% (plain), 94.6% (ACI); w_proj 0.3775 / 0.3660; Covid 80.1 / 81.5.
- atr*cal: 94.6%; w*proj 0.2889; log-vol -1.504 vs ewma's -1.749.

Per region (90%, *ewma* plain): EU 89.7, JP 89.7, UK 89.5, US 89.5. By year (ewma plain): 84.7 (2020) to 94.9 (2021). Full tables are in `summary.md` .

Reading:

- **Coverage.** The calibrated ellipsoids are on target overall (EWMA 89.6/94.5) in every region.
- **Regimes.** Every method fails the 20-session Covid window, baselines included: the shock arrived after the origins, and a 20-session outcome lag means neither the shape nor ACI can react in time. ACI lifts the worst year from 84.7 to 87.6, but costs +5% width and gives 1-5% unbounded sets.
- **Sharpness.**
- The ellipsoid has the smallest **volume** of all calibrated sets (-1.93 vs -1.71 for the ATR box: about 20% shorter sides in geometric mean).
- However, its axis projections (w_proj, the paper's headline metric) are 35% **wider** than a calibrated ATR box. The four overlapping horizons are strongly correlated (ρ(Y10, Y20) ≈ 0.7); for such coordinates a box reaches joint coverage with narrower sides, while the minimum-volume ellipsoid is long along the diagonal. The ellipsoid's per-horizon projections cover about 98.7%, which makes them simultaneous bands, not 90% bands.
- The learned GRU head is 1-2% sharper than EWMA (0.307 vs 0.313; ACI 0.325 vs 0.329) at the same coverage. That is not material, so the mvp learned head is **not** deployed.

A.2 MARGINAL SINGLE-HORIZON BANDS (EACH HORIZON CALIBRATED ALONE; THE CHART-BAND USE CASE), TARGET 90%

method	radius	cov h1/h5/h10/h20	width h1/h5/h10/h20	Covid h1/h5/h10/h20	VIX>25 h1/h20
m_ewma (EWMA scale + jump term)	plain	90.0/89.9/89.8/89.6	0.0605/0.1386/0.1979/0.2818	87.7/86.8/83.9/82.4	87.9/92.3
m_ewma	aci	90.0/89.9/89.8/89.6	0.0612/0.1412/0.2064/0.3112	89.3/88.5/85.1/83.0	88.6/93.2
m_gru	plain	89.8/89.7/89.6/89.5	0.0589/0.1361/0.1949/0.2791	87.7/86.8/83.8/82.4	88.7/93.0
m_garch	plain	89.5/89.4/89.3/89.1	0.0594/0.1354/0.1918/0.2705	83.8/82.8/80.4/78.5	85.5/89.0
m_str (ATR scale, conformal multiple)	plain	90.0/89.9/89.9/89.6	0.0585/0.1349/0.1931/0.2755	85.7/85.2/83.0/82.0	87.9/93.1
m_static	plain	89.7/89.7/89.6/89.5	0.0617/0.1388/0.1950/0.2751	65.7/66.2/65.7/64.6	78.6/83.4
atr_k2 (heuristic)	fixed	98.4/98.1/98.2/98.2	0.1035/0.2314/0.3272/0.4628	96.3/94.0/90.8/88.0	98.0/98.6

What this shows:

- The conformal step turns any scale into calibrated bands:
- a 90% band that really covers 90%;
- about 40% narrower than the heuristic 2×ATR band, which over-covers at 98%.

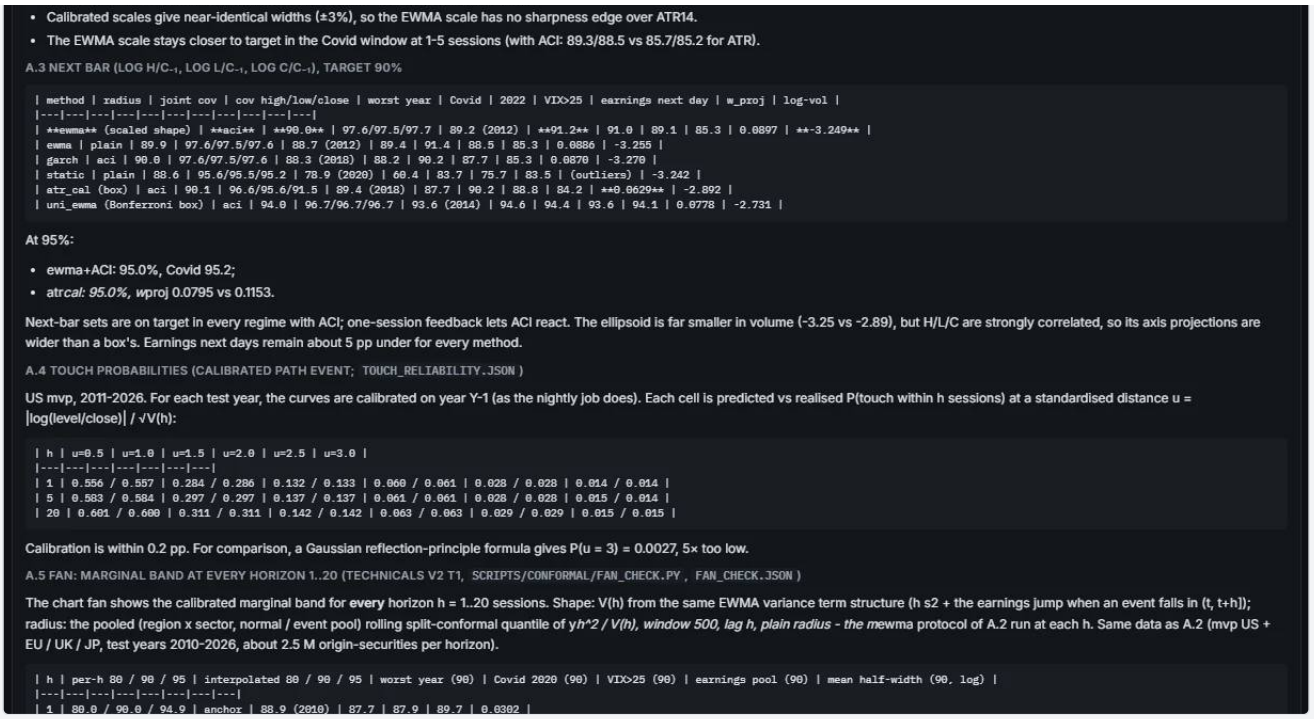


Figure 15.16: Validation report (panel 4), the validation document rendered in the app.

Controls

The page has no inputs. Scroll the Validation report; wide tables scroll horizontally inside their block. The **Validation & full audit** → link on Data Health and the "calibrated band" links on the Technicals tab lead here.

Methodology

THE GATE

The P4 gate (walk-forward 2010–2026, test years 2010 → 2026) required, per application:

1. coverage within ±1.5 pp of target (90% and 95%);
2. no regime more than 5 pp under target (Covid 2020 = origins 15 Feb – 30 Jun 2020, 2022, VIX > 25);
3. sharper than the baselines at matched coverage on **w_proj** (mean axis-projection width) — log-volume is reported too;
4. economic value in a Backtest Lab test with costs.

The verdicts shown on the page (from `GET /conformal/coverage` , field `gates`):

Application	Verdict	Shown as	Evidence (as on the page)
Peer-group ellipsoid (GCN-GRU + ACI)	PASS	descriptive anomaly percentile and pair / basket bands	Joint coverage 90.4% / 95.2%; worst regime Covid 85.3% at 90%; 10% sharper than EWMA. Economic test (fade the break-away name) null: not a trading signal.
Marginal per-horizon bands (calibrated range)	PASS (coverage)	calibrated range on the Company 360 chart	Per-horizon coverage 89.6–90.0% at 90%; about 40% narrower than the 2×ATR band (98% coverage) but not sharper than a calibrated ATR band.
Touch probabilities	PASS	chance-of-touch column, thesis target / stop	Predicted vs realised within 0.2 pp at every distance and horizon (US mvp 2011–2026).
Joint 1/5/10/20 simultaneous envelope	FAIL	not shown	Covid regime 77% at 90%; axis projections 35% wider than a calibrated ATR box.
Next-bar high / low / close bands	FAIL (sharpness)	experimental (Studio / Backtest Lab)	Coverage on target in every regime with ACI; projections 43% wider than a calibrated ATR box.
Band breakout (Backtest Lab)	FAIL	experimental signal	Not better than a naive N-day-high breakout (t = −0.4 to −3.2).
Calibrated band stops (Backtest Lab)	MIXED	experimental stop / target mode	Band stops beat ATR stops (+6%/yr, t = 2.7–3.6) but no stop beat "no stop" on 12-1 momentum.
Peer break-away fade (Backtest Lab)	FAIL	experimental signal	Negative Sharpe after costs; not better than a naive fade.

ROLLING AUDIT VERSUS WALK-FORWARD ROWS

- **Walk-forward rows** (`envelope_wf` , `nextbar_wf` , `peer_wf`) are the fixed results of the validation itself (frozen models, test years 2010–2026). They do not change nightly.
- **Rolling audit rows** (`envelope` , `nextbar` , `peer`) are recomputed every night by `conformal_envelopes` , `conformal_peer_groups` and `conformal_coverage` on the most recent realised outcomes (about the last year of origins), per region, sector, group kind, year, macro regime and high-VIX segment.

Each row has a status: **ok** if |realised − target| ≤ 1.5 pp, **under** or **over** otherwise, **thin** if n < 200.

The headline table keeps, at the 90% level and the overall segment: the rolling marginal bands at 1, 5, 10 and 20 sessions (EWMA), the walk-forward marginal bands (m_ewma), the rolling next-bar set (ACI), the rolling peer-group joint sets, and the walk-forward peer-group set (GCN-GRU + ACI).

Flag rule. A rolling-audit row is flagged when it is under target by more than 1.5 pp overall, or by more than 5 pp in any segment with n ≥ 2,000. Walk-forward rows are never flagged. The count appears under the coverage table; the list is in the Data Health panel.

Data sources and refresh

- `GET /conformal/coverage?include_doc=true` (rows, flags, gates, production configuration, the validation document).
- Nightly jobs `conformal_envelopes`, `conformal_peer_groups`, `conformal_coverage`; weekly `conformal_peer_groups_fit`.
- Gate verdicts change only when the validation scripts (`scripts/conformal/`) are re-run and the document updated.

Validation status

This page *is* the validation status of the conformal family. In short (`docs/CONFORMAL_VALIDATION.md`, Verdict and Recommendation): the joint envelope **failed** and is not in the main UI; the next-bar set **failed on sharpness** and is experimental; the marginal single-horizon bands and the touch probabilities **validated** as by-products and are shown, labelled "calibrated band"; the peer-group ellipsoid **passed** as a calibrated descriptive measure and is **not** a trading signal. Coverage is marginal over time and securities: "calibrated to X% over history", never "X% chance today".

Correct usage and worked example

Example (Figure 15.14, audit dated 22 Oct 2026). The Coverage at 90% panel showed:

Set	Horizon	Realised	n	Status
Calibrated range (rolling)	1 / 5 / 10 / 20 sessions	89.6% / 90.1% / 90.3% / 90.2%	about 5.03 M each	ok
Calibrated range (walk-forward)	1 / 5 / 10 / 20 sessions	90.0% / 89.9% / 89.8% / 89.6%	2,475,177	ok
Calibrated range (walk-forward)	joint	73.0%	2,475,177	under
Next bar (experimental, rolling)	next bar (H/L/C), ACI	90.0%	5,021,020	ok
Peer groups (rolling)	joint, GCN-GRU / GCN-GRU + ACI	89.0% / 89.5%	62,037	ok
Peer groups (walk-forward)	joint, GCN-GRU + ACI	90.4%	707,115	ok

How to read it:

- The 20-session calibrated band on a Company 360 chart can be cited as "calibrated to 90% over history (2010–2026 walk-forward: 89.6%; last year's rolling audit: 90.2%)".
- The **joint 73.0% "under"** row is expected, not a fault: it is the share of origins where all four marginal 90% bands covered at once. Marginal bands are calibrated one horizon at a time and are not simultaneous bands. Do not read the four bands of a fan as a joint 90% statement.
- 15 rolling-audit segments were flagged. Most are the experimental next-bar set on **gap days** (realised 13.0% at 90% on 26,561 days with a large overnight gap) and peer groups in specific segments (CN region 84.6% at 90%, n = 4,766; the GLOBAL contraction quadrant 82.3%, n = 2,253). These say where the sets are known to under-cover; they do not invalidate the overall result, but treat a peer-group anomaly percentile for a Chinese group, or a next-bar band on an earnings or gap day, with extra caution.

Limitations and pitfalls

- Coverage is **marginal**: averaged over all securities and days of a pool. It is not a guarantee for one stock on one day.
- Per-regime coverage is reported, not guaranteed. No method tested meets the regime criterion for the 20-session joint set in a sudden crash (Feb–Mar 2020).
- The validation used current mvp membership and current sectors (survivorship bias), and EU / JP earnings dates only from about 2014–2015.
- The rolling audit covers roughly the last year; a calm year says little about the next stress episode.
- The PASS for peer groups is statistical only: fading the break-away member had no economic value after costs.

Related mnemonics and BQL fields

- `GP` / `TECH` open the Technicals view where the calibrated bands and touch probabilities are drawn; `RV` opens the peer comparison with the conformal peer-group panel.
- BQL fields under `cf.*` (conformal) expose the bands and anomaly scores; see BQL and mnemonics.
- Related: Technicals, Methodology: conformal prediction, Gate status of every model.

QA guard

Route: panel **Regression guard** on `/system/data-health` ; summary also at `GET /api/v1/system/qa` · **Menu:** System › Data health · **Mnemonics:** none · **Refresh:** job `qa_regression` after each refresh (in the Weekly profile, inside the Saturday chain right after the snapshot is published); by hand with `python -m app.qa`

Purpose

The regression guard (`docs/OPERATIONS_ROADMAP.md` O2.3) keeps the app's pages and APIs working after every data refresh or code change. It runs three read-only suites and reports the result on Data Health and as an ops alert. For an analyst, a green guard means the pages load, the API answers in the contracted shape, and a handful of key facts (Apple revenue, the S&P 500 level, US CPI, ...) come out right through the full data path. It does not validate any model.

Screenshot

The guard is panel 2 of the Data Health page (Figure 15.7).

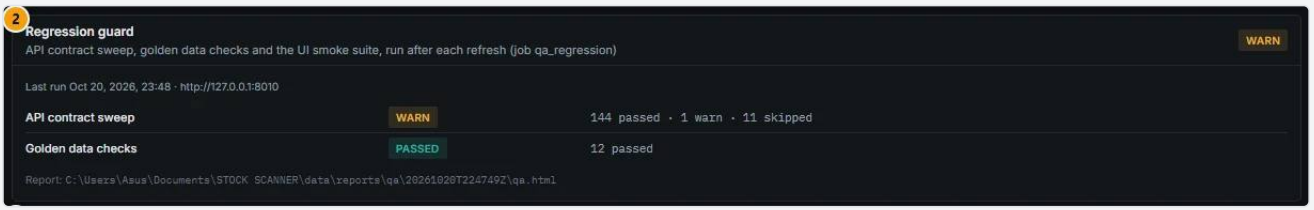


Figure 15.17: Regression guard panel on Data Health: run of 20 Oct 2026 23:48 (Lisbon) against the live API, status WARN.

The panel's elements (not numbered in the image):

#	Element	What it shows
1	Status badge	passed (green), warn (amber), failed (red), pending or skipped (grey).
2	Last run line	Finish time and whether it ran against the live API or an in-process client over the serving snapshot ("in-process API (no server running)").
3	Suite rows	**API contract sweep**, **Golden data checks**, **UI smoke suite**: status, counts ("144 passed · 1 warn · 11 skipped"), and a message (for example "API serves 1.31.0, contract 1.31.0" when they differ).
4	Failed checks	Up to ten failed items with their suite and detail, then "+N more in the report".
5	Report	Path of the HTML report of the run.

Controls

The panel has no controls. To run the guard by hand, from `backend\` with the project virtual environment:

```
python -m app.qa                                # all suites against the running API (+ UI smoke when Chrome and the UI are up)
python -m app.qa api golden                     # only the API sweep and the golden checks
python -m app.qa smoke --themes dark --no-tabs  # UI smoke, dark theme only, without clicking tabs
python -m app.qa --api http://127.0.0.1:8010 --no-write   # dry run: no report, no state written
```

Exit code 0 = passed, warn or skipped; 1 = failed. Reports go to `data\reports\qa\<run>\qa.html` (and `data\reports\smoke\<run>\` for the UI suite).

Methodology

Code in `backend/app/qa/` : `api_sweep.py` , `golden.py` + `golden.yaml` , `smoke.py` + `smoke_routes.yaml` , and `regression.py` , which orchestrates the suites and writes the summary.

Suite (CLI name)	What it checks	Fails on
API contract sweep (`api`)	Every GET operation of the served OpenAPI contract, called with real ids resolved from list endpoints or search.	Non-2xx (501 = contract operation without implementation), 5xx or timeout; a JSON response that does not match the operation's schema; NaN or ±inf values; an "as of" date in the future (1-day tolerance for time zones); snapshot or region data-as-of outside the profile SLA. Slow answers and a running API older than the contract are warnings. Operations whose id has no instance (no thesis, no saved strategy) are skipped.
Golden data checks (`golden`)	12 stable or bounded facts read through the app's own API (`golden.yaml`).	A value outside its tolerance or range, or an observation older than its maximum age. Some checks are warnings only.

Suite (CLI name)	What it checks	Fails on
UI smoke suite (`smoke`)	Headless Chrome over every route of `App.tsx` (plus extras such as `/company/7203.T`) in both themes (dark at 1440 px, light at 1280 px), clicking every tab.	Console errors or uncaught exceptions; failed requests (HTTP ≥ 400 on `/api` except an allowed list); error panels, or empty panels where data is expected; a crashed route or the 404 page; load to network-quiet over 8 s; horizontal overflow. Pending panels and slow-but-in-budget loads are warnings. On `/news` , first rows within 1,000 ms and a live update within 100 ms. Skipped, not failed, when Chrome or the UI is unavailable.

The golden checks:

Check	Expectation
Apple FY2024 revenue (10-K)	USD 391.035bn ± 0.5%
Apple FY2023 revenue	USD 383.285bn ± 0.5%
Microsoft FY2024 revenue	USD 245.122bn ± 0.5%
Microsoft shares outstanding	7.0bn – 7.8bn
Apple market cap (USD)	1.5tn – 8tn (warning only)
S&P 500 level	4,000 – 12,000
Nikkei 225 level	25,000 – 80,000 (warning only)
US CPI-U y/y	−1% to +8%, observation ≤ 75 days old
EUR/USD (Fed H.10)	0.90 – 1.40, ≤ 21 days old
USD/JPY (Fed H.10)	100 – 200, ≤ 21 days old (warning only)
US 10Y Treasury yield	1% – 7%, ≤ 14 days old
US unemployment rate	2% – 12%, ≤ 75 days old (warning only)

A golden failure means the served data path is broken (units, FX, period mapping, stitching), not that the market moved.

Overall status: failed if any suite failed, else warn if any warned, else passed; skipped suites do not change the result. **Read-only:** the smoke suite blocks every non-GET API call in the browser except a short list of compute-only POSTs (screener query, scenario run) and uses a fresh browser profile, so your saved app state is never written. The job `qa_regression` ends **partial** when a suite fails (the run itself worked; nothing downstream waits for it) and raises the ops alert `qa_regression` .

Data sources and refresh

- Summary: `data/state/ops/qa_regression.json` , served by `GET /system/qa` and inside `GET /system/data-health` .
- Job `qa_regression` (nightly DAG, soft-depends on `alerts_eval` ; timeout 60 min, no retries).
- Before the first run the panel says "Not run yet. It runs after the next refresh, or by hand with python -m app.qa (backend/)."

Validation status

Not a model; no gate. The guard is itself the regression gate for pages and APIs. At the time of writing `docs/OPERATIONS_ROADMAP.md` lists the code as done and "First scheduled run observed" as still open.

Correct usage and worked example

Example: run QA20261020T224749Z (Figure 15.17; finished 20 Oct 2026 22:48 UTC, 23:48 Lisbon, live API on :8010). Status **warn**: API contract sweep warn (153 operations: 144 passed, 1 warn, 11 skipped; served and contract version both 1.31.0; latency median 66 ms, p95 412 ms); Golden data checks passed (12 of 12). This run did not include the UI smoke suite. The ops alert reads "Regression guard warnings — api: warn (0 failed)".

What it means for you:

Guard result	What to do
passed	Pages, API shapes and key facts are fine. Data freshness is a separate question (see Data Health).
warn	Usable. Typical causes: a slow endpoint, an API older than the contract (restart due), a pending panel, a warning-only golden check.
failed: golden	Treat the numbers of that area as suspect (units, FX, period mapping) until fixed. For example a failed Apple revenue check means financial statements may be mis-scaled.
failed: api	Some panels may show errors or wrong shapes; the failed item names the endpoint.
failed: smoke	A page or tab crashed, showed an error or empty panel, or overflowed; the item names the route, tab and theme.
skipped (smoke)	Chrome or the UI was not reachable when the job ran; nothing was tested in the browser.

Limitations and pitfalls

- The guard tests that pages render and that a few facts are right. It does **not** check that models are calibrated or that every number is correct.
- Golden checks are deliberately wide ranges; they catch unit and scale errors, not small mistakes.
- Empty panels are allowed on pages whose content is user-created (`/watchlist` , `/theses` , `/studio` , `/backtest` , …); an empty watchlist does not fail the suite.
- The smoke suite needs Chrome and a running UI. On a scheduled run without them it is skipped, and the summary still reads passed or warn.

Related mnemonics and BQL fields

- No command-bar function or BQL fields.
- Related: Data Health, System overview, Troubleshooting and FAQ.

16. Methodology appendix

This appendix explains the statistical methods that recur across the app, in one place, so that each page chapter can refer to them. It ends with a summary table of the gate status of every model. The numbers quoted here come from the validation documents in `docs/` and from the gate files the jobs read (`data/reports/.../gate.json`); where a model has no formal gate, the table says so.

Two rules apply to everything below:

- **Model-based means ours.** Anything labelled model-based is a transformation or forecast made by the app, not a published statistic or a market consensus.
- **Calibrated over history, never "chance today".** A probability or a band that is "calibrated to 90%" was right 90% of the time on average over many past stocks and days. It is not a statement that the event has a 90% chance for this stock today.

Point-in-time data

A backtest, a model or a historical screen is only honest if, at each past date D, it uses what was **known** at D. The app enforces this with a knowledge date on every versioned fact (`docs/ARCHITECTURE.md` §7).

Knowledge dates by source

Source	Knowledge date
SEC filings	`filed` date (acceptance time stored as well)
EDINET / DART / CVM	submission / receipt date of the filing
ESEF	the filing's added or processed date (conservative), or its publication date
FRED macro	ALFRED `realtime_start` (the vintage date)
GDELT	`DATEADDED`
Form 4 insider trades	the filing date, never the transaction date
13F holdings	the filing deadline (quarter end + 45 days, next SEC business day) + 1 business day
CFTC Commitments of Traders	the release (normally Friday 15:30 ET), not the Tuesday as-of date

Where a source publishes no date (yfinance fundamentals, some AKShare data, macro without vintages), the knowledge date is **imputed** and flagged `knowledge_date_estimated = true` :

Case	Lag after period end
US quarterly	+45 days
US annual	+75 days
Non-US quarterly / half-year	+60 days
Non-US annual	+120 days
Macro	the series' `release_lag_days`

Rules

- **As-of query.** For a date D, take, per natural key, the row with the largest knowledge date ≤ D.
- **Restatements** are kept as new versions. A restatement of an older period never overwrites a newer one; a same-period restatement counts from its own filing date.
- **Labels.** Forward returns run from the close of D + 1 to the close of D + 1 + h, so information released after the close on D cannot leak into the label.
- **Prices.** A date D sees only bars dated ≤ D.

Known deviations (stated in the validation documents):

- **Sector classification is current only.** A company reclassified in the past appears in its current sector throughout.

- **Universe membership history starts on 2026-10-10.** Earlier, the mvp universes are today's names (survivorship bias). See Good practice: survivorship.
- In the technicals and conformal validations, **past earnings dates are treated as if pre-announced**.
- W4 macro states in meta-labelling use today's vintages and parameters, lagged two months, not a true real-time history.

The BQL console exposes the same rules: `with(as_of=...)` sets the knowledge cut-off, and snapshot-only fields are refused with `dates=` or an earlier `as_of` (see BQL and mnemonics).

Purged and embargoed cross-validation

Financial labels overlap in time: a 20-session forward return that starts on Monday shares 19 sessions with the one that starts on Tuesday. If a training row's label window overlaps the test period, ordinary cross-validation leaks the answer. The app uses **walk-forward** validation with **purging** and an **embargo** everywhere a model is fitted.

```
train  ──────────────────┤purge├┤ test block ├┤embargo├── next train
        rows whose label window ends inside the test block are removed (purge)
        rows just after the test block are skipped before training resumes (embargo)
```

Model	Scheme
ML excess-return model (Model Lab)	Expanding window, minimum 5 years of training; training rows whose `label_end_date` ≥ test start are purged; embargo of 10 trading days after each test block. The production 3M run used 9 folds, 24-month test blocks, a 63-day purge and the 10-day embargo.
Technicals T2 first-passage surface	For test year Y, fitted on origin quarters < Q(Y) − 1 (a one-quarter purge).
Technicals T3 zone hold, v3 breakouts	2-year test blocks; training origins before block start − 425 days; isotonic calibration on [start − 365, start − 60) days.
Meta-labelling	Fit block = origins before 1 Jan of Y − 1 minus an embargo of 1.5 H sessions in calendar days + 7 (28 / 49 / 91 days for H = 10 / 20 / 40); inner validation on the fit block's last year; calibration and threshold on [1 Jan Y − 1, 1 Jan Y − embargo). Sample weights = average uniqueness per security.
Conformal sets	Filters fitted on data before 1 Jan of Y − 1 and then frozen; calibration on the last 500 realised scores from origins on or after 1 Jan Y − 1.
v3 technical factors	No fitted parameter: definitions and signs pre-registered from the literature, so the whole sample is out of sample. The composite is built walk-forward (factors selected on data up to 30 days before 1 Jan of Y).

Leakage tests used by the gates:

- **Label permutation:** refit on shuffled labels; out-of-sample skill must vanish. The FT-Transformer failed this test (see the gate table).
- **Causal invariance:** replace every bar after a cut with noise; every feature before the cut must be bit-for-bit unchanged.
- **Peeking control:** a deliberately leaky feature must be caught.
- **Shift-by-one:** the factor dated one session earlier must keep most of its information content.

Conformal prediction

Conformal prediction turns any forecast of scale or shape into a set (a band, a box or an ellipsoid) with a guaranteed long-run **coverage**: the share of times the realised outcome falls inside the set. The app's implementation follows Limmer, *Filtered Conformal Ellipsoids for Graph-Native Time Series* (`docs/CONFORMAL_VALIDATION.md`).

Split conformal

1. A **filter** produces, before the outcome is known, a centre and a scale or covariance (for prices: an EWMA variance, λ = 0.94, plus an earnings jump term when an earnings day falls inside the horizon).
2. Each past outcome gets a **nonconformity score**, for example the squared standardised return of horizon h:

```
R_t = y_h² / V(h)          V(h) = h·s²_t (+ J_t if an earnings day falls in (t, t+h])
```

3. The **radius** is the empirical (1 – α) quantile of the most recent calibration scores:

```
q̂ = the ⌈(n + 1)(1 − α)⌉-th smallest of the last n = 500 realised scores
set: { y : R(y) ≤ q̂ }      → band  ±√(q̂ · V(h))  around the centre
```

If the scores are exchangeable, `P(outcome in set) ≥ 1 − α` on average. Because scores are only known once the outcome is realised, the calibration window uses a lag equal to the horizon (20 sessions for a 20-session band).

Pooled calibration

A single stock has too few scores for a stable 90% or 95% quantile. Scores are therefore **pooled** across the securities of a **region × sector** (separate pools for windows containing an earnings event), re-sorted every 5 sessions. Each pool holds at least 30 securities × 500 origins. The resulting coverage is a property of the pool, not of the individual stock.

Adaptive conformal inference (ACI)

ACI (Gibbs & Candès, 2021) adjusts the working level online after each miss:

```
α_{t+1} = α_t + γ · (α − err_t)        err_t = 1{ R_t > q̂(α_t) }        γ = 0.005, restarted each year
```

Long-run average miscoverage converges to α for any score sequence. The app uses delayed feedback (the error of a 20-session set is known 20 sessions later) and pooled feedback (err_t is the share of the pool that missed). When α_t ≤ 0 the set is **unbounded**; such sets count as covered but are left out of width statistics. ACI is used in production for the next-bar sets and the peer-group sets, where one-session feedback lets it react; the 20-session bands use the plain radius.

Ellipsoids versus boxes

For several coordinates at once (the 1, 5, 10 and 20-session returns; tomorrow's high, low and close; the returns of 20–50 peer-group members) the set can be:

- an **ellipsoid**, `{ y : (y − μ)ᵀ Σ⁻¹ (y − μ) ≤ q̂ }`, the minimum-volume set for a given covariance; or
- a **box**: one interval per coordinate, each calibrated so that all hold together (for example a Bonferroni box with α/N per coordinate, or a conformally calibrated ATR box).

Two size metrics compare them (paper eq. 3–4):

Metric	Definition	Favours
w_proj	mean coordinate-projection length (2 × half-width of the set's shadow on each axis), in log-return units, averaged over coordinates	boxes, when coordinates are strongly correlated
log-volume	normalised log-volume of the set (for boxes, the mean log of the side lengths)	ellipsoids (minimum volume)

The gate was evaluated on w_proj, as specified, and both are reported. The joint price envelope had the smallest log-volume of all calibrated sets but projections 35% wider than a calibrated ATR box, because overlapping horizons are highly correlated (ρ(Y10, Y20) ≈ 0.7) and the minimum-volume ellipsoid is long along the diagonal. That is why it failed on sharpness.

Marginal versus joint coverage

- **Marginal (per-horizon) coverage:** each band is calibrated alone; the 20-session band covers the 20-session outcome 90% of the time. This is what the Technicals chart shows (the **calibrated range** and the fan, h = 1..20; per-horizon coverage 89.6–90.0% at 90%).
- **Joint (simultaneous) coverage:** all coordinates inside the set at once. Four separately calibrated 90% bands do **not** cover jointly 90% of the time: in the walk-forward audit the joint coverage of the four marginal bands is 73.0%. Conversely, the projections of a joint 90% ellipsoid cover each horizon about 98.7% of the time, so they are simultaneous bands, not 90% bands.

Coverage in every case is **marginal over time and securities**. Per-regime coverage is reported, not guaranteed: in the Covid window (origins 15 Feb – 30 Jun 2020) the 20-session sets of every method covered only about 77–78% at 90%, because the shock arrived after the origins and a 20-session lag gives no feedback in time.

Touch probabilities

The probability that the price **touches** a level within h sessions is read from calibrated curves rather than a formula. For each pool, the share of past origins whose standardised running high (or low) reached a standardised distance

```
u = | log(level / close) | / √V(h)
```

is stored per horizon. Validated horizons are 1, 5, 10 and 20 sessions (predicted equals realised within 0.2 pp, US mvp 2011–2026). Longer horizons (21–252 sessions) reuse the 20-session curve with the variance scaled by h and are flagged extrapolated, **not validated**. A Gaussian reflection-principle formula is about 5× too low in the tail (P(u = 3) = 0.0027 against a realised 0.014).

First-passage probabilities

The Trade setup card asks: starting from today's close, which comes first within H sessions, the target (T1) or the invalidation (stop)? This is a **two-barrier first-passage** problem (`docs/TECHNICALS_V2_VALIDATION.md` , T2).

Event definition

- Barriers: T1 above the close and the invalidation below it (mirrored for a short).
- Over the next H sessions, the **high** reaching T1 is a win, the **low** reaching the invalidation is a loss; a bar that touches both counts as a **loss** (intrabar order unknown); otherwise **neither** by H.
- Distances are standardised by the horizon volatility:

```
a = log(T1 / close) / σ_H        b = log(close / invalidation) / σ_H        σ_H = √V(H)
```

Grid estimator (production)

1. For every daily origin of the gate universe (3.2 million origins × 5 horizons), record the outcome of **every** (a, b) pair on a 20 × 20 logarithmic grid from 0.1 to 8 σ_H. This uses all paths, not only the origins that happened to have a setup.
2. Count wins, losses and neither per cell, per pool: **region × volatility regime × earnings window**.
3. **Shrinkage.** Thin cells borrow strength from the parent pool, pool → region → all regions, with m = 500 pseudo-origins:

```
p̂_cell = (n_cell · p_cell + m · p_parent) / (n_cell + m)
```

4. **Monotone smoothing (PAV).** P(win) must fall as T1 moves away and rise as the stop moves away (and the reverse for P(loss)). Iterated pool-adjacent-violators passes along both grid axes enforce this.
5. **Interpolation.** Bilinear in (log a, log b) between grid nodes.
6. For each cell the mean terminal move of the "neither" paths is kept, so the **expected R** can mark unresolved trades to market:

```
E[R] ≈ P(win) · R:R − P(loss) · 1 + P(neither) · (mean move of "neither" paths, in R) − costs (in R)
```

Costs are the Backtest Lab's (commission 1 bp, ADV-bucket half spread, square-root impact on a $200k position, borrow for shorts).

Baselines it must beat: the driftless **Brownian** two-barrier formula (too pessimistic on P(win) at H ≤ 10 by 1.2–2.7 pp) and the naive **R:R-only** rule (badly miscalibrated at short horizons: ECE 15.8 pp at H = 5, because it ignores "neither by H"). A gradient-boosted candidate was better calibrated at 40 and 60 sessions but worse on Brier at 5–20 and still worse than Brownian at 40 / 60, so it was not deployed. On simulated Brownian paths the grid estimator matches the analytic formula within 1.5 pp.

Verdict in short

Only **H = 20** passed the full gate and is shown by default. H = 5 and 10 failed on the Covid 2020 window (5.1–5.6 pp more stop-outs than predicted); H = 40 and 60 failed because the grid no longer beats the Brownian baseline. The economic check of filtering setups by E[R] > 0 was a **null result** (+1.0%/yr at H = 20, t = 1.33).

Meta-labelling

Meta-labelling (Lopez de Prado, *Advances in Financial Machine Learning*, ch. 3–4) separates **direction** from **sizing**. A primary model proposes a trade; a secondary (meta) model estimates how likely that trade is to succeed given the context, and can decide to take or skip it and how much to risk (`docs/TECHNICALS_V3_VALIDATION.md` , "Meta-labelling (#2)").

- **Primary model:** the existing v2 setup generator (direction, entry, T1, invalidation) with its T2 probabilities.
- **Label (triple barrier):** win = T1 before the invalidation within H sessions; loss = invalidation first (same-bar touch of both = loss); vertical barrier at H (20 primary; 10 and 40 secondary).
- **Features (point in time, 89):** T2's own probabilities and E[R]; geometry in σ units; setup sub-scores; technical context (squeeze percentile, ADX, trend alignment, ...); market context (VIX level and change, cross-asset risk gauge, macro regime states lagged two months); peer-group break-aways; insider, short-interest and smart-money signals; earnings proximity; region / sector; the v3 factors and breakouts.
- **Learners:** logistic regression and XGBoost, both using T2's logit as an **offset** (base margin), so a model that finds no usable context reproduces T2. Calibration by identity, Platt or isotonic on a held-out year. A take / skip threshold and a capped quarter-Kelly size are chosen in sample.
- **Baselines:** T2 alone, and a **context-free** logistic recalibration of T2 (T2 + geometry + setup descriptors, no context).

Verdict. H = 10 **passed**: better calibrated than T2 (ECE 0.50 / 0.78 pp for win / loss; Covid loss gap +2.5 pp against T2's +5.1 pp) with a Brier improvement over T2 of −0.00040 [−0.00071, −0.00006]. But it is **not** significantly better than the context-free model (−0.00022 [−0.00050, +0.00013]). The pass therefore means "a better-calibrated P(win) than T2", not "context has a demonstrated edge". H = 20 and 40 failed, and the **take / skip and size decisions failed the economic check** (US meta-filtered minus all setups −4.3%/yr, t = −2.55 at H = 20).

Calibration metrics

A probability model is **calibrated** if, among all cases where it said 30%, the event happened about 30% of the time. The gates measure this in several ways.

Expected calibration error (ECE). Sort predictions into B bins (15 equal-count bins in the technicals gates), then

```
ECE = Σ_b (n_b / N) · | mean predicted p in bin b − realised frequency in bin b |
```

Reported in percentage points. Gate thresholds: ECE ≤ 2.5 pp (technicals v2, v3 and meta-labelling).

Brier score. Mean squared error of the probabilities; for K outcomes (win / loss / neither):

```
Brier = (1/N) Σ_i Σ_k (p_ik − y_ik)²          lower is better; compared with baselines
```

Log-loss. `−(1/N) Σ_i log p_i,observed` . Punishes confident mistakes heavily. Gates require the model to beat its baselines on both Brier and log-loss, usually with a month-block bootstrap confidence interval of the difference that excludes zero.

Reliability diagram. Predicted probability (x) against realised frequency (y) per bin; a calibrated model lies on the diagonal. The validation runs write them as `reliability_*.svg` under `data/reports/` .

Regime gaps (calibration in the large). Within a regime (Covid 2020 = origins 15 Feb – 30 Jun 2020, the year 2020, 2022, VIX > 25, each region, side, volatility regime, earnings window), the gap is

```
gap = realised frequency − mean predicted probability        gate: |gap| ≤ 5 pp for every regime with n ≥ 2,000
```

Overall calibration can hide a regime failure: T3 P(hold | touch) has ECE 1.45 pp overall but a −8.7 pp gap in Covid 2020, and failed.

Coverage (for conformal sets) is the analogue for intervals: realised share inside the set against the target; the conformal gate allowed ±1.5 pp overall and no regime more than 5 pp under.

Multiple testing

When many hypotheses or strategy variants are tried, some look significant by chance. The validation documents correct for this, and the Backtest Lab reports the same statistics for your own strategies (see Backtest Lab).

Holm (family-wise error). Sort the m p-values ascending, $p_{(1)} \le \dots \le p_{(m)}$; the adjusted values are

$$\tilde{p}_{(i)} = \max_{j \le i} \min(1, (m - j + 1) \cdot p_{(j)})$$

Used for the per-factor verdict of the v3 technical factors over the 7 primary cells (hi52 0.017, strev 0.002, MAX 9e-10, IVOL 6e-13 passed; mom 0.30, rmom 0.56, indmom 0.56 did not).

Benjamini–Hochberg (false discovery rate). Controls the expected share of false discoveries:

$$q_{(i)} = \min_{j \ge i} \min(1, m \cdot p_{(j)} / j)$$

Over the 105 v3 factor cells, 49 were significant at q = 0.05, every one with the literature's sign. BH is also applied to Model Studio result tables.

Probabilistic Sharpe ratio (PSR). The probability that the true Sharpe ratio exceeds a benchmark SR*, given track-record length n and the skewness γ_3 and kurtosis γ_4 of returns (Bailey & Lopez de Prado, 2012):

$$PSR(SR^*) = \Phi\left((SR - SR^*) \cdot \sqrt{n - 1} \,/\, \sqrt{1 - \gamma_3 \cdot SR + (\gamma_4 - 1)/4 \cdot SR^2} \right) \qquad \text{(per-period SRs)}$$

Deflated Sharpe ratio (DSR). PSR against the Sharpe ratio expected from the **best of N unskilled trials** (Bailey & Lopez de Prado, 2014):

$$SR^* = E[\max SR] = \sqrt{V[SR_{trials}]} \cdot \left((1 - \gamma)\cdot\Phi^{-1}(1 - 1/N) + \gamma\cdot\Phi^{-1}(1 - 1/(N\cdot e)) \right) \qquad \gamma \approx 0.5772 \text{ (Euler–Mascheroni)}$$
$$DSR = PSR(SR^*)$$

When the trial Sharpe ratios hardly differ, V[SR] is small and DSR stays near 1 for every configuration; the validations note that DSR "does not discriminate" in that case (about 0.99 in the conformal and v2 economic checks).

Probability of backtest overfitting (PBO) via CSCV. Combinatorially symmetric cross-validation (Bailey, Borwein, Lopez de Prado & Zhu, 2017) with S = 10 time blocks:

1. For every split of the blocks into two halves (C(10, 5) = 252 combinations), pick the configuration with the best in-sample Sharpe.
2. Find its relative rank w among all configurations out of sample, and its logit $\lambda = \ln(w / (1 - w))$.
3. PBO = share of splits with $\lambda \le 0$, i.e. how often the in-sample winner is below the out-of-sample median.

PBO near 0 means the in-sample choice holds up; near 0.5 or above means the selection is no better than chance. Examples: conformal breakout configurations 0.54; v2 E[R] configurations 0.07; meta-labelling 0.35 over 62 configurations.

Rank IC and factor evaluation

A **rank information coefficient (IC)** measures whether a signal orders stocks correctly, without assuming a linear relation.

$$IC_t = \text{Spearman}(\text{signal}_{i,t} , \text{forward return}_{i,t \to t+h}) \qquad \text{across stocks } i \text{ on date } t$$
$$\text{mean IC} = \text{average of } IC_t \text{ over dates}$$

Element	How the app does it
Labels	Sector-neutral: the forward log return from the close of t + 1 minus the equal-weight mean of liquid names in the same region × sector × date (region × date for thin or unclassified sectors). ML labels are cross-sectional ranks of 1M / 3M / 6M forward excess return vs sector × region.
Signal	The within-sector percentile of the factor (0..100).
Pooling	Per (region, date) Spearman, pooled per date weighting regions by number of names; weekly origins (1,216 dates for the v3 factors).

Element	How the app does it
Newey–West t	The t-statistic of the mean IC with HAC standard errors, because overlapping horizons make IC_t autocorrelated: ceil(h / 5) + 1 lags for weekly origins. Non-overlapping samples are used where stated.
Decile monotonicity	Spearman correlation between decile number and decile mean return (sign-aligned); the v3 gate requires ≥ 0.7 and a top-minus-bottom spread whose 95% month-block CI is above 0.
Robustness	Aligned in ≥ 3 of 4 regions with none significantly wrong (BH), aligned in ≥ 60% of years, no regime with aligned t ≤ −2, shift-by-one keeps ≥ 50% of the IC, label-permutation null.
Economics	Reported separately: US long-short net Sharpe after costs and long-only quintile vs equal-weight universe.

Reading an IC. A rank IC of 0.02–0.05 is typical for useful equity signals but small: it is a median, cross-sectional effect, and it need not survive costs. The passing v3 factors (hi52 +0.023, MAX −0.038, IVOL −0.044 at 20 sessions) are **not profitable as stand-alone strategies after costs**; their simple-return decile spreads are much smaller than their log-return spreads because the high-volatility deciles carry variance drag. They are labelled "rank edge" in the app.

The ML model's production promotion rule is out-of-sample rank IC t > 2 and beating the incumbent (`docs/ARCHITECTURE.md` §11).

Event studies and abnormal returns

Event studies measure the price reaction to an event (earnings, a geopolitical shock, a CFTC extreme) relative to what the market alone would predict.

Market model. Estimated by OLS on trading days [−250, −30] relative to the event day 0:

```
r_i,t = α_i + β_i · r_m,t + ε_i,t
AR_i,t = r_i,t − (α̂_i + β̂_i · r_m,t)                    abnormal return
CAR_i[t1, t2] = Σ_{t=t1..t2} AR_i,t                       cumulative abnormal return
BHAR_i[t1, t2] = Π(1 + r_i,t) − Π(1 + α̂_i + β̂_i · r_m,t)   buy-and-hold abnormal return
```

Use	Benchmark r_m	Windows	Notes
Earnings reactions (Company 360 › Earnings, Earnings season)	the issuer's country index (ACWI fallback), on the security's own trading days	[−1, +1], [0, +5], [+1, +20]	At least 60 valid estimation days; daily moves above 50% treated as data errors. "Move vs normal" = the [−1, +1] move against residual σ × √3. Day 0 = first trading day on or after the reaction date.
Event-study library (Global Events)	country index vs ACWI; sector SPDR vs SPY; issuer vs its country index	[0, 1], [0, 5], [−1, 5], [0, 10]	Shocks: attention z ≥ 2.5 or tone z ≤ −2.5 with at least 20 items, no shock of the same kind in the previous 10 days. "Expected reaction" = the historical mean CAR of similar labelled shocks (model-based).
Model Studio `event_car` method	user choice (market model or constant mean)	user choice	Reports AAR / CAAR paths, cross-sectional t and Patell Z.

Significance. The cross-sectional t uses the dispersion of CARs across events; the Patell Z standardises each AR by its estimation-period residual σ (with the forecast-error adjustment).

Pitfalls. CAR sums simple abnormal returns and understates compounding over long windows; BHAR compounds but is skewed. Overlapping events in the same market are not independent. Earnings reactions use the official release time where available, otherwise the provider date or the filing date (labelled), so a wrong date shifts day 0.

Model-based versus official data

The macro data policy of the app (`docs/MACRO_V2_PLAN.md` , `docs/MACRO_V3_PLAN.md`):

1. **Inputs come from official sources only:** central banks, statistical offices, treasuries, regulators, the NY Fed and other Fed banks, OECD, BIS, IMF, Eurostat. FRED and ALFRED count as official (St. Louis Fed). No Yahoo or scraped commercial sites for macro. (Market prices for equities, futures and ETFs are market data, not macro.)
2. **Expectations are model-based** and labelled everywhere they appear: "Model-based expectation (not consensus)". There is no consensus feed. Surprises are measured against these expectations.

3. **Approximations are acceptable but labelled**, with their method and caveats (for example the central-bank implied path).

How the labels read in the app:

Output	Label (as served by the API)
Release expectations	"Model-based expectation (not consensus)"
Surprise indices	Economic surprise indices vs model-based expectations (not consensus)
Central-bank implied path	"Approximation from official curves (not futures/OIS-implied probabilities)"
Term-premium-adjusted path	"Term-premium-adjusted with NY Fed ACM term premia (model-based, official NY Fed estimates)"
Taylor rule	"Model-based (Taylor rule; official inputs)"
Expected inflation	"Model-based (Cleveland Fed model for the US; our AR(1) elsewhere)"
Central-bank tone	"Model-based (local RoBERTa hawkish/dovish classifier + lexicon)"
Rate cycle	"Rule-based: most recent policy-rate change within 183 days decides hiking/cutting, else pausing"
Recession panel	each method labelled: NY Fed yield-curve probability (official, published); Sahm rule (official); our probit and HMM P(contraction) (model-based, ours)
Macro betas	"Model-based: rolling weekly OLS betas (156 weeks, >= 52), Newey-West HAC(4) standard errors"

The same principle extends beyond macro: earnings expectations are model-based (a seasonal time-series model on the company's own history, 80% band), not consensus; the smart-money definitions (hedge-fund-like filers, smart cohort, crowding, squeeze risk, FTD spike, the combined score) are model-based; supply-chain statistical neighbours are labelled statistical and never shown as disclosed relationships; the cross-asset risk composite and trend signals are MODEL-BASED transformations of market and official data.

Gate status of every model

One row per model or output. **Gate** quotes the verdict and its key numbers from the source document; "no formal gate" means no pre-specified pass / fail test exists, and the row says how the app labels the output instead. "Experimental" outputs are hidden unless **Show experimental outputs** is on (Technicals tab) or appear only in Model Studio / the Backtest Lab.

Technicals and conformal sets

Model / output	Where shown	Gate verdict and key numbers	Label in the app	Source
Joint 1/5/10/20-session conformal envelope (ellipsoid)	Not in the main UI; Model Studio, Backtest Lab	**FAIL**: coverage 89.6 / 94.5% but Covid 2020 77–78% at 90%; w_proj +35% vs a calibrated ATR box	experimental; "not shown" on the validation page	CONFORMAL_VALIDATION Verdict, A.1
Next-bar high / low / close set	Model Studio, Backtest Lab	**FAIL (sharpness)**: coverage 90.0 / 95.0%, every regime on target with ACI; w_proj +43% vs calibrated ATR box	experimental	CONFORMAL_VALIDATION A.3
Marginal single-horizon bands (qm_*) and fan h = 1..20	Technicals chart (calibrated range, fan), BQL `cf.*`	**PASS (coverage)**: 89.6–90.0% at 90% per horizon; ~40% narrower than 2×ATR (which covers 98%); not sharper than a calibrated ATR band	"calibrated band", model-based; "calibrated to 90% over history"	CONFORMAL_VALIDATION A.2, A.5
Touch probabilities, h = 1 / 5 / 10 / 20	Technicals (chance of touch), thesis target / stop	**PASS**: predicted = realised within 0.2 pp at every distance and horizon	model-based, calibrated	CONFORMAL_VALIDATION A.4
Touch probabilities, h = 21–252	Thesis tracker, API	not validated (20-session curve with variance scaled by h)	`extrapolated = true` , NOT validated	api.yaml (conformal touch)
Peer-group ellipsoid (GCN-GRU + ACI)	Peers tab conformal panel, RV	**PASS** as a calibrated descriptive measure: 90.4 / 95.2%; worst regime Covid 85.3% at 90%; −10% w_proj vs EWMA	anomaly percentile, pair / basket bands, model-based; **not** a trading signal	CONFORMAL_VALIDATION B, Verdict
Band breakout signal	Backtest Lab	**FAIL**: not better than a naive N-day high (paired t −0.43 to −3.23)	experimental signal	CONFORMAL_VALIDATION Economic (i)

Model / output	Where shown	Gate verdict and key numbers	Label in the app	Source
Calibrated band stops	Backtest Lab	**MIXED**: beat ATR stops (+6.2%/yr, t 3.56) but worse than no stop (−2.2%/yr, t −2.64)	experimental stop / target mode	CONFORMAL_VALIDATION Economic (ii)
Peer break-away fade	Backtest Lab	**FAIL (null)**: Sharpe −0.09 / −0.15; vs naive fade t ≤ 1.17	experimental signal	CONFORMAL_VALIDATION Economic (iii)
Level-type ranking (confluence zones)	Technicals levels / zones	no formal gate; out of sample, top-ranked level types beat random levels by about 1.5 pp bounce rate	entry filter, not a stand-alone strategy	USER_GUIDE §3
T2 P(win / loss), H = 20	Trade setup card	**PASS**: ECE 1.13 / 1.47 pp; worst regime Covid win −4.6 / loss +4.0 pp; Brier 0.3741 vs Brownian 0.3759 / R:R-only 0.3834	production, model-based, "calibrated on history"	TECHNICALS_V2 Verdict
T2, H = 5 and 10	Trade setup card (toggle)	**FAIL**: Covid 2020 loss gap +5.6 / +5.1 pp	experimental (hidden by default)	TECHNICALS_V2 Verdict
T2, H = 40 and 60	Trade setup card (toggle)	**FAIL**: no better than Brownian (Brier 0.3485 vs 0.3482; 0.3453 vs 0.3429); H = 60 ECE 2.68 / 2.73 pp	experimental (hidden by default)	TECHNICALS_V2 Verdict
Expected R filter (E[R] > 0)	Trade setup card, Backtest Lab	economic check **null**: +1.0%/yr at H = 20 (t 1.33), −0.6%/yr at H = 10 (t −0.75)	E[R] shown; no edge claimed	TECHNICALS_V2 Economic check
T3 P(hold given touch)	Zones (toggle)	**FAIL**: ECE 1.45 pp but Covid 2020 −8.7 pp	experimental	TECHNICALS_V2 Verdict, T3
T3 per-zone edge vs random	Zone flag	**PASS**: bucket predicted vs realised within 1.9 pp; level features beat the random control (Brier −0.00045 [−0.00066, −0.00025]); on average touched zones hold −0.63 pp **less** often than random bands	"holds more / less often than random" or "no reliable edge"	TECHNICALS_V2 Verdict, T3
T4 calibrated technical score → outcomes	Technical score card (toggle)	**FAIL**: rank IC −0.003 / −0.012 / −0.014 at 5 / 20 / 60 d, n.s.	experimental; card says "no out-of-sample edge"	TECHNICALS_V2 Verdict, T4
v3 factors hi52, MAX, IVOL	Technicals factors, BQL `tech.*`, Backtest Lab	**PASS**: 20-day IC +0.023 (t 2.9), −0.038 (t −6.4), −0.044 (t −7.5); none profitable stand-alone after costs	"rank edge" with economic note, shown by default	TECHNICALS_V3 Verdict #1
v3 factors mom, rmom, strev, indmom	Technicals factors (toggle)	**FAIL**: mom, rmom, indmom fail G1 (Holm p 0.30 / 0.56 / 0.56); strev passes G1 (0.002) but fails monotonicity (0.35)	experimental	TECHNICALS_V3 Verdict #1
v3 factors outside US / EU / UK / JP	Technicals factors	not validated	"region not validated"	TECHNICALS_V3 Production configuration
v3 composite factor score	not served	**FAIL**: G6 vs the v2 score (t 1.66) and G7 economics (US L/S net Sharpe −0.46)	no composite value served	TECHNICALS_V3 Verdict #1
Calibrated P(follow-through), H = 10 / 20	Breakouts (toggle)	**FAIL**: ECE 1.23 / 1.20 pp but Covid 2020 +18.3 / +17.2 pp	experimental	TECHNICALS_V3 Verdict #5
Volume-confirmation edge (state chip, flag, score)	Breakouts	**PASS**: geometry-adjusted +1.6 pp [+1.2, +2.0]; vs matched control +3.5 pp [+2.7, +4.6]; no tradable edge (confirmed vs random subset −1.0%/yr, t −0.6)	deterministic chip and flag shown	TECHNICALS_V3 Verdict #5
Meta-labelled P(win), H = 10	Trade setup "Context-adjusted (meta)" row, thesis meta table	**PASS**: ECE 0.50 / 0.78 pp; Brier vs T2 −0.00040 [−0.00071, −0.00006]; not better than context-free (−0.00022, CI includes 0)	default row, model-based	TECHNICALS_V3 Meta-labelling Verdict
Meta P(win), H = 20 and 40	Same (toggle)	**FAIL**: not better than T2 / context-free	experimental	TECHNICALS_V3 Meta-labelling Verdict
Meta take / skip and size	Same (toggle), Backtest Lab `meta_setups`	**FAIL** (economic): US −4.3%/yr (t −2.55) at H = 20	experimental at every horizon	TECHNICALS_V3 Meta-labelling Economics
Meta outputs outside the validated domain (ETFs, indices, regions other than US / EU / UK / JP)	Trade setup card	not validated	always experimental	TECHNICALS_V3 Meta-labelling Production configuration

Scores, forecasts and alternative data

Model / output	Where shown	Gate verdict and key numbers	Label in the app	Source
ML excess-return model (XGBoost, 1M / 3M / 6M)	Company 360 › Model, Model Lab, composite ML blend profile	no formal pass / fail beyond the promotion rule (OOS rank IC t > 2 and beat the incumbent); OOS rank IC about 0.035 (production runs: 1M 0.0352, t 11.7; 3M 0.0349, t 12.2; 6M 0.0212, t 5.7)	"prod" badge in Model Lab; one input, not proven alpha; backtests before 2026-10-10 survivorship-biased	USER_GUIDE §3; ARCHITECTURE §11; `/models/runs`
FT-Transformer (3M and 1M) and the ensemble	Model Lab	**FAIL** label-permutation leakage test (IC +0.06–0.09 on shuffled labels) → demoted; not in production. Its high Model Lab rank IC (0.126 for 1M) is not evidence of skill	shown without a "prod" badge; no leakage note in the UI	PHASE3_TODO (ML leakage audit)
Insider XGBoost (P(3M excess > 0))	not used	no out-of-sample edge (latest runs: 3M rank IC −0.015, t −1.17; 1M −0.0005)	only the rules-based insider score is used	USER_GUIDE §6; `/models/runs`
Insider signal score and anomaly score (rules + IsolationForest)	Insiders tab, Insider Radar, Insider factor	no formal gate	descriptive score 0–100 (50 = no activity)	ARCHITECTURE §11
Composite score, 10 factor scores, sector KPI scores	Overview, Screener, Sector Explorer	no formal gate (peer percentiles, own-history z-scores, size-adjusted valuation)	descriptive score 0–100, grade A–E; 50 = neutral or no data	ARCHITECTURE §11
Forecast, DCF and blended fair value	Forecast tab, BQL `pe_fwd`	no formal gate	model-based, not consensus; valuation confidence < 40 = indicative only and excluded from scoring	USER_GUIDE §3
Moat rating	Moat tab	no formal gate (rule-based evidence components)	Wide / Narrow / None / Insufficient data	USER_GUIDE §3
Beneish M-score screen	Company 360 alerts	no formal gate	"MODEL-BASED, not an accounting finding" (recalibrated screen)	api.yaml; AUDIT F15
Earnings expectations and surprises	Earnings tab, Earnings season, calendar	no formal gate (seasonal model chosen by trailing MAPE; 80% band)	model-based, not consensus	USER_GUIDE §3; api.yaml
Projected earnings dates	Earnings calendar	no formal gate	"Projection" with ± window (vs "Announced", "Provider est.")	USER_GUIDE §3
Post-earnings drift signal (PEAD)	Earnings tab, Backtest Lab	no formal gate	model-based	api.yaml
News sentiment (FinBERT English, XLM-R other languages)	News & Sentiment, News Terminal	no formal gate	sentiment is model-based	api.yaml; ARCHITECTURE §11
"Lazy Prices" text-change flag	News & Sentiment	no formal gate	flag on large year-over-year risk-factor text changes	USER_GUIDE §3
Smart-money combined score	Smart Money, Ownership tab	sanity check only: no significant rank IC (21d −0.004, t −0.67; 63d −0.005, t −0.56; 2023-06 → 2026-10)	MODEL-BASED; descriptive screen, not an alpha signal; optional scoring input off by default	SMART_MONEY §4
Hedge-fund-like filers, smart cohort, crowding, squeeze risk, FTD spike	Smart Money	no formal gate	model-based (our definitions)	SMART_MONEY §2
CFTC COT index, z-scores, extremes, crowding, divergence, event study	Positioning, Cross-asset	no formal gate; source data official	official data; extremes / crowding / divergence / event study model-based (labelled)	CFTC_PLAN; CROSS_ASSET_DEFINITIONS §2b; api.yaml
Cross-asset trend signal, curve shape	Cross-asset dashboard	no formal gate	MODEL-BASED	CROSS_ASSET_DEFINITIONS §2
Risk-on / risk-off composite	Cross-asset, Markets overview	no formal gate; sanity checks 2008-10-15 −2.8, 2020-03-20 −2.5 (risk_off), 2017-06-15 +0.3	MODEL-BASED	CROSS_ASSET_DEFINITIONS §3

Model / output	Where shown	Gate verdict and key numbers	Label in the app	Source
Supply-chain links (disclosed)	Network & Geo	no formal gate (rule-based extraction with provenance)	disclosed customer / supplier with source	SUPPLY_CHAIN §1
Statistical economic neighbours	Network & Geo	no formal gate	labelled statistical; co-movement, not a relationship	SUPPLY_CHAIN §3
Country risk and shock propagation	Network & Geo, Supply chain	no formal gate	model-based	SUPPLY_CHAIN §4–5; api.yaml

Macro and events

Model / output	Where shown	Gate verdict and key numbers	Label in the app	Source
Release expectations (SARIMA / seasonal naive)	Calendar, release pages, cockpit	no formal gate; SARIMA beat naive on 120 of 168 series	"Model-based expectation (not consensus)"	MACRO_V2_PLAN W2.2
Surprise indices	Cockpit, themes, countries	no formal gate	vs model-based expectations (not consensus)	MACRO_V2_PLAN W2.3
GDP nowcasts (DynamicFactorMQ; US, EA, GB, JP)	Nowcast page	no formal gate; track record shown next to GDPNow and NY Fed; 2026Q2 error +1.04 pp vs GDPNow +0.04 pp	model-based, with official benchmarks	MACRO_V2_PLAN W2.4; AUDIT F27
Recession panel	Cockpit	no formal gate	each method labelled official or model-based (ours)	MACRO_V2_PLAN W2.5; `/macro/v2/recession`
HMM macro regime (4 states)	Cockpit, Markets, macro factor	no formal gate; the audit found over-confident posteriors (99.99% "Expansion")	model-based	ARCHITECTURE §11; AUDIT F07, F08, F38
Cycle clock	Cockpit, countries	no formal gate (documented rule, cycle_v1)	method shown	MACRO_V2_PLAN W1.2
Central-bank implied path	Central banks, WIRP	no formal gate	"Approximation from official curves (not futures/OIS-implied probabilities)"	`/macro/v3/central-banks` labels
TP-adjusted path, Taylor rule, expected inflation, tone	Central banks	no formal gate	model-based (see the labels in Model-based versus official data)	`/macro/v3/central-banks` labels
r* estimates	Central banks	official (NY Fed Laubach–Williams, HLW)	official	MACRO_V3_PLAN W3.4
Rate cycle	Central banks	no formal gate	rule-based	`/macro/v3/central-banks` labels
Macro exposures (betas)	Macro tab, Exposures, Screener columns	no formal gate	"Model-based: rolling weekly OLS betas (156 weeks, >= 52), Newey-West HAC(4)"	`/macro/v4/exposures`
Regime-conditional performance	Regime performance	no formal gate (historical averages with hit rates, t-stats, n)	descriptive	MACRO_V3_PLAN W4.3
Scenario builder and episode replay	Scenarios, Episodes	no formal gate	model-based, 90% bands	MACRO_V3_PLAN W4.4
Our geopolitical-risk index (GPR)	Global Events	compared with the published Caldara–Iacoviello index, no pass / fail: daily global correlation 0.32 (7-day average 0.75); monthly country cross-section Spearman median 0.88 (2024-09 → 2026-10)	model-based	EVENTS_MODELS_REALTIME C3; `gpr_validation_20261020.json`
Escalation probabilities (Hawkes)	Global Events	no formal gate	model-based	EVENTS_MODELS_REALTIME C1
Change points (BOCPD)	Global Events	calibrated on synthetic shifts: about 0.13 false alarms per 200 days, 97% detection of 1.5 sd shifts	model-based	EVENTS_MODELS_REALTIME C2
Hotspots, spillovers, connectedness, impulse responses	Global Events	no formal gate (permutation inference for Moran / LISA)	model-based	EVENTS_MODELS_REALTIME B1–B4

Model / output	Where shown	Gate verdict and key numbers	Label in the app	Source
Event-study library and expected reaction	Global Events, News Terminal impact	no formal gate	model-based	EVENTS_MODELS_REALTIME C4
Company event exposure	Company 360 › Event risk	no formal gate	model-based	EVENTS_MODELS_REALTIME C5
GDELT event labels	Global Events, News	spot check: security links 82% (49 / 60), country 84%, theme 68%, sector about 68%	labels shown with sources	EVENTS_V2_PLAN E5
Thesis automatic status	Thesis Tracker	no formal gate	"Automatic status (model-based rules)"	api.yaml
Model Studio methods, your own studies	Model Studio	no gate; guardrails (point in time, sample-size checks, multiple-testing warnings)	your result; method diagnostics	MODEL_STUDIO_PLAN
Your Backtest Lab strategies	Backtest Lab	no gate; DSR, PSR and PBO reported for you to judge	experimental signals marked	Backtest Lab

17. Good practice and pitfalls

BiG Equity 360 puts a large amount of data and many models on one screen. That makes it fast to form a view, and equally fast to over-read a number. This chapter collects the cautions that apply across the app. It ends with a checklist for using the app in an investment decision and a template for citing the app's numbers.

What not to infer

- **A score is not a forecast.** The composite score, the factor scores and the sector KPI scores are percentiles against peers and against the company's own history. A grade A says "looks strong on these metrics today", not "will outperform". None of them has passed an out-of-sample return test (see the gate table).
- **A validated model is validated for one thing.** The peer-group conformal ellipsoid passed as a *calibrated descriptive measure*; fading the stock that breaks away from its group had **no** economic value. The 52-week-high, MAX and IVOL factors have a *rank edge* in median, cross-sectional terms, but none is profitable as a stand-alone strategy after costs. A passed calibration gate does not make a trading signal.
- **The ML model is one input.** Its out-of-sample rank IC is about 0.035. That is a weak, noisy ordering of many stocks, not a reliable call on any single one.
- **"No reliable edge" means what it says.** When a zone, factor or card states it, do not treat the number next to it as informative.
- **Fair value is a model.** The headline fair value blends a DCF and peer multiples and is partly anchored to the market price. Below a valuation confidence of 40 it is indicative only.
- **Expected R is not expected profit.** The economic check of filtering setups by expected R > 0 was a null result after costs.
- **Correlation is not a relationship.** Statistical neighbours on the Network & Geo tab are co-movements, labelled statistical; only disclosed links are customer or supplier relationships.
- **An event-study "expected reaction" is a historical average** of similar labelled shocks, with wide dispersion around it.

How to read probabilities and bands

Every probability and band in the app is **calibrated over history**: across many past stocks and days in the same pool (region × sector, or region × volatility regime × earnings window), the event happened about as often as stated.

The app says	Read it as	Not as
20-session P(win) 38%	"Of past setups with this geometry and pool, about 38% reached the target first within 20 sessions."	"This trade has a 38% chance of working."
90% calibrated range	"Bands built this way contained the 20-session outcome 90% of the time over 2010–2026 (walk-forward 89.6%)."	"The price will be inside this band with 90% probability."
Chance of touch 25%	"Historically, levels this far away (in volatility units) were touched within h sessions about 25% of the time."	"There is a 25% chance of touching this level."

Marginal, not conditional. Coverage and calibration are averages over time and securities. A specific stock on a specific day can be in a situation the average hides (a pending takeover, a binary trial result, a trading halt).

Marginal, not joint. Each horizon of the fan is calibrated on its own. The four 90% bands at 1, 5, 10 and 20 sessions contained all four outcomes at once only 73% of the time. Do not read a fan as one joint 90% statement.

Regime failures. Calibration holds on average, not in every regime. In the Covid crash window (origins 15 Feb – 30 Jun 2020):

- the 20-session joint conformal set covered only 77–78% at a 90% target;
- 5- and 10-session first-passage probabilities under-predicted stop-outs by 5.1–5.6 pp (which is why they are experimental);
- zone-hold probabilities were 8.7 pp too optimistic;
- breakout follow-through was under-predicted by 17–18 pp.

A sudden shock arrives after the forecast origin, and models with 10- or 20-session outcome lags cannot react in time. In a stress episode, widen your own error bars.

Thresholds at the extremes. Touch probabilities beyond the calibrated tail show "<1%" and near-certain ones ">99%"; touch probabilities beyond 20 sessions are extrapolated and not validated.

Model-based, experimental and approximation labels

Label	Meaning	How to use it
Model-based	Our own estimate or transformation, not a published figure or a consensus (macro expectations, earnings expectations, smart-money score, risk composite, country risk, ...).	Fine to use as an input; say "model-based" when you quote it.
Model-based expectation (not consensus)	The "expected" value of a macro release or earnings figure comes from a time-series model, not from analysts.	A "surprise" is a surprise against our model, not against the market's consensus. The market may have expected something else.
Approximation	A simplified method from official inputs (central-bank implied path from official curves, not futures or OIS probabilities).	Directionally useful; do not quote it as "the market prices X cuts" without the caveat.
Experimental (flask icon, hidden unless **Show experimental outputs** is on)	Failed its validation gate or was never validated.	Research only. Do not cite in an investment recommendation.
Rank edge	Passed a rank-IC gate but is not profitable stand-alone after costs.	A tilt or tie-breaker, not a strategy.
Labelled statistical	Derived from price co-movement.	Not a business relationship.
Projection / **Provider est.** / **Announced** (earnings dates)	Our cadence model / Yahoo date / company or exchange schedule.	Only "Announced" is confirmed.
Region not validated	A model run outside the regions of its validation (US, EU, UK, JP for the technicals).	Treat as experimental.

Survivorship

- **Universe membership history starts on 2026-10-10.** Before that date, the mvp universes (`mvp_us` , `mvp_global`) are today's names applied to the past. Any backtest, historical screen or model evaluation on them before 2026-10-10 includes only companies that survived to 2026, which flatters returns and Sharpe ratios.

- The technicals and conformal validations state this bias explicitly (current mvp membership throughout). The v3 factor evaluation includes securities that stopped trading while they traded, but listing coverage is better for current names.
- The `full` universe keeps 753 delisted US stocks with prices; they are in no universe, so universe-based backtests do not hold them.
- **Sectors are current.** A company reclassified in the past appears in its current sector throughout its history, so sector-relative history and sector-neutral labels carry a small look-ahead.
- Peer groups and supply-chain groups are the current versions; supply-chain edges exist point in time only from 2023.

What to do: prefer the `full` universe and dates after 2026-10-10 for live tracking; for history, compare results with and without the most recent winners; treat any backtest Sharpe as an upper bound.

Point-in-time versus restated data

- Fundamentals in the app are point in time: each value is shown as known at its filing date, and restatements are kept as new versions. The Financials tab shows the latest version of each period; BQL with `as_of=` or `dates=` shows what was known at the time.
- **Quoting a historical ratio:** say whether it is "as first reported" or "as restated". A backtest that uses restated numbers at past dates has look-ahead.
- **Imputed knowledge dates.** Where a source has no filing date (yfinance fundamentals, some Chinese data, macro without vintages), the app assumes publication 45–120 days after period end and flags it. The real release may have been earlier or later.
- **Macro revisions.** Where ALFRED vintages exist, the series explorer shows "first release" against "latest". A GDP or payrolls figure in a chart is usually the latest revised value, not what the market saw on the day.
- **13F** positions are as of the quarter end but known only after the filing deadline (up to 45 days later). **CFTC** positions are as of Tuesday but released on Friday. **Form 4** trades count from the filing date.
- Snapshot-only fields (`pe_fwd` , `grade` , `beta_1y` , ...) have no history; BQL refuses them for past dates.

Small samples

Small samples make estimates noisy and significance tests fragile. Watch for:

- **Peer groups** that widen because fewer than 8 peers exist at the subsector × region × size level; the percentile is then against a broader, less comparable group.
- **Short histories:** news sentiment from about August 2026; GDELT event history still being backfilled; 13F history from 2023 Q1 (the smart cohort needs 4 quarters of track record); short-interest and smart-money context from 2023; insider context from 2015–16. The smart-money score's IC test used only 33 non-overlapping dates.
- **Moat** needs about 7 years of history; with fewer it shows "Insufficient data".
- **Regime-conditional statistics** (regime performance, event-study cells, CFTC extreme event studies) often rest on a handful of episodes. Always read the n and the t-statistic shown next to the average.
- **Conformal audit segments** with n < 200 are marked thin; the coverage flags ignore segments below n = 2,000.
- **Your own studies.** Model Studio warns on minimum sample sizes; respect the warning.

Multiple testing when screening

Screening 20,000 securities with several filters, or trying many strategy variants, guarantees that some results look excellent by chance.

- **Expect false positives.** With 100 independent tests at the 5% level, about 5 are "significant" with no effect at all.
- **The validations corrected for this** (Holm per factor, Benjamini–Hochberg across 105 factor cells, PBO and DSR across strategy configurations; see Multiple testing). Your own screens are not corrected.
- **In the Backtest Lab,** look at DSR (adjusted for the number of trials) and PBO, not only the Sharpe ratio. PBO near 0.5 means your chosen configuration is no better than a random pick among those you tried. Count every variant you tried, including the ones you discarded.
- **Decide before you look.** Fix the universe, filters, horizon and holding rule before running; keep a hold-out period you do not touch until the end.
- **Extreme values are often data errors.** A stock at the top of a momentum or valuation screen deserves a check for ticker reuse, unit flips and missing sessions (see the Data Health gap queue, filter **Manual**).

Licensing

The app was built for **internal research use by one analyst**. `docs/it/DATA_LICENSING.md` is an engineering assessment, not legal advice; BiG Legal / Compliance and the market-data team must confirm each source before the app is used more widely.

- **No external redistribution** of data is assumed to be allowed for any source. Do not paste raw data tables from the app into client material.
- **Yahoo Finance (yfinance)** supplies non-US prices, benchmarks, futures curves, some fundamentals and earnings dates. It is **not licensed for commercial use** (personal use only, no redistribution). In a bank deployment these jobs must be replaced by a licensed vendor or the bank's market-data service.
- **Polygon (Massive)** US prices and news are plan-dependent: internal multi-user display needs a business plan and exchange agreements.
- **ICE BofA OAS** series obtained through FRED are third-party copyrighted; the local archive must be deleted or licensed before bank use.
- AKShare, BaoStock, Google News RSS and mindicador.cl are marked "switch off" for bank use; exchange-website data (TDnet, HKEX, KRX, JPX, ...) is mostly under personal-use terms.
- Official sources (SEC, ECB, Eurostat, OECD, IMF, World Bank, BIS, national statistics offices, CFTC, short-sale registers), GDELT and Natural Earth are usable **with attribution**. The Caldara–Iacoviello GPR index is academic data: cite it and confirm before commercial use.
- Common Crawl news is stored as derived metadata only; publishers keep the copyright of headlines and descriptions.
- Sentiment models: confirm the FinBERT and XLM-R model licences before bank use.

In practice: quote derived figures (a ratio, a percentile, a model output) with attribution of the underlying source; do not export or forward raw price or news data.

Coverage gaps by region

Coverage is deepest in the US and thinner elsewhere. Check the coverage before comparing across regions.

Area	Situation (October 2026)
Fundamentals	5,022 of 24,961 securities have fundamentals. US is deepest (SEC). Japan (EDINET / TDnet), the UK and LATAM are good; the EU comes from ESEF and national filing systems where available, else yfinance. Korea is thin without the OpenDART key (not configured); China, Taiwan and India are thin or absent. Screens of the `full` universe are mostly price-only outside the US, Japan and Europe.
Prices	All regions, but regions update on different dates (on 23 Oct 2026: US 16 Oct, EU and UK 19 Oct, JP 13 Oct). 424 Shanghai names lack the BaoStock feed; Korean full-market prices need the KRX key.
Technicals models	Validated on US, EU, UK and JP only; elsewhere "region not validated" or experimental.
Macro models	Nowcasts only for US, euro area, UK and Japan; regional regime models are not available for every region.
Insiders	US, UK, Germany, Netherlands, Hong Kong and Brazil.
Ownership	13F covers US-listed holdings of US-reporting managers; about 40% of recent 13F rows have no mapped security.
Short positions	FINRA (US) and EU / UK registers; Germany and Italy are not available.
Consensus estimates	Not available anywhere (no free source); expectations are model-based.

A missing value is shown as "—" or as a neutral 50 in scores. A neutral score from missing data is not the same as an average company: check each factor's coverage.

ADRs versus home listings

- For companies with an ADR and a home listing (for example TM and 7203.T Toyota, PBR and PETR4.SA), **the composite score, technicals and fair value are computed on the home listing by design**. The ADR page shows a banner with **See home listing →**.
- Prices of the ADR and the home share differ by the ADR ratio and the exchange rate; returns differ by FX moves and by trading-hour mismatches.
- Valuation ratios are computed USD-on-USD (market cap = shares × close × FX of the day, fundamentals in USD), which removes the currency mismatch between an ADR's trading currency and the company's reporting currency.
- Technical levels, setups and conformal bands of the ADR and the home listing are not interchangeable: they come from different price series, sessions and calibration pools.

FX conversions

- **Daily rates** are stored as USD per unit. Precedence: ECB reference rates > Fed H.10 > yfinance; gaps are forward-filled for up to 5 days (flagged as filled).
- **Balance-sheet items** convert at the period-end rate; **income and cash-flow items** at the period-average rate; **TTM in USD** is the sum of the four quarters in USD, not the local TTM times one rate. A USD revenue growth rate therefore mixes business growth with currency moves; use the reporting-currency view for operating trends.
- **Prices** are kept in the trading currency; GBp and ZAc quotes are divided to GBP and ZAR. Residual 100× unit flips on some listings are tracked in Data Health (audit F32).
- **Known FX gaps:** TWD comes only from the weekly Fed H.10 and can lag up to 7 days; BGN ends on 2026-01-05 (euro adoption).
- BQL `with(currency=...)` converts at the rate of each date; `currency=local` keeps the trading currency.

Stale data

The default refresh profile is **weekly** (Saturday 08:00 Lisbon). Between refreshes the data can be up to a week old by design, and under that profile a dataset counts as fresh up to 9 days behind its expected date.

- Look at the header: **DATA AS OF** (hover for the date per region), the **Data stale** and **Scheduler off** chips, and the live-layer strip ("NOT LIVE · LAST HEARTBEAT …" when the real-time layer is stopped).
- Regions sit on different dates. A cross-region comparison of one-day returns may compare different sessions.
- Many panels show their own as-of date (macro observations, yield curves, CFTC report and release dates, 13F quarter and filing date). Quote the panel's date, not the header's.
- A snapshot build time (System › Backend panel) is not a data date. A freshly built snapshot can contain old data.
- When in doubt, open Data Health and check the dataset's status, data as of and Lag / SLA.

Checklist for an investment decision

Before a number from the app goes into an investment decision or a recommendation:

1. **Freshness.** Header data as-of for the relevant region; Data Health status of the datasets you rely on; no red alert on the System page.
2. **Coverage.** Does the company have fundamentals from an official filing (not only yfinance)? Are the factor coverages adequate? Is it an ADR whose analytics live on the home listing?
3. **Data sanity.** Any outlier checked against the source (filing, exchange) and against the Data Health gap queue (ticker reuse, unit flips, missing sessions)?
4. **Model status.** Every model output you use is **validated** for this use (see the gate table); nothing experimental, nothing from an unvalidated region.
5. **Wording of probabilities.** Stated as "calibrated over history", with the horizon and the pool, never as "X% chance".
6. **Labels.** Model-based expectations and surprises are not consensus; approximations are named as such.
7. **Point in time.** Historical comparisons use as-first-reported data where it matters; backtests start after 2026-10-10 or are flagged as survivorship-biased.
8. **Multiple testing.** If the idea came from a screen or a backtest, how many variants were tried? DSR and PBO checked?
9. **Independent evidence.** The thesis rests on more than one model output and on fundamentals you have read in the filing.
10. **Licensing.** No raw third-party data leaves the bank; sources attributed.
11. **Record.** Snapshot id, data as-of date and the BQL query or page used are written down (see below), and a thesis is opened in the Thesis Tracker so the KPIs are monitored.

Citing the app's numbers in research notes

A citation should let a colleague reproduce the number. Include:

- the tool and its status: "BiG Equity 360 (internal research tool)";
- the **serving snapshot id** and the **data as-of date** (System › Overview › Backend panel, and the header; per region if relevant);
- the measure, with its horizon, universe or peer group, and currency;

- the **label**: model-based, approximation, calibrated over history, and the validation status with the document section;
- the underlying data source (SEC 10-K, ECB, FRED, ...), which carries the attribution;
- where possible, the BQL query with `as_of=` that returns the number.

Templates

```
Source: BiG Equity 360 (internal research tool), snapshot R20261022T2056130T13, data as of 19 Oct 2026 (US prices 16 Oct 2026). Underlying data: SEC EDGAR filings; prices from the app's price lake.
```

```
"The 20-session calibrated range is 172–205 (90%; model-based, calibrated over 2010–2026 walk-forward, 89.6% realised coverage; docs/CONFORMAL_VALIDATION.md A.2). This is a historical calibration, not a probability for this stock."
```

```
"US CPI surprise +0.2 pp vs the app's model-based expectation (not consensus)."
```

```
"Implied Fed path: about two 25 bp cuts priced over 12 months (approximation from official Treasury curves, not futures/OIS-implied probabilities)."
```

(The values in the last three templates are illustrative; replace them with the numbers you are citing.)

Do not cite experimental outputs, the composite score as a forecast, or any probability as "the chance that ...". When a figure comes from Model Studio or the Backtest Lab, give the run id and state the number of variants tried.

18. Troubleshooting and FAQ

This chapter lists the problems users and the app owner are most likely to meet, what causes them and how to fix them, followed by answers to frequent questions. Inside the app, the **?** icon next to each page title and the **Guide** menu item open this guide at the relevant section. Most problems are about **freshness** (the data is older than expected) or about **the background app not running**, not about wrong numbers.

Commands in this chapter are run in PowerShell from the `backend` folder of the project or install folder. `$py` stands for the app's Python:

```
cd "C:\Users\Asus\Documents\STOCK SCANNER\backend"
$py = "..\.venv\Scripts\python.exe"     # project folder (dev tree)
# $py = "..\python\python.exe"          # installed or portable build
```

Quick checks

Before anything else, these four checks locate most problems in under a minute.

Check	How	Healthy
Is the app running?	Look for the BiG icon in the Windows notification area; right-click › **Status**	"Supervisor: running", a *Data as of* date, "scheduler on"
Does the API answer?	Open `http://127.0.0.1:8010/health` in the browser	`{"status":"ok", ...}`
How old is the data?	Top bar **DATA AS OF** (hover for per-region dates) and the freshness dot	Within a week for the weekly profile
Are background jobs healthy?	**System › Data health**	No red datasets, gap queue draining, no "error" banners

Troubleshooting

The app does not open

Symptoms. You double-click `BiG360.exe` and no browser window appears, or the browser shows "This site can't be reached" at `127.0.0.1:8010`.

Causes and fixes.

1. **It is still starting.** The first start after a reboot takes 10 to 60 seconds; the notification *Starting... the dashboard opens at http://127.0.0.1:8010/* appears first. Wait, then double-click the tray icon.
2. **It is already running.** A second start of `BiG360.exe` does not start a second copy; it only opens the dashboard. If nothing opens, double-click the tray icon or go to `http://127.0.0.1:8010/` directly.
3. **The supervisor is not running.** Tray › **Status** shows "Supervisor: not running" or "Dashboard API ... not reachable". Quit from the tray and start `BiG360.exe` again. The tray also restarts the supervisor by itself if it dies.
4. **The API keeps crashing.** The supervisor restarts it with back-off (5 s, 15 s, 60 s, then 5 minutes), so after several crashes it can take minutes to come back. Check `data\logs\app\api.out.log` for the traceback (often an out-of-memory error during a heavy refresh) and send it to IT (see Logs to send to IT).
5. **Another program uses port 8010.** See Port 8010 is busy.

For IT: `& $py -m app.ops.supervisor --status` lists the children (API, worker, filler, live news) with their state, process id, restarts and memory; `& $py -m app.desktop.tray --status` prints the tray's status lines and menu.

Port 8010 is busy

Symptoms. The dashboard does not load, and the supervisor log says *api already running outside the supervisor*; or the System page reports the API as "external".

Cause. Something else listens on 8010, usually a development API started by hand.

Fix.

1. Find the process: `Get-NetTCPConnection -LocalPort 8010 | Select-Object OwningProcess`, then `Get-Process -Id <pid>`.
2. Stop it if it is a leftover development API, or quit BiG360 from the tray.
3. If 8010 must stay with another program, IT changes `api_port` in `config\app.yaml` (or sets `BIG360_SUPERVISOR_API_PORT`) and restarts the app. The dashboard address changes accordingly.

The supervisor re-checks an "external" API every minute and takes over when the port is free.

The dashboard is blank or shows "Page not found"

Symptom	Cause	Fix
White or dark page with nothing on it	The browser holds an old copy of the dashboard after an update, or the API is still warming up	Reload with `Ctrl F5`. Check `http://127.0.0.1:8010/health`.
"Page not found" with a **Back to Macro** button	The address does not match a page (for example a mistyped route or an old bookmark)	Use the menu or `Ctrl K`
"Something went wrong" with a **Reload** button	A page crashed in the browser	Click **Reload**. If it repeats, note the message and the address and send them to IT.
`/` shows a JSON error while `/health` works	The dashboard is not being served: either the API was started in development mode on the same port, or the built dashboard (`frontend\dist_app`) is missing	See Dev mode and the app on the same port
Pages wait a long time, especially with several app tabs open	Each app tab holds a live connection; browsers allow only 6 connections per site	Close unused app tabs. A tab hidden for more than 60 seconds releases its connection automatically.

The top bar shows "Data stale"

What it means. At least one market section of the published snapshot (prices of a region, FX, macro, filings, insider, events, news or CFTC) is behind, or core data is older than the profile's service level (9 days for weekly, 3 for daily; 6 days for prices). The numbers are correct as of their date, but newer data exists. See Data-as-of and freshness indicators.

What to do.

1. Hover **DATA AS OF** to see which region is behind, and hover the freshness dot for the list of stale sections (for example `prices:JP (stale)`).
2. Click the **Data stale** chip to open **Data health**. Find the dataset in the summary, the freshness heatmap and the gap queue:
   - if the gap filler is already *filling* it, wait;
   - if it is *failed* or *manual*, read the error under **SLA breaches & recent failures**.
3. If you need fresh data now, use tray › **Run refresh now**. A full refresh takes one to several hours; the chip clears when the new snapshot is published.
4. For IT: open the failing job's log (`data\logs\jobs\<job>\<run>.log`), fix the cause, then resume the refresh with `& $py -m app.cli jobs run-dag nightly --resume`. Prices are published even when an analytics job fails.

Related chips: **Analytics stale** / **Analytics failed** mean the market data is fine but scores, valuation, technicals or models are behind; **Prices failed** means a price section failed and is shown in red.

The refresh failed or is overdue

Symptoms. A Windows notification *BiG360: refresh weekly_refresh failed* or *partial*; a banner "Last nightly run failed" or "Refresh overdue" on Data health; **DATA AS OF** has not moved for more than a week.

Cause	Fix
The PC was asleep or off at Saturday 08:00 and did not come back within the catch-up window (6 days for weekly, 2 for daily)	Run tray › **Run refresh now**. Allow wake timers in the Windows power plan (*Power Options › Advanced › Sleep › Allow wake timers: Enable*) and leave the PC plugged in on Saturday morning.
One job failed and the run is marked *failed* ("1 of 102 jobs failed")	Usually harmless: the rest was published and the failed job retries on the next run. Check **SLA breaches & recent failures** on Data health for the job and its error.
A job ran out of memory (`MemoryError` , "Orphaned (process died)")	Avoid running other heavy work during the Saturday refresh. IT can re-run the job: `& $py -m app.cli jobs run <job_id>` , or resume the whole refresh with `jobs run-dag nightly --resume` .
A job timed out ("timed out after 60.00 min")	Slow source or rate limit. It is cancelled and retried automatically; if it keeps happening IT raises its `timeout_min` in `contracts/jobs.yaml` .
The scheduler is off	See "Scheduler off" or "Worker off".

"Refresh overdue" is raised when the last completed nightly refresh is older than the profile period plus one day (8 days for weekly).

"Scheduler off" or "Worker off"

Chip	Meaning	Fix
Worker off (red)	The worker process is not running: no scheduled refresh and no queued job will run	Quit and restart BiG360 from the tray. IT: `& $py -m app.ops.supervisor --status` .
Scheduler off (amber, red after 24 hours)	The worker runs but without its schedule, typically because it was started with `--no-schedule` or `schedule: false` in `config\app.yaml`	Restart BiG360 normally (the tray app starts the scheduler unless `schedule: false`). The System page **Scheduler** panel shows the reason.

The gap filler is paused

Symptoms. Data health shows the gap filler as *paused*, or the tray **Status** shows "Gap filler: paused (...)" with reasons.

Reason shown	Cause	Fix
manual pause	You or someone used tray › **Pause background jobs** (it creates `data\state\ops\filler.PAUSE`)	Tray › **Resume background jobs**
on battery	The laptop runs on battery	Plug in
metered network	Phone hotspot or a connection marked as metered	Use a normal network, or accept the pause
low free RAM	Less than 3 GB of free memory	Close heavy programs; it resumes above 3.5 GB
low disk	Less than 15 GB free on the data drive	Free disk space (a warning starts at 25 GB)
refresh running	A nightly, weekly or monthly refresh is running	Nothing: it resumes after the refresh

The filler is a background repair service; pausing it never affects the scheduled refreshes or the dashboard.

A panel says "Could not load data"

What it means. The panel's request to the API failed. The message under the title gives the reason (for example a timeout or an HTTP error).

What to do.

1. Click **Retry**.
2. If several panels fail at once, the API is probably restarting: check `http://127.0.0.1:8010/health` and wait a minute.
3. If one panel fails consistently for one company or one setting, note the page address and the message, and send them to IT with the time. The API log (`data\logs\api.log`) has the full error under the same request.

A panel is empty or says "pipeline pending"

Message	Meaning	What to do
"No data" (with a reason)	The query worked but there is nothing for this selection: for example no insider trades for a Japanese company in the window, or a filter that matches nothing	Widen the filter, the date range or the region
"Not yet available — pipeline pending"	The page exists, but the job that produces it has not run yet (for example a model that is still training, or a weekly job before its first run)	Wait for the next refresh; check Data health for the job
"Partial data — some inputs for this view are missing or stale"	The figures are what the pipeline published, but some inputs are missing	Use with care; check the per-panel date
"No regime model for {region}" and similar	No model exists for that scope, deliberately	Use a region that has the model, or the global view

Empty panels outside the US, Japan and Europe are often a **coverage** matter, not a fault: see Coverage by region and known gaps.

A company is not found

1. Search with `Ctrl K` by ticker (`7203.T` or `7203`), name (`Toyota`) or ISIN. Names in Latin script work for Asian companies; the local name is shown as secondary.
2. Try the home ticker with its exchange suffix (`SAP.DE` , `ASML.AS` , `PETR4.SA` , `0700.HK`) or the Bloomberg form (`SAP GY` , `7203 JT`).
3. If the company still does not appear, it is not in the app's universe: smaller companies (US below $300M market cap or $5 share price, European issuers below €300M) and markets without an exchange list (for example India beyond the curated names) are not covered.
4. Opening an unknown id directly (`/company/XYZ`) shows "Could not load data" with an "unknown security_id" message.

To request an addition, ask the app owner; additions go through the universe seeds and the monthly universe rebalance.

ADR pages show no scores or technicals

For an American Depositary Receipt (for example TM for Toyota, PBR for Petrobras, ASML on Nasdaq), Company 360 shows an **ADR** badge with the note *"Composite, technicals and fair value are computed on the home listing by design."* and a link **See home listing → 7203.T**. Follow the link: the home listing carries the scores, technicals and valuation. If the note says *"Home listing not found in the universe."*, the home listing is not covered and the ADR page shows only what is available for the ADR itself. Market cap on ADR pages uses a checked ADR ratio. See Company 360.

Figure 18.1: Header of the Toyota Motor ADR page (`/company/TM`): the ADR note, the link "See home listing → 7203.T (Tokyo Stock Exchange)" and an empty composite gauge.

The first page load after a refresh is slow

After a new snapshot is published (or after the API restarts), the API warms its caches by replaying the main pages' requests in the background, one at a time, for up to about 3 minutes. During that time the first visit to a page can take a few seconds; after it, pages load in under a second. If a page stays slow, close other app tabs (see The dashboard is blank or shows "Page not found") and check whether a refresh is running (tray › Status).

A "Mock data" badge appears

MOCK DATA in the top bar means the dashboard runs on synthetic data: it was started in development mode with `VITE_USE_MOCK=true` . **None of the figures are real.** The Windows app never uses mock data. Close that tab and open `http://127.0.0.1:8010/` .

MOCK FALLBACK · n means a development build replaced n failing endpoints with synthetic data. Hover the badge for the list. Again, use the Windows app for real work.

Dev mode and the app on the same port

The Windows app serves the dashboard and the API together on 8010. A development setup runs the API separately (often also on 8010) and the dashboard on `http://localhost:5173`.

Situation	Effect	Fix
A development API runs on 8010, then BiG360.exe starts	The supervisor treats the API as "external" and the dashboard is **not** served at `/`	Stop the development API (close its window or `Ctrl C`), or run it on another port (`--port 8000` and matching `VITE_API_URL` in `frontend\.env`)
Development dashboard on 5173 pointing at the app's API on 8010	Works, but you see the development build of the dashboard	Use `http://127.0.0.1:8010/` for the released build

The live strip says "Not live"

"NOT LIVE · LAST HEARTBEAT … AGO · LAST ITEMS RECEIVED:" means the realtime service that collects live filings, news and macro releases is stopped; the items shown are the last ones received. The rest of the app is not affected. Turn it on with tray › **Live news on/off**; the strip turns green (**LIVE**) within a minute. On a build without the live news service, IT enables the realtime layer with `realtime: true` in `config\app.yaml`.

Notifications

Question	Answer
Which notifications does the app send?	Windows notifications from the tray for new operational alerts (refresh failed or overdue, data stale beyond the service level, scheduler off, worker down) and when a scheduled refresh finishes. Inside the browser, toasts for alerts fired by your alert rules.
I do not see any	Check that Windows *Focus assist* / *Do not disturb* is off and that notifications are allowed for the app. IT may have set `notifications: false` in `config\app.yaml`.
I get too many	Ask IT to set `notify_severities: [error]` in `config\app.yaml` to keep only errors, or `notifications: false`. News alert chimes are switched in News Terminal › Alerts › Alert sound.

"Update available" in the tray

When IT places a newer build in the update folder (`update_dir` in `config\app.yaml`), the tray menu shows **Update available: x.y.z** and a notification appears (checked once a day). Click it to open the folder, then run the installer. The installer stops BiG360 gracefully, replaces the program files and **keeps your data** and keys. IT backs up `data\db\app_state.duckdb` (your watchlists, theses and alerts) before upgrading.

A data source reports "missing key"

A job whose API key is not installed skips that source and Data health shows it as *idle* or "missing key". Keys are optional except FRED and Polygon. To add one: `& $py -m app.desktop.credentials set <key name>` (input is hidden), then restart BiG360. See First run for key names.

Corporate network errors (for IT)

Symptom	Fix
SSL / certificate errors on every source	The bank's TLS inspection root is not trusted by Python: set `SSL_CERT_FILE` to a PEM bundle with the bank roots, and `HTTPS_PROXY`
SEC returns 403	Set `SEC_USER_AGENT="BiG Equity 360 <team mailbox>"`
Polygon returns 429	Free tier limit of 5 calls per minute; do not run a backfill and the refresh in parallel

Logs and diagnostics

Logs to send to IT

All logs are under `data\logs\` (tray › **Open logs folder**). They are JSON lines with a timestamp, level and request or job id. Secrets are redacted automatically.

File	Contains	Send it when
`supervisor.log`	Starts, exits, restarts and health checks of every part	The app does not start or keeps restarting
`app\supervisor.json`	Live status: each child, process id, restarts, memory	Same
`app\api.out.log`, `app\worker.out.log`, `app\filler.out.log`	Raw output of each part; crash tracebacks land here	Anything crashed
`api.log`	API errors and the access log (every request with its id and duration)	A panel fails or is slow
`worker.log`	Scheduler and refresh runs	A refresh did not run or failed
`jobs\<job>\<run>.log`	One file per job run (path shown on Data health and the System page)	A specific job failed
`filler.log`, `filler_tasks.log`	Gap filler decisions and governor state	The gap filler misbehaves
`tray.log`	Tray app	The tray icon or menu misbehaves

Include the time of the problem, the page address, and the text of any error message.

Command-line checks (doctor, jobs status, dq summary)

For the app owner and IT, run from `backend\`:

```
& $py -m app.cli doctor --offline        # keys PRESENT / MISSING (names only), disk, packages
& $py -m app.cli doctor                  # plus reachability of each data source
& $py -m app.cli jobs status             # the 30 most recent job runs (--job <id>, --status failed, --limit n)
& $py -m app.cli dq summary              # data-quality status of the last check run
& $py -m app.ops.supervisor --status     # supervisor children and their state
& $py -m app.desktop.credentials status  # which keys are stored (yes / no only)
```

Other useful commands (IT): `jobs list` (every job and its last status), `jobs run-dag nightly --resume` (resume a failed refresh), `db build-serving` (rebuild and publish the snapshot), `python -m app.ops.gaps` (run the gap detector now), `python -m app.qa` (regression guard). The HTTP probes are `/health` (liveness), `/ready` (200 when a snapshot is published and the worker is alive, otherwise 503 with the failing checks) and `/metrics`. The full runbook is `docs/it/RUNBOOK.md`.

FAQ

Why are there no consensus estimates?

There is no free, licensable source of analyst consensus. Wherever an expectation is needed, the app uses its **own model** and labels it: earnings expectations come from a seasonal time-series model on the company's own reported history (with an 80% band), and macro expectations from the app's macro models. Surprises are therefore "vs our model", not "vs consensus". `EE` in the command bar opens the forecast tab with the note "model-based, not consensus".

Why is there no options data?

The app has no options source, so there are no implied volatilities, option-implied moves or option monitors. `OMON` is listed in the command bar as unavailable. Where a "normal move" is needed (for example around earnings), the app compares the move with the stock's usual 3-day volatility instead of an option-implied move.

Why do numbers differ from Bloomberg?

Several deliberate differences, in rough order of frequency:

Area	BiG Equity 360	Typical terminal convention
Date	Data as of the last refresh (weekly by default), and regions can differ by a few days	Live or previous close
Prices	Adjusted close (splits and dividends); non-US prices from Yahoo	Exchange feed, often unadjusted by default
Fundamentals	Point-in-time from filings, latest period known at the date, TTM preferred	Vendor-standardised, possibly restated history
Valuation ratios	Computed in USD on both sides, market cap = point-in-time shares × close × FX	Local-currency conventions
Forward ratios (`pe_fwd`)	The app's base-case forecast (model-based)	Consensus (BEst)
Fair value	Blend of DCF and peer multiples, partly anchored to the market price	Analyst target prices
Sectors	The app's own GICS-like taxonomy, current classification only	GICS / BICS with history
Index membership	`'S&P 500'` / `'STOXX 600'` approximated by the app's universes	Official constituents
Policy-rate paths (`WIRP`)	Approximation from official curves	OIS / futures-implied probabilities
Macro surprises (`ECO`)	Release vs the app's model	Release vs survey median

If a difference cannot be explained by one of these, check the per-region date and the panel's source line, then report it with the ticker, field and date.

How often does the data update?

By default once a week, on **Saturday at 08:00 Lisbon time** (weekly profile): a full refresh of the latest sessions, then the weekly jobs, then the monthly jobs when due. CFTC positioning updates on Friday evening after its release, and live news (when switched on) every hour. The app owner can switch to the **daily** profile (Tuesday to Saturday at 01:30) from the tray or the System page. See Refresh schedule.

What does "model-based" mean?

A model-based number is estimated by the app's own statistical model from historical data. It is not an official statistic and not analyst consensus. Being model-based says nothing about quality by itself: each model has a validation status, and outputs that failed their gate are marked **experimental** and hidden by default. See Reading the labels and the gate summary in Methodology appendix.

Can I trust the probabilities?

Trust them for what they are: **frequencies calibrated over history**. A "37% chance the target is reached before the stop" means that, in walk-forward testing on past data, setups like this one reached the target first about 37% of the time, and that the model's stated probabilities matched realised frequencies. It does not mean "a 37% chance today", it does not account for news the model has not seen, and calibration was tested on liquid US, European, UK and Japanese names. Only the outputs that passed their gate are shown by default (for example the 20-session first-passage probabilities and the calibrated range fan); outputs that failed are hidden behind **show experimental**. Several validated outputs are described as having "no tradable edge" on their own: they describe risk, they do not predict returns.

Can I export data?

Yes, where a page offers it:

- **Screener: CSV** button (current filters and columns).
- **Company 360 › Financials:** CSV export of the statements.
- **BQL console:** CSV, XLSX and JSON exports of any query result.
- **Backtest Lab** and **Model Studio:** export of results.
- **Thesis Tracker:** export of a thesis.

For bulk extracts (for example into SQL Server or Power BI), IT can run the snapshot export (`python -m app.ops.export`). Remember the licensing caveats before sharing exported data.

Where are my watchlists and theses stored?

On this PC, in `data\db\app_state.duckdb` : watchlists, alert rules, theses and their journal, Model Studio studies, backtest definitions, scenarios and model promotions. This file cannot be rebuilt from the sources, so it should be backed up (IT procedure in `docs/it/RUNBOOK.md` §5.1). The watchlist and alert rules are also cached in your browser and resynchronised with the file when the dashboard loads. Display preferences (theme, region, locale, collapsed menu, experimental switch, recent commands, BQL history) live only in your browser.

Is any of my input sent outside the PC?

No. The dashboard talks only to the local API on `127.0.0.1` . The app contacts external sites only to download data (official sources, Polygon, Yahoo, GDELT and so on) and, on first use, the language-model weights for sentiment.

Can several people use the app, or run it on a server?

The current setup is single-user on one PC: the API listens only on `127.0.0.1` and there is no login. A department-server deployment is designed and partly implemented: Windows service, gateway (YARP or IIS) with corporate TLS, Entra ID / AD sign-in with viewer / analyst / admin roles, and shared state in SQL Server. A licence review of the data sources is required first. IT documentation: `docs/it/DEPLOYMENT_SERVER.md` , `docs/it/ARCHITECTURE.md` , `docs/it/SECURITY.md` , `docs/it/QUESTIONS_FOR_IT.md` , and the .NET integration guide `docs/it/INTEGRATION_DOTNET.md` .

Why does a backtest look better than I expected?

Universe membership history starts on 10 Oct 2026. Backtests over earlier periods use today's company lists, which exclude companies that later failed or were taken over (survivorship bias), so they flatter results. Treat them as an upper bound. See Survivorship and history depth and Backtest Lab.

Glossary

Terms and acronyms used across BiG Equity 360, in alphabetical order. Each entry links to the guide section that uses or explains it. Terms in *italics* inside a definition have their own entry.

A

- **ACI (adaptive conformal inference)** — An online correction of a *conformal* set: after each miss the working level is adjusted (`α_{t+1} = α_t + γ(α − err_t)` , γ = 0.005), so long-run coverage converges to target even when conditions change. Used for the next-bar and peer-group sets. See Conformal prediction.
- **ACM term premium** — The NY Fed's Adrian–Crump–Moench estimate of the extra yield investors demand for holding long Treasuries; used to adjust the US implied policy path. See Model-based versus official data.
- **Activism** — New Schedule 13D filings, 13D amendments and 13G → 13D switches (a passive holder becoming active). See Smart Money.
- **ADR (American depositary receipt)** — A US-listed certificate representing shares of a foreign company. Scores, technicals and fair value are computed on the home listing. See ADRs versus home listings.
- **ADV (average daily volume)** — Average traded value per day, usually over 20 sessions in USD; used for liquidity filters and the cost model. See Backtest Lab.
- **Approximation** — A label for a simplified method from official inputs, such as the central-bank implied path. See Model-based, experimental and approximation labels.
- **AR (abnormal return)** — A stock's return minus the return the *market model* predicts from the benchmark that day. See Event studies.
- **As of** — The date a value refers to (data as of) or the knowledge cut-off of a query (`as_of=` in BQL). See Point-in-time data.
- **ATR (average true range)** — The average daily high-low range including gaps, over 14 sessions (ATR14); a volatility unit for levels, zones and the baseline "2×ATR" band. See Technicals.

B

- **Backfill campaign** — A long, resumable history download run by the *gap filler* (GDELT history, 13F history, …), with progress on Data Health. See Data Health.
- **Benjamini–Hochberg (BH)** — A multiple-testing correction that controls the expected share of false discoveries. See Multiple testing.
- **Beneish M-score** — An accounting ratio model that screens for possible earnings manipulation; in the app a recalibrated, model-based screen, not an accounting finding. See Gate status.
- **BHAR (buy-and-hold abnormal return)** — The compounded return of the stock minus the compounded return the *market model* predicts over a window. See Event studies.
- **BOCPD (Bayesian online change-point detection)** — A method that flags when the level of a stream (attention, tone) has shifted. See Global Events.
- **BQL** — The app's Bloomberg-style query language (`get / for / with / where`), compiled to point-in-time SQL. See BQL and mnemonics.
- **Breakout (20 / 55)** — A close above the highest high (or below the lowest low) of the previous 20 or 55 sessions. See Technicals.
- **Brier score** — Mean squared error of probability forecasts; lower is better. See Calibration metrics.
- **Brownian baseline** — The textbook probability that a driftless random walk hits one barrier before another; a baseline the first-passage model must beat. See First-passage probabilities.

C

- **Calibrated band / calibrated range** — The conformal band around the price on the Technicals chart for a given horizon and level (80 / 90 / 95%), validated per horizon. See Conformal prediction.
- **Calibration** — Agreement between predicted probabilities and realised frequencies. "Calibrated over history" means on average over many past cases, not for this case today. See How to read probabilities.
- **CAR (cumulative abnormal return)** — The sum of daily *abnormal returns* over an event window such as [−1, +1]. See Event studies.
- **Catch-up** — A scheduled refresh that runs late because the PC was asleep or off at its scheduled time. See System overview.
- **CFTC COT (Commitments of Traders)** — The weekly CFTC report of futures positions by trader category, as of Tuesday and released on Friday. See Positioning.
- **COT index** — Where a trader category's net position sits within its own range, `(net − min) / (max − min)` from 0 to 1, over the last 52 reports (1 year) or 156 reports (3 years). See Positioning.
- **Composite score** — The 0–100 score with a grade A–E that combines the 10 factor scores (Quality, Profitability, Growth, Value, Balance sheet, Momentum, Insider, Sentiment, Macro sensitivity, Moat) under a weight profile. Descriptive; no formal gate. See Screener.
- **Confluence zone** — A price area where several technical levels cluster (a peak of a weighted density of levels), with entry, invalidation and target. See Technicals.
- **Conformal prediction** — A method that turns a forecast into a set with guaranteed long-run coverage, using the empirical quantile of past forecast errors. See Conformal prediction.
- **Context-free model** — The meta-labelling baseline that recalibrates T2 without any market or stock context. See Meta-labelling.
- **Coverage (conformal)** — The realised share of outcomes inside a set, compared with its target (for example 90%). See Conformal validation.
- **Coverage registry** — `contracts/coverage.yaml` : the list of datasets the app should hold, with cadence, entities and SLA. See Data Health.
- **Crowding** — The number of hedge-fund-like 13F filers holding a stock among their top-20 positions; model-based. See Smart Money.
- **CSCV (combinatorially symmetric cross-validation)** — The resampling scheme behind *PBO*. See Multiple testing.
- **Cycle clock** — A country's position in growth momentum × inflation momentum: recovery, expansion, slowdown or contraction. See Macro.

D

- **DAG** — A chain of dependent jobs run together (`nightly` , `weekly` , `monthly` , `cftc_weekly` , `hourly`). See System overview.
- **Data as of** — The latest market date in the serving snapshot, per region. See Stale data.
- **DCF (discounted cash flow)** — Valuation as the present value of projected free cash flow to the firm, discounted at *WACC*, plus a terminal value. See Company 360 fundamentals.
- **Decile monotonicity** — How consistently mean returns rise (or fall) from the bottom to the top decile of a signal, measured by a Spearman correlation. See Rank IC.
- **DQ (data quality)** — The result of the automatic data-quality checks on a load: ok, info, no data, warn or error. A blocking error stops publication. See System overview.
- **DSR (deflated Sharpe ratio)** — The probability that a strategy's Sharpe ratio beats the best Sharpe expected from the number of variants tried by chance alone. See Multiple testing.

E

- **E[R] (expected R)** — The model's expected outcome of a setup in *R* units, after costs, with unresolved trades marked to market. See First-passage probabilities.
- **ECE (expected calibration error)** — The weighted average gap between predicted probability and realised frequency across bins; gate ≤ 2.5 pp for the technicals models. See Calibration metrics.
- **Ellipsoid (conformal)** — A multi-dimensional conformal set shaped by a covariance matrix; minimum volume, but wide axis projections when coordinates are correlated. See Ellipsoids versus boxes.
- **Embargo** — A gap after a test block from which no training data is taken, to stop leakage through overlapping labels. See Purged and embargoed cross-validation.
- **ERP (equity risk premium)** — The extra return demanded for equities over the risk-free rate, used in the cost of equity; by default backed out per region from current prices (market-calibrated). See Company 360 fundamentals.
- **ESEF** — The EU's European Single Electronic Format for annual reports (tagged iXBRL); the main official source of EU fundamentals. See Data and coverage.
- **EWMA** — Exponentially weighted moving average; the app's daily volatility estimate uses λ = 0.94 (RiskMetrics). See Conformal prediction.
- **Experimental** — An output that failed its validation gate or was never validated; hidden unless **Show experimental outputs** is on, or available only in Model Studio / the Backtest Lab. See Labels.

F

- **Fan** — The calibrated band drawn for every horizon from 1 to 20 sessions on the Technicals chart (nested 80 / 90 / 95%). See Marginal versus joint coverage.
- **FinBERT** — The English financial-sentiment language model used for news; *XLM-R* covers other languages. See News & Sentiment.
- **First passage** — Which of two barriers (target or stop) the price reaches first within a horizon. See First-passage probabilities.
- **FT-Transformer** — A neural-network model in the Model Lab; it failed the label-permutation leakage test and is not in production. See Gate status.
- **FTD (fails to deliver)** — Shares not delivered on settlement date, published by the SEC; an "FTD spike" is a model-based flag for unusually large fails. See Smart Money.

G

- **Gap (data gap)** — Data the coverage registry expects but the lake lacks or holds in suspect form; classed missing, stale, partial, suspicious or unfillable. See Data Health.
- **Gap filler** — The low-priority background process that repairs gaps and runs backfill campaigns under a resource governor; paused with the tray menu (`filler.PAUSE`). See Data Health.
- **Gate (validation gate)** — A set of pass / fail criteria fixed before a model is evaluated (calibration, regimes, baselines, leakage, economics). See Gate status of every model.
- **GCN-GRU** — A graph convolutional recurrent neural network; the production filter of the peer-group conformal ellipsoids. See Conformal validation.
- **GDELT** — The Global Database of Events, Language and Tone: an open dataset of world news events and articles, updated every 15 minutes. See Global Events.
- **Golden data checks** — A small set of stable or bounded facts (Apple revenue, the S&P 500 level, US CPI, ...) checked through the app's API after each refresh. See QA guard.
- **Guide (Help page)** — This user guide inside the app at `/help` (menu **Guide**, command-bar `HELP`), with a chapter sidebar, full-text search and Print / PDF; the "?" icon in a page header opens the matching section.

See Getting started.

- **GPR (geopolitical risk index)** — The app's model-based index of geopolitical risk per country from GDELT, compared with the published Caldara–Iacoviello index. See Global Events.

H

- **hi52 (52-week-high proximity)** — Close divided by the highest close of the last 252 sessions; a validated "rank edge" factor. See Technicals.
- **HHI (Herfindahl–Hirschman index)** — The sum of squared shares; a concentration measure (holders, customers). See Smart Money.
- **HMM regime** — A 4-state hidden Markov model of the macro environment; model-based. See Macro.
- **Holm correction** — A multiple-testing adjustment of p-values that controls the chance of any false positive. See Multiple testing.

I

- **IC (information coefficient)** — The correlation between a signal and later returns across stocks; the app uses the rank (Spearman) version. See Rank IC.
- **Implied path** — The policy-rate path implied by official yield or OIS curves; an approximation, not futures-implied probabilities. See Macro.
- **IVOL (idiosyncratic volatility)** — The annualised standard deviation of a stock's returns not explained by its market and sector over 21 sessions; low IVOL is a validated "rank edge". See Technicals.

J

- **Joint coverage** — The share of times all coordinates of a multi-dimensional set (several horizons, or high / low / close) are inside it at once. See Marginal versus joint coverage.

K

- **Kelly fraction (quarter-Kelly)** — The bet size that maximises long-run growth given win probability and payoff; the meta-labelling size uses a capped quarter of it (experimental). See Meta-labelling.
- **Knowledge date** — The date a fact became publicly known (filing, release, publication); the basis of point-in-time queries. See Point-in-time data.

L

- **Lazy Prices** — A flag for large year-over-year changes in the text of a company's 10-K risk factors. See Company 360 fundamentals.
- **Level type** — A family of technical levels (pivots, period highs and lows, anchored VWAPs, volume nodes, ...), ranked per stock by how well it held historically. See Technicals.
- **Log-loss** — The negative mean log-probability assigned to what actually happened; punishes confident errors. See Calibration metrics.
- **Log-volume** — The normalised log of a conformal set's volume; favours ellipsoids. See Ellipsoids versus boxes.

M

- **Marginal coverage** — Coverage of one horizon (or coordinate) at a time, averaged over time and securities. See Marginal versus joint coverage.
- **Market model** — The regression `r_stock = α + β·r_market + ε` estimated before an event to define the expected return. See Event studies.
- **MAX** — The largest daily return of the last 21 sessions; stocks with high MAX tend to do worse (validated "rank edge", negative sign). See Technicals.
- **Meta-labelling** — A second model that estimates whether a proposed trade will succeed given its context, and may decide take / skip and size. See Meta-labelling.
- **Mnemonic** — A Bloomberg-style command-bar function (`DES` , `GP` , `ECO` , `WIRP` , ...). See BQL and mnemonics.
- **Moat** — A rating of durable competitive advantage (Wide, Narrow, None, Insufficient data) from evidence such as ROIC minus WACC and its persistence, margin stability and market share. See Company 360 fundamentals.
- **Model-based** — Our own estimate or transformation, not a published statistic or consensus. See Model-based versus official data.
- **mvp universes** — `mvp_us` (top 500 US companies by market cap) and `mvp_global` (curated global names); today's members, so history before 2026-10-10 is survivorship-biased. See Survivorship.

N

- **Newey–West t** — A t-statistic with standard errors robust to autocorrelation and heteroskedasticity, needed when horizons overlap. See Rank IC.
- **Next-bar set** — The conformal set for tomorrow's high, low and close together; failed on sharpness, experimental. See Conformal validation.
- **Nowcast** — A model estimate of current-quarter GDP growth updated as monthly data arrive (US, euro area, UK, Japan), shown next to official nowcasts such as GDPNow. See Macro.

O

- **OAS (option-adjusted spread)** — The yield spread of a bond index over Treasuries after adjusting for embedded options (ICE BofA indices). See Licensing.
- **Ops alert** — An operational warning (scheduler off, refresh overdue, data stale beyond SLA, regression guard failed, ...) shown on the System pages and as a Windows notification. See System overview.

P

- **PAV (pool-adjacent-violators)** — An algorithm that makes a sequence monotone with the least change; used to smooth the first-passage surface. See First-passage probabilities.
- **PBO (probability of backtest overfitting)** — How often the best in-sample strategy variant ranks below the median out of sample (via *CSCV*). See Multiple testing.
- **PEAD (post-earnings-announcement drift)** — The tendency of prices to keep moving in the direction of an earnings surprise; the app's PEAD signal is model-based. See Earnings & Events.
- **Peer group (conformal)** — A group of 20–50 related stocks (industry × region, or supply-chain / statistical ego groups) whose joint daily moves are tracked by the peer-group ellipsoid. See Conformal validation.
- **PIT (point in time)** — Using at each past date only what was known on that date. See Point-in-time data.

- **PSR (probabilistic Sharpe ratio)** — The probability that the true Sharpe ratio exceeds a benchmark, given track-record length, skewness and kurtosis. See Multiple testing.
- **Purging** — Removing training rows whose label window overlaps the test period. See Purged and embargoed cross-validation.

R

- **R (risk unit)** — The distance from entry to the invalidation (stop); outcomes are measured in multiples of it. See First-passage probabilities.
- **r-star (`r*`)** — The neutral real interest rate; the app shows official NY Fed estimates (Laubach–Williams, HLW). See Macro.
- **R:R (reward-to-risk)** — Distance to the target divided by distance to the stop. See Technicals.
- **Rank edge** — The label of a technical factor that passed its rank-IC gate but is not profitable stand-alone after costs. See Labels.
- **Rank IC** — The Spearman rank correlation between a signal and later returns across stocks, averaged over dates. See Rank IC.
- **Regime gap** — Realised minus predicted frequency within a regime (Covid 2020, VIX > 25, ...); gate $|\text{gap}| \le 5$ pp. See Calibration metrics.
- **Regression guard** — The QA suites run after each refresh (API contract sweep, golden data checks, UI smoke). See QA guard.
- **Reliability diagram** — A chart of predicted probability against realised frequency; a calibrated model lies on the diagonal. See Calibration metrics.
- **Risk composite (risk-on / risk-off gauge)** — A model-based average of z-scored cross-asset components (equity trend, VIX, credit, copper / gold, AUD/JPY, USD); above +0.5 risk-on, below −0.5 risk-off. See Cross-asset.

S

- **Schedule profile** — The refresh plan: Weekly (default), Daily or Intraday (placeholder). See System overview.
- **Shrinkage** — Pulling a thin estimate toward a broader parent estimate in proportion to sample size. See First-passage probabilities.
- **sigma_H (σ_H)** — The volatility over horizon H, `√V(H)` , from the EWMA daily variance plus an earnings jump term; distances to targets and stops are expressed in σ_H. See First-passage probabilities.
- **SLA (freshness SLA)** — The maximum lag a dataset may have before it counts as stale (9 days under the Weekly profile, 3 under Daily). See Data Health.
- **Smart cohort** — 13F hedge-fund-like filers whose past quarter-end books beat their peers; model-based. See Smart Money.
- **Smart-money score** — A model-based composite of 13F flows, smart-cohort flows, insider buying and short-interest signals; a descriptive screen with no significant IC. See Smart Money.
- **Snapshot (serving snapshot)** — The read-only database every page reads, rebuilt after each refresh and identified by a run id (for example `R20261022T2056130T13`). See Citing the app's numbers.
- **Split conformal** — Calibrating a set's radius as a quantile of scores on data not used to fit the forecast. See Split conformal.
- **Squeeze risk** — A model-based flag for high short interest (≥ 15% of shares, ≥ 5 days to cover) with a strong recent rise in a smaller company. See Smart Money.
- **Statistical neighbours** — Companies whose returns co-move strongly; labelled statistical, not disclosed relationships. See Company 360.
- **Surprise (macro)** — Actual release minus the model-based expectation, divided by the historical forecast-error standard deviation. See Macro.

- **Survivorship bias** — Flattering results from studying only companies that still exist today. See Survivorship.

T

- **T1** — The first target of a technical setup. See Technicals.
- **T2 / T3 / T4** — The Technicals v2 models: two-barrier first-passage probabilities (T2), zone hold given touch and its edge versus random (T3), and the calibrated technical score (T4). See Gate status.
- **Taylor rule** — A rule of thumb for the policy rate from the neutral rate, inflation's gap to target and the output gap; model-based with official inputs. See Macro.
- **Term premium** — The extra yield on long bonds over the expected path of short rates; the app uses the NY Fed ACM estimates. See Macro.
- **Touch probability** — The calibrated historical frequency of reaching a price level within h sessions. See Touch probabilities.
- **Triple barrier** — A label with an upper barrier (target), a lower barrier (stop) and a time limit. See Meta-labelling.

U

- **Unfillable** — A data gap the source cannot fill; recorded so nothing retries it. See Data Health.
- **Universe** — A named list of securities (`mvp_us` , `mvp_global` , `eu_broad` , `full` , watchlists, screens). See Data and coverage.

V

- **Valuation confidence** — A 0–100 measure of how reliable the fair value is; below 40 it is indicative only and excluded from scoring. See Company 360 fundamentals.
- **VIX** — The CBOE S&P 500 implied volatility index; VIX > 25 is one of the stress regimes in the validation gates. See Calibration metrics.
- **Volume confirmation** — A breakout on high relative volume with a close near the day's extreme; confirmed breakouts followed through more often (validated edge, not a tradable one). See Technicals.

W

- **w_proj** — The mean axis-projection width of a conformal set; the sharpness metric of the conformal gate. See Ellipsoids versus boxes.
- **WACC (weighted average cost of capital)** — The blended cost of equity (risk-free rate + beta × ERP) and after-tax debt used to discount free cash flow. See Company 360 fundamentals.
- **Walk-forward** — Repeated fitting on the past and testing on the next unseen period, rolling forward through time. See Purged and embargoed cross-validation.
- **WIRP** — The command-bar function for the central-bank monitor and implied path (`WIRP US`). See BQL and mnemonics.

X

- **XGBoost** — A gradient-boosted tree library; used for the production ML excess-return model and in meta-labelling. See Model Lab.
- **XLM-R** — A multilingual language model used for non-English news sentiment. See News Terminal.

Z

- **z-score** — A value's distance from a mean in standard deviations, for example against a company's own 5–10-year history or a country's 10-year history. See Screener.